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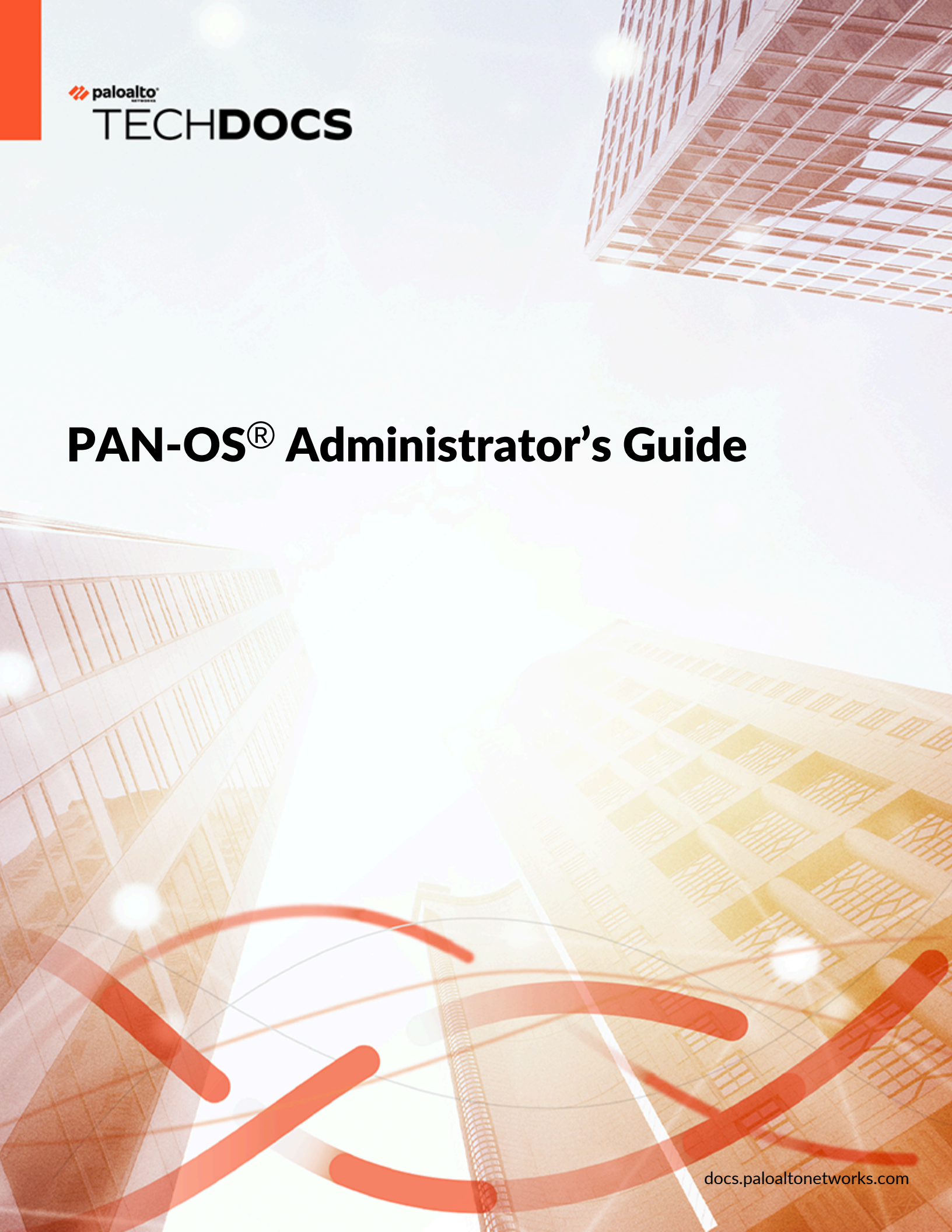
PAN-OS® Administrator's Guide



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Subscriptions

Learn about all the subscriptions and services that work with the firewall, and get started by activating subscription licenses:

- Subscriptions You Can Use With the Firewall
- Activate Subscription Licenses
- What Happens When Licenses Expire?
- Enhanced Application Logs for Palo Alto Networks Cloud Services

Subscriptions You Can Use With the Firewall

The following Palo Alto Networks subscriptions unlock certain firewall features or enable the firewall to leverage a Palo Alto Networks cloud-delivered service (or both). Here you can read more about each service or feature that requires a subscription to work with the firewall. To enable a subscription, you must first Activate Subscription Licenses; once active, most subscription services can use Dynamic Content Updates to provide new and updated functionality to the firewall.

Subscriptions You Can Use With the Firewall	
Strata Cloud Manager	Manage your Palo Alto Networks Next-Generation Firewalls (NGFW) from Strata Cloud Manager. This cloud-delivered, AI-powered security solution allows seamless management of your advanced ML-powered NGFWs, alongside Prisma Access deployments, through a single, streamlined user interface. Strata Cloud Manager has two licensing tiers: Strata Cloud Manager Essentials and Strata Cloud Manager Pro. This unified structure streamlines the deployment of network security offerings, including AIOps for NGFW, Autonomous Digital Experience Management (ADEM), cloud management functionality, and Strata Logging Service. • Get Started with Strata Cloud Manager • Strata Cloud Manager License • Cloud Management for NGFWs
IoT Security	The IoT Security solution works with next-generation firewalls to dynamically discover and maintain a real-time inventory of the IoT devices on your network. Through AI and machine-learning algorithms, the IoT Security solution achieves a high level of accuracy, even classifying IoT device types encountered for the first time. And because it's dynamic, your IoT device inventory is always up to date. IoT Security also provides the automatic generation of policy recommendations to control IoT device traffic, as well as the automatic creation of IoT device attributes for use in firewall policies. • Get Started with IoT Security.
SD-WAN	Provides intelligent and dynamic path selection on top of the industry-leading security that PAN-OS software already delivers. Managed by Panorama, the SD-WAN implementation includes: • Centralized configuration management • Automatic VPN topology creation • Traffic distribution

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Subscriptions You Can Use With the Firewall	
	• Monitoring and troubleshooting • Get Started with SD-WAN
Threat Prevention	Threat Prevention provides: • Antivirus, anti-spyware (command-and-control), and vulnerability protection. • Built-in external dynamic lists that you can use to secure your network against malicious hosts. • Ability to identify infected hosts that try to connect to malicious domains. • Get Started with Threat Prevention
Advanced Threat Prevention	In addition to all of the features included with Threat Prevention, the Advanced Threat Prevention subscription provides an inline cloud-based threat detection and prevention engine, leveraging deep learning models trained on high fidelity threat intelligence gathered by Palo Alto Networks, to defend your network from evasive and unknown command-and-control (C2) threats by inspecting all network traffic. • Get Started with Advanced Threat Prevention
DNS Security	Provides enhanced DNS sinkholing capabilities by querying DNS Security, an extensible cloud-based service capable of generating DNS signatures using advanced predictive analytics and machine learning. This service provides full access to the continuously expanding DNS-based threat intelligence produced by Palo Alto Networks. To set up DNS Security, you must first purchase and install a Threat Prevention license. • Get Started with DNS Security
Advanced DNS Security	In addition to all of the features included with DNS Security, the Advanced DNS Security subscription provides access to the Advanced DNS Security cloud, which operates cloud-based domain detection engines that inspect changes to DNS responses. This enables NGFWs to detect and categorize hijacked and misconfigured domains in real-time to block malicious activity. • Get Started with Advanced DNS Security
URL Filtering	Provides the ability to not only control web-access, but how users interact with online content based on dynamic URL categories. You can also prevent credential theft by controlling the sites to which users can submit their corporate credentials.

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Subscriptions You Can Use With the Firewall	
	To set up URL Filtering, you must purchase and install a subscription for the supported URL filtering database, PAN-DB. With PAN-DB, you can set up access to the PAN-DB public cloud or to the PAN-DB private cloud. *URL Filtering is no longer available as a standalone subscription. All URL Filtering features are included with the Advanced URL Filtering subscription.* • Get Started with URL Filtering
Advanced URL Filtering	Advanced URL Filtering uses a cloud-based ML-powered web security engine to perform ML-based inspection of web traffic in real-time. This reduces reliance on URL databases and out-of-band web crawling to detect and prevent advanced, file-less web-based attacks including targeted phishing, web-delivered malware and exploits, command-and-control, social engineering, and other types of web attacks. • Get Started with Advanced URL Filtering
WildFire	Although basic WildFire® support is included as part of the Threat Prevention license, the WildFire subscription service provides enhanced services for organizations that require immediate coverage for threats, frequent WildFire signature updates, advanced file type forwarding (APK, PDF, Microsoft Office, and Java Applet), as well as the ability to upload files using the WildFire API. A WildFire subscription is also required if your firewalls will be forwarding files to an on-premise WF-500 appliance. • Get Started with WildFire
Advanced WildFire	Advanced WildFire is a subscription offering that provides access to Intelligent Run-time Memory Analysis: a cloud-based advanced analysis engine that complements static and dynamic analysis, to detect and prevent evasive malware threats. By leveraging a cloud-based detection infrastructure, Intelligent Run-time Memory Analysis detection engines operate a wide array of detection mechanisms to target these highly-evasive malware. • Get Started with Advanced WildFire
AutoFocus	Provides a graphical analysis of firewall traffic logs and identifies potential risks to your network using threat intelligence from the AutoFocus portal. With an active license, you can also open an AutoFocus search based on logs recorded on the firewall.

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Subscriptions You Can Use With the Firewall	
	• Get Started with AutoFocus
Strata Logging Service	Provides cloud-based, centralized log storage and aggregation. The Strata Logging Service is required or highly-recommended to support several other cloud-delivered services, including Cortex XDR, IoT Security, and Prisma Access, and Traps management service. • Get Started with Strata Logging Service
GlobalProtect Gateway	Provides mobility solutions and/or large-scale VPN capabilities. By default, you can deploy GlobalProtect portals and gateways (without HIP checks) without a license. If you want to use advanced GlobalProtect features (HIP checks and related content updates, the GlobalProtect Mobile App, IPv6 connections, or a GlobalProtect Clientless VPN) you will need a GlobalProtect Gateway license for each gateway. • Get Started with GlobalProtect
Virtual Systems	This is a perpetual license, and is required to enable support for multiple virtual systems on PA-3200 Series firewalls. In addition, you must purchase a Virtual Systems license if you want to increase the number of virtual systems beyond the base number provided by default on PA-400 Series, PA-3400 Series, PA-5200 Series, PA-5400 Series, and PA-7000 Series firewalls (the base number varies by platform). The PA-220 and PA-800 Series firewalls do not support virtual systems. (PAN-OS 11.1.2 and earlier releases)The multiple virtual systems are not supported on VM- Series firewalls. (PAN-OS 11.1.3 and later releases)The multiple virtual systems are supported on VM-Series firewalls. • Get Started with Virtual Systems
Enterprise Data Loss Prevention (DLP)	Provides cloud-based protection against unauthorized access, misuse, extraction, and sharing of sensitive information. Enterprise DLP provides a single engine for accurate detection and consistent policy enforcement for sensitive data at rest and in motion using machine learning-based data classification, hundreds of data patterns using regular expressions or keywords, and data profiles using Boolean logic to scan for collective types of data. • Get Started with Enterprise DLP
SaaS Security Inline	The SaaS Security solution works with Strata Logging Service to discover all of the SaaS applications in use on your network. SaaS Security Inline can discover thousands of Shadow IT

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Subscriptions You Can Use With the Firewall	
	applications and their users and usage details. SaaS Security Inline also enforces SaaS policy rule recommendations seamlessly across your existing Palo Alto Networks firewalls. App-ID Cloud Engine (ACE) also requires SaaS Security Inline. • Get Started with SaaS Security Inline

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Activate Subscription Licenses

Follow these steps to activate a new license on the NGFW.

The Decryption Mirroring feature requires you to activate a free license to unlock feature functionality. To request the free license and activate this feature, follow the steps in Configure Decryption Port Mirroring.

STEP 1 | Locate the activation codes for the licenses you purchased.

When you purchased your subscriptions you should have received an email from Palo Alto Networks customer service listing the activation code associated with each subscription. If you cannot locate this email, contact Customer Support to obtain your activation codes before you proceed.

STEP 2 | Activate your Support license.

You will not be able to update your PAN-OS software if you do not have a valid Support license.

1. Log in to the web interface and then select **Device** > **Support**.
2. Click **Activate support using authorization code**.
3. Enter your **Authorization Code** and then click **OK**.

STEP 3 | Activate each license you purchased.

Select **Device** > **Licenses** and then activate your licenses and subscriptions in one of the following ways:

- **Retrieve license keys from license server**—Use this option if you activated your license on the Customer Support portal.
- **Activate feature using authorization code**—Use this option to enable purchased subscriptions using an authorization code for licenses that have not been previously activated on the support portal. When prompted, enter the **Authorization Code** and then click **OK**.
- **Manually upload license key**—Use this option if your firewall does not have connectivity to the Palo Alto Networks Customer Support Portal. In this case, you must download a license key file from the support site on an internet-connected computer and then upload to the firewall.

To automate activation using the Customer Support Portal API, see the process to Activate Licenses. This process works for both the hardware and VM-Series firewalls.

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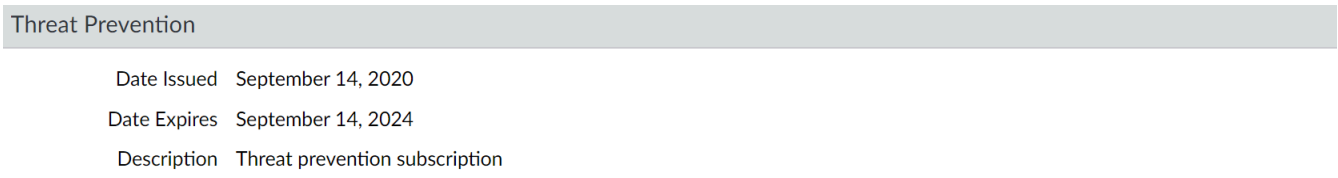
STEP 4 | Verify that the license is successfully activated

On the **Device** > **Licenses** page, verify that the license is successfully activated. For example, after activating the WildFire license, you should see that the license is valid:



STEP 5 | (WildFire, Advanced URL Filtering, and DNS Security subscriptions only) **Commit** configuration changes to complete subscription activation.

After activating a WildFire, Advanced URL Filtering, or DNS Security subscription license, a commit is required for the firewall to begin processing their corresponding traffic and data types based on the security profile configurations. You should:

- Commit any pending changes. If you do not have pending changes, which prevents you from committing any configuration updates, you can: issue a commit force command through the CLI or make an update that writes to the candidate configuration, which enables the commit option.

  Use the following CLI configuration mode command to initiate a commit force:

  ```
  username@hostname> configure
  Entering configuration mode
  [edit]
  username@hostname# commit force
  ```

  A commit force bypasses some of the validation checks that normally occur with a normal commit operation. Make sure your configuration is valid and is semantically and syntactically correct before issuing a commit force update.

- WildFire only Check that the WildFire Analysis profile rules include the advanced file types that are now supported with the WildFire subscription. If no change to any of the rules is required, make a minor edit to a rule description and perform a commit.

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What Happens When Licenses Expire?

Palo Alto Networks subscriptions provide the firewall with added functionality and/or access to a Palo Alto Networks cloud-delivered service. When a license is within 30 days of expiration, a warning message displays in the system log daily until the subscription is renewed or expires. Upon license expiration, some subscriptions continue to function in a limited capacity, and others stop operating completely. Here you can find out what happens when each subscription expires.

The precise moment of license expiry is at the beginning of the following day at 12:00 AM (GMT). For example, if your license is scheduled to end on 1/20 you will have functionality for the remainder of that day. At the start of the new day on 1/21 at 12:00 AM (GMT), the license will expire. All license-related functions operate on Greenwich Mean Time (GMT), regardless of the configured time zone on the firewall.

(Panorama license) If the support license expires, Panorama can still manage firewalls and collect logs, but software and content updates will be unavailable. The software and content versions on Panorama must be the same as or later than the versions on the managed firewalls, or else errors will occur. For details, see Panorama, Log Collector, Firewall, and WildFire Version Compatibility.

Subscription	Expiry Behavior
Advanced Threat Prevention / Threat Prevention	Alerts appear in the System Log indicating that the license has expired. **You can still:** • Use signatures that were installed at the time the license expired, unless you install a new Applications-only content update either manually or as part of an automatic schedule. If you do, the update will delete your existing threat signatures and you will no longer receive protection against them. • Use and modify Custom App-ID™ and threat signatures. **You can no longer:** • Install new signatures. • Roll signatures back to previous versions. • Detect and prevent unknown threats using real-time, ML-based detection engines provided by Advanced Threat Prevention.
DNS Security	**You can still:** • Use local DNS signatures if you have an active Threat Prevention license. **You can no longer:**

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Subscription	Expiry Behavior
	• Get new DNS signatures.
Advanced URL Filtering / URL Filtering	**You can still:** • Enforce policy using custom URL categories. **You can no longer:** • Get updates to cached PAN-DB categories. • Connect to the PAN-DB URL filtering database. • Get PAN-DB URL categories. • Analyze URL requests in real-time using advanced URL filtering.
WildFire	**You can still:** • Forward PEs for analysis. • Get signature updates every 24-48 hours if you have an active Threat Prevention subscription. **You can no longer:** • Get five-minute updates through the WildFire public and private clouds. • Forward advanced file types such as APKs, Flash files, PDFs, Microsoft Office files, Java Applets, Java files (.jar and .class), and HTTP/HTTPS email links contained in SMTP and POP3 email messages. • Use the WildFire API. • Use the WildFire appliance to host a WildFire private cloud or a WildFire hybrid cloud.
AutoFocus	**You can still:** • Use an external dynamic list with AutoFocus data for a grace period of three months. **You can no longer:** • Access the AutoFocus portal. • View the AutoFocus Intelligence Summary for Monitor log or ACC artifacts.
Strata Logging Service	**You can still:** • Access the stored data for a 30-day grace period provided for license renewal, after which the data is deleted. **You can no longer:**

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Subscription	Expiry Behavior
	• Ingest data to Strata Logging Service during 30 day grace period. • Forward logs to Strata Logging Service during 30-day grace period.
GlobalProtect	**You can still:** • Use the app for endpoints running Windows and macOS. • Configure single or multiple internal/external gateways. **You can no longer:** • Access the Linux OS app and mobile app for iOS, Android, Chrome OS, and Windows 10 UWP. • Use IPv6 for external gateways. • Run HIP checks. • Use Clientless VPN. • Enforce split tunneling based on destination domain, client process, and video streaming application.
VM-Series	See the VM-Series Deployment Guide.
Support	**You can no longer:** • Receive software updates. • Download VM images. • Benefit from technical support.

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Enhanced Application Logs for Palo Alto Networks Cloud Services

The firewall can collect data that increases visibility into network activity for Palo Alto Networks apps and services, like Cortex XDR, Device Security. These Enhanced Application logs are designed strictly for Palo Alto Networks apps and services to consume and process; you cannot view Enhanced Application logs on the firewall or Panorama. Only firewalls sending logs to the logging service can generate Enhanced Application logs.

Enhanced Application Logging can cause undesired logging behavior, such as the generation of URL Filtering logs for traffic to allowed categories. This can impact storage capacity and performance. Disable this feature to reduce undesired logs.

Follow these procedures to enable log forwarding for Enhanced Application logs for Cortex XDR and IoT Security:

- Cortex XDR
- IoT Security

Cortex XDR

The types of data that Enhanced Application logs gather include records of DNS queries, the HTTP header User Agent field that specifies the web browser or tool used to access a URL, and information about DHCP automatic IP address assignment. With DHCP information, for example, Cortex XDR™ can alert on unusual activity based on hostname instead of IP address. This allows the security analyst using Cortex XDR to meaningfully assess whether the user's activity is within the scope of their role, and if not, to more quickly take action to stop the activity.

To benefit from the most comprehensive set of Enhanced Application logs, enable User-ID. Deployments for the Windows-based User-ID agent and the PAN-OS integrated User-ID agent both collect some data that isn't reflected in the firewall User-ID logs but is useful for associating network activity with specific users.

To start forwarding Enhanced Application logs to Strata Logging Service, turn on Enhanced Application Logging globally, and then enable it on a per-security rule basis (using a Log Forwarding profile). The global setting is required and captures data for traffic that isn't session-based (ARP requests, for example). The per-security policy rule setting is strongly recommended; the majority of Enhanced Application logs are gathered from the session-based traffic that your Security policy rules enforce.

STEP 1 | Enhanced Application Logging requires a Strata Logging Service subscription. User-ID is also recommended. Here are steps to get started with Strata Logging Service and enable User-ID.

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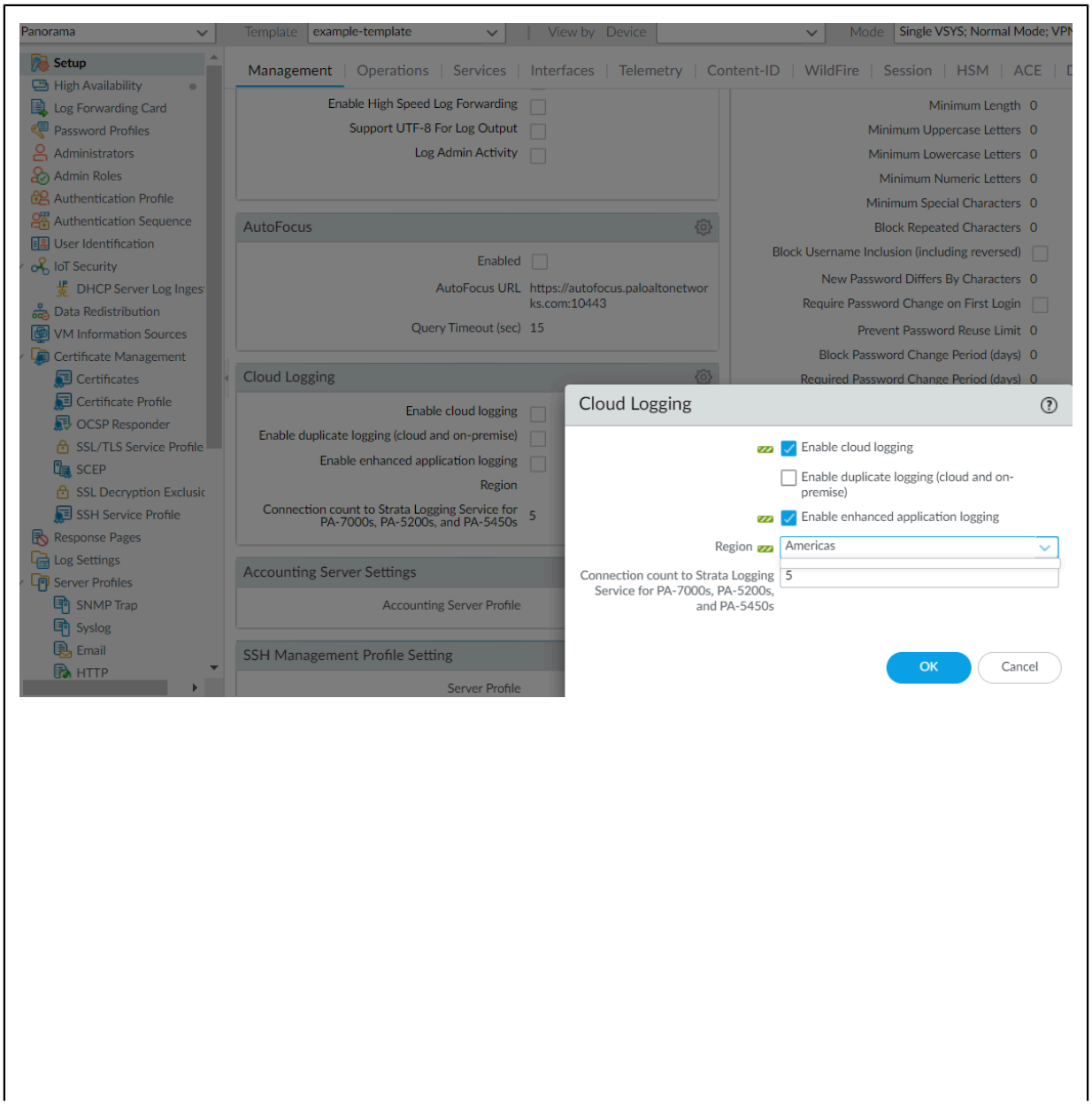
STEP 2 | To **Enable enhanced application logging** on the firewall, select **Device** > **Setup** > **Management** > **Cloud Logging** and edit the Cloud Logging Settings.



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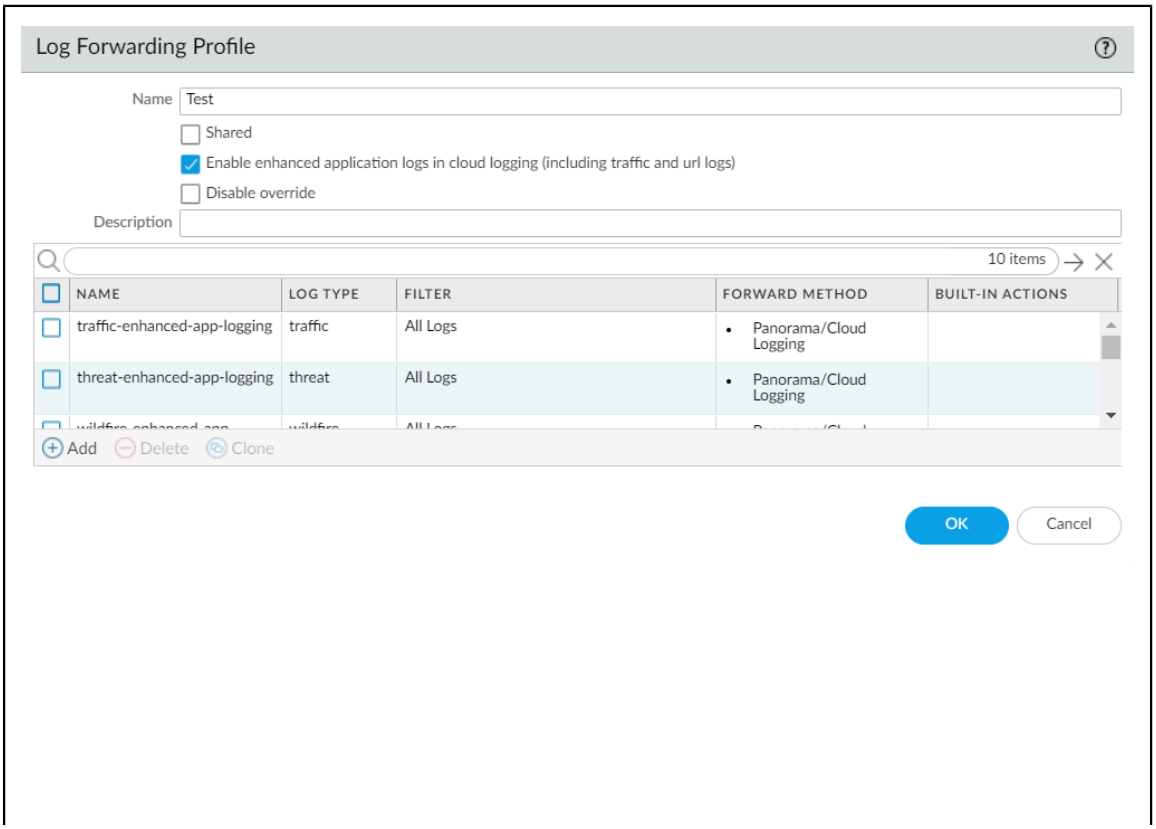
STEP 3 | Continue to enable Enhanced Application Logging for the Security policy rules that control the traffic into which you want extended visibility.

1. Select **Objects** > **Log Forwarding** and **Add** or modify a Log Forwarding profile.
2. Update the profile to **Enable enhanced application logs in cloud logging (including traffic and url logs)**.



Notice that when you enable Enhanced Application Logging in a Log Forwarding profile, match lists that specify the log types required for Enhanced Application Logging are automatically added to the profile.

3. Click **OK** to save the profile and continue to update as many profiles as needed.
4. Ensure that the Log Forwarding profile that you've updated is attached to a Security policy rule, to trigger log generation and forwarding for the traffic matched to the rule.
   1. To view the profiles attached to each Security policy rule, select **Policies** > **Security**.
   2. To update the Log Forwarding profile attached to a rule, **Add** or edit a rule and select **Policies** > **Security** > **Actions** > **Log Forwarding** and select the Log Forwarding profile enabled with Enhanced Application Logging.

IoT Security

One part of the firewall setup for IoT Security involves creating a Log Forwarding profile and applying it to Security policy rules. Although you can apply a profile to each rule individually, a simpler approach is to select a predefined Log Forwarding profile and apply it to as many rules as you like in bulk. The following steps explain this approach to adding the predefined Log Forwarding profile to Security policy rules in bulk.

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To use this workflow, you must have already configured security policy rules, enabled logging on the rules, and enabled logging services with Enhanced Application Logging.

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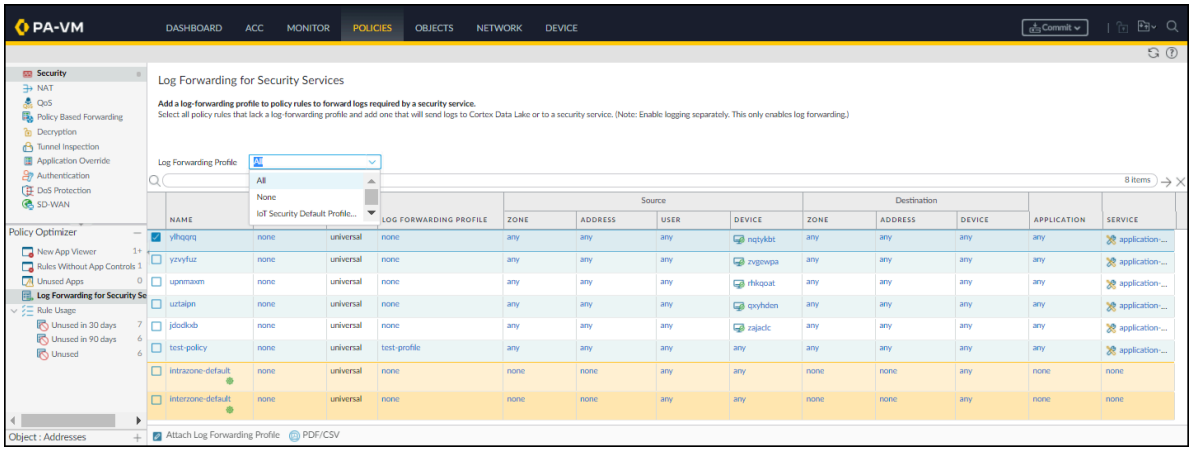
STEP 1 | Apply a Log Forwarding profile for IoT Security to Security policy rules.

1. Log in to your next-generation firewall and select **Policies** > **Log Forwarding for Security Services** in the Policy Optimizer section.
2. To view all your Security policy rules, including those with a Log Forwarding profile and those without it, select **All** for **Log Forwarding Profile**.



3. Select the rules for which you want to forward logs to the logging service.
4. **Attach Log Forwarding Profile** at the bottom of the page.
5. To apply the default Log Forwarding profile to your rules, choose **IoT Security Default Profile - EAL Enabled** and **OK**.

   The default profile is preconfigured to provide IoT Security with all the log types it requires, including Enhanced Application logs.

   *You don't have to **Enable Enhanced IoT Logging** because Enhanced Application Logging (EAL) is already enabled on IoT Security Default Profile.*

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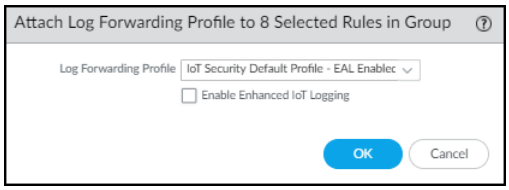

or

To add the forwarding of EALs to an existing Log Forwarding profile that doesn't already have it, choose it from the Log Forwarding Profile list, select **Enable Enhanced IoT Logging** and then **OK**.

*When you **Enable Enhanced IoT Logging**, PAN-OS updates the chosen Log Forwarding profile itself and thereby enables enhanced log forwarding on all rules that use the same Log Forwarding profile.*

PAN-OS adds the chosen Log Forwarding profile to those rules that don't already have one and replaces previously assigned profiles with this one.

STEP 2 | **Commit** your changes.

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Firewall Administration

Administrators can configure, manage, and monitor Palo Alto Networks firewalls using the web interface, CLI, and API management interface. You can customize role-based administrative access to the management interfaces to delegate specific tasks or permissions to certain administrators.

You can use the following user interfaces to manage the Palo Alto Networks firewall:

Do not enable management access from the internet or from other untrusted zones inside your enterprise security boundary. Follow the Administrative Access Best Practices to ensure that you are properly securing your firewall.

- Use the Web Interface to perform configuration and monitoring tasks with relative ease. This graphical interface allows you to access the firewall using HTTPS (recommended) or HTTP and it is the best way to perform administrative tasks.
- Use the Command Line Interface (CLI) to perform a series of tasks by entering commands in rapid succession over SSH (recommended), Telnet, or the console port. The CLI is a no-frills interface that supports two command modes, operational and configure, each with a distinct hierarchy of commands and statements. When you become familiar with the nesting structure and syntax of the commands, the CLI provides quick response times and administrative efficiency.
- Use the XML API to streamline your operations and integrate with existing, internally developed applications and repositories. The XML API is a web service implemented using HTTP/HTTPS requests and responses.
- Use Panorama to perform web-based management, reporting, and log collection for multiple firewalls. The Panorama web interface resembles the firewall web interface but with additional functions for centralized management.

Use the Reference: Web Interface Administrator Access to understand the different access that can be provided for different admin users. The Reference: Port Number Usage provides insight into the different ports and protocols available for different functionality.

Launch the Web Interface

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

The following web browsers are supported for access to the web interface:

- Google Chrome 104+
- Microsoft Edge 104+
- Mozilla Firefox 103+
- Safari 15+

For detailed information about specific tabs and fields in the web interface, refer to the Web Interface Reference Guide

Perform the following tasks to launch the web interface.

STEP 1 | Launch an Internet browser and enter the IP address of the firewall in the URL field (https://<IP address>).

*By default, the management (MGT) interface allows only HTTPS access to the web interface. To enable other protocols, select **Device** > **Setup** > **Interfaces** and edit the **Management** interface.*

STEP 2 | Log in to the firewall according to the type of authentication used for your account. If logging in to the firewall for the first time, use the default value **admin** for your username and password.

- **SAML**—Click **Use Single Sign-On** (SSO). If the firewall performs authorization (role assignment) for administrators, enter your **Username** and **Continue**. If the SAML identity provider (IdP) performs authorization, **Continue** without entering a **Username**. In both cases, the firewall redirects you to the IdP, which prompts you to enter a username and password. After you authenticate to the IdP, the firewall web interface displays.
- **Any other type of authentication**—Enter your user **Name** and **Password**. Read the login banner and select **I Accept and Acknowledge the Statement Below** if the login page has the banner and check box. Then click **Login**.

STEP 3 | Read and **Close** the messages of the day.

Configure Banners, Message of the Day, and Logos

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager)	❑ AIOps for NGFW Premium license (use the Strata Cloud Manager app)

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Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	

A *login banner* is optional text that you can add to the login page so that administrators will see information they must know before they log in. For example, you could add a message to notify users of restrictions on unauthorized use of the firewall.

You can add colored bands that highlight overlaid text across the top (*header banner*) and bottom (*footer banner*) of the web interface to ensure administrators see critical information, such as the classification level for firewall administration.

A *message of the day* dialog automatically displays after you log in. The dialog displays messages that Palo Alto Networks embeds to highlight important information associated with a software or content release. You can also add one custom message to ensure administrators see information, such as an impending system restart, that might affect their tasks.

You can replace the default logos that appear on the login page and in the header of the web interface with the logos of your organization.

- Strata Cloud Manager
- PAN-OS & Panorama

Configure Banners and Message of the Day (Strata Cloud Manager)

STEP 1 | Configure the login banner.

1. Log in to Strata Cloud Manager.
2. Select **Manage** > **Configuration** > **NGFW and Prisma Access** > **Device Settings** > **Device Setup**, and customize the General Settings.
3. Enter the **Login Banner** (up to 3,200 characters).
4. (Optional) Select **Force Admins to Acknowledge Login Banner** to force administrators to select an **I Accept and Acknowledge the Statement Below** check box above the banner text to activate the **Login** button.
5. Click **Save**.

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STEP 2 | Set the message of the day.

1. Select **Device Settings** > **Device Setup**, and customize the Banners and Messages settings.
2. Enable the **Message of the Day**.
3. Enter the **Message of the Day** (up to 3,200 characters).

   *After you enter the message and click **OK**, administrators who subsequently log in, and active administrators who refresh their browsers, see the new or updated message immediately; a commit isn't necessary. This enables you to inform other administrators of an impending commit that might affect their configuration changes. Based on the commit time that your message specifies, the administrators can then decide whether to complete, save, or undo their changes.*
4. (Optional) Select **Allow Do Not Display Again** (default is disabled) to give administrators the option to suppress a message of the day after the first login session. Each administrator can suppress messages only for his or her own login sessions. In the message of the day dialog, each message will have its own suppression option.
5. (Optional) Enter a header **Title** for the message of the day dialog (default is `Messageof the Day`).
6. Set the background color and Icon.

   A bright background color and contrasting text color can increase the likelihood that administrators will notice and read a banner. You can also use colors that correspond to classification levels in your organization.

STEP 3 | Configure the header and footer banners.

1. Enter the **Header Banner** (up to 3,200 characters).
2. Configure the **Header Color** and **Header Text Color**.
3. (Optional) Clear **Same Banner Header and Footer** (enabled by default) to use different header and footer banners.
4. Enter the **Footer Banner** (up to 3,200 characters) if the header and footer banners differ.
5. Configure the **Footer Color** and **Footer Text Color**.
6. Click **Save**.

STEP 4 | **Push Config** to push your configuration changes.

STEP 5 | Verify that the banners and message of the day display as expected.

1. Enter your login credentials, review the banner, select **I Accept and Acknowledge the Statement Below** to enable the **Login** button, and then **Login**.

   A dialog displays the message of the day. Messages that Palo Alto Networks embedded display on separate pages in the same dialog. To navigate the pages, click the right or left

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arrows along the sides of the dialog or click a page selector ●○○○ at the bottom of the dialog.

2. (Optional) You can select **Do not show again** for the message you configured and for any messages that Palo Alto Networks embedded.
3. **Close** the message of the day dialog to access the web interface.

   Header and footer banners display in every web interface page with the text and colors that you configured.

Configure Banners, Message of the Day, and Logos (PAN-OS)

STEP 1 | Configure the login banner.

1. Select **Device** > **Setup** > **Management** and edit the General Settings.
2. Enter the **Login Banner** (up to 3,200 characters).
3. (Optional) Select **Force Admins to Acknowledge Login Banner** to force administrators to select an **I Accept and Acknowledge the Statement Below** check box above the banner text to activate the **Login** button.
4. Click **OK**.

STEP 2 | Set the message of the day.

1. Select **Device** > **Setup** > **Management** and edit the Banners and Messages settings.
2. Enable the **Message of the Day**.
3. Enter the **Message of the Day** (up to 3,200 characters).

   *After you enter the message and click **OK**, administrators who subsequently log in, and active administrators who refresh their browsers, see the new or updated message immediately; a commit isn't necessary. This enables you to inform other administrators of an impending commit that might affect their configuration changes. Based on the commit time that your message specifies, the administrators can then decide whether to complete, save, or undo their changes.*
4. (Optional) Select **Allow Do Not Display Again** (default is disabled) to give administrators the option to suppress a message of the day after the first login session. Each administrator can suppress messages only for his or her own login sessions. In the message of the day dialog, each message will have its own suppression option.
5. (Optional) Enter a header **Title** for the message of the day dialog (default is `Messageof the Day`).

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STEP 3 | Configure the header and footer banners.

A bright background color and contrasting text color can increase the likelihood that administrators will notice and read a banner. You can also use colors that correspond to classification levels in your organization.

1. Enter the **Header Banner** (up to 3,200 characters).
2. (Optional) Clear **Same Banner Header and Footer** (enabled by default) to use different header and footer banners.
3. Enter the **Footer Banner** (up to 3,200 characters) if the header and footer banners differ.
4. Click **OK**.

STEP 4 | Replace the logos on the login page and in the header.

The maximum size for any logo image is 128KB. The supported file types are png and jpg. The firewall does not support image files that are interlaced, images that contain alpha channels, and gif file types because such files interfere with PDF generation.

1. Select **Device** > **Setup** > **Operations** and click **Custom Logos** in the Miscellaneous section.
2. Perform the following steps for both the **Login Screen** logo and the **Main UI** (header) logo:
   1. Click upload .
   2. Select a logo image and click **Open**.

      You can preview the image to see how PAN-OS will crop it to fit by clicking the magnifying glass icon.
   3. Click **Close**.
3. **Commit** your changes.

STEP 5 | Verify that the banners, message of the day, and logos display as expected.

1. Log out to return to the login page, which displays the new logos you selected.
2. Enter your login credentials, review the banner, select **I Accept and Acknowledge the Statement Below** to enable the **Login** button, and then **Login**.

   A dialog displays the message of the day. Messages that Palo Alto Networks embedded display on separate pages in the same dialog. To navigate the pages, click the right or left arrows along the sides of the dialog or click a page selector at the bottom of the dialog.
3. (Optional) You can select **Do not show again** for the message you configured and for any messages that Palo Alto Networks embedded.
4. **Close** the message of the day dialog to access the web interface.

   Header and footer banners display in every web interface page with the text and colors that you configured. The new logo you selected for the web interface displays below the header banner.

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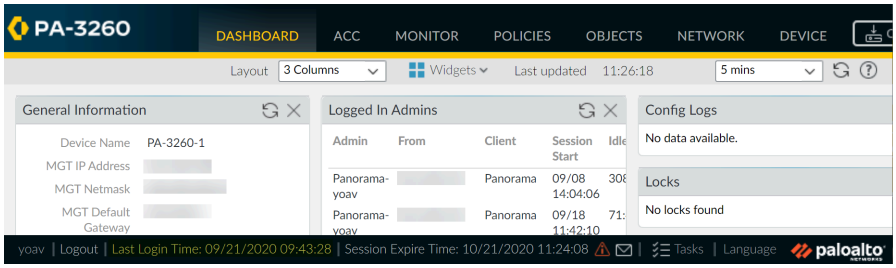
Use the Administrator Login Activity Indicators to Detect Account Misuse

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

The last login time and failed login attempts indicators provide a visual way to detect misuse of your administrator account on a Palo Alto Networks firewall or Panorama management server. Use the last login information to determine if someone else logged in using your credentials and use the failed login attempts indicator to determine if your account is being targeted in a brute-force attack.

STEP 1 | View the login activity indicators to monitor recent activity on your account.

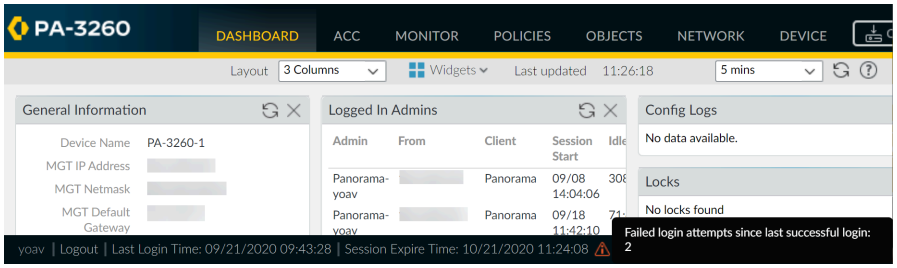
1. Log in to the web interface on your firewall or Panorama management server.
2. View the last login details located at the bottom left of the window and verify that the timestamp corresponds to your last login.



3. Look for a caution symbol to the right of the last login time information for failed login attempts.

   The failed login indicator appears if one or more failed login attempts occurred using your account since the last successful login.

   1. If you see the caution symbol, hover over it to display the number of failed login attempts.



   2. Click the caution symbol to view the failed login attempts summary. Details include the admin account name, the reason for the login failure, the source IP address, and the date and time.

> *After you successfully log in and then log out, the failed login counter resets to zero so you will see new failed login details, if any, the next time you log in.*

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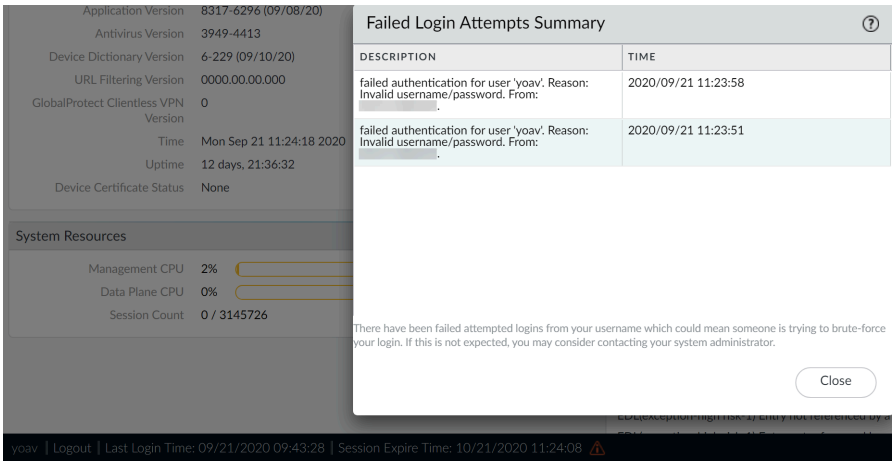
STEP 2 | Locate hosts that are continually attempting to log in to your firewall or Panorama management server.

1. Click the failed login caution symbol to view the failed login attempts summary.
2. Locate and record the source IP address of the host that attempted to log in. For example, the following figure shows multiple failed login attempts.



3. Work with your network administrator to locate the user and host that is using the IP address that you identified.

   If you cannot locate the system that is performing the brute-force attack, consider renaming the account to prevent future attacks.

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STEP 3 | Take the following actions if you detect an account compromise.

1. Select **Monitor** > **Logs** > **Configuration** and view the configuration changes and commit history to determine if your account was used to make changes without your knowledge.
2. Select **Device** > **Config Audit** to compare the current configuration and the configuration that was running just prior to the configuration you suspect was changed using your credentials. You can also do this using Panorama.

   If your administrator account was used to create a new account, performing a configuration audit helps you detect changes that are associated with any unauthorized accounts, as well.
3. Revert the configuration to a known good configuration if you see that logs were deleted or if you have difficulty determining if improper changes were made using your account.

   Before you commit to a previous configuration, review it to ensure that it contains the correct settings. For example, the configuration that you revert to may not contain recent changes, so apply those changes after you commit the backup configuration.

   Use the following best practices to help prevent brute-force attacks on privileged accounts.

   - *Limit the number of failed attempts allowed before the firewall locks a privileged account by setting the number of Failed Attempts and the Lockout Time (min) in the authentication profile or in the Authentication Settings for the Management interface (**Device** > **Setup** > **Management** > **Authentication Settings**).*
   - Use Interface Management Profiles to Restrict Access.
   - *Enforce complex passwords for privileged accounts.*

Manage and Monitor Administrative Tasks

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

The Task Manager displays details about all the operations that you and other administrators initiated (such as manual commits) or that the firewall initiated (such as scheduled report generation) since the last firewall reboot. You can use the Task Manager to troubleshoot failed operations, investigate warnings associated with completed commits, view details about queued commits, or cancel pending commits.

You can also view System Logs to monitor system events on the firewall or view Config Logs to monitor firewall configuration changes.

STEP 1 | Click **Tasks** at the bottom of the web interface.

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STEP 2 | **Show** only **Running** tasks (in progress) or **All** tasks (default). Optionally, filter the tasks by type:

- **Jobs**—Administrator-initiated commits, firewall-initiated commits, and software or content downloads and installations.
- **Reports**—Scheduled reports.
- **Log Requests**—Log queries that you trigger by accessing the **Dashboard** or a **Monitor** page.

STEP 3 | Perform any of the following actions:

- **Display or hide task details**—By default, the Task Manager displays the Type, Status, Start Time, and Messages for each task. To see the End Time and Job ID for a task, you must manually configure the display to expose those columns. To display or hide a column, open the drop-down in any column header, select **Columns**, and select or deselect the column names as needed.
- **Investigate warnings or failures**—Read the entries in the Messages column for task details. If the column says `Too many messages`, click the corresponding entry in the Type column to see more information.
- **Display a commit description**—If an administrator entered a description when configuring a commit, you can click **Commit Description** in the Messages column to display the description.
- **Check the position of a commit in the queue**—The Messages column indicates the queue position of commits that are in progress.
- **Cancel pending commits**—Click **Clear Commit Queue** to cancel all pending commits (available only to predefined administrative roles). To cancel an individual commit, click **x** in the Action column for that commit (the commit remains in the queue until the firewall dequeues it). You cannot cancel commits that are in progress.

Configure Response Pages

Where Can I Use This?	What Do I Need?
• NGFW • Prisma Access	One of these licenses for Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Essentials ☐ Strata Cloud Manager Pro Other licenses needed: ☐ Prisma Access license

Custom response pages display an HTML message when a user tries to access a URL. You can add provide a custom HTML message that is downloaded and displayed instead of the requested page or file. Each virtual system can have its own customized response pages.

Each virtual system can have its own custom response pages. The following table describes the types of custom response pages that support customer messages.

You can perform any of the following functions for **Response Pages.**

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- PAN-OS & Panorama
- Strata Cloud Manager

Configure Response Pages (PAN-OS)

- To import a custom HTML response page, click the link of the page type you would like to change and then click import/export. Browse to locate the page. A message is displayed to indicate whether the import succeeded. For the import to be successful, the file must be in HTML format.
- To export a custom HTML response page, click Export HTML Template for the type of page. Select whether to open the file or save it to disk and, if appropriate, select Always use the same option.
- To enable or disable the Application Block page or SSL Decryption Opt-out pages, click Enable for the type of page. Select or deselect Enable, as appropriate.
- To use the default response page instead of a previously uploaded custom page, delete the custom block page and commit. This will set the default block page as the new active page.

You can configure the following response pages in PAN-OS:

Custom Response Page Types	Description
Antivirus Block Page	Access blocked due to a virus infection.
Application Block Page	Access blocked because the application is blocked by a Security policy rule.
Authentication Portal Comfort Page	The firewall displays this page so that users can enter login credentials to access services that are subject to Authentication policy rules. Enter a message that tells users how to respond to this authentication challenge. The firewall authenticates users based on the **Authentication Profile** specified in the authentication enforcement object assigned to an Authentication rule. *You can display unique authentication instructions for each Authentication rule by entering a **Message** in the associated authentication enforcement object. The message defined in the object overrides the message defined in the Authentication Portal Comfort Page.*
Data Filtering Block Page	Content was matched against a data filtering profile and blocked because sensitive information was detected.
File Blocking Continue Page	Page for users to confirm that downloading should continue. This option is available only if Continue functionality is enabled in the security profile.

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Custom Response Page Types	Description
File Blocking Block Page	Access blocked because access to the file is blocked.
GlobalProtect App Help Page	Custom help page for GlobalProtect users (accessible from the settings menu on the GlobalProtect status panel).
GlobalProtect Portal Login Page	Login page for users who attempt to authenticate to the GlobalProtect portal webpage.
GlobalProtect Portal Home Page	Home page for users who successfully authenticate to the GlobalProtect portal webpage.
GlobalProtect App Welcome Page	Welcome page for users who successfully connect to GlobalProtect.
MFA Login Page	The firewall displays this page so that users can respond to multi-factor authentication (MFA) challenges when accessing services that are subject to Authentication policy rules. Enter a message that tells users how to respond to the MFA challenges.
SAML Auth Internal Error Page	Page to inform users that SAML authentication failed. The page includes a link for the user to retry authentication.
SSL Certificate Errors Notify Page	Notification that an SSL certificate has been revoked.
SSL Decryption Opt-out Page	User warning page indicating that the firewall will decrypt SSL sessions for inspection.
URL Filtering and Category Match Block Page	Access blocked by a URL filtering profile or because the URL category is blocked by a Security policy rule.
URL Filtering Continue and Override Page	Page with initial block policy that allows users to bypass the block. For example, a user who thinks the page was blocked inappropriately can click **Continue** to proceed to the page. With the override page, a password is required for the user to override the policy that blocks this URL.
URL Filtering Safe Search Enforcement Block Page	Access blocked by a Security policy rule with a URL filtering profile that has the **Safe Search Enforcement** option enabled. The user sees this page if a search is performed using Bing, Google, Yahoo, Yandex, or YouTube and their browser or search engine account setting for Safe Search is not set to strict. The block page will instruct the user to set the Safe Search setting to strict.

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Custom Response Page Types	Description
Anti Phishing Block Page	Displays to users when they attempt to enter valid corporate credentials (usernames or passwords) on a web page for which credential submissions are blocked. The user can continue to access the site but remains unable to submit valid corporate credentials to any associated web forms.
Anti Phishing Continue Page	This page warns users against submitting corporate credentials (usernames and passwords) to a web site. Warning users against submitting credentials can help to discourage them from reusing corporate credentials and to educate them about possible phishing attempts. Users see this page when they attempt to submit credentials to a site for which the **User Credential Submission** permissions are set to **continue**. They must select **Continue** to enter credentials on the site.

Configure Response Pages (Strata Cloud Manager)

You can configure the following response pages in:

Custom Response Page Types	Description
GlobalProtect App Help Page	Custom help page for GlobalProtect users (accessible from the settings menu on the GlobalProtect status panel). (**Configuration** > **Device Settings Services** > **Agent Settings** > **Response Pages**)
GlobalProtect Portal Login Page	Login page for users who attempt to authenticate to the GlobalProtect portal webpage. (**Configuration** > **Device Settings Services** > **Agent Settings** > **Response Pages**)
GlobalProtect Portal Home Page	Home page for users who successfully authenticate to the GlobalProtect portal webpage. (**Configuration** > **Device Settings Services** > **Agent Settings** > **Response Pages**)
GlobalProtect App Welcome Page	Welcome page for users who successfully connect to GlobalProtect. (**Configuration** > **Device Settings Services** > **Agent Settings** > **Response Pages**)

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Custom Response Page Types	Description
MFA Login Page	The firewall displays this page so that users can respond to multi-factor authentication (MFA) challenges when accessing services that are subject to Authentication policy rules. Enter a message that tells users how to respond to the MFA challenges. (**Configuration** > **Identity Services** > **Authentication** > **Response Pages**)
SAML Auth Internal Error Page	Page to inform users that SAML authentication failed. The page includes a link for the user to retry authentication. (**Configuration** > **Identity Services** > **Authentication** > **Response Pages**)
Captive Portal Comfort Page	A message that confirms to the user that they have successfully completed the authentication process. (**Configuration** > **Identity Services** > **Authentication** > **Response Pages**)
SSL Certificate Errors Notify Page	Notification that an SSL certificate has been revoked. (**Configuration** > **Security Services** > **Decryption** > **Response Pages**)
SSL Decryption Opt-out Page	User warning page indicating that the firewall will decrypt SSL sessions for inspection. (**Configuration** > **Security Services** > **Decryption** > **Response Pages**)
URL Access Management Block Page	Access blocked by a URL filtering profile or because the URL category is blocked by a Security policy rule. (**Configuration** > **Security Services** > **Settings** > **Response Pages**)
URL Access Management Continue and Override Page	Page with initial block policy that allows users to bypass the block. For example, a user who thinks the page was blocked inappropriately can click Continue to proceed to the page. With the override page, a password is required for the user to override the policy that blocks this URL. (**Configuration** > **Security Services** > **Settings** > **Response Pages**)
URL Access Management Safe Search Enforcement Block Page	Access blocked by a Security policy rule with a URL filtering profile that has the Safe Search Enforcement option enabled.

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Custom Response Page Types	Description
	The user sees this page if a search is performed using Bing, Google, Yahoo, Yandex, or YouTube and their browser or search engine account setting for Safe Search is not set to strict. The block page will instruct the user to set the Safe Search setting to strict. (**Configuration** > **Security Services** > **Settings** > **Response Pages**)
Anti Phishing Block Page	Displays to users when they attempt to enter valid corporate credentials (usernames or passwords) on a web page for which credential submissions are blocked. The user can continue to access the site but remains unable to submit valid corporate credentials to any associated web forms. (**Configuration** > **Security Services** > **Settings** > **Response Pages**)
Anti Phishing Continue Page	This page warns users against submitting corporate credentials (usernames and passwords) to a web site. Warning users against submitting credentials can help to discourage them from reusing corporate credentials and to educate them about possible phishing attempts. Users see this page when they attempt to submit credentials to a site for which the User Credential Submission permissions are set to continue. They must select Continue to enter credentials on the site. (**Configuration** > **Security Services** > **Settings** > **Response Pages**)

- To import a custom HTML response page, click the link of the page type you would like to change and then click**Import HTML Template**. For the import to be successful, the file must be in HTML format.
- To export a custom HTML response page, click **Export HTML Template** for the type of page.
- To edit a custom response page, click the link of the page, browse and choose the required file, click **Save**, and then click **Push Config**.

Commit, Validate, and Preview Firewall Configuration Changes

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

A commit is the process of activating pending changes to the firewall configuration. You can filter pending changes by administrator or *location* and then preview, validate, or commit only those changes. The locations can be specific virtual systems, shared policies and objects, or shared device and network settings.

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The firewall queues commit requests so that you can initiate a new commit while a previous commit is in progress. The firewall performs the commits in the order they are initiated but prioritizes auto-commits that are initiated by the firewall (such as FQDN refreshes). However, if the queue already has the maximum number of administrator-initiated commits, you must wait for the firewall to finish processing a pending commit before initiating a new one. To cancel pending commits or view details about commits of any status, see Manage and Monitor Administrative Tasks.

When you initiate a commit, the firewall checks the validity of the changes before activating them. The validation output displays conditions that either block the commit (errors) or that are important to know (warnings). For example, validation could indicate an invalid route destination that you need to fix for the commit to succeed. The validation process enables you to find and fix errors before you commit (it makes no changes to the running configuration). This is useful if you have a fixed commit window and want to be sure the commit will succeed without errors.

When enabled and managed by a Panorama™ management server, managed firewalls locally test the configuration committed locally or pushed from Panorama to verify that the new changes do not break the connection between Panorama and the managed firewall. If the committed configuration breaks the connection between Panorama and a managed firewall, then the firewall automatically fails the commit and the configuration is reverted to the previous running configuration. Additionally, firewalls managed by a Panorama management server test their connection to Panorama every 60 minutes and if a managed firewalls detects that it can no longer successfully connect to Panorama, then it reverts its configuration to the previous running configuration.

The commit, validate, preview, save, and revert operations apply only to changes made after the last commit. To restore configurations to the state they were in before the last commit, you must load a previously backed up configuration.

To prevent multiple administrators from making configuration changes during concurrent sessions, see Manage Locks for Restricting Configuration Changes.

STEP 1 | Configure the scope of configuration changes that you will commit, validate, or preview.

1. Click **Commit** at the top of the web interface.
2. Select one of the following options:
   - **Commit All Changes** (default)—Applies the commit to all changes for which you have administrative privileges. You cannot manually filter the commit scope when you

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select this option. Instead, the administrator role assigned to the account you used to log in determines the commit scope.

- **Commit Changes Made By**—Enables you to filter the commit scope by administrator or location. The administrative role assigned to the account you used to log in determines which changes you can filter.

*To commit the changes of other administrators, the account you used to log in must be assigned the Superuser role or an Admin Role profile with the **Commit For Other Admins** privilege enabled.*

3. (Optional) To filter the commit scope by administrator, select **Commit Changes Made By**, click the adjacent link, select the administrators, and click **OK**.
4. (Optional) To filter by location, select **Commit Changes Made By** and clear any changes that you want to exclude from the Commit Scope.

If dependencies between the configuration changes you included and excluded cause a validation error, perform the commit with all the changes included. For example, when you commit changes to a virtual system, you must include the changes of all administrators who added, deleted, or repositioned rules for the same rulebase in that virtual system.

STEP 2 | Preview the changes that the commit will activate.

This can be useful if, for example, you don't remember all your changes and you're not sure you want to activate all of them.

The firewall compares the configurations you selected in the Commit Scope to the running configuration. The preview window displays the configurations side-by-side and uses color coding to indicate which changes are additions (green), modifications (yellow), or deletions (red).

Preview Changes and select the **Lines of Context**, which is the number of lines from the compared configuration files to display before and after each highlighted difference. These additional lines help you correlate the preview output to settings in the web interface. Close the preview window when you finish reviewing the changes.

Because the preview results display in a new browser window, your browser must allow pop-ups. If the preview window does not open, refer to your browser documentation for the steps to allow pop-ups.

STEP 3 | Preview the individual settings for which you are committing changes.

This can be useful if you want to know details about the changes, such as the types of settings and who changed them.

1. Click **Change Summary**.
2. (Optional) **Group By** a column name (such as the **Type** of setting).
3. **Close** the Change Summary dialog when you finish reviewing the changes.

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STEP 4 | Validate the changes before you commit to ensure the commit will succeed.

1. **Validate Changes**.

   The results display all the errors and warnings that an actual commit would display.
2. Resolve any errors that the validation results identify.

STEP 5 | Commit your configuration changes.

Commit your changes to validate and activate them.

To view details about commits that are pending (which you can still cancel), in progress, completed, or failed, see Manage and Monitor Administrative Tasks.

Commit Selective Configuration Changes

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

Configuration changes occur often and are typically made by multiple administrators who are not aware of what other configuration changes were made. It is vital to be able to control which configuration objects are committed and prevent incomplete configurations from being committed to your firewall. Rather than committing all pending configuration changes, you can instead select configuration objects to commit. A system log is generated after a successful selective commit.

The ability to select specific objects commit allows multiple administrators to effectively make configuration changes without disrupting other administrators who make configuration changes that are not ready to be committed. Leveraging the ability to selectively commit configuration changes allows you to maintain your defined operational procedure while still being able to successfully make independent configuration changes that are not defined within your operational scope.

STEP 1 | Log in to the firewall web interface.

STEP 2 | Perform configuration changes on the firewall and **Commit**.

STEP 3 | Change the commit scope to **Commit Changes Made By** to select configuration changes to commit.

The push scope displays the name of the admin currently logged in. Click the admin name to view a list of admins who have made configuration changes that have not been committed.

STEP 4 | (Optional) Preview and validate your pending configuration changes to ensure you want to commit the selected configuration objects.

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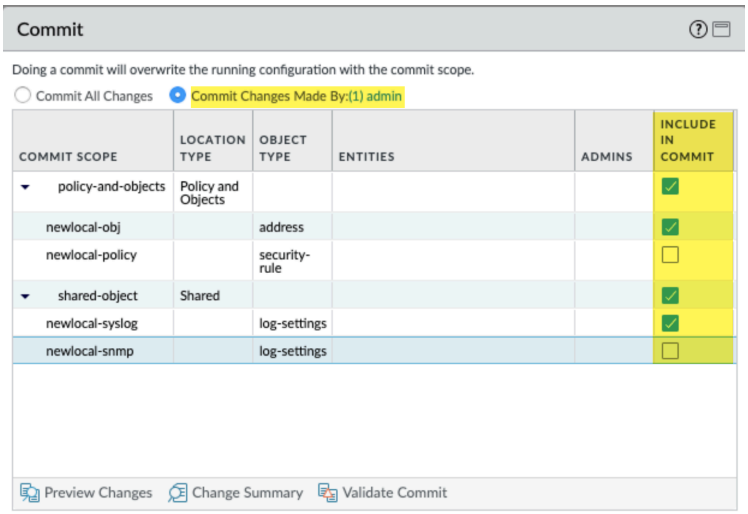
STEP 5 | **Commit**.

The Commit Status page displays the administrators that made configuration changes that were committed and the location of the committed configuration changes.



Export Configuration Table Data

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

Export policy rules, configuration objects, and IPS signatures from Panorama™ and firewalls to demonstrate regulatory compliance to external auditors, to conduct periodic reviews of the firewall configuration, and to generate reports on firewall policies. This prevents you from having to give auditors direct access to your firewalls and appliances, to take screen shots or to access the XML API to generate configuration reports. From the web interface, you can export the configuration table data for policies, objects, network, firewall, and Panorama configurations, as well as Signature exceptions in the Antivirus, Anti-Spyware, and Vulnerability Protection Security profiles, in either a PDF or CSV file.

Exporting to a PDF file supports only English descriptions.

Configuration table export works like a print function—you cannot import generated files back into Panorama or the firewall. When you export data as a PDF file and the table data exceeds 50,000 rows, the data is split in to multiple PDF files (for example, <report-name>_part1.pdf and <report-name>_part2.pdf) When you export data as a CSV file, the data is exported as a single file. These export formats allow you to apply filters that match your report criteria and search within PDF reports to quickly find specific data. Additionally, when you export the configuration table data, a system log is generated to record the event.

STEP 1 | Launch the Web Interface and identify the configuration data you need to export.

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STEP 2 | Apply filters as needed to produce the configuration data you need to export and click **PDF/CSV**.

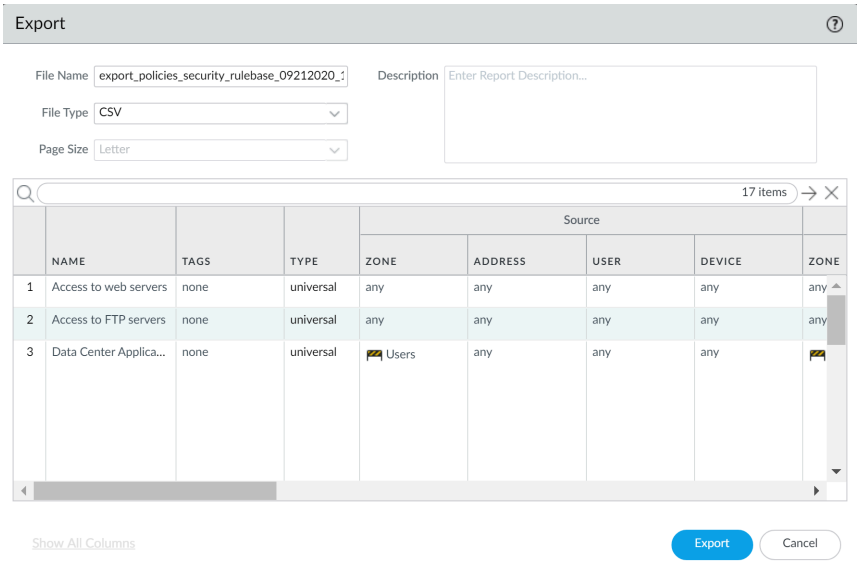
STEP 3 | Configure the Configuration Table Export report:

1. Enter a **File Name**.
2. Select the **File Type**.
3. (Optional) Enter a report Description.
4. Confirm the configuration table data matches the filters you applied.

*Select **Show All Columns** to show all filters applied.*

STEP 4 | **Export** the configuration table data.

Configuration table export works like a print function—you cannot import generated files back in to Panorama or the firewall.



STEP 5 | Select a location to save the exported file.

Use Global Find to Search the Firewall or Panorama Management Server

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

Global Find enables you to search the candidate configuration on a firewall or on Panorama for a particular string, such as an IP address, object name, policy rule name, threat ID, UUID, or application name. In addition to searching for configuration objects and settings, you can search by job ID or job type for manual commits that administrators performed or auto-commits that

the firewall or Panorama performed. You can also use Global Find to search for IP addresses within external dynamic lists. The search results are grouped by category and provide links to the configuration location in the web interface, so that you can easily find all of the places where the string is referenced. The search results also help you identify other objects that depend on or make reference to the search term or string. For example, when deprecating a security profile enter the profile name in Global Find to locate all instances of the profile and then click each instance to navigate to the configuration page and make the necessary change. After all references are removed, you can then delete the profile. You can do this for any configuration item that has dependencies.

With (12.1.2 and later releases) Global Find is optimized to provide a more targeted and enhanced search experience. It significantly improves responsiveness when searching for a large database or configuration, and when multiple administrators work simultaneously on the system.

The Optimized Search prioritizes and searches for the most relevant records based on admin-usage patterns. The new usage-based search returns results in batches, preventing the user interface from freezing during intensive searches. This substantially reduces search times across large configurations. You can also choose to exclusively search for UUIDs or Template References by selecting the Search UUIDs and Include Template References options respectively.

Watch the video.

Global Find does not search dynamic content (such as logs, address ranges, or allocated DHCP addresses). In the case of DHCP, you can search on a DHCP server attribute, such as the DNS entry, but you cannot search for individual addresses allocated to users. Global Find also does not search for individual user or group names identified by User-ID unless the user/group is defined in a policy. In general, you can only search content that the firewall writes to the configuration.

- PAN-OS 11.1 and earlier
- PAN-OS 12.1.2 and later

PAN-OS 11.1 and earlier

- Launch Global Find by clicking the **Search** icon located on the upper right of the web interface.

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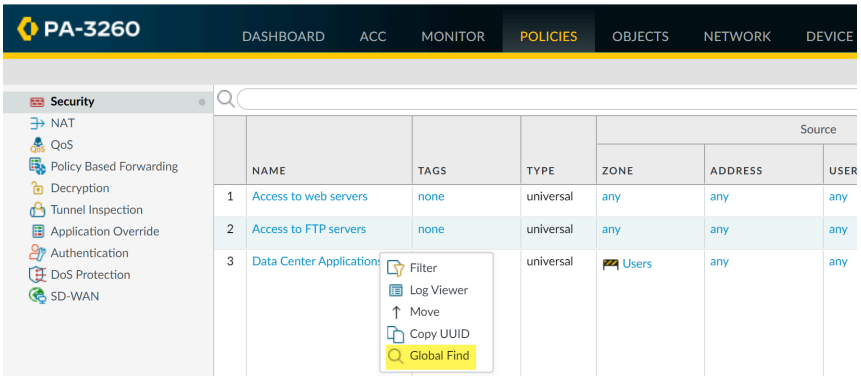
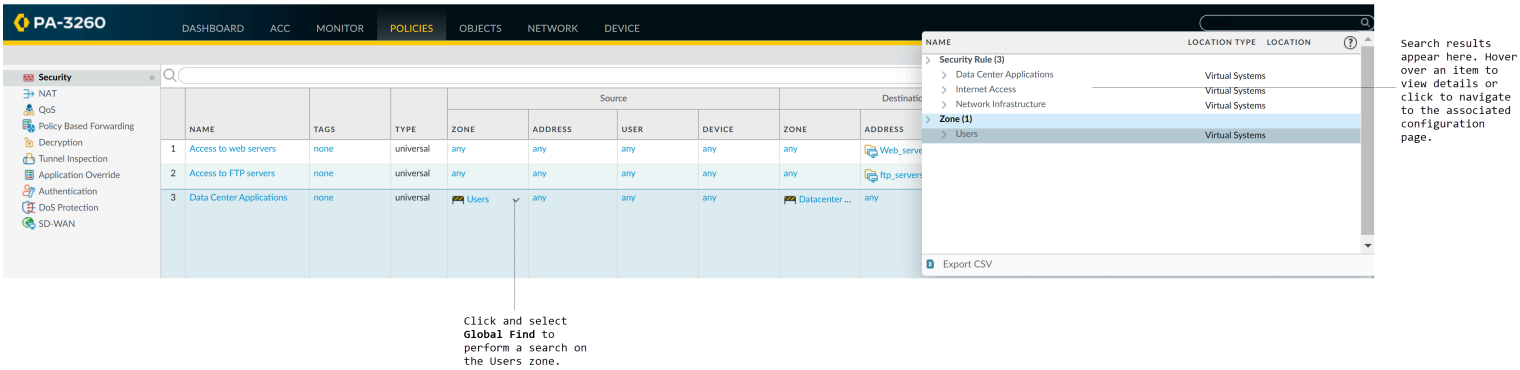
- To access the Global Find from within a configuration area, click the drop-down next to an item and select **Global Find**:



For example, click **Global Find** on a zone named **Users** to search the candidate configuration for each location where the zone is referenced. The following screen capture shows the search results for the zone Users:



Search tips:

- If you initiate a search on a firewall that has multiple virtual systems enabled or if custom administrative role types are defined, Global Find will only return results for areas of the firewall in which the administrator has permissions. The same applies to Panorama device groups.
- Spaces in search terms are handled as AND operations. For example, if you search on **`corp policy`**, the search results include instances where corp and policy exist in the configuration.
- To find an exact phrase, enclose the phrase in quotation marks.
- Enter no more than five keywords or use an exact phrase match with quotation marks.
- To rerun a previous search, click Search (located on the upper right of the web interface) to see a list of the last 20 searches. Click an item in the list to rerun that search. Search history is unique to each administrator account.
- To search for a UUID, you must copy and paste the UUID.

PAN-OS 12.1.2 and later

- Launch Global Find by clicking the **Search** icon located on the upper right of the web interface.

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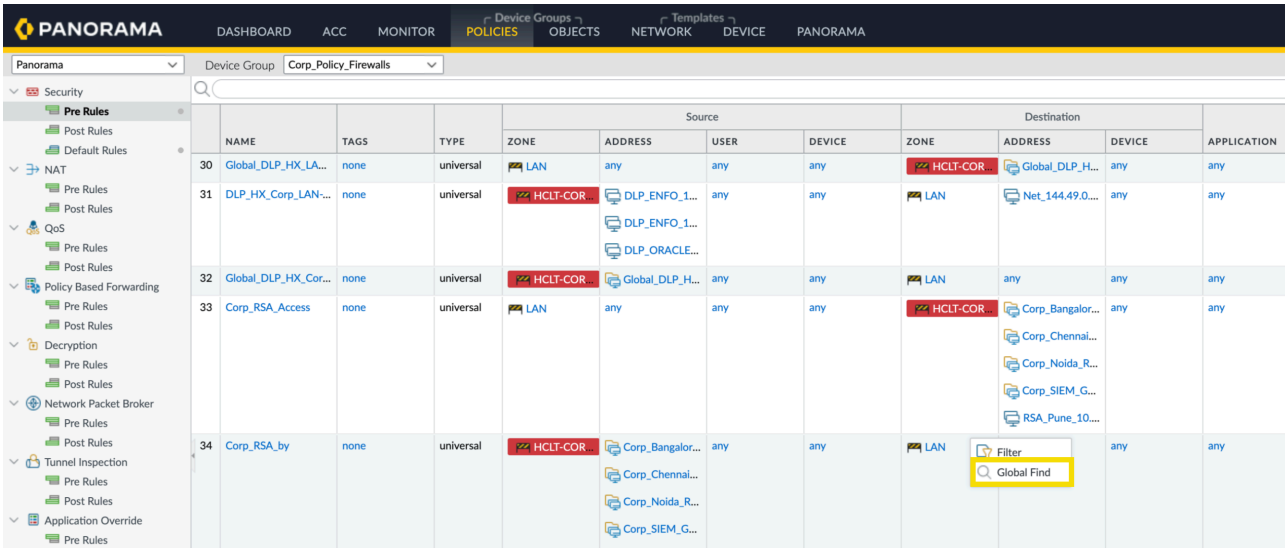
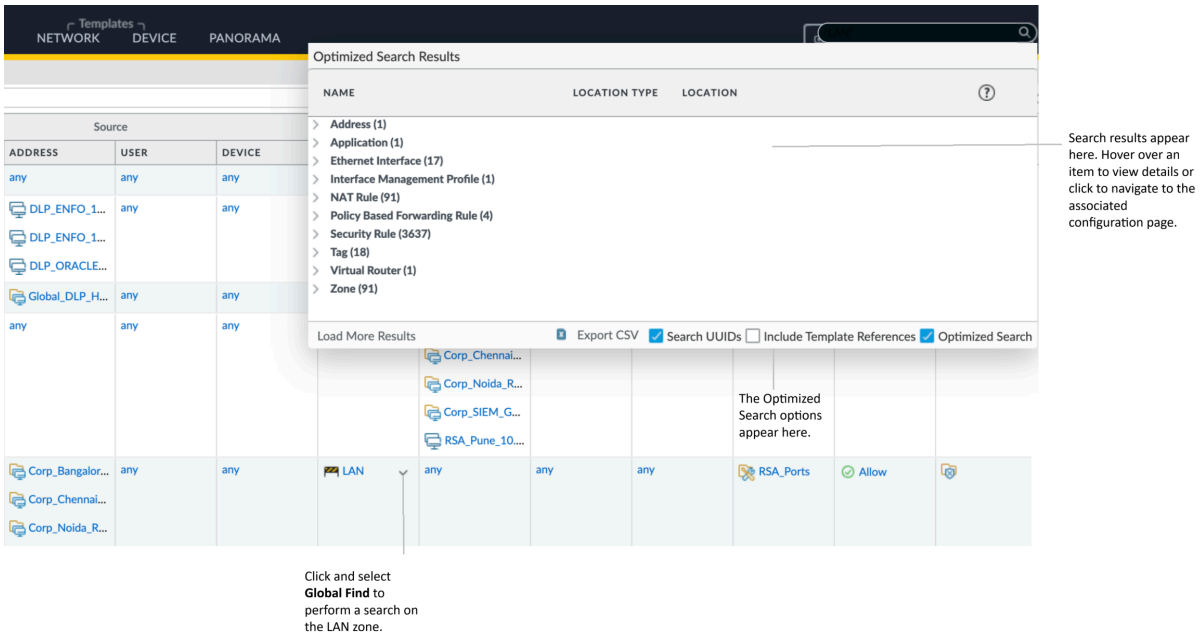
- To access the Global Find from within a configuration area, click the drop-down next to an item and select **Global Find**:



For example, click **Global Find** on a zone named **LAN** to search the candidate configuration for each location where the zone is referenced.



The **Optimized Search Results** dialog provides the following options:

- **Search UUIDs**: Use this option to perform targeted search exclusively for UUIDs.
- **Include Template References**: Use this option to perform targeted search exclusively for template references.
- **Optimized Search**: Use this option to prioritize search results based on admin usage patterns. The system returns the most relevant results in batches. This substantially reduces overall search times.

Search tips:

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- If you initiate a search on a firewall that has multiple virtual systems enabled or if custom administrative role types are defined, Global Find will only return results for areas of the firewall in which the administrator has permissions. The same applies to Panorama device groups.
- Spaces in search terms are handled as AND operations. For example, if you search on **`corp policy`**, the search results include instances where corp and policy exist in the configuration.
- To find an exact phrase, enclose the phrase in quotation marks.
- Enter no more than five keywords or use an exact phrase match with quotation marks.
- To rerun a previous search, click Search (located on the upper right of the web interface) to see a list of the last 20 searches. Click an item in the list to rerun that search. Search history is unique to each administrator account.
- To search for a UUID, you must copy and paste the UUID.

Manage Locks for Restricting Configuration Changes

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

You can use configuration locks to prevent other administrators from changing the candidate configuration or from committing configuration changes until you manually remove the lock or the firewall automatically removes it (after a commit). Locks ensure that administrators don't make conflicting changes to the same settings or interdependent settings during concurrent login sessions.

The firewall queues commit requests and performs them in the order that administrators initiate the commits. For details, see Commit, Validate, and Preview Firewall Configuration Changes. *To view the status of queued commits, see* Manage and Monitor Administrative Tasks.

- View details about current locks.

  For example, you can check whether other administrators have set locks and read comments they entered to explain the locks.

  Click the lock at the top of the web interface. An adjacent number indicates the number of current locks.

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- Lock a configuration.
  1. Click the lock at the top of the web interface.

     The lock image varies based on whether existing locks are *or are not* *set.*
  2. **Take a Lock** and select the lock **Type**:
     - **Config**—Blocks other administrators from changing the candidate configuration.
     - **Commit**—Blocks other administrators from committing changes made to the candidate configuration.
  3. (Firewall with multiple virtual systems only) Select a **Location** to lock the configuration for a specific virtual system or the **Shared** location.
  4. (Optional) As a best practice, enter a **Comment** so that other administrators will understand the reason for the lock.
  5. Click **OK** and **Close**.

- Unlock a configuration.

  Only a superuser or the administrator who locked the configuration can manually unlock it. However, the firewall automatically removes a lock after completing the commit operation.
  1. Click the lock at the top of the web interface.
  2. Select the lock entry in the list.
  3. Click **Remove Lock**, **OK**, and **Close**.

- Configure the firewall to automatically apply a commit lock when you change the candidate configuration. This setting applies to all administrators.
  1. Select **Device** > **Setup** > **Management** and edit the General Settings.
  2. Select **Automatically Acquire Commit Lock** and then click **OK** and **Commit**.

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Manage Configuration Backups

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

The running configuration on the firewall comprises all settings you have committed and that are therefore active, such as policy rules that currently block or allow various types of traffic in your network. The candidate configuration is a copy of the running configuration plus any inactive changes that you made after the last commit. Saving backup versions of the running or candidate configuration enables you to later restore those versions. For example, if a commit validation shows that the current candidate configuration has more errors than you want to fix, you can restore a previous candidate configuration. You can also revert to the current running configuration without saving a backup first. If you need to export specific parts of the configuration for internal review or audit, you can Export Configuration Table Data.

See Commit, Validate, and Preview Firewall Configuration Changes for details about commit operations.

Perform a Config Audit

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

Perform a config audit to assess and document impact of configuration changes, trace back changes in case of an outage, and perform regular audits in order to adhere to security compliance standards. For firewall in an active/passive high availability (HA) configuration, you can only perform a config audit on the `active` HA peer. Performing a config audit on the `secondary` HA peer is not supported.

The Config Audit **Change Summary** supports config changes up to 25 MB in size. You can use the **XML Diff** if the config change size for the selected config versions is larger than 25 MB. A warning message is displayed when you view the **Change Summary** in the event one of the selected config versions has config changes larger than 25 MB.

STEP 1 | Log in to the firewall web interface.

STEP 2 | Select **Device** > **Config Audit**.

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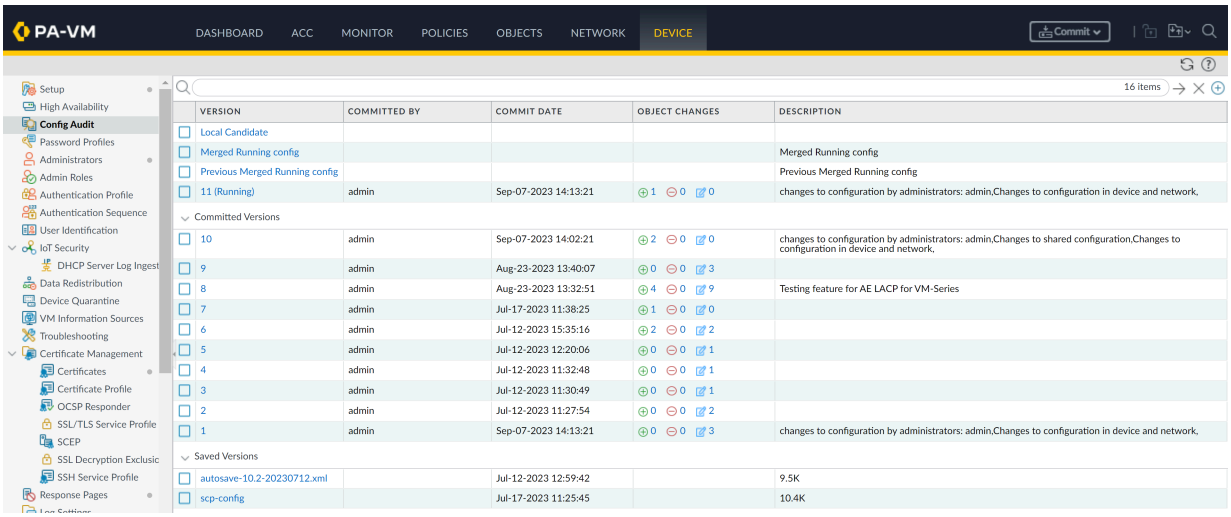
STEP 3 | A summary audit of the local and running conf version, previous config versions, and saved config versions is displayed.

- **Versions**—The commit version for a particular commit. The version is assigned to a config commit by default and is sequential.

  *(Panorama managed firewalls) The **Previous Merged Running Config** option supports both a successfully merged and failed merged configuration pushed from Panorama.*

- **Committed By**—Administrator that committed the config change.
- **Commit Date**—Date and time the config was committed.
- **Objects**—Summary of config changes that occurred in the commit version. The config change summary displayed is relative to the running config at the time of the commit.
  - ⊕—New configuration objects or policy rules were created as part of the commit.
  - ⊖—Existing configuration objects or policy rules were deleted as part of the commit.
  - —Existing configuration objects or policy rules were modified as part of the commit.
- **Description**—Commit description if added.



STEP 4 | Select up two config versions and **Compare Versions**.

When you select two versions that have multiple commit versions between them, the config audit displays the total sum of changes between the oldest and most recent config version. For example, when you compare versions 1 and 7, the config audit also displays all changes made in commit versions 2 through 6.

STEP 5 | The **XML Diff** displays a side-by-side comparison of the XML file differences between the two selected config versions.

The left-hand XML is the older version and the right-hand XML is the newer version. Objects highlighted in green newly added configuration objects. Objects highlighted in red are deleted

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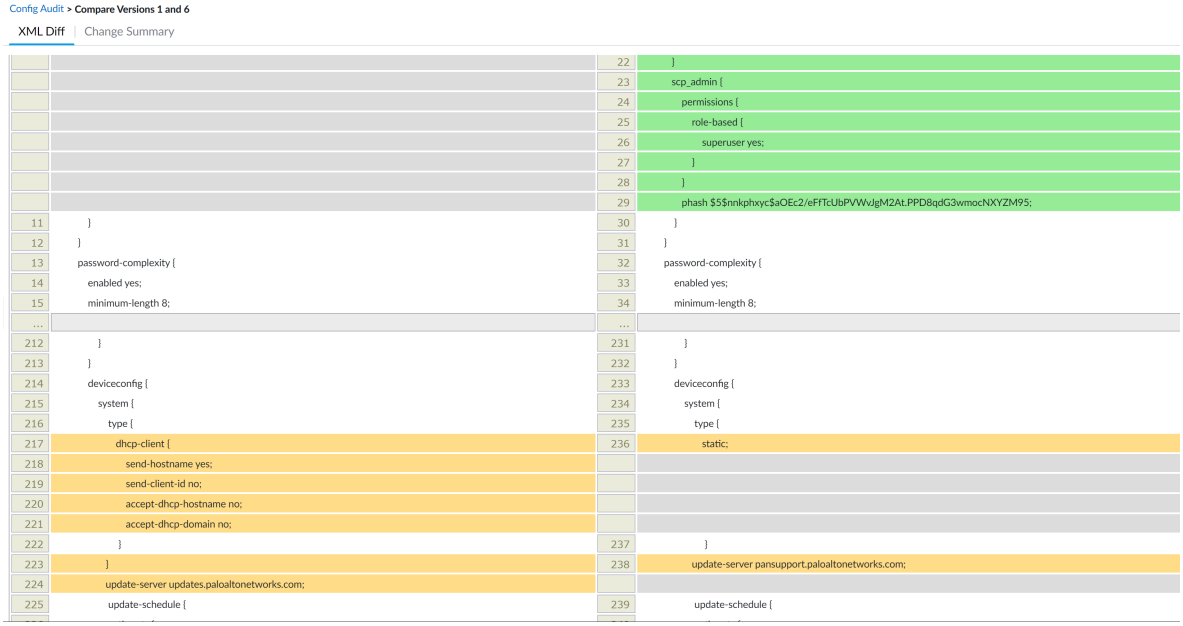
configuration objects. Objects highlighted in yellow are existing configuration objects that were modified.



		22	}
		23	scp_admin {
		24	permissions {
		25	role-based {
		26	superuser yes;
		27	}
		28	}
		29	phash 5nnkphxyc$aOEc2/eFfTcUbPVWvJgM2At.PPD8qdG3wmocNXYZM95;
11	}	30	}
12	}	31	}
13	password-complexity {	32	password-complexity {
14	enabled yes;	33	enabled yes;
15	minimum-length 8;	34	minimum-length 8;
...		...	
212	}	231	}
213	}	232	}
214	deviceconfig {	233	deviceconfig {
215	system {	234	system {
216	type {	235	type {
217	dhcp-client {	236	static;
218	send-hostname yes;		
219	send-client-id no;		
220	accept-dhcp-hostname no;		
221	accept-dhcp-domain no;		
222	}	237	}
223	}	238	update-server pansupport.paloaltonetworks.com;
224	update-server updates.paloaltonetworks.com;		
225	update-schedule {	239	update-schedule {

STEP 6 | The **Change Summary** displays a detailed list of the configuration objects associated with the selected config versions.

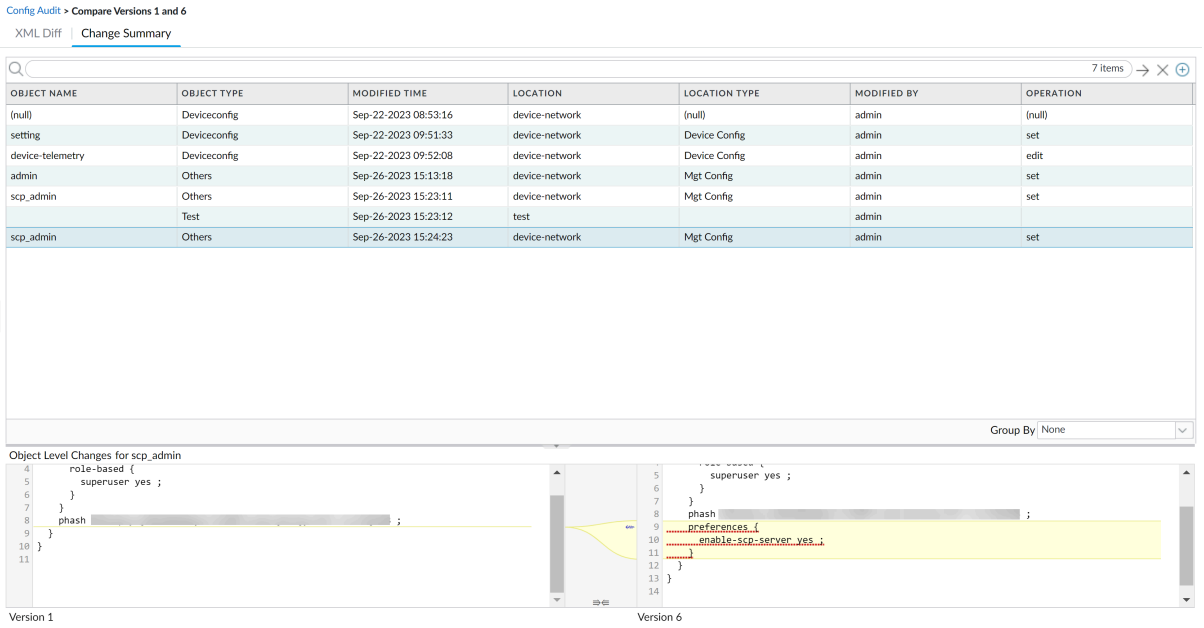
Review the change summary details to understand where and what configuration changes were made. Specifically, the `Operation` column displays what specific action was taken for the impacted configuration objects.

Select a specific **Object Name** to view the **Object Level Changes** for the configuration object between the selected config versions. This shows you an XML snippet highlighting what changed.

- **Set**—New configuration object was added.
- **Edit**—Existing configuration object was modified.
- **Rename**—Existing configuration object was renamed.
- **Move**—Reordering or moving policy rules within a rulebase.
- **Delete**—Configuration object was deleted.

The following operations may display as either two separate operations, or may not be displayed at all.

- *Renaming an existing configuration object is displayed as two separate changes. The first is a `delete rename` operation of the object with the old name. The second is a `edit create` for the same object with the new name.*
- *The **Move** operation is displayed only in the Change Summary. Policy rules that are moved are not displayed in the **XML Diff**.*
- *Config Audit cannot capture Load and Revert operations.*
- *(HA only) For firewalls in an Active/Passive high availability (HA) configuration, a config audit is supported only on the primary HA peer. You cannot perform a config audit from the web interface of the secondary HA peer.*



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Save and Export Firewall Configurations

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

Saving a backup of the candidate configuration to persistent storage on the firewall enables you to later revert to that backup (see Revert Firewall Configuration Changes). This is useful for preserving changes that would otherwise be lost if a system event or administrator action causes the firewall to reboot. After rebooting, PAN-OS automatically reverts to the current version of the running configuration, which the firewall stores in a file named running-config.xml. Saving backups is also useful if you want to revert to a firewall configuration that is earlier than the current running configuration. The firewall does not automatically save the candidate configuration to persistent storage. You must manually save the candidate configuration as a default snapshot file (.snapshot.xml) or as a custom-named snapshot file. The firewall stores the snapshot file locally but you can export it to an external host.

*You don't have to save a configuration backup to revert the changes made since the last commit or reboot; just select **Config** > **Revert Changes** (see Revert Firewall Configuration Changes).*

*When you edit a setting and click **OK**, the firewall updates the candidate configuration but does not save a backup snapshot.*

Additionally, saving changes does not activate them. To activate changes, perform a commit (see Commit, Validate, and Preview Firewall Configuration Changes).

Palo Alto Networks recommends that you back up any important configuration to a host external to the firewall.

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STEP 1 | Save a local backup snapshot of the candidate configuration if it contains changes that you want to preserve in the event the firewall reboots.

These are changes you are not ready to commit—for example, changes you cannot finish in the current login session.

To overwrite the default snapshot file (.snapshot.xml) with all the changes that all administrators made, perform one of the following steps:

- Select **Device** > **Setup** > **Operations** and **Save candidate configuration**.
- Log in to the firewall with an administrative account that is assigned the Superuser role or an Admin Role profile with the **Save For Other Admins** privilege enabled. Then select **Config** > **Save Changes** at the top of the web interface, select **Save All Changes** and **Save**.

To create a snapshot that includes all the changes that all administrators made but without overwriting the default snapshot file:

1. Select **Device** > **Setup** > **Operations** and **Save named configuration snapshot**.
2. Specify the **Name** of a new or existing configuration file.
3. Click **OK** and **Close**.

To save only specific changes to the candidate configuration without overwriting any part of the default snapshot file:

1. Log in to the firewall with an administrative account that has the role privileges required to save the desired changes.
2. Select **Config** > **Save Changes** at the top of the web interface.
3. Select **Save Changes Made By**.
4. To filter the Save Scope by administrator, click **<administrator-name>**, select the administrators, and click **OK**.
5. To filter the Save Scope by location, clear any locations that you want to exclude. The locations can be specific virtual systems, shared policies and objects, or shared device and network settings.
6. Click **Save**, specify the **Name** of a new or existing configuration file, and click **OK**.

STEP 2 | Export a candidate configuration, a running configuration, or the firewall state information to a host external to the firewall.

Select **Device** > **Setup** > **Operations** and click an export option:

- **Export named configuration snapshot**—Export the current running configuration, a named candidate configuration snapshot, or a previously imported configuration (candidate or running). The firewall exports the configuration as an XML file with the **Name** you specify.
- **Export configuration version**—Select a **Version** of the running configuration to export as an XML file. The firewall creates a version whenever you commit configuration changes.
- **Export device state**—Export the firewall state information as a bundle. Besides the running configuration, the state information includes device group and template settings pushed from Panorama. If the firewall is a GlobalProtect portal, the information also includes certificate information, a list of satellites, and satellite authentication information. If you replace a firewall or portal, you can restore the exported information on the replacement by importing the state bundle.

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Revert Firewall Configuration Changes

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

Revert operations replace settings in the current candidate configuration with settings from another configuration. Reverting changes is useful when you want to undo changes to multiple settings as a single operation instead of manually reconfiguring each setting.

You can revert pending changes that were made to the firewall configuration since the last commit. The firewall provides the option to filter the pending changes by administrator or *location*. The locations can be specific virtual systems, shared policies and objects, or shared device and network settings. If you saved a snapshot file for a candidate configuration that is earlier than the current running configuration (see Save and Export Firewall Configurations), you can also revert to that snapshot. Reverting to a snapshot enables you to restore a candidate configuration that existed before the last commit. The firewall automatically saves a new version of the running configuration whenever you commit changes, and you can restore any of those versions.

- Revert to the current running configuration (file named running-config.xml).

  This operation undoes changes you made to the candidate configuration since the last commit.

  To revert all the changes that all administrators made, perform one of the following steps:

  - Select **Device** > **Setup** > **Operations, Revert to running configuration**, and click **Yes** to confirm the operation.
  - Log in to the firewall with an administrative account that is assigned the Superuser role or an Admin Role profile with the **Commit For Other Admins** privilege enabled. Then select **Config** > **Revert Changes** at the top of the web interface, select **Revert All Changes** and **Revert**.

  To revert only specific changes to the candidate configuration:

  1. Log in to the firewall with an administrative account that has the role privileges required to revert the desired changes.

     The privileges that control commit operations also control revert operations.

  2. Select **Config** > **Revert Changes** at the top of the web interface.
  3. Select **Revert Changes Made By**.
  4. To filter the Revert Scope by administrator, click **<administrator-name>**, select the administrators, and click **OK**.
  5. To filter the Revert Scope by location, clear any locations that you want to exclude.
  6. **Revert** the changes.

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- Revert to the default snapshot of the candidate configuration.

  This is the snapshot that you create or overwrite when you click **Config** > **Save Changes** at the top of the web interface.

  1. Select **Device** > **Setup** > **Operations** and **Revert to last saved configuration**.
  2. Click **Yes** to confirm the operation.
  3. (Optional) Click **Commit** to overwrite the running configuration with the snapshot.

- Revert to a previous version of the running configuration that is stored on the firewall.

  The firewall creates a version whenever you commit configuration changes.

  1. Select **Device** > **Setup** > **Operations** and **Load configuration version**.
  2. Select a configuration **Version** and click **OK**.
  3. (Optional) Click **Commit** to overwrite the running configuration with the version you just restored.

- Revert to one of the following:
  - Custom-named version of the running configuration that you previously imported
  - Custom-named candidate configuration snapshot (instead of the default snapshot)

  1. Select **Device** > **Setup** > **Operations** and click **Load named configuration snapshot**.
  2. Select the snapshot **Name** and click **OK**.
  3. (Optional) Click **Commit** to overwrite the running configuration with the snapshot.

- Revert to a running or candidate configuration that you previously exported to an external host.
  1. Select **Device** > **Setup** > **Operations**, click **Import named configuration snapshot**, **Browse** to the configuration file on the external host, and click **OK**.
  2. Click **Load named configuration snapshot**, select the **Name** of the configuration file you just imported, and click **OK**.
  3. (Optional) Click **Commit** to overwrite the running configuration with the snapshot you just imported.

- Restore state information that you exported from a firewall.

  Besides the running configuration, the state information includes device group and template settings pushed from Panorama. If the firewall is a GlobalProtect portal, the information also includes certificate information, a list of satellites, and satellite authentication information. If you replace a firewall or portal, can you can restore the information on the replacement by importing the state bundle.

  Import state information:

  1. Select **Device** > **Setup** > **Operations**, click **Import device state**, **Browse** to the state bundle, and click **OK**.
  2. (Optional) Click **Commit** to apply the imported state information to the running configuration.

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Manage Firewall Administrators

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

Administrative accounts specify roles and authentication methods for the administrators of Palo Alto Networks firewalls. Every Palo Alto Networks firewall has a predefined default administrative account (admin) that provides full read-write access (also known as superuser access) to the firewall.

As a best practice, create a separate administrative account for each person who needs access to the administrative or reporting functions of the firewall. This enables you to better protect the firewall from unauthorized configuration and enables logging of the actions of individual administrators. Make sure you are following the Administrative Access Best Practices to ensure that you are securing administrative access to your firewalls and other security devices in a way that prevents successful attacks.

Administrative Role Types

A *role* defines the type of access that an administrator has to the firewall. The Administrator Types are:

- **Role Based**—Custom roles you can configure for more granular access control over the functional areas of the web interface, CLI, and XML API. For example, you can create an Admin Role profile for your operations staff that provides access to the firewall and network configuration areas of the web interface and a separate profile for your security administrators that provides access to security policy definitions, logs, and reports. On a firewall with multiple virtual systems, you can select whether the role defines access for all virtual systems or specific virtual systems. When new features are added to the product, you must update the roles with corresponding access privileges: the firewall does not automatically add new features to custom role definitions. For details on the privileges you can configure for custom administrator roles, see Reference: Web Interface Administrator Access.
- **Dynamic**—Built-in roles that provide access to the firewall. When new features are added, the firewall automatically updates the definitions of dynamic roles; you never need to manually update them. The following table lists the access privileges associated with dynamic roles.

Dynamic Role	Privileges
Superuser	Full access to the firewall, including defining new administrator accounts and virtual systems. You must have Superuser privileges to create an administrative user with Superuser privileges.
Superuser (read-only)	Read-only access to the firewall (enables the XML API in a read-only state).

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Dynamic Role	Privileges
Device administrator	Full access to all firewall settings except for defining new accounts or virtual systems.
Device administrator (read-only)	Read-only access to all firewall settings except password profiles (no access) and administrator accounts (only the logged in account is visible).
Virtual system administrator	Access to selected virtual systems on the firewall to create and manage specific aspects of virtual systems. A virtual system administrator doesn't have access to network interfaces, VLANs, virtual wires, virtual routers, IPSec tunnels, GRE tunnels, DHCP, DNS Proxy, QoS, LLDP, or network profiles.
Virtual system administrator (read-only)	Read-only access to selected virtual systems on the firewall and specific aspects of virtual systems. A virtual system administrator with read-only access doesn't have access to network interfaces, VLANs, virtual wires, virtual routers, IPSec tunnels, GRE tunnels, DHCP, DNS Proxy, QoS, LLDP, or network profiles.

Configure an Admin Role Profile

Where Can I Use This?	What Do I Need?
• NGFW	One of these licenses for Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Essentials ❑ Strata Cloud Manager Pro

Admin Role profiles enable you to define granular administrative access privileges to ensure protection for sensitive company information and privacy for end users.

- **Defining Access Levels:** Admin Role profiles function as a foundational element of security, allowing you to define specific access levels and permissions for different user types. By creating distinct roles, you can control what administrators can access from full access to read-only access. This ensures that users only have the privileges necessary for their responsibilities.

  Follow the principle of least privilege access to create Admin Role profiles that enable administrators to access only the areas of the management interface that they need to access to perform their jobs and follow Administrative Access Best Practices.

- **Configuring Interface Privileges:** Each role is highly customizable and can be configured to manage access across various device interfaces, including the **Web UI**, **REST API**, **XML API**, and **Command Line (CLI)**. This flexibility lets you tailor permissions precisely to your operational

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needs. For example, you can create a role for a junior administrator with full access to the Web UI but no access to the CLI, or for a monitoring team with only read-only access across all interfaces.

- **Role-Based Management:** Once a role is defined, it is assigned to one or more administrator accounts instead of configuring permissions for each individual user.
- PAN-OS & Panorama
- Strata Cloud Manager

Configure an Admin Role Profile (PAN-OS)

You can create an Admin Role profile, specify that the role applies to Virtual System, and then select Web UI, for example, and choose the part of the configuration that the administrator can control within a virtual system. Click OK to save the Admin Role Profile. Then select **Device** > **Administrators**, name the role, select Role Based, enter the name of the Admin Role Profile, and select the virtual system that the administrator can control. The MGT interface doesn't give full access to the firewall; access is controlled by the Admin Role.

If the Admin Role Profile is based on Virtual System, that administrator won't have control over a virtual router. Only a subset of the Network options are available in a Virtual System role, and virtual router isn't one of the included options. If you want virtual router available in an Admin Role Profile, the role must be Device, not Virtual System. (You can define a superuser Administrator to have both Virtual System and Virtual Router access.)

You can create a second Admin Role Profile, specify that the role applies to Device, and then select portions under Network, such as Virtual Routers. Name the Admin Role Profile, and then apply it to a different administrator.

You might have different departments that have different functions. Based on the login, the administrator gets the right to control the objects enabled in the Admin Role Profile.

In summary, you can't define a Virtual System Admin Role profile that includes routing (Virtual Router). You can create two accounts to have these separate roles and assign them to two different users. An Administrator account can have only one Admin Role profile.

The MGT interface can have role-based access; it doesn't strictly provide full access to the device. The login account (Admin Role) is what gives a user rights or limited access to the objects, not the MGT interface.

STEP 1 | Select **Device** > **Admin Roles** and click **Add**.

STEP 2 | Enter a **Name** to identify the role.

STEP 3 | For the scope of the **Role**, select **Device** or **Virtual System**.

STEP 4 | In the **Web UI** and **REST API** tabs, click the icon for each functional area to toggle it to the desired setting: Enable, Read Only or Disable. For the **XML API** tab select, Enable or Disable. For details on the **Web UI** options, see Web Interface Access Privileges.

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STEP 5 | Select the **Command Line** tab and select a CLI access option. The **Role** scope controls the available options:

- **Device** role:
  - **None**—CLI access is not permitted (default).
  - **superuser**—Full access. Can define new administrator accounts and virtual systems. Only a superuser can create administrator users with superuser privileges.
  - **superreader**—Full read-only access.
  - **deviceadmin**—Full access to all settings except defining new accounts or virtual systems.
  - **devicereader**—Read-only access to all settings except password profiles (no access) and administrator accounts (only the logged in account is visible).
- **Virtual System** role:
  - **None**—Access is not permitted (default).
  - **vsysadmin**—Access to specific virtual systems to create and manage specific aspects of virtual systems. Does not enable access to firewall-level or network-level functions including static and dynamic routing, interface IP addresses, IPSec tunnels, VLANs, virtual wires, virtual routers, GRE tunnels, DCHP, DNS Proxy, QoS, LLDP, or network profiles.
  - **vsysreader**—Read-only access to specific virtual systems to specific aspects of virtual systems. Does not enable access to firewall-level or network-level functions including static and dynamic routing, interface IP addresses, IPSec tunnels, VLANs, virtual wires, virtual routers, GRE tunnels, DCHP, DNS Proxy, QoS, LLDP, or network profiles.

STEP 6 | Click **OK** to save the profile.

STEP 7 | Assign the role to an administrator. See Configure a Firewall Administrator Account.

Configure an Admin Role Profile (Strata Cloud Manager)

In Strata Cloud Manager, you can create and customize admin role profiles to define granular access permissions. You can control which parts of the firewall configuration an administrator can manage across the web UI, REST API, XML API, and command line interfaces.

STEP 1 | Select **Device Settings** > **Admin Roles** and click **Add Admin Role**.

STEP 2 | Enter a **Name** to identify the role.

STEP 3 | In the **Web UI** and **REST API** tabs, select the required feature to toggle it to the desired setting: Enable, Read Only or Disable. For the **XML API** tab select, Enable or Disable. For details on the **Web UI** options, see Web Interface Access Privileges.

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STEP 4 | Select the **Command Line** tab and select a CLI access option.

- **None**—CLI access is not permitted (default).
- **superuser**—Full access. Can define new administrator accounts and virtual systems. Only a superuser can create administrator users with superuser privileges.
- **superreader**—Full read-only access.
- **deviceadmin**—Full access to all settings except defining new accounts or virtual systems.
- **devicereader**—Read-only access to all settings except password profiles (no access) and administrator accounts (only the logged in account is visible).

STEP 5 | Click **OK** to save the profile.

STEP 6 | Assign the role to an administrator. See Configure a Firewall Administrator Account.

Example Admin Role Profile Construction

This example (in PAN-OS) shows an Admin Role profile for a Security Operations Center (SOC) manager who needs access to investigate potential issues. The SOC Manager needs read access to many areas of the firewall, but generally doesn't need write access. The example covers all four of the Admin Role Profile's tabs and each step describes why the profile enables or disables a particular area of access to the SOC manager.

This is an example profile for a fictional SOC manager. Configure Admin Role profiles for your administrators based on the functions they manage and the access required to do their job. Do not enable unnecessary access. Create separate profiles for each administrative group that shares the same duties and for administrators who have unique duties. Each administrator should have the exact level of access required to perform their duties and no access beyond that.

STEP 1 | Configure Web UI access permissions. Each snip of the Web UI screen shows a different area of Web UI permissions. Permissions are listed by firewall tab, in the order you see the tabs in the Web UI, followed by permissions for other actions.

The **Dashboard**, **ACC**, and **Monitor** > **Logs** areas of the firewall don't contain configuration elements—all of the objects are informational (you can only toggle them between enable and

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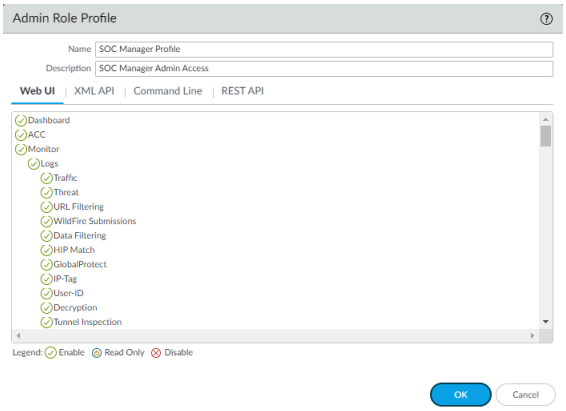
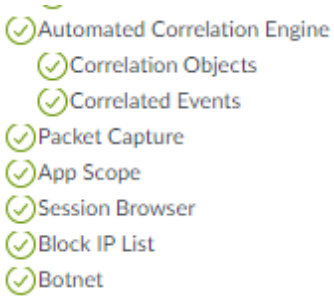
disable because they are already read only). Because the SOC Manager needs to investigate potential issues, the SOC Manager needs access to the information on these tabs.

The profile name and description make it easy to understand the profile's objective. This snip doesn't show all of the **Logs** permissions, but all of them are enabled for this profile.



The next snip shows permissions for more informational objects on the **Monitor** tab. The SOC Manager uses these tools to investigate potential issues and therefore requires access.



The next two snips show permissions for PDF Reports, Custom Reports, and predefined reports on the **Monitor** tab. While the SOC Manager needs access to PDF reports to gather information, in this example, the SOC Manager does not need to configure reports, so access is set to read-only (summary reports are not configurable). However, the SOC Manager needs to manage custom reports to investigate specific potential issues, so full access permissions are

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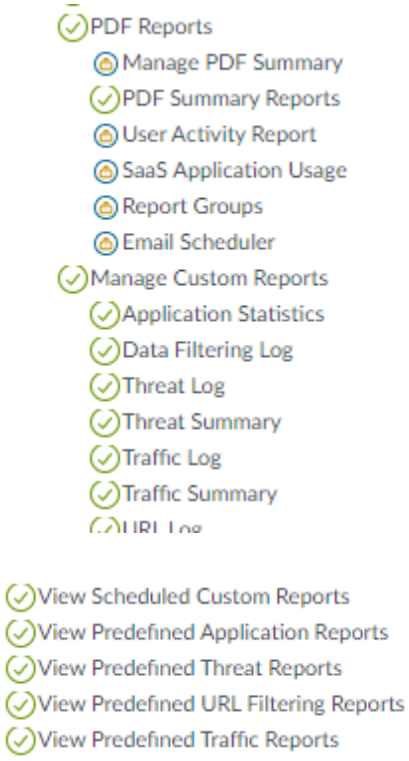
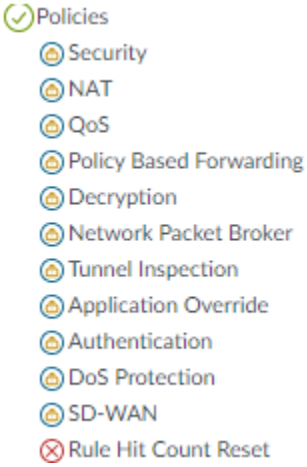
granted for all custom reports (including those not shown in the snip). Finally the SOC Manager requires access to predefined reports for investigating potential issues.



Because the SOC Manager is an investigator and not an administrator who configures the firewall, permissions for the **Policies** tab are read-only, with the exception of resetting the rule hit count. Resetting the rule hit count is not one of the SOC Manager's duties (and changing the hit count could adversely affect or confuse other administrators), so access is disabled. Read access enables the SOC Manager to investigate the construction of a policy that the SOC Manager suspects may have caused an issue.



Permissions for the **Objects** tab are also read-only for the same reason—the SOC Manager's job doesn't require configuration, so no configuration permissions are assigned. For areas that aren't included in the SOC Manager's duties, access is disabled. In this example, the SOC Manager has read-only access to investigate objects configurations for all objects except **URL**

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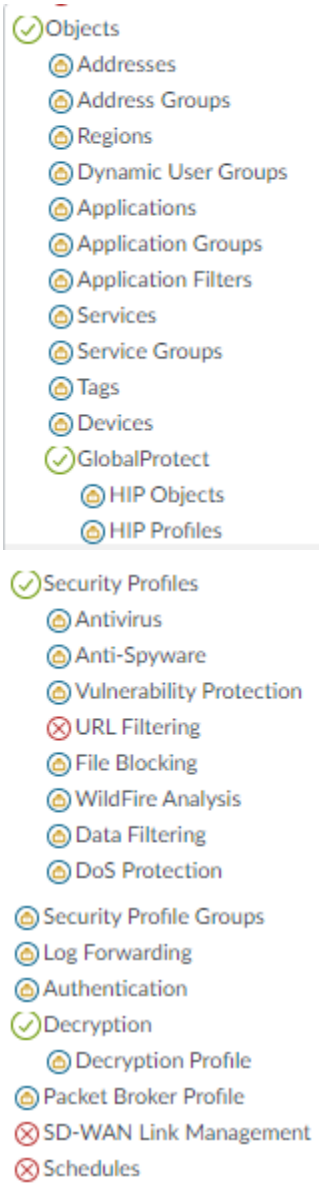
Filtering, **SD-WAN Link Management** and **Schedules**, which are under the control of different administrators in this example.



For **Network** tab permissions, the scenario is similar: the SOC Manager doesn't need to configure any of the objects, but may need information to investigate issues, so read-only access is assigned to the areas that the SOC Manager may need to investigate. In this example,

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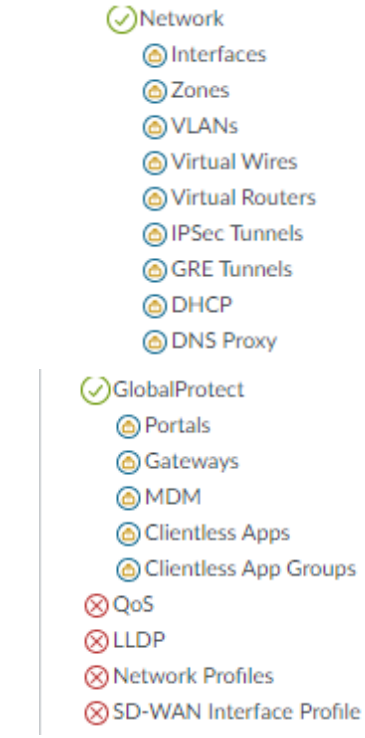
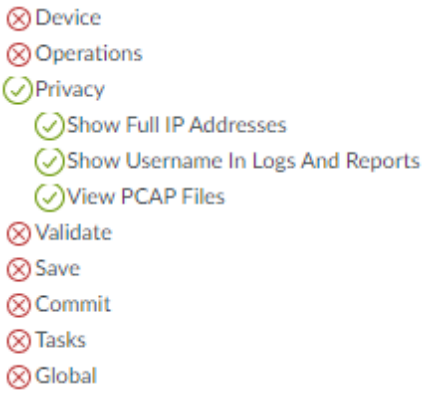
access is disabled for QoS, LLDP, Network Profiles, or SD-WAN Interface profiles because these items are not part of the SOC Manager's duties.



In this example, the SOC Manager needs no access to the **Device** tab capabilities for investigative purposes, so all **Device** tab permissions are blocked. In addition, investigation doesn't require commit actions or access to any of the remaining actions, so those permissions are also blocked.



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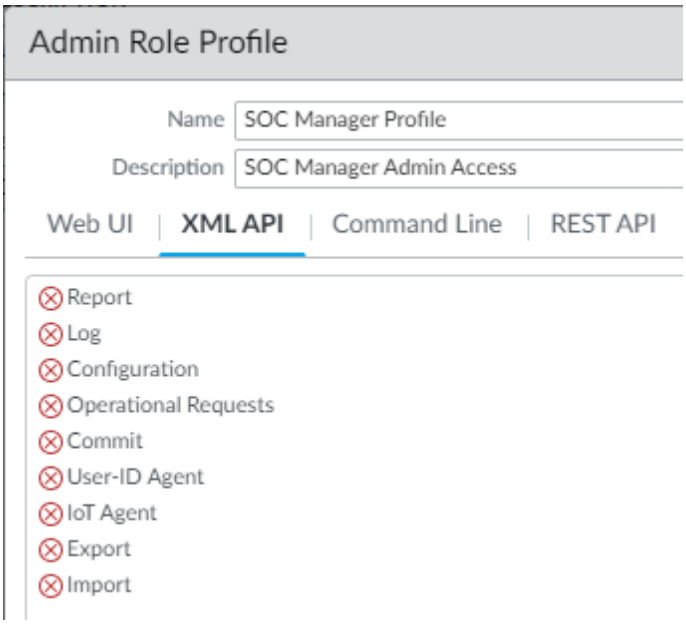
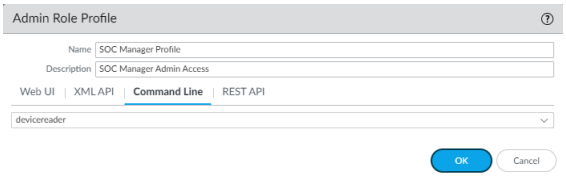
STEP 2 | Configure XML API access permissions.

The following snip shows that all XML API permissions are disabled for the SOC Manager because the SOC Manager doesn't access the firewall using XML API commands.



STEP 3 | Configure Command Line (CLI) access permissions.

CLI access permissions are read-only for the SOC Manager because the SOC Manager needs access to logs and other monitoring tools and also needs to be able to see certain configurations in order to investigate potential issues. However, the SOC Manager doesn't configure the firewall, so no configuration permissions are assigned. The access level is set to **devicereader** instead of to **superreader** because the SOC Manager doesn't need access to password profiles or to other administrative accounts.



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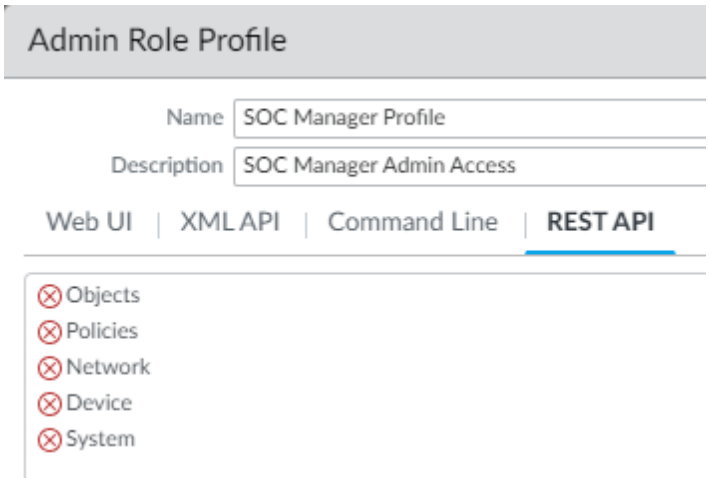
STEP 4 | Configure REST API access permissions.

The SOC Manager doesn't access the firewall using REST API commands, so all REST API access is disabled.



Administrative Authentication

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

You can configure the following types of authentication and authorization (role and access domain assignment) for firewall administrators:

Authentication Method	Authorization Method	Description
Local	Local	The administrative account credentials and authentication mechanisms are local to the firewall. You can define the accounts with or without a user database that is local to the firewall—see Local Authentication for the advantages and disadvantages of using a local database. You use the firewall to manage role assignments but access domains are not supported. For details, see Configure Local or External Authentication for Firewall Administrators.
SSH Keys	Local	The administrative accounts are local to the firewall, but authentication to the CLI is based on SSH keys. You use the firewall to manage role assignments but access domains are not supported. For details, see Configure SSH Key-Based Administrator Authentication to the CLI.
Certificates	Local	The administrative accounts are local to the firewall, but authentication to the web interface is based on client certificates.

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Authentication Method	Authorization Method	Description
		You use the firewall to manage role assignments but access domains are not supported. For details, see Configure Certificate-Based Administrator Authentication to the Web Interface.
External service	Local	The administrative accounts you define locally on the firewall serve as references to the accounts defined on an external Multi-Factor Authentication, SAML, Kerberos, TACACS+, RADIUS, or LDAP server. The external server performs authentication. You use the firewall to manage role assignments but access domains are not supported. For details, see Configure Local or External Authentication for Firewall Administrators.
External service	External service	The administrative accounts are defined on an external SAML, TACACS+, or RADIUS server. The server performs both authentication and authorization. For authorization, you define Vendor-Specific Attributes (VSAs) on the TACACS+ or RADIUS server, or SAML attributes on the SAML server. PAN-OS maps the attributes to administrator roles, access domains, user groups, and virtual systems that you define on the firewall. For details, see: • Configure SAML Authentication • Configure TACACS+ Authentication • Configure RADIUS Authentication

Configure Administrative Accounts and Authentication

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

If you have already configured an authentication profile (see Configure an Authentication Profile and Sequence) or you don't require one to authenticate administrators, you are ready to Configure a Firewall Administrator Account. Otherwise, perform one of the other procedures in this section to configure administrative accounts for specific types of authentication.

Configure a Firewall Administrator Account

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

Administrative accounts specify roles and authentication methods for firewall administrators. The service that you use to assign roles and perform authentication determines whether you add

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the accounts on the firewall, on an external server, or both (see Administrative Authentication). If the authentication method relies on a local firewall database or an external service, you must configure an authentication profile before adding an administrative account (see Configure Administrative Accounts and Authentication). If you already configured the authentication profile or you will use Local Authentication without a firewall database, perform the following steps to add an administrative account on the firewall.

Create a separate administrative account for each person who needs access to the administrative or reporting functions of the firewall. This enables you to better protect the firewall from unauthorized configuration and enables logging of the actions of individual administrators.

Make sure you are following the Adminstrative Access Best Practices to ensure that you are securing administrative access to your firewalls and other security devices in a way that prevents successful attacks.

STEP 1 | Modify the number of supported administrator accounts.

Configure the total number of supported concurrent administrative accounts sessions for a firewall in the normal operational mode or in FIPS-CC mode. You can allow up to four concurrent administrative account sessions or configure the firewall to support an unlimited number of concurrent administrative account sessions.

1. Select **Device** > **Setup** > **Management** and edit the Authentication Settings.
2. Edit the **Max Session Count** to specify the number of supported concurrent sessions (range is **0** to **4**) allowed for all administrator and user accounts.

   Enter **0** to configure the firewall to support an unlimited number of administrative accounts.
3. Edit the **Max Session Time** in minutes for an administrative account. Default is 720 minutes.
4. Click **OK**.
5. **Commit**.

You can also configure the total number of supported concurrent sessions by logging in to the firewall CLI.

```
admin> configure
```

```
admin# set deviceconfig setting management admin-session
 max-session-count <0-4>
```

```
admin# set deviceconfig setting management admin-session
 max-session-time <0, 60-1499>
```

```
admin# commit
```

STEP 2 | Select **Device** > **Administrators** and **Add** an account.

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STEP 3 | Enter a user **Name**.

If the firewall uses a local user database to authenticate the account, enter the name that you specified for the account in the database (see Add the user group to the local database.)

STEP 4 | Select an **Authentication Profile** or sequence if you configured either for the administrator.

If the firewall uses Local Authentication without a local user database for the account, select **None** (default) and enter a **Password**.

STEP 5 | Select the **Administrator Type**.

If you configured a custom role for the user, select **Role Based** and select the Admin Role **Profile**. Otherwise, select **Dynamic** (default) and select a dynamic role. If the dynamic role is **virtual system administrator**, add one or more virtual systems that the virtual system administrator is allowed to manage.

STEP 6 | (Optional) Select a **Password Profile** for administrators that the firewall authenticates locally without a local user database. For details, see Define a Password Profile.

STEP 7 | Click **OK** and **Commit**.

Configure Local or External Authentication for Firewall Administrators

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

You can use Local Authentication and External Authentication Services to authenticate administrators who access the firewall. These authentication methods prompt administrators to respond to one or more authentication challenges, such as a login page for entering a username and password.

If you use an external service to manage both authentication and authorization (role and access domain assignments), see:

- Configure SAML Authentication
- Configure TACACS+ Authentication
- Configure RADIUS Authentication

To authenticate administrators without a challenge-response mechanism, you can Configure Certificate-Based Administrator Authentication to the Web Interface *and* Configure SSH Key-Based Administrator Authentication to the CLI.

STEP 1 | (External authentication only) Enable the firewall to connect to an external server for authenticating administrators.

Configure a server profile:

- Add a RADIUS server profile.

  If the firewall integrates with a Multi-Factor Authentication (MFA) service through RADIUS, you must add a RADIUS server profile. In this case, the MFA service provides all the

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authentication factors (challenges). If the firewall integrates with an MFA service through a vendor API, you can still use a RADIUS server profile for the first factor but MFA server profiles are required for additional factors.

- Add an MFA server profile.
- Add a TACACS+ server profile.
- Add a SAML IdP server profile. You cannot combine Kerberos single sign-on (SSO) with SAML SSO; you can use only one type of SSO service.
- Add a Kerberos server profile.
- Add an LDAP server profile.

STEP 2 | (Local database authentication only) Configure a user database that is local to the firewall.

1. Add the user account to the local database.
2. (Optional) Add the user group to the local database.

STEP 3 | (Local authentication only) Define password complexity and expiration settings.

These settings help protect the firewall against unauthorized access by making it harder for attackers to guess passwords.

1. Define global password complexity and expiration settings for all local administrators. The settings don't apply to local database accounts for which you specified a password hash instead of a password (see Local Authentication).
   1. Select **Device** > **Setup** > **Management** and edit the Minimum Password Complexity settings.
   2. Select **Enabled**.
   3. Define the password settings and click **OK**.
2. Define a Password Profile.

   You assign the profile to administrator accounts for which you want to override the global password expiration settings. The profiles are available only to accounts that are not associated with a local database (see Local Authentication).
   1. Select **Device** > **Password Profiles** and **Add** a profile.
   2. Enter a **Name** to identify the profile.
   3. Define the password expiration settings and click **OK**.

STEP 4 | (Kerberos SSO only) Create a Kerberos keytab.

A keytab is a file that contains Kerberos account information for the firewall. To support Kerberos SSO, your network must have a Kerberos infrastructure.

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STEP 5 | Configure an authentication profile.

If your administrative accounts are stored across multiple types of servers, you can create an authentication profile for each type and add all the profiles to an authentication sequence.

Configure an Authentication Profile and Sequence. In the authentication profile, specify the **Type** of authentication service and related settings:

- **External service**—Select the **Type** of external service and select the **Server Profile** you created for it.
- **Local database authentication**—Set the **Type** to **Local Database**.
- **Local authentication without a database**—Set the **Type** to **None**.
- **Kerberos SSO**—Specify the **Kerberos Realm** and **Import** the **Kerberos Keytab**.

STEP 6 | Assign the authentication profile or sequence to an administrator account.

1. Configure a Firewall Administrator Account.
   - Assign the **Authentication Profile** or sequence that you configured.
   - (Local database authentication only) Specify the **Name** of the user account you added to the local database.
2. **Commit** your changes.
3. (Optional) Test Authentication Server Connectivity to verify that the firewall can use the authentication profile to authenticate administrators.

Configure Certificate-Based Administrator Authentication to the Web Interface

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

As a more secure alternative to password-based authentication to the firewall web interface, you can configure certificate-based authentication for administrator accounts that are local to the firewall. Certificate-based authentication involves the exchange and verification of a digital signature instead of a password.

Configuring certificate-based authentication for any administrator disables the username/password logins for all administrators on the firewall; administrators thereafter require the certificate to log in.

STEP 1 | Generate a certificate authority (CA) certificate on the firewall.

You will use this CA certificate to sign the client certificate of each administrator.

Create a Self-Signed Root CA Certificate.

Alternatively, Import a Certificate and Private Key *from your enterprise CA or a third-party CA.*

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STEP 2 | Configure a certificate profile for securing access to the web interface.

Configure a Certificate Profile.

- Set the **Username Field** to **Subject**.
- In the CA Certificates section, **Add** the **CA Certificate** you just created or imported.

STEP 3 | Configure the firewall to use the certificate profile for authenticating administrators.

1. Select **Device** > **Setup** > **Management** and edit the Authentication Settings.
2. Select the **Certificate Profile** you created for authenticating administrators and click **OK**.

STEP 4 | Configure the administrator accounts to use client certificate authentication.

For each administrator who will access the firewall web interface, Configure a Firewall Administrator Account and select **Use only client certificate authentication**.

If you have already deployed client certificates that your enterprise CA generated, skip to Step 8. Otherwise, go to Step 5.

STEP 5 | Generate a client certificate for each administrator.

Generate a Certificate. In the **Signed By** drop-down, select a self-signed root CA certificate.

STEP 6 | Export the client certificate.

1. Export a Certificate and Private Key.
2. **Commit** your changes. The firewall restarts and terminates your login session. Thereafter, administrators can access the web interface only from client systems that have the client certificate you generated.

STEP 7 | Import the client certificate into the client system of each administrator who will access the web interface.

Refer to your web browser documentation.

STEP 8 | Verify that administrators can access the web interface.

1. Open the firewall IP address in a browser on the computer that has the client certificate.
2. When prompted, select the certificate you imported and click **OK**. The browser displays a certificate warning.
3. Add the certificate to the browser exception list.
4. Click **Login**. The web interface should appear without prompting you for a username or password.

Configure SSH Key-Based Administrator Authentication to the CLI

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

For administrators who use Secure Shell (SSH) to access the CLI of a Palo Alto Networks firewall, SSH keys provide a more secure authentication method than passwords. SSH keys almost

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eliminate the risk of brute-force attacks, provide the option for two-factor authentication (key and passphrase), and don't send passwords over the network. SSH keys also enable automated scripts to access the CLI.

STEP 1 | Use an SSH key generation tool to create an asymmetric keypair on the client system of the administrator.

The supported key formats are IETF SECSH and Open SSH. The supported algorithms are Ed25519 (256 bits) and RSA (768-4,096 bits).

For the commands to generate the keypair, refer to your SSH client documentation.

The public key and private key are separate files. Save both to a location that the firewall can access. For added security, enter a passphrase to encrypt the private key. The firewall prompts the administrator for this passphrase during login.

STEP 2 | Configure the administrator account to use public key authentication.

1. Configure a Firewall Administrator Account.
   - Configure the authentication method to use as a fallback if SSH key authentication fails. If you configured an **Authentication Profile** for the administrator, select it in the drop-down. If you select **None**, you must enter a **Password** and **Confirm Password**.
   - Select **Use Public Key Authentication (SSH)**, then **Import Key**, **Browse** to the public key you just generated, and click **OK**.
2. **Commit** your changes.

STEP 3 | Configure the SSH client to use the private key to authenticate to the firewall.

Perform this task on the client system of the administrator. For the steps, refer to your SSH client documentation.

STEP 4 | Verify that the administrator can access the firewall CLI using SSH key authentication.

1. Use a browser on the client system of the administrator to go to the firewall IP address.
2. Log in to the firewall CLI as the administrator. After entering a username, you will see the following output (the key value is an example):

```
Authenticating with public key "rsa-key-20130415"
```

3. If prompted, enter the passphrase you defined when creating the keys.

Configure API Key Lifetime

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

The API keys on the firewall and Panorama enable you to authenticate API calls to the XML API and REST API. Because these keys grant access to the firewall and Panorama that are critical elements of your security posture, as a best practice, specify an API key lifetime to enforce regular key rotation. After you specify the key lifetime, when you regenerate an API key, each key is unique.

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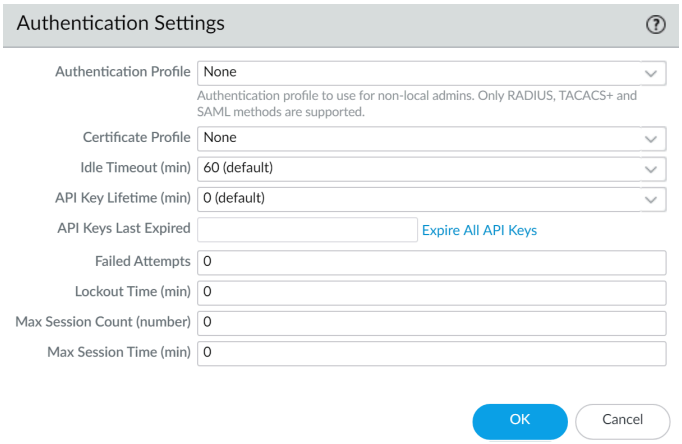
In addition to setting a key lifetime that prompts you to regenerate new keys periodically, you can also revoke all currently valid API keys in the event one or more keys are compromised. Revoking keys is a way to expire all currently valid keys.

STEP 1 | Select **Device** > **Setup** > **Management**.

STEP 2 | Edit Authentication Settings to specify the **API Key Lifetime (min)**.

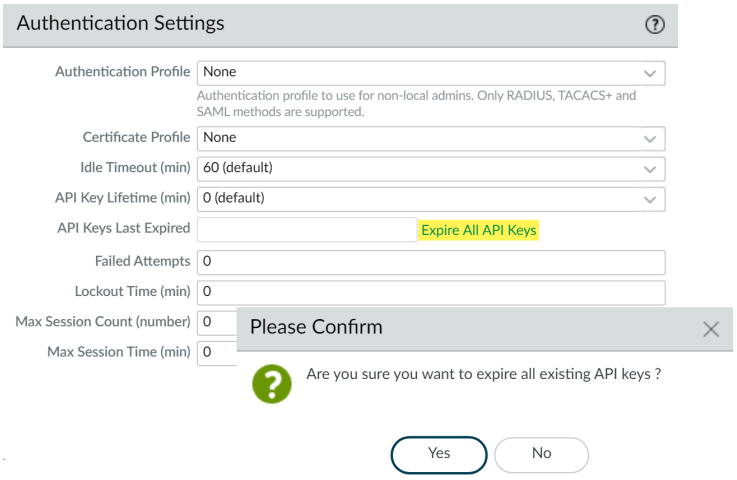


Set the API key lifetime to protect against compromise and to reduce the effects of an accidental exposure. By default, the API key lifetime is set to 0, which means that the keys will never expire. To ensure that your keys are frequently rotated and each key is unique when regenerated, you must specify a validity period that ranges between 1—525600 minutes. Refer to the audit and compliance policies for your enterprise to determine how you should specify the lifetime for which your API keys are valid.

STEP 3 | **Commit** the changes.

STEP 4 | (To revoke all API keys) Select **Expire all API Keys** to reset currently valid API keys.

If you have just set a key lifetime and want to reset all API keys to adhere to the new term, you can expire all existing keys.



On confirmation, the keys are revoked and you can view the timestamp for when the **API Keys Last Expired**.

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Enable SCP Uploads for an Administrator

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

Enable the use of Secure Copy Protocol (SCP) for Superuser administrators on Palo Alto Networks Next-Generation Firewalls to upload supported files, such as PAN-OS software updates, dynamic content updates, and configuration files, from a local device to a Palo Alto Networks firewall. This allows you to automate supported file uploads using the CLI rather than uploading using the firewall web interface.

A system log generates when you successfully use SCP to upload files to your firewall or if an SCP upload fails for any reason.

Palo Alto Networks supports SCP uploads of PAN-OS software versions, PAN-OS software changes, dynamic content updates, PAN-OS plugin versions, configuration files, and license key files.

STEP 1 | (Optional) Configure a firewall administrator with Superuser privileges for SCP functionality.

In this example, a Superuser firewall administrator named `scp_admin` was created.

STEP 2 | Log in to the firewall CLI.

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STEP 3 | Enable SCP functionality for a Superuser admin.

The admin initiating SCP must have Superuser privileges.

In this example, SCP functionality is enabled for the dedicated Superuser `scp_admin` created in the previous step.

1. Enter configuration mode.

   ```
   admin>configure
   ```

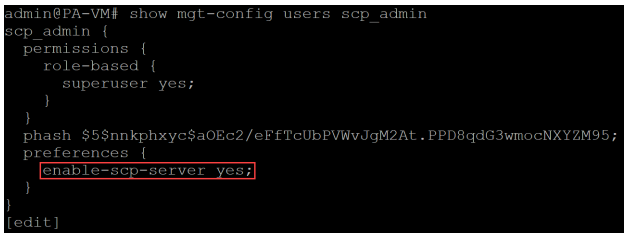
2. Enable SCP functionality for a Superuser admin.

   ```
   admin#set mgt-config users <admin_name> preferences enable-scp-server yes
   ```

3. Verify that SCP functionality was successfully enabled for the Superuser admin.

   ```
   admin#show mgt-config users <admin_name>
   ```

   In the `permissions`, verify that `enable-scp-server` displays `yes`.

4. Commit.

   ```
   admin# commit
   ```

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STEP 4 | Perform an SCP upload to your firewall.

This step assumes you already have the file you want to upload to your firewall available on your local device.

This example demonstrates how to upload an Application & Threats content update to your firewall. The predefined target directories for SCP uploads are:

- **PAN-OS Software Versions—`/scp/software/`**
- **PAN-OS Software Patches—`/scp/patch/`**
- **Application & Threats Content Updates—`/scp/content/`**
- **WildFire Content Updates—`/scp/wildfire/`**
- **Antivirus Content Updates—`/scp/anti-virus/`**
- **PAN-OS Plugin Versions—`/scp/plugin/`**
- **XML Configuration Files—`/scp/config/`**

  All PAN-OS configuration files must have the `.xml` extension appended to the file name for SCP uploads to succeed.

- **License Key Files—`/scp/license/`**

1. Open a CLI terminal and use the **`cd`** command to navigate to the folder or directory where the file you want to SCP is located.

   After navigating to the correct folder or directory, enter **`ls`** to view the folder or directory contents.

   In this example, you can see the `panupv2-all-contents-8765-8342` file we will upload to the firewall.
2. Upload a file to the firewall using the SCP-enabled Superuser admin.

   SCP applications like WinSCP and FileZilla are not supported. You must run the SCP command from the device command line.

   - **Operating System running OpenSSH 8 or earlier**

     ```
      scp <file_name> <scp_superuser>@<firewall_IP>:/scp/
     <file_type>/<file_name>
     ```

     Example of the SCP command to upload the Application & Threats content update using the `scp_admin`.

     ```
     scp panupv2-all-contents-8765-8342 scp_admin@<firewall_IP>:/
     scp/content/panupv2-all-contents-8765-8342
     ```

   - **Operating System running OpenSSH 9 or later**

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Starting from OpenSSH 9.0, the scp program defaults to the SFTP protocol for file transfers. This default behavior is incompatible with PAN-OS SCP support, resulting in the following error message in system logs: `Invalid number of arguments for SCP command`.

*Add the **`-O`** option to the SCP command to force the system to use the legacy SCP protocol instead of the default SFTP protocol.*

```
 scp -O <file_name> <scp_superuser>@<firewall_IP>:/scp/
<file_type>/<file_name>
```

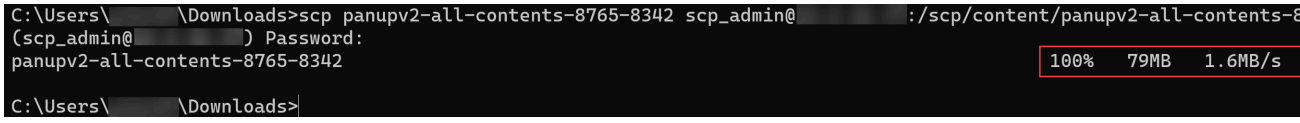
Example of the SCP command to upload the Application & Threats content update using the `scp_admin`.

```
scp -O panupv2-all-contents-8765-8342
 scp_admin@<firewall_IP>:/scp/content/panupv2-all-
contents-8765-8342
```

3. Enter **`yes`** when prompted to verify the authenticity of the firewall.

   You're not prompted to verify authenticity if you have already connected to the firewall using SSH from this device and can skip this step.
4. Enter the SCP admin `Password` when prompted and click Enter to continue.
5. The SCP upload progress displays.

   The SCP upload is complete when the progress status displays `100%` and the CLI command prompt is becomes available.



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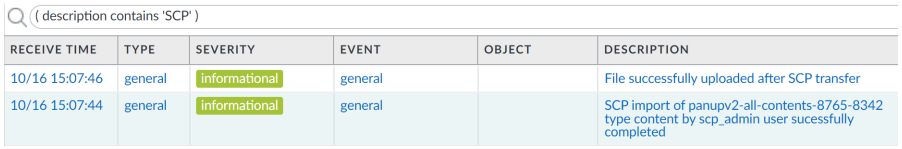
STEP 5 | Review the system logs to verify the SCP upload was successful.

You can verify that the SCP upload was successful by reviewing the generated system log and confirm that the uploaded file is available. In this example, we review the system log for the SCP upload of Application & Threats content update version 8765-8342.

1. Log in to the firewall web interface.
2. Select **Monitor** > **Logs** > **System** and filter for SCP uploads.

   Filter for SCP uploads using the following query:

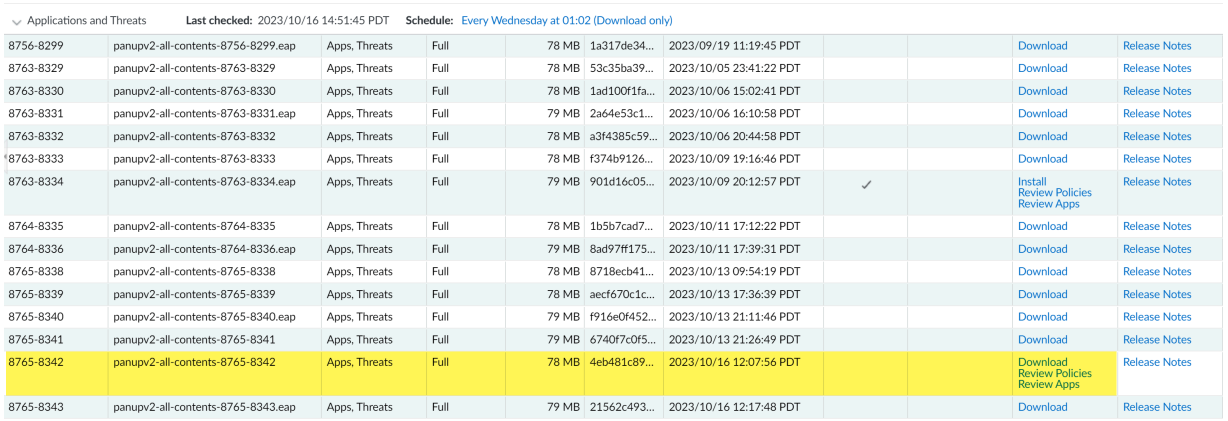
   ```
   ( description contains 'SCP' )
   ```



(description contains 'SCP')

RECEIVE TIME	TYPE	SEVERITY	EVENT	OBJECT	DESCRIPTION
10/16 15:07:46	general	informational	general		File successfully uploaded after SCP transfer
10/16 15:07:44	general	informational	general		SCP import of panupv2-all-contents-8765-8342 type content by scp_admin user sucessfully completed

3. Select **Device** > **Dynamic Updates** and confirm the uploaded content update is available to **Download**.



Applications and Threats **Last checked:** 2023/10/16 14:51:45 PDT **Schedule:** Every Wednesday at 01:02 (Download only)

8756-8299	panupv2-all-contents-8756-8299.eap	Apps, Threats	Full	78 MB	1a317de34...	2023/09/19 11:19:45 PDT			Download	Release Notes
8763-8329	panupv2-all-contents-8763-8329	Apps, Threats	Full	78 MB	53c35ba39...	2023/10/05 23:41:22 PDT			Download	Release Notes
8763-8330	panupv2-all-contents-8763-8330	Apps, Threats	Full	78 MB	1ad100f1fa...	2023/10/06 15:02:41 PDT			Download	Release Notes
8763-8331	panupv2-all-contents-8763-8331.eap	Apps, Threats	Full	79 MB	2a64e53c1...	2023/10/06 16:10:58 PDT			Download	Release Notes
8763-8332	panupv2-all-contents-8763-8332	Apps, Threats	Full	78 MB	a3f4385c59...	2023/10/06 20:44:58 PDT			Download	Release Notes
8763-8333	panupv2-all-contents-8763-8333	Apps, Threats	Full	78 MB	f374b9126...	2023/10/09 19:16:46 PDT			Download	Release Notes
8763-8334	panupv2-all-contents-8763-8334.eap	Apps, Threats	Full	79 MB	901d16c05...	2023/10/09 20:12:57 PDT	✓		Install Review Policies Review Apps	Release Notes
8764-8335	panupv2-all-contents-8764-8335	Apps, Threats	Full	78 MB	1b5b7cad7...	2023/10/11 17:12:22 PDT			Download	Release Notes
8764-8336	panupv2-all-contents-8764-8336.eap	Apps, Threats	Full	79 MB	8ad97ff175...	2023/10/11 17:39:31 PDT			Download	Release Notes
8765-8338	panupv2-all-contents-8765-8338	Apps, Threats	Full	78 MB	8718ecb41...	2023/10/13 09:54:19 PDT			Download	Release Notes
8765-8339	panupv2-all-contents-8765-8339	Apps, Threats	Full	78 MB	aecf670c1c...	2023/10/13 17:36:39 PDT			Download	Release Notes
8765-8340	panupv2-all-contents-8765-8340.eap	Apps, Threats	Full	79 MB	f916e0f452...	2023/10/13 21:11:46 PDT			Download	Release Notes
8765-8341	panupv2-all-contents-8765-8341	Apps, Threats	Full	79 MB	6740f7c0f5...	2023/10/13 21:26:49 PDT			Download	Release Notes
8765-8342	panupv2-all-contents-8765-8342	Apps, Threats	Full	78 MB	4eb481c89...	2023/10/16 12:07:56 PDT			Download Review Policies Review Apps	Release Notes
8765-8343	panupv2-all-contents-8765-8343.eap	Apps, Threats	Full	79 MB	21562c493...	2023/10/16 12:17:48 PDT			Download	Release Notes

Configure Tracking of Administrator Activity

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

Track administrator activity on the firewall web interface and CLI to achieve real time reporting of activity across your firewall. If you have reason to believe an administrator account is compromised, you have a full history of where this administrator account navigated throughout the web interface or what operational commands they executed so you can analyze in detail and respond to all actions the compromised administrator took.

When an event occurs, an audit log is generated and forwarded to the specified syslog server each time an administrator navigates through the web interface or when an operational command is executed in the CLI. An audit log is generated for each navigation or commend executed. Take for example if you want to create a new address object. An audit log is generated when you click on **Objects**, and a second audit log is generated when you then click on Addresses.

Audit logs are only visible as syslogs forwarded to your syslog server and cannot be viewed in the firewall web interface. Audit logs can only be forwarded to a syslog server, cannot be forwarded to Strata Logging Service, and are not stored locally on the firewall.

STEP 1 | Configure a syslog server profile to forward audit logs of administrator activity on the firewall.

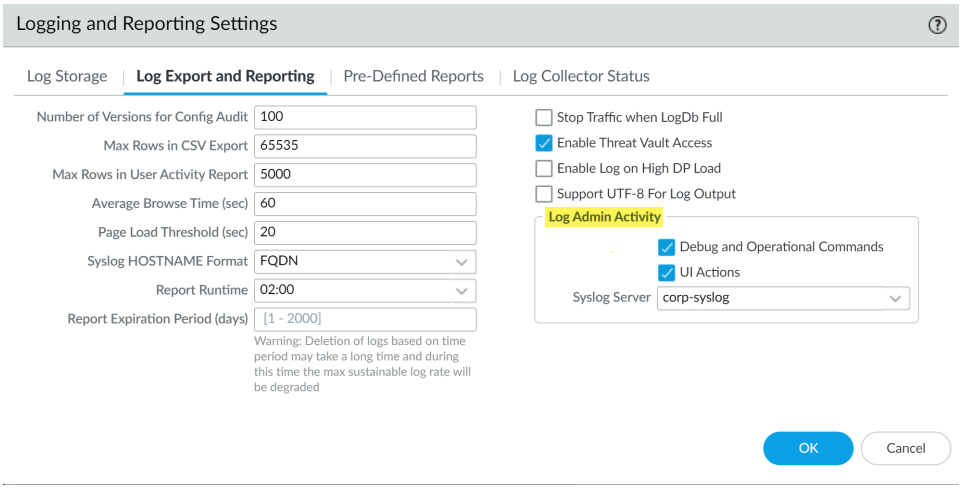
This step is required to successfully store audit logs for tracking administrator activity on the firewall.

1. Log in to the firewall web interface.
2. Configure a syslog server profile.

STEP 2 | Configure tracking of administrator activity.

1. Select **Device** > **Setup** > **Management** and edit the Logging and Reporting Settings.
2. Select **Log Export and Reporting**.
3. In the Log Admin Activity section, configure what administrator activity to track.
   - **Operational Commands**—Generate an audit log when an administrator executes an operational or debug command in the CLI or an operational command triggered from the web interface. See the CLI Operational Command Hierarchy for a full list of PAN-OS operational and debug commands.
   - **UI Actions**—Generate an audit log when an administrator navigates throughout the web interface. This includes navigation between configuration tabs, as well as individual objects within a tab.

     For example, an audit log is generated when an administrator navigates from the **ACC** to the **Policies** tab. Additionally, an audit log is generated when an administrator navigates from **Objects** > **Addresses** to **Objects** > **Tags**.
   - **Syslog Server**—Select a target syslog server profile to forward audit logs.
4. Click **OK**
5. Select **Commit**.



Reference: Web Interface Administrator Access

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

You can configure privileges for an entire firewall or for one or more virtual systems (on platforms that support multiple virtual systems). Within that **Device** or **Virtual System** designation, you can configure privileges for custom administrator roles, which are more granular than the fixed privileges associated with a dynamic administrator role.

Configuring privileges at a granular level ensures that lower level administrators cannot access certain information. You can create custom roles for firewall administrators (see Configure a Firewall Administrator Account), Panorama administrators, or Device Group and Template administrators (refer to the Panorama Administrator's Guide). You apply the admin role to a custom role-based administrator account where you can assign one or more virtual systems. The topics in this section describe the privileges you can configure for custom administrator roles.

Web Interface Access Privileges

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

If you want to prevent a role-based administrator from accessing specific tabs on the web interface, you can disable the tab and the administrator will not see it when logging in using the associated role-based administrative account. For example, you could create an Admin Role Profile for your operations staff that provides access to the **Device** and **Network** tabs only and a separate profile for your security administrators that provides access to the **Object**, **Policy**, and **Monitor** tabs.

An admin role can apply at the **Device** level or **Virtual System** level as defined by the **Device** or **Virtual System** radio button. If you select **Virtual System**, the admin assigned this profile is restricted to the virtual system(s) they are assigned to. Furthermore, only the **Device** > **Setup** > **Services** > **Virtual Systems** tab is available to that admin, not the **Global** tab.

Define Access to the Web Interface Tabs

The following table describes the top-level access privileges you can assign to an admin role profile (**Device** > **Admin Roles**). You can enable, disable, or define read-only access privileges at the top-level tabs in the web interface.

Access Level	Description	Enable	Read Only	Disable
Dashboard	Controls access to the **Dashboard** tab. If you disable this privilege, the	Yes	No	Yes

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Access Level	Description	Enable	Read Only	Disable
	administrator will not see the tab and will not have access to any of the Dashboard widgets.			
ACC	Controls access to the Application Command Center (ACC). If you disable this privilege, the **ACC** tab will not display in the web interface. Keep in mind that if you want to protect the privacy of your users while still providing access to the ACC, you can disable the **Privacy** > **Show Full IP Addresses** option and/or the **Show User Names In Logs And Reports** option.	Yes	No	Yes
Monitor	Controls access to the **Monitor** tab. If you disable this privilege, the administrator will not see the **Monitor** tab and will not have access to any of the logs, packet captures, session information, reports or to App Scope. For more granular control over what monitoring information the administrator can see, leave the Monitor option enabled and then enable or disable specific nodes on the tab as described in Provide Granular Access to the Monitor Tab.	Yes	No	Yes
Policies	Controls access to the **Policies** tab. If you disable this privilege, the administrator will not see the **Policies** tab and will not have access to any policy information. For more granular control over what policy information the administrator can see, for example to enable access to a specific type of policy or to enable read-only access to policy information, leave the **Policies** option enabled and then enable or disable specific nodes on the tab as described in Provide Granular Access to the Policy Tab.	Yes	No	Yes
Objects	Controls access to the **Objects** tab. If you disable this privilege, the administrator will not see the **Objects** tab and will not have access to any objects, security profiles, log forwarding profiles, decryption profiles, or schedules. For	Yes	No	Yes

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Access Level	Description	Enable	Read Only	Disable
	more granular control over what objects the administrator can see, leave the **Objects** option enabled and then enable or disable specific nodes on the tab as described in Provide Granular Access to the Objects Tab.			
Network	Controls access to the **Network** tab. If you disable this privilege, the administrator will not see the **Network** tab and will not have access to any interface, zone, VLAN, virtual wire, virtual router, IPsec tunnel, DHCP, DNS Proxy, GlobalProtect, or QoS configuration information or to the network profiles. For more granular control over what objects the administrator can see, leave the **Network** option enabled and then enable or disable specific nodes on the tab as described in Provide Granular Access to the Network Tab.	Yes	No	Yes
Device	Controls access to the **Device** tab. If you disable this privilege, the administrator will not see the **Device** tab and will not have access to any firewall-wide configuration information, such as User-ID, high availability, server profile or certificate configuration information. For more granular control over what objects the administrator can see, leave the **Objects** option enabled and then enable or disable specific nodes on the tab as described in Provide Granular Access to the Device Tab. *You cannot enable access to the **Admin Roles** or **Administrators** nodes for a role-based administrator even if you enable full access to the **Device** tab.*	Yes	No	Yes

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Provide Granular Access to the Monitor Tab

In some cases you might want to enable the administrator to view some but not all areas of the **Monitor** tab. For example, you might want to restrict operations administrators to the Config and System logs only, because they do not contain sensitive user data. Although this section of the administrator role definition specifies what areas of the **Monitor** tab the administrator can see, you can also couple privileges in this section with privacy privileges, such as disabling the ability to see usernames in logs and reports. One thing to keep in mind, however, is that any system-generated reports will still show usernames and IP addresses even if you disable that functionality in the role. For this reason, if you do not want the administrator to see any of the private user information, disable access to the specific reports as detailed in the following table.

The following table lists the **Monitor** tab access levels and the administrator roles for which they are available.

Device Group and Template roles can see log data only for the device groups that are within the access domains assigned to those roles.

Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
Monitor	Enables or disables access to the **Monitor** tab. If disabled, the administrator will not see this tab or any of the associated logs or reports.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
Logs	Enables or disables access to all log files. You can also leave this privilege enabled and then disable specific logs that you do not want the administrator to see. Keep in mind that if you want to protect the privacy of your users while still providing access to one or more of the logs, you can disable the **Privacy** > **Show Full IP Addresses** option and/or the **Show User Names In Logs And Reports** option.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
Traffic	Specifies whether the administrator can see the traffic logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
Threat	Specifies whether the administrator can see the threat logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
URL Filtering	Specifies whether the administrator can see the URL filtering logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
WildFire Submissions	Specifies whether the administrator can see the WildFire logs. These logs are only available if you have a WildFire subscription.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
Data Filtering	Specifies whether the administrator can see the data filtering logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
HIP Match	Specifies whether the administrator can see the HIP Match logs. HIP Match logs are available only if you have a GlobalProtect license (subscription).	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
GlobalProtect	Specifies whether the administrator can see the GlobalProtect logs. These logs are available only if you have a GlobalProtect license (subscription).	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
User-ID	Specifies whether the administrator can see the User-ID logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
GTP	Specifies whether the mobile network operator can see GTP logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
Tunnel Inspection	Specifies whether the administrator can see the Tunnel Inspection logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
SCTP	Specifies whether the mobile network operator can see Stream Control Transmission Protocol (SCTP) logs. *You must enable SCTP on Panorama (**Device** > **Setup** > **Management**) before you can control Administrator access to SCTP logs, custom reports, or predefined reports for Panorama and Device Group/ Template.*	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
Configuration	Specifies whether the administrator can see the configuration logs.	Firewall: Yes Panorama: Yes Device Group/ Template: No	Yes	No	Yes
System	Specifies whether the administrator can see the system logs.	Firewall: Yes Panorama: Yes Device Group/ Template: No	Yes	No	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
Alarms	Specifies whether the administrator can see system-generated alarms.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
Authentication	Specifies whether the administrator can see the Authentication logs.	Firewall: Yes Panorama: Yes Device Group/ Template: No	Yes	No	Yes
Automated Correlation Engine	Enables or disables access to the correlation objects and correlated event logs generated on the firewall.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
Correlation Objects	Specifies whether the administrator can view and enable/disable the correlation objects.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
Correlated Events	Specifies whether the administrator can view and enable/disable the correlation events.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
Packet Capture	Specifies whether the administrator can see packet captures (pcaps) from the **Monitor** tab. Keep in mind that packet captures are raw flow data and as such may contain user IP addresses. Disabling the **Show Full IP Addresses** privileges will not obfuscate the IP address in the pcap and you should therefore disable the Packet Capture privilege if you are concerned about user privacy.	Firewall: Yes Panorama: No Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
App Scope	Specifies whether the administrator can see the App Scope visibility and analysis tools. Enabling App Scope enables access to all of the **App Scope** charts.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
Session Browser	Specifies whether the administrator can browse and filter current running sessions on the firewall. Keep in mind that the session browser shows raw flow data and as such may contain user IP addresses. Disabling the **Show Full IP Addresses** privileges will not obfuscate the IP address in the session browser and you should therefore disable the **Session Browser** privilege if you are concerned about user privacy.	Firewall: Yes Panorama: No Device Group/ Template: No	Yes	No	Yes
Block IP List	Specifies whether the administrator can view the block list (Enable or Read Only) and delete entries from the list (Enable). If you disable the setting, the administrator won't be able to view or delete entries from the block list.	Firewall: Yes Panorama: under Context Switch UI: Yes Template: Yes	Yes	Yes	Yes
Botnet	Specifies whether the administrator can generate and view botnet analysis reports or view botnet reports in read-only mode. Disabling the **Show Full IP Addresses** privileges will not obfuscate the IP address in scheduled botnet reports and you should therefore disable the **Botnet** privilege if you are concerned about user privacy.	Firewall: Yes Panorama: No Device Group/ Template: No	Yes	Yes	Yes
PDF Reports	Enables or disables access to all PDF reports. You can also leave this privilege enabled	Firewall: Yes Panorama: Yes	Yes	No	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	and then disable specific PDF reports that you do not want the administrator to see. Keep in mind that if you want to protect the privacy of your users while still providing access to one or more of the reports, you can disable the **Privacy** > **Show Full IP Addresses** option and/or the **Show User Names In Logs And Reports** option.	Device Group/ Template: Yes			
Manage PDF Summary	Specifies whether the administrator can view, add or delete PDF summary report definitions. With read-only access, the administrator can see PDF summary report definitions, but not add or delete them. If you disable this option, the administrator can neither view the report definitions nor add/delete them.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	Yes	Yes
PDF Summary Reports	Specifies whether the administrator can see the generated PDF Summary reports in **Monitor** > **Reports**. If you disable this option, the **PDF Summary Reports** category will not display in the **Reports** node.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
User Activity Report	Specifies whether the administrator can view, add or delete User Activity report definitions and download the reports. With read-only access, the administrator can see User Activity report definitions, but not add, delete, or download them. If you disable this option, the administrator cannot see this category of PDF report.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	Yes	Yes
SaaS Application	Specifies whether the administrator can view, add or delete a SaaS application usage	Firewall: Yes Panorama: Yes	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
Usage Report	report. With read-only access, the administrator can see the SaaS application usage report definitions, but cannot add or delete them. If you disable this option, the administrator can neither view the report definitions nor add or delete them.	Device Group/ Template: Yes			
Report Groups	Specifies whether the administrator can view, add or delete report group definitions. With read-only access, the administrator can see report group definitions, but not add or delete them. If you disable this option, the administrator cannot see this category of PDF report.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	Yes	Yes
Email Scheduler	Specifies whether the administrator can schedule report groups for email. Because the generated reports that get emailed may contain sensitive user data that is not removed by disabling the **Privacy** > **Show Full IP Addresses** option and/ or the **Show User Names In Logs And Reports** options and because they may also show log data to which the administrator does not have access, you should disable the **Email Scheduler** option if you have user privacy requirements.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	Yes	Yes
Manage Custom Reports	Enables or disables access to all custom report functionality. You can also leave this privilege enabled and then disable specific custom report categories that you do not want the administrator to be able to access. Keep in mind that if you want to protect the privacy of your users while	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	still providing access to one or more of the reports, you can disable the **Privacy** > **Show Full IP Addresses** option and/or the **Show User Names In Logs And Reports** option. *Reports that are scheduled to run rather than run on demand will show IP address and user information. In this case, be sure to restrict access to the corresponding report areas. In addition, the custom report feature does not restrict the ability to generate reports that contain log data contained in logs that are excluded from the administrator role.*				
Application Statistics	Specifies whether the administrator can create a custom report that includes data from the application statistics database.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
Data Filtering Log	Specifies whether the administrator can create a custom report that includes data from the Data Filtering logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
Threat Log	Specifies whether the administrator can create a custom report that includes data from the Threat logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
Threat Summary	Specifies whether the administrator can create a custom report that includes data from the Threat Summary database.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
Traffic Log	Specifies whether the administrator can create a custom report that includes data from the Traffic logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
Traffic Summary	Specifies whether the administrator can create a custom report that includes data from the Traffic Summary database.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
URL Log	Specifies whether the administrator can create a custom report that includes data from the URL Filtering logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
URL Summary	Specifies whether the administrator can create a custom report that includes data from the URL Summary database.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
HIP Match	Specifies whether the administrator can create a custom report that includes data from the HIP Match logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
GlobalProtect	Specifies whether the administrator can create a custom report that includes data from the GlobalProtect logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
WildFire Log	Specifies whether the administrator can create a	Firewall: Yes Panorama: Yes	Yes	No	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	custom report that includes data from the WildFire logs.	Device Group/ Template: Yes			
GTP Log	Specifies whether the mobile network operator can create a custom report that includes data from GTP logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
GTP Summary	Specifies whether the mobile network operator can create a custom report that includes data from GTP logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
Tunnel Log	Specifies whether the administrator can create a custom report that includes data from tunnel inspection logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
Tunnel Summary	Specifies whether the administrator can create a custom report that includes data from the Tunnel Summary database.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
SCTP Log	Specifies whether the mobile network operator can create a custom report that includes data from SCTP logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
SCTP Summary	Specifies whether the mobile network operator can create a custom report that includes data from the SCTP Summary database.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
User-ID	Specifies whether the administrator can create a custom report that includes data from the User-ID logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
Authentication	Specifies whether the administrator can create a custom report that includes data from the Authentication logs.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
View Scheduled Custom Reports	Specifies whether the administrator can view a custom report that has been scheduled to generate.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
View Predefined Application Reports	Specifies whether the administrator can view Application Reports. Privacy privileges do not impact reports available on the **Monitor** > **Reports** node and you should therefore disable access to the reports if you have user privacy requirements.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
View Predefined Threat Reports	Specifies whether the administrator can view Threat Reports. Privacy privileges do not impact reports available on the **Monitor** > **Reports** node and you should therefore disable access to the reports if you have user privacy requirements.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
View Predefined URL Filtering Reports	Specifies whether the administrator can view URL Filtering Reports. Privacy privileges do not impact reports available on the **Monitor** > **Reports** node and you should therefore disable access to the reports if you have user privacy requirements.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
View Predefined Traffic Reports	Specifies whether the administrator can view Traffic Reports. Privacy privileges do not impact reports available on the **Monitor** > **Reports** node and	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	you should therefore disable access to the reports if you have user privacy requirements.				
View Predefined GTP Reports	Specifies whether the mobile network operator can view GTP Reports. Privacy privileges do not impact reports available on the **Monitor** > **Reports** node and you should therefore disable access to the reports if you have user privacy requirements.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes
View Predefined SCTP Reports	Specifies whether the mobile network operator can view SCTP Reports. Privacy privileges do not impact reports available on the **Monitor** > **Reports** node and you should therefore disable access to the reports if you have user privacy requirements.	Firewall: Yes Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes

Provide Granular Access to the Policy Tab

If you enable the Policy option in the Admin Role profile, you can then enable, disable, or provide read-only access to specific nodes within the tab as necessary for the role you are defining. By enabling access to a specific policy type, you enable the ability to view, add, or delete policy rules. By enabling read-only access to a specific policy, you enable the administrator to view the corresponding policy rule base, but not add or delete rules. Disabling access to a specific type of policy prevents the administrator from seeing the policy rule base.

Because policy that is based on specific users (by username or IP address) must be explicitly defined, privacy settings that disable the ability to see full IP addresses or usernames do not apply to the Policy tab. Therefore, you should only allow access to the Policy tab to administrators that are excluded from user privacy restrictions.

Access Level	Description	Enable	Read Only	Disable
Security	Enable this privilege to allow the administrator to view, add, and/or delete security rules. Set the privilege to read-only if you want the administrator to be able to see the rules, but not modify them. To prevent the administrator from	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
	seeing the security rulebase, disable this privilege.			
NAT	Enable this privilege to allow the administrator to view, add, and/or delete NAT rules. Set the privilege to read-only if you want the administrator to be able to see the rules, but not modify them. To prevent the administrator from seeing the NAT rulebase, disable this privilege.	Yes	Yes	Yes
QoS	Enable this privilege to allow the administrator to view, add, and/or delete QoS rules. Set the privilege to read-only if you want the administrator to be able to see the rules, but not modify them. To prevent the administrator from seeing the QoS rulebase, disable this privilege.	Yes	Yes	Yes
Policy Based Forwarding	Enable this privilege to allow the administrator to view, add, and/or delete Policy-Based Forwarding (PBF) rules. Set the privilege to read-only if you want the administrator to be able to see the rules, but not modify them. To prevent the administrator from seeing the PBF rulebase, disable this privilege.	Yes	Yes	Yes
Decryption	Enable this privilege to allow the administrator to view, add, and/or delete decryption rules. Set the privilege to read-only if you want the administrator to be able to see the rules, but not modify them. To prevent the administrator from seeing the decryption rulebase, disable this privilege.	Yes	Yes	Yes
Network Packet Broker	Enable this privilege to allow the administrator to view, add, and/or delete Network Packet Broker policy rules. Set the privilege to read-only if you want the administrator to be able to see the rules, but not modify them. To prevent the administrator from seeing the Network Packet Broker rulebase in the interface, disable this privilege.	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
Tunnel Inspection	Enable this privilege to allow the administrator to view, add, and/or delete Tunnel Inspection rules. Set the privilege to read-only if you want the administrator to be able to see the rules, but not modify them. To prevent the administrator from seeing the Tunnel Inspection rulebase, disable this privilege.	Yes	Yes	Yes
Application Override	Enable this privilege to allow the administrator to view, add, and/or delete application override policy rules. Set the privilege to read-only if you want the administrator to be able to see the rules, but not modify them. To prevent the administrator from seeing the application override rulebase, disable this privilege.	Yes	Yes	Yes
Authentication	Enable this privilege to allow the administrator to view, add, and/or delete Authentication policy rules. Set the privilege to read-only if you want the administrator to be able to see the rules, but not modify them. To prevent the administrator from seeing the Authentication rulebase, disable this privilege.	Yes	Yes	Yes
DoS Protection	Enable this privilege to allow the administrator to view, add, and/or delete DoS protection rules. Set the privilege to read-only if you want the administrator to be able to see the rules, but not modify them. To prevent the administrator from seeing the DoS protection rulebase, disable this privilege.	Yes	Yes	Yes
SD-WAN	Enable this privilege to allow the administrator to view, add, and/or delete SD-WAN policy rules. Set the privilege to read-only if you want the administrator to be able to see the rules, but not modify them. To prevent the administrator from seeing the SD-WAN policy rulebase, disable this privilege.	Yes	Yes	Yes

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Provide Granular Access to the Objects Tab

An *object* is a container that groups specific policy filter values—such as IP addresses, URLs, applications, or services—for simplified rule definition. For example, an address object might contain specific IP address definitions for the web and application servers in your DMZ zone.

When deciding whether to allow access to the objects tab as a whole, determine whether the administrator will have policy definition responsibilities. If not, the administrator probably does not need access to the tab. If, however, the administrator will need to create policy, you can enable access to the tab and then provide granular access privileges at the node level.

By enabling access to a specific node, you give the administrator the privilege to view, add, and delete the corresponding object type. Giving read-only access allows the administrator to view the already defined objects, but not create or delete any. Disabling a node prevents the administrator from seeing the node in the web interface.

Access Level	Description	Enable	Read Only	Disable
Addresses	Specifies whether the administrator can view, add, or delete address objects for use in security policy.	Yes	Yes	Yes
Address Groups	Specifies whether the administrator can view, add, or delete address group objects for use in security policy.	Yes	Yes	Yes
Regions	Specifies whether the administrator can view, add, or delete regions objects for use in security, decryption, or DoS policy.	Yes	Yes	Yes
Applications	Specifies whether the administrator can view, add, or delete application objects for use in policy.	Yes	Yes	Yes
Application Groups	Specifies whether the administrator can view, add, or delete application group objects for use in policy.	Yes	Yes	Yes
Application Filters	Specifies whether the administrator can view, add, or delete application filters for simplification of repeated searches.	Yes	Yes	Yes
Services	Specifies whether the administrator can view, add, or delete service objects for use in creating policy rules that limit the port numbers an application can use.	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
Service Groups	Specifies whether the administrator can view, add, or delete service group objects for use in security policy.	Yes	Yes	Yes
Tags	Specifies whether the administrator can view, add, or delete tags that have been defined on the firewall.	Yes	Yes	Yes
GlobalProtect	Specifies whether the administrator can view, add, or delete HIP objects and profiles. You can restrict access to both types of objects at the GlobalProtect level, or provide more granular control by enabling the GlobalProtect privilege and restricting HIP Object or HIP Profile access.	Yes	No	Yes
HIP Objects	Specifies whether the administrator can view, add, or delete HIP objects, which are used to define HIP profiles. HIP Objects also generate HIP Match logs.	Yes	Yes	Yes
Clientless Apps	Specifies whether the administrator can view, add, modify, or delete GlobalProtect VPN Clientless applications.	Yes	Yes	Yes
Clientless App Groups	Specifies whether the administrator can view, add, modify, or delete GlobalProtect VPN Clientless application groups.	Yes	Yes	Yes
HIP Profiles	Specifies whether the administrator can view, add, or delete HIP Profiles for use in security policy and/or for generating HIP Match logs.	Yes	Yes	Yes
External Dynamic Lists	Specifies whether the administrator can view, add, or delete external dynamic lists for use in security policy.	Yes	Yes	Yes
Custom Objects	Specifies whether the administrator can see the custom spyware and vulnerability signatures. You can restrict access to either enable or disable access to all custom signatures at this level, or provide more granular control by enabling the Custom Objects privilege	Yes	No	Yes

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Access Level	Description	Enable	Read Only	Disable
	and then restricting access to each type of signature.			
Data Patterns	Specifies whether the administrator can view, add, or delete custom data pattern signatures for use in creating custom Vulnerability Protection profiles.	Yes	Yes	Yes
Spyware	Specifies whether the administrator can view, add, or delete custom spyware signatures for use in creating custom Vulnerability Protection profiles.	Yes	Yes	Yes
Vulnerability	Specifies whether the administrator can view, add, or delete custom vulnerability signatures for use in creating custom Vulnerability Protection profiles.	Yes	Yes	Yes
URL Category	Specifies whether the administrator can view, add, or delete custom URL categories for use in policy.	Yes	Yes	Yes
Security Profiles	Specifies whether the administrator can see security profiles. You can restrict access to either enable or disable access to all security profiles at this level, or provide more granular control by enabling the Security Profiles privilege and then restricting access to each type of profile.	Yes	No	Yes
Antivirus	Specifies whether the administrator can view, add, or delete antivirus profiles.	Yes	Yes	Yes
Anti-Spyware	Specifies whether the administrator can view, add, or delete Anti-Spyware profiles.	Yes	Yes	Yes
Vulnerability Protection	Specifies whether the administrator can view, add, or delete Vulnerability Protection profiles.	Yes	Yes	Yes
URL Filtering	Specifies whether the administrator can view, add, or delete URL filtering profiles.	Yes	Yes	Yes
File Blocking	Specifies whether the administrator can view, add, or delete file blocking profiles.	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
WildFire Analysis	Specifies whether the administrator can view, add, or delete WildFire analysis profiles.	Yes	Yes	Yes
Data Filtering	Specifies whether the administrator can view, add, or delete data filtering profiles.	Yes	Yes	Yes
DoS Protection	Specifies whether the administrator can view, add, or delete DoS protection profiles.	Yes	Yes	Yes
GTP Protection	Specifies whether the mobile network operator can view, add, or delete GTP Protection profiles.	Yes	Yes	Yes
SCTP Protection	Specifies whether the mobile network operator can view, add, or delete Stream Control Transmission Protocol (SCTP) Protection profiles.	Yes	Yes	Yes
Security Profile Groups	Specifies whether the administrator can view, add, or delete security profile groups.	Yes	Yes	Yes
Log Forwarding	Specifies whether the administrator can view, add, or delete log forwarding profiles.	Yes	Yes	Yes
Authentication	Specifies whether the administrator can view, add, or delete authentication enforcement objects.	Yes	Yes	Yes
Decryption Profile	Specifies whether the administrator can view, add, or delete decryption profiles.	Yes	Yes	Yes
SD-WAN Link Management	Specifies whether the administrator can add or delete Path Quality, SaaS Quality, Traffic Distribution, and Error Correction profiles.	Yes	No	Yes
Path Quality Profile	Specifies whether the administrator can view, add, or delete SD-WAN Path Quality profiles.	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
SaaS Quality Profile	Specifies whether the administrator can view, add, or delete SD-WAN SaaS Quality profiles.	Yes	Yes	Yes
Traffic Distribution Profile	Specifies whether the administrator can view, add, or delete SD-WAN Traffic Distribution profiles.	Yes	Yes	Yes
Error Correction Profile	Specifies whether the administrator can view, add, or delete SD-WAN Error Correction profiles.	Yes	Yes	Yes
Packet Broker Profile	Specifies whether the administrator can view, add, or delete Packet Broker profiles.	Yes	Yes	Yes
Schedules	Specifies whether the administrator can view, add, or delete schedules for limiting a security policy to a specific date and/or time range.	Yes	Yes	Yes

Provide Granular Access to the Network Tab

When deciding whether to allow access to the **Network** tab as a whole, determine whether the administrator will have network administration responsibilities, including GlobalProtect administration. If not, the administrator probably does not need access to the tab.

You can also define access to the **Network** tab at the node level. By enabling access to a specific node, you give the administrator the privilege to view, add, and delete the corresponding network configurations. Giving read-only access allows the administrator to view the already-defined configuration, but not create or delete any. Disabling a node prevents the administrator from seeing the node in the web interface.

A number of Routing access levels are visible and apply only when **Advanced Routing** is enabled for the device, in which case logical routers replace virtual routers.

Access Level	Description	Enable	Read Only	Disable
Interfaces	Specifies whether the administrator can view, add, or delete interface configurations.	Yes	Yes	Yes
Zones	Specifies whether the administrator can view, add, or delete zones.	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
VLANs	Specifies whether the administrator can view, add, or delete VLANs.	Yes	Yes	Yes
Virtual Wires	Specifies whether the administrator can view, add, or delete virtual wires.	Yes	Yes	Yes
Virtual Routers	Specifies whether the administrator can view, add, modify or delete virtual routers.	Yes	Yes	Yes
Routing	(Advanced Routing Engine) Specifies whether the administrator can view, add, modify or delete any of the routing fields for an Advanced Routing Engine.	Yes	Yes	Yes
Logical Routers	(Advanced Routing Engine) Specifies whether the administrator can view, add, modify or delete logical routers.	Yes	Yes	Yes
Routing Profiles	(Advanced Routing Engine) Specifies whether the administrator can view, add, modify or delete routing profiles.	Yes	Yes	Yes
BGP	(Advanced Routing Engine) Specifies whether the administrator can view, add, modify or delete BGP routing profiles.	Yes	Yes	Yes
BFD	(Advanced Routing Engine) Specifies whether the administrator can view, add, modify or delete BFD routing profiles.	Yes S	Yes	Yes
OSPF	(Advanced Routing Engine) Specifies whether the administrator can view, add, modify or delete OSPFv2 routing profiles.	Yes	Yes	Yes
OSPFv3	(Advanced Routing Engine) Specifies whether the administrator can view, add, modify or delete OSPFv3 routing profiles.	Yes	Yes	Yes
RIPv2	(Advanced Routing Engine) Specifies whether the administrator can view, add, modify or delete RIPv2 routing profiles.	Yes	Yes	Yes
Filters	(Advanced Routing Engine) Specifies whether the administrator can view, add, modify or delete filters.	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
Multicast	(Advanced Routing Engine) Specifies whether the administrator can view, add, modify or delete IPv4 multicast routing profiles.	Yes	Yes	Yes
IPSec Tunnels	Specifies whether the administrator can view, add, modify, or delete IPSec Tunnel configurations.	Yes	Yes	Yes
GRE Tunnels	Specifies whether the administrator can view, add, modify, or delete GRE Tunnel configurations.	Yes	Yes	Yes
DHCP	Specifies whether the administrator can view, add, modify, or delete DHCP server and DHCP relay configurations.	Yes	Yes	Yes
DNS Proxy	Specifies whether the administrator can view, add, modify, or delete DNS proxy configurations.	Yes	Yes	Yes
GlobalProtect	Specifies whether the administrator can view, add, modify GlobalProtect portal and gateway configurations. You can disable access to the GlobalProtect functions entirely, or you can enable the GlobalProtect privilege and then restrict the role to either the portal or gateway configuration areas.	Yes	No	Yes
Portals	Specifies whether the administrator can view, add, modify, or delete GlobalProtect portal configurations.	Yes	Yes	Yes
Gateways	Specifies whether the administrator can view, add, modify, or delete GlobalProtect gateway configurations.	Yes	Yes	Yes
MDM	Specifies whether the administrator can view, add, modify, or delete GlobalProtect MDM server configurations.	Yes	Yes	Yes
Device Block List	Specifies whether the administrator can view, add, modify, or delete device block lists.	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
Clientless Apps	Specifies whether the administrator can view, add, modify, or delete GlobalProtect Clientless VPN applications.	Yes	Yes	Yes
Clientless App Groups	Specifies whether the administrator can view, add, modify, or delete GlobalProtect Clientless VPN application groups.	Yes	Yes	Yes
QoS	Specifies whether the administrator can view, add, modify, or delete QoS configurations.	Yes	Yes	Yes
LLDP	Specifies whether the administrator can view add, modify, or delete LLDP configurations.	Yes	Yes	Yes
Network Profiles	Sets the default state to enable or disable for all of the Network settings described below.	Yes	No	Yes
GlobalProtect IPSec Crypto	Controls access to the **Network Profiles** > **GlobalProtect IPSec Crypto** node. If you disable this privilege, the administrator will not see that node, or configure algorithms for authentication and encryption in VPN tunnels between a GlobalProtect gateway and clients. If you set the privilege to read-only, the administrator can view existing GlobalProtect IPSec Crypto profiles but cannot add or edit them.	Yes	Yes	Yes
IKE Gateways	Controls access to the **Network Profiles** > **IKE Gateways** node. If you disable this privilege, the administrator will not see the **IKE Gateways** node or define gateways that include the configuration information necessary to perform IKE protocol negotiation with peer gateway. If the privilege state is set to read-only, you can view the currently configured IKE Gateways but cannot add or edit gateways.	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
IPSec Crypto	Controls access to the **Network Profiles** > **IPSec Crypto** node. If you disable this privilege, the administrator will not see the **Network Profiles** > **IPSec Crypto** node or specify protocols and algorithms for identification, authentication, and encryption in VPN tunnels based on IPSec SA negotiation. If the privilege state is set to read-only, you can view the currently configured IPSec Crypto configuration but cannot add or edit a configuration.	Yes	Yes	Yes
IKE Crypto	Controls how devices exchange information to ensure secure communication. Specify the protocols and algorithms for identification, authentication, and encryption in VPN tunnels based on IPsec SA negotiation (IKEv1 Phase-1).	Yes	Yes	Yes
Monitor	Controls access to the **Network Profiles** > **Monitor** node. If you disable this privilege, the administrator will not see the **Network Profiles** > **Monitor** node or be able to create or edit a monitor profile that is used to monitor IPSec tunnels and monitor a next-hop device for policy-based forwarding (PBF) rules. If the privilege state is set to read-only, you can view the currently configured monitor profile configuration but cannot add or edit a configuration.	Yes	Yes	Yes
Interface Mgmt	Controls access to the **Network Profiles** > **Interface Mgmt** node. If you disable this privilege, the administrator will not see the **Network Profiles** > **Interface Mgmt** node or be able to specify the protocols that are used to manage the firewall. If the privilege state is set to read-only, you can view the currently configured Interface management profile	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
	configuration but cannot add or edit a configuration.			
Zone Protection	Controls access to the **Network Profiles** > **Zone Protection** node. If you disable this privilege, the administrator will not see the **Network Profiles** > **Zone Protection** node or be able to configure a profile that determines how the firewall responds to attacks from specified security zones. If the privilege state is set to read-only, you can view the currently configured Zone Protection profile configuration but cannot add or edit a configuration.	Yes	Yes	Yes
QoS Profile	Controls access to the **Network Profiles** > **QoS** node. If you disable this privilege, the administrator will not see the **Network Profiles** > **QoS** node or be able to configure a QoS profile that determines how QoS traffic classes are treated. If the privilege state is set to read-only, you can view the currently configured QoS profile configuration but cannot add or edit a configuration.	Yes	Yes	Yes
LLDP Profile	Controls access to the **Network Profiles** > **LLDP** node. If you disable this privilege, the administrator will not see the **Network Profiles** > **LLDP** node or be able to configure an LLDP profile that controls whether the interfaces on the firewall can participate in the Link Layer Discovery Protocol. If the privilege state is set to read-only, you can view the currently configured LLDP profile configuration but cannot add or edit a configuration.	Yes	Yes	Yes
BFD Profile	Controls access to the **Network Profiles** > **BFD Profile** node. If you disable this privilege, the administrator will not see the **Network Profiles** > **BFD Profile** node or be able to configure a BFD profile. A Bidirectional Forwarding Detection	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
	(BFD) profile allows you to configure BFD settings to apply to one or more static routes or routing protocols. Thus, BFD detects a failed link or BFD peer and allows an extremely fast failover. If the privilege state is set to read-only, you can view the currently configured BFD profile but cannot add or edit a BFD profile.			
SD-WAN Interface Profile	Controls access to the **SD-WAN Interface Profile** node. If you disable this privilege, the administrator will not see the **SD-WAN Interface Profile** node or be able to configure an SD-WAN Interface Profile. An SD-WAN Interface Profile defines the characteristics of ISP connections and specifies the link speed and how frequently the firewall monitors the link. If the privilege state is set to read-only, you can view the currently configured SD-WAN Interface Profile but cannot add or edit one.	Yes	Yes	Yes

Provide Granular Access to the Device Tab

To define granular access privileges for the **Device** tab, when creating or editing an admin role profile (**Device** > **Admin Roles**), scroll down to the **Device** node on the **WebUI** tab.

Access Level	Description	Enable	Read Only	Disable
Setup	Controls access to the **Setup** node. If you disable this privilege, the administrator will not see the **Setup** node or have access to firewall-wide setup configuration information, such as Management, Operations, Service, Content-ID, WildFire or Session setup information. If the privilege state is set to read-only, you can view the current configuration but cannot make any changes.	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
Management	Controls access to the **Management** node. If you disable this privilege, the administrator will not be able to configure settings such as the hostname, domain, timezone, authentication, logging and reporting, Panorama connections, banner, message, and password complexity settings, and more. If the privilege state is set to read-only, you can view the current configuration but cannot make any changes.	Yes	Yes	Yes
Operations	Controls access to the **Operations** and **Telemetry and Threat Intelligence** nodes. If you disable this privilege, the administrator cannot: • Load firewall configurations. • Save or revert the firewall configuration. *This privilege applies only to the **Device** > **Operations** options. The Save and Commit privileges control whether the administrator can save or revert configurations through the **Config** > **Save** and **Config** > **Revert** options.* • Create custom logos. • Configure SNMP monitoring of firewall settings. • Configure the Statistics Service feature. • Configure **Telemetry and Threat Intelligence** settings. Only administrators with the predefined Superuser role can export or import firewall configurations and shut down the firewall. Only administrators with the predefined Superuser or Device Administrator role	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
	can reboot the firewall or restart the dataplane. Administrators with a role that allows access only to specific virtual systems cannot load, save, or revert firewall configurations through the **Device** > **Operations** options.			
Services	Controls access to the **Services** node. If you disable this privilege, the administrator will not be able to configure services for DNS servers, an update server, proxy server, or NTP servers, or set up service routes. If the privilege state is set to read-only, you can view the current configuration but cannot make any changes.	Yes	Yes	Yes
Content-ID	Controls access to the **Content-ID** node. If you disable this privilege, the administrator will not be able to configure URL filtering or Content-ID. If the privilege state is set to read-only, you can view the current configuration but cannot make any changes.	Yes	Yes	Yes
WildFire	Controls access to the **WildFire** node. If you disable this privilege, the administrator will not be able to configure WildFire settings. If the privilege state is set to read-only, you can view the current configuration but cannot make any changes.	Yes	Yes	Yes
Session	Controls access to the **Session** node. If you disable this privilege, the administrator will not be able to configure session settings or timeouts for TCP, UDP or ICMP, or configure decryption or VPN session settings. If the privilege state is set to read-only, you can view the current configuration but cannot make any changes.	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
HSM	Controls access to the **HSM** node. If you disable this privilege, the administrator will not be able to configure a Hardware Security Module. If the privilege state is set to read-only, you can view the current configuration but cannot make any changes.	Yes	Yes	Yes
High Availability	Controls access to the **High Availability** node. If you disable this privilege, the administrator will not see the **High Availability** node or have access to firewall-wide high availability configuration information such as General setup information or Link and Path Monitoring. If you set this privilege to read-only, the administrator can view High Availability configuration information for the firewall but is not allowed to perform any configuration procedures.	Yes	Yes	Yes
Config Audit	Controls access to the **Config Audit** node. If you disable this privilege, the administrator will not see the **Config Audit** node or have access to any firewall-wide configuration information.	Yes	No	Yes
Administrators	Controls access to the **Administrators** node. This function can only be allowed for read-only access. If you disable this privilege, the administrator will not see the **Administrators** node or have access to information about their own administrator account. If you set this privilege to read-only, the administrator can view the configuration information for their own administrator account. They will not see any information about other administrator accounts configured on the firewall.	No	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
Admin Roles	Controls access to the **Admin Roles** node. This function can only be allowed for read-only access. If you disable this privilege, the administrator will not see the **Admin Roles** node or have access to any firewall-wide information concerning Admin Role profiles configuration. If you set this privilege to read-only, you can view the configuration information for all administrator roles configured on the firewall.	No	Yes	Yes
Authentication Profile	Controls access to the **Authentication Profile** node. If you disable this privilege, the administrator will not see the **Authentication Profile** node or be able to create or edit authentication profiles that specify RADIUS, TACACS +, LDAP, Kerberos, SAML, multi-factor authentication (MFA), or local database authentication settings. PAN-OS uses authentication profiles to authenticate firewall administrators and Authentication Portal or GlobalProtect end users. If you set this privilege to read-only, the administrator can view the **Authentication Profile** information but cannot create or edit authentication profiles.	Yes	Yes	Yes
Authentication Sequence	Controls access to the **Authentication Sequence** node. If you disable this privilege, the administrator will not see the **Authentication Sequence** node or be able to create or edit an authentication sequence. If you set this privilege to read-only, the administrator can view the **Authentication Profile** information but cannot create or edit an authentication sequence.	Yes	Yes	Yes
Virtual Systems	Controls access to the **Virtual Systems** node. If you disable this privilege, the	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
	administrator will not see or be able to configure virtual systems. If the privilege state is set to read-only, you can view the currently configured virtual systems but cannot add or edit a configuration.			
Shared Gateways	Controls access to the **Shared Gateways** node. Shared gateways allow virtual systems to share a common interface for external communications. If you disable this privilege, the administrator will not see or be able to configure shared gateways. If the privilege state is set to read-only, you can view the currently configured shared gateways but cannot add or edit a configuration.	Yes	Yes	Yes
User Identification	Controls access to the **User Identification** node. If you disable this privilege, the administrator will not see the **User Identification** node or have access to firewall-wide User Identification configuration information, such as User Mapping, Connection Security, User-ID Agents, Terminal Server Agents, Group Mappings Settings, or Authentication Portal Settings. If you set this privilege to read-only, the administrator can view configuration information for the firewall but is not allowed to perform any configuration procedures.	Yes	Yes	Yes
VM Information Source	Controls access to the **VM Information Source** node that allows you to configure the firewall/Windows User-ID agent to collect VM inventory automatically. If you disable this privilege, the administrator will not see the **VM Information Source** node. If you set this privilege to read-only, the administrator can view the VM	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
	information sources configured but cannot add, edit, or delete any sources. *This privilege is not available to Device Group and Template administrators.*			
Certificate Management	Sets the default state to enable or disable for all of the Certificate settings described below.	Yes	No	Yes
Certificates	Controls access to the **Certificates** node. If you disable this privilege, the administrator will not see the **Certificates** node or be able to configure or access information regarding Device Certificates or Default Trusted Certificate Authorities. If you set this privilege to read-only, the administrator can view Certificate configuration information for the firewall but is not allowed to perform any configuration procedures.	Yes	Yes	Yes
Certificate Profile	Controls access to the **Certificate Profile** node. If you disable this privilege, the administrator will not see the **Certificate Profile** node or be able to create certificate profiles. If you set this privilege to read-only, the administrator can view Certificate Profiles that are currently configured for the firewall but is not allowed to create or edit a certificate profile.	Yes	Yes	Yes
OCSP Responder	Controls access to the **OCSP Responder** node. If you disable this privilege, the administrator will not see the **OCSP Responder** node or be able to define a server that will be used to verify the revocation status of certificates issues by the firewall. If you set this privilege to read-only, the administrator can view the **OCSP Responder** configuration for the firewall	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
	but is not allowed to create or edit an OCSP responder configuration.			
SSL/TLS Service Profile	Controls access to the **SSL/TLS Service Profile** node. If you disable this privilege, the administrator will not see the node or configure a profile that specifies a certificate and a protocol version or range of versions for firewall services that use SSL/TLS. If you set this privilege to read-only, the administrator can view existing SSL/TLS Service profiles but cannot create or edit them.	Yes	Yes	Yes
SCEP	Controls access to the **SCEP** node. If you disable this privilege, the administrator will not see the node or be able to define a profile that specifies simple certificate enrollment protocol (SCEP) settings for issuing unique device certificates. If you set this privilege to read-only, the administrator can view existing SCEP profiles but cannot create or edit them.	Yes	Yes	Yes
SSL Decryption Exclusion	Controls access to the **SSL Decryption Exclusion** node. If you disable this privilege, the administrator will not see the node or be able to add custom exclusions. If you set this privilege to read-only, the administrator can view existing SSL decryption exceptions but cannot create or edit them.	Yes	Yes	Yes
SSH Service Profile	Controls access to the **SSH Service Profile** node. If you disable this privilege, the administrator will be unable to see the node or configure a profile to specify parameters for SSH connections to your Palo Alto Networks management and high availability (HA) appliances.	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
	If you set this privilege to read-only, the administrator can view existing SSH service profiles but cannot edit or create them.			
Response Pages	Controls access to the **Response Pages** node. If you disable this privilege, the administrator will not see the **Response Page** node or be able to define a custom HTML message that is downloaded and displayed instead of a requested web page or file. If you set this privilege to read-only, the administrator can view the **Response Page** configuration for the firewall but is not allowed to create or edit a response page configuration.	Yes	Yes	Yes
Log Settings	Sets the default state to enable or disable for all of the Log settings described below.	Yes	No	Yes
System	Controls access to the **Log Settings** > **System** node. If you disable this privilege, the administrator cannot see the **Log Settings** > **System** node or specify which System logs the firewall forwards to Panorama or external services (such as a syslog server). If you set this privilege to read-only, the administrator can view the **Log Settings** > **System** settings for the firewall but cannot add, edit, or delete the settings.	Yes	Yes	Yes
Configuration	Controls access to the **Log Settings** > **Configuration** node. If you disable this privilege, the administrator cannot see the **Log Settings** > **Configuration** node or specify which Configuration logs the firewall forwards to Panorama or external services (such as a syslog server). If you set this privilege to read-only, the administrator can view the **Log Settings** > **Configuration** settings for the firewall but cannot add, edit, or delete the settings.	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
User-ID	Controls access to the **Log Settings** > **User-ID** node. If you disable this privilege, the administrator cannot see the **Log Settings** > **User-ID** node or specify which User-ID logs the firewall forwards to Panorama or external services (such as a syslog server). If you set this privilege to read-only, the administrator can view the **Log Settings** > **User-ID** settings for the firewall but cannot add, edit, or delete the settings.	Yes	Yes	Yes
HIP Match	Controls access to the **Log Settings** > **HIP Match** node. If you disable this privilege, the administrator cannot see the **Log Settings** > **HIP Match** node or specify which Host Information Profile (HIP) match logs the firewall forwards to Panorama or external services (such as a syslog server). HIP match logs provide information on Security policy rules that apply to GlobalProtect endpoints. If you set this privilege to read-only, the administrator can view the **Log Settings** > **HIP** settings for the firewall but cannot add, edit, or delete the settings.	Yes	Yes	Yes
GlobalProtect	Controls access to the **Log Settings** > **GlobalProtect** node. If you disable this privilege, the administrator cannot see the **Log Settings** > **GlobalProtect** node or specify which GlobalProtect logs the firewall forwards to Panorama or external services (such as a syslog server). If you set this privilege to read-only, the administrator can view the **Log Settings** > **GlobalProtect** settings for the firewall but cannot add, edit, or delete the settings.	Yes	Yes	Yes
Correlation	Controls access to the **Log Settings** > **Correlation** node. If you disable this privilege, the administrator cannot see the **Log Settings** > **Correlation** node or add, delete, or modify correlation log	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
	forwarding settings or tag source or destination IP addresses. If you set this privilege to read-only, the administrator can view the **Log Settings** > **Correlation** settings for the firewall but cannot add, edit, or delete the settings.			
Alarm Settings	Controls access to the **Log Settings** > **Alarm Settings** node. If you disable this privilege, the administrator cannot see the **Log Settings** > **Alarm Settings** node or configure notifications that the firewall generates when a Security policy rule (or group of rules) is hit repeatedly within a configurable time period. If you set this privilege to read-only, the administrator can view the **Log Settings** > **Alarm Settings** for the firewall but cannot edit the settings.	Yes	Yes	Yes
Manage Logs	Controls access to the **Log Settings** > **Manage Logs** node. If you disable this privilege, the administrator cannot see the **Log Settings** > **Manage Logs** node or clear the indicated logs. If you set this privilege to read-only, the administrator can view the **Log Settings** > **Manage Logs** information but cannot clear any of the logs.	Yes	Yes	Yes
Server Profiles	Sets the default state to enable or disable for all of the Server Profiles settings described below.	Yes	No	Yes
SNMP Trap	Controls access to the **Server Profiles** > **SNMP Trap** node. If you disable this privilege, the administrator will not see the **Server Profiles** > **SNMP Trap** node or be able to specify one or more SNMP trap destinations to be used for system log entries. If you set this privilege to read-only, the administrator can view the **Server Profiles**	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
	> **SNMP Trap Logs** information but cannot specify SNMP trap destinations.			
Syslog	Controls access to the **Server Profiles** > **Syslog** node. If you disable this privilege, the administrator will not see the **Server Profiles** > **Syslog** node or be able to specify one or more syslog servers. If you set this privilege to read-only, the administrator can view the **Server Profiles** > **Syslog** information but cannot specify syslog servers.	Yes	Yes	Yes
Email	Controls access to the **Server Profiles** > **Email** node. If you disable this privilege, the administrator will not see the **Server Profiles** > **Email** node or be able to configure an email profile that can be used to enable email notification for system and configuration log entries. If you set this privilege to read-only, the administrator can view the **Server Profiles** > **Email** information but cannot configure an email server profile.	Yes	Yes	Yes
HTTP	Controls access to the **Server Profiles** > **HTTP** node. If you disable this privilege, the administrator will not see the **Server Profiles** > **HTTP** node or be able to configure an HTTP server profile that can be used to enable log forwarding to HTTP destinations any log entries. If you set this privilege to read-only, the administrator can view the **Server Profiles** > **HTTP** information but cannot configure an HTTP server profile.	Yes	Yes	Yes
Netflow	Controls access to the **Server Profiles** > **Netflow** node. If you disable this privilege, the administrator will not see the **Server Profiles** > **Netflow** node or be able to define a NetFlow server profile, which specifies the frequency of the export along with the NetFlow servers that will receive the exported data.	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
	If you set this privilege to read-only, the administrator can view the **Server Profiles** > **Netflow** information but cannot define a Netflow profile.			
RADIUS	Controls access to the **Server Profiles** > **RADIUS** node. If you disable this privilege, the administrator will not see the **Server Profiles** > **RADIUS** node or be able to configure settings for the RADIUS servers that are identified in authentication profiles. If you set this privilege to read-only, the administrator can view the **Server Profiles** > **RADIUS** information but cannot configure settings for the RADIUS servers.	Yes	Yes	Yes
SCP	Controls access to the **Server Profiles** > **SCP** node. If you disable this privilege, the administrator will not see the node or configure settings for SCP servers. If you set this privilege to read-only, the administrator can view existing SCP server profiles but cannot add or edit them.	Yes	Yes	Yes
TACACS+	Controls access to the **Server Profiles** > **TACACS+** node. If you disable this privilege, the administrator will not see the node or configure settings for the TACACS + servers that authentication profiles reference. If you set this privilege to read-only, the administrator can view existing TACACS + server profiles but cannot add or edit them.	Yes	Yes	Yes
LDAP	Controls access to the **Server Profiles** > **LDAP** node. If you disable this privilege, the administrator will not see the **Server Profiles** > **LDAP** node or be able to configure settings for the LDAP servers	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
	to use for authentication by way of authentication profiles. If you set this privilege to read-only, the administrator can view the **Server Profiles** > **LDAP** information but cannot configure settings for the LDAP servers.			
Kerberos	Controls access to the **Server Profiles** > **Kerberos** node. If you disable this privilege, the administrator will not see the **Server Profiles** > **Kerberos** node or configure a Kerberos server that allows users to authenticate natively to a domain controller. If you set this privilege to read-only, the administrator can view the **Server Profiles** > **Kerberos** information but cannot configure settings for Kerberos servers.	Yes	Yes	Yes
SAML Identity Provider	Controls access to the **Server Profiles** > **SAML Identity Provider** node. If you disable this privilege, the administrator cannot see the node or configure SAML identity provider (IdP) server profiles. If you set this privilege to read-only, the administrator can view the **Server Profiles** > **SAML Identity Provider** information but cannot configure SAML IdP server profiles.	Yes	Yes	Yes
Multi Factor Authentication	Controls access to the **Server Profiles** > **Multi Factor Authentication** node. If you disable this privilege, the administrator cannot see the node or configure multi-factor authentication (MFA) server profiles. If you set this privilege to read-only, the administrator can view the **Server Profiles** > **SAML Identity Provider** information but cannot configure MFA server profiles.			

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Access Level	Description	Enable	Read Only	Disable
Local User Database	Sets the default state to enable or disable for all of the Local User Database settings described below.	Yes	No	Yes
Users	Controls access to the **Local User Database** > **Users** node. If you disable this privilege, the administrator will not see the **Local User Database** > **Users** node or set up a local database on the firewall to store authentication information for remote access users, firewall administrators, and Authentication Portal users. If you set this privilege to read-only, the administrator can view the **Local User Database** > **Users** information but cannot set up a local database on the firewall to store authentication information.	Yes	Yes	Yes
User Groups	Controls access to the **Local User Database** > **Users** node. If you disable this privilege, the administrator will not see the **Local User Database** > **Users** node or be able to add user group information to the local database. If you set this privilege to read-only, the administrator can view the **Local User Database** > **Users** information but cannot add user group information to the local database.	Yes	Yes	Yes
Access Domain	Controls access to the **Access Domain** node. If you disable this privilege, the administrator will not see the **Access Domain** node or be able to create or edit an access domain. If you set this privilege to read-only, the administrator can view the **Access Domain** information but cannot create or edit an access domain.	Yes	Yes	Yes
Scheduled Log Export	Controls access to the **Scheduled Log Export** node. If you disable this privilege, the administrator will not see the **Scheduled Log Export** node or be able	Yes	No	Yes

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Access Level	Description	Enable	Read Only	Disable
	schedule exports of logs and save them to a File Transfer Protocol (FTP) server in CSV format or use Secure Copy (SCP) to securely transfer data between the firewall and a remote host. If you set this privilege to read-only, the administrator can view the **Scheduled Log Export Profile** information but cannot schedule the export of logs.			
Software	Controls access to the **Software** node. If you disable this privilege, the administrator will not see the **Software** node or view the latest versions of the PAN-OS software available from Palo Alto Networks, read the release notes for each version, and select a release to download and install. If you set this privilege to read-only, the administrator can view the **Software** information but cannot download or install software.	Yes	Yes	Yes
GlobalProtect Client	Controls access to the **GlobalProtect Client** node. If you disable this privilege, the administrator will not see the **GlobalProtect Client** node or view available GlobalProtect releases, download the code or activate the GlobalProtect app. If you set this privilege to read-only, the administrator can view the available **GlobalProtect Client** releases but cannot download or install the app software.	Yes	Yes	Yes
Dynamic Updates	Controls access to the **Dynamic Updates** node. If you disable this privilege, the administrator will not see the **Dynamic Updates** node or be able to view the latest updates, read the release notes for each update, or select an update to upload and install. If you set this privilege to read-only, the administrator can view the available **Dynamic Updates** releases, read the	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
	release notes but cannot upload or install the software.			
Licenses	Controls access to the **Licenses** node. If you disable this privilege, the administrator will not see the **Licenses** node or be able to view the licenses installed or activate licenses. If you set this privilege to read-only, the administrator can view the installed **Licenses**, but cannot perform license management functions.	Yes	Yes	Yes
Support	Controls access to the **Support** node. If you disable this privilege, the administrator cannot see the **Support** node, activate support, or access production and security alerts from Palo Alto Networks. If you set this privilege to read-only, the administrator can see the **Support** node and access production and security alerts but cannot activate support.	Yes	Yes	Yes
Master Key and Diagnostics	Controls access to the **Master Key and Diagnostics** node. If you disable this privilege, the administrator will not see the **Master Key and Diagnostics** node or be able to specify a master key to encrypt private keys on the firewall. If you set this privilege to read-only, the administrator can view the **Master Key and Diagnostics** node and view information about master keys that have been specified but cannot add or edit a new master key configuration.	Yes	Yes	Yes
Policy Recommendation	Controls access to **IoT** and **SaaS** policy rule recommendations. If you disable these privileges, the administrator can't see the **Policy Recommendation** > **IoT** node, the **Policy Recommendation** > **SaaS** node, or both, depending on which privileges you disable.	Yes	Yes	Yes

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Access Level	Description	Enable	Read Only	Disable
	If you set these privileges to read-only, the administrator can view the nodes but cannot import policy rules or edit information.			

Define User Privacy Settings in the Admin Role Profile

To define what private end user data an administrator has access to, when creating or editing an admin role profile (**Device** > **Admin Roles**), scroll down to the **Privacy** option on the **WebUI** tab.

Access Level	Description	Enable	Read Only	Disable
Privacy	Sets the default state to enable or disable for all of the privacy settings described below.	Yes	N/A	Yes
Show Full IP addresses	When disabled, full IP addresses obtained by traffic running through the Palo Alto firewall are not shown in logs or reports. In place of the IP addresses that are normally displayed, the relevant subnet is displayed. *Scheduled reports that are displayed in the interface through **Monitor** > **Reports** and reports that are sent via scheduled emails will still display full IP addresses. Because of this exception, we recommend that the following settings within the **Monitor** tab be set to disable: Custom Reports, Application Reports, Threat Reports, URL Filtering Reports, Traffic Reports and Email Scheduler.*	Yes	N/A	Yes
Show User Names in Logs and Reports	When disabled, usernames obtained by traffic running through the Palo Alto Networks firewall are not shown in logs or reports. Columns where the usernames would normally be displayed are empty.	Yes	N/A	Yes

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Access Level	Description	Enable	Read Only	Disable
	*Scheduled reports that are displayed in the interface through **Monitor** > **Reports** or reports that are sent via the email scheduler will still display usernames. Because of this exception, we recommend that the following settings within the Monitor tab be set to disable: Custom Reports, Application Reports, Threat Reports, URL Filtering Reports, Traffic Reports and Email Scheduler.*			
View PCAP Files	When disabled, packet capture files that are normally available within the Traffic, Threat and Data Filtering logs are not displayed.	Yes	N/A	Yes

Restrict Administrator Access to Commit and Validate Functions

To restrict access to commit (and revert), save, and validate functions when creating or editing an Admin Role profile (**Device** > **Admin Roles**), scroll down to the **Commit**, **Save**, and **Validate** options on the **WebUI** tab.

Access Level	Description	Enable	Read Only	Disable
Commit	Sets the default state to enabled or disabled for all of the commit and revert privileges described below.	Yes	N/A	Yes
Device	When disabled, an administrator cannot commit or revert changes that any administrator made to the firewall configuration, including his or her own changes.	Yes	N/A	Yes
Commit For Other Admins	When disabled, an administrator cannot commit or revert changes that other administrators made to the firewall configuration.	Yes	N/A	Yes

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Access Level	Description	Enable	Read Only	Disable
Save	Sets the default state to enabled or disabled for all of the save operation privileges described below.	Yes	N/A	Yes
Partial save	When disabled, an administrator cannot save changes that any administrator made to the firewall configuration, including his or her own changes.	Yes	N/A	Yes
Save For Other Admins	When disabled, an administrator cannot save changes that other administrators made to the firewall configuration.	Yes	N/A	Yes
Validate	When disabled, an administrator cannot validate a configuration.	Yes	N/A	Yes

Provide Granular Access to Global Settings

To define what global settings and administrator has access to, when creating or editing an admin role profile (**Device** > **Admin Roles**), scroll down to the **Global** option on the **WebUI** tab.

Access Level	Description	Enable	Read Only	Disable
Global	Sets the default state to enable or disable for all of the global settings described below. In effect, this setting is only for System Alarms at this time.	Yes	N/A	Yes
System Alarms	When disabled, an administrator cannot view or acknowledge alarms that are generated.	Yes	N/A	Yes

Provide Granular Access to the Panorama Tab

The following table lists the **Panorama** tab access levels and the custom Panorama administrator roles for which they are available. Firewall administrators cannot access any of these privileges.

Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
Setup	Specifies whether the administrator can view or edit Panorama setup information,	Panorama: Yes	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	including **Management**, **Operations and Telemetry**, **Services**, Content-ID, **WildFire**, Session, or **HSM**. If you set the privilege to: • read-only, the administrator can see the information but cannot edit it. • disable this privilege, the administrator cannot see or edit the information.	Device Group/ Template: No			
High Availability	Specifies whether the administrator can view and manage high availability (HA) settings for the Panorama management server. If you set this privilege to read-only, the administrator can view HA configuration information for the Panorama management server but can't manage the configuration. If you disable this privilege, the administrator can't see or manage HA configuration settings for the Panorama management server.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes
Config Audit	Specifies whether the administrator can run Panorama configuration audits. If you disable this privilege, the administrator can't run Panorama configuration audits.	Panorama: Yes Device Group/ Template: No	Yes	No	Yes
Firewall Clusters	Specifies whether the administrator can create and configure CN-Series and PA-Series firewall clusters. If you set this privilege to read-only, the administrator can view firewall cluster information for the Panorama	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	management server but can't manage the configuration. If you disable this privilege, the administrator can't see or manage firewall clusters for the Panorama management server.				
Administrators	Specifies whether the administrator can view Panorama administrator account details. You can't enable full access to this function: just read-only access. (Only Panorama administrators with a dynamic role can add, edit, or delete Panorama administrators.) With read-only access, the administrator can see information about his or her own account but no other Panorama administrator accounts. If you disable this privilege, the administrator can't see information about any Panorama administrator account, including his or her own.	Panorama: Yes Device Group/ Template: No	No	Yes	Yes
Admin Roles	Specifies whether the administrator can view Panorama administrator roles. You can't enable full access to this function: just read-only access. (Only Panorama administrators with a dynamic role can add, edit, or delete custom Panorama roles.) With read-only access, the administrator can see Panorama administrator role configurations but can't manage them.	Panorama: Yes Device Group/ Template: No	No	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	If you disable this privilege, the administrator can't see or manage Panorama administrator roles.				
Access Domain	Specifies whether the administrator can view, add, edit, delete, or clone access domain configurations for Panorama administrators. (This privilege controls access only to the configuration of access domains, not access to the device groups, templates, and firewall contexts that are assigned to access domains.) If you set this privilege to read-only, the administrator can view Panorama access domain configurations but can't manage them. If you disable this privilege, the administrator can't see or manage Panorama access domain configurations.	Panorama: Yes Device Group/ Template: No *You assign access domains to Device Group and Template administrators so they can access the configuration and monitoring data within the device groups, templates, and firewall contexts that are assigned to those access domains.*	Yes	Yes	Yes
Authentication Profile	Specifies whether the administrator can view, add, edit, delete, or clone authentication profiles for Panorama administrators. If you set this privilege to read-only, the administrator can	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	view Panorama authentication profiles but can't manage them. If you disable this privilege, the administrator can't see or manage Panorama authentication profiles.				
Authentication Sequence	Specifies whether the administrator can view, add, edit, delete, or clone authentication sequences for Panorama administrators. If you set this privilege to read-only, the administrator can view Panorama authentication sequences but can't manage them. If you disable this privilege, the administrator can't see or manage Panorama authentication sequences.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes
User Identification	Specifies whether the administrator can configure User-ID connection security and view, add, edit, or delete data redistribution points (such as User-ID agents). If you set this privilege to read-only, the administrator can view settings for User-ID connection security and redistribution points but can't manage the settings. If you disable this privilege, the administrator can't see or manage settings for User-ID connection security or redistribution points.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes
Managed Devices	Specifies whether the administrator can view, add, edit, or delete firewalls as managed devices, and install	Panorama: Yes Device Group/ Template: Yes	Yes (No for Device	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	software or content updates on them. If you set this privilege to read-only, the administrator can see managed firewalls but can't add, delete, tag, or install updates on them. If you disable this privilege, the administrator can't view, add, edit, tag, delete, or install updates on managed firewalls. *An administrator with Device Deployment privileges can still select **Panorama** > **Device Deployment** to install updates on managed firewalls.*		Group and Template roles)		
Templates	Specifies whether the administrator can view, edit, add, or delete templates and template stacks. If you set the privilege to read-only, the administrator can see template and stack configurations but can't manage them. If you disable this privilege, the administrator can't see or manage template and stack configurations.	Panorama: Yes Device Group/ Template: Yes *Device Group and Template administrators can see only the templates and stacks that are within the access domains assigned to those administrators.*	Yes (No for Device Group and Template admins)	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
Device Groups	Specifies whether the administrator can view, edit, add, or delete device groups. If you set this privilege to read-only, the administrator can see device group configurations but can't manage them. If you disable this privilege, the administrator can't see or manage device group configurations.	Panorama: Yes Device Group/ Template: Yes *Device Group and Template administrators can access only the device groups that are within the access domains assigned to those administrators.*	Yes	Yes	Yes
Managed Collectors	Specifies whether the administrator can view, edit, add, or delete managed collectors. If you set this privilege to read-only, the administrator can see managed collector configurations but can't manage them. If you disable this privilege, the administrator can't view, edit, add, or delete managed collector configurations.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	*An administrator with Device Deployment privileges can still use the **Panorama** > **Device Deployment** options to install updates on managed collectors.*				
Collector Groups	Specifies whether the administrator can view, edit, add, or delete Collector Groups. If you set this privilege to read-only, the administrator can see Collector Groups but can't manage them. If you disable this privilege, the administrator can't see or manage Collector Groups.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes
VMware Service Manager	Specifies whether the administrator can view and edit VMware Service Manager settings. If you set this privilege to read-only, the administrator can see the settings but can't perform any related configuration or operational procedures. If you disable this privilege, the administrator can't see the settings or perform any related configuration or operational procedures.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes
Certificate Management	Sets the default state, enabled or disabled, for all of the Panorama certificate management privileges.	Panorama: Yes Device Group/ Template: No	Yes	No	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
Certificates	Specifies whether the administrator can view, edit, generate, delete, revoke, renew, or export certificates. This privilege also specifies whether the administrator can import or export HA keys. If you set this privilege to read-only, the administrator can see Panorama certificates but can't manage the certificates or HA keys. If you disable this privilege, the administrator can't see or manage Panorama certificates or HA keys.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes
Certificate Profile	Specifies whether the administrator can view, add, edit, delete or clone Panorama certificate profiles. If you set this privilege to read-only, the administrator can see Panorama certificate profiles but can't manage them. If you disable this privilege, the administrator can't see or manage Panorama certificate profiles.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes
SSL/TLS Service Profile	Specifies whether the administrator can view, add, edit, delete or clone SSL/TLS Service profiles. If you set this privilege to read-only, the administrator can see SSL/TLS Service profiles but can't manage them. If you disable this privilege, the administrator can't see or manage SSL/TLS Service profiles.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
Log Settings	Sets the default state, enabled or disabled, for all the log setting privileges.	Panorama: Yes Device Group/ Template: No	Yes	No	Yes
System	Specifies whether the administrator can see and configure the settings that control the forwarding of System logs to external services (syslog, email, SNMP trap, or HTTP servers). If you set this privilege to read-only, the administrator can see the System log forwarding settings but can't manage them. If you disable this privilege, the administrator can't see or manage the settings.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	This privilege pertains only to System logs that Panorama and Log Collectors generate. The Collector Groups *privilege (**Panorama** > **Collector Groups**) controls forwarding for System logs that Log Collectors receive from firewalls. The **Device** > **Log Settings** >* System *privilege controls log forwarding from firewalls directly to external services (without aggregation on Log Collectors).*				
Config	Specifies whether the administrator can see and configure the settings that control the forwarding of Config logs to external services (syslog, email, SNMP trap, or HTTP servers). If you set this privilege to read-only, the administrator can see the Config log forwarding settings but can't manage them. If you disable this privilege, the administrator can't see or manage the settings.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	This privilege pertains only to Config logs that Panorama and Log Collectors generate. The Collector Groups *privilege (**Panorama** > **Collector Groups**) controls forwarding for Config logs that Log Collectors receive from firewalls. The **Device** > **Log Settings** >* Configuration *privilege controls log forwarding from firewalls directly to external services (without aggregation on Log Collectors).*				
User-ID	Specifies whether the administrator can see and configure the settings that control the forwarding of User-ID logs to external services (syslog, email, SNMP trap, or HTTP servers). If you set this privilege to read-only, the administrator can see the Config log forwarding settings but can't manage them. If you disable this privilege, the administrator can't see or manage the settings.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	*This privilege pertains only to User-ID logs that Panorama generates. The Collector Groups privilege (**Panorama > Collector Groups**) controls forwarding for User-ID logs that Log Collectors receive from firewalls. The **Device > Log Settings** > User-ID privilege controls log forwarding from firewalls directly to external services (without aggregation on Log Collectors).*				
HIP Match	Specifies whether the administrator can see and configure the settings that control the forwarding of HIP Match logs from a Panorama virtual appliance in Legacy mode to external services (syslog, email, SNMP trap, or HTTP servers). If you set this privilege to read-only, the administrator can see the forwarding settings of HIP Match logs but can't manage them. If you disable this privilege, the administrator can't see or manage the settings.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	*The Collector Groups privilege (**Panorama** > **Collector Groups**) controls forwarding for HIP Match logs that Log Collectors receive from firewalls. The **Device** > **Log Settings** > HIP Match privilege controls log forwarding from firewalls directly to external services (without aggregation on Log Collectors).*				
GlobalProtect	Specifies whether the administrator can see and configure the settings that control the forwarding of GlobalProtect logs from a Panorama virtual appliance in Legacy mode to external services (syslog, email, SNMP trap, or HTTP servers). If you set this privilege to read-only, the administrator can see the forwarding settings of GlobalProtect logs but can't manage them. If you disable this privilege, the administrator can't see or manage the settings.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	*The Collector Groups privilege (**Panorama > Collector Groups**) controls forwarding for GlobalProtect logs that Log Collectors receive from firewalls. The **Device > Log Settings > GlobalProtect** privilege controls log forwarding from firewalls directly to external services (without aggregation on Log Collectors).*				
Correlation	Specifies whether the administrator can see and configure the settings that control the forwarding of Correlation logs from a Panorama virtual appliance in Legacy mode to external services (syslog, email, SNMP trap, or HTTP servers). If you set this privilege to read-only, the administrator can see the Correlation log forwarding settings but can't manage them. If you disable this privilege, the administrator can't see or manage the settings.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	*The Collector Groups privilege (**Panorama > Collector Groups**) controls forwarding of Correlation logs from a Panorama M-Series appliance or Panorama virtual appliance in Panorama mode.*				
Traffic	Specifies whether the administrator can see and configure the settings that control the forwarding of Traffic logs from a Panorama virtual appliance in Legacy mode to external services (syslog, email, SNMP trap, or HTTP servers). If you set this privilege to read-only, the administrator can see the forwarding settings of Traffic logs but can't manage them. If you disable this privilege, the administrator can't see or manage the settings.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	*The Collector Groups privilege (**Panorama > Collector Groups**) controls forwarding for Traffic logs that Log Collectors receive from firewalls. The Log Forwarding privilege (**Objects > Log Forwarding**) controls forwarding from firewalls directly to external services (without aggregation on Log Collectors).*				
Threat	Specifies whether the administrator can see and configure the settings that control the forwarding of Threat logs from a Panorama virtual appliance in Legacy mode to external services (syslog, email, SNMP trap, or HTTP servers). If you set this privilege to read-only, the administrator can see the forwarding settings of Threat logs but can't manage them. If you disable this privilege, the administrator can't see or manage the settings.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	*The Collector Groups privilege (**Panorama > Collector Groups**) controls forwarding for Threat logs that Log Collectors receive from firewalls. The Log Forwarding privilege (**Objects > Log Forwarding**) controls forwarding from firewalls directly to external services (without aggregation on Log Collectors).*				
WildFire	Specifies whether the administrator can see and configure the settings that control the forwarding of WildFire logs from a Panorama virtual appliance in Legacy mode to external services (syslog, email, SNMP trap, or HTTP servers). If you set this privilege to read-only, the administrator can see the forwarding settings of WildFire logs but can't manage them. If you disable this privilege, the administrator can't see or manage the settings.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	*The Collector Groups privilege (**Panorama** > **Collector Groups**) controls the forwarding for WildFire logs that Log Collectors receive from firewalls. The Log Forwarding privilege (**Objects** > **Log Forwarding**) controls forwarding from firewalls directly to external services (without aggregation on Log Collectors).*				
Server Profiles	Sets the default state, enabled or disabled, for all the server profile privileges.	Panorama: Yes Device Group/ Template: No	Yes	No	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	*These privileges pertain only to the server profiles that are used for forwarding logs from Panorama or Log Collectors and the server profiles that are used for authenticating Panorama administrators. The **Device** > Server Profiles privileges control access to the server profiles that are used for forwarding logs directly from firewalls to external services and for authenticating firewall administrators.*				
SNMP Trap	Specifies whether the administrator can see and configure SNMP trap server profiles. If you set this privilege to read-only, the administrator can see SNMP trap server profiles but can't manage them. If you disable this privilege, the administrator can't see or manage SNMP trap server profiles.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes
Syslog	Specifies whether the administrator can see and configure Syslog server profiles.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	If you set this privilege to read-only, the administrator can see Syslog server profiles but can't manage them. If you disable this privilege, the administrator can't see or manage Syslog server profiles.				
Email	Specifies whether the administrator can see and configure email server profiles. If you set this privilege to read-only, the administrator can see email server profiles but can't manage them. If you disable this privilege, the administrator can't see or manage email server profiles.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes
RADIUS	Specifies whether the administrator can see and configure the RADIUS server profiles that are used to authenticate Panorama administrators. If you set this privilege to read-only, the administrator can see the RADIUS server profiles but can't manage them. If you disable this privilege, the administrator can't see or manage the RADIUS server profiles.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes
TACACS+	Specifies whether the administrator can see and configure the TACACS+ server profiles that are used to authenticate Panorama administrators. If you disable this privilege, the administrator can't see the node or configure settings for the TACACS+ servers	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	that authentication profiles reference. If you set this privilege to read-only, the administrator can view existing TACACS+ server profiles but can't add or edit them.				
LDAP	Specifies whether the administrator can see and configure the LDAP server profiles that are used to authenticate Panorama administrators. If you set this privilege to read-only, the administrator can see the LDAP server profiles but can't manage them. If you disable this privilege, the administrator can't see or manage the LDAP server profiles.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes
Kerberos	Specifies whether the administrator can see and configure the Kerberos server profiles that are used to authenticate Panorama administrators. If you set this privilege to read-only, the administrator can see the Kerberos server profiles but can't manage them. If you disable this privilege, the administrator can't see or manage the Kerberos server profiles.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes
SAML Identity Provider	Specifies whether the administrator can see and configure the SAML Identity Provider (IdP) server profiles that are used to authenticate Panorama administrators.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	If you set this privilege to read-only, the administrator can see the SAML IdP server profiles but can't manage them. If you disable this privilege, the administrator can't see or manage the SAML IdP server profiles.				
Scheduled Config Export	Specifies whether the administrator can view, add, edit, delete, or clone scheduled Panorama configuration exports. If you set this privilege to read-only, the administrator can view the scheduled exports but can't manage them. If you disable this privilege, the administrator can't see or manage the scheduled exports.	Panorama: Yes Device Group/ Template: No	Yes	No	Yes
Software	Specifies whether the administrator can: view information about software updates installed on the Panorama management server; download, upload, or install the updates; and view the associated release notes. If you set this privilege to read-only, the administrator can view information about Panorama software updates and view the associated release notes but can't perform any related operations. If you disable this privilege, the administrator can't see Panorama software updates, see the associated release notes, or perform any related operations.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	*The **Panorama** > Device Deployment > Software privilege controls access to PAN-OS software deployed on firewalls and Panorama software deployed on Dedicated Log Collectors.*				
Dynamic Updates	Specifies whether the administrator can: view information about content updates installed on the Panorama management server (for example, WildFire updates); download, upload, install, or revert the updates; and view the associated release notes. If you set this privilege to read-only, the administrator can view information about Panorama content updates and view the associated release notes but can't perform any related operations. If you disable this privilege, the administrator can't see Panorama content updates, see the associated release notes, or perform any related operations.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	*The **Panorama** > Device Deployment > Dynamic Updates privilege controls access to content updates deployed on firewalls and Dedicated Log Collectors.*				
Support	Specifies whether the administrator can: view Panorama support license information, product alerts, and security alerts; activate a support license, and manage cases. Only a superuser admin can generate Tech Support files. If you set this privilege to read-only, the administrator can view Panorama support information, product alerts, and security alerts, but can't activate a support license, generate Tech Support files, or manage cases. If you disable this privilege, the administrator can't: see Panorama support information, product alerts, or security alerts; activate a support license, generate Tech Support files, or manage cases.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes
Device Deployment	Sets the default state, enabled or disabled, for all the privileges associated with deploying licenses and software or content updates to firewalls and Log Collectors.	Panorama: Yes Device Group/ Template: Yes	Yes	No	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	*The **Panorama** > Software and **Panorama** > Dynamic Updates privileges control the software and content updates installed on a Panorama management server.*				
Software	Specifies whether the administrator can: view information about the software updates installed on firewalls and Log Collectors; download, upload, or install the updates; and view the associated release notes. If you set this privilege to read-only, the administrator can see information about the software updates and view the associated release notes but can't deploy the updates to firewalls or dedicated Log Collectors. If you disable this privilege, the administrator can't see information about the software updates, see the associated release notes, or deploy the updates to firewalls or Dedicated Log Collectors.	Panorama: Yes Device Group/ Template: Yes	Yes	Yes	Yes
GlobalProtect Client	Specifies whether the administrator can: view information about GlobalProtect app software updates on firewalls; download, upload, or activate the updates; and view the associated release notes.	Panorama: Yes Device Group/ Template: Yes	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	If you set this privilege to read-only, the administrator can see information about GlobalProtect app software updates and view the associated release notes but can't activate the updates on firewalls. If you disable this privilege, the administrator can't see information about GlobalProtect app software updates, see the associated release notes, or activate the updates on firewalls.				
Dynamic Updates	Specifies whether the administrator can: view information about the content updates (for example, Applications updates) installed on firewalls and Dedicated Log Collectors; download, upload, or install the updates; and view the associated release notes. If you set this privilege to read-only, the administrator can see information about the content updates and view the associated release notes but can't deploy the updates to firewalls or Dedicated Log Collectors. If you disable this privilege, the administrator can't see information about the content updates, see the associated release notes, or deploy the updates to firewalls or Dedicated Log Collectors.	Panorama: Yes Device Group/ Template: Yes	Yes	Yes	Yes
Licenses	Specifies whether the administrator can view, refresh, and activate firewall licenses.	Panorama: Yes Device Group/ Template: Yes	Yes	Yes	Yes

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Access Level	Description	Administrator Role Availability	Enable	Read Only	Disable
	If you set this privilege to read-only, the administrator can view firewall licenses but can't refresh or activate those licenses. If you disable this privilege, the administrator can't view, refresh, or activate firewall licenses.				
Master Key and Diagnostics	Specifies whether the administrator can view and configure a master key by which to encrypt private keys on Panorama. If you set this privilege to read-only, the administrator can view the Panorama master key configuration but can't change it. If you disable this privilege, the administrator can't see or edit the Panorama master key configuration.	Panorama: Yes Device Group/ Template: No	Yes	Yes	Yes

Provide Granular Access to Operations Settings

To define which operations settings an administrator has access to, when creating or editing an admin role profile for a firewall (**Device** > **Admin Roles**), scroll down to the **Operations** option on the **Web UI** tab.

Access Level	Description	Enable	Read Only	Disable
Reboot	Restart the firewall. The firewall logs out all users, reloads the PAN-OS software and active configuration, closes and logs existing sessions, and creates a system log entry that shows the name of the administrator that initiated the reboot. This access also affects Shutdown operations.	Yes	N/A	Yes

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Access Level	Description	Enable	Read Only	Disable
Generate Tech Support File	Generate a tech support system file that the Palo Alto Networks support team can use to troubleshoot issues that you may be experiencing with the firewall.	Yes	N/A	Yes
Generate Stats Dump File	Generate and download a set of XML reports that summarizes network traffic over the last seven days for the firewall.	Yes	N/A	Yes
Download Core Files	If the firewall experiences a system process failure, a core file is automatically generated that contains details about the process and why it failed. You can download this core file to upload to your Palo Alto Networks support case to obtain further assistance in resolving the issue.	Yes	N/A	Yes
Download Debug and Management Pcap Files	If your firewall experiences a packet capture failure, it generates a packet capture (pcap) file that contains debug and management details for why it failed. You can download this pcap file to upload it to a Palo Alto Networks support case to obtain assistance in resolving the issue.	Yes	N/A	Yes

Panorama Web Interface Access Privileges

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

The custom Panorama administrator roles allow you to define access to the options on Panorama and the ability to only allow access to Device Groups and Templates (**Policies**, **Objects**, **Network**, **Device** tabs).

The administrator roles you can create are **Panorama** and **Device Group and Template**. You can't assign CLI access privileges to a **Device Group and Template** Admin Role profile. If you assign superuser privileges for the CLI to a **Panorama** Admin Role profile, administrators with that role can access all features regardless of the web interface privileges you assign.

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Access Level	Description	Enable	Read Only	Disable
Dashboard	Controls access to the **Dashboard** tab. If you disable this privilege, the administrator will not see the tab and will not have access to any of the Dashboard widgets.	Yes	No	Yes
ACC	Controls access to the Application Command Center (ACC). If you disable this privilege, the **ACC** tab will not display in the web interface. Keep in mind that if you want to protect the privacy of your users while still providing access to the ACC, you can disable the **Privacy** > **Show Full IP Addresses** option and/or the **Show User Names In Logs And Reports** option.	Yes	No	Yes
Monitor	Controls access to the **Monitor** tab. If you disable this privilege, the administrator will not see the **Monitor** tab and will not have access to any of the logs, packet captures, session information, reports or to App Scope. For more granular control over what monitoring information the administrator can see, leave the Monitor option enabled and then enable or disable specific nodes on the tab as described in Provide Granular Access to the Monitor Tab.	Yes	No	Yes
Policies	Controls access to the **Policies** tab. If you disable this privilege, the administrator will not see the **Policies** tab and will not have access to any policy information. For more granular control over what policy information the administrator can see, for example to enable access to a specific type of policy or to enable read-only access to policy information, leave the **Policies** option enabled and then enable or disable specific nodes on the tab as described in Provide Granular Access to the Policy Tab.	Yes	No	Yes
Objects	Controls access to the **Objects** tab. If you disable this privilege, the administrator will not see the **Objects** tab and will	Yes	No	Yes

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Access Level	Description	Enable	Read Only	Disable
	not have access to any objects, security profiles, log forwarding profiles, decryption profiles, or schedules. For more granular control over what objects the administrator can see, leave the **Objects** option enabled and then enable or disable specific nodes on the tab as described in Provide Granular Access to the Objects Tab.			
Network	Controls access to the **Network** tab. If you disable this privilege, the administrator will not see the **Network** tab and will not have access to any interface, zone, VLAN, virtual wire, virtual router, IPsec tunnel, DHCP, DNS Proxy, GlobalProtect, or QoS configuration information or to the network profiles. For more granular control over what objects the administrator can see, leave the **Network** option enabled and then enable or disable specific nodes on the tab as described in Provide Granular Access to the Network Tab.	Yes	No	Yes
Device	Controls access to the **Device** tab. If you disable this privilege, the administrator will not see the **Device** tab and will not have access to any firewall-wide configuration information, such as User-ID, High Availability, server profile or certificate configuration information. For more granular control over what objects the administrator can see, leave the **Device** option enabled and then enable or disable specific nodes on the tab as described in Provide Granular Access to the Device Tab. *You can't enable access to the **Admin Roles** or **Administrators** nodes for a role-based administrator even if you enable full access to the **Device** tab.*	Yes	No	Yes

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Access Level	Description	Enable	Read Only	Disable
Panorama	Controls access to the **Panorama** tab. If you disable this privilege, the administrator will not see the **Panorama** tab and will not have access to any Panorama-wide configuration information, such as Managed Devices, Managed Collectors, or Collector Groups. For more granular control over what objects the administrator can see, leave the **Panorama** option enabled and then enable or disable specific nodes on the tab as described in Provide Granular Access to the Panorama Tab.	Yes	No	Yes
Privacy	Controls access to the privacy settings described in Define User Privacy Settings in the Admin Role Profile.	Yes	No	Yes
Validate	When disabled, an administrator cannot validate a configuration.	Yes	No	Yes
Save	Sets the default state (enabled or disabled) for all the save privileges described below (Partial Save and Save For Other Admins).	Yes	No	Yes
• Partial Save	When disabled, an administrator cannot save changes that any administrator made to the Panorama configuration.	Yes	No	Yes
• Save For Other Admins	When disabled, an administrator cannot save changes that other administrators made to the Panorama configuration.	Yes	No	Yes
Commit	Sets the default state (enabled or disabled) for all the commit, push, and revert privileges described below (Panorama, Device Groups, Templates, Force Template Values, Collector Groups, WildFire Appliance Clusters).	Yes	No	Yes
• Panorama	When disabled, an administrator cannot commit or revert configuration changes that any administrators made, including his or her own changes.	Yes	No	Yes

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Access Level	Description	Enable	Read Only	Disable
• Commit for Other Admins	When disabled, an administrator cannot commit or revert configuration changes that other administrators made.	Yes	No	Yes
• Push All Changes	When disabled, an administrator cannot push all configuration changes made by admins.	Yes	No	Yes
• Push For Other Admins	When disabled, an administrator cannot select and push configuration changes made by another admin.	Yes	No	Yes
• Object Level Changes	When disabled, an administrator cannot select individual configuration objects to push.	Yes	No	Yes
Device Groups	When disabled, an administrator cannot push changes to device groups.	Yes	No	Yes
Templates	When disabled, an administrator cannot push changes to templates.	Yes	No	Yes
Force Template Values	This privilege controls access to the **Force Template Values** option in the Push Scope Selection dialog. When disabled, an administrator cannot replace overridden settings in local firewall configurations with settings that Panorama pushes from a template. *If you push a configuration with **Force Template Values** enabled, all overridden values on the firewall are replaced with values from the template. Before you use this option, check for overridden values on the firewalls to ensure your commit does not result in any unexpected network outages or issues caused by replacing those overridden values.*	Yes	No	Yes

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Access Level	Description	Enable	Read Only	Disable
Collector Groups	When disabled, an administrator cannot push changes to Collector Groups.	Yes	No	Yes
WildFire Appliance Clusters	When disabled, an administrator cannot push changes to WildFire appliance clusters.	Yes	No	Yes
Tasks	When disabled, an administrator cannot access the Task Manager.	Yes	No	Yes
Global	Controls access to the global settings (system alarms) described in Provide Granular Access to Global Settings.	Yes	No	Yes

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Reference: Port Number Usage

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

The tables in this reference list the ports that firewalls and Panorama use to communicate with each other, or with other services on the network.

Ports Used for Management Functions

The firewall and Panorama use the following ports for management functions.

Destination Port	Protocol	Description
22	TCP	Used for communication from a client system to the firewall CLI interface.
80	TCP	The port the firewall listens on for Online Certificate Status Protocol (OCSP) updates when acting as an OCSP responder. *Port 80 is also used for OCSP verification if specified in the server certificate.*
123	UDP	Port the firewall uses for NTP updates.
443	TCP	Used for communication from a client system to the firewall web interface. This is also the port the firewall and User-ID agent listens on for updates when you Enable VM Monitoring to Track Changes on the Virtual Network. Used for outbound communications from the firewall to the Palo Alto Networks Update Server. For monitoring an AWS environment, this is the only port that is used. For monitoring a VMware vCenter/ESXi environment, the listening port defaults to 443, but it is configurable.
4443	TCP	Used as an alternative SSL port for HTTPS.
162	UDP	Port the firewall, Panorama, or a Log Collector uses to Forward Traps to an SNMP Manager.

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Destination Port	Protocol	Description
		This port doesn't need to be open on the Palo Alto Networks firewall. You must configure the Simple Network Management Protocol (SNMP) manager to listen on this port. For details, refer to the documentation of your SNMP management software.
161	UDP TCP	Port the firewall listens on for polling requests (GET messages) from the SNMP manager.
514 514 6514	TCP UDP SSL	Port that the firewall, Panorama, or a Log Collector uses to send logs to a syslog server if you Configure Syslog Monitoring, and the ports that the PAN-OS integrated User-ID agent or Windows-based User-ID agent listens on for authentication syslog messages.
2055	UDP	Default port the firewall uses to send NetFlow records to a NetFlow collector if you Configure NetFlow Exports, but this is configurable.
5008	TCP	Port the GlobalProtect Mobile Security Manager listens on for HIP requests from the GlobalProtect gateways. If you are using a third-party MDM system, you can configure the gateway to use a different port as required by the MDM vendor.
6080 6081 6082	TCP TLS 1.2 TCP	Ports used for User-ID™ Authentication Portal: • 6080 for NT LAN Manager (NTLM) authentication • 6081 for Authentication Portal without an SSL/TLS Server Profile • 6082 for Authentication Portal with an SSL/TLS Server Profile
10443	SSL	Port that the firewall and Panorama use to provide contextual information about a threat or to seamlessly shift your threat investigation to the Threat Vault and AutoFocus.
9300 9301 9302	TCP	Port used by log collectors to listen to ElasticSearch clustering.

Ports Used for HA

Firewalls configured as High Availability (HA) peers must be able to communicate with each other to maintain state information (HA1 control link) and synchronize data (HA2 data link). In

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Active/Active HA deployments the peer firewalls must also forward packets to the HA peer that owns the session. The HA3 link is a Layer 2 (MAC-in-MAC) link and it does not support Layer 3 addressing or encryption.

Destination Port	Protocol	Description
28764	TCP	Port used for HA1 sysd ssh tunneled communication.
28765	TCP	Port used for backup HA1 sysd ssh tunneled communication.
28766	TCP	Port used for HA1 ssh tunneled communication.
28767	TCP	Port used for backup HA1 ssh tunneled communication.
28769 28260	TCP TCP	Used for the HA1 control link for clear text communication between the HA peer firewalls. The HA1 link is a Layer 3 link and requires an IP address.
28	TCP	Used for the HA1 control link for encrypted communication (SSH over TCP) between the HA peer firewalls.
28770	TCP	Listening port for HA1 backup links.
28771	TCP	Used for heartbeat backups. Palo Alto Networks recommends enabling heartbeat backup on the MGT interface if you use an in-band port for the HA1 or the HA1 backup links.
99 29281	IP UDP	Used for the HA2 link to synchronize sessions, forwarding tables, IPSec security associations and ARP tables between firewalls in an HA pair. Data flow on the HA2 link is always unidirectional (except for the HA2 keep-alive); it flows from the active firewall (Active/Passive) or active-primary (Active/Active) to the passive firewall (Active/Passive) or active-secondary (Active/Active). The HA2 link is a Layer 2 link, and it uses ether type 0x7262 by default. The HA data link can also be configured to use either IP (protocol number 99) or UDP (port 29281) as the transport, and thereby allow the HA data link to span subnets.

Ports Used for Clustering

The firewall and Panorama use the following ports for clustering.

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Destination Port	Protocol	Description
5404–5406	UDP	Used for Corosync (HSCI port communication).
28829	TCP	Used for C3 cluster streaming proxy service, low latency (HSCI port communication).
28830	TCP	Used for C3 cluster streaming proxy service, normal latency (HSCI port communication).
28840	TCP	Used to listen for the management port heartbeat.

Hyperscale security fabric (HSF) uses the following ports for clustering.

Destination Port	Protocol	Description
28884	UDP	Used over the HSCI interface for CI keepalive.
28885	UDP	Used to listen to the management interface heartbeat messages among cluster nodes.
29284	UDP	TI interface for cluster traffic keepalives
28832	TCP	Used for C3 cluster secure streaming proxy service.

Ports Used for Panorama

Panorama uses the following ports.

Destination Port	Protocol	Description
22	TCP	Used for communication from a client system to the Panorama CLI interface.
443	TCP	Used for communication from a client system to the Panorama web interface. Used for outbound communications from Panorama to the Palo Alto Networks Update Server.
444	TCP	Used for communication between Panorama and Strata Logging Service.

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Destination Port	Protocol	Description
3978	TCP	Used for communication between Panorama and managed firewalls or managed collectors, as well as for communication among managed collectors in a Collector Group: • For communication between Panorama and firewalls. This connection is initiated from the managed firewall to Panorama and facilitates a bi-directional data exchange on which the firewalls forward logs to Panorama and Panorama pushes configuration changes to the firewalls. Context switching commands are sent over the same connection. • Log Collectors use this destination port to forward logs to Panorama. • For communication with the default Log Collector on an M-Series appliance in Panorama mode and with Dedicated Log Collectors.
28443	TCP	Used for managed devices (firewalls and Log Collectors) to retrieve software and content updates from Panorama. *Only devices that run PAN-OS 8.x and later releases retrieve updates from Panorama over this port. For devices running earlier releases, Panorama pushes the update packages over port 3978.*
28769 28260	TCP TCP	Used for the HA connectivity and synchronization between Panorama HA peers using clear text communication. Communication can be initiated by either peer. *ICMP must be allowed on the network for successful Panorama HA peer connection and synchronization. Additionally, ICMP is required to monitor the failover metrics used to detect whether an HA failover is required.*
28	TCP	Used for the HA connectivity and synchronization between Panorama HA peers using encrypted communication (SSH over TCP). Communication can be initiated by either peer. Used for communication between Log Collectors in a Collector Group for log distribution.

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Destination Port	Protocol	Description
28270 9300 to 9302 (11.1 and later)	TCP	Used for communication among Log Collectors in a Collector Group for log distribution.
2049	TCP	Used by the Panorama virtual appliance to write logs to the NFS datastore.
10443	SSL	Port that Panorama uses to provide contextual information about a threat or to seamlessly shift your threat investigation to the Threat Vault and AutoFocus.
23000 to 23999	TCP, UDP, or SSL	Used for Syslog communication between Panorama and the Traps ESM components.
8765	HTTP	Used as an offline license server to distribute licenses in air-gapped environments. Requests to 443 are routed to 8765 via `nginx`.

Ports Used for GlobalProtect

GlobalProtect uses the following ports.

Destination Port	Protocol	Description
443	TCP	Used for communication between GlobalProtect apps and portals, or GlobalProtect apps and gateways and for SSL tunnel connections. GlobalProtect gateways also use this port to collect host information from GlobalProtect apps and perform host information profile (HIP) checks.
4501	UDP	Used for IPSec tunnel connections between GlobalProtect apps and gateways.

For tips on how to use a loopback interface to provide access to GlobalProtect on different ports and addresses, refer to Can GlobalProtect Portal Page be Configured tobe Accessed on any Port?

Ports Used for User-ID

User-ID is a feature that enables mapping of user IP addresses to usernames and group memberships, enabling user- or group-based policy and visibility into user activity on your network (for example, to be able to quickly track down a user who may be the victim of a threat).

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To perform this mapping, the firewall, the User-ID agent (either installed on a Windows-based system or the PAN-OS integrated agent running on the firewall), and/or the Terminal Server agent must be able to connect to directory services on your network to perform Group Mapping and User Mapping. Additionally, if the agents are running on systems external to the firewall, they must be able to connect to the firewall to communicate the IP address to username mappings to the firewall. The following table lists the communication requirements for User-ID along with the port numbers required to establish connections.

Destination Port	Protocol	Description
389	TCP	Port the firewall uses to connect to an LDAP server (plaintext or Start Transport Layer Security (Start TLS) to Map Users to Groups.
3268	TCP	Port the firewall uses to connect to an Active Directory global catalog server (plaintext or Start TLS) to Map Users to Groups.
636	TCP	Port the firewall uses for LDAP over SSL connections with an LDAP server to Map Users to Groups.
3269	TCP	Port the firewall uses for LDAP over SSL connections with an Active Directory global catalog server to Map Users to Groups.
514 6514	TCP UDP SSL	Port the User-ID agent listens on for authentication syslog messages if you Configure User-ID to Monitor Syslog Senders for User Mapping. The port depends on the type of agent and protocol: • PAN-OS integrated User-ID agent—Port 6514 for SSL and port 514 for UDP. • Windows-based User-ID agent—Port 514 for both TCP and UDP.
5007	TCP	Port the firewall listens on for user mapping information. The agent sends the IP address and username mapping along with a timestamp whenever it learns of a new or updated mapping. In addition, it refreshes known mappings.
5006	TCP	Port the User-ID agent listens on for XML API requests. The source for this communication is typically the system running a script that invokes the API.
88	UDP/TCP	Port the User-ID agent uses to authenticate to a Kerberos server. The firewall tries UDP first and falls back to TCP.
1812	UDP	Port the User-ID agent uses to authenticate to a RADIUS server.

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Destination Port	Protocol	Description
49	TCP	Port the User-ID agent uses to authenticate to a TACACS+ server.
135	TCP	Port the User-ID agent uses to establish TCP-based WMI connections with the Microsoft Remote Procedure Call (RPC) Endpoint Mapper. The Endpoint Mapper then assigns the agent a randomly assigned port in the 49152-65535 port range. The agent uses this connection to make RPC queries for Exchange Server or AD server security logs, session tables. This is also the port used to access Terminal Servers. The User-ID agent also uses this port to connect to client systems to perform Windows Management Instrumentation (WMI) probing.
139	TCP	Port the User-ID agent uses to establish TCP-based NetBIOS connections to the AD server so that it can send RPC queries for security logs and session information.
445	TCP	Port the User-ID agent uses to connect to the Active Directory (AD) using TCP-based SMB connections to the AD server for access to user logon information (print spooler and Net Logon).
5985	HTTP	Port the User-ID agent uses to monitor security logs and session information with the WinRM protocol over HTTP.
5986	HTTPS	Port the User-ID agent uses to monitor security logs and session information with the WinRM protocol over HTTPS.
5009	TCP	Port the firewall uses to connect to the Terminal Server Agent.

Ports Used for IPSec

The firewall and Panorama use the following ports for IPSec functions.

Destination Port	Protocol	Description
500	UDP	Port used by IKE on the management plane to connect with remote IKE peers.
4500	UDP	Port used by IKE on the management plane to connect with remote IKE peers.
4510	UDP	Port used by the dataplane to send requests to IKE.

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Destination Port	Protocol	Description
4511	UDP	Port used by the dataplane to send requests to `keymgr`.

Ports Used for Routing

The firewall and Panorama use the following ports for routing functions.

Destination Port	Protocol	Description
179	TCP	Port used by BGP to connect to peers.
3784 3785 4784	UDP	Ports used by BGP to connect to peers.
520	UDP	Port used for RIPv2.
89	IP	Port used for OSPF and OSPFv3.
103	IP	Port used for Protocol Independent Multicast (PIM).
639	TCP	Port used by MSDP to connect to peers.

Ports Used for DHCP

The firewall and Panorama use the following ports for DHCP functions.

Destination Port	Protocol	Description
67 68 546 547	UDP	Ports used as DHCP server listening ports.

Ports Used for Infrastructure

The firewall and Panorama use the following ports for infrastructure functions.

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Destination Port	Protocol	Description
111	TCP/UDP	Port used as a port mapper.
23	TCP/UDP	Port used for the Telnet application protocol.
69	TCP/UDP	Port used for TFTP.
2049	TCP/UDP	Port used for the Network File System (NFS).
28260	TCP	Port used by internal sysd IPC communication for internal processes.
28261	TCP	Port used by internal md applications to manage internal processes.
Dynamic	TCP/UDP	Dynamic port used by NFS operations to a host dataplane file system in the management plane.

Ports Used for DNS Proxy

The firewall and Panorama use the following ports for DNS proxy functions.

Destination Port	Protocol	Description
53	TCP/UDP	Port used to listen on interfaces associated with the DNS proxy feature.
443	TCP	Port used to listen to DNS-over-HTTPS (Encrypted DNS Proxy feature).
853	TCP	Port used to listen to DNS-over-TLS (Encrypted DNS Proxy feature).

Ports Used for Web Proxy

Network ports used by firewalls and Panorama for web proxy functions.

Destination Port	Protocol	Description
8080	TCP	Used to listen for explicit proxy. Exposed on the listening interface.

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Destination Port	Protocol	Description
6081	UDP	Used by Geneve protocol to encapsulate all user traffic.

Ports Used for CNGFW

Network ports used by Cloud NGFW to forward logs to a logging endpoint.

Destination Port	Protocol	Description
601	TCP	Used to listen for Fluent Bit, and forward logs to a logging endpoint.

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Reset the Firewall to Factory Default Settings

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

Resetting the firewall to factory defaults will result in the loss of all configuration settings and logs.

For hardware NGFWs running PAN-OS 12.2.2 and PAN-OS 12.1.11 or later, you can also perform a complete wipe using a bootable USB drive without requiring TAC assistance. The USB-based method is recommended when the device may be compromised and you want to ensure a cryptographically verified recovery. See Perform a Self-Service Enhanced Factory Reset

STEP 1 | Set up a console connection to the firewall.

1. Connect a serial cable from your computer to the Console port and connect to the firewall using terminal emulation software (9600-8-N-1).

   If your computer does not have a 9-pin serial port, use a USB-to-serial port connector.

2. Enter your login credentials.
3. Enter the following CLI command:

   `debug system maintenance-mode`

   The firewall will reboot in the maintenance mode.

STEP 2 | Reset the system to factory default settings.

1. When the firewall reboots, press **`Enter`** to continue to the maintenance mode menu.
2. Select **`Factory Reset`** and press **`Enter`**.
3. Select **`Factory Reset`** and press **`Enter`** again.

   The firewall will reboot without any configuration settings. The default username and password to log in to the firewall is admin/admin.

   To perform initial configuration on the firewall and to set up network connectivity, see Integrate NGFWs into you management network.

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Self-Service Enhanced Factory Reset (EFR)

Where Can I Use This?	What Do I Need?
• Supported NGFWs (see list of supported platforms)	❑ PAN-OS 12.2.2 and PAN-OS 12.1.11 or later ❑ USB flash drive (16 GB minimum) ❑ EFR image and PAN-OS images downloaded from the Customer Support Portal

Advanced cyberattacks can compromise network security infrastructure. The self-service Enhanced Factory Reset (EFR) provides a way to perform a remediation of a compromised NGFW by performing a complete cleanup of the system. This procedure is recommended when you are concerned about potential persistent compromise.

EFR wipes system disk and formats log disk. Line cards are not affected.

Chain of trust: All EFR boot components are cryptographically signed with a Palo Alto Networks signing key. The chain of trust is maintained during the self-service EFR process starting from BIOS. If any component fails signature verification, the boot stops. This ensures that only authentic Palo Alto Networks® recovery tools can execute on the system.

Interactive process: EFR runs as an interactive process accessed from the physical console. Before starting the disk wipe, the firewall reads the PAN-OS images you placed on the USB drive, checks their integrity and compatibility with your specific firewall model and PAN-OS version, and displays the results. You must confirm before the wipe begins. If an error is detected, you can abort or retry without rebooting.

Recovery logging: EFR logs the complete recovery process to the firewall (

```
less panrepo-log usb-efr.log
```

). After a successful EFR, a record is also appended to history logs (

```
less panrepo-log
                    history.log
```

) (keyword: `efr_images`). These logs support forensic review after an incident.

The USB port protection setting does not impact USB-based self-service factory reset (EFR), which requires enabling boot time access to the USB port.

Supported Platforms

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Self-service EFR requires an updated BIOS that supports USB boot. This BIOS update is embedded in PAN-OS 12.2.2 and PAN-OS 12.1.11 and is automatically installed when you upgrade to PAN-OS 12.2.2, 12.1.11 or later.

The following hardware NGFWs support self-service EFR for PAN-OS 12.2.2 or later:

Platform	EFR Image File
• PA-440 • PA-450 • PA-460 • PA-445 • PA-455 • PA-450R • PA-450R-5G • PA-455-5G • PA-455R-5G	`PanOS_400-EFR-1.0.0.img.gz`
• PA-505 • PA-510	`PanOS_500s-EFR-1.0.0.img.gz`
• PA-1410 • PA-1420	`PanOS_1400-EFR-1.0.0.img.gz`
• PA-3410 • PA-3420 • PA-3430 • PA-3440	`PanOS_3400-EFR-1.0.0.img.gz`
• PA-5450	`PanOS_5400-EFR-1.0.0.img.gz`
• PA-5410 • PA-5420 • PA-5430	`PanOS_5400f-EFR-1.0.0.img.gz`
• PA-5540 • PA-5550 • PA-5560 • PA-5570 • PA-5580	`PanOS_5500-EFR-1.0.0.img.gz`

The following hardware NGFWs support self-service EFR for PAN-OS 12.1.11 or later:

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Platform	EFR Image File
• PA-410 • PA-415 • PA-415-5G • PA-410R • PA-410R-5G	`PanOS_400-EFR-1.0.0.img.gz`

Self-service EFR is not supported on VM-Series, CN-Series, Cloud NGFW, or M-Series appliances. For virtual NGFWs and Panorama, use the standard option of replacing or redeploying the instance with a clean image.

Prepare a USB Drive for Enhanced Factory Reset

Where Can I Use This?	What Do I Need?
• Supported NGFWs (see list of supported platforms)	❑ PAN-OS 12.2.2 and PAN-OS 12.1.11 or later ❑ USB flash drive (16 GB minimum) ❑ A Windows, Linux, or macOS computer with a USB port ❑ EFR image and PAN-OS images downloaded from the Customer Support Portal

Before performing a self-service Enhanced Factory Reset (EFR), you must prepare a bootable USB flash drive that contains the EFR image for your firewall platform family and the PAN-OS images you want to reinstall.

When preparing the USB drive, follow these rules for the PAN-OS images you place in the IMAGES folder:

- The USB drive can contain a maximum of two PAN-OS images: one base image and one maintenance release (MR) image.
- A base image is required. An MR image is optional but if included, the corresponding base image must also be present.
- The base image and MR image must be from the same release train (same major.minor version). For example, PAN-OS 12.2.2 and PAN-OS 12.2.5 for the same model are compatible, but PAN-OS 12.1.2 and PAN-OS 12.2.2 are not.
- Both images must match the platform family of the firewall being recovered.
- EFR cannot be used to upgrade to a new major.minor release. For example, if the firewall is running 12.2.x, you can only reinstall 12.2.x images.

Before you begin:

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- Save all firewall configuration, HA state, and cluster configurations before the EFR process. The EFR wipes the system disk completely. You must manually reapply your configuration after the firewall restores.
- Identify your firewall's platform family and the PAN-OS version currently running on the device (review

```
show system info
```

or **Dashboard** > **General Information**).

STEP 1 | Download the EFR image from the Customer Support Portal.

1. Go to support.paloaltonetworks.com and sign in.
2. Select **Updates** > **Software Updates**.
3. For **Content Type**, choose **PAN-OS Enhanced Factory Reset (EFR)**.
4. Download the EFR image that matches your firewall platform family (for example, `PanOS_1400-EFR-1.0.0.img.gz` for PA-1400 Series firewalls).

STEP 2 | Download the PAN-OS images from the Customer Support Portal.

1. Select **Updates** > **Software Updates**.
2. For **Content Type**, choose **PAN-OS** for the *PA-<platform>* platform.
3. Download the base image for your firewall platform and the PAN-OS version you want to restore (for example, `PanOS_1400-12.2.2`). If you also want to install a maintenance release, download the corresponding MR image (for example, `PanOS_1400-12.2.5`).
4. Verify the SHA-256 checksum of each downloaded image against the checksum displayed on the Customer Support Portal.

> *EFR cannot be used to upgrade to a new major.minor release. The PAN-OS images you download must be from the same release train as the version currently running on the firewall.*

STEP 3 | Write the EFR image to the USB drive using the instructions for your operating system.

- Linux
- macOS
- Windows

STEP 4 | Copy PAN-OS images to the USB drive.

After writing the EFR image to the USB drive, copy your PAN-OS images into the USB drive's IMAGES folder.

On Linux and macOS, copy the images directly from the terminal:

Linux:

```
sudo cp ~/USBEFR_STAGING/PanOS_1400-12.2.2 /mnt/usbefr/IMAGES/
sudo cp ~/USBEFR_STAGING/PanOS_1400-12.2.5 /mnt/usbefr/IMAGES/
```

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```
sync
```

macOS:

```
sudo cp ~/USBEFR_STAGING/PanOS_1400-12.2.2 /Volumes/USBEFR/IMAGES
sudo cp ~/USBEFR_STAGING/PanOS_1400-12.2.5 /Volumes/USBEFR/IMAGES
sync
```

*Run **sync** after copying to prevent data loss from removing the drive before the copy completes.*

On Windows, use your partition explorer tool to navigate to the IMAGES folder of the USB drive, then copy the PAN-OS image files into it.

Verify the IMAGES folder contents:

```
IMAGES/
  README
  PanOS_1400-12.2.2
  PanOS_1400-12.2.5
```

STEP 5 | Cleanly unmount and eject the USB drive before removing it from the computer.

Linux:

```
sudo umount /mnt/usbefr
```

macOS:

```
diskutil eject /dev/diskX
```

On Windows, use the **Safely Remove Hardware** option in the system tray before unplugging the USB drive.

Linux

STEP 1 | Open a terminal and create a staging directory:

```
mkdir ~/USBEFR_STAGING
cd ~/USBEFR_STAGING
```

STEP 2 | Copy the downloaded EFR image to the staging directory.

STEP 3 | Verify the SHA-256 checksum of the EFR image:

```
sha256sum <EFR_IMAGE_FILE.img.gz>
```

Compare the output against the checksum from the Customer Support Portal.

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STEP 4 | Extract the compressed EFR image:

```
gunzip <EFR_IMAGE_FILE.img.gz>
```

This produces an uncompressed file: <EFR_IMAGE_FILE.img> (approximately 8 GB).

STEP 5 | Connect the USB flash drive to your Linux computer.

STEP 6 | Identify the USB drive device path:

```
lsblk -S -o NAME,TYPE,VENDOR,MODEL,SIZE,TRAN
```

Identify your USB drive by its size and note the device name (for example, `/dev/sda`).

STEP 7 | Unmount any partitions on the USB drive that are automatically mounted:

```
sudo umount /dev/sdX*
```

Replace *sdX* with your device name.

STEP 8 | Write the EFR image to the USB drive:

*The **dd** command will overwrite all data on the target USB drive. Double-check the device identifier before running the command. Targeting the wrong device can permanently overwrite your file system.*

```
sudo dd if=<EFR_IMAGE_FILE.img> of=/dev/sdX bs=4M status=progress
 conv=fdatasync
```

Replace *sdX* with your device name. The operation completes when **dd** displays a summary of records transferred.

STEP 9 | Verify the USB drive file structure by mounting it and listing its contents:

```
sudo mkdir -p /mnt/usbefr
sudo mount -L USBEFR /mnt/usbefr
ls -R /mnt/usbefr
```

The USB drive should contain the following directory structure:

```
EFI/
  BOOT/
    BOOTX64.EFI
    grubx64.efi
  centos/
    SHIMX64.EFI
    grubx64.efi
  fonts/
    unicode.pf2
boot/
  resusys.vmlinuz
  resusys.cpio
  resusys.cpio.sig
```

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```
IMAGES/
  README
```

macOS

STEP 1 | Open Terminal and create a staging directory:

```
mkdir ~/USBEFR_STAGING
cd ~/USBEFR_STAGING
```

STEP 2 | Copy the downloaded EFR image to the staging directory.

STEP 3 | Verify the SHA-256 checksum of the EFR image:

```
shasum -a 256 <EFR_IMAGE_FILE.img.gz>
```

Compare the output against the checksum from the Customer Support Portal.

STEP 4 | Extract the compressed EFR image:

```
gunzip <EFR_IMAGE_FILE.img.gz>
```

This produces an uncompressed file: `<EFR_IMAGE_FILE.img>` (approximately 8 GB).

STEP 5 | Connect the USB flash drive to your Mac.

STEP 6 | Identify the USB drive device identifier:

```
diskutil list
```

Identify your USB drive by its size and note the disk identifier (for example, `/dev/disk4`).

STEP 7 | Unmount all partitions on the USB drive:

```
diskutil unmountDisk /dev/diskX
```

Replace *diskX* with your disk identifier.

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STEP 8 | Write the EFR image to the USB drive:

> *The **dd** command will overwrite all data on the target USB drive. Double-check the disk identifier before running the command. Targeting the wrong disk can permanently overwrite your file system.*

```
sudo dd if=<EFR_IMAGE_FILE.img> of=/dev/rdiskX bs=4m
```

Use `/dev/rdiskX` (raw disk) rather than `/dev/diskX` for faster write speeds. Replace *diskX* with your disk identifier. The operation may take several minutes. When it completes, **dd** displays a summary of records transferred.

After writing completes, run:

```
sync
```

STEP 9 | If macOS displays a dialog that the disk is unreadable, click **Ignore** — do not click **Erase**. The EFI partition written by **dd** is not recognized by macOS.

STEP 10 | Verify the USB drive file structure by mounting the USBEFR volume and listing its contents:

```
ls -R /Volumes/USBEFR
```

The USB drive should contain the following directory structure:

```
EFI/
  BOOT/
    BOOTX64.EFI
    grubx64.efi
  centos/
    SHIMX64.EFI
    grubx64.efi
  fonts/
    unicode.pf2
boot/
  resusys.vmlinuz
  resusys.cpio
  resusys.cpio.sig
IMAGES/
  README
```

Windows

STEP 1 | Create a staging folder on your Windows computer (for example, `C:\USBEFR_STAGING`) and copy the downloaded EFR image and PAN-OS images into it.

STEP 2 | Verify the SHA-256 checksum of the EFR image using PowerShell:

```
Get-FileHash C:\USBEFR_STAGING\<EFR_IMAGE_FILE.img.gz> -Algorithm
 SHA256
```

Compare the output against the checksum from the Customer Support Portal.

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STEP 3 | Extract the compressed EFR image.

Use a tool that supports `.gz` extraction, such as 7-Zip or a similar utility. Right-click the `.gz` file, choose your extraction tool, and extract to the staging folder.

This produces an uncompressed file: <EFR_IMAGE_FILE.img> (approximately 8 GB).

STEP 4 | Connect the USB flash drive to your Windows computer.

STEP 5 | Write the EFR image to the USB drive using Rufus or a similar raw disk imaging tool.

1. Download and launch Rufus.
2. For **Device**, choose your USB drive. Confirm the drive letter and size carefully — writing to the wrong device will overwrite its contents.
3. For **Boot selection**, choose **Disk or ISO image**, then click **SELECT** and navigate to your extracted <EFR_IMAGE_FILE.img> file.
4. Leave all other settings at their defaults and click **START**.
5. When prompted, choose **Write in DD Image mode**, then click **OK** to confirm.

   All data on the USB drive will be erased. Confirm you have selected the correct device before proceeding.
6. Wait for Rufus to complete. The status bar displays **READY** when finished.

STEP 6 | After Rufus completes, use Windows File Explorer to verify the USB drive contains an IMAGES folder with a README file.

Windows may only show one partition of the USB drive. This is expected — the EFI and boot partitions are not visible in File Explorer but are present and required for the EFR boot process.

Perform a Self-Service Enhanced Factory Reset

Where Can I Use This?	What Do I Need?
• Supported NGFWs (see list of supported platforms)	☐ PAN-OS 12.2.2 and PAN-OS 12.1.11 or later ☐ Prepared USB flash drive (16 GB minimum)

The Enhanced Factory Reset (EFR) wipes the firewall's system disk and reinstalls PAN-OS from the images you placed on the prepared USB drive. Before starting, ensure you have saved all configuration, HA state, and cluster information — EFR does not preserve firewall configuration. You must manually restore your configuration after the reset completes.

EFR wipes the system disk and formats the log disk. This action is irreversible. Line cards are not affected. Do not power off the firewall during the EFR process.

STEP 1 | Connect a console cable to the firewall's console port and open a terminal session.

Use console settings: 9600 baud, 8 data bits, no parity, 1 stop bit (9600 8N1).

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STEP 2 | Insert the prepared USB flash drive into one of the firewall's USB ports.

STEP 3 | Reboot the firewall.

From the CLI:

```
request restart system
```

Or power-cycle the firewall using the power switch.

STEP 4 | When the BIOS POST screen appears, press **F11** to enter the boot menu.

*The boot menu prompt appears briefly during POST. Be ready to press **F11** as soon as the BIOS screen displays. If you miss the prompt, power-cycle and try again.*

STEP 5 | In the boot menu, select your USB drive as the boot device.

The USB drive typically appears as a removable device or by its vendor name. Use the arrow keys to highlight it and press **Enter** to boot.

STEP 6 | At the GRUB menu, select **Palo Alto Networks Enhanced Factory Reset** and press **Enter**.

STEP 7 | Wait while EFR reads and verifies the PAN-OS images on the USB drive.

EFR checks the integrity, signatures, and compatibility of each image with your specific firewall model and PAN-OS version. The results are displayed on the console. If all checks pass, EFR prompts you to confirm before the wipe begins.

If EFR reports an error:

- Review the error message and correct the issue (for example, wrong platform image or missing base image).
- Choose **Abort** or **Retry** without rebooting — you do not need to power-cycle the firewall to correct image errors.

STEP 8 | When prompted, confirm that you want to proceed with the factory reset.

Confirming starts the disk wipe immediately. This action cannot be undone. All data on the system disk will be permanently erased.

STEP 9 | Monitor the EFR progress on the console.

EFR displays progress messages as it:

- Wipes the system disk
- Formats the log disk
- Installs the base PAN-OS image
- Installs the maintenance release image (if present on the USB drive)

Do not power off the firewall during this process. The reset takes approximately 20–40 minutes depending on your firewall model.

STEP 10 | When EFR completes and the firewall reboots, remove the USB flash drive before the firewall finishes its first boot into PAN-OS.

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STEP 11 | After the firewall boots, verify the installed PAN-OS version from the console:

```
show system info | match version
```

STEP 12 | Restore your firewall configuration.

The firewall boots to a factory-default configuration. Load your saved configuration backup and reapply your HA, cluster, or management settings as required.

STEP 13 | (Optional) Verify the EFR recovery log on the firewall:

```
less panrepo-log usb-efr.log
```

The log records the complete EFR process for forensic review. A summary record is also appended to the history log:

```
less panrepo-log history.log
```

Search for keyword `efr_images` to locate the EFR entry in the history log.

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Configure USB Port Access

Where Can I Use This?	What Do I Need?
• Supported NGFWs (see list of supported platforms)	❑ PAN-OS 12.2.2 or later ❑ Administrator access

PAN-OS 12.2.2 introduces USBGuard, a software framework that controls which USB devices can connect to the firewall. By default, the USB port is enabled and restricted to USB mass storage devices. You can disable the USB port entirely using the CLI when USB access is not required.

The USB port setting does not persist through a factory reset or private data reset. After either of these operations, the USB port returns to its default state (enabled, storage devices only).

- Enable the USB port (allow USB mass storage devices only):

  ```
  configure
  set deviceconfig system usb-port state enable
  commit
  ```

  When enabled, USBGuard allows only USB mass storage devices (USB class 08, subclass 06, protocol 50). All other USB device types are blocked.

- Disable the USB port (block all USB devices):

  ```
  configure
  set deviceconfig system usb-port state disable
  commit
  ```

  When disabled, all USB devices are blocked. A system log entry is generated when the USB port setting changes.

  Even when the USB port is disabled through the OS, the physical USB port can still be used during early boot (before PAN-OS loads) to perform an Enhanced Factory Reset via the BIOS menu. USB port access control at the OS level does not prevent BIOS-level USB boot.

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Bootstrap the Firewall

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

Bootstrapping speeds up the process of configuring and licensing the firewall to make it operational on the network with or without Internet access. Bootstrapping allows you to choose whether to configure the firewall with a basic configuration file (init-cfg.txt) so that it can connect to Panorama and obtain the complete configuration or to fully configure the firewall with the basic configuration and the optional bootstrap.xml file.

USB Flash Drive Support

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

The USB flash drive that bootstraps a hardware-based Palo Alto Networks firewall must support one of the following:

- File Allocation Table 32 (FAT32)
- Third Extended File System (ext3)

The firewall can bootstrap from the following flash drives with USB2.0 or USB3.0 connectivity:

Supported USB Flash Drives
Kingston • Kingston SE9 8GB (2.0) • Kingston SE9 16GB (3.0) • Kingston SE9 32GB (3.0)
SanDisk • SanDisk Cruzer Fit CZ33 8GB (2.0) • SanDisk Cruzer Fit CZ33 16GB (2.0) • SanDisk Cruzer CZ36 16GB (2.0) • SanDisk Cruzer CZ36 32GB (2.0) • SanDisk Extreme CZ80 32GB (3.0)
Silicon Power • Silicon Power Jewel 32GB (3.0)

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Supported USB Flash Drives
• Silicon Power Blaze 16GB (3.0)
PNY • PNY Attache 16GB (2.0) • PNY Turbo 32GB (3.0)

Sample init-cfg.txt Files

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

An init-cfg.txt file is required for the bootstrap process; this file is a basic configuration file that you create using a text editor. To create this file, see 5 The following sample init-cfg.txt files show the parameters that are supported in the file; the parameters that you must provide are in bold.

Sample init-cfg.txt (Static IP Address)

```
type=static
ip-address=10.5.107.19
default-gateway=10.5.107.1
netmask=255.255.255.0
ipv6-address=2001:400:f00::1/64
ipv6-default-
gateway=2001:400:f00::2
hostname=Ca-FW-DC1
panorama-server=10.5.107.20
panorama-server-2=10.5.107.21
tplname=FINANCE_TG4
dgname=finance_dg
dns-primary=10.5.6.6
dns-secondary=10.5.6.7
op-command-modes=multi-
vsys,jumbo-frame
dhcp-send-hostname=no
dhcp-send-client-id=no
dhcp-accept-server-hostname=no
dhcp-accept-server-domain=no
```

Sample init-cfg.txt (DHCP Client)

```
type=dhcp-client
ip-address=
default-gateway=
netmask=
ipv6-address=
ipv6-default-gateway=
hostname=Ca-FW-DC1
panorama-server=10.5.107.20
panorama-server-2=10.5.107.21
tplname=FINANCE_TG4
dgname=finance_dg
dns-primary=10.5.6.6
dns-secondary=10.5.6.7
op-command-modes=multi-
vsys,jumbo-frame
dhcp-send-hostname=yes
dhcp-send-client-id=yes
dhcp-accept-server-
hostname=yes
dhcp-accept-server-domain=yes
```

The following table describes the fields in the init-cfg.txt file. The type is required; if the type is static, the IP address, default gateway and netmask are required, or the IPv6 address and IPv6 default gateway are required.

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Field	Description
type	(Required) Type of management IP address: static or dhcp-client.
ip-address	(Required for IPv4 static management address) IPv4 address. The firewall ignores this field if the type is dhcp-client.
default-gateway	(Required for IPv4 static management address) IPv4 default gateway for the management interface. The firewall ignores this field if the type is dhcp-client.
netmask	(Required for IPv4 static management address) IPv4 netmask. The firewall ignores this field if the type is dhcp-client.
ipv6-address	(Required for IPv6 static management address) IPv6 address and / prefix length of the management interface. The firewall ignores this field if the type is dhcp-client.
ipv6-default-gateway	(Required for IPv6 static management address) IPv6 default gateway for the management interface. The firewall ignores this field if the type is dhcp-client.
hostname	(Optional) Host name for the firewall.
panorama-server	(Recommended) IPv4 or IPv6 address of the primary Panorama server.
panorama-server-2	(Optional) IPv4 or IPv6 address of the secondary Panorama server.
tplname	(Recommended) Panorama template name.
dgname	(Recommended) Panorama device group name.
dns-primary	(Optional) IPv4 or IPv6 address of the primary DNS server.
dns-secondary	(Optional) IPv4 or IPv6 address of the secondary DNS server.
pa-auth-key	(Mandatory for USB bootstrapping) Hardware device authentication key used for SC3 registration.
vm-auth-key	(VM-Series firewalls only) Virtual machine authentication key.
op-command-modes	(Optional) Enter multi-vsys, jumbo-frame, or both separated by a comma only. Enables multiple virtual systems and jumbo frames while bootstrapping.
dhcp-send-hostname	(DHCP client type only) The DHCP server determines a value of yes or no. If yes, the firewall sends its hostname to the DHCP server.

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Field	Description
dhcp-send-client-id	(DHCP client type only) The DHCP server determines a value of yes or no. If yes, the firewall sends its client ID to the DHCP server.
dhcp-accept-server-hostname	(DHCP client type only) The DHCP server determines a value of yes or no. If yes, the firewall accepts its hostname from the DHCP server.
dhcp-accept-server-domain	(DHCP client type only) The DHCP server determines a value of yes or no. If yes, the firewall accepts its DNS server from the DHCP server.

Prepare a USB Flash Drive for Bootstrapping a Firewall

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

You can use a USB flash drive to bootstrap a physical firewall. However, to do so you must be running a PAN-OS 7.1.0 or later image and Reset the Firewall to Factory Default Settings. For security reasons, you can bootstrap a firewall only when it is in factory default state or has all private data deleted.

STEP 1 | Obtain serial numbers (S/Ns) and auth codes for support subscriptions from your order fulfillment email.

STEP 2 | Register S/Ns of new firewalls on the Customer Support portal.

1. Go to support.paloaltonetworks.com, log in, and select **Assets** > **Devices** > **Register New Device** > **Register device using Serial Number or Authorization Code**.
2. Follow the steps to register a firewall.
3. Click **Submit**.

STEP 3 | Activate authorization codes on the Customer Support portal, which creates license keys.

1. Go to support.paloaltonetworks.com, log in, and select the **Assets** > **Devices** on the left-hand navigation pane.
2. For each device S/N you just registered, click the **Action** link (the pencil icon).
3. Under Activate Licenses, select **Activate Auth-Code**.
4. Enter the **Authorization code** and click **Agree** and **Submit**.

STEP 4 | Add the S/Ns in Panorama.

Complete Step 1 in Add a Firewall as a Managed Device in the Panorama Administrator's Guide.

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STEP 5 | Create the init-cfg.txt file.

Create the init-cfg.txt file, a mandatory file that provides bootstrap parameters. The fields are described in Sample init-cfg.txt Files.

If the init-cfg.txt file is missing, the bootstrap process will fail and the firewall will boot up with the default configuration in the normal boot-up sequence.

There are no spaces between the key and value in each field; do not add spaces because they cause failures during parsing on the management server side.

You can have multiple init-cfg.txt files—one each for different remote sites—by prepending the S/N to the file name. For example:

0008C200105-init-cfg.txt

0008C200107-init-cfg.txt

If no prepended filename is present, the firewall uses the init-cfg.txt file and proceeds with bootstrapping.

STEP 6 | (Optional) Create the bootstrap.xml file.

The optional bootstrap.xml file is a complete firewall configuration that you can export from an existing production firewall.

1. Select **Device** > **Setup** > **Operations** > **Export named configuration snapshot**.
2. Select the **Name** of the saved or the running configuration.
3. Click **OK**.
4. Rename the file as **`bootstrap.xml`**.

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STEP 7 | Create and download the bootstrap bundle from the Customer Support portal.

For a physical firewall, the bootstrap bundle requires only the /license and /config directories.

Use one of the following methods to create and download the bootstrap bundle:

- Use **Method 1** to create a bootstrap bundle specific to a remote site (you have only one init-cfg.txt file).
- Use **Method 2** to create one bootstrap bundle for multiple sites.

Method 1

1. On your local system, go to support.paloaltonetworks.com and log in.
2. Select **Assets**.
3. Select the S/N of the firewall you want to bootstrap.
4. Select **Bootstrap Container**.
5. Click **Select**.
6. Upload and **Open** the init-cfg.txt file you created.
7. (Optional) Select the bootstrap.xml file you created and **Upload Files**.

   You must use a bootstrap.xml file from a firewall of the same model and PAN-OS version.
8. Select **Bootstrap Container Download** to download a tar.gz file named **`bootstrap_<S/N>_<date>.tar.gz`** to your local system. This bootstrap container includes the license keys associated with the S/N of the firewall.

Method 2

Create a tar.gz file on your local system with two top-level directories: /license and /config. Include all licenses and all init-cfg.txt files with S/Ns prepended to the filenames.

The license key files you download from the Customer Support portal have the S/N in the license file name. PAN-OS checks the S/N in the file name against the firewall S/N while executing the bootstrap process.

STEP 8 | Import the tar.gz file you created (to a firewall running a PAN-OS 7.1.0 or later image) using Secure Copy (SCP) or TFTP.

Access the CLI and enter one of the following commands:

- **`tftp import bootstrap-bundle file <path and filename> from <host IP address>`**

  For example:

  `tftp import bootstrap-bundle file /home/userx/bootstrap/devices/pa5000.tar.gz from 10.1.2.3`
- **`scp import bootstrap-bundle from <<user>@<host>:<path to file>>`**

  For example:

  `scp import bootstrap-bundle from userx@10.1.2.3:/home/userx/bootstrap/devices/pa200_bootstrap_bundle.tar.gz`

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STEP 9 | Prepare the USB flash drive.

1. Insert the USB flash drive into the firewall that you used in the prior step.
2. Enter the following CLI operational command, using your tar.gz filename in place of "**pa5000.tar.gz**". This command formats the USB flash drive, unzips the file, and validates the USB flash drive:

   ```
   request system bootstrap-usb prepare from pa5000.tar.gz
   ```
3. Press **y** to continue. The following message displays when the USB drive is ready:

   ```
   USB prepare completed successfully.
   ```
4. Remove the USB flash drive from the firewall.
5. You can prepare as many USB flash drives as needed.

STEP 10 | Deliver the USB flash drive to your remote site.

If you used Method 2 to create the bootstrap bundle, you can use the same USB flash drive content for bootstrapping firewalls at multiple remote sites. You can translate the content into multiple USB flash drives or a single USB flash drive used multiple times.

Bootstrap a Firewall Using a USB Flash Drive

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

After you receive a new Palo Alto Networks firewall and a USB flash drive loaded with bootstrap files, you can bootstrap the firewall.

Microsoft Windows and Apple Mac operating systems are unable to read the bootstrap USB flash drive because the drive is formatted using an ext4 file system. You must install third-party software or use a Linux system to read the USB drive.

STEP 1 | The firewall must be in a factory default state or must have all private data deleted.

STEP 2 | To ensure connectivity with your corporate headquarters, cable the firewall by connecting the management interface (MGT) using an Ethernet cable to one of the following:

- An upstream modem
- A port on the switch or router
- An Ethernet jack in the wall

STEP 3 | Insert the USB flash drive into the USB port on the firewall and power on the firewall. The factory default firewall bootstraps itself from the USB flash drive.

The firewall Status light turns from yellow to green when the firewall is configured; autocommit is successful.

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STEP 4 | Verify bootstrap completion. You can see basic status logs on the console during the bootstrap and you can verify that the process is complete.

1. If you included Panorama values (panorama-server, tplname, and dgname) in your init-cfg.txt file, check Panorama managed devices, device group, and template name.
2. Verify the general system settings and configuration by accessing the web interface and selecting **Dashboard** > **Widgets** > **System** or by using the CLI operational commands **`show system info`** and **`show config running`**.
3. Verify the license installation by selecting **Device** > **Licenses** or by using the CLI operational command **`request license info`**.
4. If you have Panorama configured, manage the content versions and software versions from Panorama. If you do not have Panorama configured, use the web interface to manage content versions and software versions.

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STEP 5 | (Panorama managed firewalls only) Create a device registration authentication key and add it to the firewall.

This is required to successfully add a bootstrapped firewall to Panorama management. The device registration authentication key has a finite lifetime and including the device registration authentication key in the init-cfg.txt file is not supported.

1. Log in to the Panorama web interface.
2. Select **Panorama** > **Device Registration Auth Key** and **Add** a new authentication key.
3. Configure the authentication key.
   - **Name**—Add a descriptive name for the authentication key.
   - **Lifetime**—Specify the key lifetime to limit how long you can use the authentication key to onboard new firewalls.
   - **Count**—Specify how many times you can use the authentication key to onboard new firewalls.
   - **Device Type**—Specify that this authentication key is used to authenticate only a **Firewall**.

     *You can select **Any** to use the device registration authentication key to onboard firewalls, Log Collectors, and WildFire appliances.*
   - (Optional) **Devices**—Enter one or more device serial numbers to specify for which firewalls the authentication key is valid.
4. Click **OK**.

   When prompted, **Copy Auth Key** and **Close**.
5. Log in to the firewall web interface.

   You can also log in to the firewall CLI to add the device registration authentication key.

   ```
   admin> request authkey set <auth key>
   ```
6. Select **Device** > **Setup** > **Management** and edit the Panorama Settings.
7. Paste the device registration authentication key you copied in the previous step and click **OK**.
8. **Commit**.
9. Log in to the Panorama web interface and select **Panorama** > **Managed Devices** > **Summary** to verify the firewall is `Connected` to Panorama

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Device Telemetry

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Essentials ❑ Strata Cloud Manager Pro

Device telemetry collects data about your next-generation firewall or Panorama, and shares it with Palo Alto Networks by uploading the data to Strata Logging Service. This data is used to power telemetry apps, and for sharing threat intelligence.

- Device Telemetry Overview
- Device Telemetry Collection and Transmission Intervals
- Manage Device Telemetry
- Monitor Device Telemetry
- Sample the Data that Device Telemetry Collects

Device Telemetry Overview

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Essentials ☐ Strata Cloud Manager Pro Telemetry autoenablement feature requires PAN-OS 10.2.17, 11.1.11, 11.2.8, 12.1.2, and later releases.

Device telemetry collects data about your next-generation firewall or Panorama and shares it with Palo Alto Networks by uploading the data to Strata Logging Service. This data is used to power telemetry apps, which are cloud-based applications that make it easy to monitor and manage your next-generation firewalls and Panoramas. These apps improve your visibility into device health, performance, capacity planning, and configuration. Through these apps, you can maximize the benefits you enjoy from the products and services that Palo Alto Networks delivers.

Telemetry data is also used for sharing threat intelligence, providing enhanced intrusion prevention, evaluation of threat signatures, as well as improved malware detection within PAN-DB URL filtering, DNS-based command-and-control (C2) signatures, WildFire, and to further improve Palo Alto Networks products and services. Review the PAN-OS Privacy information data sheet for details about the data that Palo Alto Networks collects.

Palo Alto Networks automatically selects recommended settings when you configure telemetry. When you commit the settings, PAN-OS begins collecting and sending telemetry data.

Telemetry data is collected and stored locally on your device for a limited period of time. This data is shared with Palo Alto Networks only if you configure a destination region for the data. If your organization has a Strata Logging Service license, then you can only send the data to the same region as where your Strata Logging Service instance resides. If your organization does not have a Strata Logging Service license, then you must install a device certificate in order to share this data. In this case, you can choose any available region, although you must conform to all applicable local laws regarding privacy and data storage.

Telemetry data is collected and shared with Palo Alto Networks on predefined collection intervals starting from the time when the firewall is turned on. These predefined intervals are set by the PAN-OS analytics engine, however you can control whether data is collected and shared by enabling/disabling categories of data. You can also monitor the current status of data collection and transmission.

The size of each bundle of telemetry data depends on the features enabled on your firewall, the number of metrics collected, and the model of the firewall. PAN-OS collects metrics related to operational health and performance, such as CPU and memory, more frequently.

You can obtain a live sample of the data that your firewall is collecting for telemetry purposes. For a complete description of all the telemetry metrics that can be shared with Palo Alto Networks, including the privacy implication for each metric, see the PAN-OS Device Telemetry Metrics Reference Guide.

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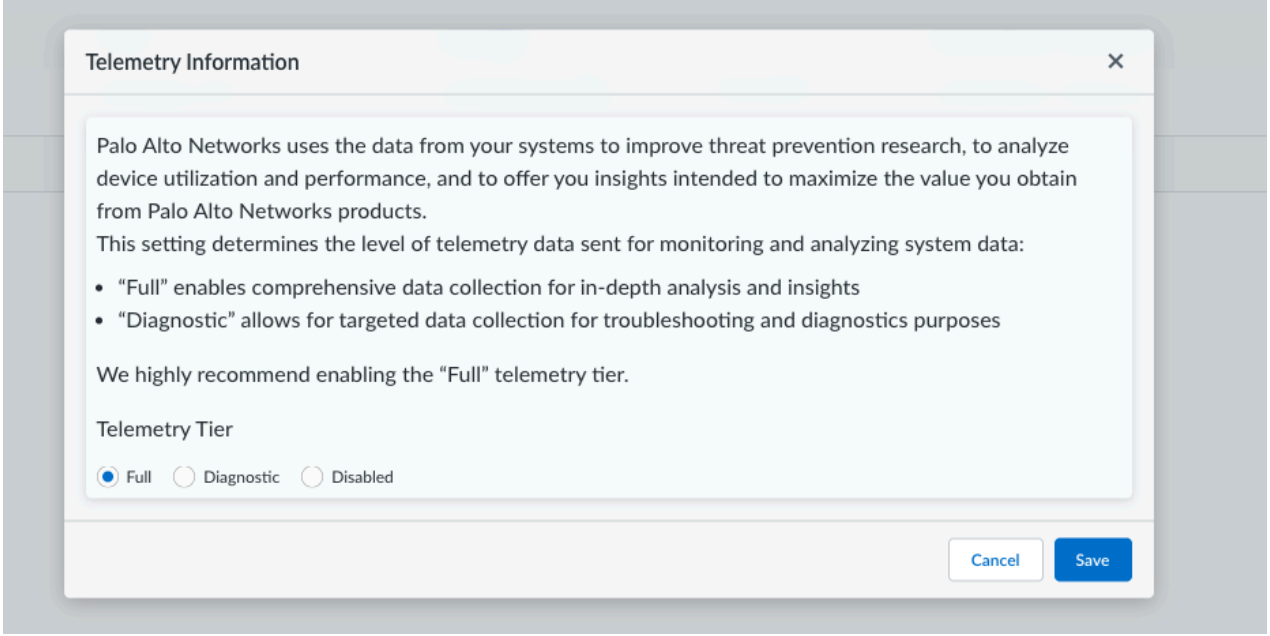
*The automatically created user **_cliuser** or **__telemetryuser** may appear under **Logged in Admins** on the dashboard while telemetry is enabled. This user is created only for telemetry collection. See __telemetryuser System Account.*

Telemetry Autoenablement

Beginning with PAN-OS 10.2.17, 11.1.11, 11.2.8, 12.1.2, and later releases, the telemetry autoenablement feature configures telemetry to be enabled by default on your devices. Upon onboarding a new device (Panorama or firewall), telemetry is automatically enabled with settings centrally controlled through Strata Cloud Manager. This centralized approach ensures consistent telemetry settings across your entire environment. Metrics are automatically streamed to your data residency region, eliminating the need for manual configuration.

A key distinction from previous configurations is that telemetry settings no longer reside on individual Panorama or firewall devices. Instead, these settings are retrieved from Strata Cloud Manager that store information for all devices with a tenant service group (TSG). For devices without a TSG, the system maintains backward compatibility by supporting direct configurations on Panorama or firewalls. By centralizing and automating telemetry configuration, telemetry autoenablement removes operational barriers to adoption. This enables you to fully leverage the benefits of telemetry while maintaining control over your data sharing preferences.

Telemetry autoenablement includes tiered telemetry options (Full and Diagnostic), providing control over the level of data shared while streamlining the enablement process. These tiers dictate the specific metrics collected and transmitted to Palo Alto Networks. Each device independently polls the Device Discovery Service endpoint daily using its device certificate to obtain region information and tier settings. This method applies uniformly across all device types, including Panorama, log collectors, Panorama managed firewalls, and standalone firewalls. After telemetry autoenablement activates on your devices, you retain full control over your settings and can edit telemetry tier or disable telemetry at any time through the hub.



Telemetry autoenablement determines the appropriate data residency region based on your existing configurations. If you're utilizing Strata Logging Service, your telemetry data is automatically synchronized with your Strata Logging Service region. For customers without Strata

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Logging Service, the system determines your region based on information in your Customer Support Portal account. You also have the option to manually select a region if needed. You retain full control of your telemetry settings and can adjust the telemetry tier from Full to Diagnostic as needed through the hub or Strata Cloud Manager interface. This tiered approach ensures you can select the level of information shared while adhering to data privacy requirements.

Telemetry autoenablement automatically provisions TSGs, associates your Next-Generation Firewalls (NGFWs), and activates telemetry as soon as your firewalls become available through your customer support portal (CSP). By automating this lifecycle, your firewalls begin forwarding telemetry data with minimal delay, giving you access to threat visibility, logging, and analytics capabilities without manual configuration.

This feature handles TSG and telemetry enablement dynamically based on your current environment:

- No existing TSG and telemetry is disabled—Creates a TSG on your behalf using a standardized naming convention derived from your CSP ID and tenant, associates all discovered NGFWs to it, and activates telemetry using the region tied to your CSP.
- Existing TSG with inactive telemetry—Associates your devices with the existing TSG and enables telemetry automatically once those devices appear in your CSP.
- Multiple TSGs—Creates a new, distinctly named TSG, consolidates your NGFWs under it, and enables telemetry to avoid ambiguity across groups.
- Existing TSG with associated NGFWs but no telemetry—Provisions telemetry using your billing country code and completes the activation without further action on your part.

See Common Services: Tenant Management for more information about how to manage the automatically created tenants.

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__telemetryuser System Account

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Essentials ☐ Strata Cloud Manager Pro Telemetry autoenablement feature requires PAN-OS 10.2.17, 11.1.11, 11.2.8, 12.1.2, and later releases.

The `__telemetryuser` account is a permanent, system-generated local administrator account that PAN-OS uses to execute internal commands required for Device Telemetry data collection. You do not need to create, manage, or configure this account.

The `__telemetryuser` account is present on the system regardless of whether you enable or disable Device Telemetry. If you delete the account, PAN-OS recreates it during the next system reboot. To manage telemetry settings, select **Tenant Management** on the hub, or select **Settings** > **Tenants** in Strata Cloud Manager.

Telemetry Account Visibility

You can view the `__telemetryuser` account in the following locations in the web interface:

- **Device** > **Administrators**
- The **Logged in Admins** section of the dashboard during active collection cycles

Both behaviors are expected. To protect your system security, the account cannot log in externally, execute interactive commands, or perform configuration changes.

Telemetry Account Version History

PAN-OS introduced the `__telemetryuser` account to permanently replace the legacy transient `_cliuser` account, which PAN-OS created and deleted during each collection cycle. Transitioning to a permanent account improves telemetry collection reliability.

The account is available in the following releases:

- PAN-OS 10.2.18, PAN-OS 10.2.20, and later releases
- PAN-OS 11.1.10-h12, PAN-OS 11.1.12, and later releases
- PAN-OS 11.2.4-h14, PAN-OS 11.2.7-h4, PAN-OS 11.2.10, PAN-OS 11.2.11, and later releases
- PAN-OS 12.1.4-h3, PAN-OS 12.1.5, PAN-OS 12.1.8, and later releases
- PAN-OS 12.2.0 and later releases

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Telemetry Collection Behavior

To collect telemetry data, the `__telemetryuser` account executes predefined, read-only command-line interface (CLI) commands and queries, compresses the output into a `.tgz` bundle, and uploads the bundle to the secure Palo Alto Networks® endpoint.

PAN-OS uploads bundles at the following predefined intervals:

- **Minute interval**: Executes every five minutes in PAN-OS 11.0 and later releases, or every 20 minutes in releases earlier than PAN-OS 11.0.
- **Hour interval**: Executes every 60 minutes.
- **Day interval**: Executes every 24 hours.

Security Considerations

The `__telemetryuser` account incorporates the following security design controls:

- **Privileges**: The account operates with limited, read-only privileges required strictly for telemetry data collection and script execution. It does not possess superuser or elevated administrative permissions.
- **Interactive access**: The account does not support interactive login. You cannot use it to log in through Secure Shell (SSH), the web interface, or an application programming interface (API).
- **Configuration changes**: The account is architecturally incapable of making configuration changes.
- **Disabled telemetry**: The `__telemetryuser` account remains on the system even if you disable Device Telemetry. If you manually delete the account, PAN-OS recreates it during the next reboot to maintain consistent system behavior.

Frequently Asked Questions

- Is `__telemetryuser` a security risk?

  No. The account is restricted to internal, read-only command execution. It cannot log in externally or make configuration changes.

- Can you delete `__telemetryuser`?

  Do not delete the account. If you delete the account while Device Telemetry is enabled, the device sends empty bundles, which causes issues in telemetry-related services. Additionally, PAN-OS® recreates the account during the next system reboot. To stop data collection, select **Device** > **Setup** > **Telemetry** and disable Device Telemetry. The `__telemetryuser` account remains on the system regardless of your telemetry configuration.

- Why did this account appear after a PAN-OS upgrade?

  When you upgrade to a PAN-OS® release that contains the fix for PAN-292447, the upgrade permanently adds the `__telemetryuser` account. In PAN-OS releases earlier than PAN-OS 10.2.18, PAN-OS 11.1.10-h12, PAN-OS 11.2.4-h14, and PAN-OS 12.1.4-h3, the legacy transient `_cliuser` account performed the same telemetry collection functions.

- Why does `__telemetryuser` appear in Logged in Admins?

  The account appears briefly during active collection cycles. This is expected behavior and does not indicate unauthorized external access.

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To learn more about Device Telemetry data categories, privacy, and encryption, see Device Telemetry Overview.

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Device Telemetry Collection and Transmission Intervals

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Essentials ☐ Strata Cloud Manager Pro

PAN-OS collects and sends telemetry data on fixed intervals. Collection is defined on a metric by metric basis, and can be one of:

- (Default) Every 5 minutes.
- Every hour.
- Daily.

Telemetry is collected into data bundles. Each bundle is an aggregation of all the data collected up to the point of data transmission. These bundles are stored on the device until a transmission event. When a bundle has been successfully sent to Palo Alto Networks, the bundle is deleted from the device.

If an error occurs while sending a bundle to Palo Alto Networks, the firewall waits 10 minutes and then tries again. The firewall will continue to try to send the bundle until it is either successful, or it needs the storage space to collect new telemetry data.

At every regular transmission interval, the firewall begins by sending the bundles scheduled for that event. After a successful transfer of those bundles, the firewall sends any failed bundles that it might have stored from previous transmission events.

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Manage Device Telemetry

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Essentials ☐ Strata Cloud Manager Pro

To manage device telemetry you can:

- Enable Device Telemetry
- Enable Service Routes for Telemetry
- Manage the Data that Device Telemetry Collects
- Manage Historical Device Telemetry

Enable Device Telemetry

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Essentials ☐ Strata Cloud Manager Pro

Beginning with PAN-OS 10.2.17, 11.1.11, 11.2.8, 12.1.2, and later releases, the telemetry autoenablement feature configures telemetry to be enabled by default on your devices.

By default, your device does not share data with Palo Alto Networks. If sharing is enabled, you can stop sharing all device telemetry by: **Device** > **Setup** > **Telemetry**, uncheck the **Enable Telemetry** box, and then commit your change.

To enable Device Telemetry so that data is shared with Palo Alto Networks:

STEP 1 | Enable Strata Logging Service.

1. If your organization does not have a Strata Logging Service license, install a device certificate if one is not already installed on your device.

   If your organization does have a Strata Logging Service license, make sure it is activated.
2. Make sure that your network is properly configured so that the firewall can send data to Strata Logging Service.

STEP 2 | Navigate to **Device** > **Setup** > **Telemetry**

STEP 3 | Edit the **Telemetry** widget.

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STEP 4 | In **Telemetry Destination**, select your region, if it is not automatically selected. If your organization is using Strata Logging Service, you must use the region that your Strata Logging Service is configured to use.

STEP 5 | Click **OK**, and then commit your changes.

> *`_cliuser` appears as a logged in admin whenever the firewall sends the telemetry file to it's destination.*

Disable Device Telemetry

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Essentials ☐ Strata Cloud Manager Pro

If your next-generation firewall is configured to share data with Palo Alto Networks, you can disable this sharing by:

- NGFWs With Telemetry Autoenablement
- NGFWs Without Telemetry Autoenablement

NGFWs With Telemetry Autoenablement

Applies to NGFWs managed by Strata Cloud Manager with PAN-OS 10.2.17+, 11.1.11+, 11.2.8+, or 12.1.2+.

Do one of the following:

- To disable telemetry for your tenant, navigate to **Hub** > **Common Services** > **Tenant Management** > **Telemetry Bar** > **Actions** > **Telemetry Information** > **Disabled** > **Save**.

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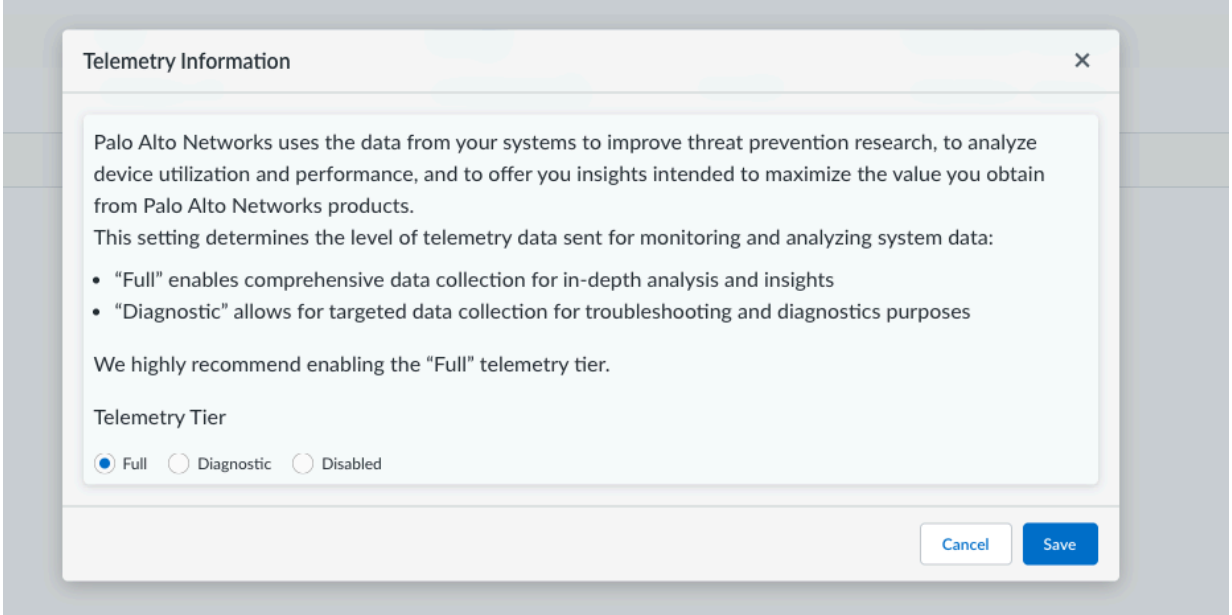
- To reduce telemetry tier to Diagnostic-only instead of fully disabling, navigate to **Hub** > **Common Services** > **Tenant Management** > **Telemetry Bar** > **Actions** > **Telemetry Information** > **Diagnostic** > **Save**.



NGFWs Without Telemetry Autoenablement

STEP 1 | Navigate to **Device** > **Setup** > **Telemetry**

STEP 2 | Edit the **Telemetry** widget.

STEP 3 | Uncheck the **Enable Telemetry** box.

STEP 4 | Click **OK**, and then commit your changes.

STEP 5 | Any telemetry data currently stored in Strata Logging Service is automatically purged one year after your firewall uploaded it. Optionally, if you do not want the data to reside in Strata Logging Service for this amount of time after you disable telemetry, open a support ticket and ask Palo Alto Networks to purge your telemetry data.

Enable Service Routes for Telemetry

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Essentials ☐ Strata Cloud Manager Pro

You can configure specific configuration requirements for device telemetry that collects data about your next-generation firewall or Panorama. For each virtual system, you can configure

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service routes to use specific interfaces for outbound telemetry data and share it by uploading to Strata Logging Service.

STEP 1 | Select **Device** > **Setup** > **Services**.

STEP 2 | Click the **Service Route Configuration** link under **Services Features**.

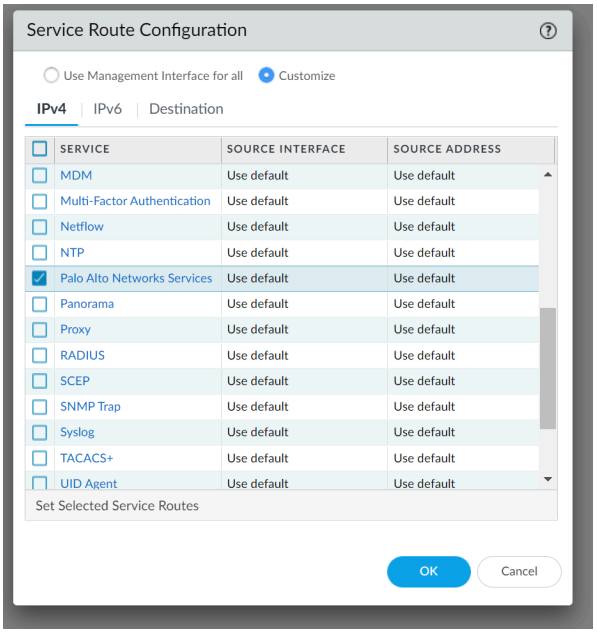
STEP 3 | Select **Customize**.

STEP 4 | Select **IPv4**.

STEP 5 | Select the **Palo Alto Networks Service**.

Choose the custom **Source Interface** you want to use as the interface for telemetry.

Choose the custom **Source Address** associated with the interface.



STEP 6 | **Commit** the configuration.

Manage the Data the Device Telemetry Collects

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Essentials ☐ Strata Cloud Manager Pro

Beginning with PAN-OS 10.2.17, 11.1.11, 11.2.8, 12.1.2, and later releases, the telemetry autoenablement feature configures telemetry to be enabled by default on your devices.

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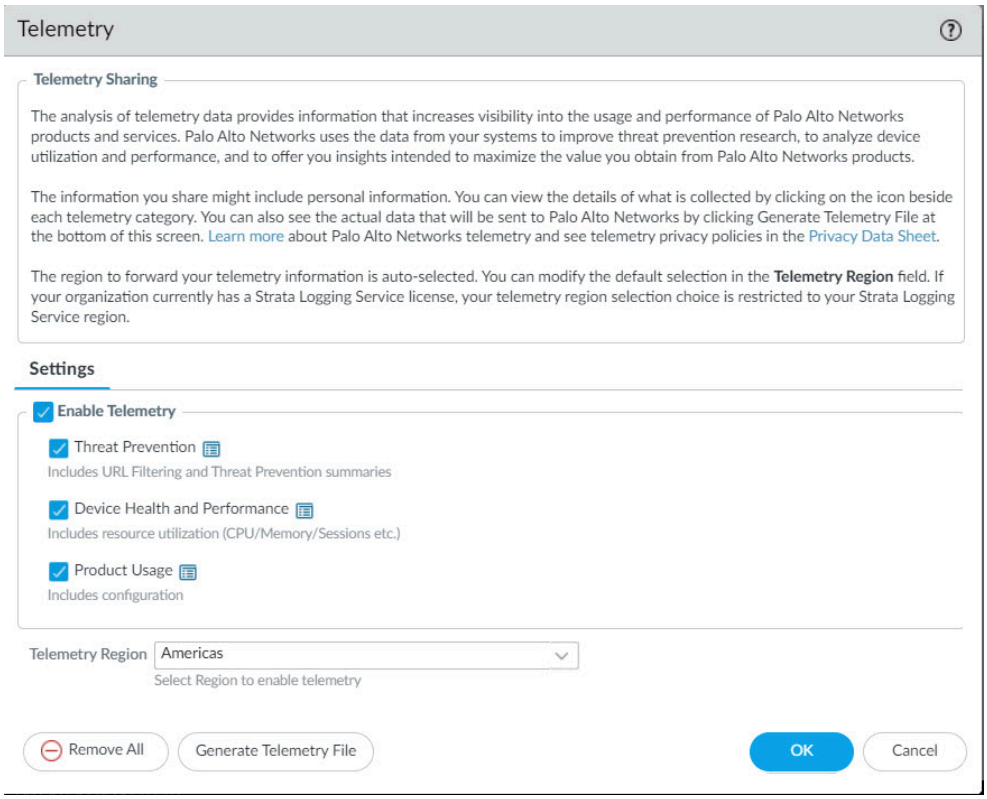
Select **Device** > **Setup** > **Telemetry** to see the currently collected telemetry categories. To change these categories, edit the Telemetry widget. Deselect any categories that you don't want the firewall to collect, **OK**, and then commit the change.

(PAN-OS version 11.0.1 and later versions of 11.0) The telemetry region is auto-selected.



*To stop sharing all device telemetry, uncheck the **Enable Telemetry** box, and then commit your change.*

Manage Historical Device Telemetry

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Essentials ☐ Strata Cloud Manager Pro

Prior to 10.0, telemetry data was mostly of interest for threat intelligence purposes. As of 10.0, threat intelligence metrics are still a large portion the data collected by the device, but a great deal more data involving the health, performance, and configuration of the device is collected as well.

If you have an existing next-generation firewall, and you have any of the historical telemetry data categories enabled, then when you upgrade to PAN-OS 11.1 your firewall will continue to collect

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and share this information. If you want to turn this telemetry data sharing off, use the following CLI commands:

```
set deviceconfig system update-schedule statistics-service
 application-reports no
set deviceconfig system update-schedule statistics-service threat-
prevention-reports no
set deviceconfig system update-schedule statistics-service threat-
prevention-information no
set deviceconfig system update-schedule statistics-service threat-
prevention-pcap no
set deviceconfig system update-schedule statistics-service passive-
dns-monitoring no
set deviceconfig system update-schedule statistics-service url-
reports no
set deviceconfig system update-schedule statistics-service health-
performance-reports no
set deviceconfig system update-schedule statistics-service file-
identification-reports no
```

If you have a 11.1 firewall and this telemetry sharing is turned off, but you want to share this data with Palo Alto Networks, then you can turn it on using:

```
set deviceconfig system update-schedule statistics-service
 application-reports yes
set deviceconfig system update-schedule statistics-service threat-
prevention-reports yes
set deviceconfig system update-schedule statistics-service threat-
prevention-information yes
set deviceconfig system update-schedule statistics-service threat-
prevention-pcap yes
set deviceconfig system update-schedule statistics-service passive-
dns-monitoring yes
set deviceconfig system update-schedule statistics-service url-
reports yes
set deviceconfig system update-schedule statistics-service health-
performance-reports yes
set deviceconfig system update-schedule statistics-service file-
identification-reports yes
```

You can see whether your device is collecting and sharing this historical telemetry data using the following CLI command:

```
show deviceconfig system update-schedule statistics-service
```

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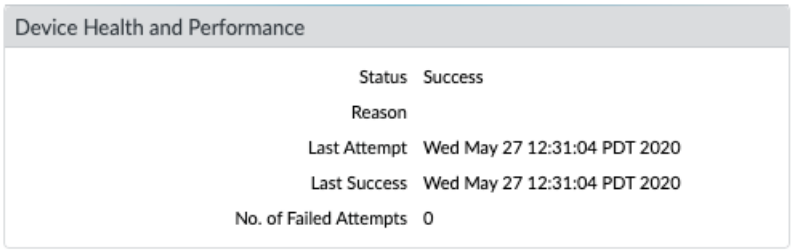
Monitor Device Telemetry

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Essentials ☐ Strata Cloud Manager Pro

PAN-OS shows you the sharing status for each telemetry category. Widgets for each metrics category are available at **Device** > **Setup** > **Telemetry**.



In the event of a failure, your device will retry the send attempt at the next transmission time. If the problem persists, check to make sure that your devices are properly configured to send data to Strata Logging Service:

- If your organization has a Strata Logging Service license, then make sure your Strata Logging Service license has been activated, and that your firewall is configured to use Strata Logging Service.
- If your organization does not have a Strata Logging Service license, then make sure you have installed a device certificate, and that your network is configured to allow traffic to Strata Logging Service.

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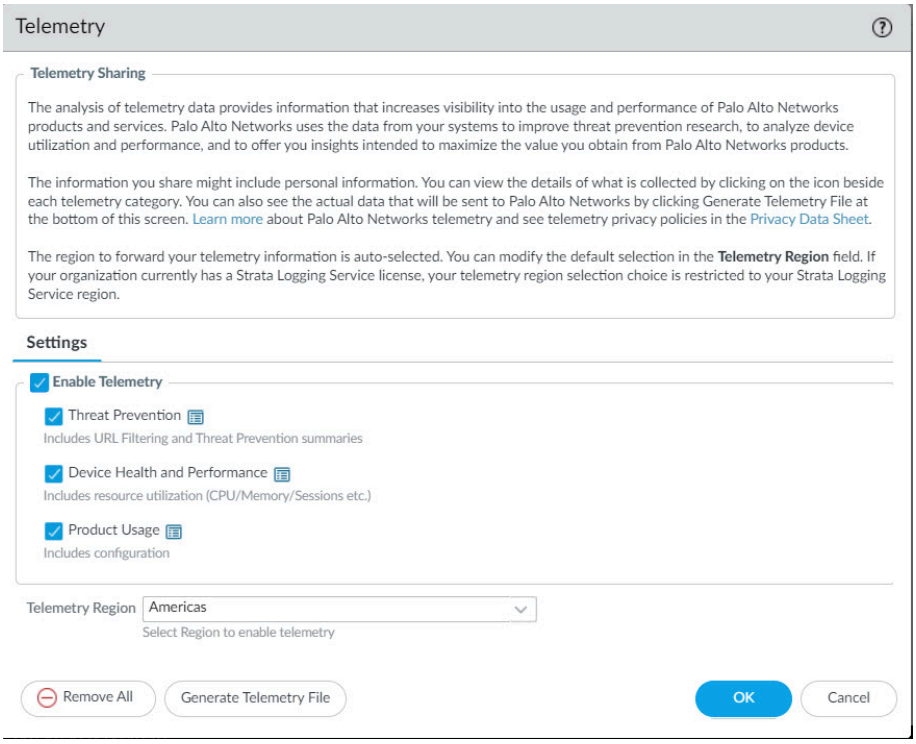
Sample the Data that Device Telemetry Collects

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Essentials ☐ Strata Cloud Manager Pro

You can download a live example of the data that device telemetry collects and shares with Palo Alto Networks. To do this, go to **Device** > **Setup** > **Telemetry**, and edit the **Telemetry** widget. Then click **Generate Telemetry File**.

(PAN-OS version 11.0.1 and later versions of 11.0)



The data collection will take a few minutes, depending on the speed of your firewall. When the process completes, click **Download Device Telemetry Data**. The telemetry bundle is a compressed tar ball, and it is placed in your default browser download directory.

For a description of every metric that device telemetry collects and shares with Palo Alto Networks, see the PAN-OS Device Telemetry Metrics Reference Guide.

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Security Settings

Where Can I Use This?	What Do I Need?
• NGFW	❑ PAN-OS 12.1.2

PAN-OS 12.1 Orion includes several enhancements and new features that improve the security of PAN-OS against attacks on the platform. A majority of these features are implemented behind the scenes as part of the platform's built-in protections. These features are designed to prevent successful exploits, reduce the impact of exploits, detect attempted exploits, and enable the ability to respond to attacks on PAN-OS. The features described here either have settings that can be configured or that generate logs to provide more information on PAN-OS security.

Integrity Measurement Architecture (IMA) runs in enforcement mode by default, and only allows execution of binaries and programs cryptographically signed by Palo Alto Networks. This prevents the execution of malware that might be dropped by an attacker and blocks attempts to modify existing PAN-OS binaries, effectively extending the secure boot into the run-time environment. You can monitor IMA violations through system logs using the CLI or the web interface. When IMA detects an attempted violation, it logs a critical severity system that you can use for investigation.

When IMA detects violations or attempted violations, PAN-OS can be configured to either continue running (collect logs and alerts for investigation), which is the default, or reboot to maintenance mode to disrupt the attacker and facilitate a more thorough investigation.

Software integrity checks run at boot time and then daily at a set time. You can now schedule the checks to run at a different time to better accommodate your requirements.

Configure Device Security

Where Can I Use This?	What Do I Need?
• NGFW	❑ PAN-OS 12.1 and later

Configure your device to manage how security violations are handled via the user interface or CLI.

User Interface

STEP 1 | Select **Device** > **Setup** > **PAN-OS Security** or **Panorama** > **Device** > **Setup** > **PAN-OS Security**.

STEP 2 | Edit **Device Security Settings**.

STEP 3 | Select **continue-running** (default) or **maintenance-mode** to configure the appliance or firewall response to a security violation.

1. **CLI Settings**

   Use the `continue-running` (default) or `maintenance-mode` to configure the appliance or firewall response to a security violation.

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PAN-OS Shield

Where Can I Use This?	What Do I Need?
• NGFW	❑ PAN-OS 12.2.2 and later

PAN-OS Shield provides vulnerability protection for firewalls with GlobalProtect gateway or portal. When you enable PAN-OS Shield, the firewall scans inbound control traffic using threat prevention signatures and takes action against detected exploits before they reach the management plane. This protects PAN-OS services from vulnerabilities—including critical exploits such as those targeting privilege escalation after initial compromise—by inspecting traffic at the data plane before it is forwarded to management plane processes.

PAN-OS Shield requires PAN-OS 12.2.2 or a later release. An Advanced Threat Prevention license provides signatures through regular content packages, but the license is not strictly required—signatures are also delivered through application-only content updates.

You can enable PAN-OS Shield without GlobalProtect gateway or portal configured, but the feature doesn't protect anything until GlobalProtect control traffic is present.

PAN-OS Shield operates through a dedicated internal virtual system (`panos-shield-vsys`) that the firewall creates automatically when you enable the feature. This internal vsys hosts a vulnerability protection profile and a security policy that are delivered and updated through content packages. You don't need to configure multi-vsys mode or allocate additional vsys licenses—the internal vsys is created independently of your licensed vsys capacity.

The vulnerability protection profile and security policy that PAN-OS Shield uses are read-only—they are delivered through content updates and you cannot modify them directly. However, you can add threat exceptions to override the action for specific threat IDs if you encounter false positives that impact your environment. When a signature triggers, the firewall generates a threat log with the matched threat ID, action taken, and relevant session details.

PAN-OS Shield is disabled by default. After you enable or disable the feature, a commit and reboot is required for the change to take effect. If you disable PAN-OS Shield, a confirmation warning explains the security implications of removing the protection.

When PAN-OS Shield is enabled, a read-only vulnerability protection profile appears under **Objects** > **PAN-OS Shield** > **Vulnerability Protection**. You can add threat exceptions to this profile to change the action for specific threat IDs.

Enable PAN-OS Shield

Where Can I Use This?	What Do I Need?
• NGFW	❑ PAN-OS 12.2.2 and later

PAN-OS Shield requires PAN-OS 12.2.2 or a later release. When you enable PAN-OS Shield, the firewall creates a dedicated internal virtual system that inspects inbound control traffic using

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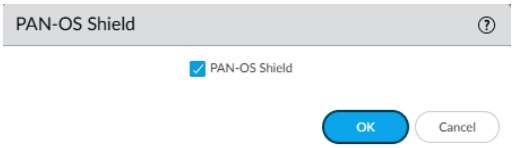
threat prevention signatures delivered through content updates. If you have an Advanced Threat Prevention license, signatures are delivered through regular content packages. If you don't have an Advanced Threat Prevention license (or if it has expired), the signatures are delivered through application-only content packages instead.

You can enable PAN-OS Shield without GlobalProtect gateway or portal configured, but the feature doesn't protect anything until GlobalProtect control traffic is present.

STEP 1 | Select **Device** > **Setup** > **PAN-OS Security**.

STEP 2 | In the **PAN-OS Shield** section, enable **PAN-OS Shield**.



When enabled, the firewall scans inbound control traffic for vulnerability exploits before it reaches the management plane. A threat log is generated each time a signature triggers.

STEP 3 | Commit your changes and reboot the firewall.

After the commit and reboot complete, the firewall creates the internal PAN-OS Shield vsys and begins inspecting control traffic using the vulnerability protection profile and security policy delivered through content updates.

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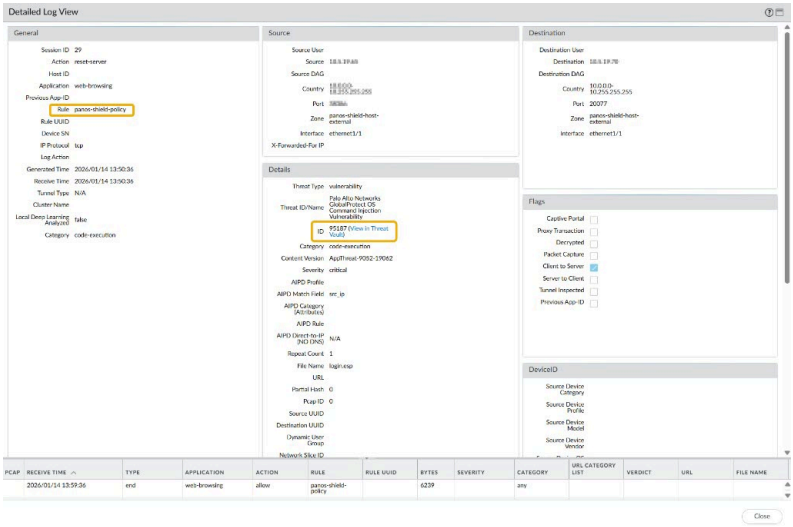
STEP 4 | (Optional) Select **Monitor** > **Logs** > **Threat** to view threats detected by PAN-OS Shield.

Threats detected by PAN-OS Shield display **panos-shield-policy** in the **Rule** column.

Click a log entry to open the **Detailed Log View**, which shows the threat type, threat ID, severity, action taken, and session details. Click **View in Threat Vault** next to the threat ID to view the associated CVE and additional threat intelligence.



Manage PAN-OS Shield Threat Exceptions

Where Can I Use This?	What Do I Need?
• NGFW	☐ PAN-OS 12.2.2 and later

The PAN-OS Shield vulnerability protection profile is read-only—you cannot modify the signatures or their default actions directly. However, if a signature triggers a false positive that impacts your environment, you can add a threat exception to override the action for that specific threat ID. Threat exceptions are stored separately from the content-delivered profile and persist across content updates.

STEP 1 | Select **Objects** > **PAN-OS Shield** > **Vulnerability Protection**.

STEP 2 | Click the PAN-OS Shield vulnerability protection profile.

NAME	LOCATION	COUNT	RULE NAME	THREAT NAME	HOST TYPE	SEVERITY	ACTION	PACKET CAPTURE
panos-shield-vul-profile		Rules: 1 Exceptions: 59	allow-any	any	any	any	allow	disable

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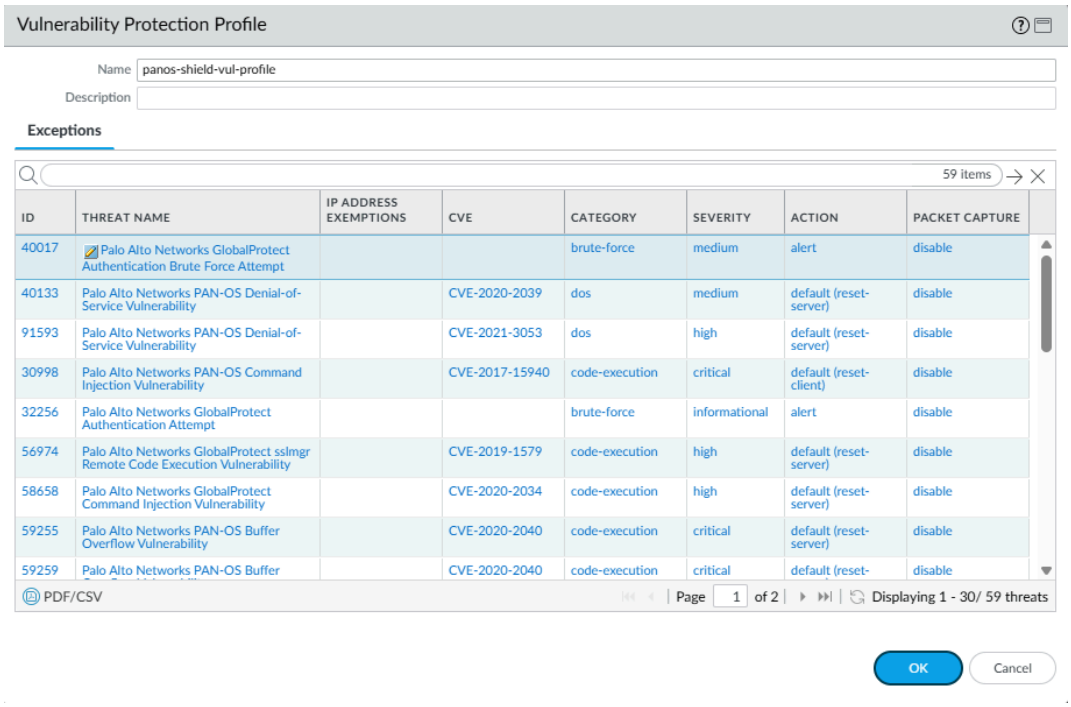
STEP 3 | Select the **Exceptions** tab.

The tab displays a table of threat signatures with the following columns: **ID**, **Threat Name**, **IP Address Exemptions**, **CVE**, **Category**, **Severity**, **Action**, and **Packet Capture**.

Clicking **ID**, **CVE**, **Category**, or **Severity** for an entry adds it as a search parameter. Clicking the **Threat Name** opens a **Threat Details** window showing the name, ID, description, severity level, and associated CVE.



STEP 4 | Locate the threat ID that you want to override.

You can find the threat ID in the threat logs generated when PAN-OS Shield detects a match.

STEP 5 | For the target entry, click **Action** and choose the action to apply for this threat ID.

- **alert**—Generate a threat log but allow the traffic.
- **allow**—Allow the traffic without taking action.
- **Block IP**—Block traffic from the source IP for a specified duration.
- **Default (Allow)**—Use the default action defined by the content-delivered profile, which allows the traffic.
- **Drop**—Silently drop the traffic.
- **Reset Both**—Send a TCP reset to both client and server.
- **Reset Client**—Send a TCP reset to the client.
- **Reset Server**—Send a TCP reset to the server.

STEP 6 | (Optional) For the target entry, click **Packet Capture** and choose **single-packet** or **extended-capture** to capture traffic when this threat ID triggers.

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STEP 7 | (Optional) Click **IP Address Exemptions** to add exemptions that exclude specific source addresses from this threat exception.

STEP 8 | Click **OK**.

STEP 9 | **Commit** your changes and reboot the firewall.

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Configure Software Integrity Checks

Where Can I Use This?	What Do I Need?
• NGFW	❑ PAN-OS 12.1 and later

Software integrity checks run automatically at boot time and periodically then every day at 1:30 am in the device's local timezone. You can configure the checks to run at a time of your preference. After configured, the time you set replaces the default 1:30 am run.

STEP 1 | Schedule a software integrity check using the CLI.

```
set deviceconfig setting management common-criteria self-test-
schedule software-integrity start-time <time in hh:mm format>
```

STEP 2 | Commit your changes.

STEP 3 | Select **Monitor** > **Logs** > **System**.

STEP 4 | View the completed software integrity check.

1. A failed system integrity check will result in PAN-OS booting into maintenance mode. If SELinux or IMA violations or attempted violations are detected, contact your security forensics team and create a customer support case with a technical support file (TSF).

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View IMA Logs

Where Can I Use This?	What Do I Need?
• NGFW	❑ PAN-OS 12.1 and later

With IMA in enforcement mode, any attempts to modify PAN-OS binaries are blocked. Attempts to execute an unknown file not signed by Palo Alto Networks are also blocked. This prevents malware from executing or modifying PAN-OS.

Logs generated by IMA might indicate attacker activity or false positives. This will generate logs of critical severity, and you can view these logs to determine if you need to take additional action.

STEP 1 | Select **Monitor** > **Logs** > **System**.

STEP 2 | Identify logs with an EventID of **ima** (`eventid eq ima`).

The description column displays a denial message indicating that an executable file was blocked from running. Review the key fields for additional information.

- **pid:** The numeric identifier of the process performing an operation on a system file.
- **subj:** The third component of this field indicates the type of the process performing the operation.
- **op:** Specifies the operation performed on a system file.
- **cause:** Indicates that the operation failed and why it failed.
- **name:** Refers to the full path of the system file on which the operation was performed.

For example, the following log indicates that an appraisal operation (op=appraise_data) was performed on the file /var/cache/pan/device/frr/frr_start_syslog_ng.sh. The process that triggered this appraisal was initiated by the sh command and has PID 2959 and type panw_routed_t. The appraisal failed (cause=IMA-signature-required) because an IMA signature was required but missing or invalid, possibly due to file tampering. This failure resulted in access to the file being blocked because IMA is in enforcement mode.

```
type=INTEGRITY_DATA msg=audit(1731707334.275:106):
pid=2959 uid=0 auid=4294967295 ses=4294967295
subj=system_u:system_r:panw_routed_t:s0 op=appraise_data cause=IMA-
signature-required comm="sh" name="/var/cache/pan/device/frr/
frr_start_syslog_ng.sh" dev="nvme0n1p2" ino=803911 res=0 errno=0
```

STEP 3 | Your incident response (IR) or forensics team should review these logs for further action. You can also report the issue on the Customer Support Portal (CSP). Be sure to include the entire IMA log message including the timestamp. It is also recommended to generate and export the technical support file (TSF) immediately, and provide it with the support case.

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Authentication

Authentication is a method for protecting services and applications by verifying the identities of users so that only legitimate users have access. Several firewall and Panorama features require authentication. Administrators authenticate to access the web interface, CLI, or XML API of the firewall and Panorama. End users authenticate through Authentication Portal or GlobalProtect to access various services and applications. You can choose from several authentication services to protect your network and to accommodate your existing security infrastructure while ensuring a smooth user experience.

If you have a public key infrastructure, you can deploy certificates to enable authentication without users having to manually respond to login challenges (see Certificate Management). Alternatively, or in addition to certificates, you can implement interactive authentication, which requires users to authenticate using one or more methods. The following topics describe how to implement, test, and troubleshoot the different types of interactive authentication:

- Authentication Types
- Plan Your Authentication Deployment
- Configure Multi-Factor Authentication
- Configure SAML Authentication
- Configure Kerberos Single Sign-On
- Configure Kerberos Server Authentication
- Configure TACACS+ Authentication
- Configure TACACS Accounting
- Configure RADIUS Authentication
- Configure LDAP Authentication
- Connection Timeouts for Authentication Servers
- Configure Local Database Authentication
- Configure an Authentication Profile and Sequence
- Test Authentication Server Connectivity
- Authentication Policy
- Troubleshoot Authentication Issues

Authentication Types

- External Authentication Services
- Multi-Factor Authentication
- SAML
- Kerberos
- TACACS+
- RADIUS
- LDAP
- Local Authentication

External Authentication Services

The firewall and Panorama can use external servers to control administrative access to the web interface and end user access to services or applications through Authentication Portal and GlobalProtect. In this context, any authentication service that is not local to the firewall or Panorama is considered external, regardless of whether the service is internal (such as Kerberos) or external (such as a SAML identity provider) relative to your network. The server types that the firewall and Panorama can integrate with include Multi-Factor Authentication (MFA), SAML, Kerberos, TACACS+, RADIUS, and LDAP. Although you can also use the Local Authentication services that the firewall and Panorama support, usually external services are preferable because they provide:

- Central management of all user accounts in an external identity store. All the supported external services provide this option for end users and administrators.
- Central management of account authorization (role and access domain assignments). SAML, TACACS+, and RADIUS support this option for administrators.
- Single sign-on (SSO), which enables users to authenticate only once for access to multiple services and applications. SAML and Kerberos support SSO.
- Multiple authentication challenges of different types (factors) to protect your most sensitive services and applications. MFA services support this option.

Authentication through an external service requires a server profile that defines how the firewall connects to the service. You assign the server profile to authentication profiles, which define settings that you customize for each application and set of users. For example, you can configure one authentication profile for administrators who access the web interface and another profile for end users who access a GlobalProtect portal. For details, see Configure an Authentication Profile and Sequence.

Multi-Factor Authentication

You can Configure Multi-Factor Authentication (MFA) to ensure that each user authenticates using multiple methods (factors) when accessing highly sensitive services and applications. For example, you can force users to enter a login password and then enter a verification code that they receive by phone before allowing access to important financial documents. This approach helps to prevent attackers from accessing every service and application in your network just by

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stealing passwords. Of course, not every service and application requires the same degree of protection, and MFA might not be necessary for less sensitive services and applications that users access frequently. To accommodate a variety of security needs, you can Configure Authentication Policy rules that trigger MFA or a single authentication factor (such as login credentials or certificates) based on specific services, applications, and end users.

When choosing how many and which types of authentication factors to enforce, it's important to understand how policy evaluation affects the user experience. When a user requests a service or application, the firewall first evaluates Authentication policy. If the request matches an Authentication policy rule with MFA enabled, the firewall displays a Authentication Portal web form so that users can authenticate for the first factor. If authentication succeeds, the firewall displays an MFA login page for each additional factor. Some MFA services prompt the user to choose one factor out of two to four, which is useful when some factors are unavailable. If authentication succeeds for all factors, the firewall evaluates Security policy for the requested service or application.

To reduce the frequency of authentication challenges that interrupt the user workflow, configure the first factor to use Kerberos or SAML single sign-on (SSO) authentication.

To implement MFA for GlobalProtect, refer to Configure GlobalProtect to facilitate multi-factor authentication notifications.

You cannot use MFA authentication profiles in authentication sequences.

For end-user authentication via Authentication Policy, the firewall directly integrates with several MFA platforms (Duo v2, Okta Adaptive, PingID, and RSA SecurID), as well as integrating through RADIUS or SAML for all other MFA platforms. For remote user authentication to GlobalProtect portals and gateways and for administrator authentication to the Panorama and PAN-OS web interface, the firewall integrates with MFA vendors using RADIUS and SAML only.

The firewall supports the following MFA factors:

Factor	Description
Push	An endpoint device (such as a phone or tablet) prompts the user to allow or deny authentication.
Short message service (SMS)	An SMS message on the endpoint device prompts the user to allow or deny authentication. In some cases, the endpoint device provides a code that the user must enter in the MFA login page.
Voice	An automated phone call prompts the user to authenticate by pressing a key on the phone or entering a code in the MFA login page.
One-time password (OTP)	An endpoint device provides an automatically generated alphanumeric string, which the user enters in the MFA login page to enable authentication for a single transaction or session.

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SAML

You can use Security Assertion Markup Language (SAML) 2.0 to authenticate administrators who access the firewall or Panorama web interface and end users who access web applications that are internal or external to your organization. In environments where each user accesses many applications and authenticating for each one would impede user productivity, you can configure SAML single sign-on (SSO) to enable one login to access multiple applications. Likewise, SAML single logout (SLO) enables a user to end sessions for multiple applications by logging out of just one session. SSO is available to administrators who access the web interface and to end users who access applications through GlobalProtect or Authentication Portal. SLO is available to administrators and GlobalProtect end users, but not to Authentication Portal end users. When you configure SAML authentication on the firewall or on Panorama, you can specify SAML attributes for administrator authorization. SAML attributes enable you to quickly change the roles, access domains, and user groups of administrators through your directory service, which is often easier than reconfiguring settings on the firewall or Panorama.

Administrators cannot use SAML to authenticate to the CLI on the firewall or Panorama.

You cannot use SAML authentication profiles in authentication sequences.

SAML authentication requires a *service provider* (the firewall or Panorama), which controls access to applications, and an *identity provider* (IdP) such as PingFederate, which authenticates users. When a user requests a service or application, the firewall or Panorama intercepts the request and redirects the user to the IdP for authentication. The IdP then authenticates the user and returns a *SAML assertion*, which indicates authentication succeeded or failed. SAML Authentication for Authentication Portal End Users illustrates SAML authentication for an end user who accesses applications through Authentication Portal.

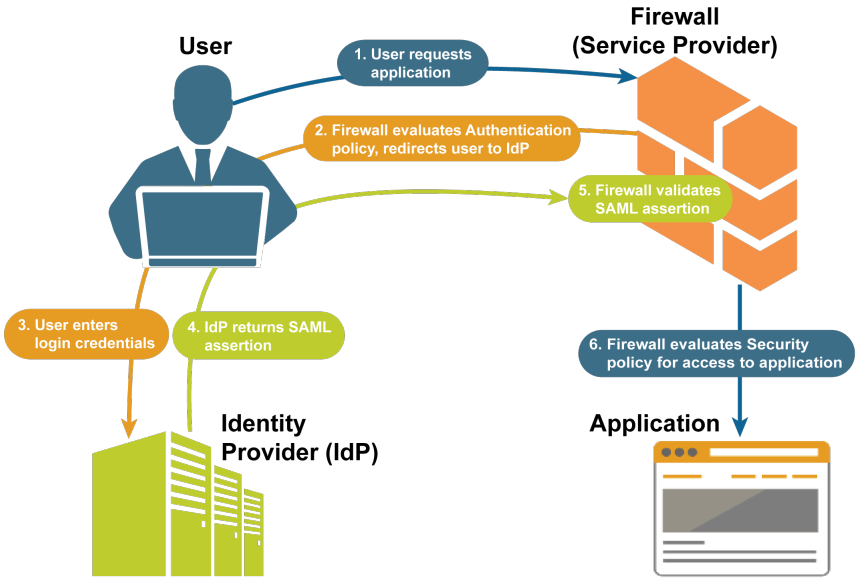

Figure 1: SAML Authentication for Authentication Portal End Users

Kerberos

Kerberos is an authentication protocol that enables a secure exchange of information between parties over an insecure network using unique keys (called tickets) to identify the parties. The

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firewall and Panorama support two types of Kerberos authentication for administrators and end users:

- **Kerberos server authentication**—A Kerberos server profile enables users to natively authenticate to an Active Directory domain controller or a Kerberos V5-compliant authentication server. This authentication method is interactive, requiring users to enter usernames and passwords. For the configuration steps, see Configure Kerberos Server Authentication.
- **Kerberos single sign-on (SSO)**—A network that supports Kerberos V5 SSO prompts a user to log in only for initial access to the network (such as logging in to Microsoft Windows). After this initial login, the user can access any browser-based service in the network (such as the firewall web interface) without having to log in again until the SSO session expires. (Your Kerberos administrator sets the duration of SSO sessions.) If you enable both Kerberos SSO and another external authentication service (such as a TACACS+ server), the firewall first tries SSO and, only if that fails, falls back to the external service for authentication. To support Kerberos SSO, your network requires:
  - A Kerberos infrastructure, including a key distribution center (KDC) with an authentication server (AS) and ticket-granting service (TGS).
  - A Kerberos account for the firewall or Panorama that will authenticate users. An account is required to create a Kerberos keytab, which is a file that contains the principal name and hashed password of the firewall or Panorama. The SSO process requires the keytab.

  For the configuration steps, see Configure Kerberos Single Sign-On.

  Kerberos SSO is available only for services and applications that are internal to your Kerberos environment. To enable SSO for external services and applications, use SAML.

TACACS+

Terminal Access Controller Access-Control System Plus (TACACS+) is a family of protocols that enable authentication and authorization through a centralized server. TACACS+ encrypts usernames and passwords, making it more secure than RADIUS, which encrypts only passwords. TACACS+ is also more reliable because it uses TCP, whereas RADIUS uses UDP. You can configure TACACS+ authentication for end users or administrators on the firewall and for administrators on Panorama. Optionally, you can use TACACS+ Vendor-Specific Attributes (VSAs) to manage administrator authorization. TACACS+ VSAs enable you to quickly change the roles, access domains, and user groups of administrators through your directory service instead of reconfiguring settings on the firewall and Panorama.

The firewall and Panorama support the following TACACS+ attributes and VSAs. Refer to your TACACS+ server documentation for the steps to define these VSAs on the TACACS+ server.

Name	Value
service	This attribute is required to identify the VSAs as specific to Palo Alto Networks. You must set the value to **`PaloAlto`**.

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Name	Value
protocol	This attribute is required to identify the VSAs as specific to Palo Alto Networks devices. You must set the value to **`firewall`**.
PaloAlto-Admin-Role	A default (dynamic) administrative role name or a custom administrative role name on the firewall.
PaloAlto-Admin-Access-Domain	The name of an access domain for firewall administrators (configured in the **Device** > **Access Domains** page). Define this VSA if the firewall has multiple virtual systems.
PaloAlto-Panorama-Admin-Role	A default (dynamic) administrative role name or a custom administrative role name on Panorama.
PaloAlto-Panorama-Admin-Access-Domain	The name of an access domain for Device Group and Template administrators (configured in the **Panorama** > **Access Domains** page).
PaloAlto-User-Group	The name of a user group in the Allow List of an authentication profile.

RADIUS

Remote Authentication Dial-In User Service (RADIUS) is a broadly supported networking protocol that provides centralized authentication and authorization. You can configure RADIUS authentication for end users or administrators on the firewall and for administrators on Panorama. Optionally, you can use RADIUS Vendor-Specific Attributes (VSAs) to manage administrator authorization. RADIUS VSAs enable you to quickly change the roles, access domains, and user groups of administrators through your directory service instead of reconfiguring settings on the firewall and Panorama. You can also configure the firewall to use a RADIUS server for:

- Collecting VSAs from GlobalProtect endpoints.
- Implementing Multi-Factor Authentication.

When sending authentication requests to a RADIUS server, the firewall and Panorama use the authentication profile name as the network access server (NAS) identifier, even if the profile is assigned to an authentication sequence for the service (such as administrative access to the web interface) that initiates the authentication process.

The firewall and Panorama support the following RADIUS VSAs. To define VSAs on a RADIUS server, you must specify the vendor code (25461 for Palo Alto Networks firewalls or Panorama) and the VSA name and number. Some VSAs also require a value. Refer to your RADIUS server documentation for the steps to define these VSAs.

Alternatively, you can download the Palo Alto Networks RADIUS dictionary, which defines the authentication attributes that the Palo Alto Networks firewall and a RADIUS server use to

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communicate with each other, and install it on your RADIUS server to map the attributes to the RADIUS binary data.

When you predefine dynamic administrator roles for users on the server, use lower-case to specify the role (for example, enter ***superuser****, not* ***SuperUser****).*

When configuring the advanced vendor options on a Cisco Secure Access Control Server (ACS), you must set both the ***Vendor Length Field Size*** *and* ***Vendor Type Field Size*** *to* ***1****. Otherwise, authentication will fail.*

Name	Number	Value
VSAs for administrator account management and authentication		
PaloAlto-Admin-Role	1	A default (dynamic) administrative role name or a custom administrative role name on the firewall.
PaloAlto-Admin-Access-Domain	2	The name of an access domain for firewall administrators (configured in the **Device** > **Access Domains** page). Define this VSA if the firewall has multiple virtual systems.
PaloAlto-Panorama-Admin-Role	3	A default (dynamic) administrative role name or a custom administrative role name on Panorama.
PaloAlto-Panorama-Admin-Access-Domain	4	The name of an access domain for Device Group and Template administrators (configured in the **Panorama** > **Access Domains** page).
PaloAlto-User-Group	5	The name of a user group that an authentication profile references.
VSAs forwarded from GlobalProtect endpoints to the RADIUS server		
PaloAlto-User-Domain	6	Don't specify a value when you define these VSAs.
PaloAlto-Client-Source-IP	7	
PaloAlto-Client-OS	8	
PaloAlto-Client-Hostname	9	
PaloAlto-GlobalProtect-Client-Version	10	

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LDAP

Lightweight Directory Access Protocol (LDAP) is a standard protocol for accessing information directories. You can Configure LDAP Authentication for end users and for firewall and Panorama administrators.

Configuring the firewall to connect to an LDAP server also enables you to define policy rules based on users and user groups instead of just IP addresses. For the steps, see Map Users to Groups and Enable User- and Group-Based Policy.

Local Authentication

Although the firewall and Panorama provide local authentication for administrators and end users, External Authentication Services are preferable in most cases because they provide central account management. However, you might require special user accounts that you don't manage through the directory servers that your organization reserves for regular accounts. For example, you might define a superuser account that is local to the firewall so that you can access the firewall even if the directory server is down. In such cases, you can use the following local authentication methods:

- **(Firewall only) Local database authentication**—To Configure Local Database Authentication, you create a database that runs locally on the firewall and contains user accounts (usernames and passwords or hashed passwords) and user groups. This type of authentication is useful for creating user accounts that reuse the credentials of existing Unix accounts in cases where you know only the hashed passwords, not the plaintext passwords. Because local database authentication is associated with authentication profiles, you can accommodate deployments where different sets of users require different authentication settings, such as Kerberos single sign-on (SSO) or Multi-Factor Authentication (MFA). (For details, see Configure an Authentication Profile and Sequence). For administrator accounts that use an authentication profile, password complexity and expiration settings are not applied. This authentication method is available to administrators who access the firewall (but not Panorama) and end users who access services and applications through Authentication Portal or GlobalProtect.
- **Local authentication without a database**—You can configure firewall administrative accounts or Panorama administrative accounts without creating a database of users and user groups that runs locally on the firewall or Panorama. Because this method is not associated with authentication profiles, you cannot combine it with Kerberos SSO or MFA. However, this is the only authentication method that allows password profiles, which enable you to associate individual accounts with password expiration settings that differ from the global settings. (For details, see Define password complexity and expiration settings)

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Plan Your Authentication Deployment

The following are key questions to consider before you implement an authentication solution for administrators who access the firewall and end users who access services and applications through Authentication Portal.

For both end users and administrators, consider:

- How can you leverage your existing security infrastructure? Usually, integrating the firewall with an existing infrastructure is faster and cheaper than setting up a new, separate solution just for firewall services. The firewall can integrate with Multi-Factor Authentication, SAML, Kerberos, TACACS+, RADIUS, and LDAP servers. If your users access services and applications that are external to your network, you can use SAML to integrate the firewall with an identity provider (IdP) that controls access to both external and internal services and applications.
- How can you optimize the user experience? If you don't want users to authenticate manually and you have a public key infrastructure, you can implement certificate authentication. Another option is to implement Kerberos or SAML single sign-on (SSO) so that users can access multiple services and applications after logging in to just one. If your network requires additional security, you can combine certificate authentication with interactive (challenge-response) authentication.
- Do you require special user accounts that you don't manage through the directory servers that your organization reserves for regular accounts? For example, you might define a superuser account that is local to the firewall so that you can access the firewall even if the directory server is down. You can configure Local Authentication for these special-purpose accounts.

  External Authentication Services *are usually preferable to local authentication because they provide central account management, reliable authentication services, and usually logging and troubleshooting features.*
- Are the user names for your user accounts properly formatted? Leveraging SAML, Kerberos, TACACS+, RADIUS, and LDAP authentication requires all user names adhere to the regular expression Linux login name rule. User names must have the format **`[a-zA-Z0-9_.][a-zA-Z0-9_.-]{0,30}[a-zA-Z0-9_.$-]`**.

  This means that:

  - The first character of the user name must be an upper or lower case alphabetical letter, a number (0-9), or either _ (underscore) or **.** (period).
  - Other than the first and last characters, the user name may contain upper or lower case alphabetical characters, numbers (0-9), and _ (underscore), **.** (period), or **-** (dash). The maximum length is 30 characters excluding the first and last characters.
  - The last character of the user name may be an upper or lower case alphabetical letter, a number (0-9), or _ (underscore), **.** (period), **$**, or **-** (dash).

  Adhering to the regular expression Linux login name rule is required for PAN-OS administrators only. It is not required for GlobalProtect and Captive Portal users.

For end users only, consider:

- Which services and applications are more sensitive than others? For example, you might want stronger authentication for key financial documents than for search engines. To protect

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your most sensitive services and applications, you can configure Multi-Factor Authentication (MFA) to ensure that each user authenticates using multiple methods (factors) when accessing those services and applications. To accommodate a variety of security needs, Configure Authentication Policy rules that trigger MFA or single factor authentication (such as login credentials or certificates) based on specific services, applications, and end users. Other ways to reduce your attack surface include network segmentation and user groups for allowed applications.

For administrators only, consider:

- Do you use an external server to centrally manage authorization for all administrative accounts? By defining Vendor-Specific Attributes (VSAs) on the external server, you can quickly change administrative role assignments through your directory service instead of reconfiguring settings on the firewall. VSAs also enable you to specify access domains for administrators of firewalls with multiple virtual systems. SAML, TACACS+, and RADIUS support external authorization.

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Configure Multi-Factor Authentication

To use Multi-Factor Authentication (MFA) for protecting sensitive services and applications, you must configure Authentication Portal to display a web form for the first authentication factor and to record Authentication Timestamps. The firewall uses the timestamps to evaluate the timeouts for Authentication Policy rules. To enable additional authentication factors, you can integrate the firewall with MFA vendors through RADIUS or vendor APIs. After evaluating Authentication policy, the firewall evaluates Security policy, so you must configure rules for both policy types.

Palo Alto Networks provides support for MFA vendors through Applications content updates. This means that if you use Panorama to push device group configurations to firewalls, you must install the same Applications updates on the firewalls as on Panorama to avoid mismatches in vendor support.

MFA vendor API integrations are supported for end-user authentication through Authentication Policy only. For remote user authentication to GlobalProtect portals or gateways or for administrator authentication to the PAN-OS or Panorama web interface, you can only use MFA vendors supported through RADIUS or SAML; MFA services through vendor APIs are not supported in these use cases.

STEP 1 | Configure Authentication Portal in **Redirect** mode to display a web form for the first authentication factor, to record authentication timestamps, and to update user mappings.

STEP 2 | Configure one of the following server profiles to define how the firewall will connect to the service that authenticates users for the first authentication factor.

- Add a RADIUS server profile. This is required if the firewall integrates with an MFA vendor through RADIUS. In this case, the MFA vendor provides the first and all additional authentication factors, so you can skip the next step (configuring an MFA server profile). If the firewall integrates with an MFA vendor through an API, you can still use a RADIUS server profile for the first factor but MFA server profiles are required for the additional factors.
- Add a SAML IdP server profile.
- Add a Kerberos server profile.
- Add a TACACS+ server profile.
- Add an LDAP server profile.

In most cases, an external service is recommended for the first authentication factor. However, you can configure Configure Local Database Authentication as an alternative.

STEP 3 | Add an MFA server profile.

The profile defines how the firewall connects to the MFA server. Add a separate profile for each authentication factor after the first factor. The firewall integrates with these MFA

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servers through vendor APIs. You can specify up to three additional factors. Each MFA vendor provides one factor, though some vendors let users choose one factor out of several.

1. Select **Device** > **Server Profiles** > **Multi Factor Authentication** and **Add** a profile.
2. Enter a **Name** to identify the MFA server.
3. Select the **Certificate Profile** that the firewall will use to validate the MFA server certificate when establishing a secure connection to the MFA server.
4. Select the **MFA Vendor** you deployed.
5. Configure the **Value** of each vendor attribute.

   The attributes define how the firewall connects to the MFA server. Each vendor **Type** requires different attributes and values; refer to your vendor documentation for details.
6. Click **OK** to save the profile.

STEP 4 | Configure an authentication profile.

The profile defines the order of the authentication factors that users must respond to.

1. Select **Device** > **Authentication Profile** and **Add** a profile.
2. Enter a **Name** to identify the authentication profile.
3. Select the **Type** for the first authentication factor and select the corresponding **Server Profile**.
4. Select **Factors**, **Enable Additional Authentication Factors**, and **Add** the MFA server profiles you configured.

   The firewall will invoke each MFA service in the listed order, from top to bottom.
5. Click **OK** to save the authentication profile.

STEP 5 | Configure an authentication enforcement object.

The object associates each authentication profile with an Authentication Portal method. The method determines whether the first authentication challenge (factor) is transparent or requires a user response.

Select the **Authentication Profile** you configured and enter a **Message** that tells users how to authenticate for the first factor. The message displays in the Authentication Portal web form.

*If you set the **Authentication Method** to **browser-challenge**, the Authentication Portal web form displays only if Kerberos SSO authentication fails. Otherwise, authentication for the first factor is automatic; users won't see the web form.*

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STEP 6 | Configure an Authentication policy rule.

The rule must match the services and applications you want to protect and the users who must authenticate.

1. Select **Policies** > **Authentication** and **Add** a rule.
2. Enter a **Name** to identify the rule.
3. Select **Source** and **Add** specific zones and IP addresses or select **Any** zones or IP addresses.

   The rule applies only to traffic coming from the specified IP addresses or from interfaces in the specified zones.
4. Select **User** and select or **Add** the source users and user groups to which the rule applies (default is **any**).
5. Select **Destination** and **Add** specific zones and IP addresses or select **any** zones or IP addresses.

   The IP addresses can be resources (such as servers) for which you want to control access.
6. Select **Service/URL Category** and select or **Add** the services and service groups for which the rule controls access (default is **service-http**).
7. Select or **Add** the URL categories for which the rule controls access (default is **any**). For example, you can create a custom URL category that specifies your most sensitive internal sites.
8. Select **Actions** and select the **Authentication Enforcement** object you created.
9. Specify the **Timeout** period in minutes (default 60) during which the firewall prompts the user to authenticate only once for repeated access to services and applications.

   ***Timeout** is a tradeoff between tighter security (less time between authentication prompts) and the user experience (more time between authentication prompts). More frequent authentication is often the right choice for access to critical systems and sensitive areas such as a data center. Less frequent authentication is often the right choice at the network perimeter and for businesses for which the user experience is key.*
10. Click **OK** to save the rule.

STEP 7 | Customize the MFA login page.

The firewall displays this page to tell users how to authenticate for MFA factors and to indicate the authentication status (in progress, succeeded, or failed).

1. Select **Device** > **Response Pages** and select **MFA Login Page**.
2. Select the **Predefined** response page and **Export** the page to your client system.
3. On your client system, use an HTML editor to customize the downloaded response page and save it with a unique filename.
4. Return to the MFA Login Page dialog on the firewall, **Import** your customized page, **Browse** to select the **Import File**, select the **Destination** (virtual system or **shared** location), click **OK**, and click **Close**.

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STEP 8 | Configure a Security policy rule that allows users to access the services and applications that require authentication.

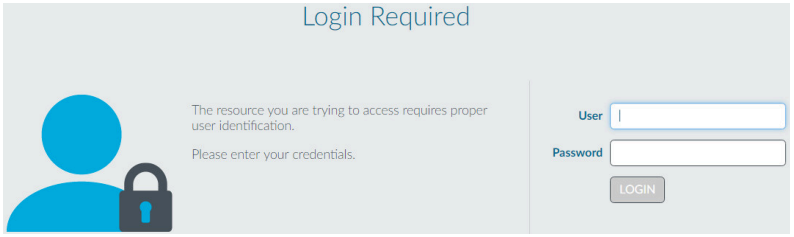
1. Create a Security Policy Rule.
2. **Commit** your changes.

   *The automated correlation engine on the firewall uses several correlation objects to detect events on your network that could indicate credential abuse relating to MFA. To review the events, select **Monitor** > **Automated Correlation Engine** > **Correlated Events**.*

STEP 9 | Verify that the firewall enforces MFA.

1. Log in to your network as one of the source users specified in the Authentication rule.
2. Request a service or application that matches one of the services or applications specified in the rule.

   The firewall displays the Authentication Portal web form for the first authentication factor. The page contains the message you entered in the authentication enforcement object. For example:



3. Enter your user credentials for the first authentication challenge.

   The firewall then displays an MFA login page for the next authentication factor. For example, the MFA service might prompt you to select the Voice, SMS, push, or PIN code

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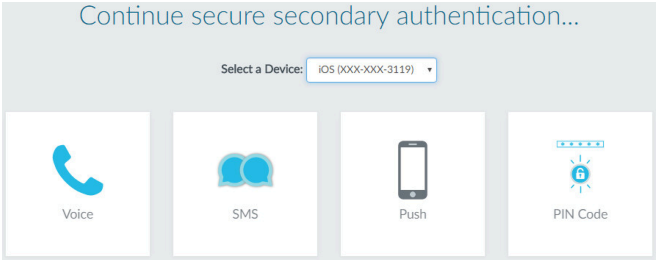
(OTP) authentication method. If you select push, your phone prompts you to approve the authentication.



4. Authenticate for the next factor.

   The firewall displays an authentication success or failure message. If authentication succeeded, the firewall displays an MFA login page for the next authentication factor, if any.

   Repeat this step for each MFA factor. After you authenticate for all the factors, the firewall evaluates Security policy to determine whether to allow access to the service or application.
5. End the session for the service or application you just accessed.
6. Start a new session for the same service or application. Be sure to perform this step within the **Timeout** period you configured in the Authentication rule.

   The firewall allows access without re-authenticating.
7. Wait until the **Timeout** period expires and request the same service or application.

   The firewall prompts you to re-authenticate.

Configure MFA Between RSA SecurID and the Firewall

Multi-factor authentication allows you to protect company assets by using multiple factors to verify a user's identity before allowing them to access network resources. To enable multi-factor authentication (MFA) between the firewall and the RSA SecurID Access Cloud Authentication Service, you must first configure the RSA SecurID Service so that you have the details that you need to configure the firewall to authenticate users using multiple factors. After you have performed the required configuration on the RSA SecurID Access Console, you can configure the firewall to integrate with RSA SecurID.

The Palo Alto Networks next-generation firewall integrates with the RSA SecurID Access Cloud Authentication Service. The MFA API integration with RSA SecurID is supported for cloud-based services only and does not support two-factor authentication for the on-premise Authentication Manager when the second factor uses the Vendor Specific API. The minimum content version required for this integration is 752 and PAN-OS 8.0.2.

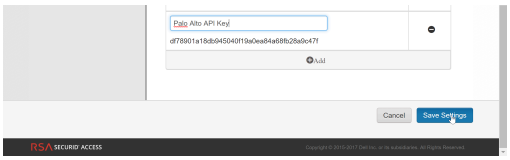
- Get the RSA SecurID Access Cloud Authentication Service Details
- Configure the Firewall for MFA with RSA SecurID

Get the RSA SecurID Access Cloud Authentication Service Details

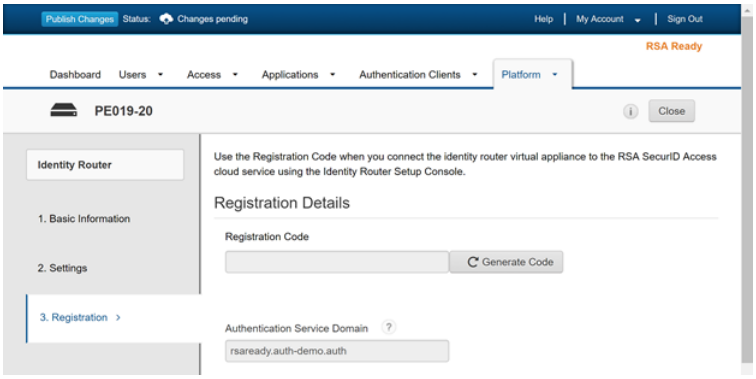
In order to securely pass user authentication requests to and from the firewall and the RSA SecurID Access Cloud Authentication Service, you must first go to the RSA SecurID Access

Console and configure the RSA Access ID, the authentication service URL, and the client API key that the firewall needs to authenticate to and interact with the service. The firewall also needs the Access Policy ID that uses either the RSA Approve or RSA Tokencode authentication method to authenticate to the identity source.

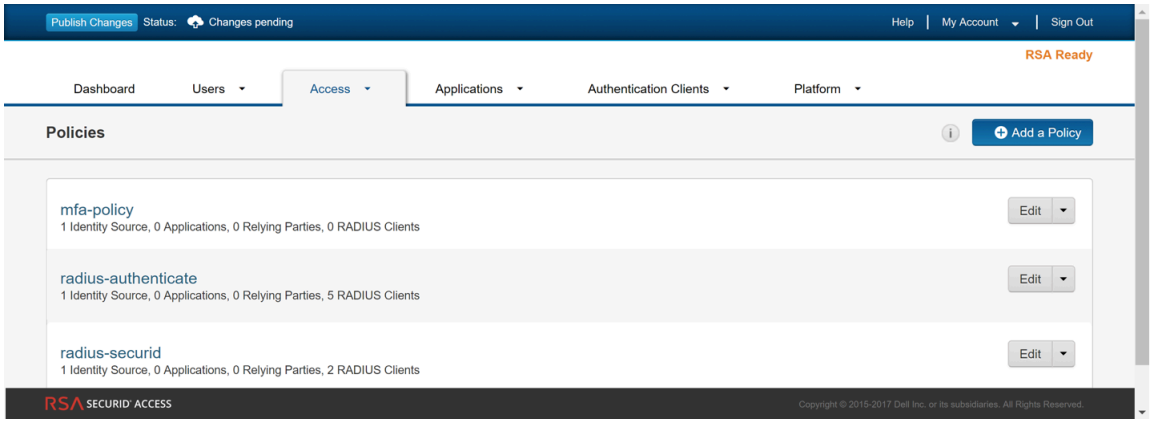
- **Generate the RSA SecurID API key**—Log on to RSA SecurID Access Console and select **My Account** > **Company Settings** > **Authentication API Keys**. **Add** a new key and then **Save Settings** and **Publish Changes**.



- **Get the RSA SecurID Access endpoint API (Authentication Service Domain) to which the firewall must connect**—Select **Platform** > **Identity Routers**, pick an Identity Router to **Edit** and jot down the **Authentication Service Domain**. In this example it is https://rsaready.auth-demo.auth.



- **Get the Access Policy ID**—Select **Access** > **Policies** and jot down the name of the access policy that will allow the firewall to act as an authentication client to the RSA SecurID service. The policy must be configured to use either the RSA Approve or the RSA Tokencode authentication methods only.



Configure the Firewall for MFA with RSA SecurID

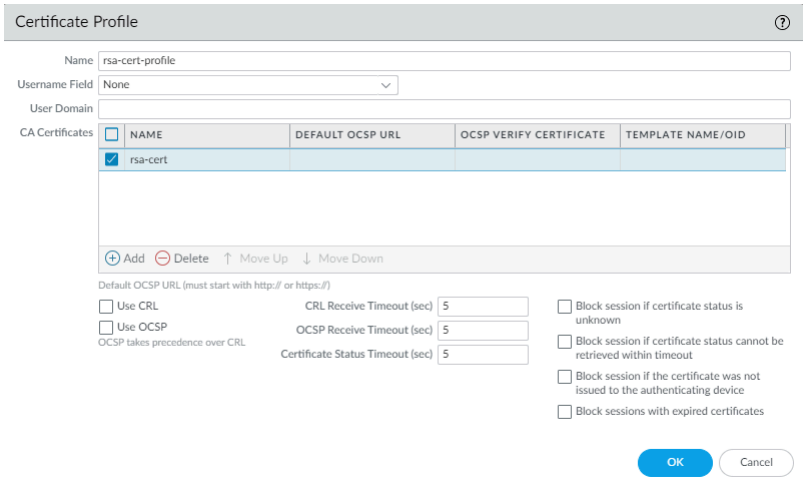
After you Get the RSA SecurID Access Cloud Authentication Service Details, you can configure the firewall to prompt users for an RSA SecurID token when MFA is invoked.

STEP 1 | Configure the firewall to trust the SSL certificate provided by the RSA SecurID Access endpoint API.

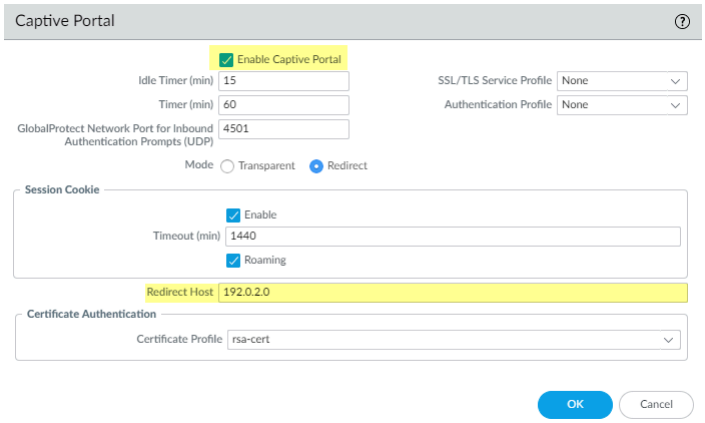
1. Export the SSL certificate from the RSA SecurID Access endpoint and import it into the firewall.

   To enable trust between the firewall and the RSA SecurID Access endpoint API, you must either import a self-signed certificate, or the CA certificate used to sign the certificate.

2. Configure a Certificate Profile (**Device** > **Certificate Management** > **Certificate Profile** and click **Add**).

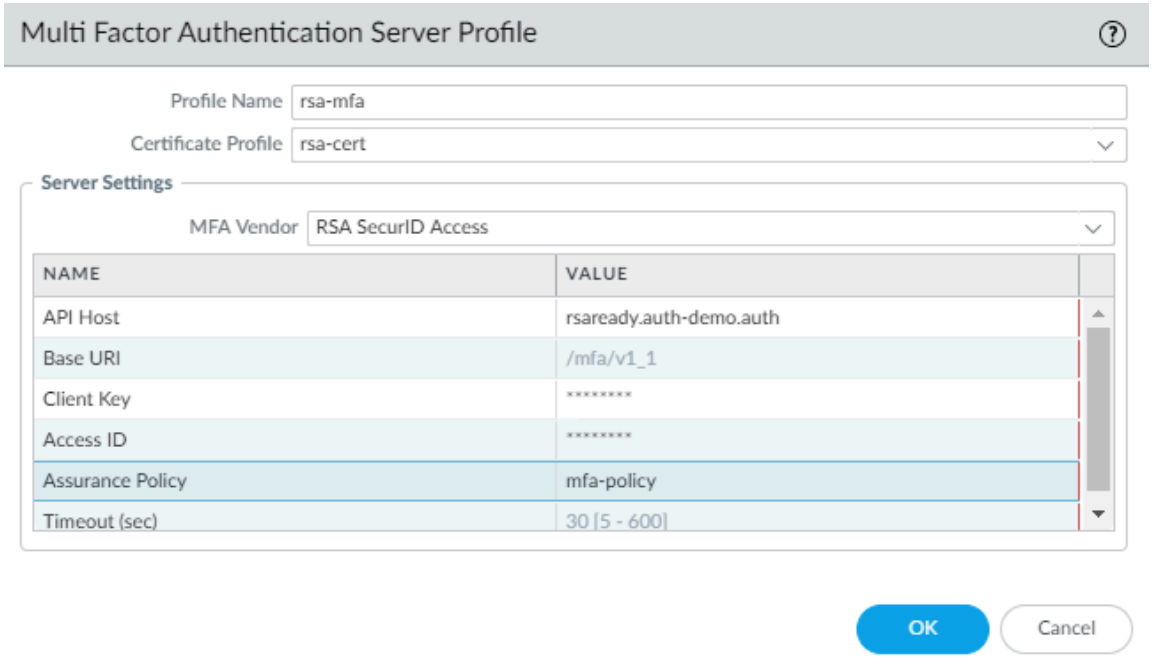


STEP 2 | Configure Authentication Portal (**Device** > **User Identification** > **Authentication Portal Settings**) in Redirect mode to display a web form for authenticating to RSA SecureID. Make sure to specify the Redirect Host as an IP address or a hostname (with no periods in its name) that resolves to the IP address of the Layer 3 interface on the firewall to which web requests are redirected.



STEP 3 | Configure a multi-factor authentication server profile to specify how the firewall must connect with the RSA SecurID cloud service **(Device** > **Server Profiles** > **Multi Factor Authentication** and click **Add**).

1. Enter a **Name** to identify the MFA server profile.
2. Select the **Certificate Profile** that you created earlier, rsa-cert-profile in this example. The firewall will use this certificate when establishing a secure connection with RSA SecurID cloud service.
3. In the **MFA Vendor** drop-down, select **RSA SecurID Access**.
4. Configure the **Value** for each attribute that you noted in Get the RSA SecurID Access Cloud Authentication Service Details:
   - **API Host**—Enter the hostname or IP address of the RSA SecurID Access API endpoint to which the firewall must connect, rsaready.auth-demo.auth in this example.
   - **Base URI** —Do not modify the default value (/mfa/v1_1)
   - **Client Key**—Enter the RSA SecurID Client Key.
   - **Access ID**—Enter the RSA SecurID Access ID.
   - **Assurance Policy**—Enter the RSA SecurID Access Policy name, mfa-policy in this example.
   - **Timeout**—The default is 30 seconds.



5. Save the profile.

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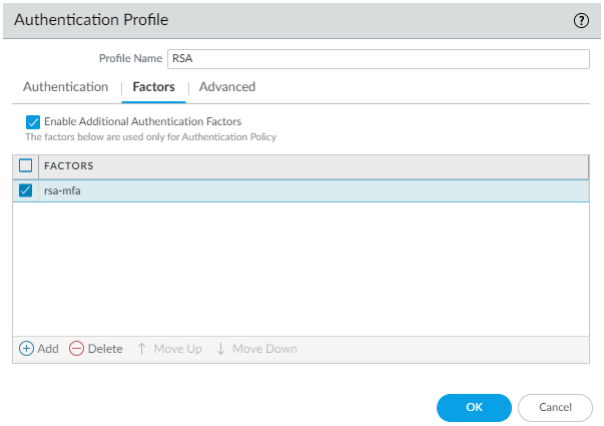
STEP 4 | Configure an authentication profile (**Device** > **Authentication Profile** and click **Add**).

The profile defines the order of the authentication factors that users must respond to.

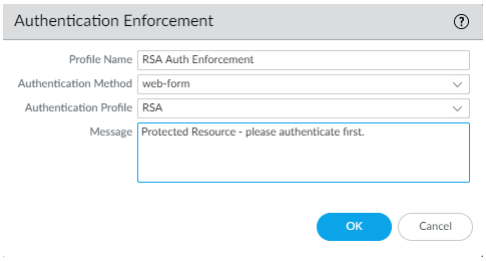
1. Select the **Type** for the first authentication factor and select the corresponding **Server Profile**.
2. Select **Factors**, **Enable Additional Authentication Factors**, and **Add** the rsa-mfa server profile you created earlier in this example.



3. Click **OK** to save the authentication profile.

STEP 5 | Configure an authentication enforcement object. (**Objects** > **Authentication** and click **Add**).

Make sure to select the authentication profile you just defined called RSA in this example.



STEP 6 | Configure an Authentication policy rule. (**Policies** > **Authentication** and click **Add**)

Your authentication policy rule must match the services and applications you want to protect, specify the users who must authenticate, and include the authentication enforcement object that triggers the authentication profile. In this example, RSA SecurID Access authenticates all users who accessing HTTP, HTTPS, SSH, and VNC traffic with the authentication enforcement

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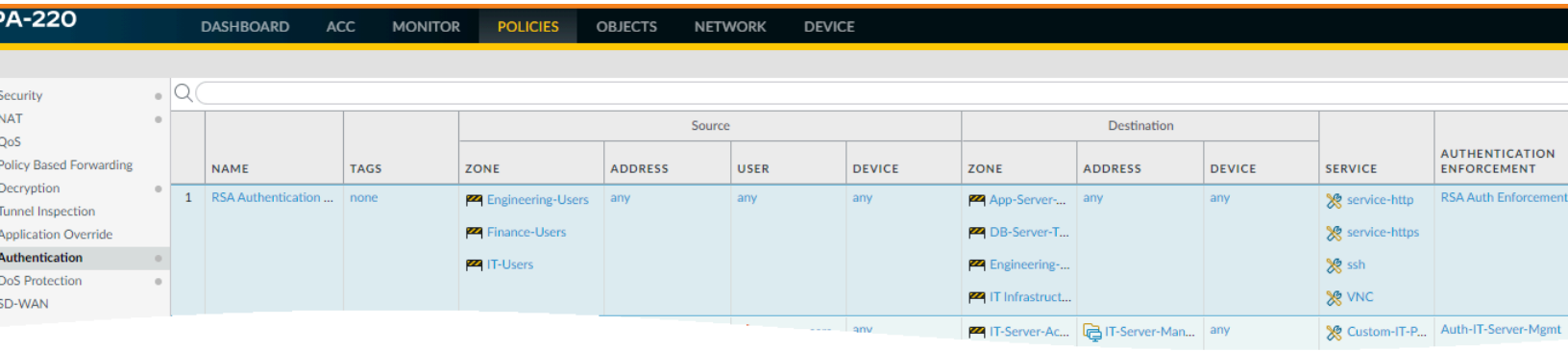
object called RSA Auth Enforcement (in **Actions**, select the **Authentication Enforcement** object).



STEP 7 | **Commit** your changes on the firewall.

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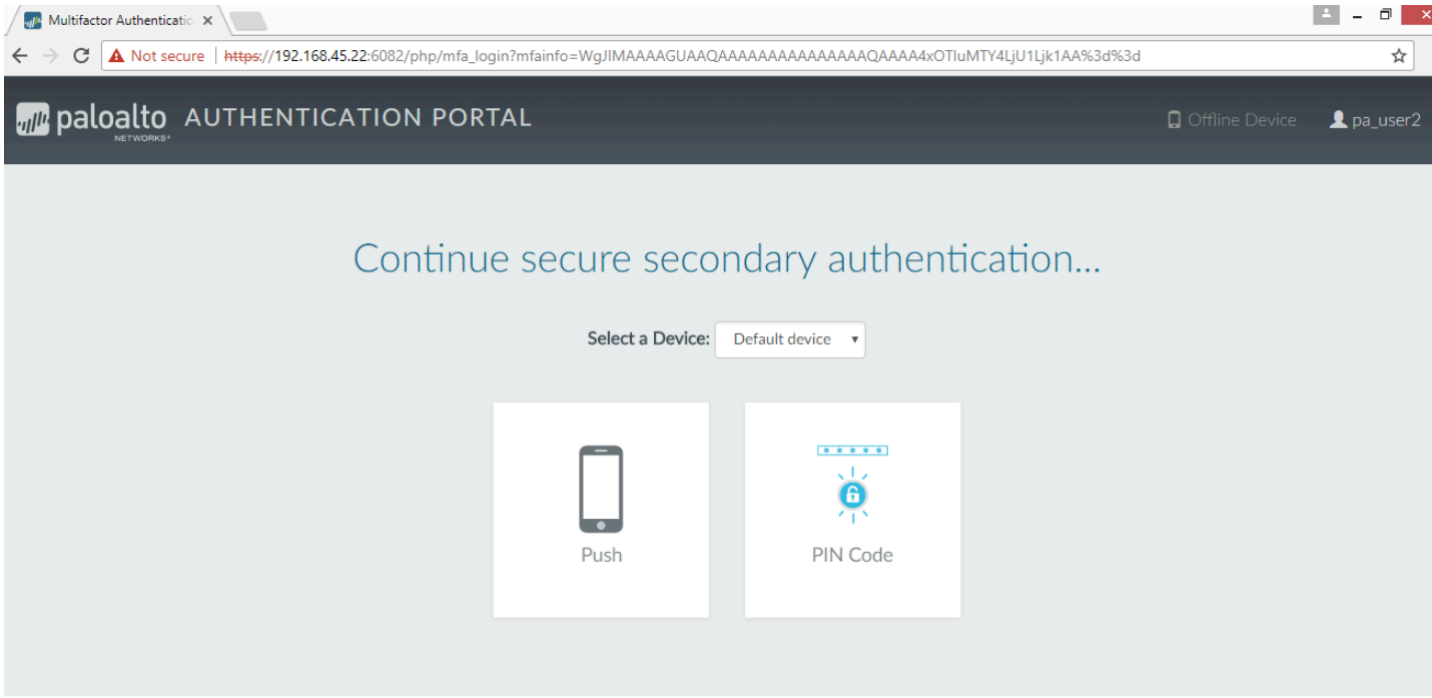
STEP 8 | Verify that users on your network are being secured using RSA SecurID using the Push or PIN Code authentication method you enabled.

1. **Push authentication**

   1. Ask a user on your network to launch a web browser and access a website. The Authentication Portal page with the IP address or hostname for the Redirect Host you defined earlier should display.
   2. Verify that the user enters the credentials for the first authentication factor and then continues to the secondary authentication factor, and selects **Push**.



   3. Check for a **Sign-In request** on the RSA SecurID Access application on the user's mobile device.
   4. Ask the user to **Accept** the Sign-In Request on the mobile device, and wait for a few seconds for the firewall to receive the notification of successful authentication. The user should be able to access the requested website.

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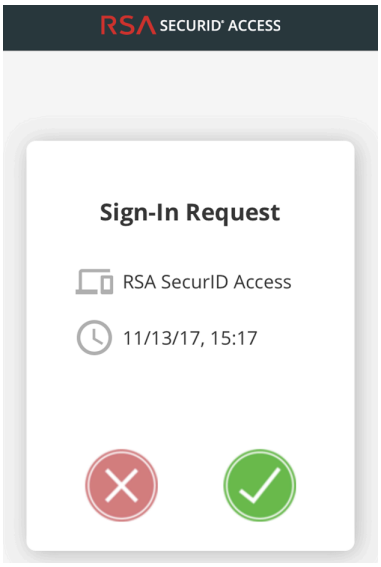
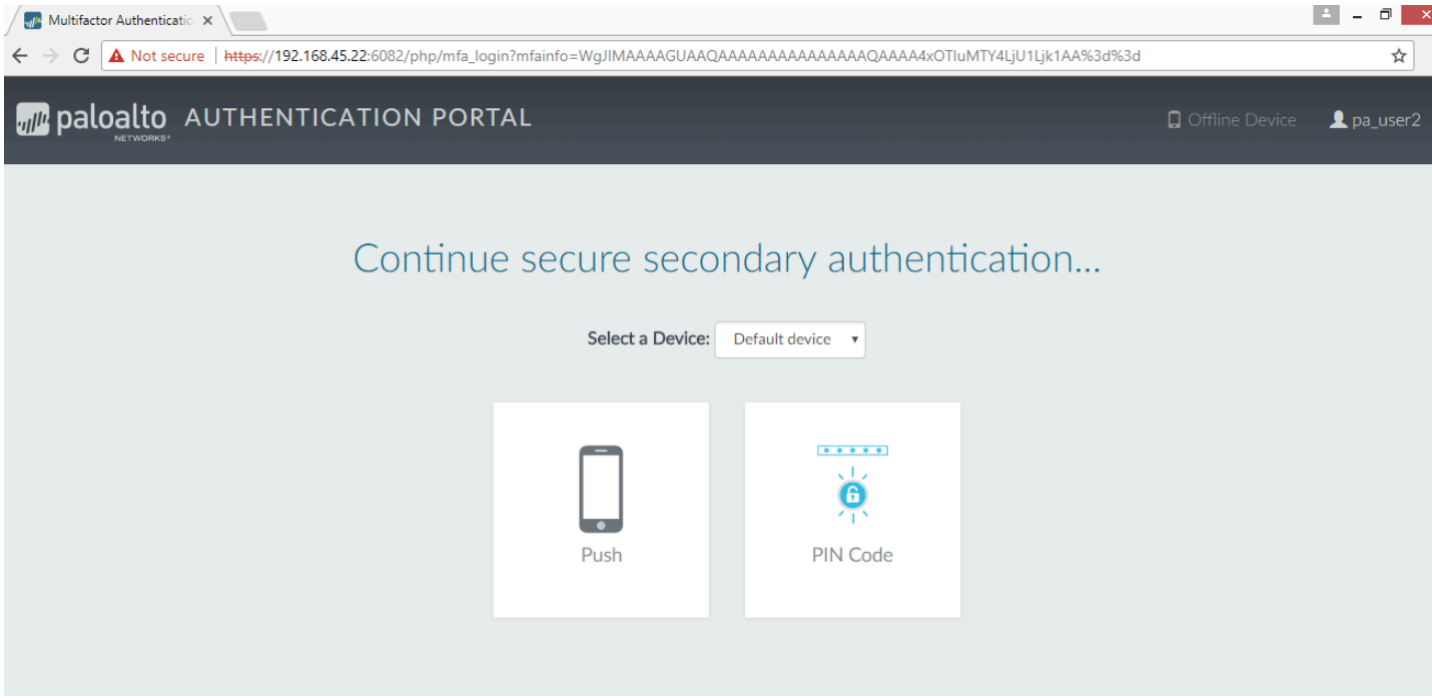


*To test an authentication failure, **Decline** the sign-in request on the mobile device.*

2. **PIN Code authentication**

   1. Ask a user on your network to launch a web browser and access a website. The Authentication Portal page with the IP address or hostname for the redirect host you defined earlier should display.
   2. Verify that the user enters the credentials for the first authentication factor and then continues to the secondary authentication factor, and selects **PIN Code**.



   3. Check that a **PIN Code** displays on the RSA SecurID Access application on the user's mobile device.

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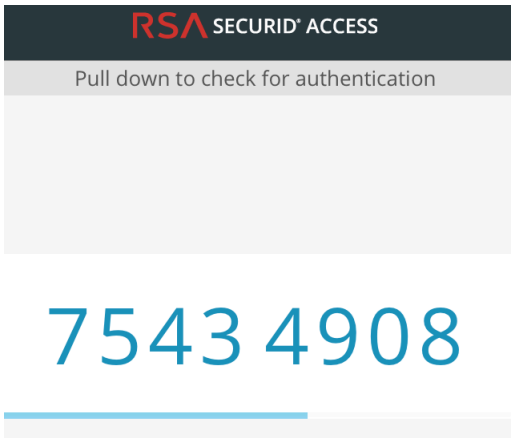

4. Ask the user to copy the PIN code in the **Enter the PIN...** prompt of the web browser and click **Submit**. Wait for a few seconds for the firewall to receive the notification of successful authentication. The user should be able to access the requested website.

Configure MFA Between Okta and the Firewall

Multi-factor authentication allows you to protect company assets by using multiple factors to verify the identity of users before allowing them to access network resources.

To enable multi-factor authentication (MFA) between the firewall and the Okta identity management service:

- Configure Okta
- Configure the firewall to integrate with Okta
- Verify MFA with Okta

Configure Okta

Log in to the Okta Admin Portal to create your user accounts, define your Okta MFA policy, and obtain the token information required to configure MFA with Okta on the firewall.

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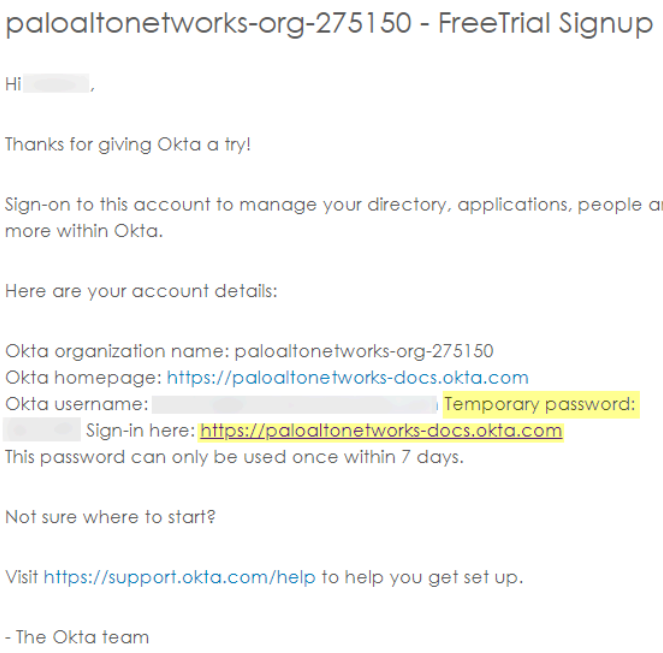
STEP 1 | Create your Okta Admin user account.

1. Submit your email address and name, then click **Get Started**.
2. Click the link in the confirmation email and use the included temporary password to log in to the Okta Admin Portal.



paloaltonetworks-org-275150 - FreeTrial Signup

Hi [redacted],

Thanks for giving Okta a try!

Sign-on to this account to manage your directory, applications, people and more within Okta.

Here are your account details:

Okta organization name: paloaltonetworks-org-275150
Okta homepage: https://paloaltonetworks-docs.okta.com
Okta username: [redacted] Temporary password: [redacted]
Sign-in here: https://paloaltonetworks-docs.okta.com
This password can only be used once within 7 days.

Not sure where to start?

Visit https://support.okta.com/help to help you get set up.

- The Okta team

3. Create a new password that includes at least 8 characters, one lowercase letter, one uppercase letter, a number, and does not include any part of your username.
4. Select a password reminder question and enter the answer.
5. Select a security image, then **Create My Account**.

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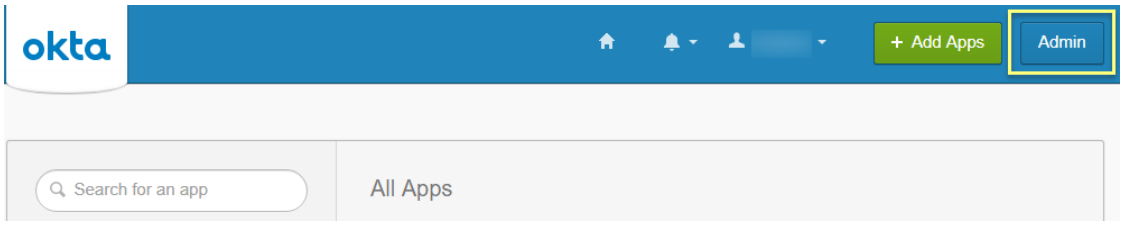
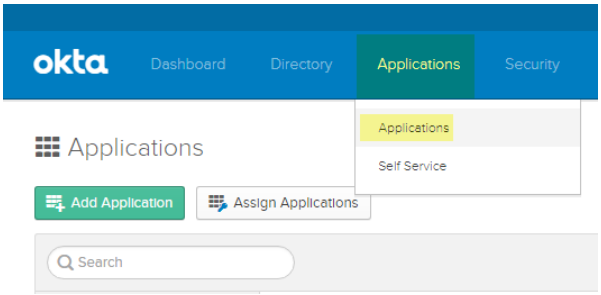
STEP 2 | Configure your Okta service.

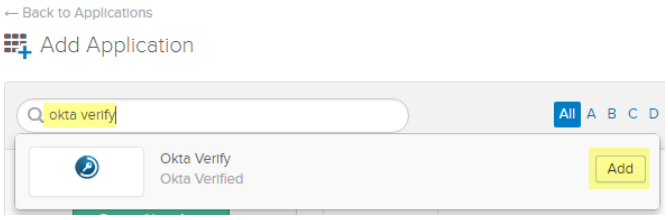
*If you log in and are not redirected to the Okta Admin Portal, select **Admin** at the upper right.*



1. From the Okta Dashboard, log in with your Okta Admin credentials, then select **Applications** > **Applications**.



2. Select **Add Application**.
3. Search for **`Okta Verify`**.
4. Select **Add**, then **Done**.



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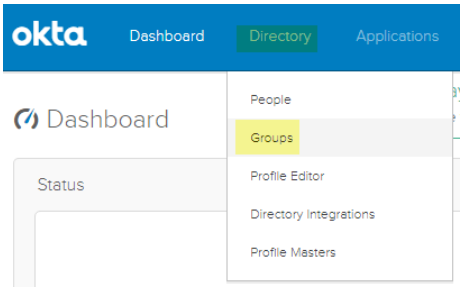
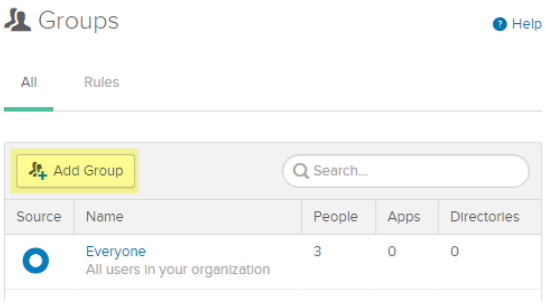
STEP 3 | Create one or more user groups to categorize your users (for example, by device, by policy, or by department) and assign the Okta Verify application.

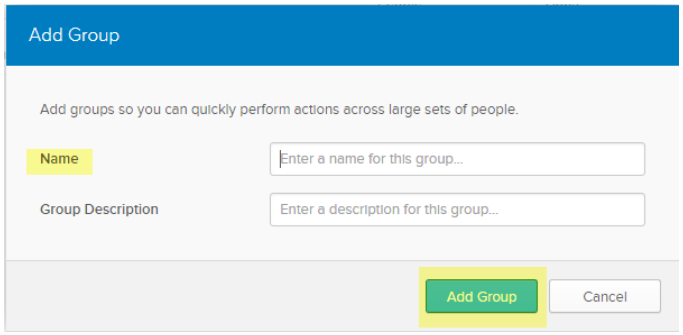
1. Select **Directory** > **Groups**.



2. Click **Add Group**.

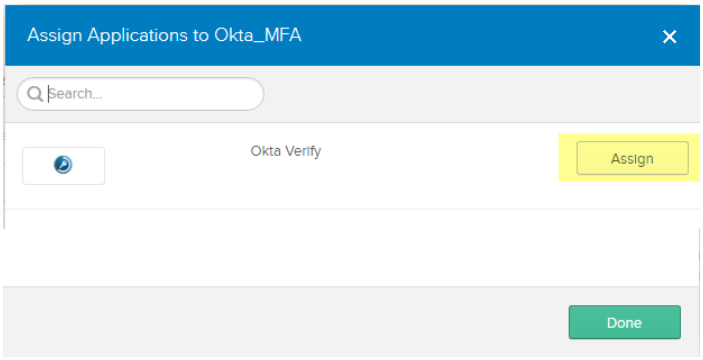


3. Enter a group **Name** and optionally a **Group Description**, then **Add Group**.



*The default group **Everyone** includes all users configured for your organization during the first step in Configure Okta.*

4. Select the group you created, then select **Manage Apps**.
5. **Assign** the Okta Verify application you added in Step 2.



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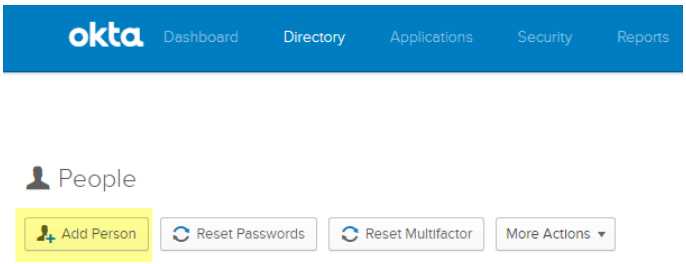
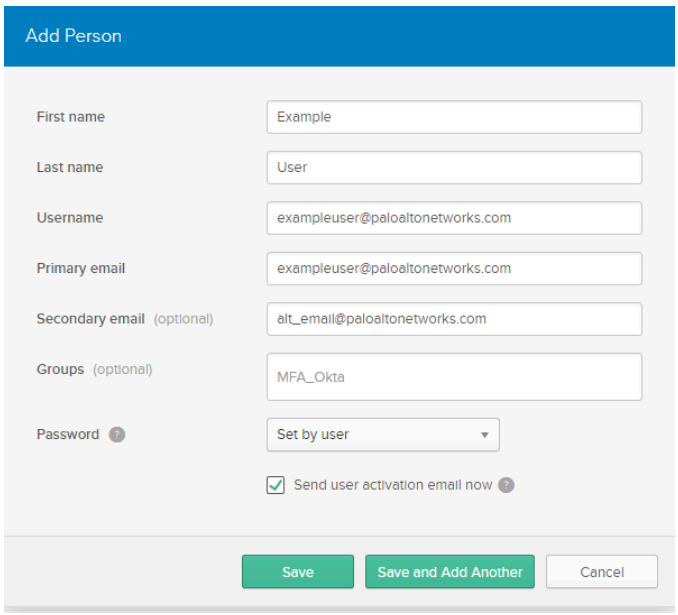
6. After the application has been **Assigned**, click **Done**.
7. Repeat this process for all groups that will use the Okta Verify application for MFA.

STEP 4 | Add users and assign them to a group.

1. From the Okta Dashboard, select **Directory** > **People** > **Add Person**.



2. Enter the user's **First Name**, **Last Name**, and **Username**. The username must match the **Primary email**, which populates automatically, and the username entered on the firewall. You can optionally enter an alternate email address for the user as the **Secondary Email**.



3. Enter the name of the group or **Groups** to associate with this user. When you start typing, the group name populates automatically.
4. Check **Send user activation email now**, then **Save** to add a single user or **Save and Add Another** to continue adding users.

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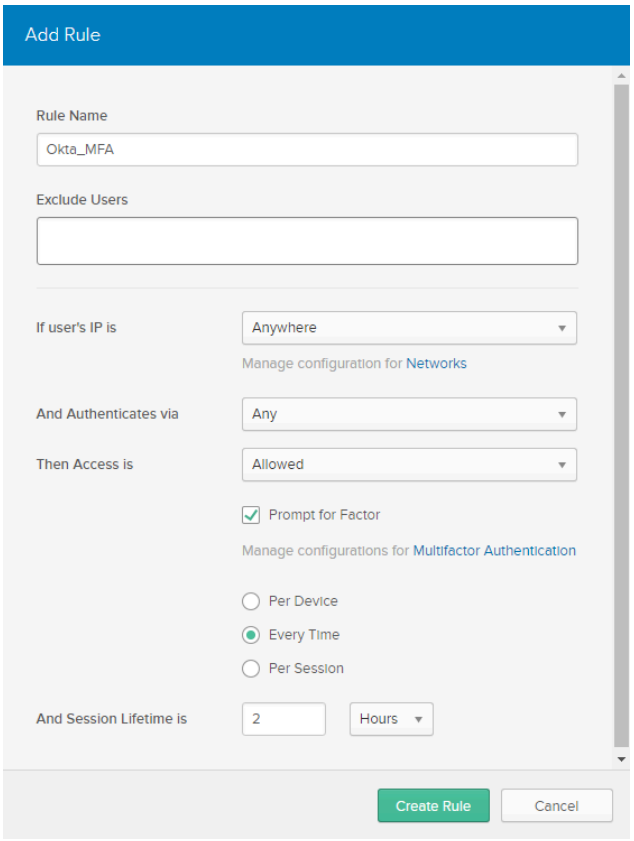
STEP 5 | Assign a test policy to users.

1. Select **Security** > **Authentication** > **Sign On**.

   There is a **Default Policy** with a **Default Rule** that does not prompt users to log in with MFA.

2. Enter the **Rule Name** and check **Prompt for Factor** to enforce the MFA prompt, and select the type of prompt (**Per Device**, **Every Time**, or **Per Session**), then **Create Rule**.



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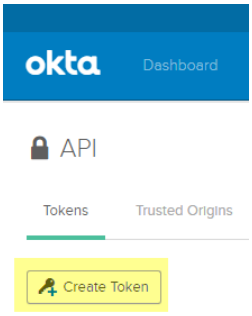
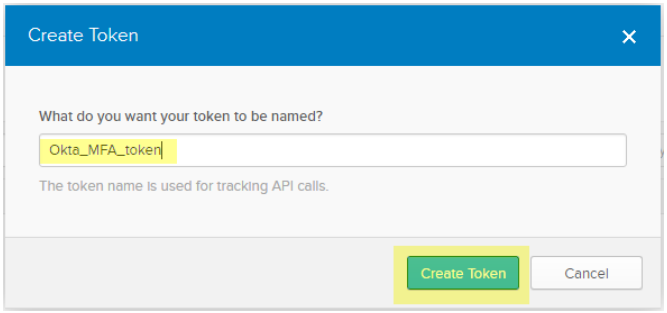
STEP 6 | Record the Okta authentication token information in a safe place because it is only displayed once.

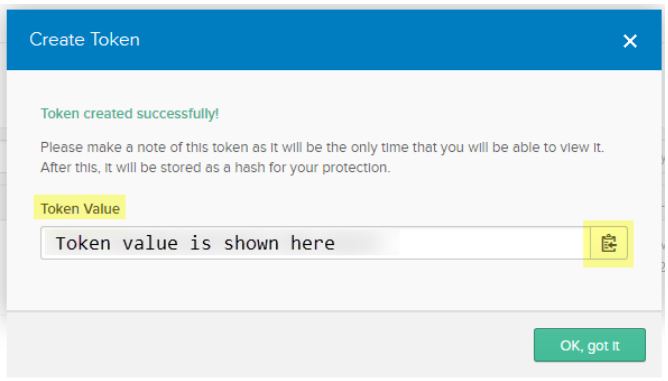
1. Select **Security** > **API** > **Tokens**.
2. Select **Create Token**.



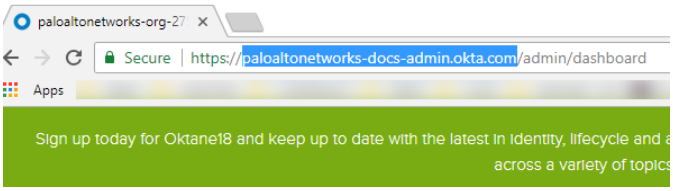
3. Enter a name for the token, then **Create Token**.



4. Copy the **Token Value**.
   You can click the **Copy to clipboard** button to copy the Token Value to your clipboard.



5. In the URL for the Okta Admin Dashboard, copy the portion of the URL after **`https://`** up to **`/admin`** to use as the **API host**.



6. Omit the domain **`okta.com`** from this URL to use as the **Organization**.

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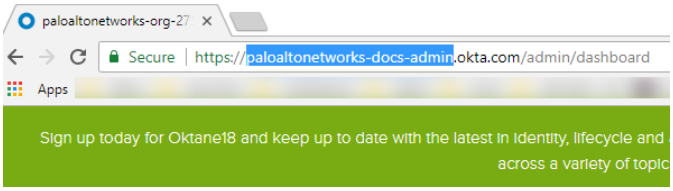


For example, in the example Okta Admin Dashboard URL above, **`https://paloaltonetworks-doc-admin.okta.com/admin/dashboard`**:

- The API hostname is **`paloaltonetworks-doc-admin.okta.com`**.
- The Organization is **`paloaltonetworks-doc-admin`**.

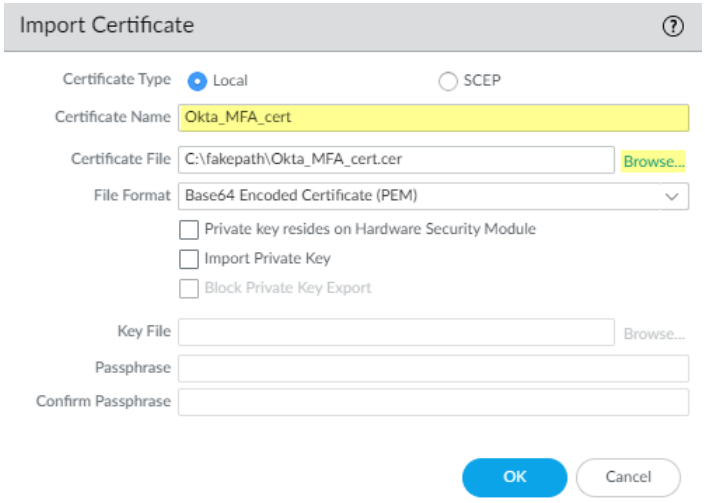
STEP 7 | Export all certificates in the certificate chain using Base-64 encoding:

1. Depending on your browser, use one of the following methods to export all certificates in the chain.
   - **Chrome**—Press **F12**, then select **Security** > **View Certificate** > **Details** > **Copy to File**.
   - **Firefox**—Select **Options** > **Privacy & Security** > **View Certificates** > **Export**.
   - **Internet Explorer**—Select **Settings** > **Internet Options** > **Content** > **Certificates** > **Export**.
2. Use the Certificate Export Wizard to export all certificates in the chain and select **Base-64 encoded X.509** as the format.

Configure the firewall to integrate with Okta

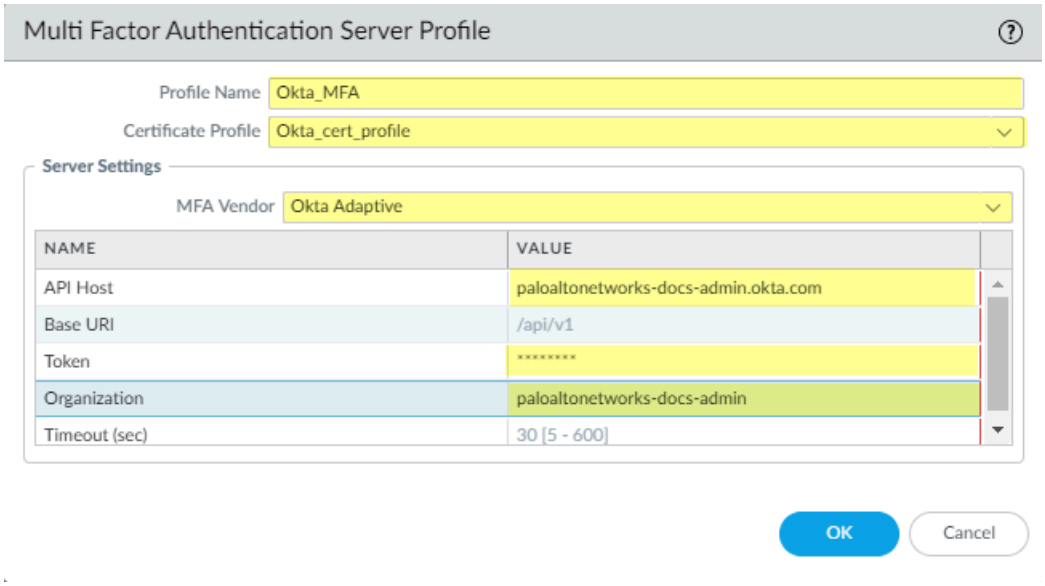
As a prerequisite, confirm that you have mapped all users that you want to authenticate using Okta.

STEP 1 | Import all certificates in the certificate chain on the firewall and add the imported CA certificates (root and intermediate) to a Certificate Profile.



STEP 2 | Add a **Multi Factor Authentication Server Profile** for Okta.

1. Select **Device** > **Server Profiles** > **Multi Factor Authentication**.
2. **Add** an MFA server profile.



3. Enter a **Profile Name**.
4. Select the **Certificate Profile** you created in Step 1 in Configure the firewall to integrate with Okta.
5. Select **Okta Adaptive** as the **MFA Vendor**.
6. Enter the **API Host**, **Token**, and **Organization** from Step 4 in Configure the firewall to integrate with Okta.

STEP 3 | Configure Authentication Portal using **Redirect Mode** to redirect users to the MFA vendor's challenge.

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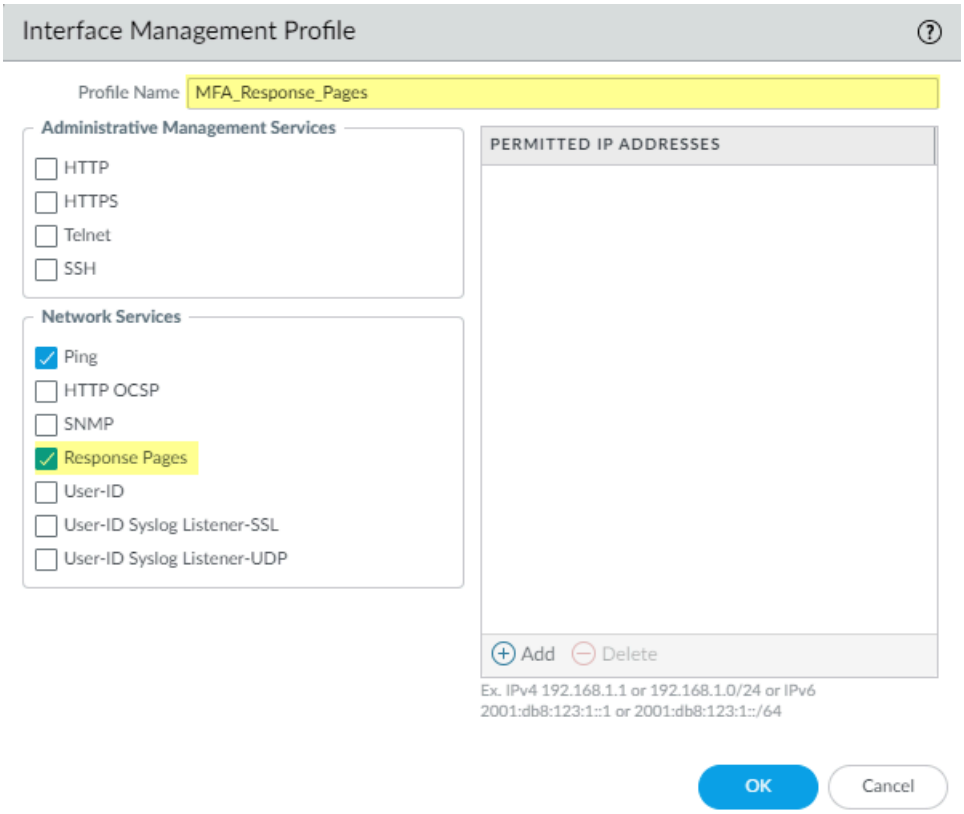
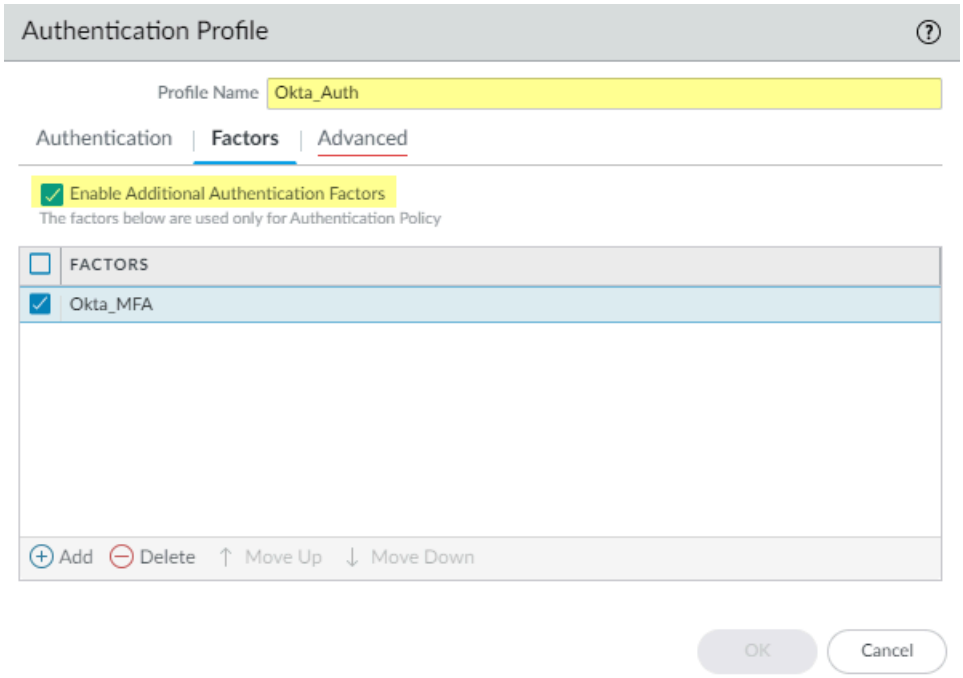
STEP 4 | Enable response pages on the Interface Management Profile to redirect users to the response page challenge.



STEP 5 | Create an Authentication Profile and add the MFA vendor as a **Factor** (see Configure Multi-Factor Authentication, Step 3.)



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STEP 6 | Enable User-ID on the source zone to require identified users to respond to the challenge using your MFA vendor.

STEP 7 | Create an Authentication Enforcement Object to use the MFA vendor and create an Authentication policy rule (see Configure Authentication Policy, Steps 4 and 5).

STEP 8 | **Commit** your changes.

Verify MFA with Okta

STEP 1 | Verify your users received their enrollment emails, have activated their accounts, and have downloaded the Okta Verify app on their devices.

STEP 2 | Go to a website that will prompt the response page challenge.

> *If you are using a self-signed certificate instead of a PKI-assigned certificate from your organization, a security warning displays that users must click through to access the challenge.*

STEP 3 | Log in to the response page using your Okta credentials.

STEP 4 | Confirm the device receives the challenge push notification.

STEP 5 | Confirm users can successfully access the page after authenticating the challenge by accepting the push notification on their devices.

Configure MFA Between Duo and the Firewall

Multi-factor authentication (MFA) allows you to protect company assets by using multiple factors to verify the identity of users before allowing them to access network resources. There are multiple ways to use the Duo identity management service to authenticate with the firewall:

- Two-factor authentication for VPN logins using the GlobalProtect Gateway and a RADIUS server profile (supported on PAN-OS 7.0 and later).
- API-based integration using Authentication Portal and an MFA server profile (does not require a Duo Authentication Proxy or SAML IdP - supported on PAN-OS 8.0 and later).
- SAML integration for on-premise servers (supported on PAN-OS 8.0 and later).

To enable SAML MFA between the firewall and Duo to secure administrative access to the firewall:

- Configure Duo for SAML MFA with Duo Access Gateway
- Configure the Firewall to Integrate with Duo
- Verify MFA with Duo

Configure Duo for SAML MFA with Duo Access Gateway

Before you begin, verify that you have deployed the DuoAccessGateway (DAG) on an on-premise server in your DMZ zone.

Create your Duo administrator account and configure the Duo Access Gateway to authenticate your users before they can access resources.

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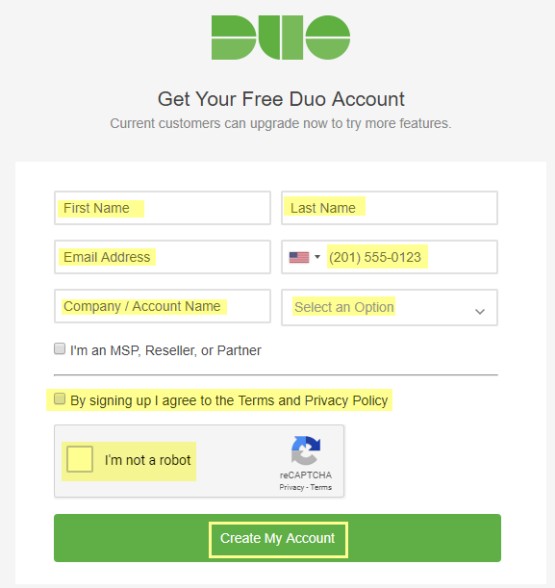
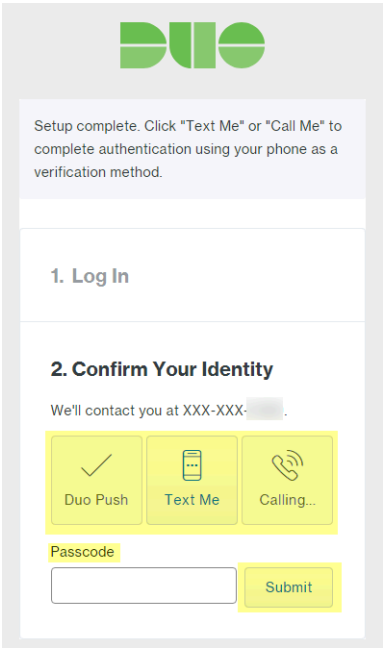
STEP 1 | Create your Duo administrator account.

1. On the Duo account creation page, enter your **First Name**, **Last Name**, **Email Address**, **Cell Phone Number**, **Company / Account Name**, and select the number of employees in the organization.
2. Agree to the Terms and Privacy Policy and respond to the reCAPTCHA challenge to **Create My Account**.



STEP 2 | Verify your Duo administrator account.

1. Select the authentication verification method (**Duo Push**, **Text Me**, or **Calling...**).
2. Enter the **Passcode** you receive and **Submit** it to verify your account.



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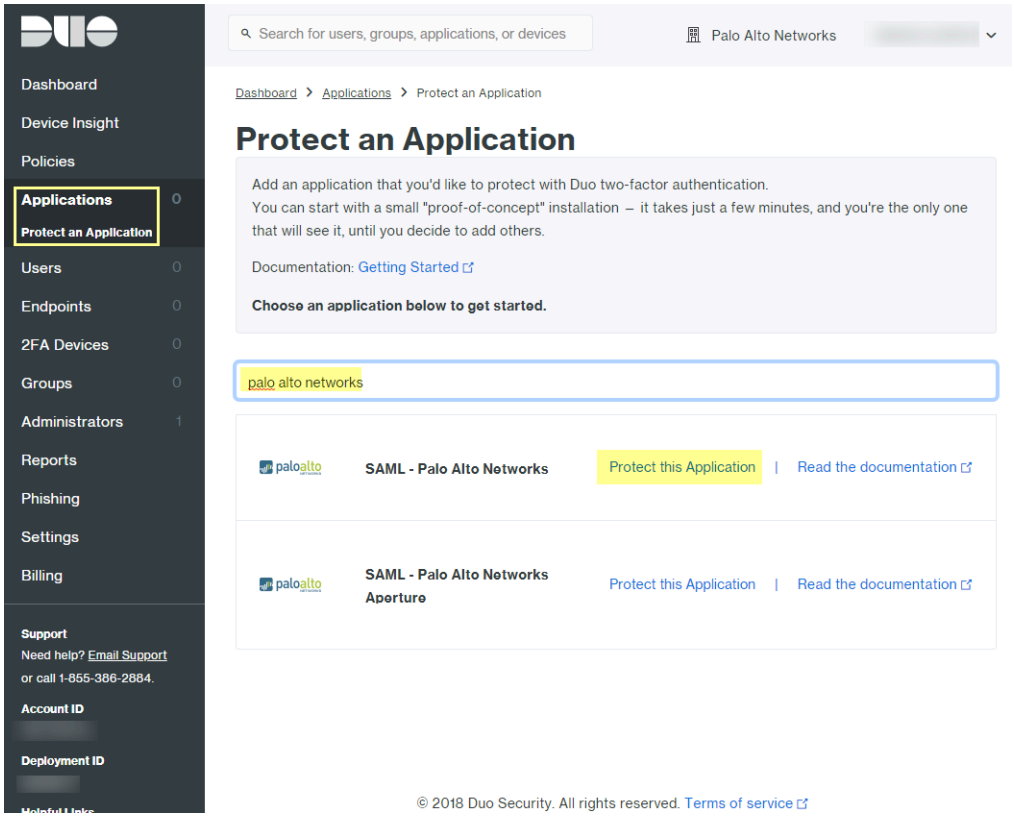
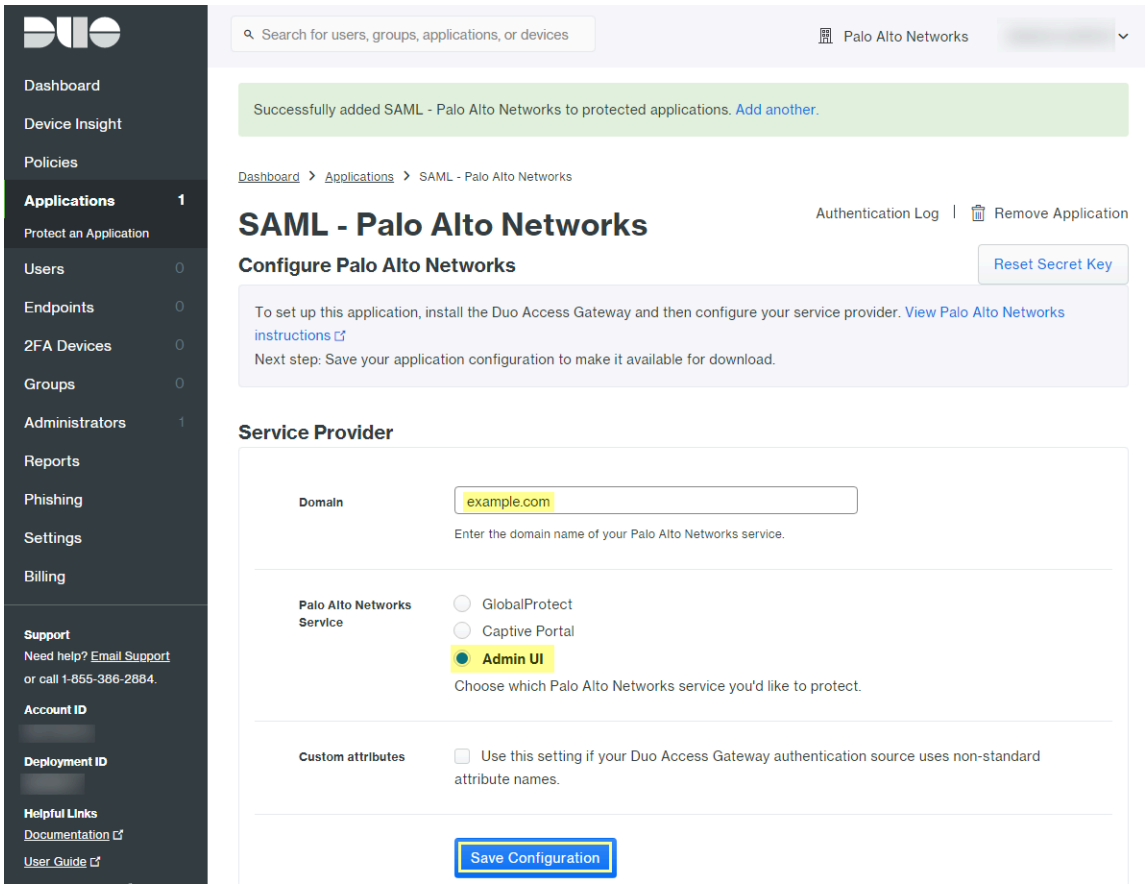
STEP 3 | Configure your Duo service for SAML.

After creating your configuration, download the configuration file at the top of the page.

1. In the Duo Admin Panel, select **Applications** > **Protect an Application**.
2. Enter **`Palo Alto Networks`** to search the applications.
3. Locate **SAML - Palo Alto Networks** in the list of results, then **Protect this Application**.

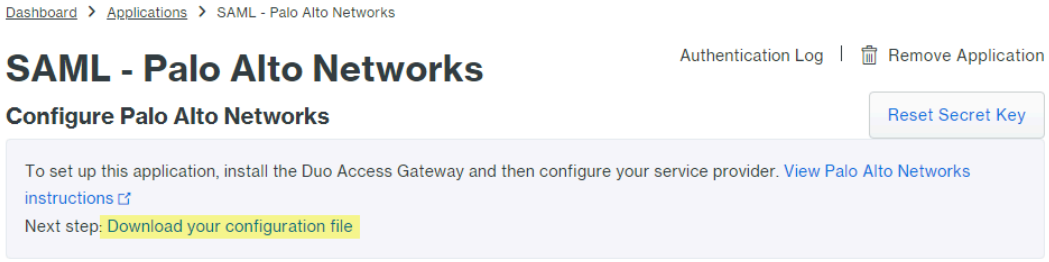


4. Enter the **Domain**.
5. Select **Admin UI** as the **Palo Alto Networks Service**.
6. Configure your **Policy** and other **Settings**, and **Save Configuration**.



7. **Download your configuration file.**

   The link to download the file is at the top of the page.



STEP 4 | Upload the configuration file to the Duo Access Gateway (DAG).

1. In the DAG admin console, select **Applications**.
2. Click **Choose File** and select the configuration file you downloaded, then **Upload** it.
3. In **Settings** > **Session Management**, disable **User agent binding**, then **Save Settings**.

STEP 5 | In the DAG admin console, configure your Active Directory or OpenLDAP server as the authentication source and download the metadata file.

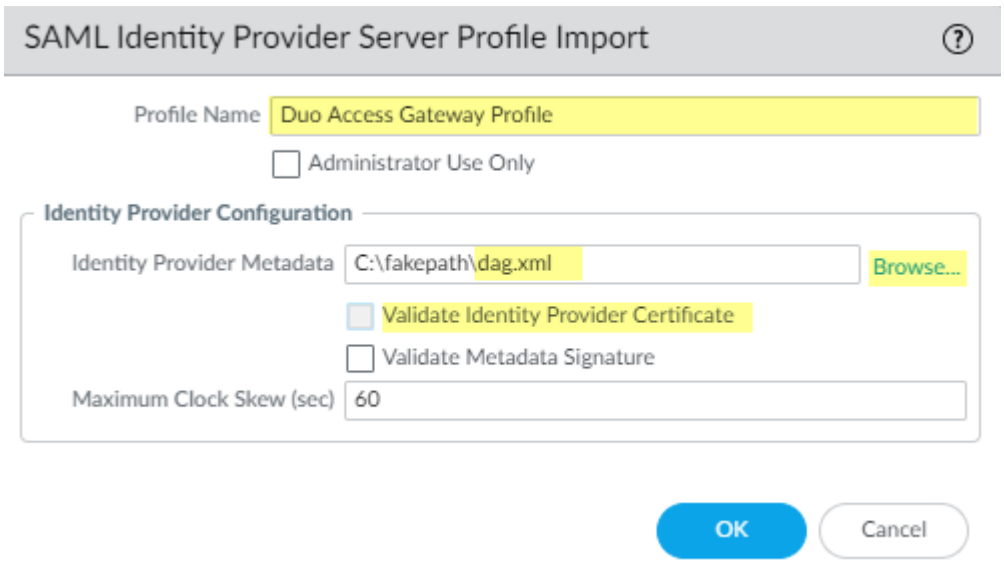
1. Log in to the DAG admin console.
2. In **Authentication Source** > **Set Active Source**, select your **Source type** (Active Directory or OpenLDAP) and **Set Active Source**.
3. In **Configure Sources**, enter the **Attributes**.
   - For Active Directory, enter **`mail,sAMAccountName,userPrincipalName,objectGUID`**.
   - For OpenLDAP, enter **`mail,uid`**.
   - For any custom attributes, append them to the end of the list and separate each attribute with a comma. Do not delete any existing attributes.
4. **Save Settings** to save the configuration.
5. Select **Applications** > **Metadata**, then click **Download XML metadata** to download the XML metadata you will need to import into the firewall.

   The file will be named dag.xml. Because this file includes sensitive information to authenticate your Duo account with the firewall, make sure to keep the file in a secure location to avoid the risk of compromising this information.

Configure the Firewall to Integrate with Duo

STEP 1 | Import the Duo metadata.

1. Log on to the firewall web interface.
2. On the firewall, select **Device** > **Server Profiles** > **SAML Identity Provider** > **Import**.
3. Enter the **Profile Name**.
4. **Browse** to the **Identity Provider Metadata** file (**dag.xml**).
5. If the Duo Access Gateway provides a self-signed certificate as the signing certificate for the IdP, you cannot **Validate Identity Provider Certificate**. In this case, ensure that you are using PAN-OS 11.1 to mitigate exposure to CVE-2020-2021.



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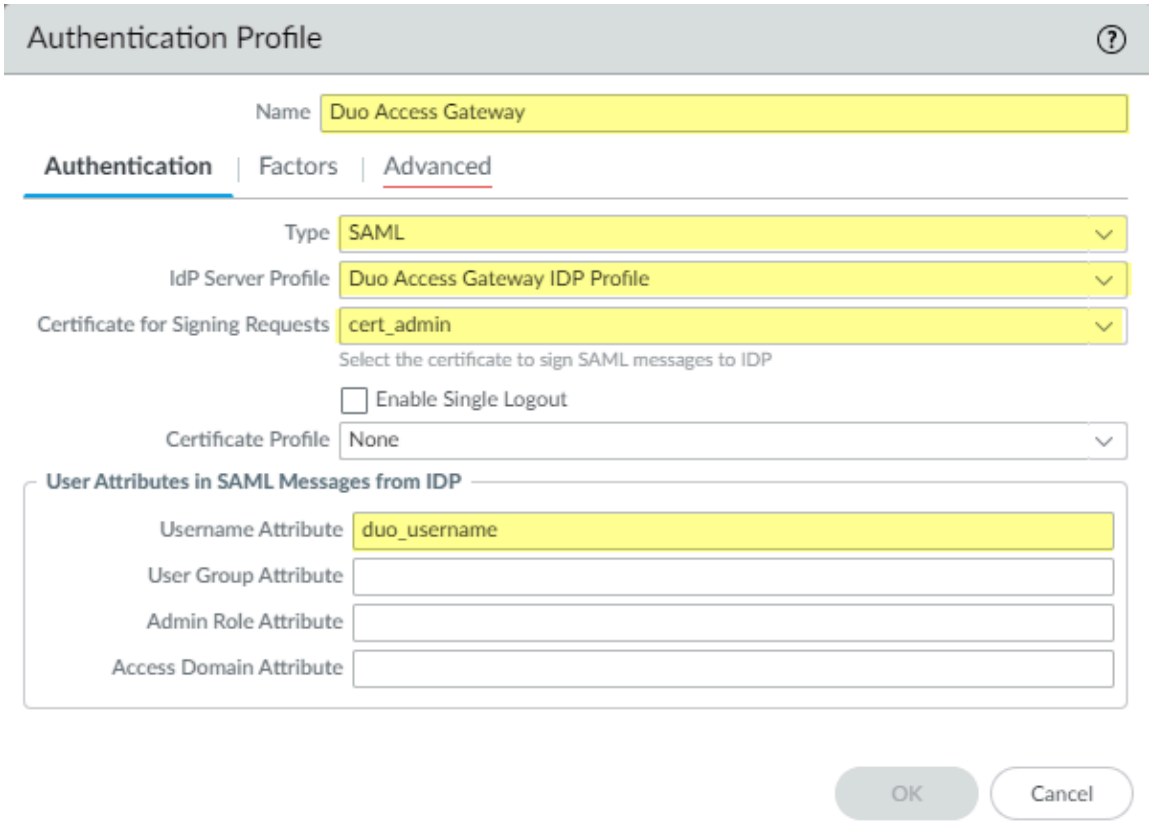
STEP 2 | Add an authentication profile.

The authentication profile allows Duo as the identity provider that validates administrator login credentials.

1. **Add** an **Authentication Profile**.
2. Enter the profile **Name**.
3. Select **SAML** as the authentication **Type**.
4. Select **Duo Access Gateway Profile** as the **IdP Server Profile**.
5. Select the certificate you want to use for SAML communication with the Duo Access Gateway for the **Certificate for Signing Requests**.
6. Enter **`duo_username`** as the **Username Attribute**.



7. Select **Advanced** to **Add** an allow list.
8. Select **all**, then click **OK**.
9. **Commit** the changes.

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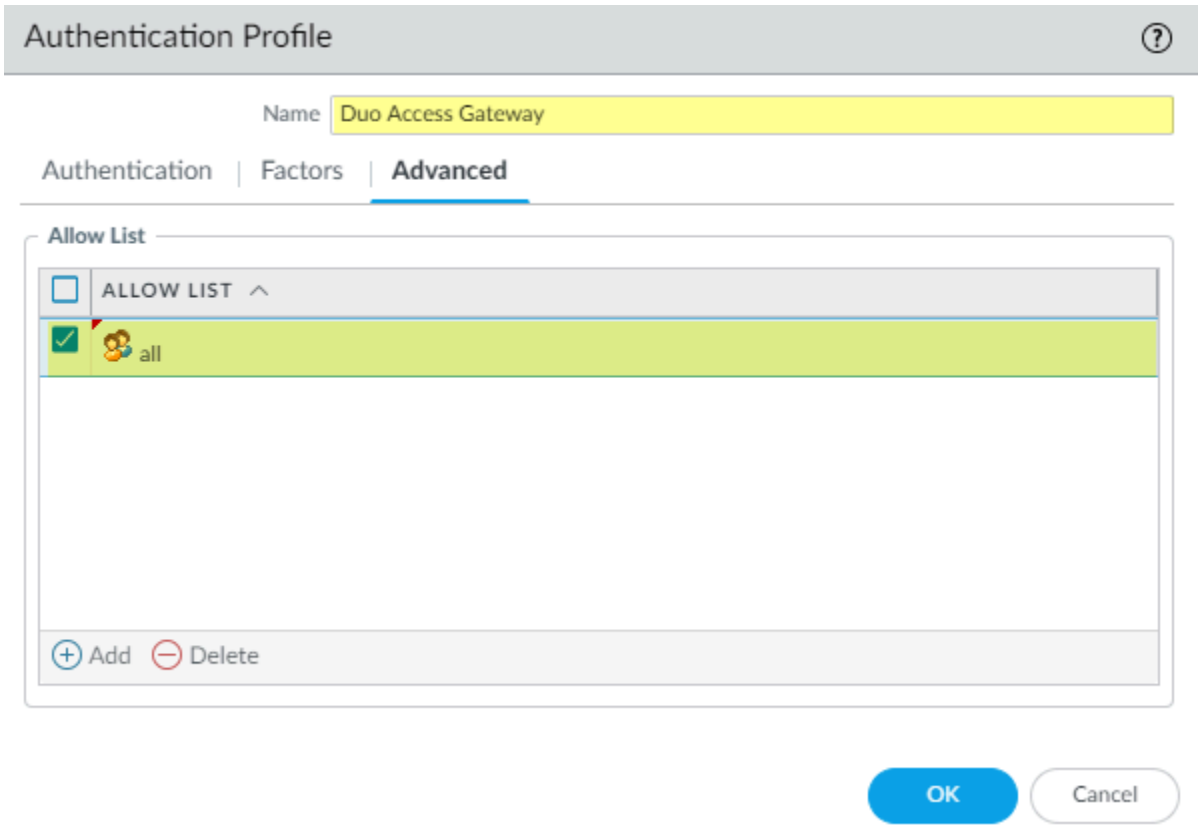

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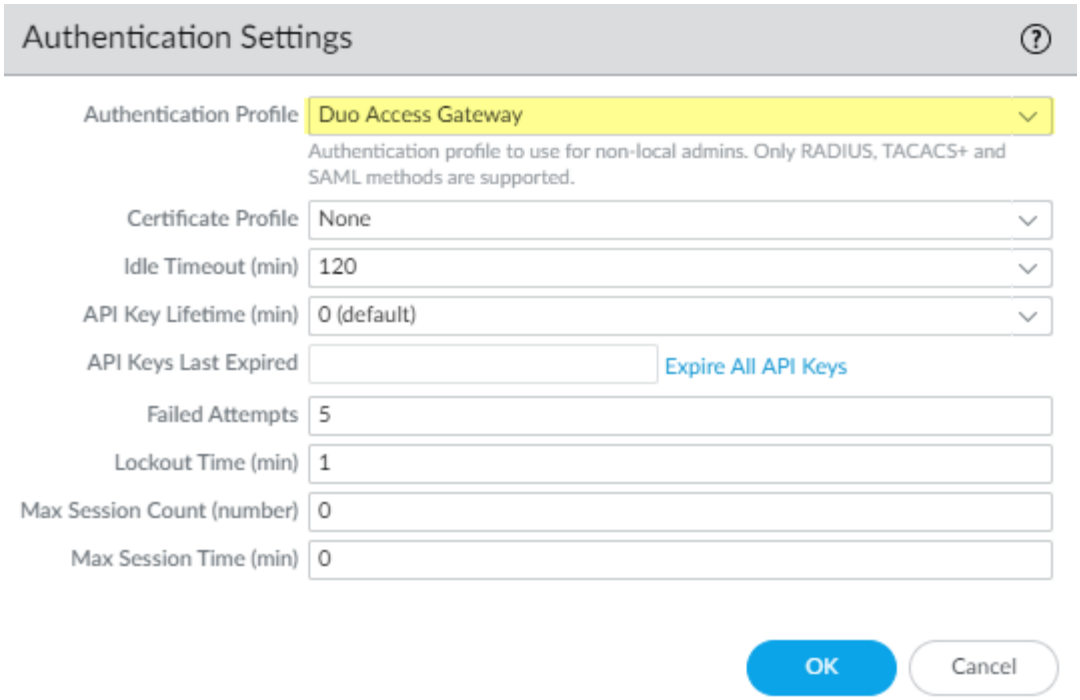
STEP 3 | Specify the authentication settings that the firewall uses for SAML authentication with Duo.

1. Select **Device** > **Setup** > **Management** and edit the **Authentication Settings**.
2. Select **Duo Access Gateway** as the **Authentication Profile**, then click **OK**.



3. **Commit** your changes.

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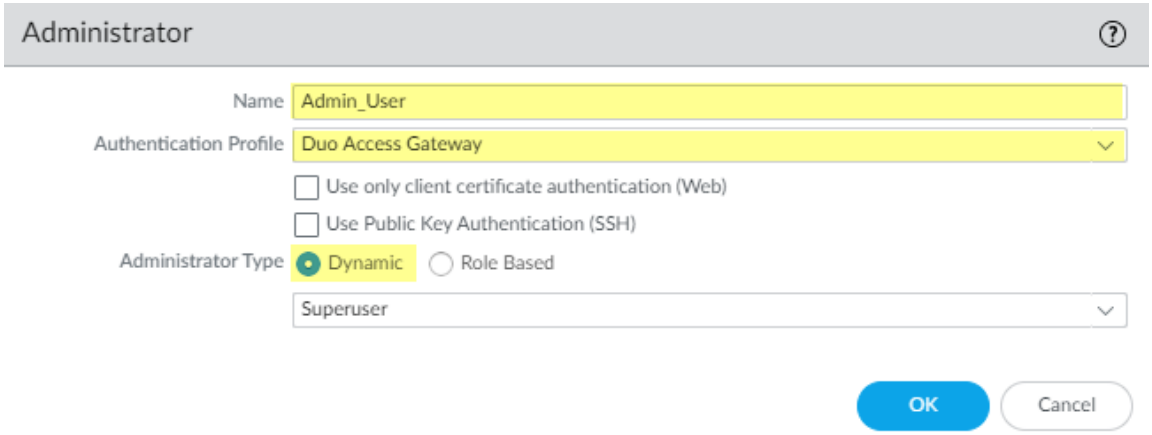
STEP 4 | Add accounts for administrators who will authenticate to the firewall using Duo.

1. Select **Device** > **Administrators** and **Add** an account.
2. Enter a user **Name**.
3. Select **Duo Access Gateway** as the **Authentication Profile**.
4. Select the **Administrator Type**, then click **OK**.

   Select **Role Based** if you want to use a custom role for the user. Otherwise, select **Dynamic**. To require administrators to log in using SSO with Duo, assign the authentication profile to all current administrators.



Verify MFA with Duo

STEP 1 | Log in to the web interface on the firewall.

STEP 2 | Select **Use Single Sign-On** and **Continue**.

STEP 3 | Enter your login credentials on the Duo Access Gateway login page.

STEP 4 | Select an authentication method (push notification, phone call, or passcode entry).

When you authenticate successfully, you will be redirected to the firewall web interface.

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Pre-Logon for SAML Authentication

Where Can I Use This?	What Do I Need?
• GlobalProtect Subscription	❑ GlobalProtect App 5.0 with PAN-OS 8.0 and later releases ❑ macOS 10.9 and later releases ❑ Windows 7 and 10 ❑ GlobalProtect endpoints running on Windows and macOS

Pre-logon is a connect method that establishes a VPN tunnel before a user logs in. The GlobalProtect app for Windows and Mac endpoints now supports pre-logon followed by SAML authentication for user login. When an endpoint boots up and Internet is readily available, GlobalProtect establishes a pre-logon tunnel using the machine certificate on the endpoint. After the pre-logon tunnel is established, the user can log in to the endpoint and authenticate to GlobalProtect using the configured SAML identity provider (IDP). If SAML authentication is successful on Windows endpoints, the pre-logon tunnel is seamlessly renamed to User tunnel, and the GlobalProtect connection is established. If SAML authentication is successful on Mac endpoints, a new tunnel is created, and the GlobalProtect connection is established.

A pre-logon VPN tunnel has no username association because the user has not logged in. In order to grant access to resources, it is necessary to establish security policies that are compatible with the pre-logon user. These policies should permit access to essential services required for system startup, such as DHCP, DNS, specific Active Directory services, antivirus, and operating system update services. After the user authenticates to the gateway, the GlobalProtect app reassigns the VPN tunnel to the authenticated user, resulting in a change in the IP address mapping on the firewall from the pre-logon endpoint to the authenticated user.

Use the following steps to configure the GlobalProtect app to use pre-logon followed by SAML authentication for user login:

1. Remote Access VPN with Pre-Logon
2. Set Up SAML Authentication

After you configure pre-logon for SAML authentication, you're ready to Configure SAML Authentication.

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Configure SAML Authentication

To configure SAML single sign-on (SSO) and single logout (SLO), you must register the firewall and the IdP with each other to enable communication between them. If the IdP provides a metadata file containing registration information, you can import it onto the firewall to register the IdP and to create an IdP server profile. The server profile defines how to connect to the IdP and specifies the certificate that the IdP uses to sign SAML messages. You can also use a certificate for the firewall to sign SAML messages. Using certificates is a requirement to secure communications between the firewall and the IdP.

Palo Alto Networks requires HTTPS to ensure the confidentiality of all SAML transactions instead of alternative approaches such as encrypted SAML assertions. To ensure the integrity of all messages processed in a SAML transaction, Palo Alto Networks requires digital certificates to cryptographically sign all messages.

The following procedure describes how to configure SAML authentication for end users and firewall administrators. You can also configure SAML authentication for Panorama administrators.

SSO is available to administrators and to GlobalProtect and Authentication Portal end users. SLO is available to administrators and GlobalProtect end users, but not to Authentication Portal end users.

Administrators can use SAML to authenticate to the firewall web interface, but not to the CLI.

STEP 1 | Obtain the certificates that the IdP and firewall will use to sign SAML messages.

If the certificates don't specify key usage attributes, all usages are allowed by default, including signing messages. In this case, you can Obtain Certificates by any method.

If the certificates do specify key usage attributes, one of the attributes must be Digital Signature, which is not available on certificates that you generate on the firewall or Panorama. In this case, you must import the certificates:

- **Certificate the firewall uses to sign SAML messages**—Import the certificate from your enterprise certificate authority (CA) or a third-party CA.
- **Certificate the IdP uses to sign SAML messages** (Required for all deployments)—Import a metadata file containing the certificate from the IdP (see the next step). The IdP certificate is limited to the following algorithms:

  Public key algorithms—RSA (1,024 bits or larger) and ECDSA (all sizes). A firewall in FIPS/CC mode supports RSA (2,048 bits or larger) and ECDSA (all sizes).

  Signature algorithms—SHA1, SHA256, SHA384, and SHA512. A firewall in FIPS/CC mode supports SHA256, SHA384, and SHA512.

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STEP 2 | Add a SAML IdP server profile.

The server profile registers the IdP with the firewall and defines how they connect.

In this example, you import a SAML metadata file from the IdP so that the firewall can automatically create a server profile and populate the connection, registration, and IdP certificate information.

*If the IdP doesn't provide a metadata file, select **Device** > **Server Profiles** > **SAML Identity Provider**, **Add** the server profile, and manually enter the information (consult your IdP administrator for the values).*

1. Export the SAML metadata file from the IdP to a client system from which you can upload the metadata to the firewall.

   The certificate specified in the file must meet the requirements listed in the preceding step. Refer to your IdP documentation for instructions on exporting the file.
2. Select **Device** > **Server Profiles** > **SAML Identity Provider** or **Panorama** > **Server Profiles** > **SAML Identity Provider** on Panorama™ and **Import** the metadata file onto the firewall.
3. Enter a **Profile Name** to identify the server profile.
4. **Browse** to the **Identity Provider Metadata** file.
5. Select **Validate Identity Provider Certificate** (default) to validate the chain of trust and optionally the revocation status of the IdP certificate.

   To enable this option, a Certificate Authority (CA) must issue your IdP's signing certificate. You must create a Certificate Profile that has the CA that issued the IdP's signing certificate. In the Authentication Profile, select the SAML Server profile and Certificate Profile to validate the IdP certificate.

   If your IdP signing certificate is a self-signed certificate, there is no chain of trust; as a result, you cannot enable this option. The firewall always validates the signature of the SAML Responses or Assertions against the Identity Provider certificate that you configure whether or not you enable the **Validate Identity Provider Certificate** option. If your IdP provides a self-signed certificate, ensure that you are using PAN-OS 11.1 to mitigate exposure to CVE-2020-2021.

   Validate the certificate to ensure it hasn't been compromised and to improve security.
6. Enter the **Maximum Clock Skew**, which is the allowed difference in seconds between the system times of the IdP and the firewall at the moment when the firewall validates IdP messages (default is 60; range is 1 to 900). If the difference exceeds this value, authentication fails.
7. Click **OK** to save the server profile.
8. Click the server profile Name to display the profile settings. Verify that the imported information is correct and edit it if necessary.
9. Whether you import the IdP metadata or manually enter the IdP information, always ensure that the signing certificate of your SAML identity provider is the **Identity Provider Certificate** for your server profile and your IdP sends signed SAML Responses, Assertions, or both.

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STEP 3 | Configure an authentication profile.

The profile defines authentication settings that are common to a set of users.

1. Select **Device** > **Authentication Profile** and **Add** a profile.
2. Enter a **Name** to identify the profile.
3. Set the **Type** to **SAML**.
4. Select the **IdP Server Profile** you configured.
5. Select the **Certificate for Signing Requests**.

   The firewall uses this certificate to sign messages it sends to the IdP. You can import a certificate generated by your enterprise CA or you can generate a certificate using the root CA that was generated on the firewall or Panorama.
6. (Optional) **Enable Single Logout** (disabled by default).
7. Select the **Certificate Profile** that the firewall will use to validate the **Identity Provider Certificate**.
8. Enter the **Username Attribute** that IdP messages use to identify users (default **`username`**).

   *When you predefine dynamic administrator roles for users, use lower-case to specify the role (for example, enter **`superreader`**, not **`SuperReader`**). If you manage administrator authorization in the IdP identity store, specify the **Admin Role Attribute** and **Access Domain Attribute** also.*
9. Select **Advanced** and **Add** the users and user groups that are allowed to authenticate with this authentication profile.
10. Click **OK** to save the authentication profile.

STEP 4 | Assign the authentication profile to firewall applications that require authentication.

1. Assign the authentication profile to:
   - Administrator accounts that you manage locally on the firewall. In this example, Configure a Firewall Administrator Account before you verify the SAML configuration later in this procedure.
   - Administrator accounts that you manage externally in the IdP identity store. Select **Device** > **Setup** > **Management**, edit the Authentication Settings, and select the **Authentication Profile** you configured.
   - Authentication policy rules that secure the services and applications that end users access through Authentication Portal. See Configure Authentication Policy.
   - GlobalProtect portals and gateways that end users access.
2. **Commit** your changes.

   The firewall validates the **Identity Provider Certificate** that you assigned to the SAML IdP server profile.

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STEP 5 | Create a SAML metadata file to register the firewall application (management access, Authentication Portal, or GlobalProtect) on the IdP.

1. Select **Device** > **Authentication Profile** and, in the Authentication column for the authentication profile you configured, click **Metadata**.
2. In the **Service** drop-down, select the application you want to register:
   - **management** (default)—Administrative access to the web interface.
   - **authentication-portal**—End user access to services and applications through Authentication Portal.
   - **global-protect**—End user access to services and applications through GlobalProtect.
3. (Authentication Portal or GlobalProtect only) for the **Vsysname Combo**, select the virtual system in which the Authentication Portal settings or GlobalProtect portal are defined.
4. Enter the interface, IP address, or hostname based on the application you will register:
   - **management**—For the **Management Choice**, select **Interface** (default) and select an interface that is enabled for management access to the web interface. The default selection is the IP address of the MGT interface.
   - **authentication-portal**—For the **IP Hostname**, enter the IP address or hostname of the **Redirect Host** (see **Device** > **User Identification** > **Authentication Portal Settings**).
   - **global-protect**—For the **IP Hostname**, enter the hostname or IP address of the GlobalProtect portal or gateway.
5. Click **OK** and save the metadata file to your client system.
6. Import the metadata file into the IdP server to register the firewall application. Refer to your IdP documentation for instructions.

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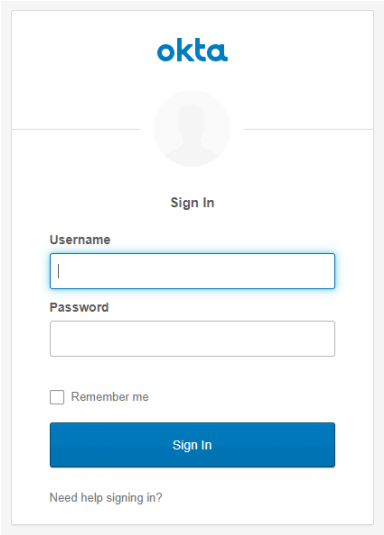
STEP 6 | Verify that users can authenticate using SAML SSO.

For example, to verify that SAML is working for access to the web interface using a local administrator account:

1. Go to the URL of the firewall web interface.
2. Click **Use Single Sign-On**.
3. Enter the username of the administrator.
4. Click **Continue**.

   The firewall redirects you to authenticate to the IdP, which displays a login page. For example:



5. Log in using your SSO username and password.

   After you successfully authenticate on the IdP, it redirects you back to the firewall, which displays the web interface.
6. Use your firewall administrator account to request access to another SSO application.

   Successful access indicates SAML SSO authentication succeeded.

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Configure Kerberos Single Sign-On

Palo Alto Networks firewalls and Panorama support Kerberos V5 single sign-on (SSO) to authenticate administrators to the web interface and end users to Authentication Portal. With Kerberos SSO enabled, the user needs to log in only for initial access to your network (such as logging in to Microsoft Windows). After this initial login, the user can access any browser-based service in the network (such as the firewall web interface) without having to log in again until the SSO session expires.

STEP 1 | Create a Kerberos keytab.

The keytab is a file that contains the principal name and password of the firewall, and is required for the SSO process. When you configure Kerberos in your Authentication Profile and Sequence, the firewall first checks for a Kerberos SSO hostname. If you provide a hostname, the firewall searches the keytabs for a service principal name that matches the hostname and uses only that keytab for decryption. If you do not provide a hostname, the firewall tries

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each keytab in the authentication sequence until it is able to successfully authenticate using Kerberos.

If the Kerberos SSO hostname is included in the request sent to the firewall, then the hostname must match the service principal name of the keytab; otherwise, the Kerberos authentication request is not sent.

1. Log in to the Active Directory server and open a command prompt.
2. Enter the following command to register the service principal name (SPN) for GlobalProtect or Authentication Portal, where <*portal_fqdn*> and <*service_account_username*> are variables.

   setspn -s HTTP/<*portal_fqdn*> <*service_account_username*>
3. Create Kerberos account for the firewall. Refer to your Kerberos documentation for the steps.
4. Log in to the KDC and open a command prompt.
5. Enter the following command, where <*portal_fqdn*>, <*kerberos_realm*>, <*netbios_name*>, <*service_account_username*>, <*password*>, <*filename*>, and <*algorithm*> are variables.

   ktpass /princ HTTP <*portal_fqdn*>@<*kerberos_realm*> /mapuser <*netbios_name*>\<*service_account_username*> /pass <*password*>/out <*filename*>.keytab /ptype KRB5_NT_PRINCIPAL /crypto <*algorithm*>

   *The <kerberos_realm> value must be in all uppercase characters (for example, enter **AD1.EXAMPLE.COM**, not **ad1.example.com**).*

   *If the firewall is in FIPS/CC mode, the algorithm must be **aes128-cts-hmac-sha1-96** or **aes256-cts-hmac-sha1-96**. Otherwise, you can also use **des3-cbc-sha1 or arcfour-hmac**. To use an Advanced Encryption Standard (AES) algorithm, the functional level of the KDC must be Windows Server 2012 or later and you must enable AES encryption for the firewall account.*

   The algorithm in the keytab must match the algorithm in the service ticket that the TGS issues to clients. Your Kerberos administrator determines which algorithms the service tickets use.

STEP 2 | Configure an Authentication Profile and Sequence to define Kerberos settings and other authentication options that are common to a set of users.

- Enter the **Kerberos Realm** (usually the DNS domain of the users, except that the realm is uppercase).
- **Import** the **Kerberos Keytab** that you created for the firewall.

STEP 3 | Assign the authentication profile to the firewall application that requires authentication.

- Administrative access to the web interface—Configure a Firewall Administrator Account and assign the authentication profile you configured.
- End user access to services and applications—Assign the authentication profile you configured to an authentication enforcement object. When configuring the object, set the **Authentication Method** to **browser-challenge**. Assign the object to Authentication

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policy rules. For the full procedure to configure authentication for end users, see Configure Authentication Policy.

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Configure Kerberos Server Authentication

You can use Kerberos to natively authenticate end users and firewall or Panorama administrators to an Active Directory domain controller or a Kerberos V5-compliant authentication server. This authentication method is interactive, requiring users to enter usernames and passwords.

To use a Kerberos server for authentication, the server must be accessible over an IPv4 address. IPv6 addresses are not supported.

STEP 1 | Add a Kerberos server profile.

The profile defines how the firewall connects to the Kerberos server.

1. Select **Device** > **Server Profiles** > **Kerberos** or **Panorama** > **Server Profiles** > **Kerberos** on Panorama™ and **Add** a server profile.
2. Enter a **Profile Name** to identify the server profile.
3. **Add** each server and specify a **Name** (to identify the server), IPv4 address or FQDN of the **Kerberos Server**, and optional **Port** number for communication with the server (default 88).

   If you use an FQDN address object to identify the server and you subsequently change the address, you must commit the change in order for the new server address to take effect.
4. Click **OK** to save your changes to the profile.

STEP 2 | Assign the server profile to an Configure an Authentication Profile and Sequence.

The authentication profile defines authentication settings that are common to a set of users.

STEP 3 | Assign the authentication profile to the firewall application that requires authentication.

- Administrative access to the web interface—Configure a Firewall Administrator Account and assign the authentication profile you configured.
- End user access to services and applications—Assign the authentication profile you configured to an authentication enforcement object and assign the object to Authentication policy rules. For the full procedure to configure authentication for end users, see Configure Authentication Policy.

STEP 4 | Verify that the firewall can Test Authentication Server Connectivity to authenticate users.

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Configure TACACS+ Authentication

You can configure TACACS+ authentication for end users as well as firewall or Panorama administrators. You can also use a TACACS+ server to manage administrator authorization (role and access domain assignments) by defining Vendor-Specific Attributes (VSAs). For all users, you must configure a TACACS+ server profile that defines how the firewall or Panorama™ connects to the server. You then assign the server profile to an authentication profile for each set of users who require common authentication settings. What you do with the authentication profile depends on which users the TACACS+ server authenticates:

- **End users**—Assign the authentication profile to an authentication enforcement object and assign the object to Authentication policy rules. For the full procedure, see Configure Authentication Policy.
- **Administrative accounts with authorization managed locally on the firewall or Panorama**—Assign the authentication profile to firewall administrator or Panorama administrator accounts.
- **Administrative accounts with authorization managed on the TACACS+ server**—The following procedure describes how to configure TACACS+ authentication and authorization for firewall administrators. For Panorama administrators, refer to Configure TACACS+ Authentication for Panorama Administrators.

STEP 1 | Add a TACACS+ server profile.

The profile defines how the firewall connects to the TACACS+ server.

1. Select **Device** > **Server Profiles** > **TACACS+** or **Panorama** > **Server Profiles** > **TACACS+** on Panorama and **Add** a profile.
2. Enter a **Profile Name** to identify the server profile.
3. (Optional) Select **Administrator Use Only** to restrict access to administrators.
4. Enter a **Timeout** interval in seconds after which an authentication request times out (default is 3; range is 1–20).
5. Select the **Authentication Protocol** (default is **CHAP**) that the firewall uses to authenticate to the TACACS+ server.

   *Select **CHAP** if the TACACS+ server supports that protocol; it's more secure than **PAP**.*
6. **Add** each TACACS+ server and enter the following:
   - A **Name** to identify the server.
   - The **TACACS+ Server** IP address or FQDN. If you use an FQDN address object to identify the server and you subsequently change the address, you must commit the change for the new server address to take effect.
   - A **Secret** and **Confirm Secret** to encrypt usernames and passwords.
   - The server **Port** for authentication requests (default is 49).
7. Click **OK** to save the server profile.

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STEP 2 | Assign the TACACS+ server profile to an authentication profile.

The authentication profile defines the authentication settings that are common to a set of users.

1. Select **Device** > **Authentication Profile** and **Add** a profile.
2. Enter a **Name** to identify the profile.
3. Set the **Type** to **TACACS+**.
4. Select the **Server Profile** you configured.
5. Select **Retrieve user group from TACACS+** to collect user group information from VSAs defined on the TACACS+ server.

   The firewall matches the group information using the groups you specify in the **Allow List** of the authentication profile.
6. Select **Advanced** and in the **Allow List**, **Add** the users and groups that can authenticate with this authentication profile.
7. Click **OK** to save the authentication profile.

STEP 3 | Configure the firewall to use the authentication profile for all administrators.

1. Select **Device** > **Setup** > **Management** and edit the Authentication Settings.
2. Select the **Authentication Profile** you configured and click **OK**.

STEP 4 | Configure the roles and access domains that define authorization settings for administrators.

If you already defined TACACS+ VSAs on the TACACS+ server, the names you specify for roles and access domains on the firewall must match the VSA values.

1. Configure an Admin Role Profile if the administrator will use a custom role instead of a predefined (dynamic) role.
2. If the firewall has more than one virtual system, configure an access domain by selecting **Device** > **Access Domain**, then **Add** an access domain, enter a **Name** to identify the access domain, and **Add** each virtual system that the administrator will access, and then click **OK**.

STEP 5 | **Commit** your changes to activate them on the firewall.

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STEP 6 | Configure the TACACS+ server to authenticate and authorize administrators.

Refer to your TACACS+ server documentation for the specific instructions to perform these steps:

1. Add the firewall IP address or hostname as the TACACS+ client.
2. Add the administrator accounts.

   *If you selected **CHAP** as the **Authentication Protocol**, you must define accounts with reversibly encrypted passwords. Otherwise, CHAP authentication will fail.*
3. Define TACACS+ VSAs for the role, access domain, and user group of each administrator.

   *When you predefine dynamic administrator roles for users, use lower-case to specify the role (for example, enter **`superuser`**, not **`SuperUser`**).*

STEP 7 | Verify that the TACACS+ server performs authentication and authorization for administrators.

1. Log in the firewall web interface using an administrator account that you added to the TACACS+ server.
2. Verify that you can access only the web interface pages that are allowed for the role you associated with the administrator.
3. In the **Monitor**, **Policies**, and **Objects** tabs, verify that you can access only the virtual systems that are allowed for the access domain you associated with the administrator.

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Configure TACACS Accounting

TACACS+ is designed to use the Authentication, Authorization, and Accounting (AAA) framework for device administration. Authentication confirms the user's identity while authorization determines access to resources. To keep track of the services provided during the user's session, accounting keeps a record of when services were initiated and terminated, as well as any services in progress.

TACACS+ Accounting can be used as an auditing tool. The TACACS+ accounting records contain all information used in authorization records, as well as accounting-specific information such as start and stop times and resource usage information. There are three types of TACACS+ accounting records:

- Start records to indicate a service is about to begin
- Stop records to indicate termination of the service
- Update records to indicate service is still in progress

The firewall uses the TACACS+ accounting client to communicate with your TACACS+ accounting server using the RFC 8907 protocol for connection and session establishment with the accounting server as well as the accounting packets that are sent or received from the server. The TACACS + accounting client uses an accounting server profile that you select and the server profile can contain more than one accounting server.

The client selects the first server in the list where it can successfully establish a connection and receive accounting records. The connection is cached until the client is unable to send accounting records to the connected accounting server. If the client cannot connect to the cached server, it attempts to connect to the next accounting server in the server profile. If the client is unable to connect to any of the servers in the server profile, it logs the failure as an error.

You can configure a TACACS+ server profile for either authentication or accounting but you cannot use the same profile for both authentication and accounting. If you need to use the same TACACS+ server for accounting and authentication, you must configure two profiles containing the same server information, then configure one profile for authentication and the second profile for accounting.

The TACACS+ accounting client supports the following arguments:

task_id	timezone
start_time	event
stop_time	reason
elapsed_time	err_msg

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If you use TACACS+ for Device Administration, all TACACS+ client devices must be configured to send an accounting start packet for each command that is entered regardless of how the commands were authorized. The Command Accounting packet must include the ***service*** *and* ***cmd*** *arguments and if necessary, the* ***cmd-arg*** *arguments described in Section 8.2 of RFC 8907.*

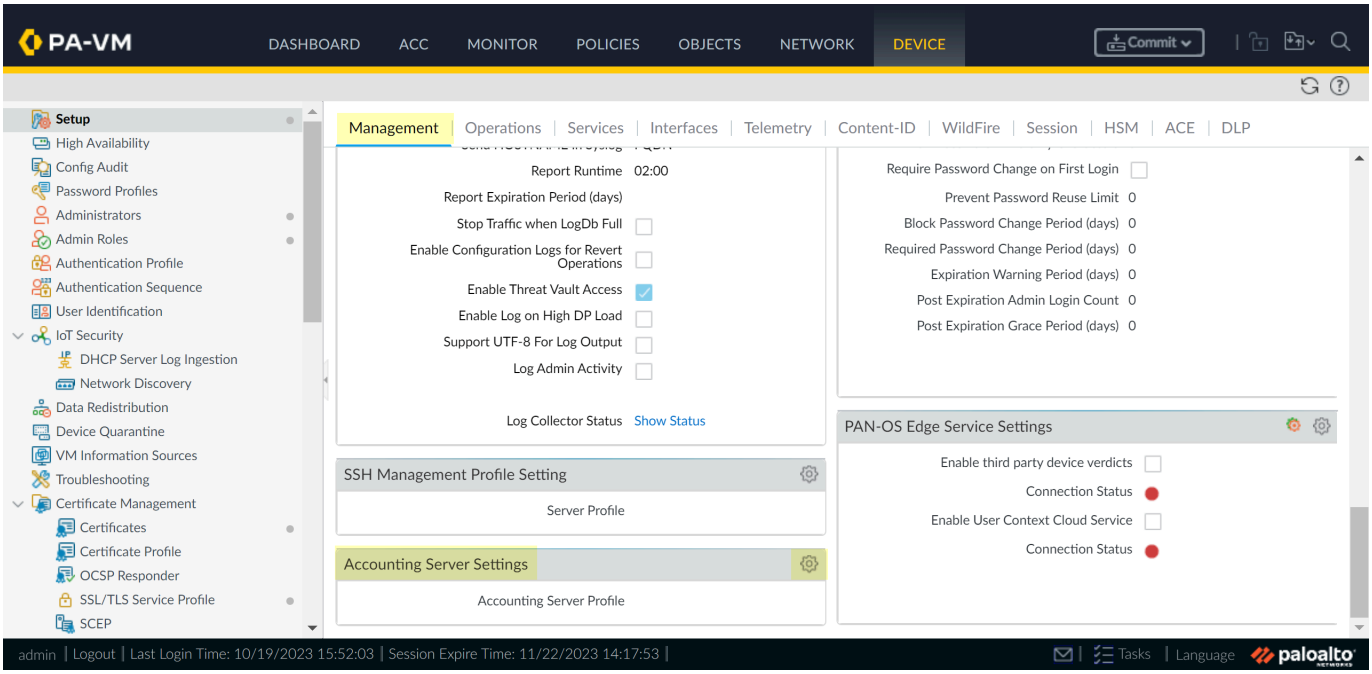
STEP 1 | Create a TACACS+ server profile on the firewall or Panorama for each TACACS+ accounting server that you want to include.

STEP 2 | (Panorama only) Add the TACACS+ server profile to a template or template stack.

You only need to complete this step if you want to push the TACACS+ server profile to one or more firewalls. If you only want to configure accounting for TACACS+ for Panorama, you do not need to complete this step.

STEP 3 | **Edit** the **Accounting Settings** depending on your configuration.

- For firewalls, select **Device** > **Setup** > **Management**.
- For Panorama, select **Panorama** > **Setup** > **Management**.



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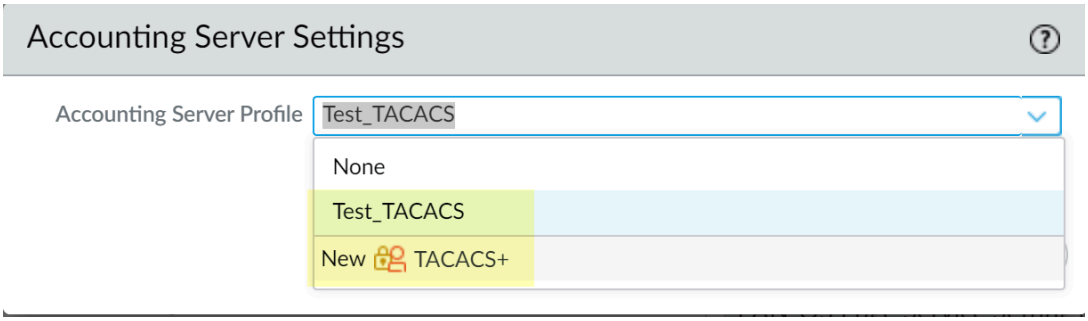
STEP 4 | Select the TACACS+ server profile or create a **New TACACS+** server profile as the **Accounting Server Profile** and click **OK**.

Palo Alto Networks recommends using port number 49 for accounting. This is the default port the firewall uses for accounting.



STEP 5 | **Commit** the changes.

The TACACS+ accounting client attempts to connect to the first server in the TACACS+ server profile when the commit is successful. After it successfully connects, it logs the information in the TACACS+ accounting records.

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Configure RADIUS Authentication

You can configure RADIUS authentication for end users and firewall or Panorama administrators. For administrators, you can use RADIUS to manage authorization (role and access domain assignments) by defining Vendor-Specific Attributes (VSAs). You can also use RADIUS to implement Multi-Factor Authentication (MFA) for administrators and end users. To enable RADIUS authentication, you must configure a RADIUS server profile that defines how the firewall or Panorama connects to the server (see Step 1 below). You then assign the server profile to an authentication profile for each set of users who require common authentication settings (see Step 5 below). What you do with the authentication profile depends on which users the RADIUS server authenticates:

- **End users**—Assign the authentication profile to an authentication enforcement object and assign the object to Authentication policy rules. For the full procedure, see Configure Authentication Policy.

  You can also configure client systems to send RADIUS Vendor-Specific Attributes (VSAs) to the RADIUS server by assigning the authentication profile to a GlobalProtect portal or gateway. RADIUS administrators can then perform administrative tasks based on those VSAs.

- **Administrative accounts with authorization managed locally on the firewall or Panorama**—Assign the authentication profile to firewall administrator or Panorama administrator accounts.
- **Administrative accounts with authorization managed on the RADIUS server**—The following procedure describes how to configure RADIUS authentication and authorization for firewall administrators. For Panorama administrators, refer to Configure RADIUS Authentication for PanoramaAdministrators.

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STEP 1 | Add a RADIUS server profile.

The profile defines how the firewall connects to the RADIUS server.

1. Select **Device** > **Server Profiles** > **RADIUS** or **Panorama** > **Server Profiles** > **RADIUS** on Panorama™and **Add** a profile.
2. Enter a **Profile Name** to identify the server profile.
3. (Optional) Select **Administrator Use Only** to restrict access to administrators.
4. Enter a **Timeout** interval in seconds after which an authentication request times out (default is 3; range is 1–120).

   If you use the server profile to integrate the firewall with an MFA service, enter an interval that gives users enough time to authenticate. For example, if the MFA service prompts for a one-time password (OTP), users need time to see the OTP on their endpoint device and then enter the OTP in the MFA login page.
5. Enter the number of **Retries**.
6. Select the **Authentication Protocol** (default is **PEAP-MSCHAPv2**) that the firewall uses to authenticate to the RADIUS server.

   Depending on which factors you want to use to authenticate users within your multi-factor authentication (MFA) environment, select the appropriate authentication protocol:

   - **Username, password, and push (an automatically triggered out-of-band request)**: Supported with all authentication protocols
   - **Push, password, token, and PIN (when password or token or PIN are provided together)**: Supported with PAP, PEAP with GTC, and EAP-TTLS with PAP
   - **Username, password, token, and PIN, and challenge-response (when password or token or PIN are provided together)**: Supported with PAP and PEAP with GTC

   If you select an EAP authentication method (PEAP-MSCHAPv2, PEAP with GTC, or EAP-TTLS with PAP), confirm that your RADIUS server supports Transport Layer Security (TLS) 1.1 or higher and that the root and intermediate certificate authorities (CAs) for your RADIUS server are included in the certificate profile associated with the RADIUS server profile. If you select an EAP method and you do not associate a correctly configured certificate profile with the RADIUS profile, authentication fails.
7. **Add** each RADIUS server and enter the following:
   - **Name** to identify the server
   - **RADIUS Server** IP address or FQDN. If you use an FQDN to identify the server and you subsequently change the address, you must commit the change for the new server address to take effect.
   - **Secret**/**Confirm Secret** is a key to encrypt passwords and can be up to 64 characters in length.
   - Server **Port** for authentication requests (default is 1812)
8. Click **OK** to save the server profile.

For redundancy, add multiple RADIUS servers in the sequence you want the firewall to use. If you have selected an EAP method, configure an authentication sequence to ensure that users will be able to successfully respond to the authentication challenge. There is no alternate authentication method with EAP: if the user fails the authentication challenge and you have

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not configured an authentication sequence that allows another authentication method, authentication fails.

STEP 2 | If you are using PEAP-MSCHAPv2 with GlobalProtect, select **Allow users to change passwords after expiry** to allow GlobalProtect users to changed expired passwords to log in.

STEP 3 | (PEAP-MSCHAPv2, PEAP with GTC, or EAP-TTLS with PAP only) To anonymize the user's identity in the outer tunnel that is created after authenticating with the server, select **Make Outer Identity Anonymous**.

You must configure the RADIUS server so that the entire chain allows access for anonymous users. Some RADIUS server configurations may not support anonymous outer IDs, and you may need to clear the option. When cleared, the RADIUS server transmits usernames in cleartext.

STEP 4 | If you select an EAP authentication method, select a Certificate Profile.

STEP 5 | Assign the RADIUS server profile to an authentication profile.

The authentication profile defines authentication settings that are common to a set of users.

1. Select **Device** > **Authentication Profile** and **Add** a profile.
2. Enter a **Name** to identify the authentication profile.
3. Set the **Type** to **RADIUS**.
4. Select the **Server Profile** you configured.
5. Select **Retrieve user group from RADIUS** to collect user group information from VSAs defined on the RADIUS server.

   The firewall matches the group information against the groups you specify in the Allow List of the authentication profile.
6. Select **Advanced** and, in the Allow List, **Add** the users and groups that are allowed to authenticate with this authentication profile.
7. Click **OK** to save the authentication profile.

STEP 6 | Configure the firewall to use the authentication profile for all administrators.

1. Select **Device** > **Setup** > **Management** and edit the Authentication Settings.
2. Select the **Authentication Profile** you configured and click **OK**.

STEP 7 | Configure the roles and access domains that define authorization settings for administrators.

If you already defined RADIUS VSAs on the RADIUS server, the names you specify for roles and access domains on the firewall must match the VSA values.

1. Configure an Admin Role Profile if the administrator uses a custom role instead of a predefined (dynamic) role.
2. Configure an access domain if the firewall has more than one virtual system:
   1. Select **Device** > **Access Domain**, **Add** an access domain, and enter a **Name** to identify the access domain.
   2. **Add** each virtual system that the administrator will access, and then click **OK**.

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STEP 8 | **Commit** your changes to activate them on the firewall.

STEP 9 | Configure the RADIUS server to authenticate and authorize administrators.

Refer to your RADIUS server documentation for the specific instructions to perform these steps:

1. Add the firewall IP address or hostname as the RADIUS client.
2. Add the administrator accounts.

   *If the RADIUS server profile specifies **CHAP** as the **Authentication Protocol**, you must define accounts with reversibly encrypted passwords. Otherwise, CHAP authentication will fail.*
3. Define the vendor code for the firewall (25461) and define the RADIUS VSAs for the role, access domain, and user group of each administrator.

   When you predefine dynamic administrator roles for users, use lower-case to specify the role (for example, enter **`superuser`**, not **`SuperUser`**).

   *When configuring the advanced vendor options on the ACS, you must set both the **Vendor Length Field Size** and **Vendor Type Field Size** to **1**. Otherwise, authentication will fail.*
4. If you have selected an EAP method, the firewall validates the server but not the client. To ensure client validity, restrict clients by IP address or subdomain.

STEP 10 | Verify that the RADIUS server performs authentication and authorization for administrators.

1. Log in the firewall web interface using an administrator account that you added to the RADIUS server.
2. Verify that you can access only the web interface pages that are allowed for the role you associated with the administrator.
3. In the **Monitor**, **Policies**, and **Objects** tabs, verify that you can access only the virtual systems that are allowed for the access domain you associated with the administrator.
4. In **Monitor** > **Authentication**, verify the **Authentication Protocol**.
5. Test the connection and the validity of the certificate profile using the following CLI command:

```
admin@PA-220 > test authentication authentication-profile
 auth-profile username <username> password <password>
```

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Configure LDAP Authentication

You can use LDAP to authenticate end users who access applications or services through Authentication Portal and authenticate firewall or Panorama administrators who access the web interface.

You can also connect to an LDAP server to define policy rules based on user groups. For details, see Map Users to Groups.

STEP 1 | Add an LDAP server profile.

The profile defines how the firewall connects to the LDAP server.

1. Select **Device** > **Server Profiles** > **LDAP** or **Panorama** > **Server Profiles** > **LDAP** on Panorama™ and **Add** a server profile.
2. Enter a **Profile Name** to identify the server profile.
3. (Multi-vsys only) Select the **Location** in which the profile is available.
4. (Optional) Select **Administrator Use Only** to restrict access to administrators.
5. **Add** the LDAP servers (up to four). For each server, enter a **Name** (to identify the server), **LDAP Server** IP address or FQDN, and server **Port** (default 389).

   If you use an FQDN address object to identify the server and you subsequently change the address, you must commit the change for the new server address to take effect.
6. Select the server **Type**.
7. Select the **Base DN**.

   To identify the Base DN of your directory, open the **Active Directory Domains and Trusts** Microsoft Management Console snap-in and use the name of the top-level domain.
8. Enter the **Bind DN** and **Password** to enable the authentication service to authenticate the firewall.

   The Bind DN account must have permission to read the LDAP directory.
9. Enter the **Bind Timeout** and **Search Timeout** in seconds (default is 30 for both).
10. Enter the **Retry Interval** in seconds (default is 60).
11. Enable the option to **Require SSL/TLS secured connection** (enabled by default). The protocol that the endpoint uses depends on the server port:
    - 389 (default)—TLS (Specifically, the device uses the StartTLS operation, which upgrades the initial plaintext connection to TLS.)
    - 636—SSL
    - Any other port—The device first attempts to use TLS. If the directory server doesn't support TLS, the device falls back to SSL.
12. (Optional) For additional security, enable to the option to **Verify Server Certificate for SSL sessions** so that the endpoint verifies the certificate that the directory server

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presents for SSL/TLS connections. To enable verification, you must also enable the option to **Require SSL/TLS secured connection**. For verification to succeed, the certificate must meet one of the following conditions:

- It is in the list of device certificates: **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later). If necessary, import the certificate into the device.
- The certificate signer is in the list of trusted certificate authorities: **Device** > **Certificate Management** > **Certificates** > **Default Trusted Certificate Authorities**.

13. Click **OK** to save the server profile.

STEP 2 | Assign the server profile to Configure an Authentication Profile and Sequence to define various authentication settings.

STEP 3 | Assign the authentication profile to the firewall application that requires authentication.

- **Administrative access to the web interface**—Configure a Firewall Administrator Account and assign the authentication profile you configured.
- **End user access to services and applications**—For the full procedure to configure authentication for end users, see Configure Authentication Policy.

STEP 4 | Verify that the firewall can Test Authentication Server Connectivity to authenticate users.

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Connection Timeouts for Authentication Servers

You can configure the firewall to use External Authentication Services for authenticating administrators who access the firewall or Panorama and end users who access services or applications through Authentication Portal. To ensure that the firewall does not waste resources by continuously trying to reach an authentication server that is unreachable, you can set a timeout interval after which the firewall stops trying to connect. You set the timeout in the server profiles that define how the firewall connects to the authentication servers. When choosing timeout values, your goal is to strike a balance between the need to conserve firewall resources and to account for normal network delays that affect how quickly authentication servers respond to the firewall.

- Guidelines for Setting Authentication Server Timeouts
- Modify the PAN-OS Web Server Timeout
- Modify the Authentication Portal Session Timeout

Guidelines for Setting Authentication Server Timeouts

The following are some guidelines for setting the timeouts for firewall attempts to connect with External Authentication Services.

- In addition to the timeouts you set in server profiles for specific servers, the firewall has a global PAN-OS web server timeout. This global timeout applies when the firewall connects to any external server for authenticating administrative access to the firewall web interface or PAN-OS XML API and end user access to applications or services through Authentication Portal. The global timeout is 30 seconds by default (range is 3 to 125). It must be the same as or greater than the total time that any server profile allows for connection attempts. The total time in a server profile is the timeout value multiplied by the number of retries and the number of servers. For example, if a RADIUS server profile specifies a 3-second timeout, 3 retries, and 4 servers, the total time that the profile allows for connection attempts is 36 seconds (3 x 3 x 4). Modify the PAN-OS Web Server Timeout if necessary.

  Do not change the PAN-OS web server timeout unless you see authentication failures. Setting the timeout too high could degrade the performance of the firewall or cause it to drop authentication requests. You can review authentication failures in Authentication logs.

- The firewall applies an Authentication Portal session timeout that defines how long end users can take to respond to the authentication challenge in a Authentication Portal web form. The web form displays when users request services or applications that match an Authentication policy rule. The session timeout is 30 seconds by default (range is 1 to 1,599,999). It must be the same as or greater than the PAN-OS web server timeout. Modify the Authentication Portal Session Timeout if necessary. Keep in mind that increasing the PAN-OS web server and Authentication Portal session timeouts might degrade the performance of the firewall or cause it to drop authentication requests.

  The Authentication Portal session timeout is not related to the timers that determine how long the firewall retains IP address-to-username mappings.

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- Timeouts are cumulative for authentication sequences. For example, consider the case of an authentication sequence with two authentication profiles. One authentication profile specifies a RADIUS server profile with a 3-second timeout, 3 retries, and 4 servers. The other authentication profile specifies a TACACS+ server profile with a 3-second timeout and 2 servers. The longest possible period in which the firewall can try to authenticate user accounts with that authentication sequence is 42 seconds: 36 seconds for the RADIUS server profile plus 6 seconds for the TACACS+ server profile.
- The non-configurable timeout for Kerberos servers is 17 seconds for each server specified in the Kerberos server profile.
- To configure the timeouts and related settings for other server types, see:
  - Add an MFA server profile.
  - Add a SAML IdP server profile.
  - Add a TACACS+ server profile.
  - Add a RADIUS server profile.
  - Add an LDAP server profile.

Modify the PAN-OS Web Server Timeout

The PAN-OS web server timeout must be the same as or greater than the timeout in any authentication server profile multiplied by the number of retries and the number of servers in that profile.

Do not change the PAN-OS web server timeout unless you see authentication failures. Setting the timeout too high could degrade the performance of the firewall or cause it to drop authentication requests. You can review authentication failures in Authentication logs.

STEP 1 | Access the firewall CLI.

STEP 2 | Set the PAN-OS web server timeout by entering the following commands, where *<value>* is the number of seconds (default is 30; range is 3 to 125).

```
> configure
# set deviceconfig setting l3-service timeout <value>
# commit
```

Modify the Authentication Portal Session Timeout

The Authentication Portal session timeout must be the same as or greater than the PAN-OS web server timeout. For details, see Connection Timeouts for Authentication Servers.

The more you raise the PAN-OS web server and Authentication Portal session timeouts, the slower Authentication Portal will respond to users.

STEP 1 | Select **Device** > **Setup** > **Session** and edit the Session Timeouts.

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STEP 2 | Enter a new **Authentication Portal** value in seconds (default is 30; range is 1 to 1,599,999) and click **OK**.

STEP 3 | **Commit** your changes.

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Configure Local Database Authentication

You can configure a user database that is local to the firewall to authenticate administrators who access the firewall web interface and to authenticate end users who access applications through Authentication Portal or GlobalProtect. Perform the following steps to configure Local Authentication with a local database.

*Configuring new minimum password complexity settings (**Device** > **Setup**) or modifying an existing minimum password complexity settings does not apply retroactively to existing local data base user accounts.*

If you create or modify the minimum password complexity settings, you must re-add the existing local database administrator accounts so the passwords comply with the minimum password complexity settings.

External Authentication Services *are usually preferable to local authentication because they provide the benefit of central account management.*

You can also configure local authentication without a database, but only for firewall *or* Panorama *administrators.*

STEP 1 | Add the user account to the local database.

1. Select **Device** > **Local User Database** > **Users** and click **Add**.
2. Enter a user **Name** for the administrator.
3. Enter a **Password** and **Confirm Password** or enter a **Password Hash**.
4. **Enable** the account (enabled by default) and click **OK**.

STEP 2 | Add the user group to the local database.

Required if your users require group membership.

1. Select **Device** > **Local User Database** > **User Groups** and click **Add**.
2. Enter a **Name** to identify the group.
3. **Add** each user who is a member of the group and click **OK**.

STEP 3 | Configure an authentication profile.

The authentication profile defines authentication settings that are common to a set of users. Set the authentication **Type** to **Local Database**.

STEP 4 | Assign the authentication profile to an administrator account or to an Authentication policy rule for end users.

- **Administrators**—Configure a Firewall Administrator Account:

  Specify the **Name** of a user you defined earlier in this procedure.

  Assign the **Authentication Profile** that you configured for the account.

- **End users**—For the full procedure to configure authentication for end users, see Configure Authentication Policy.

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STEP 5 | Verify that the firewall can Test Authentication Server Connectivity to authenticate users.

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Configure an Authentication Profile and Sequence

An authentication profile defines the authentication service that validates the login credentials of administrators who access the firewall web interface and end users who access applications through Authentication Portal or GlobalProtect. The service can be Local Authentication that the firewall provides or External Authentication Services. The authentication profile also defines options such as Kerberos single sign-on (SSO).

Some networks have multiple databases (such as TACACS+ and LDAP) for different users and user groups. To authenticate users in such cases, configure an *authentication sequence*—a ranked order of authentication profiles that the firewall matches a user against during login. By default, the firewall checks against each profile in sequence until one successfully authenticates the user and a user is denied access only if authentication fails for all the profiles in the sequence. The sequence can specify authentication profiles that are based on any authentication service that the firewall supports excepts Multi-Factor Authentication (MFA) and SAML.

STEP 1 | (External service only) Enable the firewall to connect to an external server for authenticating users:

1. Set up the external server. Refer to your server documentation for instructions.
2. Configure a server profile for the type of authentication service you use.
   - Add a RADIUS server profile.

     If the firewall integrates with an MFA service through RADIUS, you must add a RADIUS server profile. In this case, the MFA service provides all the authentication factors. If the firewall integrates with an MFA service through a vendor API, you can still use a RADIUS server profile for the first factor but MFA server profiles are required for additional factors.

   - Add an MFA server profile.
   - Add a SAML IdP server profile.
   - Add a Kerberos server profile.
   - Add a TACACS+ server profile.
   - Add an LDAP server profile.

STEP 2 | (Local database authentication only) Configure a user database that is local to the firewall.

Perform these steps for each user and user group for which you want to configure Local Authentication based on a user identity store that is local to the firewall:

1. Add the user account to the local database.
2. (Optional) Add the user group to the local database.

STEP 3 | (Kerberos SSO only) Create a Kerberos keytab for the firewall if Kerberos single sign-on (SSO) is the primary authentication service.

Create a Kerberos keytab. A keytab is a file that contains Kerberos account information for the firewall. To support Kerberos SSO, your network must have a Kerberos infrastructure.

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STEP 4 | Configure an authentication profile.

Define one or both of the following:

- **Kerberos SSO**—The firewall first tries SSO authentication. If that fails, it falls back to the specified authentication **Type**.
- **External authentication or local database authentication**—The firewall prompts the user to enter login credentials, and uses an external service or local database to authenticate the user.

1. Select **Device** > **Authentication Profile** and **Add** the authentication profile.
2. Enter a **Name** to identify the authentication profile.
3. Select the **Type** of authentication service.
   - If you use Multi-Factor Authentication, the selected type applies only to the first authentication factor. You select services for additional MFA factors in the **Factors** tab.
   - If you select **RADIUS**, **TACACS+**, **LDAP**, or **Kerberos**, select the **Server Profile**.
   - If you select **LDAP**, select the **Server Profile** and define the **Login Attribute**. For Active Directory, enter **sAMAccountName** as the value.
   - If you select **SAML**, select the **IdP Server Profile**.
   - If you select **Cloud Authentication Service**, configure a Cloud Identity Engine instance to communicate with the firewall. For more information on the Cloud Identity Engine, see the Cloud Identity Engine Getting Started guide.
4. If you want to enable Kerberos SSO, enter the **Kerberos Realm** (usually the DNS domain of the users, except that the realm is UPPERCASE) and **Import** the **Kerberos Keytab** that you created for the firewall or Panorama.
5. (MFA only) Select **Factors**, **Enable Additional Authentication Factors**, and **Add** the MFA server profiles you configured.

   The firewall will invoke each MFA service in the listed order, from top to bottom.
6. Select **Advanced** and **Add** the users and groups that can authenticate with this profile.

   You can select users and groups from the local database or, if you configured the firewall to Map Users to Groups, from an LDAP-based directory service such as Active Directory. By default, the list is empty, meaning no users can authenticate.

   You can also select custom groups defined in a group mapping configuration.
7. (Optional) To modify the user information before the firewall sends the authentication request to the server, configure a **Username Modifier**.
   - **%USERDOMAIN%\%USERINPUT%**—If the source does not include the domain (for example, it uses the sAMAccountName), the firewall adds the **User Domain** you specify before the username. If the source includes the domain, the firewall replaces that domain with the **User Domain**. If the **User Domain** is empty, the firewall removes

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the domain from the user information that the firewall receives from source before the firewall sends the request to the authentication server.

Because LDAP servers don't support backslashes in the sAMAccountName, don't use this option to authenticate with an LDAP server.

- **%USERINPUT%**—(Default) The firewall sends the user information to the authentication server in the format it receives from the source.
- **%USERINPUT%@%USERDOMAIN%**—If the source does not include the domain, the firewall adds the **User Domain** value after the username. If the source includes domain, the firewall replaces that domain with the **User Domain** value. If the **User Domain** is empty, the firewall removes the domain from the user information that the firewall receives from the source before the firewall sends the request to the authentication server.
- **None**—If you manually enter **None**:
  - For LDAP and Kerberos server profiles, the firewall uses the domain it receives from the source to select the appropriate authentication profile, then removes the domain when it sends the authentication request to the server. This allows you to include the **User Domain** during the authentication sequence but remove the domain before the firewall sends the authentication request to the server. For example, if you are using an LDAP server profile and the samAccountName as the attribute, use this option so that the firewall does not send the domain to the authentication server that expects only a username and not a domain.
  - For RADIUS server profiles:
    - If the source sends the user information in **domain\username** format, the firewall sends the user information to the server in the same format.
    - If the source sends the user information in **username@domain** format, the firewall normalizes the user information to the **domain\username** format before sending it to the server.
    - If the source sends only the username, the firewall adds the **User Domain** you specify before sending the information to the server in **domain\username** format.
  - For local databases, TACACS+, and SAML, the firewall sends the user information to the authentication server in the format it receives from the source.

8. Click **OK** to save the authentication profile.

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STEP 5 | Configure an authentication sequence.

Required if you want the firewall to try one or more authentication profiles to authenticate users.

1. Select **Device** > **Authentication Sequence** and **Add** the authentication sequence.
2. Enter a **Name** to identify the authentication sequence.
3. (Optional but recommended) To expedite the authentication process and avoid the computational load of running the entire authentication sequence if it is not necessary, you can have the firewall **Exit the sequence on failed authentication**.

   This option supports the following authentication methods:

   - *Kerberos*
   - *RADIUS*
   - *TACACS+*
   - *LDAP*
   - *Local database*

   If you select this option, the authentication sequence ends when the firewall successfully authenticates the authentication profile or if the authentication fails (for example, due to an incorrect password). If the attempt times out or if the firewall does not find a matching user on the allow list, the authentication sequence proceeds to the next authentication profile in the sequence.

   If you do not select this option, the firewall attempts authentication with all authentication profiles in the sequence and ends the sequence only when an authentication profile authenticates successfully or if all authentication attempts with the authentication profiles in the sequence fail.
4. (Optional but recommended) To expedite the authentication process, select **Use domain to determine authentication profile**. When you select this option, the firewall matches the domain name that a user enters during login with an authentication profile in the sequence then uses that profile to authenticate the user. If the firewall does not find a match or if you disable the option, the firewall tries the profiles in the top-to-bottom sequence.
5. (Optional but recommended) To normalize the domain name that the user enters during login before applying the authentication sequence, select **Use User-ID domain to determine authentication profile**. If you do not select this option, the firewall does not normalize the domain name that the user enters during login before applying the authentication profile sequence.
6. **Add** each authentication profile. To change the evaluation order of the profiles, select a profile and **Move Up** or **Move Down**.
7. Click **OK** to save the authentication sequence.

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STEP 6 | Assign the authentication profile or sequence to an administrative account for firewall administrators or to Authentication policy for end users.

- **Administrators**—Assign the authentication profile based on how you manage administrator authorization:

  Authorization managed locally on the firewall—Configure a Firewall Administrator Account.

  Authorization managed on a SAML, TACACS+, or RADIUS server—Select **Device** > **Setup** > **Management**, edit the Authentication Settings, and select the **Authentication Profile**.

- **End users**—For the full procedure to configure authentication for end users, see Configure Authentication Policy.

STEP 7 | Verify that the firewall can Test Authentication Server Connectivity to authenticate users.

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Configure Passwordless Authentication

Where Can I Use This?	What Do I Need?

• Supported on the following platforms:
  • PA-7500
  • PA-5500
  • PA-5450
  • PA-5400 Series
  • PA-3400
  • PA-1400
  • PA-500
  • PA-400
  • VM-Series
	❑ License (perpetual)
	❑ URL Filtering
	❑ User-ID
	❑ Decryption

To simplify authentication and reduce user friction, you can enable passwordless authentication.

Passwordless authentication delegates a ticket on the behalf of the authenticated user, so that users have to log in to authenticate only once. After successfully authenticating, they can access any apps managed by the authentication server until the authentication period expires without having to log in again.

First, create a delegation profile to delegate the ticket on behalf of the authenticated user. Associate the delegation profile with your authentication server profile (in this case Kerberos) and configure an authentication object to use the delegation profile for user authentication.

Next, create a custom URL filter for the URLs of the apps, and then create a Security policy rule that references the custom URL filter. As the final step, create an HTTP header insertion that references the custom URL category.

Now, users just have to log in once to authenticate, and then they can easily switch between apps without having to reauthenticate immediately.

Passwordless authentication supports HTTPS traffic only.

STEP 1 | Prepare to deploy passwordless authentication.

1. Enable User-ID in the Kerberos traffic zone.
2. Enable the URL filter feature.
3. Enable decryption.

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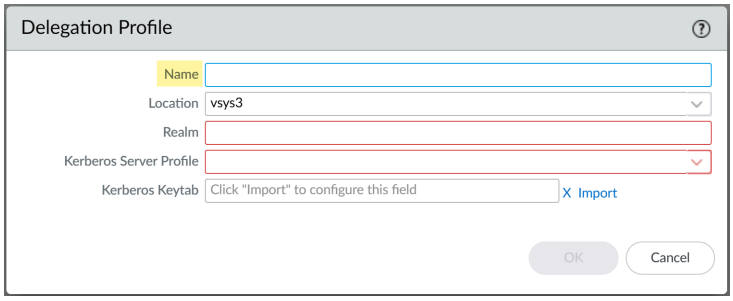
STEP 2 | Configure the Kerberos server profile.

1. Select **Device** > **Server Profiles** > **Kerberos**.
2. **Add** the **Name** and IP address of your **Kerberos Server**.
3. Specify the **Port** number (typically 88).
4. Click **OK**.

STEP 3 | Define the delegation profile.

1. Select **Device** > **Delegation Profile**.
2. **Add** a new delegation profile.
3. Specify a **Name** for the delegation profile.

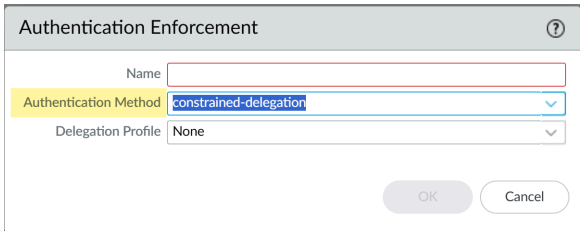


4. (Panorama only) Select the virtual system **Location** for the delegation profile.
5. Specify the Kerberos **Realm** (for example, **`example.com`**).
6. Select the **Kerberos Server Profile** you created in step 2.
7. **Import** the **Kerberos Keytab**.

   Only include the users who require passwordless authentication in the keytab. Passwordless authentication supports keytabs in the AES 256 format.
8. Click **OK**.

STEP 4 | Configure the authentication object.

1. Select **Objects** > **Authentication**.
2. **Add** a new **Authentication Enforcement** object and specify a **Name**.
3. Select **constrained-delegation** as the **Authentication Method**.



4. Select the **Delegation Profile** you created in step 3.
5. Click **OK**.

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STEP 5 | Define the URL category for the login URLs of the applications that require Kerberos for authentication.

1. Select **Objects** > **Custom Objects** > **URL Category**.
2. **Add** a new **URL Category** and enter a **Name**.
3. **Add** the login URLs for the **Sites** of all the applications that currently require Kerberos authentication for which you want to enable passwordless authentication.
4. Click **OK**.

STEP 6 | Create the URL filter profile for the login URLs of the allowed applications that require Kerberos for authentication.

1. Select **Objects** > **Security Profiles** > **URL Filtering**.
2. **Add** a new **URL Filtering Profile** and enter a **Name**.
3. Select **HTTP Header Insertion** then click **Add**.
4. **Add** a new header insertion and either **Add** the **Categories** of the allowed URLs or enter their **Domains**.
5. **Add** a **Header** with the **Name** **`"Authorization"`** and the **Value** **`Negotiate ($apreq)`**

   You must enter the header exactly as displayed above.
6. Click **OK** twice.

STEP 7 | Enable strip ALPN to disable HTTPS traffic.

1. Select **Objects** > **Decryption** > **Decryption Profile**.
2. **Add** a **Decryption Profile** and enter a **Name**.
3. Select **Strip ALPN**.
4. Click **OK**.

STEP 8 | Configure a Security policy rule and associate it with the URL Filtering profile.

1. Select **Policies**.
2. **Add** a new Security policy rule.
3. Enter a **Name** for the rule.
4. Select **Actions**.
5. Select the **URL Filtering** profile you created in step 6
6. Click **OK**.

STEP 9 | Create a decryption rule to decrypt the necessary traffic.

1. Select **Policies** > **Decryption**.
2. **Add** a rule to decrypt the traffic between the firewall and Kerberos.

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STEP 10 | Configure an authentication policy rule and associate it with the URL category.

1. Select **Policies** > **Authentication**.
2. **Add** a new Security policy rule.
3. Enter a **Name** for the rule.
4. For the source zone, source address, and source user, specify the information for the users for whom you want to enable passwordless authentication.
5. For the destination zone and destination address, specify as needed.
6. For the application, specify **web-browsing** or other relevant applications.
7. Select the **URL category** you created in step 5.
8. Select the **Service type** (typically **application-default**).
9. Select **Actions** then select **Authentication enforcement** object you created in step 4.
10. Click **OK**.

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Test Authentication Server Connectivity

The test authentication feature enables you to verify whether the firewall or Panorama can communicate with the authentication server specified in an authentication profile and whether an authentication request succeeds for a specific user. You can test authentication profiles that authenticate administrators who access the web interface or that authenticate end users who access applications through GlobalProtect or Authentication Portal. You can perform authentication tests on the candidate configuration to verify the configuration is correct before committing.

STEP 1 | Configure an authentication profile. You do not need to commit the authentication profile or server profile configuration before testing.

STEP 2 | Log into the firewall CLI.

STEP 3 | (Firewalls with multiple virtual systems) Define the target virtual system that the test command will access.

This is required on firewalls with multiple virtual systems so that the test authentication command can locate the user you will test.

Define the target virtual system by entering:

```
admin@PA-325060> set system setting target-vsys <vsys-name>
```

For example, if the user is defined in vsys2, enter:

```
admin@PA-3250> set system setting target-vsys vsys2
```

*The **target-vsys** option is per login session; the firewall clears the option when you log off.*

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STEP 4 | Test the authentication profile by entering the following command:

```
admin@PA-3250> test authentication authentication-
profile <authentication-profile-name> username <username> password
```

For example, to test an authentication profile named **my-profile** for a user named **bsimpson**, enter:

```
admin@PA-3250> test authentication authentication-profile my-
profile username bsimpson password
```

*When running the **test** command, the names of authentication profiles and server profiles are case sensitive. Also, if an authentication profile has a username modifier defined, you must enter the modifier with the username. For example, if you add the username modifier **%USERINPUT%@%USERDOMAIN%** for a user named **bsimpson** and the domain name is **mydomain.com**, enter **bsimpson@mydomain.com** as the username. This ensures that the firewall sends the correct credentials to the authentication server. In this example, mydomain.com is the domain that you define in the **User Domain** field in the authentication profile.*

STEP 5 | View the test output.

If the authentication profile is configured correctly, the output displays `Authentication succeeded`. If there is a configuration issue, the output displays information to help you troubleshoot the configuration.

The output results vary based on several factors related to the authentication type that you are testing as well as the type of issue. For example, RADIUS and TACACS + use different underlying libraries, so the same issue that exists for both of these types will produce different errors. Also, if there is a network problem, such as using an incorrect port or IP address in the authentication server profile, the output error is not specific. This is because the test command cannot perform the initial handshake between the firewall and the authentication server to determine details about the issue.

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Authentication Policy

Authentication policy enables you to authenticate end users before they can access services and applications. Whenever a user requests a service or application (such as by visiting a web page), the firewall evaluates Authentication policy. Based on the matching Authentication policy rule, the firewall then prompts the user to authenticate using one or more methods (factors), such as login and password, Voice, SMS, Push, or One-time Password (OTP) authentication. For the first factor, users authenticate through a Authentication Portal web form. For any additional factors, users authenticate through a Multi-Factor Authentication (MFA) login page.

To implement Authentication policy for GlobalProtect, refer to Configure GlobalProtect to facilitate multi-factor authentication notifications.

After the user authenticates for all factors, the firewall evaluates Security Policy to determine whether to allow access to the service or application.

To reduce the frequency of authentication challenges that interrupt the user workflow, you can specify a timeout period during which a user authenticates only for initial access to services and applications, not for subsequent access. Authentication policy integrates with Authentication Portal to record the timestamps used to evaluate the timeout and to enable user-based policies and reports.

Based on user information that the firewall collects during authentication, User-ID creates a new IP address-to-username mapping or updates the existing mapping for that user (if the mapping information has changed). The firewall generates User-ID logs to record the additions and updates. The firewall also generates an Authentication log for each request that matches an Authentication rule. If you favor centralized monitoring, you can configure reports based on User-ID or Authentication logs and forward the logs to Panorama or external services as you would for any other log types.

- Authentication Timestamps
- Configure Authentication Policy

Authentication Timestamps

When configuring an Authentication policy rule, you can specify a timeout period during which a user authenticates only for initial access to services and applications, not for subsequent access. Your goal is to specify a timeout that strikes a balance between the need to secure services and applications and the need to minimize interruptions to the user workflow. When a user authenticates, the firewall records a timestamp for the first authentication challenge (factor) and a timestamp for any additional Multi-Factor Authentication (MFA) factors. When the user subsequently requests services and applications that match an Authentication rule, the firewall evaluates the timeout specified in the rule relative to each timestamp. This means the firewall reissues authentication challenges on a per-factor basis when timeouts expire. If you Redistribute User Mappings and Authentication Timestamps, all your firewalls will enforce Authentication policy timeouts consistently for all users.

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The firewall records a separate timestamp for each MFA vendor. For example, if you use Duo v2 and PingID servers to issue challenges for MFA factors, the firewall records one timestamp for the response to the Duo factor and one timestamp for the response to the PingID factor.

Within the timeout period, a user who successfully authenticates for one Authentication rule can access services or applications that other rules protect. However, this portability applies only to rules that trigger the same authentication factors. For example, a user who successfully authenticates for a rule that triggers TACACS+ authentication must authenticate again for a rule that triggers SAML authentication, even if the access requests are within the timeout period for both rules.

When evaluating the timeout in each Authentication rule and the global timer defined in the Authentication Portal settings (see Configure Authentication Portal), the firewall prompts the user to re-authenticate for whichever setting expires first. Upon re-authenticating, the firewall records new authentication timestamps for the rules and resets the time count for the Authentication Portal timer. Therefore, to enable different timeout periods for different Authentication rules, set the Authentication Portal timer to a value that is the same as or higher than the timeout in any rule.

Configure Authentication Policy

Perform the following steps to configure Authentication policy for end users who access services through Authentication Portal. Before starting, ensure that your Security Policy allows users to access the services and URL categories that require authentication.

Before you configure an Authentication policy rule, make sure you understand that the set of IPv4 addresses is treated as a subset of the set of IPv6 addresses, as described in detail in Policy.

STEP 1 | Configure Authentication Portal. If you use Multi-Factor Authentication (MFA) services to authenticate users, you must set the **Mode** to **Redirect**.

STEP 2 | Configure the firewall to use one of the following services to authenticate users.

- External Authentication Services—Configure a server profile to define how the firewall connects to the service.
- Local database authentication—Add each user account to the local user database on the firewall.
- Kerberos single sign-on (SSO)—Create a Kerberos keytab for the firewall. Optionally, you can configure the firewall to use Kerberos SSO as the primary authentication service and, if SSO failures occur, fall back to an external service or local database authentication.

STEP 3 | Configure an Authentication Profile and Sequence for each set of users and Authentication policy rules that require the same authentication services and settings.

Select the **Type** of authentication service and related settings:

- **External service**—Select the **Type** of external server and select the **Server Profile** you created for it.
- **Local database authentication**—Set the **Type** to **Local Database**. In the **Advanced** settings, **Add** the Authentication Portal users and user groups you created.
- **Kerberos SSO**—Specify the **Kerberos Realm** and **Import** the **Kerberos Keytab**.

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STEP 4 | Configure an authentication enforcement object.

The object associates each authentication profile with an Authentication Portal method. The method determines whether the first authentication challenge (factor) is transparent or requires a user response.

1. Select **Objects** > **Authentication** and **Add** an object.
2. Enter a **Name** to identify the object.
3. Select an **Authentication Method** for the authentication service **Type** you specified in the authentication profile:
   - **browser-challenge**—Select this method if you want the client browser to respond to the first authentication factor instead of having the user enter login credentials. For this method, you must configure Kerberos SSO in the authentication profile. If the browser challenge fails, the firewall falls back to the **web-form** method.
   - **web-form**—Select this method if you want the firewall to display a Authentication Portal web form for users to enter login credentials.
4. Select the **Authentication Profile** you configured.
5. Enter the **Message** that the Authentication Portal web form will display to tell users how to authenticate for the first authentication factor.
6. Click **OK** to save the object.

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STEP 5 | Configure an Authentication policy rule.

Create a rule for each set of users, services, and URL categories that require the same authentication services and settings.

*The firewall does not apply the Authentication Portal timeout if your authentication policy uses default authentication enforcement objects (for example, **default-browser-challenge**).To require users to re-authenticate after the Authentication Portal timeout, clone the rule for the default authentication object and move it before the existing rule for the default authentication object.*

1. Select **Policies** > **Authentication** and **Add** a rule.
2. Enter a **Name** to identify the rule.
3. Select **Source** and **Add** specific zones and IP addresses or select **Any** zones or IP addresses.

   The rule applies only to traffic coming from the specified IP addresses or from interfaces in the specified zones.
4. Select **User** and select or **Add** the source users and user groups to which the rule applies (default is **any**).
5. Select or **Add** the Host Information Profiles to which the rule applies (default is **any**).
6. Select **Destination** and **Add** specific zones and IP addresses or select **any** zones or IP addresses.

   The IP addresses can be resources (such as servers) for which you want to control access.
7. Select **Service/URL Category** and select or **Add** the services and service groups for which the rule controls access (default is **service-http**).
8. Select or **Add** the URL categories for which the rule controls access (default is **any**). For example, you can create a custom URL category that specifies your most sensitive internal sites.
9. Select **Actions** and select the **Authentication Enforcement** object you created.
10. Specify the **Timeout** period in minutes (default 60) during which the firewall prompts the user to authenticate only once for repeated access to services and applications.

    ***Timeout** is a tradeoff between tighter security (less time between authentication prompts) and the user experience (more time between authentication prompts). More frequent authentication is often the right choice for access to critical systems and sensitive areas such as a data center. Less frequent authentication is often the right choice at the network perimeter and for businesses for which the user experience is key.*
11. Click **OK** to save the rule.

STEP 6 | (MFA only) Customize the MFA login page.

The firewall displays this page so that users can authenticate for any additional MFA factors.

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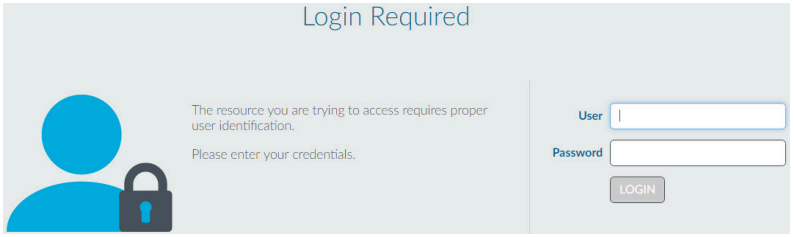
STEP 7 | Verify that the firewall enforces Authentication policy.

1. Log in to your network as one of the source users specified in an Authentication policy rule.
2. Request a service or URL category that matches one specified in the rule.

   The firewall displays the Authentication Portal web form for the first authentication factor. For example:



   If you configured the firewall to use one or more MFA services, authenticate for the additional authentication factors.
3. End the session for the service or URL you just accessed.
4. Start a new session for the same service or application. Be sure to perform this step within the **Timeout** period you configured in the Authentication rule.

   The firewall allows access without re-authenticating.
5. Wait until the **Timeout** period expires and request the same service or application.

   The firewall prompts you to re-authenticate.

STEP 8 | (Optional) Redistribute Data and Authentication Timestamps to other firewalls that enforce Authentication policy to ensure they all apply the timeouts consistently for all users.

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Troubleshoot Authentication Issues

When users fail to authenticate to a Palo Alto Networks firewall or Panorama, or the Authentication process takes longer than expected, analyzing authentication-related information can help you determine whether the failure or delay resulted from:

- **User behavior**—For example, users are locked out after entering the wrong credentials or a high volume of users are simultaneously attempting access.
- **System or network issues**—For example, an authentication server is inaccessible.
- **Configuration issues**—For example, the Allow List of an authentication profile doesn't have all the users it should have.

The following CLI commands display information that can help you troubleshoot these issues:

Task	Command

Display the number of locked user accounts associated with the authentication profile (**auth-profile**), authentication sequence (**is-seq**), or virtual system (**vsys**).
 	
To unlock users, use the following operational command:
 	
`> request authentication [unlock-admin | unlock-user]`
	`PA-220> show authentication locked-users`
	`{`
	`vsys <value> |`
	`auth-profile <value> |`
	`is-seq`
	`{yes | no}`
	`{auth-profile | vsys} <value>`
	`}`

Use the **debug authentication** command to troubleshoot authentication events.
 	
Use the **show** options to display authentication request statistics and the current debugging level:
• **show** displays the current debugging level for the authentication service (authd).
• **show-active-requests** displays the number of active checks for authentication requests, allow lists, locked user accounts, and Multi-Factor Authentication (MFA) requests.
• **show-pending-requests** displays the number of pending checks for authentication requests, allow lists, locked user accounts, and MFA requests.
	`PA-220> debug authentication`
	`{`
	`on {debug | dump | error | info | warn} |`
	`show |`
	`show-active-requests |`
	`show-pending-requests |`
	`connection-show |`
	`{`
	`connection-id |`
	`protocol-type`
	`{`
	`Kerberos connection-id <value> |`
	`LDAP connection-id <value> |`
	`RADIUS connection-id <value> |`
	`TACACS+ connection-id <value> |`
	`}`

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Task	Command
• **connection-show** displays authentication request and response statistics for all authentication servers or for a specific protocol type. Use the **connection-debug** options to enable or disable authentication debugging: • Use the **on** option to enable or the **off** option to disable debugging for authd. • Use the **connection-debug-on** option to enable or the **connection-debug-off** option to disable debugging for all authentication servers or for a specific protocol type.	**connection-debug-on |** **{** **connection-id |** **debug-prefix |** **protocol-type** **{** **Kerberos connection-id <value> |** **LDAP connection-id <value> |** **RADIUS connection-id <value> |** **TACACS+ connection-id <value> |** **}** **connection-debug-off |** **{** **connection-id |** **protocol-type** **{** **Kerberos connection-id <value> |** **LDAP connection-id <value> |** **RADIUS connection-id <value> |** **TACACS+ connection-id <value> |** **}** **connection-debug-on** **}**
Test the connection and validity of the certificate profile.	PA-220> test authentication authentication-profile auth-profile username <*username*>password <*password*>
Troubleshoot a specific authentication using the **Authentication ID** displayed in **Monitor** > **Logs** > **Authentication**.	PA-220> grep <*Authentication ID*>

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Certificate Management

Certificate management is the management of digital certificates throughout their lifecycle to maintain secure network communications. This critical process involves the generation, storage, protection, deployment, renewal, and revocation of digital certificates. Monitoring certificate status, receiving alerts for expiring certificates, and more are key. Effective certificate management ensures that only authorized users can access resources, minimal downtime and continuity of service.

You can set up certificates, add certificate authorities, add OCSP responders, and define certificate checks from a single administrative interface. The certificates and settings you set up in the Certificate Management section on the firewall secure features like decryption, the Authentication Portal, and the GlobalProtect™ app.

Configure different keys and certificates for each application.

Palo Alto Networks firewalls and Panorama use certificates in the following applications:

- User authentication for Authentication Portal, multi-factor authentication (MFA), and web interface access to a firewall or Panorama
- Device authentication for GlobalProtect VPN (remote user-to-site or large scale) and IPSec site-to-site VPN with IKE
- External dynamic list validation
- User-ID agent and TS agent access
- Decrypting inbound and outbound SSL traffic

  A firewall decrypts the traffic to apply policy rules, then re-encrypts it before forwarding the traffic to the final destination. For outbound traffic, the firewall acts as a forward proxy server, establishing an SSL/TLS connection to the destination server. To secure a connection between itself and the client, the firewall uses a *signing certificate* to automatically generate a copy of the destination server certificate.

To manage certificates, select **Device** > **Certificate Management**.

For more details on core components of certificates and certificate management, see Keys and Certificates.

Handled incorrectly, certificate management can lead to major costs for your organization and major frustration for end users. If you have an Enterprise PKI, generate the Forward Trust CA certificate for forward proxy traffic from your Enterprise Root CA and import it into the certificate store on your Next-Generation Firewall. Since the certificate is part of the root CA, your users' endpoints trust it automatically, and end users won't get frustrating error messages.

Keys and Certificates

Understanding how certificates and keys work together is essential for network security. While certificates verify identities and contain public keys, private keys (securely stored on your device) provide cryptographic proof of certificate ownership and decrypt incoming communications.

Cryptographic keys are numerical strings typically generated through mathematical operations involving random numbers and large primes. These keys transform data, such as passwords and shared secrets, from unencrypted plaintext to encrypted ciphertext and from encrypted ciphertext to unencrypted plaintext. Keys can be symmetric (the same key encrypts and decrypts) or asymmetric (one key encrypts and a mathematically related key decrypts). This mathematical relationship enables secure communications without requiring the exchange of secret keys beforehand.

The SSL/TLS protocol use public-key cryptography (or asymmetric cryptography) to establish secure web communications. Each key pair in this system consists of a *public key* and a *private key*. A website's SSL/TLS certificate contains the public key, which is shared freely. The private key is kept hidden. Private keys can be stored on firewalls, in keystores, or in hardware security modules (HSMs). Together with certificates, keys ensure that sensitive data remains protected during transmission.

Key management is critical to your NGFWs security posture. Private keys require safeguarding, as compromise could allow attackers to impersonate legitimate services or decrypt sensitive communications. Palo Alto Networks NGFWs provide secure storage for private keys and robust key management features to help you maintain the integrity of your encryption infrastructure.

The following table describes the various keys and certificates that Palo Alto Networks firewalls and Panorama use.

For maximum security, use different keys and certificates for each application or use case.

Table 1: Palo Alto Networks Device Keys and Certificates

Certificate or Key Usage	Description
Administrative Access	Secure access to firewall or Panorama administration interfaces (HTTPS access to the web interface) requires a server certificate for the MGT interface (or a designated interface on the dataplane if the firewall or Panorama does not use MGT) and, optionally, a certificate to authenticate the administrator.
Authentication Portal	In deployments where Authentication policy rules identify users who access HTTPS resources, designate a server certificate for the Authentication Portal interface. If you configure Authentication Portal to use certificates for user identification (instead of, or in addition to, interactive authentication), deploy client certificates as well. For

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Certificate or Key Usage	Description
	more information on Authentication Portal, see Map IP Addresses to Usernames Using Authentication Portal.
Forward Trust	For outbound SSL/TLS traffic, if a firewall acting as a forward proxy trusts the CA that signed the certificate of the destination server, the firewall uses the forward trust CA certificate to generate a copy of the destination server certificate to present to the client. To set the private key size, see Configure the Key Size for SSL Forward Proxy Server Certificates. For added security, store the private key on a hardware security module (HSM).
Forward Untrust	For outbound SSL/TLS traffic, if a firewall acting as a forward proxy doesn't trust the CA that signed the certificate of the destination server, the firewall uses the forward untrust CA certificate to generate a copy of the destination server certificate it presents to the client.
SSL Inbound Inspection	The keys that decrypt inbound SSL/TLS traffic for inspection and policy enforcement. For this application, import a private key for each server that is subject to SSL/TLS inbound inspection. See Configure SSL Inbound Inspection. *Firewalls use the Elliptic-Curve Diffie-Hellman Ephemeral (ECDHE) algorithm to perform strict certificate checking. This means that if the firewall uses an intermediate certificate, you must reimport the certificate from your web server to the firewall and combine the server certificate with the intermediate certificate (install a chained certificate). Otherwise, SSL Inbound Inspection sessions that have an intermediate certificate in the chain will fail. To install a chained certificate:* 1. *Open each certificate (.cer) file in a plain-text editor such as Notepad.* 2. *Paste each certificate end-to-end with the Server Certificate at the top with each signer included below.* 3. *Save the file as a text (.txt) or certificate (.cer) file (the name of the file cannot contain blank spaces).* 4. *Import the combined (chained) certificate into the firewall.*
GlobalProtect	All interaction among GlobalProtect components occurs over SSL/TLS connections. Therefore, as part of the GlobalProtect deployment, deploy server certificates for all GlobalProtect portals, gateways, and Mobile Security Managers. Optionally, deploy certificates for authenticating users also.

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Certificate or Key Usage	Description
	The GlobalProtect Large Scale VPN (LSVPN) feature requires a CA signing certificate.
Site-to-Site VPNs (IKE)	In a site-to-site IPSec VPN deployment, peer devices use Internet Key Exchange (IKE) gateways to establish a secure channel. IKE gateways use certificates or preshared keys to authenticate the peers to each other. You can configure and assign the certificates or keys when defining an IKE gateway on a firewall.
Master Key	The firewall uses a master key to encrypt all private keys and passwords. If your network requires a secure location for storing private keys, you can use an encryption (wrapping) key stored on an HSM to encrypt the master key. For details, see Encrypt and Refresh Master Keys Using an HSM.
Secure Syslog	The certificate to enable secure connections between the firewall and a syslog server. See Syslog Field Descriptions.
Trusted Root CA	The designation for a root certificate issued by a CA that the firewall trusts. The firewall can use a self-signed root CA certificate to automatically issue certificates for other applications (for example, SSL Forward Proxy). Also, if a firewall must establish secure connections with other firewalls, the root CA that issues their certificates must be in the list of trusted root CAs on the firewall. (Panorama managed firewalls) The **Trusted Root CA** setting for a CA must be configured as part of the template configuration, and not part of the template stack configuration. If you configure the **Trusted Root CA** setting for a CA as part of the template stack configuration, the associated templates do not inherit the setting for the CA.
Inter-Device Communication	By default, Panorama, firewalls, and Log Collectors use a set of predefined certificates for the SSL/TLS connections used for management and log forwarding. However, you can enhance these connections by deploying custom certificates to the devices in your deployment. These certificates can also be used to secure the SSL/TLS connection between Panorama HA peers.

Certificates

Digital certificates (also known as X.509 certificates or SSL/TLS certificates) contain a cryptographic key to encrypt plaintext or decrypt ciphertext. Each certificate includes a digital signature to authenticate the identity of the issuer. Certificate authorities (CAs) issue these certificates to verify the identity of entities such as clients or servers.

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On Palo Alto Networks firewalls, certificates play a crucial role in device and user authentication and securing communications. When properly implemented and managed, certificates help ensure that traffic remains encrypted, endpoints are authenticated, and communications are secure from eavesdropping and tampering.

A digital certificate contains several basic fields or components:

- **Version**–The version of the encoded certificate.
- **Serial Number**–A unique integer given to the certificate (unique for all certificates issued from the same CA)
- **Signature**–The identifier for the cryptographic algorithm the CA used to sign the certificate
- **Issuer**–The CA that signed and issued the certificate
- **Validity**–The time window a certificate is valid between
- **Subject**–The entity to which the certificate was issued
- **Subject Public Key Info**–Contains the public key and identifies the algorithm with which the key is used (for example, RSA, DSA, or Diffie-Hellman)
- **Unique Identifiers**–Present only in version 2 or 3 certificates, these identifiers handle potential reuse of subject and issuer names over time

An organization that issues certificates can establish a hierarchy of CAs. The root CA has a self-signed certificate. Each subordinate CA has a certificate that is signed by the next highest CA in the hierarchy. A *certificate chain* is the certificate of a particular CA, plus the certificates of any higher CAs up through the root CA.

A certificate issuer must be in the authenticating party's list of trusted certificate authorities. Optionally, the authenticating party verifies the issuer did not revoke the certificate (see Certificate Revocation).

Palo Alto Networks supports several certificate types to accommodate different security needs:

- **Enterprise CA certificates** (unlike most certificates purchased from a trusted, third-party CA) leverage your organization's existing public key infrastructure (PKI) and can automatically issue certificates for applications such as SSL/TLS decryption or large scale VPN deployments.
- **External CA certificates** come from trusted third-party CAs–either well-known public CA or enterprise CAs–and provide broader recognition outside your organization.
- **Device certificates** enable secure access to certain cloud services.
- **Self-signed certificates** offer a way to quickly generate certificates without external validation, which is useful for testing or internal applications. The owner and issuer of self-signed certificates are the same. You can obtain, provision, and renew these certificates through Palo Alto Networks firewall interfaces.

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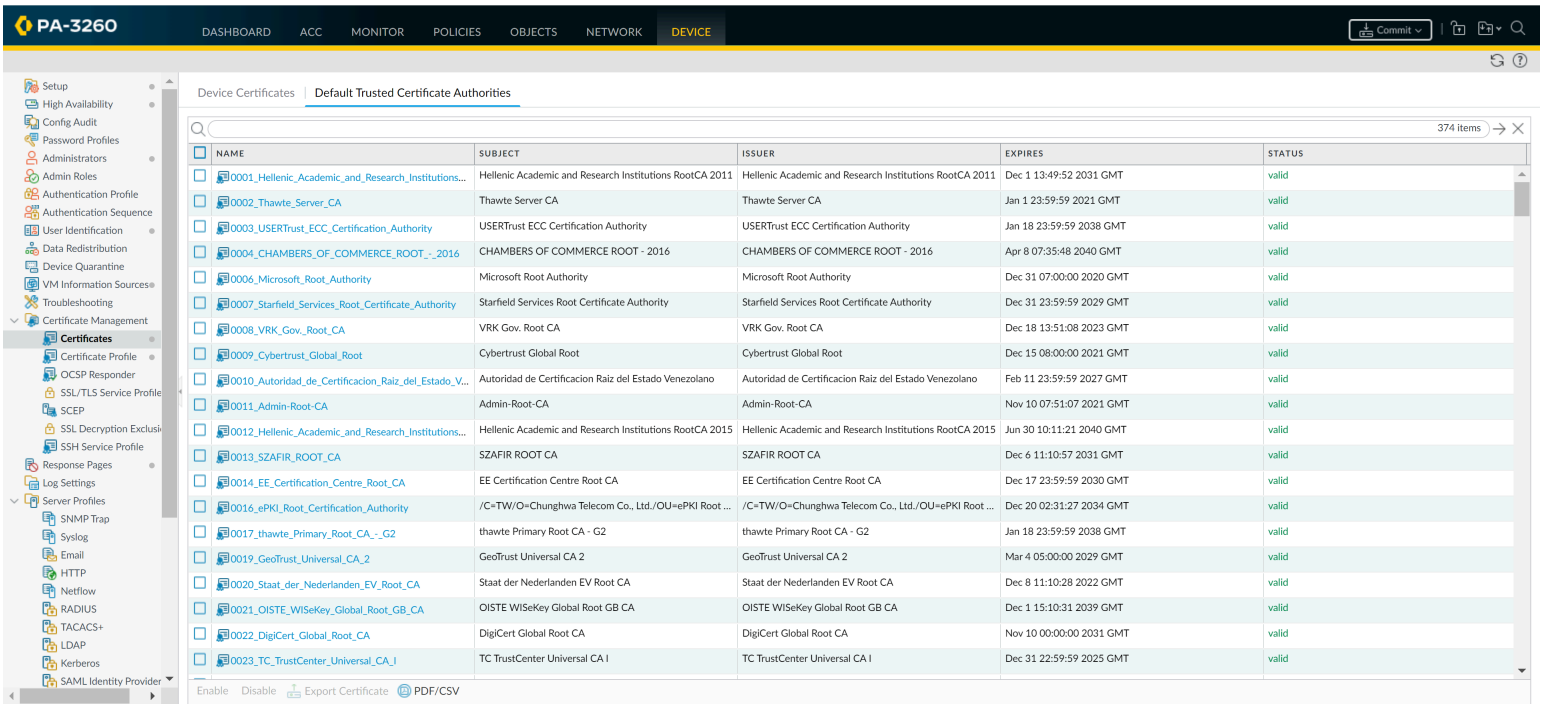
Default Trusted Certificate Authorities (CAs)

The Default Trusted Certificate Authorities store (**Device** > **Certificate Management** > **Certificates** > **Default Trusted Certificate Authorities**) contains certificates issued by the most common and trusted certificate authorities (CAs). Palo Alto Networks Next-Generation Firewalls use these preinstalled certificates to secure connections to the internet. The store displays the name, subject, issuer, expiration date, and revocation status of each certificate.

The Default Trusted Certificate Authorities store is updated with major PAN-OS releases.

You can enable, disable, or export CA certificates from the store. To add additional enterprise CA certificates to your firewall, obtain the certificates and import them to Device Certificates or Custom Certificates (**Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later)).



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Certificate Revocation

Next-Generation Firewalls (NGFWs) and Panorama use digital certificates to establish trust between parties during a secure communication session. A digital certificate verifies the identity of the entity to which it was issued. When issued, X.509 certificates are assigned a validity period—a specific start date and inclusive expiration date. Certificates are considered valid and can secure communications during this period. However, a certificate can become invalid before its expiration date for several reasons:

- Compromised private key (known or suspected)
- Change in certificate information (for example, a company name change or a domain name change)
- Discontinued operation of the service belonging to the certificate (for example, because there is a new service under a different name)
- Change of association between subject and certificate authority (for example, an employee terminates employment)

Under these circumstances, the certificate authority (CA) that issued the certificate must revoke it. *Certificate revocation* is the process of invalidating a certificate before it expires. A revoked certificate is no longer trustworthy and can't be used to establish secure connections.

When a certificate is part of a chain, an NGFW or Panorama checks the validity of every certificate in that chain, except for the root CA certificate. This process, known as *certificate revocation checking*, prevents potential security breaches and protects users from untrustworthy websites or services. Some web browsers also perform this check and, if a website's certificate is revoked, might display a warning or refuse the connection.

By default, certificate revocation checking is disabled. As a security best practice, enable this feature to verify the certificates used for device or user authentication and SSL/TLS decryption. NGFWs and Panorama support two methods for verifying certificate revocation status:

- Certificate Revocation List (CRL)—A CRL is a list of certificates, identified by serial number, that have been issued and subsequently revoked by a CA. CAs typically publish CRLs periodically.
- Online Certificate Status Protocol (OCSP)—OCSP is an Internet Protocol that enables clients, like a firewall or web browser, to obtain the revocation status of an X.509 certificate in real time. The protocol governs the communication between an OCSP client, which requests the revocation status, and an OCSP responder, which provides it.

  If your enterprise has its own public key infrastructure (PKI), you can configure an NGFW to function as the OCSP responder.

  ***Proxy Server Support**: OCSP works with network deployments that include a web proxy. To route OCSP requests and responses through a proxy server, configure the HTTP proxy for OCSP verification.*

If you configure both methods, the NGFW or Panorama first tries the OCSP method. The appliance uses the CRL method only if the OCSP server is unavailable.

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Certificate Revocation List (CRL)

Each certificate authority (CA) periodically issues a certificate revocation list (CRL) to a public repository. The CRL is a time-stamped list that identifies revoked certificates by their serial numbers. After the CA revokes a certificate, the next CRL update includes that certificate's serial number. The firewall supports CRLs in Distinguished Encoding Rules (DER) and Privacy Enhanced Mail (PEM) formats.

Browsers, applications, and other parties that rely on certificates for authentication purposes, not only check both the certificate's signature and validity, but also retrieve a recent CRL to confirm that a certificate's serial number is not listed. Palo Alto Networks firewall downloads and caches the last-issued CRL for every CA listed in the trusted CA list of the firewall. Caching only applies to validated certificates; if a firewall never validated a certificate, the firewall cache does not store the CRL for the issuing CA. Also, the cache only stores a CRL until it expires.

If you configure multiple CRL distribution points (CDPs) and the firewall cannot reach the first CDP, the firewall does not check the remaining CDPs. To redirect invalid CRL requests, configure a DNS proxy as an alternate server.

An advantage of the CRL revocation method is that CRLs can be distributed through the same channels as certificates themselves, through untrusted servers and untrusted communications. One limitation is the period nature of updates. There's a chance that a client could download a CRL before it's updated, missing recent revocations.

To use CRLs for verifying the revocation status of certificates that authenticate users and devices, configure a certificate profile and assign it to the interfaces that are specific to the application: Authentication Portal, GlobalProtect (remote user-to-site or large scale), site-to-site IPSec VPN, or web interface access to Palo Alto Networks firewalls or Panorama. For details, see Configure Revocation Status Verification of Certificates.

To use CRLs for verifying the revocation status of certificates used for the decryption of inbound and outbound SSL/TLS traffic, see Configure Revocation Status Verification of Certificates Used for SSL/TLS Decryption.

An alternative to CRL is the Online Certificate Status Protocol (OCSP), which provides real-time certificate status.

Online Certificate Status Protocol (OCSP)

Palo Alto Networks firewalls can use the Online Certificate Status Protocol (OCSP) to check the revocation status of X.509 digital certificates (SSL/TLS certificates). The advantages of using OCSP instead of or in addition to certificate revocation lists (CRLs) are real-time certificate status responses and usage of fewer network and client resources. Certificate status can be `good`, `revoked`, or `unknown`.

After you enable certificate verification using OCSP, the firewall verifies the status of a certificate when establishing an SSL/TLS session. First, an authenticating client (firewall) sends an status request to an OCSP responder (server). The request includes the serial number of the target certificate. Certificates are not accepted until the responder provides a status response. Next, the OCSP responder uses the serial number to search the database of the CA that issued the certificate for its revocation status. Then, the OCSP responder returns the certificate status to the client. The firewall drops sessions with revoked certificates.

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If your network deployment consists of a web proxy, the OCSP request workflow differs. OCSP requests and responses pass through your proxy server first. The procedure to enable an HTTP proxy for OCSP status checks describes the workflow in more detail.

Palo Alto Networks firewalls download and cache OCSP responses for every CA in the trusted CA list of the firewall. The cache includes OCSP responses for an issuing CA only if the firewall has already validated a certificate. Caching OCSP responses speeds up the response time and minimizes OCSP traffic to the responder.

The following applications use certificates to authenticate users and devices: Authentication Portal, GlobalProtect (remote user-to-site or large scale), site-to-site IPSec VPN, and web interface access to Palo Alto Networks firewalls or Panorama. To use OCSP to verify the revocation status of certificates that authenticate users and devices, perform the following steps:

If your firewall functions as an SSL forward proxy, you'll need to configure decryption certificate revocation settings.

- Configure an OCSP responder.
- Enable HTTP OCSP service on the firewall (if you configure the firewall as an OCSP responder).
- Create or obtain a certificate for each application.
- Configure a certificate profile for each application.
- Assign the certificate profile to the relevant application.

Configure CRL as a fall-back method to cover situations where the OCSP responder is unavailable. For details, see Configure Revocation Status Verification of Certificates.

Configure an OCSP Responder

To use Online Certificate Status Protocol (OCSP) for verifying the revocation status of certificates, you must configure the firewall to access an OCSP responder (server). The entity that manages the OCSP responder can be a third-party certificate authority (CA). If your enterprise has its own public key infrastructure (PKI), you can use external OCSP responders or you can configure the firewall itself as an OCSP responder. For details on OCSP, see Certificate Revocation.

*Configure an OCSP responder Certificate Profile only when you generate a new certificate (**Device** > **Certificate Management** > **Certificates**). Specify the **OCSP Responder** when you generate a new certificate so that the firewall populates the Authority Information Access (AIA) field with the appropriate URL and then specify the new certificate in the Certificate Profile. Configuring a Certificate Profile does not override the Certificate Profile for existing certificates or Root CAs.*

You can enable OCSP validation or override the AIA field of certificate in the Certificate Profile. The Certificate Profile configuration determines which certificate validation mechanisms are used on certificates that authenticate to services hosted on the firewall, such as GlobalProtect.

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STEP 1 | Define an external OCSP responder or configure the firewall itself as an OCSP responder.

1. Select **Device** > **Certificate Management** > **OCSP Responder** and click **Add**.
2. Enter a **Name** to identify the responder (up to 31 characters). The name is case-sensitive. It must be unique and use only letters, numbers, spaces, hyphens, and underscores.
3. If the firewall has more than one virtual system (vsys), select a **Location** (vsys or **Shared**) for the certificate.
4. In the **Host Name** field, enter the host name (recommended) or IP address of the OCSP responder. You can enter an IPv4 or IPv6 address. From this value, PAN-OS automatically derives a URL and adds it to the certificate being verified.

   If you configure the firewall itself as an OCSP responder, the host name must resolve to an IP address in the interface that the firewall uses for OCSP services.
5. Click **OK**.

STEP 2 | If you want the firewall to use the management interface for the OCSP responder interface, enable OCSP communication on the firewall. Otherwise, continue to the next step to configure an alternate interface.

1. Select **Device** > **Setup** > **Interfaces** > **Management**.
2. In the Network Services section, select the **HTTP OCSP** check box, then click **OK**.

STEP 3 | To use an alternate interface as the OCSP responder interface, add an Interface Management Profile to the interface used for OCSP services.

1. Select **Network** > **Network Profiles** > **Interface Mgmt**.
2. Click **Add** to create a new profile or click the name of an existing profile.
3. Select the **HTTP OCSP** check box and click **OK**.
4. Select **Network** > **Interfaces** and click the name of the interface that the firewall will use for OCSP services. The OCSP **Host Name** specified in Step 1 must resolve to an IP address in this interface.
5. Select **Advanced** > **Other info** and select the Interface Management Profile you configured.
6. Click **OK** and **Commit**.

Configure Revocation Status Verification of Certificates Used for Device or User Authentication

Before you begin: Obtain CA certificates to assign to the certificate profile. You can generate a certificate or import a certificate.

The firewall and Panorama use certificates to authenticate users and devices for such applications as Authentication Portal, GlobalProtect™, site-to-site IPSec VPN, and web interface access to a next-generation firewall or Panorama. You can configure verification of the revocation status of certificates used for device or user authentication to these applications.

For details on the certificates that various applications use, see Keys and Certificates.

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STEP 1 | Configure a certificate profile for each application.

Assign one or more root CA certificates to the profile, configure certificate revocation checking, and block sessions for various reasons. For example, you can block sessions if the certificate status is unknown or if the certificate status can't be retrieved within specified timeout limits.

If you configure both the CRL and OCSP certificate revocation checking methods, the NGFW or Panorama first tries the OCSP method. If the OCSP server is unavailable, it uses the CRL method.

STEP 2 | Assign the certificate profiles to the relevant applications.

The steps to assign a certificate profile depend on the application that requires it.

Configure Revocation Status Verification of Certificates Used for SSL/TLS Decryption

Where Can I Use This?	What Do I Need?
• NGFW • Prisma Access	Depending on the products you're using, you need at least one of... • For Strata Cloud Manager, you must have one of the following licenses: ☐ Prisma Access license ☐ AIOps for NGFW Premium license (use the Strata Cloud Manager app) → The features and capabilities available to you in Strata Cloud Manager depend on which license(s) you are using. • Software NGFW Credits (*for VM-Series software NGFWs*) ☐ AIOps for NGFW Free (use the AIOps for NGFW Free app)

Next-Generation Firewalls (NGFWs) decrypt inbound and outbound SSL/TLS traffic to inspect the traffic for threats. After you create a Security policy rule that allows traffic and apply Security profiles to the rule, create an analogous decryption policy rule to decrypt that traffic. Decryption provides visibility into the traffic, which enables NGFWs to inspect and enforce the Security profiles on the traffic. The NGFW re-encrypts the traffic before forwarding it to its destination. (For details on how this works, see SSL Inbound Inspection and SSL Forward Proxy.)

You can verify certificate revocation status using the Online Certificate Status Protocol (OCSP) and Certificate Revocation List (CRL) methods. If you configure both methods, the NGFW or Panorama first tries the OCSP method. If the OCSP server is unavailable, it uses the CRL method.

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Enabling revocation status verification for SSL/TLS decryption certificates adds time to the process of establishing the session. The first attempt to access a site might fail if the verification does not finish before the session times out. For these reasons, verification is disabled by default.

- Strata Cloud Manager
- PAN-OS & Panorama

Configure Revocation Status Verification of Certificates Used for SSL/TLS Decryption (Strata Cloud Manager)

STEP 1 | Define the service-specific timeout intervals for revocation status checks.

1. Select **Configuration** > **NGFW and Prisma Access** > **Security Services** > **Decryption**, and edit the Decryption Settings (⚙). A Decryption Settings window appears.
2. Under the Certificate Settings for proxying an untrusted site, click **Advanced**. An Advanced Decryption Settings overlay appears.
3. Perform one or both of the following steps, depending on whether the firewall will use the Online Certificate Status Protocol (OCSP) or Certificate Revocation List (CRL) method to verify the revocation status of certificates.

   If the firewall will use both, it first tries OCSP; if the OCSP responder is unavailable, the firewall then tries the CRL method.

   - In the OCSP section, select **Use OCSP to check certificate status**, and specify a **Receive Timeout (sec)** value. This is the interval (1-60 seconds) after which the firewall stops waiting for a response from the OCSP responder.
   - In the CRL section, select **Use CRL to check certificate status**, and specify a **Receive Timeout (sec)** value. This is the interval (1-60 seconds) after which the firewall stops waiting for a response from the CRL service.

   Depending on the **Certificate Status Timeout** value you specify in the next step, the firewall might register a timeout before either or both of the **Receive Timeout (sec)** intervals pass.
4. **Save** the Advanced Decryption settings.

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STEP 2 | Define the total timeout interval for revocation status requests.

This is the interval (1-60 seconds) after which the firewall stops waiting for a response from any certificate status service and applies the session-blocking logic you optionally define in step 3.

1. For **Certificate Status Timeout (sec)**, specify a value between 1-60.

   The **Certificate Status Timeout** relates to the OCSP/CRL **Receive Timeout (sec)** as follows:

   - If you enable both OCSP and CRL: The firewall registers a request timeout after the lesser of two intervals passes: the **Certificate Status Timeout (sec)** value or the aggregate of the two **Receive Timeout (sec)** values.
   - If you enable only OCSP: The firewall registers a request timeout after the lesser of two intervals passes: the **Certificate Status Timeout (sec)** value or the OCSP **Receive Timeout (sec)** value.
   - If you enable only CRL: The firewall registers a request timeout after the lesser of two intervals passes: the **Certificate Status Timeout (sec)** value or the CRL **Receive Timeout (sec)** value.
2. **Save** the Advanced Decryption Settings.
3. **Save** the Decryption Settings.

STEP 3 | Define the blocking behavior for a certificate status of “unknown” or a revocation status request timeout.

1. Select **Configuration** > **NGFW and Prisma Access** > **Security Services** > **Decryption**, and select an existing Decryption Profile or create a new one.
2. Edit the **Server Certificate Verification** settings for SSL/TLS Decryption.

   Select **Advanced** under the Bypass Checks option. An Advanced SSL Forward Proxy Settings overlay shows the block settings of interest.

   - To block SSL/TLS sessions when the OCSP or CRL service returns a certificate revocation status of “unknown,” select **Block sessions with unknown certificate status**. Otherwise, the firewall proceeds with the session.
   - To block SSL/TLS sessions after the firewall registers a request timeout, select **Block sessions on certificate status check timeout**. Otherwise, the firewall proceeds with the session.
3. **Save** the Advanced SSL Forward Proxy settings.
4. **Save** the Decryption Profile.

STEP 4 | To commit your changes, click **Push Config** > **Push**.

Configure Revocation Status Verification of Certificates Used for SSL/TLS Decryption (PAN-OS & Panorama)

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STEP 1 | Define the service-specific timeout intervals for revocation status requests.

1. Select **Device** > **Setup** > **Session** and, in the Session Features section, select **Decryption Certificate Revocation Settings**.
2. Perform one or both of the following steps, depending on whether the firewall will use Online Certificate Status Protocol (OCSP) or the Certificate Revocation List (CRL) method to verify the revocation status of certificates. If the firewall will use both, it first tries OCSP; if the OCSP responder is unavailable, the firewall then tries the CRL method.
   - In the CRL section, select the **Enable** check box and enter the **Receive Timeout**. This is the interval (1-60 seconds) after which the firewall stops waiting for a response from the CRL service.
   - In the OCSP section, select the **Enable** check box and enter the **Receive Timeout**. This is the interval (1-60 seconds) after which the firewall stops waiting for a response from the OCSP responder.

   Depending on the **Certificate Status Timeout** value you specify in step 2, the firewall might register a timeout before either or both of the **Receive Timeout** intervals pass.

STEP 2 | Define the total timeout interval for revocation status requests.

1. Enter the **Certificate Status Timeout**.

   This is the interval (1-60 seconds) after which the firewall stops waiting for a response from any certificate status service and applies the session-blocking logic you optionally define in step 3. The **Certificate Status Timeout** relates to the OCSP/CRL **Receive Timeout** as follows:
   - If you enable both OCSP and CRL—The firewall registers a request timeout after the lesser of two intervals passes: the **Certificate Status Timeout** value or the aggregate of the two **Receive Timeout** values.
   - If you enable only OCSP—The firewall registers a request timeout after the lesser of two intervals passes: the **Certificate Status Timeout** value or the OCSP **Receive Timeout** value.
   - If you enable only CRL—The firewall registers a request timeout after the lesser of two intervals passes: the **Certificate Status Timeout** value or the CRL **Receive Timeout** value.
2. Click **OK**.

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STEP 3 | (Optional) Define the blocking behavior for a certificate status of "unknown" or a revocation status request timeout.

1. Select **Objects** > **Decryption** > **Decryption Profile**, and select an existing profile or create a new one.
2. Edit the SSL Forward Proxy **Server Certificate Verification** settings.
   - To block SSL/TLS sessions when the OCSP or CRL service returns a certificate revocation status of "unknown," select **Block sessions with unknown certificate status**. Otherwise, the firewall proceeds with the session.
   - To block SSL/TLS sessions after the firewall registers a request timeout, select **Block sessions on certificate status check timeout**. Otherwise, the firewall proceeds with the session.
3. Click **OK**.

STEP 4 | **Commit** your changes.

Enable an HTTP Proxy for OCSP Status Checks

If your network deployment consists of a web proxy, you can configure Online Certificate Status Protocol (OCSP) to validate certificates. All OCSP requests and responses will pass through your proxy server. The benefits of checking certificate status using OCSP instead of or in addition to certificate revocation lists (CRLs) include real-time status responses and reduced usage of network and client resources.

The workflow of OCSP certificate validation through a web proxy is as follows:

1. An authenticating client (firewall) forwards an OCSP request to the proxy. The request contains the serial number for the certificate the client wants to validate.
2. The proxy validates the request and identifies the OCSP responder for the certificate authority (CA) that issued the certificate.
3. The proxy forwards the OCSP request to the responder, and the OCSP responder looks up the revocation status for the certificate in the CA database.
4. The OCSP responder sends the certificate status (`good`, `revoked`, or `unknown`) to the proxy.
5. The proxy forwards the certificate status to the client.

The following procedure assumes you have not set up a web proxy.

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STEP 1 | Configure a proxy server.

1. Go to **Device** > **Setup** > **Services**, and edit the Services settings.
2. Edit the Proxy Server settings.
   - For **Server**, enter the IP address or host name of the proxy server.
   - Enter a **Port**.
   - For **User**, enter a username that an administrator enters to access the proxy server.
   - Enter and confirm a **Password** that an administrator enters to access the proxy server.

   You can also use the following CLI commands to configure your proxy server for OCSP status checks (and CRL downloads).
   - **set deviceconfig system secure-proxy-server <*value*>**
   - **set deviceconfig system secure-proxy-port <*1-65535*>**
   - **set deviceconfig system secure-proxy-user <*value*>**
   - **set deviceconfig system secure-proxy-password <*value*>**

STEP 2 | Configure an OCSP responder.

STEP 3 | Configure revocation status verification of certificates.

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Obtain and Import Certificates

You can obtain certificates from your enterprise PKI, external CA or third-party, or generate one on firewall.

- **Obtain certificates from a trusted third-party CA**—You can obtain certificates from trusted third-party certificate authorities (CAs) through a formal request process. This process includes submitting a certificate signing request (CSR) with a server's public key, identifying information about your organization, and the Common Name of the server or website.

  The benefit of obtaining a certificate from a trusted third-party certificate authority (CA) such as VeriSign or GoDaddy is that end clients will already trust the certificate because common browsers include root CA certificates from well-known CAs in their trusted root certificate stores. For applications requiring end clients to establish secure connections with the firewall or Panorama, purchase a certificate from a CA that end clients trust to avoid predeploying root CA certificates to the end clients. Applications this applies to are GlobalProtect™ portal or GlobalProtect Mobile Security Manager. However, most third-party CAs can't issue signing certificates, making this type of certificate inappropriate for applications, such as SSL/TLS decryption and Large Scale VPN, that require the firewall to issue certificates. See Obtain a Certificate from an External CA.
- **Obtain certificates from an enterprise CA**—If your organization maintains its own public key infrastructure (PKI), you can import certificates and private keys directly from your enterprise certificate authority (CA). The benefit is that end clients probably already trust the enterprise CA.

  Enterprise CA certificates offer the advantage of automatically issuing certificates for applications such as SSL/TLS decryption or GlobalProtect Large Scale VPN deployments, unlike most third-party commercial certificates. You can either generate the needed certificates and import them onto the firewall, or generate a certificate signing request (CSR) on the firewall and send it to the enterprise CA for signing. A benefit of this method is that the private key doesn't leave the firewall. See Import a Certificate and Private Key.

  If you have a Simple Certificate Enrollment Protocol (SCEP) server in your enterprise PKI, you can automate the generation and distribution of unique client certificates using SCM. See Deploy Certificates Using SCEP.
- **Generate self-signed certificates**—A self-signed root CA certificate sits at the top of a certificate chain hierarchy. Firewalls can use these certificates to automatically issue subordinate certificates for various purposes, including SSL/TLS decryption and GlobalProtect Large Scale VPN satellites. Before generating a certificate, import or create a self-signed root CA certificate to sign it.

  When you use this method to generate certificates for an application that requires an end client to trust the certificate, end users will see a certificate error because the root CA certificate is not in their trusted root certificate store. To prevent this, deploy the self-signed root CA certificate to all end-user systems. You can deploy the certificates manually or use a centralized deployment method such as an Active Directory Group Policy Object (GPO).
- **Acquire a device certificate**–For certain cloud services, you'll need to install a device certificate onto your NGFW. Device certificates expire after 90 days. A firewall with the device certificate

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installed automatically attempts to reinstall the device certificate 15 days before the certificate expires. However, you can manually restore an expired device certificate if it fails to do so automatically.

Create a Self-Signed Root CA Certificate

A self-signed root certificate authority (CA) certificate is the top-most certificate in a certificate chain. A firewall can use this certificate to automatically issue certificates for other uses. For example, the firewall issues certificates for SSL/TLS decryption and for satellites in a GlobalProtect large-scale VPN.

When establishing a secure connection with the firewall, the remote client must trust the root CA that issued the certificate. Otherwise, the client browser will display a warning that the certificate is invalid and might (depending on security settings) block the connection. To prevent this, after generating the self-signed root CA certificate, import it into the client systems.

On a Palo Alto Networks firewall or Panorama, you can generate self-signed certificates only if they are CA certificates.

STEP 1 | Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later).

STEP 2 | If the firewall has more than one virtual system (vsys), select a **Location** (vsys or **Shared**) for the certificate.

STEP 3 | Click **Generate**.

STEP 4 | Enter a **Certificate Name**, such as **`GlobalProtect_CA`**. The name is case-sensitive and can have up to 63 characters on the firewall or up to 31 characters on Panorama. It must be unique and use only letters, numbers, hyphens, and underscores.

STEP 5 | In the **Common Name** field, enter the FQDN (recommended) or IP address of the interface where you will configure the service that will use this certificate.

STEP 6 | If the firewall has more than one vsys and you want the certificate to be available to every vsys, select the **Shared** check box.

STEP 7 | Leave the **Signed By** field blank to designate the certificate as self-signed.

STEP 8 | (Required) Select the **Certificate Authority** check box.

STEP 9 | Leave the **OCSP Responder** field blank; revocation status verification doesn't apply to root CA certificates.

STEP 10 | Click **Generate** and **Commit**.

Generate a Certificate

Palo Alto Networks firewalls and Panorama use certificates to authenticate clients, servers, users, and devices in several applications, including SSL/TLS decryption, Authentication Portal, GlobalProtect™, site-to-site IPSec VPN, and web interface access to the firewall or Panorama. Generate certificates for each usage: for details, see Keys and Certificates.

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To generate a certificate, first create a self-signed root CA certificate or import one (Import a Certificate and Private Key) to sign it. To use the Online Certificate Status Protocol (OCSP) to verify certificate revocation status, configure an OCSP responder before generating the certificate.

- Strata Cloud Manager
- PAN-OS & Panorama

Generate Certificate (Strata Cloud Manager)

STEP 1 | Log in to Strata Cloud Manager.

STEP 2 | Select **Configuration** > **NGFW and Prisma Access** > **Objects** > **Certificate Management**.

STEP 3 | In the Custom Certificates section, click **Generate**.

STEP 4 | Enter a unique **Certificate Name**.

The name is case-sensitive and can have up to 63 characters. Use only letters, numbers, hyphens, and underscores.

STEP 5 | For **Common Name**, enter the FQDN (recommended) or IP address of the interface where you will configure the service that uses this certificate.

STEP 6 | For **Signed By**, select the root CA certificate that will issue the certificate.

STEP 7 | To allow an NGFW to issue the certificate, enable **Certificate Authority**.

Marking this certificate as a CA grants it the ability to sign other certificates.

STEP 8 | For **Certificate Use for**, select **Forward Trust Certificate**, **Forward UnTrust Certificate**, or **Trusted Root CA**.

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STEP 9 | Configure the Cryptographic Settings.

1. Select a key generation **Algorithm**:
   - **RSA** (default)
   - **Elliptic Curve DSA** (ECDSA)

     The Thales CipherTrust Manager integration with PAN-OS does not support ECDSA keys.

     ECDSA is recommended for client browsers and operating systems that support it.
   - **ML-DSA** (PAN-OS 12.1.2 and later)
   - **SLH-DSA** (PAN-OS 12.1.2 and later)
2. (SLH-DSA only) Select **Algorithm Parameters**.

   These parameters define the hash function family (SHA2 or SHAKE), NIST security level (128 bits, 192 bits, or 256 bits), and digital signature size of the scheme.
3. Select the **Number of Bits** (certificate key length):

   Higher numbers are more secure but require more processing time.

   ***SLH-DSA** bits are preselected based on the **Algorithm Parameters**.*

   - For **RSA**, select **512** bits, **1024** bits, **2048** bits, **3072** bits, or **4096** bits.

     *If a managed NGFW is in FIPS-CC mode, RSA keys must be either **2048** or **3072** bits.*
   - For **Elliptic Curve DSA**, select either **256** bits or **384** bits.
   - For **ML-DSA**, select **10496** bits, **15616** bits, or **20736** bits.
4. Select a **Digest** algorithm.

   From most to least secure, the options are: **sha512**, **sha384**, **sha256** (default), **sha1**, and **md5**.

   *If you use client certificates for firewall services that rely on TLSv1.2, do not select **sha512**. Use **sha384** or lower. If **sha512** is required, set **Max Version** to **TLSv1.1** in the SSL/TLS service profiles for the services.*

   - For **RSA**, select **md5**, **sha1**, **sha256**, **sha384**, or **sha512**.

     *If a managed NGFW is in FIPS-CC mode, you must select **sha256**, **sha384**, or **sha512**.*
   - For **Elliptic Curve DSA**, **ML-DSA**, or **SLH-DSA**, select **sha256**, **sha384**, or **sha512**.
5. For **Expiration**, enter the number of days (default is 365) the certificate is valid.

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STEP 10 | (Optional) **Add (+)** Certificate Attributes to uniquely identify the NGFW and service that uses the certificate.

> *If you add a **Host Name** (DNS name) attribute, match it to the **Common Name**, because the hostname populates the Subject Alternate Name (SAN) field of the certificate and some browsers require the SAN to specify the domains the certificate protects; in addition, the **Host Name** matching the **Common Name** is mandatory for GlobalProtect.*

STEP 11 | (Optional) Select an **OCSP Responder**.

STEP 12 | **Save** the certificate.

The certificate displays in the Custom Certificates list.

STEP 13 | To commit your changes, select **Push Config** > **Push**.

Generate Certificate (PAN-OS and Panorama)

STEP 1 | Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later).

STEP 2 | For **Certificate Type**, select **Local** (default) unless you want to deploy **SCEP** certificates to GlobalProtect endpoints.

STEP 3 | Enter a unique **Certificate Name**.

Names are case-sensitive and can use up to 63 characters on the firewall or 31 characters on Panorama. Use only letters, numbers, hyphens, and underscores.

STEP 4 | If the firewall has more than one virtual system (vsys), select a **Location** (vsys or **Shared**) for the certificate.

To share the certificate across all vsys, enable **Shared**.

STEP 5 | For **Common Name**, enter the FQDN (recommended) or IP address of the interface where you configure the service using this certificate.

STEP 6 | For **Signed By**, select the root CA certificate that will issue the certificate.

STEP 7 | To allow the firewall to issue the certificate, enable **Certificate Authority**.

Marking this certificate as a CA grants it the ability to sign other certificates on the firewall.

STEP 8 | (Optional) Enable **Block Private Key Export**.

Enable this setting to prevent the private key from being exported when you export the certificate.

> *If you enable this setting, you must manually import the associated private key if you import the certificate to Panorama or to other firewalls. For firewalls managed by Panorama, the private key is required to successfully push configuration changes to managed firewalls that you imported the certificate to.*

STEP 9 | (Optional) Select an **OCSP Responder**.

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STEP 10 | Select a key generation **Algorithm**:

- **RSA** (default)
- **Elliptic Curve DSA** (ECDSA)

  The Thales CipherTrust Manager integration with PAN-OS does not support ECDSA keys.

  ECDSA is recommended for client browsers and operating systems that support it.
- (PAN-OS 12.1.2 and later) **ML-DSA**
- (PAN-OS 12.1.2 and later) **SLH-DSA**

STEP 11 | (SLH-DSA only) Select **Algorithm Parameters** for the hash-based signature scheme.

These parameters define the hash function family (SHA2 or SHAKE), NIST security level (128 bits, 192 bits, or 256 bits), and digital signature size of the scheme.

STEP 12 | Select the **Number of Bits** to define the certificate key length.

Higher numbers are more secure but require more processing time.

***SLH-DSA** bits are preselected based on the **Algorithm Parameters**.*

- For **RSA**, select **512** bits, **1024** bits, **2048** bits, **3072** bits, or **4096** bits.

  *If the NGFW is in FIPS-CC mode, the RSA keys generated must be either **2048** or **3072** bits.*
- For **Elliptic Curve DSA**, select either **256** bits or **384** bits.
- (PAN-OS 12.1.2 and later) For **ML-DSA**, select **10496** bits, **15616** bits, or **20736** bits.

STEP 13 | Select a **Digest** algorithm.

From most to least secure, the options are: **sha512**, **sha384**, **sha256** (default), **sha1**.

*If you use client certificates for firewall services that rely on TLSv1.2, do not select **sha512**. Use **sha384** or lower. If **sha512** is required, set **Max Version** to **TLSv1.1** in the SSL/TLS service profiles for the services.*

- For **RSA**, select **sha1**, **sha256**, **sha384**, or **sha512**.

  *If your NGFW is in FIPS-CC mode, you must select **sha256**, **sha384**, or **sha512**.*
- For **Elliptic Curve DSA**, select **sha256**, **sha384**, or **sha512**.
- (PAN-OS 12.1.2 and later) For **ML-DSA** and **SLH-DSA**, select **sha256**, **sha384**, or **sha512**.

STEP 14 | For **Expiration**, enter the number of days (default is 365) the certificate is valid.

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STEP 15 | (Optional) **Add Certificate Attributes** to uniquely identify the firewall and the service that uses the certificate.

*If you add a **Host Name** (DNS name) attribute, match it to the **Common Name**, because the hostname populates the Subject Alternate Name (SAN) field of the certificate and some browsers require the SAN to specify the domains the certificate protects; in addition, the **Host Name** matching the **Common Name** is mandatory for GlobalProtect.*

STEP 16 | Click **Generate** and, in the Device Certificates page, click the certificate Name.

Regardless of the time zone on the firewall, it always displays the corresponding Greenwich Mean Time (GMT) for certificate validity and expiration dates and times.

STEP 17 | Specify the intended use of the certificate.

For example, if the firewall will use this certificate to secure forwarding of syslogs to an external syslog server, enable **Certificate for Secure Syslog**.

STEP 18 | Click **OK** and **Commit**.

Obtain a Certificate from an External CA

The advantage of obtaining a certificate from an external certificate authority (CA) is that the private key does not leave the firewall. To obtain a certificate from an external CA, generate a certificate signing request (CSR) and submit it to the CA. After the CA issues a certificate with the specified attributes, import it onto the firewall. The CA can be a well-known, public CA or an enterprise CA.

To use Online Certificate Status Protocol (OCSP) for verifying the revocation status of the certificate, configure an OCSP responder before generating a CSR.

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STEP 1 | Request the certificate from an external CA.

1. Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later).
2. If the firewall has more than one virtual system (vsys), select a **Location** (vsys or **Shared**) for the certificate.
3. Click **Generate**.
4. Enter a **Certificate Name**. The name is case-sensitive and can have up to 63 characters on the firewall or up to 31 characters on Panorama. It must be unique and use only letters, numbers, hyphens, and underscores.
5. In the **Common Name** field, enter the FQDN (recommended) or IP address of the interface where you will configure the service that will use this certificate.
6. If the firewall has more than one vsys and you want the certificate to be available to every vsys, select the **Shared** check box.
7. In the **Signed By** field, select **External Authority (CSR)**.
8. If applicable, select an **OCSP Responder**.
9. (Optional) **Add** the **Certificate Attributes** to uniquely identify the firewall and the service that will use the certificate.

   *If you add a **Host Name** attribute, it should match the **Common Name** (this is mandatory for GlobalProtect). The host name populates the Subject Alternative Name field of the certificate.*
10. Click **Generate**. The **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later) tab displays the CSR with a Status of `pending`.

STEP 2 | Submit the CSR to the CA.

1. Select the CSR and click **Export** to save the .csr file to a local computer.
2. Upload the .csr file to the CA.

STEP 3 | Import the certificate.

1. After the CA sends a signed certificate in response to the CSR, return to the **Device Certificates** tab and click **Import**.
2. Enter the **Certificate Name** used to generate the CSR.
3. Enter the path and name of the PEM **Certificate File** that the CA sent, or **Browse** to it.
4. Click **OK**. The **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later) tab displays the certificate with a Status of `valid`.

STEP 4 | Configure the certificate.

1. Click the certificate **Name**.
2. Select the check boxes that correspond to the intended use of the certificate on the firewall. For example, if the firewall will use this certificate to secure forwarding of syslogs to an external syslog server, select the **Certificate for Secure Syslog** check box.
3. Click **OK** and **Commit**.

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Certificate Deployment

Certificate deployment is the installation of certificates and configuration of settings for use with applications or services. You deploy certificates after you have obtained them.

Depending on how you obtain a certificate, the installation and configuration looks different.

- **Obtain certificates from a trusted third-party CA**—You can obtain certificates from trusted third-party certificate authorities (CAs) through a formal request process. This process includes submitting a certificate signing request (CSR) with a server's public key, identifying information about your organization, and the Common Name of the server or website.

  The benefit of obtaining a certificate from a trusted third-party certificate authority (CA) such as VeriSign or GoDaddy is that end clients will already trust the certificate because common browsers include root CA certificates from well-known CAs in their trusted root certificate stores. For applications requiring end clients to establish secure connections with the firewall or Panorama, purchase a certificate from a CA that end clients trust to avoid predeploying root CA certificates to the end clients. Applications this applies to are GlobalProtect™ portal or GlobalProtect Mobile Security Manager. However, most third-party CAs can't issue signing certificates, making this type of certificate inappropriate for applications, such as SSL/TLS decryption and Large Scale VPN, that require the firewall to issue certificates. See Obtain a Certificate from an External CA.

- **Obtain certificates from an enterprise CA**—If your organization maintains its own public key infrastructure (PKI), you can import certificates and private keys directly from your enterprise certificate authority (CA). The benefit is that end clients probably already trust the enterprise CA.

  Enterprise CA certificates offer the advantage of automatically issuing certificates for applications such as SSL/TLS decryption or GlobalProtect Large Scale VPN deployments, unlike most third-party commercial certificates. You can either generate the needed certificates and import them onto the firewall, or generate a certificate signing request (CSR) on the firewall and send it to the enterprise CA for signing. A benefit of this method is that the private key doesn't leave the firewall. See Import a Certificate and Private Key.

  If you have a Simple Certificate Enrollment Protocol (SCEP) server in your enterprise PKI, you can automate the generation and distribution of unique client certificates using SCM. See Deploy Certificates Using SCEP.

- **Generate self-signed certificates**—A self-signed root CA certificate sits at the top of a certificate chain hierarchy. Firewalls can use these certificates to automatically issue subordinate certificates for various purposes, including SSL/TLS decryption and GlobalProtect Large Scale VPN satellites. To generate a certificate, first create a self-signed root CA certificate, and then generate a certificate.

  When you use this method to generate certificates for an application that requires an end client to trust the certificate, end users will see a certificate error because the root CA certificate is not in their trusted root certificate store. To prevent this, deploy the self-signed root CA certificate to all end-user systems. You can deploy the certificates manually or use a centralized deployment method such as an Active Directory Group Policy Object (GPO).

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For certain cloud services, you must install a device certificate onto your NGFW. (See Install or Restore a Device Certificate.) Device certificates expire after 90 days. A firewall with a device certificate installed automatically attempts to reinstall the device certificate 15 days before it expires. However, you can reinstall an expired device certificate manually if it fails to do so automatically.

Import a Certificate and Private Key

If your enterprise has its own public key infrastructure (PKI), you can import a certificate and private key into the firewall from your enterprise certificate authority (CA). Enterprise CA certificates (unlike most certificates purchased from a trusted, third-party CA) can automatically issue CA certificates for applications such as SSL/TLS decryption or large-scale VPN.

On a Palo Alto Networks firewall or Panorama, you can import self-signed certificates only if they are CA certificates.

Instead of importing a self-signed root CA certificate into all the client systems, it is a best practice to import a certificate from the enterprise CA because the clients will already have a trust relationship with the enterprise CA, which simplifies the deployment.

If the certificate you will import is part of a certificate chain, it is a best practice to import the entire chain.

STEP 1 | From the enterprise CA, export the certificate and private key that the firewall will use for authentication.

When exporting a private key, you must enter a passphrase to encrypt the key for transport. Ensure the management system can access the certificate and key files. When importing the key onto the firewall, you must enter the same passphrase to decrypt it.

STEP 2 | Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later).

STEP 3 | If the firewall has more than one virtual system (vsys), select a **Location** (vsys or **Shared**) for the certificate.

STEP 4 | Click **Import** and enter a **Certificate Name**. The name is case-sensitive and can have up to 63 characters on the firewall or up to 31 characters on Panorama. It must be unique and use only letters, numbers, hyphens, and underscores.

STEP 5 | To make the certificate available to all virtual systems, select the **Shared** check box. This check box appears only if the firewall supports multiple virtual systems.

STEP 6 | Enter the path and name of the **Certificate File** received from the CA, or **Browse** to find the file.

STEP 7 | Select a **File Format**:

- **Encrypted Private Key and Certificate (PKCS12)**—This is the default and most common format, in which the key and certificate are in a single container (**Certificate File**). If a

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hardware security module (HSM) will store the private key for this certificate, select the **Private key resides on Hardware Security Module** check box.

- **Base64 Encoded Certificate (PEM)**—You must import the key separately from the certificate. If a hardware security module (HSM) stores the private key for this certificate, select the **Private key resides on Hardware Security Module** check box and skip the next step. Otherwise, select the **Import Private Key** check box, enter the **Key File** or **Browse** to it, then continue to the next step.

  *(Panorama managed firewalls) You are required to **Import Private Key** if you enabled **Block Private Key Export** when the certificate was generated to successfully push configuration changes from the Panorama management server to managed firewalls.*

- (SD-WAN IKEv2 Certificate Authentication Support) (Beginning with SD-WAN 3.2.0 Release) **Multiple Certificates (.tar)**—Contains multiple certificates archived in a tar format.

  Use CSV file to bulk import the certificates into the Panorama management server. Follow these steps if you select **Multiple Certificates (.tar)**.

  1. **Download Sample CSV** to download and save the **Certificates.CSV** template. Populate the template with certificate related information: certificate name, format, passphrase, block private key, and file name.

     Following is a sample **Certificates.CSV**:

	A	B	C	D	E
1	#"certificate-name"	format	passphrase	block-priv-key	file-name
2	Root-Cert	pkcs12	paloalto	no	cert_Root-CA.p12
3	Hub-Cert	pkcs12	paloalto	no	cert_HubCertificate.p12
4	Branch-Cert	pkcs12	paloalto	yes	cert_BranchCertificate.p12

     - *Enter the certificate **file name** with an extension. The **file name** should exactly match the certificate that you will be uploading to the device.*
     - *All the certificate fields are case-sensitive except the **format** and **passphrase** field.*

  2. To import multiple certificates, archive all the certificates and the populated **Certificates.CSV** file in .tar format. **Browse** and select a **.tar file** to bulk import the **Certificate File**. Certificate format .PKCS12 is supported. The size of the archive(.tar) file should be less than 10MB. Ensure that the extension of the tar file is **.tar**.

     All the certificates must be signed by the same CA within the SD-WAN VPN cluster. You must import the CA certificate along with the device certificate. Device certificate must be directly signed by the root CA (no intermediate certificates are allowed).

  3. You must commit after bulk import for the certificates to be available for further configuration.

STEP 8 | Enter and re-enter (confirm) the **Passphrase** used to encrypt the private key.

STEP 9 | Click **OK**. The Device Certificates page displays the imported certificate.

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Install or Restore a Device Certificate

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ Outbound internet access ❑ Customer Support Portal (CSP) account with one of the following user roles: Super User, Standard User, Limited User, Threat Researcher, AutoFocus Trial Role, Group Super User, Group Standard User, Group Limited User, Group Threat Researcher, Authorized Support Center (ASC) User, and ASC Full Service User. ❑ Superuser privileges to the firewall

You must install the device certificate on your Next-Generation Firewall (NGFW) to use one or more cloud services. You only need to install a device certificate once. The device certificate has a 90-day lifetime. The firewall reinstalls the device certificate 15 days before the certificate expires.

For successful installation, the firewall must have outbound internet access and the following Fully Qualified Domain Names (FQDN) and ports must be allowed on your network to reach to the Customer Support Portal.

For Panorama managed firewalls, you can install the device certificate for managed firewalls from the Panorama management server. This enables you to install the device certificate for multiple managed firewalls at once.

FQDN	Ports
• http://ocsp.paloaltonetworks.com • http://crl.paloaltonetworks.com • http://ocsp.godaddy.com	TCP 80
• https://api.paloaltonetworks.com • https://apitrusted.paloaltonetworks.com • https://certificatetrusted.paloaltonetworks.com • https://certificate.paloaltonetworks.com	TCP 443

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The following Palo Alto Networks NGFW models automatically install the device certificate when they first connect to the Customer Support Portal during the initial registration process. You don't need to manually install the device certificate for these models:

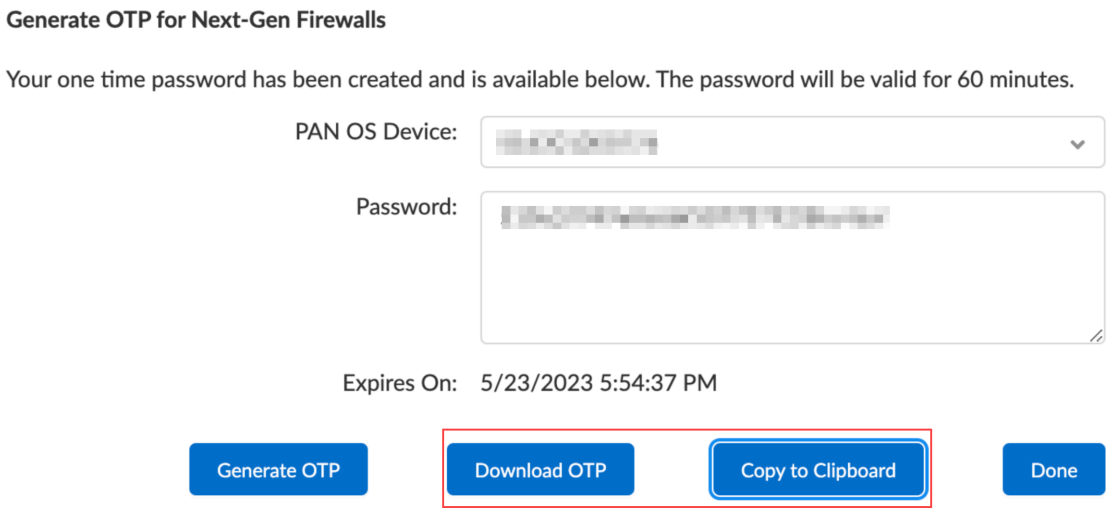
- *PA-400 Series firewalls*
- *PA-500 Series firewalls*
- *PA-1400 Series firewalls*
- *PA-3400 Series firewalls*
- *PA-5400 Series firewalls*
- *PA-5450 firewall*
- *PA-5500 Series firewalls*
- *PA-7500 Series firewalls*

STEP 1 | Generate a one-time password (OTP).

An OTP lifetime is 60 minutes and expires if not used within the 60-minute lifetime.

The firewall may only attempt to retrieve the OTP from the Customer Support Portal one time. If it fails to fetch the OTP for any reason, the OTP expires and you must generate a new OTP.

1. Log in to the Customer Support Portal with a user role that has permission to generate an OTP.
2. Select **Products** > **Device Certificates**, and then click **Generate OTP**.
3. For the **Device Type**, select **Generate OTP for a Next-Gen Firewall (PanOS)**, and then click **Next**.
4. Select your **PAN-OS Device** serial number, and then **Generate OTP**.
5. **Download OTP** or **Copy to Clipboard**.



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STEP 2 | Log in to the firewall web interface as a Superuser.

An admin with Superuser access privileges is required to apply the OTP used to install the device certificate.

STEP 3 | Configure the Network Time Protocol (NTP) server.

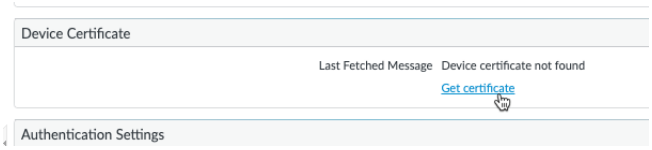
An NTP server is required to validate the device certification expiration date and ensure the device certificate does not expire early or become invalid.

1. Select **Device** > **Setup** > **Services** and edit the Services section.
2. Select **NTP** and enter the hostname or IP address of the **Primary NTP Server**.
3. (Optional) Enter the hostname or IP address of the **Secondary NTP Server**.
4. (Optional) To authenticate time updates from the NTP servers, for **Authentication Type**, select one of the following for each server:
   - **None** (default)—Disables NTP authentication.
   - **Symmetric Key**—Firewall uses symmetric key exchange (shared secrets) to authenticate time updates.
     - **Key ID**—Enter the Key ID (1-65534)
     - **Algorithm**—Select an algorithm to use for NTP authentication: **MD5**, **SHA1**, (PAN-OS 12.1.2 & later) **SHA256**, or **SHA512**.
   - **Autokey**—Firewall uses autokey (public key cryptography) to authenticate time updates.
5. Click **OK** to save your changes.
6. **Commit** your changes.

STEP 4 | Select **Device** > **Setup** > **Management** > **Device Certificate** and **Get certificate**.

You can also install the device certificate from the firewall CLI using the command:

```
admin>request certificate fetch otp <otp_value>
```



STEP 5 | Paste the **One-time Password** you generated and click **OK**.

STEP 6 | Your next-generation firewall successfully retrieves and installs the certificate.

You may need to refresh the page to verify that device certificate installation was successful.

STEP 7 | (WildFire and Advanced WildFire) Log in to the firewall CLI and refresh the firewall settings to establish a connection to the Advanced WildFire cloud with the updated device certificate.

```
admin>request wildfire registration channel public
```

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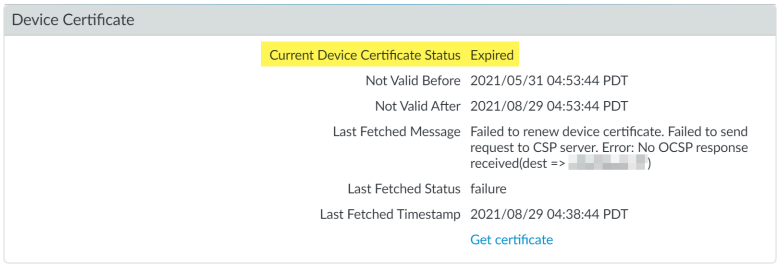
Restore an Expired Device Certificate

The device certificate installed on your firewall has a 90-day lifetime. A firewall with the device certificate installed automatically attempts to reinstall the device certificate 15 days before the certificate expires. However, reinstallation can fail for multiple reasons, including network connectivity issues. When a device certificate expires, the `Current Device Certificate Status` displays `Expired`. If this happens, you can manually reinstall the device certificate.

STEP 1 | Log in to the firewall web interface.

STEP 2 | Verify that the device certificate has expired.

Select **Device** > **Setup** > **Management**, and then check that the `Current Device Certificate Status` is `Expired`.



STEP 3 | Install a device certificate.

If the ***`request certificate fetch otp <otp_value>`*** *command isn't available, it means the firewall is a Trusted Platform Module (TPM) device.*

To restore the device certificate for a TPM device, run the following command:

`request certificate fetch`

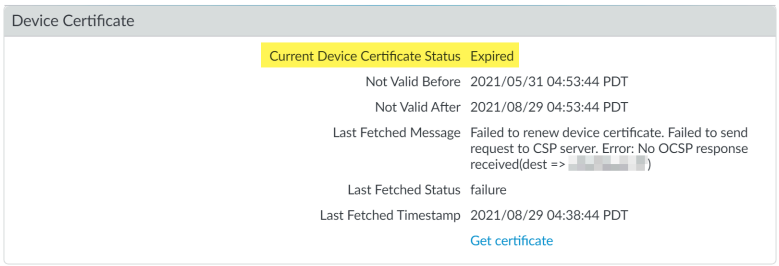
Restore an Expired Device Certificate

The device certificate installed on your firewall has a 90 day lifetime. A firewall with the device certificate installed automatically attempts to reinstall the device certificate 15 days before the certificate expires. However, you have the ability to manually reinstall the device certificate if it fails to reinstall automatically.

STEP 1 | Log in to the firewall web interface.

STEP 2 | Select **Device** > **Setup** > **Management** and review the `Current Device Certificate Status` in the Device Certificate Section.

The `Current Device Certificate Status` displays `Expired`.



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STEP 3 | Install a device certificate.

> *If the* ***request certificate fetch otp <otp_value>*** *command is not available, it means the firewall is a Trusted Platform Module (TPM) device.*
>
> *To restore the device certificate for a TPM device, run the following command:*
>
> ***request certificate fetch***

Deploy Certificates Using SCEP

If you have a Simple Certificate Enrollment Protocol (SCEP) server in your enterprise PKI, you can configure a SCEP profile to automate the generation and distribution of unique client certificates. SCEP operation is dynamic in that the enterprise PKI generates a user-specific certificate when the SCEP client requests it and sends the certificate to the SCEP client. The SCEP client then transparently deploys the certificate to the client device.

You can use a SCEP profile with GlobalProtect to assign user-specific client certificates to each GlobalProtect user. In this use case, the GlobalProtect portal acts as a SCEP client to the SCEP server in your enterprise PKI. Additionally, you can use a SCEP profile to assign client certificates to Palo Alto Networks devices for mutual authentication with other Palo Alto Networks devices for management access and inter-device communication.

STEP 1 | Create a SCEP profile.

1. Select **Device** > **Certificate Management** > **SCEP** and then **Add** a new profile.
2. Enter a **Name** to identify the SCEP profile.
3. If this profile is for a firewall with multiple virtual systems capability, select a virtual system or **Shared** as the **Location** where the profile is available.

STEP 2 | (**Optional**) To make the SCEP-based certificate generation more secure, configure a SCEP challenge-response mechanism between the PKI and portal for each certificate request.

After you configure this mechanism, its operation is invisible, and no further input from you is necessary.

To comply with the U.S. Federal Information Processing Standard (FIPS), use a **Dynamic** SCEP challenge and specify a **Server URL** that uses HTTPS.

Select one of the following options:

- **None**—(**Default**) The SCEP server does not challenge the portal before it issues a certificate.
- **Fixed**—Obtain the enrollment challenge password from the SCEP server in the PKI infrastructure and then enter the password into the Password field.
- **Dynamic**—Enter a username and password of your choice (possibly the credentials of the PKI administrator) and the SCEP **Server URL** where the portal-client submits these credentials. The uses the credentials to authenticate with the SCEP server which transparently generates an OTP password for the portal upon each certificate request. (You can see this OTP change after a screen refresh in `The enrollment challengepassword is` field after each certificate request.) The PKI transparently passes each new password to the portal, which then uses the password for its certificate request.

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STEP 3 | Specify the settings for the connection between the SCEP server and the portal to enable the portal to request and receive client certificates.

You can include additional information about the client device or user by specifying tokens in the **Subject** name of the certificate.

The portal includes the token value and host ID in the CSR request to the SCEP server.

1. Configure the **Server URL** that the portal uses to reach the SCEP server in the PKI (for example, `http://10.200.101.1/certsrv/mscep/`).
2. Enter a string (up to 255 characters in length) in the **CA-IDENT Name** field to identify the SCEP server.
3. Enter the **Subject** name to use in the certificates generated by the SCEP server. The subject must be a distinguished name in the **<attribute>=<value>** format and must include a common name (CN) attribute (`CN=<variable>`). The CN supports the following dynamic tokens:
   - `$USERNAME`—Use this token to enable the portal to request certificates for a specific user. To use this variable with GlobalProtect, you must also Enable Group Mapping. The username entered by the user must match the name in the user-group mapping table.
   - `$EMAILADDRESS`—Use this token to request certificates associated with a specific email address. To use this variable, you must also Enable Group Mapping and configure the **Mail Attributes** in the Mail Domains section of the Server Profile. If GlobalProtect cannot identify an email address for the user, it generates a unique ID and populates the CN with that value.
   - `$HOSTID`—To request certificates for the device only, specify the host ID token. When a user attempts to log in to the portal, the endpoint sends identifying information that includes its host ID value. The host ID value varies by device type, either GUID (Windows) MAC address of the interface (Mac), Android ID (Android devices), UDID (iOS devices), or a unique name that GlobalProtect assigns (Chrome).
   - `$UDID`—Use the UDID common name attribute to request certificates based on the client's device UDID for GlobalProtect or device serial number for mutual authentication between Palo Alto Networks devices.

   When the GlobalProtect portal pushes the SCEP settings to the agent, the CN portion of the subject name is replaced with the actual value (username, host ID, or email address) of the certificate owner (for example, **`O=acme,CN=johndoe`**).
4. Select the **Subject Alternative Name Type**:

   *Use static entries for the Subject Alternative Name Type. The firewall does not support dynamic tokens such as **$USERNAME**.*

   - **RFC 822 Name**—Enter the email name in a certificate's subject or Subject Alternative Name extension.
   - **DNS Name**—Enter the DNS name used to evaluate certificates.
   - **Uniform Resource Identifier**—Enter the name of the resource from which the client will obtain the certificate.
   - **None**—Do not specify attributes for the certificate.

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STEP 4 | (**Optional**) Configure cryptographic settings for the certificate.

- Select the key length (**Number of Bits**) for the certificate.

  If the firewall is in FIPS-CC mode and the key generation algorithm is RSA. The RSA keys must be 2,048 bits or larger.
- Select the **Digest for CSR** which indicates the digest algorithm for the certificate signing request (CSR): sha1, sha256, or sha384.

STEP 5 | (**Optional**) Configure the permitted uses of the certificate, either for signing or encryption.

- To use this certificate for signing, select the **Use as digital signature** check box. This enables the endpoint use the private key in the certificate to validate a digital signature.
- To use this certificate for encryption, select the **Use for key encipherment** check box. This enables the client use the private key in the certificate to encrypt data exchanged over the HTTPS connection established with the certificates issued by the SCEP server.

STEP 6 | (**Optional**) To ensure that the portal is connecting to the correct SCEP server, enter the **CA Certificate Fingerprint**. Obtain this fingerprint from the SCEP server interface in the Thumbprint field.

1. Enter the URL for the SCEP server's administrative UI (for example, **`http://<hostname or IP>/CertSrv/mscep_admin/`**).
2. Copy the thumbprint and enter it in the **CA Certificate Fingerprint** field.

STEP 7 | Enable mutual SSL authentication between the SCEP server and the firewall. This is required to comply with the U.S. Federal Information Processing Standard (FIPS).

FIPS-CC operation is indicated on the firewall login page and in its status bar.

Select the SCEP server's root **CA Certificate**. Optionally, you can enable mutual SSL authentication between the SCEP server and the firewall by selecting a **Client Certificate**.

STEP 8 | Save and commit the configuration.

1. Click **OK** to save the settings and close the SCEP configuration.
2. **Commit** the configuration.

The portal attempts to request a CA certificate using the settings in the SCEP profile and saves it to the firewall hosting the portal. If successful, the CA certificate is shown in **Device** > **Certificate Management** > **Certificates**.

STEP 9 | (**Optional**) If after saving the SCEP profile, the portal fails to obtain the certificate, you can manually generate a certificate signing request (CSR) from the portal.

1. Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later). Select **Generate**.
2. Enter a **Certificate Name**. This name cannot contain spaces.
3. Select the **SCEP Profile** to use to submit a CSR to your enterprise PKI.
4. Click **OK** to submit the request and generate the certificate.

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Export a Certificate and Private Key

Palo Alto Networks recommends that you use your enterprise public key infrastructure (PKI) to distribute a certificate and private key in your organization. However, if necessary, you can also export a certificate and private key from the firewall or Panorama. You can use an exported certificate and private key in the following cases:

- Configure Certificate-Based Administrator Authentication to the Web Interface
- Enable SSL Between GlobalProtect LSVPN Components to configure GlobalProtect agent/app authentication to portals and gateways
- SSL Forward Proxy decryption
- Obtain a Certificate from an External CA

STEP 1 | Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later).

STEP 2 | If the firewall has more than one virtual system (vsys), select a **Location** (a specific vsys or **Shared**) for the certificate.

STEP 3 | Select the certificate, click **Export**, and select a **File Format**:

- **Base64 Encoded Certificate (PEM)**—This is the default format. It is the most common and has the broadest support on the Internet. If you want the exported file to include the private key, select the **Export Private Key** check box.
- **Encrypted Private Key and Certificate (PKCS12)**—This format is more secure than PEM but is not as common or as broadly supported. The exported file will automatically include the private key.
- **Binary Encoded Certificate (DER)**—More operating system types support this format than the others. You can export only the certificate, not the key: ignore the **Export Private Key** check box and passphrase fields.

STEP 4 | Enter a **Passphrase** and **Confirm Passphrase** to encrypt the private key if the **File Format** is PKCS12 or if it is PEM and you selected the **Export Private Key** check box. You will use this passphrase when importing the certificate and key into client systems.

> *(Panorama managed firewalls) If you enabled **Block Private Key Export** when you generated or imported the certificate, you must be sure to **Import Private Key** and add the **key File** when you import the exported certificate. This is required to successfully push configuration changes from Panorama to managed firewalls that you imported the certificate to.*

STEP 5 | Click **OK** and save the certificate/key file to your computer.

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Blocking Private Key Export

You can permanently block the export of private keys for certificates that you generate or import onto PAN-OS appliances or Panorama. Blocking the export of private keys from your PAN-OS devices hardens your security posture because it prevents rogue administrators or other bad actors from misusing keys. Administrators with roles that include certificate management can block the export of private keys. You can't block keys that already exist on a device; you can only block keys at the time that you generate them in or import them into PAN-OS.

When an administrator blocks the export of a private key, no administrator can export that key, not even Superuser administrators. If you need to export a private key from a PAN-OS appliance, regenerate the certificate and the key without selecting the option to block private key export.

Blocking the export of private keys is supported on PAN-OS version 10.1 or later releases.

To downgrade to an earlier version of PAN-OS, first delete the certificates whose private keys are blocked. If you don't delete the certificates whose private keys are blocked before you attempt to downgrade, an error message asks you to delete those certificates. You can't downgrade until you delete them. After you downgrade, reimport or regenerate the deleted certificates if you need them.

If you use an enterprise Public Key Infrastructure (PKI) to generate certificates and private keys, block the export of private keys because you can install them on new firewalls and Panoramas from your enterprise certificate authority (CA), so there is no reason to export them from PAN-OS.

If you generate self-signed certificates on the firewall or Panorama and apply the block private key export option, you can't export the certificate and key to other PAN-OS appliances.

You can export and import the device state (**Device** > **Setup** > **Operations**) even if you block the export of private keys. We include the private keys in device state imports and exports, but administrators can't read or decode them.

You can import or load the configuration of one firewall on another firewall if the master key is the same on both firewalls. If the master key is different on the firewalls, then importing or loading the configuration doesn't work and the commit fails while reading the certificates.

Generate a Private Key and Block It

To prevent the misuse of a private key after generating a certificate, you can permanently block the export of the corresponding private key. You can only enable the **Block Private Key Export** option at the time of generating or importing a certificate onto PAN-OS.

If you generate self-signed certificates on the firewall or Panorama and apply the block private key export option, you can't export the certificate and key to other PAN-OS appliances.

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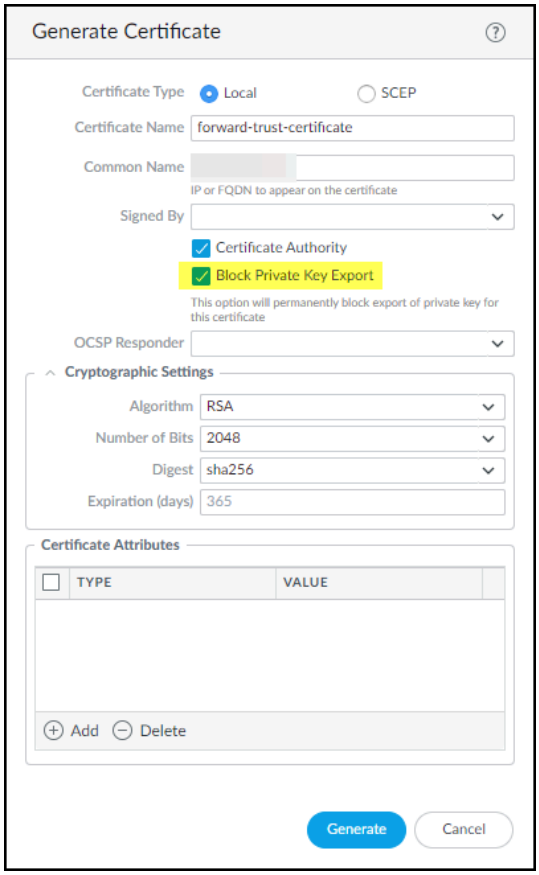
STEP 1 | Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later).

If there is more than one virtual system, select a **Location** or **Shared** for the certificate.

STEP 2 | **Generate** the certificate.

STEP 3 | Select **Block Private Key Export** to prevent anyone from exporting the certificate.

See Generate a Certificate for information about the other certificate fields.



STEP 4 | **Generate** the new certificate.

You can also generate a certificate and block its private key from export using the operational CLI command:

```
admin@pa-220> request certificate generate block-private-keys yes
```

The preceding CLI command can also include the certificate and other parameters that are not shown.

STEP 5 | Verify that you can't export private keys,

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Import a Private Key and Block It

To prevent the misuse of a private key after importing a certificate, you can permanently block the export of the corresponding private key. You can only enable the **Block Private Key Export** option at the time of generating or importing a certificate onto PAN-OS.

STEP 1 | Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later).

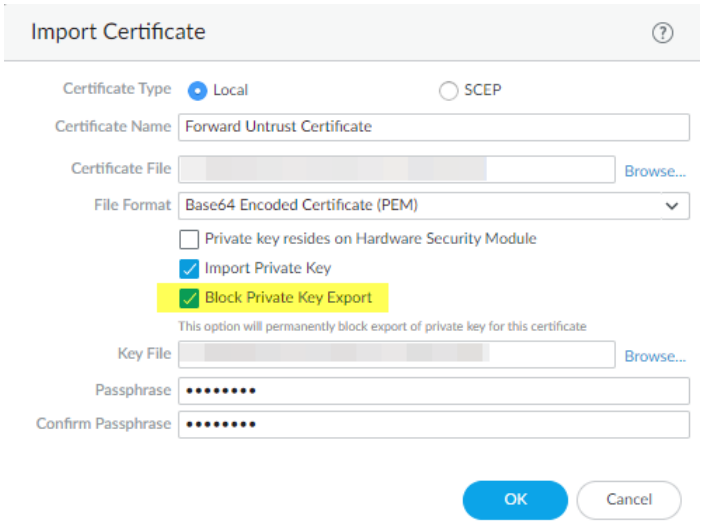
If there is more than one virtual system, select a **Location** or **Shared** for the certificate.

STEP 2 | **Import** the certificate.

STEP 3 | Select **Import Private Key** to activate the option to block private key export.

STEP 4 | Select **Block Private Key Export** to prevent anyone from exporting the certificate.

See Import a Certificate and Private Key for information about the other certificate import fields.



STEP 5 | Click **OK** to import the certificate.

If you use the SCP operational CLI command to import a certificate or to import a private key for a certificate, you can still block export of the private key:

- ***admin@pa-220> scp import private-key block-private-key ...***

Each of the preceding CLI commands can also include keywords to specify the source, the certificate name, and other parameters that are not shown.

*If you use the SCP operational CLI command to export a certificate and include its private key (**scp export certificate passphrase <phrase> remote-port <1-65536> to <destination> certificate-name <name> include-key <yes | no> format <der | pem | pkcs10 | pkcs12>**), and if the certificate's private key is blocked, the command fails and returns an error message because you cannot export a blocked private key.*

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STEP 6 | Verify that you can't export private keys,

Import a Private Key for IKE Gateway and Block It

Block the export of a private key to prevent its misuse after generating a certificate for IKE Gateway authentication.

STEP 1 | Select **Network** > **Network Profiles** > **IKE Gateways**.

STEP 2 | **Add** a new IKE Gateway.

STEP 3 | On the **General** tab, for **Authentication**, select **Certificate**.

STEP 4 | For **Local Certificate** select **Import** or **Generate** depending on whether you want to import an existing certificate or create a certificate.

STEP 5 | Enter the certificate information. If you are importing the certificate, select **Import Private Key** to activate the **Block Private Key Export** checkbox.

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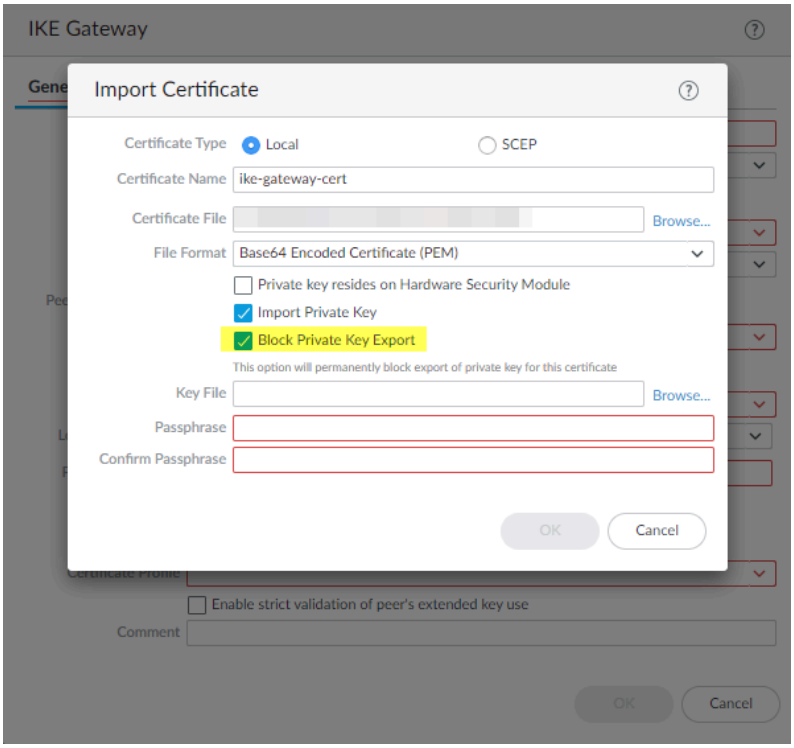
STEP 6 | Select **Block Private Key Export** to prevent anyone from exporting the key.

For importing a certificate, enter and confirm the **Passphrase** and then click **OK**



For generating a certificate, click **Generate**.

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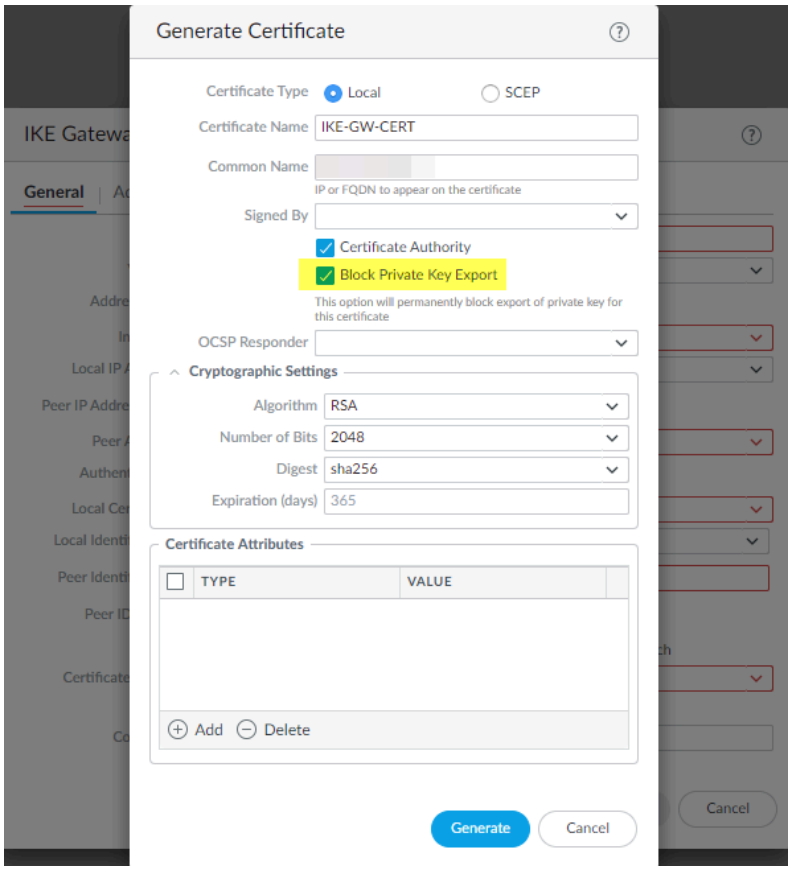

STEP 7 | Enter the **Passphrase**, confirm it, and then click **OK**.

STEP 8 | Verify that you can't export private keys,

Verify Private Key Blocking

Blocking the export of private keys from your PAN-OS devices hardens your security posture because it prevents rogue administrators or other bad actors from misusing private keys. As an extra precaution, after enabling **Block Private Key Export**, verify that the feature works. Use the following methods to confirm that you can't export the private key for the certificates to which you enabled this option:

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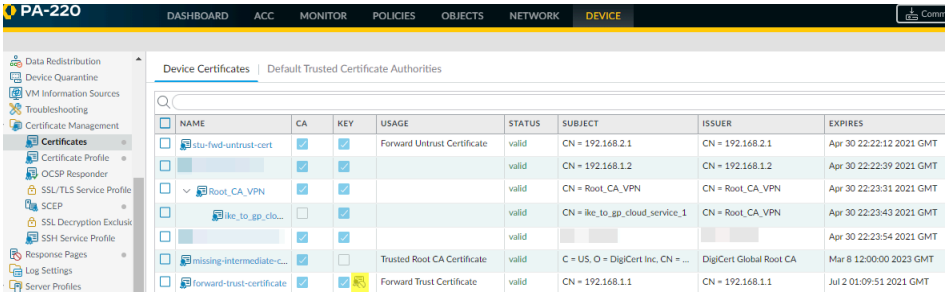
- Check the **Key** column in the **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later tab.

  Select **Device** > **Certificate Management** > **Certificates**, and then view **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later).

  In this example, the forward-trust-certificate is blocked:



- Check if the **Export Private Key** is available when you attempt to export a certificate whose private key is blocked from export.

  If the checkbox isn't available, you can't export the key, but you can export only the certificate, then you can be sure this feature is active.

- Use the following operational CLI command to list all certificates on the device or in a particular Vsys that have private keys blocked from export:

```
admin@pa-220> request certificate show-blocked <shared | vsys>
```

- Use the following operational CLI command to check whether a particular certificate's private key is blocked from export:

```
admin@pa-220> request certificate is-blocked certificate-name
 <name>
```

  If the certificate's private key is blocked from export, the command returns `yes`. If the certificate's private key is not blocked from export, the command returns `no`.

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Configure a Certificate Profile

Certificate profiles define user and device authentication for Authentication Portal, multi-factor authentication (MFA), GlobalProtect, site-to-site IPSec VPN, external dynamic list validation, dynamic DNS (DDNS), User-ID agent and TS agent access, and web interface access to Palo Alto Networks firewalls or Panorama. The profiles specify which certificates to use, how to verify certificate revocation status, and how that status constrains access. Configure a certificate profile for each application.

Enable Online Certificate Status Protocol (OCSP) and certificate revocation list (CRL) status verification in certificate profiles to verify that a certificate hasn't been revoked. Enable both OCSP and CRL so that if the OCSP server isn't available, the firewall uses CRL. For details on these methods, see Certificate Revocation.

STEP 1 | Obtain the certificate authority (CA) certificates you will assign to the profile.

Assign at least one certificate to a certificate profile. Choose one of the following options to obtain the CA certificates you will assign to the profile.

- Generate a certificate.
- Export a certificate from your enterprise CA, then import it onto the firewall (refer to step 3.2).

STEP 2 | Identify the certificate profile.

1. Select **Device** > **Certificate Management** > **Certificate Profile**, and then **Add** a profile.
2. Enter a **Name** to identify the profile.

   The name is case-sensitive, must be unique, and can use up to 63 characters on the firewall or up to 31 characters on Panorama. Only letters, numbers, spaces, hyphens, and underscores are allowed.
3. If the firewall has more than one virtual system (vsys), select a **Location** (vsys or **Shared**) for the certificate.

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STEP 3 | Assign one or more certificates to the profile.

Repeat the following steps for each CA certificate:

1. In the CA Certificates table, click **Add**.
2. Select a **CA Certificate**.

   Alternatively, you can **Import** a certificate. To import a certificate, enter a **Certificate Name**, **Browse** for a **Certificate File** you exported from an enterprise CA, and then click **OK**.
3. (**Optional**) If the firewall uses OCSP to verify certificate revocation status, configure the following fields to override the default behavior. For most deployments, these fields do not apply.
   - By default, the firewall uses the Authority Information Access (AIA) information from the certificate to extract the OCSP responder information. To override the AIA information, enter a **Default OCSP URL** (starting with **`http://`** or **`https://`**).
   - By default, the firewall uses the certificate selected in the **CA Certificate** field to validate OCSP responses. To use a different certificate for validation, select it in the **OCSP Verify CA Certificate** field.
4. Click **OK**. The CA Certificates table displays the assigned certificate.

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STEP 4 | Define the methods for verifying certificate revocation status and the associated blocking behavior.

1. Select **Use CRL** or **Use OCSP**. If you select both, the firewall first tries OCSP and falls back to the CRL method only if the OCSP responder is unavailable.
2. Depending on the verification method, specify a **CRL Receive Timeout** or **OCSP Receive Timeout** value in seconds (range is 1 to 60).

   After these intervals, the firewall stops waiting for a response from the CRL or OCSP service.
3. Specify a **Certificate Status Timeout** in seconds (range is 1 to 60).

   After this interval, the firewall stops waiting for a response from either certificate status service and applies any session-blocking logic you define. The **Certificate Status Timeout** relates to the OCSP or CRL **Receive Timeout** setting as follows:

   - If you enable both OCSP and CRL, the firewall registers a request timeout after the lesser of two intervals passes: the **Certificate Status Timeout** value or the aggregate of the two **Receive Timeout** values.
   - If you enable only OCSP, the firewall registers a request timeout after the lesser of two intervals passes: the **Certificate Status Timeout** value or the OCSP **Receive Timeout** value.
   - If you enable only CRL, the firewall registers a request timeout after the lesser of two intervals passes: the **Certificate Status Timeout** value or the CRL **Receive Timeout** value.
4. **Block sessions if certificate status is unknown**.

   If you select this option, the firewall blocks sessions when the OCSP or CRL service returns a certificate revocation status of unknown. Otherwise, the firewall allows these sessions.
5. **Block sessions if certificate status cannot be retrieved within timeout**.

   If you select this option, the firewall blocks sessions after the firewall registers the timeout of an OCSP or CRL request. Otherwise, the firewall allows these sessions.
6. (GlobalProtect only) **Block sessions if the certificate was not issued to the authenticating device**.

   If you select this option, the firewall blocks sessions when the serial number attribute in the subject of the client certificate does not match the host ID that the GlobalProtect app reports for the endpoint.
7. **Block sessions with expired certificates**.

STEP 5 | Click **OK**, then **Commit** your changes.

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Configure an SSL/TLS Service Profile

Where Can I Use This?	What Do I Need?
• NGFW (Cloud Managed) • NGFW (PAN-OS or Panorama Managed)	For cloud managed NGFWs: Strata Cloud Manager license

Palo Alto Networks firewalls and Panorama appliances use SSL/TLS to secure connections to the Authentication Portal, GlobalProtect portals and gateways, the management interface, HTTPS websites that require password access (URL Admin Override), and the User-ID™ syslog listening service. You can create an SSL/TLS service profile to define the server certificate, SSL/TLS protocol versions, and ciphers supported for connections to these services. Cipher suites are automatically selected based on the protocol versions chosen. However, you can disable individual ciphers as needed. If a service request involves a protocol version outside the specified range, the firewall or Panorama appliance downgrades or upgrades the connection to a supported version. To activate an SSL/TLS service profile, attach the profile to the settings for a specific service.

In the client systems that request firewall services, the certificate trust list (CTL) must include the certificate authority (CA) certificate that issued the certificate specified in the SSL/TLS service profile. Otherwise, users will see a certificate error when requesting firewall services. Most third-party CA certificates are present by default in client browsers. If an enterprise or firewall-generated CA certificate is the issuer, you must deploy that CA certificate to the CTL in client browsers.

TLSv1.3 support is limited to administrative access to management interfaces, GlobalProtect portals and gateways, and Authentication Portal. You can only attach SSL/TLS service profiles that allow TLSv1.3 to the settings for these services.

Post-quantum cryptography (PQC) certificates are not yet available for selection in SSL/TLS service profiles as they are for testing purposes only.

- Strata Cloud Manager
- PAN-OS & Panorama

Configure an SSL/TLS Service Profile (Strata Cloud Manager)

STEP 1 | Log in to Strata Cloud Manager.

STEP 2 | For each desired service, generate or import a certificate.

1. Select **Configuration** > **NGFW and Prisma Access** > **Objects** > **Certificate Management**.
2. In the Custom Certificates pane, **Generate** or **Import** a certificate.
3. **Save** the certificate.

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STEP 3 | Configure an SSL/TLS service profile.

1. Select **Configuration** > **NGFW and Prisma Access** > **Objects** > **Certificate Management**.
2. In the SSL/TLS Service Profiles pane, click **Add Profile**.
3. Enter a **Name** for the profile.
4. Select or **Import** a **Certificate**.

   PQC certificates are not available for selection or import.

5. For Protocol Settings, define the range of TLS versions that the service can use.

   TLSv1.3 support is limited to administrative access to management interfaces and GlobalProtect portals and gateways. You can only attach SSL/TLS service profiles that allow TLSv1.3 to the settings for these services.

   Administrative Access and GlobalProtect Portals and Gateways:

   *Set the **Min Version** and **Max Version** to **TLSv1.3**.*

   - For the **Min Version**, select the earliest allowed TLS version: **TLSv1.0**, **TLSv1.1**, **TLSv1.2**, or **TLSv1.3**.
   - For the **Max Version**, select the latest allowed TLS version: **TLSv1.0**, **TLSv1.1**, **TLSv1.2**, or **TLSv1.3**.

   All Other Services:

   *Set the **Min Version** and **Max Version** to **TLSv1.2**.*

   - For the **Min Version**, select the earliest allowed TLS version: **TLSv1.0**, **TLSv1.1**, or **TLSv1.2**.
   - For the **Max Version**, select the latest allowed TLS version: **TLSv1.0**, **TLSv1.1**, or **TLSv1.2**.

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STEP 4 | (Optional) Select or clear **Key Exchange Algorithms**, **Encryption Algorithms**, or **Authentication Algorithms**.

*Starting in PAN-OS 12.1.2, you can enable post-quantum key exchange algorithms for TLSv1.3 sessions. You must enable **TLSv1.3** in the Protocol Settings.*

To configure **Key Exchange Algorithms**, select the algorithm type:

- For classical algorithms (**RSA**, **DHE**, or **ECDHE**): Select the **Classical** tab, and then select or clear algorithms.

  *By default, **RSA**, **DHE**, and **ECDHE** are enabled.*

- (TLSv1.3 only) For PQC algorithms:
  1. Select the **Post Quantum Cryptography (PQC)** tab, and then click **Add**.
  2. For **Algorithm**, select **ML-KEM** (Module-Lattice-based Key Encapsulation Mechanism).
  3. Select at least one **Security Level**:

     Each security level corresponds to one of three ML-KEM parameter sets specified in FIPS 203. Higher security levels offer greater protection but reduced performance.

     - **Level 1**—ML-KEM-512
     - **Level 3**—ML-KEM-768
     - **Level 5**—ML-KEM-1024
  4. Select one or more **PQC Supported Groups**.

     *The available curve groups change based on **Algorithm** and **Security Level**. You can generate session keys using post-quantum or hybrid post-quantum key exchange. Hybrid key exchange pairs Elliptic Curve Cryptography (ECC) with ML-KEM to protect against both classical and quantum threats. The following curves are supported for hybrid key agreement: x25519, x448, p256, p384, and p512.*
  5. **Save** the PQC algorithms.

STEP 5 | **Save** the profile.

STEP 6 | To commit your changes, click **Push Config** > **Push**.

Configure an SSL/TLS Service Profile (PAN-OS & Panorama)

STEP 1 | For each desired service, generate or import a certificate on the firewall (see Obtain Certificates).

Use only signed certificates, not CA certificates, in SSL/TLS service profiles.

STEP 2 | Select **Device** > **Certificate Management** > **SSL/TLS Service Profile**, and then click **Add**.

STEP 3 | Enter a **Name** for the profile.

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STEP 4 | If the firewall has more than one virtual system (vsys), select the **Location** (vsys or **Shared**) where the profile is available.

STEP 5 | Select the **Certificate** you obtained in step one.

PQC certificates are not available for selection.

STEP 6 | Under **Protocol Settings**, define the range of TLS versions the service can use.

TLSv1.3 support is limited to administrative access to management interfaces, GlobalProtect portals and gateways, and Authentication Portal. You can only attach SSL/TLS service profiles that allow TLSv1.3 to the settings for these services.

- Administrative Access, GlobalProtect Portals and Gateways, and Authentication Portal:

  *Set the **Min Version** and **Max Version** to **TLSv1.3**.*

  - For **Min Version**, select the earliest allowed TLS version: **TLSv1.0**, **TLSv1.1**, **TLSv1.2**, or **TLSv1.3**.
  - For **Max Version**, select the latest allowed TLS version: **TLSv1.0**, **TLSv1.1**, **TLSv1.2**, or **TLSv1.3**.
- All Other Services:

  *Set the **Min Version** and **Max Version** to **TLSv1.2**.*

  - For the **Min Version**, select the earliest allowed TLS version: **TLSv1.0**, **TLSv1.1**, or **TLSv1.2**.
  - For the **Max Version**, select the latest allowed TLS version: **TLSv1.0**, **TLSv1.1**, or **TLSv1.2**.

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STEP 7 | (Optional) Configure Key Exchange Algorithms, Encryption Algorithms, and Authentication Algorithms.

*Starting in PAN-OS 12.1.2, you can enable post-quantum key exchange algorithms for TLSv1.3 sessions. You must enable **TLSv1.3** in the Protocol Settings.*

- To configure classical key exchange algorithms (**RSA**, **DHE**, and **ECDHE**):

  *By default, **RSA**, **DHE**, and **ECDHE** are enabled.*

  - (PAN-OS 11.2 and earlier) Enable or disable algorithms as needed.
  - (PAN-OS 12.1.2 and later) Select the **Classical** tab, and then enable or disable algorithms as needed.
- (PAN-OS 12.1.2 and later) To specify PQC key exchange algorithms for TLSv1.3 sessions:
  1. Select the **Post-quantum Cryptography (PQC)** tab, and then click **Add**.
  2. For **Algorithm**, select **ML-KEM** (Module-Lattice-based Key Encapsulation Mechanism).
  3. For each algorithm, select at least one **Security Level**:

     Each security level corresponds to one of three ML-KEM parameter sets specified in FIPS 203. Higher security levels offer greater protection but reduced performance.

     - **Level 1**—ML-KEM-512
     - **Level 3**—ML-KEM-768
     - **Level 5**—ML-KEM-1024
  4. For each algorithm, select one or more **PQC Supported Groups**.

     *The available curve groups change based on **Algorithm** and **Security Level**. You can generate session keys using post-quantum or hybrid post-quantum key exchange. Hybrid key exchange pairs Elliptic Curve Cryptography (ECC) with ML-KEM to protect against both classical and quantum threats. The following curves are supported for hybrid key agreement: x25519, x448, p256, p384, and p512.*

STEP 8 | Click **OK** and **Commit** your changes.

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Configure an SSH Service Profile

SSH service profiles enable you to customize SSH parameters to enhance the security and integrity of SSH connections to your Palo Alto Networks management and high availability (HA) appliances. By default, SSH supports all ciphers, key exchange algorithms, and message authentication codes, which leaves your connection vulnerable to attack. In an SSH service profile, you can restrict the algorithms your SSH server supports. You can also generate a new host key and specify data volume, time, and packet-based thresholds for SSH session key regeneration and exchange.

Depending on the SSH server instance, configure either a management or HA SSH service profile. You can configure the profiles from your firewall, Panorama™ web interface (if applying settings to multiple firewalls or appliances), or CLI.

You can configure a maximum of four management and four HA server profiles.

To use the same SSH connection settings for each Dedicated Log Collector (M-series or Panorama virtual appliance in Log Collector mode) in a Collector Group*, configure an SSH service profile from the Panorama management server,* ***Commit*** *your changes to Panorama, and then* ***Push*** *the configuration to the Log Collectors. You can also perform these steps from the CLI using* ***`set log-collector-group <name> general-setting management ssh`*** *commands.*

- Create an SSH Management Profile
- Create an SSH HA Profile

Create an SSH Management Profile

To customize SSH settings for management connections, create an SSH management profile.

You can configure or update an existing management profile from your CLI.

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STEP 1 | Create a Management - Server Profile.

1. Select **Device** > **Certification Management** > **SSH Service Profile**.
2. **Add** a Management - Server Profile.

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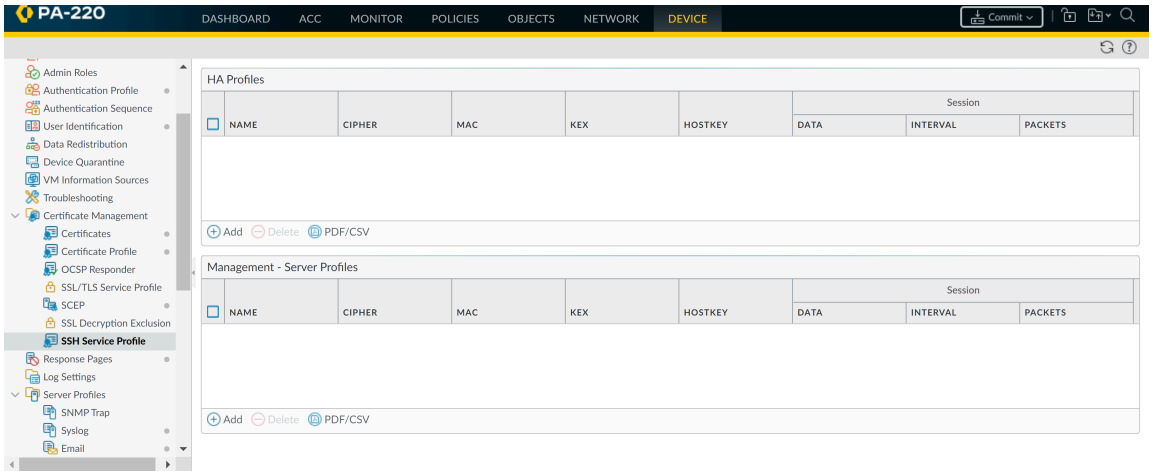

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3. Enter a **Name** to identify the profile.
4. (Optional) **Add** the ciphers, message authentication codes, or key exchange algorithms the profile supports.
5. (Optional) Select a **Hostkey** and key length.
6. (Optional) Enter values for the SSH session rekey parameters: **Data**, **Interval**, and **Packets**.

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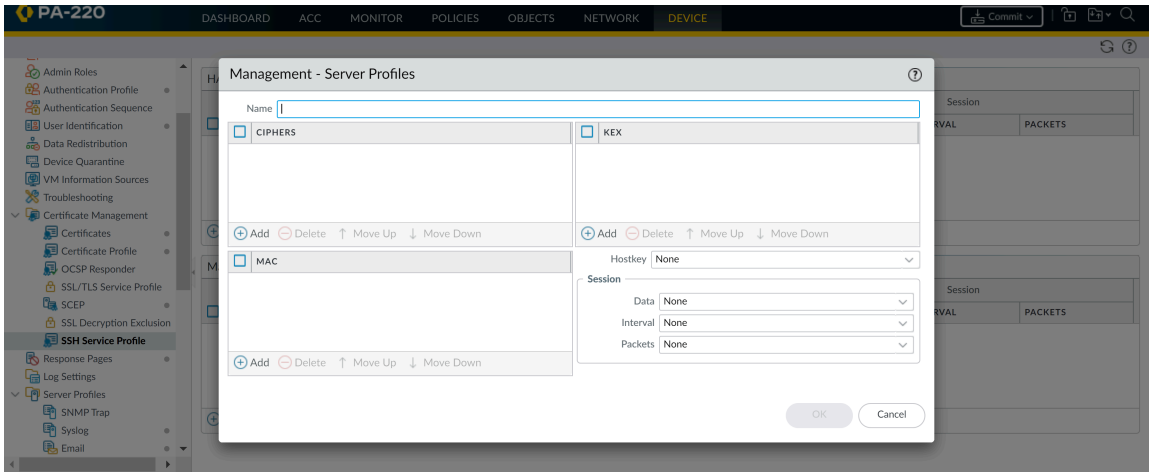

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7. Click **OK** and **Commit** your changes.

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STEP 2 | Select a management profile to apply.

1. Select **Device** > **Setup** > **Management**.
2. Under SSH Management Profiles Settings, select an existing profile.

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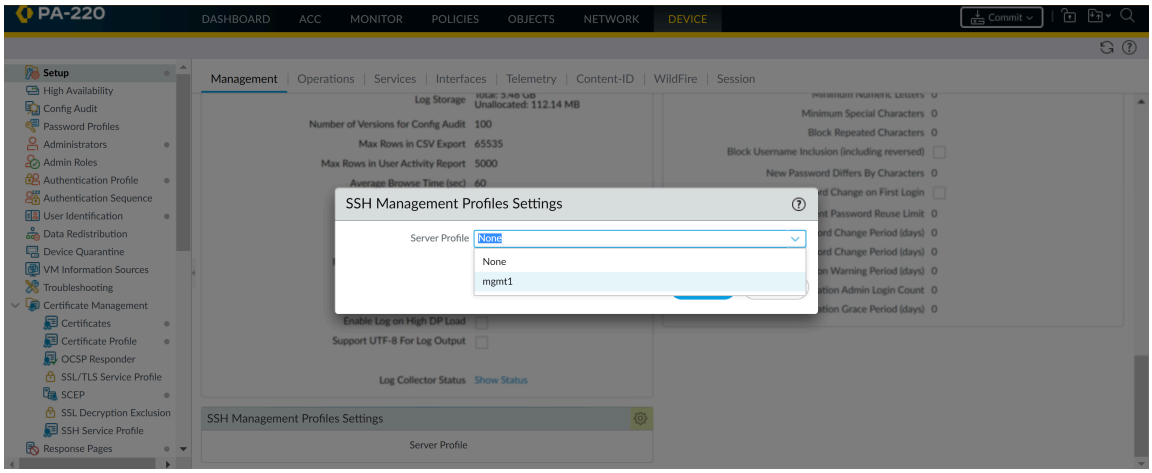

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3. Click **OK** and **Commit** your changes.

STEP 3 | Restart management SSH service from the CLI to apply the profile.

You need to restart the connection each time you apply a new profile or make changes to a profile in use. Configuration changes do not affect active sessions, and new profiles apply to subsequent connections (or sessions).

Use the **`set ssh service-restart mgmt`** CLI command.

Create an SSH HA Profile

To secure SSH communications between appliances in an HA pair, create an SSH HA profile. Before you create a profile, establish an HA connection between the HA peers. To establish an HA connection, you'll need to enable encryption on the control link connection, export the HA key to a network location, and import the HA key on the peer. (See Configure Active/Passive HA or Configure Active/Active HA.)

You can configure or update an existing HA profile from your CLI.

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STEP 1 | Create an HA Profile.

1. Select **Device** > **Certification Management** > **SSH Service Profile**.
2. **Add** an HA Profile.

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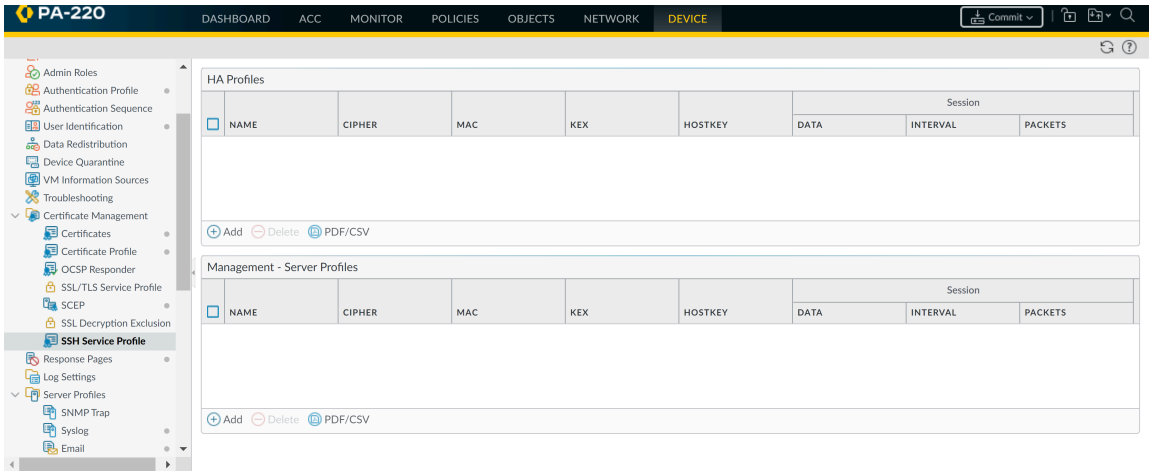

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3. Enter a **Name** to identify the profile.
4. (Optional) **Add** the ciphers, message authentication codes, or key exchange algorithms the profile supports.
5. (Optional) Select a **Hostkey** and key length.
6. (Optional) Enter values for the SSH session rekey parameters: **Data**, **Interval**, and **Packets**.

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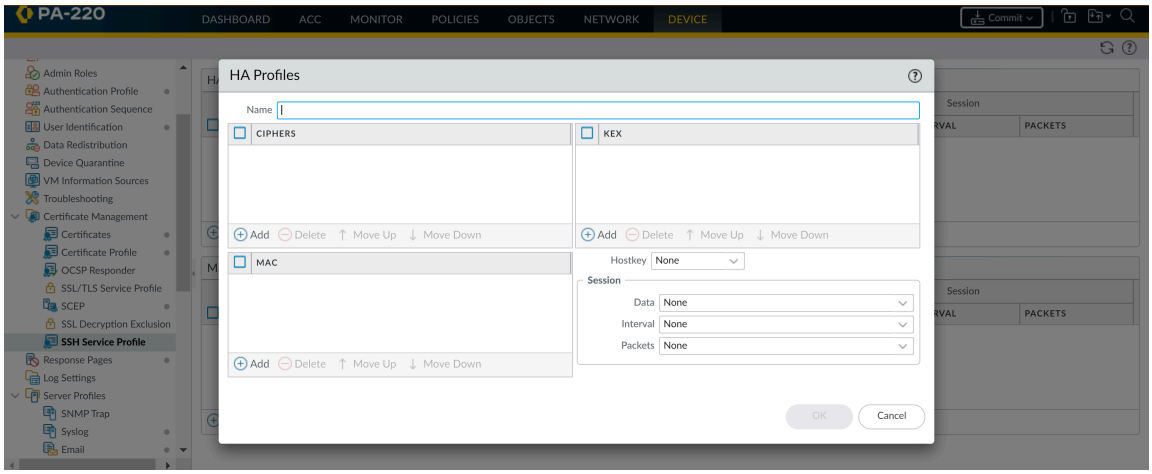

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7. Click **OK** and **Commit** your changes.

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STEP 2 | Select an HA Profile to apply.

1. Select **Device** > **High Availability** > **General**.
2. Under SSH HA Profile Setting, select an existing profile.

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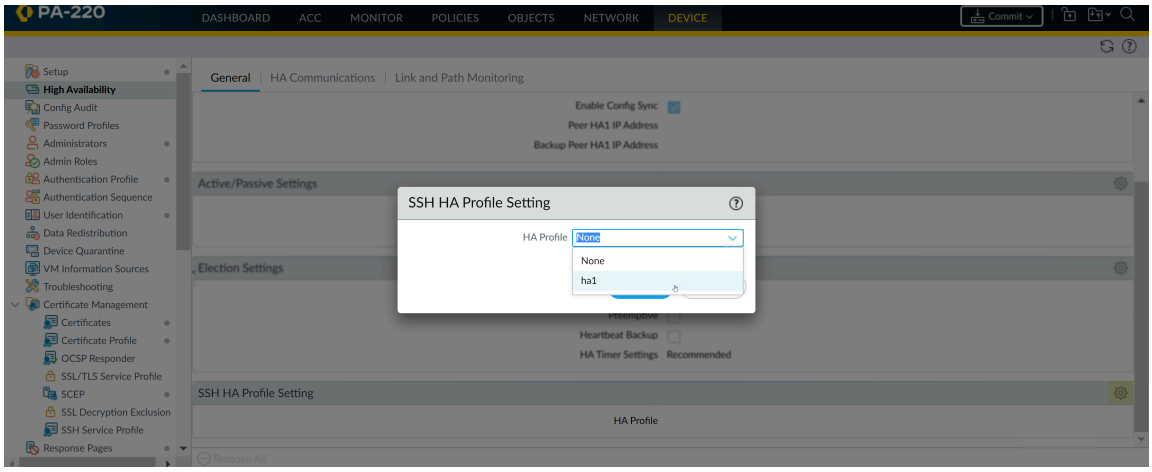

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3. Click **OK** and **Commit** your changes.

STEP 3 | Restart HA1 SSH service from the CLI to apply the profile.

You need to restart the connection each time you apply a new profile or make changes to a profile in use. Configuration changes do not affect active sessions, and new profiles apply to subsequent connections (or sessions).

Use the **`set ssh service-restart ha`** CLI command.

If a connection exists between the appliances in the HA pair, you can use the following commands to minimize the downtime that accompanies an SSH service restart.

- *(HA1 Backup is configured) admin@PA-3260>* ***`request high-availability session-reestablish`***
- *(No HA1 Backup is configured or HA1 Backup link is down) admin@PA-3260>* ***`request high-availability session-reestablish force`***

  You can force the firewall to reestablish HA1 sessions if there is no HA1 backup. However, this causes a brief split-brain condition, where the HA peers cannot detect each other and assume an active role as a result. (Using the ***`force`*** *option when an HA1 backup is configured has no effect.)*

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Replace the Certificate for Inbound Management Traffic

When you first boot up the firewall or Panorama, it automatically generates a default certificate that enables HTTPS access to the web interface and XML API over the management (MGT) interface and (on the firewall only) over any other interface that supports HTTPS management traffic (for details, see Use Interface Management Profiles to Restrict Access). To improve the security of inbound management traffic, replace the default certificate with a new certificate issued specifically for your organization.

You cannot view, modify, or delete the default certificate.

To secure management traffic, you must also Configure Administrative Accounts and Authentication.

STEP 1 | Obtain the certificate that will authenticate the firewall or Panorama to the client systems of administrators.

You can simplify your Certificate Deployment by using a certificate that the client systems already trust. Therefore, we recommend that you Import a Certificate and Private Key from your enterprise certificate authority (CA) or Obtain a Certificate from an External CA; the trusted root certificate store of the client systems is likely to already have the associated root CA certificate that ensures trust.

If you Generate a Certificate *on the firewall or Panorama, administrators will see a certificate error because the root CA certificate is not in the trusted root certificate store of client systems. To prevent this, deploy the self-signed root CA certificate to all client systems.*

Regardless of how you obtain the certificate, we recommend a ***Digest*** *algorithm of* ***sha256*** *or higher for enhanced security.*

STEP 2 | Configure an SSL/TLS Service Profile.

Select the **Certificate** you just obtained.

For enhanced security, we recommend that you set the ***Min Version*** *(earliest allowed TLS version) to* ***TLSv1.2*** *for inbound management traffic. We also recommend that you use a different SSL/TLS Service Profile for each firewall or Panorama service instead of reusing this profile for all services.*

STEP 3 | Apply the SSL/TLS Service Profile to inbound management traffic.

1. Select **Device** > **Setup** > **Management** and edit the General Settings.
2. Select the **SSL/TLS Service Profile** you just configured.
3. Click **OK** and **Commit**.

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Configure the Key Size for SSL Forward Proxy Server Certificates

When responding to a client in an SSL Forward Proxy session, the firewall creates a copy of the certificate that the destination server presents and uses the copy to establish a connection with the client. By default, the firewall generates certificates with the same key size as the certificate that the destination server presented. However, you can change the key size for the firewall-generated certificate.

Changing key size settings clears the current certificate cache.

STEP 1 | Select **Device** > **Setup** > **Session** and, in the Decryption Settings section, click **SSL Forward Proxy Settings**.

STEP 2 | Edit the Forward Proxy Server Certificate Settings.

1. Select an **RSA Key Size**:
   - **Defined by destination host** (default)—The firewall determines the key size and the hashing algorithm for the certificates it generates to establish SSL proxy sessions with clients based on the destination server certificate.
     - If the destination server uses a 1,024-bit RSA key, the firewall generates a certificate with a 1,024-bit RSA key.
     - If the destination server uses a key size larger than 1,024 bits (for example, 2,048 or 4,096 bits), the firewall generates a certificate with a 2,048-bit RSA key.
     - If the destination server uses the SHA-1 hashing algorithm, the firewall generates a certificate with the SHA-1 hashing algorithm.
     - If the destination server uses a hashing algorithm stronger than SHA-1, the firewall generates a certificate with the SHA-256 algorithm.
   - **1024-bit RSA**—The firewall generates certificates that use a 1,024-bit RSA key and the SHA-256 hashing algorithm regardless of the key size of the destination server certificates.

     As of December 31, 2013, public certificate authorities (CAs) and popular browsers have limited support for X.509 certificates that use keys of fewer than 2,048 bits. In the future, depending on security settings, when presented with such keys the browser might warn the user or block the SSL/TLS session entirely.
   - **2048-bit RSA**—The firewall generates certificates that use a 2,048-bit RSA key and the SHA-256 hashing algorithm regardless of the key size of the destination server

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certificates. Public CAs and popular browsers support 2,048-bit keys, which provide better security than 1,024-bit keys.

- **3072-bit RSA**—The firewall generates certificates that use a 3,072-bit RSA key and the SHA-256 hashing algorithm regardless of the key size of the destination server certificates.
- **4096-bit RSA**—The firewall generates certificates that use a 4,096-bit RSA key and the SHA-256 hashing algorithm regardless of the key size of the destination server certificates.

2. Select an **ECDSA Key Size**:
   - **Defined by destination host** (default)—The firewall generates certificates based on the key size that the destination server uses.
     - If the destination server uses an ECDSA 256-bit or 384-bit key, the firewall generates a certificate with that key size.
     - If the destination server uses a key size larger than 384 bits, the firewall generates a certificate with a 521-bit key.
   - **256-bit ECDSA**—The firewall generates certificates with an ECDSA 256-bit key regardless of the key size that the destination server uses.
   - **384-bit ECDSA**—The firewall generates certificates with an ECDSA 384-bit key regardless of the key size that the destination server uses.

STEP 3 | Click **OK** and **Commit** your changes.

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Revoke and Renew Certificates

Revoke a Certificate

Various circumstances can invalidate a certificate before the expiration date. Some examples are a change of name, change of association between subject and certificate authority (for example, an employee terminates employment), and compromise (known or suspected) of the private key. Under such circumstances, the certificate authority (CA) that issued the certificate must revoke it. The following task describes how to revoke a certificate for which the firewall is the CA.

STEP 1 | Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later).

STEP 2 | If the firewall supports multiple virtual systems, the tab displays a **Location** drop-down. Select the virtual system to which the certificate belongs.

STEP 3 | Select the certificate to revoke.

STEP 4 | Click **Revoke**. PAN-OS immediately sets the status of the certificate to revoked and adds the serial number to the Online Certificate Status Protocol (OCSP) responder cache or certificate revocation list (CRL). You need not perform a commit.

Renew a Certificate

If a certificate expires, or soon will, you can reset the validity period. If an external certificate authority (CA) signed the certificate and the firewall uses the Online Certificate Status Protocol (OCSP) to verify certificate revocation status, the firewall uses the OCSP responder information to update the certificate status (see Configure an OCSP Responder). If the firewall is the CA that issued the certificate, the firewall replaces it with a new certificate that has a different serial number but the same attributes as the old certificate.

STEP 1 | Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later).

STEP 2 | If the firewall has more than one virtual system (vsys), select a **Location** (vsys or **Shared**) for the certificate.

STEP 3 | Select a certificate to renew and click **Renew**.

STEP 4 | Enter a **New Expiration Interval** (in days).

STEP 5 | Click **OK** and **Commit**.

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Master Key Encryption

Where Can I Use This?	What Do I Need?
• NGFW	One of these licenses for Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Essentials ❑ Strata Cloud Manager Pro

Master key encryption uses a master key to encrypt other cryptographic keys. These subordinate keys, in turn, encrypt sensitive data such as passwords or other keys. This layered approach enhances security. If the master key is compromised, for example, only the keys and the data it protects are at risk.

On physical and virtual Palo Alto Networks devices, you can configure the master key to use either the AES-256-CBC or AES-256-GCM encryption algorithm. Both algorithms map to an *encryption level*. AES-256-CBC is level 1 and AES-256-GCM is level 2. The default encryption algorithm is AES-256-CBC. However, AES-256-GCM provides stronger security and more efficient encryption. AES-GCM provides built-in authentication and integrity checks, and performance is typically faster than AES-CBC due to support for parallel processing and hardware acceleration. The master key uses the configured encryption algorithm to encrypt sensitive data stored on next-generation firewalls (NGFWs) and Panorama.

If you set the encryption algorithm to AES-256-GCM, you can still use an HSM to encrypt the master key with an encryption key that is stored on the HSM.

Use AES 256-GCM level 2 for master key encryption.

Use the same encryption level on Panorama and its managed devices, and use the same encryption level on NGFWs in a high availability (HA) pair. Upgrade devices to use the strongest possible encryption level. The configuration of managed or paired devices that use different encryption levels may become out of sync.

When you change the encryption level, you can specify whether to:

- Re-encrypt existing encrypted data with the new algorithm.
- Leave existing data encrypted with the old encryption algorithm and use the new algorithm only for new (future) encryptions.

By default, when you change the encryption algorithm, the device uses the new algorithm to re-encrypt existing encrypted data as well as to encrypt new data. If you manage devices with Panorama, they may be on different versions of PAN-OS and may not support the newest encryption algorithms. Be sure you understand which encryption algorithms Panorama and its managed devices support before you change the encryption algorithm or re-encrypt data that has already been encrypted.

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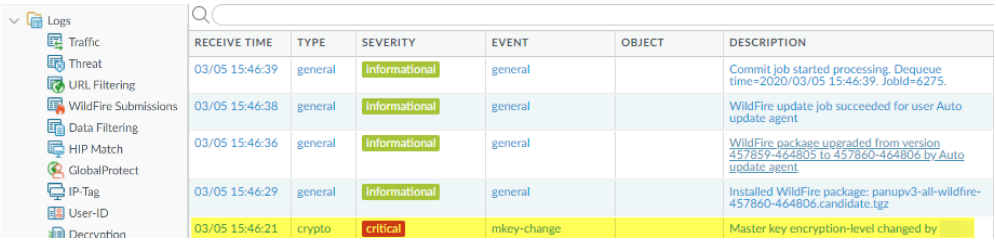
High Availability (HA) Pair Considerations

- The master key on each NGFW in a high availability (HA) pair must be identical. You'll need to manually enter the master key on each NGFW. For more information, see Configure a Master Key.
- Both NGFWs in an HA pair must use the same encryption algorithm to avoid becoming out of sync.

Use AES-256-GCM encryption on both NGFWs in the HA pair.

Master Key Encryption Logs

The firewall generates System logs (**Monitor** > **Logs** > **System**) when you change the master key encryption algorithm (level) and for reencryption events.



To view all of the System Logs for master key encryption, create a filter that shows all logs of the **Type** crypto: **`(subtype eq crypto)`**.

Configure a Master Key

Where Can I Use This?	What Do I Need?
• NGFW	One of these licenses for Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Essentials ☐ Strata Cloud Manager Pro Role requirements: • Superuser role (to update the master key) • Admin role or higher (to synchronize and view)

A master key encrypts all private keys and passwords in a configuration to secure them (such as the private key used for SSL Forward Proxy decryption). Every Next-Generation Firewall (NGFW), Panorama appliance, log collector, and WF-500 appliance has a default master key.

Master key requirements vary by deployment:

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- In a high availability (HA) configuration, both NGFWs or Panorama appliances must use the same master key since keys are not synchronized across HA peers. Otherwise, HA synchronization will not work properly.
- If you're using Panorama to manage your NGFWs, you can either configure the same master key on Panorama and the managed NGFWs or configure a unique master key for each managed NGFW. The most secure option is to configure a unique master key for Panorama and each managed NGFW. This limits the security impact of a compromised master key. See Manage the Master Key from Panorama if your NGFWs are managed by a Panorama appliance.

  Unique master keys are supported only for Panorama and managed NGFWs. Log collectors and WF-500 appliances must share the same master key as Panorama.

Store master keys in a safe location such as a hardware security module (HSM). Lost master keys cannot be recovered. The only way to restore the default master key is to reset the NGFW to factory default settings.

Change the default master key as soon as possible to ensure that you use a unique master key for encryption.

*(PAN-OS 12.2.2 and later releases) PAN-OS 12.2.2 enforces replacement of the default master key with a custom master key within a specific grace period. Configure a custom master key before the **60-day** default grace period expires. See **Default Master Key Replacement** below for full details.*

Default Master Key Replacement

(PAN-OS 12.2.2 and later releases)

The default master key is publicly known and poses a critical security risk. Starting with PAN-OS 12.2.2, the system enforces replacement of the default master key with a custom master key within a configurable grace period.

When the system detects the default master key, a 60-day grace period begins automatically. If you need additional time, you can extend the grace period to a maximum of 120 days, but you must configure this extension before the current grace period expires. This 120-day limit is absolute and measured from the original trigger event — an upgrade, factory reset, or first power-on. The extension does not add a new window from the date you configure it. The system always calculates the final deadline from the date of the original trigger event, regardless of when you configure the extension. You cannot extend the grace period beyond 120 days under any circumstances.

After the grace period expires, the system blocks all standard configuration commits, commit-all jobs, and HA synchronization until you configure a custom master key. Auto-commits and dynamic updates, including content and antivirus deployments, continue to operate normally during this blocked state.

Palo Alto Networks strongly recommends configuring a custom master key immediately after upgrading to PAN-OS 12.2.2. Do not wait for the default grace period of 60 days to approach expiry.

The countdown begins when any of the following trigger events are detected:

- You upgrade a device to PAN-OS 12.2.2 or a later version.

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- You restore a device running PAN-OS 12.2.2 or a later version to factory default settings.
- You power on a new factory-default device running PAN-OS 12.2.2 or a later version for the first time.

If a grace period timer is already running, upgrading or downgrading between PAN-OS 12.2.2 and later versions does not restart the timer. If the grace period has already expired and you downgrade to another PAN-OS 12.2.2 or later release, the timer does not restart and commits fail immediately on the downgraded version.

If your firewalls are managed by Strata Cloud Manager, you can configure a per-tenant custom master key and synchronize it to all managed firewalls centrally, without configuring the key on each device individually. Strata Cloud Manager also automatically deploys the custom master key to newly onboarded firewalls during the bootstrap process, ensuring they never operate with the default key beyond first connect. See the **Strata Cloud Manager** tab below for details.

- PAN-OS & Panorama
- Strata Cloud Manager

Configure the Master Key Panorama and PAN-OS

Ensure there are no pending configuration changes before starting this procedure. Commit any pending changes prior to configuring the master key.

STEP 1 | Backup the configuration.

STEP 2 | (HA only) Disable configuration synchronization.

This step is required before deploying a new master key to any NGFW HA pair.

Before deploying a new master key to any NGFW in an HA pair, disable Config Sync.

If Panorama is in a passive HA configuration, a failover to make Panorama active is required before proceeding. Configure the master key only on the active Panorama peer.

1. Select **Device** > **High Availability** > **General** and edit the **Setup**.
2. Disable (clear) **Enable Config Sync** and then click **OK**.
3. **Commit** your configuration changes.

STEP 3 | Select **Device** > **Master Key and Diagnostics** and edit the Master Key section.

In the **Master Key** dialog, select the **Master Key** checkbox to activate the master key configuration fields. The fields are greyed out until the checkbox is selected.

STEP 4 | Enter the **Current Master Key** if one exists.

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STEP 5 | Define a new **New Master Key**, and then **Confirm New Master Key**. The key must contain exactly 16 characters.

Record and store your master key in a secure, access-controlled location. You will need this key to perform future key rotations, restore configurations, or recover from HA failover scenarios. If the master key is lost, the device cannot be recovered without a full factory reset, resulting in complete loss of your running configuration.

For devices participating in a high availability (HA) configuration, you must configure the same master key on both HA peers. HA synchronization fails if the master keys do not match.

STEP 6 | To specify the master key **Lifetime**, enter the number of **Days** or **Hours** after which the key expires.

Configure a new master key before the current key expires. If the master key expires, the NGFW or Panorama automatically reboots in Maintenance mode causing a network traffic outage. Then, you must reset the NGFW to factory default settings.

*Set the **Lifetime** to two years or less, depending on how many encryptions the device performs. The more encryptions a device performs, the shorter the **Lifetime** you should set. The critical consideration is to not run out of unique encryptions before you change the master key. Each master key can provide up to 2^{32} unique encryptions based on the master key value and the Initialization Vector (IV) value. After 2^{32} unique encryptions, encryptions repeat (are no longer unique), which is a security risk.*

*Set a **Time for Reminder** value (see next step) for the master key and when the reminder notification occurs, change the master key.*

STEP 7 | Enter a **Time for Reminder** that specifies the number of **Days** and **Hours** before the master key expires when the NGFW generates an expiration alarm. The NGFW automatically opens the System Alarms dialog to display the alarm.

*Set the reminder so that it gives you plenty of time to configure a new master key before it expires in a scheduled maintenance window. When the **Time for Reminder** expires and the NGFW or Panorama sends a notification log, change the master key, don't wait for the **Lifetime** to expire. For grouped devices, track every device (for example, NGFWs that Panorama manages and NGFW HA pairs) and when the reminder value expires for any device in the group, change the master key.*

*To ensure the expiration alarm displays, select **Device** > **Log Settings**, edit the Alarm Settings, and **Enable Alarms**.*

STEP 8 | Enable **Auto Renew Master Key** to configure the NGFW to automatically renew the master key. To configure **Auto Renew With Same Master Key**, specify the number of **Days** or **Hours** to renew the same master key. The key extension enables the NGFW to remain operational and continue securing your network; it is not a replacement for configuring a new key if the existing master key lifetime expires soon.

Automatically renewing the master key has benefits and risks. The benefit is that extending the master key **Lifetime** protects against failure to change the master key before its lifetime expires. The risk is that encryptions will repeat and cause a security risk if the number

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of encryptions the device performs with the master key exceeds the number of unique encryptions the master key can generate (2^{32} unique encryptions).

If the master key expires (you don't automatically renew or replace it in a timely manner), the device reboots into maintenance mode, causing a network traffic outage. The only recovery method is to factory reset the device.

*If you enable **Auto Renew Master Key**, set it so that the total time (lifetime plus the auto renew time) does not cause the device to run out of unique encryptions. For example, if you believe the device will consume the master key's number of unique encryptions in two and a half years, you could set the **Lifetime** for two years, set the **Time for Reminder** to 60 days, and set the **Auto Renew Master Key** for 60-90 days to provide the extra time to configure a new master key before the **Lifetime** expires. However, the best practice is still to change the master key before the lifetime expires to ensure that no device repeats encryptions.*

Consider the number of days until your next available maintenance window when configuring the master key to automatically renew after the lifetime of the key expires.

STEP 9 | (12.2.2 and later releases) Configure the **Default Master Key Grace Period** in the **Master Key** dialog.

1. In the **Master Key** dialog, locate the **Default Master Key Grace Period** field at the bottom of the dialog.
   - This field is only active and editable while the device is still using the default master key. Once a custom master key is configured on the device, this field is hidden — it is not applicable and the option to configure it is removed from the dialog.
   - Enter a value between 60 and 120 days. If you do not configure a value, the grace period defaults to 60 days.
   - 120-day limit is cumulative from the original trigger event (upgrade, factory reset, or first power-on) — it is not an additional extension. For example, if 60 days have already elapsed since upgrade, you cannot gain a further 120 days. The maximum total time available is 120 days from the original start of the timer.
   - The dialog displays a warning indicator showing the number of days remaining to configure a new master key — for example, *96 Days Remaining to Configure New Master Key*.

     During the grace period, each configuration commit generates an on-screen warning prompting you to configure a custom master key. In addition, a critical severity system log is generated on every commit. Monitor these warnings and act before the grace period expires — once it does, commits fail and HA sync is suspended until a custom master key is configured.

   If this firewall is managed by Panorama, the grace period can also be configured centrally. See Manage the Master Key from Panorama.
2. Click **OK**.

STEP 10 | (Optional) For added security, select whether to use an **HSM** to encrypt the master key. For details, see Encrypt and Refresh Master Keys Using an HSM.

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STEP 11 | Click **OK** and select **Commit** > **Commit to Panorama** > **Commit All Changes**.

STEP 12 | (HA only) Re-enable configuration synchronization.

1. Select **Device** > **High Availability** > **General** and edit the **Setup**.
2. **Enable Config Sync**, and then click **OK**.
3. Select **Commit** > **Commit to Panorama** > **Commit All Changes**.

Configure the Master Key Strata Cloud Manager

Strata Cloud Manager provides two methods to configure and manage master keys on your cloud-managed firewalls:

- **Sync to SCM Master Key (recommended)**—Configure a per-tenant master key in Strata Cloud Manager and synchronize it to all managed firewalls. Strata Cloud Manager stores the key, tracks versions, handles rotations without requiring the previous key, and automatically deploys the key to newly bootstrapped firewalls.
- **Deploy Master Key (legacy)**—Deploy a master key directly to individual firewalls. Keys deployed through this method are not stored or tracked by Strata Cloud Manager, and you must provide the current key for subsequent rotations.

Master key configuration status never blocks device onboarding. If you have not configured a master key for the tenant, the Device Management page displays a warning banner. Devices onboard successfully with the default key, and you can configure and synchronize a custom key at any time.

Role requirements: Updating the Strata Cloud Manager master key (creating a new key version) requires the Superuser role. Synchronizing the master key to devices and viewing master key state requires the Admin role or higher.

This feature is available on request. Contact your account team to enable the feature.

Sync to SCM Master Key

Use the centralized workflow to configure a single master key for your tenant and synchronize it to all managed firewalls. This is the recommended approach for replacing the default master key before the grace period expires.

1. Log in to Strata Cloud Manager.

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2. Select **System Settings** > **Device Management**.

   The **Cloud Managed Devices** tab displays the device table with master key status columns including **Status**, **Master Key Managed By SCM**, **Grace Period**, **Updated At**, **Last Sync Status**, **Device Status**, and **Bootstrap Status**.

   The Device Management toolbar displays the following master key actions depending on what is enabled for your tenant:

   - **Master Key Actions** dropdown (includes **Update SCM Master Key** and **Sync to SCM Master Key**)—Appears when centralized master key management is enabled for your tenant.
   - **Deploy Master Key** button—This button is disabled (grayed out) when centralized master key management is active and the selected devices are already managed by Strata Cloud Manager.

3. Select **Master Key Actions** > **Update SCM Master Key**.

   This creates a new master key version for the tenant. Firewalls that are already connected continue to use their current key until you synchronize the new key to them.

4. Choose a generation method:

   - **Auto-generate (Recommended)**—Strata Cloud Manager generates a secure 16-character key and stores it in a vault. The master key lifetime is set to the maximum value. Use this option when you do not need to know the key value and plan to keep the firewalls managed by Strata Cloud Manager.
   - **Provide Custom Key**—Manually enter a 16-character alphanumeric key (A-Z, a-z, 0-9). Use this option when you need to know the key value, for example, if you plan to offboard firewalls from Strata Cloud Manager in the future.

     If you use the auto-generate option and later offboard a firewall from Strata Cloud Manager, you will not have the key value needed to change the master key on that firewall. Before offboarding a firewall, update the master key to a known custom key and synchronize it to the firewall.

5. (Custom Key only) Configure the following fields:

   - Enter the 16-character key value in **Enter Custom Master Key**. The key must be exactly 16 alphanumeric characters (A-Z, a-z, 0-9).
   - Set the **Master Key Lifetime** — the duration (in hours) before the key must be rotated.
   - Set the **Update Reminder** — the number of hours before expiry that Strata Cloud Manager displays a reminder to rotate the key.

6. Select **Save**.

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7. Select one or more firewalls in the device table, then select **Master Key Actions** > **Sync to SCM Master Key**.

   The sync dialog adapts based on each device's master key status:

   - *If a device is on the default master key or its key is already managed by Strata Cloud Manager, no key input is required. The **Current Master Key Input** column displays "Not required (known by SCM)."*
   - *If a device has a locally configured custom key that is not managed by Strata Cloud Manager, you must enter the device's current master key value. The **Current Master Key Input** column displays a text field.*

   The dialog displays the selected firewalls with the following columns:

   - **Name**—The firewall serial number.
   - **Status**—The current master key status.
   - **Master Key Managed By SCM**—Whether Strata Cloud Manager currently manages the key on this device.
   - **Stored on HSM**—Select this checkbox if the firewall stores the master key on a hardware security module (HSM).
   - **Current Master Key Input**—If the firewall has a locally-configured custom key that is unknown to Strata Cloud Manager, enter the current key value. If Strata Cloud Manager already knows the key (because it previously deployed it), this field displays "Not required (known by SCM)."
8. Select **Sync to SCM Master Key**.

   Strata Cloud Manager pushes the master key to each selected firewall and commits the change. Monitor the **Last Sync Status** and **Updated At** columns to verify the operation completed successfully.
9. Verify the sync was successful.

   After a successful sync:

   - The **Status** column shows **Custom Key** (green).
   - The **Master Key Managed By SCM** column shows **Yes**.
   - The **Last Sync Status** column shows **Success**.
   - The **Updated At** column shows the sync timestamp.

   If the **Last Sync Status** shows **Failed**, select the status link to view error details. Common causes include:

   - **Invalid current master key**—The key value you entered for a locally-managed device does not match the key on the firewall.
   - **Pending configuration changes**—The firewall has uncommitted changes. Revert pending changes on the firewall and retry the sync.

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***Master Key on First Connect (Bootstrap)**—When you configure a Strata Cloud Manager master key before onboarding a new firewall, the firewall automatically receives the custom master key during the bootstrap process on first connect. After bootstrapping, the **Status** column shows **Custom Key** and **Master Key Managed By SCM** shows **Yes**. No additional sync action is required for newly bootstrapped firewalls.*

Configure Grace Period and Auto-Renew Settings

Configure the default master key grace period and auto-renew lifetime as part of your device configuration. These settings are pushed to firewalls using the standard Push Config workflow. These fields appear under **General Settings** when the master key management feature is enabled for your tenant.

1. Select **Configuration** > **NGFW and Prisma Access**.
2. Select the appropriate **Configuration Scope**:
   - **All Firewalls**—Applies to all managed firewalls (lowest precedence).
   - **Folder**—Applies to firewalls in a specific folder.
   - **Snippet**—Applies to firewalls associated with the snippet.
   - **Device**—Applies to a specific firewall (highest precedence).
3. Select **Network & Device** > **Device Setup**.
4. Under **General Settings**, select the settings icon and configure:
   - **Auto Renew With Same Master Key**—The interval (in hours or days) at which the firewall automatically renews the master key using the same value. This prevents the key from expiring and the firewall from entering maintenance mode. Set to **0** to disable auto-renewal (recommended when Strata Cloud Manager manages key rotation explicitly). A non-zero value means the firewall automatically renews the key lifetime at the specified interval without generating a new key — the device remains synchronized with Strata Cloud Manager.
   - **Default Master Key Grace Period (days)**—The number of days (60 to 120) before a firewall using the default key triggers a commit failure (PAN-OS 12.2.2 and later). Defaults to 60 days if not configured.
5. Select **OK** to save the settings.
6. Select **Push Config** to push the configuration to the target firewalls.

Master key settings follow the Strata Cloud Manager configuration scope hierarchy. If you configure settings at a higher scope (for example, All Firewalls), lower scopes (Device) inherit those values — the General Settings panel displays "Inherited from [scope name]." If you then configure the settings at an intermediate scope (for example, Snippet), the Device scope inherits from the Snippet instead. Settings configured at a higher scope are inherited by firewalls at lower scopes unless overridden. Precedence order (lowest to highest): All Firewalls, Folder, Snippet, Device.

Load a Local Configuration Version

The **Local Config Management** option is available in the per-device Actions menu when this feature is enabled for your tenant. This feature allows you to load a previous local configuration version snapshot on a firewall from Strata Cloud Manager.

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Local configuration version snapshots represent the local firewall running configuration only — they do not include the SCM-pushed configuration. If you pushed a master key from Strata Cloud Manager to a device, any secrets in the local configuration are encrypted using that master key version. When you load a previous configuration version, Strata Cloud Manager must supply the correct key to decrypt those secrets.

Strata Cloud Manager handles this automatically based on the device's master key status:

- **Key managed by Strata Cloud Manager**—If Strata Cloud Manager knows the master key version that was active when the configuration snapshot was captured, it automatically sends that key version as part of the load operation. No user action is required.
- **Locally configured key unknown to Strata Cloud Manager**—If the device has a locally configured master key that Strata Cloud Manager does not know about, you must enter the key value before loading the configuration.

This capability requires PAN-OS 12.2.2 or later. For firewalls running earlier versions, load the configuration locally using the CLI command `load config key <value>`.

1. Select **System Settings** > **Device Management** > **Cloud Managed Devices**.
2. Locate the firewall and select the Actions menu (three dots).
3. Select **Local Config Management**.
4. In the **Local Config Version Snapshots** dialog, locate the version you want to restore.
5. Select **Load** for the desired version.

   A confirmation dialog box appears.
6. Select **Load**.

   Strata Cloud Manager automatically supplies the correct master key version if the key was managed by Strata Cloud Manager when the snapshot was captured.

Deploy a Master Key to Individual Firewalls (Legacy)

Use this per-device method only if you need to manage individual device keys independently of the centralized Strata Cloud Manager workflow. Keys deployed through this action are not stored or tracked by Strata Cloud Manager.

For centralized key lifecycle management, use the ***Sync to SCM Master Key*** *workflow instead.*

1. Log in to Strata Cloud Manager.
2. Select **System Settings** > **Device Management**.

   The **Device Management** page displays the firewalls.
3. Select the required firewalls, click **Deploy Master Key** and edit the Deploy Master Key section.
4. Enter the **Current Master Key** if one exists.
5. Define a new **New Master Key**, and then **Confirm New Master Key**. The key must contain exactly 16 characters.

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6. To specify the master key **Lifetime**, enter the number of **Days** or **Hours** after which the key expires.

   Configure a new master key before the current one expires. You can set the lifetime of the master key from 1 to 18,250 days. If the master key expires, the firewall automatically reboots in Maintenance mode. Then, you must reset the NGFW to factory default settings.

   *Set the **Lifetime** to two years or less, depending on how many encryptions the device performs. The more encryptions a device performs, the shorter the **Lifetime** you should set. The critical consideration is to not run out of unique encryptions before you change the master key. Each master key can provide up to 2^{32} unique encryptions based on the master key value and the Initialization Vector (IV) value. After 2^{32} unique encryptions, encryptions repeat (are no longer unique), which is a security risk.*

   *Set a **Time for Reminder** value (see next step) for the master key and when the reminder notification occurs, change the master key.*
7. Enter a **Time for Reminder** that specifies the number of **Days** and **Hours** before the master key expires when the firewall generates an expiration alarm. The firewall automatically opens the System Alarms dialog to display the alarm.

   *Set the reminder so that it gives you plenty of time to configure a new master key before it expires in a scheduled maintenance window. When the **Time for Reminder** expires and the firewall sends a notification log, change the master key, do not wait for the **Lifetime** to expire.*
8. (Optional) For added security, select whether to use an **HSM** to encrypt the master key. For details, see Encrypt and Refresh Master Keys Using an HSM.
9. Click **Deploy Master Key**.

   *Keys deployed using this method are not tracked as Strata Cloud Manager-managed. The **Master Key Managed By SCM** column displays **No** for these devices and the **Updated At** and **Last Sync Status** columns are not populated.*

Configure Master Key Encryption Level

You configure the master key encryption algorithm level and whether to re-encrypt all currently encrypted data with a new encryption algorithm level using the CLI. Depending on the order of the keywords, you can change the encryption level or you can change the encryption level and also specify whether to re-encrypt previously encrypted data.

The following operational CLI command changes the encryption level and automatically re-encrypts all currently encrypted data with the specified encryption level:

```
admin@PA-NGFW>request encryption-level level <0|1|2>
```

The following operational CLI command changes the encryption level and specifies whether to re-encrypt all currently encrypted data with the new encryption level:

```
admin@PA-NGFW>request encryption-level re-encrypt <yes|no> level <0|
1|2>
```

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Keyword	Options
level	**0** = Use the default algorithm (AES-256-CBC) to encrypt data **1** = Use the AES-256-CBC algorithm to encrypt data **2** = Use the AES-256-GCM algorithm to encrypt data The firewall re-encrypts all currently encrypted data and encrypts new sensitive data using the specified algorithm. If you don't want to re-encrypt existing encrypted data with the new algorithm, specify **re-encrypt no** in the command string. This prevents the firewall from automatically re-encrypting data that the firewall has already encrypted. *Only use AES-256-GCM when Panorama and all of its managed devices (or both devices in an HA pair) run PAN-OS 11.1 or greater and configure all of the devices to use AES-256-GCM. Managed or paired devices that use different encryption levels may become out of sync.*
re-encrypt	**no** = Do not re-encrypt currently encrypted data. The firewall does not re-encrypt currently encrypted data. Currently encrypted data remains encrypted with whichever algorithm the firewall originally used to encrypt the data. The firewall uses the specified algorithm only to encrypt sensitive data in the future. **yes** = Re-encrypt currently encrypted data with the specified algorithm and use that algorithm to encrypt sensitive data in the future.

Use the operational CLI command **show system masterkey-properties** to verify the encryption algorithm (level) currently configured on the device, for example:

```
admin@PA-NGFW>show system masterkey-properties
```

```
Master key expires at: unspecified
Reminders will begin at: unspecified
Master key on hsm: no
Automatically renew master key lifetime: 0
Encryption Level: 1
```

The output shows that the current encryption level is 1, which is AES-256-CBC.

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If you downgrade to an earlier version of PAN-OS, the device automatically reverts the encryption algorithm to a level that the downgraded PAN-OS version supports and automatically re-encrypts encrypted data using that level so that the device can decrypt and use the data as needed. For example, if your device is on PAN-OS 11.1 and uses AES-256-GCM as the encryption algorithm (which is not supported on earlier versions of PAN-OS), and you downgrade to PAN-OS 9.1, then the device re-encrypts the encrypted data to AES-256-CBC, which is supported in PAN-OS 9.1.

Unique Master Key Encryption with AES-256-GCM

The master key can only generate a finite number of unique encryptions before it runs out of unique combinations and must repeat encryptions. The firewall creates unique encryptions using the AES-256-GCM encryption algorithm with an Initialization Vector (IV). An IV is an arbitrary number that should only be used one time to create an encryption to ensure that each encryption is unique.

Each encryption using the master key and IV must be unique to prevent forgery attacks. The firewall meets the uniqueness requirement that the probability that the authenticated encryption is ever created with the same IV and the same key on two or more distinct sets of input data is no greater than 2^{32}.

When the IV runs through all of its unique values, the IV value repeats. When the IV value repeats, using the same master key and the repeated IV value to encrypt data means that the encryption is the same as an encryption previously used on other data. Change the Master Key before the system runs out of unique encryptions to prevent the firewall from using the same encryption (master key and IV value combination) on more than one piece of sensitive data. Unique encryption combinations should never be repeated or reused.

To track when you need to change the master key, set the master key **Lifetime** and **Reminder** values on each appliance (**Device** > **Master Key and Diagnostics** and edit the master key). Set the values conservatively, based on the expected volume of master key encryptions, to ensure that all encryptions are unique and no encryption combinations are repeated or reused.

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Secure Keys with a Hardware Security Module

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ A supported HSM

A hardware security module (HSM) is a physical device that securely generates, stores, and manages cryptographic keys. HSMs provide both logical and physical protection of cryptographic materials against unauthorized use and tampering. Cryptographic operations (for example, certificate signing) occur exclusively inside the device at the request of a trusted application or entity. This protection helps you comply with strict industry and security requirements, such as the Payment Card Industry Data Security Standard (PCI DSS) or Federal Information Processing Standards (FIPS).

You can store and generate private keys used for TLS decryption on an HSM integrated with a Next-Generation Firewall (NGFW) or Panorama. A master key encrypts the private keys stored on the HSM. The NGFW or Panorama acts as a client, establishing a secure TLS connection with an external HSM server. When the client requests a cryptographic operation, the HSM server performs the operation and sends the result. This process ensures private keys are never exposed. For enhanced security, you can also store the master key on an HSM instead of the NGFW.

NGFWs integrate with the following HSM and key management products:

- Thales Luna Network (formerly SafeNet Network)
- Thales CipherTrust Manager
- Entrust nShield

To secure keys with an HSM, you need to set up connectivity between your NGFW and the HSM server, register the NGFW as an HSM client with the HSM server, then configure HSM settings for master keys and private keys, and any additional steps specific to the HSM product.

Set Up Connectivity with an HSM

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ HSM set up and ready for deployment per vendor documentation ❑ Static IP address assigned to NGFW management interface unless connecting through service route. *The HSM authenticates firewalls by IP address. If the address changes at runtime, HSM operations fail.*

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Before your Next-Generation Firewall (NGFW) or Panorama can offload cryptographic operations and protect its keys using a hardware security module (HSM), it must establish a secure connection to the HSM server. While the specific steps to establish trust vary by HSM product, the high-level process is the same:

- Define connection settings for the HSM server
- Authenticate the NGFW to the HSM
- Register the NGFW as an HSM client
- Verify connectivity

PAN-OS does not sync HSM configurations between high availability (HA) firewall peers. You must configure the HSM separately on each peer. In active/passive HA configurations, you must manually perform one failover to configure and authenticate each HA peer to the HSM. After this initial failover, subsequent failovers don't require user interaction.

PAN-OS integrates HSM client software with PA-3200 Series, PA-3400 Series, PA-5200 Series, PA-5400 Series, PA-7000 Series, PA-7500 and VM-Series firewalls and the Panorama management server (virtual and M-Series appliances).

The following table lists supported HSM client versions by PAN-OS release.

HSM Products	HSM Client Versions	PAN-OS Versions
Entrust nShield	13.6.3	12.1.2 and later
	12.40.2 (backward compatible up to client version 11.50 for older appliances)	10.2, 11.1, 11.2
Thales Luna Network	10.9.2	12.1.8 and later
	5.4.2 and 7.2.0	10.2, 11.1, 11.2 through 12.1.7
Thales CipherTrust Manager	8.14.1	11.1 and later

HSM client and server software must be compatible. Compatibility mismatches can result in connection failures. Refer to vendor documentation for a client-server compatibility matrix.

Downgrading HSM servers might not be possible after you upgrade them.

- Thales Luna Network HSM
- Entrust nShield HSM
- Thales CipherTrust Manager

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Set Up Connectivity with a Thales Luna Network HSM

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Luna Network HSM ready for deployment ❑ Luna HSM Client version compatible with HSM firmware and appliance version

STEP 1 | Install the Luna HSM Client RPM Package Manager.

(Thales Luna Network prerequisite) On the firewall or Panorama, select the Luna HSM Client version that is compatible with your Luna Network HSM server.

1. Select **Device** > **Setup** > **HSM**.
2. From the and **Select HSM Client Version** (Hardware Security Operations settings).
3. Select **Version 5.4.2** (default) or **7.2** as appropriate for your HSM server version.
4. Click **OK**.
5. (Required only if you change the HSM version on the firewall) If the version change succeeds, the firewall prompts you to reboot to change to the new HSM version. If prompted, click **Yes**.
6. If the master key isn't on the firewall, the client version upgrade will fail. **Close** the message and make the master key local to the firewall:
   - Edit the Hardware Security Module Provider and disable (clear) the **Master Key Secured by HSM** option.
   - Click **OK**.
   - Select **Device** > **Master Key and Diagnostics** to edit the Master Key.
   - Enter the **Current Master Key**; you can then enter that same key to be the **New Master Key** and then **Confirm New Master Key**.
   - Click **OK**.
   - Repeat the first four steps to **Select HSM Client Version** and reboot again.

STEP 2 | Log in to the firewall web interface.

STEP 3 | Define connection settings for the Thales Luna Network HSM.

1. Select **Device** > **Setup** > **HSM**, and **Edit** the Hardware Security Module Details settings.
2. For **Provider Configured**, select **Thales Luna Network HSM**.
3. **Add** each HSM server.

   A high-availability HSM configuration requires at least two servers; you can cluster up to 16 HSM servers. All servers in the cluster must run the same Luna HSM version and

authenticate separately. Use a Luna Network cluster only when you want to replicate the keys across the cluster.

Alternatively, you can add up to 16 Luna Network HSM servers that function independently.

*If you configure two or more HSM servers, enable **High Availability**; otherwise, the firewall does not use the additional HSM servers.*

1. Enter a **Module Name** (up to 31 ASCII characters).
2. Enter the IPv4 **Server Address**.

4. (HA only) Enable **High Availability**, and then configure the following settings:
   - **Auto Recovery Retry**—Enter the maximum number of times the HSM client tries to recover its connection to an HSM server before failing over to a peer server (range is 0-500; default is 0)
   - **High Availability Group Name**—Enter a name for the HA group (up to 31 ASCII characters)
5. Click **OK** and **Commit** your changes.

STEP 4 | (PAN-OS 12.1.7 and earlier) Select an **HSM Client Version**:

- **5.4.2**
- **7.2.0**

STEP 5 | Click **OK** and **Commit** your changes.

STEP 6 | (Optional) Configure a service route to connect to the HSM if you don't want the firewall to connect through the management interface (default).

*If you configure a service route for the HSM, running the **`clear session all`** CLI command clears all existing HSM sessions, which brings all HSM states down and then up again. During the several seconds required for recovery, all SSL/TLS operations fail.*

1. Select **Device** > **Setup** > **Services** and click **Service Route Configuration**.
2. **Customize** a service route. The **IPv4** tab is active by default.
3. Click **HSM** in the Service column.
4. Select a **Source Interface** for the HSM.
5. Click **OK** and **Commit** your changes.

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STEP 7 | Configure the firewall to authenticate to the HSM.

1. Select **Device** > **Setup** and **Setup Hardware Security Module**.
2. Select the HSM **Server Name**.
3. Select **Automatic** or **Manual** for your authentication and trust certificate.
4. Enter the **Administrator Password** to authenticate the firewall to the HSM.
5. Click **OK**.

   The firewall tries to authenticate to the HSM and displays a status message.
6. Click **OK** again.

STEP 8 | Register the firewall as an HSM client and assign it to a partition on the HSM server.

If a client (firewall) with the same <cl-name> is already registered on the HSM, you must delete the duplicate registration before registering the new client. Run the ***client delete -client <cl-name>*** *command, where <cl-name> is the name of the registered client you want to delete.*

1. Log in to the HSM from a remote system.
2. Register the firewall using the **client register -c <*cl-name*> -ip <*fw-ip-addr*>** CLI command, where *<cl-name>* is the name you assign to the firewall for use on the HSM and *<fw-ip-addr>* is the IP address for that firewall.
3. Assign a partition to the firewall using the **client assignpartition -c <*cl-name*> -p <*partition-name*>** CLI command, where *<cl-name>* is the name you assigned to the firewall using the **client register** command and *<partition-name>* is the name of a previously configured partition that you want to assign to this firewall.

STEP 9 | Configure the firewall to connect to the HSM partition.

1. Select **Device** > **Setup** > **HSM** and refresh ( ) the display.
2. **Setup HSM Partition** (Hardware security operations settings).
3. Enter the **Partition Password** to authenticate the firewall to the partition on the HSM.
4. Click **OK**.

STEP 10 | (HA only) Repeat the previous authentication, registration, and partition connection steps to add another HSM to the existing HA group.

If you remove an HSM from your configuration, repeat the previous partition connection step to remove the deleted HSM from the HA group.

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STEP 11 | Verify firewall connectivity and authentication with the HSM.

1. Select **Device** > **Setup** > **HSM** and check the authentication and connection Status:
   - **Green**—The firewall successfully authenticated and connected to the HSM.
   - **Red**—The firewall failed to authenticate or connect to the HSM.
2. View the following columns in Hardware Security Module Status to determine the authentication status:
   - **Serial Number**—The serial number of the HSM partition if the firewall successfully authenticated to the HSM.
   - **Partition**—The partition name on the HSM assigned to the firewall.
   - **Module State**—The current state of the HSM connection. This value is always `Authenticated` if the Hardware Security Module Status displays the HSM.

Set Up Connectivity with an Entrust nShield HSM

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Entrust nShield HSM installed and enrolled in a Security World ❑ Remote file system (RFS) configured on a host running nShield Security World software (Windows or Linux)

ECDSA certificates are not supported for Entrust nShield HSMs.

STEP 1 | Define connection settings for each nShield HSM.

1. Log in to the firewall web interface and select **Device** > **Setup** > **HSM**.
2. Edit the hardware security module provider settings and set the **Provider Configured** to **Entrust nShield HSM**.
3. **Add** each HSM server as follows. An HA HSM configuration requires two servers:
   1. Enter a **Module Name** for the HSM server. This can be any ASCII string of up to 31 characters.
   2. Enter an IPv4 address for the HSM **Server Address**.
4. Enter an IPv4 address for the **Remote Filesystem Address**.
5. Click **OK** and **Commit** your changes.

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STEP 2 | (Optional) Configure a service route to connect to the HSM if you don't want the firewall to connect through the management interface (default).

> *If you configure a service route for the HSM, running the* ***`clear session all`*** *CLI command clears all existing HSM sessions, which brings all HSM states down and then up again. During the several seconds required for HSM to recover, all SSL/TLS operations will fail.*

1. Select **Device** > **Setup** > **Services** and click **Service Route Configuration**.
2. **Customize** a service route. The **IPv4** tab is active by default.
3. Click **HSM** in the Service column.
4. Select a **Source Interface** for the HSM.
5. Click **OK** and **Commit** your changes.

STEP 3 | Register the firewall as an HSM client with the HSM server.

This step describes the procedure for using the front panel interface of an nShield HSM. For more details, refer to nShield documentation.

1. Log in to the front panel display of the nShield HSM.
2. Use the right-hand navigation button to select **System** > **System configuration** > **Client config** > **New client**.
3. Enter the firewall IP address.
4. Select **System** > **System configuration** > **Client config** > **Remote file system** and enter the IP address of the client computer where you set up the RFS.

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STEP 4 | Configure the RFS to accept connections from the firewall.

1. Log in to the RFS from a Linux client.
2. Obtain the electronic serial number (ESN) and the hash of the K_{NETI} key, which authenticates the HSM to clients, by running the **anonkneti *<ip-address>*** CLI command, where *<ip-address>* is the HSM IP address.

   For example:

   ```
   anonkneti 192.0.2.1
   ```

   ```
   B1E2-2D4C-E6A2 5a2e5107e70d525615a903f6391ad72b1c03352c
   ```

   In this example, `B1E2-2D4C-E6A2` is the ESN and `5a2e5107e70d525615a903f6391ad72b1c03352c` is the hash of the K_{NETI} key.
3. Use the following command from a superuser account to set up the RFS:

   ```
   rfs-setup --force <ip-address> <ESN> <hash-Kneti-key>
   ```

   The *<ip-address>* is the IP address of the HSM, *<ESN>* is the electronic serial number, and *<hash-Kneti-key>* is the hash of the K_{NETI} key.

   The following example uses the values obtained in this procedure:

   ```
   rfs-setup --force 192.0.2.1 B1E2-2D4C-E6A2
    5a2e5107e70d525615a903f6391ad72b1c03352c
   ```
4. Use the following command to permit HSM client submissions on the RFS:

   ```
   rfs-setup --gang-client --write-noauth <FW-IPaddress>
   ```

   where *<FW-IPaddress>* is the firewall IP address.

STEP 5 | Authenticate the firewall to the HSM.

1. In the firewall web interface, select **Device** > **Setup** > **HSM** and **Setup Hardware Security Module**.
2. Click **OK**.

   The firewall tries to authenticate to the HSM and displays a status message.
3. Click **OK**.

STEP 6 | Synchronize the firewall with the RFS by selecting **Device** > **Setup** > **HSM** and **Synchronize with Remote Filesystem**.

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STEP 7 | Verify firewall connectivity and authentication with the HSM.

1. Select **Device** > **Setup** > **HSM** and check the authentication and connection Status:
   - **Green**—The firewall is successfully authenticated and connected to the HSM.
   - **Red**—The firewall failed to authenticate to the HSM or network connectivity to the HSM is down.
2. Check the Hardware Security Module Status to determine the authentication status.
   - **Name**—The name of the HSM.
   - **IP address**—The IP address of the HSM.
   - **Module State**—The current state of the HSM connection: `Authenticated` or `NotAuthenticated`.

Set Up Connectivity with a Thales CipherTrust Manager HSM

STEP 1 | Define connection settings for each Thales CipherTrust Manager HSM.

1. Log in to the firewall web interface and select **Device** > **Setup** > **HSM**.
2. Edit the hardware security module provider settings and set the **Provider Configured** to **Thales CipherTrust Manager**.
3. **Add** each HSM server as follows. An HA HSM configuration requires two servers.
   1. Enter a **Module Name** for the HSM server. This can be any ASCII string of up to 31 characters.
   2. Enter an IPv4 address for the HSM **Server Address**.
4. Click **OK** and **Commit** your changes.

STEP 2 | **Set Up HSM Connectivity Account**.

1. Enter the **Server Name**. This should match the Module Name from the connection setting.
2. Import the certificates you generated in Thales CipherTrust Manager.
   - HSM Server CA Certificate—Import a Base64 encoded certificate (PEM).
   - HSM Client Certificate—Import a Base64 encoded certificate (PEM).
   - HSM Client Private Key—Import a Base64 encoded certificate (PEM) and enter a **Passphrase** fewer than 32 characters.
3. Click **OK**.

STEP 3 | **Restart HSM Connection** to refresh the PAN-OS state. This removes the old certificates and adds the new certificates.

1. Click **OK**.
2. Wait for the module state to display as Reachable.

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STEP 4 | **Set Up HSM Crypto User Account** to match the Thales CipherTrust Manager account you want to use.

1. Enter a **Username**.
2. Enter a **Password**.
3. Click **OK**.

The success dialog displays and the Status changes to green in the dashboard.

STEP 5 | **Show Detailed Information** to view the new fields.

STEP 6 | Confirm that your certificate is imported and valid.

1. Select **Device** > **Certification Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later).
2. Confirm that the **Key** displays a lock and the **Status** is valid.

Store Private Keys on an HSM

For added security, you can use a hardware security module (HSM) to secure the private keys used in SSL/TLS decryption for:

- SSL Forward Proxy—The HSM can store the private key of the Forward Trust certificate that signs certificates in SSL/TLS forward proxy operations. The firewall will then send the certificates that it generates during such operations to the HSM for signing before forwarding the certificates to the client.
- SSL Inbound Inspection—The HSM can store the private keys for the internal servers for which you are performing SSL/TLS inbound inspection.

If you use the DHE or ECDHE key exchange algorithms to enable perfect forward secrecy (PFS) support for SSL decryption, you can use an HSM to store the private keys for SSL Inbound Inspection. You can also use an HSM to store ECDSA keys used for SSL Forward Proxy or SSL Inbound Inspection.

- (PAN-OS 11.1 and earlier) PAN-OS supports SSL Forward Proxy decryption with HSM for TLSv1.3 sessions. SSL Inbound Inspection occurs over TLSv1.2 even if both client and server support TLSv1.3.
- (PAN-OS 11.2 and later) PAN-OS supports SSL Forward Proxy and SSL Inbound Inspection with HSMs for TLSv1.3 sessions. To turn on this support for SSL Inbound Inspection, use the **`set ssl inbound-inspection tls1.3-with-hsm enable yes`** CLI command.

STEP 1 | On the HSM, import or generate the certificate and private key used in your decryption deployment.

For instructions on importing or generating a certificate and private key on the HSM, refer to your HSM documentation.

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STEP 2 | (Entrust nShield) Synchronize the key data from the nShield remote file system to the firewall.

Synchronization with the Thales Luna Network HSM is automatic.

1. Access the firewall web interface and select **Device** > **Setup** > **HSM**.
2. **Synchronize with Remote Filesystem** (Hardware Security Operations settings).

STEP 3 | Import the certificate that corresponds to the HSM-stored key.

1. Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later). Select **Import**.
2. Enter the **Certificate Name**.
3. **Browse** to the **Certificate File** on the HSM.
4. Select a **File Format**.
5. Select **Private Key resides on Hardware Security Module**.
6. Click **OK** and **Commit** your changes.

STEP 4 | (Forward Trust certificates only) Enable the certificate for use in SSL/TLS Forward Proxy.

1. Open the certificate you imported in Step 3 for editing.
2. Select **Forward Trust Certificate**.
3. Click **OK** and **Commit** your changes.

STEP 5 | Verify that you successfully imported the certificate onto the firewall.

Locate the certificate you imported in Step 3 and check the icon in the Key column:

- **Lock icon**—The private key for the certificate is on the HSM.
- **Error icon**—The private key is not on the HSM or the HSM is not properly authenticated or connected.

Encrypt and Refresh Master Keys Using an HSM

A master key encrypts all private keys and passwords on the firewall and Panorama. If you have security requirements to store your private keys in a secure location, you can encrypt the master key using an encryption key that is stored on an hardware security module (HSM). The firewall or Panorama then requests the HSM to decrypt the master key whenever it is required to decrypt a password or private key on the firewall. Typically, the HSM is in a highly secure location that is separate from the firewall or Panorama for greater security.

The HSM encrypts the master key using a wrapping key. To maintain security, you must occasionally change (refresh) this wrapping key.

If you have not previously encrypted the master key on a firewall, use the following procedure to encrypt it. Use this procedure for first time encryption of a key, or if you define a new master key and you want to encrypt it. If you want to refresh the encryption on a previously encrypted key, see Refresh the Master Key Encryption.

STEP 1 | Select **Device** > **Master Key and Diagnostics**.

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STEP 2 | Specify the key that is currently used to encrypt all of the private keys and passwords on the firewall in the **Master Key** field.

STEP 3 | If changing the master key, enter the new master key and confirm.

STEP 4 | Select the **HSM** check box.

- **Life Time**—The number of days and hours after which the master key expires (range 1-730 days).
- **Time for Reminder**—The number of days and hours before expiration when the user is notified of the impending expiration (range 1–365 days).

STEP 5 | Click **OK**.

*To disable storing the master key on the HSM, deselect **Stored on HSM**, then configure a new master key. Enter the **Current Master Key** and define a new **New Master Key**.*

Alternatively, you can issue the following CLI command:

```
request master-key new-master-key <value> current-master-
key <value> lifetime <1-438000> reminder <1-8760> on-hsm
 <yes|no> no-commit <yes|no>
```

***Commit** the configuration afterwards.*

Refresh the Master Key Encryption

As a best practice, periodically refresh the master key encryption by rotating the wrapping key that encrypts it. The frequency of the rotation depends on your application. The wrapping key resides on your HSM. The following command is the same for Thales Luna Network and Entrust nShield HSMs.

STEP 1 | Log in to the firewall CLI.

STEP 2 | Use the following CLI command to rotate the wrapping key for the master key on an HSM:

```
> request hsm mkey-wrapping-key-rotation
```

If the master key is encrypted on the HSM, the CLI command will generate a new wrapping key on the HSM and encrypt the master key with the new wrapping key.

If the master key is not encrypted on the HSM, the CLI command will generate new wrapping key on the HSM for future use.

The old wrapping key is not deleted by this command.

Manage the HSM Deployment

You can perform the following tasks to manage your HSM deployment:

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- View HSM configuration settings.

  This view displays the HSM provider, firewall source address, HSM client version on the firewall, whether the master key is secured by the HSM, and if the firewall is connected and authenticated to the HSM. The fields that display vary based on the configured provider.

  Select **Device** > **Setup** > **HSM**.

- Display detailed HSM information.

  This view displays information regarding the HSM servers, HSM high availability status, and HSM hardware.

  Select **Show Detailed Information** from the Hardware Security Operations section.

- Export the Support file.

  A test file is created to help customer support when addressing a problem with an HSM configuration on the firewall.

  Select **Export Support File** from the Hardware Security Operations section.

- Reset the HSM configuration.

  This option removes all HSM connections to the firewall. All authentication procedures must be repeated after resetting the HSM configuration.

  Select **Reset HSM Configuration** from the Hardware Security Operations section.

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High Availability

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

High availability (HA) is a deployment in which two firewalls are placed in a group or up to 16 firewalls are placed in an HA cluster and their configuration is synchronized to prevent a single point of failure on your network. A heartbeat connection between the firewall peers ensures seamless failover in the event that a peer goes down. Setting up HA provides redundancy and allows you to ensure business continuity.

You can configure two Palo Alto Networks firewalls as an HA pair or configure up to 16 firewalls as peer members of an HA cluster. The peers in the cluster can be HA pairs or standalone firewalls. HA allows you to minimize downtime by making sure that an alternate firewall is available in the event that a peer firewall fails. The firewalls in an HA pair or cluster use dedicated or in-band HA ports on the firewall to synchronize data—network, object, and policy configurations—and to maintain state information. Firewall-specific configuration such as management interface IP address or administrator profiles, HA specific configuration, log data, and the Application Command Center (ACC) information is not shared between peers.

For a consolidated application and log view across an HA pair, you must use Panorama, the Palo Alto Networks centralized management system. See Context Switch—Firewall or Panorama in the Panorama Administrator's Guide. Consult the Set Up Active/Passive HA and Set Up Active/Active HA. It is highly recommended that you use Panorama to provision HA cluster members. Consult the HA Clustering Best Practices and Provisioning.

When a failure occurs on a firewall in an HA pair or HA cluster and a peer firewall takes over the task of securing traffic, the event is called a Failover. The conditions that trigger a failover are:

- One or more of the monitored interfaces fail. (Link Monitoring)
- One or more of the destinations specified on the firewall cannot be reached. (Path Monitoring)
- The firewall does not respond to heartbeat polls. (Heartbeat Polling and Hello messages)
- A critical chip or software component fails, known as packet path health monitoring.

Palo Alto Networks firewalls support stateful active/passive or active/active high availability with session and configuration synchronization with a few exceptions:

- The VM-Series firewall on Azure and VM-Series firewall on AWS support active/passive HA only.

  On AWS, when you deploy the firewall with the Amazon Elastic Load Balancing (ELB) service, it does not support HA (in this case, ELB service provides the failover capabilities).
- The VM-Series firewall on Google Cloud Platform does not support HA.

HA Modes

You can set up the firewalls in an HA pair in one of two modes:

- **Active/Passive**— One firewall actively manages traffic while the other is synchronized and ready to transition to the active state, should a failure occur. In this mode, both firewalls share the same configuration settings, and one actively manages traffic until a path, link, system, or network failure occurs. When the active firewall fails, the passive firewall transitions to the active state and takes over seamlessly and enforces the same policies to maintain network security. Active/passive HA is supported in the virtual wire, Layer 2, and Layer 3 deployments.
- **Active/Active**— Both firewalls in the pair are active and processing traffic and work synchronously to handle session setup and session ownership. Both firewalls individually maintain session tables and routing tables and synchronize to each other. Active/active HA is supported in virtual wire and Layer 3 deployments.

  In active/active HA mode, the firewall does not support DHCP client. Furthermore, only the active-primary firewall can function as a DHCP Relay. If the active-secondary firewall receives DHCP broadcast packets, it drops them.

  An active/active configuration does not load-balance traffic. Although you can load-share by sending traffic to the peer, no load balancing occurs. Ways to load share sessions to both firewalls include using ECMP, multiple ISPs, and load balancers.

When deciding whether to use active/passive or active/active mode, consider the following differences:

- Active/passive mode has simplicity of design; it is significantly easier to troubleshoot routing and traffic flow issues in active/passive mode. Active/passive mode supports a Layer 2 deployment; active/active mode does not.
- Active/active mode requires advanced design concepts that can result in more complex networks. Depending on how you implement active/active HA, it might require additional configuration such as activating networking protocols on both firewalls, replicating NAT pools, and deploying floating IP addresses to provide proper failover. Because both firewalls are actively processing traffic, the firewalls use additional concepts of session owner and session setup to perform Layer 7 content inspection. Active/active mode is recommended if each firewall needs its own routing instances and you require full, real-time redundancy out of both firewalls all the time. Active/active mode has faster failover and can handle peak traffic flows better than active/passive mode because both firewalls are actively processing traffic.

  In active/active mode, the HA pair can be used to temporarily process more traffic than what one firewall can normally handle. However, this should not be the norm because a failure of one firewall causes all traffic to be redirected to the remaining firewall in the HA pair. Your design must allow the remaining firewall to process the maximum capacity of your traffic loads with content inspection enabled. If the design oversubscribes the capacity of the remaining firewall, high latency and/or application failure can occur.

For information on setting up your firewalls in active/passive mode, see Set Up Active/Passive HA. For information on setting up your firewalls in active/active mode, see Set Up Active/Active HA.

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In an HA cluster, all members are considered active; there is no concept of passive firewalls except for HA pairs in the clusters, which can keep their active/passive relationship after you add them to an HA cluster.

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HA Links and Backup Links

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

The firewalls in an HA pair use HA links to synchronize data and maintain state information. Some models of the firewall have dedicated HA ports—Control link (HA1) and Data link (HA2), while others require you to use the in-band ports or management ports as HA links. On PA-3500 Series and PA-5510, PA-5520, and PA-5530 firewalls, the HSCI port serves as both the HA1 control link and the HA2 data link in a single raw Layer 1 connection. PA-5540, PA-5550, PA-5560, PA-5570, PA-5580, and PA-7500 firewalls use NGFW clustering instead of traditional HA and connect using the HSCI port as the Inter Firewall Link (IFL).

- For firewalls with dedicated HA ports, use these ports to manage communication and synchronization between the firewalls. For details, see HA Ports on Palo Alto Networks Firewalls.
- For firewalls without dedicated HA ports such as the PA-220 and PA-220R firewalls, as a best practice use the management port for the HA1 port, and use the dataplane port for the HA1 backup. If you use the management port and configured a Permitted IP list, you must add the peer HA1 IP address to the Permitted IP list. (If you didn't configure a Permitted IP list, you don't need to add the peer HA1 IP address.)

You can configure data ports as both dedicated HA interfaces and as dedicated backup HA interfaces. For firewalls without dedicated HA interfaces, such as the PA-200 and PA-400 Series, it is required to configure a data port as a HA interface.

Data ports configured as HA1, HA2, or HA3 interfaces can be connected directly to each HA interface on the firewall or connected through a Layer2 switch. For data ports configured as an HA3 interface, you must enable jumbo frames as HA3 messages exceed 1,500 bytes.

HA peers in an HA cluster can be a combination of standalone members and HA pairs. HA cluster members use an HA4 link and HA4 backup link to perform session state syncronization. HA1 (control link), HA2 (data link), and HA3 (packet-forwarding link) are not supported between cluster members that aren’t HA pairs.

Any filtering device between your HA pair members must allow the protocols and ports that the HA pair uses, as described in the following table.

HA Links and Backup Links	Description
Control Link	The HA1 link is used to exchange hellos, heartbeats, and HA state information, and management plane sync for routing, and User-

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HA Links and Backup Links	Description
	ID information. The firewalls also use this link to synchronize configuration changes with its peer. The HA1 link is a Layer 3 link and requires an IP address. ICMP is used to exchange heartbeats between HA peers. Ports used for HA1—TCP port 28769 and 28260 for clear text communication; port 28 for encrypted communication (SSH over TCP). If you enable encryption on the HA1 link, you can also Refresh HA1 SSH Keys and Configure Key Options. *On PA-3500 Series and PA-5510, PA-5520, and PA-5530 firewalls, the HSCI port serves as both the HA1 control link and the HA2 data link. The HSCI port carries raw Layer 1 traffic and does not use an IP address.*
Data Link	The HA2 link is used to synchronize sessions, forwarding tables, IPSec security associations and ARP tables between firewalls in an HA pair. Data flow on the HA2 link is always unidirectional (except for the HA2 keep-alive); it flows from the active or active-primary firewall to the passive or active-secondary firewall. The HA2 link is a Layer 2 link, and it uses ether type 0x7261 by default. Ports used for HA2—The HA data link can be configured to use either IP (protocol number 99) or UDP (port 29281) as the transport, and thereby allow the HA data link to span subnets. *On PA-3500 Series and PA-5510, PA-5520, and PA-5530 firewalls, there is no separate HA2 link. The HSCI port serves as both the HA1 control link and the HA2 data link.*
HA1 and HA2 Backup Links	Provide redundancy for the HA1 and the HA2 links. In-band ports can be used for backup links for both HA1 and HA2 connections when dedicated backup links are not available. Consider the following guidelines when configuring backup HA links: • The IP addresses of the primary and backup HA links must not overlap each other. • HA backup links must be on a different subnet from the primary HA links. • HA1-backup and HA2-backup ports must be configured on separate physical ports. The HA1-backup link uses port 28770 and 28260. • PA-3200 Series firewalls don't support an IPv6 address for the HA1-backup link; use an IPv4 address.

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HA Links and Backup Links	Description
	Palo Alto Networks recommends enabling heartbeat backup (uses port 28771 on the MGT interface) if you use an in-band port for the HA1 or the HA1 backup links. *PA-3500 Series and PA-5510, PA-5520, and PA-5530 firewalls do not use separate HA1-backup or HA2-backup links. Connecting both HSCI ports between the firewalls provides redundancy for the HA connection.*
Packet-Forwarding Link	In addition to HA1 and HA2 links, an active/active deployment also requires a dedicated HA3 link. The firewalls use this link for forwarding packets to the peer during session setup and asymmetric traffic flow. The HA3 link is a Layer 2 link that uses MAC-in-MAC encapsulation. It does not support Layer 3 addressing or encryption. PA-7000 Series firewalls synchronize sessions across the NPCs one-for-one. On PA-800 Series, PA-3200 Series, PA-3400 Series, PA-5200 Series, and PA-5400 Series firewalls, you can configure aggregate interfaces as an HA3 link. The aggregate interfaces can also provide redundancy for the HA3 link; you cannot configure backup links for the HA3 link. On PA-1400 Series, PA-1500 Series, PA-3200 Series, PA-3400 Series, PA-5200 Series, PA-5400 Series, and PA-7000 Series firewalls, the dedicated HSCI ports support the HA3 link. The firewall adds a proprietary packet header to packets traversing the HA3 link, so the MTU over this link must be greater than the maximum packet length forwarded. *PA-3500 Series and PA-5510, PA-5520, and PA-5530 firewalls support active/passive HA only and do not use an HA3 link. PA-5540, PA-5550, PA-5560, PA-5570, PA-5580, and PA-7500 firewalls use NGFW clustering and do not use an HA3 link.*
HA4 Link and HA4 Backup Link	The HA4 link and HA4 backup link perform session cache synchronization among all HA cluster members having the same cluster ID. The HA4 link between cluster members detects connectivity failures between cluster members by sending and receiving Layer 2 keepalive messages. View the status of the HA4 and HA4 backup links on the firewall dashboard.

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HA Ports on Palo Alto Networks Firewalls

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

When connecting two Palo Alto Networks® firewalls in a high availability (HA) configuration, we recommend that you use the dedicated HA ports for HA Links and Backup Links. These dedicated ports include: the HA1 ports labeled HA1, HA1-A, and HA1-B used for HA control and synchronization traffic; and HA2 and the High Speed Chassis Interconnect (HSCI) ports used for HA session setup traffic. The PA-5200 Series firewalls have multipurpose auxiliary ports labeled AUX-1 and AUX-2 that you can configure for HA1 traffic.

You can also configure the HSCI port for HA3, which is used for packet forwarding to the peer firewall during session setup and asymmetric traffic flow (active/active HA only). The HSCI port can be used for HA2 traffic, HA3 traffic, or both.

The HA1 and AUX links provide synchronization for functions that reside on the management plane. Using the dedicated HA interfaces on the management plane is more efficient than using the in-band ports as this eliminates the need to pass the synchronization packets over the dataplane.

You can configure data ports as both dedicated HA interfaces and as dedicated backup HA interfaces. For firewalls without dedicated HA interfaces, such as the PA-200 and PA-400 Series, it is required to configure a data port as a HA interface.

Data ports configured as HA1, HA2, or HA3 interfaces can be connected directly to each HA interface on the firewall or connected through a Layer2 switch. For data ports configured as an HA3 interface, you must enable jumbo frames as HA3 messages exceed 1,500 bytes.

Whenever possible, connect HA ports directly between the two firewalls in an HA pair (not through a switch or router) to avoid HA link and communications problems that could occur if there is a network issue.

Use the following table to learn about dedicated HA ports and how to connect the HA Links and Backup Links:

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Model	Front-Panel Dedicated Port(s)

PA-800 Series Firewalls	• **HA1 and HA2**—Ethernet 10Mbps/100Mbps/1000Mbps ports used for HA1 and HA2 in both HA Modes.
	  • **For HA1 traffic**—Connect the HA1 port on the first firewall directly to the HA1 port on the second firewall in the pair or connect these ports together through a switch or router.
	  • **For HA2 traffic**—Connect the HA2 port on the first firewall directly to the HA2 port on the second firewall in the pair or connect these ports together through a switch or router.

PA-1400 Series Firewalls	• **HA1-A and HA1-B**—Ethernet 10Mbps/100Mbps/1000Mbps ports used for HA1 traffic in both HA Modes.
	  • **For HA1 traffic**—Connect the HA1-A port on the first firewall directly to the HA1-A port on the second firewall in the pair or connect them together through a switch or router.
	  • **For a backup to the HA1-A connection**—Connect the HA1-B port on the first firewall directly to the HA1-B port on the second firewall in the pair or connect them together through a switch or router.
	
	  If the firewall dataplane restarts due to a failure or manual restart, the HA1-B link will also restart. If this occurs and the HA1-A link is not connected and configured, then a split brain condition occurs. Therefore, we recommend that you connect and configure the HA1-A ports and the HA1-B ports to provide redundancy and to avoid split brain issues.
	• **HSCI**—The HSCI port is a Layer 1 SFP+ interface that connects two PA-1400 Series firewalls in an HA configuration. Use this port for an HA2 connection, HA3 connection, or both.
	
	  The traffic carried on the HSCI port is raw Layer 1 traffic, which is not routable or switchable. Therefore, you must connect the HSCI ports directly to each other (from the HSCI port on the first firewall to the HSCI port on the second firewall).

PA-1500 Series Firewalls	• **HA1-A and HA1-B**—Ethernet 10Mbps/100Mbps/1000Mbps ports used for HA1 traffic in both HA Modes.
	  • **For HA1 traffic**—Connect the HA1-A port on the first firewall directly to the HA1-A port on the second firewall in the pair or connect them together through a switch or router.
	  • **For a backup to the HA1-A connection**—Connect the HA1-B port on the first firewall directly to the HA1-B port on the

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Model	Front-Panel Dedicated Port(s)
	second firewall in the pair or connect them together through a switch or router. *If the firewall dataplane restarts due to a failure or manual restart, the HA1-B link will also restart. If this occurs and the HA1-A link is not connected and configured, then a split brain condition occurs. Therefore, we recommend that you connect and configure the HA1-A ports and the HA1-B ports to provide redundancy and to avoid split brain issues.* • **HSCI**—The HSCI port is a Layer 1 interface that connects two PA-1500 Series firewalls in an HA configuration. Use this port for an HA2 connection, HA3 connection, or both. *The HSCI port on the PA-1510 firewall is an SFP+/SFP (10Gbps/1Gbps) port and the HSCI port on the PA-1520 and PA-1530 firewalls is an SFP28/SFP+/SFP (25Gbps/10Gbps/1Gbps) port.* The traffic carried on the HSCI port is raw Layer 1 traffic, which is not routable or switchable. Therefore, you must connect the HSCI ports directly to each other (from the HSCI port on the first firewall to the HSCI port on the second firewall).

PA-3200 Series Firewalls	• **HA1-A and HA1-B**—Ethernet 10Mbps/100Mbps/1000Mbps ports used for HA1 traffic in both HA Modes.
	  • **For HA1 traffic**—Connect the HA1-A port on the first firewall directly to the HA1-A port on the second firewall in the pair or connect them together through a switch or router.
	  • **For a backup to the HA1-A connection**—Connect the HA1-B port on the first firewall directly to the HA1-B port on the second firewall in the pair or connect them together through a switch or router.
	
	If the firewall dataplane restarts due to a failure or manual restart, the HA1-B link will also restart. If this occurs and the HA1-A link is not connected and configured, then a split brain condition occurs. Therefore, we recommend that you connect and configure the HA1-A ports and the HA1-B ports to provide redundancy and to avoid split brain issues.
	
	You can remap the firewall's SFP ports as HA1-A and HA1-B ports via PAN-OS or Panorama.

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Model	Front-Panel Dedicated Port(s)
	• **HSCI**—The HSCI port is a Layer 1 SFP+ interface that connects two PA-3200 Series firewalls in an HA configuration. Use this port for an HA2 connection, HA3 connection, or both. The traffic carried on the HSCI ports is raw Layer 1 traffic, which is not routable or switchable. Therefore, you must connect the HSCI ports directly to each other (from the HSCI port on the first firewall to the HSCI port on the second firewall).

PA-3400 Series Firewalls	• **HA1-A and HA1-B**—Ethernet 10Mbps/100Mbps/1000Mbps ports used for HA1 traffic in both HA Modes.
	  • **For HA1 traffic**—Connect the HA1-A port on the first firewall directly to the HA1-A port on the second firewall in the pair or connect them together through a switch or router.
	  • **For a backup to the HA1-A connection**—Connect the HA1-B port on the first firewall directly to the HA1-B port on the second firewall in the pair or connect them together through a switch or router.
	
	If the firewall dataplane restarts due to a failure or manual restart, the HA1-B link will also restart. If this occurs and the HA1-A link is not connected and configured, then a split brain condition occurs. Therefore, we recommend that you connect and configure the HA1-A ports and the HA1-B ports to provide redundancy and to avoid split brain issues.
	
	• **HSCI**—The HSCI port is a Layer 1 SFP+ interface that connects two PA-3400 Series firewalls in an HA configuration. Use this port for an HA2 connection, HA3 connection, or both.
	
	The traffic carried on the HSCI port is raw Layer 1 traffic, which is not routable or switchable. Therefore, you must connect the HSCI ports directly to each other (from the HSCI port on the first firewall to the HSCI port on the second firewall).
	
	The management interface cannot be configured as a HA port.

PA-3500 Series Firewalls	• **HSCI**—Two HSCI ports that offer 40Gbps/100Gbps connectivity and connect two PA-3500 Series firewalls in an active/passive HA configuration. The HSCI ports serve as both the HA1 control link and the HA2 data link. The traffic carried on the HSCI ports is raw Layer 1 traffic, which is not routable or switchable. Therefore, you must connect the HSCI ports directly to each other (from the HSCI port on the first firewall to the HSCI port on the second firewall).

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Model	Front-Panel Dedicated Port(s)

PA-5200 Series Firewalls	• **HA1-A and HA1-B**—Ethernet 10Mbps/100Mbps/1000Mbps ports used for HA1 traffic in both HA Modes.
	  • **For HA1 traffic**—Connect the HA1-A port on the first firewall directly to the HA1-A port on the second firewall in the pair or connect them together through a switch or router.
	  • **For a backup to the HA1-A connection**—Connect the HA1-B port on the first firewall directly to the HA1-B port on the second firewall in the pair or connect them together through a switch or router.
	• **HSCI**—The HSCI port is a Layer 1 interface that connects two PA-5200 Series firewalls in an HA configuration. Use this port for an HA2 connection, HA3 connection, or both.
	 The HSCI port on the PA-5220 firewall is a QSFP+ port and the HSCI port on the PA-5250, PA-5260, and PA-5280 firewalls is a QSFP28 port.
	The traffic carried on the HSCI port is raw Layer 1 traffic, which is not routable or switchable. Therefore, you must connect the HSCI ports directly to each other (from the HSCI port on the first firewall to the HSCI port on the second firewall).
	• **AUX-1 and AUX-2**—The auxiliary SFP+ ports are multipurpose ports that you can configure for HA1, management functions, or log forwarding to Panorama. Use these ports when you need a fiber connection for one of these functions.
	  • **For HA1 traffic**—Connect the AUX-1 port on the first firewall directly to the AUX-1 port on the second firewall in the pair or connect them together through a switch or router.
	  • **For a backup to the AUX-1 connection**—Connect the AUX-2 port on the first firewall directly to the AUX-2 port on the second firewall in the pair or connect them together through a switch or router.

PA-5400 Series Firewalls (PA-5410, PA-5420, PA-5430, and PA-5440)	• **HA1-A and HA1-B**—SFP/SFP+ 1Gbps/10Gbps ports used for HA1 traffic in both HA Modes.
	  • **For HA1 traffic**—Connect the HA1-A port on the first firewall directly to the HA1-A port on the second firewall in the pair or connect them together through a switch or router.
	  • **For a backup to the HA1-A connection**—Connect the HA1-B port on the first firewall directly to the HA1-B port on the second firewall in the pair or connect them together through a switch or router.

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Model	Front-Panel Dedicated Port(s)
	• **HSCI**—The HSCI port is a Layer 1 QSFP+ interface that connects two PA-5400 Series firewalls in an HA configuration. Use this port for an HA2 connection, HA3 connection, or both. The traffic carried on the HSCI port is raw Layer 1 traffic, which is not routable or switchable. Therefore, you must connect the HSCI ports directly to each other (from the HSCI port on the first firewall to the HSCI port on the second firewall). • **For HA2 and HA3 traffic**—Connect the HSCI-A port on the first firewall directly to the HSCI-A port on the second firewall. *You can use the firewall data ports for HA2 or HA3 traffic as well; however, the same data port cannot be used for both HA2 and HA3 at the same time. To have both HA2 and HA3 connections, you must use separate data ports.*
PA-5450 Firewall	• **HA1-A and HA1-B**—SFP/SFP+ 1Gbps/10Gbps ports used for HA1 traffic in both HA Modes. • **For HA1 traffic**—Connect the HA1-A port on the first firewall directly to the HA1-A port on the second firewall in the pair or connect them together through a switch or router. • **For a backup to the HA1-A connection**—Connect the HA1-B port on the first firewall directly to the HA1-B port on the second firewall in the pair or connect them together through a switch or router. • **HSCI-A and HSCI-B**—The HSCI ports are Quad-SFP+ (QSFP+/QSFP28) interfaces that connect two PA-5450 firewalls in an

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Model	Front-Panel Dedicated Port(s)
	HA configuration. Use these ports for an HA2 connection, HA3 connection, or both. The traffic carried on the HSCI ports is raw Layer 1 traffic, which is not routable or switchable. Therefore, you must connect these ports as follows: • **For HA2 and HA3 traffic**—Connect the HSCI-A port on the first firewall directly to the HSCI-A port on the second firewall. *You can configure HA2 (data link) on the HSCI ports or on NC data ports. When configuring on dataplane ports, you must ensure that both the HA2 and HA2-Backup links are configured on dataplane interfaces. A mix of a dataplane port and an HSCI port for either HA2 or HA2-Backup will result in a commit failure.* • **For a backup to the HSCI-A connection**—Connect the HSCI-B port on the first firewall directly to the HSCI-B port on the second firewall.
PA-5500 Series Firewalls (PA-5510, PA-5520, and PA-5530)	• **HSCI**—Two HSCI ports that offer 40Gbps/100Gbps connectivity and connect two PA-5510, PA-5520, or PA-5530 firewalls in an active/passive HA configuration. The HSCI ports serve as both the HA1 control link and the HA2 data link. The traffic carried on the HSCI ports is raw Layer 1 traffic, which is not routable or switchable. Therefore, you must connect the HSCI ports directly to each other (from the HSCI port on the first firewall to the HSCI port on the second firewall).
PA-5500 Series Firewalls (PA-5540, PA-5550, PA-5560, PA-5570, and PA-5580)	• **HSCI**—Two HSCI ports that offer 100Gbps/400Gbps connectivity and connect two PA-5540, PA-5550, PA-5560, PA-5570, or PA-5580 firewalls in an NGFW clustering configuration. The HSCI ports create the Inter Firewall Link (IFL) between the firewalls. The traffic carried on the HSCI ports is raw Layer 1 traffic, which is not routable or switchable. Therefore, you must connect the HSCI ports directly to each other (from the HSCI port on the first firewall to the HSCI port on the second firewall).
PA-7000 Series Firewalls	• **HA1-A and HA1-B**—Ethernet 10Mbps/100Mbps/1000Mbps ports used for HA1 traffic in both HA Modes. • **For HA1 traffic**—Connect the HA1-A port on the first firewall directly to the HA1-A port on the second firewall in the pair or connect them together through a switch or router. • **For a backup to the HA1-A connection**—Connect the HA1-B port on the first firewall directly to the HA1-B port on the

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Model	Front-Panel Dedicated Port(s)
	second firewall in the pair or connect them together through a switch or router. *You cannot configure an HA1 connection on the NPC data ports or the management (MGT) port.* • **HSCI-A and HSCI-B**—The HSCI ports are Layer 1 QSFP+ interfaces that connect two PA-7000 Series firewalls in an HA configuration. Use these ports for an HA2 connection, HA3 connection, or both. The traffic carried on the HSCI ports is raw Layer 1 traffic, which is not routable or switchable. Therefore, you must connect these ports as follows: • **For HA2 and HA3 traffic**—Connect the HSCI-A port on the first firewall directly to the HSCI-A port on the second firewall. *For HA2 or HA2/HA3 traffic, the PA-7000 Series firewalls synchronize sessions across the NPCs one-for-one.* • **For a backup to the HSCI-A connection**—Connect the HSCI-B port on the first firewall directly to the HSCI-B port on the second firewall. *HA2 and HA2-Backup links can be configured to use a dataplane interface instead of the HSCI ports. However, if configured this way, both the HA2 and HA2-Backup links need to use dataplane interfaces. A mix of a dataplane port and an HSCI port for either HA2 or HA2-Backup will result in a commit failure. This applies to the PA-7050-SMC, PA-7080-SMC, PA-7050-SMC-B, and PA-7080-SMC-B.*
PA-7500 Firewall	• **HSCI-A and HSCI-B**—QSFP-DD interfaces on the Management Processing Card (MPC) that offer 100Gbps/400Gbps connectivity and connect two PA-7500 firewalls in an NGFW clustering

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Model	Front-Panel Dedicated Port(s)
	configuration. The HSCI ports create the Inter Firewall Link (IFL) between the firewalls. The traffic carried on the HSCI ports is raw Layer 1 traffic, which is not routable or switchable. Therefore, you must connect these ports as follows: • **For the IFL connection**—Connect the HSCI-A port on the first firewall directly to the HSCI-A port on the second firewall. • **For a backup to the HSCI-A connection**—Connect the HSCI-B port on the first firewall directly to the HSCI-B port on the second firewall. *Do not assign an IP address to the HSCI ports. The ports carry raw Layer 1 traffic that is not routable or switchable and does not support IP addressing. Assigning an IP address to an HSCI port causes the firewall to enter a permanent suspended state.*

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Device Priority and Preemption

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

In high-availability (HA) firewall pairs, device priority and preemption work together to determine which firewall acts as the primary, or active, unit. These settings are crucial for managing failover behavior and ensuring consistent traffic flow

The firewalls in an Active-Passive HA pair can be assigned a *device priority* value to indicate a preference for which firewall should assume the active role. If you need to use a specific firewall in the HA pair for actively securing traffic, you must enable the preemptive behavior on both the firewalls and assign a device priority value for each firewall. The firewall with the lower numerical value, and therefore *higher priority*, is designated as active. The other firewall is the passive firewall. When both firewalls are healthy and online, the one with the highest priority will assume the active role. If both firewalls have the same priority, other factors like the MAC address of the HA link may be used as a tie-breaker.

The same is true for an Active-Active HA pair; however, the *device ID* is used to assign a device priority value. Similarly, the lower numerical value in device ID corresponds to a higher priority. The firewall with the higher priority becomes active-primary and the paired firewall becomes active-secondary.

By default, preemption is disabled on the firewalls and must be enabled on both firewalls. When enabled, the preemptive behavior allows the firewall with the *higher priority* (lower numerical value) to resume as active or active-primary after it recovers from a failure. When preemption occurs, the event is logged in the system logs.

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Failover

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Pro

When a failure occurs on one firewall and the peer in the HA pair (or a peer in the HA cluster) takes over the task of securing traffic, the event is called a *failover*. A failover is triggered, for example, when a monitored metric on a firewall in the HA pair fails. The metrics that the firewall monitors for detecting a firewall failure are:

- **Heartbeat Polling and Hello messages**

  The firewalls use hello message and heartbeats to verify that the peer firewall is responsive and operational. Hello messages are sent from one peer to the other at the configured *Hello Interval* to verify the state of the firewall. The heartbeat is an ICMP ping to the HA peer over the control link, and the peer responds to the ping to establish that the firewalls are connected and responsive. By default, the interval for the heartbeat is 1,000 milliseconds. A ping is sent every 1,000 milliseconds and if there are three consecutive heartbeat losses, a failover occurs. For details on the HA timers that trigger a failover, see HA Timers.

- **Link Monitoring**

  You can specify a group of physical interfaces that the firewall will monitor (a link group) and the firewall monitors the state of each link in the group (link up or link down). You determine the failure condition for the link group: **Any** link down or **All** links down in the group constitutes a link group failure (but not necessarily a failover).

  You can create multiple link groups. Therefore, you also determine the failure condition of the set of link groups: **Any** link group fails or **All** link groups fail, which determines when a failover is triggered. The default behavior is that failure of **Any** one link in **Any** link group causes the firewall to change the HA state to non-functional (or to tentative state in active/active mode) to indicate a failure of a monitored object.

- **Path Monitoring**

  You can specify a destination IP group of IP address that the firewall will monitor. The firewall monitors the full path through the network to mission-critical IP addresses using ICMP pings to verify reachability of the IP address. The default interval for pings is 200ms. An IP address is considered unreachable when 10 consecutive pings (the default value) fail. You specify the failure condition for the IP addresses in a destination IP group: **Any** IP address unreachable or **All** IP addresses unreachable in the group. You can specify multiple destination IP groups for a path group for a virtual wire, VLAN, or virtual router; you specify the failure condition of destination IP groups in a path group: **Any** or **All**, which constitutes a path group failure.

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You can configure multiple virtual wire path groups, VLAN path groups, and virtual router path groups.

You also determine the global failure condition: **Any** path group fails or **All** path groups fail, which determines when a failover is triggered. The default behavior is that **Any** one of the IP addresses becoming unreachable in **Any** destination IP group in **Any** virtual wire, VLAN, or virtual router path group causes the firewall to change the HA state to non-functional (or to tentative state in active/active mode) to indicate a failure of a monitored object.

In an HA pair, link monitoring failures or path monitoring failures won't take down the so-called "last device standing." Instead, when the only failure is link or path monitoring or a slot mismatch failure and the peer is in a non-active state, the active state device continues to log link failures and path failures and won't fault or move out of active state due to these failures. We refer to this behavior as the "last device standing."

States that are functional are active, passive, active-primary, and active-secondary. States that are not functional are initial, non-functional, tentative, and suspended.

The preceding behavior for link or path monitoring can result in a loop if the link or path continues to be down, especially for path monitoring in an active/passive HA pair. In this case, a device must come back to active state to monitor the links or path because these are generally down (links can be up, depending on the Passive Link State setting). Therefore, an HA pair loops through a failure, especially if both devices have path monitoring enabled and both are failing. The loop is broken by the Flap Max setting, which puts the device into a suspended state to break the loop. When this looping behavior results in the device being in a suspended state, the other (peer) device in the HA pair remains in an active state. This is one example of the "last device standing."

An HA cluster does not support the "last device standing" behavior.

Keep in mind that a device in a suspended state can't move to an HA functional state without your [user] intervention.

In addition to the failover triggers listed above, a failover also occurs when the administrator suspends the firewall or when preemption occurs.

Depending on the firewall model, a failover can occur when an internal health check fails. This health check isn't configurable and is enabled to monitor the critical components, such as the FPGA and CPUs. Additionally, general health checks occur on any platform, causing failover.

The following describes what occurs in the event of a failure of a Network Processing Card (NPC) on a PA-7000 Series firewall or a PA-5450 firewall that is a member of an HA cluster:

- If the NPC that's being used to hold the HA clustering session cache (a copy of the other members' sessions) goes down, the firewall goes non-functional. When this occurs, the session distribution device (such as a load balancer) must detect that the firewall is down and distribute session load to the other members of the cluster.
- If the NPC of a cluster member goes down and no link monitoring or path monitoring is enabled on that NPC, the PA-7000 Series firewall or PA-5450 firewall member will stay up, but with a lower capacity because one NPC is down.
- If the NPC of a cluster member goes down and link monitoring or path monitoring is enabled on that NPC, the PA-7000 Series firewall or PA-5450 firewall will go non-functional and the

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session distribution device (such as a load balancer) must detect that the firewall is down and distribute session load to the other members of the cluster.

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Floating IP Address and Virtual MAC Address

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

In a Layer 3 deployment of HA active/active mode, you can assign floating IP addresses, which move from one HA firewall to the other if a link or firewall fails. The interface on the firewall that owns the floating IP address responds to ARP requests with a virtual MAC address.

Floating IP addresses are recommended when you need functionality such as Virtual Router Redundancy Protocol (VRRP). Floating IP addresses can also be used to implement VPNs and source NAT, allowing for persistent connections when a firewall offering those services fails.

As shown in the figure below, each HA firewall interface has its own IP address and floating IP address. The interface IP address remains local to the firewall, but the floating IP address moves between the firewalls upon firewall failure. You configure the end hosts to use a floating IP address as its default gateway, allowing you to load balance traffic to the two HA peers. You can also use external load balancers to load balance traffic.

If a link or firewall fails or a path monitoring event causes a failover, the floating IP address and virtual MAC address move over to the functional firewall. (In the figure below, each firewall has two floating IP addresses and virtual MAC addresses; they all move over if the firewall fails.) The functioning firewall sends a gratuitous ARP to update the MAC tables of the connected switches to inform them of the change in floating IP address and MAC address ownership to redirect traffic to itself.

After the failed firewall recovers, by default the floating IP address and virtual MAC address move back to firewall with the Device ID [0 or 1] to which the floating IP address is bound. More specifically, after the failed firewall recovers, it comes on line. The currently active firewall determines that the firewall is back online and checks whether the floating IP address it is handling belongs natively to itself or the other firewall. If the floating IP address was originally bound to the other Device ID, the firewall automatically gives it back. (For an alternative to this default behavior, see Use Case: Configure Active/Active HA with Floating IP Address Bound to Active-Primary Firewall.)

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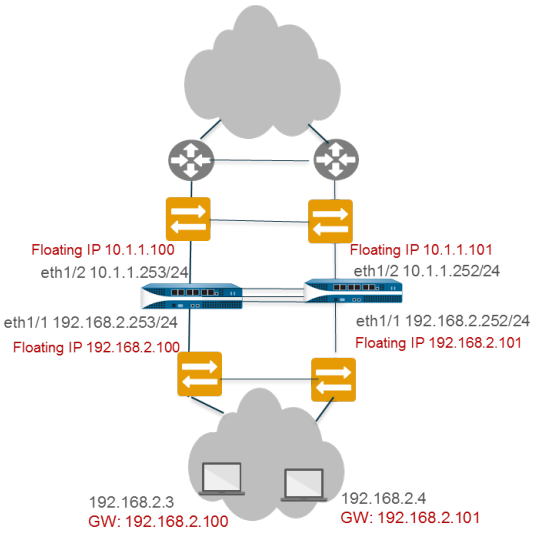

Each firewall in the HA pair creates a virtual MAC address for each of its interfaces that has a floating IP address or ARP Load-Sharing IP address.

The format of the virtual MAC address on PA-7000, PA-7000b, PA-5400, PA-5200, PA-3200 Series, and CN-Series firewalls is B4-0C-25-XX-YY-ZZ, where B4-0C-25 is the vendor ID (of Palo Alto Networks in this case), and the next 24 bits indicate the Device ID, Group ID and Interface ID as follows:

765	4	321076	5432	1076543210
111	Device-ID	Group-ID	0000	Interface-ID

The following graphic provides an example. Suppose the HA firewall has an Interface ID of 66. The number 66 in binary is 01000010. The Firewall Info row of the pink section shows the rightmost ten bit positions have a 1 in the 64 (binary) column and a 1 in the 2 (binary) column, totaling 66, and two leading zeroes. The green section contains fixed zeroes. Now suppose the firewall Group ID is 58. The number 58 in binary is 111010, as shown in the Firewall Info row of the purple section. Finally, suppose the Device ID is 1, as shown in the Firewall Info row of the blue section. The Firewall Info row of the yellow section contains fixed ones. When you look at the full string of bits, starting from the left, the orange octet totals 254 (decimal), the pale blue octet totals 128 (decimal), and the bright green octet totals 66 (decimal). Converting decimal to hex, we have FE-80-42. Therefore, the full virtual MAC address including the Palo Alto Networks vendor ID is B4-0C-25-FE-80-42.

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				Device-ID	Group ID											Interface ID							
Binary	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
Bit Position	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
Firewall Info	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0
	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0
Decimal	XX			254					YY			128					ZZ			66				
Hex				FE								80								42				
B4:0C:25:XX:YY:ZZ		B4:0C:25:FE:80:42																						

The format of the virtual MAC address on the remaining firewall models is 00-1B-17-00-xx-yy, where 00-1B-17 is the vendor ID (of Palo Alto Networks in this case), 00 is fixed, xx indicates the Device ID and Group ID as shown in the following figure, and yy is the Interface ID:

7	6	5 4 3 2 1 0	7 6 5 4 3 2 1 0
Device-ID	0	Group-ID	Interface-ID

When a new active firewall takes over, it sends gratuitous ARPs from each of its connected interfaces to inform the connected Layer 2 switches of the new location of the virtual MAC address. To configure floating IP addresses, see Use Case: Configure Active/Active HA with Floating IP Addresses.

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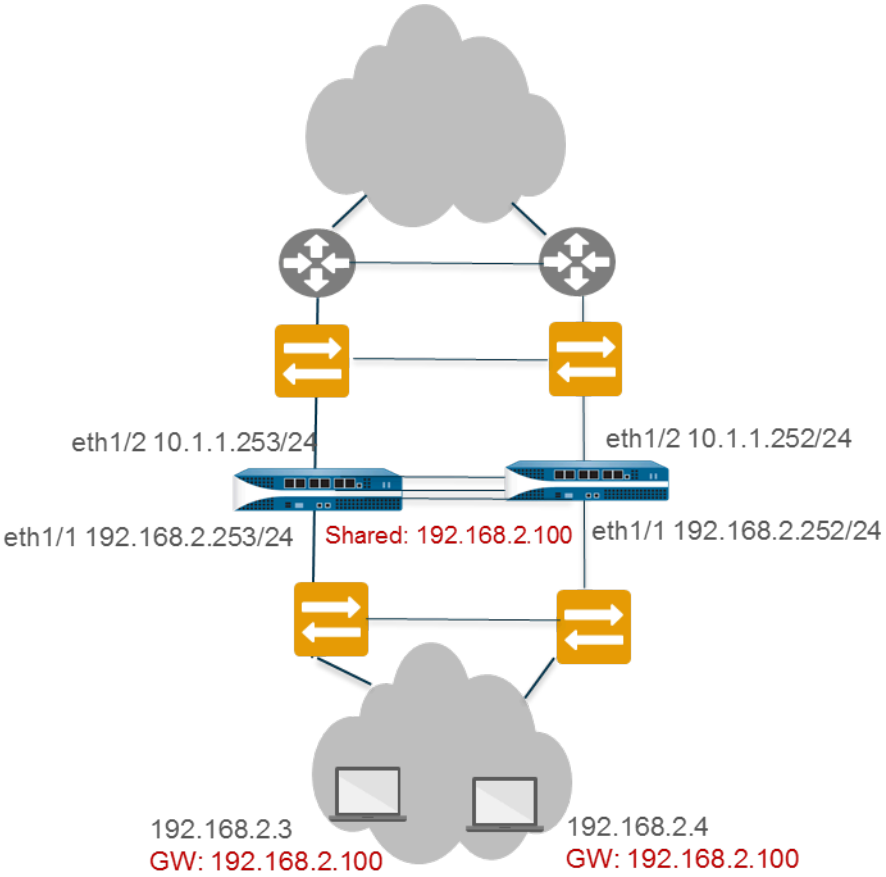
ARP Load-Sharing

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Pro

In a Layer 3 interface deployment and active/active HA configuration, ARP load-sharing allows the firewalls to share an IP address and provide gateway services. Use ARP load-sharing only when no Layer 3 device exists between the firewall and end hosts, that is, when end hosts use the firewall as their default gateway.



In such a scenario, all hosts are configured with a single gateway IP address. One of the firewalls responds to ARP requests for the gateway IP address with its virtual MAC address. Each firewall has a unique virtual MAC address generated for the shared IP address. The load-sharing algorithm that controls which firewall will respond to the ARP request is configurable; it is determined by computing the hash or modulo of the source IP address of the ARP request.

After the end host receives the ARP response from the gateway, it caches the MAC address and all traffic from the host is routed via the firewall that responded with the virtual MAC address for

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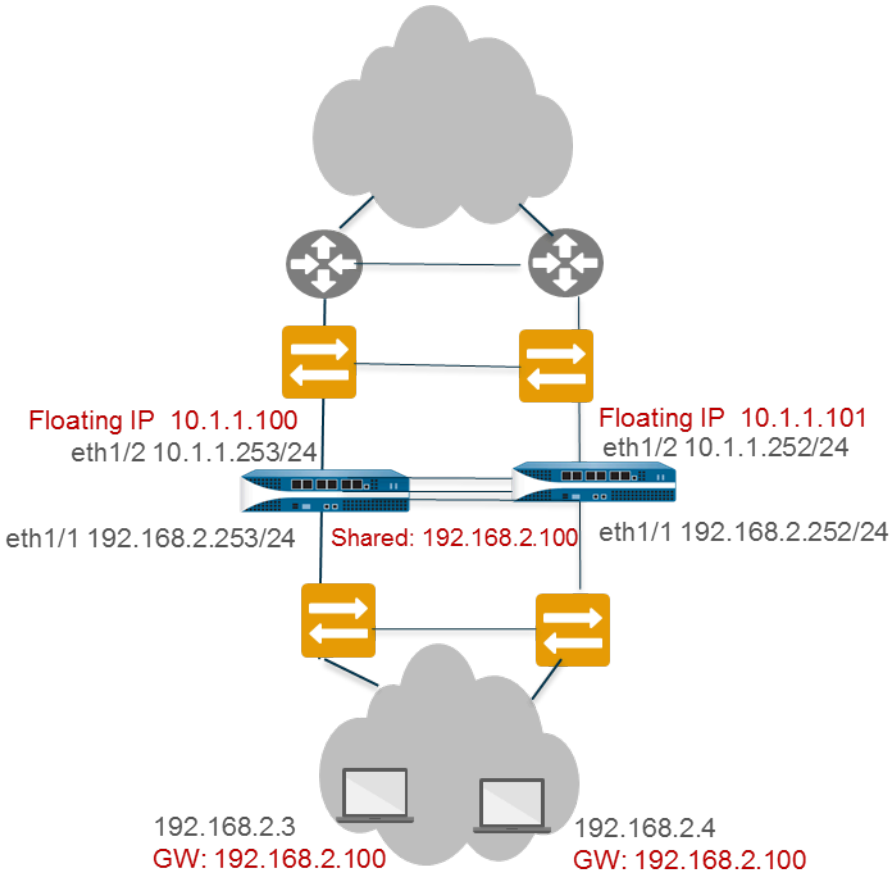
the lifetime of the ARP cache. The lifetime of the ARP cache depends on the end host operating system.

If a link or firewall fails, the floating IP address and virtual MAC address move over to the functional firewall. The functional firewall sends gratuitous ARPs to update the MAC table of the connected switches to redirect traffic from the failed firewall to itself. See Use Case: Configure Active/Active HA with ARP Load-Sharing.

You can configure interfaces on the WAN side of the HA firewalls with floating IP addresses, and configure interfaces on the LAN side of the HA firewalls with a shared IP address for ARP load-sharing. For example, the figure below illustrates floating IP addresses for the upstream WAN edge routers and an ARP load-sharing address for the hosts on the LAN segment.



As illustrated in the floating IP address scenario, the firewall supports a shared IP address for ARP load-sharing only on the LAN side of the firewall; the shared IP address cannot be on the WAN side.

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HA Timers

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

High availability (HA) timers facilitate a firewall to detect a firewall failure and trigger a failover. To reduce the complexity in configuring timers for an HA pair, you can select from three profiles: **Recommended**, **Aggressive** and **Advanced**. These profiles auto-populate the optimum HA timer values for the specific firewall platform to enable a speedier HA deployment.

Use the **Recommended** profile for typical failover timer settings and the **Aggressive** profile for faster failover timer settings. The **Advanced** profile allows you to customize the timer values to suit your network requirements.

The following table describes each timer included in the profiles and the current preset values (Recommended/Aggressive) across the different hardware models; these values are for current reference only and can change in a subsequent release.

Timers that affect members of an HA cluster are described in Configure HA Clustering.

Timers	Description	PA-7500 PA-7000 Series PA-5500 Series PA-5400 Series PA-5200 Series PA-3400 Series PA-3200 Series PA-1400 Series	PA-800 Series PA-500 Series PA-400 Series PA-400R Series PA-220 VM-Series	Panorama Virtual Appliance Panorama M-Series
Monitor Fail Hold Up Time (ms)	Interval during which the firewall will remain active following a path monitor or link monitor failure. This setting is recommended to avoid an HA failover due to the occasional flapping of neighboring devices.	0/0	0/0	0/0

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Timers	Description	PA-7500 PA-7000 Series PA-5500 Series PA-5400 Series PA-5200 Series PA-3400 Series PA-3200 Series PA-1400 Series	PA-800 Series PA-500 Series PA-400 Series PA-400R Series PA-220 VM-Series	Panorama Virtual Appliance Panorama M-Series
Preemption Hold Time (min)	Time that a passive or active-secondary firewall will wait before taking over as the active or active-primary firewall.	1/1	1/1	1/1
Heartbeat Interval (ms)	Frequency at which the HA peers exchange heartbeat messages in the form of an ICMP (ping).	1000/1000	2000/1000	2000/1000
Promotion Hold Time (ms)	Time that the passive firewall (in active/passive mode) or the active-secondary firewall (in active/active mode) will wait before taking over as the active or active-primary firewall after communications with the HA peer have been lost. This hold time will begin only after the peer failure declaration has been made.	2000/500	2000/500	2000/500
Additional Master Hold Up Time (ms)	Time interval in milliseconds that is applied to the same event as Monitor Fail Hold Up Time (range is 0 to 60,000; default is 500). The additional time interval	500/500	500/500	7000/5000

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Timers	Description	PA-7500 PA-7000 Series PA-5500 Series PA-5400 Series PA-5200 Series PA-3400 Series PA-3200 Series PA-1400 Series	PA-800 Series PA-500 Series PA-400 Series PA-400R Series PA-220 VM-Series	Panorama Virtual Appliance Panorama M-Series
	is applied only to the active firewall in active/passive mode and to the active-primary firewall in active/active mode. This timer is recommended to avoid a failover when both firewalls experience the same link/path monitor failure simultaneously.			
Hello Interval (ms)	Interval in milliseconds between hello packets that are sent to verify that the HA functionality on the other firewall is operational (range is 8,000 to 60,000; default is 8,000).	8000/8000	8000/8000	8000/8000
Flap Max	A flap is counted when one of the following occurs: • A preemption-enabled firewall leaves the active state within 20 minutes after becoming active. • A link or path fails to stay up for 10 minutes after becoming functional.	3/3	3/3	Not Applicable

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Timers	Description	PA-7500 PA-7000 Series PA-5500 Series PA-5400 Series PA-5200 Series PA-3400 Series PA-3200 Series PA-1400 Series	PA-800 Series PA-500 Series PA-400 Series PA-400R Series PA-220 VM-Series	Panorama Virtual Appliance Panorama M-Series
	In the case of a failed preemption or non-functional loop, this value indicates the maximum number of flaps that are permitted before the firewall is suspended (range 0 to 16; default is 3).			

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Session Owner

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

In an HA active/active configuration, both firewalls are active simultaneously, which means packets can be distributed between them. Such distribution requires the firewalls to fulfill two functions: session ownership and session setup. Typically, each firewall of the pair performs one of these functions, thereby avoiding race conditions that can occur in asymmetrically routed environments.

You configure the session owner of sessions to be either the firewall that receives the First Packet of a new session from the end host or the firewall that is in active-primary state (the Primary device). If Primary device is configured, but the firewall that receives the first packet is not in active-primary state, the firewall forwards the packet to the peer firewall (the session owner) over the HA3 link.

The session owner performs all Layer 7 processing, such as App-ID, Content-ID, and threat scanning for the session. The session owner also generates all traffic logs for the session.

If the session owner fails, the peer firewall becomes the session owner. The existing sessions fail over to the functioning firewall and no Layer 7 processing is available for those sessions. When a firewall recovers from a failure, by default, all sessions it owned before the failure revert back to that original firewall; Layer 7 processing does not resume.

If you configure session ownership to be Primary device, the session setup defaults to Primary device also.

Palo Alto Networks recommends setting the Session Owner to First Packet and the Session Setup to IP Modulo unless otherwise indicated in a specific use case. Setting the Session Owner to First Packet reduces traffic across the HA3 link and helps distribute the dataplane load across peers.

Setting Session Owner and Session Setup to Primary Device causes the active-primary firewall to perform all traffic processing. You might want to configure this for one of these reasons:

- *You are troubleshooting and capturing logs and pcaps, so that packet processing is not split between the firewalls.*
- *You want to force the active/active HA pair to function like an active/passive HA pair. See* Use Case: Configure Active/Active HA with Floating IP Address Bound to Active-Primary Firewall.

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Session Setup

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

The session setup firewall performs the Layer 2 through Layer 4 processing necessary to set up a new session. The session setup firewall also performs NAT using the NAT pool of the session owner. You determine the session setup firewall in an active/active configuration by selecting one of the following session setup load sharing options.

Session Setup Option	Description
IP Modulo	The firewall distributes the session setup load based on parity of the source IP address. This is a deterministic method of sharing the session setup.
IP Hash	The firewall uses a hash of the source and destination IP addresses to distribute session setup responsibilities.
Primary Device	The active-primary firewall always sets up the session; only one firewall performs all session setup responsibilities.
First Packet	The firewall that receives the first packet of a session performs session setup.

- *If you want to load-share the session owner and session setup responsibilities, set session owner to First Packet and session setup to IP modulo. These are the recommended settings.*
- *If you want to do troubleshooting or capture logs or pcaps, or if you want an active/active HA pair to function like an active/passive HA pair, set both the session owner and session setup to Primary device so that the active-primary device performs all traffic processing. See* Use Case: Configure Active/Active HA with Floating IP Address Bound to Active-Primary Firewall.

The firewall uses the HA3 link to send packets to its peer for session setup if necessary. The following figure and text describe the path of a packet that firewall FW1 receives for a new session. The red dotted lines indicate FW1 forwarding the packet to FW2 and FW2 forwarding the packet back to FW1 over the HA3 link.

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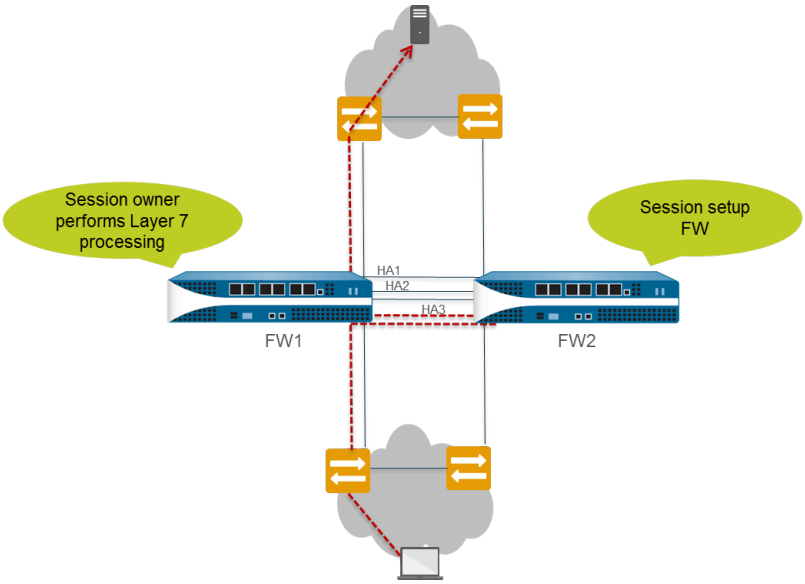
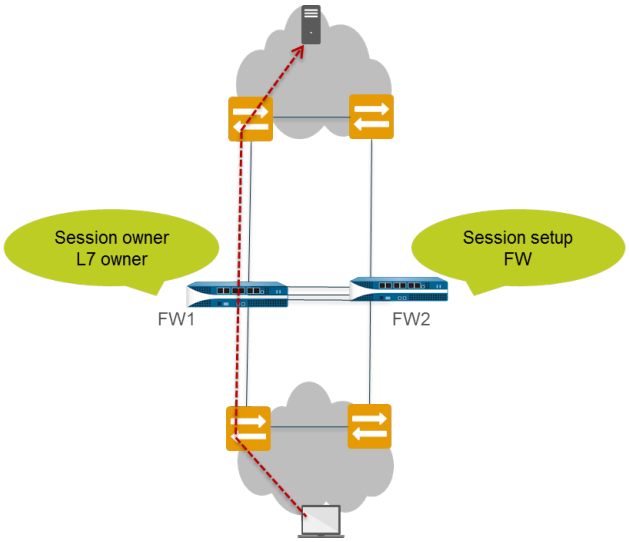


- The end host sends a packet to FW1.
- FW1 examines the contents of the packet to match it to an existing session. If there is no session match, FW1 determines that it has received the first packet for a new session and therefore becomes the session owner (assuming **Session Owner Selection** is set to **First Packet**).
- FW1 uses the configured session setup load-sharing option to identify the session setup firewall. In this example, FW2 is configured to perform session setup.
- FW1 uses the HA3 link to send the first packet to FW2.
- FW2 sets up the session and returns the packet to FW1 for Layer 7 processing, if any.
- FW1 then forwards the packet out the egress interface to the destination.

The following figure and text describe the path of a packet that matches an existing session:



- The end host sends a packet to FW1.

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- FW1 examines the contents of the packet to match it to an existing session. If the session matches an existing session, FW1 processes the packet and sends the packet out the egress interface to the destination.

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Set Up Active/Passive HA

Where Can I Use This?	What Do I Need?
• NGFW	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro ❑ Strata Cloud Manager Essentials

To set up high availability on your Palo Alto Networks firewalls, you need a pair of firewalls that meet the following requirements:

- ❑ **The same model**—Both the firewalls in the pair must be of the same hardware model or virtual machine model. (Verify that by viewing Dashboard, General Information, Model.)
- ❑ **The same PAN-OS version**—Both the firewalls should be running the same PAN-OS version and must each be up-to-date on the application, URL, and threat databases. (Verify that by viewing Dashboard, General Information, Software Version.)
- ❑ The same multi virtual system capability—Both firewalls must have **Multi Virtual System Capability** either enabled or not enabled. When enabled, each firewall requires its own multiple virtual systems licenses. (Verify that by viewing Device > Setup > Management, General Settings, Multi Virtual System Capability enabled or disabled.)

  (Cloud Managed NGFWs Only)—Both firewalls must have the multi-vsys capability disabled.
- ❑ **The same type of interfaces**—Dedicated HA links, or a combination of the management port and in-band ports that are set to *interface type* HA. (Verify the following on Device > High Availability > HA Communications.)
  - **(Cloud Managed NGFWs Only)**—Strata Cloud Manager supports IPv4 addresses only.
  - Determine the IP address for the HA1 (control) connection between the HA peers. The HA1 IP address for both peers must be on the same subnet if they are directly connected or are connected to the same switch.

    For firewalls without dedicated HA ports, you can use the management port for the control connection. Using the management port provides a direct communication link between the management planes on both firewalls. However, because the management ports will not be directly cabled between the peers, make sure that you have a route that connects these two interfaces across your network.
  - If you use Layer 3 as the transport method for the HA2 (data) connection, determine the IP address for the HA2 link. Use Layer 3 only if the HA2 connection must communicate over a routed network. The IP subnet for the HA2 links must not overlap with that of the HA1 links or with any other subnet assigned to the data ports on the firewall.
- ❑ **The same set of licenses**—Licenses are unique to each firewall and cannot be shared between the firewalls. Therefore, you must license both firewalls identically. If both firewalls do not have an identical set of licenses, they cannot synchronize configuration information and

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maintain parity for a seamless failover. (Verify that the licenses match by comparing Device > Licenses.)

As a best practice, if you have an existing firewall and you want to add a new firewall for HA purposes and the new firewall has an existing configuration Reset the Firewall to Factory Default Settings on the new firewall. This ensures that the new firewall has a clean configuration. After HA is configured, you will then sync the configuration on the primary firewall to the newly introduced firewall with the clean configuration.

- ☐ **(Cloud Managed NGFWs Only)**—Both firewalls in the HA pair must be added to the same folder.

  Firewalls in an HA pair cannot be moved to a new folder. To move them, you must first break the HA configuration, move both firewalls to the new folder, and then reconfigure HA

LACP and LLDP Pre-Negotiation for Active/Passive HA

If a firewall uses LACP or LLDP, negotiation of those protocols upon failover prevents sub-second failover. However, you can enable an interface on a passive firewall to negotiate LACP and LLDP prior to failover. Thus, a firewall in Passive or Non-functional HA state can communicate with neighboring devices using LACP or LLDP. Such pre-negotiation speeds up failover.

All firewall models except VM-Series firewalls support a pre-negotiation configuration, which depends on whether the Ethernet or AE interface is in a Layer 2, Layer 3, or virtual wire deployment. An HA passive firewall handles LACP and LLDP packets in one of two ways:

- **Active**—The firewall has LACP or LLDP configured on the interface and actively participates in LACP or LLDP pre-negotiation, respectively.
- **Passive**—LACP or LLDP is not configured on the interface and the firewall does not participate in the protocol, but allows the peers on either side of the firewall to pre-negotiate LACP or LLDP, respectively.

The following table displays which deployments are supported on Aggregate Ethernet (AE) and Ethernet interfaces.

Interface Deployment	AE Interface	Ethernet Interface
LACP in Layer 2	Active	Not supported
LACP in Layer 3	Active	Not supported
LACP in Virtual Wire	Not supported	Passive
LLDP in Layer 2	Active	Active
LLDP in Layer 3	Active	Active
LLDP in Virtual Wire	Active	• Active if LLDP itself is configured. • Passive if LLDP itself is not configured.

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Pre-negotiation is not supported on subinterfaces or tunnel interfaces.

To configure LACP or LLDP pre-negotiation, see the step (Optional) Enable LACP and LLDP Pre-Negotiation for Active/Passive HA for faster failover in your network uses LACP or LLAP.

Configuration Guidelines for Active/Passive HA

Where Can I Use This?	What Do I Need?
• NGFW	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro ❑ Strata Cloud Manager Essentials

To set up an active (PeerA) passive (PeerB) pair in HA, you must configure some options identically on both firewalls and some independently (non-matching) on each firewall. These HA settings are not synchronized between the firewalls. For details on what is/is not synchronized, see Reference: HA Synchronization.

The following checklist details the settings that you must configure identically on both firewalls:

- ❑ You must enable HA on both firewalls.
- ❑ You must configure the same Group ID value on both firewalls. The firewall uses the Group ID value to create a virtual MAC address for all the configured interfaces. See Floating IP Address and Virtual MAC Address for information about virtual MAC addresses. When a new active firewall takes over, it sends Gratuitous ARP messages from each of its connected interfaces to inform the connected Layer 2 switches of the virtual MAC address' new location.
- ❑ If you are using in-band ports as HA links, you must set the interfaces for the HA1 and HA2 links to type HA.
- ❑ Set the HA Mode to Active Passive on both firewalls.
- ❑ If required, enable preemption on both firewalls. The device priority value, however, must not be identical.
- ❑ If required, configure encryption on the HA1 link (for communication between the HA peers) on both firewalls.

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- Based on the combination of HA1 and HA1 Backup ports you are using, use the following recommendations to decide whether you should enable heartbeat backup:

  *HA functionality (HA1 and HA1 backup) is not supported on the management interface if it's configured for DHCP addressing (**IP Type** set to **DHCP Client**). The exceptions are AWS and Azure, where the management interface is configured as DHCP Client and it supports HA1 and HA1 Backup links.*

  - HA1: Dedicated HA1 port

    HA1 Backup: Dedicated HA1 port

    Recommendation: Enable Heartbeat Backup
  - HA1: Dedicated HA1 port

    HA1 Backup: In-band port

    Recommendation: Enable Heartbeat Backup
  - HA1: Dedicated HA1 port

    HA1 Backup: Management port

    Recommendation: Do not enable Heartbeat Backup
  - HA1: In-band port

    HA1 Backup: In-band port

    Recommendation: Enable Heartbeat Backup
  - HA1: Management port

    HA1 Backup: In-band port

    Recommendation: Do not enable Heartbeat Backup

The following table lists the HA settings that you must configure independently on each firewall. See Reference: HA Synchronization for more information about other configuration settings are not automatically synchronized between peers.

Independent Configuration Settings	PeerA	PeerB
Control Link	IP address of the HA1 link configured on this firewall (PeerA).	IP address of the HA1 link configured on this firewall (PeerB).
	For firewalls without dedicated HA ports, use the management port IP address for the control link.	
Data Link The data link information is synchronized	By default, the HA2 link uses Ethernet/Layer 2.	By default, the HA2 link uses Ethernet/Layer 2.

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Independent Configuration Settings	PeerA	PeerB
between the firewalls after HA is enabled and the control link is established between the firewalls.	If using a Layer 3 connection, configure the IP address for the data link on this firewall (PeerA).	If using a Layer 3 connection, configure the IP address for the data link on this firewall (PeerB).
Device Priority (required, if preemption is enabled)	The firewall you plan to make active must have a lower numerical value than its peer. So, if Peer A is to function as the active firewall, keep the default value of 100 and increment the value on PeerB. If the firewalls have the same device priority value, they use the MAC address of their HA1 as the tie-breaker.	If PeerB is passive, set the device priority value to a number larger than the setting on PeerA. For example, set the value to 110.
Link Monitoring—Monitor one or more physical interfaces that handle vital traffic on this firewall and define the failure condition.	Select the physical interfaces on the firewall that you would like to monitor and define the failure condition (all or any) to trigger a failover.	Pick a similar set of physical interfaces that you would like to monitor on this firewall and define the failure condition (all or any) to trigger a failover.
Path Monitoring—Monitor one or more destination IP addresses that the firewall can use ICMP pings to ascertain responsiveness.	Define the failure condition (all or any), ping interval and the ping count. This is particularly useful for monitoring the availability of other interconnected networking devices. For example, monitor the availability of a router that connects to a server, connectivity to the server itself, or some other vital device that is in the flow of traffic. Make sure that the node/device that you are monitoring is not likely to be unresponsive, especially when it comes under load, as this could cause a a path monitoring failure and trigger a failover.	Pick a similar set of devices or destination IP addresses that can be monitored for determining the failover trigger for PeerB. Define the failure condition (all or any), ping interval and the ping count.

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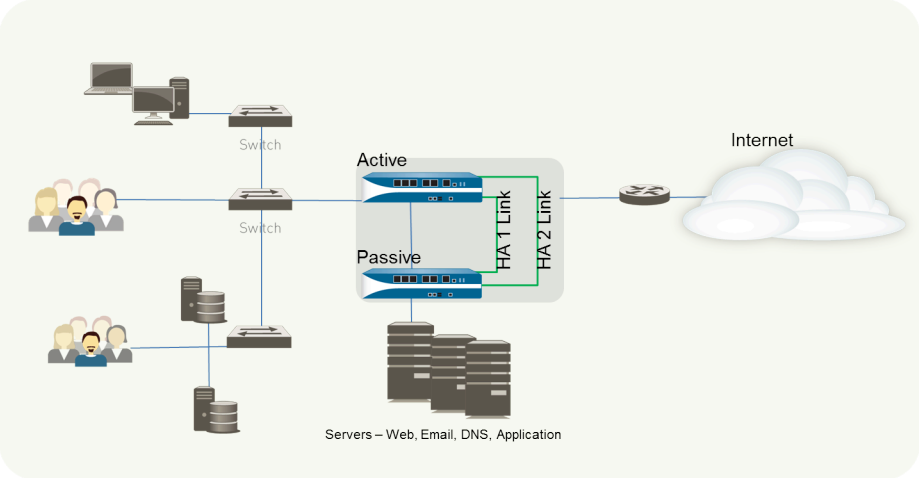
Configure Active/Passive HA

Where Can I Use This?	What Do I Need?
• NGFW	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

The following procedure shows how to configure a pair of firewalls in an active/passive deployment as depicted in the following example topology.



Active/Passive High Availability (HA) provides firewall redundancy by maintaining two firewalls in a primary-secondary relationship, where one firewall actively processes traffic while the other remains in standby mode, ready to assume control if the primary firewall fails. In an Active/Passive HA deployment, the active firewall handles all network traffic and maintains the runtime state, while the passive firewall continuously synchronizes its configuration and session information with the active unit. When a failover event occurs—whether due to hardware failure, software issues, or network connectivity problems—the passive firewall seamlessly transitions to the active role, ensuring minimal service disruption and maintaining network security policies. This configuration is ideal for environments that require high availability but do not need the increased throughput that Active/Active HA provides, making it a cost-effective solution for ensuring business continuity while maintaining a single point of traffic processing.

HA Active/Passive Support for PA-7500 and PA-5500 Series Firewalls

High Availability active/passive mode, which provides firewall redundancy and failover capabilities, is supported on PA-7500 series firewalls beginning with PAN-OS 12.1.5 and on PA-5500 series firewalls beginning with PAN-OS 12.1.7.

The firewalls support only HA active/passive mode; HA active/active mode is not supported. HA active/passive includes comprehensive network integration features such as LACP, link monitoring, and session synchronization functionality. LACP is supported on both Layer 2 and Layer 3 aggregate interfaces, including LACP pre-negotiation capabilities that enable the passive device to establish link aggregation readiness and reduce failover times. Link and path monitoring is supported for network health assessments. HA active/passive also maintains full runtime synchronization of session state between active and passive devices, ensuring that active sessions

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can continue uninterrupted during failover events and that the standby device maintains current awareness of all network connections and their associated state information.

On PA-7500 and PA-5500 series firewalls, HA active/passive uses dedicated High-Speed Chassis Interconnect (HSCI) interfaces for all HA communications and maintains session state synchronization between the active and passive devices.

- **HA1 Control Link**: Carries heartbeat messages, configuration synchronization, and state information between HA peers.
- **HA2 Data Link**: Synchronizes session tables and forwarding information between firewalls.
- **Layer 2 Transport**: All HA traffic uses Layer 2 connectivity over HSCI interfaces.

You cannot configure HA and NGFW clustering simultaneously on the same firewall.

The HSCI interface assignments for HA links follow a specific configuration pattern that ensures proper communication between HA peers. All HSCI traffic is transmitted in cleartext format. HA1 control link and HA2 data link are supported only on layer 2. The primary HA1 control link can only be configured on HSCI-A, while the HA1-backup link is restricted to HSCI-B, providing redundancy for critical control plane communications. Similarly, the HA2 data synchronization link can only be configured on HSCI-A, with the HA2-backup link assigned to HSCI-B for failover protection. While HSCI interfaces handle the primary HA communications, management interfaces can be configured to support heartbeat backup functionality, which provides an additional layer of split-brain prevention by maintaining minimal connectivity between HA peers through the management network when HSCI links become unavailable.

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(PA-7500 and PA-5500 series firewalls)

- *HA active/passive mode on PA-7500 and PA-5500 series firewalls does not support encryption on HA1, HA2, or HA1/HA2 backup links. Therefore the HSCI-A and HSCI-B interfaces do not support encryption.*
- *HA3 and HA4 interfaces are not supported and cannot be configured.*
- *HA2 and HA2 backup interfaces only support Layer 2 (L2) connectivity.*
- *MACsec functionality is not supported (unlike the clustering feature).*
- *The HA1 and HA2 functions are aggregated on the HSCI-A and HSCI-B interface. The administrator cannot assign the front port interfaces to the HA1 or HA2 role.*
- *The HSCI-A and HSCI-B operate exclusively at Layer 2 only and although you have to configure an IP address on these interfaces, they must reside on the same network.*
- *IPv6 is not supported for HA1/HA2 interfaces. IPv4 addresses for HA1/HA2 is used exclusively for internal node-to-node communication. Both HA nodes must be configured on the same subnet.*
- *PA-7500and PA-5500 firewalls require trunk mode (tagged traffic) for HSCI port connections when a switch is present in the connection path; access mode (untagged) is not supported in this configuration. If you are using a switch between HSCI connections, you must enable jumbo frames on the firewall and configure the switch ports as tagged ports. You should enable jumbo frames first, which will require a reboot if you're activating jumbo frames for the first time, and only then proceed to enable HA.*
- *Panorama web interface may display configuration options that are not actually supported by PA-7500 and PA-5500 series firewalls. Always follow these documented constraints regardless of what appears in the interface.*
  - *Data ports cannot be configured for HA1, HA2, or HA1 and HA2 backup interfaces. Panorama may display these as selectable options, but they are not functional.*
  - *HSCI-A must be used for HA1 and HA2 primary interfaces. Other interface options may appear in Panorama but are not supported.*
  - *HSCI-B must be used for HA1/HA2 backup interfaces. Other interface options may appear in Panorama but are not supported.*

The PA-7500 and PA-5500 Series firewalls utilize Condor 3 architecture, which introduces several key changes to the traditional High Availability (HA) active/passive configuration:

PA-7500 and PA-5500 Series firewalls uses management processor relay for HA communications. When restarting dataplanes on the firewall HA configurations, be aware that individual dataplane restarts will trigger a restart of the entire chassis to maintain HA consistency between peers.

(PA-7500 and PA-5500 series firewalls) The following additional parameters are required when configuring HA on active/passive mode:

- (Mandatory) **Device ID**: A unique identifier (0 or 1) that must be configured differently on each HA peer to ensure proper HSCI port MAC address generation.
- (Mandatory) **Group ID**: A numeric identifier ranging from 0 to 63 that must match on both HA peers. This ensures the devices recognize each other as part of the same HA pair. If the group IDs don't match, HA formation will fail and may result in split-brain scenarios.

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- (Optional) **VLAN Configuration**: VLAN tags (2-4094) for HSCI traffic, with VLAN 2 used as default for back-to-back connections.

  (PA-7500 and PA-5500 series firewalls) Both the Device ID and Group ID are essential to ensure proper HSCI port MAC address generation. Misconfiguring either parameter will cause HA communication to fail. If there is a Device ID overlap, the devices will enter a split-brain state. If there is a Group ID mismatch, the devices will also enter a split-brain state without transitioning to a 'non-functional' state.

- PAN-OS & Panorama
- Strata Cloud Manager

Configure Active/Passive HA (PAN-OS)

To configure an active/passive HA pair, first verify that any filtering devices between your HA pair members allow the protocols and ports used by the HA links, then complete the following workflows on the first firewall and repeat the steps on the second firewall.

STEP 1 | Connect the HA ports to set up a physical connection between the firewalls.

- For firewalls with dedicated HA ports, use an Ethernet cable to connect the dedicated HA1 ports and the HA2 ports on peers. Use a crossover cable if the peers are directly connected to each other.
- For firewalls without dedicated HA ports, select two data interfaces for the HA2 link and the backup HA1 link. Then, use an Ethernet cable to connect these in-band HA interfaces across both firewalls.

Use the management port for the HA1 link and ensure that the management ports can connect to each other across your network.

STEP 2 | Enable ping on the management port.

Enabling ping allows the management port to exchange heartbeat backup information.

1. Select **Device** > **Setup** > **Interfaces** > **Management**.
2. Select **Ping** as a service that is permitted on the interface.

STEP 3 | If the firewall does not have dedicated HA ports, set up the data ports to function as HA ports.

For firewalls with dedicated HA ports continue to the next step.

1. Select **Network** > **Interfaces**.
2. Confirm that the link is up on the ports that you want to use.
3. Select the interface and set **Interface Type** to **HA**.
4. Set the **Link Speed** and **Link Duplex** settings, as appropriate.

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STEP 4 | Set the HA mode and group ID.

1. Select **Device** > **High Availability** > **General** and edit the Setup section.
2. Set a **Group ID** and optionally a **Description** for the pair. The Group ID uniquely identifies each HA pair on your network. If you have multiple HA pairs that share the same broadcast domain you must set a unique Group ID for each pair.
3. Set the mode to **Active Passive**.

STEP 5 | Set up the control link connection.

This example shows an in-band port that is set to interface type HA.

For firewalls that use the management port as the control link, the IP address information is automatically pre-populated.

1. In **Device** > **High Availability** > **HA Communications**, edit Control Link (HA1).
2. Select the **Port** that you have cabled for use as the HA1 link.
3. Set the **IPv4/IPv6 Address** and **Netmask**.

   If the HA1 interfaces are on separate subnets, enter the IP address of the **Gateway**. Do not add a gateway address if the firewalls are directly connected or are on the same VLAN.

STEP 6 | (Optional) Enable encryption for the control link connection.

This is typically used to secure the link if the two firewalls are not directly connected, that is if the ports are connected to a switch or a router.

PA-7500 and PA-5500 Series firewalls use HSCI fabric for HA links that do not support SSH-based HA1 control link encryption. Because of this limitation, FIPS-CC mode and HA Active/Passive cannot be enabled together on these firewalls.

You must choose one:

- FIPS-CC mode *(where HA Active/Passive will not be available), or*
- *HA Active/Passive (where FIPS-CC mode will not be available).*

1. Export the HA key from one firewall and import it into the peer firewall.
   1. Select **Device** > **Certificate Management** > **Certificates**.
   2. Select **Export HA key**. Save the HA key to a network location that the peer can access.
   3. On the peer firewall, select **Device** > **Certificate Management** > **Certificates**, and select **Import HA key** to browse to the location that you saved the key and import it in to the peer.
   4. Repeat this process on the second firewall to exchange HA keys on both devices.
2. Select **Device** > **High Availability** > **HA Communications**, edit the Control Link (HA1) section.
3. Select **Encryption Enabled**.

   If you enable encryption, after you finish configuring the HA firewalls, you can Refresh HA1 SSH Keys and Configure Key Options.

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STEP 7 | Set up the backup control link connection.

1. In **Device** > **High Availability** > **HA Communications**, edit Control Link (HA1 Backup).
2. Select the HA1 backup interface and set the **IPv4/IPv6 Address** and **Netmask**.

   PA-3200 Series firewalls don't support an IPv6 address for the HA1 backup control link; use an IPv4 address.

STEP 8 | Set up the data link connection (HA2) and the backup HA2 connection between the firewalls.

1. In **Device** > **High Availability** > **HA Communications**, edit the Data Link (HA2) section.
2. Select the **Port** to use for the data link connection.
3. Select the **Transport** method. The default is **ethernet**, and will work when the HA pair is connected directly or through a switch. If you need to route the data link traffic through the network, select **IP** or **UDP** as the transport mode.

   UDP is the only supported transport mode in Azure environments. UDP is the preferred transport mode for PA-1400 Series and PA-3400 Series firewalls.
4. If you use IP or UDP as the transport method, enter the **IPv4/IPv6 Address** and **Netmask**.
5. Verify that **Enable Session Synchronization** is selected.
6. Select **HA2 Keep-alive** to enable monitoring on the HA2 data link between the HA peers. If a failure occurs based on the threshold that is set (default is 10000 ms), the defined action will occur. For active/passive configuration, a critical system log message is generated when an HA2 keep-alive failure occurs.

   You can configure the HA2 keep-alive option on both firewalls, or just one firewall in the HA pair. If the option is only enabled on one firewall, only that firewall will send the keep-alive messages. The other firewall will be notified if a failure occurs.
7. Edit the **Data Link (HA2 Backup)** section, select the interface, and add the **IPv4/IPv6 Address** and **Netmask**.

STEP 9 | Enable heartbeat backup if your control link uses a dedicated HA port or an in-band port.

You do not need to enable heartbeat backup if you are using the management port for the control link.

1. In **Device** > **High Availability** > **General**, edit the Election Settings.
2. Select **Heartbeat Backup**.

   To allow the heartbeats to be transmitted between the firewalls, you must verify that the management port across both peers can route to each other.

   Enabling heartbeat backup also allows you to prevent a split-brain situation. Split brain occurs when the HA1 link goes down causing the firewall to miss heartbeats, although the firewall is still functioning. In such a situation, each peer believes that the other is down and attempts to start services that are running, thereby causing a split brain. When the heartbeat backup link is enabled, split brain is prevented because redundant heartbeats and hello messages are transmitted over the management port.

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STEP 10 | Set the device priority and enable preemption.

This setting is only required if you wish to make sure that a specific firewall is the preferred active firewall. For information, see Device Priority and Preemption.

1. In **Device** > **High Availability** > **General**, edit the Election Settings.
2. Set the numerical value in **Device Priority**. Make sure to set a lower numerical value on the firewall that you want to assign a higher priority to.

   If both firewalls have the same device priority value, the firewall with the lowest MAC address on the HA1 control link will become the active firewall.
3. Select **Preemptive**.

   You must enable preemptive on both the active firewall and the passive firewall.

STEP 11 | (Optional) Modify the HA Timers.

By default, the HA timer profile is set to the **Recommended** profile and is suited for most HA deployments.

1. In **Device** > **High Availability** > **General**, edit the Election Settings.
2. Select the **Aggressive** profile for triggering failover faster; select **Advanced** to define custom values for triggering failover in your set up.

   *To view the preset value for an individual timer included in a profile, select **Advanced** and click **Load Recommended** or **Load Aggressive**. The preset values for your hardware model will be displayed on screen.*

STEP 12 | (Optional) Modify the link status of the HA ports on the passive firewall.

*The passive link state is **shutdown**, by default. After you enable HA, the link state for the HA ports on the active firewall will be green and those on the passive firewall will be down and display as red.*

Setting the link state to **Auto** allows for reducing the amount of time it takes for the passive firewall to take over when a failover occurs and it allows you to monitor the link state.

To enable the link status on the passive firewall to stay up and reflect the cabling status on the physical interface:

1. In **Device** > **High Availability** > **General**, edit the Active Passive Settings.
2. Set the **Passive Link State** to **Auto**.

   The auto option decreases the amount of time it takes for the passive firewall to take over when a failover occurs.

   Although the interface displays green (as cabled and up) it continues to discard all traffic until a failover is triggered.

   When you modify the passive link state, make sure that the adjacent devices do not forward traffic to the passive firewall based only on the link status of the firewall.

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STEP 13 | Enable HA.

1. Select **Device** > **High Availability** > **General** and edit the Setup section.
2. Select **Enable HA**.
3. Select **Enable Config Sync**. This setting enables the synchronization of the configuration settings between the active and the passive firewall.
4. Enter the IP address assigned to the control link of the peer in **Peer HA1 IP Address**.

   For firewalls without dedicated HA ports, if the peer uses the management port for the HA1 link, enter the management port IP address of the peer.
5. Enter the **Backup HA1 IP Address**.

STEP 14 | (PA-7500 and PA-5500 series firewalls) Configure unique **Device ID** as 0 or 1 on each HA peer.

STEP 15 | (Optional) (PA-7500 and PA-5500 series firewalls) You must specify the VLAN tag for L2 headers to ensure the configuration matches the supported VLAN tags on HSCI switches. Configure **HSCI-A VLAN** and **HSCI-B VLAN**.

If you do not specify a value, the default VLAN tag of 2 will be applied to both ports. This default setting allows you to skip this configuration when devices are connected back-to-back, keeping your setup minimal.

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STEP 16 | (Optional) Enable LACP and LLDP Pre-Negotiation for Active/Passive HA for faster failover if your network uses LACP or LLDP.

Enable LACP and LLDP before configuring HA pre-negotiation for the protocol if you want pre-negotiation to function in active mode.

1. Ensure that in Step 12you set the link state to **Auto**.
2. Select **Network** > **Interfaces** > **Ethernet**.
3. To enable LACP active pre-negotiation:
   1. Select an AE interface in a Layer 2 or Layer 3 deployment.
   2. Select the **LACP** tab.
   3. Select **Enable in HA Passive State**.
   4. Click **OK**.

   *You cannot also select **Same System MAC Address for Active-Passive HA** because pre-negotiation requires unique interface MAC addresses on the active and passive firewalls.*
4. To enable LACP passive pre-negotiation:
   1. Select an Ethernet interface in a virtual wire deployment.
   2. Select the **Advanced** tab.
   3. Select the **LACP** tab.
   4. Select **Enable in HA Passive State**.
   5. Click **OK**.
5. To enable LLDP active pre-negotiation:
   1. Select an Ethernet interface in a Layer 2, Layer 3, or virtual wire deployment.
   2. Select the **Advanced** tab.
   3. Select the **LLDP** tab.
   4. Select **Enable in HA Passive State**.
   5. Click **OK**.

   If you want to allow LLDP passive pre-negotiation for a virtual wire deployment, perform Step d but do not enable LLDP itself.

STEP 17 | Save your configuration changes.

Click **Commit**.

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STEP 18 | After you finish configuring both firewalls, verify that the firewalls are paired in active/passive HA.

1. Access the **Dashboard** on both firewalls, and view the High Availability widget.
2. On the active firewall, click the **Sync to peer** link.
3. Confirm that the firewalls are paired and synced, as shown as follows:
   - On the passive firewall: the state of the local firewall should display **passive** and the Running Config should show as **synchronized**.
   - On the active firewall: The state of the local firewall should display **active** and the Running Config should show as **synchronized**.

Configure Active/Passive HA (SCM)

This procedure assumes you already onboarded the firewalls you want to configure in an active/passive HA configuration to Strata Cloud Manager and have added them the same folder. Also verify that any filtering devices between your HA pair members allow the protocols and ports used by the HA links.

STEP 1 | Log in to Strata Cloud Manager.

STEP 2 | Configure your HA Ethernet interfaces if you intend to use dedicated interfaces for the HA1 control links and HA2 data links.

STEP 3 | Select **Configuration** > **NGFW and Prisma Access** and select the Folder Configuration Scope that the managed firewalls are associated with.

Selecting the folder that the managed firewalls are associated with allows you to find and select the managed firewalls you want to configure in an active/passive HA configuration.

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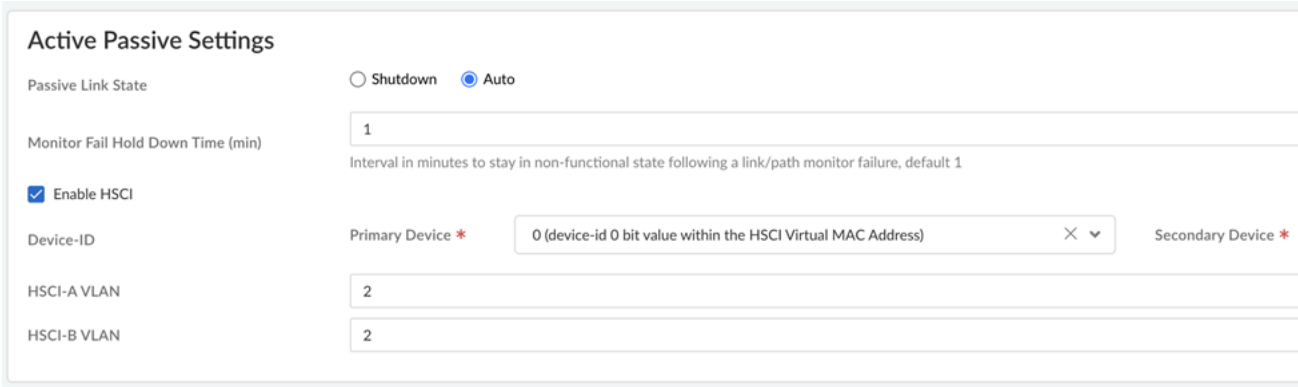
STEP 4 | In the HA Setup section, **Create HA**.

1. Set the **HA Mode** of deployment as **Active - Passive**.
2. Select **Enable Config Sync**. This setting enables the synchronization of the configuration settings between the HA peers.
3. Set a **Group ID** and optionally a **Description** for the pair. The Group ID uniquely identifies each HA pair on your network. If you have multiple HA pairs that share the same broadcast domain you must set a unique Group ID for each pair.
4. Select the **SSH HA Profile** to apply to the SSH sessions for the HA peers on your network.
5. (Active-Passive Only) Active-Passive settings.
   - (Optional) Modify the link status of the HA ports on the passive firewall.

     *The **passive link state** is **shutdown**, by default. After you enable HA, the link state for the HA ports on the active firewall will be green and those on the passive firewall will be down and display as red.*

     Setting the link state to **Auto** allows for reducing the amount of time it takes for the passive firewall to take over when a failover occurs and it allows you to monitor the link state.
   - (Optional) Set the **Monitor Fail Hold Down Time** (min), which is the number of minutes after which a down link is retested to see if it is back up; range is 1 to 60; default is 1.
   - (PA-7500 Firewalls only) Enable HSCI to configure **Device ID** and HSCI VLANs.
     - (Mandatory) Configure unique **Device ID** as 0 or 1 on each HA peer.
     - (Optional) You must specify the VLAN tag for L2 headers to ensure the configuration matches the supported VLAN tags on HSCI switches. Configure **HSCI-A VLAN** and **HSCI-B VLAN**.

       If you do not specify a value, the default VLAN tag of 2 will be applied to both ports. This default setting allows you to skip this configuration when devices are connected back-to-back, keeping your setup minimal.



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STEP 5 | Select the managed firewalls to configure in an active/passive HA configuration.

- **Select Primary Device**—Select the firewall to act as the primary active HA peer.
- **Select Secondary Device**—Select the firewall to act as the secondary passive HA peer.

Click **Next** to continue.

STEP 6 | Configure **Election Settings**.

1. Set the numerical value in **Device Priority**. Make sure to set a lower numerical value on the firewall that you want to assign a higher priority to.

   If both firewalls have the same device priority value, the firewall with the lowest MAC address on the HA1 control link will become the active firewall.

   This setting is only required if you wish to make sure that a specific firewall is the preferred active firewall. For information, see Device Priority and Preemption.
2. Enable **Preemptive**.

   You must enable preemptive on both the active firewall and the passive firewall.
3. Enable **Heartbeat Backup**.

   To allow the heartbeats to be transmitted between the firewalls, you must verify that the management port across both peers can route to each other.

   Enabling heartbeat backup also allows you to prevent a split-brain situation. Split brain occurs when the HA1 link goes down causing the firewall to miss heartbeats, although the firewall is still functioning. In such a situation, each peer believes that the other is down and attempts to start services that are running, thereby causing a split brain. When the heartbeat backup link is enabled, split brain is prevented because redundant heartbeats and hello messages are transmitted over the management port.
4. (Optional) Modify the HA Timers.

   By default, the HA timer profile is set to the **Recommended** profile and is suited for most HA deployments.

   Select the **Aggressive** profile for triggering failover faster; select **Advanced** to define custom values for triggering failover in your set up.

   *To view the preset value for an individual timer included in a profile, select **Advanced** and then **Load Recommended** or **Load Aggressive**. The preset values for your hardware model will be displayed on screen.*

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STEP 7 | Configure the HA1 Control Link Settings.

The HA1 control link is used to exchange hellos, heartbeats, HA state information, and management plane synchronization for routing. This link is also used to synchronize configuration changes with its peer.

1. Configure the HA1 control link settings for the Primary Device.
   1. Select the Ethernet **Port** for the HA1 control link on the primary HA peer.

      Strata Cloud Manager interface may display configuration options that are not actually supported by PA-7500 series firewalls. Always follow these documented constraints regardless of what appears in the interface.

      - *Data ports cannot be configured for HA1, HA2, or HA1 and HA2 backup interfaces. Strata Cloud Manager may display these as selectable options, but they are not functional.*
      - *HSCI-A must be used for HA1 and HA2 primary interfaces. Other interface options may appear in Strata Cloud Manager but are not supported.*
      - *HSCI-B must be used for HA1 and HA2 backup interfaces. Other interface options may appear in Strata Cloud Manager but are not supported.*
   2. Configure the **IPv4 Address**, **Netmask**, and **Gateway** of the HA1 control link for the primary HA peer.
   3. (Optional) Expand the **Advanced Settings** and enter the **Monitor Hold Time (ms)** that the firewall waits before declaring a peer failure due to a control link failure. Range is 1,000 to 60,000; default is 3,000.
2. Configure the HA1 control link settings for the Secondary Device.
   1. Select the Ethernet **Port** for the HA1 control link on the secondary HA peer.
   2. Configure the **IPv4 Address**, **Netmask**, and **Gateway** of the HA1 control link for the secondary HA peer.
3. (Optional) Configure the HA1 Backup control link for both the Primary Device and Secondary Device.

   Configuring the HA1 Backup control link provides redundancy for the HA1 link. Consider the following guidelines when configuring back HA links.

   - The IP addresses of the primary and backup HA links must not overlap each other.
   - HA backup links must be on a different subnet from the primary HA links.
   - HA1-backup and HA2-backup ports must be configured on separate physical ports. The HA1-backup link uses port 28770 and 28260.
4. Click **Next**.

STEP 8 | Configure the HA2 Data Link Settings.

The HA2 data link is used to synchronize sessions, forwarding tables, IPSec security associations, and ARP tables. Data flow on the HA2 link is always unidirectional except for the

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HA2 keep alive. It flows from the `active` or `active-primary` HA peer to the `secondary` or `secondary-active` HA peer. The HA2 link is a Layer 2 link.

1. Configure the HA2 data link settings for the Primary Device.
   1. Select the Ethernet **Port** for the HA2 data link on the primary HA peer.
   2. Configure the **IPv4 Address**, **Netmask**, and **Gateway** of the HA2 data link for the primary HA peer.
   3. (Optional) Expand the **Advanced Settings** and verify that **Enable Session Synchronization** is enabled.

      Enable session synchronization so that the secondary device has the session in its dataplane, which allows the firewall to match packets to the synchronized session and quickly forward packets. If you don't enable session synchronization, the firewall must create the session again, which introduces latency and could drop connections.

      1. Select the **Transport** method.
         - **Layer2 transport via Ethernet**—Use when the firewalls are connected back-to-back or through a switch (EtherType 0x7262).
         - **Layer3 transport via IP protocol 99**—Use when Layer 3 transport is required (IP protocol number 99).
         - **Layer4 transport via UDP/29281**—Use to take advantage of the fact that the checksum is calculated on the entire packet rather than just the header, as in the IP option (UDP port 29281). The benefit of using UDP mode is the presence of the UDP checksum to verify the integrity of a session sync message.
      2. (Best Practices) Enable and configure **HA2 Keep-alive** to monitor the health of the HA2 data link between the HA peers.
         1. Specify the **Keep-alive Action** as **Log Only**.

            Logs the failure of the HA2 interface in the system log as a critical event. Select this option for active/passive deployments because the active peer is the only firewall forwarding traffic. The passive peer is in a backup state and isn't forwarding traffic; therefore a split datapath isn't required. If you haven't configured any HA2 Backup links, state synchronization will be turned off. If the HA2 path recovers, an informational log will be generated.
         2. Configure the **Keep-alive Threshold (ms)** to specify the duration in which keep-alive messages have failed before the **Keep-alive Action** is triggered. Range is `5,000` to `60,000`; default is `10,000`.
2. Configure the HA2 data link settings for the Secondary Device.
   1. Select the Ethernet **Port** for the HA2 data link on the secondary HA peer.
   2. Configure the **IPv4 Address**, **Netmask**, and **Gateway** of the HA2 data link for the secondary HA peer.
3. (Optional) Configure the HA2 Backup data link for both the Primary Device and Secondary Device.

   When an HA2 backup link is configured, failover to the backup link will occur if there's a physical link failure. With the HA2 keep-alive option enabled, the failover will also occur if the HA keep-alive messages fail based on the defined threshold.
4. Click **Next**.

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STEP 9 | Define HA Failover Conditions.

STEP 10 | **Push Config** to push your configuration changes.

Define HA Failover Conditions

Where Can I Use This?	What Do I Need?
• NGFW	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Pro

Perform the following task to use link monitoring or path monitoring to define Failover conditions and thus establish what will cause a firewall in an HA pair to fail over, an event where the task of securing traffic passes from the previously active firewall to its HA peer.

You can monitor multiple IP path groups per virtual router, VLAN, or virtual wire. You can enable each path group with one or more IP addresses and give each its own peer failure conditions. Additionally, you can set these failure conditions at both the path-group level and the broader virtual router or VLAN or virtual wire group level using “any” or “all” fail checks to determine the status of the active firewall.

When you upgrade to PAN-OS 10.0, the firewall automatically transfers your currently monitored destination IP addresses to a newly created destination group and gives that group a default path-monitoring name. The new destination group retains your previous failover condition at the path-group level.

Ensure that you delete all VLAN path monitoring configurations in active/active HA before you upgrade to PAN-OS 11.1 because VLAN path monitoring is not compatible with active/active HA pairing in PAN-OS 10.0; retaining an earlier active/active HA configuration results in an autocommit failure.

Before you enable path monitoring, you must set up your logical routers, virtual routers, VLAN, or virtual wires or a combination of these logical networking components. Path monitoring in virtual routers and virtual wires is compatible with both active/active and active/passive HA deployments; however, path monitoring in VLANs is supported only on active/passive pairs.

Before you enable path monitoring, you must also:

- Check reachability for destination IP groups in your virtual routers.
- Ensure that the VLANs (for which you intend to enable path monitoring) include configured interfaces.
- Obtain the source IP address that you will use to receive pings from the appropriate destination IP address.

If you are using SNMPv3 to monitor the firewalls, note that the SNMPv3 Engine ID is synchronized between the HA pair. For information on setting up SNMP, see Forward Traps to an SNMP Manager. Because the EngineID is generated using the firewall serial number, on the VM-Series firewall you must apply a valid license in order to obtain a unique EngineID for each firewall.

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- PAN-OS & Panorama
- Strata Cloud Manager

Define HA Failover Conditions (PAN-OS)

STEP 1 | To configure HA link monitoring, specify a group of physical interfaces for the firewall to monitor (link up or link down).

1. Select **Device** > **High Availability** > **Link and Path Monitoring**.
2. In the Link Monitoring section, **Add** a link group by **Name**.
3. Select **Enabled** to enable the link group.
4. Select the **Failure Condition** for the interfaces in the link group: **Any** (default) or **All**.
5. **Add** the **Interface**(s) to monitor.
6. Click **OK**.

STEP 2 | (Optional) Modify the failure condition for the set of Link Groups configured on the firewall.

By default, the firewall triggers a failover when any monitored Link Group fails.

1. Edit the **Link Monitoring** section.
2. Set the **Failure Condition** to **Any** (default) or **All**.
3. Click **OK**.

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STEP 3 | To configure HA path monitoring for a virtual wire, VLAN, or virtual router (or logical router for an Advanced Routing Engine), specify the destination IP addresses that the firewall will ping to verify network connectivity.

1. In the Path Monitoring section, select **Add Virtual Wire Path**, **Add VLAN Path**, or **Add Virtual Router Path** (or **Add Logical Router Path** for Advanced Routing Engine).
2. Enter a **Name** for the virtual wire, VLAN, virtual router path group, or logical routero path group.
3. (Virtual Wire Path or VLAN Path only) Enter the **Source IP** address to use to ping the destination IP address through the virtual wire or VLAN.
4. Select **Enabled** to enable the path group.
5. Select the **Failure Condition** that results in a failure for this path group: **Any** (default) to issue a failure when one or more Destination IP groups in this path group fail or **All** to issue a failure when all Destination IP groups in this path group fail.
6. Enter the **Ping Interval** in milliseconds; the interval between ICMP messages sent to the Destination IP address (range is 200 to 60,000; default is 200).
7. Enter the **Ping Count** of pings that must fail before declaring a failure (range is 3 to 10; default is 10).
8. **Add** and enter a **Destination IP Group** name.
9. **Add** one or more **Destination IP** addresses to ping.
10. Select **Enabled** to enable path monitoring for the Destination IP group.
11. Select the **Failure Condition** that results in a failure for this Destination IP group: **Any** (default) to issue a failure when one or more listed IP addresses is unreachable or **All** to issue a failure when all listed IP addresses are unreachable.
12. Click **OK** twice.
13. (Panorama only) Select the appropriate Panorama template to push the path monitoring configuration to your appliance.

    You can push HA path monitoring for a virtual wire, VLAN, or virtual router only to firewalls running PAN-OS 10.0 or a later releases. If you try to push the configuration to firewalls running a release earlier than PAN-OS 10.0 (such as 9.1.x or 9.0.x), the commit may fail or the commit may remove destination IP addresses from the path group.

    Only HA Path Groups containing one Destination IP Group are supported for managed firewalls running PAN-OS 9.1 and earlier releases.

    To manage the destination IP addresses from Panorama for managed firewalls running different PAN-OS releases, create a separate template for managed firewalls running PAN-OS 10.0 and later releases and a separate template for managed firewalls running PAN-OS 9.1 and earlier releases. This allows you to more accurately control the destination IP address configuration if you created multiple destination IP groups and ensures your managed firewall successfully fails over.

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STEP 4 | (Optional) Modify the failure condition for the set of Path Groups configured on the firewall.

By default, the firewall triggers a failover when any monitored Path Group fails.

1. Edit the **Path Monitoring** section.
2. Select **Enabled** to enable path monitoring on the appliance.
3. Set the **Failure Condition** to **Any** (default) to issue a failure for this firewall when one or more monitored virtual routers, VLANs, or virtual wires is down. Select **All** to issue a failure for this firewall when all monitored virtual routers, VLANs, or virtual wires are down.
4. Click **OK**.

STEP 5 | **Commit**.

Define HA Failover Conditions (SCM)

STEP 1 | Log in to Strata Cloud Manager.

STEP 2 | Configure a Logical Router or Configure a VLAN to establish the Destination IP addresses you want to monitor.

Before you enable path monitoring, you must set up your logical routers, VLAN, or a combination of these logical networking components.

STEP 3 | Configure Active/Passive HA.

STEP 4 | Select **Manage** > **Configuration** > **NGFW and Prisma Access** > **Overview** and select the **Folder** Configuration Scope that the HA peers are associated with.

STEP 5 | In the HA Devices section, edit the HA pair for which you want to define the HA failover conditions.

STEP 6 | Select the **Failover Condition Settings**.

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STEP 7 | Configure the failover conditions settings for the Primary Device.

1. Configure the Link Monitoring Failure Condition.
   1. Select the **Failure Condition**.
      - **All of Link Group** (default)—Failover occurs when all Link Groups fail.
      - **Any of Link Group**—Failover occurs when one or more Link Groups fail.
   2. Click **+Link Group** and select **Link Group Failure Condition** for the logical routers you want to monitor.
      - **All of Link Group** (default)—Failure for a Link Group occurs when the firewall is unable to connect to all Destination IP addresses associated with the Link Group.
      - **Any of Link Group**—Failure for a Link Group occurs when the firewall is unable to connect to any Destination IP addresses associated with the Link Group.
   3. Select the link groups to monitor.
2. Configure the Path Monitor Failure Condition.
   1. Select the **Failure Condition**.
      - **All of Path Group** (default)—Failover occurs when all Path Groups fail.
      - **Any of Path Group**—Failover occurs when one or more Path Groups fail.
   2. Click **+Path Group** and select **Link Group Failure Condition** for the logical routers you want to monitor.
      - **All of Link Group** (default)—Failure for a Path Group occurs when the firewall is unable to connect to all Destination IP addresses associated with the Path Group.
      - **Any of Link Group**—Failure for a Path Group occurs when the firewall is unable to connect to any Destination IP addresses associated with the Path Group.
   3. Select the path groups to monitor.

STEP 8 | Configure the Link Monitoring Failure Condition and Path Monitoring Failure Condition for the Secondary Device.

STEP 9 | **Save**.

STEP 10 | **Push Config** to push your configuration changes.

Verify Failover or Suspend HA

Where Can I Use This?	What Do I Need?
• NGFW	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Pro

Verifying failover functionality through controlled testing ensures that your HA deployment will perform as expected during actual failure scenarios. By intentionally triggering failover events and monitoring the transition process, you can validate that session synchronization, IP address

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failover, and traffic handling occur seamlessly, confirming that your high availability configuration will provide the expected redundancy and business continuity when real failures occur.

Additionally, suspending High Availability temporarily disables the HA functionality on a firewall, effectively taking it out of the HA pair while allowing you to perform maintenance, troubleshooting, or testing activities without triggering an unintended failover. When you suspend HA on the active firewall, it remains active but stops sending heartbeat messages and ceases synchronization with its peer, while suspending HA on the passive firewall prevents it from becoming active even if the current active firewall fails. This administrative control is essential for planned maintenance windows, configuration testing, or when you need to isolate one firewall to diagnose issues without affecting the HA relationship permanently.

- PAN-OS & Panorama
- Strata Cloud Manager

Verify Failover or Suspend HA(PAN-OS)

STEP 1 | Suspend the active firewall.

Select **Device** > **High Availability** > **Operational Commands** and click the **Suspend local device** link.

STEP 2 | Verify that the passive firewall has taken over as active.

On the **Dashboard**, verify that the state of the passive firewall changes to **active** in the High Availability widget.

STEP 3 | Restore the suspended firewall to a functional state. Wait for a couple of minutes, and then verify that preemption has occurred, if **Preemptive** is enabled.

1. On the firewall you previously suspended, select **Device** > **High Availability** > **Operational Commands** and click the **Make local device functional** link.
2. In the High Availability widget on the **Dashboard**, confirm that the firewall has taken over as the active firewall and that the peer is now in a passive state.

Verify Failover or Suspend HA (SCM)

STEP 1 | Log in to Strata Cloud Manager.

STEP 2 | Configure Active/Passive HA.

STEP 3 | Suspend HA for the HA peer.

1. Select **Workflows** > **NGFW Setup** > **Device ManagementSystem Settings** > **Device Management** > **Cloud Managed Devices** and locate the HA peer for which you want to suspend HA.
2. In the Actions column, expand the menu and **Suspend HA**.

   Click **OK** to confirm suspending HA on the HA peer.
3. Verify the HA status for the now suspended HA peer displays `suspended`.

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STEP 4 | Restore HA for the suspended HA peer.

1. Select **Workflows** > **NGFW Setup** > **Device ManagementSystem Settings** > **Device Management** > **Cloud Managed Devices** and locate the suspended HA.
2. In the Actions column, expand the menu and **Restore HA**.
   Click **OK** to confirm restoring HA on the suspended HA peer.
3. Verify the HA status for the restored HA peer displays `active`.

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Set Up Active/Active HA

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

To set up active/active HA on your firewalls, you need a pair of firewalls that meet the following requirements:

- ❑ **The same model**—The firewalls in the pair must be of the same hardware model.
- ❑ **The same PAN-OS version**—The firewalls must be running the same PAN-OS version and must each be up-to-date on the application, URL, and threat databases.
- ❑ **The same multi virtual system capability**—Both firewalls must have **Multi Virtual System Capability** either enabled or not enabled. When enabled, each firewall requires its own multiple virtual systems licenses.
- ❑ **The same type of interfaces**—Dedicated HA links, or a combination of the management port and in-band ports that are set to *interface type* HA.
  - The HA interfaces must be configured with static IP addresses only, not IP addresses obtained from DHCP (except AWS can use DHCP addresses). Determine the IP address for the HA1 (control) connection between the HA peers. The HA1 IP address for the peers must be on the same subnet if they are directly connected or are connected to the same switch.

    For firewalls without dedicated HA ports, you can use the management port for the control connection. Using the management port provides a direct communication link between the management planes on both firewalls. However, because the management ports will not be directly cabled between the peers, make sure that you have a route that connects these two interfaces across your network.
  - If you use Layer 3 as the transport method for the HA2 (data) connection, determine the IP address for the HA2 link. Use Layer 3 only if the HA2 connection must communicate over a routed network. The IP subnet for the HA2 links must not overlap with that of the HA1 links or with any other subnet assigned to the data ports on the firewall.
  - Each firewall needs a dedicated interface for the HA3 link. The PA-7000 Series, PA-5400 Series, PA-3400 Series, PA-3200, and PA-1400 Series firewalls use the HSCI port for HA3. The PA-5200 Series firewalls can use the HSCI port for HA3 or you can configure aggregate interfaces on the dataplane ports for HA3 for redundancy. On the remaining platforms, you can configure aggregate interfaces on dataplane ports as the HA3 link for redundancy.
- ❑ **The same set of licenses**—Licenses are unique to each firewall and cannot be shared between the firewalls. Therefore, you must license both firewalls identically. If both firewalls do not

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have an identical set of licenses, they cannot synchronize configuration information and maintain parity for a seamless failover.

If you have an existing firewall and you want to add a new firewall for HA purposes and the new firewall has an existing configuration, it is recommended that you Reset the Firewall to Factory Default Settings on the new firewall. This will ensure that the new firewall has a clean configuration. After HA is configured, you will then sync the configuration on the primary firewall to the newly introduced firewall with the clean config. You will also have to configure local IP addresses.

ECMP in Active/Active HA

When an active/active HA peer fails, its sessions transfer to the new active-primary firewall, which tries to use the same egress interface that the failed firewall was using. If the firewall finds that interface among the ECMP paths, the transferred sessions will take the same egress interface and path. This behavior occurs regardless of the ECMP algorithm in use; using the same interface is desirable.

Only if no ECMP path matches the original egress interface will the active-primary firewall select a new ECMP path.

If you did not configure the same interfaces on the active/active peers, upon failover the active-primary firewall selects the next best path from the FIB table. Consequently, the existing sessions might not be distributed according to the ECMP algorithm.

NAT in Active/Active HA

In an active/active HA configuration:

- You must bind each Dynamic IP (DIP) NAT rule and Dynamic IP and Port (DIPP) NAT rule to either Device ID 0 or Device ID 1.
- You must bind each static NAT rule to either Device ID 0, Device ID 1, both Device IDs, or the firewall in active-primary state.

Thus, when one of the firewalls creates a new session, the Device ID **0** or Device ID **1** binding determines which NAT rules match the firewall. The device binding must include the session owner firewall to produce a match.

The session setup firewall performs the NAT policy match, but the NAT rules are evaluated based on the session owner. That is, the session is translated according to NAT rules that are bound to the session owner firewall. While performing NAT policy matching, a firewall skips all NAT rules that are not bound to the session owner firewall.

For example, suppose the firewall with Device ID 1 is the session owner and session setup firewall. When the firewall with Device ID 1 tries to match a session to a NAT rule, it skips all rules bound to Device ID 0. The firewall performs the NAT translation only if the session owner and the Device ID in the NAT rule match.

You will typically create device-specific NAT rules when the peer firewalls use different IP addresses for translation.

If one of the peer firewalls fails, the active firewall continues to process traffic for synchronized sessions from the failed firewall, including NAT traffic. In a source NAT configuration, when one firewall fails:

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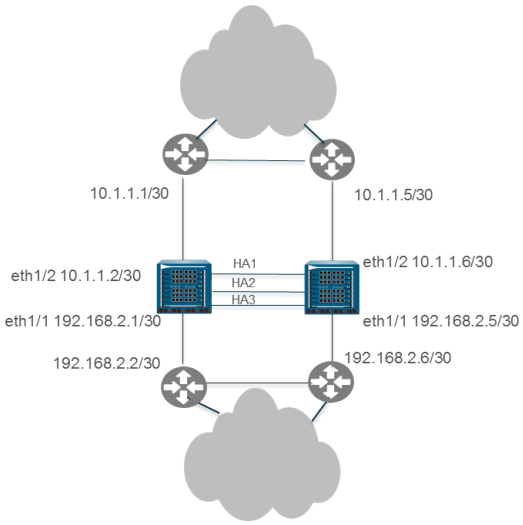
- The floating IP address that is used as the Translated IP address of the NAT rule transfers to the surviving firewall. Hence, the existing sessions that fail over will still use this IP address.
- All new sessions will use the device-specific NAT rules that the surviving firewall naturally owns. That is, the surviving firewall translates new sessions using only the NAT rules that match its Device ID; it ignores any NAT rules bound to the failed Device ID.

Route-Based Redundancy

In a Layer 3 interface deployment and active/active HA configuration, the firewalls are connected to routers, not switches. The firewalls use dynamic routing protocols to determine the best path (asymmetric route), ensure continuous service, minimize downtime, and to load share between the HA pair. In such a scenario, no floating IP addresses are necessary. If a link, monitored path, or firewall fails, or if Bidirectional Forwarding Detection (BFD) detects a link failure, the routing protocol (RIP, OSPF, or BGP) handles the rerouting of traffic to the functioning firewall. You configure each firewall interface with a unique IP address. The IP addresses remain local to the firewall where they are configured; they do not move between devices when a firewall fails. See Use Case: Configure Active/Active HA with Route-Based Redundancy.



Determine Your Active/Active Use Case

Determine which type of use case you have and then select the corresponding procedure to configure active/active HA.

If you are using route-based redundancy, Floating IP Address and Virtual MAC Address, or ARP Load-Sharing, select the corresponding procedure:

- Use Case: Configure Active/Active HA with Route-Based Redundancy
- Use Case: Configure Active/Active HA with Floating IP Addresses
- Use Case: Configure Active/Active HA with ARP Load-Sharing

If you want a Layer 3 active/active HA deployment that behaves like an active/passive deployment, select the following procedure:

- Use Case: Configure Active/Active HA with Floating IP Address Bound to Active-Primary Firewall

If you are configuring NAT in Active/Active HA, see the following procedures:

- Use Case: Configure Active/Active HA with Source DIPP NAT Using Floating IP Addresses
- Use Case: Configure Separate Source NAT IP Address Pools for Active/Active HA Firewalls
- Use Case: Configure Active/Active HA for ARP Load-Sharing with Destination NAT
- Use Case: Configure Active/Active HA for ARP Load-Sharing with Destination NAT in Layer 3

Configure Active/Active HA

The following procedure describes the basic workflow for configuring your firewalls in an active/active configuration. Before you begin, verify that any filtering devices between your HA pair members allow the protocols and ports used by the HA links and review the Determine Your Active/Active Use Case for configuration examples more tailored to your specific network environment.

You can configure data ports as both dedicated HA interfaces and as dedicated backup HA interfaces. For firewalls without dedicated HA interfaces, such as the PA-200 and PA-400 Series, it is required to configure a data port as a HA interface.

Data ports configured as HA1, HA2, or HA3 interfaces can be connected directly to each HA interface on the firewall or connected through a Layer2 switch. For data ports configured as an HA3 interface, you must enable jumbo frames as HA3 messages exceed 1,500 bytes.

To configure active/active, first complete the following steps on one peer and then complete them on the second peer, ensuring that you set the Device ID to different values (0 or 1) on each peer.

STEP 1 | Connect the HA ports to set up a physical connection between the firewalls.

For each use case, the firewalls could be any hardware model; choose the HA3 step that corresponds with your model.

- For firewalls with dedicated HA ports, use an Ethernet cable to connect the dedicated HA1 ports and the HA2 ports on peers. Use a crossover cable if the peers are directly connected to each other.
- For firewalls without dedicated HA ports, select two data interfaces for the HA2 link and the backup HA1 link. Then, use an Ethernet cable to connect these in-band HA interfaces

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across both firewalls. Use the management port for the HA1 link and ensure that the management ports can connect to each other across your network.

- For HA3:
  - On PA-7000 Series firewalls, connect the High Speed Chassis Interconnect (HSCI-A) on the first chassis to the HSCI-A on the second chassis, and the HSCI-B on the first chassis to the HSCI-B on the second chassis.
  - On the PA-5450 firewall, connect the HSCI-A on the first chassis to the HSCI-A on the second chassis, and the HSCI-B on the first chassis to the HSCI-B on the second chassis.
  - On the PA-5400 Series firewalls (which have one HSCI port), connect the HSCI port on the first chassis to the HSCI port on the second chassis.
  - On PA-5200 Series firewalls (which have one HSCI port), connect the HSCI port on the first chassis to the HSCI port on the second chassis. You can also use data ports for HA3 on PA-5200 Series firewalls.
  - On PA-3400 Series firewalls (which have one HSCI port), connect the HSCI port on the first chassis to the HSCI port on the second chassis.
  - On PA-3200 Series firewalls (which have one HSCI port), connect the HSCI port on the first chassis to the HSCI port on the second chassis.
  - On any other hardware model, use dataplane interfaces for HA3.

STEP 2 | Enable ping on the management port.

Enabling ping allows the management port to exchange heartbeat backup information.

1. Select **Device** > **Setup** > **Interfaces** > **Management**.
2. Select **Ping** as a service that is permitted on the interface.

STEP 3 | If the firewall does not have dedicated HA ports, set up the data ports to function as HA ports.

For firewalls with dedicated HA ports continue to the next step.

1. Select **Network** > **Interfaces**.
2. Confirm that the link is up on the ports that you want to use.
3. Select the interface and set **Interface Type** to **HA**.
4. Set the **Link Speed** and **Link Duplex** settings, as appropriate.

STEP 4 | Enable active/active HA and set the group ID.

1. In **Device** > **High Availability** > **General**, edit Setup.
2. Select **Enable HA**.
3. Enter a **Group ID**, which must be the same for both firewalls. The firewall uses the Group ID to calculate the virtual MAC address (range is 1-63).
4. (Optional) Enter a **Description**.
5. For **Mode**, select **Active Active**.

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STEP 5 | Set the Device ID, enable synchronization, and identify the control link on the peer firewall

1. In **Device** > **High Availability** > **General**, edit Setup.
2. Select **Device ID** as follows:
   - When configuring the first peer, set the **Device ID** to **0**.
   - When configuring the second peer, set the **Device ID** to **1**.
3. Select **Enable Config Sync**. This setting is required to synchronize the two firewall configurations (enabled by default).
4. Enter the **Peer HA1 IP Address**, which is the IP address of the HA1 control link on the peer firewall.
5. (Optional) Enter a **Backup Peer HA1 IP Address**, which is the IP address of the backup control link on the peer firewall.
6. Click **OK**.

STEP 6 | Determine whether or not the firewall with the lower Device ID preempts the active-primary firewall upon recovery from a failure.

1. In **Device** > **High Availability** > **General**, edit Election Settings.
2. Select **Preemptive** to cause the firewall with the lower Device ID to automatically resume active-primary operation after either firewall recovers from a failure. Both firewalls must have **Preemptive** selected for preemption to occur.

   Leave **Preemptive** unselected if you want the active-primary role to remain with the current firewall until you manually make the recovered firewall the active-primary firewall.

STEP 7 | Enable heartbeat backup if your control link uses a dedicated HA port or an in-band port.

You need not enable heartbeat backup if you are using the management port for the control link.

1. In **Device** > **High Availability** > **General**, edit Election Settings.
2. Select **Heartbeat Backup**.

   To allow the heartbeats to be transmitted between the firewalls, you must verify that the management port across both peers can route to each other.

   Enabling heartbeat backup allows you to prevent a split-brain situation. Split brain occurs when the HA1 link goes down, causing the firewall to miss heartbeats, although the firewall is still functioning. In such a situation, each peer believes the other is down and attempts to start services that are running, thereby causing a split brain. Enabling heartbeat backup prevents split brain because redundant heartbeats and hello messages are transmitted over the management port.

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STEP 8 | (Optional) Modify the HA Timers.

By default, the HA timer profile is set to the **Recommended** profile and is suited for most HA deployments.

1. In **Device** > **High Availability** > **General**, edit Election Settings.
2. Select **Aggressive** to trigger faster failover. Select **Advanced** to define custom values for triggering failover in your setup.

   *To view the preset value for an individual timer included in a profile, select **Advanced** and click **Load Recommended** or **Load Aggressive**. The preset values for your hardware model will be displayed on screen.*

STEP 9 | Set up the control link connection.

This example uses an in-band port that is set to interface type HA.

For firewalls that use the management port as the control link, the IP address information is automatically pre-populated.

1. In **Device** > **High Availability** > **HA Communications**, edit Control Link (HA1).
2. Select the **Port** that you have cabled for use as the HA1 link.
3. Set the **IPv4/IPv6 Address** and **Netmask**.

   If the HA1 interfaces are on separate subnets, enter the IP address of the **Gateway**. Do not add a gateway address if the firewalls are directly connected.

STEP 10 | (Optional) Enable encryption for the control link connection.

This is typically used to secure the link if the two firewalls are not directly connected, that is if the ports are connected to a switch or a router.

1. Export the HA key from one firewall and import it into the peer firewall.
   1. Select **Device** > **Certificate Management** > **Certificates**.
   2. Select **Export HA key**. Save the HA key to a network location that the peer can access.
   3. On the peer firewall, select **Device** > **Certificate Management** > **Certificates**, and select **Import HA key** to browse to the location that you saved the key and import it in to the peer.
2. In **Device** > **High Availability** > **General**, edit the Control Link (HA1).
3. Select **Encryption Enabled**.

   If you enable encryption, after you finish configuring the HA firewalls, you can Refresh HA1 SSH Keys and Configure Key Options.

STEP 11 | Set up the backup control link connection.

1. In **Device** > **High Availability** > **HA Communications**, edit Control Link (HA1 Backup).
2. Select the HA1 backup interface and set the **IPv4/IPv6 Address** and **Netmask**.

   PA-3200 Series firewalls don't support an IPv6 address for the HA1 backup control link; use an IPv4 address.

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STEP 12 | Set up the data link connection (HA2) and the backup HA2 connection between the firewalls.

1. In **Device** > **High Availability** > **General**, edit Data Link (HA2).
2. Select the **Port** to use for the data link connection.
3. Select the **Transport** method. The default is **ethernet**, and will work when the HA pair is connected directly or through a switch. If you need to route the data link traffic through the network, select **IP** or **UDP** as the transport mode.
4. If you use IP or UDP as the transport method, enter the **IPv4/IPv6 Address** and **Netmask**.
5. Verify that **Enable Session Synchronization** is selected.
6. Select **HA2 Keep-alive** to enable monitoring on the HA2 data link between the HA peers. If a failure occurs based on the threshold that is set (default is 10000 ms), the defined action will occur. When an HA2 Keep-alive failure occurs, the system either generates a critical system log message or causes a split dataplane depending on your configuration.

   You can configure the HA2 Keep-alive option on both firewalls, or just one firewall in the HA pair. If the option is only enabled on one firewall, only that firewall sends the Keep-alive messages. The other firewall is notified if a failure occurs.

   A split dataplane causes the dataplanes of both peers to operate independently while leaving the high-available state as Active-Primary and Active-Secondary. If only one firewall is configured to split dataplane, then split dataplane applies to the other device as well.
7. Edit the **Data Link (HA2 Backup)** section, select the interface, and add the **IPv4/IPv6 Address** and **Netmask**.
8. Click **OK**.

STEP 13 | Configure the HA3 link for packet forwarding.

1. In **Device** > **High Availability** > **HA Communications**, edit Packet Forwarding.
2. For **HA3 Interface**, select the interface you want to use to forward packets between active/active HA peers. It must be a dedicated interface capable of Layer 2 transport and set to **Interface Type HA**.
3. Select **VR Sync** to force synchronization of all virtual routers configured on the HA peers. Select when the virtual router is not configured for dynamic routing protocols. Both peers must be connected to the same next-hop router through a switched network and must use static routing only.
4. Select **QoS Sync** to synchronize the QoS profile selection on all physical interfaces. Select when both peers have similar link speeds and require the same QoS profiles on all physical interfaces. This setting affects the synchronization of QoS settings on the **Network** tab. QoS policy is synchronized regardless of this setting.

STEP 14 | (Optional) Modify the Tentative Hold time.

1. In **Device** > **High Availability** > **HA Communications**, edit Packet Forwarding.
2. For **Tentative Hold Time (sec)**, enter the number of seconds that a firewall stays in Tentative state after it recovers post-failure (range is 10-600, default is 60).

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STEP 15 | Configure Session Owner and Session Setup.

1. In **Device** > **High Availability** > **HA Communications**, edit Packet Forwarding.
2. For **Session Owner Selection**, select one of the following:
   - **First Packet**—The firewall that receives the first packet of a new session is the session owner (recommended setting). This setting minimizes traffic across HA3 and load shares traffic across peers.
   - **Primary Device**—The firewall that is in active-primary state is the session owner.
3. For **Session Setup**, select one of the following:
   - **IP Modulo**—The firewall performs an XOR operation on the source and destination IP addresses from the packet and based on the result, the firewall chooses which HA peer will set up the session.
   - **Primary Device**—The active-primary firewall sets up all sessions.
   - **First Packet**—The firewall that receives the first packet of a new session performs session setup (recommended setting).

     Start with First Packet for Session Owner and Session Setup, and then based on load distribution, you can change to one of the other options.
   - **IP Hash**—The firewall uses a hash of either the source IP address or a combination of the source and destination IP addresses to distribute session setup responsibilities.
4. Click **OK**.

STEP 16 | Configure an HA virtual address.

You need a virtual address to use a Floating IP Address and Virtual MAC Address or ARP Load-Sharing.

1. In **Device** > **High Availability** > **Active/Active Config**, **Add** a Virtual Address.
2. Enter or select an **Interface**.
3. Select the **IPv4** or **IPv6** tab and click **Add**.
4. Enter an **IPv4 Address** or **IPv6 Address**.
5. For **Type**:
   - Select **Floating** to configure the virtual IP address to be a floating IP address.
   - Select **ARP Load Sharing** to configure the virtual IP address to be a shared IP address and skip to the step to configure ARP load-sharing.

STEP 17 | Configure the floating IP address.

1. Do not select **Floating IP bound to the Active-Primary device** unless you want the active/active HA pair to behave like an active/passive HA pair.
2. For **Device 0 Priority** and **Device 1 Priority**, enter a priority for the firewall configured with Device ID 0 and Device ID 1, respectively. The relative priorities determine which

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peer owns the floating IP address you just configured (range is 0-255). The firewall with the lowest priority value (highest priority) owns the floating IP address.

3. Select **Failover address if link state is down** to cause the firewall to use the failover address when the link state on the interface is down.
4. Click **OK**.

STEP 18 | Configure ARP Load-Sharing.

The device selection algorithm determines which HA firewall responds to the ARP requests to provide load sharing.

1. For **Device Selection Algorithm**, select one of the following:
   - **IP Modulo**—The firewall that will respond to ARP requests is based on the parity of the ARP requester's IP address.
   - **IP Hash**—The firewall that will respond to ARP requests is based on a hash of the ARP requester's IP address.
2. Click **OK**.

STEP 19 | Define HA Failover Conditions.

STEP 20 | **Commit** the configuration.

Use Case: Configure Active/Active HA with Route-Based Redundancy

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Pro

Route-based redundancy enables you to implement high availability configurations using dynamic routing protocols instead of traditional floating IP addresses. When you deploy this approach, each device in your HA pair maintains its own unique IP address on the Layer 3 network and uses protocols like OSPF or BGP to advertise routes and manage traffic distribution. You can leverage this method to create active-active deployments where both devices simultaneously process traffic, providing automatic load distribution and seamless failover capabilities when device or link failures occur.

Consider implementing route-based redundancy when you need scalable high availability solutions that can adapt to complex network topologies without the limitations of shared IP addressing schemes. This approach becomes particularly valuable when you require load sharing capabilities to improve network performance and resource utilization across your HA pair. You will find route-based redundancy especially beneficial in environments where asymmetric routing is acceptable or desired, as traffic can enter through one device and exit through another without disrupting operations.

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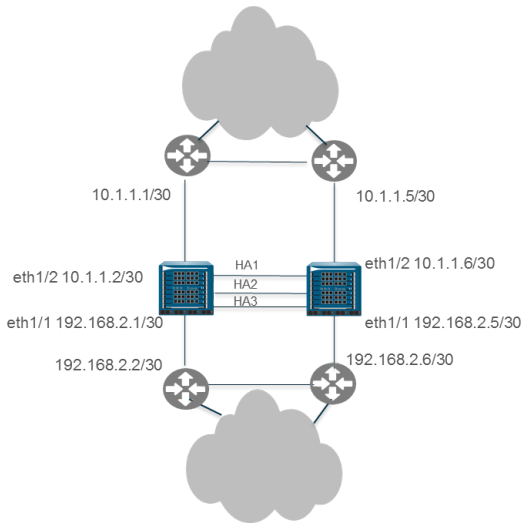
You might choose route-based redundancy over other active/active HA methods when you want to eliminate single points of failure associated with floating IP addresses and when your network infrastructure already supports dynamic routing protocols. While this method requires more sophisticated configuration and troubleshooting skills compared to simpler HA setups, you gain the advantage of automatic traffic rerouting and improved network resilience.

The following Layer 3 topology illustrates two PA-7050 firewalls in an active/active HA environment that use Route-Based Redundancy. The firewalls belong to an OSPF area. When a link or firewall fails, OSPF handles the redundancy by redirecting traffic to the functioning firewall.



STEP 1 | Configure Active/Active HA.

Perform Step 1 through Step 15.

STEP 2 | Configure OSPF.

See OSPF.

STEP 3 | Define HA failover conditions.

Define HA Failover Conditions.

STEP 4 | **Commit** the configuration.

STEP 5 | Configure the peer firewall in the same way, except in Step 5, if you selected Device ID **0** for the first firewall, select Device ID **1** for the peer firewall.

Use Case: Configure Active/Active HA with Floating IP Addresses

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Pro

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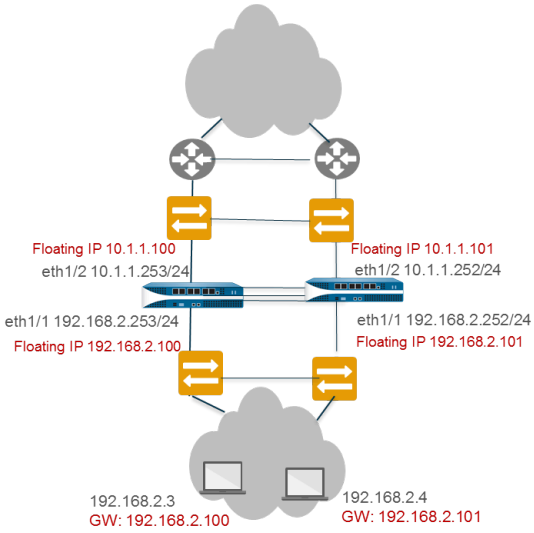
In this Layer 3 interface example, the HA firewalls connect to switches and use floating IP addresses to handle link or firewall failures. The end hosts are each configured with a gateway, which is the floating IP address of one of the HA firewalls. See Floating IP Address and Virtual MAC Address.



STEP 1 | Configure Active/Active HA.

Perform Step 1 through Step 15.

STEP 2 | Configure an HA virtual address.

You need a virtual address to use a Floating IP Address and Virtual MAC Address.

1. In **Device** > **High Availability** > **Active/Active Config**, **Add** a Virtual Address.
2. Enter or select an **Interface**.
3. Select the **IPv4** or **IPv6** tab and click **Add**.
4. Enter an **IPv4 Address** or **IPv6 Address**.
5. For **Type**, select **Floating** to configure the virtual IP address to be a floating IP address.

STEP 3 | Configure the floating IP address.

1. Do not select **Floating IP bound to the Active-Primary device**.
2. For **Device 0 Priority** and **Device 1 Priority**, enter a priority for the firewall configured with Device ID 0 and Device ID 1, respectively. The relative priorities determine which peer owns the floating IP address you just configured (range is 0 to 255). The firewall with the lowest priority value (highest priority) owns the floating IP address.
3. Select **Failover address if link state is down** to cause the firewall to use the failover address when the link state on the interface is down.
4. Click **OK**.

STEP 4 | Enable jumbo frames on firewalls other than PA-7000 Series firewalls.

Perform Step 19 of Configure Active/Active HA.

STEP 5 | Define HA Failover Conditions

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STEP 6 | **Commit** the configuration.

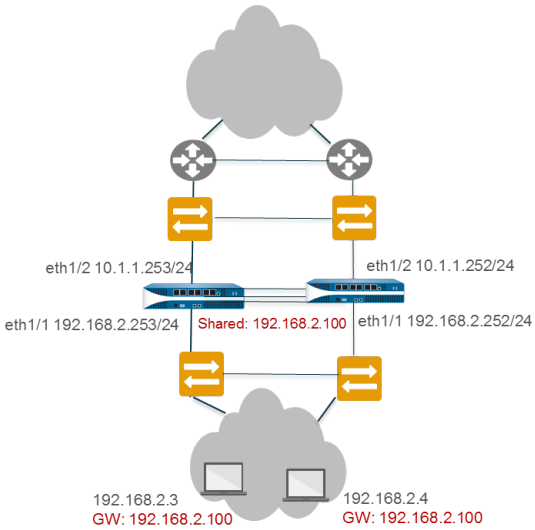
STEP 7 | Configure the peer firewall in the same way, except selecting a different Device ID.

For example, if you selected Device ID **0** for the first firewall, select Device ID **1** for the peer firewall.

Use Case: Configure Active/Active HA with ARP Load-Sharing

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Pro

In this example, hosts in a Layer 3 deployment need gateway services from the HA firewalls. The firewalls are configured with a single shared IP address, which allows ARP Load-Sharing. The end hosts are configured with the same gateway, which is the shared IP address of the HA firewalls.



STEP 1 | Configure Active/Active HA.

Perform Step 1 through Step 15.

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STEP 2 | Configure an HA virtual address.

The virtual address is the shared IP address that allows ARP Load-Sharing.

1. Select **Device** > **High Availability** > **Active/Active Config** > **Virtual Address** and click **Add**.
2. Enter or select an **Interface**.
3. Select the **IPv4** or **IPv6** tab and click **Add**.
4. Enter an **IPv4 Address** or **IPv6 Address**.
5. For **Type**, select **ARP Load Sharing**, which allows both peers to use the virtual IP address for ARP Load-Sharing.

STEP 3 | Configure ARP Load-Sharing.

The device selection algorithm determines which HA firewall responds to the ARP requests to provide load sharing.

1. For **Device Selection Algorithm**, select one of the following:
   - **IP Modulo**—The firewall that will respond to ARP requests is based on the parity of the ARP requester's IP address.
   - **IP Hash**—The firewall that will respond to ARP requests is based on a hash of the ARP requester's IP address.
2. Click **OK**.

STEP 4 | Enable jumbo frames on firewalls other than PA-7000 Series firewalls.

STEP 5 | Define HA Failover Conditions

STEP 6 | **Commit** the configuration.

STEP 7 | Configure the peer firewall in the same way, except selecting a different Device ID.

For example, if you selected Device ID **0** for the first firewall, select Device ID **1** for the peer firewall.

Use Case: Configure Active/Active HA with Floating IP Address Bound to Active-Primary Firewall

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

In mission-critical data centers, you may want both Layer 3 HA firewalls to participate in path monitoring so that they can detect path failures upstream from both firewalls. Additionally, you prefer to control if and when the floating IP address returns to the recovered firewall after it

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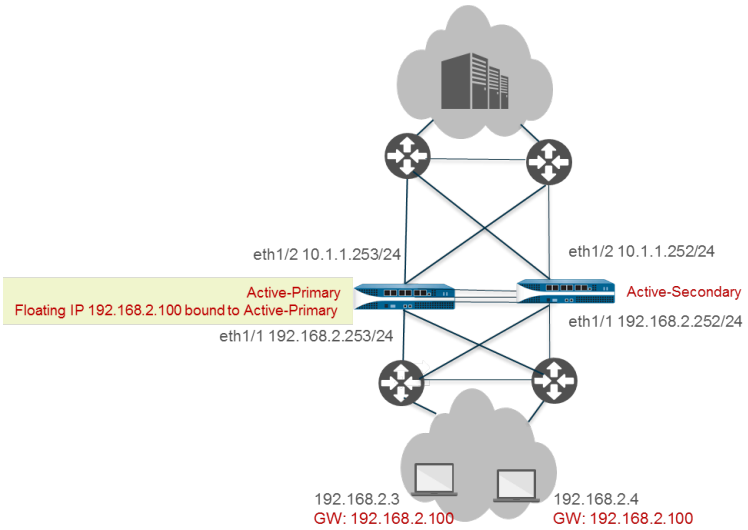
comes back up, rather than the floating IP address returning to the device ID to which it is bound. (That default behavior is described in Floating IP Address and Virtual MAC Address.)

In this use case, you control when the floating IP address and therefore the active-primary role move back to a recovered HA peer. The active/active HA firewalls share a single floating IP address that you bind to whichever firewall is in the active-primary state. With only one floating IP address, network traffic flows predominantly to a single firewall, so this active/active deployment functions like an active/passive deployment.

In this use case, Cisco Nexus 7010 switches with virtual PortChannels (vPCs) operating in Layer 3 connect to the firewalls. You must configure the Layer 3 switches (router peers) north and south of the firewalls with a route preference to the floating IP address. That is, you must design your network so the route tables of the router peers have the best path to the floating IP address. This example uses static routes with the proper metrics so that the route to the floating IP address uses a lower metric (the route to the floating IP address is preferred) and receives the traffic. An alternative to using static routes would be to design the network to redistribute the floating IP address into the OSPF routing protocol (if you are using OSPF).

The following topology illustrates the floating IP address bound to the active-primary firewall, which is initially Peer A, the firewall on the left.



Upon a failover, when the active-primary firewall (Peer A) goes down and the active-secondary firewall (Peer B) takes over as the active-primary peer, the floating IP address moves to Peer B (shown in the following figure). Peer B remains the active-primary firewall and traffic continues to go to Peer B, even when Peer A recovers and becomes the active-secondary firewall. You decide if and when to make Peer A the active-primary firewall again.

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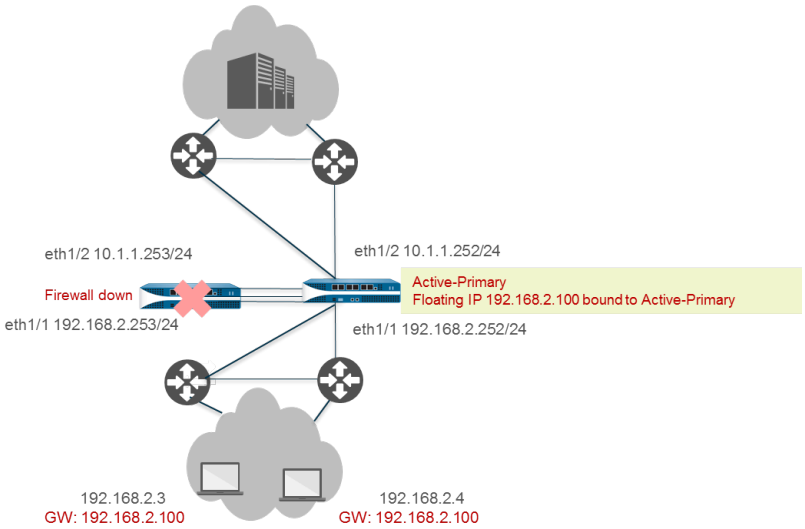

Binding the floating IP address to the active-primary firewall provides you with more control over how the firewalls determine floating IP address ownership as they move between various HA Firewall States. The following advantages result:

- You can have an active/active HA configuration for path monitoring out of both firewalls, but have the firewalls function like an active/passive HA configuration because traffic directed to the floating IP address always goes to the active-primary firewall.

When you disable preemption on both firewalls, you have the following additional benefits:

- The floating IP address does not move back and forth between HA firewalls if the active-secondary firewall flaps up and down.
- You can review the functionality of the recovered firewall and the adjacent components before manually directing traffic to it again, which you can do at a convenient down time.
- You have control over which firewall owns the floating IP address so that you keep all flows of new and existing sessions on the active-primary firewall, thereby minimizing traffic on the HA3 link.

- *We strongly recommended you configure HA link monitoring on the interface(s) that support the floating IP address(es) to allow each HA peer to quickly detect a link failure and fail over to its peer. Both HA peers must have link monitoring for it to function.*
- *We strongly recommend you configure HA path monitoring to notify each HA peer when a path has failed so a firewall can fail over to its peer. Because the floating IP address is always bound to the active-primary firewall, the firewall cannot automatically fail over to the peer when a path goes down and path monitoring is not enabled.*

You cannot configure NAT for a floating IP address that is bound to an active-primary firewall.

STEP 1 | Perform Step 1 through Step 5 of Configure Active/Active HA.

STEP 2 | (Optional) Disable preemption.

Disabling preemption allows you full control over when the recovered firewall becomes the active-primary firewall.

1. In **Device** > **High Availability** > **General**, edit the Election Settings.
2. Clear **Preemptive** if it is enabled.
3. Click **OK**.

STEP 3 | Perform Step 7 through Step 14 of Configure Active/Active HA.

STEP 4 | Configure Session Owner and Session Setup.

1. In **Device** > **High Availability** > **Active/Active Config**, edit Packet Forwarding.
2. For **Session Owner Selection**, we recommend you select **Primary Device**. The firewall that is in active-primary state is the session owner.

   Alternatively, for **Session Owner Selection** you can select **First Packet** and then for **Session Setup**, select **Primary Device** or **First Packet**.
3. For **Session Setup**, select **Primary Device**—The active-primary firewall sets up all sessions. This is the recommended setting if you want your active/active configuration to behave like an active/passive configuration because it keeps all activity on the active-primary firewall.

   *You must also engineer your network to eliminate the possibility of asymmetric traffic going to the HA pair. If you don't do so and traffic goes to the active-secondary firewall, setting **Session Owner Selection** and **Session Setup** to **Primary Device** causes the traffic to traverse HA3 to get to the active-primary firewall for session ownership and session setup.*
4. Click **OK**.

STEP 5 | Configure an HA virtual address.

1. Select **Device** > **High Availability** > **Active/Active Config** > **Virtual Address** and click **Add**.
2. Enter or select an **Interface**.
3. Select the **IPv4** or **IPv6** tab and **Add** an **IPv4 Address** or **IPv6 Address**.
4. For **Type**, select **Floating**, which configures the virtual IP address to be a floating IP address.
5. Click **OK**.

STEP 6 | Bind the floating IP address to the active-primary firewall.

1. Select **Floating IP bound to the Active-Primary device**.
2. Select **Failover address if link state is down** to cause the firewall to use the failover address when the link state on the interface is down.
3. Click **OK**.

STEP 7 | Enable jumbo frames on firewalls other than PA-7000 Series firewalls.

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STEP 8 | **Commit** the configuration.

STEP 9 | Configure the peer firewall in the same way, except selecting a different Device ID.

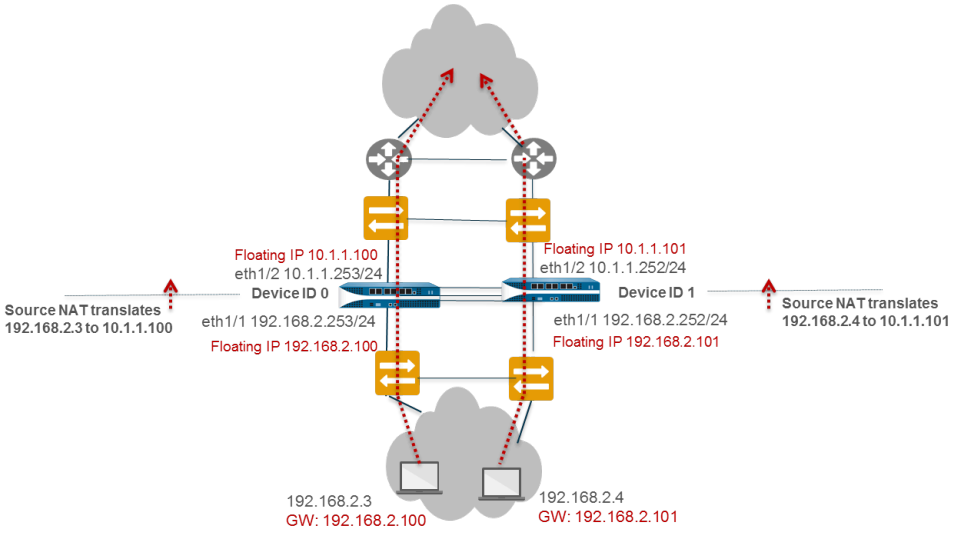
For example, if you selected Device ID **0** for the first firewall, select Device ID **1** for the peer firewall.

Use Case: Configure Active/Active HA with Source DIPP NAT Using Floating IP Addresses

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Pro

This Layer 3 interface example uses source NAT in Active/Active HA. The Layer 2 switches create broadcast domains to ensure users can reach everything north and south of the firewalls.

PA-3050-1 has Device ID 0 and its HA peer, PA-3050-2, has Device ID 1. In this use case, NAT translates the source IP address and port number to the floating IP address configured on the egress interface. Each host is configured with a default gateway address, which is the floating IP address on Ethernet1/1 of each firewall. The configuration requires two source NAT rules, one bound to each Device ID, although you configure both NAT rules on a single firewall and they are synchronized to the peer firewall.



STEP 1 | On PA-3050-2 (Device ID 1), perform Step 1 through Step 3 of Configure Active/Active HA.

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STEP 2 | Enable active/active HA.

1. In **Device** > **High Availability** > **General**, edit Setup.
2. Select **Enable HA**.
3. Enter a **Group ID**, which must be the same for both firewalls. The firewall uses the Group ID to calculate the virtual MAC address (range is 1-63).
4. For **Mode**, select **Active Active**.
5. Set the **Device ID** to **1**.
6. Select **Enable Config Sync**. This setting is required to synchronize the two firewall configurations (enabled by default).
7. Enter the **Peer HA1 IP Address**, which is the IP address of the HA1 control link on the peer firewall.
8. (Optional) Enter a **Backup Peer HA1 IP Address**, which is the IP address of the backup control link on the peer firewall.
9. Click **OK**.

STEP 3 | Configure Active/Active HA.

Complete Step 6 through Step 14.

STEP 4 | Configure Session Owner and Session Setup.

1. In **Device** > **High Availability** > **Active/Active Config**, edit Packet Forwarding.
2. For **Session Owner Selection**, select **First Packet**—The firewall that receives the first packet of a new session is the session owner.
3. For **Session Setup**, select **IP Modulo**—Distributes session setup load based on parity of the source IP address.
4. Click **OK**.

STEP 5 | Configure an HA virtual address.

1. Select **Device** > **High Availability** > **Active/Active Config** > **Virtual Address** and click **Add**.
2. Select **Interface** eth1/1.
3. Select **IPv4** and **Add** an **IPv4 Address** of 10.1.1.101.
4. For **Type**, select **Floating**, which configures the virtual IP address to be a floating IP address.

STEP 6 | Configure the floating IP address.

1. Do not select **Floating IP bound to the Active-Primary device**.
2. Select **Failover address if link state is down** to cause the firewall to use the failover address when the link state on the interface is down.
3. Click **OK**.

STEP 7 | Enable jumbo frames on firewalls other than the PA-7000 Series.

STEP 8 | Define HA Failover Conditions.

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STEP 9 | **Commit** the configuration.

STEP 10 | Configure the peer firewall, PA-3050-1 with the same settings, except for the following changes:

- Select **Device ID 0**.
- Configure an HA virtual address of 10.1.1.100.
- For **Device 1 Priority**, enter 255. For **Device 0 Priority**, enter 0.

In this example, Device ID 0 has a lower priority value so a higher priority; therefore, the firewall with Device ID 0 (PA-3050-1) owns the floating IP address 10.1.1.100.

STEP 11 | Still on PA-3050-1, create the source NAT rule for Device ID 0.

1. Select **Policies** > **NAT** and click **Add**.
2. Enter a **Name** for the rule that in this example identifies it as a source NAT rule for Device ID 0.
3. For **NAT Type**, select **ipv4** (default).
4. On the **Original Packet**, for **Source Zone**, select **Any**.
5. For **Destination Zone**, select the zone you created for the external network.
6. Allow **Destination Interface**, **Service**, **Source Address**, and **Destination Address** to remain set to **Any**.
7. For the **Translated Packet**, select **Dynamic IP And Port** for **Translation Type**.
8. For **Address Type**, select **Interface Address**, in which case the translated address will be the IP address of the interface. Select an **Interface** (eth1/1 in this example) and an **IP Address** of the floating IP address 10.1.1.100.
9. On the **Active/Active HA Binding** tab, for **Active/Active HA Binding**, select **0** to bind the NAT rule to Device ID 0.
10. Click **OK**.

STEP 12 | Create the source NAT rule for Device ID 1.

1. Select **Policies** > **NAT** and click **Add**.
2. Enter a **Name** for the policy rule that in this example helps identify it as a source NAT rule for Device ID 1.
3. For **NAT Type**, select **ipv4** (default).
4. On the **Original Packet**, for **Source Zone**, select **Any**. For **Destination Zone**, select the zone you created for the external network.
5. Allow **Destination Interface**, **Service**, **Source Address**, and **Destination Address** to remain set to **Any**.
6. For the **Translated Packet**, select **Dynamic IP And Port** for **Translation Type**.
7. For **Address Type**, select **Interface Address**, in which case the translated address will be the IP address of the interface. Select an **Interface** (eth1/1 in this example) and an **IP Address** of the floating IP address 10.1.1.101.
8. On the **Active/Active HA Binding** tab, for the **Active/Active HA Binding**, select **1** to bind the NAT rule to Device ID 1.
9. Click **OK**.

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STEP 13 | **Commit** the configuration.

Use Case: Configure Separate Source NAT IP Address Pools for Active/Active HA Firewalls

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Pro

If you want to use IP address pools for source NAT in Active/Active HA, each firewall must have its own pool, which you then bind to a Device ID in a NAT rule.

Address objects and NAT rules are synchronized (in both active/passive and active/active mode), so they need to be configured on only one of the firewalls in the HA pair.

This example configures an address object named Dyn-IP-Pool-dev0 containing the IP address pool 10.1.1.140-10.1.1.150. It also configures an address object named Dyn-IP-Pool-dev1 containing the IP address pool 10.1.1.160-10.1.1.170. The first address object is bound to Device ID 0; the second address object is bound to Device ID 1.

STEP 1 | On one HA firewall, create address objects.

1. Select **Objects** > **Addresses** and **Add** an address object **Name**, in this example, Dyn-IP-Pool-dev0.
2. For **Type**, select **IP Range** and enter the range 10.1.1.140-10.1.1.150.
3. Click **OK**.
4. Repeat this step to configure another address object named Dyn-IP-Pool-dev1 with the **IP Range** of 10.1.1.160-10.1.1.170.

STEP 2 | Create the source NAT rule for Device ID 0.

1. Select **Policies** > **NAT** and **Add** a NAT policy rule with a **Name**, for example, Src-NAT-dev0.
2. For **Original Packet**, for **Source Zone**, select **Any**.
3. For **Destination Zone**, select the destination zone for which you want to translate the source address, such as Untrust.
4. For **Translated Packet**, for **Translation Type**, select **Dynamic IP and Port**.
5. For **Translated Address**, **Add** the address object you created for the pool of addresses belonging to Device ID 0: Dyn-IP-Pool-dev0.
6. For **Active/Active HA Binding**, select **0** to bind the NAT rule to Device ID 0.
7. Click **OK**.

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STEP 3 | Create the source NAT rule for Device ID 1.

1. Select **Policies** > **NAT** and **Add** a NAT policy rule with a **Name**, for example, Src-NAT-dev1.
2. For **Original Packet**, for **Source Zone**, select **Any**.
3. For **Destination Zone**, select the destination zone for which you want to translate the source address, such as Untrust.
4. For **Translated Packet**, for **Translation Type**, select **Dynamic IP and Port**.
5. For **Translated Address**, **Add** the address object you created for the pool of addresses belonging to Device ID 1: Dyn-IP-Pool-dev1.
6. For **Active/Active HA Binding**, select **1** to bind the NAT rule to Device ID 1.
7. Click **OK**.

STEP 4 | **Commit** the configuration.

Use Case: Configure Active/Active HA for ARP Load-Sharing with Destination NAT

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

This Layer 3 interface example uses NAT in Active/Active HA and ARP Load-Sharing with destination NAT. Both HA firewalls respond to an ARP request for the destination NAT address with the ingress interface MAC address. Destination NAT translates the public, shared IP address (in this example, 10.1.1.200) to the private IP address of the server (in this example, 192.168.2.200).

When the HA firewalls receive traffic for the destination 10.1.1.200, both firewalls could possibly respond to the ARP request, which could cause network instability. To avoid the potential issue, configure the firewall that is in active-primary state to respond to the ARP request by binding the destination NAT rule to the active-primary firewall.

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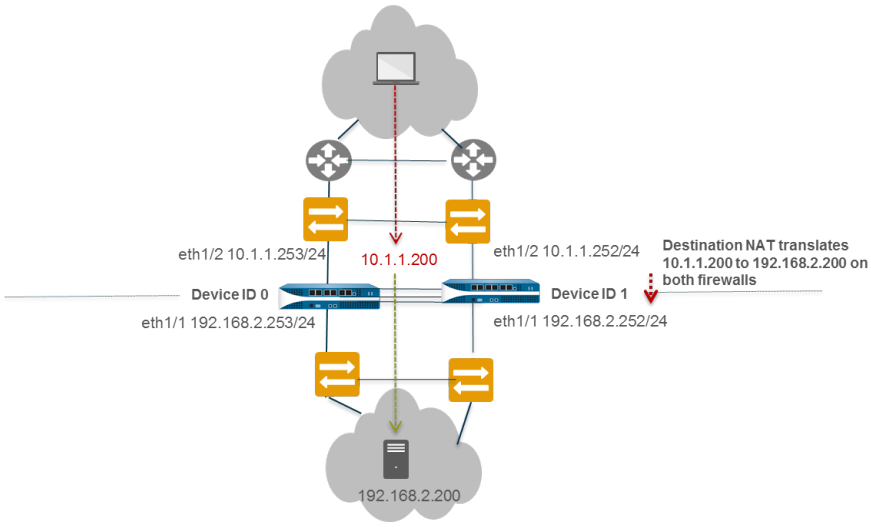

STEP 1 | On PA-3050-2 (Device ID 1), perform Step 1 through Step 3 of Configure Active/Active HA.

STEP 2 | Enable active/active HA.

1. In **Device** > **High Availability** > **General**, edit Setup.
2. Select **Enable HA**.
3. Enter a **Group ID**, which must be the same for both firewalls. The firewall uses the Group ID to calculate the virtual MAC address (range is 1 to 63).
4. (Optional) Enter a **Description**.
5. For **Mode**, select **Active Active**.
6. Select **Device ID** to be **1**.
7. Select **Enable Config Sync**. This setting is required to synchronize the two firewall configurations (enabled by default).
8. Enter the **Peer HA1 IP Address**, which is the IP address of the HA1 control link on the peer firewall.
9. (Optional) Enter a **Backup Peer HA1 IP Address**, which is the IP address of the backup control link on the peer firewall.
10. Click **OK**.

STEP 3 | Perform Step 6 through Step 15 in Configure Active/Active HA.

STEP 4 | Configure an HA virtual address.

1. Select **Device** > **High Availability** > **Active/Active Config** > **Virtual Address** and click **Add**.
2. Select **Interface** eth1/1.
3. Select **IPv4** and **Add** an **IPv4 Address** of 10.1.1.200.
4. For **Type**, select **ARP Load Sharing**, which configures the virtual IP address to be for both peers to use for ARP Load-Sharing.

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STEP 5 | Configure ARP Load-Sharing.

The device selection algorithm determines which HA firewall responds to the ARP requests to provide load sharing.

1. For **Device Selection Algorithm**, select **IP Modulo**. The firewall that will respond to ARP requests is based on the parity of the ARP requester's IP address.
2. Click **OK**.

STEP 6 | Enable jumbo frames on firewalls other than the PA-7000 Series.

STEP 7 | Define HA Failover Conditions.

STEP 8 | **Commit** the configuration.

STEP 9 | Configure the peer firewall, PA-3050-1 (Device ID 0), with the same settings, except in Step 2 select **Device ID 0**.

STEP 10 | Still on PA-3050-1 (Device ID 0), create the destination NAT rule so that the active-primary firewall responds to ARP requests.

1. Select **Policies** > **NAT** and click **Add**.
2. Enter a **Name** for the rule that, in this example, identifies it as a destination NAT rule for Layer 2 ARP.
3. For **NAT Type**, select **ipv4** (default).
4. On the **Original Packet**, for **Source Zone**, select **Any**.
5. For **Destination Zone**, select the Untrust zone you created for the external network.
6. Allow **Destination Interface**, **Service**, and **Source Address** to remain set to **Any**.
7. For **Destination Address**, specify 10.1.1.200.
8. For the **Translated Packet**, Source Address Translation remains **None**.
9. For **Destination Address Translation**, enter the private IP address of the destination server, in this example, 192.168.1.200.
10. On the **Active/Active HA Binding** tab, for **Active/Active HA Binding**, select **primary** to bind the NAT rule to the firewall in active-primary state.
11. Click **OK**.

STEP 11 | **Commit** the configuration.

Use Case: Configure Active/Active HA for ARP Load-Sharing with Destination NAT in Layer 3

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Pro

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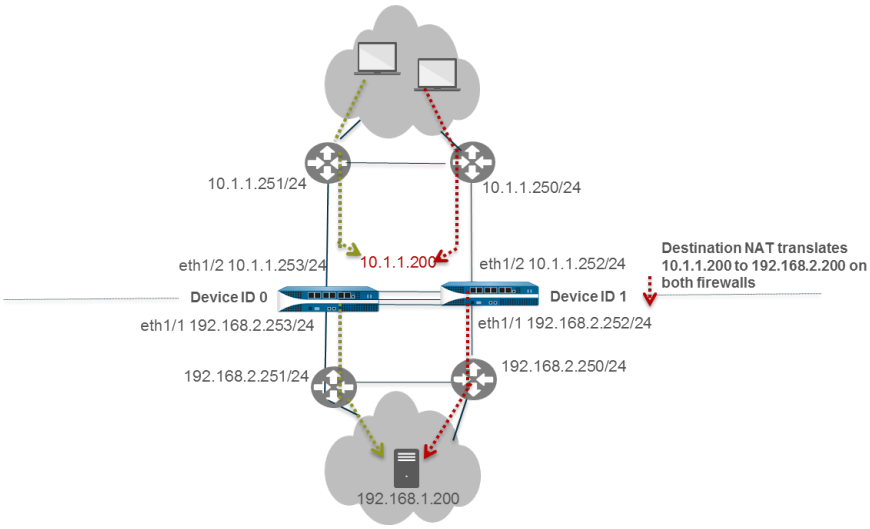
This Layer 3 interface example uses NAT in Active/Active HA and ARP Load-Sharing. PA-3050-1 has Device ID 0 and its HA peer, PA-3050-2, has Device ID 1.

In this use case, both of the HA firewalls must respond to an ARP request for the destination NAT address. Traffic can arrive at either firewall from either WAN router in the untrust zone. Destination NAT translates the public-facing, shared IP address to the private IP address of the server. The configuration requires one destination NAT rule bound to both Device IDs so that both firewalls can respond to ARP requests.



STEP 1 | On PA-3050-2 (Device ID 1), perform Step 1 through Step 3 of Configure Active/Active HA.

STEP 2 | Enable active/active HA.

1. Select **Device** > **High Availability** > **General** > **Setup** and edit.
2. Select **Enable HA**.
3. Enter a **Group ID**, which must be the same for both firewalls. The firewall uses the Group ID to calculate the virtual MAC address (range is 1-63).
4. (Optional) Enter a **Description**.
5. For **Mode**, select **Active Active**.
6. Select **Device ID** to be **1**.
7. Select **Enable Config Sync**. This setting is required to synchronize the two firewall configurations (enabled by default).
8. Enter the **Peer HA1 IP Address**, which is the IP address of the HA1 control link on the peer firewall.
9. (Optional) Enter a **Backup Peer HA1 IP Address**, which is the IP address of the backup control link on the peer firewall.
10. Click **OK**.

STEP 3 | Perform Step 6 through Step 15 in Configure Active/Active HA.

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STEP 4 | Configure an HA virtual address.

1. Select **Device** > **High Availability** > **Active/Active Config** > **Virtual Address** and click **Add**.
2. Select **Interface** eth1/2.
3. Select **IPv4** and **Add** an **IPv4 Address** of 10.1.1.200.
4. For **Type**, select **ARP Load Sharing**, which configures the virtual IP address to be for both peers to use for ARP Load-Sharing.

STEP 5 | Configure ARP Load-Sharing.

The device selection algorithm determines which HA firewall responds to the ARP requests to provide load sharing.

1. For **Device Selection Algorithm**, select one of the following
   - **IP Modulo**—The firewall that will respond to ARP requests is based on the parity of the ARP requester's IP address.
   - **IP Hash**—The firewall that will respond to ARP requests is based on a hash of the ARP requester's source IP address and destination IP address.
2. Click **OK**.

STEP 6 | Enable jumbo frames on firewalls other than PA-7000 Series firewalls.

STEP 7 | Define HA Failover Conditions.

STEP 8 | **Commit** the configuration.

STEP 9 | Configure the peer firewall, PA-3050-1 (Device ID 0), with the same settings, except set the **Device ID** to **0** instead of **1**.

STEP 10 | Still on PA-3050-1 (Device ID 0), create the destination NAT rule for both Device ID 0 and Device ID 1.

1. Select **Policies** > **NAT** and click **Add**.
2. Enter a **Name** for the rule that in this example identifies it as a destination NAT rule for Layer 3 ARP.
3. For **NAT Type**, select **ipv4** (default).
4. On the **Original Packet**, for **Source Zone**, select **Any**.
5. For **Destination Zone**, select the Untrust zone you created for the external network.
6. Allow **Destination Interface**, **Service**, and **Source Address** to remain set to **Any**.
7. For **Destination Address**, specify 10.1.1.200.
8. For the **Translated Packet**, Source Address Translation remains None.
9. For **Destination Address Translation**, enter the private IP address of the destination server, in this example 192.168.1.200.
10. On the **Active/Active HA Binding** tab, for **Active/Active HA Binding**, select **both** to bind the NAT rule to both Device ID 0 and Device ID 1.
11. Click **OK**.

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STEP 11 | **Commit** the configuration.

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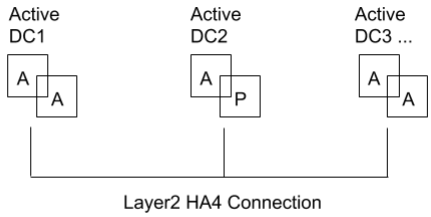
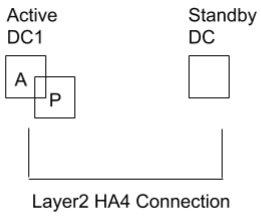
HA Clustering Overview

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

A number of Palo Alto Networks® firewall models now support session state synchronization among firewalls in a high availability (HA) cluster of up to 16 firewalls. The HA cluster peers synchronize sessions to protect against failure of the data center or a large security inspection point with horizontally scaled firewalls. In the case of a network outage or a firewall going down, the sessions fail over to a different firewall in the cluster. Such synchronization is especially helpful in the following use cases.

One use case is when HA peers are spread across multiple data centers so that there is no single point of failure within or between data centers. A second multi-data center use case is when one data center is active and the other is standby.





A third HA clustering use case is horizontal scaling, in which you add HA cluster members to a single data center to scale security and ensure session survivability.

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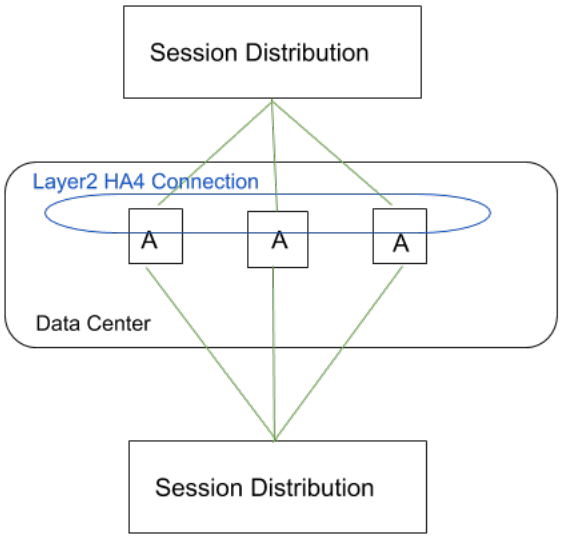

HA clusters support a Layer 3 or virtual wire deployment. HA peers in the cluster can be a combination of HA pairs and standalone cluster members. In an HA cluster, all members are considered active; there is no concept of passive firewalls except for HA pairs, which can keep their active/passive relationship after you add them to an HA cluster.

Understand that in the event of a link monitoring or path monitoring failure, an HA cluster does not support the "last device standing" behavior that occurs in a simple HA pair outside of a cluster. Instead, one of the devices in an HA pair within the HA cluster can go into a non-functional loop and suspended state, while the other device of the pair continues to loop on the path monitoring failures. After this non-suspended device reaches its Flap Max setting (which also applies to HA clustering), this device can also go into suspended state. Hence, the "last device standing" behavior does not occur.

Keep in mind that a device in a suspended state can't move to an HA functional state without your [user] intervention.

All cluster members share session state. When a new firewall joins an HA cluster, that triggers all firewalls in the cluster to synchronize all existing sessions. HA4 and HA4 backup connections are the dedicated cluster links that synchronize session state among all cluster members having the same cluster ID. The HA4 link between cluster members detects connectivity failures between cluster members. HA1 (control link), HA2 (data link), and HA3 (packet-forwarding link) are not supported between cluster members that are not HA pairs.

For a normal session that has not failed over, only the firewall that is the session owner creates a traffic log. For a session that failed over, the new session owner (the firewall that receives the failed over traffic) creates the traffic log.

The firewall models that support HA clustering and the maximum number of members supported per cluster are as follows:

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Firewall Model	Number of Members Supported Per Cluster
PA-3200 Series	6
PA-3400 Series	6
PA-5200 Series	16
PA-5410, PA-5420, PA-5430, PA-5440, PA-5445, and PA-5450	8
PA-7000 Series firewalls that have at least one of the following cards: PA-7000-100G-NPC, PA-7000-20GQXM-NPC, PA-7000-20GXM-NPC	PA-7080: 4 PA-7050: 6
VM-300	6
VM-500	6
VM-700	16

HA clustering isn't supported in public cloud deployments. Consider the HA Clustering Best Practices and Provisioning before you start to Configure HA Clustering.

How an HA Pair State Affects an HA Cluster State

First, let's understand how an HA pair state maps to an HA cluster state:

HA Pair State	HA Cluster State
unknown	cluster-unknown*
suspended	cluster-suspended*
non-functional	cluster-non-functional*
initial	cluster-initial*
passive	cluster-non-functional*
active	cluster-active. The cluster uses its own state machine; state isn't tied to the HA pair state.
active-primary	
active-secondary	
tentative	

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Now, we can look at the following factors that determine how an HA pair state affects an HA cluster state:

- The HA cluster states marked with an asterisk (*) in the preceding table will adopt the HA pair local device state if the device is in both an HA pair and an HA cluster. For example, if the HA pair state is suspended, the cluster state is cluster-suspended.
- For an HA pair state of active, active-primary, active-secondary, or tentative, the cluster state will follow its own state machine and can be in any cluster state. Other HA pair states will all result in a cluster-non-functional state unless noted otherwise in the preceding table.
- The HA local device pair state that maps to the cluster-active state can still enter a non-functional cluster state because all faults considered by cluster state are still relevant when an HA local device pair state is active.
  - This means that a member of an HA pair with active state can be in an HA cluster non-functional state, especially if the HA pair active state member is there due to "last device standing."
  - Soft faults (such as link monitoring failures, path monitoring failures, or slot mismatch errors) would result in a "last device standing" in an HA pair. However, if that HA pair is in an HA cluster, then the HA pair local device continues to honor the faults that are normally ignored when the HA pair peer device state is in one of the states that isn't functional.
  - The state machine in an HA pair device behaves differently when the pair is in an HA cluster versus when the pair isn't in an HA cluster. When the pair is in an HA cluster, the pair behaves more like a standalone cluster device. Therefore, both devices in an HA pair in a data center can behave like a standalone HA cluster device, allowing the entire data center to go into failure if soft faults trigger.

*In the Panorama® Managed Devices Summary, the **HA CLUSTER STATE** column displays **cluster-unknown** for firewalls running PAN-OS 10.0 or later that are configured as HA pairs but not as HA clusters. Before PAN-OS 10.0, this column appeared blank for the same configuration. This is expected behavior—because no cluster is configured, the cluster state is unknown. To monitor the status of your HA pair, use the **HA HA Pair Status** column. You can also hide the **HA CLUSTER STATE** column.*

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HA Clustering Best Practices and Provisioning

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

These are the provisioning requirements and best practices for HA clustering.

- Provisioning Requirements and Best Practices
  - HA cluster members must be the same firewall model and run the same PAN-OS® version.

    When upgrading, firewall members will continue to synchronize sessions with one member at a different version.
  - It is highly recommended and a best practice to use Panorama to provision HA cluster members to keep all configuration and policies synchronized among all cluster members.
  - HA cluster members must be licensed for the same components to ensure consistent policy enforcement and content inspection capabilities.
  - The licenses must expire at the same time to prevent mismatched licenses and loss of functionality.
  - All cluster members should be running with the same version of dynamic Content Updates for consistent security enforcement.
  - HA cluster members must share the same zone names in order for sessions to successfully fail over to another cluster member. For example, suppose sessions going to an ingress zone named **`internal`** are dropped because the link is down. For those sessions to fail over to an HA firewall peer in the cluster, that peer must also have a zone named **`internal`**.
  - Client-to-server and server-to-client flows must go back to the same firewall under normal (non-failure) conditions in order for security content scanning to occur. Asymmetric traffic won't be dropped, but it cannot be scanned for security purposes.
- Session Synchronization Best Practices
  - Dedicated HA communication interfaces should be used over dataplane interfaces. HSCI interfaces aren't used for HA4. This allows separation of HA pair and cluster session synchronization to ensure maximum bandwidth and reliability for session syncing.
  - HA4 should be adequately sized if you use dataplane interfaces. This ensures best effort session state synchronizing between cluster members.
  - Best practice is to have a dedicated cluster network for the HA4 communications link to ensure adequate bandwidth and non-congested, low-latency connections between cluster members.
  - Architect your networks and perform traffic engineering to avoid possible race conditions, in which a network steers traffic from the session owner to a cluster member before the session is successfully synced between the firewalls. Layer2 HA4 connections must have

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sufficient bandwidth and low latency to allow timely synchronization between HA members. The HA4 latency must be lower than the latency incurred when the peering devices switch traffic between cluster members.

- Architect your networks to minimize asymmetric flows. Session setup requires one cluster member to see the complete TCP three-way handshake.

- Health Check Best Practices
  - On HA pairs in a cluster, configure an Active/Passive pair with HA backup communication links for HA1, HA2, and HA4. Configure an Active/Active pair with HA backup communications links for HA1, HA2, HA3, and HA4.
  - Configure HA4 backup links on all cluster members.

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Configure HA Clustering

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

Learn about HA clustering and follow the HA Clustering Best Practices and Provisioning before you configure HA firewalls as members of a cluster.

STEP 1 | Establish an interface as an HA interface (to later assign as the HA4 link).

1. Select **Network** > **Interfaces** > **Ethernet** and select an interface; for example, ethernet1/1.
2. Select the **Interface Type** to be **HA**.
3. Click **OK**.
4. Repeat this step to configure another interface to use as the HA4 backup link.

STEP 2 | Enable HA clustering.

1. Select **Device** > **High Availability** > **General** and edit the Clustering Settings.
2. **Enable Cluster Participation**.
3. Enter the **Cluster ID**, a unique numeric ID for an HA cluster in which all members can share session state; range is 1 to 99.
4. Enter a short, helpful **Cluster Description**.
5. (Optional) Change **Cluster Synchronization Timeout (min)**, which is the maximum number of minutes that the local firewall waits before going to Active state when another cluster member (for example, in unknown state) is preventing the cluster from fully synchronizing; range is 0 to 30; default is 0.
6. (Optional) Change **Monitor Fail Hold Down Time (min)**, which is the number of minutes after which a down link is retested to see if it is back up; range is 1 to 60; default is 1.
7. Click **OK**.

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STEP 3 | Configure the HA4 link.

1. Select **HA Communications** and in the Clustering Links section, edit the HA4 section.
2. Select the interface you configured in the first step as an **HA** interface to be the **Port** for the HA4 link; for example, ethernet1/1.
3. Enter the **IPv4/IPv6 Address** of the local HA4 interface.
4. Enter the **Netmask**.
5. (Optional) Change the **HA4 Keep-alive Threshold (ms)** to specify the timeframe within which the firewall must receive keepalives from a cluster member to know that the cluster member is functional; range is 5,000 to 60,000; default is 10,000.
6. Click **OK**.

STEP 4 | Configure the HA4 Backup link.

1. Edit the HA4 Backup section.
2. Select the other interface you configured in the first step as an **HA** interface to be the **Port** for the HA4 backup link.
3. Enter the **IPv4/IPv6 Address** of the local HA4 backup interface.
4. Enter the **Netmask**.
5. Click **OK**.

STEP 5 | Specify all members of the HA cluster, including the local member and both HA peers in any HA pair.

1. Select **Cluster Config**.
2. (On a supported firewall) **Add** a peer member's **Device Serial Number**.
3. (On Panorama) **Add** and select a **Device** from the dropdown and enter a **Device Name**.
4. Enter the **HA4 IP Address** of the HA peer in the cluster.
5. Enter the **HA4 Backup IP Address** of the HA peer in the cluster.
6. Enable **Session Synchronization** with the peer you identified.
7. (Optional) Enter a helpful **Description**.
8. Click **OK**.
9. Select the device and **Enable** it.

STEP 6 | Define HA Failover Conditions with link and path monitoring.

STEP 7 | **Commit**.

STEP 8 | (Panorama only) Refresh the list of HA firewalls in the HA cluster.

1. Under Templates, select **Device** > **High Availability** > **Cluster Config**.
2. Click **Refresh** at the bottom of the screen.

STEP 9 | View HA cluster information in the UI.

1. Select **Dashboard**.
2. View the HA cluster fields. The top section displays cluster state and HA4 connections to provide cluster health at a glance. The HA4 and HA4 Backup indicators will be one of the following: Green indicates the link status of the cluster members is Up. Red indicates

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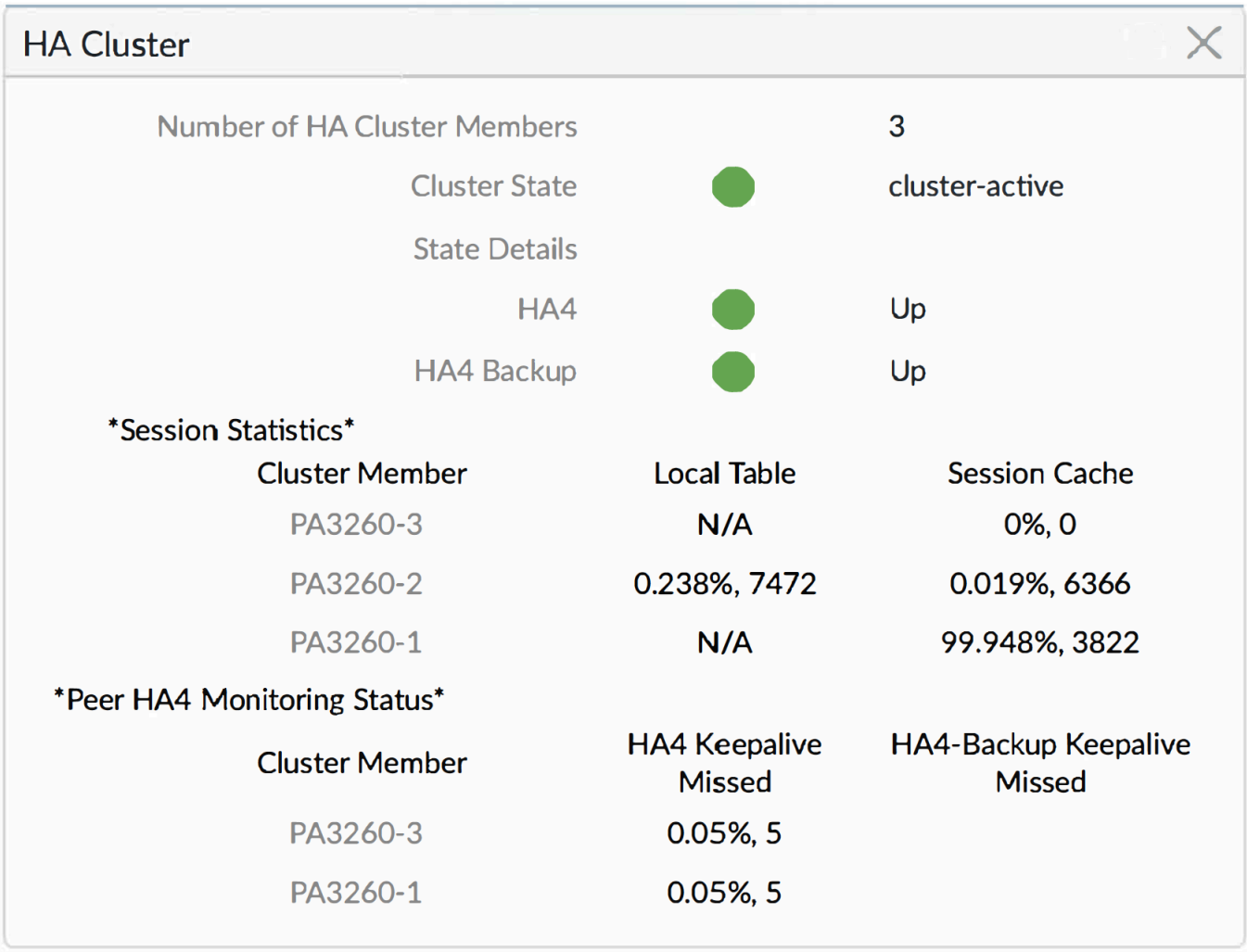
the link status of all the cluster members is Down. Yellow indicates the link status of some cluster members is Up while the status of other cluster members is Down. Grey indicates not configured. The center section displays the capacity of the local session table and session cache table so you can monitor how full the tables are and plan for firewall upgrades. The lower section displays communication errors on the HA4 and HA4 backup links, signifying possible problems with synchronizing information between members.



STEP 10 | Access the CLI to view HA cluster and HA4 link information and perform other HA clustering tasks.

You can view HA cluster flap statistics. The cluster flap count is reset when the HA device moves from suspended to functional and vice versa. The cluster flap count also resets when the non-functional hold time expires.

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Refresh HA1 SSH Keys and Configure Key Options

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Pro

All Palo Alto Networks firewalls come with Secure Shell (SSH) pre-configured, and the high availability (HA) firewalls can act as SSH server and SSH client simultaneously. When you configure active/passive or active/active HA, you can enable encryption for the HA1 (control link) connection between the HA firewalls. We recommend you secure the HA1 traffic between the HA peers with encryption, particularly if the firewalls aren't located in the same site. After you enable encryption on the HA1 control link, you can use the CLI to create an SSH service profile and secure the connection between the HA firewalls.

SSH service profiles enable you to change the default host key type, generate a new pair of public and private SSH host keys for the HA1 control link, and configure other SSH HA1 settings. You can apply the new host keys and configured settings to the firewalls without restarting the HA peers. The firewall will reestablish HA1 sessions with its peer to synchronize the configuration changes. It also generates system logs (subtype is ha) for reestablishing HA1 and HA1-backup sessions.

The following examples show how to configure various SSH settings for your HA1 after you enable encryption and access the CLI. (See Refresh SSH Keys and Configure Key Options for Management Interface Connection for SSH management server profile examples.)

You must enable encryption and it must be functioning properly on an HA pair before you can perform the following tasks.

If you are configuring the HA1 control link in FIPS-CC mode, you must set automatic rekeying parameters for session keys.

*To use the same SSH connection settings for each Dedicated Log Collector (M-series or Panorama virtual appliance in Log Collector mode) in a Collector Group, configure an SSH service profile from the Panorama management server, **Commit** the changes to Panorama, and then **Push** the configuration to the Log Collectors. You can use the* ***`set log-collector-group <name> general-setting management ssh`*** *commands.*

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- Create an SSH service profile to exercise greater control over SSH connections between your HA firewalls.

  This example creates an HA profile without configuring any settings.

  1. admin@PA-3250> **configure**
  2. admin@PA-3250# **set deviceconfig system ssh profiles ha-profiles *<name>***
  3. admin@PA-3250# **commit**
  4. admin@PA-3250# **exit**
  5. To verify that the new profile has been created and view the settings for any existing profiles:

     admin@PA-3250> **configure**

     admin@PA-3250# **show deviceconfig system ssh profiles**

- (Optional) Set the SSH server to use only the specified encryption ciphers for the HA1 sessions.

  By default, HA1 SSH allows all supported ciphers for encryption of CLI HA sessions. When you set one or more ciphers, the SSH server advertises only those ciphers while connecting, and

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if the SSH client (HA peer) tries to connect using a different cipher, the server terminates the connection.

1. admin@PA-3250> **configure**
2. admin@PA-3250# **set deviceconfig system ssh profiles ciphers ha-profiles <*name*> ciphers <*cipher*>**

   aes128-cbc—AES 128-bit cipher with Cipher Block Chaining

   aes128-ctr—AES 128-bit cipher with Counter Mode

   aes128-gcm—AES 128-bit cipher with GCM (Galois/Counter Mode)

   aes192-cbc—AES 192-bit cipher with Cipher Block Chaining

   aes192-ctr—AES 192-bit cipher with Counter Mode

   aes256-cbc—AES 256-bit cipher with Cipher Block Chaining

   aes256-ctr—AES 256-bit cipher with Counter Mode

   aes256-gcm—AES 256-bit cipher with GCM
3. admin@PA-3250# **commit**
4. admin@PA-3250# **exit**
5. (HA1 Backup is configured) admin@PA-3250> **request high-availability session-reestablish**
6. (No HA1 Backup is configured or HA1 Backup link is down) admin@PA-3250> **request high-availability session-reestablish force**

   *You can force the firewall to reestablish HA1 sessions if there is no HA1 backup, which causes a brief split-brain condition between the HA peers. (Using the **force** option when an HA1 backup is configured has no effect.)*
7. To verify the ciphers have been updated:

   admin@PA-3250> **configure**

   admin@PA-3250# **show deviceconfig system ssh profiles ha-profiles ciphers**

- (Optional) Set the default host key type.

  If you enable encryption on the HA1 control link, the firewall uses a default host key type of RSA 2048 unless you change it. The HA1 SSH connection uses only the `default host key type` to authenticate the HA peers (before an encrypted session is established between them). You can change the default host key type; the choices are ECDSA 256, 384, or 521, or RSA 2048, 3072, or 4096. Change the default host key type if you prefer a longer RSA key length or if you prefer ECDSA rather than RSA. This example sets the default host key type to

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an ECDSA key of 256 bits. It also re-establishes the HA1 connection using the new host key without restarting the HA peers.

1. admin@PA-3250> **configure**
2. admin@PA-3250# **set deviceconfig system ssh profiles ha-profiles <*name*> default-hostkey key-type ECDSA key-length 256**
3. admin@PA-3250# **commit**
4. admin@PA-3250# **exit**
5. admin@PA-3250> **request high-availability sync-to-remote ssh-key**

   An HA connection must already be established between the HA firewalls. If the firewalls have not yet established an HA connection, you must enable encryption on the control link connection, export the HA key to a network location and import the HA key on the peer. See Configure Active/Passive HA *or* Configure Active/Active HA.
6. (HA1 Backup is configured) admin@PA-3250> **request high-availability session-reestablish**
7. (No HA1 Backup is configured or HA1 Backup link is down) admin@PA-3250> **request high-availability session-reestablish force**

   You can force the firewall to reestablish HA1 sessions if there is no HA1 backup, which causes a brief split-brain condition between the two HA peers. (Using the ***force*** *option when an HA1 backup is configured has no effect.)*
8. To verify the host key has been updated:

   admin@PA-3250> **configure**

   admin@PA-3250# **show deviceconfig system ssh profiles ha-profiles <*name*> default-hostkey**

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- (Optional) Delete a cipher from the set of ciphers you selected for SSH over the HA1 control link.

  This example deletes the AES CBC cipher with 128-bit key.

  1. admin@PA-3250> **configure**
  2. admin@PA-3250# **delete deviceconfig system ssh profiles ha-profiles <*name*> ciphers aes128-cbc**
  3. admin@PA-3250# **commit**
  4. admin@PA-3250# **exit**
  5. (HA1 Backup is configured) admin@PA-3250> **request high-availability session-reestablish**
  6. (No HA1 Backup is configured or HA1 Backup link is down) admin@PA-3250> **request high-availability session-reestablish force**

     *You can force the firewall to reestablish HA1 sessions if there is no HA1 backup, which causes a brief split-brain condition between the two HA peers. (Using the **force** option when an HA1 backup is configured has no effect.*
  7. To verify the cipher has been deleted:

     admin@PA-3250> **configure**

     admin@PA-3250# **show deviceconfig system ssh profiles ha-profiles <*name*> ciphers**

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- (Optional) Set the session key exchange algorithms the HA1 SSH server will support.

  By default, the SSH server (HA firewall) advertises all the key exchange algorithms to the SSH client (HA peer firewall).

  If you are using an ECDSA default key type, the best practice is to use an ECDH key algorithm.

  1. admin@PA-3250> **configure**
  2. admin@PA-3250# **set deviceconfig system ssh profiles ha-profiles <*name*> kex <*value*>**

     diffie-hellman-group14-sha1—Diffie-Hellman group 14 with SHA1 hash

     ecdh-sha2-nistp256—Elliptic-Curve Diffie-Hellman over National Institute of Standards and Technology (NIST) P-256 with SHA2-256 hash

     ecdh-sha2-nistp384—Elliptic-Curve Diffie-Hellman over NIST P-384 with SHA2-384 hash

     ecdh-sha2-nistp521—Elliptic-Curve Diffie-Hellman over NIST P-521 with SHA2-521 hash
  3. admin@PA-3250# **commit**
  4. admin@PA-3250# **exit**
  5. (HA1 Backup is configured) admin@PA-3250> **request high-availability session-reestablish**
  6. (No HA1 Backup is configured or HA1 Backup link is down) admin@PA-3250> **request high-availability session-reestablish force**

     *You can force the firewall to reestablish HA1 sessions if there is no HA1 backup, which causes a brief split-brain condition between the two HA peers. (Using the **force** option when an HA1 backup is configured has no effect.*
  7. To verify the key exchange algorithms have been updated:

     admin@PA-3250> **configure**

     admin@PA-3250# **show deviceconfig system ssh profiles ha-profiles**

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- (Optional) Set the message authentication codes (MAC) the HA1 SSH server will support.

  By default, the server advertises all of the MAC algorithms to the client.

  1. admin@PA-3250> **configure**
  2. admin@PA-3250# **set deviceconfig system ssh profiles ha-profiles <*name*> mac <*value*>**

     hmac-sha1—MAC with SHA1 cryptographic hash

     hmac-sha2-256—MAC with SHA2-256 cryptographic hash

     hmac-sha2-512—MAC with SHA2-512 cryptographic hash
  3. admin@PA-3250# **commit**
  4. admin@PA-3250# **exit**
  5. (HA1 Backup is configured) admin@PA-3250> **request high-availability session-reestablish**
  6. (No HA1 Backup is configured or HA1 Backup link is down) admin@PA-3250> **request high-availability session-reestablish force**

     *You can force the firewall to reestablish HA1 sessions if there is no HA1 backup, which causes a brief split-brain condition between the two HA peers. (Using the **force** option has no effect when an HA1 backup is configured.*
  7. To verify the MAC algorithms have been updated:

     admin@PA-3250> **configure**

     admin@PA-3250# **show deviceconfig system ssh profiles ha-profiles**

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- (Optional) Regenerate ECDSA or RSA host keys for HA1 SSH to replace the existing keys, and re-establish HA1 sessions between HA peers using the new keys without restarting the HA peers.

  The HA peers use the host keys to authenticate each other. This example regenerates the ECDSA 256 default host key.

  Regenerating a host key does not change your default host key type. To regenerate the default host key you are using, you must specify your default host key type and length when you regenerate. Regenerating a host key that isn't your default host key type simply regenerates a key that you aren't using and therefore has no effect.

  1. admin@PA-3250> **configure**
  2. admin@PA-3250# **set deviceconfig system ssh regenerate-hostkeys ha key-type ECDSA key-length 256**
  3. admin@PA-3250# **commit**
  4. admin@PA-3250# **exit**
  5. admin@PA-3250> **request high-availability sync-to-remote ssh-key**

     An HA connection must already be established between the HA firewalls. If the firewalls have not yet established an HA connection, you must enable encryption on the control link connection, export the HA key to a network location, and import the HA key on the peer. See Configure Active/Passive HA *or* Configure Active/Active HA.
  6. (HA1 Backup is configured) admin@PA-3250> **request high-availability session-reestablish**
  7. (No HA1 Backup is configured or HA1 Backup link is down) admin@PA-3250> **request high-availability session-reestablish force**

     You can force the firewall to reestablish HA1 sessions if there is no HA1 backup, which causes a brief split-brain condition between the two HA peers. (Using the ***force*** *option when an HA1 backup is configured has no effect.)*

- (Optional) Set rekey parameters to establish when automatic rekeying of the session keys occurs for SSH over the HA1 control link.

  The session keys are used to encrypt the traffic between the HA peers. The parameters you can set are data volume (in megabytes), time interval (seconds), and packet count. After any one rekey parameter reaches its configured value, SSH initiates a key exchange.

  You can set a second or third parameter if you aren't sure the parameter you configured will reach its value as soon as you want rekeying to occur. The first parameter to reach its configured value will prompt a rekey, then the firewall will reset all rekey parameters.

  1. admin@PA-3250> **configure**
  2. admin@PA-3250# **set deviceconfig system ssh profiles ha-profiles <*name*> session-rekey data 32**

     Rekeying occurs after the volume of data (in megabytes) is transmitted following the previous rekey. The default is based on the cipher you use and ranges from 1GB to

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4GB; the range is 10MB to 4,000MB. Alternatively, you can enter **set deviceconfig system ssh profiles ha-profiles <*name*> session-rekey data default** command, which sets the data parameter to the default value of the individual cipher you are using.

3. admin@PA-3250# **set deviceconfig system ssh profiles ha-profiles <*name*> session-rekey interval 3600**

   Rekeying occurs after the specified time interval (in seconds) passes following the previous rekeying. By default, time-based rekeying is disabled (set to none). The range is 10 to 3,600.

4. admin@PA-3250# **set deviceconfig system ssh profiles ha-profiles <*name*> session-rekey packets 27**

   Rekeying occurs after the defined number of packets (2^n) are transmitted following the previous rekey. For example, 14 configures that a maximum of 2^{14} packets are transmitted before a rekey occurs. The default is 2^{28}. The range is 12 to 27 (2^{12} to 2^{27}). Alternatively, you can enter **set deviceconfig system ssh profiles ha-profiles <*name*> session-rekey packets default**, which sets the packets parameter to 2^{28}.

   Choose rekeying parameters based on your type of traffic and network speeds (in addition to FIPS-CC requirements if they apply to you). Don't set the parameters so low that they affect SSH performance.

5. admin@PA-3250# **commit**
6. admin@PA-3250# **exit**
7. (HA1 Backup is configured) admin@PA-3250> **request high-availability session-reestablish**
8. (No HA1 Backup is configured or HA1 Backup link is down) admin@PA-3250> **request high-availability session-reestablish force**

   *You can force the firewall to reestablish HA1 sessions if there is no HA1 backup, which causes a brief split-brain condition between the two HA peers. (Using the **force** option when an HA1 backup is configured has no effect.)*

9. To verify the changes:

   admin@PA-3250> **configure**

   admin@PA-3250# **show deviceconfig system ssh profiles ha-profiles <*name*> session-rekey**

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- Activate the profile by selecting the profile and restarting HA1 SSH service.
  1. admin@PA-3250> **configure**
  2. admin@PA-3250# **set deviceconfig system ssh ha ha-profile *<name>***
  3. admin@PA-3250# **commit**
  4. admin@PA-3250# **exit**
  5. admin@PA-3250> **set ssh service-restart ha**
  6. To verify the correct profile is in use:

     admin@PA-3250> **configure**

     admin@PA-3250# **show deviceconfig system ssh ha**

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Reset the ID Manager After Replacing a Firewall in an HA Cluster

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

When a replacement firewall joins an HA cluster as the passive peer, PAN-OS merges the replacement unit's factory ID manager database with the active peer's operational database. This merge can inject stale entries — application or policy IDs scoped to virtual systems that don't exist in the running configuration. If the replacement unit then becomes active while these stale entries are present, traffic can be misidentified and dropped.

Resetting the ID manager on the **active** peer clears the merged database on both peers. A subsequent **`commit force`** repopulates both peers from the running configuration, removing any stale entries. Run the reset on the active peer — running it on the passive peer doesn't resolve the issue because the next HA sync cycle re-injects stale entries from the active peer's intact database.

STEP 1 | On the active firewall, verify which ID manager types are out of sync with the passive peer:

```
username@hostname(active)> debug device-server dump idmgr high-
availability state
```

Any type listed as `has difference` indicates a sync inconsistency.

STEP 2 | On the active firewall, reset the ID manager for all types:

```
username@hostname(active)> debug device-server reset id-manager
 type all
```

Run this command only on the active firewall. Running it on the passive peer doesn't resolve the issue — the next HA sync cycle re-injects stale entries from the active peer.

STEP 3 | Perform a commit force on the active firewall:

```
username@hostname(active)> commit force
```

This resets the passive peer's ID manager first, then the active peer's, and repopulates both from the running configuration.

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STEP 4 | Verify that the ID manager is in sync on both peers:

```
username@hostname(active)> debug device-server dump idmgr high-
availability state
```

The expected output is:

```
id manager compared with HA peer; id manager has no difference
```

Run the same command on the passive peer to confirm.

STEP 5 | Verify that no stale application entries remain in unexpected virtual systems on either peer:

```
username@hostname(active)> debug device-server dump idmgr type
 vsys-application all
```

Confirm that no custom application entries appear in virtual systems where they are not configured.

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HA Firewall States

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

An HA firewall can be in one of the following states:

HA Firewall State	Occurs In	Description
Initial	A/P or A/A	Transient state of a firewall when it joins the HA pair. The firewall remains in this state after boot-up until it discovers a peer and negotiations begins. After a timeout, the firewall becomes active if HA negotiation has not started.
Active	A/P	State of the active firewall in an active/passive configuration.
Passive	A/P	State of the passive firewall in an active/passive configuration. The passive firewall is ready to become the active firewall with no disruption to the network. Although the passive firewall is not processing other traffic: • If passive link state auto is configured, the passive firewall is running routing protocols, monitoring link and path state, and the passive firewall will pre-negotiate LACP and LLDP if LACP and LLDP pre-negotiation are configured, respectively. • The passive firewall is synchronizing flow state, runtime objects, and configuration. • The passive firewall is monitoring the status of the active firewall using the hello protocol.
Active-Primary	A/A	In an active/active configuration, state of the firewall that connects to User-ID agents, runs DHCP server and DHCP relay, and matches NAT and PBF rules with the Device ID of the active-primary firewall. A firewall in this state can own sessions and set up sessions.
Active-Secondary	A/A	In an active/active configuration, state of the firewall that connects to User-ID agents, runs DHCP server, and matches NAT and PBF rules with the Device ID of the active-secondary firewall. A firewall in active-secondary state does not support

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HA Firewall State	Occurs In	Description
		DHCP relay. A firewall in this state can own sessions and set up sessions.
Tentative	A/A	State of a firewall (in an active/active configuration) caused by one of the following: • Failure of a firewall. • Failure of a monitored object (a link or path). • The firewall leaves suspended or non-functional state. A firewall in tentative state synchronizes sessions and configurations from the peer. • In a virtual wire deployment, when a firewall enters tentative state due to a path failure and receives a packet to forward, it sends the packet to the peer firewall over the HA3 link for processing. The peer firewall processes the packet and sends it back over the HA3 link to the firewall to be sent out the egress interface. This behavior preserves the forwarding path in a virtual wire deployment. • In a Layer 3 deployment, when a firewall in tentative state receives a packet, it sends that packet over the HA3 link for the peer firewall to own or set up the session. Depending on the network topology, this firewall either sends the packet out to the destination or sends it back to the peer in tentative state for forwarding. After the failed path or link clears or as a failed firewall transitions from tentative state to active-secondary state, the **Tentative Hold Time** is triggered and routing convergence occurs. The firewall attempts to build routing adjacencies and populate its route table before processing any packets. Without this timer, the recovering firewall would enter active-secondary state immediately and would silently discard packets because it would not have the necessary routes. When a firewall leaves suspended state, it goes into tentative state for the **Tentative Hold Time** after links are up and able to process incoming packets. **Tentative Hold Time range (sec)** can be disabled (which is 0 seconds) or in the range 10-600; default is 60.
Non-functional	A/P or A/A	Error state due to a dataplane failure or a configuration mismatch, such as only one firewall configured for packet forwarding, VR sync or QoS sync. In active/passive mode, all of the causes listed for Tentative state cause non-functional state.

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HA Firewall State	Occurs In	Description
Suspended	A/P or A/A	The device is disabled so won't pass data traffic and although HA communications still occur, the device doesn't participate in the HA election process. It can't move to an HA functional state without user intervention.

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Reference: HA Synchronization

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager) • NGFW (Managed by PAN-OS or Panorama)	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Pro

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If you have enabled configuration synchronization on both peers in an HA pair, most of the configuration settings you configure on one peer will automatically sync to the other peer upon commit. To avoid configuration conflicts, always make configuration changes on the active (active/passive) or active-primary (active/active) peer and wait for the changes to sync to the peer before making any additional configuration changes.

Only committed configurations synchronize between HA peers. Any configuration in the commit queue at the time of an HA sync won't be synchronized.

- What Doesn't Sync in Active/Passive HA?
- What Doesn't Sync in Active/Active HA?
- System Runtime Information Synchronized Between HA Peers
- CLI Commands for HA Synchronization
- Reasons That Synchronization Doesn't Happen

What Doesn't Sync in Active/Passive HA?

The following table identifies which configuration settings don't synchronize in active/passive HA. You must configure the settings on each firewall in the HA pair; the settings do not sync from one peer to another.

Configuration Item	What Doesn't Sync in Active/Passive?
Management Interface Settings	All management configuration settings must be configured individually on each firewall, including: • **Device** > **Setup** > **Management** > **General Settings**—Hostname, Domain, Login Banner, SSL/TLS Service Profile (and associated certificates), Time Zone, Locale, Date, Time, Latitude, Longitude. • **Device** > **Setup** > **Management** > **Management Interface Settings**—IP Type, IP Address, Netmask, Default Gateway, IPv6 Address/ Prefix Length, Default IPv6 Gateway, Speed, MTU, and Services (HTTP, HTTP OCSP, HTTPS, Telnet, SSH, Ping, SNMP, User-ID, User-ID Syslog Listener-SSL, User-ID Syslog Listener-UDP)

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Configuration Item	What Doesn't Sync in Active/Passive?
Multi-vsys Capability	You must activate the Virtual Systems license on each firewall in the pair to increase the number of virtual systems beyond the base number provided by default on PA-400 Series, PA-3200 Series, PA-3400 Series, PA-5200 Series, PA-5400 Series, and PA-7000 Series firewalls. You must also enable **Multi Virtual System Capability** on each firewall (**Device** > **Setup** > **Management** > **General Settings**).
SNMP	**Device** > **Setup** > **Operations** > **SNMP Setup**
Services	**Device** > **Setup** > **Services**
Global Service Routes	**Device** > **Setup** > **Services** > **Service Route Configuration**
Data Protection	**Device** > **Setup** > **Content-ID** > **Manage Data Protection**
Jumbo Frames	**Device** > **Setup** > **Session** > **Session Settings** > **Enable Jumbo Frame**
Packet Buffer Protection	**Device** > **Setup** > **Session** > **Session Settings** > **Packet Buffer Protection** **Network** > **Zones** > **Enable Packet Buffer Protection**
Forward Proxy Server Certificate Settings	**Device** > **Setup** > **Session** > **Decryption Settings** > **SSL Forward Proxy Settings**
Master Key Secured by HSM	**Device** > **Setup** > **HSM** > **Hardware Security Module Provider** > **Master Key Secured by HSM**
Log Export Settings	**Device** > **Scheduled Log Export**
Software Updates	With software updates, you can either download and install them separately on each firewall, or download them on one peer and sync the update to the other peer. You must install the update on each peer (**Device** > **Software**).
GlobalProtect Agent Package	With GlobalProtect app updates, you can either download and install them separately on each firewall, or download them to one peer and sync the update to the other peer. You must activate separately on each peer (**Device** > **GlobalProtect Client**).
Content Updates	With content updates, you can either download and install them separately on each firewall, or download them on one peer and sync the update to the other peer. You must install the update on each peer (**Device** > **Dynamic Updates**).

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Configuration Item	What Doesn't Sync in Active/Passive?
Licenses/ Subscriptions	**Device** > **Licenses**
Support Subscription	**Device** > **Support**
Master Key	The master key must be identical on each firewall in the HA pair, but you must manually enter it on each firewall (**Device** > **Master Key and Diagnostics**). Before changing the master key, you must disable config sync on both peers (**Device** > **High Availability** > **General** > **Setup** and clear the **Enable Config Sync** check box) and then re-enable it after you change the keys.
Reports, logs, and Dashboard Settings	Log data, reports, and Dashboard data and settings (column display, widgets) are not synced between peers. Report configuration settings, however, are synced. • *Panorama supports log forwarding for User-ID logs from active firewalls only. It does not support log forwarding for User-ID logs from passive firewalls.* • *Panorama supports log forwarding for IP-tag logs from both active and passive firewalls.* • *If you have an active/passive configuration and you search the User-ID logs using Panorama for a specific IP address, Panorama displays the mappings from the active firewall. You can use the* `show user ip-user-mapping ip <ip-address>` *CLI command (where <ip-address> is the IP address) to confirm the mapping exists on both firewalls.* • *If you have an active/passive configuration and you search the IP-Tag logs on the same configuration for an IP address, Panorama displays a record for the mappings from both firewalls with the Source Type for the active firewall as* `xml-api` *and the Source Type for the passive firewall as* `ha`.
HA settings	**Device** > **High Availability**
Decryption	After a failover, firewalls do not support HA sync for decrypted SSL sessions.
Rule Usage Data	Rule usage data, such as hit count, Created, and Modified Dates, are not synced between peers. You need to log in to the each firewall to

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Configuration Item	What Doesn't Sync in Active/Passive?
	view the policy rule hit count data for each firewall or use Panorama to view information on the HA firewall peers.
Certificates for Device Management and Syslog Communication over SSL only	**Device** > **Certificate Management** > **Certificates** Certificates used for device management or for syslog communication over SSL don't synchronize with an HA peer. *Although the certificates used for the management interface are not synchronized (and can be different), the name of the certificate entry should be the same for the active and passive devices.*
Certificates in a Certificate Profile	**Device** > **Certificate Management** > **Certificate Profile**
SSL/TLS Service Profile for Device Management only	**Device** > **Certificate Management** > **SSL/TLS Service Profile** SSL/TLS Service Profile for Device Management doesn't synchronize with an HA peer.
Device-ID and IoT Security	IP address-to-device mappings and policy rule recommendations don't synchronize with an HA peer.

What Doesn't Sync in Active/Active HA?

The following table identifies which configuration settings don't synchronize in active/active HA. You must configure the settings on each firewall in the HA pair; the settings do not sync from one peer to another.

Configuration Item	What Doesn't Sync in Active/Active?
Management Interface Settings	You must configure all management settings individually on each firewall, including: • **Device** > **Setup** > **Management** > **General Settings**—Hostname, Domain, Login Banner, SSL/TLS Service Profile (and associated certificates), Time Zone, Locale, Date, Time, Latitude, Longitude. • **Device** > **Setup** > **Management** > **Management Interface Settings**—IP Address, Netmask, Default Gateway, IPv6 Address/Prefix Length, Default IPv6 Gateway, Speed, MTU, and Services (HTTP, HTTP OCSP, HTTPS, Telnet, SSH, Ping, SNMP, User-ID, User-ID Syslog Listener-SSL, User-ID Syslog Listener-UDP)
Multi-vsys Capability	You must activate the Virtual Systems license on each firewall in the pair to increase the number of virtual systems beyond the base number provided by default on PA-400 Series, PA-3200 Series,

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Configuration Item	What Doesn't Sync in Active/Active?
	PA-3400 Series, PA-5200 Series, PA-5400 Series, and PA-7000 Series firewalls. You must also enable **Multi Virtual System Capability** on each firewall (**Device** > **Setup** > **Management** > **General Settings**).
SNMP	**Device** > **Setup** > **Operations** > **SNMP Setup**
Services	**Device** > **Setup** > **Services**
Global Service Routes	**Device** > **Setup** > **Services** > **Service Route Configuration**
Telemetry and Threat Intelligence Settings	**Device** > **Setup** > **Telemetry and Threat Intelligence**
Data Protection	**Device** > **Setup** > **Content-ID** > **Manage Data Protection**
Jumbo Frames	**Device** > **Setup** > **Session** > **Session Settings** > **Enable Jumbo Frame**
Packet Buffer Protection	**Device** > **Setup** > **Session** > **Session Settings** > **Packet Buffer Protection** **Network** > **Zones** > **Enable Packet Buffer Protection**
Forward Proxy Server Certificate Settings	**Device** > **Setup** > **Session** > **Decryption Settings** > **SSL Forward Proxy Settings**
HSM Configuration	**Device** > **Setup** > **HSM**
Log Export Settings	**Device** > **Scheduled Log Export**
Software Updates	With software updates, you can either download and install them separately on each firewall, or download them on one peer and sync the update to the other peer. You must install the update on each peer (**Device** > **Software**).
GlobalProtect Agent Package	With GlobalProtect app updates, you can either download and install them separately on each firewall, or download them to one peer and sync the update to the other peer. You must activate separately on each peer (**Device** > **GlobalProtect Client**).
Content Updates	With content updates, you can either download and install them separately on each firewall, or download them on one peer and sync the update to the other peer. You must install the update on each peer (**Device** > **Dynamic Updates**).
Licenses/ Subscriptions	**Device** > **Licenses**

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Configuration Item	What Doesn't Sync in Active/Active?
Support Subscription	**Device** > **Support**
Ethernet Interface IP Addresses	All Ethernet interface configuration settings sync except for the IP address (**Network** > **Interface** > **Ethernet**).
Loopback Interface IP Addresses	All Loopback interface configuration settings sync except for the IP address (**Network** > **Interface** > **Loopback**).
Tunnel Interface IP Addresses	All Tunnel interface configuration settings sync except for the IP address (**Network** > **Interface** > **Tunnel**).
LACP System Priority	Each peer must have a unique LACP System ID in an active/active deployment (**Network** > **Interface** > **Ethernet** > **Add Aggregate Group** > **System Priority**).
VLAN Interface IP Address	All VLAN interface configuration settings sync except for the IP address (**Network** > **Interface** > **VLAN**).
Virtual Routers	Virtual router configuration synchronizes only if you have enabled VR Sync (**Device** > **High Availability** > **Active/Active Config** > **Packet Forwarding**). Whether or not to do this depends on your network design, including whether you have asymmetric routing.
IPSec Tunnels	IPSec tunnel configuration synchronization is dependent on whether you have configured the Virtual Addresses to use Floating IP addresses (**Device** > **High Availability** > **Active/Active Config** > **Virtual Address**). If you have configured a floating IP address, these settings sync automatically. Otherwise, you must configure these settings independently on each peer.
GlobalProtect Portal Configuration	GlobalProtect portal configuration synchronization is dependent on whether you have configured the Virtual Addresses to use Floating IP addresses (**Network** > **GlobalProtect** > **Portals**). If you have configured a floating IP address, the GlobalProtect portal configuration settings sync automatically. Otherwise, you must configure the portal settings independently on each peer.
GlobalProtect Gateway Configuration	GlobalProtect gateway configuration synchronization is dependent on whether you have configured the Virtual Addresses to use Floating IP addresses (**Network** > **GlobalProtect** > **Gateways**). If you have configured a floating IP address, the GlobalProtect gateway configuration settings sync automatically. Otherwise, you must configure the gateway settings independently on each peer.
QoS	QoS configuration synchronizes only if you have enabled **QoS Sync** (**Device** > **High Availability** > **Active/Active Config** > **Packet Forwarding**). You might choose not to sync QoS setting if, for example,

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Configuration Item	What Doesn't Sync in Active/Active?
	you have different bandwidth on each link or different latency through your service providers.
LLDP	No LLDP state or individual firewall data is synchronized in an active/active configuration (**Network** > **Network Profiles** > **LLDP**).
BFD	No BFD configuration or BFD session data is synchronized in an active/active configuration (**Network** > **Network Profiles** > **BFD Profile**).
IKE Gateways	IKE gateway configuration synchronization is dependent on whether you have configured the Virtual Addresses to use floating IP addresses (**Network** > **IKE Gateways**). If you have configured a floating IP address, the IKE gateway configuration settings sync automatically. Otherwise, you must configure the IKE gateway settings independently on each peer.
Master Key	The master key must be identical on each firewall in the HA pair, but you must manually enter it on each firewall (**Device** > **Master Key and Diagnostics**). Before changing the master key, you must disable config sync on both peers (**Device** > **High Availability** > **General** > **Setup** and clear the **Enable Config Sync** check box) and then re-enable it after you change the keys.
Reports, logs, and Dashboard Settings	Log data, reports, and dashboard data and settings (column display, widgets) are not synced between peers. Report configuration settings, however, are synced.
HA settings	• **Device** > **High Availability** • (The exception is **Device** > **High Availability** > **Active/Active Configuration** > **Virtual Addresses**, which do sync.)
Decryption	After a failover, firewalls do not support HA sync for decrypted SSL sessions.
Rule Usage Data	Rule usage data, such as hit count, Created, and Modified Dates, are not synced between peers. You need to log in to the each firewall to view the policy rule hit count data for each firewall or use Panorama to view information on the HA firewall peers.
Certificates for Device Management and Syslog Communication over SSL only	**Device** > **Certificate Management** > **Certificates** Certificates used for device management or for syslog communication over SSL don't synchronize with an HA peer.

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Configuration Item	What Doesn't Sync in Active/Active?
Certificates in a Certificate Profile	**Device** > **Certificate Management** > **Certificate Profile**
SSL/TLS Service Profile for Device Management only	**Device** > **Certificate Management** > **SSL/TLS Service Profile** SSL/TLS Service Profile for Device Management doesn't synchronize with an HA peer.
Device-ID and IoT Security	IP address-to-device mappings and policy rule recommendations don't synchronize with an HA peer.

System Runtime Information Synchronized Between HA Peers

The following table summarizes the system runtime information that is synchronized between HA peers.

Runtime Information	Information Synced?		HA Link	Details
	A/P	A/A		
Management Plane				
User to Group Mappings	Yes	Yes	HA1	
User Mappings across Virtual Systems	Yes	Yes	HA1	
User to IP Address Mappings	Yes	Yes	HA1	In an A/A configuration, only the Active-Primary peer connects to User-ID Servers or Agents, and not the Active-Secondary peer. If the Active-Primary peer is Suspended or offline, the Active-Secondary peer connects to the User-ID Servers or Agents.
DHCP Lease (as server)	Yes	Yes	HA1	If the PAN-OS versions on the HA peers don't match, the DHCP Lease (as server) config information won't sync.
DNS Cache	No	No	N/A	

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Runtime Information	Information Synced?		HA Link	Details
	A/P	A/A		
FQDN Refresh	No	No	N/A	
IKE SAs [Security Associations] (phase 1)	No	No	N/A	
Forward Information Base (FIB)	Yes	No	HA1	
Multicast FIB (MFIB)	Yes	No	HA1	
PAN-DB URL Cache	Yes	No	HA1	This is synchronized upon database backup to disk (every eight hours, when URL database version updates), or when the firewall reboots.
Content (manual sync)	Yes	Yes	HA1	
PPPoE, PPPoE Lease	Yes	Yes	HA1	
DHCP Client Settings and Lease	Yes	Yes	HA1	If the PAN-OS versions on the HA peers don't match, the DHCP Client Settings and Lease config information won't sync.
SSL VPN Logged in User List	Yes	Yes	HA1	
Dataplane				
Session Table	Yes	Yes	HA2	• Active/passive peers do not sync ICMP or host session information. • Active/active peers do not sync host session, multicast

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Runtime Information	Information Synced?		HA Link	Details
	A/P	A/A		
				session, or BFD session information. *A host session is a session terminated on one of the firewall interfaces, such as an ICMP session pinging one of the firewall interfaces or a GP tunnel.*
ARP Table	Yes	No	HA2	
Multicast Session Table	Yes	No	HA2	
Neighbor Discovery (ND) Table	Yes	No	HA2	
MAC Table	Yes	No	HA2	
IPSec SAs [Security Associations] (phase 2)	Yes	Yes	HA2	
IPSec Sequence Number (anti-replay)	Yes	Yes	HA2	
DoS Block List Entries	No	No	N/A	
Virtual MAC	Yes	Yes	HA2	
SCTP Associations	Yes	No	HA2	

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CLI Commands for HA Synchronization

You can use the following CLI operational command and options to synchronize HA peers:

```
username@hostname>request high-availability sync-to-remote
>candidate-config     Sync candidate configuration to peer
>clock                Sync the local time and date to the peer
>id-manager           id-manager
>running-config       Sync running configuration to peer
>ssh-key              Sync ha ssh key to peer
```

After you replace a firewall in an HA cluster, the replacement unit's factory ID manager database can merge stale entries into the active peer. Use Reset the ID Manager After Replacing a Firewall in an HA Cluster to clear stale entries before making the replacement unit active.

A configuration pushed from Panorama isn't synchronized between firewalls. If you use Panorama to manage firewalls, you could decide, for example, to use the no form of the following CLI configuration command to disable configuration synchronization on the firewalls:

```
username@hostname#set deviceconfig high-availability group
 configuration-synchronization enabled
no    no
yes   yes
```

Use the no form of the following CLI configuration command to disable state (session) synchronization on the firewalls:

```
username@hostname#set deviceconfig high-availability group state-
synchronization enabled
no    no
yes   yes
```

Reasons That Synchronization Doesn't Happen

Sessions won't synchronize for the following reasons:

- If you disable session (state) synchronization.
- If the HA2 link or connection is down.

The HA configurations won't synchronize for the following reasons:

- If you disabled configuration synchronization on either HA peer.
- If the PAN-OS versions are incompatible on HA peers.
- If configurations on HA peers are not already synchronized.
- If Multi Virtual System Capability is enabled on one HA peer and not the other.
- If GTP is enabled on one HA peer and not the other.
- If SCTP is enabled on one HA peer and not the other.
- If VPN is enabled on one HA peer and not the other.
- If the same features aren't enabled on both HA peers.

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- If the dataplane and slots aren't ready on an HA peer.
- If the URL databases are incompatible on the HA peers.
- If the licenses aren't the same on the HA peers.
- Additionally, a plugin mismatch might (not always) prevent configurations from synchronizing.

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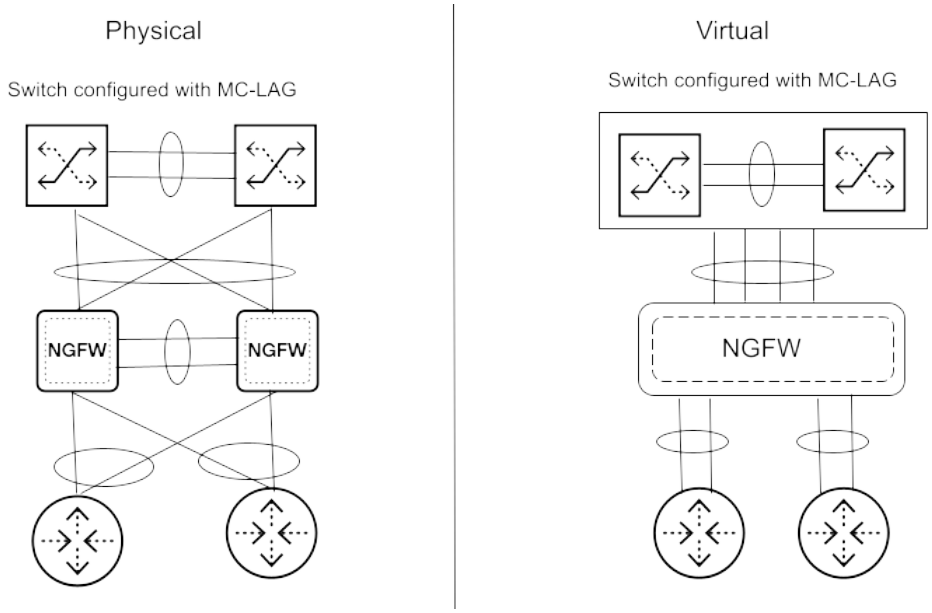
NGFW Clustering

Where Can I Use This?	What Do I Need?
• NGFW	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro One of the following for Panorama managed NGFWs: ❑ Panorama PAN-OS 12.1.2 (Panorama Clustering Plugin is bundled with Panorama) ❑ Panorama PAN-OS 11.1.5 and Panorama Clustering Plugin 2.0.0 ❑ Panorama PAN-OS 11.1.3 and Panorama Clustering Plugin 2.0.0

Data centers need very high levels of network bandwidth and reliability. Beginning with PAN-OS 11.1.3 and later 11.1 releases and beginning with PAN-OS 12.1.2 and later 12.1 releases, PA-7500 Series firewalls support an NGFW cluster of two firewalls that provide redundancy in the event of a link failure, card failure, or chassis failure.Beginning with PAN-OS 12.1.2 and later 12.1 releases, PA-5540, PA-5550, PA-5560, PA-5570, and PA-5580 firewalls require an NGFW cluster of two firewalls.

The two firewalls in the NGFW cluster function in a new mode of operation to provide high availability. The NGFW cluster blends the legacy HA active/active and active/passive solutions into a single HA solution, reducing the complexity of multiple HA connections (HA1, HA2, and HA3) to a single High Speed Chassis Interconnect (HSCI) connection. The firewalls maintain a dual active data plane with a single active control plane. Neighboring devices see the NGFW cluster as a single Layer 2 or Layer 3 device. The NGFW cluster solution reduces failover time (compared to legacy HA), increases resiliency, and supports a multichassis link aggregation group (MC-LAG). The graphic illustrates a physical topology compared to a virtual topology.



The firewalls in a cluster are as easy to configure as an HA active/passive pair was, while they provide the benefits of an active/active solution with extremely fast failover time (less than one second). Configuration through Panorama contributes to the ease of implementation. The pair of firewalls in the NGFW cluster increase port availability, require fewer IP addresses (there are no floating IP addresses), and they rely on open standards. NGFW clustering easily integrates with Layer 3 and virtual wire devices, including those running in a Cisco VPC, Arista MLAG, and Juniper QFX.

The goal of the two firewalls in the NGFW cluster is redundancy; the supported capacity of the pair is one node, not two nodes. The session capacity and all control plane functions remain the same as a single standalone device. The firewall hardware references provide hardware information for the devices that support NGFW clustering.

Learn about the elements of an NGFW cluster.

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NGFW Clusters

Where Can I Use This?	What Do I Need?
• NGFW	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro One of the following: ❑ Panorama PAN-OS 12.1.2 (Panorama Clustering Plugin is bundled with Panorama) ❑ Panorama PAN-OS 11.1.5 and Panorama Clustering Plugin 2.0.0 ❑ Panorama PAN-OS 11.1.3 and Panorama Clustering Plugin 2.0.0

Learn about NGFW clustering before creating a cluster:

- Benefits and Structure of an NGFW Cluster
- Inter-Firewall Links (IFLs)
- MACsec (PAN-OS 11.1.5 and later releases)
- MC-LAGs, Orphan Ports, and Orphan LAGs
- Node States Determine the Cluster State
- System Faults
- Role of the Leader Node
- Multiple Virtual Systems in an NGFW Cluster (PAN-OS 11.1.7 and later releases)
- Layer 7 Support and Graceful Failover

Benefits and Structure of an NGFW Cluster

Before describing an NGFW cluster, let's review the two legacy HA modes:

- Active/passive (A/P): One firewall actively manages traffic while the other is synchronized to the first firewall (in configuration and states) and is ready to transition to the active state if a failure occurs.
- Active/active (A/A): Both firewalls in the pair are active, process traffic, and work synchronously to handle session setup and session ownership. Both firewalls individually maintain session tables and synchronize to each other. The two firewalls support two separate routing domains.

For the firewalls in an NGFW cluster, the control planes (management planes) of the two firewalls are in active/passive mode where the passive firewall is fully synchronized to the active firewall in configuration and states. The data planes of the two firewalls are in active/active mode where the firewall with the active management plane incorporates its own interfaces with all of the interfaces of the passive chassis. The PA-7500 firewall in particular has a maximum of seven data

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processing cards (DPC); each DPC has six data planes. All of the interfaces of both chassis are representable and controllable on a single, centralized chassis, which is the leader or leader node. One node is elected the leader; the other firewall is a nonleader node. In an NGFW cluster, the firewalls are logically merged into one logical firewall from the control plane and management perspective. They have a single routing domain. The NGFW cluster replaces legacy HA pairs and legacy HA clustering, which are not available on these firewalls (whether in an NGFW cluster or not).

Use Panorama to configure the firewalls for an NGFW cluster. The first node you assign to the cluster automatically becomes Node 1. After you assign a node to a cluster, you can't configure the node locally, but must use Panorama. Place all the firewalls in a cluster in the same template stack.

NGFW cluster nodes must have the Advanced Routing Engine enabled; they don't support the legacy routing engine.

Unlike interfaces outside of an NGFW cluster, interfaces on firewalls in a cluster include the Node ID at the beginning of the interface name. For example, node1:ethernet2/1 or node2:ethernet1/19.

NGFW clustering is independent of legacy HA pairs and legacy HA clusters. On PA-5500 Series firewalls, both NGFW clustering and Active/Passive HA are supported in Strata Cloud Manager. However, the firewalls in the NGFW cluster use the Group ID that the legacy HA uses. The Group ID helps differentiate MAC addresses when two HA pairs (or an HA pair and an NGFW cluster) in the same Layer 2 network share MAC addresses. The Group ID is in the same location within the virtual MAC address for an NGFW firewall node as it is for a legacy HA firewall node.

Prior to PAN-OS 11.1.7, firewalls in an NGFW cluster didn't support multiple virtual systems (multi-vsys). Even if you had a multi-vsys license installed on the PA-7500 Series firewalls in an NGFW cluster, those firewalls ignored the license. (The multi-vsys license was usable if the firewall became standalone.) For support beginning with PAN-OS 11.1.7, see Multiple Virtual Systems in an NGFW Cluster.

Inter-Firewall Links (IFLs)

Firewalls in an NGFW cluster have a chassis interconnection through their HSCI interfaces, which function at Layer 2. The firewalls are back-to-back; HSCI-A ports on each firewall connect to each other, and HSCI-B ports on each firewall connect to each other. (The firewalls are directly connected to each other; there can't be a switch or other device between them.) The connections between the HSCI interfaces are high-bandwidth, inter-firewall links (IFLs) that handle asymmetric traffic, along with cluster synchronization at a control and dataplane level. The two IFLs function in active/backup mode. HSCI-A is the default active link; the active and backup roles are not configurable. If HSCI-A goes down, HSCI-B becomes the active link.

MACsec

The IFL connection utilizes the HSCI ports on the NGFW exclusively. You can configure the HSCI-A and HSCI-B for redundancy using an active/backup model.

Beginning with PAN-OS 11.1.5, NGFW clustering allows you to configure Media Access Control Security (MACsec) on the HSCI interface. MACsec is a standard (802.1AE) that operates at Layer 2. MACsec provides data confidentiality and integrity between endpoints by adding a security tag and an integrity check value to each Ethernet frame. It provides confidentiality through encryption of the data so that only endpoints with the correct encryption key can access the

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data. MACsec provides integrity through a cryptographic mechanism, ensuring that no one has tampered with data in transit. MACsec also provides authentication by ensuring that only known endpoints can communicate on the Ethernet segment.

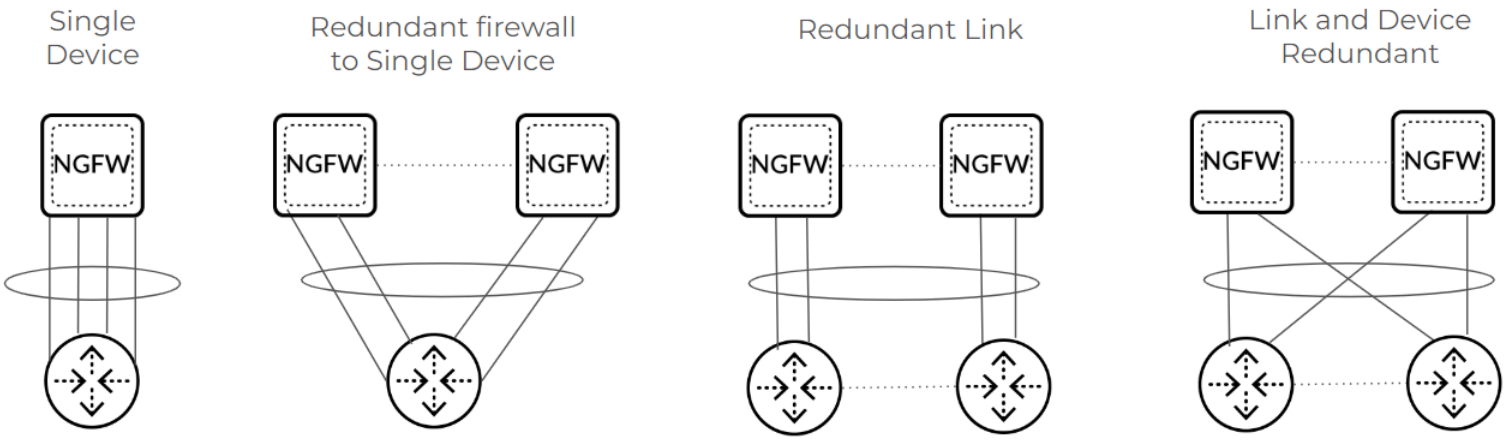
You can configure MACsec for the HSCI-A and HSCI-B ports (the active and backup ports) to protect the Layer 2 connections between the cluster peers. MACsec is disabled by default; it requires configuration to enable it. (Once enabled, you can remove the configuration items to disable MACsec.) MACsec runs over each HSCI port. The firewalls are connected back-to-back, therefore MACsec can establish a tunnel between the two firewalls. (The firewalls are directly connected to each other; there can't be a switch or other device between them.) MACsec has an associated pre-shared key (PSK) that must match on the two HSCI-A ports, and a PSK that must match on the two HSCI-B ports. Each session confirms that the PSKs match.

On the Communications tab, configuring MACsec involves a field and two profiles for each node: **Key Server Priority** (a ranged integer), **Crypto Profile** (a drop-down for profiles), and **Pre-Shared Key Profile** (a drop-down for profiles).

MC-LAGs, Orphan Ports, and Orphan LAGs

The NGFW cluster supports an MC-LAG, which is a type of LAG. Recall that a LAG (also known as an aggregated Ethernet (AE) link or aggregate interface group) is a group of links that appear as one link to provide link redundancy. Links in a LAG that connect to endpoints on multiple chassis are configured as part of an MC-LAG. The multiple chassis appear to be a single firewall, providing node redundancy for Layer 3 and virtual wire interfaces only. (Layer 2 does not support MC-LAGs.)

An MC-LAG is an AE interface group that has members spread across both firewalls (also referred to as nodes or chassis). The illustration below represents different MC-LAG scenarios. MC-LAGs are controlled by a single control plane and seen as a single system with a backup management plane. MC-LAG supports redundancy in the event of link failure, card failure, or chassis failure. Each MC-LAG supports a maximum of eight members. A pair of firewalls in an NGFW cluster support a maximum of 64 AE interface groups; the AE interface groups support Layer 3 within the cluster. (You can have an AE interface group using Layer 2 on a single device, but not an AE interface group using Layer 2 on an MC-LAG in the cluster.)



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Even a single device scenario can use an MC-LAG, which protects the firewall against failure, but does not protect against failure of the connected third-party device.

An orphan port is a single Layer 2, Layer 3, or virtual wire link used for connection from one node. Instead of using a floating IP address, an orphan port has its own IP address. The first packet of a session determines the dataplane that owns that session flow. Return data take the reverse path over the same hops back to the source.

An orphan LAG is an AE interface group that has all members originating or terminating on a single firewall (similar to a standalone AE interface group). The single device in the graphic illustrates an example of an orphan LAG. Firewalls supported orphan ports and orphan LAGs prior to the introduction of NGFW clustering.

Aggregate Ethernet local bias is a behavior where traffic forwarding prefers a local member over remote ports. Because an MC-LAG has members on both nodes, local bias enforces traffic egress from the local node, instead of forwarding traffic over an HSCI link to the remote node. (The typical behavior of a LAG is to hash to forward traffic to any members.)

Node States Determine the Cluster State

The combined node states of the nodes in an NGFW cluster determine the cluster state. First, let's consider the state of a cluster node, which can be one of these states:

- UNKNOWN—Clustering isn't enabled. The node remains in this state until a cluster configuration push from Panorama or a commit enables clustering.
- INIT—Node transitions from UNKNOWN to INIT state after clustering is enabled. Node remains in INIT state until cluster initialization of node is complete. The node transitions to ONLINE state if INIT criteria are met. If INIT criteria fail to be met, the node transitions to ONLINE state after a timeout.
- ONLINE—Node is passing traffic and working as expected.
- DEGRADED—Node transitions to DEGRADED state when a soft fault occurs. DEGRADED state allows Layer 7 continuity for sessions that the DEGRADED state device owns. Traffic links are down in DEGRADED state. The node can transition from DEGRADED to INIT state if all the faults are resolved.
- FAILED—Node transitions to FAILED state when a hard fault occurs. FAILED state has traffic ports down and does not allow Layer 7 continuity. The node can transition from FAILED to INIT state if all the faults are resolved.
- SUSPENDED—Triggered by administrator. Another cause of SUSPENDED state is if a node state flaps to DEGRADED or FAILED state repeatedly; the node is SUSPENDED after six flaps. An administrator can unsuspend the node. SUSPENDED state has traffic ports down and does not allow Layer 7 continuity.

Because the collective states of the nodes in an NGFW cluster determine the cluster state, the cluster state will be:

- OK—If all nodes are in ONLINE state.
- IMPACTED—If at least one node is in ONLINE state and another node isn't in ONLINE state.
- ERROR—If there isn't a single node in ONLINE state.

The System Monitoring portion of the NGFW cluster configuration allows you to specify the minimum number of network cards and data processing cards that must be functional. If the node

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drops below that minimum, the node state transitions to DEGRADED or FAILED (whichever you configured).

System Faults

Soft faults and hard faults affect whether the node state is DEGRADED or FAILED. Soft faults result in a node state of DEGRADED; hard faults result in a node state of FAILED. Causes of a soft fault are:

- ID Map synchronization has failed.
- IPSec VPN security association (SA) synchronization has failed.
- The system reported an out-of-memory (OOM) fault.
- Chassis does not have minimum capacity and degraded state is configured.
- The cluster node is in degraded state waiting to be suspended.

Causes of a hard fault are:

- Cluster infrastructure service has failed.
- The system reported a disk fault.
- The chassis has no DPC slots up.
- FIB synchronization has failed.
- The chassis does not have minimum capacity and failed state is configured.
- Cluster node configuration is incompatible with other nodes.
- Cluster messaging service has failed.
- Cluster node is avoiding a split brain.
- Cluster node is recovering from a split brain.

Role of the Leader Node

As mentioned, one node of the NGFW cluster is elected the leader node, and the other node is a nonleader node. The leader and nonleader nodes synchronize the following system runtime information:

- Management Plane:
  - User to Group Mappings
  - User to IP Address Mappings
  - DHCP Lease (as server)
  - Forwarding Information Base (FIB)
  - PAN-DB URL Cache
  - Content (manual sync)
  - PPPoE and PPPoE Lease
  - DHCP Client Settings and Lease
  - SSL VPN Logged in User List

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- Dataplane:
  - ARP Table
  - Neighbor Discovery (ND) Table
  - MAC Table
  - IPSec SAs [security associations] (phase 2)
  - IPSec Sequence Number (anti-replay)
  - Virtual MAC

Upon a leader node failover, the following protocols and functions are renegotiated:

- BGP
- OSPF
- OSPFv3
- RIP
- PIM
- BFD
- DHCP client
- PPPoE client
- Static route path monitor

Multiple Virtual Systems in an NGFW Cluster

To efficiently use firewalls in a cluster, customers need the cluster to support multiple Virtual Systems, which are an important capability of an NGFW. Beginning with PAN-OS 11.1.7, an NGFW cluster supports multiple virtual systems (multi-VSYS). An NGFW node (and therefore the cluster) supports a maximum of 25 virtual systems. Configure the firewalls in the cluster with the same multi-VSYS support (either enabled or disabled). The cluster does not support inter-VSYS traffic or shared gateway.

Layer 7 Support and Graceful Failover

In general, NGFW clustering aims to provide parity with HA regarding Layer 7 support. The following table lists Layer 7 features and whether a standalone PA-7500 Series firewall supports them, whether NGFW cluster nodes support them, and whether they support graceful failover.

Layer 7 Feature	Standalone PA-7500 Series Firewall	NGFW Cluster Node	Graceful Failover
Proxy			
Decryption (Forward Proxy/Inbound Inspection)	Yes	Yes	No
Hardware Security Module (HSM)	Yes	Yes	N/A (HSM must be configured per node)

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Layer 7 Feature	Standalone PA-7500 Series Firewall	NGFW Cluster Node	Graceful Failover
Certificate Revocation (CRL/ OCSP)	Yes	Yes	Service Route
Decryption Port Mirror	Yes	Yes	N/A
Network Packet Broker	Yes	Yes, but no redundancy support	No
SSH Decryption	Yes	Yes	No
GlobalProtect™— IPSec Tunnel	Yes	Yes	No
GlobalProtect— SSLVPN Tunnel	Yes	Yes	No
Clientless SSLVPN	Yes	No	No
LSVPN (Satellite)	Yes	Yes	Yes
IDmgr	Yes	Yes	Yes
Content Threat Detection (CTD)			
Application-Level Gateway (ALG)	Yes	Yes	No
dnsproxy	Yes	Yes	No
varrcvr	Yes	Yes	No
threat detection	Yes	Yes	No
Advance Data Loss Prevention (DLP)	Yes	Yes	No
URL Filtering	Yes	Yes	No
WIF features	Yes	Yes	No
App-ID Cloud Engine (ACE)	Yes	Yes	No
User-ID and Configuration			

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Layer 7 Feature	Standalone PA-7500 Series Firewall	NGFW Cluster Node	Graceful Failover
User Identification	Yes	Yes	Yes
Device Identification	Yes	Yes	Yes
CUID	Yes	Yes	Yes
Device Quarantine	Yes	Yes	Yes
Dynamic Address Group (DAG)	Yes	Yes	Yes
DUG	Yes	Yes	Yes

ID Map Sync—Management planes of NGFW cluster nodes individually generate IDs for various types of objects (such as IP address objects and security profile objects) in a firewall configuration. These are known as Management Global IDs. Each node maps its own set of IDs with the set of IDs of the peer cluster node. Some IDs are used in session flow data. Upon a chassis failover, Layer 4 sessions are marked as orphaned and sent to the peer chassis for continued processing. During this process, the IDs in the session flow data are updated to match the set of IDs that the peer's management plane provided. NGFW cluster nodes must complete their ID map synchronization before leaving INIT state.

Additionally, Panorama generates a subset of IDs (zone, logical interface, and virtual router IDs) and pushes them to each cluster node. These are Cluster Global IDs.

If you need to migrate from a non-PA-7500 Series firewall, proceed to Migrate to NGFW Clustering.

If you're using a PA-7500 Series firewall, proceed to Configure an NGFW Cluster and then view the NGFW Cluster Summary and Monitoring information.

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Migrate to NGFW Clustering

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Panorama PAN-OS 11.1.5 and Panorama Clustering Plugin 2.0.0

In preparation for configuring NGFW clustering on a PA-7500 Series firewall, this task describes how to migrate from a non-PA-7500 Series firewall with an existing Panorama non-clustering template (that has an Ethernet interface reference) to PA-7500 Series firewall with a Panorama clustering template (that has a cluster Ethernet interface reference). The migration process also converts a device group that references an Ethernet interface to a device group that references a cluster Ethernet interface.

This migration process is supported in PAN-OS 11.1.5 and later releases.

The system does not support the migration of multi-vsys regular templates to multi-vsys cluster templates.

Perform the following prerequisites before you begin migration:

- In order to migrate a template to a clustering template, the local firewall configuration must be available as a template and device group in Panorama. If that isn't the case, migrate all firewall local configurations to Panorama by following the procedure in Migrate a Firewall to Panorama Management.
- NGFW clustering doesn't support multiple virtual systems. If your original configuration has **Multi Virtual System Capability** enabled, perform the following steps:
  1. Convert the multi-vsys configuration to a single virtual system and remove the other virtual systems.

     *If you have a local configuration in **Device** > **Server Profile** > **DNS**, remove the configuration.*
  2. Disable **Multi Virtual System Capability** in the template configuration.
  3. On the firewall, from **Device** > **Setup** > **Panorama Settings**, select **Disable Panorama Policy and Objects** and **Disable Device and Network Template** to remove the Panorama pushed configuration. (You don't need to commit.)
  4. In the Panorama Template Stack for the firewall, set Default VSYS to **`vsys1`**.
  5. In Panorama, push the device group to the firewall, and select **`include Device and Network Templates`**. (This will push both the device group and template to the firewall.)

After completing the prerequisites, you're ready to perform the migration using Panorama. This task illustrates an example migration.

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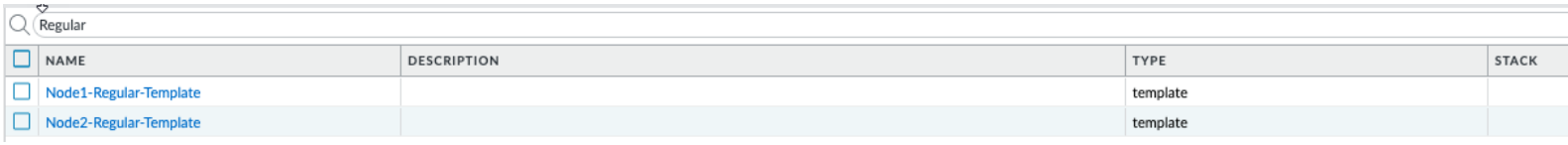
STEP 1 | Select **Panorama** > **Templates**. Our example has two preexisting non-clustering templates:

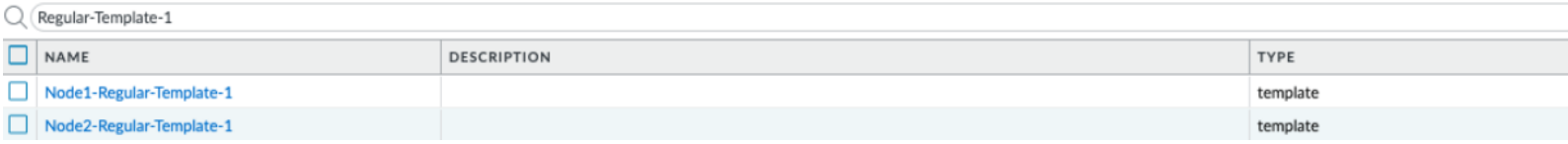
- Node1-Regular-Template
- Node2-Regular-Template



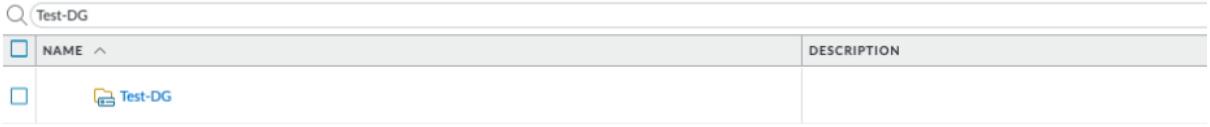
STEP 2 | Select the checkbox to the left of one of the templates and select **Clone** at the bottom of the screen.

This adds a new template with -1 appended to the end of the template name. Clone both templates. The cloned templates are named:

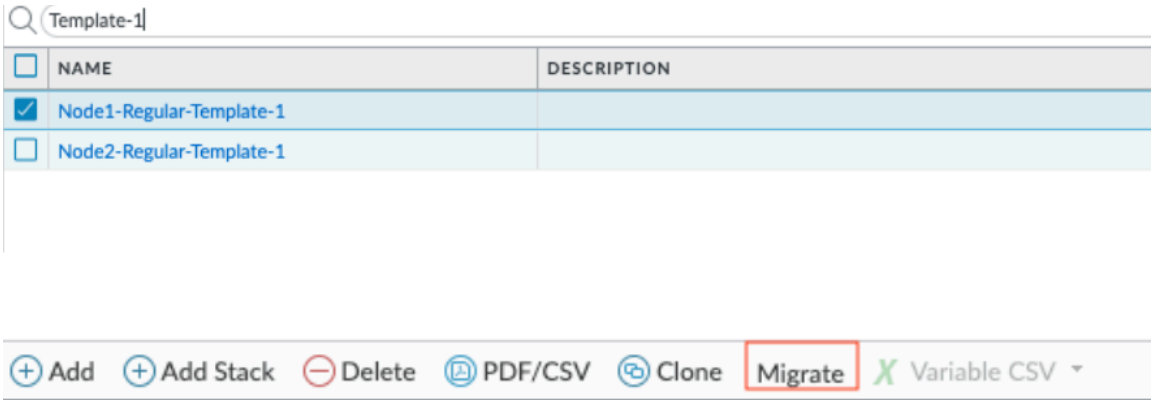
- Node1-Regular-Template-1
- Node2-Regular-Template-1



STEP 3 | Add the non-clustering templates from the prior step to a device group under **Panorama** > **Device Groups**.



STEP 4 | Select the cloned Node1-Regular-Template-1 and select **Migrate** at the bottom of the screen.



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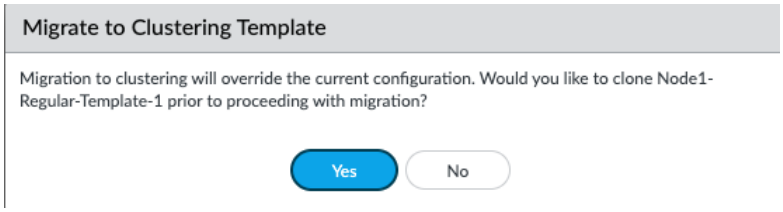
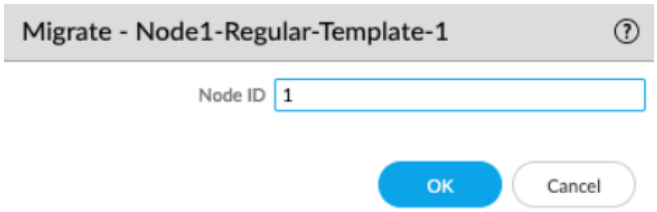
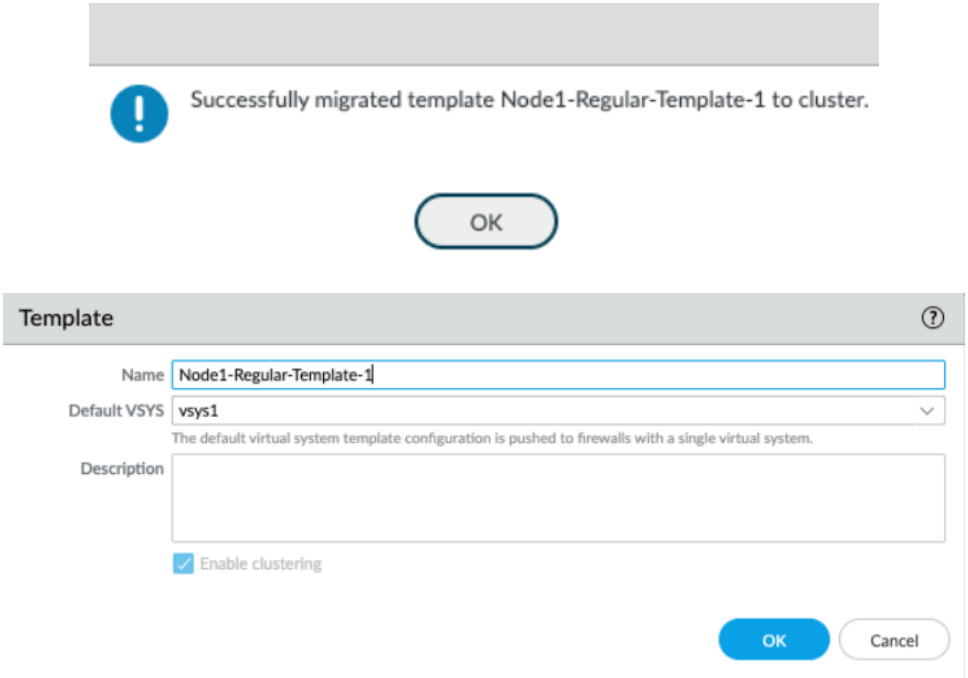
STEP 5 | Select **Yes**, which creates a backup of the cloned non-clustering template (which you can ignore).



STEP 6 | Enter the Node ID (1 or 2). For our example, enter 1.



STEP 7 | The cloned non-clustering template (Node1-Regular-Template-1) is overridden to become the migrated clustering template by the same name. You can give the template a new **Name**. **Enable clustering** is selected by default.



STEP 8 | Repeat Step 4 through Step 6 for the other non-clustering template (Node2-Regular-Template-1), using the other Node ID (Node ID 2) to override that non-clustering template and migrate it into a clustering template (Node2-Regular-Template-1). Again, you can give the template a new **Name**.

STEP 9 | Create a clustering template stack and associate the created templates from Steps 7 and 8 for common configuration.

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STEP 10 | Any non-cluster templates that get migrated to cluster templates and have references in Device Group are also converted. You can keep clustering templates or change to clustering template stacks in the device group.

Keep in mind:

- *Template stacks cannot be cloned.*
- *A template stack that has templates present in it cannot be migrated successfully.*

STEP 11 | Learn about NGFW Clustering and learn about inter-firewall links, MC-LAGs, node states, and more in NGFW Clusters.

STEP 12 | In Panorama, from **Panorama** > **Firewall Cluster**, create a cluster of cluster type: PA, and add the two PA-7500 Series firewalls. Refer to Configure an NGFW Cluster.

STEP 13 | Push the device group and cluster template to the PA-7500 nodes.

1. Review the device group and template configuration for interface references and unsupported features.
2. In **Network** > **IKE Gateways**, if IKEv1 is configured, change it to IKEv2. (An NGFW cluster doesn't support IKEv1.)
3. In Panorama, push the cluster template to the PA-7500 Series nodes.
4. In Panorama, push the device group to the PA-7500 Series firewall; select **include Device and Network Templates**. (This will push both the device group and template to the PA-7500 Series firewall.)

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Configure an NGFW Cluster

Where Can I Use This?	What Do I Need?
• NGFW	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro One of the following for Panorama managed NGFWs: ❑ Panorama PAN-OS 12.1.2 (Panorama Clustering Plugin is bundled with Panorama) ❑ Panorama PAN-OS 11.1.5 and Panorama Clustering Plugin 2.0.0 ❑ Panorama PAN-OS 11.1.3 and Panorama Clustering Plugin 2.0.0

Before you configure the firewalls to an NGFW cluster, perform the following prerequisites:

- Confirm that Panorama and the firewalls that you will assign to an NGFW cluster are all running the same software version.
- Install Panorama Clustering plugin 2.0.0 if you're using PAN-OS 11.1.3; refer to Install Panorama Plugins. For subsequent PAN-OS releases, you must install a compatible Panorama Clustering plugin release. Also refer to the Panorama plugin for Clustering in the VM-Series and Panorama Plugin Release Notes.
- Be familiar with Panorama tasks, such as managing devices and creating templates, template stacks, and device groups.
- Add the two firewalls as managed devices of Panorama so that they are communicating with each other over the management interface. Confirm by selecting **Panorama** > **Managed Devices** > **Summary**, and verify the two firewalls (nodes) have an IP address on their management interface and the Device State is Connected.
- Connect the firewalls back-to-back with 100G or 400G HSCI-A and HSCI-B links. (Verify the connections on each firewall by using the CLI command **`show interface all`** to see the hsci-a and hsci-b interfaces.)
- Familiarize yourself with NGFW Clusters.

The steps in the example task to configure an NGFW cluster are based on this topology example of two MC-LAGs. The orange links connected to Node 1 and Node 2 on the client side belong to AE1 (an MC-LAG). The orange links connected to Node 1 and Node 2 on the server side belong to AE2 (another MC-LAG). Traffic from the client at 10.1.7.100 goes to the switch and is then divided between the two ingress AE1 interfaces and then egresses the two AE2 interfaces to the switch, and then goes over the orange link to the server at 10.1.8.200.

The gray links connected Node 1 and Node 2 are orphan ports. Traffic from the client at 10.1.1.100 goes to the switch, to Node 1, across an HSCI interface to Node 2, egresses Node 2 to the switch, and then to the server at 10.1.2.100.

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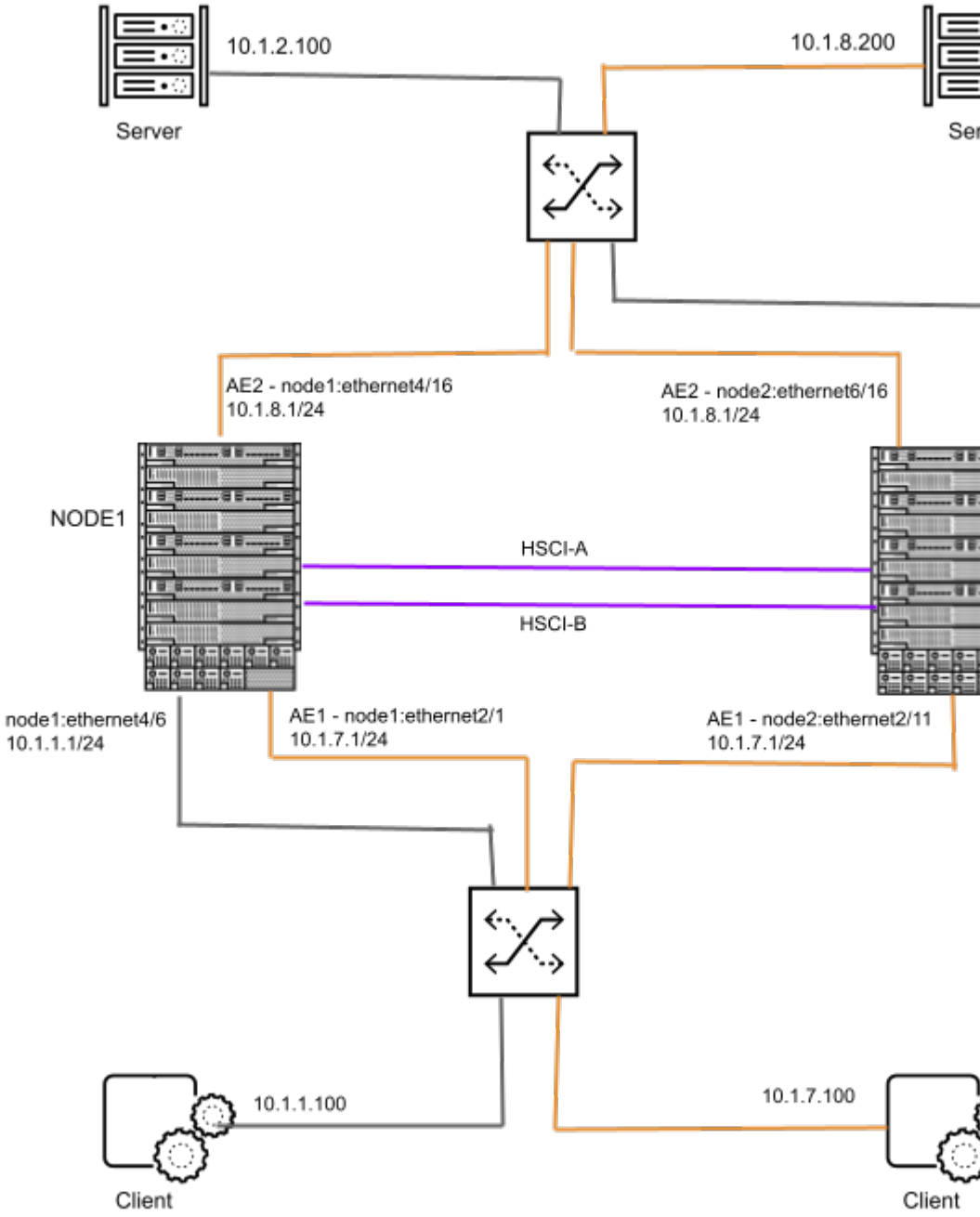

- PAN-OS
- Strata Cloud Manager

STEP 1 | Log in to the Panorama Web Interface.

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STEP 2 | Create an NGFW cluster.

1. Select **Panorama** > **Firewall Clusters** > **Create Cluster**.
2. Enter a **Cluster Name** containing zero or more alphanumeric characters, underscores (_), hyphens (-), dots (.), or spaces.
3. Select **Cluster Type** as **PA**.
4. Click **OK**.

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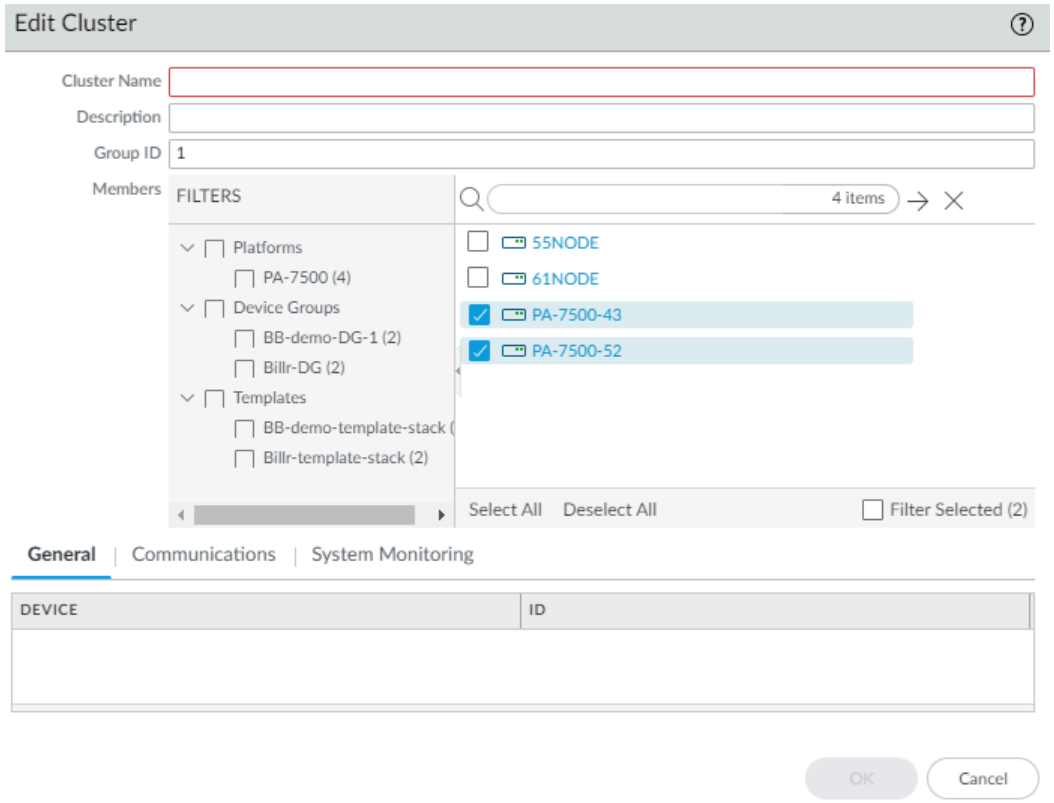
STEP 3 | Add the firewalls to the cluster.

1. Select **Panorama** > **Firewall Clusters** > **Summary View** and select the cluster you created.
2. Enter a **Group ID** in the range 1 to 63; the default is 1.
3. Select the two firewalls to assign to the cluster. The first firewall you select automatically becomes Node 1.



4. Click **OK**.
5. View the firewalls in the cluster by selecting **Panorama** > **Firewall Clusters** > **Summary View** and selecting the cluster you created. The **General** tab displays nonconfigurable fields: Device serial number and Node ID (1 or 2). Click **OK** to close the window.

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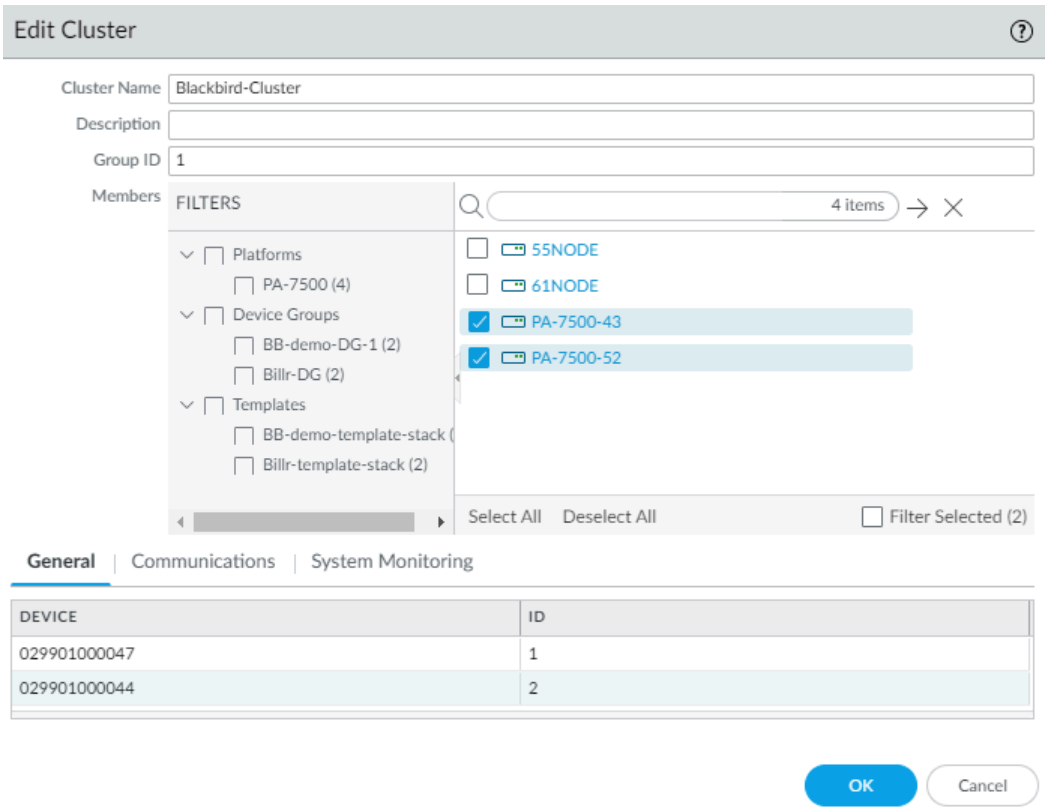

6. Select **Commit to Panorama** and **Commit**. Both firewalls are rebooted and get assigned a Node ID, the existing configuration (policy, network, etc.) is cleared, and the firewalls are connected back to Panorama.
7. Select **Push to Devices**, select **Push All Changes**, select the newly created firewall cluster, and **Push**.

STEP 4 | (Optional) Verify the cluster state and node state.

1. Access the CLI.
2. > **show cluster leader**
3. > **show cluster local node-id**
4. > **show cluster local state**
5. > **show cluster all**
6. > **exit**

STEP 5 | Log in to the Panorama Web Interface.

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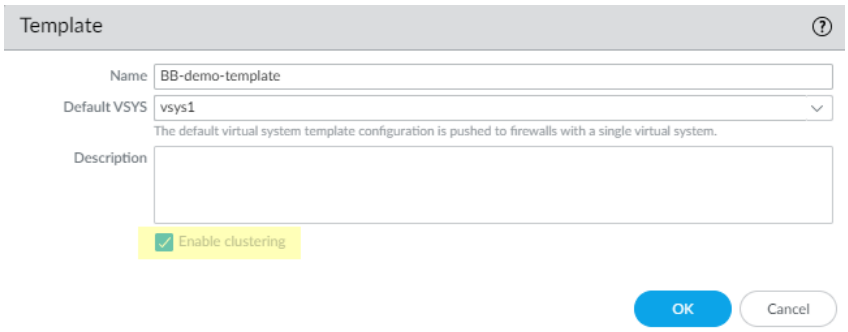
STEP 6 | View the Config Sync Status.

1. Select **Panorama** > **Firewall Clusters** > **Summary View** and select **PA-Series**.
2. View the Config Sync Status. After the Commit All is successful, the firewalls have a Config Sync Status of In Sync.

STEP 7 | Add a cluster template.

1. Select **Panorama** > **Templates** and **Add** a template.
2. Enter a **Name** and **Description** of the template.
3. **Enable clustering**. You must enable clustering when you first create the template; you can't go back to enable clustering later.
4. Click **OK**.



STEP 8 | Create a template stack.

1. Select **Panorama** > **Templates** > **Add Stack**.
2. Enter a **Name** for the template stack.
3. In the Devices section, select the two firewalls in the cluster.
4. In the Templates section, **Add** the template you created.
5. **Enable clustering**. You must enable clustering when you first create the template stack; you can't go back to enable clustering later. (If you don't **Enable clustering** for the

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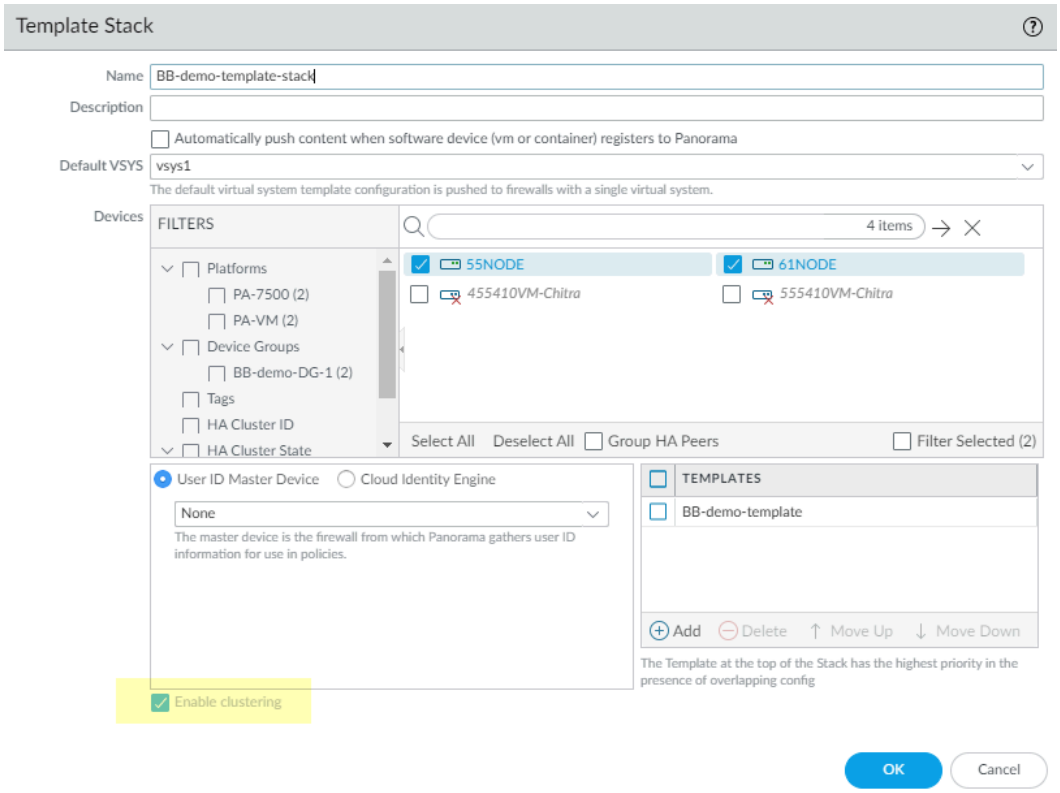
template stack now, it won't match the template and an Operation Failed message appears.)

6. Click **OK**.



STEP 9 | Add a device group.

1. Select **Panorama** > **Device Groups** > **Add**.
2. Enter a **Name** for the device group.
3. In the Devices section, select the Names of the two firewalls in the cluster.
4. In the Reference Templates section, **Add** the template stack that you created.
5. Click **OK**.

STEP 10 | **Commit to Panorama** and **Commit** to apply your configuration in Panorama.

STEP 11 | (Two-node cluster only) Verify that the management interface permits the same IP addresses on the two nodes. The firewalls use the management interface to exchange heartbeats to detect and avoid a split-brain situation.

1. Select **Device** > **Setup** > **Interfaces**.
2. In the Template field, select your template.
3. Select the **Management** interface.
4. If you added **Permitted IP Addresses** on one of the nodes, you must also permit the same IP addresses on the peer node. Each node must be able to reach the management

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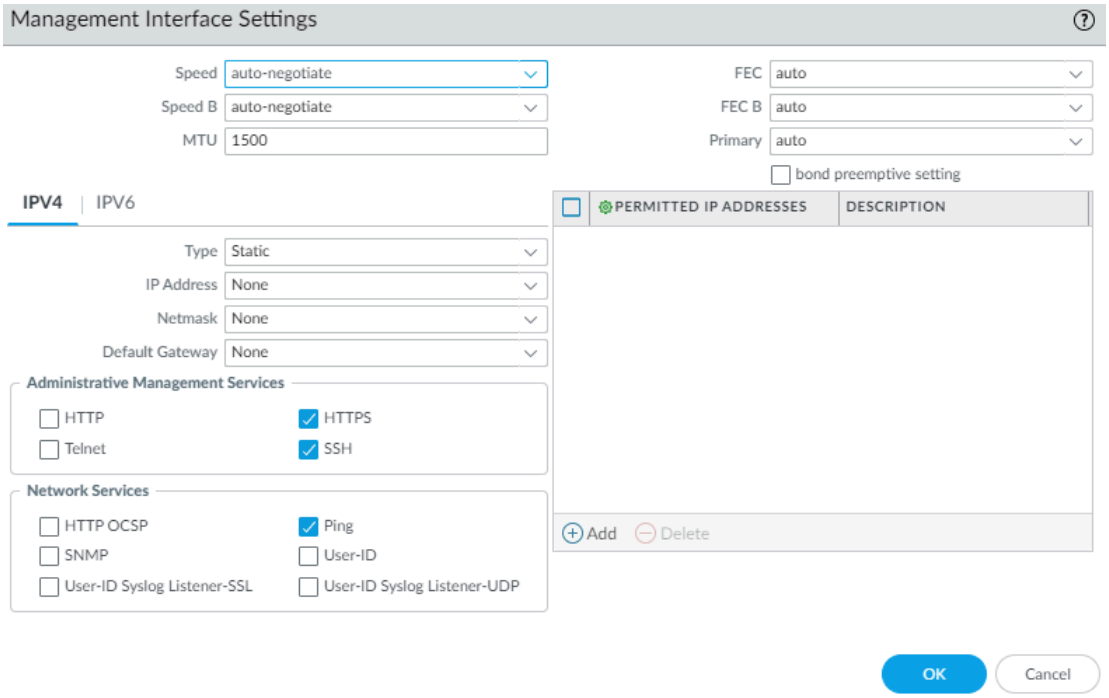
interface of the peer node. (If you deny the IP address or network of the peer, split brain detection won't work.)

5. Click **OK**.



STEP 12 | Configure the template.

1. Select **Network**
2. In the Template field, select your template.

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STEP 13 | To configure the orphan port, configure a Layer 3 interface on the firewall that connects to the client.

1. Select **Network** > **Interfaces** > **Cluster Ethernet** and **Add Interface**.
2. Select the **Node ID** (1 for this example).
3. Select the **Slot** (Slot 4).
4. Select the **Interface Name**, for example, node1:ethernet4/6.
5. Select the **Interface Type** as **Layer3**.
6. On the **Config** tab, create a **Logical Router** by adding a **Name**; click **OK**.
7. Create a **Security Zone**, such as client.
8. Select **IPv4**, select the **Type** as **Static**, for example, and **Add** the IPv4 address with netmask (10.1.1.1/24 in this example).

   *Alternatively, select **IPv6**, **Enable IPv6 on the interface**, select the **Type** as **Static**, and **Add** the IPv6 Address with network prefix length. This task example uses static IP addressing; NGFW clustering also supports DHCP, DHCPv6, PPPoE, and PPPoEv6.*
9. Select the **Advanced** tab and then the **Other Info** tab.
10. For **Management Profile**, create a new Interface Management profile to allow access to the interface. Select the management and network services allowed (such as HTTPS, SSH, and Ping) and click **OK**.
11. Click **OK**.

STEP 14 | Configure the other orphan port on node 2, Slot 2, node2:ethernet2/19, using IP address 10.1.2.1/24 for the interface that faces the server. Create a different security zone. The gray path in the example topology is configured.

STEP 15 | Configure the AE interface (MC-LAG) for the client-facing interface on Node 1.

1. Select **Network** > **Interfaces** > **Cluster Ethernet** and **Add Aggregate Group**.
2. For **Interface Name**, next to ae, enter the interface number (in this example, 1).
3. Select **Interface Type** as **Layer3**.
4. On the **Config** tab, select the same **Logical Router**.
5. Select the **Security Zone**.
6. Click **OK**.
7. Select **IPv4**, select the **Type** as **Static**, and **Add** the IPv4 address with netmask (10.1.7.1/24 in this example).
8. (Optional) Configure LACP settings if you want to enable LACP for the aggregate group.
9. Select the **Advanced** tab and then the **Other Info** tab.
10. For **Management Profile**, create a new Interface Management profile to allow access to the interface. Select the services allowed and click **OK**.
11. Click **OK**.

STEP 16 | Configure the AE interface (MC-LAG) for the server-facing interface for Node 1. For this interface, follow similar substeps as in the prior step, but configure AE2, add the IP address (10.1.8.1/24), the same logical router, and a server **Security zone**.

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STEP 17 | Add an interface member to the MC-LAG on the client side.

1. Select **Network** > **Interfaces** > **Cluster Ethernet** and **Add Interface**.
2. On the Cluster Ethernet Interface, select the **Node ID** Node 1.
3. Select the **Slot** (Slot 2).
4. Select the **Interface Name**, for example, node1:ethernet2/1.
5. Select the **Interface Type** as **Aggregate Ethernet**.
6. Select the **Aggregate Group**, such as ae1.
7. Click **OK**.

STEP 18 | Add a second interface member to the MC-LAG on the client side, assigning **Node ID** as Node 2, Slot 2, node2:ethernet2/11. Select the same **Aggregate Group** (ae1).

STEP 19 | Add an interface member to the MC-LAG on the server side. Select Node 1, Slot 4, node1:ethernet4/16. Select the **Aggregate Group**, ae2.

STEP 20 | Add a second interface member to the MC-LAG on the server side, assigning it to Node 2, Slot 6, node2:ethernet6/16. Select the **Aggregate Group**, such as ae2.

STEP 21 | (PAN-OS 11.1.5 and later 11.1 releases) Prepare to configure MACsec for the HSCI ports by creating a MACsec profile.

IMPORTANT: When you migrate from a non-MACsec cluster to a MACsec cluster, you must do so during a maintenance window.

1. Select **Network** > **Network Profiles** > **MACsec Profile** and select **MACsec Policy**.
2. **Add** a MACsec crypto profile by **Name**, which contains zero or more alphanumeric characters, underscores (_), hyphens (-), or dots (.). The Name is a maximum of 31 characters.
3. Select the **Encryption** type: **AES-128-GCM** (default) or **AES-256-GCM**.
4. Select the **Confidentiality Offset** to specify a number of bytes (starting from the frame header), after which MACsec encrypts the bytes in a frame. Values are **0** (default), **30**, or **50**.
5. **Enable SCI Include** to include the Secure Channel Identifier (SCI) tag in the Security Tag field of the MACsec header. The default is disabled.
6. **Enable Anti Replay** to enable replay protection. This allows a MACsec port to accept frames out of order if they are within the Anti Replay Window. The default is disabled.
7. Specify the **Anti Replay Window** size in the range 0 to 65,535; the default is 16,384. This value determines the range of packet numbers that the port will accept, for packets that might be out of order. The port will accept packets with a packet number greater than or equal to the last packet number minus the window size. For example, after the

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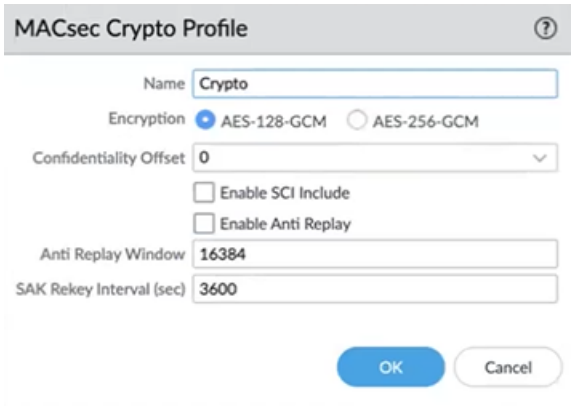
port receives packet number 12, if the window is 5, the port will subsequently accept only packets numbered 7 or higher.

Specify an Anti Replay window size based on the traffic rate because the HSCI ports will likely experience packets arriving out of order due to prioritization or load balancing in the network.

8. Specify the **SAK Rekey Interval (sec)** in seconds; range is 60 to 86,400; default is 3,600. Connectivity Association participants negotiate a Secure Association agreement, which includes a cipher suite and keys. A Key Server generates a Secure Association Key (SAK) from the Connectivity Association Key (CAK); the SAK is refreshed at the SAK rekey interval. The end-to-end nodes use the SAK to encrypt traffic for a given session. To avoid the SAKs of all interfaces being refreshed at the same time, MACsec might spread the rekeying actions evenly by subtracting a small random number from the SAK rekey interval.
9. Click **OK**.



STEP 22 | (PAN-OS 11.1.5 and later 11.1 releases) Create a MACsec Pre Shared Key Profile.

1. Select **Network** > **Network Profiles** > **MACsec Profile** and select **Pre Shared Key**.
2. **Add** a MACsec Pre Shared Key Profile by **Name**, which contains zero or more alphanumeric characters, underscores (_), hyphens (-), dots (.), or spaces. The Name is a maximum of 31 characters.
3. Enter the **CKN** (Connectivity Association Key Name) that identifies the Connectivity Association Key. The CKN is 1 byte to 32 bytes of hexadecimal string (2 to 64 hex digits) with an even number of digits; for example, 1234.
4. Enter the **CAK** (Connectivity Association Key), which is a 16-byte hexadecimal string (32 hex digits) for AES-128-GCM, or a 32-byte hexadecimal string (64 hex digits) for AES-256-GCM. The CAK is used to generate all other keys used for MACsec. An

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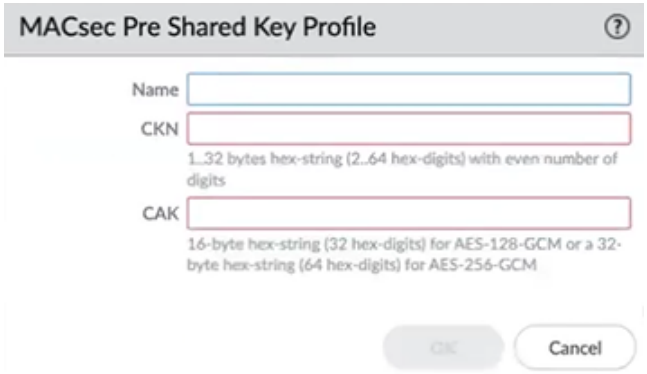
example CAK for AES-128-GCM is 12345678912345678912345678912345. A CAK for AES-256-GCM is twice as long as a CAK for AES-128-GCM.

The Pre Shared Key, CKN, and CAK must match on the two ends of the HSCI-A link. Likewise, the Pre Shared Key, CKN, and CAK must match on the two ends of the HSCI-B link.

5. Click **OK**.

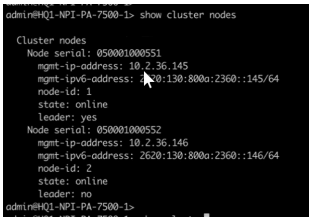


STEP 23 | (PAN-OS 11.1.5 and later 11.1 releases) **Commit** to Panorama.

STEP 24 | (PAN-OS 11.1.5 and later 11.1 releases) **Push to Devices** and **Push Templates** so that the devices have the MACsec profiles, thus allowing you to later push the cluster configuration with MACsec enabled.

STEP 25 | (PAN-OS 11.1.5 and later 11.1 releases) Identify the leader node and nonleader node.

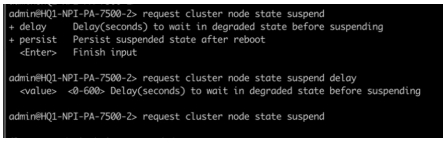
1. Access the CLI on one of the firewall nodes. (If you already know which node is the nonleader, access the CLI on the nonleader node.)
2. **>show cluster nodes**



STEP 26 | (PAN-OS 11.1.5 and later 11.1 releases) Suspend the nonleader node.

1. Access the CLI on the nonleader node.
2. **>request cluster node state suspend**

The suspension takes approximately 120 seconds (the default delay value).



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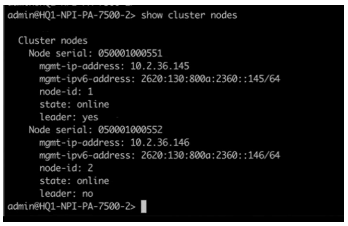
STEP 27 | (PAN-OS 11.1.5 and later 11.1 releases) Confirm the nonleader node is suspended.

1. **>show cluster nodes**



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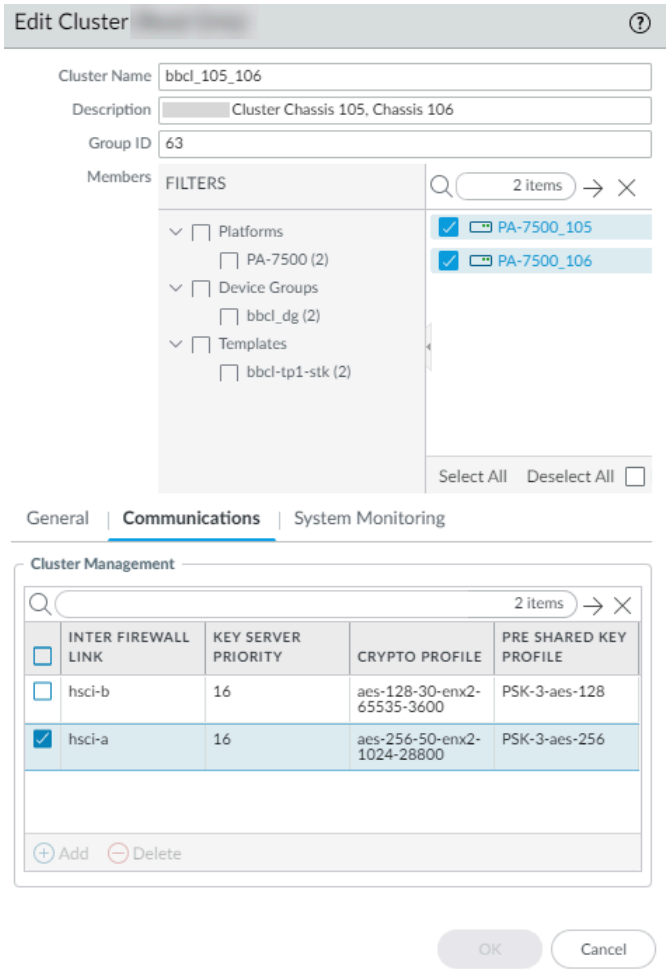
STEP 28 | (PAN-OS 11.1.5 and later 11.1 releases) Apply your MACsec profiles to the HSCI-A and HSCI-B ports.

1. Select **Panorama** > **Firewall Clusters** and select **Summary View**.
2. Select **PA-Series** or **All Clusters**.
3. Select a **Cluster Name** to edit the cluster.
4. Select **Communications**.
5. Select **hsci-a** and enter the **Key Server Priority** in the range from 0 to 255; the default is 16. The lower the value, the higher the priority of the Key Server.

   If the priority values for the HSCI-A links on the two nodes are equal, the node with the lower MAC address is the Key Server. The Key Server (one of the nodes in the cluster) selects and advertises a cipher suite, and also generates the SAK from the CAK.
6. For **Crypto Profile**, select the MACsec crypto profile you created or select the **default** profile.
7. For **Pre Shared Key Profile**, select the profile you created.
8. Select **hsci-b** and perform the same steps to assign a Key Server Priority and apply the crypto profile and pre-shared key Profile to the port.
9. Click **OK**.



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STEP 29 | (PAN-OS 11.1.5 and later 11.1 releases) **Commit**, **Commit to Panorama**, and **Commit**.

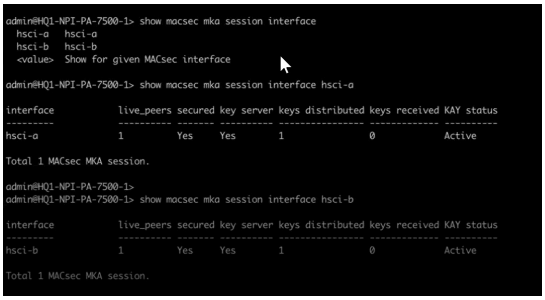
STEP 30 | (PAN-OS 11.1.5 and later 11.1 releases) **Push to Devices** and **Push** , which enables MACsec on the ports.

STEP 31 | (PAN-OS 12.1.2 and later 12.1 releases) (Optional) If you renamed the MACsec key profile or replaced the MACsec pre-shared key and your commit fails, select **Edit Selections** and perform the following substeps:

1. If you need to push a device group configuration, perform these next three substeps. Select the **Device Groups** tab. Select the affected device group or groups. Select **Include Device and Network Templates**. Select **Include Firewalls Clusters**.
2. Select the **Templates** tab and deselect the affected template.
3. Select the **Firewall Clusters** tab. Deselect **Filter Selected**. Deselect the affected cluster.
4. If you need to push the template configuration, perform these next three substeps. Select the **Device Groups** tab and deselect the affected device group.
5. Select the **Templates** tab and select the affected template. Select **Include Firewall Clusters**.
6. Select the **Firewall Clusters** tab. Deselect **Filter Selected**. Deselect the affected cluster.
7. Now that you have edited the selections, repeat Steps 29 and 30 to commit and push your changes.

STEP 32 | (PAN-OS 11.1.5 and later 11.1 releases) Confirm MACsec is enabled on both ports.

1. Access the CLI.
2. **>show macsec mka session interface hsci-a**
3. **>show macsec mka session interface hsci-b**



STEP 33 | (PAN-OS 11.1.5 and later 11.1 releases) Unsuspend the nonleader node.

1. Access the CLI on the nonleader node.
2. **>request cluster node state unsuspend**

It takes a few minutes to unsuspend the nonleader node; the length of time depends on the traffic level.

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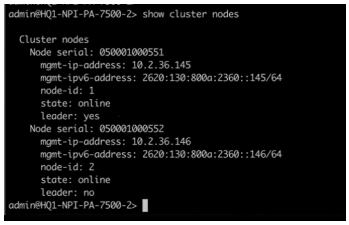
STEP 34 | (PAN-OS 11.1.5 and later 11.1 releases) Verify that both nodes are online.

1. **>show cluster nodes**



STEP 35 | (PAN-OS 11.1.7 and later 11.1 releases) (Optional) Enable multiple virtual systems for the firewalls in the template.

1. Select **Device** > **Virtual Systems** and select the **Template** by name.
2. For **Mode**, select **Multi VSYS**.

STEP 36 | Create Security policy rules.

1. Select **Policies** and select the Device Group.
2. Create Security policy rules to control access, such as allowing a specific source zone, destination zone, address, user, device, application, and service.

STEP 37 | Configure routing and other features your firewalls require (except for HA). Refer to the PAN-OS Administrator's Guide and the PAN-OS Networking Administrator's Guide.

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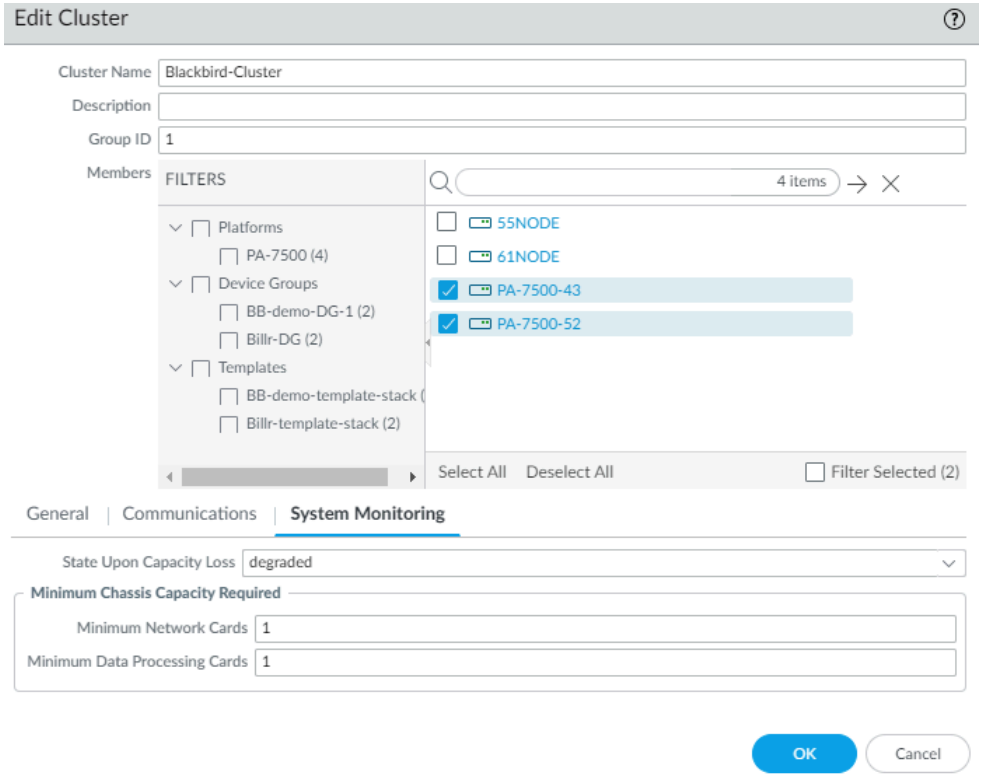
STEP 38 | Configure system monitoring for the NGFW cluster.

1. Select **Panorama** > **Firewall Clusters** > **Summary View** and select a cluster or a single firewall in the cluster.
2. Select **System Monitoring**.



3. Select the **State Upon Capacity Loss**:
   - **degraded**—Specifies that the firewall will be in a DEGRADED state if the count of functional network cards or data processing cards goes below the configured Minimum Network Cards or Minimum Data Processing Cards, respectively. Furthermore:
     - A cluster node in DEGRADED state has traffic ports down, but is still a part of the sharded (fragmented) member table.
     - Data processing card resources of a DEGRADED cluster node can be used to process traffic and for Layer 7 processing.
     - A cluster node in INIT or ONLINE state transitions to DEGRADED state when a soft fault is reported to the cluster node state machine.
     - A cluster node in ONLINE state transitions to DEGRADED state (and a node in DEGRADED state remains in DEGRADED state) if you suspend the cluster node after a delay (using the CLI operational command: **`request cluster node`**

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`state suspend`). A delay allows data planes to gracefully complete Layer 7 processing or other processes that were underway.

- If all soft faults are cleared, the cluster node transitions to INIT state.
- If a hard fault occurs, a cluster node in DEGRADED state transitions to FAILED state.
- A cluster node in DEGRADED state transitions to SUSPENDED state if the maximum number of state flaps is seen or you suspend the cluster node.
- If a chassis has no functional data processing cards remaining (all DPC slots are powered off or in a FAILED state), the cluster node state will be FAILED, even if you configured **degraded**, because having no functional DPC is a hard fault.

- **failed**—Specifies that the firewall will be in a FAILED state if the count of functional network cards or data processing cards goes below the configured Minimum Network Cards or Minimum Data Processing Cards, respectively. Furthermore:
  - A cluster node in FAILED state has traffic ports down and isn't part of the sharded member table.
  - Data processing card resources of a FAILED cluster node can't be used to process traffic or for Layer 7 processing.
  - A cluster node in INIT, ONLINE, or DEGRADED state transitions to FAILED state when a hard fault is reported to the cluster node state machine.
  - A cluster node in FAILED state transitions to INIT state if all hard faults are cleared.
  - A cluster node in FAILED state transitions to SUSPENDED state if the maximum number of state flaps is seen or you suspend the cluster node.

NGFW Clusters *provides a list of hard faults and soft faults.*

4. In the Minimum Chassis Capacity Required section, enter the **Minimum Network Cards** required; range is 1 to 7, default is 1. When fewer Network Cards than the configured value are functional, the firewall in the cluster goes to the State Upon Capacity Loss that you configured (degraded or failed).
5. Enter the **Minimum Data Processing Cards** required; range is 1 to 7, default is 1. When fewer Data Processing Cards than the configured value are functional, the firewall in the cluster goes to the State Upon Capacity Loss that you configured (degraded or failed).
6. Click **OK**.

STEP 39 | (Optional) Configure Log Forwarding for the PA-7500 Series firewall.

STEP 40 | Push the configuration from Panorama to the firewalls in the cluster.

1. **Commit** and **Commit and Push** and **Commit**.
2. **Commit** and **Push to Devices**, **Push All Changes**, select the cluster, and **Push** the configuration to both nodes.

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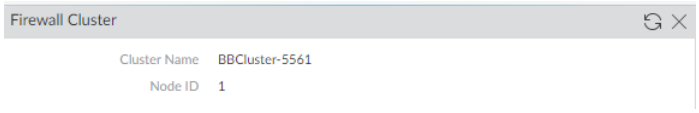
STEP 41 | (PAN-OS 11.1.7 and later 11.1 releases) If you enabled multiple virtual systems for the Panorama device template, enable multiple virtual systems for an individual firewall.

1. Log on to an individual firewall (not Panorama).
2. Select **Device** > **Setup** > **Management**.
3. In General Settings, select **Multi Virtual System Capability**.
4. Click **OK**.
5. Repeat this step for the other firewall in the cluster.

STEP 42 | View NGFW cluster information for an individual firewall on its Dashboard.

1. Log on to an individual firewall (not Panorama).
2. Select **Dashboard** and for Widgets, select **System** > **Firewall Cluster** to view the Firewall Cluster card, which displays the Cluster Name and Node ID.



STEP 43 | View NGFW Cluster Summary and Monitoring information and health for the cluster.

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PAN-OS

Use this process to configure an NGFW cluster.

STEP 1 | Log in to the Panorama Web Interface.

STEP 2 | Create an NGFW cluster.

1. Select **Panorama** > **Firewall Clusters** > **Create Cluster**.
2. Enter a **Cluster Name** containing zero or more alphanumeric characters, underscores (_), hyphens (-), dots (.), or spaces.
3. Select **Cluster Type** as **PA**.
4. Click **OK**.

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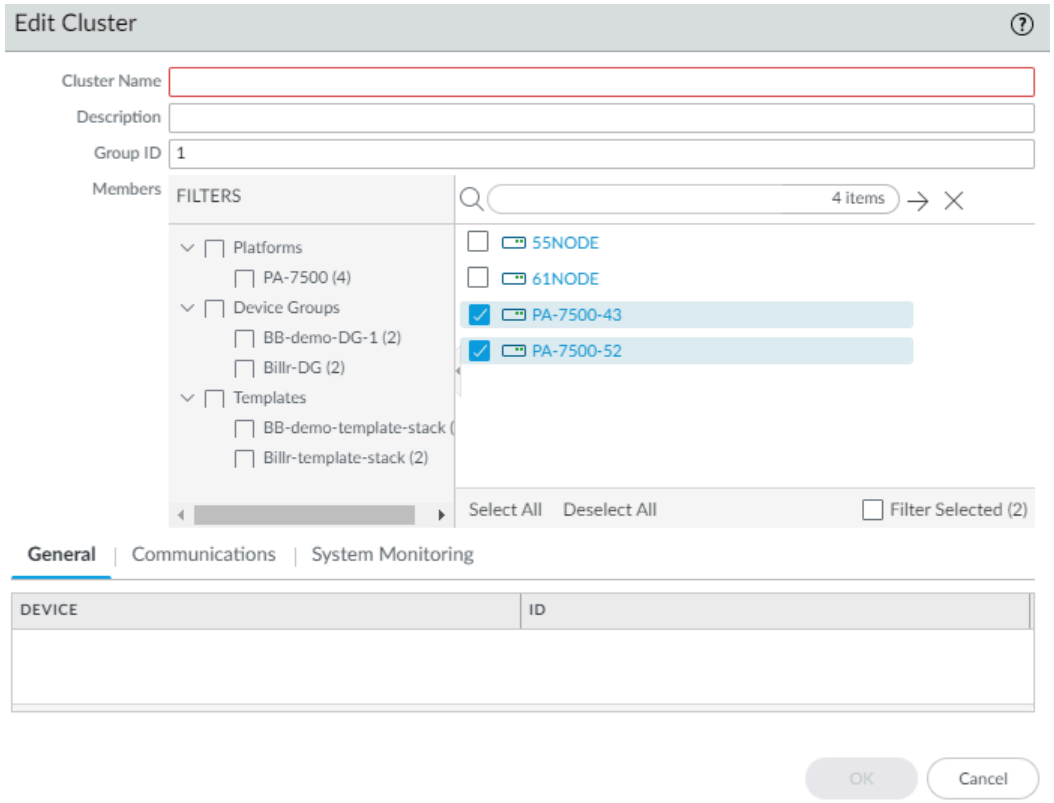
STEP 3 | Add the firewalls to the cluster.

1. Select **Panorama** > **Firewall Clusters** > **Summary View** and select the cluster you created.
2. Enter a **Group ID** in the range 1 to 63; the default is 1.
3. Select the two firewalls to assign to the cluster. The first firewall you select automatically becomes Node 1.



4. Click **OK**.
5. View the firewalls in the cluster by selecting **Panorama** > **Firewall Clusters** > **Summary View** and selecting the cluster you created. The **General** tab displays nonconfigurable fields: Device serial number and Node ID (1 or 2). Click **OK** to close the window.

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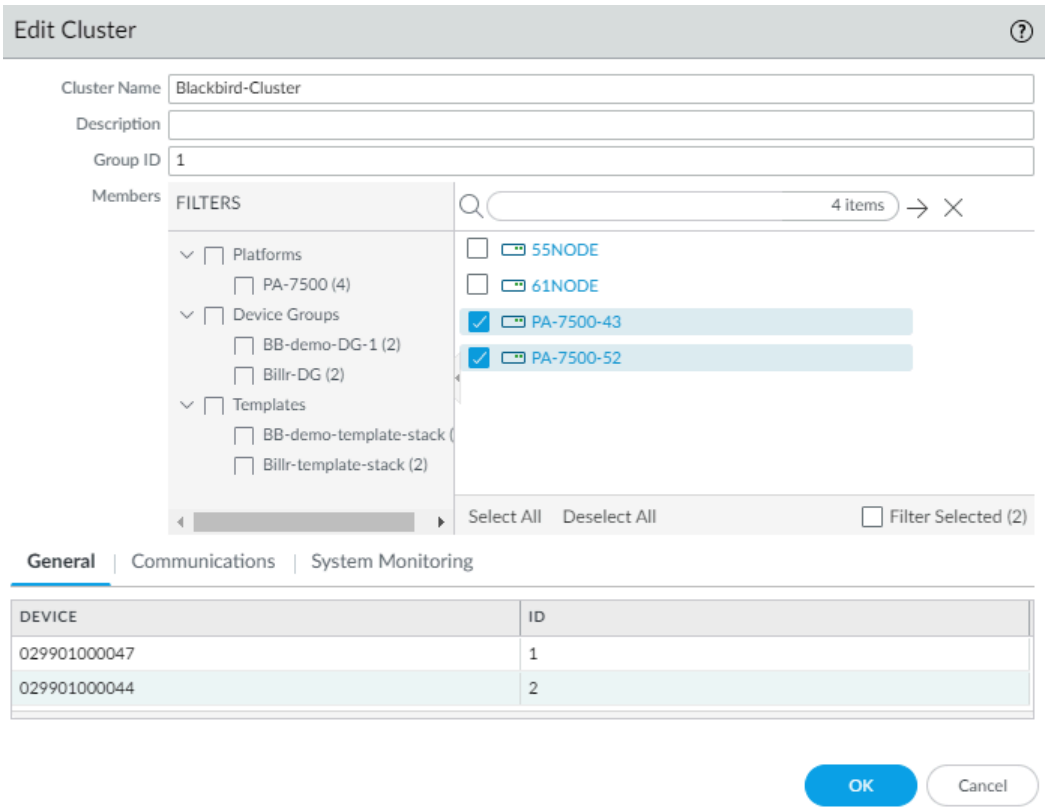

6. Select **Commit to Panorama** and **Commit**. Both firewalls are rebooted and get assigned a Node ID, the existing configuration (policy, network, etc.) is cleared, and the firewalls are connected back to Panorama.
7. Select **Push to Devices**, select **Push All Changes**, select the newly created firewall cluster, and **Push**.

STEP 4 | (Optional) Verify the cluster state and node state.

1. Access the CLI.
2. > **show cluster leader**
3. > **show cluster local node-id**
4. > **show cluster local state**
5. > **show cluster all**
6. > **exit**

STEP 5 | Log in to the Panorama Web Interface.

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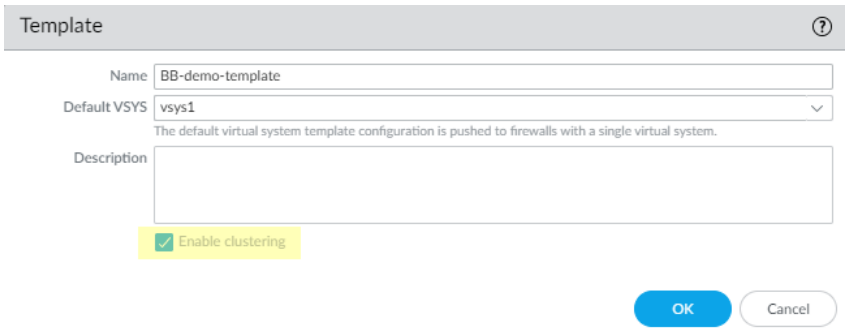
STEP 6 | View the Config Sync Status.

1. Select **Panorama** > **Firewall Clusters** > **Summary View** and select **PA-Series**.
2. View the Config Sync Status. After the Commit All is successful, the firewalls have a Config Sync Status of In Sync.

STEP 7 | Add a cluster template.

1. Select **Panorama** > **Templates** and **Add** a template.
2. Enter a **Name** and **Description** of the template.
3. **Enable clustering**. You must enable clustering when you first create the template; you can't go back to enable clustering later.
4. Click **OK**.



STEP 8 | Create a template stack.

1. Select **Panorama** > **Templates** > **Add Stack**.
2. Enter a **Name** for the template stack.
3. In the Devices section, select the two firewalls in the cluster.
4. In the Templates section, **Add** the template you created.
5. **Enable clustering**. You must enable clustering when you first create the template stack; you can't go back to enable clustering later. (If you don't **Enable clustering** for the

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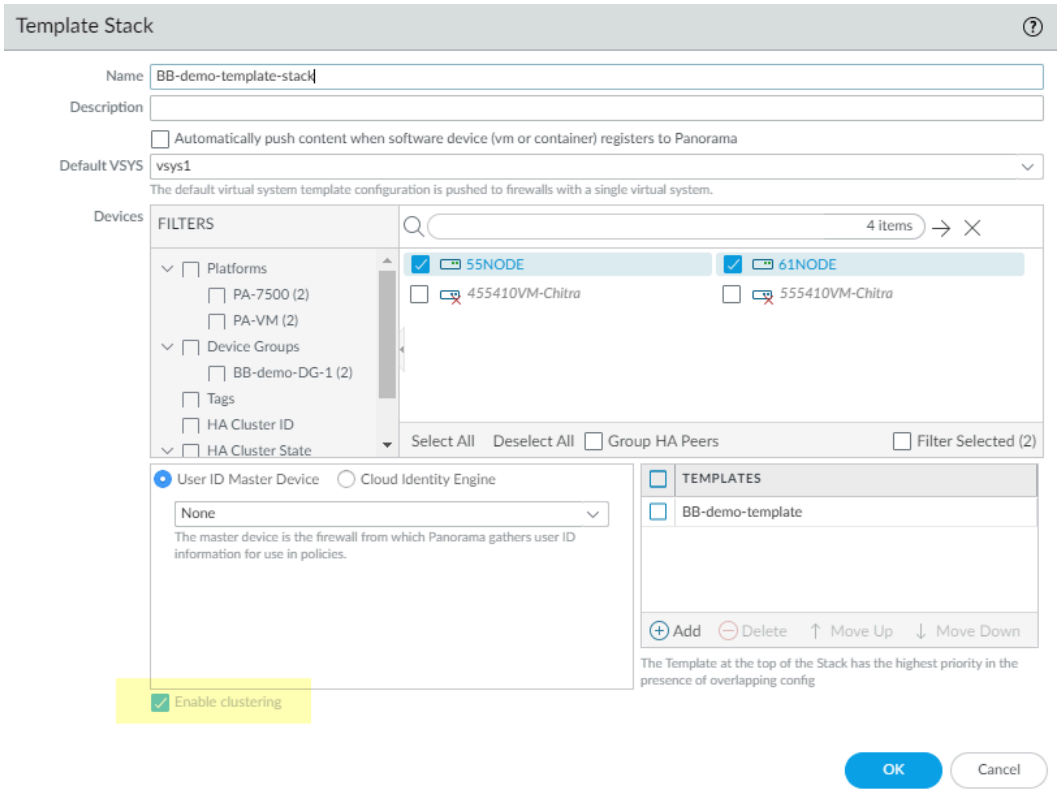
template stack now, it won't match the template and an Operation Failed message appears.)

6. Click **OK**.



STEP 9 | Add a device group.

1. Select **Panorama** > **Device Groups** > **Add**.
2. Enter a **Name** for the device group.
3. In the Devices section, select the Names of the two firewalls in the cluster.
4. In the Reference Templates section, **Add** the template stack that you created.
5. Click **OK**.

STEP 10 | **Commit to Panorama** and **Commit** to apply your configuration in Panorama.

STEP 11 | (Two-node cluster only) Verify that the management interface permits the same IP addresses on the two nodes. The firewalls use the management interface to exchange heartbeats to detect and avoid a split-brain situation.

1. Select **Device** > **Setup** > **Interfaces**.
2. In the Template field, select your template.
3. Select the **Management** interface.
4. If you added **Permitted IP Addresses** on one of the nodes, you must also permit the same IP addresses on the peer node. Each node must be able to reach the management

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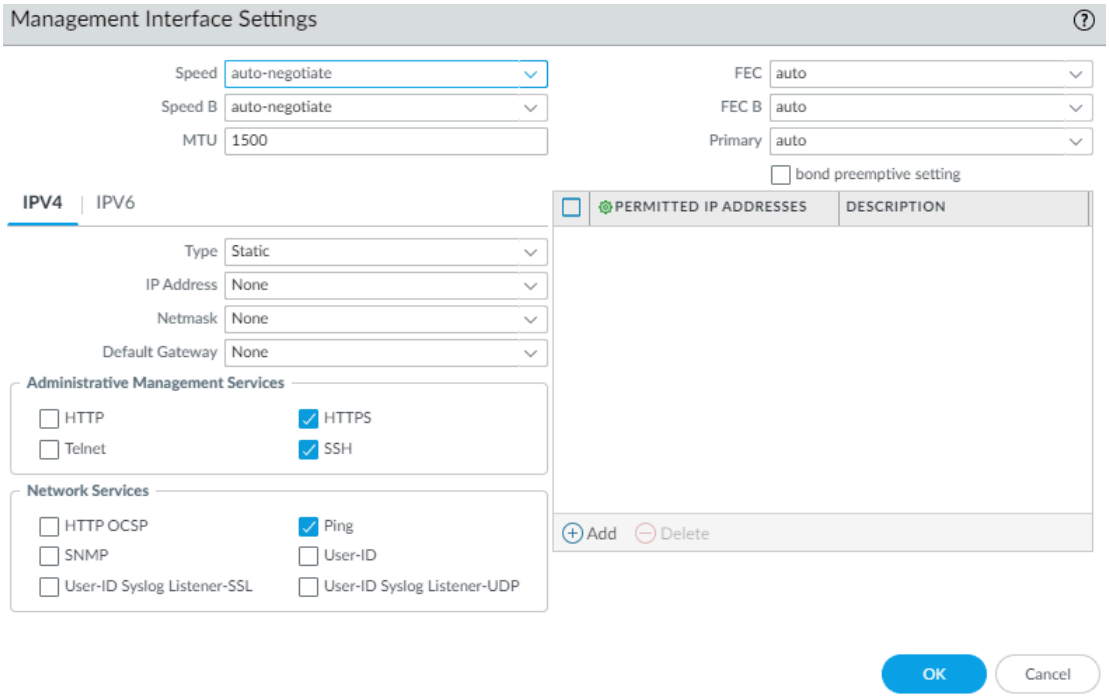
interface of the peer node. (If you deny the IP address or network of the peer, split brain detection won't work.)

5. Click **OK**.



STEP 12 | Configure the template.

1. Select **Network**
2. In the Template field, select your template.

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STEP 13 | To configure the orphan port, configure a Layer 3 interface on the firewall that connects to the client.

1. Select **Network** > **Interfaces** > **Cluster Ethernet** and **Add Interface**.
2. Select the **Node ID** (1 for this example).
3. Select the **Slot** (Slot 4).
4. Select the **Interface Name**, for example, node1:ethernet4/6.
5. Select the **Interface Type** as **Layer3**.
6. On the **Config** tab, create a **Logical Router** by adding a **Name**; click **OK**.
7. Create a **Security Zone**, such as client.
8. Select **IPv4**, select the **Type** as **Static**, for example, and **Add** the IPv4 address with netmask (10.1.1.1/24 in this example).

   *Alternatively, select **IPv6**, **Enable IPv6 on the interface**, select the **Type** as **Static**, and **Add** the IPv6 Address with network prefix length. This task example uses static IP addressing; NGFW clustering also supports DHCP, DHCPv6, PPPoE, and PPPoEv6.*
9. Select the **Advanced** tab and then the **Other Info** tab.
10. For **Management Profile**, create a new Interface Management profile to allow access to the interface. Select the management and network services allowed (such as HTTPS, SSH, and Ping) and click **OK**.
11. Click **OK**.

STEP 14 | Configure the other orphan port on node 2, Slot 2, node2:ethernet2/19, using IP address 10.1.2.1/24 for the interface that faces the server. Create a different security zone. The gray path in the example topology is configured.

STEP 15 | Configure the AE interface (MC-LAG) for the client-facing interface on Node 1.

1. Select **Network** > **Interfaces** > **Cluster Ethernet** and **Add Aggregate Group**.
2. For **Interface Name**, next to ae, enter the interface number (in this example, 1).
3. Select **Interface Type** as **Layer3**.
4. On the **Config** tab, select the same **Logical Router**.
5. Select the **Security Zone**.
6. Click **OK**.
7. Select **IPv4**, select the **Type** as **Static**, and **Add** the IPv4 address with netmask (10.1.7.1/24 in this example).
8. (Optional) Configure LACP settings if you want to enable LACP for the aggregate group.
9. Select the **Advanced** tab and then the **Other Info** tab.
10. For **Management Profile**, create a new Interface Management profile to allow access to the interface. Select the services allowed and click **OK**.
11. Click **OK**.

STEP 16 | Configure the AE interface (MC-LAG) for the server-facing interface for Node 1. For this interface, follow similar substeps as in the prior step, but configure AE2, add the IP address (10.1.8.1/24), the same logical router, and a server **Security zone**.

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STEP 17 | Add an interface member to the MC-LAG on the client side.

1. Select **Network** > **Interfaces** > **Cluster Ethernet** and **Add Interface**.
2. On the Cluster Ethernet Interface, select the **Node ID** Node 1.
3. Select the **Slot** (Slot 2).
4. Select the **Interface Name**, for example, node1:ethernet2/1.
5. Select the **Interface Type** as **Aggregate Ethernet**.
6. Select the **Aggregate Group**, such as ae1.
7. Click **OK**.

STEP 18 | Add a second interface member to the MC-LAG on the client side, assigning **Node ID** as Node 2, Slot 2, node2:ethernet2/11. Select the same **Aggregate Group** (ae1).

STEP 19 | Add an interface member to the MC-LAG on the server side. Select Node 1, Slot 4, node1:ethernet4/16. Select the **Aggregate Group**, ae2.

STEP 20 | Add a second interface member to the MC-LAG on the server side, assigning it to Node 2, Slot 6, node2:ethernet6/16. Select the **Aggregate Group**, such as ae2.

STEP 21 | (PAN-OS 11.1.5 and later 11.1 releases) Prepare to configure MACsec for the HSCI ports by creating a MACsec profile.

IMPORTANT: When you migrate from a non-MACsec cluster to a MACsec cluster, you must do so during a maintenance window.

1. Select **Network** > **Network Profiles** > **MACsec Profile** and select **MACsec Policy**.
2. **Add** a MACsec crypto profile by **Name**, which contains zero or more alphanumeric characters, underscores (_), hyphens (-), or dots (.). The Name is a maximum of 31 characters.
3. Select the **Encryption** type: **AES-128-GCM** (default) or **AES-256-GCM**.
4. Select the **Confidentiality Offset** to specify a number of bytes (starting from the frame header), after which MACsec encrypts the bytes in a frame. Values are **0** (default), **30**, or **50**.
5. **Enable SCI Include** to include the Secure Channel Identifier (SCI) tag in the Security Tag field of the MACsec header. The default is disabled.
6. **Enable Anti Replay** to enable replay protection. This allows a MACsec port to accept frames out of order if they are within the Anti Replay Window. The default is disabled.
7. Specify the **Anti Replay Window** size in the range 0 to 65,535; the default is 16,384. This value determines the range of packet numbers that the port will accept, for packets that might be out of order. The port will accept packets with a packet number greater than or equal to the last packet number minus the window size. For example, after the

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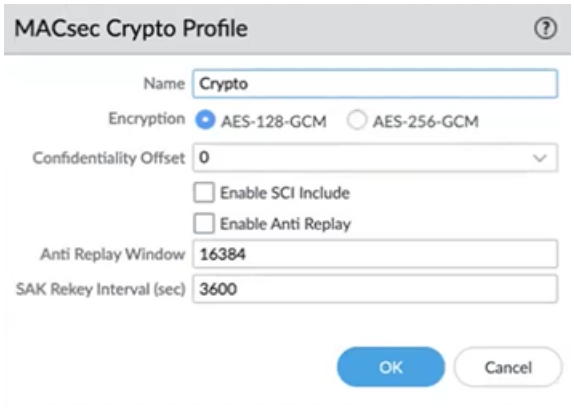
port receives packet number 12, if the window is 5, the port will subsequently accept only packets numbered 7 or higher.

Specify an Anti Replay window size based on the traffic rate because the HSCI ports will likely experience packets arriving out of order due to prioritization or load balancing in the network.

8. Specify the **SAK Rekey Interval (sec)** in seconds; range is 60 to 86,400; default is 3,600. Connectivity Association participants negotiate a Secure Association agreement, which includes a cipher suite and keys. A Key Server generates a Secure Association Key (SAK) from the Connectivity Association Key (CAK); the SAK is refreshed at the SAK rekey interval. The end-to-end nodes use the SAK to encrypt traffic for a given session. To avoid the SAKs of all interfaces being refreshed at the same time, MACsec might spread the rekeying actions evenly by subtracting a small random number from the SAK rekey interval.
9. Click **OK**.



STEP 22 | (PAN-OS 11.1.5 and later 11.1 releases) Create a MACsec Pre Shared Key Profile.

1. Select **Network** > **Network Profiles** > **MACsec Profile** and select **Pre Shared Key**.
2. **Add** a MACsec Pre Shared Key Profile by **Name**, which contains zero or more alphanumeric characters, underscores (_), hyphens (-), dots (.), or spaces. The Name is a maximum of 31 characters.
3. Enter the **CKN** (Connectivity Association Key Name) that identifies the Connectivity Association Key. The CKN is 1 byte to 32 bytes of hexadecimal string (2 to 64 hex digits) with an even number of digits; for example, 1234.
4. Enter the **CAK** (Connectivity Association Key), which is a 16-byte hexadecimal string (32 hex digits) for AES-128-GCM, or a 32-byte hexadecimal string (64 hex digits) for AES-256-GCM. The CAK is used to generate all other keys used for MACsec. An

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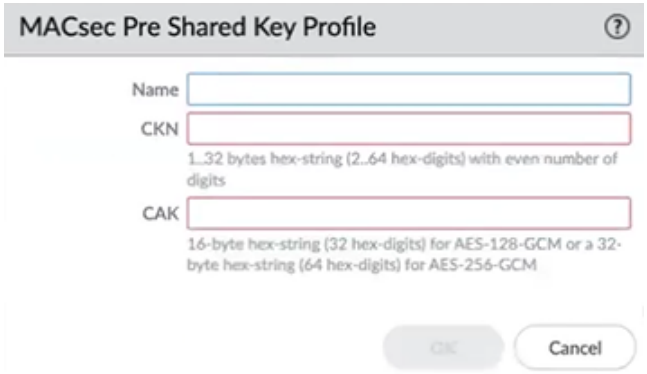
example CAK for AES-128-GCM is 12345678912345678912345678912345. A CAK for AES-256-GCM is twice as long as a CAK for AES-128-GCM.

The Pre Shared Key, CKN, and CAK must match on the two ends of the HSCI-A link. Likewise, the Pre Shared Key, CKN, and CAK must match on the two ends of the HSCI-B link.

5. Click **OK**.

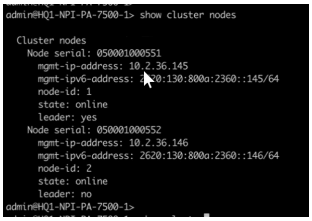


STEP 23 | (PAN-OS 11.1.5 and later 11.1 releases) **Commit** to Panorama.

STEP 24 | (PAN-OS 11.1.5 and later 11.1 releases) **Push to Devices** and **Push Templates** so that the devices have the MACsec profiles, thus allowing you to later push the cluster configuration with MACsec enabled.

STEP 25 | (PAN-OS 11.1.5 and later 11.1 releases) Identify the leader node and nonleader node.

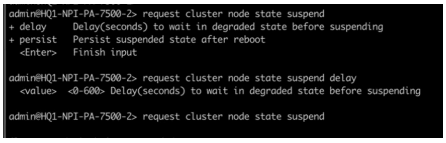
1. Access the CLI on one of the firewall nodes. (If you already know which node is the nonleader, access the CLI on the nonleader node.)
2. **>show cluster nodes**



STEP 26 | (PAN-OS 11.1.5 and later 11.1 releases) Suspend the nonleader node.

1. Access the CLI on the nonleader node.
2. **>request cluster node state suspend**

The suspension takes approximately 120 seconds (the default delay value).



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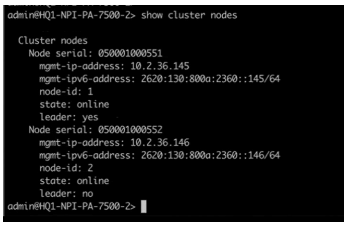
STEP 27 | (PAN-OS 11.1.5 and later 11.1 releases) Confirm the nonleader node is suspended.

1. **>show cluster nodes**



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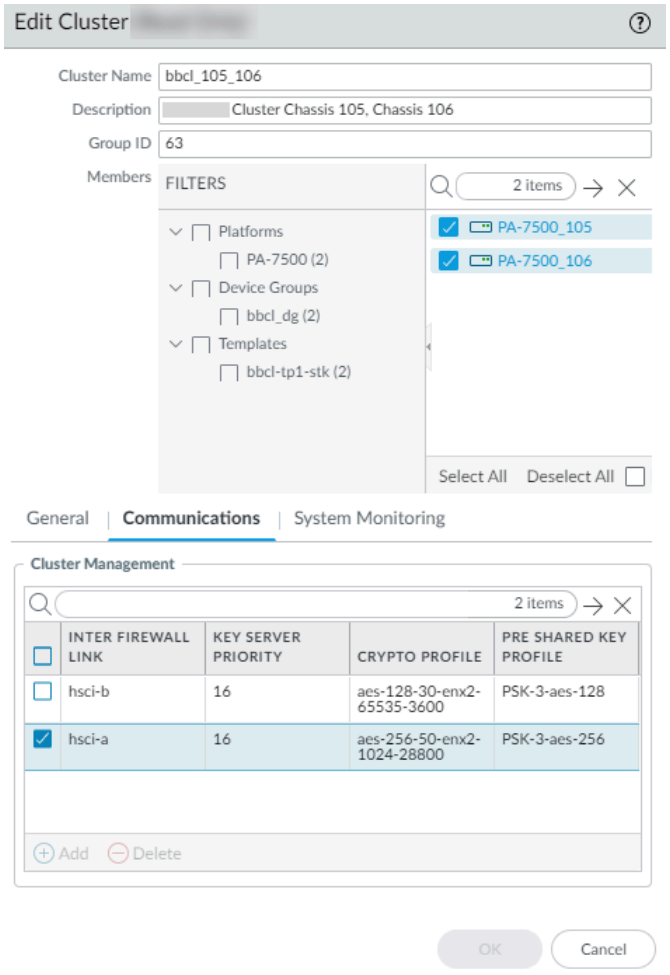
STEP 28 | (PAN-OS 11.1.5 and later 11.1 releases) Apply your MACsec profiles to the HSCI-A and HSCI-B ports.

1. Select **Panorama** > **Firewall Clusters** and select **Summary View**.
2. Select **PA-Series** or **All Clusters**.
3. Select a **Cluster Name** to edit the cluster.
4. Select **Communications**.
5. Select **hsci-a** and enter the **Key Server Priority** in the range from 0 to 255; the default is 16. The lower the value, the higher the priority of the Key Server.

   If the priority values for the HSCI-A links on the two nodes are equal, the node with the lower MAC address is the Key Server. The Key Server (one of the nodes in the cluster) selects and advertises a cipher suite, and also generates the SAK from the CAK.
6. For **Crypto Profile**, select the MACsec crypto profile you created or select the **default** profile.
7. For **Pre Shared Key Profile**, select the profile you created.
8. Select **hsci-b** and perform the same steps to assign a Key Server Priority and apply the crypto profile and pre-shared key Profile to the port.
9. Click **OK**.



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STEP 29 | (PAN-OS 11.1.5 and later 11.1 releases) **Commit**, **Commit to Panorama**, and **Commit**.

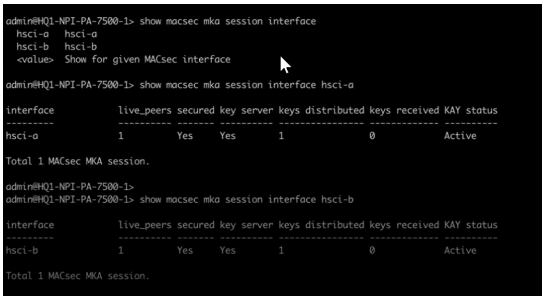
STEP 30 | (PAN-OS 11.1.5 and later 11.1 releases) **Push to Devices** and **Push** , which enables MACsec on the ports.

STEP 31 | (PAN-OS 12.1.2 and later 12.1 releases) (Optional) If you renamed the MACsec key profile or replaced the MACsec pre-shared key and your commit fails, select **Edit Selections** and perform the following substeps:

1. If you need to push a device group configuration, perform these next three substeps. Select the **Device Groups** tab. Select the affected device group or groups. Select **Include Device and Network Templates**. Select **Include Firewalls Clusters**.
2. Select the **Templates** tab and deselect the affected template.
3. Select the **Firewall Clusters** tab. Deselect **Filter Selected**. Deselect the affected cluster.
4. If you need to push the template configuration, perform these next three substeps. Select the **Device Groups** tab and deselect the affected device group.
5. Select the **Templates** tab and select the affected template. Select **Include Firewall Clusters**.
6. Select the **Firewall Clusters** tab. Deselect **Filter Selected**. Deselect the affected cluster.
7. Now that you have edited the selections, repeat Steps 29 and 30 to commit and push your changes.

STEP 32 | (PAN-OS 11.1.5 and later 11.1 releases) Confirm MACsec is enabled on both ports.

1. Access the CLI.
2. **>show macsec mka session interface hsci-a**
3. **>show macsec mka session interface hsci-b**



STEP 33 | (PAN-OS 11.1.5 and later 11.1 releases) Unsuspend the nonleader node.

1. Access the CLI on the nonleader node.
2. **>request cluster node state unsuspend**

It takes a few minutes to unsuspend the nonleader node; the length of time depends on the traffic level.

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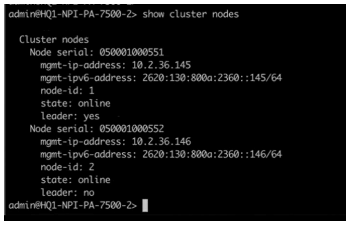
STEP 34 | (PAN-OS 11.1.5 and later 11.1 releases) Verify that both nodes are online.

1. **>show cluster nodes**



STEP 35 | (PAN-OS 11.1.7 and later 11.1 releases) (Optional) Enable multiple virtual systems for the firewalls in the template.

1. Select **Device** > **Virtual Systems** and select the **Template** by name.
2. For **Mode**, select **Multi VSYS**.

STEP 36 | Create Security policy rules.

1. Select **Policies** and select the Device Group.
2. Create Security policy rules to control access, such as allowing a specific source zone, destination zone, address, user, device, application, and service.

STEP 37 | Configure routing and other features your firewalls require (except for HA). Refer to the PAN-OS Administrator's Guide and the PAN-OS Networking Administrator's Guide.

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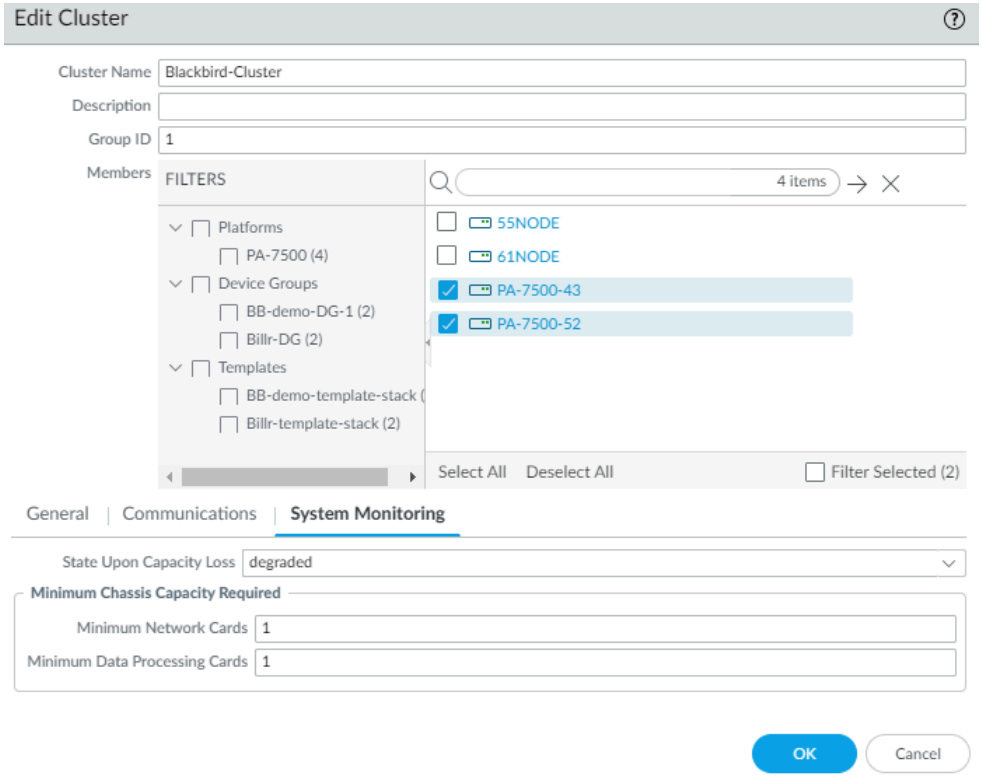
STEP 38 | Configure system monitoring for the NGFW cluster.

1. Select **Panorama** > **Firewall Clusters** > **Summary View** and select a cluster or a single firewall in the cluster.
2. Select **System Monitoring**.



3. Select the **State Upon Capacity Loss**:
   - **degraded**—Specifies that the firewall will be in a DEGRADED state if the count of functional network cards or data processing cards goes below the configured Minimum Network Cards or Minimum Data Processing Cards, respectively. Furthermore:
     - A cluster node in DEGRADED state has traffic ports down, but is still a part of the sharded (fragmented) member table.
     - Data processing card resources of a DEGRADED cluster node can be used to process traffic and for Layer 7 processing.
     - A cluster node in INIT or ONLINE state transitions to DEGRADED state when a soft fault is reported to the cluster node state machine.
     - A cluster node in ONLINE state transitions to DEGRADED state (and a node in DEGRADED state remains in DEGRADED state) if you suspend the cluster node after a delay (using the CLI operational command: **`request cluster node`**

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`state suspend`). A delay allows data planes to gracefully complete Layer 7 processing or other processes that were underway.

- If all soft faults are cleared, the cluster node transitions to INIT state.
- If a hard fault occurs, a cluster node in DEGRADED state transitions to FAILED state.
- A cluster node in DEGRADED state transitions to SUSPENDED state if the maximum number of state flaps is seen or you suspend the cluster node.
- If a chassis has no functional data processing cards remaining (all DPC slots are powered off or in a FAILED state), the cluster node state will be FAILED, even if you configured **degraded**, because having no functional DPC is a hard fault.

- **failed**—Specifies that the firewall will be in a FAILED state if the count of functional network cards or data processing cards goes below the configured Minimum Network Cards or Minimum Data Processing Cards, respectively. Furthermore:
  - A cluster node in FAILED state has traffic ports down and isn't part of the sharded member table.
  - Data processing card resources of a FAILED cluster node can't be used to process traffic or for Layer 7 processing.
  - A cluster node in INIT, ONLINE, or DEGRADED state transitions to FAILED state when a hard fault is reported to the cluster node state machine.
  - A cluster node in FAILED state transitions to INIT state if all hard faults are cleared.
  - A cluster node in FAILED state transitions to SUSPENDED state if the maximum number of state flaps is seen or you suspend the cluster node.

NGFW Clusters *provides a list of hard faults and soft faults.*

4. In the Minimum Chassis Capacity Required section, enter the **Minimum Network Cards** required; range is 1 to 7, default is 1. When fewer Network Cards than the configured value are functional, the firewall in the cluster goes to the State Upon Capacity Loss that you configured (degraded or failed).
5. Enter the **Minimum Data Processing Cards** required; range is 1 to 7, default is 1. When fewer Data Processing Cards than the configured value are functional, the firewall in the cluster goes to the State Upon Capacity Loss that you configured (degraded or failed).
6. Click **OK**.

STEP 39 | (Optional) Configure Log Forwarding for the PA-7500 Series firewall.

STEP 40 | Push the configuration from Panorama to the firewalls in the cluster.

1. **Commit** and **Commit and Push** and **Commit**.
2. **Commit** and **Push to Devices**, **Push All Changes**, select the cluster, and **Push** the configuration to both nodes.

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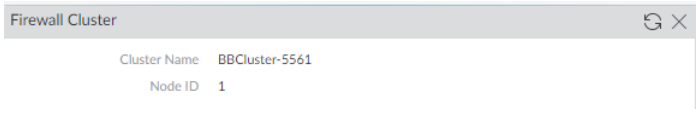
STEP 41 | (PAN-OS 11.1.7 and later 11.1 releases) If you enabled multiple virtual systems for the Panorama device template, enable multiple virtual systems for an individual firewall.

1. Log on to an individual firewall (not Panorama).
2. Select **Device** > **Setup** > **Management**.
3. In General Settings, select **Multi Virtual System Capability**.
4. Click **OK**.
5. Repeat this step for the other firewall in the cluster.

STEP 42 | View NGFW cluster information for an individual firewall on its Dashboard.

1. Log on to an individual firewall (not Panorama).
2. Select **Dashboard** and for Widgets, select **System** > **Firewall Cluster** to view the Firewall Cluster card, which displays the Cluster Name and Node ID.



STEP 43 | View NGFW Cluster Summary and Monitoring information and health for the cluster.

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Strata Cloud Manager

Use this process to configure an NGFW Cluster of up to two firewalls using Strata Cloud Manager.

STEP 1 | Select **Configuration** > **NGFW and Prisma Access**.

STEP 2 | For the **Configuration Scope**, select a folder.

The folder selection cannot be more than four levels deep.

STEP 3 | Enable **Device Scope Configuration** for the Node 1 and Node 2 devices.

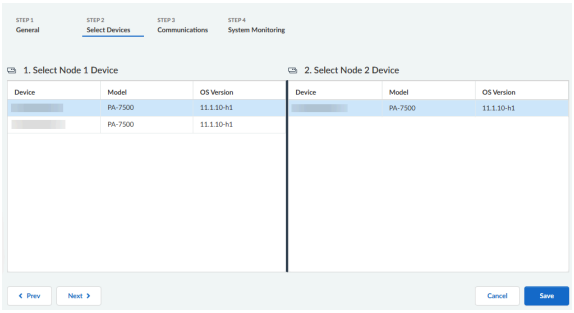
*This step is mandatory. You must enable **Device Scope Configuration** for both the Node 1 and Node 2 devices before initiating clustering. Without this setting enabled on both devices, the Strata Cloud Manager pushed clustering configuration will not apply.*

STEP 4 | Create an NGFW cluster.

1. Select **Overview** and in the **Cluster Setup** pane, select the gear icon or click on the number of clusters already configured.
2. Select **Clusters** and **Add Cluster**.
3. On the **General** tab, enter the **Cluster Name** containing zero or more alphanumeric characters, underscores (_), hyphens (-), dots (.), or spaces.
4. (Optional) Enter the **Group ID** (in the range 1 to 63; the default is 1).

STEP 5 | Add devices to the cluster.

1. On the **Select Devices** tab, select the Node 1 Device (a firewall to assign to the cluster).
2. Select the Node 2 Device (the second firewall to assign to the cluster).



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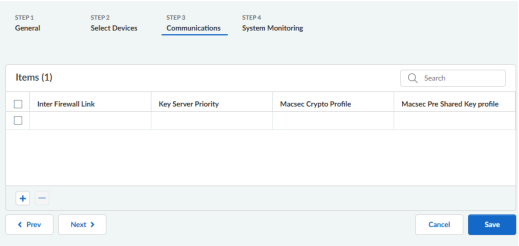
STEP 6 | Configure the HSCI-A and HSCI-B links.

If the firewalls are directly connected via fibre, then MACSec configuration is optional. You may click **Next** to skip this step.

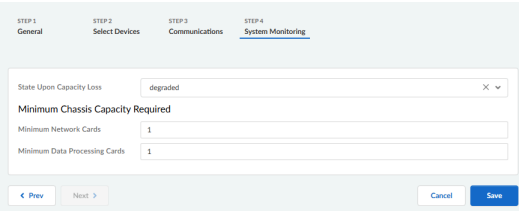
If the firewalls are connected over a dark-fibre network between sites, MACSec configuration is mandatory.

1. On the **Communications** tab, select the **+** and add an **Inter Firewall Link** by selecting **hsci-a**.
2. Enter the **Key Server Priority** in the range 1 to 255.
3. Select a **MACsec Crypto Profile** to apply to the link.
4. Select a **MACsec Pre Shared Key Profile** to apply to the link.
5. Repeat this step to configure the **hsci-b** link.



STEP 7 | Configure system monitoring for the NGFW cluster.

1. On the **System Monitoring** tab, select the **State Upon Capacity Loss**: **degraded** or **failed**.
2. Enter the **Minimum Network Cards** required; range is 1 to 7; default is 1.
3. Enter the **Minimum Data Processing Cards** required; range is 1 to 7; default is 1.
4. **Save** the configuration.



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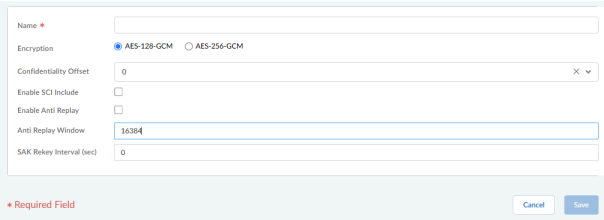
STEP 8 | Create a MACsec policy to apply to the HSCI ports.

1. On the **MACsec Profiles** tab, **Add MACsec Policy**.
2. Enter a **Name** for the profile, which contains zero or more alphanumeric characters, underscores (_), hyphens (-), or dots (.). The name is a maximum of 31 characters.
3. Select the **Encryption** type: **AES-128-GCM** (default) or **AES-256-GCM**.
4. Select the **Confidentiality Offset** to specify a number of bytes (starting from the frame header), after which MACsec encrypts the bytes in a frame. The value is **0** (default), **30**, or **50**.
5. **Enable SCI Include** to include the Secure Channel Identifier (SCI) tag in the Security Tag field of the MACsec header. The default is disabled.
6. **Enable Anti Replay** to enable replay protection. This allows a MACsec port to accept frames out of order if they are within the Anti Replay Window. The default is disabled.
7. Enter the **Anti Replay Window** size in the range 0 to 65,535; the default is 16,384. This value determines the range of packet numbers that the port will accept, for packets that might be out of order. The port will accept packets with a packet number greater than or equal to the last packet number minus the window size. For example, after the port receives packet number 12, if the window is 5, the port will subsequently accept only packets numbered 7 or higher.

   Specify an Anti Replay window size based on the traffic rate because the HSCI ports will likely experience packets arriving out of order due to prioritization or load balancing in the network.
8. Enter the **SAK Rekey Interval (sec)** in seconds; range is 60 to 86,400; default is 3,600. Connectivity Association participants negotiate a Secure Association agreement, which includes a cipher suite and keys. A Key Server generates a Secure Association Key (SAK) from the Connectivity Association Key (CAK); the SAK is refreshed at the SAK rekey interval. The end-to-end nodes use the SAK to encrypt traffic for a given session. To avoid the SAKs of all interfaces being refreshed at the same time, MACsec might spread the rekeying actions evenly by subtracting a small random number from the SAK rekey interval.



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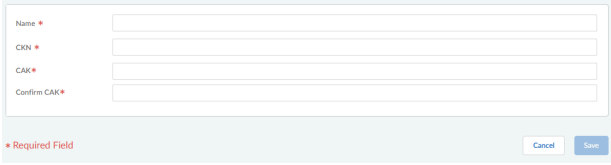
STEP 9 | Create a MACsec pre-shared key profile.

1. On the **MACsec Profiles** tab, **Add Pre Shared Key**.
2. Enter a **Name** for the pre shared key profile, which contains zero or more alphanumeric characters, underscores (_), hyphens (-), dots (.), or spaces. The name is a maximum of 31 characters.
3. Enter the **CKN** (Connectivity Association Key Name) that identifies the Connectivity Association Key. The CKN is 1 byte to 32 bytes of hexadecimal string (2 to 64 hex digits) with an even number of digits; for example, 1234.
4. Enter the **CAK** (Connectivity Association Key), which is a 16-byte hexadecimal string (32 hex digits) for AES-128-GCM, or a 32-byte hexadecimal string (64 hex digits) for AES-256-GCM. The CAK is used to generate all other keys used for MACsec. An example CAK for AES-128-GCM is 12345678912345678912345678912345. A CAK for AES-256-GCM is twice as long as a CAK for AES-128-GCM.
5. **Confirm CAK**.
6. **Save** the profiles.



STEP 10 | (PA-7500 only) Configure System Monitoring. Enter the parameters of the NGFW Clusters if components of the PA-7500 fail.

1. State Upon Capacity Loss" is when the components such as NPCs, DPCs, PSUs fail, with two options being available.
   - "degraded" means the node is put into a degraded state <define with this means and the affect to traffic flow>
   - "failed" means the node is put into a failed state <define with this means and the affect to traffic flow>
2. Define the "Minimum Network Cards" required to be active for the node to continue to participate in the NGFW Cluster, with the default value being 1.
3. Define the "Minimum Data Processing Cards" required to be active for the node to continue to participate in the NGFW Cluster, with the default value being 1.

STEP 11 | Click "Save" to complete the workflow.

STEP 12 | Once the workflow is saved, it will take the operator back to the "Cluster Setup" screen. At this point.

1. Choose the new cluster, click "Initiate Clustering"
2. At the validation menu, click "Yes"

   Strata Cloud Manager sends OP command to the devices to make them node 1 and node 2 from the default node 0 (i.e. unclustered) state. This should trigger an automatic reboot of both devices, however in some circumstances a manual reboot will be required.

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STEP 13 | Once the devices are up and active after their reboot, check each node that there are cluster mode by connecting to each NGFW device:

Remove a Device from an NGFW Cluster

Use this procedure to remove a single firewall from an NGFW Cluster using Strata Cloud Manager.

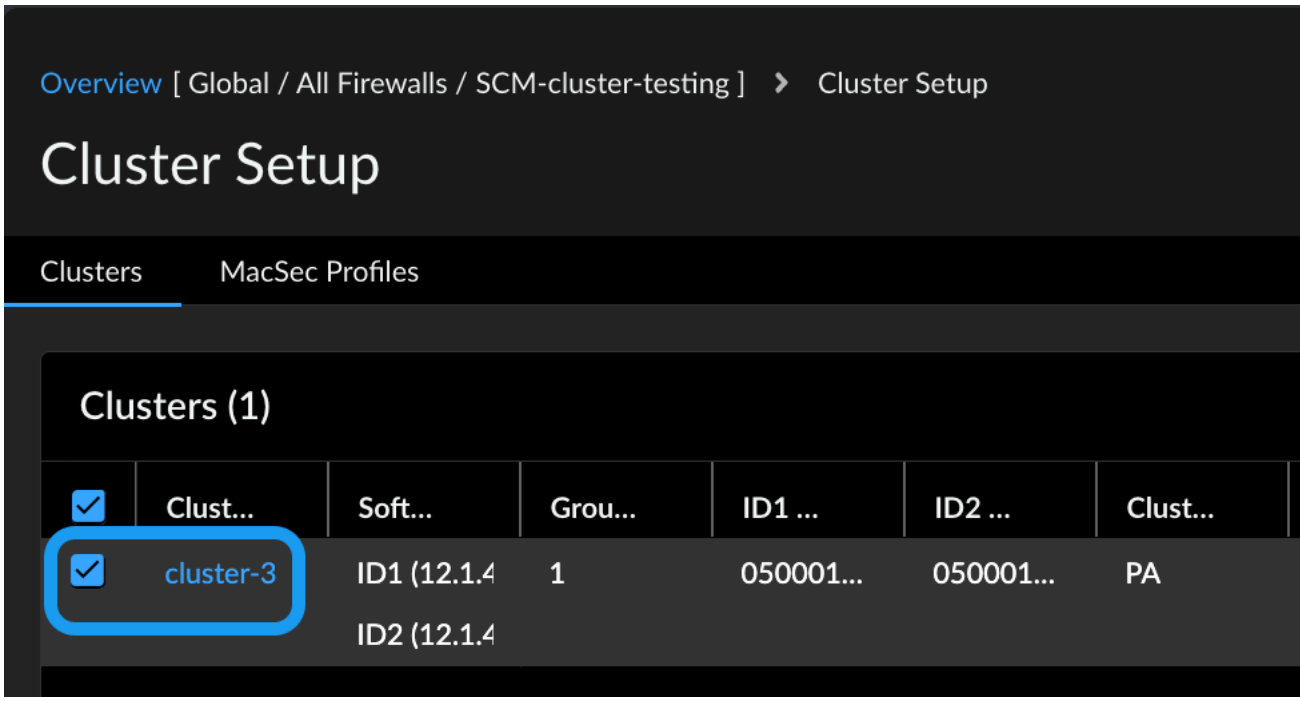
STEP 1 | Select **Configuration** > **NGFW and Prisma Access**.

STEP 2 | For the **Configuration Scope**, select a folder.

The folder selection cannot be more than four levels deep.

STEP 3 | Remove a device from an NGFW cluster.

1. Select **Overview** and in the **Cluster Setup** pane, click the cluster name to open the workflow.



2. Select **Clusters** and **Add Cluster**.
3. On the **General** tab, enter the **Cluster Name** containing zero or more alphanumeric characters, underscores (_), hyphens (-), dots (.), or spaces.
4. (Optional) Enter the **Group ID** (in the range 1 to 63; the default is 1).

STEP 4 | Click the “Cluster Setup” under the cluster folder for the device and click on the cluster name to open up the workflow.

STEP 5 | Click to the “Step 2 Select Devices” screen and unselect the device in the “Select Node 2 Device” panel. For this, you may need to toggle the devices in “Select Node 1 Device” panel to unselect/unhighlight the device in Node 2

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STEP 6 | Click “Save”.

STEP 7 | After you click “Save”, the workflow will be closed and then the device will be removed from the cluster folder.

Strata Cloud Manager sends OP command to the node 2 device to make to node 0 (i.e. unclustered) state. This should trigger an automatic reboot of the node 2 device, however in some circumstances a manual reboot will be required.

STEP 8 | Once the devices are up and active after their reboot, connect to node 2 directly and check that it is no longer in cluster mode:

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NGFW Cluster Summary and Monitoring

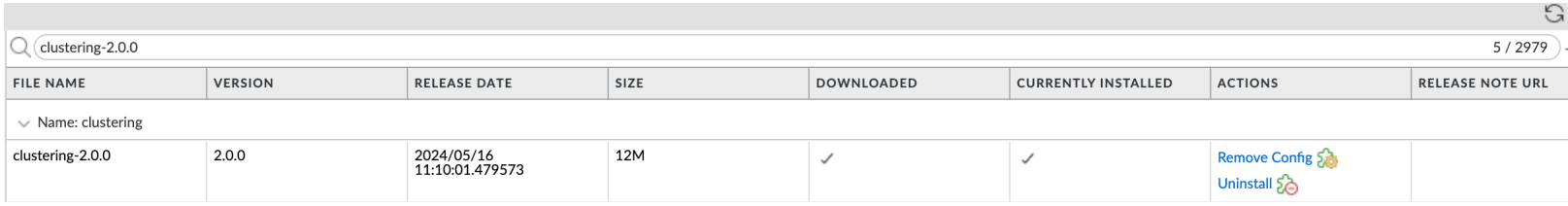
Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	One of the following: ❑ Panorama PAN-OS 12.1.2 (Panorama Clustering Plugin is bundled with Panorama) ❑ Panorama PAN-OS 11.1.5 and Panorama Clustering Plugin 2.0.0 ❑ Panorama PAN-OS 11.1.3 and Panorama Clustering Plugin 2.0.0

After you Configure an NGFW Cluster, you can view the summary and monitoring information about the cluster.

The CN-Series and PA-Series cluster plugin visibility data is not in real time; it's delayed by five minutes.

The prerequisites for cluster summary and monitoring are:

- You must **Enable** > **Firewall Clusters** from the **Panorama** > **Admin Roles** > **Web UI** list (enabled by default). For more information, see Configure an Admin Role Profile.
- You must install the Panorama Clustering Plugin (a version that is compatible with the Panorama version you're running) from **Panorama** > **Plugins**. Scroll to **clustering** plugin.



Summary View

View the cluster summary under **Panorama** > **Firewall Clusters** > **Summary View**. In the **Clusters** field, select **PA-Series** (or **All Clusters**).

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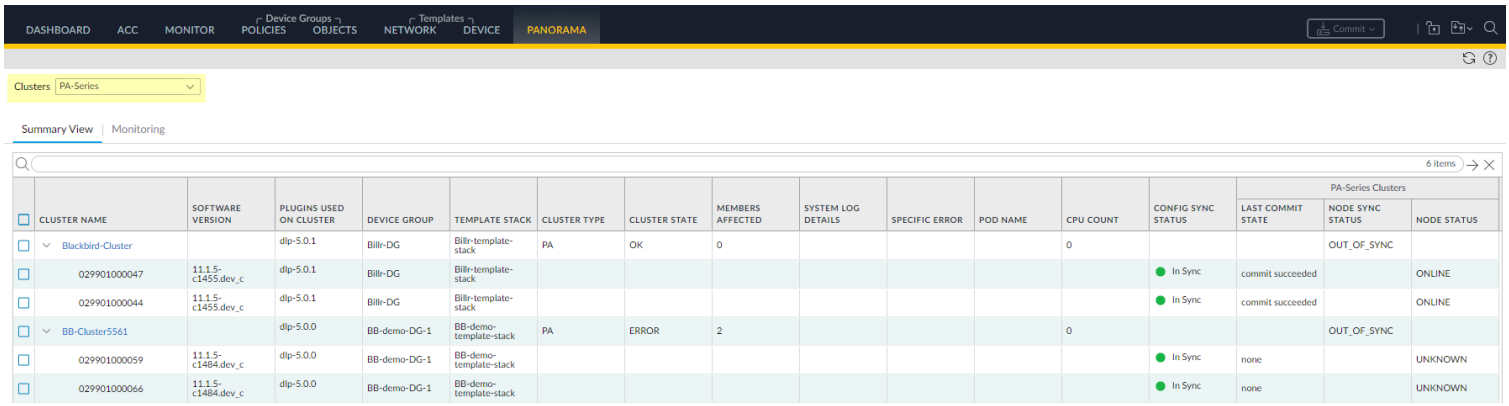

Field	Description
Cluster Name	Name of the firewall cluster.
Software Version	PAN-OS version.
Plugins Used on Cluster	List of plugins used on the cluster.
Device Group	Name of the device group associated with the cluster.
Template Stack	Name of the template stack associated with the cluster.
Cluster Type	Type of cluster, such as PA or CN.
Cluster State	Displays the health of the cluster, which is derived from node status of all nodes in the cluster. Cluster state will be: • OK if all nodes are in ONLINE state. • IMPACTED if there is at least one node in ONLINE state and another node isn't in ONLINE state. • ERROR if there is not a single node in ONLINE state.
Members Affected	Number of impacted cluster members and their names.
System Log Details	Displays the details of the system events.
Specific Error	List of specific errors in the cluster. Click the link to view more details about the error under **Monitor** > **Logs** > **System** where you can view logs.
Pod Name	(CN-Series cluster only) Name of the pod.
CPU Count	Number of CPUs used.
Config Sync Status	(PA-Series Clusters only) Config synchronization status between Panorama and the firewalls in the PA cluster. Status can be In Sync or Out of Sync. After you successfully add firewalls to the

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Field	Description
	cluster, commit, and push, the Config Sync Status displays as In Sync.
Last Commit State	(PA-Series cluster only) State of the last attempted commit (not the actual state of the cluster): • commit failed • commit succeeded • commit succeeded with warnings • commit reverted
Node Sync Status	(PA-Series cluster only) Synchronization status of the Node Flow Table: • IN_SYNC • UPDATING • OUT_OF_SYNC
Node Status	(PA-Series Clusters only) Possible status of a cluster node: • UNKNOWN—Clustering is not enabled. Node remains in this state until a cluster configuration push from Panorama or a commit enables clustering. • INIT—Node transitions from UNKNOWN to INIT after clustering is enabled. Node remains in INIT state until cluster initialization of node is complete. Node transitions to ONLINE after a timeout. • ONLINE—Node is passing traffic and working as expected. • DEGRADED—Node transitions to DEGRADED when a soft fault occurs. Node can transition back to INIT if all the faults are resolved. • FAILED—Node transitions to FAILED state when a hard fault occurs. • SUSPENDED—Triggered by administrator. Another cause of SUSPENDED state is if a node state flaps to DEGRADED or FAILED state repeatedly; the node is SUSPENDED after six flaps. An administrator can unsuspend the node. SUSPENDED state has traffic ports down and doesn't allow L7 continuity.

Monitoring

Monitor the health information of the PA-Series (NGFW) cluster under **Panorama** > **Firewall Clusters** > **Monitoring**. In the **Clusters** field, select **PA-Series** (or **All Clusters**).

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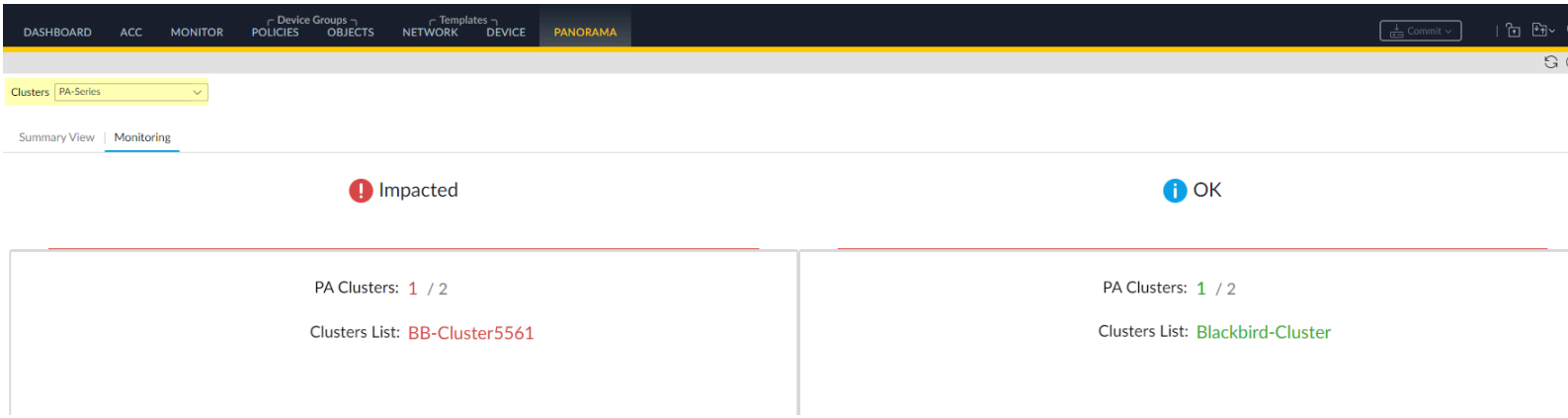
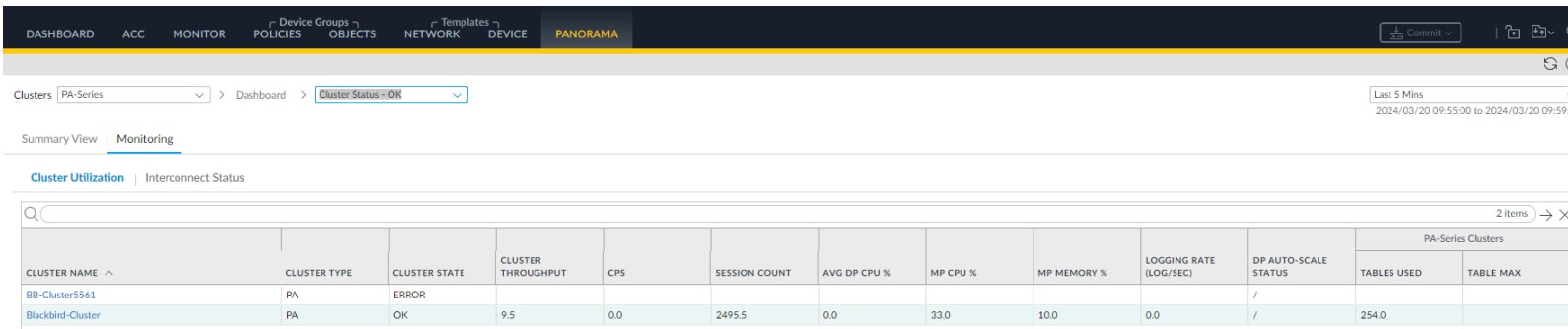
The CN-Series and PA-Series cluster plugin visibility data is not in real time; it's delayed by five minutes.



Field	Description
Clusters	Select the Cluster type **PA-Series**.
Impacted	List of impacted clusters. • **PA Clusters**—Number of impacted PA-Series clusters out of total PA-Series Clusters, • **Clusters List**—Displays the list of clusters that are impacted. Click to view detailed information about the clusters in the **Cluster Utilization** and **Interconnect Status** dashboards.
OK	List of clusters that aren't impacted. • **PA Clusters**—Number of PA-Series clusters that aren't impacted out of total PA-Series Clusters. • **Clusters List**—Displays the list of clusters that aren't impacted. Click to view detailed information about the clusters in the **Cluster Utilization** and **Interconnect Status** dashboards.

Click in the **Monitoring** window or select a cluster to view **Cluster Utilization**.



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Field	Description
Dashboard	Select to switch views between **Cluster Status - OK** and **Cluster Status - Impacted**.
Timeframe	Select timeframe of data displayed: • Last 5 Mins • Last Hr • Last 6 Hrs • Last 12 Hrs • Last 24 Hrs • Last Calendar Day • Custom—Select **Start Date** and time, **End Date** and time, and click **OK**. *The CN-Series and PA-Series cluster plugin visibility data is not in real time; it's delayed by five minutes.*
Cluster Name	Name of the firewall cluster.
Cluster Type	Type of cluster (CN or PA).
Cluster State	Displays the health of the cluster, which is derived from node status of all nodes in the cluster. Cluster state will be: • OK if all nodes are in ONLINE state. • IMPACTED if there is at least one node in ONLINE state and another node isn't in ONLINE state. • ERROR if there is not a single node in ONLINE state.
Cluster Throughput	Firewall cluster throughput in Gbps.
CPS	Number of connections per second.
Session Count	Number of sessions.
Avg DP CPU %	Average DP CPU utilization over the selected time period.
MP CPU %	Management plan CPU utilization in percentage.
MP Memory %	Management plan memory utilization in percentage.
Logging Rate (Log/Sec)	Rate at which the logs are being generated on the cluster.
DP Auto-Scale Status	Dataplane autoscale details.

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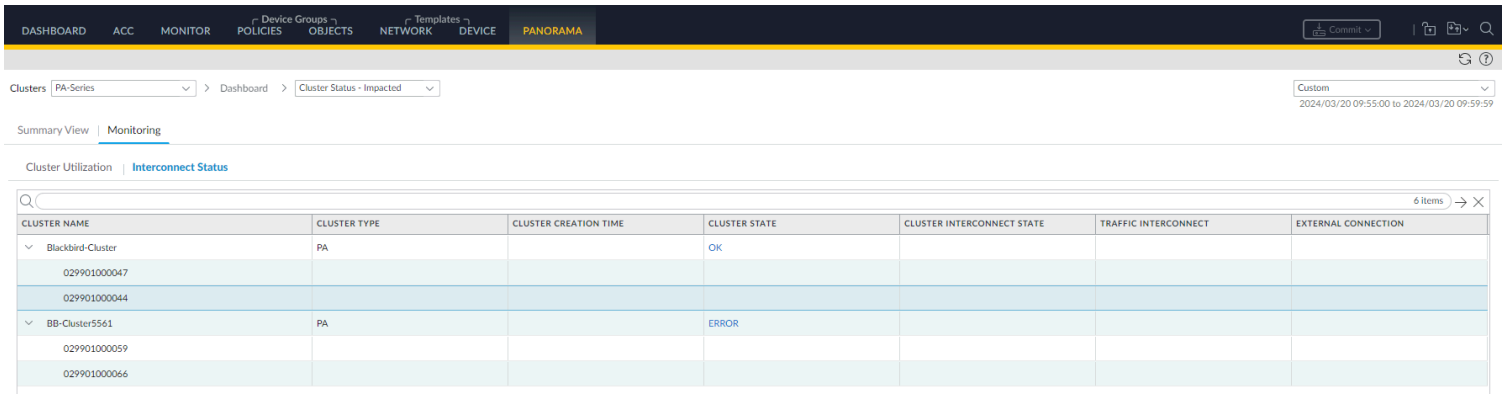
Field	Description
Tables Used	(PA-Series cluster only) Entries in the node flow table that are in use.
Table Max	(PA-Series cluster only) Total possible number of entries in the node flow table.

Click in the **Monitoring** window or select a cluster to view **Interconnect Status**.



Field	Description
Dashboard	Select to switch views between **Cluster Status - OK** and **Cluster Status - Impacted**.
Timeframe	Select timeframe of data displayed: • Last 5 Mins • Last Hr • Last 6 Hrs • Last 12 Hrs • Last 24 Hrs • Last Calendar Day • Custom—Select **Start Date** and time, **End Date** and time, and click **OK**. *The CN-Series and PA-Series cluster plugin visibility data is not in real time; it's delayed by five minutes.*
Cluster Name	Name of the firewall cluster.
Cluster Type	Type of cluster (CN or PA).
Cluster Creation Time	The time of cluster creation.

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Field	Description
Cluster State	Displays the health of the cluster, which is derived from node status of all nodes in the cluster. Cluster state will be: • OK if all nodes are in ONLINE state. • IMPACTED if there is at least one node in ONLINE state and another node isn't in ONLINE state. • ERROR if there is not a single node in ONLINE state. Click the cluster state link to view more details about the impacted cluster.
Cluster Interconnect State	Displays cluster interconnectivity. • Click the interconnect state link to view more details about the impacted cluster.
Traffic Interconnect	Status of traffic interconnectivity.
External Connection	Status of external connectivity.
Impacted Links	(CN-Series cluster only) Number of impacted links.
Management Connectivity	(CN-Series cluster only) Number of management connections.
Impacted Members	(CN-Series cluster only) List of impacted cluster members.
Time Stamp Uptime	(CN-Series cluster only) Uptime time stamp.
Time Stamp Downtime	(CN-Series cluster only) Downtime time stamp.

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Monitoring

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama) • NGFW (Managed by Strata Cloud Manager)	❑ Support license ❑ (Panorama) Device management license ❑ AIOps for NGFW Premium license (use the Strata Cloud Manager app)

Monitoring capabilities in Palo Alto Networks firewalls provide you with comprehensive visibility into network traffic, threats, and system performance. When you use these monitoring features, you gain critical insights that accelerate incident response and enable proactive network management.

The Dashboard serves as your quick reference for firewall status, displaying key metrics like software version, interface status, and resource utilization. You can customize widgets to show the information most relevant to your security operations and adjust refresh intervals to maintain current data. This immediate visibility helps you quickly identify potential issues before they impact your network.

For deeper analysis, the Application Command Center (ACC) offers interactive, graphical summaries of network activity derived from firewall logs. You can apply filters to examine specific traffic segments, customize tab layouts to match your monitoring priorities, and export views as PDFs for reporting. The "Compromised Hosts" widget is particularly valuable, aggregating correlated events to help you identify potentially infected systems that require immediate attention.

Reports provide detailed insights into long-term network behavior. You can generate App Scope reports to track application usage changes, bandwidth consumption, and emerging threats. Custom reports allow you to tailor analysis to your organization's specific security requirements, while specialized Botnet, SaaS Application Usage, and User Activity reports help identify security gaps. You can schedule reports for automatic email delivery, ensuring stakeholders receive regular security updates.

The Automated Correlation Engine (ACE) enhances your threat detection capabilities by analyzing logs for suspicious patterns. When investigating specific incidents, Packet Captures (Pcaps) provide detailed traffic data for troubleshooting. For comprehensive audit trails, the various log types (Traffic, Threat, URL Filtering, etc.) can be filtered, exported, and forwarded to external monitoring systems.

To implement effective monitoring, you must configure appropriate log storage settings, establish external log forwarding if needed, and customize dashboards and reports to align with your security objectives. These monitoring capabilities enable you to identify security incidents, track compliance, optimize network performance, and demonstrate security effectiveness to management.

To use the monitoring functionality with the PA-410 you must manage PA-410 firewalls through a Panorama management server.

Use the Dashboard

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama) • NGFW (Managed by Strata Cloud Manager)	❑ Support license ❑ (Panorama) Device management license ❑ AIOps for NGFW Premium license (use the Strata Cloud Manager app)

The **Dashboard** tab widgets show general firewall information, such as the software version, the operational status of each interface, resource utilization, and up to 10 of the most recent entries in the threat, configuration, and system logs. All of the available widgets are displayed by default, but each administrator can remove and add individual widgets, as needed. Click the refresh icon to update the dashboard or an individual widget. To change the automatic refresh interval, select an interval from the drop-down (**1 min**, **2 mins**, **5 mins**, or **Manual**). To add a widget to the dashboard, click the widget drop-down, select a category and then the widget name. To delete a widget, click in the title bar. The following table describes the dashboard widgets.

Dashboard Charts	Descriptions
Top Applications	Displays the applications with the most sessions. The block size indicates the relative number of sessions (mouse-over the block to view the number), and the color indicates the security risk—from green (lowest) to red (highest). Click an application to view its application profile.
Top High Risk Applications	Similar to Top Applications, except that it displays the highest-risk applications with the most sessions.
General Information	Displays the firewall name, model, PAN-OS software version, the application, threat, and URL filtering definition versions, the current date and time, and the length of time since the last restart.
Interface Status	Indicates whether each interface is up (green), down (red), or in an unknown state (gray).
Threat Logs	Displays the threat ID, application, and date and time for the last 10 entries in the Threat log. The threat ID is a malware description or URL that violates the URL filtering profile.
Config Logs	Displays the administrator username, client (Web or CLI), and date and time for the last 10 entries in the Configuration log.
Data Filtering Logs	Displays the description and date and time for the last 60 minutes in the Data Filtering log.

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Dashboard Charts	Descriptions
URL Filtering Logs	Displays the description and date and time for the last 60 minutes in the URL Filtering log.
System Logs	Displays the description and date and time for the last 10 entries in the System log. *A **`Config installed`** entry indicates configuration changes were committed successfully.*
System Resources	Displays the Management CPU usage, Data Plane usage, and the Session Count, which displays the number of sessions established through the firewall.
Logged In Admins	Displays the source IP address, session type (Web or CLI), and session start time for each administrator who is currently logged in.
ACC Risk Factor	Displays the average risk factor (1 to 5) for the network traffic processed over the past week. Higher values indicate higher risk.
High Availability	If high availability (HA) is enabled, indicates the HA status of the local and peer firewall—green (active), yellow (passive), or black (other). For more information about HA, see High Availability.
Locks	Shows configuration locks taken by administrators.

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Use the Application Command Center

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

The Application Command Center (ACC) is an interactive, graphical summary of the applications, users, URLs, threats, and content traversing your network. The ACC uses the firewall logs to provide visibility into traffic patterns and actionable information on threats. The ACC layout includes a tabbed view of network activity, threat activity, and blocked activity and each tab includes pertinent widgets for better visualization of network traffic. The graphical representation allows you to interact with the data and visualize the relationships between events on the network, so that you can uncover anomalies or find ways to enhance your network security rules. For a personalized view of your network, you can also add a custom tab and include widgets that allow you to drill down into the information that is most important to you.

*ACC data, including ACC widgets and exported ACC reports, use security policy rules data that you enabled to **Log at Session End**. If some data you expect to view in the ACC is not displayed, view your Traffic and Threat logs to determine the correct Security policy rule to modify as needed so all new logs generated that match the Security policy rule are viewable in the ACC.*

- ACC—First Look
- ACC Tabs
- ACC Widgets (Widget Descriptions)
- ACC Filters
- Interact with the ACC
- Use Case: ACC—Path of Information Discovery

ACC—First Look

Take a quick tour of the ACC.

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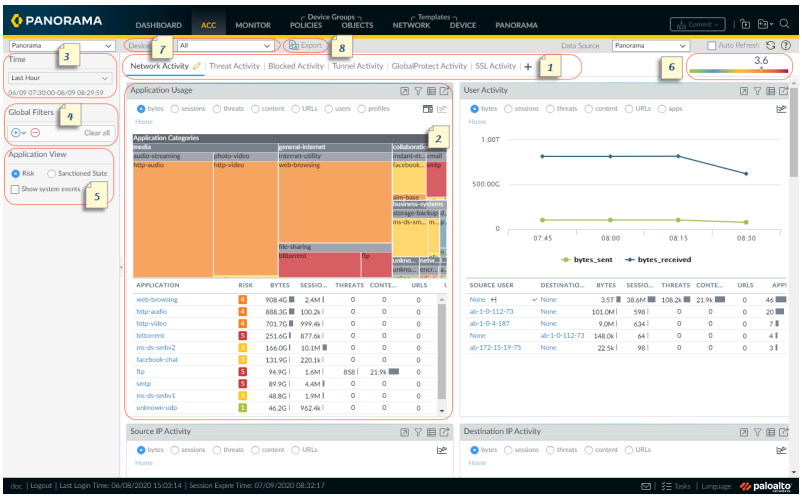

ACC—First Look		
1	**Tabs**	The ACC includes three predefined tabs that provide visibility into network traffic, threat activity, and blocked activity. For information on each tab, see ACC Tabs.
2	**Widgets**	Each tab includes a default set of widgets that best represent the events/trends associated with the tab. The widgets allow you to survey the data using the following filters: • bytes (in and out) • sessions • content (files and data) • URL categories • threats (and count) For information on each widget, see ACC Widgets.
3	**Time**	The charts or graphs in each widget provide a summary and historic view. You can choose a custom range or use the predefined time periods that range from the last 15 minutes up to the last 90 days or last 30 calendar days. The selected time period applies across all tabs in the ACC. The time period used to render data, by default, is the **Last Hour** updated in 15 minute intervals. The date and time interval are displayed onscreen, for example at 11:40, the time range is 01/12 10:30:00-01/12 11:29:59.

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ACC—First Look		
	Global Filters	The Global Filters allow you to set the filter across all widgets and all tabs. The charts/graphs apply the selected filters before rendering the data. For information on using the filters, see ACC Filters.
	Application View	The application view allows you filter the ACC view by either the sanctioned and unsanctioned applications in use on your network, or by the risk level of the applications in use on your network. Green indicates sanctioned applications, blue unsanctioned applications, and yellow indicates applications that are partially sanctioned. Partially sanctioned applications are those that have a mixed sanctioned state; it indicates that the application is inconsistently tagged as sanctioned, for example it might be sanctioned on one or more virtual systems on a firewall enabled for multiple virtual systems or across one or more firewalls within a device group on Panorama.
	Risk Factor	The risk factor (1=lowest to 5=highest) indicates the relative risk based on the applications used on your network. The risk factor uses a variety of factors to assess the associated risk levels, such as whether the application can share files, is it prone to misuse or does it try to evade firewalls, it also factors in the threat activity and malware as seen through the number of blocked threats, compromised hosts or traffic to malware hosts/ domains.
	Source	The data used for the ACC display. The options vary on the firewall and on Panorama. On the firewall, if enabled for multiple virtual systems, you can use the **Virtual System** drop-down to change the ACC display to include data from all virtual systems or just a selected virtual system. On Panorama, you can select the **Device Group** drop-down to change the ACC display to include data from all device groups or just a selected device group. Additionally, on Panorama, you can change the **Data Source** as **Panorama** data or **Remote Device Data**. **Remote Device Data** is only available when all the managed firewalls are on PAN-OS 7.0.0

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ACC—First Look		
		or later. When you filter the display for a specific device group, **Panorama** data is used as the data source.
	Export	You can export the widgets displayed in the currently selected tab as a PDF. The PDF is downloaded and saved to the downloads folder associated with your web browser, on your computer.

ACC Tabs

The ACC includes the following predefined tabs for viewing network activity, threat activity, and blocked activity.

Tab	Description
Network Activity	Displays an overview of traffic and user activity on your network including: • Top applications in use • Top users who generate traffic (with a drill down into the bytes, content, threats or URLs accessed by the user) • Most used security rules against which traffic matches occur In addition, you can also view network activity by source or destination zone, region, or IP address, ingress or egress interfaces, and GlobalProtect host information such as the operating systems of the devices most commonly used on the network.
Threat Activity	Displays an overview of the threats on the network, focusing on the top threats: vulnerabilities, spyware, viruses, hosts visiting malicious domains or URLs, top WildFire submissions by file type and application, and applications that use non-standard ports. The Compromised Hosts widget in this tab (the widget is supported on some platforms only), supplements detection with better visualization techniques; it uses the information from the correlated events tab (**Automated Correlation Engine** > **Correlated Events**) to present an aggregated view of compromised hosts on your network by source users/IP addresses and sorted by severity.
Blocked Activity	Focuses on traffic that was prevented from coming into the network. The widgets in this tab allow you to view activity denied by application name, username, threat name, blocked content—files and data that were blocked by a file blocking profile. It also lists the

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Tab	Description
	top security rules that were matched on to block threats, content, and URLs.
Tunnel Activity	Displays the activity of tunnel traffic that the firewall inspected based on your tunnel inspection policies. Information includes tunnel usage based on tunnel ID, monitor tag, user, and tunnel protocols such as Generic Routing Encapsulation (GRE), General Packet Radio Service (GPRS) Tunneling Protocol for User Data (GTP-U), and non-encrypted IPSec.
GlobalProtect Activity	Displays an overview of user activity in your GlobalProtect deployment. Information includes the number of users and number of times users connected, the gateways to which users connected, the number of connection failures and the failure reason, a summary of authentication methods and GlobalProtect app versions used, and the number of endpoints that are quarantined. In addition, this tab displays a chart view summary of devices that have been quarantined. Use the toggle at the top of the chart to view the quarantined devices by the actions that caused GlobalProtect to quarantine the device, the reason GlobalProtect quarantined the device, and the location of the quarantined devices.
SSL Activity	Displays an overview of TLS/SSL decryption activity on the firewall. Information includes successful and unsuccessful decryption activity in your network, decryption failure reasons such as protocol, certificate, and version issues, TLS versions, key exchange algorithms, and the amount and type of decrypted and undecrypted traffic. Use the ACC information to evaluate how decryption is working on your network and then use the decryption logs to drill down into details.

You can also Interact with the ACC to create customized tabs with custom layout and widgets that meet your network monitoring needs, export the tab and share with another administrator.

ACC Widgets

The widgets on each tab are interactive; you can set the ACC Filters and drill down into the details for each table or graph, or customize the widgets included in the tab to focus on the information you need. For details on what each widget displays, see Widget Descriptions.

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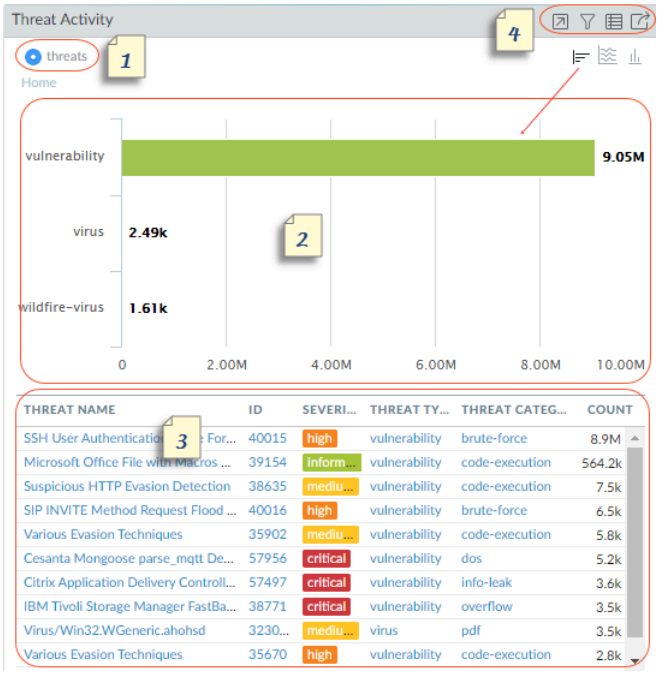

Widgets		
1	**View**	You can sort the data by bytes, sessions, threats, count, content, URLs, malicious, benign, files, applications, data, profiles, objects, users. The available options vary by widget.
2	**Graph**	The graphical display options are treemap, line graph, horizontal bar graph, stacked area graph, stacked bar graph, and map. The available options vary by widget; the interaction experience also varies with each graph type. For example, the widget for Applications using Non-Standard Ports allows you to choose between a treemap and a line graph. To drill down into the display, click into the graph. The area you click into becomes a filter and allows you to zoom into the selection and view more granular information on the selection.
3	**Table**	The detailed view of the data used to render the graph is provided in a table below the graph. You can interact with the table in several ways: • Click and set a local filter for an attribute in the table. The graph is updated and the table is sorted using the local filter. The information displayed in the graph and the table are always synchronized.

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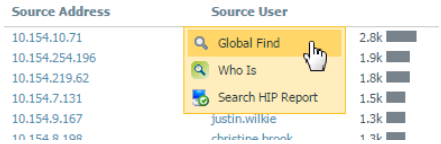
Widgets		
		• Hover over the attribute in the table and use the options available in the drop-down. 
4	**Actions**	**Maximize view**— Allows you enlarge the widget and view the table in a larger screen space and with more viewable information. **Set up local filters**—Allows you to add ACC Filters to refine the display within the widget. Use these filters to customize the widgets; these customizations are retained between logins. **Jump to logs**—Allows you to directly navigate to the logs (**Monitor** > **Logs** > **<log-type>** tab). The logs are filtered using the time period for which the graph is rendered. If you have set local and global filters, the log query concatenates the time period and the filters and only displays logs that match the combined filter set. **Export**—Allows you to export the graph as a PDF. The PDF is downloaded and saved on your computer. It is saved in the Downloads folder associated with your web browser.

Widget Descriptions

Each tab on the ACC includes a different set of widgets.

Widget	Description
Network Activity—Displays an overview of traffic and user activity on your network.	
Application Usage	The table displays the top ten applications used on your network, all the remaining applications used on the network are aggregated and displayed as other. The graph displays all applications by application category, sub category, and application. Use this widget to scan

Widget	Description
	for applications being used on the network, it informs you about the predominant applications using bandwidth, session count, file transfers, triggering the most threats, and accessing URLs. Sort attributes: bytes, sessions, threats, content, URLs Charts available: treemap, area, column, line (the charts vary by the sort by attribute selected)
User Activity	Displays the top ten most active users on the network who have generated the largest volume of traffic and consumed network resources to obtain content. Use this widget to monitor top users on usage sorted on bytes, sessions, threats, content (files and patterns), and URLs visited. Sort attributes: bytes, sessions, threats, content, URLs Charts available: area, column, line (the charts vary by the sort by attribute selected)
Source IP Activity	Displays the top ten IP addresses or hostnames of the devices that have initiated activity on the network. All other devices are aggregated and displayed as other. Sort attributes: bytes, sessions, threats, content, URLs Charts available: area, column, line (the charts vary by the sort by attribute selected)
Destination IP Activity	Displays the IP addresses or hostnames of the top ten destinations that were accessed by users on the network. Sort attributes: bytes, sessions, threats, content, URLs Charts available: area, column, line (the charts vary by the sort by attribute selected)
Source Regions	Displays the top ten regions (built-in or custom defined regions) around the world from where users initiated activity on your network. Sort attributes: bytes, sessions, threats, content, URLs Charts available: map, bar
Destination Regions	Displays the top ten destination regions (built-in or custom defined regions) on the world map from where content is being accessed by users on the network. Sort attributes: bytes, sessions, threats, content, URLs Charts available: map, bar

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Widget	Description
GlobalProtect Host Information	Displays information on the state of the hosts on which the GlobalProtect agent is running; the host system is a GlobalProtect endpoint. This information is sourced from entries in the HIP match log that are generated when the data submitted by the GlobalProtect app matches a HIP object or a HIP profile you have defined on the firewall. If you do not have HIP Match logs, this widget is blank. To learn how to create HIP objects and HIP profiles and use them as policy match criteria, see Configure HIP-Based Policy Enforcement. Sort attributes: profiles, objects, operating systems Charts available: bar
Rule Usage	Displays the top ten rules that have allowed the most traffic on the network. Use this widget to view the most commonly used rules, monitor the usage patterns, and to assess whether the rules are effective in securing your network. Sort attributes: bytes, sessions, threats, content, URLs Charts available: line
Ingress Interfaces	Displays the firewall interfaces that are most used for allowing traffic into the network. Sort attributes: bytes, bytes sent, bytes received Charts available: line
Egress Interfaces	Displays the firewall interfaces that are most used by traffic exiting the network. Sort attributes: bytes, bytes sent, bytes received Charts available: line
Source Zones	Displays the zones that are most used for allowing traffic into the network. Sort attributes: bytes, sessions, threats, content, URLs Charts available: line
Destination Zones	Displays the zones that are most used by traffic going outside the network. Sort attributes: bytes, sessions, threats, content, URLs Charts available: line
Threat Activity—Displays an overview of the threats on the network	

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Widget	Description
Compromised Hosts	Displays the hosts that are likely compromised on your network. This widget summarizes the events from the correlation logs. For each source user/IP address, it includes the correlation object that triggered the match and the match count, which is aggregated from the match evidence collated in the correlated events logs. For details see Use the Automated Correlation Engine. Available on the PA-5200 Series, PA-7000 Series, and Panorama. Sort attributes: severity (by default)
Hosts Visiting Malicious URLs	Displays the frequency with which hosts (IP address/hostnames) on your network have accessed malicious URLs. These URLs are known to be malware based on categorization in PAN-DB. Sort attributes: count Charts available: line
Hosts Resolving Malicious Domains	Displays the top hosts matching DNS signatures; hosts on the network that are attempting to resolve the hostname or domain of a malicious URL. This information is gathered from an analysis of the DNS activity on your network. It utilizes passive DNS monitoring, DNS traffic generated on the network, activity seen in the sandbox if you have configured DNS sinkhole on the firewall, and DNS reports on malicious DNS sources that are available to Palo Alto Networks customers. Sort attributes: count Charts available: line
Threat Activity	Displays the threats seen on your network. This information is based on signature matches in Antivirus, Anti-Spyware, and Vulnerability Protection profiles and viruses reported by WildFire. Sort attributes: threats Charts available: bar, area, column
WildFire Activity by Application	Displays the applications that generated the most WildFire submissions. This widget uses the malicious and benign verdict from the WildFire Submissions log. Sort attributes: malicious, benign Charts available: bar, line
WildFire Activity by File Type	Displays the threat vector by file type. This widget displays the file types that generated the most WildFire submissions and uses the malicious and benign verdict from the WildFire Submissions log. If this data is unavailable, the widget is empty. Sort attributes: malicious, benign

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Widget	Description
	Charts available: bar, line
Applications using Non Standard Ports	Displays the applications that are entering your network on non-standard ports. If you have migrated your firewall rules from a port-based firewall, use this information to craft policy rules that allow traffic only on the default port for the application. Where needed, make an exception to allow traffic on a non-standard port or create a custom application. Sort attributes: bytes, sessions, threats, content, URLs Charts available: treemap, line
Rules Allowing Applications On Non Standard Ports	Displays the security policy rules that allow applications on non-default ports. The graph displays all the rules, while the table displays the top ten rules and aggregates the data from the remaining rules as other. This information helps you identify gaps in network security by allowing you to assess whether an application is hopping ports or sneaking into your network. For example, you can validate whether you have a rule that allows traffic on any port except the default port for the application. Say for example, you have a rule that allow DNS traffic on its *application-default* port (port 53 is the standard port for DNS). This widget will display any rule that allows DNS traffic into your network on any port except port 53. Sort attributes: bytes, sessions, threats, content, URLs Charts available: treemap, line
Blocked Activity—Focuses on traffic that was prevented from coming into the network	
Blocked Application Activity	Displays the applications that were denied on your network, and allows you to view the threats, content, and URLs that you kept out of your network. Sort attributes: threats, content, URLs Charts available: treemap, area, column
Blocked User Activity	Displays user requests that were blocked by a match on an Antivirus, Anti-spyware, File Blocking or URL Filtering profile attached to Security policy rule. Sort attributes: threats, content, URLs Charts available: bar, area, column
Blocked Threats	Displays the threats that were successfully denied on your network. These threats were matched on antivirus signatures, vulnerability

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Widget	Description
	signatures, and DNS signatures available through the dynamic content updates on the firewall. Sort attributes: threats Charts available: bar, area, column
Blocked Content	Displays the files and data that was blocked from entering the network. The content was blocked because security policy denied access based on criteria defined in a File Blocking security profile or a Data Filtering security profile. Sort attributes: files, data Charts available: bar, area, column
Security Policies Blocking Activity	Displays the security policy rules that blocked or restricted traffic into your network. Because this widget displays the threats, content, and URLs that were denied access into your network, you can use it to assess the effectiveness of your policy rules. This widget does not display traffic that blocked because of deny rules that you have defined in policy. Sort attributes: threats, content, URLs Charts available: bar, area, column
GlobalProtect Activity—Displays information of user activity in your GlobalProtect deployment.	
Successful GlobalProtect Connection Activity	Displays a chart view of GlobalProtect connection activity over the selected time period. Use the toggle at the top of the chart to switch between connection statistics by users, portals and gateways, and location. Sort attributes: users, portals/gateways, location Charts available: bar, line
Unsuccessful GlobalProtect Connection Activity	Displays a chart view of unsuccessful GlobalProtect connection activity over the selected time period. Use the toggle at the top of the chart to switch between connection statistics by users, portals and gateways, and location. To help you identify and troubleshoot connection issues, you can also view the reasons chart or graph. For this chart, the ACC indicates the error, source user, public IP address and other information to help you identify and resolve the issue quickly. Sort attributes: users, portals/gateways, reasons, location Charts available: bar, line

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Widget	Description
GlobalProtect Deployment Activity	Displays a chart view summary of your deployment. Use the toggle at the top of the chart to view the distribution of users by authentication method, GlobalProtect app version, and operating system version. Sort attributes: auth method, globalprotect app version, os Charts available: bar, line
GlobalProtect Quarantine Activity	Displays a chart view summary of devices that have been quarantined. Use the toggle at the top of the chart to view the quarantined devices by the actions that caused GlobalProtect to quarantine the device, the reason GlobalProtect quarantined the device, and the location of the quarantined devices. Sort attributes: actions, reason, location Charts available: bar, line
SSL Activity—Displays information about SSL/TLS activity in your network.	
Traffic Activity	Shows SSL/TLS activity compared to non-SSL/TLS activity by total number of sessions or bytes.
SSL/TLS Activity	Shows successful TLS connections by TLS version and application or SNI. This widget helps you understand how much risk you are taking on by allowing weaker TLS protocol versions. Identifying applications and SNIs that use weak protocols enables you to evaluate each one and decide whether you need to allow access to it for business reasons. If you don't need the application for business purposes, you may want to block the traffic instead of allowing it. Click an application or an SNI to drill down and see detailed information.
Decryption Failure Reasons	Shows the reasons for decryption failures, such as certificate or protocol issues, by SNI. Use this information to detect problems caused by Decryption policy or profile misconfiguration or by traffic that uses weak protocols or algorithms. Click a failure reason to drill down and isolate the number of sessions per SNI or click an SNI to see the failures for that SNI.
Successful TLS Version Activity	Shows the amount of decrypted and non-decrypted traffic by sessions or bytes. Traffic that was not decrypted may be excepted from decryption by policy, policy misconfiguration, or by being on the Decryption Exclusion List (**Device** > **Certificate Management** > **SSL Decryption Exclusion**).
Successful Key Exchange Activity	Shows successful key exchange activity per algorithm, by application or by SNI. Click a key exchange algorithm to see the activity for just

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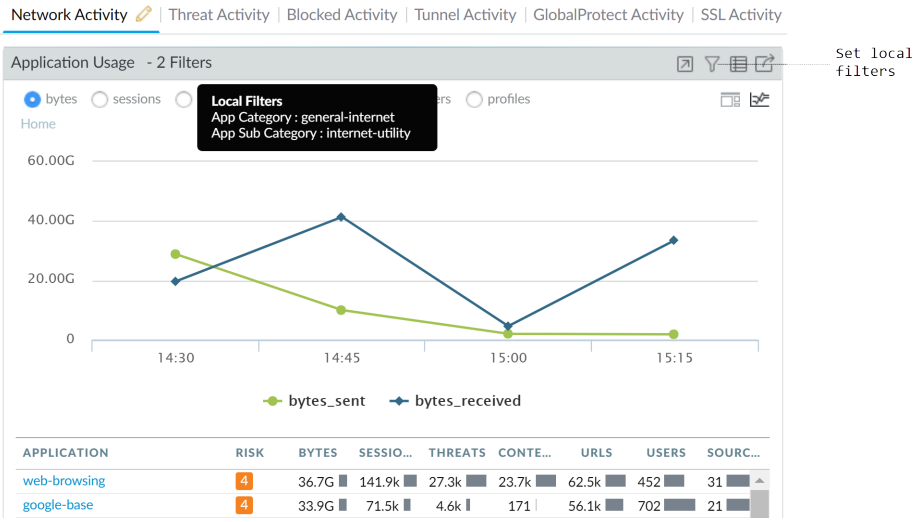
Widget	Description
	that algorithm or click an application or SNI to view the key exchange activity for that application or SNI.

ACC Filters

The graphs and tables on the ACC widgets allow you to use filters to narrow the scope of data that is displayed, so that you can isolate specific attributes and analyze information you want to view in greater detail. The ACC supports the simultaneous use of widget and global filters.

- **Widget Filters**—Apply a widget filter, which is a filter that is *local* to a specific widget. A widget filter allows you to interact with the graph and customize the display so that you can drill down in to the details and access the information you want to monitor on a specific widget. To create a widget filter that is persistent across reboots, you must use the **Set Local Filter** option.



- **Global filters**—Apply global filters across all the tabs in the ACC. A global filter allows you to pivot the display around the details you care about right now and exclude the unrelated information from the current display. For example, to view all events relating to a specific user and application, you can apply the username and the application as a global filter and view only

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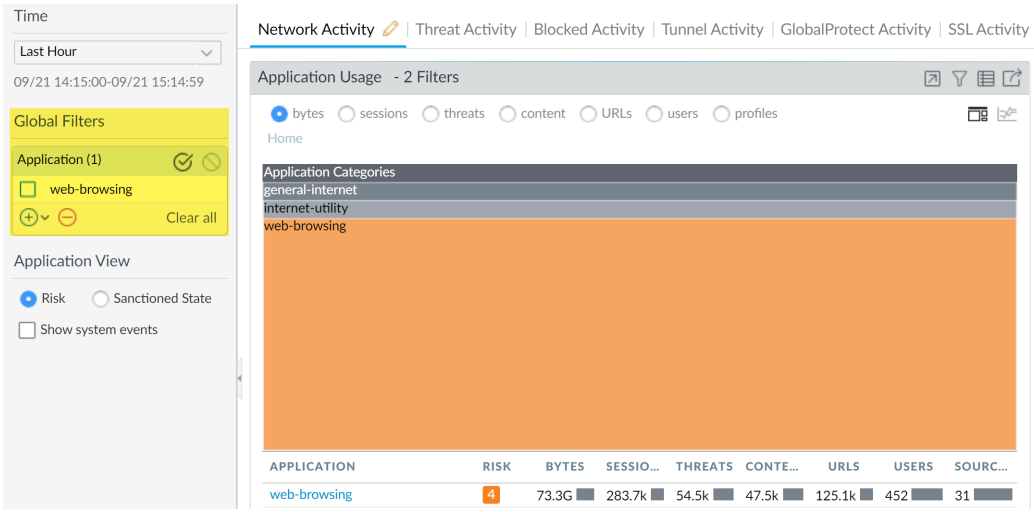
information pertaining to the user and the application through all the tabs and widgets on the ACC. Global filters are not persistent.



You can apply global filters in three ways:

- **Set a global filter from a table**—Select an attribute from a table in any widget and apply the attribute as a global filter.
- **Add a widget filter to a global filter**—Hover over the attribute and click the arrow icon to the right of the attribute. This option allows you to elevate a local filter used in a widget, and apply the attribute globally to update the display across all the tabs on the ACC.
- **Define a global filter**—Define a filter using the **Global Filters** pane on the ACC.

See Interact with the ACC for details on using these filters.

Interact with the ACC

To customize and refine the ACC display, you can add, delete, export and import tabs, add and delete widgets, set local and global filters, and interact with the widgets.

- Add a tab.
  1. Select the + icon along the list of tabs.
  2. Add a **View Name**. This name will be used as the name for the tab. You can add up to five tabs.

- Edit a tab.

  Select the tab, and click the pencil icon next to the tab name, to edit the tab. For example Threat Activity .

  Editing a tab allows you to add or delete or reset the widgets that are displayed in the tab. You can also change the widget layout in the tab.

  To save the tab as the default tab, select .

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- Export and Import tabs.
  1. Select the tab, and click the pencil icon next to the tab name, to edit the tab.
  2. Select the icon to export the current tab as a .txt file. You can share this .txt file with another administrator.
  3. To import the tab as a new tab on another firewall, select the + icon along the list of tabs, and add a name and click the import icon, browse to select the .txt file.

- See what widgets are included in a tab.
  1. Select the tab, and click on the pencil icon to edit it.
  2. Select the **Add Widget** drop-down and verify the widgets that have the check boxes selected.

- Add a widget or a widget group.
  1. Add a new tab or edit a predefined tab.
  2. Select **Add Widget**, and then select the check box that corresponds to the widget you want to add. You can select up to a maximum of 12 widgets.
  3. (Optional) To create a 2-column layout, select **Add Widget Group**. You can drag and drop widgets into the 2-column display. As you drag the widget into the layout, a placeholder will display for you to drop the widget.

     You cannot name a widget group.

- Delete a tab or a widget group/ widget.
  1. To delete a custom tab, select the tab and click the X icon. Custom_threat_user_activity

     You cannot delete a predefined tab.

  2. To delete a widget group/widget, edit the tab and in the workspace section, click the [X] icon on the right. You cannot undo a deletion.

- Reset the default widgets in a tab.

  On a predefined tab, such as the **Blocked Activity** tab, you can delete one or more widgets. If you want to reset the layout to include the default set of widgets for the tab, edit the tab and click **Reset View**.

- Zoom in on the details in an area, column, or line graph.

  Watch how the zoom-in capability works.

  Click and drag an area in the graph to zoom in. For example, when you zoom into a line graph, it triggers a re-query and the firewall fetches the data for the selected time period. It is not a mere magnification.

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- Use the table drop-down to find more information on an attribute.
  1. Hover over an attribute in a table to see the drop-down.
  2. Click into the drop-down to view the available options.
     - **Global Find**—#unique_296 for references to the attribute (username/IP address, object name, policy rule name, threat ID, or application name) anywhere in the candidate configuration.
     - **Value**—Displays the details of the threat ID, or application name, or address object.
     - **Who Is**—Performs a domain name (*WHOIS*) lookup for the IP address. The lookup queries databases that store the registered users or assignees of an Internet resource.
     - **Search HIP Report**—Uses the username or IP address to find matches in a HIP Match report.

- Set a widget filter.

  You can also click an attribute in the table (below the graph) to apply it as a widget filter.

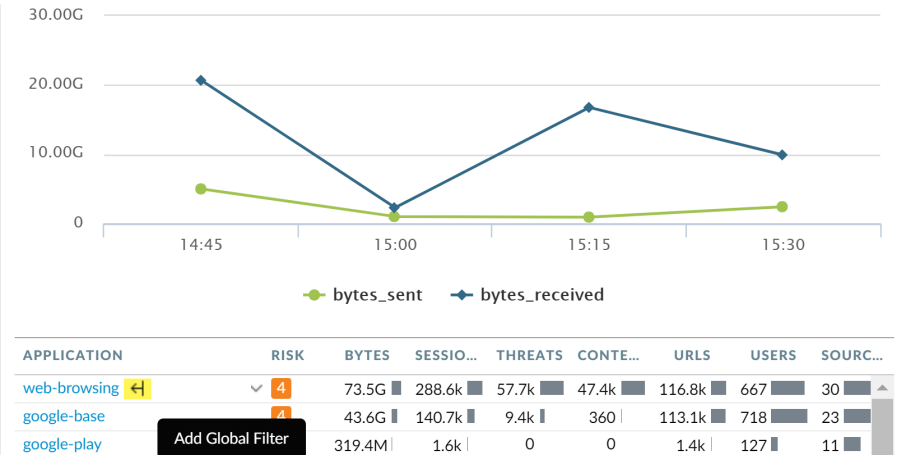
  1. Select a widget and click the icon.
  2. Click the icon to add the filters you want to apply.
  3. Click **Apply**. These filters are persistent across reboots.

     The active widget filters are indicated next to the widget name.

- Negate a widget filter
  1. Click the icon to display the Setup Local Filters dialog.
  2. Add a filter, and then click the negate icon.

- Set a global filter from a table.

  Hover over an attribute in the table below the chart and click the arrow icon to the right of the attribute.



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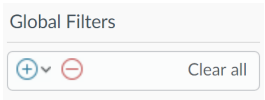
- Set a global filter using the Global Filters pane.

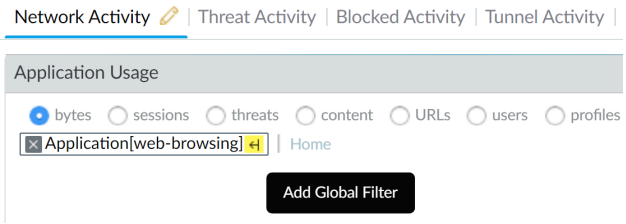
  Watch global filters in action.

  1. Locate the **Global Filters** pane on the left side of the ACC.

  2. Click the icon to view the list of filters you can apply.

- Promote a widget filter to a global filter.
  1. On any table in a widget, click the link for an attribute. This sets the attribute as a widget filter.
  2. To promote the filter to be a global filter, select the arrow to the right of the filter.

- Remove a filter.

  Click the icon to remove a filter.

  - For global filters: It is located in the Global Filters pane.
  - For widget filters: Click the icon to display the Setup Local Filters dialog, then select the filter, and click the icon.

- Clear all filters.
  - For global filters: Click the **Clear All** button under Global Filters.
  - For widget filters: Select a widget and click the icon. Then click the **Clear All** button in the Setup Local Filters dialog.

- See what filters are in use.
  - For global filters: The number of global filters applied are displayed on the left pane under Global Filters.
  - For widget filters: The number of widget filters applied on a widget are displayed next to the widget name. To view the filters, click the icon.

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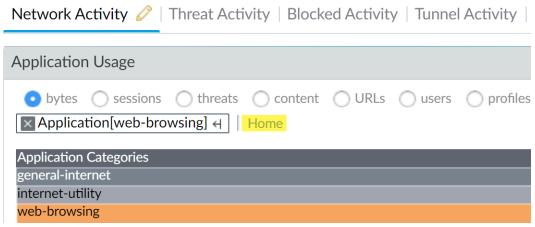
- Reset the display on a widget.
  - If you set a widget filter or drill into a graph, click the **Home** link to reset the display in the widget.



Use Case: ACC—Path of Information Discovery

The ACC has a wealth of information that you can use as a starting point for analyzing network traffic. Let's look at an example on using the ACC to uncover events of interest. This example illustrates how you can use the ACC to ensure that legitimate users can be held accountable for their actions, detect and track unauthorized activity, and detect and diagnose compromised hosts and vulnerable systems on your network.

The widgets and filters in the ACC give you the capability to analyze the data and filter the views based on events of interest or concern. You can trace events that pique your interest, directly export a PDF of a tab, access the raw logs, and save a personalized view of the activity that you want to track. These capabilities make it possible for you to monitor activity and develop policies and countermeasures for fortifying your network against malicious activity. In this section, you will Interact with the ACC widgets across different tabs, drill down using widget filters, and pivot the ACC views using global filters, and export a PDF for sharing with incidence response or IT teams.

At first glance, you see the Application Usage and User Activity widgets in the **ACC** > **Network Activity** tab. The User Activity widget shows that user Marsha Wirth has transferred 154 Megabytes of data during the last hour. This volume is nearly six times more than any other user on the network. To see the trend over the past few hours, expand the **Time** period to the **Last 6 Hrs**, and now Marsha's activity has been 1.7 Gigabytes over 1,500 sessions and has triggered 455 threats signatures.

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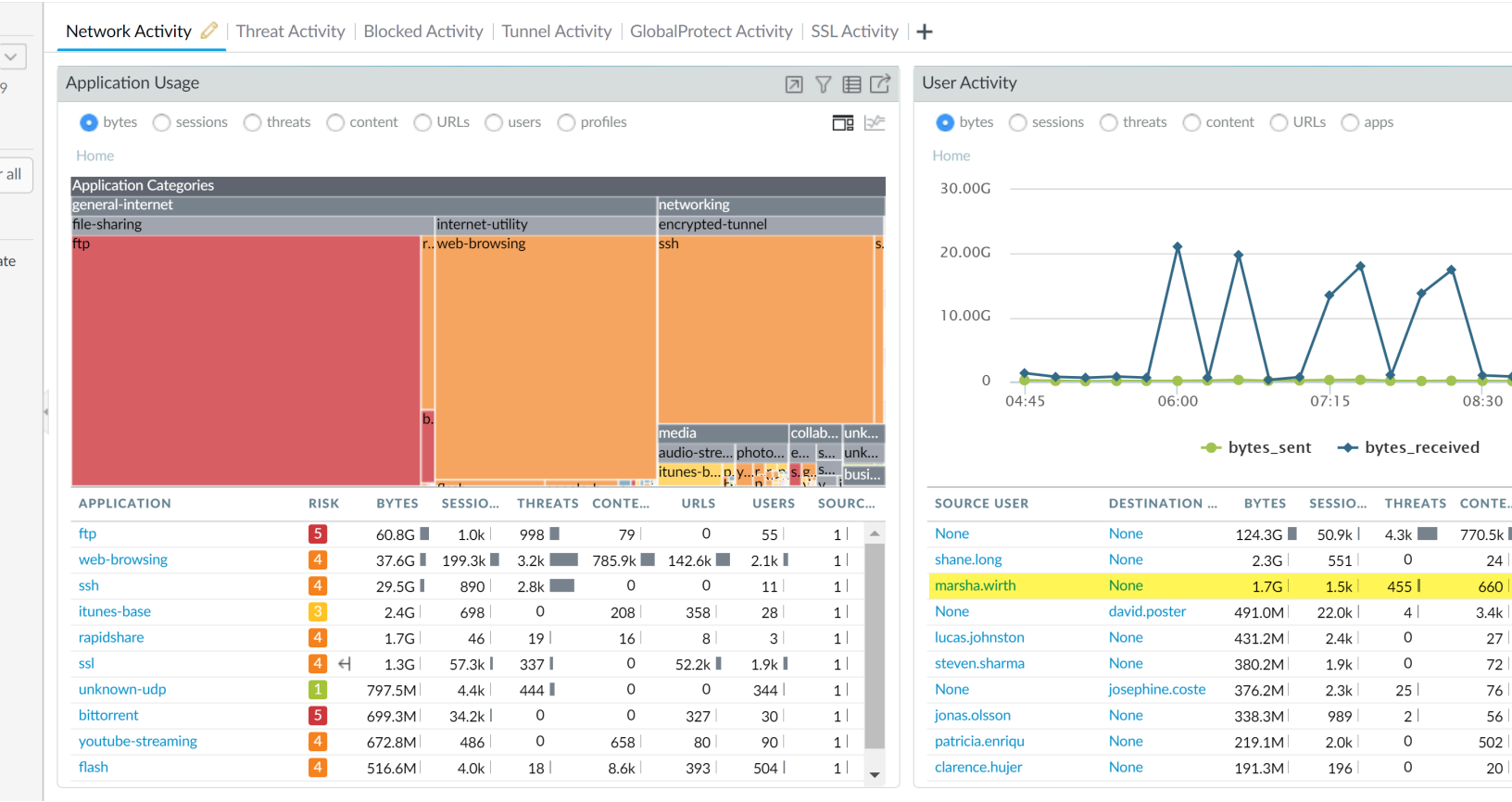
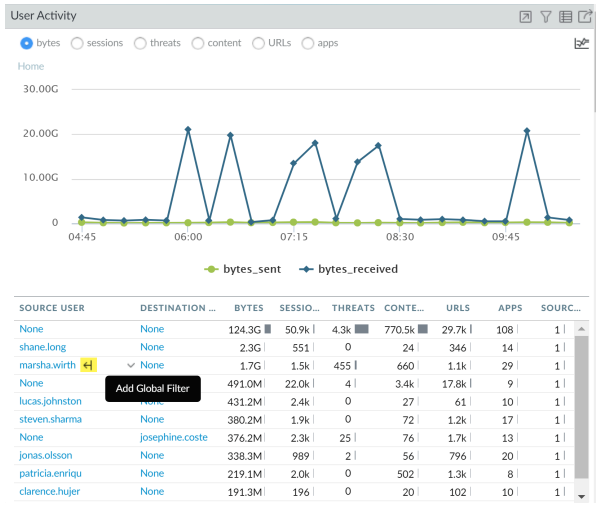


Because Marsha has transferred a large volume of data, apply her username as a global filter (ACC Filters) and pivot all the views in the ACC to Marsha's traffic activity.



The Application Usage tab now shows that the top application that Martha used was rapidshare, a Swiss-owned file-hosting site that belongs to the file-sharing URL category. For further investigation, add rapidshare as a global filter, and view Marsha's activity in the context of rapidshare.

Consider whether you want to sanction rapidshare for company use. Should you allow uploads to this site and do you need a QoS policy to limit bandwidth?

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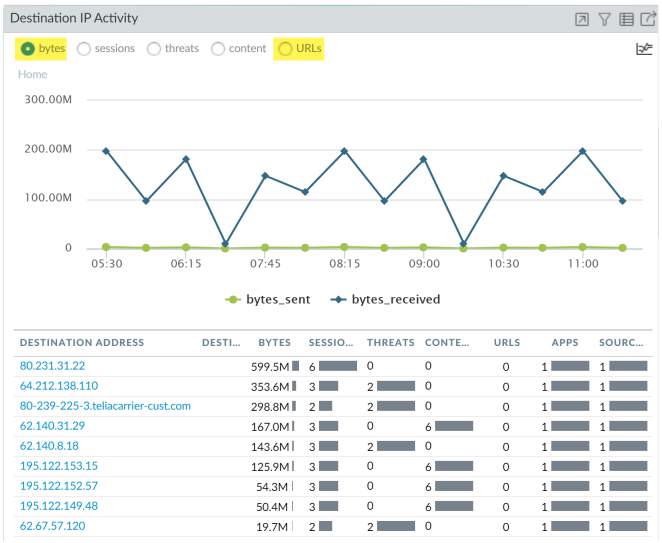
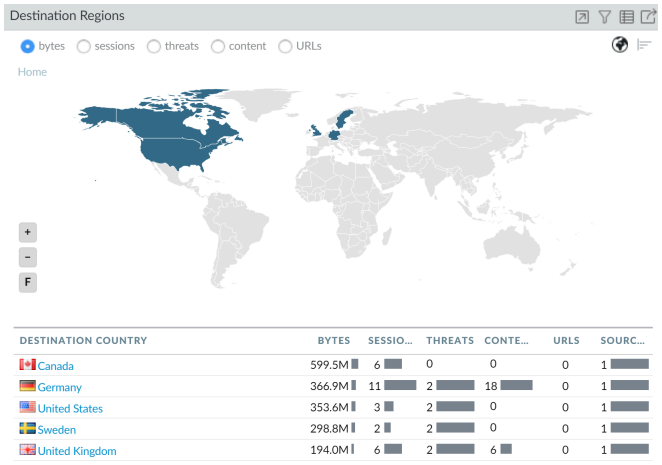
To view which IP addresses Marsha has communicated with, check the **Destination IP Activity** widget, and view the data by bytes and by URLs.

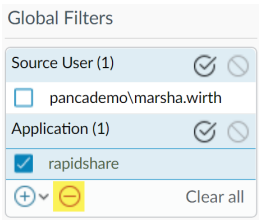


To find out which countries Marsha communicated with, sort on **sessions** in the **Destination Regions** widget.



From this data, you can confirm that Marsha, a user on your network, has established sessions in Canada, Germany, Sweden, United Kingdom, and the United States. She logged 2 threats in her sessions with each destination country.

To look at Marsha's activity from a threat perspective, remove the global filter for rapidshare.



In the **Threat Activity** widget on the **Threat Activity** tab, view the threats. The widget displays that her activity had triggered a match for 452 vulnerabilities in the brute force, information leak,

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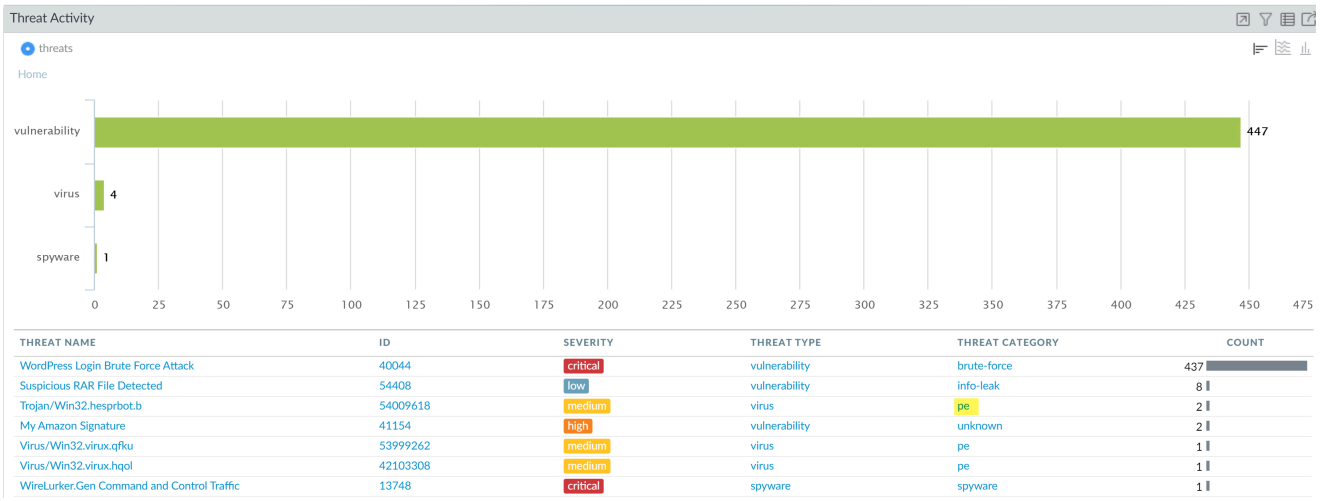
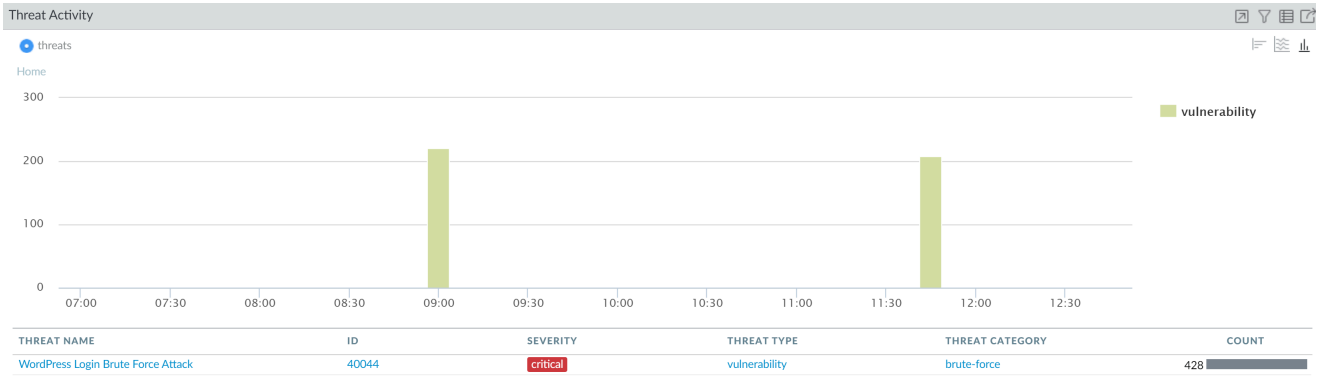
portable executable (PE) and spyware threat category. Several of these vulnerabilities are of critical severity.



THREAT NAME	ID	SEVERITY	THREAT TYPE	THREAT CATEGORY	COUNT
WordPress Login Brute Force Attack	40044	critical	vulnerability	brute-force	437
Suspicious RAR File Detected	54408	low	vulnerability	info-leak	8
Trojan/Win32.hesprbot.b	54009618	medium	virus	pe	2
My Amazon Signature	41154	high	vulnerability	unknown	2
Virus/Win32.virux.qfku	53999262	medium	virus	pe	1
Virus/Win32.virux.hqol	42103308	medium	virus	pe	1
WireLurker.Gen Command and Control Traffic	13748	critical	spyware	spyware	1

To further drill-down into each vulnerability, click into the graph and narrow the scope of your investigation. Each click automatically applies a local filter on the widget.



THREAT NAME	ID	SEVERITY	THREAT TYPE	THREAT CATEGORY	COUNT
WordPress Login Brute Force Attack	40044	critical	vulnerability	brute-force	428

To investigate each threat by name, you can create a global filter for say, **WordPress Login Brute Force Attack**. Then, view the **User Activity widget** in the **Network Activity** tab. The tab is automatically filtered to display threat activity for Marsha (notice the global filters in the screenshot).

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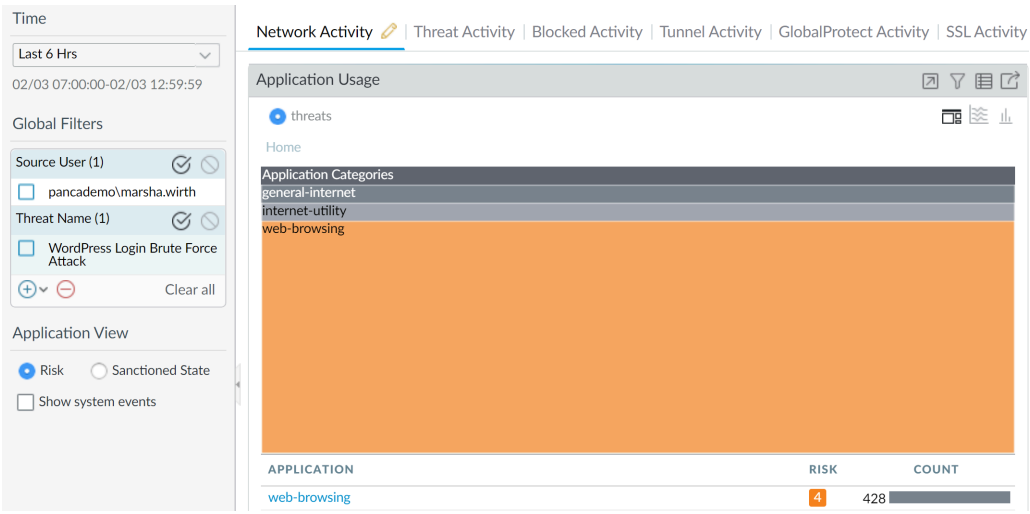
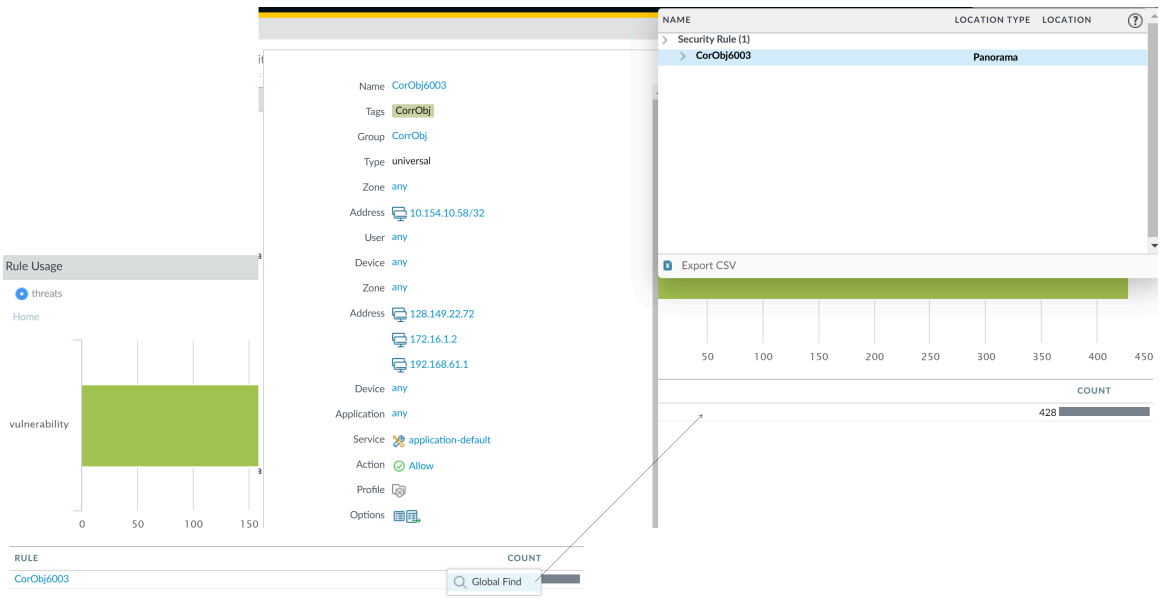


Notice that this Microsoft code-execution vulnerability was triggered over email, by the imap application. You can now establish that Martha has IE vulnerabilities and email attachment vulnerabilities, and perhaps her computer needs to be patched. You can now either navigate to the **Blocked Threats** widget in the **Blocked Activity** tab to check how many of these vulnerabilities were blocked.

Or, you can check the **Rule Usage** widget on the **Network Activity** tab to discover how many vulnerabilities made it into your network and which security rule allowed this traffic, and navigate directly to the security rule using the **Global Find** capability.



Then, drill into the attackers using web-browsing to attack target destination. Consider modifying the security policy rule to restrict these malicious IP addresses or more narrowly defining which IP addresses can access your network resources.

To review if any threats were logged over web-browsing, check Marsha's activity in the **WildFire Activity by Application** widget in the **Threat Activity** tab. You can confirm that Marsha had no malicious activity, but to verify that other no other user was compromised by the web-browsing

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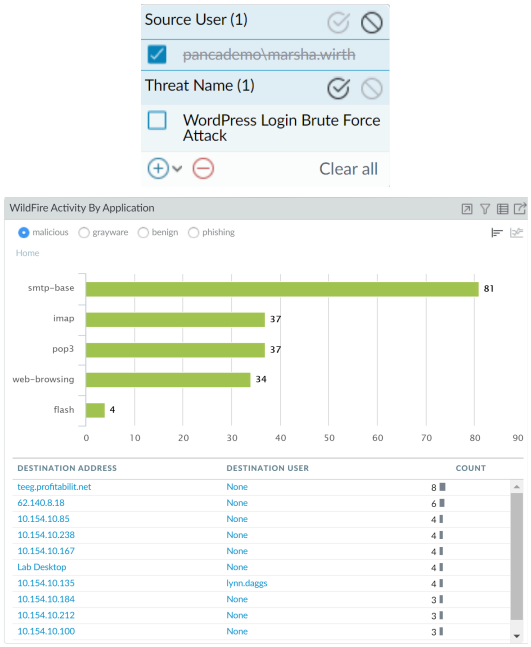
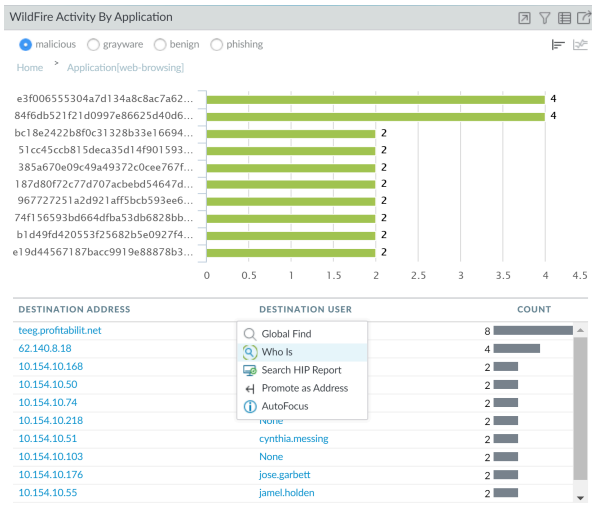
application, negate Marsha as a global filter and look for other users who triggered threats over web-browsing.



Click into the bar for imap in the graph and drill into the inbound threats associated with the application. To find out who an IP address is registered to, hover over the attacker IP address and select the **Who Is** link in the drop-down.



Because the session count from this IP address is high, check the **Blocked Content** and **Blocked Threats** widgets in the **Blocked Activity** tab for events related to this IP address. The **Blocked Activity** tab allows you to validate whether or not your policy rules are effective in blocking content or threats when a host on your network is compromised.

Use the **Export PDF** capability on the ACC to export the current view (create a snapshot of the data) and send it to an incidence response team. To view the threat logs directly from the widget, you can also click the icon to jump to the logs; the query is generated automatically and only the relevant logs are displayed onscreen (for example in **Monitor** > **Logs** > **Threat Logs**).

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You have now used the ACC to review network data/trends to find which applications or users are generating the most traffic, and how many application are responsible for the threats seen on the network. You were able to identify which application(s), user(s) generated the traffic, determine whether the application was on the default port, and which policy rule(s) allowed the traffic into the network, and determine whether the threat is spreading laterally on the network. You also identified the destination IP addresses, geo-locations with which hosts on the network are communicating with. Use the conclusions from your investigation to craft goal-oriented policies that can secure users and your network.

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Use the App Scope Reports

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

The App Scope reports provide visibility and analysis tools to help pinpoint problematic behavior, helping you understand changes in application usage and user activity, users and applications that take up most of the network bandwidth, and identify network threats.

With the App Scope reports, you can quickly see if any behavior is unusual or unexpected. Each report provides a dynamic, user-customizable window into the network; hovering the mouse over and clicking either the lines or bars on the charts opens detailed information about the specific application, application category, user, or source on the ACC. The App Scope charts on **Monitor** > **App Scope** give you the ability to:

- Toggle the attributes in the legend to only view chart details that you want to review. The ability to include or exclude data from the chart allows you to change the scale and review details more closely.
- Click into an attribute in a bar chart and drill down to the related sessions in the ACC. Click into an Application name, Application Category, Threat Name, Threat Category, Source IP address or Destination IP address on any bar chart to filter on the attribute and view the related sessions in the ACC.
- Export a chart or map to PDF or as an image. For portability and offline viewing, you can Export charts and maps as PDFs or PNG images.

The following App Scope reports are available:

- Summary Report
- Change Monitor Report
- Threat Monitor Report
- Threat Map Report
- Network Monitor Report
- Traffic Map Report

Summary Report

The App Scope Summary report (**Monitor** > **App Scope** > **Summary**) displays charts for the top five gainers, losers, and bandwidth consuming applications, application categories, users, and sources.

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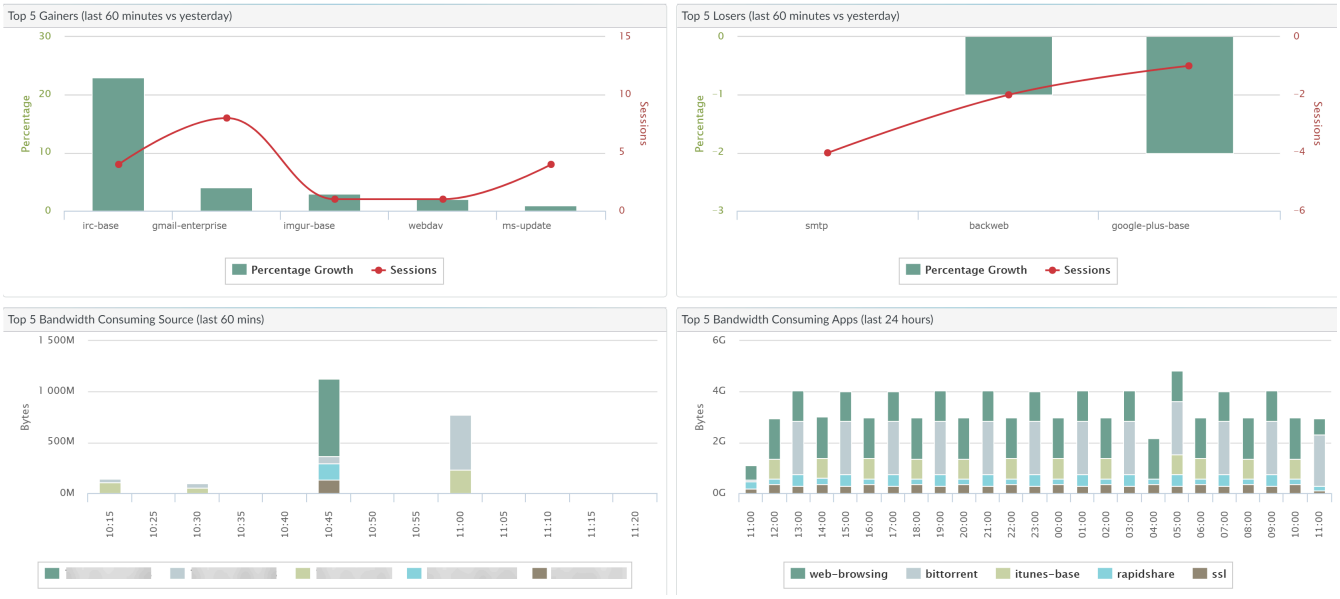
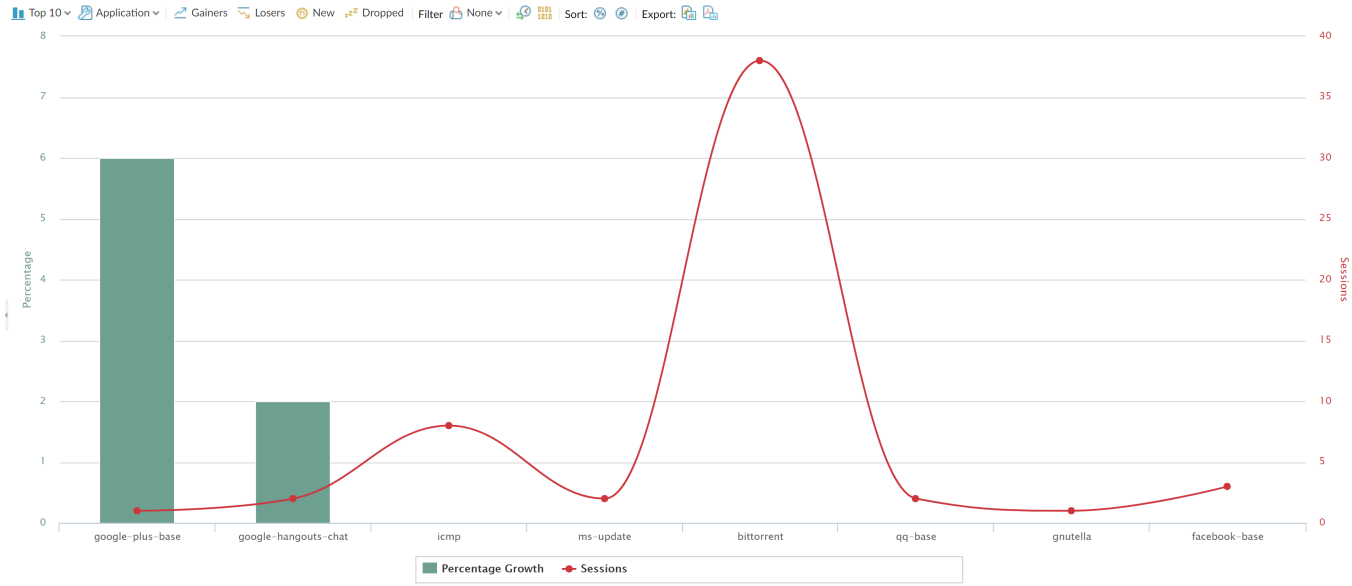


Change Monitor Report

The App Scope Change Monitor report (**Monitor** > **App Scope** > **Change Monitor**) displays changes over a specified time period. For example, the following chart displays the top applications that gained in use over the last hour as compared with the last 24-hour period. The top applications are determined by session count and sorted by percent.



The Change Monitor Report contains the following buttons and options.

Button	Description
Top 10	Determines the number of records with the highest measurement included in the chart.

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Button	Description
Application	Determines the type of item reported: Application, Application Category, Source, or Destination.
Gainers	Displays measurements of items that have increased over the measured period.
Losers	Displays measurements of items that have decreased over the measured period.
New	Displays measurements of items that were added over the measured period.
Dropped	Displays measurements of items that were discontinued over the measured period.
Filter	Applies a filter to display only the selected item. None displays all entries.
	Determines whether to display session or byte information.
Sort	Determines whether to sort entries by percentage or raw growth.
Export	Exports the graph as a .png image or as a PDF.
Compare	Specifies the period over which the change measurements are taken.

Threat Monitor Report

The App Scope Threat Monitor report (**Monitor** > **App Scope** > **Threat Monitor**) displays a count of the top threats over the selected time period. For example, the following figure shows the top 10 threat types over the last 6 hours.

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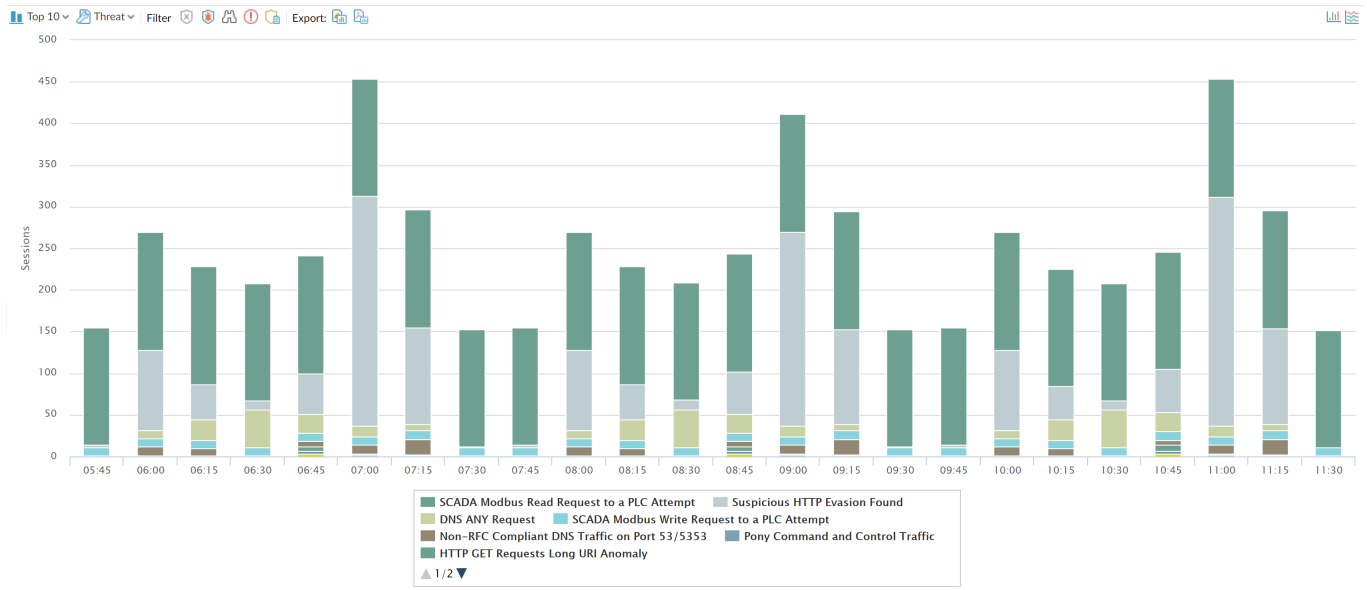

Each threat type is color-coded as indicated in the legend below the chart. The Threat Monitor report contains the following buttons and options.

Button	Description
Top 10	Determines the number of records with the highest measurement included in the chart.
Threats	Determines the type of item measured: Threat, Threat Category, Source, or Destination.
Filter	Applies a filter to display only the selected type of items.
	Determines whether the information is presented in a stacked column chart or a stacked area chart.
Export	Exports the graph as a .png image or as a PDF.
	Specifies the period over which the measurements are taken.

Threat Map Report

The App Scope Threat Map report (**Monitor** > **App Scope** > **Threat Map**) shows a geographical view of threats, including severity. Each threat type is color-coded as indicated in the legend below the chart.

The firewall uses geolocation for creating threat maps. The firewall is placed at the bottom of the threat map screen, if you have not specified the geolocation coordinates (**Device** > **Setup** > **Management**, General Settings section) on the firewall.

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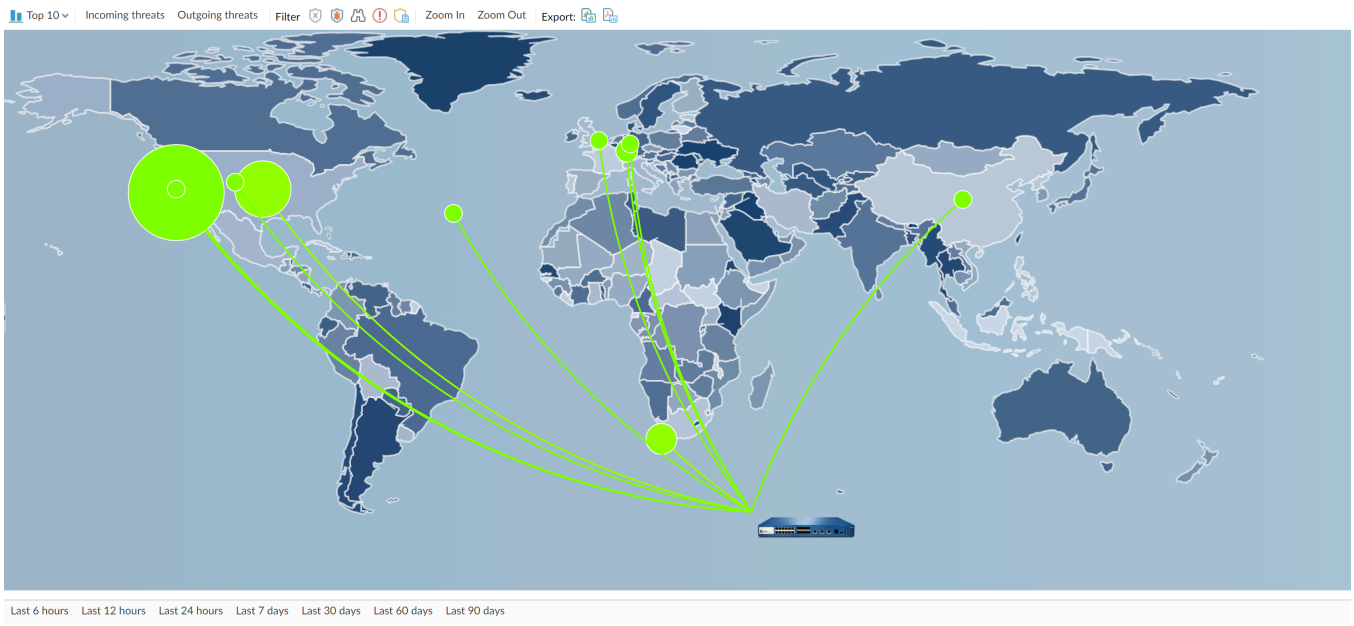
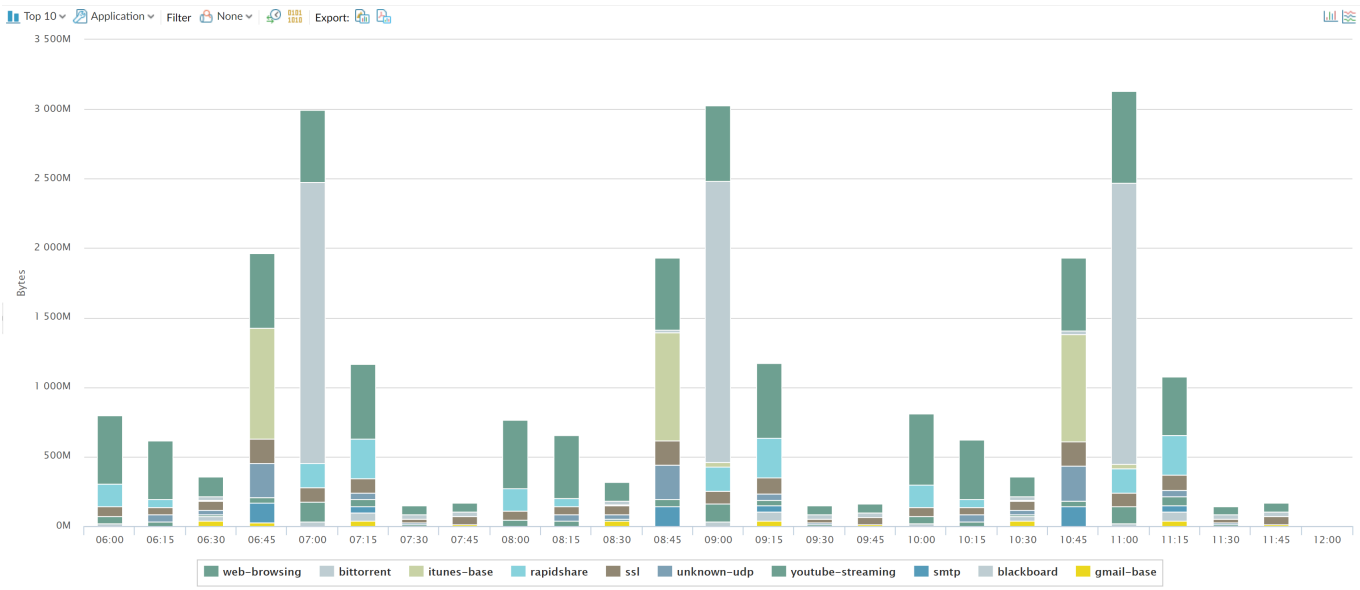


The Threat Map report contains the following buttons and options.

Button	Description
Top 10	Determines the number of records with the highest measurement included in the chart.
Incoming threats	Displays incoming threats.
Outdoing threats	Displays outgoing threats.
Filer	Applies a filter to display only the selected type of items.
Zoom In and Zoom Out	Zoom in and zoom out of the map.
Export	Exports the graph as a .png image or as a PDF.
Last 6 hours Last 12 hours Last 24 hours Last 7 days Last 30 days Last 60 days Last 90 da	Indicates the period over which the measurements are taken.

Network Monitor Report

The App Scope Network Monitor report (**Monitor** > **App Scope** > **Network Monitor**) displays the bandwidth dedicated to different network functions over the specified period of time. Each network function is color-coded as indicated in the legend below the chart. For example, the image below shows application bandwidth for the past 7 days based on session information.



The Network Monitor report contains the following buttons and options.

Button	Description
Top 10	Determines the number of records with the highest measurement included in the chart.
Application	Determines the type of item reported: Application, Application Category, Source, or Destination.
Filter	Applies a filter to display only the selected item. **None** displays all entries.
	Determines whether to display session or byte information.
Export	Exports the graph as a .png image or as a PDF.
	Determines whether the information is presented in a stacked column chart or a stacked area chart.
	Indicates the period over which the change measurements are taken.

Traffic Map Report

The App Scope Traffic Map (**Monitor** > **App Scope** > **Traffic Map**) report shows a geographical view of traffic flows according to sessions or flows.

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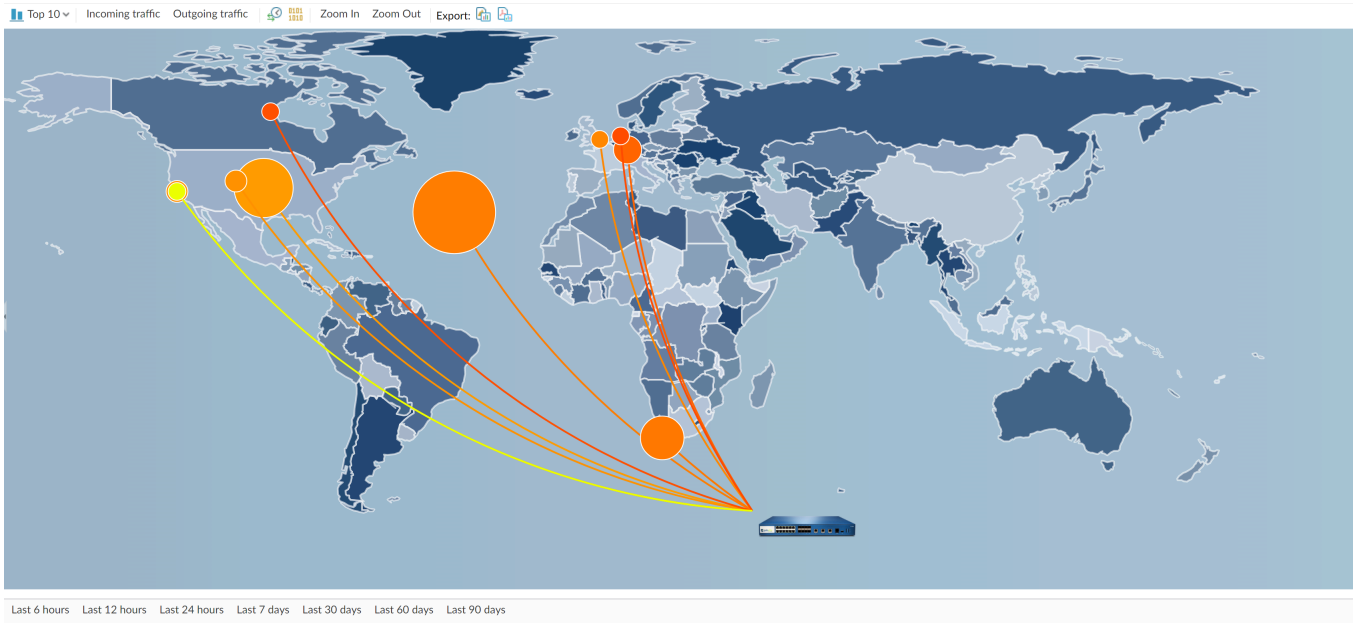
The firewall uses geolocation for creating traffic maps. The firewall is placed at the bottom of the traffic map screen, if you have not specified the geolocation coordinates (**Device** > **Setup** > **Management,** General Settings section) on the firewall.



Each traffic type is color-coded as indicated in the legend below the chart. The Traffic Map report contains the following buttons and options.

Buttons	Description
Top 10	Determines the number of records with the highest measurement included in the chart.
Incoming threats	Displays incoming threats.
Outgoing threats	Displays outgoing threats.
	Determines whether to display session or byte information.
Zoom In and Zoom Out	Zoom in and zoom out of the map.
Export	Exports the graph as a .png image or as a PDF.
Last 6 hours Last 12 hours Last 24 hours Last 7 days Last 30 days Last 60 days Last 90 days	Indicates the period over which the change measurements are taken.

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Use the Automated Correlation Engine

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

The automated correlation engine is an analytics tool that uses the logs on the firewall to detect actionable events on your network. The engine correlates a series of related threat events that, when combined, indicate a likely compromised host on your network or some other higher level conclusion. It pinpoints areas of risk, such as compromised hosts on the network, allows you to assess the risk and take action to prevent exploitation of network resources.

The following models support the automated correlation engine:

- *Panorama—M-Series appliances and virtual appliances*
- *PA-7000 Series firewalls*
- *PA-5400 Series firewall*
- *PA-5200 Series firewalls*
- *PA-3400 Series firewalls*
- *PA-3200 Series firewalls*

The automated correlation engine uses *correlation objects* to analyze the logs for patterns and when a match occurs, it generates a *correlated event*

Correlation Object

A correlation object is a definition file that specifies patterns to match against, the data sources to use for the lookups, and time period within which to look for these patterns. A pattern is a boolean structure of conditions that queries the following data sources (or logs) on the firewall: application statistics, traffic, traffic summary, threat summary, threat, data filtering, and URL filtering. Each pattern has a severity rating, and a threshold for the number of times the pattern match must occur within a defined time limit to indicate malicious activity. When the match conditions are met, a correlated event is logged.

A correlation object can connect isolated network events and look for patterns that indicate a more significant event. These objects identify suspicious traffic patterns and network anomalies, including suspicious IP activity, known command-and-control activity, known vulnerability exploits, or botnet activity that, when correlated, indicate with a high probability that a host on the network has been compromised. Correlation objects are defined and developed by the Palo Alto Networks Threat Research team, and are delivered with the weekly dynamic updates to the firewall and Panorama. To obtain new correlation objects, the firewall must have a Threat Prevention license. Panorama requires a support license to get the updates.

The patterns defined in a correlation object can be static or dynamic. Correlated objects that include patterns observed in WildFire are dynamic, and can correlate malware patterns detected by WildFire with command-and-control activity initiated by a host that was targeted with the

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malware on your network or activity seen by a Traps protected endpoint on Panorama. For example, when a host submits a file to the WildFire cloud and the verdict is malicious, the correlation object looks for other hosts or clients on the network that exhibit the same behavior seen in the cloud. If the malware sample had performed a DNS query and browsed to a malware domain, the correlation object will parse the logs for a similar event. When the activity on a host matches the analysis in the cloud, a high severity correlated event is logged.

Correlated Events

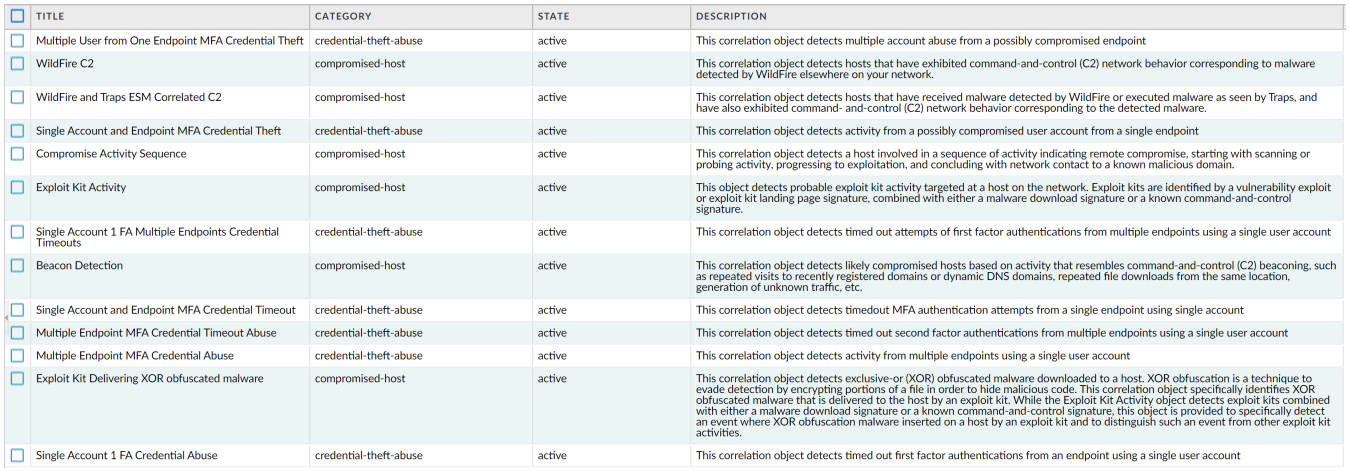
A correlated event is logged when the patterns and thresholds defined in a correlation object match the traffic patterns on your network. To Interpret Correlated Events and to view a graphical display of the events, see Use the Compromised Hosts Widget in the ACC.

View the Correlated Objects

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	☐ Support license ☐ (Panorama) Device management license

You can view the correlation objects that are currently available on the firewall.

STEP 1 | Select **Monitor** > **Automated Correlation Engine** > **Correlation Objects**. All the objects in the list are enabled by default.



TITLE	CATEGORY	STATE	DESCRIPTION
Multiple User from One Endpoint MFA Credential Theft	credential-theft-abuse	active	This correlation object detects multiple account abuse from a possibly compromised endpoint
WildFire C2	compromised-host	active	This correlation object detects hosts that have exhibited command-and-control (C2) network behavior corresponding to malware detected by WildFire elsewhere on your network.
WildFire and Traps ESM Correlated C2	compromised-host	active	This correlation object detects hosts that have received malware detected by WildFire or executed malware as seen by Traps, and have also exhibited command- and-control (C2) network behavior corresponding to the detected malware.
Single Account and Endpoint MFA Credential Theft	credential-theft-abuse	active	This correlation object detects activity from a possibly compromised user account from a single endpoint
Compromise Activity Sequence	compromised-host	active	This correlation object detects a host involved in a sequence of activity indicating remote compromise, starting with scanning or probing activity, progressing to exploitation, and concluding with network contact to a known malicious domain.
Exploit Kit Activity	compromised-host	active	This object detects probable exploit kit activity targeted at a host on the network. Exploit kits are identified by a vulnerability exploit or exploit kit landing page signature, combined with either a malware download signature or a known command-and-control signature.
Single Account 1 FA Multiple Endpoints Credential Timeouts	credential-theft-abuse	active	This correlation object detects timed out attempts of first factor authentications from multiple endpoints using a single user account
Beacon Detection	compromised-host	active	This correlation object detects likely compromised hosts based on activity that resembles command-and-control (C2) beaconing, such as repeated visits to recently registered domains or dynamic DNS domains, repeated file downloads from the same location, generation of unknown traffic, etc.
Single Account and Endpoint MFA Credential Timeout	credential-theft-abuse	active	This correlation object detects timedout MFA authentication attempts from a single endpoint using single account
Multiple Endpoint MFA Credential Timeout Abuse	credential-theft-abuse	active	This correlation object detects timed out second factor authentications from multiple endpoints using a single user account
Multiple Endpoint MFA Credential Abuse	credential-theft-abuse	active	This correlation object detects activity from multiple endpoints using a single user account
Exploit Kit Delivering XOR obfuscated malware	compromised-host	active	This correlation object detects exclusive-or (XOR) obfuscated malware downloaded to a host. XOR obfuscation is a technique to evade detection by encrypting portions of a file in order to hide malicious code. This correlation object specifically identifies XOR obfuscated malware that is delivered to the host by an exploit kit. While the Exploit Kit Activity object detects exploit kits combined with either a malware download signature or a known command-and-control signature, this object is provided to specifically detect an event where XOR obfuscation malware inserted on a host by an exploit kit and to distinguish such an event from other exploit kit activities.
Single Account 1 FA Credential Abuse	credential-theft-abuse	active	This correlation object detects timed out first factor authentications from an endpoint using a single user account

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STEP 2 | View the details on each correlation object. Each object provides the following information:

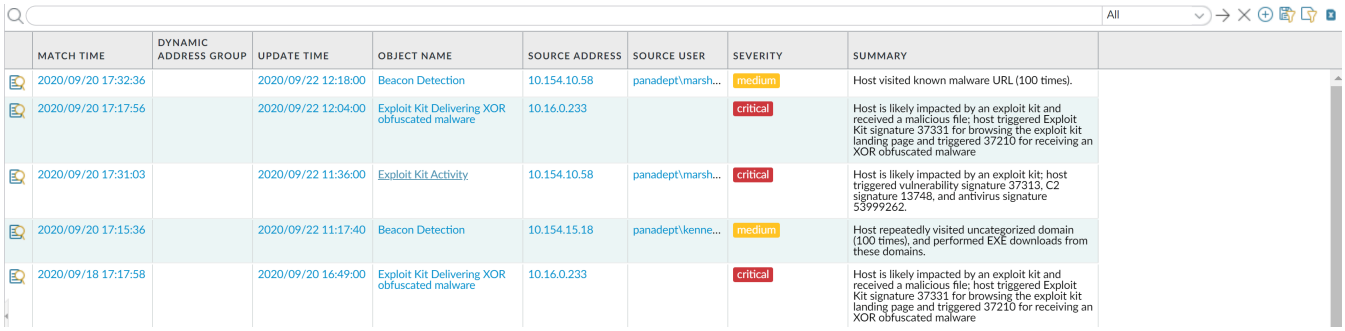
- **Name** and **Title**—The name and title indicate the type of activity that the correlation object detects. The name column is hidden from view, by default. To view the definition of the object, unhide the column and click the name link.
- **ID**— A unique number that identifies the correlation object; this column is also hidden by default. The IDs are in the 6000 series.
- **Category**—A classification of the kind of threat or harm posed to the network, user, or host. For now, all the objects identify compromised hosts on the network.
- **State**—Indicates whether the correlation object is enabled (active) or disabled (inactive). All the objects in the list are enabled by default, and are hence active. Because these objects are based on threat intelligence data and are defined by the Palo Alto Networks Threat Research team, keep the objects active in order to track and detect malicious activity on your network.
- **Description**—Specifies the match conditions for which the firewall or Panorama will analyze logs. It describes the sequence of conditions that are matched on to identify acceleration or escalation of malicious activity or suspicious host behavior. For example, the **Compromise Lifecycle** object detects a host involved in a complete attack lifecycle in a three-step escalation that starts with scanning or probing activity, progressing to exploitation, and concluding with network contact to a known malicious domain.

For more information, see Correlation Object and Use the Automated Correlation Engine.

Interpret Correlated Events

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	☐ Support license ☐ (Panorama) Device management license

You can view and analyze the logs generated for each correlated event in the **Monitor** > **Automated Correlation Engine** > **Correlated Events** tab.



MATCH TIME	DYNAMIC ADDRESS GROUP	UPDATE TIME	OBJECT NAME	SOURCE ADDRESS	SOURCE USER	SEVERITY	SUMMARY
2020/09/20 17:32:36		2020/09/22 12:18:00	Beacon Detection	10.154.10.58	panadept\marsh...	medium	Host visited known malware URL (100 times).
2020/09/20 17:17:56		2020/09/22 12:04:00	Exploit Kit Delivering XOR obfuscated malware	10.16.0.233		critical	Host is likely impacted by an exploit kit and received a malicious file; host triggered Exploit Kit signature 37331 for browsing the exploit kit landing page and triggered 37210 for receiving an XOR obfuscated malware
2020/09/20 17:31:03		2020/09/22 11:36:00	Exploit Kit Activity	10.154.10.58	panadept\marsh...	critical	Host is likely impacted by an exploit kit; host triggered vulnerability signature 37313, C2 signature 13748, and antivirus signature 53999262.
2020/09/20 17:15:36		2020/09/22 11:17:40	Beacon Detection	10.154.15.18	panadept\kenne...	medium	Host repeatedly visited uncategorized domain (100 times), and performed EXE downloads from these domains.
2020/09/18 17:17:58		2020/09/20 16:49:00	Exploit Kit Delivering XOR obfuscated malware	10.16.0.233		critical	Host is likely impacted by an exploit kit and received a malicious file; host triggered Exploit Kit signature 37331 for browsing the exploit kit landing page and triggered 37210 for receiving an XOR obfuscated malware

Correlated Events includes the following details:

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Field	Description
Match Time	The time the correlation object triggered a match.
Update Time	The time when the event was last updated with evidence on the match. As the firewall collects evidence on pattern or sequence of events defined in a correlation object, the time stamp on the correlated event log is updated.
Object Name	The name of the correlation object that triggered the match.
Source Address	The IP address of the user/device on your network from which the traffic originated.
Source User	The user and user group information from the directory server, if User-ID is enabled.
Severity *To configure the firewall or Panorama to send alerts using email, SNMP or syslog messages for a desired severity level, see Use External Services for Monitoring.*	A rating that indicates the urgency and impact of the match. The severity level indicates the extent of damage or escalation pattern, and the frequency of occurrence. Because correlation objects are primarily for detecting threats, the correlated events typically relate to identifying compromised hosts on the network and the severity implies the following: • **Critical**—Confirms that a host has been compromised based on correlated events that indicate an escalation pattern. For example, a critical event is logged when a host that received a file with a malicious verdict by WildFire exhibits the same command-and-control activity that was observed in the WildFire sandbox for that malicious file. • **High**—Indicates that a host is very likely compromised based on a correlation between multiple threat events, such as malware detected anywhere on the network that matches the command-and-control activity generated by a particular host. • **Medium**—Indicates that a host is likely compromised based on the detection of one or multiple suspicious events, such as repeated visits to known malicious URLs, which suggests a scripted command-and-control activity. • **Low**—Indicates that a host is possibly compromised based on the detection of one or multiple suspicious events, such as a visit to a malicious URL or a dynamic DNS domain. • **Informational**—Detects an event that may be useful in aggregate for identifying suspicious activity, but the event is not necessarily significant on its own.
Summary	A description that summarizes the evidence gathered on the correlated event.

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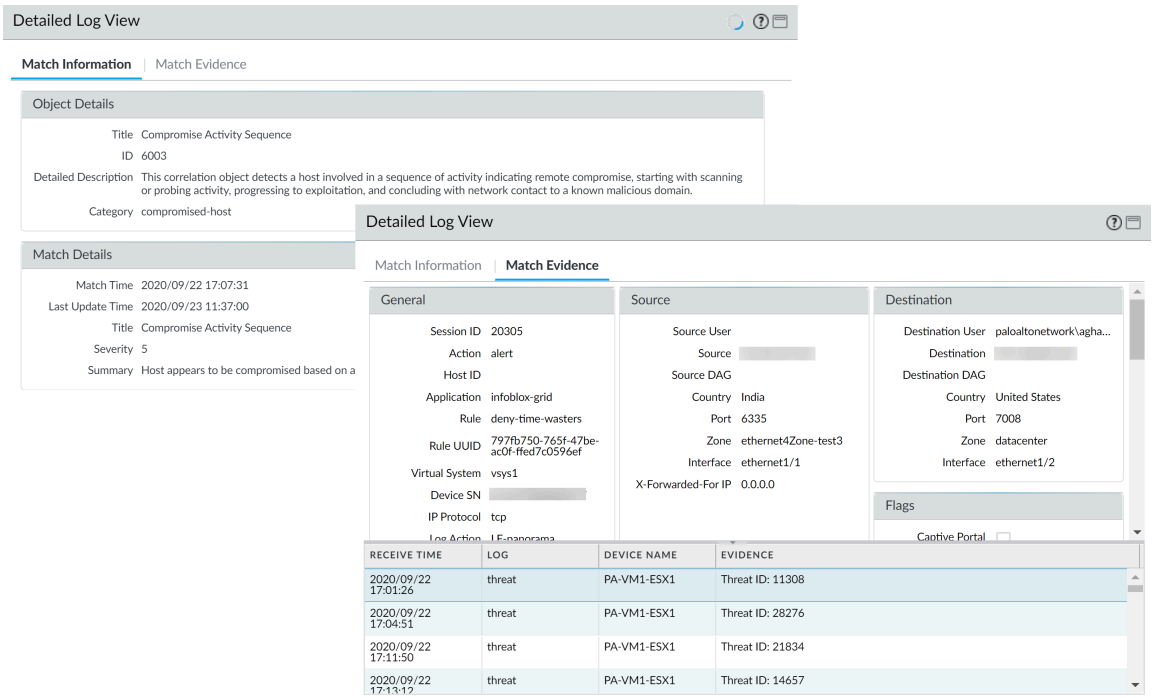
Click the icon to see the detailed log view, which includes all the evidence on a match:

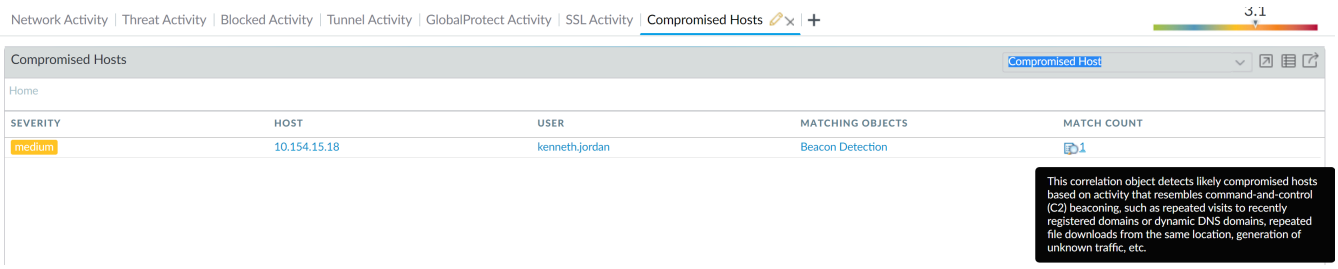


Tab	Description
Match Information	Object Details: Presents information on the Correlation Object that triggered the match.
	Match Details: A summary of the match details that includes the match time, last update time on the match evidence, severity of the event, and an event summary.
Match Evidence	Presents all the evidence that corroborates the correlated event. It lists detailed information on the evidence collected for each session.

Use the Compromised Hosts Widget in the ACC

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	☐ Support license ☐ (Panorama) Device management license

The compromised hosts widget on **ACC** > **Threat Activity**, aggregates the Correlated Events and sorts them by severity. It displays the source IP address/user who triggered the event, the correlation object that was matched and the number of times the object was matched. Use the match count link to jump to the match evidence details.



For more details, see Use the Automated Correlation Engine and Use the Application Command Center.

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Take Packet Captures

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

All Palo Alto Networks firewalls allow you to take packet captures (pcaps) of traffic that traverses the management interface and network interfaces on the firewall. When taking packet captures on the dataplane, you may need to Disable Hardware Offload to ensure that the firewall captures all traffic.

Packet capture is a troubleshooting feature that is rate limited in order to lower the impact on regular packet processing. If the firewall reaches the packet capture rate limit, you can view the number of packets that haven't been captured using the global counter **`flow_host_vardata_rate_limit_reached`**.

*Due to the way packets are processed in multi-core CPU platforms, packets captured in the **received** stage may not always appear in the same order as they were received by the network.*

Packet capture can be very CPU intensive and can degrade firewall performance. Only use this feature when necessary and make sure you turn it off after you have collected the required packets.

When troubleshooting performance issues or out-of-order related issues, it is recommended that you perform external packet captures on neighboring devices, such as switch SPAN ports.

Palo Alto Networks firewalls offer various types of packet captures to suit different troubleshooting needs:

- **Disable Hardware Offload:** This is a crucial step for ensuring comprehensive captures on the dataplane.
- **Take a Custom Packet Capture:** This allows a highly granular control over the capture parameters, such as specifying interfaces, filters (IP addresses, ports, protocols), and capture duration.
- **Take a Threat Packet Capture:** This specialized capture focuses on traffic related to detected threats, providing forensic data for security investigations.
- **Take an Application Packet Capture:** This type of capture helps in understanding application behavior and troubleshooting application-specific issues by focusing on traffic associated with particular applications.
- **Take a Packet Capture on the Management Interface:** This is useful for troubleshooting issues related to management plane access, such as GUI connectivity or syslog forwarding.

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Types of Packet Captures

There are different types of packet captures you can enable, depending on what you need to do:

- **Custom Packet Capture**—The firewall captures packets for all traffic or for specific traffic based on filters that you define. For example, you can configure the firewall to only capture packets to and from a specific source and destination IP address or port. You then use the packet captures for troubleshooting network#related issues or for gathering application attributes to enable you to write custom application signatures or to request an application signature from Palo Alto Networks. See Take a Custom Packet Capture.
- **Threat Packet Capture**—The firewall captures packets when it detects a virus, spyware, or vulnerability. You enable this feature in Antivirus, Anti-Spyware, and Vulnerability Protection security profiles. A link to view or export the packet captures will appear in the second column of the Threat log. These packet captures provide context around a threat to help you determine if an attack is successful or to learn more about the methods used by an attacker. You can also submit this type of pcap to Palo Alto Networks to have a threat re-analyzed if you feel it's a false-positive or false-negative. See Take a Threat Packet Capture.
- **Application Packet Capture**—The firewall captures packets based on a specific application and filters that you define. A link to view or export the packet captures will appear in the second column of the Traffic logs for traffic that matches the packet capture rule. See Take an Application Packet Capture.
- **Management Interface Packet Capture**—The firewall captures packets on the management interface (MGT). The packet captures are useful when troubleshooting services that traverse the interface, such as firewall management authentication to External Authentication Services, software and content updates, log forwarding, communication with SNMP servers, and authentication requests for GlobalProtect and Authentication Portal. See Take a Packet Capture on the Management Interface.
- **GTP Event Packet Capture**—The firewall captures a single GTP event, such as GTP-in-GTP, end user IP spoofing, and abnormal GTP messages, to make GTP troubleshooting easier for mobile network operators. Enable packet capture in a Mobile Network Protection profile.

Disable Hardware Offload

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	☐ Support license ☐ (Panorama) Device management license

Packet captures for traffic passing through the network data ports on a Palo Alto Networks firewall are performed by the dataplane CPU. To capture traffic that passes through the management interface, you must Take a Packet Capture on the Management Interface, in which case the packet capture is performed on the management plane.

When a packet capture is performed on the dataplane, the packet capture filter is used differently by the ingress stage, compared to the firewall, drop, and egress capture stages. The ingress stage uses the packet capture filter to copy individual packets that match the filter to the capture file. Packets that fail packet-parsing checks are dropped before being captured. The firewall, drop, and egress capture stages use the same packet capture filter to mark all new sessions that match the

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filter. Because each session, as recorded in the session tables, identifies both client-to-server and server-to-client connections, any traffic, in either direction, that matches to the flagged session will be copied to the firewall-stage and transmit-stage capture files. Likewise, any dropped traffic (post receive stage) in either direction that matches to a flagged session will be copied to the drop-stage capture file.

On firewall models that include a network processor, traffic that meets certain pre-determined criteria by Palo Alto Networks may be offloaded for handling by the network processor. Such offloaded traffic will not reach the dataplane CPU and will, therefore, not be captured. To capture offloaded traffic, you must use the CLI to turn off the hardware offload feature.

Common types of traffic that may be offloaded include non-decrypted SSL and SSH traffic (which being encrypted cannot be usefully inspected beyond the initial SSL/SSH session setup), network protocols (such as OSPF, BGP, RIP), and traffic that matches an application-override policy. Some types of traffic will never be offloaded, such as ARP, all non-IP traffic, IPSec, and VPN sessions. Individual SYN, FIN, and RST packets, even for session traffic that has been offloaded, will never be offloaded, and will always be passed through to the dataplane CPU, once recognized as such by the network processor.

Hardware offload is supported on the PA-3200 Series, PA-5200 Series, PA-5450, PA-7000 Series, and PA-7500 Series firewalls.

Disabling hardware offload may increase the dataplane CPU usage. If dataplane CPU usage is already high, you may want to schedule a maintenance window before disabling hardware offload.

(PAN-OS 11.1.0 and later releases) As an alternative to the below procedure, you can disable session offload only for capture traffic using the following command: ***debug dataplane packet-diag set filter offload no****. This is recommended if you don't want to disable offload on the whole firewall.*

STEP 1 | Disable hardware offload by running the following CLI command:

```
admin@PA-7050>set session offload no
```

STEP 2 | After the firewall captures the required traffic, enable hardware offload by running the following CLI command:

```
admin@PA-7050>set session offload yes
```

Take a Custom Packet Capture

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

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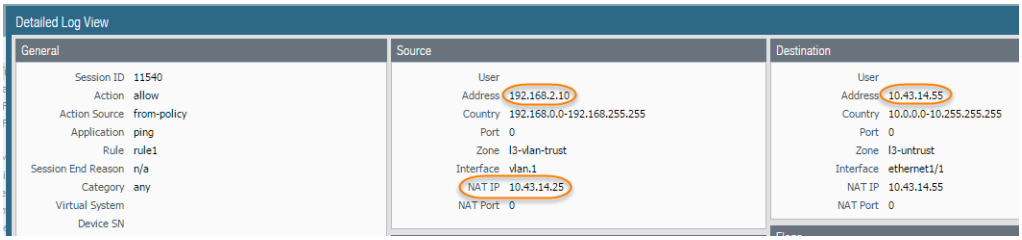
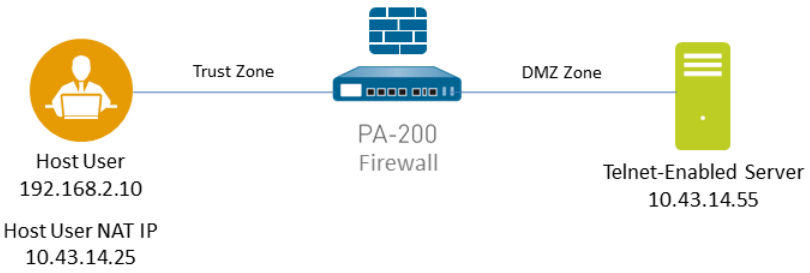
Custom packet captures allow you to define the traffic that the Next-Generation Firewall will capture. To ensure that you capture all traffic, you may need to Disable Hardware Offload.

STEP 1 | Before you start a packet capture, identify the attributes of the traffic that you want to capture.

For example, to determine the source IP address, source NAT IP address, and the destination IP address for traffic between two systems, perform a ping from the source system to the to the destination system. After the ping is complete, go to **Monitor** > **Traffic** and locate the traffic log for the two systems. Click the **Detailed Log View** icon located in the first column of the log and note the source address, source NAT IP, and the destination address.



The following example shows how to use a packet capture to troubleshoot a Telnet connectivity issue from a user in the Trust zone to a server in the DMZ zone.



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STEP 2 | Set packet capture filters, so the Next-Generation Firewall only captures traffic you are interested in.

Using filters makes it easier for you to locate the information you need in the packet capture and will reduce the processing power required by the Next-Generation Firewall to take the packet capture. To capture all traffic, do not define filters and leave the filter option off.

For example, if you configured NAT on the Next-Generation Firewall, you will need to apply two filters. The first one filters on the pre-NAT source IP address to the destination IP address and the second one filters traffic from the destination server to the source NAT IP address.

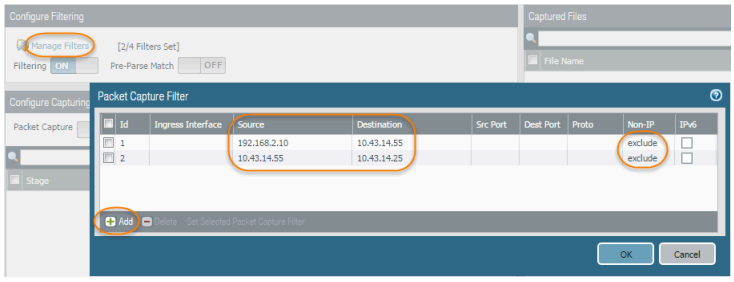
1. Select **Monitor** > **Packet Capture**.
2. Click **Clear All Settings** at the bottom of the window to clear any existing capture settings.
3. Click **Manage Filters** and click **Add**.
4. Select **Id** 1 and in the **Source** field enter the source IP address you are interested in and in the **Destination** field enter a destination IP address.

   For example, enter the source IP address **`192.168.2.10`** and the destination IP address **`10.43.14.55`**.

   (12.1.2 and later releases) You can enter a source IP address range and a destination IP address range to capture traffic within the specified range. Use dash delimiters to separate the range of values. For example, enter the source IP address range **`192.168.2.10-192.168.2.20`** and the destination IP address range **`10.43.14.55-10.43.14.65`**.

   To further filter the capture, set **Non-IP** to **exclude** non-IP traffic, such as broadcast traffic.
5. **Add** the second filter and select **Id** 2.

   For example, in the **Source** field enter **`10.43.14.55`** and in the **Destination** field enter **`10.43.14.25`**. In the **Non-IP** drop-down menu select **exclude**.



6. Click **OK**.

STEP 3 | Set **Filtering** to **On**.

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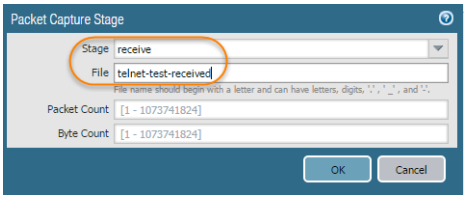
STEP 4 | Specify the traffic stage(s) that trigger the packet capture and the filename(s) to use to store the captured content. For a definition of each stage, click the **Help** icon on the packet capture page.

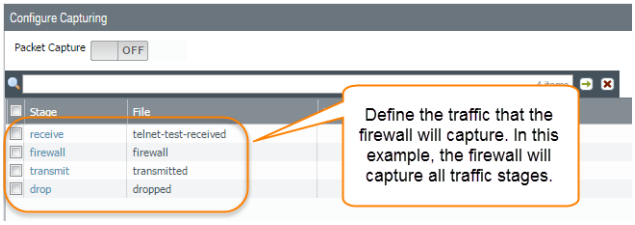
For example, to configure all packet capture stages and define a filename for each stage, perform the following procedure:

1. **Add** a **Stage** to the packet capture configuration and define a **File** name for the resulting packet capture.

   For example, select **receive** as the **Stage** and set the **File** name to telnet-test-received.

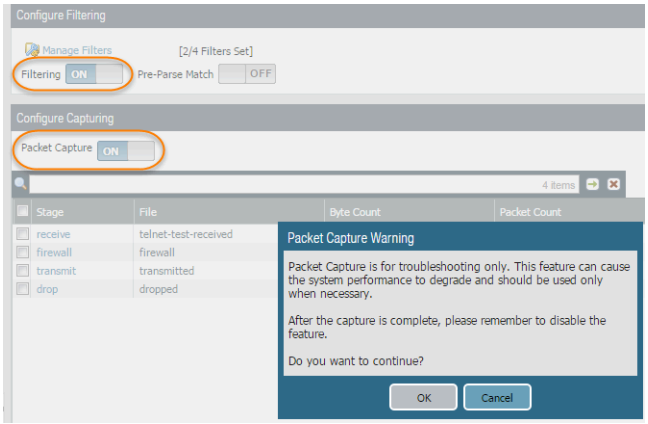


2. Continue to **Add** each **Stage** you want to capture (**receive, firewall**, **transmit**, and **drop**) and set a unique **File** name for each stage.



STEP 5 | Set **Packet Capture** to **ON**.

The Next-Generation Firewall or appliance warns you that system performance can be degraded; acknowledge the warning by clicking **OK**. If you define filters, the packet capture should have little impact on performance, but you should always turn **Off** packet capture after the Next-Generation Firewall captures the data that you want to analyze.



STEP 6 | Generate traffic that matches the filters that you defined.

For this example, generate traffic from the source system to the Telnet-enabled server by running the following command from the source system (192.168.2.10):

```
telnet 10.43.14.55
```

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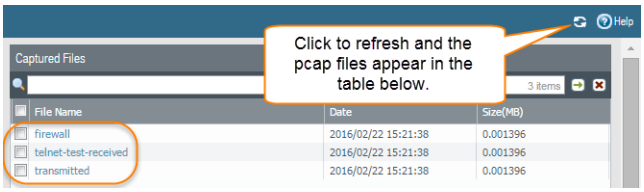
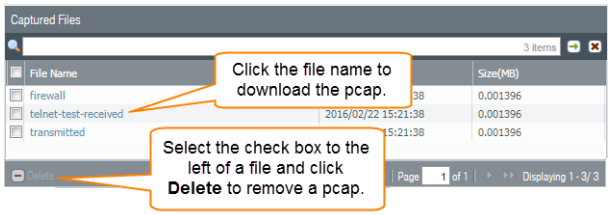
STEP 7 | Turn packet capture **OFF** and then click the refresh icon to see the packet capture files.



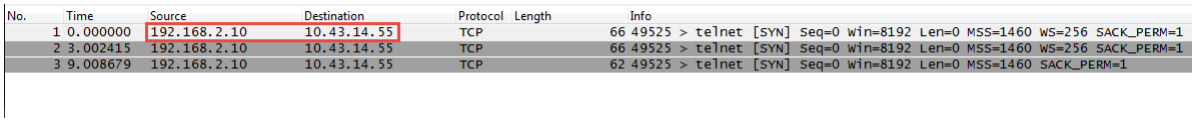
Notice that in this case, there were no dropped packets, so the Next-Generation Firewall did not create a file for the drop stage.

STEP 8 | Download the packet captures by clicking the filename in the File Name column.



STEP 9 | View the packet capture files using a network packet analyzer.

In this example, the received.pcap packet capture shows a failed Telnet session from the source system at 192.168.2.10 to the Telnet-enabled server at 10.43.14.55. The source system sent the Telnet request to the server, but the server did not respond. In this example, the server may not have Telnet enabled, so check the server.



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.2.10	10.43.14.55	TCP		66 49525 > telnet [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
2	3.002415	192.168.2.10	10.43.14.55	TCP		66 49525 > telnet [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
3	9.008679	192.168.2.10	10.43.14.55	TCP		62 49525 > telnet [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM=1

STEP 10 | Enable the Telnet service on the destination server (10.43.14.55) and turn on packet capture to take a new packet capture.

STEP 11 | Generate traffic that will trigger the packet capture.

Run the Telnet session again from the source system to the Telnet-enabled server

```
telnet 10.43.14.55
```

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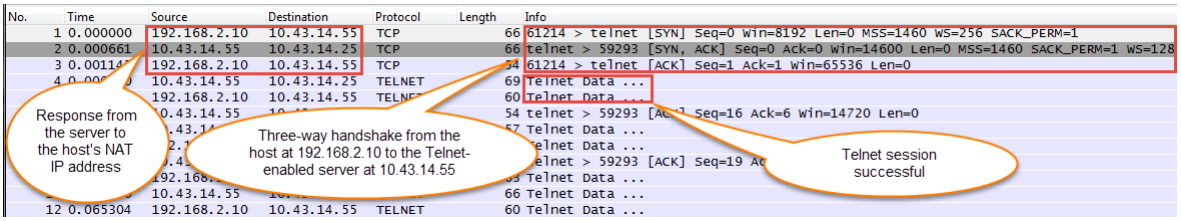
STEP 12 | Download and open the received.pcap file and view it using a network packet analyzer.

The following packet capture now shows a successful Telnet session from the host user at 192.168.2.10 to the Telnet-enabled server at 10.43.14.55.

You also see the NAT address 10.43.14.25. When the server responds, it does so to the NAT address. You can see the session is successful as indicated by the three-way handshake between the host and the server and then you see Telnet data.



Take a Threat Packet Capture

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	☐ Support license ☐ (Panorama) Device management license

To configure the firewall to take a packet capture (pcap) when it detects a threat, enable packet capture on Antivirus, Anti-Spyware, and Vulnerability Protection security profiles.

Threats that are detected using the advanced Inline Cloud Analysis engines do not generate packet capture data.

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STEP 1 | Enable the packet capture option in the security profile.

Some security profiles allow you to define a single-packet capture or an extended-capture. If you choose extended-capture, define the capture length. This will allow the firewall to capture more packets to provide additional context related to the threat.

If the action for a given threat is allow, the firewall does not trigger a Threat log and does not capture packets. If the action is alert, you can set the packet capture to single-packet or extended-capture. All blocking actions (drop, block, and reset actions) capture a single packet. The content package on the device determines the default action.

1. Select **Objects** > **Security Profiles** and enable the packet capture option for the supported profiles as follows:
   - **Antivirus**—Select a custom antivirus profile and in the **Antivirus** tab select the **Packet Capture** check box.
   - **Anti-Spyware**—Select a custom Anti-Spyware profile, click **Signature Policies**, **Signature Exceptions**, or the **DNS Policies** tab and in the Packet Capture drop-down, select **single-packet** or **extended-capture**.

     ***Signature Policies** packet captures apply to multiple signatures across a specified category or matching threat name, while **Signature Exceptions** packet captures apply to a specific signature.*
   - **Vulnerability Protection**—Select a custom Vulnerability Protection profile and in the **Rules** tab, click **Add** to add a new rule, or select an existing rule. Set **Packet Capture** to **single-packet** or **extended-capture**.

     *If the profile has signature exceptions defined, click the **Exceptions** tab and in the **Packet Capture** column for a signature, set **single-packet** or **extended-capture**.*
2. (Optional) If you selected **extended-capture** for any of the profiles, define the extended packet capture length.
   1. Select **Device** > **Setup** > **Content-ID** and edit the Content-ID Settings.
   2. In the **Extended Packet Capture Length (packets)** section, specify the number of packets that the firewall will capture (range is 1-50; default is 5).
   3. Click **OK**.

STEP 2 | Add the security profile (with packet capture enabled) to a Security Policy rule.

1. Select **Policies** > **Security** and select a rule.
2. Select the **Actions** tab.
3. In the Profile Settings section, select a profile that has packet capture enabled.

   For example, click the **Antivirus** drop-down and select a profile that has packet capture enabled.

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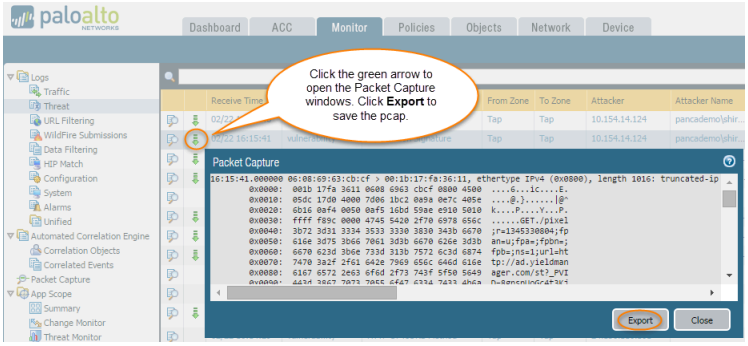
STEP 3 | View/export the packet capture from the Threat logs.

1. Select **Monitor** > **Logs** > **Threat**.
2. In the log entry that you are interested in, click the green packet capture icon in the second column. View the packet capture directly or **Export** it to your system.



Take an Application Packet Capture

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	☐ Support license ☐ (Panorama) Device management license

To configure the firewall for application packet captures, you have two primary methods:

- Take a Packet Capture for Unknown Applications
- Take a Custom Application Packet Capture

Take a Packet Capture for Unknown Applications

Palo Alto Networks firewalls automatically generate a packet capture for sessions that contain an application that the firewall cannot identify. Typically, the only applications that are classified as unknown traffic—tcp, udp, or non-syn-tcp—are commercially available applications that do not yet have App-ID signatures, are internal or custom applications on your network, or potential threats. You can use these packet captures to gather more context related to the unknown application or use the information to analyze the traffic for potential threats. You can also Manage Custom or Unknown Applications by controlling them through security policy or by writing a custom application signature and then creating a security rule based on the custom signature. If the application is a commercial application, you can submit the packet capture to Palo Alto Networks to have an App-ID signature created.

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STEP 1 | Verify that unknown application packet capture is enabled (this option is enabled by default).

1. To view the unknown application capture setting, run the following CLI command:

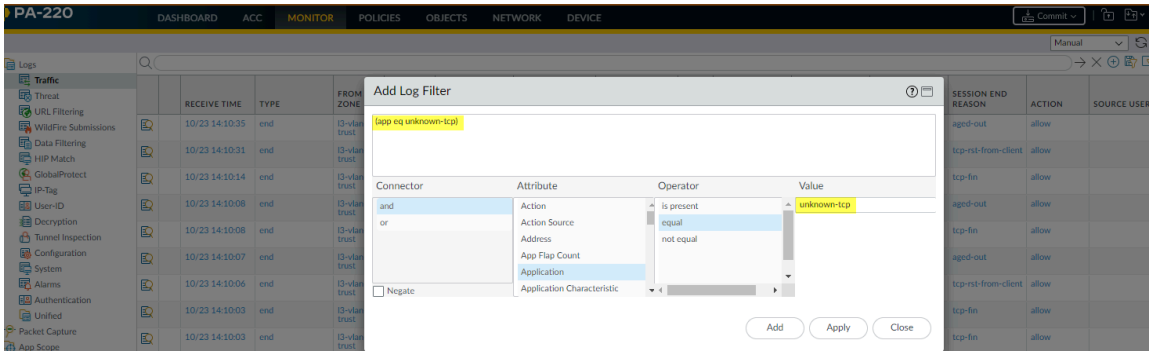
```
admin@PA-220>show running application setting | match “Unknown
 capture”
```

2. If the unknown capture setting option is off, enable it:

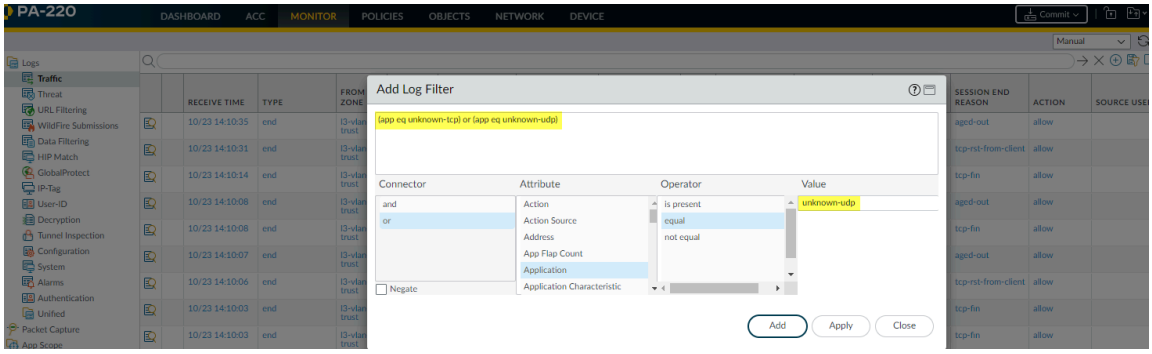
```
admin@PA-220>set application dump-unknown yes
```

STEP 2 | Locate unknown TCP and UDP applications by filtering the traffic logs.

1. Select **Monitor** > **Logs** > **Traffic**.
2. Click **Add Filter**, create the unknown TCP portion of the filter (**Connector** = “and”, **Attribute** = “Application”, **Operator** = “equal”, and enter “unknown-tcp” as the **Value**), and then click **Add** to add the query to the filter.



3. Create the unknown UDP portion of the filter (**Connector** = “or”, **Attribute** = “Application”, **Operator** = “equal”, and enter “unknown-udp” as the **Value**), and then click **Add** to add the query to the filter.



4. Click **Apply** to place the filter in the log screen query field.

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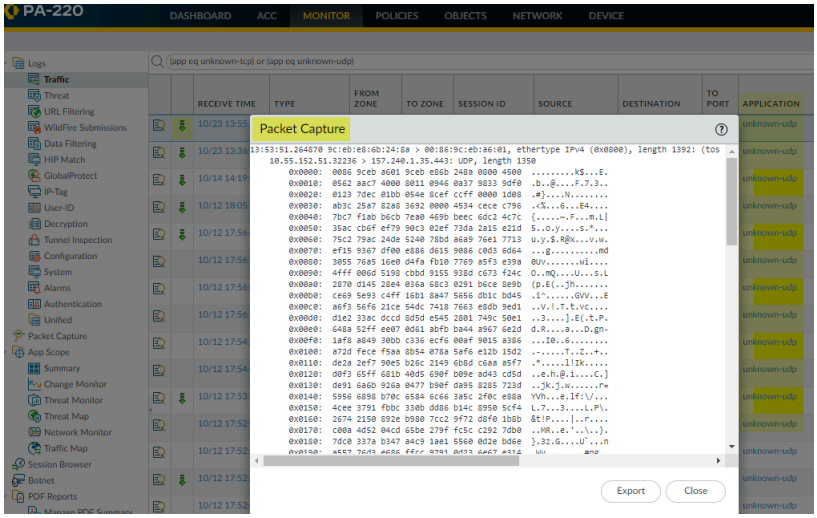
STEP 3 | Click the **Apply Filter** arrow next to the query field to run the filter and then click the packet capture icon to view the packet capture or **Export** it to your local system.



Take a Custom Application Packet Capture

You can configure a Palo Alto Networks firewall to take a packet capture based on an application name and filters that you define. You can then use the packet capture to troubleshoot issues with controlling an application. When configuring an application packet capture, you must use the application name defined in the App-ID database. You can view a list of all App-ID applications using Applipedia or from the web interface on the firewall in **Objects** > **Applications**.

STEP 1 | Using a terminal emulation application, such as PuTTY, launch an SSH session to the firewall.

STEP 2 | Turn on the application packet capture and define filters.

```
admin@PA-220>set application dump on application <application-name>
 rule <rule-name>
```

For example, to capture packets for the linkedin-base application that matches the security rule named Social Networking Apps, run the following CLI command:

```
admin@PA-220>set application dump on application linkedin-base rule
 "Social Networking Apps"
```

You can also apply other filters, such as source IP address and destination IP address.

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STEP 3 | View the packet capture output to ensure that the correct filters are applied. The output displays after you enable the packet capture.

The following output confirms that application capture filtering is now based on the linkedin-base application for traffic that matches the Social Networking Apps rule.

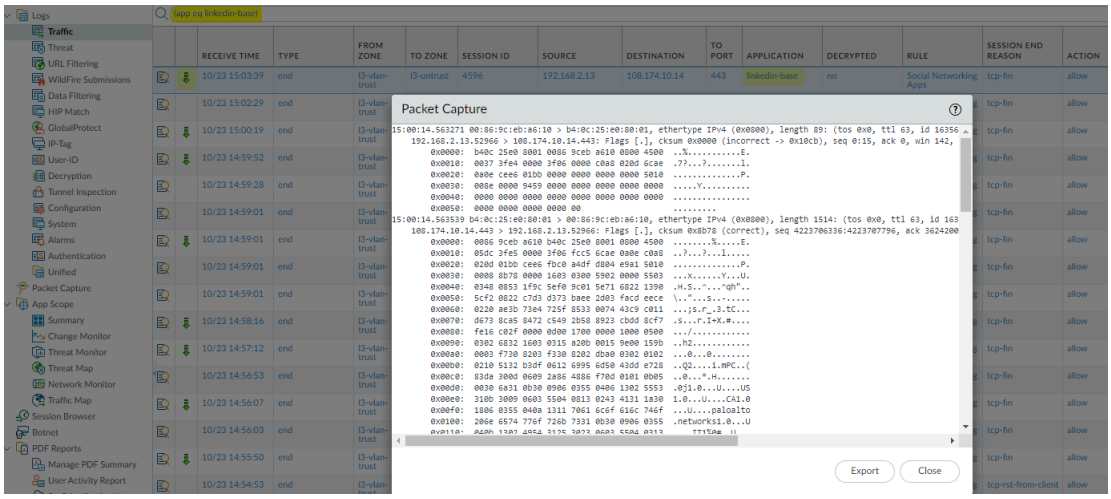
```
Application setting:
Application cache              : yes
Supernode                      : yes
Heuristics                     : yes
Cache Threshold                : 16
Bypass when exceeds queue limit: no
Traceroute appid               : yes
Traceroute TTL threshold       : 30
Use cache for appid            : no
Use simple appsigs for ident   : yes
Use AppID cache on SSL/SNI     : no
Unknown capture                : on
Max. unknown sessions          : 5000
Current unknown sessions       : 7
Application capture            : on
Max. application sessions      : 5000
Current application sessions   : 0
Application filter setting:
    Rule                       : Social Networking Apps
    From                       : any
    To                         : any
    Source                     : any
    Destination                : any
    Protocol                   : any
    Source Port                : any
    Dest. Port                 : any
    Application                : linkedin-base

Current APPID Signature
   Memory Usage                : 16768  KB (Actual 16440  KB)
      TCP 1 C2S                : regex 11898  states
      TCP 1 S2C                : regex 4549   states
      UDP 1 C2S                : regex 4234   states
      UDP 1 S2C                : regex 1605   states

Alternate APPID Signature
   Memory Usage                : 16768  KB (Actual 16425  KB)
      TCP 1 C2S                : regex 11878  states
      TCP 1 S2C                : regex 4549   states
      UDP 1 C2S                : regex 4233   states
      UDP 1 S2C                : regex 1604   states
```

STEP 4 | Access linkedin.com from a web browser and perform some LinkedIn tasks to generate LinkedIn traffic, and then run the following CLI command to turn off application packet capture:

```
admin@PA-220>set application dump off
```

STEP 5 | View/export the packet capture.

1. Log in to the web interface on the firewall and select **Monitor** > **Logs** > **Traffic**.
2. In the log entry that you are interested in, click the green packet capture icon .
3. View the packet capture directly or **Export** it to your computer. The following screen capture shows the linkedin-base packet capture.



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Take a Packet Capture on the Management Interface

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

The **tcpdump** CLI command enables you to capture packets that traverse the management interface (MGT) on a Palo Alto Networks firewall.

*Each platform has a default number of bytes that **tcpdump** captures. The PA-220 firewalls capture 68 bytes of data from each packet and anything over that is truncated. The PA-7000 Series firewalls and VM-Series firewalls capture 96 bytes of data from each packet. To define the number of packets that **tcpdump** will capture, use the **snaplen** (snap length) option (range 0-65535). Setting the **snaplen** to 0 will cause the firewall to use the maximum length required to capture whole packets.*

STEP 1 | Using a terminal emulation application, such as PuTTY, launch an SSH session to the firewall.

STEP 2 | To start a packet capture on the MGT interface, run the following command:

```
admin@PA-220>tcpdump filter “<filter-option> <IP-address>” snaplen
 length
```

For example, to capture the traffic that is generated when and administrator authenticates to the firewall using RADIUS, filter on the destination IP address of the RADIUS server (10.5.104.99 in this example):

```
admin@PA-220>tcpdump filter “dst 10.5.104.99” snaplen 0
```

You can also filter on src (source IP address), host, net, and you can exclude content. For example, to filter on a subnet and exclude all SCP, SFTP, and SSH traffic (which uses port 22), run the following command:

```
admin@PA-220>tcpdump filter “net 10.5.104.0/24 and not port 22”
 snaplen 0
```

*Each time **tcpdump** takes a packet capture, it stores the content in a file named mgmt.pcap. This file is overwritten each time you run **tcpdump**.*

STEP 3 | After the traffic you are interested in has traversed the MGT interface, press Ctrl + C to stop the capture.

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STEP 4 | View the packet capture by running the following command:

```
admin@PA-220> view-pcap mgmt-pcap mgmt.pcap
```

The following output shows the packet capture from the MGT port (10.5.104.98) to the RADIUS server (10.5.104.99):

```
09:55:29.139394 IP 10.5.104.98.43063 > 10.5.104.99.radius: RADIUS,
 Access Request (1), id: 0x00 length: 89
 09:55:29.144354 arp reply 10.5.104.98 is-at 00:25:90:23:94:98 (oui
 Unknown)
 09:55:29.379290 IP 10.5.104.98.43063 > 10.5.104.99.radius: RADIUS,
 Access Request (1), id: 0x00 length: 70
 09:55:34.379262 arp who-has 10.5.104.99 tell 10.5.104.98
```

STEP 5 | (Optional) Export the packet capture from the firewall using SCP (or TFTP). For example, to export the packet capture using SCP, run the following command:

```
admin@PA-220>scp export mgmt-pcap from mgmt.pcap
 to <username@host:path>
```

For example, to export the pcap to an SCP enabled server at 10.5.5.20 to a temp folder named temp-SCP, run the following CLI command:

```
admin@PA-220>scp export mgmt-pcap from mgmt.pcap to
 admin@10.5.5.20:c:/temp-SCP
```

Enter the login name and password for the account on the SCP server to enable the firewall to copy the packet capture to the c:\temp-SCP folder on the SCP-enabled.

STEP 6 | You can now view the packet capture files using a network packet analyzer, such as Wireshark.

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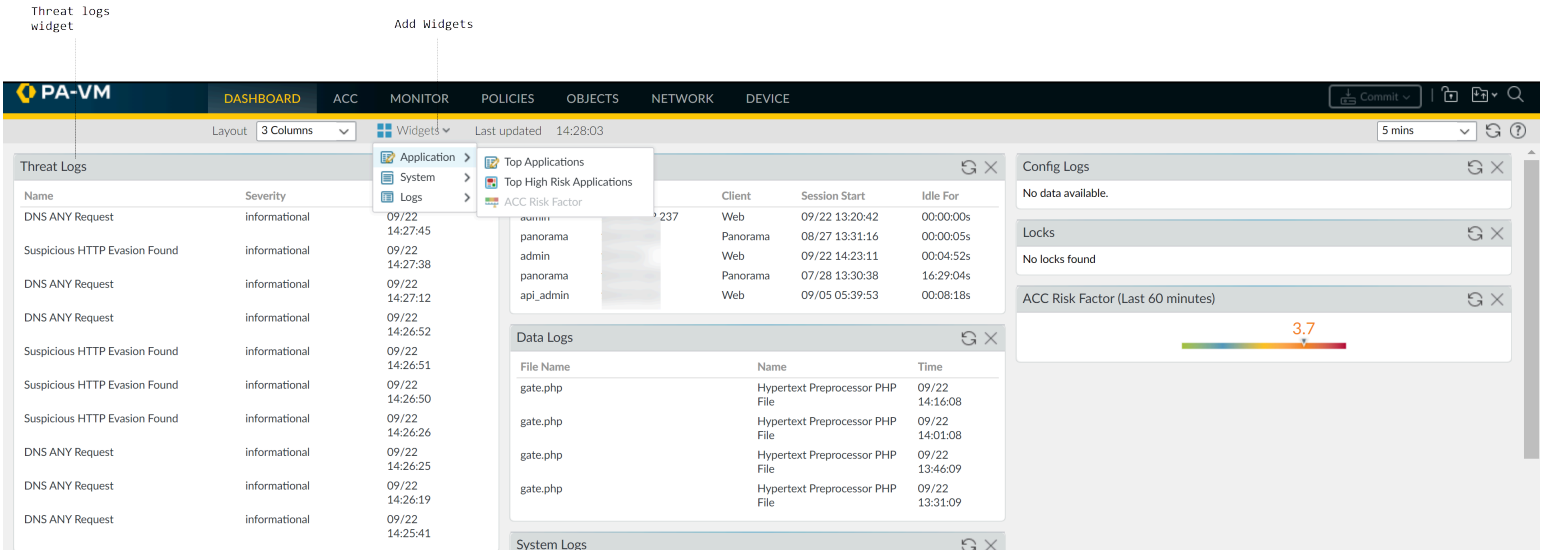
Monitor Applications and Threats

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

Palo Alto Networks next-generation firewalls come equipped with the App-ID technology, which identifies the applications traversing your network, irrespective of protocol, encryption, or evasive tactic. This deep application awareness helps you manage applications effectively using the Application Command Center (ACC). The ACC offers a graphical summary of data from various log databases, highlighting which applications are in use, by whom, and their potential security implications. ACC is dynamically updated, using the continuous traffic classification that App-ID performs; if an application changes ports or behavior, App-ID continues to see the traffic, displaying the results in ACC. Beyond applications, the ACC enriches this visibility with insights into URL categories, threats, and data patterns, painting a comprehensive picture of the network landscape and enabling the creation of more informed security policies.

To enhance monitoring capabilities, you can also leverage the Dashboard feature on the firewall. This customizable dashboard provides a quick overview of key network metrics and security posture.



To proactively assess security risks, review the Content Delivery Network (CDN) infrastructure. This involves examining logged events on the firewall that might indicate suspicious activity related to CDN traffic. Additionally, the AutoFocus intelligence summary provides context on the prevalence of specific properties, activities, or behaviors associated with logs in your network and on a global scale. This summary also includes the WildFire verdict and relevant AutoFocus tags. With an active AutoFocus subscription, you can leverage this information to configure customized AutoFocus Alerts, enabling proactive tracking of specific threats and behaviors pertinent to your network environment.

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View and Manage Logs

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

A log is an automatically generated, time-stamped file that provides an audit trail for system events on the firewall or network traffic events that the firewall monitors. Log entries contain *artifacts*, which are properties, activities, or behaviors associated with the logged event, such as the application type or the IP address of an attacker. Each log type records information for a separate event type, ensuring a structured approach to data collection. For example, the firewall generates a Threat log. This specialized log captures entries whenever network traffic aligns with a configured spyware, vulnerability, or virus signature. It also records instances of Denial-of-Service (DoS) attacks that meet predefined thresholds, such as those configured for port scan or host sweep activities, signaling potential malicious intent. This segregation by log type allows administrators to focus on specific security or operational aspects.

You can utilize various functions to manage these logs. This includes reviewing different Log Types and Severity Levels, and accessing them through View Logs. To refine their analysis, you can Filter Logs based on specific criteria. For record-keeping or offline analysis, the system supports Exporting Logs, with a dedicated Use Case: Export Traffic Logs for a Date Range provided. Furthermore, firewalls allow you to Configure Log Storage Quotas and Expiration Periods to manage disk space, and Schedule Log Exports to an SCP or FTP Server for long-term archiving and integration with external systems.

View Logs

You can view the different log types on the firewall in a tabular format. The firewall locally stores all log files and automatically generates Configuration and System logs by default. To learn more about the security rules that trigger the creation of entries for the other types of logs, see Log Types and Severity Levels.

To configure the firewall to forward logs as syslog messages, email notifications, or Simple Network Management Protocol (SNMP) traps, Use External Services for Monitoring.

STEP 1 | Select a log type to view.

1. Select **Monitor** > **Logs**.
2. Select a log type from the list.

The firewall displays only the logs you have permission to see. For example, if your administrative account does not have permission to view WildFire Submissions logs, the firewall does not display that log type when you access the logs pages. Administrator Role Types *define the permissions.*

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STEP 2 | (Optional) Customize the log column display.

1. Click the arrow to the right of any column header, and select **Columns**.
2. Select columns to display from the list. The log updates automatically to match your selections.

STEP 3 | View additional details about log entries.

- Click the spyglass ( ) for a specific log entry. The Detailed Log View has more information about the source and destination of the session, as well as a list of sessions related to the log entry.
- (Threat log only) Click next to an entry to access local packet captures of the threat. To enable local packet captures, see Take Packet Captures.
- (Traffic, Threat, URL Filtering, WildFire Submissions, Data Filtering, and Unified logs only) View AutoFocus threat data for a log entry.

  1. Enable AutoFocus.

     *Enable AutoFocus in Panorama to view AutoFocus threat data for all Panorama log entries, including those from firewalls that are not connected to AutoFocus and/or are running PAN-OS 7.0 and earlier release versions (**Panorama** > **Setup** > **Management** > **AutoFocus**).*

  2. Hover over an IP address, URL, user agent, threat name (subtype: virus and wildfire-virus only), filename, or SHA-256 hash.
  3. Click the drop-down ( ) and select **AutoFocus**.
  4. Content Delivery Network Infrastructure.

Next Steps...

- Filter Logs.
- Export Logs.
- Configure Log Storage Quotas and Expiration Periods.

Filter Logs

Each log has a filter area that allows you to set a criteria for which log entries to display. The ability to filter logs is useful for focusing on events on your firewall that possess particular properties or attributes. Filter logs by artifacts that are associated with individual log entries.

For example, filtering by the rule UUID makes it easier to pinpoint the specific rule you want to locate, even among many similarly-named rules. If your ruleset is very large and contains many rules, using the rule's UUID as a filter spotlights the particular rule you need to find without having to navigate through pages of results.

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STEP 1 | (Unified logs only) Select the log types to include in the Unified log display.

1. Click Effective Queries ( ).
2. Select one or more log types from the list (**traffic**, **threat**, **url**, **data**, and **wildfire**).
3. Click **OK**. The Unified log updates to show only entries from the log types you have selected.

STEP 2 | Add a filter to the filter field.

*If the value of the artifact matches the operator (such as **has** or **in**), enclose the value in quotation marks to avoid a syntax error. For example, if you filter by destination country and use IN as a value to specify INDIA, enter the filter as **(dstloc eq "IN")**.*

- Click one or more artifacts (such as the application type associated with traffic and the IP address of an attacker) in a log entry. For example, click the Source **10.0.0.25** and Application **web-browsing** of a log entry to display only entries that contain both artifacts in the log (AND search).
- To specify artifacts to add to the filter field, click Add Filter ( ).
- To add a previously saved filter, click Load Filter ( ).

STEP 3 | Apply the filter to the log.

Click Apply Filter ( ). The log will refresh to display only log entries that match the current filter.

STEP 4 | (Optional) Save frequently used filters.

1. Click Save Filter ( ).
2. Enter a **Name** for the filter.
3. Click **OK**. You can view your saved filters by clicking Load Filter ( ).

Next Steps...

- View Logs.
- Export Logs.

Export Logs

You can export the contents of a log type to a comma-separated value (CSV) formatted report. By default, the report contains up to 2,000 rows of log entries.

Although the PAN-OS firewall exports logs successfully, large log exports might appear truncated when opened in some spreadsheet types, even though the underlying file is complete. Check the limitations for the spreadsheet application to determine if those limits are affecting the export view. Use a different application for large log analysis.

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STEP 1 | Set the number of rows to display in the report.

1. Select **Device** > **Setup** > **Management**, then edit the Logging and Reporting Settings.
2. Click the **Log Export and Reporting** tab.
3. Edit the number of **Max Rows in CSV Export** (up to 1048576 rows).
4. Click **OK**.

STEP 2 | Download the log.

1. Click Export to CSV ( ). A progress bar showing the status of the download appears.
2. When the download is complete, click **Download file** to save a copy of the log to your local folder. For descriptions of the column headers in a downloaded log, refer to Syslog Field Descriptions.

Next Step...

Schedule Log Exports to an SCP or FTP Server.

Use Case: Export Traffic Logs for a Date Range

This example provides information and tips for filtering and exporting traffic logs for a specific date range. Examples of date range filters for Traffic logs are:

- All Traffic for a specific date (yyyy/mm/dd) and time (hh:mm:ss)
- All Traffic received on or before the date (yyyy/mm/dd) and time (hh:mm:ss)
- All Traffic received on or after the date (yyyy/mm/dd) and time (hh:mm:ss)
- All Traffic received between the date-time range of yyyy/mm/dd hh:mm:ss and yyyy/mm/dd hh:mm:ss (this use case)

To filter for traffic received between a date and time range,

STEP 1 | Select **Monitor** > **Logs**.

STEP 2 | Select the **Traffic** log type.

STEP 3 | Add the filter to the filter field.

For example, to export Traffic logs from 08/03/2023 to 08/04/2023, add **`(receive_time geq '2023/08/03 00:00:00') and (receive_time leq '2023/08/04 23:59:59')`** to the filter field and **Apply Filter**.

STEP 4 | **Export to CSV**.

*Use smaller date ranges or reduce the **Max Rows in CSV Export** if your exported log file does not include the complete results expected.*

STEP 5 | Download the exported file.

Configure Log Storage Quotas and Expiration Periods

The firewall automatically deletes logs that exceed the expiration period. When the firewall reaches the storage quota for a log type, it automatically deletes older logs of that type to create space even if you don't set an expiration period.

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*If you want to manually delete logs, select **Device** > **Log Settings** and, in the Manage Logs section, click the links to clear logs by type.*

STEP 1 | Select **Device** > **Setup** > **Management** and edit the Logging and Reporting Settings.

STEP 2 | Select **Log Storage** and enter a **Quota (%)** for each log type. When you change a percentage value, the dialog refreshes to display the corresponding absolute value (Quota GB/MB column).

STEP 3 | Enter the **Max Days** (expiration period) for each log type (range is 1-2,000). The fields are blank by default, which means the logs never expire.

The firewall synchronizes expiration periods across high availability (HA) pairs. Because only the active HA peer generates logs, the passive peer has no logs to delete unless failover occurs and it starts generating logs.

STEP 4 | Click **OK** and **Commit**.

Schedule Log Exports to an SCP or FTP Server

You can schedule exports of Traffic, Threat, URL Filtering, Data Filtering, HIP Match, and WildFire Submission logs to a Secure Copy (SCP) server or File Transfer Protocol (FTP) server. Perform this task for each log type you want to export.

You can use Secure Copy (SCP) commands from the CLI to export the entire log database to an SCP server and import it to another firewall. Because the log database is too large for an export or import to be practical on the following platforms, they do not support these options: PA-7000 Series firewalls (all PAN-OS releases), Panorama virtual appliance running Panorama 6.0 or later releases, and Panorama M-Series appliances (all Panorama releases).

STEP 1 | Select **Device** > **Scheduled Log Export** and click **Add**.

STEP 2 | Enter a **Name** for the scheduled log export and **Enable** it.

STEP 3 | Select the **Log Type** to export.

STEP 4 | Select the daily **Scheduled Export Start Time**. The options are in 15-minute increments for a 24-hour clock (00:00 - 23:59).

STEP 5 | Select the **Protocol** to export the logs: **SCP** (secure) or **FTP**.

STEP 6 | Enter the **Hostname** or IP address of the server.

STEP 7 | Enter the **Port** number. By default, FTP uses port 21 and SCP uses port 22.

STEP 8 | Enter the **Path** or directory in which to save the exported logs.

STEP 9 | Enter the **Username** and, if necessary, the **Password** (and **Confirm Password**) to access the server.

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STEP 10 | **(**FTP only**)** Select **Enable FTP Passive Mode** if you want to use FTP passive mode, in which the firewall initiates a data connection with the FTP server. By default, the firewall uses FTP active mode, in which the FTP server initiates a data connection with the firewall. Choose the mode based on what your FTP server supports and on your network requirements.

STEP 11 | **(**SCP only**)** Click **Test SCP server connection**.

(PAN-OS 10.2.4 and later releases) A pop-up window is displayed requiring you to enter a clear text **Password**and then to **Confirm Password** in order to test the SCP server connection and enable the secure transfer of data. The firewall does not establish and test the SCP server connection until you enter and confirm the SCP server password. If the firewall is in an HA configuration, perform this step on each HA peer so that each one can successfully connect to the SCP server. If the firewall can successfully connect to the SCP server, it creates and uploads the test file named `ssh-export-test.txt`.

If you use a Panorama template to configure the log export schedule, you must perform this step after committing the template configuration to the firewalls. After the template commit, log in to each firewall, open the log export schedule, and click ***Test SCP server connection****.*

STEP 12 | Click **OK** and **Commit**.

Log Types and Severity Levels

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

You can see the following log types in the **Monitor** > **Logs** pages.

Log Types and Severity Levels

Use the **Monitor** > **Logs** pages to view logs. You can view the following logs:

Traffic Logs

Traffic logs display an entry for the start and end of each session. Each entry includes the following information: date and time; source and destination zones, source and destination dynamic address groups, addresses and ports; application name; security rule applied to the traffic flosecw; rule action (allow, deny, or drop); ingress and egress interface; number of bytes; and session end reason.

A dynamic address group only appears in a log if the rule the traffic matches includes a dynamic address group. If an IP address appears in more than one dynamic address group, the firewall displays up to five dynamic address groups in logs along with the source IP address

The Type column indicates whether the entry is for the start or end of the session. The Action column indicates whether the firewall allowed, denied, or dropped the session. A drop indicates

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the security rule that blocked the traffic specified any application, while a deny indicates the rule identified a specific application. If the firewall drops traffic before identifying the application, such as when a rule drops all traffic for a specific service, the Application column displays not-applicable.

Click beside an entry to view additional details about the session, such as whether an ICMP entry aggregates multiple sessions between the same source and destination (in which case the Count column value is greater than one).

When the Decryption log introduced in PAN-OS 11.1 is disabled, the firewall sends HTTP/2 logs as Traffic logs. However, when the Decryption logs are enabled, the firewall sends HTTP/2 logs as Tunnel Inspection logs (when Decryption logs are disabled, HTTP/2 logs are sent as Traffic logs), so you need to check the Tunnel Inspection logs instead of the Traffic logs for HTTP/2 events.

Threat Logs

Threat logs display entries when traffic matches one of the Security Profiles attached to a security rule on the firewall. Each entry includes the following information: date and time; type of threat (such as virus or spyware); threat description or URL (Name column); source and destination zones, addresses, source and destination dynamic address groups, and ports; application name; alarm action (such as allow or block); and severity level.

A dynamic address group only appears in a log if the rule the traffic matches includes a dynamic address group. If an IP address appears in more than one dynamic address group, the firewall displays up to five dynamic address groups in logs along with the source IP address

To see more details on individual Threat log entries:

- Click beside a threat entry to view details such as whether the entry aggregates multiple threats of the same type between the same source and destination (in which case the Count column value is greater than one).
- If you configured the firewall to Take Packet Captures, click beside an entry to access the captured packets.

The following table summarizes the Threat severity levels:

Severity	Description
Critical	Serious threats, such as those that affect default installations of widely deployed software, result in root compromise of servers, and the exploit code is widely available to attackers. The attacker usually does not need any special authentication credentials or knowledge about the individual victims and the target does not need to be manipulated into performing any special functions.
High	Threats that have the ability to become critical but have mitigating factors; for example, they may be difficult to exploit, do not result in elevated privileges, or do not have a large victim pool.

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Severity	Description
	WildFire Submissions log entries with a malicious verdict and an action set to allow are logged as High.
Medium	Minor threats in which impact is minimized, such as DoS attacks that do not compromise the target or exploits that require an attacker to reside on the same LAN as the victim, affect only non-standard configurations or obscure applications, or provide very limited access. • Threat log entries with a malicious verdict and an action of block or alert, based on the existing WildFire signature severity, are logged as Medium.
Low	Warning-level threats that have very little impact on an organization's infrastructure. They usually require local or physical system access and may often result in victim privacy or DoS issues and information leakage. • Data Filtering profile matches are logged as Low. • WildFire Submissions log entries with a grayware verdict and any action are logged as Low.
Informational	Suspicious events that do not pose an immediate threat, but that are reported to call attention to deeper problems that could possibly exist. • URL Filtering log entries are logged as Informational. • WildFire Submissions log entries with a benign verdict and any action are logged as Informational. • WildFire Submissions log entries with any verdict and an action set to block and forward are logged as Informational. • Log entries with any verdict and an action set to block are logged as Informational.

URL Filtering Logs

URL filtering logs (**Monitor** > **Logs** > **URL Filtering**) display comprehensive information about traffic to URL categories monitored in Security policy rules. Attributes or properties recorded for each session include `receive time, category, URL, from zone, to zone, source, and source user`. You can customize your log view so that only the attributes you are most interested in display. The firewall generates URL filtering log entries in the following cases, with exceptions noted:

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- Traffic matches a Security policy rule with a URL category as match criteria. The rule enforces one of the following actions on the traffic: `deny, drop, or reset (client, server, both)`.

  URL filtering logs are only generated when an action results from a URL category match. If you have Security policy rules with applications as match criteria, a URL can be blocked due to an application (App-ID) rather than a URL category match. This behavior depends on how packets in the session are parsed.

  For example, suppose you have a Security policy rule that blocks the social-networking category and another rule that blocks a specific social media application. Traffic to the social media website could result in a Security policy lookup that hits an App-ID rule instead of a URL filtering rule. In this case, a URL filtering log isn't generated.

- Traffic matches a Security policy rule with a URL Filtering profile attached. Site Access for categories in the profile is set to `alert, block, continue, or override`.

*By default, categories set to **allow** do not generate URL filtering log entries. The exception is if you configure log forwarding.*

*If you want the firewall to log traffic to categories that you allow but would like more visibility into, set **Site Access** for these categories to **alert** in your URL Filtering profiles.*

WildFire Submissions Logs

The firewall forwards samples (files and emails links) to the WildFire cloud for analysis based on WildFire Analysis profiles settings (**Objects** > **Security Profiles** > **WildFire Analysis**). The firewall generates WildFire Submissions log entries for each sample it forwards after WildFire completes static and dynamic analysis of the sample. WildFire Submissions log entries include the firewall Action for the sample (allow or block), the WildFire verdict for the submitted sample, and the severity level of the sample.

The following table summarizes the WildFire verdicts:

Verdict	Description
Benign	Indicates that the entry received a WildFire analysis verdict of benign. Files categorized as benign are safe and do not exhibit malicious behavior.
Grayware	Indicates that the entry received a WildFire analysis verdict of grayware. Files categorized as grayware do not pose a direct security threat, but might display otherwise obtrusive behavior. Grayware can include, adware, spyware, and Browser Helper Objects (BHOs).
Phishing	Indicates that WildFire assigned a link an analysis verdict of phishing. A phishing verdict indicates that the site to which the link directs users displayed credential phishing activity.
Malicious	Indicates that the entry received a WildFire analysis verdict of malicious. Samples categorized as malicious are can pose a security threat. Malware can include

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Verdict	Description
	viruses, C2 (command-and-control), worms, Trojans, Remote Access Tools (RATs), rootkits, and botnets. For samples that are identified as malware, the WildFire cloud generates and distributes a signature to prevent against future exposure. *C2 samples are classified as C2 in the WildFire analysis report and other Palo Alto Networks products that rely on WildFire analysis data; however, that verdict is translated and categorized as malicious by the firewall.*

Data Filtering Logs

Data Filtering logs display entries for the security rules that help prevent sensitive information such as credit card numbers from leaving the area that the firewall protects. See Data Filtering for information on defining Data Filtering profiles.

This log type also shows information for File Blocking Profiles. For example, if a rule blocks .exe files, the log shows the blocked files.

Correlation Logs

The firewall logs a correlated event when the patterns and thresholds defined in a Correlation Object match the traffic patterns on your network. To Interpret Correlated Events and view a graphical display of the events, see Use the Compromised Hosts Widget in the ACC.

The following table summarizes the Correlation log severity levels:

Severity	Description
Critical	Confirms that a host has been compromised based on correlated events that indicate an escalation pattern. For example, a critical event is logged when a host that received a file with a malicious verdict by WildFire, exhibits the same command-and control activity that was observed in the WildFire sandbox for that malicious file.
High	Indicates that a host is very likely compromised based on a correlation between multiple threat events, such as malware detected anywhere on the network that matches the command and control activity being generated from a particular host.
Medium	Indicates that a host is likely compromised based on the detection of one or multiple suspicious events, such as repeated visits to known malicious URLs that suggests a scripted command-and-control activity.
Low	Indicates that a host is possibly compromised based on the detection of one or multiple suspicious events, such as a visit to a malicious URL or a dynamic DNS domain.

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Severity	Description
Informational	Detects an event that may be useful in aggregate for identifying suspicious activity; each event is not necessarily significant on its own.

Tunnel Inspection Logs

Tunnel inspection logs are like traffic logs for tunnel sessions; they display entries of non-encrypted tunnel sessions. To prevent double counting, the firewall saves only the inner flows in traffic logs, and sends tunnel sessions to the tunnel inspection logs. The tunnel inspection log entries include Receive Time (date and time the log was received), the tunnel ID, monitor tag, session ID, the Security rule applied to the tunnel session, number of bytes in the session, parent session ID (session ID for the tunnel session), source address, source user and source zone, destination address, destination user, and destination zone.

When the decryption logs introduced in PAN-OS 11.1 are enabled, the firewall sends HTTP/2 logs as Tunnel Inspection logs (when decryption logs are disabled, HTTP/2 logs are sent as Traffic logs), so you need to check the Tunnel Inspection logs instead of the Traffic logs for HTTP/2 events.

Click the Detailed Log view to see details for an entry, such as the tunnel protocol used, and the flag indicating whether the tunnel content was inspected or not. Only a session that has a parent session will have the Tunnel Inspected flag set, which means the session is in a tunnel-in-tunnel (two levels of encapsulation). The first outer header of a tunnel will not have the Tunnel Inspected flag set.

Config Logs

Config logs display entries for changes to the firewall configuration. Each entry includes the date and time, the administrator username, the IP address from where the administrator made the change, the type of client (Web, CLI, or Panorama), the type of command executed, the command status (succeeded or failed), the configuration path, and the values before and after the change.

System Logs

System logs display entries for each system event on the firewall. Each entry includes the date and time, event severity, and event description. The following table summarizes the System log severity levels. For a partial list of System log messages and their corresponding severity levels, refer to System Log Events.

Severity	Description
Critical	Hardware failures, including high availability (HA) failover and link failures.
High	Serious issues, including dropped connections with external devices, such as LDAP and RADIUS servers.
Medium	Mid-level notifications, such as antivirus package upgrades.

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Severity	Description
Low	Minor severity notifications, such as user password changes.
Informational	Log in/log off, administrator name or password change, any configuration change, and all other events not covered by the other severity levels.

HIP Match Logs

The GlobalProtect Host Information Profile (HIP) matching enables you to collect information about the security status of the end devices accessing your network (such as whether they have disk encryption enabled). The firewall can allow or deny access to a specific host based on adherence to the HIP-based security rules you define. HIP Match logs display traffic flows that match a HIP Object or HIP Profile that you configured for the rules.

GlobalProtect Logs

GlobalProtect logs display the following logs related to GlobalProtect:

- GlobalProtect system logs.

  GlobalProtect authentication event logs remain in **Monitor** > **Logs** > **System**; however, the **Auth Method** column of the GlobalProtect logs display the authentication method used for logins.
- LSVPN/satellite events.
- GlobalProtect portal and gateway logs.
- Clientless VPN logs.

IP-Tag Logs

IP-Tag logs display how and when a source IP address is registered or unregistered on the firewall, along with the tag applied to the address. Each log entry also displays the mapping timeout (if applicable) and the source of the IP address-to-tag mapping. These sources include:

- User-ID agent
- Virtual machine (VM) Information Source
- Panorama™ plugin
- Autotag (log-based tagging)
- XML API (includes cURL requests and XML API calls without a specified origin)

For more information, see Register IP Address and Tags Dynamically.

User-ID Logs

User-ID logs display information about IP address-to-username mappings and Authentication Timestamps, such as the sources of the mapping information and the times when users authenticated. You can use this information to help troubleshoot User-ID and authentication issues. For example, if the firewall is applying the wrong policy rule for a user, you can view the logs to verify whether that user is mapped to the correct IP address and whether the group associations are correct.

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Decryption Logs

Decryption logs provide a detailed record of failed or successful decryption sessions on your network. By default, your Next-Generation Firewall (NGFW) only generates decryption logs for unsuccessful TLS handshakes. For full visibility into decryption activity, you can log successful TLS handshakes. However, ensure your system has enough resources (log space) to handle the increased volume of logs.

Monitoring decryption logs helps you understand decryption activity and troubleshoot decryption issues. These logs are comprehensive, with over 62 columns of information that fall into the following categories:

- **Session and Policy Rule Details**—Information about the traffic, including the source and destination IP addresses, the user who initiated the session, the specific decryption policy rule that was applied to the traffic, and the type of decryption performed.
- **Certificate Details**—Information about the certificate used in the session, including the subject common name, issuer common name, root common name, root status, certificate key type and size, certificate start and end date, certificate serial number, and certificate fingerprint.
- **TLS Connection Details**—Information about the parameters used to establish the session, including the negotiated TLS version, key exchange algorithm, encryption algorithm, authentication algorithms, elliptic curve (EC), and Server Name Indication (SNI).
- **Error Details**—Error information related to certificates, ciphers, feature, hardware security modules (HSM), protocols, resources, session resumption, and TLS versions. Error indexes (codes) are also provided for easy lookup of more detailed error information.

You can click the magnifying glass ( ) for any log entry to view detailed session information in a consolidated view.

NGFWs don't generate decryption logs for web traffic blocked during an SSL/TLS handshake. These sessions don't appear in decryption logs because the NGFW resets the SSL/TLS connection, which ends the handshake and prevents decryption. You can find details for these blocked sessions in your URL Filtering logs instead.

Decryption logs are not supported for SSH Proxy traffic. In addition, certificate information isn't available for session resumption logs.

Starting with PAN-OS 12.1.2, decryption logs provide clearer insights into proxied connections. The logs differentiate between the client-side and server-side attributes of a session using "Client" and "Server" column headers. For example, the TLS version negotiated between the client and the firewall displays under the "Client" header, while the TLS version negotiated between the server and the firewall displays under the "Server" header.

Alarms Logs

An alarm is a firewall-generated message indicating that the number of events of a particular type (for example, encryption and decryption failures) has exceeded the threshold configured for that event type. To enable alarms and configure alarm thresholds, select **Device** > **Log Settings** and edit the Alarm Settings.

When generating an alarm, the firewall creates an Alarm log and opens the System Alarms dialog to display the alarm. After you **Close** the dialog, you can reopen it anytime by clicking **Alarms**

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( ) at the bottom of the web interface. To prevent the firewall from automatically opening the dialog for a particular alarm, select the alarm in the Unacknowledged Alarms list and **Acknowledge** the alarm.

Authentication Logs

Authentication logs display information about authentication events that occur when end users try to access network resources for which access is controlled by Authentication Policy rules. You can use this information to help troubleshoot access issues and to adjust your Authentication policy as needed. In conjunction with correlation objects, you can also use Authentication logs to identify suspicious activity on your network, such as brute force attacks.

Optionally, you can configure Authentication rules to log timeout events. These timeouts relate to the period when a user need authenticate for a resource only once but can access it repeatedly. Seeing information about the timeouts helps you decide if and how to adjust them (for details, see Authentication Timestamps).

System logs record authentication events relating to GlobalProtect and to administrator access to the web interface.

Unified Logs

Unified logs are entries from the Traffic, Threat, URL Filtering, WildFire Submissions, and Data Filtering logs displayed in a single view. Unified log view enables you to investigate and filter the latest entries from different log types in one place, instead of searching through each log type separately. Click Effective Queries ( ) in the filter area to select which log types will display entries in Unified log view.

The Unified log view displays only entries from logs that you have permission to see. For example, an administrator who does not have permission to view WildFire Submissions logs will not see WildFire Submissions log entries when viewing Unified logs. Administrative Role Types define these permissions.

When you Set Up Remote Search in AutoFocus to perform a targeted search on the firewall, the search results are displayed in Unified log view.

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Monitor Block List

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

There are two ways you can cause the firewall to place an IP address on the block list:

- Configure a Vulnerability Protection profile with a rule to Block IP connections and apply the profile to a Security policy, which you apply to a zone.
- Configure a DoS Protection policy rule with the Protect action and a Classified DoS Protection profile, which specifies a maximum rate of connections per second allowed. When incoming packets match the DoS Protection policy and exceed the Max Rate, and if you specified a Block Duration and a Classified policy rule to include source IP address, the firewall puts the offending source IP address on the block list.

In the cases described above, the firewall automatically blocks that traffic in hardware before those packets use CPU or packet buffer resources. If attack traffic exceeds the blocking capacity of the hardware, the firewall uses IP blocking mechanisms in software to block the traffic.

The firewall automatically creates a hardware block list entry based on your Vulnerability Protection profile or DoS Protection policy rule; the source address from the rule is the source IP address in the hardware block list.

Entries on the block list indicate in the Type column whether they were blocked by hardware (hw) or software (sw). The bottom of the screen displays:

- Count of **Total Blocked IPs** out of the number of blocked IP addresses the firewall supports.
- Percentage of the block list that the firewall has used.

To view details about an address on the block list, hover over a Source IP address and click the down arrow link. Click the Who Is link, which displays the Network Solutions Who Is feature, providing information about the address.

For information on configuring a Vulnerability Protection profile, customize the action and trigger conditions for a brute force signature. For more information on block list and DoS Protection profiles, see DoS Protection Against Flooding of New Sessions.

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View and Manage Reports

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

The reporting capabilities on the firewall allow you to keep a pulse on your network, validate your policies, and focus your efforts on maintaining network security for keeping your users safe and productive.

The firewall includes predefined reports that you can use as-is, or you can build custom reports that meet your needs for specific data and actionable tasks. You can also combine predefined and custom reports to compile information you need. The firewall provides the following types of reports:

- **Predefined Reports**—Allow you to view a quick summary of the traffic on your network. A suite of predefined reports are available in four categories—Applications, Traffic, Threat, and URL Filtering. See View Reports.
- **User or Group Activity Reports**—Allow you to schedule or create an on-demand report on the application use and URL activity for a specific user or for a user group. The report includes the URL categories and an estimated browse time calculation for individual users. See Generate User/Group Activity Reports.
- **Custom Reports**—Create and schedule custom reports that show exactly the information you want to see by filtering on conditions and columns to include. You can also include query builders for more specific drill down on report data. See Generate Custom Reports.
- **PDF Summary Reports**—Aggregate up to 18 predefined or custom reports/graphs from Threat, Application, Trend, Traffic, and URL Filtering categories into one PDF document. See Manage PDF Summary Reports.
- **Botnet Reports**—Allow you to use behavior-based mechanisms to identify potential botnet-infected hosts in the network. See Generate Botnet Reports.
- **Report Groups**—Combine custom and predefined reports into report groups and compile a single PDF that is emailed to one or more recipients. See Manage Report Groups.

Reports can be generated on demand, on a recurring schedule, and can be scheduled for email delivery.

View Reports

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

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The firewall provides an assortment of over 40 predefined reports that it generates every day. You can view these reports directly on the firewall. You can also view custom reports and summary reports.

About 200 MB of storage is allocated for saving reports on the firewall. This limit can be reconfigured for PA-7000 series and PA-5200 series firewalls only. For all other firewall models, you can Configure the Expiration Period and Run Time for Reports to allow the firewall to delete reports that exceed the period. Keep in mind that when the firewall reaches its storage limit, it automatically deletes older reports to create space even if you don't set an expiration period. Another way to conserve system resources on the firewall is to Disable Predefined Reports. For long-term retention of reports, you can export the reports (as described below) or Schedule Reports for Email Delivery.

Unlike other reports, you can't save User/Group Activity reports on the firewall. You must Generate User/Group Activity Reports on demand or schedule them for email delivery.

STEP 1 | (VM-50, VM-50 Lite, and PA-200 firewalls only) Enable generation of predefined reports.

By default, predefined reports are disabled on VM-50, VM-50 Lite, and PA-200 firewalls to save resources.

1. Select **Device** > **Setup** > **Management** and edit **Logging and Reporting**.
2. Select **Pre-Defined Reports** and enable (check) **Pre-Defined Reports**.
3. Check (enable) the predefined reports you want to generate and click **OK**
4. **Commit** your configuration changes.
5. Access the firewall CLI to enable predefined reports.

   This step is required for local predefined reports and predefined reports pushed from a Panorama™ management server.

   ```
   admin> debug predefined-default enable
   ```

STEP 2 | Select **Monitor** > **Reports**.

The reports are grouped into sections (types) on the right-hand side of the page: **Custom Reports**, **Application Reports**, **Traffic Reports**, **Threat Reports**, **URL Filtering Reports**, and **PDF Summary Reports**.

STEP 3 | Select a report to view. The reports page then displays the report for the previous day.

To view reports for other days, select a date in the calendar at the bottom right of the page and select a report. If you select a report in another section, the date selection resets to the current date.

STEP 4 | To view a report offline, you can export the report to PDF, CSV or to XML formats. Click **Export to PDF**, **Export to CSV**, or **Export to XML** at the bottom of the page, then print or save the file.

If PDF export fails or does not complete, try exporting to CSV first, then export to PDF. This two-step process can work around issues with direct PDF generation for large or complex reports.

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Configure the Expiration Period and Run Time for Reports

The expiration period and run time are global settings that apply to all Report Types. After running new reports, the firewall automatically deletes reports that exceed the expiration period.

STEP 1 | Select **Device** > **Setup** > **Management**, edit the Logging and Reporting Settings, and select the **Log Export and Reporting** tab.

STEP 2 | Set the **Report Runtime** to an hour in the 24-hour clock schedule (default is 02:00; range is 00:00 [midnight] to 23:00).

STEP 3 | Enter the **Report Expiration Period** in days (default is no expiration; range is 1 is 2,000).

*You can't change the storage that the firewall allocates for saving reports: it is predefined at about 200 MB. When the firewall reaches the storage maximum, it automatically deletes older reports to create space even if you don't set a **Report Expiration Period**.*

STEP 4 | Click **OK** and **Commit**.

Disable Predefined Reports

To manage system resources efficiently on your firewall, you can disable predefined reports that are not being utilized. The firewall typically generates about 40 such reports daily, and deactivating the ones that are not needed helps conserve valuable system resources.

Important Considerations Before Disabling Reports

Before you disable predefined reports, it is crucial to ensure that no report group or PDF summary report report includes the specific predefined reports you intend to disable. If they are included, those report groups or PDF summaries will be rendered without any data, leading to incomplete or blank reports. Therefore, ensure that you perform a quick check of your existing report configurations.

STEP 1 | Select **Device** > **Setup** > **Management** and edit the Logging and Reporting Settings.

STEP 2 | Select the **Pre-Defined Reports** tab and clear the check box for each report you want to disable. To disable all predefined reports, click **Deselect All**.

STEP 3 | Click **OK** and **Commit**.

Custom Reports

In order to create purposeful custom reports, you must consider the attributes or key pieces of information that you want to retrieve and analyze, such as threats, as well as the best way to categorize the information, such as grouping by rule UUID, which will allow you to see the rule that applies to each threat type. This consideration guides you in making the following selections in a custom report:

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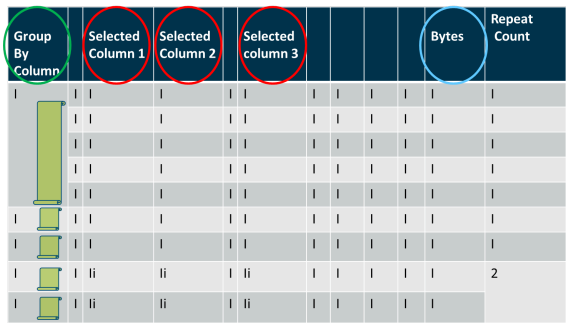
Selection	Description
Database	You can base the report on one of the following database types: • **Summary databases**—These databases are available for Application Statistics, Traffic, Threat, URL Filtering, and Tunnel Inspection logs. The firewall aggregates the detailed logs at 15-minute intervals. To enable faster response time when generating reports, the firewall condenses the data: duplicate sessions are grouped and incremented with a repeat counter, and some attributes (columns) are excluded from the summary. • **Detailed logs**—These databases itemize the logs and list all the attributes (columns) for each log entry. *Reports based on detailed logs take much longer to run and are not recommended unless absolutely necessary.*
Attributes	The columns that you want to use as the match criteria. The attributes are the columns that are available for selection in a report. From the list of **Available Columns**, you can add the selection criteria for matching data and for aggregating the details (the **Selected Columns**).
Sort By/ Group By	The **Sort By** and the **Group By** criteria allow you to organize/segment the data in the report; the sorting and grouping attributes available vary based on the selected data source. The Sort By option specifies the attribute that is used for aggregation. If you do not select an attribute to sort by, the report will return the first N number of results without any aggregation. The Group By option allows you to select an attribute and use it as an anchor for grouping data; all the data in the report is then presented in a set of top 5, 10, 25 or 50 groups. For example, when you select Hour as the Group By selection and want the top 25 groups for a 24-hr time period, the results of the report will be generated on an hourly basis over a 24-hr period. The first column in the report will be the hour and the next set of columns will be the rest of your selected report columns.
	The following example illustrates how the **Selected Columns** and **Sort By**/**Group By** criteria work together when generating reports: 

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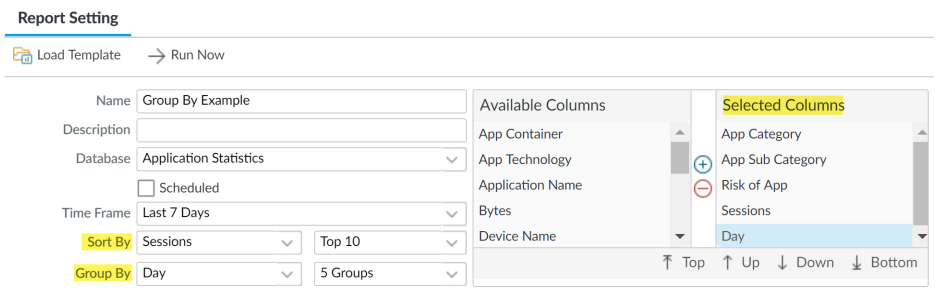
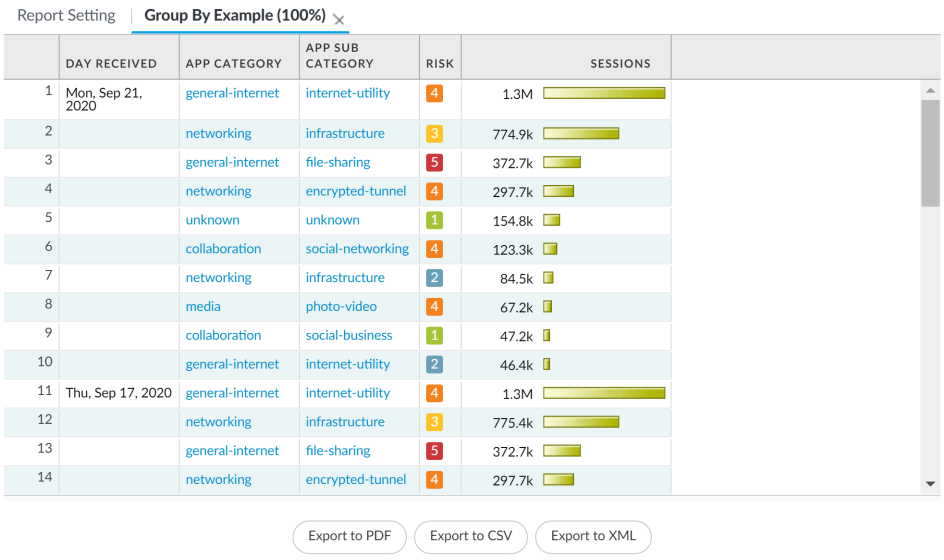
Selection	Description
	The columns circled in red (above) depict the columns selected, which are the attributes that you match against for generating the report. Each log entry from the data source is parsed and these columns are matched on. If multiple sessions have the same values for the selected columns, the sessions are aggregated and the repeat count (or sessions) is incremented. The column circled in blue indicates the chosen sort order. When the sort order (**Sort By**) is specified, the data is sorted (and aggregated) by the selected attribute. The column circled in green indicates the **Group By** selection, which serves as an anchor for the report. The **Group By** column is used as a match criteria to filter for the top N groups. Then, for each of the top N groups, the report enumerates the values for all the other selected columns.
	For example, if a report has the following selections:  The output will display as follows:  The report is anchored by **Day** and sorted by **Sessions**. It lists the 5 days (**5 Groups**) with maximum traffic in the **Last 7 Days** time frame. The

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Selection	Description
	data is enumerated by the **Top 5** sessions for each day for the selected columns—**App Category**, **App Subcategory** and **Risk**.
Time Frame	The date range for which you want to analyze data. You can define a custom range or select a time period ranging from the last 15 minutes to the last 30 days. The reports can be run on demand or scheduled to run at a daily or weekly cadence.
Query Builder	The query builder allows you to define specific queries to further refine the selected attributes. It allows you see just what you want in your report using **and** and **or** operators and a match criteria, and then include or exclude data that matches or negates the query in the report. Queries enable you to generate a more focused collation of information in a report.

Generate Custom Reports

You can configure custom reports that the firewall generates immediately (on demand) or on schedule (each night). To understand the selections available to create a purposeful custom report, see Custom Reports.

After the firewall has generated a scheduled custom report, you risk invalidating the past results of that report if you modify its configuration to change its future output. If you need to modify a scheduled report configuration, the best practice is to create a new report.

STEP 1 | Select **Monitor** > **Manage Custom Reports**.

STEP 2 | Click **Add** and then enter a **Name** for the report.

*To base a report on an predefined template, click **Load Template** and choose the template. You can then edit the template and save it as a custom report.*

STEP 3 | Select the **Database** to use for the report.

Each time you create a custom report, a log view report is automatically created. This report show the logs that were used to build the custom report. The log view report uses the same name as the custom report, but appends the phrase (Log View) to the report name.

When creating a report group, you can include the log view report with the custom report. For more information, see Manage Report Groups.

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STEP 4 | Select the **Scheduled** check box to run the report each night. The report is then available for viewing in the **Reports** column on the side.

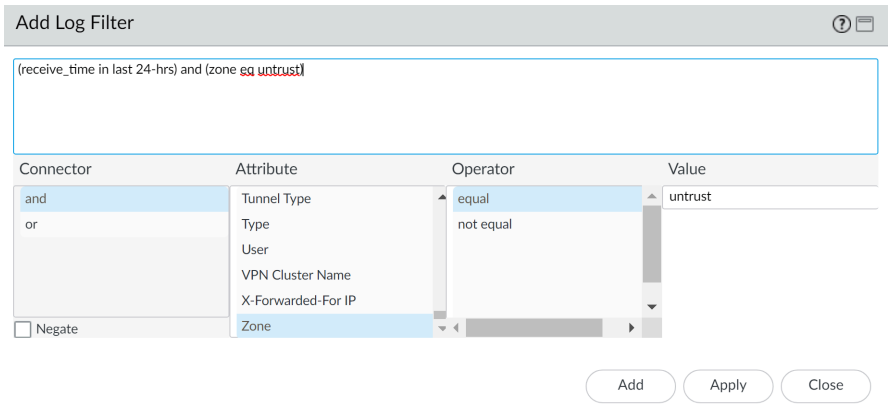
> *To generate a scheduled custom report using logs stored in Strata Logging Service on the Panorama™ management server, Cloud Service plugin 1.8 or later release must be installed on Panorama.*

STEP 5 | Define the filtering criteria. Select the **Time Frame**, the **Sort By** order, **Group By** preference, and select the columns that must display in the report.

STEP 6 | (Optional) Select the **Query Builder** attributes if you want to further refine the selection criteria. To build a report query, specify the following and click **Add**. Repeat as needed to construct the full query.

- **Connector**—Choose the connector (and/or) to precede the expression you are adding.
- **Negate**—Select the check box to interpret the query as a negation. If, for example, you choose to match entries in the last 24 hours and/or are originating from the untrust zone, the negate option causes a match on entries that are not in the past 24 hours and/or are not from the untrust zone.
- **Attribute**—Choose a data element. The available options depend on the choice of database.
- **Operator**—Choose the criterion to determine whether the attribute applies (such as =). The available options depend on the choice of database.
- **Value**—Specify the attribute value to match.

For example, the following figure (based on the `Traffic Log` database) shows a query that matches if the Traffic log entry was received in the past 24 hours and is from the untrust zone.



STEP 7 | To test the report settings, select **Run Now**. Modify the settings as required to change the information that is displayed in the report.

> *After running the report, you can export it to PDF, CSV, or XML format. If PDF export fails or does not complete, try exporting to CSV first, then export to PDF. This two-step process can work around issues with direct PDF generation for large or complex reports.*

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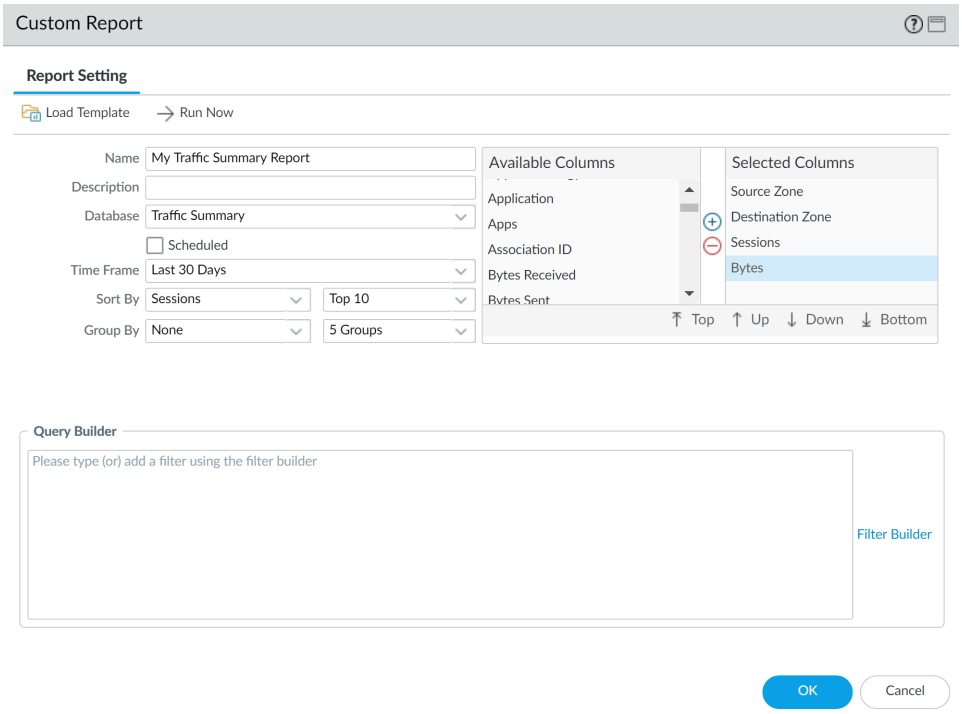
STEP 8 | Click **OK** to save the custom report.

Examples of Custom Reports

If you want to set up a simple report in which you use the traffic summary database from the last 30 days, and sort the data by the top 10 sessions and these sessions are grouped into 5 groups by day of the week. You would set up the custom report to look like this:



And the PDF output for the report would look as follows:

My Traffic Summary Report

ca1demo.paloaltonetworks.com : 2016/01/25 10:34:39 - 2016/02/24 10:34:38

Source Zone	Destination Zone	App Category	Application	Sessions	Bytes
Tap	Tap	general-internet	web-browsing	74.54 M	2.47 T
Tap	Tap	networking	dns	52.03 M	28.93 G
Tap	Tap	networking	ssl	18.01 M	678.13 G
Tap	Tap	general-internet	bittorrent	9.80 M	1.62 T
Tap	Tap	general-internet	google-base	4.48 M	168.99 G
Tap	Tap	unknown	insufficient-data	4.45 M	31.30 G
Tap	Tap	collaboration	facebook-base	4.09 M	99.14 G
Tap	Tap	networking	ntp	4.07 M	3.29 G
Tap	Tap	collaboration	blackboard	2.84 M	186 G
Tap	Tap	collaboration	smtp	1.92 M	172.57 G
Tap	Tap	networking	icmp	1.36 M	320.49 M
Tap	Tap	general-internet	gnutella	1.17 M	17.84 G
Tap	Tap	collaboration	myspace-base	1.10 M	35.22 G
Tap	Tap	general-internet	ping	1.06 M	86.21 M
Tap	Tap	general-internet	flash	1.01 M	168.14 G

Now, if you want to use the query builder to generate a custom report that represents the top consumers of network resources within a user group, you would set up the report to look like this:

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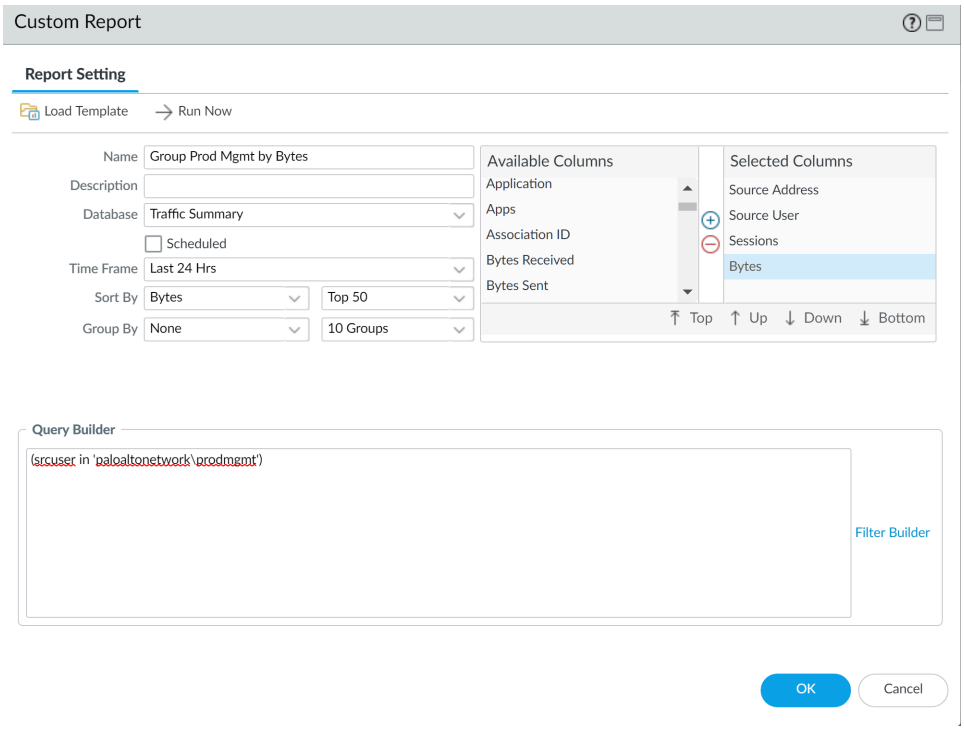

The report would display the top users in the product management user group sorted by bytes.

Generate Botnet Reports

The botnet report enables you to use heuristic and behavior-based mechanisms to identify potential malware- or botnet-infected hosts in your network. To evaluate botnet activity and infected hosts, the firewall correlates user and network activity data in Threat, URL, and Data Filtering logs with the list of malware URLs in PAN-DB, known dynamic DNS domain providers, and domains registered within the last 30 days. You can configure the report to identify hosts that visited those sites, as well as hosts that communicated with Internet Relay Chat (IRC) servers or that used unknown applications. Malware often use dynamic DNS to avoid IP blocking, while IRC servers often use bots for automated functions.

The firewall requires Threat Prevention and URL Filtering licenses to use the botnet report. You can Use the Automated Correlation Engine to monitor suspicious activities based on additional indicators besides those that the botnet report uses. However, the botnet report is the only tool that uses newly registered domains as an indicator.

- Configure a Botnet Report
- Interpret Botnet Report Output

Configure a Botnet Report

You can schedule a botnet report or run it on demand. The firewall generates scheduled botnet reports every 24 hours because behavior-based detection requires correlating traffic across multiple logs over that timeframe.

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STEP 1 | Define the types of traffic that indicate possible botnet activity.

1. Select **Monitor** > **Botnet** and click **Configuration** on the right side of the page.
2. **Enable** and define the **Count** for each type of HTTP Traffic that the report will include.

   The **Count** values represent the minimum number of events of each traffic type that must occur for the report to list the associated host with a higher confidence score (higher likelihood of botnet infection). If the number of events is less than the **Count**, the report will display a lower confidence score or (for certain traffic types) won't display an entry for the host. For example, if you set the **Count** to three for **Malware URL visit**, then hosts that visit three or more known malware URLs will have higher scores than hosts that visit less than three. For details, see Interpret Botnet Report Output.
3. Define the thresholds that determine whether the report will include hosts associated with traffic involving Unknown TCP or Unknown UDP applications.
4. Select the **IRC** check box to include traffic involving IRC servers.
5. Click **OK** to save the report configuration.

STEP 2 | Schedule the report or run it on demand.

1. Click **Report Setting** on the right side of the page.
2. Select a time interval for the report in the **Test Run Time Frame** drop-down.
3. Select the **No. of Rows** to include in the report.
4. **(**Optional**) Add** queries to the Query Builder to filter the report output by attributes such as source/destination IP addresses, users, or zones.

   For example, if you know in advance that traffic initiated from the IP address 10.3.3.15 contains no potential botnet activity, add **`not (addr.src in 10.0.1.35)`** as a query to exclude that host from the report output. For details, see Interpret Botnet Report Output.
5. Select **Scheduled** to run the report daily or click **Run Now** to run the report immediately.
6. Click **OK** and **Commit**.

Interpret Botnet Report Output

The botnet report displays a line for each host that is associated with traffic you defined as suspicious when configuring the report. For each host, the report displays a confidence score of 1 to 5 to indicate the likelihood of botnet infection, where 5 indicates the highest likelihood. The scores correspond to threat severity levels: 1 is informational, 2 is low, 3 is medium, 4 is high, and 5 is critical. The firewall bases the scores on:

- **Traffic type**—Certain HTTP traffic types are more likely to involve botnet activity. For example, the report assigns a higher confidence to hosts that visit known malware URLs than to hosts that browse to IP domains instead of URLs, assuming you defined both those activities as suspicious.
- **Number of events**—Hosts that are associated with a higher number of suspicious events will have higher confidence scores based on the thresholds (**Count** values) you define when you Configure a Botnet Report.
- **Executable downloads**—The report assigns a higher confidence to hosts that download executable files. Executable files are a part of many infections and, when combined with the

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other types of suspicious traffic, can help you prioritize your investigations of compromised hosts.

When reviewing the report output, you might find that the sources the firewall uses to evaluate botnet activity (for example, the list of malware URLs in PAN-DB) have gaps. You might also find that these sources identify traffic that you consider safe. To compensate in both cases, you can add query filters when you Configure a Botnet Report.

Generate the SaaS Application Usage Report

The SaaS Application Usage PDF report is a two-part report that allows you to easily explore SaaS application activity by risk and sanction state. A sanctioned application is an application that you formally approve for use on your network. A SaaS application is an application that has the characteristic SaaS=yes in the applications details page in **Objects** > **Applications**, all other applications are considered as non-SaaS. To indicate that you have sanctioned a SaaS or non-SaaS application, you must tag it with the predefined tag named Sanctioned. The firewall and Panorama consider any application without this predefined tag as unsanctioned for use on the network.

- The first part of the report presents the key findings for the SaaS applications on your network during the reporting period with a comparison of the sanctioned versus unsanctioned applications and lists the top applications based on sanction state by usage, compliance, and data transfers. To help you identify and explore the extent of high risk application usage, the applications with risky characteristics section of the report lists the SaaS applications with the following unfavorable hosting characteristics: certifications achieved, past data breaches, support for IP-based restrictions, financial viability, and terms of service. You can also view a comparison of sanctioned versus unsanctioned SaaS applications by total number of applications used on your network, bandwidth consumed by these applications, the number of users using these applications, top user groups that use the largest number of SaaS applications, and the top user groups that transfer the largest volume of data through sanctioned and unsanctioned SaaS applications. This first part of the report also highlights the top SaaS application subcategories listed in order by maximum number of applications used, the number of users, and the amount of data (bytes) transferred in each application subcategory.
- The second part of the report focuses on the detailed browsing information for SaaS and non-SaaS applications for each application subcategory listed in the first-part of the report. For each application in a subcategory, it also includes information about the top users who transferred data, the top blocked or alerted file types, and the top threats for each application. In addition, this section of the report tallies samples for each application that the firewall submitted for WildFire analysis, and the number of samples determined to be benign and malicious.

Use the insights from this report to consolidate the list of business-critical and approved SaaS applications and to enforce policies for controlling unsanctioned and risky applications that pose unnecessary risks for malware propagation and data leaks.

The predefined SaaS application usage report is still available as a daily View Reports that lists the top 100 SaaS applications (which means applications with the SaaS application characteristic, SaaS=yes) running on your network on a given day. This report does not give visibility into applications you have designated as sanctioned, but rather gives visibility into all of the SaaS applications in use on your network.

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Ensure that all prerequisites are met to enable this report generation on new deployments.

- **Enable logging on security policies.** If you don't enable logging, the report will be empty.
  1. Select **Policies** > **Security**.
  2. Select the rule.
  3. Select **Actions**.
  4. Enable **Log at Session End**
  5. Click **OK**.
  6. **Commit**.
- **Plan for a 24-hour period before running the report.** After logging is enabled and traffic is flowing, allow at least 24 hours before running the report. Traffic log summation runs on an hourly cycle.
- **Set the date range to match the deployment date.** The report can only show data from the date the firewall began passing live traffic. Set the report start date to the deployment date, not the first day of the calendar month. For example, if you deployed a firewall on September 7, set the report start date to that date, not September 1. Partial month coverage is expected behavior for new deployments, and is not a data loss issue.

STEP 1 | Tag applications that you approve for use on your network as Sanctioned.

For generating an accurate and informative report, you need to tag the sanctioned applications consistently across firewalls with multiple virtual systems, and across firewalls that belong to a device group on Panorama. If the same application is tagged as sanctioned in one virtual system and is not sanctioned in another or, on Panorama, if an application is unsanctioned in a parent device group but is tagged as sanctioned in a child device group (or vice versa), the SaaS Application Usage report will report the application as partially sanctioned and will have overlapping results.

Example: If Box is sanctioned on vsys1 and Google Drive is sanctioned on vsys2, Google Drive users in vsys1 will be counted as users of an unsanctioned SaaS application and Box users in vsys2 will be counted as users of an unsanctioned SaaS application. The key finding in the

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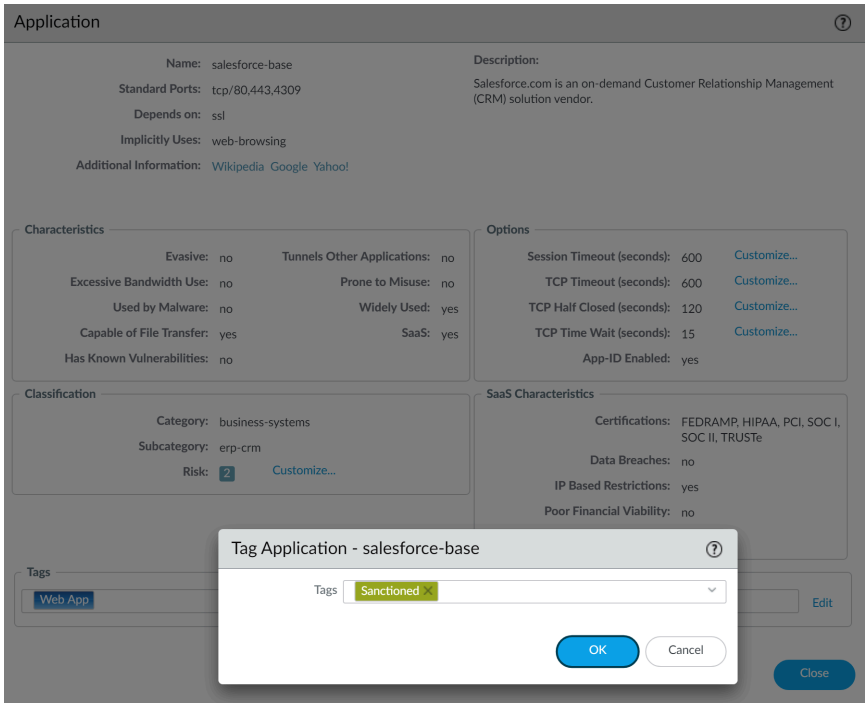
report will highlight that a total of two unique SaaS applications are discovered on the network with two sanctioned applications and two unsanctioned applications.

1. Select **Objects** > **Applications**.
2. Click the application **Name** to edit an application and select **Edit** in the Tag section.
3. Select **Sanctioned** from the **Tags** drop-down.

   You must use the predefined **Sanctioned** tag (Sanctioned). If you use any other tag to indicate that you sanctioned an application, the firewall will fail to recognize the tag and the report will be inaccurate.



4. Click **OK** and **Close** to exit all open dialogs.

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STEP 2 | Configure the SaaS Application Usage report.

1. Select **Monitor** > **PDF Reports** > **SaaS Application Usage**.
2. Click **Add**, enter a **Name**, and select a **Time Period** for the report (default is **Last 7 Days**).

   *By default, the report includes detailed information on the top SaaS and non-SaaS application subcategories, which can make the report large by page count and file size. Clear the **Include detailed application category information in report** check box if you want to reduce the file size and restrict the page count to 10 pages.*
3. Select whether you want the report to **Include logs from**:

   *In PAN-OS 10.0.2 and later releases, reports generated from logs in the Strata Logging Service only support including logs from the **Selected Zone**.*

   - **All User Groups and Zones**—The report includes data on all security zones and user groups available in the logs.

     If you want to include specific user groups in the report, select **Include user group information in the report** and click the **manage groups** link to select the groups you want to include. You must add between one and up to a maximum of 25 user groups, so that the firewall or Panorama can filter the logs for the selected user groups. If you do select the groups to include, the report will aggregate all user groups in to one group called Others.
   - **Selected Zone**—The report filters data for the specified security zone, and includes data on that zone only.

     If you want to include specific user groups in the report, select **Include user group information in the report** and click the **manage groups for selected zone** link to select the user groups within this zone that you want to include in the report. You must add between one and up to a maximum of 25 user groups, so that the firewall or Panorama can filter the logs for the selected user groups within the security zone. If

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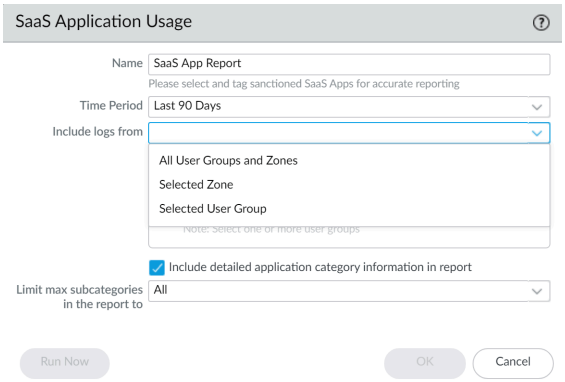
you do select the groups to include, the report will aggregate all user groups in to one group called Others.

- **Selected User Group**—The report filters data for the specified user group only, and includes SaaS application usage information for the selected user group only.



4. Select whether you want to include all the application subcategories in the report (the default) or **Limit the max subcategories in the report** to the top 10, 15, 20 or 25 categories (default is all subcategories).
5. Click **Run Now** to generate the report on-demand for the last 7-day and the last 30-day time period. Make sure that the pop-up blocker is disabled on your browser because the report opens in a new tab.
6. Click **OK** to save your changes.

STEP 3 | Schedule Reports for Email Delivery.

The last 90-days report must be scheduled for email delivery.

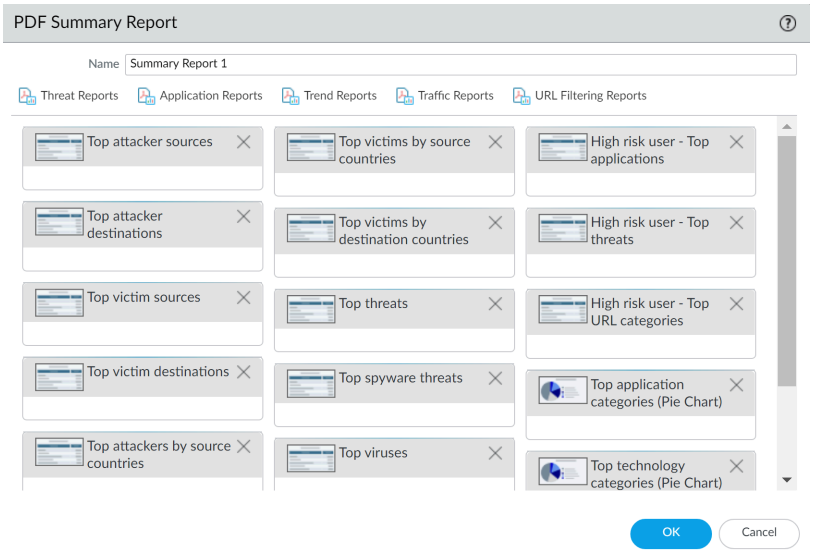
On the PA-220R and the PA-800 Series firewalls, the SaaS Application Usage report is not sent as a PDF attachment in the email. Instead, the email includes a link that you must click to open the report in a web browser.

Manage PDF Summary Reports

PDF summary reports contain information compiled from existing reports, based on data for the top 5 in each category (instead of top 50). They also contain trend charts that are not available in other reports.

STEP 1 | Set up a **PDF Summary Report**.

1. Select **Monitor** > **PDF Reports** > **Manage PDF Summary**.
2. Click **Add** and then enter a **Name** for the report.
3. Use the drop-down for each report group and select one or more of the elements to design the PDF Summary Report. You can include a maximum of 18 report elements.



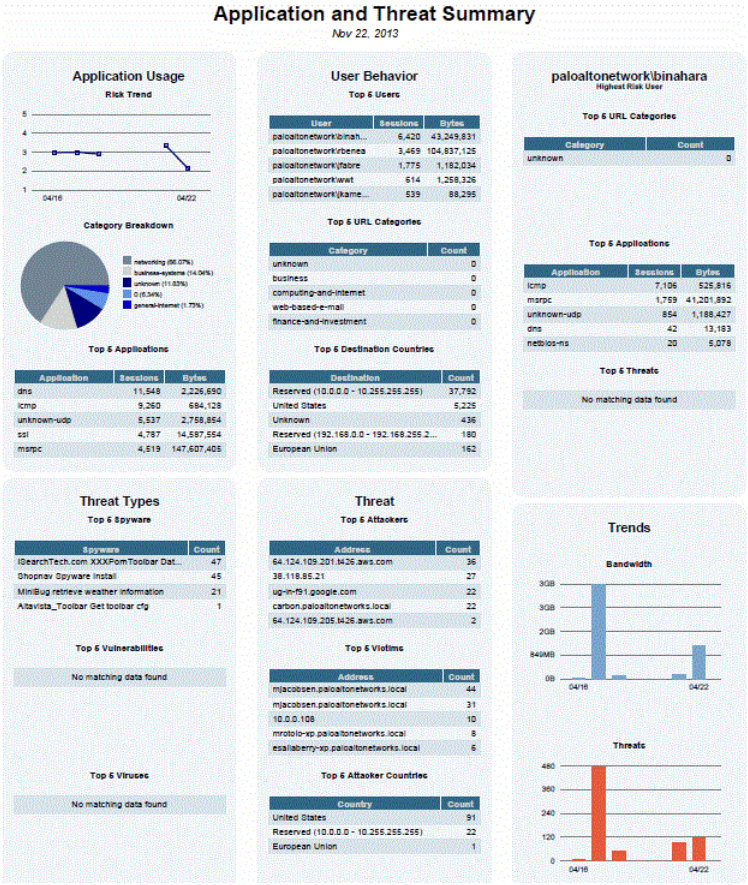
*Selecting **Top Threats** is displayed as `top-attacks` in the Predefined Widgets column for the PDF Summary Report.*

- To remove an element from the report, click the **x** icon or clear the selection from the drop-down for the appropriate report group.
- To rearrange the reports, drag and drop the element icons to another area of the report.

4. Click **OK** to save the report.
5. **Commit** the changes.

STEP 2 | View the report.

To download and view the PDF Summary Report, see View Reports.



The following summary sections refer to the following PDF Summary Report elements:

- ***Top 5 Attacks**—Refers to the **Top threats** element.*
- ***Top 5 Threats**—Refers to the **High risk user - Top threats** element.*
- *Top Threats report—Refers to the full list of threats from the **Top threats** element.*

Generate User/Group Activity Reports

User/Group Activity reports summarize the web activity of individual users or user groups. Both reports include the same information except for the **Browsing Summary by URL Category** and **Browse time calculations**, which only the User Activity report includes.

You must configure User-ID on the firewall to access the list of users and user groups.

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STEP 1 | Configure the browse times and number of logs for User/Group Activity reports.

Required only if you want to change the default values.

1. Select **Device** > **Setup** > **Management**, edit the Logging and Reporting Settings, and select the **Log Export and Reporting** tab.
2. For the **Max Rows in User Activity Report**, enter the maximum number of rows that the detailed user activity report supports (range is 1-1048576, default is 5000). This determines the number of logs that the report analyzes.
3. Enter the **Average Browse Time** in seconds that you estimate users should take to browse a web page (range is 0-300, default is 60). Any request made after the average browse time elapses is considered a new browsing activity. The calculation uses container pages (logged in the URL Filtering logs) as the basis and ignores any new web pages that are loaded between the time of the first request (start time) and the average browse time. For example, if you set the **Average Browse Time** to two minutes and a user opens a web page and views that page for five minutes, the browse time for that page will still be two minutes. This is done because the firewall can't determine how long a user views a given page. The average browse time calculation ignores sites categorized as web advertisements and content delivery networks.
4. For the **Page Load Threshold**, enter the estimated time in seconds for page elements to load on the page (default is 20). Any requests that occur between the first page load and the page load threshold are assumed to be elements of the page. Any requests that occur outside of the page load threshold are assumed to be the user clicking a link within the page.
5. Click **OK** to save your changes.

STEP 2 | Generate the User/Group Activity report.

1. Select **Monitor** > **PDF Reports** > **User Activity Report**.
2. Click **Add** and then enter a **Name** for the report.
3. Create the report:
   - User Activity Report—Select **User** and enter the **Username** or **IP address** (IPv4 or IPv6) of the user.
   - Group Activity Report—Select **Group** and select the **Group Name** of the user group.
4. Select the **Time Period** for the report.
5. (Optional) Select the **Include Detailed Browsing** check box (default is cleared) to include detailed URL logs in the report.

   The detailed browsing information can include a large volume of logs (thousands of logs) for the selected user or user group and can make the report very large.
6. To run the report on demand, click **Run Now**.
7. To save the report configuration, click **OK**. You can't save the output of User/Group Activity reports on the firewall. To schedule the report for email delivery, see Schedule Reports for Email Delivery.

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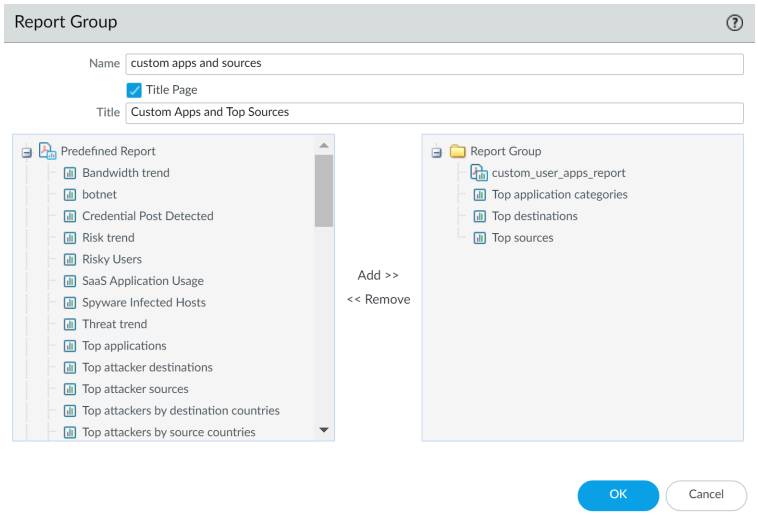
Manage Report Groups

Report groups allow you to create sets of reports that the system can compile and send as a single aggregate PDF report with an optional title page and all the constituent reports included.

Set up report groups.

You must set up a **Report Group** to email report(s).

1. Create an Email server profile.
2. Define the **Report Group**. A report group can compile predefined reports, PDF Summary reports, custom reports, and Log View report into a single PDF.
   1. Select **Monitor** > **Report Group**.
   2. Click **Add** and then enter a **Name** for the report group.
   3. (Optional) Select **Title Page** and add a **Title** for the PDF output.
   4. Select reports from the left column and click **Add** to move each report to the report group on the right.



   The **Log View** report is a report type that is automatically created each time you create a custom report and uses the same name as the custom report. This report will show the logs that were used to build the contents of the custom report.

   To include the log view data, when creating a report group, add your custom report under the **Custom Reports** list and then add the log view report by selecting the matching report name from the **Log View** list. The report will include the custom report data and the log data that was used to create the custom report.
   5. Click **OK** to save the settings.
   6. To use the report group, see Schedule Reports for Email Delivery.

Schedule Reports for Email Delivery

Reports can be scheduled for daily delivery or delivered weekly on a specified day. Scheduled reports are executed starting at 2:00 AM, and email delivery starts after all scheduled reports have been generated.

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STEP 1 | Select **Monitor** > **PDF Reports** > **Email Scheduler** and click **Add**.

STEP 2 | Enter a **Name** to identify the schedule.

STEP 3 | Select the **Report Group** for email delivery. To set up a report group; see Manage Report Groups.

STEP 4 | For the **Email Profile**, select an Email server profile to use for delivering the reports, or click the **Email Profile** link to Create an Email server profile.

STEP 5 | Select the frequency at which to generate and send the report in **Recurrence**.

STEP 6 | The **Override Email Addresses** field allows you to send this report exclusively to the specified recipients. When you add recipients to the field, the firewall does not send the report to the recipients configured in the Email server profile. Use this option for those occasions when the report is for the attention of someone other than the administrators or recipients defined in the Email server profile.

STEP 7 | Click **OK** and **Commit**.

Manage Report Storage Capacity

By default, firewalls contain 200MB of dedicated storage for reports generated by the firewall. In some instances, especially for PA-7000 series and PA-5200 series firewalls, you may need to increase the capacity of available report storage space in order to successfully generate new reports.

STEP 1 | Access the firewall CLI.

STEP 2 | Confirm the current report storage capacity of the firewall:

The command output displays the report storage size in bytes. For this procedure, the firewall has the default 200MB report storage capacity.

```
admin@ISP-CONDOR-B(active)> request report-storage-size show

209715200
```

STEP 3 | Verify you have sufficient storage across the firewall to allocate toward expanding the report storage capacity:

```
admin> show system disk-space
```

```
admin@ISP-CONDOR-B(active)> show system disk-space

Filesystem      Size  Used Avail Use% Mounted on
/dev/root        12G  8.9G  2.0G  83% /
none            7.9G   52K  7.9G   1% /dev
/dev/sda5        16G  8.5G  5.9G  59% /opt/pancfg
/dev/sda6        12G  5.8G  5.0G  54% /opt/panrepo
tmpfs           7.9G  247M  7.6G   4% /dev/shm
/dev/sda8        22G  8.7G   12G  43% /opt/panlogs
tmpfs            12M     0   12M   0% /opt/pancfg/mgmt/lcaas/ssl/private
```

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STEP 4 | Increase the report storage capacity as needed:

For example, we are increasing the report storage size to 1GB.

```
admin> request report-storage-size set size <0-4>
```

```
admin@ISP-CONDOR-B(active)> request report-storage-size set size 1

cfg.report-storage-size-gb: 1
```

STEP 5 | Verify that the report storage capacity is increased to the amount set in the previous step:

```
admin> request report-storage-size show
```

```
admin@ISP-CONDOR-B(active)> request report-storage-size show

1073741824
```

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View Policy Rule Usage

View the number of times a Security, NAT, QoS, policy-based forwarding (PBF), Decryption, Tunnel Inspection, Application Override, Authentication, or DoS protection rule matches traffic to help keep your firewall policies up to date as your environment and security needs change. To prevent attackers from exploiting over-provisioned access, such as when a server is decommissioned or when you no longer need temporary access to a service, use the policy rule hit count data to identify and remove unused rules.

Policy rule usage data enables you to validate rule additions and rule changes and to monitor the time frame when a rule was used. For example, when you migrate port-based rules to app-based rules, you create an app-based rule above the port-based rule and check for any traffic that matches the port-based rule. After migration, the hit count data helps you determine whether it is safe to remove the port-based rule by confirming whether traffic is matching the app-based rule instead of the port-based rule. The policy rule hit count helps you determine whether a rule is effective for access enforcement.

You can reset the rule hit count data to validate an existing rule or to gauge rule usage within a specified period of time. Policy rule hit count data is not stored on the firewall or Panorama so that data is no longer available after you reset (clear) the hit count.

After filtering your policy rulebase, administrators can take action to delete, disable, enable, and tag policy rules directly from the policy optimizer. For example, you can filter for unused rules and then tag them for review to determine whether they can be safely deleted or kept in the rulebase. By enabling administrators to take action directly from the policy optimizer, you reduce the management overhead required to further assist in simplifying your rule lifecycle management and ensure that your firewalls are not over-provisioned.

The rule hit count data is not synchronized across firewalls in a high availability (HA) deployment so you need to log in to each firewall to view the policy rule hit count data for each firewall or use Panorama to view information on the HA firewall peers.

Policy rule usage data is also useful when using Security Policy Rule Optimization to determine which rules to migrate or clean up first.

*(PAN-OS 12.1.2 and later versions) To ensure the best performance, customize your view to display the **Rule Usage** column, only when you need it. The system fetches data for this component on-demand, which prevents automatic data fetching for the columns when hidden and avoids system slowdowns.*

STEP 1 | Launch the Web Interface.

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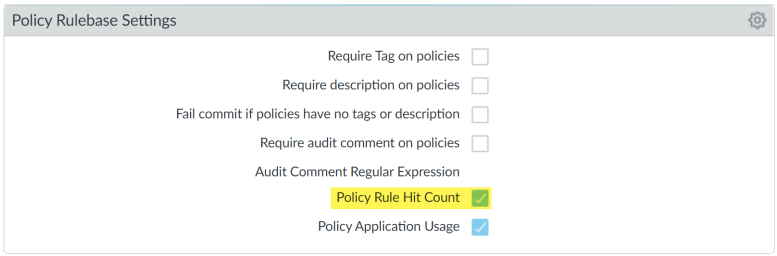
STEP 2 | Verify that **Policy Rule Hit Count** is enabled.

1. Navigate to Policy Rulebase Settings (**Device** > **Setup** > **Management**).
2. Verify that **Policy Rule Hit Count** is enabled.



STEP 3 | Select **Policies**.

STEP 4 | View the policy rule usage for each policy rule:

- **Hit Count**—The number of times traffic matched the criteria you defined in the policy rule. Persists through reboot, dataplane restarts, and upgrades unless you manually reset or rename the rule.
- **Last Hit**—The most recent timestamp for when traffic matched the rule.
- **First Hit**—The first instance when traffic was matched to this rule.
- **Modified**—The date and time the policy rule was last modified.
- **Created**—The date and time the policy rule was created.

If the rule was created when Panorama was running PAN-OS 8.1 and the Policy Rule Hit Count setting is enabled, the First Hit date and time is used as the Created date and time on upgrade to PAN-OS 9.0. If the rule was created in PAN-OS 8.1 when the Policy Rule Hit Count setting was disabled or if the rule was created when Panorama was running PAN-OS 8.0 or an earlier release, the Created date for the rule will be the date and time you successfully upgraded Panorama to PAN-OS 9.0

		Source			Rule Usage				
NAME	T...	Z...	A...	U...	HIT COUNT	LAST HIT	FIRST HIT	MODIFIED	CREATED
'ideo	n...	a...	a...	a...	2424328	2020-09-22 11:33:00	2019-07-30 10:12:57	2020-07-27 13:27:16	2019-07-30 09:50
Video Streaming	n...	a...	a...	a...	14337228	2020-09-22 16:26:58	2019-07-30 10:12:57	2020-07-27 13:27:16	2019-07-30 09:50
'cavenger	n...	a...	a...	a...	321760616	2020-09-22 16:27:10	2019-07-30 10:12:57	2020-07-27 13:27:16	2019-07-30 09:50
Web Traffic	n...	a...	a...	a...	1509584361	2020-09-22 16:27:10	2019-07-30 10:12:02	2020-07-27 13:27:16	2019-07-30 09:50
iperf	n...	a...	a...	a...	5	2019-10-15 14:54:31	2019-10-11 13:08:28	2020-07-27 13:27:16	2019-07-30 09:50

STEP 5 | In the Policy Optimizer dialog, view the **Rule Usage** filter.

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STEP 6 | Filter rules in the selected rulebase.

Use the rule usage filter to evaluate the rule usage within a specified period of time. For example, filter the selected rulebase for Unused rules within the last 30 days. You can also evaluate rule usage with other rule attributes, such as the Created and Modified dates, which enables you to filter for the correct set of rules to review. You can use this data to help manage your rule lifecycle and to determine if a rule needs to be removed to reduce your network attack surface.

1. Select the **Timeframe** you want to filter on or specify a **Custom** time frame.
2. Select the rule **Usage** on which to filter.
3. (Optional) If you have reset the rule usage data for any rules, check for **Exclude rules reset during the last *<number of days>* days** and decide when to exclude a rule based

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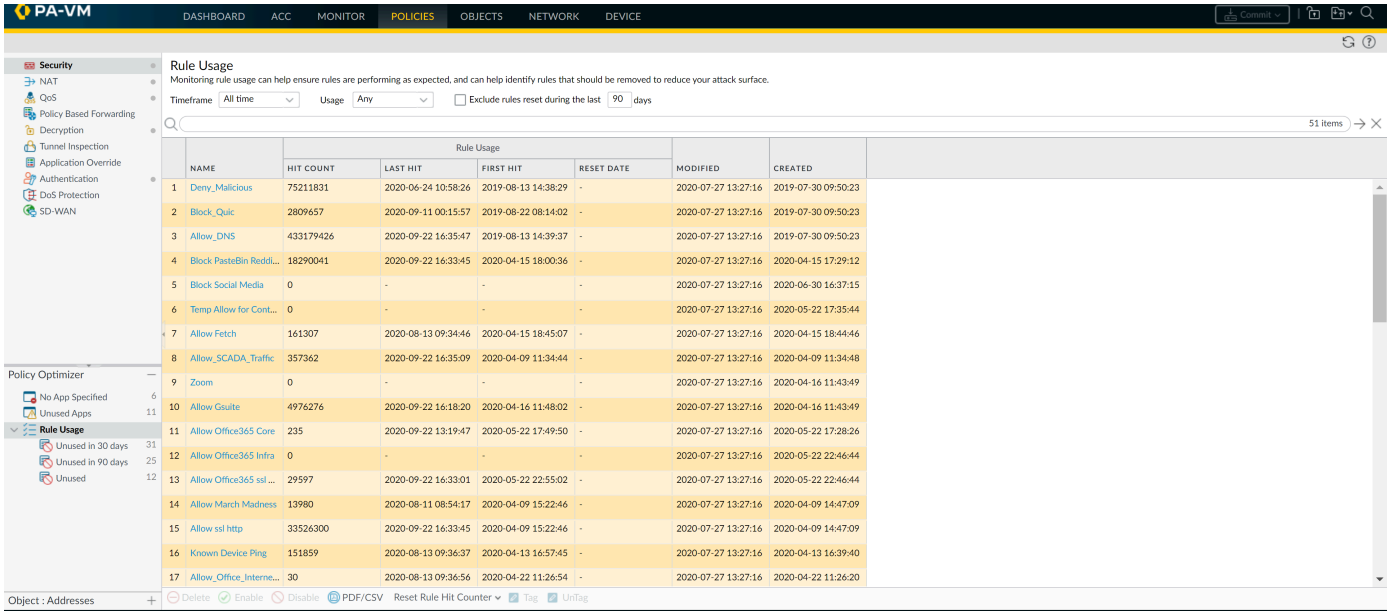
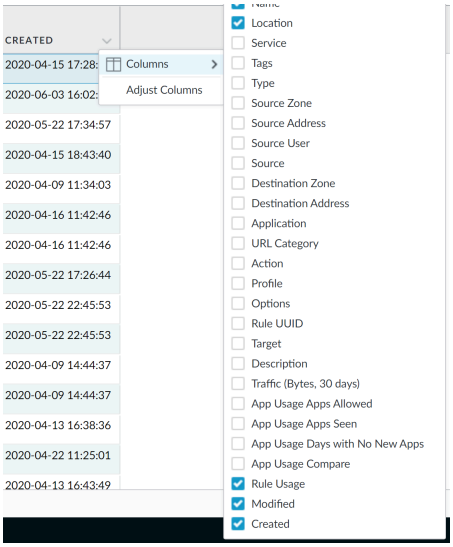
on the number of days you specify since the rule was reset. Only rules that were reset before your specified number of days are included in the filtered results.

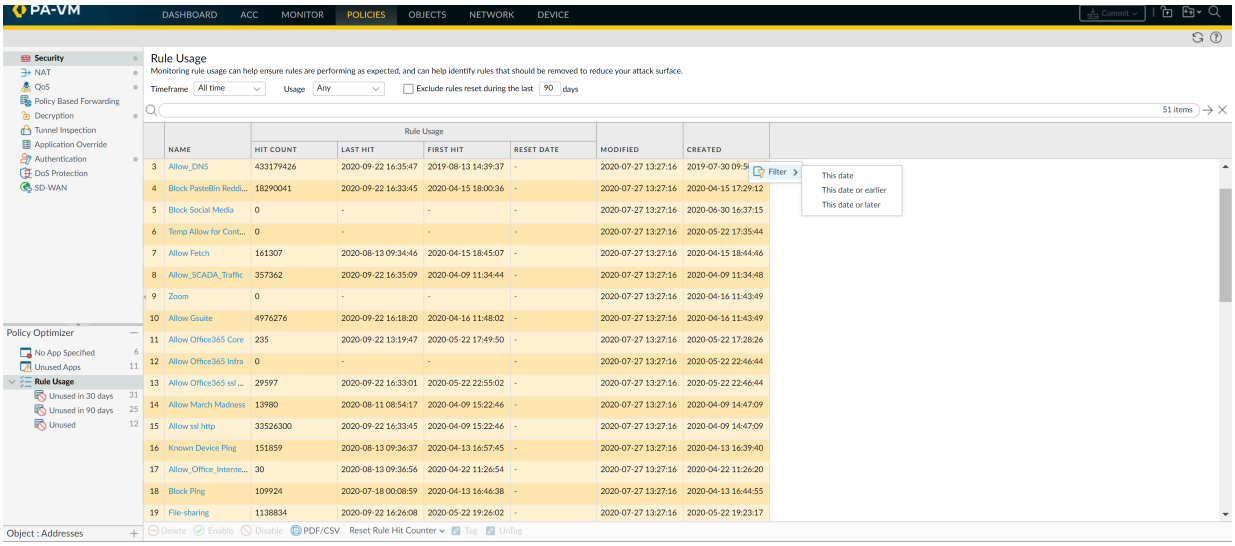


4. (Optional) Specify search filters based on rule data

   1. Hover your cursor over the column header and **Columns**.
   2. Add any additional columns you want to display or use for filter.



   3. Hover your cursor over the column data that you would like to filter on **Filter**. For data that contain dates, select whether to filter using **This date**, **This date or earlier**, or **This date or later**.
   4. **Apply Filter** (→).



STEP 7 | Take action on one or more unused policy rules.

1. Select one or more unused policy rules.
2. Perform one of the following actions:
   - **Delete**—Delete one or more selected policy rules.
   - **Enable**—Enable one or more selected policy rules when disabled.
   - **Disable**—Disable one or more selected policy rules.
   - **Tag**—Apply one or more group tags to one or more selected policy rules. The group tag must already exist in order to tag policy rule.
   - **Untag**—Remove one or more group tags from one or more selected policy rules.
3. **Commit** your changes.

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Use External Services for Monitoring

Using an external service to monitor the firewall enables you to receive alerts for important events, archive monitored information on systems with dedicated long-term storage, and integrate with third-party security monitoring tools. The following are some common scenarios for using external services:

- For immediate notification about important system events or threats, you can Monitor Statistics Using SNMP, Forward Traps to an SNMP Manager, or Configure Email Alerts.
- To send an HTTP-based API request directly to any third-party service that exposes an API to automate a workflow or an action. You can, for example, forward logs that match a defined criteria to create an incidence ticket on ServiceNow instead of relying on an external system to convert syslog messages or SNMP traps to an HTTP request. You can modify the URL, HTTP header, parameters, and the payload in the HTTP request to trigger an action based on the attributes in a firewall log. See Forward Logs to an HTTP(S) Destination.
- For long-term log storage and centralized firewall monitoring, you can Configure Syslog Monitoring to send log data to a syslog server. This enables integration with third-party security monitoring tools such as Splunk or ArcSight.
- For monitoring statistics on the IP traffic that traverses firewall interfaces, you can Configure NetFlow Exports to view the statistics in a NetFlow collector.

You can Configure Log Forwarding from the firewalls directly to external services or from the firewalls to Panorama and then configure Panorama to forwardlogs to the servers. Refer to Log Forwarding Options for the factors to consider when deciding where to forward logs.

You can't aggregate NetFlow records on Panorama; you must send them directly from the firewalls to a NetFlow collector.

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Configure Log Forwarding

Where Can I Use This?	What Do I Need?
• NGFW	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro ❑ Strata Cloud Manager Essentials

In an environment where you use multiple firewalls to control and analyze network traffic, any single firewall can display logs and reports only for the traffic it monitors. Because logging in to multiple firewalls can make monitoring a cumbersome task, you can more efficiently achieve global visibility into network activity by forwarding the logs from all firewalls to Panorama or external services. If you Use External Services for Monitoring, the firewall automatically converts the logs to the necessary format: syslog messages, SNMP traps, email notifications, or as an HTTP payload to send the log details to an HTTP(S) server. In cases where some teams in your organization can achieve greater efficiency by monitoring only the logs that are relevant to their operations, you can create forwarding filters based on any log attributes (such as threat type or source user). For example, a security operations analyst who investigates malware attacks might be interested only in Threat logs with the type attribute set to wildfire-virus.

By default, logs are forwarded over the management interface unless you configure a dedicated service route to forward logs.

Configuring a service route for Panorama log forwarding is available for PA-5200 platforms only.

Forwarded logs have a maximum log record size of 4,096 bytes. A forwarded log with a log record size larger than the maximum is truncated at 4,096 bytes while logs that do not exceed the maximum log record size are not.

Log forwarding is supported only for supported log fields. Forwarding logs that contain unsupported log fields or pseudo-fields causes the firewall to crash.

You can forward logs from the firewalls directly to external services or from the firewalls to Panorama and then configure Panorama to forward logs to the servers. Refer to Log Forwarding Options for the factors to consider when deciding where to forward logs.

You can use Secure Copy (SCP) commands from the CLI to export the entire log database to an SCP server and import it to another firewall. Because the log database is too large for an export or import to be practical on the PA-7000 Series firewall, it does not support these options. You can also use the web interface on all platforms to View and Manage Reports, but only on a per log type basis, not for the entire log database.

- PAN-OS
- Strata Cloud Manager

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Configure Log Forwarding (PAN-OS)

- 10.1 and 10.2
- 11.1 and later

Configure Log Forwarding PAN-OS 10.1 10.2

STEP 1 | Configure a server profile for each external service that will receive log information.

You can use separate profiles to send different sets of logs, filtered by log attributes, to a different server. To increase availability, define multiple servers in a single profile.

Configure one or more of the following server profiles:

- (Required for SMTP over TLS) If you have not already done so, create a certificate profile for the email server.
- To enable the SNMP manager (trap server) to interpret firewall traps, you must load the Palo Alto Networks Supported MIBs into the SNMP manager and, if necessary, compile them. For details, refer to your SNMP management software documentation.
- If the syslog server requires client authentication, you must also #unique_74/unique_74_Connect_42_id6dde9e7b-bf0e-4a24-bcd5-fc9b8af8f294
- Configure an HTTP server profile (see Forward Logs to an HTTP/S Destination).

Log forwarding to an HTTP server is designed for log forwarding at low frequencies and is not recommend for deployments with a high volume of log forwarding. You may experience log loss when forwarding to an HTTP server if your deployment generate a high volume of logs that need to be forwarded.

STEP 2 | Create a Log Forwarding profile.

The profile defines the destinations for Traffic, Threat, WildFire Submission, URL Filtering, Data Filtering, Tunnel and Authentication logs.

1. Select **Objects** > **Log Forwarding** and **Add** a profile.
2. Enter a **Name** to identify the profile.

   If you want the firewall to automatically assign the profile to new security rules and zones, enter **`default`**. If you don't want a default profile, or you want to override

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an existing default profile, enter a **Name** that will help you identify the profile when assigning it to security rules and zones.

> *If no log forwarding profile named **default** exists, the profile selection is set to **None** by default in new security rules (**Log Forwarding** field) and new security zones (**Log Setting** field), although you can change the selection.*

3. **Add** one or more *match list profiles*.

   The profiles specify log query filters, forwarding destinations, and automatic actions such as tagging. For each match list profile:

   1. Enter a **Name** to identify the profile.
   2. Select the **Log Type**.
   3. In the **Filter** drop-down, select **Filter Builder**. Specify the following and then **Add** each query:
      - **Connector** logic (and/or)
      - Log **Attribute**
      - **Operator** to define inclusion or exclusion logic
      - Attribute **Value** for the query to match
   4. Select **Panorama** if you want to forward logs to Log Collectors or the Panorama management server.
   5. For each type of external service that you use for monitoring (SNMP, Email, Syslog, and HTTP), **Add** one or more server profiles.
4. (Optional, GlobalProtect Only) If you are using a log forwarding profile with a security policy to automatically quarantine a device using GlobalProtect, select **Quarantine** in the **Built-in Actions** area.
5. Click **OK** to save the Log Forwarding profile.

STEP 3 | Assign the Log Forwarding profile to policy rules and network zones.

Security, Authentication, and DoS Protection rules support log forwarding. In this example, you assign the profile to a Security rule.

Perform the following steps for each rule that you want to trigger log forwarding:

1. Select **Policies** > **Security** and edit the rule.
2. Select **Actions** and select the **Log Forwarding** profile you created.
3. Set the **Profile Type** to **Profiles** or **Group**, and then select the security profiles or **Group Profile** required to trigger log generation and forwarding for:
   - Threat logs—Traffic must match any security profile assigned to the rule.
   - WildFire Submission logs—Traffic must match a WildFire Analysis profile assigned to the rule.
4. For Traffic logs, select **Log At Session Start** and/or **Log At Session End**.

   Log At Session Start consumes more resources than logging only at the session end. In most cases, you only **Log At Session End**. Enable both **Log At Session Start** and **Log At Session End** only for troubleshooting, for long-lived tunnel sessions such as GRE tunnels

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(you can't see these sessions in the ACC unless you log at the start of the session), and to gain visibility into Operational Technology/Industrial Control Systems (OT/ICS) sessions, which are also long-lived sessions.

5. Click **OK** to save the rule.

STEP 4 | Configure the destinations for System, Configuration, Correlation, GlobalProtect, HIP Match, and User-ID logs.

Panorama generates Correlation logs based on the firewall logs it receives, rather than aggregating Correlation logs from firewalls.

1. Select **Device** > **Log Settings**.
2. For each log type that the firewall will forward, see Step Add one or more match list profiles.

STEP 5 | (PA-7000 Series firewalls with Log Cards only) Configure a log card interface to perform log forwarding.

As of PAN-OS 10.1, you can no longer forward system logs and other Management plane logs using the Management interface or service routes. The only way to forward system logs from a PA-7000 Series firewall with a LFC running PAN-OS 10.1 or later is by configuring a log card interface

1. Select **Network** > **Interfaces** > **Ethernet** and click **Add Interface**.
2. Select the **Slot** and **Interface Name**.
3. Set the **Interface Type** to **Log Card**.
4. Enter the **IP Address**, **Default Gateway**, and (for IPv4 only) **Netmask**.
5. Select **Advanced** and specify the **Link Speed**, **Link Duplex**, and **Link State**.

   *These fields default to **auto**, which specifies that the firewall automatically determines the values based on the connection. However, the minimum recommended **Link Speed** for any connection is **1000** (Mbps).*
6. Click **OK** to save your changes.

STEP 6 | (PA-5450 firewall only) Configure a log interface to perform log forwarding.

This step is not required if you are forwarding logs to a Panorama or Strata Logging Service using the management interface. The management interface handles log forwarding by default and does not require the log interface to be configured.

- (PAN-OS 10.2.0 and 10.2.1) The management interface handles log forwarding by default unless you configure a specific service route for log forwarding.
- (PAN-OS 10.2.2 and later releases) The management interface handles log forwarding by default unless you configure the log interface or a specific service route for log forwarding.

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If a log interface is configured and committed, all internal logging, Strata Logging Service, SNMP, HTTP, and Syslog will be forwarded by the log interface.

Ensure that the log interface you are configuring is not in the same subnetwork as the management interface. Configuring both interfaces in the same subnetwork can cause connectivity issues and result in the wrong interface being used for log forwarding.

*LOG-1 and LOG-2 are bundled as a single logical interface called **bond1**. Bond1 uses LACP (link aggregation control protocol) as IEEE 802.3ad. Set the **Mode** for LACP status queries to **Active** and the **Transmission Rate** for LACP query and response exchanges to **Slow**.*

1. Select **Device** > **Setup** > **Management**.
2. Select the settings gear on the top menu bar of **Log Interface**.
3. Fill in the **IP Address**, **Netmask**, and **Default Gateway** fields.

   If your network uses IPv6, fill in the **IPv6 Address** and **IPv6 Default Gateway** fields instead.

   When the log interface is configured with an IP address, communication between the firewall and Panorama automatically switches from being handled by the management interface (default) to the log interface.
4. Specify the **Link Speed**, **Link Duplex**, and **Link State**. These fields default to **auto**, which specifies that the firewall automatically determines the values based on the connection.
5. Click **OK** to save your changes.

STEP 7 | Commit and verify your changes.

1. **Commit** your changes.
2. Verify the log destinations you configured are receiving firewall logs:
   - Panorama—If the firewall forwards logs to a Panorama virtual appliance in Panorama mode or to an M-Series appliance, you must configure a Collector Group before Panorama will receive the logs. You can then verify log forwarding.
   - Email server—Verify that the specified recipients are receiving logs as email notifications.
   - Syslog server—Refer to your syslog server documentation to verify it's receiving logs as syslog messages.
   - SNMP manager—Use an SNMP Manager to Explore MIBs and Objects to verify it's receiving logs as SNMP traps.
   - HTTP server—Forward Logs to an HTTP/S Destination.

Configure Log Forwarding PAN-OS 11.1

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STEP 1 | Configure a server profile for each external service that will receive log information.

You can use separate profiles to send different sets of logs, filtered by log attributes, to a different server. To increase availability, define multiple servers in a single profile.

Configure one or more of the following server profiles:

- (Required for SMTP over TLS) If you have not already done so, create a certificate profile for the email server.
- To enable the SNMP manager (trap server) to interpret firewall traps, you must load the Palo Alto Networks Supported MIBs into the SNMP manager and, if necessary, compile them. For details, refer to your SNMP management software documentation.
- If the syslog server requires client authentication, you must also #unique_74/unique_74_Connect_42_id6dde9e7b-bf0e-4a24-bcd5-fc9b8af8f294
- Configure an HTTP server profile (see Forward Logs to an HTTP/S Destination).

Log forwarding to an HTTP server is designed for log forwarding at low frequencies and is not recommend for deployments with a high volume of log forwarding. You may experience log loss when forwarding to an HTTP server if your deployment generate a high volume of logs that need to be forwarded.

STEP 2 | Create a Log Forwarding profile.

The profile defines the destinations for Traffic, Threat, WildFire Submission, URL Filtering, Data Filtering, Tunnel and Authentication logs.

1. Select **Objects** > **Log Forwarding** and **Add** a profile.
2. Enter a **Name** to identify the profile.

   If you want the firewall to automatically assign the profile to new security rules and zones, enter **`default`**. If you don't want a default profile, or you want to override

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an existing default profile, enter a **Name** that will help you identify the profile when assigning it to security rules and zones.

> *If no log forwarding profile named **default** exists, the profile selection is set to **None** by default in new security rules (**Log Forwarding** field) and new security zones (**Log Setting** field), although you can change the selection.*

3. **Add** one or more *match list profiles*.

   The profiles specify log query filters, forwarding destinations, and automatic actions such as tagging. For each match list profile:

   1. Enter a **Name** to identify the profile.
   2. Select the **Log Type**.
   3. In the **Filter** drop-down, select **Filter Builder**. Specify the following and then **Add** each query:
      - **Connector** logic (and/or)
      - Log **Attribute**
      - **Operator** to define inclusion or exclusion logic
      - Attribute **Value** for the query to match
   4. Select **Panorama** if you want to forward logs to Log Collectors or the Panorama management server.
   5. For each type of external service that you use for monitoring (SNMP, Email, Syslog, and HTTP), **Add** one or more server profiles.
4. (Optional, GlobalProtect Only) If you are using a log forwarding profile with a security policy to automatically quarantine a device using GlobalProtect, select **Quarantine** in the **Built-in Actions** area.
5. Click **OK** to save the Log Forwarding profile.

STEP 3 | Assign the Log Forwarding profile to policy rules and network zones.

Security, Authentication, and DoS Protection rules support log forwarding. In this example, you assign the profile to a Security rule.

Perform the following steps for each rule that you want to trigger log forwarding:

1. Select **Policies** > **Security** and edit the rule.
2. Select **Actions** and select the **Log Forwarding** profile you created.
3. Set the **Profile Type** to **Profiles** or **Group**, and then select the security profiles or **Group Profile** required to trigger log generation and forwarding for:
   - Threat logs—Traffic must match any security profile assigned to the rule.
   - WildFire Submission logs—Traffic must match a WildFire Analysis profile assigned to the rule.
4. For Traffic logs, select **Log At Session Start** and/or **Log At Session End**.

   Log At Session Start consumes more resources than logging only at the session end. In most cases, you only **Log At Session End**. Enable both **Log At Session Start** and **Log At Session End** only for troubleshooting, for long-lived tunnel sessions such as GRE tunnels

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(you can't see these sessions in the ACC unless you log at the start of the session), and to gain visibility into Operational Technology/Industrial Control Systems (OT/ICS) sessions, which are also long-lived sessions.

5. Click **OK** to save the rule.

STEP 4 | Configure the destinations for System, Configuration, Correlation, GlobalProtect, HIP Match, and User-ID logs.

Panorama generates Correlation logs based on the firewall logs it receives, rather than aggregating Correlation logs from firewalls.

1. Select **Device** > **Log Settings**.
2. For each log type that the firewall will forward, see Step Add one or more match list profiles.

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STEP 5 | (PA-7500 Series firewall only) Configure a log interface to perform log forwarding.

Management Plane logs are forwarded through the management interface to a log collector such as Panorama or Cortex Data Lake. Data plane logs are forwarded through the log ports on the PA-7500 Management Processing Card (MPC).

*LOG-1 and LOG-2 are bundled as a single logical interface called **bond1**. When connecting both log ports, use the Ethernet port channel on the switch.*

This step is not required if you are storing Data Plane logs locally, which can be configured by enabling demo mode. To enable demo mode, you must first add and enable a logging drive in the PA-7500 Management Processing Card (MPC).

After the logging drive is enabled, log in to the CLI and enter

```
admin@pa-7500> debug log-receiver mp demo_mode set dp
 <data plane number> slot <DPC slot number> duration <in
 minutes>
```

For the data plane and DPC slot number, refer to the chassis slot numbering on the PA-7500 front panel.

To view the current duration of demo mode, enter

```
admin@pa-7500> debug log-receiver mp demo_mode show
```

*To turn off demo mode, enter the following command and set the duration to **0**.*

```
admin@pa-7500> debug log-receiver mp demo_mode set dp
 <data plane number> slot <DPC slot number> duration <in
 minutes>
```

1. Select **Device** > **Setup** > **Management**.
2. Select the settings gear on the top menu bar of **Log Interface**.
3. Fill in the **IP Address**, **Netmask**, and **Default Gateway** fields.

   If your network uses IPv6, fill in the **IPv6 Address** and **IPv6 Default Gateway** fields instead.

   The log interface can be configured with either an IPv4 address or an IPv6 address; it cannot have both an IPv4 address and IPv6 address at the same time.
4. Specify the **Link Speed**, **Link Duplex**, and **Link State**. These fields default to **auto**, which specifies that the firewall automatically determines the values based on the connection.
5. Click **OK** to save your changes.

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STEP 6 | (PA-7000 Series firewalls with Log Cards only) Configure a log card interface to perform log forwarding.

As of PAN-OS 10.1, you can no longer forward system logs and other Management plane logs using the Management interface or service routes. The only way to forward system logs from a PA-7000 Series firewall with a LFC running PAN-OS 10.1 or later is by configuring a log card interface

1. Select **Network** > **Interfaces** > **Ethernet** and click **Add Interface**.
2. Select the **Slot** and **Interface Name**.
3. Set the **Interface Type** to **Log Card**.
4. Enter the **IP Address**, **Default Gateway**, and (for IPv4 only) **Netmask**.
5. Select **Advanced** and specify the **Link Speed**, **Link Duplex**, and **Link State**.

   *These fields default to **auto**, which specifies that the firewall automatically determines the values based on the connection. However, the minimum recommended **Link Speed** for any connection is **1000** (Mbps).*
6. Click **OK** to save your changes.

STEP 7 | (PA-5450 firewall only) Configure a log interface to perform log forwarding.

This step is not required if you are forwarding logs to a Panorama or Strata Logging Service using the management interface. The management interface handles log forwarding by default and does not require the log interface to be configured.

- (PAN-OS 10.2.0 and 10.2.1) The management interface handles log forwarding by default unless you configure a specific service route for log forwarding.
- (PAN-OS 10.2.2 and later releases) The management interface handles log forwarding by default unless you configure the log interface or a specific service route for log forwarding.

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If a log interface is configured and committed, all internal logging, Strata Logging Service, SNMP, HTTP, and Syslog will be forwarded by the log interface.

Ensure that the log interface you are configuring is not in the same subnetwork as the management interface. Configuring both interfaces in the same subnetwork can cause connectivity issues and result in the wrong interface being used for log forwarding.

*LOG-1 and LOG-2 are bundled as a single logical interface called **bond1**. Bond1 uses LACP (link aggregation control protocol) as IEEE 802.3ad. Set the **Mode** for LACP status queries to **Active** and the **Transmission Rate** for LACP query and response exchanges to **Slow**.*

1. Select **Device** > **Setup** > **Management**.
2. Select the settings gear on the top menu bar of **Log Interface**.
3. Fill in the **IP Address**, **Netmask**, and **Default Gateway** fields.

   If your network uses IPv6, fill in the **IPv6 Address** and **IPv6 Default Gateway** fields instead.

   When the log interface is configured with an IP address, communication between the firewall and Panorama automatically switches from being handled by the management interface (default) to the log interface.
4. Specify the **Link Speed**, **Link Duplex**, and **Link State**. These fields default to **auto**, which specifies that the firewall automatically determines the values based on the connection.
5. Click **OK** to save your changes.

STEP 8 | Commit and verify your changes.

1. **Commit** your changes.
2. Verify the log destinations you configured are receiving firewall logs:
   - Panorama—If the firewall forwards logs to a Panorama virtual appliance in Panorama mode or to an M-Series appliance, you must configure a Collector Group before Panorama will receive the logs. You can then verify log forwarding.
   - Email server—Verify that the specified recipients are receiving logs as email notifications.
   - Syslog server—Refer to your syslog server documentation to verify it's receiving logs as syslog messages.
   - SNMP manager—Use an SNMP Manager to Explore MIBs and Objects to verify it's receiving logs as SNMP traps.
   - HTTP server—Forward Logs to an HTTP/S Destination.

Configure Log Forwarding (SCM)

Create a HTTP/S server profile to forward all logs or specific logs to trigger an action on an external HTTP-based service when an event occurs. When forwarding logs to an HTTP server, configure the firewall to send an HTTP-based API request directly to a third-party service to trigger an action that is based on the attributes in a firewall log. You can configure the firewall

to work with any HTTP-based service that exposes an API and you can modify the URL, HTTP header, parameters, and the payload in the HTTP request to meet your integration needs.

STEP 1 | Log in to cloud management.

STEP 2 | Select **Manage** > **Configuration** > **NGFW and Prisma Access** > **Objects** > **Log Forwarding** > **HTTPS Server ProfileConfiguration** > **NGFW and Prisma Access** > **Objects** > **Log Forwarding** > **HTTPS Server Profile** and select the Configuration Scope where you want to create the HTTP/S server profile.

You can select a folder or firewall from your **Folders** or select **Snippets** to configure the HTTP/S server profile in a snippet.

STEP 3 | **Add HTTP**.

STEP 4 | Enter a descriptive **Name**

STEP 5 | (Optional) check (enable) **Tag Registration** to add or remove a tag on a source or destination IP address in a log entry and register the IP address and tag mapping to the User-ID agent on the firewall using HTTP/S.

You can then define Dynamic Address Groups that use these tags as a filtering criteria to determine its members, and enforce policy rules to an IP address based on tags. To register tags to the User-ID agent on Panorama, you don't need a server profile. Additionally, you can't use the HTTP server profile to register tags to a User-ID agent running on a Windows server.

STEP 6 | Add and configure the HTTP/S server profile to forward logs to an HTTP/S destination.

1. Enter a **Name** for the HTTP/S destination.
2. Enter the HTTP/S destination IP **Address**.
3. Select the **Protocol**.

   You can select **HTTP** or **HTTPS**.
4. Select the **Port** number the HTTP/S destination listens to.

   Default for HTTP is 80. Default for HTTPS is 443.
5. Select the supported **TLS Version**.

   You can select **1.0**, **1.1**, or **1.2**. HTTP has no default TLS Version. HTTPS default TLS Version is **1.2**.
6. Select the **Certificate Profile** to use for the TLS connection with the HTTP/S destination.
7. Select the **HTTP Method** that the third-party service supports.

   You can select **DELETE**, **GET**, **POST**, or **PUT**. Default is **POST**.
8. (Optional) Enter the **Username** and **Password** for authentication the HTTP/S destination, if needed.

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STEP 7 | (Optional) Select the **Payload Format** for the HTTP/S request.

1. Select the **Log Type** link for each log type for you want to define the HTTP/S request format.
2. Select the **Pre-defined Formats** or create a custom format.

   If you create a custom format, the **URI** is the resource endpoint on the HTTP/S service. The firewall appends the URI to the IP address you defined earlier to construct the URL for the HTTP/S request. Ensure that the URI and payload format matches the syntax that your third-party vendor requires. You can use any attribute supported on the selected log type within the HTTP Header, the Parameter and Value pairs, and in the request payload.

STEP 8 | Configure log forwarding to the HTTP/S destination.

1. Select **Manage** > **Configuration** > **NGFW and Prisma Access** > **Objects** > **Log Forwarding** > **Log Forwarding ProfileConfiguration** > **NGFW and Prisma Access** > **Objects** > **Log Forwarding** > **Log Forwarding Profile** and select the Configuration Scope where you want to create the Log Forwarding profile.

   You can select a folder or firewall from your **Folders** or select **Snippets** to configure the Log Forwarding profile in a snippet.
2. **Add Log Forwarding Profile**.
3. Enter a descriptive **Name**.
4. **Add** the profile match list for the Log Forwarding profile.

   A match list profile specifies the log query filter, forwarding destinations, and automatic actions to take. Multiple profile match lists can be added to the same Log Forwarding profile to allow you to add different profile match lists for different log types in the same Log Forwarding profile.

   1. Enter a descriptive **Name**.
   2. Select the **Log Type**.

      Only one log type can be added per profile match list.
   3. (Optional) Configure the log query **Filter**. Default is `All Logs`.
   4. Add the **HTTP Server Profile** you created in the previous step.
   5. **Save**.
   6. Repeat this step for all the log types you want to forward to your HTTP/S destination.
5. **Save**.

STEP 9 | Select **Push Config**.

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Configure Email Alerts

Where Can I Use This?	What Do I Need?
• NGFW	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro ❑ Strata Cloud Manager Essentials

You can configure email alerts for log types, such as System, Config, HIP Match, Correlation, Threat, WildFire Submission, and Traffic logs. For each log type, you can set up separate email profiles, allowing you to send notifications to different email servers based on the log type. To ensure high availability, you can define multiple servers (up to four) within a single profile. If one server fails or becomes unreachable, the system attempts to send the alert through the next available server.

It is a best practice to enable transport layer security (TLS). This requires the firewall to authenticate with the email server before the firewall relays email to the server. Using TLS helps prevent malicious activities, such as Simple Mail Transfer Protocol (SMTP) relay attacks. Additionally, TLS helps to prevent email spoofing, which is commonly used in phishing attacks.

- Strata Cloud Manager
- PAN-OS & Panorama

Configure Email Alerts (Strata Cloud Manager)

STEP 1 | Log in to Strata Cloud Manager.

STEP 2 | Select **Manage** > **Configuration** > **NGFW and Prisma Access** > **Objects** > **Log Forwarding** > **Email Server Profile**.

STEP 3 | Click **Add Email Server** .

STEP 4 | Enter a name, and click **Add Email Server Profile**.

STEP 5 | Enter a **Name**.

STEP 6 | (Optional) Enter an **Email Display Name** to specify the name to display in the From field of the email.

STEP 7 | Enter the email address **From** which the firewall sends emails.

STEP 8 | Enter the email address **To** which the firewall sends emails.

STEP 9 | (Optional) If you want to send emails to a second account, enter the address of the **Additional Recipient**. You can add only one additional recipient. For multiple recipients, add the email address of a distribution list.

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STEP 10 | Enter the IP address or hostname of the **Email Gateway** to use for sending emails.

STEP 11 | Select the **Type** of protocol to use to connect to the email server:

- **Unauthenticated SMTP**—Use SMTP to connect to the email server without authentication. The default **Port** is 25, but you can optionally specify a different port. This protocol does not provide the same security as SMTP over TLS, but if you select this protocol, skip the next step.
- **SMTP over TLS**—(Recommended) Use TLS to require authentication to connect to the email server. Continue to the next step to configure the TLS authentication.

STEP 12 | (SMTP over TLS only) Configure the firewall to use TLS authentication to connect to the email server.

1. (Optional) Specify the **Port** to use to connect to the email server (default is 587).
2. **TLS Version**—Specify the TLS version (**1.1** or **1.2**).

   Palo Alto Networks strongly recommends using the latest TLS version.

3. Select the **Authentication Method** for the firewall and the email server:
   - **Auto**—Allow the firewall and the email server to determine the authentication method.
   - **Login**—Use Base64 encoding for the username and password and transmit them separately.
   - **Plain**—Use Base64 encoding for the username and password and transmit them together.
4. Select a **Certificate Profile** to authenticate with the email server.
5. Enter the **Username** and **Password** of the account that sends the emails, then **Confirm Password**.

STEP 13 | Click **Add** to save the Email server profile.

STEP 14 | (Optional) Select the **Custom Log Format** tab and customize the format of the email messages. For details on how to create custom formats for the various log types, refer to the Common Event Format Configuration Guide.

STEP 15 | Configure email alerts for Traffic, Threat, and WildFire Submission logs.

1. Select **Objects** > **Log Forwarding**.
2. Click **Add Log Forwarding Profile**, and enter a **Name** to identify the profile.
3. For each log type and each severity level or WildFire verdict, select the Email server profile, and click **Save**.

STEP 16 | Create a Security policy rule.

STEP 17 | **Push Config** to push your configuration changes.

Configure Email Alerts (PAN-OS)

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STEP 1 | (Required for SMTP over TLS) If you have not already done so, create a certificate profile for the email server.

STEP 2 | Select **Device** > **Server Profiles** > **Email**.

STEP 3 | **Add** an email server profile and enter a **Name**.

STEP 4 | From the read-only window that appears, **Add** the email server and enter a **Name.**

STEP 5 | If the firewall has more than one virtual system (vsys), select the **Location** (vsys or **Shared**) where this profile is available.

STEP 6 | (Optional) Enter an **Email Display Name** to specify the name to display in the From field of the email.

STEP 7 | Enter the email address **From** which the firewall sends emails.

STEP 8 | Enter the email address **To** which the firewall sends emails.

STEP 9 | (Optional) If you want to send emails to a second account, enter the address of the **Additional Recipient**. You can add only one additional recipient. For multiple recipients, add the email address of a distribution list.

STEP 10 | Enter the IP address or hostname of the **Email Gateway** to use for sending emails.

STEP 11 | Select the **Type** of protocol to use to connect to the email server:

- **Unauthenticated SMTP**—Use SMTP to connect to the email server without authentication. The default **Port** is 25, but you can optionally specify a different port. This protocol does not provide the same security as SMTP over TLS, but if you select this protocol, skip the next step.
- **SMTP over TLS**—(Recommended) Use TLS to require authentication to connect to the email server. Continue to the next step to configure the TLS authentication.

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STEP 12 | (SMTP over TLS only) Configure the firewall to use TLS authentication to connect to the email server.

1. (Optional) Specify the **Port** to use to connect to the email server (default is 587).
2. **TLS Version**—Specify the TLS version (**1.1** or **1.2**).

   Palo Alto Networks strongly recommends using the latest TLS version.

3. Select the **Authentication Method** for the firewall and the email server:
   - **Auto**—Allow the firewall and the email server to determine the authentication method.
   - **Login**—Use Base64 encoding for the username and password and transmit them separately.
   - **Plain**—Use Base64 encoding for the username and password and transmit them together.
4. Select a **Certificate Profile** to authenticate with the email server.
5. Enter the **Username** and **Password** of the account that sends the emails, then **Confirm Password**.
6. (Optional) To confirm that the firewall can successfully authenticate with the email server, you can **Test Connection**.

STEP 13 | Click **OK** to save the Email server profile.

STEP 14 | (Optional) Select the **Custom Log Format** tab and customize the format of the email messages. For details on how to create custom formats for the various log types, refer to the Common Event Format Configuration Guide.

STEP 15 | Configure email alerts for Traffic, Threat, and WildFire Submission logs.

1. See Create a Log Forwarding profile.
   1. Select **Objects** > **Log Forwarding**, click **Add**, and enter a **Name** to identify the profile.
   2. For each log type and each severity level or WildFire verdict, select the Email server profile and click **OK**.
2. See Assign the Log Forwarding profile to policy rules and network zones.

STEP 16 | Configure email alerts for System, Config, HIP Match, and Correlation logs.

1. Select **Device** > **Log Settings**.
2. For System and Correlation logs, click each Severity level, select the **Email** server profile, and click **OK**.
3. For Config and HIP Match logs, edit the section, select the **Email** server profile, and click **OK**.
4. Click **Commit**.

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Use Syslog for Monitoring

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

Syslog is a widely adopted standard for centralizing log data from various network devices, including routers, firewalls, and printers, regardless of the vendor. It's a fundamental tool for collecting and aggregating event logs into a central repository. This consolidation allows for effective archiving, analysis, and reporting across your network infrastructure.

Palo Alto Network firewalls can leverage this standard to forward every type of log they generate to an external syslog server. This capability is vital for organizations that need to maintain a comprehensive record of network activity, security events, and system health. You can configure log forwarding to meet your specific security and reliability needs.

For secure and reliable log forwarding, you can use TCP or TLS (specifically TLSv1.2). TCP ensures that packets are delivered and reassembled in the correct order, while TLS adds an extra layer of encryption, protecting sensitive log data in transit. If security is not a primary concern and you prefer a faster, more lightweight protocol, you can opt for UDP. The choice of protocol depends on the balance you need between speed, reliability, and security.

- Configure Syslog Monitoring
- Syslog Field Descriptions
- Syslog Severity Reference Manual

Configure Syslog Monitoring

Where Can I Use This?	What Do I Need?
• NGFW	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

To Use Syslog for Monitoring a Palo Alto Networks firewall, create a Syslog server profile and assign it to the log settings for each log type. Optionally, you can configure the header format used in syslog messages and enable client authentication for syslog over TLSv1.2.

For CEF-formated syslog events collection, you must edit the default syslog configuration. The default syslog monitoring configuration is not supported for CEF syslog events collection.

- PAN-OS
- Strata Cloud Manager

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Configure Syslog Monitoring (PAN-OS)

STEP 1 | Configure a Syslog server profile.

You can use separate profiles to send syslogs for each log type to a different server. To increase availability, define multiple servers (up to four) in a single profile.

1. Select **Device** > **Server Profiles** > **Syslog**.
2. Click **Add** and enter a **Name** for the profile.
3. If the firewall has more than one virtual system (vsys), select the **Location** (vsys or **Shared**) where this profile is available.
4. For each syslog server, click **Add** and enter the information that the firewall requires to connect to it:
   - **Name**—Unique name for the server profile.
   - **Syslog Server**—IP address or fully qualified domain name (FQDN) of the syslog server.

     *If you configure an FQDN and use **UDP** transport, if the firewall cannot resolve the FQDN, the firewall uses the existing IP address resolution for the FQDN as the **Syslog Server** address.*
   - **Transport**—Select **TCP**, **UDP**, or **SSL** (TLS) as the protocol for communicating with the syslog server. For **SSL**, the firewall supports only TLSv1.2.
   - **Port**—The port number on which to send syslog messages (default is UDP on port 514); you must use the same port number on the firewall and the syslog server.
   - **Format**—Select the syslog message format to use: **BSD** (the default) or **IETF**. Traditionally, **BSD** format is over UDP and **IETF** format is over TCP or SSL/TLS.
   - **Facility**—Select a syslog standard value (default is **LOG_USER**) to calculate the priority (PRI) field in your syslog server implementation. Select the value that maps to how you use the PRI field to manage your syslog messages.
5. (**Optional**) To customize the format of the syslog messages that the firewall sends, select the **Custom Log Format** tab. For details on how to create custom formats for the various log types, refer to the Common Event Format Configuration Guide.
6. Click **OK** to save the server profile.

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STEP 2 | Configure syslog forwarding for Traffic, Threat, and WildFire Submission logs.

1. Configure the firewall to forward logs. For more information, see Step Create a Log Forwarding profile.
   1. Select **Objects** > **Log Forwarding**, click **Add**, and enter a **Name** to identify the profile.
   2. For each log type and each severity level or WildFire verdict, select the **Syslog** server profile and click **OK**.
2. Assign the log forwarding profile to a security policy to trigger log generation and forwarding. For more information, See Step Assign the Log Forwarding profile to policy rules and network zones.
   1. Select **Policies** > **Security** and select a policy rule.
   2. Select the **Actions** tab and select the **Log Forwarding** profile you created.
   3. For Traffic logs, select one or both of the **Log at Session Start** and **Log At Session End** check boxes, and click **OK**.

   For detailed information about configuring a log forwarding profile and assigning the profile to a policy rule, see Configure Log Forwarding.

STEP 3 | Configure syslog forwarding for System, Config, HIP Match, and Correlation logs.

1. Select **Device** > **Log Settings**.
2. For System and Correlation logs, click each Severity level, select the **Syslog** server profile, and click **OK**.
3. For Config, HIP Match, and Correlation logs, edit the section, select the **Syslog** server profile, and click **OK**.

STEP 4 | (Optional) Configure the header format of syslog messages.

The log data includes the unique identifier of the firewall that generated the log. Choosing the header format provides more flexibility in filtering and reporting on the log data for some Security Information and Event Management (SIEM) servers.

This is a global setting and applies to all Syslog server profiles configured on the firewall.

1. Select **Device** > **Setup** > **Management** and edit the Logging and Reporting Settings.
2. Select the **Log Export and Reporting** tab and select the Syslog HOSTNAME Format:
   - **FQDN** (default)—Concatenates the hostname and domain name defined on the sending firewall.
   - **hostname**—Uses the hostname defined on the sending firewall.
   - **ipv4-address**—Uses the IPv4 address of the firewall interface used to send logs. By default, this is the MGT interface.
   - **ipv6-address**—Uses the IPv6 address of the firewall interface used to send logs. By default, this is the MGT interface.
   - **none**—Leaves the hostname field unconfigured on the firewall. There is no identifier for the firewall that sent the logs.
3. Click **OK** to save your changes.

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STEP 5 | Create a certificate to secure syslog communication over TLSv1.2.

Required only if the syslog server uses client authentication. The syslog server uses the certificate to verify that the firewall is authorized to communicate with the syslog server.

Ensure the following conditions are met:

- The private key must be available on the sending firewall; the keys can't reside on a Hardware Security Module (HSM).
- The subject and the issuer for the certificate must not be identical.
- The syslog server and the sending firewall must have certificates that the same trusted certificate authority (CA) signed. Alternatively, you can generate a self-signed certificate on the firewall, export the certificate from the firewall, and import it in to the syslog server.
- The connection to a Syslog server over TLS is validated using the Online Certificate Status Protocol (OCSP) or using Certificate Revocation Lists (CRL) so long as each certificate in the trust chain specifies one or both of these extensions. However, you cannot bypass OCSP or CRL failures so you must ensure that the certificate chain is valid and that you can verify each certificate using OCSP or CRL.

1. Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later). Select **Generate**.
2. Enter a **Name** for the certificate.
3. In the **Common Name** field, enter the IP address of the firewall sending logs to the syslog server.
4. In **Signed by**, select the trusted CA or the self-signed CA that the syslog server and the sending firewall both trust.

   The certificate can't be a **Certificate Authority** nor an **External Authority** (certificate signing request [CSR]).
5. Click **Generate**. The firewall generates the certificate and key pair.
6. Click the certificate Name to edit it, select the **Certificate for Secure Syslog** check box, and click **OK**.

STEP 6 | Commit your changes and review the logs on the syslog server.

1. Click **Commit**.
2. To review the logs, refer to the documentation of your syslog management software. You can also review the Syslog Field Descriptions.

STEP 7 | (Optional) Configure the firewall to terminate the connection to the syslog server upon FQDN refresh.

When you configure a syslog server profile using a FQDN, the firewall maintains its connection to the syslog server by default in the event of an FQDN name change.

For example, you have replaced an existing syslog server with a new syslog server that uses a different FQDN name. If you want the firewall to connect to the new syslog server using a new FQDN name, you can configure the firewall to automatically terminate its connection to

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the old syslog server and establish a connection to the new syslog server using the new FQDN name.

1. Log in to the firewall CLI.
2. Configure the firewall to terminate the connection to the syslog server upon FQDN refresh.

```
admin> set syslogng fqdn-refresh yes
```

Configure Syslog Monitoring (SCM)

STEP 1 | Log in to Strata Cloud Manager.

STEP 2 | Select **Manage** > **Configuration** > **NGFW and Prisma Access** > **Objects** > **Log Forwarding** > **Syslog Server ProfileConfiguration** > **NGFW and Prisma Access** > **Objects** > **Log Forwarding** > **Syslog Server Profile** and select the Configuration Scope where you want to create the Syslog server profile.

You can select a folder or firewall from your **Folders** or select **Snippets** to configure the Syslog server profile in a snippet.

STEP 3 | **Add Syslog**.

STEP 4 | Configure the Syslog server profile.

1. Enter a descriptive **Name**.
2. Add a syslog server.

   Multiple syslog servers might be added to a single Syslog server profile.

   - **Name**—Unique name for the syslog server.
   - **Syslog Server**—IP address or fully qualified domain name (FQDN) of the syslog server.

     *If you configure an FQDN and use **UDP** transport, if the firewall can't resolve the FQDN, the firewall uses the existing IP address resolution for the FQDN as the **Syslog Server** address.*

   - **Transport**—Select **TCP** or **UDP** as the protocol for communicating with the syslog server.
   - **Port**—The port number on which to send syslog messages (default is UDP on port 514); you must use the same port number on the firewall and the syslog server.
   - **Format**—Select the syslog message format to use: **BSD** (default) or **IETF**. Traditionally, **BSD** format is over UDP and **IETF** format is over TCP.
   - **Facility**—Select a syslog standard value (default is **LOG_USER**) to calculate the priority (PRI) field in your syslog server implementation. Select the value that maps to how you use the PRI field to manage your syslog messages.
3. (Optional) Create a custom log/event format.

   To customize the format of the syslog messages the firewall sends, select the **Custom Log Format** tab.
4. **Save**.

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STEP 5 | Configure syslog forwarding.

1. Select **Manage** > **Configuration** > **Objects** > **Log Forwarding** > **Log Forwarding ProfileConfiguration** > **NGFW and Prisma Access** > **Objects** > **Log Forwarding** > **Log Forwarding Profile** and select the Configuration Scope where you want to create the Log Forwarding profile.

   You can select a folder or firewall from your **Folders** or select **Snippets** to configure the Log Forwarding profile in a snippet.
2. **Add Log Forwarding Profile**.
3. Enter a descriptive **Name**.
4. **Add** the profile match list for the Log Forwarding profile.

   A match list profile specifies the log query filter, forwarding destinations, and automatic actions to take. Multiple profile match lists can be added to the same Log Forwarding profile to allow you to add different profile match lists for different log types in the same Log Forwarding profile.

   1. Enter a descriptive **Name**.
   2. Select the **Log Type**.

      Only one log type can be added per profile match list.
   3. (Optional) Configure the log query **Filter**. Default is `All Logs`.
   4. Add the **Syslog Profile** you created in the previous step.
   5. **Save**.
   6. Repeat this step for all the log types that you want to forward to your syslog server.
5. **Save**.

STEP 6 | Modify the log forwarding settings for the policy rule.

- **Security Policy**—In the Actions, select **Log Settings** and select the Log Forwarding profile you created for **External Log Forwarding**.
- **Decryption**—In the Log Settings, select the Log Forwarding profile you created for **External Log Forwarding**.
- **DoS Protection**—Expand the Advanced Settings and select the Log Forwarding profile you created for **Log Forwarding**.
- **Authentication**—In the Log Settings and select the Log Forwarding profile that you created for **Log Forwarding**.

STEP 7 | **Push Config** to push your configuration changes.

Syslog Field Descriptions

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	☐ Support license ☐ (Panorama) Device management license

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Palo Alto Networks firewalls can forward various log types to an external server, with each type containing a set of standard fields. These logs are formatted as a comma-separated value (CSV) string, where a comma acts as the delimiter. This formatting facilitates parsing for syslog ingestion. The standard fields for each log type—such as Traffic, Threat, URL, WildFire, and System logs—are designed to provide detailed event information. Fields tagged with FUTURE_USE are currently reserved and are not useful for immediate syslog analysis.

Log Severity Levels

Each log is assigned a severity level to indicate the urgency and importance of the event. These levels range from Critical, for events that require immediate attention, to Informational, for general, low-priority logs. Other levels include High, Medium, and Low, each signifying a different degree of concern. These levels are crucial for prioritizing event investigation and response.

Custom Formats and Escape Sequences

For specialized log forwarding, firewalls support custom formats, allowing administrators to select and arrange specific fields. To prevent misinterpretation of commas within a field, escape sequences are used. For example, a backslash (\) can be used to escape a comma, ensuring it's treated as part of the data rather than a delimiter. These features provide flexibility and ensure data integrity during syslog ingestion and analysis.

WildFire Submissions logs are a subtype of Threat log and use the same syslog format.

- Traffic Log Fields
- Threat Log Fields
- URL Filtering Log Fields
- Data Filtering Log Fields
- HIP Match Log Fields
- GlobalProtect Log Fields
- IP-Tag Log Fields
- User-ID Log Fields
- Decryption Log Fields
- Tunnel Inspection Log Fields
- SCTP Log Fields
- Config Log Fields
- Authentication Log Fields
- System Log Fields
- Correlated Events Log Fields
- GTP Log Fields
- Audit Log Fields
- Custom Log/Event Format
- Escape Sequences

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Traffic Log Fields

Format: FUTURE_USE, Receive Time, Serial Number, Type, Threat/Content Type, FUTURE_USE, Generated Time, Source Address, Destination Address, NAT Source IP, NAT Destination IP, Rule Name, Source User, Destination User, Application, Virtual System, Source Zone, Destination Zone, Inbound Interface, Outbound Interface, Log Action, FUTURE_USE, Session ID, Repeat Count, Source Port, Destination Port, NAT Source Port, NAT Destination Port, Flags, Protocol, Action, Bytes, Bytes Sent, Bytes Received, Packets, Start Time, Elapsed Time, Category, FUTURE_USE, Sequence Number, Action Flags, Source Country, Destination Country, FUTURE_USE, Packets Sent, Packets Received, Session End Reason, Device Group Hierarchy Level 1, Device Group Hierarchy Level 2, Device Group Hierarchy Level 3, Device Group Hierarchy Level 4, Virtual System Name, Device Name, Action Source, Source VM UUID, Destination VM UUID, Tunnel ID/IMSI, Monitor Tag/IMEI, Parent Session ID, Parent Start Time, Tunnel Type, SCTP Association ID, SCTP Chunks, SCTP Chunks Sent, SCTP Chunks Received, Rule UUID, HTTP/2 Connection, App Flap Count, Policy ID, Link Switches, SD-WAN Cluster, SD-WAN Device Type, SD-WAN Cluster Type, SD-WAN Site, Dynamic User Group Name, XFF Address, Source Device Category, Source Device Profile, Source Device Model, Source Device Vendor, Source Device OS Family, Source Device OS Version, Source Hostname, Source Mac Address, Destination Device Category, Destination Device Profile, Destination Device Model, Destination Device Vendor, Destination Device OS Family, Destination Device OS Version, Destination Hostname, Destination Mac Address, Container ID, POD Namespace, POD Name, Source External Dynamic List, Destination External Dynamic List, Host ID, Serial Number, Source Dynamic Address Group, Destination Dynamic Address Group, Session Owner, High Resolution Timestamp, A Slice Service Type, A Slice Differentiator, Application Subcategory, Application Category, Application Technology, Application Risk, Application Characteristic, Application Container, Tunneled Application, Application SaaS, Application Sanctioned State, Offloaded, Flow Type, Cluster Name, AI Traffic, AI Forward Error, K8S Cluster ID, tcp_rtt_c2s, tcp_rtt_s2c, total_n_ooseq_c2s, total_n_ooseq_s2c, tcp_retransit_cnt_c2s, tcp_retransit_cnt_s2c, tcp_zero_window_cnt_c2s, tcp_zero_window_cnt_s2c, Source Adv DevID, Destination Adv DevID

The TCP Telemetry fields `tcp_rtt_c2s` through `tcp_zero_window_cnt_s2c` are intended for internal use. You can safely skip filtering or analyzing these fields for standard monitoring and reporting.

Field Name	Description
Receive Time (receive_time or cef-formatted-receive_time)	Time the log was received at the management plane.
Serial Number (serial)	Serial number of the firewall that generated the log.
Type (type)	Specifies the type of log; value is TRAFFIC.
Threat/Content Type (subtype)	Subtype of traffic log; values are start, end, drop, and deny • Start—session started • End—session ended

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Field Name	Description
	• Drop—session dropped before the application is identified and there is no rule that allows the session. • Deny—session dropped after the application is identified and there is a rule to block or no rule that allows the session.
Generated Time (time_generated or cef-formatted-time_generated)	Time the log was generated on the dataplane.
Source Address (src)	Original session source IP address.
Destination Address (dst)	Original session destination IP address.
NAT Source IP (natsrc)	If Source NAT performed, the post-NAT Source IP address.
NAT Destination IP (natdst)	If Destination NAT performed, the post-NAT Destination IP address.
Rule Name (rule)	Name of the rule that the session matched.
Source User (srcuser)	Username of the user who initiated the session.
Destination User (dstuser)	Username of the user to which the session was destined.
Application (app)	Application associated with the session.
Virtual System (vsys)	Virtual System associated with the session.
Source Zone (from)	Zone the session was sourced from.
Destination Zone (to)	Zone the session was destined to.
Inbound Interface (inbound_if)	Interface that the session was sourced from.
Outbound Interface (outbound_if)	Interface that the session was destined to.
Log Action (logset)	Log Forwarding Profile that was applied to the session.
Session ID (sessionid)	An internal numerical identifier applied to each session.
Repeat Count (repeatcnt)	Number of sessions with same Source IP, Destination IP, Application, and Subtype seen within 5 seconds.

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Field Name	Description
Source Port (sport)	Source port utilized by the session.
Destination Port (dport)	Destination port utilized by the session.
NAT Source Port (natsport)	Post-NAT source port.
NAT Destination Port (natdport)	Post-NAT destination port.
Flags (flags)	32-bit field that provides details on session; this field can be decoded by AND-ing the values with the logged value: • 0x80000000—session has a packet capture (PCAP) • 0x40000000—option is enabled to allow a client to use multiple paths to connect to a destination host • 0x20000000—indicates whether a sample has been submitted for analysis using the WildFire public or private cloud channel • 0x10000000—enterprise credential submission by end user detected • 0x08000000— source for the flow is on the allow list and not subject to recon protection • 0x02000000—IPv6 session • 0x01000000—SSL session is decrypted (SSL Proxy) • 0x00800000—session is denied via URL filtering • 0x00400000—session has a NAT translation performed • 0x00200000—user information for the session was captured through Authentication Portal • 0x00100000—application traffic is on a non-standard destination port • 0x00080000 —X-Forwarded-For value from a proxy is in the source user field • 0x00040000—log corresponds to a transaction within a http proxy session (Proxy Transaction) • 0x00020000—Client to Server flow is subject to policy based forwarding • 0x00010000—Server to Client flow is subject to policy based forwarding • 0x00008000—session is a container page access (Container Page)

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Field Name	Description
	• 0x00002000—session has a temporary match on a rule for implicit application dependency handling. Available in PAN-OS 5.0.0 and above. • 0x00000800—symmetric return is used to forward traffic for this session • 0x00000400—decrypted traffic is being sent out clear text through a mirror port • 0x00000100—payload of the outer tunnel is being inspected
IP Protocol (proto)	IP protocol associated with the session.
Action (action)	Action taken for the session; possible values are: • allow—session was allowed by policy • deny—session was denied by policy • drop—session was dropped silently • drop ICMP—session was silently dropped with an ICMP unreachable message to the host or application • reset both—session was terminated and a TCP reset is sent to both the sides of the connection • reset client—session was terminated and a TCP reset is sent to the client • reset server—session was terminated and a TCP reset is sent to the server
Bytes (bytes)	Number of total bytes (transmit and receive) for the session.
Bytes Sent (bytes_sent)	Number of bytes in the client-to-server direction of the session.
Bytes Received (bytes_received)	Number of bytes in the server-to-client direction of the session.
Packets (packets)	Number of total packets (transmit and receive) for the session.
Start Time (start)	Time of session start.
Elapsed Time (elapsed)	Elapsed time of the session.
Category (category)	URL category associated with the session (if applicable).
Sequence Number (seqno)	A 64-bit log entry identifier incremented sequentially; each log type has a unique number space.

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Field Name	Description
Action Flags (actionflags)	A bit field indicating if the log was forwarded to Panorama.
Source Country (srcloc)	Source country or Internal region for private addresses; maximum length is 32 bytes.
Destination Country (dstloc)	Destination country or Internal region for private addresses. Maximum length is 32 bytes.
Packets Sent (pkts_sent)	Number of client-to-server packets for the session.
Packets Received (pkts_received)	Number of server-to-client packets for the session.
Session End Reason (session_end_reason)	The reason a session terminated. If the termination had multiple causes, this field displays only the highest priority reason. The possible session end reason values are as follows, in order of priority (where the first is highest): • threat—The firewall detected a threat associated with a reset, drop, or block (IP address) action. • policy-deny—The session matched a security rule with a deny or drop action. • decrypt-cert-validation—The session terminated because you configured the firewall to block SSL forward proxy decryption or SSL inbound inspection when the session uses client authentication or when the session uses a server certificate with any of the following conditions: expired, untrusted issuer, unknown status, or status verification time-out. This session end reason also displays when the server certificate produces a fatal error alert of type bad_certificate, unsupported_certificate, certificate_revoked, access_denied, or no_certificate_RESERVED (SSLv3 only). • decrypt-unsupport-param—The session terminated because you configured the firewall to block SSL forward proxy decryption or SSL inbound inspection when the session uses an unsupported protocol version, cipher, or SSH algorithm. This session end reason is displays when the session produces a fatal error alert of type unsupported_extension, unexpected_message, or handshake_failure. • decrypt-error—The session terminated because you configured the firewall to block SSL forward proxy decryption or SSL inbound inspection when firewall resources or the hardware security module (HSM) were unavailable. This session end reason is also displayed when

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Field Name	Description
	you configured the firewall to block SSL traffic that has SSL errors or that produced any fatal error alert other than those listed for the decrypt-cert-validation and decrypt-unsupport-param end reasons. • tcp-rst-from-client—The client sent a TCP reset to the server. • tcp-rst-from-server—The server sent a TCP reset to the client. • resources-unavailable—The session dropped because of a system resource limitation. For example, the session could have exceeded the number of out-of-order packets allowed per flow or the global out-of-order packet queue. • tcp-fin—Both hosts in the connection sent a TCP FIN message to close the session.

	• tcp-reuse—A session is reused and the firewall closes the previous session.
	• decoder—The decoder detects a new connection within the protocol (such as HTTP-Proxy) and ends the previous connection.
	• aged-out—The session aged out.
	• unknown—This value applies in the following situations:
	  • Session terminations that the preceding reasons do not cover (for example, a `clear session all` command).
	  • For logs generated in a PAN-OS release that does not support the session end reason field (releases older than PAN-OS 6.1), the value will be **unknown** after an upgrade to the current PAN-OS release or after the logs are loaded onto the firewall.
	  • In Panorama, logs received from firewalls for which the PAN-OS version does not support session end reasons will have a value of **unknown**.
	• n/a—This value applies when the traffic log type is not **end**.

Device Group Hierarchy (dg_hier_level_1 to dg_hier_level_4)	A sequence of identification numbers that indicate the device group's location within a device group hierarchy. The firewall (or virtual system) generating the log includes the identification number of each ancestor in its device group hierarchy. The shared device group (level 0) is not included in this structure. If the log values are 12, 34, 45, 0, it means that the log was generated by a firewall (or virtual system) that belongs to device group 45, and its ancestors are 34, and 12. To view the

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Field Name	Description
	device group names that correspond to the value 12, 34 or 45, use one of the following methods: API query: `/api/?type=op&cmd=<show><dg-hierarchy></dg-hierarchy></show>`
Virtual System Name (vsys_name)	The name of the virtual system associated with the session; only valid on firewalls enabled for multiple virtual systems.
Device Name (device_name)	The hostname of the firewall on which the session was logged.
Action Source (action_source)	Specifies whether the action taken to allow or block an application was defined in the application or in policy. The actions can be allow, deny, drop, reset- server, reset-client or reset-both for the session.
Source VM UUID (src_uuid)	Identifies the source universal unique identifier for a guest virtual machine in the VMware NSX environment.
Destination VM UUID (dst_uuid)	Identifies the destination universal unique identifier for a guest virtual machine in the VMware NSX environment.
Tunnel ID/IMSI (tunnelid/ imsi)	International Mobile Subscriber Identity (IMSI) is a unique number allocated to each mobile subscriber in the GSM/ UMTS/EPS system. IMSI shall consist of decimal digits (0 through 9) only and maximum number of digits allowed are 15.
Monitor Tag/IMEI (monitortag/imei)	International Mobile Equipment Identity (IMEI) is a unique 15 or 16 digit number allocated to each mobile station equipment.
Parent Session ID (parent_session_id)	ID of the session in which this session is tunneled. Applies to inner tunnel (if two levels of tunneling) or inside content (if one level of tunneling) only.
Parent Start Time (parent_start_time)	Year/month/day hours:minutes:seconds that the parent tunnel session began.
Tunnel Type (tunnel)	Type of tunnel, such as GRE or IPSec.
SCTP Association ID (assoc_id)	Number that identifies all connections for an association between two SCTP endpoints.
SCTP Chunks (chunks)	Sum of SCTP chunks sent and received for an association.

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Field Name	Description
SCTP Chunks Sent (chunks_sent)	Number of SCTP chunks sent for an association.
SCTP Chunks Received (chunks_received)	Number of SCTP chunks received for an association.
Rule UUID (rule_uuid)	The UUID that permanently identifies the rule.
HTTP/2 Connection (http2_connection)	Identifies if traffic used an HTTP/2 Connection by displaying one of the following values: • Parent session ID—HTTP/2 connection • 0—SSL session
App Flap Count (link_change_count)	Number of link flaps that occurred during the session.
Policy ID (policy_id)	Name of the SD-WAN policy.
Link Switches (link_switches)	Contains up to four link flap entries, with each entry containing the link name, link tag, link type, physical interface, timestamp, bytes read, bytes written, link health, and link flap cause.
SD-WAN Cluster (sdwan_cluster)	Name of the SD-WAN cluster.
SD-WAN Device Type (sdwan_device_type)	Type of device (`hub` or `branch`).
SD-WAN Cluster Type (sdwan_cluster_type)	Type of cluster (`mesh` or `hub-spoke`).
SD-WAN Site (sdwan_site)	Name of the SD-WAN site.
Dynamic User Group Name (dynusergroup_name)	Name of the dynamic user group that contains the user who initiated the session.
XFF Address (xff_ip)	The IP address of the user who requested the web page or the IP address of the next to last device that the request traversed. If the request goes through one or more proxies, load balancers, or other upstream devices, the firewall displays the IP address of the most recent device. *Based on different appliance implementations, the XFF field may contain non-IP address values.*

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Field Name	Description
Source Device Category (src_category)	The category for the device that Device-ID identifies as the source of the traffic.
Source Device Profile (src_profile)	The device profile for the device that Device-ID identifies as the source of the traffic.
Source Device Model (src_model)	The model of the device that Device-ID identifies as the source of the traffic.
Source Device Vendor (src_vendor)	The vendor of the device that Device-ID identifies as the source of the traffic.
Source Device OS Family (src_osfamily)	The operating system type for the device that Device-ID identifies as the source of the traffic.
Source Device OS Version (src_osversion)	The version of the operating system for the device that Device-ID identifies as the source of the traffic.
Source Hostname (src_host)	The hostname of the device that Device-ID identifies as the source of the traffic.
Source MAC Address (src_mac)	The MAC address for the device that Device-ID identifies as the source of the traffic.
Destination Device Category (dst_category)	The category for the device that Device-ID identifies as the destination for the traffic.
Destination Device Profile (dst_profile)	The device profile for the device that Device-ID identifies as the destination for the traffic.
Destination Device Model (dst_model)	The model of the device that Device-ID identifies as the destination for the traffic.
Destination Device Vendor (dst_vendor)	The vendor of the device that Device-ID identifies as the destination for the traffic.
Destination Device OS Family (dst_osfamily)	The operating system type for the device that Device-ID identifies as the destination for the traffic.
Destination Device OS Version (dst_osversion)	The version of the operating system for the device that Device-ID identifies as the destination for the traffic.
Destination Hostname (dst_host)	The hostname of the device that Device-ID identifies as the destination for the traffic.

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Field Name	Description
Destination MAC Address (dst_mac)	The MAC address for the device that Device-ID identifies as the destination for the traffic.
Container ID (container_id)	The container ID of the PAN-NGFW pod on the Kubernetes node where the application POD is deployed.
POD Namespace (pod_namespace)	The namespace of the application POD being secured.
POD Name (pod_name)	The application POD being secured.
Source External Dynamic List (src_edl)	The name of the external dynamic list that contains the source IP address of the traffic.
Destination External Dynamic List (dst_edl)	The name of the external dynamic list that contains the destination IP address of the traffic.
Host ID (hostid)	Unique ID GlobalProtect assigns to identify the host.
User Device Serial Number (serialnumber)	Serial number of the user's machine or device.
Source Dynamic Address Group (src_dag)	Original session source dynamic address group.
Destination Dynamic Address Group (dst_dag)	Original destination source dynamic address group.
Session Owner (session_owner)	The original high availability (HA) peer session owner in an HA cluster from which the session table data was synchronized upon HA failover.
High Resolution Timestamp (high_res_timestamp)	Time in milliseconds the log was received at the management plane. The format for this new field is YYYY-MM-DDThh:ss:sssTZD: • **YYYY**—Four digit year • **MM**—Two-digit month • **DD**—Two-digit day of the month (01 through 31) • **T**—Indicator for the beginning of the timestamp • **hh**—Two-digit hour using 24-hour time (00 through 23) • **mm**—Two-digit minute (00 through 59) • **ss**—Two-digit second (00 through 60) • **sss**—One or more digits for millisecond

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Field Name	Description
	• **TZD**—Time zone designator (+hh:mm or -hh:mm) *The High Resolution Timestamp is supported for logs received from managed firewalls running PAN-OS 10.0 and later releases. Logs received from managed firewalls running PAN-OS 9.1 and earlier releases display a `1969-12-31T16:00:00:000-8:00` timestamp regardless of when the log was received.*
A Slice Service Type (nssai_sst)	The A Slice Service Type of the Network Slice ID.
A Slice Differentiator (nssai_sd)	The A Slice Differentiator of the Network Slice ID.
Application Subcategory (subcategory_of_app)	The application subcategory specified in the application configuration properties.
Application Category (category_of_app)	The application category specified in the application configuration properties. Values are: • business-systems • collaboration • general-internet • media • networking • saas
Application Technology (technology_of_app)	The application technology specified in the application configuration properties. Values are: • browser-based • client-server • network-protocol • peer-to-peer
Application Risk (risk_of_app)	Risk level associated with the application (1=lowest to 5=highest).
Application Characteristic (characteristic_of_app)	Comma-separated list of applicable characteristic of the application

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Field Name	Description
Application Container (container_of_app)	The parent application for an application.
Tunneled Application (tunneled_app)	Name of the tunneled application.
Application SaaS (is_saas_of_app)	Displays `1` if a SaaS application or `0` if not a SaaS application.
Application Sanctioned State (sanctioned_state_of_app)	Displays `1` if application is sanctioned or `0` if application is not sanctioned.
Offloaded (offloaded)	Displays `1` if traffic flow has been offloaded or `0` if traffic flow was not offloaded.
Flow Type (flow_type) (11.1 and later releases)	Identifies the type of proxy used for traffic. If a proxy is used, `Explicit Proxy` or `Transparent Proxy` is displayed. If no proxy is used, `NonProxyTraffic` is displayed.
Cluster Name (cluster_name) (11.1 and later releases)	Name of the CN-Series firewall cluster.
AI Traffic (ai_traffic) (11.1 and later releases)	Indicates whether the network session is being processed by AI-driven security services.
AI Forward Error (ai_fwd_error) (11.1 and later releases)	Indicates whether an error was encountered during traffic forwarding by the AI engine.
K8S Cluster ID (k8s_cluster_id) (11.1 and later releases)	The unique identifier of the Kubernetes cluster that is the source of the traffic. This ID establishes a link to the related Virtual Machine (VM) UUID.
Source Adv DevID (src_adv_dev_id) (12.1.2 and later releases)	The unique identifier for the endpoint initiating the session. This advanced ID provides highly precise device recognition for the security policy engine. It establishes a direct link between network traffic and the comprehensive Device-ID object used for enforcement.
Destination Adv DevID (dst_adv_dev_id) (12.1.2 and later releases)	The unique identifier for the endpoint receiving the session. This advanced ID provides highly precise device recognition for the security policy engine. It establishes a direct link between network traffic and the comprehensive Device-ID object used for enforcement.

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Threat Log Fields

Format: FUTURE_USE, Receive Time, Serial Number, Type, Threat/Content Type, FUTURE_USE, Generated Time, Source Address, Destination Address, NAT Source IP, NAT Destination IP, Rule Name, Source User, Destination User, Application, Virtual System, Source Zone, Destination Zone, Inbound Interface, Outbound Interface, Log Action, FUTURE_USE, Session ID, Repeat Count, Source Port, Destination Port, NAT Source Port, NAT Destination Port, Flags, IP Protocol, Action, URL/Filename, Threat ID, Category, Severity, Direction, Sequence Number, Action Flags, Source Location, Destination Location, FUTURE_USE, Content Type, PCAP_ID, File Digest, Cloud, URL Index, User Agent, File Type, X-Forwarded-For, Referer, Sender, Subject, Recipient, Report ID, Device Group Hierarchy Level 1, Device Group Hierarchy Level 2, Device Group Hierarchy Level 3, Device Group Hierarchy Level 4, Virtual System Name, Device Name, FUTURE_USE, Source VM UUID, Destination VM UUID, HTTP Method, Tunnel ID/IMSI, Monitor Tag/IMEI, Parent Session ID, Parent Start Time, Tunnel Type, Threat Category, Content Version, FUTURE_USE, SCTP Association ID, Payload Protocol ID, HTTP Headers, URL Category List, Rule UUID, HTTP/2 Connection, Dynamic User Group Name, XFF Address, Source Device Category, Source Device Profile, Source Device Model, Source Device Vendor, Source Device OS Family, Source Device OS Version, Source Hostname, Source MAC Address, Destination Device Category, Destination Device Profile, Destination Device Model, Destination Device Vendor, Destination Device OS Family, Destination Device OS Version, Destination Hostname, Destination MAC Address, Container ID, POD Namespace, POD Name, Source External Dynamic List, Destination External Dynamic List, Host ID, Serial Number, Domain EDL, Source Dynamic Address Group, Destination Dynamic Address Group, Partial Hash, High Resolution Timestamp, Reason, Justification, A Slice Service Type, Application Subcategory, Application Category, Application Technology, Application Risk, Application Characteristic, Application Container, Tunneled Application, Application SaaS, Application Sanctioned State, Cloud Report ID, Flow Type, Cluster Name

Field Name	Description
Receive Time (receive_time or cef-formatted-receive_time)	Time the log was received at the management plane.
Serial Number (serial #)	Serial number of the firewall that generated the log.
Type (type)	Specifies the type of log; value is THREAT.
Threat/Content Type (subtype)	Subtype of threat log. Values include the following: • data—Data pattern matching a Data Filtering profile. • file—File type matching a File Blocking profile. • flood—Flood detected via a Zone Protection profile. • packet—Packet-based attack protection triggered by a Zone Protection profile. • scan—Scan detected via a Zone Protection profile. • spyware —Spyware detected via an Anti-Spyware profile. • url—URL filtering log.

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Field Name	Description
	• ml-virus—Virus detected by WildFire Inline ML via an Antivirus profile. • virus—Virus detected via an Antivirus profile. • vulnerability —Vulnerability exploit detected via a Vulnerability Protection profile. • wildfire —A WildFire verdict generated when the firewall submits a file to WildFire per a WildFire Analysis profile and a verdict (malware, phishing, grayware, or benign, depending on what you are logging) is logged in the WildFire Submissions log. • wildfire-virus—Virus detected via an Antivirus profile.
Generate Time (time_generated or cef-formatted-time_generated)	Time the log was generated on the dataplane.
Source address (src)	Original session source IP address.
Destination address (dst)	Original session destination IP address.
NAT Source IP (natsrc)	If source NAT performed, the post-NAT source IP address.
NAT Destination IP (natdst)	If destination NAT performed, the post-NAT destination IP address.
Rule Name (rule)	Name of the rule that the session matched.
Source User (srcuser)	Username of the user who initiated the session.
Destination User (dstuser)	Username of the user to which the session was destined.
Application (app)	Application associated with the session.
Virtual System (vsys)	Virtual System associated with the session.
Source Zone (from)	Zone the session was sourced from.
Destination Zone (to)	Zone the session was destined to.
Inbound Interface (inbound_if)	Interface that the session was sourced from.
Outbound Interface (outbound_if)	Interface that the session was destined to.

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Field Name	Description
Log Action (logset)	Log Forwarding Profile that was applied to the session.
Session ID (sessionid)	An internal numerical identifier applied to each session.
Repeat Count (repeatcnt)	Number of sessions with same Source IP, Destination IP, Application, and Content/Threat Type seen within 5 seconds.
Source Port (sport)	Source port utilized by the session.
Destination Port (dport)	Destination port utilized by the session.
NAT Source Port (natsport)	Post-NAT source port.
NAT Destination Port (natdport)	Post-NAT destination port.
Flags (flags)	32-bit field that provides details on session; this field can be decoded by AND-ing the values with the logged value: • 0x80000000—session has a packet capture (PCAP) • 0x40000000—option is enabled to allow a client to use multiple paths to connect to a destination host • 0x20000000—file is submitted to WildFire for a verdict • 0x10000000—enterprise credential submission by end user detected • 0x08000000— source for the flow is on an allow list and not subject to recon protection • 0x02000000—IPv6 session • 0x01000000—SSL session is decrypted (SSL Proxy) • 0x00800000—session is denied via URL filtering • 0x00400000—session has a NAT translation performed • 0x00200000—user information for the session was captured through Authentication Portal • 0x00100000—application traffic is on a non-standard destination port • 0x00080000 —X-Forwarded-For value from a proxy is in the source user field • 0x00040000 —log corresponds to a transaction within a http proxy session (Proxy Transaction) • 0x00020000—Client to Server flow is subject to policy based forwarding

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Field Name	Description
	• 0x00010000—Server to Client flow is subject to policy based forwarding • 0x00008000 —session is a container page access (Container Page) • 0x00002000 —session has a temporary match on a rule for implicit application dependency handling. Available in PAN-OS 5.0.0 and above. • 0x00000800 —symmetric return is used to forward traffic for this session • 0x00000400—decrypted traffic is being sent out clear text through a mirror port • 0x00000010—payload of the outer tunnel is being inspected
IP Protocol (proto)	IP protocol associated with the session.
Action (action)	Action taken for the session; values are alert, allow, deny, drop, drop-all-packets, reset-client, reset-server, reset-both, block-url. • alert—threat or URL detected but not blocked • allow— flood detection alert • deny—flood detection mechanism activated and deny traffic based on configuration • drop— threat detected and session remains, but drops all packets • reset-client —threat detected and a TCP RST is sent to the client • reset-server —threat detected and a TCP RST is sent to the server • reset-both —threat detected and a TCP RST is sent to both the client and the server • block-url —URL request was blocked because it matched a URL category that was set to be blocked • block-ip—threat detected and client IP is blocked • random-drop—flood detected and packet was randomly dropped • sinkhole—DNS sinkhole activated • syncookie-sent—syncookie alert • block-continue (URL subtype only)—a HTTP request is blocked and redirected to a Continue page with a button for confirmation to proceed • continue (URL subtype only)—response to a block-continue URL continue page indicating a block-continue request was allowed to proceed

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Field Name	Description
	• block-override (URL subtype only)—a HTTP request is blocked and redirected to an Admin override page that requires a pass code from the firewall administrator to continue • override-lockout (URL subtype only)—too many failed admin override pass code attempts from the source IP. IP is now blocked from the block-override redirect page • override (URL subtype only)—response to a block-override page where a correct pass code is provided and the request is allowed • block (Wildfire only)—file was blocked by the firewall and uploaded to Wildfire
URL/Filename (misc)	Field with variable length. A Filename has a maximum of 63 characters. A URL has a maximum of 1023 characters The actual URI when the subtype is url File name or file type when the subtype is file File name when the subtype is virus File name when the subtype is wildfire-virus File name when the subtype is wildfire URL or File name when the subtype is vulnerability if applicable URL when Threat Category is `domain-edl` Spoofed SNI domain when a host header mismatch is detected (identified by a unique threat ID of 86467).
Threat/Content Name (threatid)	Palo Alto Networks identifier for known and custom threats. It is a description string followed by a 64-bit numerical identifier in parentheses for some Subtypes: • 8000 – 8099— scan detection • 8500 – 8599— flood detection • 9999— URL filtering log • 10000 – 19999 —spyware phone home detection • 20000 – 29999 —spyware download detection • 30000 – 44999 —vulnerability exploit detection • 52000 – 52999— filetype detection • 60000 – 69999 —data filtering detection If the Domain EDL field is populated, then this field is populated with the same value.

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Field Name	Description
	Threat ID ranges for virus detection, WildFire signature feed, and DNS C2 signatures used in previous releases have been replaced with permanent, globally unique IDs. Refer to the Threat/Content Type (subtype) and Threat Category (thr_category) field names to create updated reports, filter threat logs, and ACC activity.
Category (category)	For URL Subtype, it is the URL Category; For WildFire subtype, it is the verdict on the file and is either 'malware', 'phishing', 'grayware', or 'benign'; For other subtypes, the value is 'any'.
Severity (severity)	Severity associated with the threat; values are informational, low, medium, high, critical.
Direction (direction)	Indicates the direction of the attack, client-to-server or server-to-client: • 0—direction of the threat is client to server • 1—direction of the threat is server to client
Sequence Number (seqno)	A 64-bit log entry identifier incremented sequentially. Each log type has a unique number space.
Action Flags (actionflags)	A bit field indicating if the log was forwarded to Panorama.
Source Country (srcloc)	Source country or Internal region for private addresses. Maximum length is 32 bytes.
Destination Country (dstloc)	Destination country or Internal region for private addresses. Maximum length is 32 bytes.
Content Type (contenttype)	Applicable only when Subtype is URL. Content type of the HTTP response data. Maximum length 32 bytes.
PCAP ID (pcap_id)	The packet capture (pcap) ID is a 64 bit unsigned integral denoting an ID to correlate threat pcap files with extended pcaps taken as a part of that flow. All threat logs will contain either a pcap_id of 0 (no associated pcap), or an ID referencing the extended pcap file.
File Digest (filedigest)	Only for WildFire subtype; all other types do not use this field The filedigest string shows the binary hash of the file sent to be analyzed by the WildFire service.

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Field Name	Description
Cloud (cloud)	Only for WildFire subtype; all other types do not use this field. The cloud string displays the FQDN of either the WildFire appliance (private) or the WildFire cloud (public) from where the file was uploaded for analysis.
URL Index (url_idx)	Used in URL Filtering and WildFire subtypes. When an application uses TCP keepalives to keep a connection open for a length of time, all the log entries for that session have a single session ID. In such cases, when you have a single threat log (and session ID) that includes multiple URL entries, the url_idx is a counter that allows you to correlate the order of each log entry within the single session. For example, to learn the URL of a file that the firewall forwarded to WildFire for analysis, locate the session ID and the url_idx from the WildFire Submissions log and search for the same session ID and url_idx in your URL filtering logs. The log entry that matches the session ID and url_idx will contain the URL of the file that was forwarded to WildFire.
User Agent (user_agent)	Only for the URL Filtering subtype; all other types do not use this field. The User Agent field specifies the web browser that the user used to access the URL, for example Internet Explorer. This information is sent in the HTTP request to the server.
File Type (filetype)	Only for WildFire subtype; all other types do not use this field. Specifies the type of file that the firewall forwarded for WildFire analysis.
X-Forwarded-For (xff)	Only for the URL Filtering subtype; all other types do not use this field. The X-Forwarded-For field in the HTTP header contains the IP address of the user who requested the web page. It allows you to identify the IP address of the user, which is useful particularly if you have a proxy server on your network that replaces the user IP address with its own address in the source IP address field of the packet header. *Based on different appliance implementations, the XFF field may contain non-IP address values.*
Referer (referer)	Only for the URL Filtering subtype; all other types do not use this field.

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Field Name	Description
	The Referer field in the HTTP header contains the URL of the web page that linked the user to another web page; it is the source that redirected (referred) the user to the web page that is being requested.
Sender (sender)	Specifies the name of the sender of an email.
Subject (subject)	Specifies the subject of an email.
Recipient (recipient)	Specifies the name of the receiver of an email.
Report ID (reportid)	Only for Data Filtering and WildFire subtype; all other types do not use this field. Identifies the analysis request on the firewall, WildFire cloud, or the WildFire appliance.
Device Group Hierarchy (dg_hier_level_1 to dg_hier_level_4)	A sequence of identification numbers that indicate the device group's location within a device group hierarchy. The firewall (or virtual system) generating the log includes the identification number of each ancestor in its device group hierarchy. The shared device group (level 0) is not included in this structure. If the log values are 12, 34, 45, 0, it means that the log was generated by a firewall (or virtual system) that belongs to device group 45, and its ancestors are 34, and 12. To view the device group names that correspond to the value 12, 34 or 45, use one of the following methods: API query: `/api/?type=op&cmd=<show><dg-hierarchy></dg-hierarchy></show>`
Virtual System Name (vsys_name)	The name of the virtual system associated with the session; only valid on firewalls enabled for multiple virtual systems.
Device Name (device_name)	The hostname of the firewall on which the session was logged.
Source VM UUID (src_uuid)	Identifies the source universal unique identifier for a guest virtual machine in the VMware NSX environment.
Destination VM UUID (dst_uuid)	Identifies the destination universal unique identifier for a guest virtual machine in the VMware NSX environment.

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Field Name	Description
HTTP Method (http_method)	Only in URL filtering logs. Describes the HTTP Method used in the web request. Only the following methods are logged: Connect, Delete, Get, Head, Options, Post, Put.
Tunnel ID/IMSI (tunnel_id/imsi)	International Mobile Subscriber Identity (IMSI) is a unique number allocated to each mobile subscriber in the GSM/UMTS/EPS system. IMSI shall consist of decimal digits (0 through 9) only and maximum number of digits allowed are 15.
Monitor Tag/IMEI (monitortag/imei)	International Mobile Equipment Identity (IMEI) is a unique 15 or 16 digit number allocated to each mobile station equipment.
Parent Session ID (parent_session_id)	ID of the session in which this session is tunneled. Applies to inner tunnel (if two levels of tunneling) or inside content (if one level of tunneling) only.
Parent Session Start Time (parent_start_time)	Year/month/day hours:minutes:seconds that the parent tunnel session began.
Tunnel Type (tunnel)	Type of tunnel, such as GRE or IPSec.
Threat Category (thr_category)	Describes threat categories used to classify different types of threat signatures. If a domain external dynamic list generated the log, `domain-edl` populates this field.
Content Version (contentver)	Applications and Threats version on your firewall when the log was generated.
SCTP Association ID (assoc_id)	Number that identifies all connections for an association between two SCTP endpoints.
Payload Protocol ID (ppid)	ID of the protocol for the payload in the data portion of the data chunk.
HTTP Headers (http_headers)	Indicates the inserted HTTP header in the URL log entries on the firewall.
URL Category List (url_category_list)	Lists the URL filtering categories that the firewall used to enforce policy.
Rule UUID (rule_uuid)	The UUID that permanently identifies the rule.
HTTP/2 Connection (http2_connection)	Identifies if traffic used an HTTP/2 connection by displaying one of the following values: • TCP connection session ID—session is HTTP/2

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Field Name	Description
	• 0—session is not HTTP/2
Dynamic User Group Name (dynusergroup_name)	The name of the dynamic user group that contains the user who initiated the session.
XFF Address (xff_ip)	The IP address of the user who requested the web page or the IP address of the next to last device that the request traversed. If the request goes through one or more proxies, load balancers, or other upstream devices, the firewall displays the IP address of the most recent device. *Based on different appliance implementations, the XFF field may contain non-IP address values.*
Source Device Category (src_category)	The category for the device that Device-ID identifies as the source of the traffic.
Source Device Profile (src_profile)	The device profile for the device that Device-ID identifies as the source of the traffic.
Source Device Model (src_model)	The model of the device that Device-ID identifies as the source of the traffic.
Source Device Vendor (src_vendor)	The vendor of the device that Device-ID identifies as the source of the traffic.
Source Device OS Family (src_osfamily)	The operating system type for the device that Device-ID identifies as the source of the traffic.
Source Device OS Version (src_osversion)	The version of the operating system for the device that Device-ID identifies as the source of the traffic.
Source Hostname (src_host)	The hostname of the device that Device-ID identifies as the source of the traffic.
Source MAC Address (src_mac)	The MAC address for the device that Device-ID identifies as the source of the traffic.
Destination Device Category (dst_category)	The category for the device that Device-ID identifies as the destination for the traffic.
Destination Device Profile (dst_profile)	The device profile for the device that Device-ID identifies as the destination for the traffic.

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Field Name	Description
Destination Device Model (dst_model)	The model of the device that Device-ID identifies as the destination for the traffic.
Destination Device Vendor (dst_vendor)	The vendor of the device that Device-ID identifies as the destination for the traffic.
Destination Device OS Family (dst_osfamily)	The operating system type for the device that Device-ID identifies as the destination for the traffic.
Destination Device OS Version (dst_osversion)	The version of the operating system for the device that Device-ID identifies as the destination for the traffic.
Destination Hostname (dst_host)	The hostname of the device that Device-ID identifies as the destination for the traffic.
Destination MAC Address (dst_mac)	The MAC address for the device that Device-ID identifies as the destination for the traffic.
Container ID (container_id)	The container ID of the PAN-NGFW pod on the Kubernetes node where the application POD is deployed.
POD Namespace (pod_namespace)	The namespace of the application POD being secured.
POD Name (pod_name)	The application POD being secured.
Source External Dynamic List (src_edl)	The name of the external dynamic list that contains the source IP address of the traffic.
Destination External Dynamic List (dst_edl)	The name of the external dynamic list that contains the destination IP address of the traffic.
Host ID (hostid)	Unique ID GlobalProtect assigns to identify the host.
User Device Serial Number (serialnumber)	Serial number of the user's machine or device.
Domain EDL (domain_edl)	The name of the external dynamic list that contains the domain name of the traffic.
Source Dynamic Address Group (src_dag)	Original session source dynamic address group.
Destination Dynamic Address Group (dst_dag)	Original destination source dynamic address group.

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Field Name	Description
Partial Hash (partial_hash)	Machine Learning partial hash.
High Resolution Timestamp (high_res timestamp)	Time in milliseconds the log was received at the management plane. The format for this new field is YYYY-MM-DDThh:ss:sssTZD: • **YYYY**—Four digit year • **MM**—Two-digit month • **DD**—Two-digit day of the month (01 through 31) • **T**—Indicator for the beginning of the timestamp • **hh**—Two-digit hour using 24-hour time (00 through 23) • **mm**—Two-digit minute (00 through 59) • **ss**—Two-digit second (00 through 60) • **sss**—One or more digits for millisecond • **TZD**—Time zone designator (+hh:mm or -hh:mm) *The High Resolution Timestamp is supported for logs received from managed firewalls running PAN-OS 11.1 and later releases. Logs received from managed firewalls running PAN-OS 9.1 and earlier releases display a `1969-12-31T16:00:00:000-8:00` timestamp regardless of when the log was received.*
Reason (reason)	Reason for Data Filtering action.
Justification (justification)	Justification for Data Filtering action.
A Slice Service Type (nssai_sst)	The A Slice Service Type of the Network Slice ID.
Application Subcategory (subcategory_of_app)	The application subcategory specified in the application configuration properties.
Application Category (category_of_app)	The application category specified in the application configuration properties. Values are: • business-systems • collaboration • general-internet • media • networking

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Field Name	Description
	• saas
Application Technology (technology_of_app)	The application technology specified in the application configuration properties. Values are: • browser-based • client-server • network-protocol • peer-to-peer
Application Risk (risk_of_app)	Risk level associated with the application (1=lowest to 5=highest).
Application Characteristic (characteristic_of_app)	Comma-separated list of applicable characteristic of the application
Application Container (container_of_app)	The parent application for an application.
Tunneled Application (tunneled_app)	Name of the tunneled application.
Application SaaS (is_saas_of_app)	Displays `1` if a SaaS application or `0` if not a SaaS application.
Application Sanctioned State (sanctioned_state_of_app)	Displays `1` if application is sanctioned or `0` if application is not sanctioned.
Cloud Report ID (cloud_reportid)	(PAN-OS 10.2.0) Unique 32 character ID for a file scanned by the DLP cloud service sent by a firewall. (PAN-OS 10.2.1 and later releases) Unique 67 character ID for a file scanned by the DLP cloud service sent by a firewall. The same Cloud Report ID is displayed for a file the DLP cloud service has already scanned and generated a Cloud Report ID for.
Cluster Name (cluster_name)	(11.1 and later releases) Name of the CN-Series firewall cluster.
Flow Type (flow_type)	(11.1 and later releases) Identifies the type of proxy used for traffic. If a proxy is used, `Explicit Proxy` or `Transparent Proxy` is displayed. If no proxy is used, `NonProxyTraffic` is displayed.

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URL Filtering Log Fields

Format: FUTURE_USE, Receive Time, Serial Number, Type, Threat/Content Type, FUTURE_USE, Generated Time, Source Address, Destination Address, NAT Source IP, NAT Destination IP, Rule Name, Source User, Destination User, Application, Virtual System, Source Zone, Destination Zone, Inbound Interface, Outbound Interface, Log Action, FUTURE_USE, Session ID, Repeat Count, Source Port, Destination Port, NAT Source Port, NAT Destination Port, Flags, IP Protocol, Action, URL/Filename, Threat ID, Category, Severity, Direction, Sequence Number, Action Flags, Source Country, Destination Country, FUTURE_USE, Content Type, PCAP_ID, File Digest, Cloud, URL Index, User Agent, File Type, X-Forwarded-For, Referer, Sender, Subject, Recipient, Report ID, Device Group Hierarchy Level 1, Device Group Hierarchy Level 2, Device Group Hierarchy Level 3, Device Group Hierarchy Level 4, Virtual System Name, Device Name, FUTURE_USE, Source VM UUID, Destination VM UUID, HTTP Method, Tunnel ID/IMSI, Monitor Tag/IMEI, Parent Session ID, Parent Start Time, Tunnel Type, Threat Category, Content Version, FUTURE_USE, SCTP Association ID, Payload Protocol ID, Request Headers, Response Headers, HTTP Status Code, URL Category List, Rule UUID, HTTP/2 Connection, Dynamic User Group Name, XFF Address, Source Device Category, Source Device Profile, Source Device Model, Source Device Vendor, Source Device OS Family, Source Device OS Version, Source Hostname, Source MAC Address, Destination Device Category, Destination Device Profile, Destination Device Model, Destination Device Vendor, Destination Device OS Family, Destination Device OS Version, Destination Hostname, Destination MAC Address, Container ID, POD Namespace, POD Name, Source External Dynamic List, Destination External Dynamic List, Host ID, Serial Number, Domain EDL, Source Dynamic Address Group, Destination Dynamic Address Group, Partial Hash, High Resolution Timestamp, Reason, Justification, A Slice Service Type, Application Subcategory, Application Category, Application Technology, Application Risk, Application Characteristic, Application Container, Tunneled Application, Application SaaS, Application Sanctioned State, Cloud Report ID, Cluster Name, Flow Type

Field Name	Description
Receive Time (receive_time or cef-formatted-receive_time)	Time the log was received at the management plane.
Serial Number (serial #)	Serial number of the firewall that generated the log.
Type (type)	Specifies the type of log; value is THREAT.
Threat/Content Type (subtype)	Subtype of threat log; value is url.
Generate Time (time_generated or cef-formatted-time_generated)	Time the log was generated on the dataplane.
Source address (src)	Original session source IP address.
Destination address (dst)	Original session destination IP address.

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Field Name	Description
NAT Source IP (natsrc)	If source NAT performed, the post-NAT source IP address.
NAT Destination IP (natdst)	If destination NAT performed, the post-NAT destination IP address.
Rule Name (rule)	Name of the rule that the session matched.
Source User (srcuser)	Username of the user who initiated the session.
Destination User (dstuser)	Username of the user to which the session was destined.
Application (app)	Application associated with the session.
Virtual System (vsys)	Virtual System associated with the session.
Source Zone (from)	Zone the session was sourced from.
Destination Zone (to)	Zone the session was destined to.
Inbound Interface (inbound_if)	Interface that the session was sourced from.
Outbound Interface (outbound_if)	Interface that the session was destined to.
Log Action (logset)	Log Forwarding Profile that was applied to the session.
Session ID (sessionid)	An internal numerical identifier applied to each session.
Repeat Count (repeatcnt)	Number of sessions with same Source IP, Destination IP, Application, and Content/Threat Type seen within 5 seconds.
Source Port (sport)	Source port utilized by the session.
Destination Port (dport)	Destination port utilized by the session.
NAT Source Port (natsport)	Post-NAT source port.
NAT Destination Port (natdport)	Post-NAT destination port.
Flags (flags)	32-bit field that provides details on session; this field can be decoded by AND-ing the values with the logged value: • 0x80000000—session has a packet capture (PCAP)

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Field Name	Description
	• 0x40000000—option is enabled to allow a client to use multiple paths to connect to a destination host • 0x20000000—file is submitted to WildFire for a verdict • 0x10000000—enterprise credential submission by end user detected • 0x08000000— source for the flow is on an allow list and not subject to recon protection • 0x02000000—IPv6 session • 0x01000000—SSL session is decrypted (SSL Proxy) • 0x00800000—session is denied via URL filtering • 0x00400000—session has a NAT translation performed • 0x00200000—user information for the session was captured through Authentication Portal • 0x00100000—application traffic is on a non-standard destination port • 0x00080000 —X-Forwarded-For value from a proxy is in the source user field • 0x00040000 —log corresponds to a transaction within a http proxy session (Proxy Transaction) • 0x00020000—Client to Server flow is subject to policy based forwarding • 0x00010000—Server to Client flow is subject to policy based forwarding • 0x00008000 —session is a container page access (Container Page) • 0x00002000 —session has a temporary match on a rule for implicit application dependency handling. Available in PAN-OS 5.0.0 and above. • 0x00000800 —symmetric return is used to forward traffic for this session • 0x00000400—decrypted traffic is being sent out clear text through a mirror port • 0x00000010—payload of the outer tunnel is being inspected
IP Protocol (proto)	IP protocol associated with the session.
Action (action)	Action taken for the session; values are alert, allow, block-url, block-continue, continue, block-override, override-lockout, override. • alert—threat or URL detected but not blocked

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Field Name	Description
	• block-url —URL request was blocked because it matched a URL category that was set to be blocked • block-continue—a HTTP request is blocked and redirected to a Continue page with a button for confirmation to proceed • continue —response to a block-continue URL continue page indicating a block-continue request was allowed to proceed • block-override —a HTTP request is blocked and redirected to an Admin override page that requires a pass code from the firewall administrator to continue • override-lockout—too many failed admin override pass code attempts from the source IP. IP is now blocked from the block-override redirect page • override —response to a block-override page where a correct pass code is provided and the request is allowed
URL/Filename (misc)	Field with variable length. A URL has a maximum of 1023 characters. The actual URI when the subtype is url. URL when Threat Category is `domain-edl`.
Threat/Content Name (threatid)	Palo Alto Networks identifier for known and custom threats. It is a description string followed by a 64-bit numerical identifier in parentheses for some Subtypes: • 8000 – 8099— scan detection • 8500 – 8599— flood detection • 9999— URL filtering log • 10000 – 19999 —spyware phone home detection • 20000 – 29999 —spyware download detection • 30000 – 44999 —vulnerability exploit detection • 52000 – 52999— filetype detection • 60000 – 69999 —data filtering detection If the domain EDL field is populated, then this field is populated with the same value. *Threat ID ranges for virus detection, WildFire signature feed, and DNS C2 signatures used in previous releases have been replaced with permanent, globally unique IDs. Refer to the Threat/Content Type (subtype) and Threat Category (thr_category) field names to create updated reports, filter threat logs, and ACC activity.*

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Field Name	Description
Category (category)	For URL Subtype, it is the URL Category; For WildFire subtype, it is the verdict on the file and is either 'malware', 'phishing', 'grayware', or 'benign'; For other subtypes, the value is 'any'.
Severity (severity)	Severity associated with the threat; values are informational, low, medium, high, critical.
Direction (direction)	Indicates the direction of the attack: • client-to-server • server-to-client
Sequence Number (seqno)	A 64-bit log entry identifier incremented sequentially. Each log type has a unique number space.
Action Flags (actionflags)	A bit field indicating if the log was forwarded to Panorama.
Source Country (srcloc)	Source country or Internal region for private addresses. Maximum length is 32 bytes.
Destination Country (dstloc)	Destination country or Internal region for private addresses. Maximum length is 32 bytes.
Content Type (contenttype)	Content type of the HTTP response data. Maximum length 32 bytes.
PCAP ID (pcap_id)	The packet capture (pcap) ID is a 64 bit unsigned integral denoting an ID to correlate threat pcap files with extended pcaps taken as a part of that flow. All threat logs will contain either a pcap_id of 0 (no associated pcap), or an ID referencing the extended pcap file.
File Digest (filedigest)	Only for WildFire subtype; all other types do not use this field The filedigest string shows the binary hash of the file sent to be analyzed by the WildFire service.
Cloud (cloud)	Only for WildFire subtype; all other types do not use this field. The cloud string displays the FQDN of either the WildFire appliance (private) or the WildFire cloud (public) from where the file was uploaded for analysis.
URL Index (url_idx)	When an application uses TCP keepalives to keep a connection open for a length of time, all the log entries for that session have a single session ID. In such cases, when you have a single threat log (and session ID) that includes multiple URL entries, the url_idx is

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Field Name	Description
	a counter that allows you to correlate the order of each log entry within the single session. For example, to learn the URL of a file that the firewall forwarded to WildFire for analysis, locate the session ID and the url_idx from the WildFire Submissions log and search for the same session ID and url_idx in your URL filtering logs. The log entry that matches the session ID and url_idx will contain the URL of the file that was forwarded to WildFire.
User Agent (user_agent)	The User Agent field specifies the web browser that the user used to access the URL, for example Internet Explorer. This information is sent in the HTTP request to the server.
File Type (filetype)	Only for WildFire subtype; all other types do not use this field. Specifies the type of file that the firewall forwarded for WildFire analysis.
X-Forwarded-For (xff)	The X-Forwarded-For field in the HTTP header contains the IP address of the user who requested the web page. It allows you to identify the IP address of the user, which is useful particularly if you have a proxy server on your network that replaces the user IP address with its own address in the source IP address field of the packet header. *Based on different appliance implementations, the XFF field may contain non-IP address values.*
Referer (referer)	The Referer field in the HTTP header contains the URL of the web page that linked the user to another web page; it is the source that redirected (referred) the user to the web page that is being requested.
Sender (sender)	Specifies the name of the sender of an email.
Subject (subject)	Specifies the subject of an email.
Recipient (recipient)	Specifies the name of the receiver of an email.
Report ID (reportid)	Only for Data Filtering and WildFire subtype; all other types do not use this field. Identifies the analysis request on the firewall, WildFire cloud, or the WildFire appliance.

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Field Name	Description
Device Group Hierarchy (dg_hier_level_1 to dg_hier_level_4)	A sequence of identification numbers that indicate the device group's location within a device group hierarchy. The firewall (or virtual system) generating the log includes the identification number of each ancestor in its device group hierarchy. The shared device group (level 0) is not included in this structure. If the log values are 12, 34, 45, 0, it means that the log was generated by a firewall (or virtual system) that belongs to device group 45, and its ancestors are 34, and 12. To view the device group names that correspond to the value 12, 34 or 45, use one of the following methods: API query: `/api/?type=op&cmd=<show><dg-hierarchy></dg-hierarchy></show>`
Virtual System Name (vsys_name)	The name of the virtual system associated with the session; only valid on firewalls enabled for multiple virtual systems.
Device Name (device_name)	The hostname of the firewall on which the session was logged.
Source VM UUID (src_uuid)	Identifies the source universal unique identifier for a guest virtual machine in the VMware NSX environment.
Destination VM UUID (dst_uuid)	Identifies the destination universal unique identifier for a guest virtual machine in the VMware NSX environment.
HTTP Method (http_method)	Describes the HTTP Method used in the web request. Only the following methods are logged: Connect, Delete, Get, Head, Options, Post, Put.
Tunnel ID/IMSI (tunnel_id/imsi)	International Mobile Subscriber Identity (IMSI) is a unique number allocated to each mobile subscriber in the GSM/UMTS/EPS system. IMSI shall consist of decimal digits (0 through 9) only and maximum number of digits allowed are 15.
Monitor Tag/IMEI (monitortag/imei)	International Mobile Equipment Identity (IMEI) is a unique 15 or 16 digit number allocated to each mobile station equipment.
Parent Session ID (parent_session_id)	ID of the session in which this session is tunneled. Applies to inner tunnel (if two levels of tunneling) or inside content (if one level of tunneling) only.
Parent Session Start Time (parent_start_time)	Year/month/day hours:minutes:seconds that the parent tunnel session began.

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Field Name	Description
Tunnel Type (tunnel)	Type of tunnel, such as GRE or IPSec.
Threat Category (thr_category)	Describes threat categories used to classify different types of threat signatures. If a domain external dynamic list generated the log, `domain-edl` populates this field.
Content Version (contentver)	Applications and Threats version on your firewall when the log was generated.
SCTP Association ID (assoc_id)	Number that identifies all connections for an association between two SCTP endpoints.
Payload Protocol ID (ppid)	ID of the protocol for the payload in the data portion of the data chunk.
Request Headers (request_headers)	The HTTP request headers captured from client traffic. This field is populated when you select specific request headers or enable **Log All Headers** for request logging in a URL filtering profile.
Response Headers (response_headers)	The HTTP response headers captured from server traffic. This field is populated when you select specific response headers or enable **Log All Headers** for response logging in a URL filtering profile.
HTTP Status Code (http_status_code)	The HTTP response status code returned by the server for each request. This field is populated when you enable **Enable logging of response code in HTTP response** in a URL filtering profile.
URL Category List (url_category_list)	Lists the URL filtering categories that the firewall used to enforce policy.
Rule UUID (rule_uuid)	The UUID that permanently identifies the rule.
HTTP/2 Connection (http2_connection)	Identifies if traffic used an HTTP/2 connection by displaying one of the following values: • TCP connection session ID—session is HTTP/2 • 0—session is not HTTP/2
Dynamic User Group Name (dynusergroup_name)	The name of the dynamic user group that contains the user who initiated the session.
XFF Address (xff_ip)	The IP address of the user who requested the web page or the IP address of the next to last device that the request traversed. If the request goes through one or more proxies, load balancers, or other

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Field Name	Description
	upstream devices, the firewall displays the IP address of the most recent device. *Based on different appliance implementations, the XFF field may contain non-IP address values.*
Source Device Category (src_category)	The category for the device that Device-ID identifies as the source of the traffic.
Source Device Profile (src_profile)	The device profile for the device that Device-ID identifies as the source of the traffic.
Source Device Model (src_model)	The model of the device that Device-ID identifies as the source of the traffic.
Source Device Vendor (src_vendor)	The vendor of the device that Device-ID identifies as the source of the traffic.
Source Device OS Family (src_osfamily)	The operating system type for the device that Device-ID identifies as the source of the traffic.
Source Device OS Version (src_osversion)	The version of the operating system for the device that Device-ID identifies as the source of the traffic.
Source Hostname (src_host)	The hostname of the device that Device-ID identifies as the source of the traffic.
Source MAC Address (src_mac)	The MAC address for the device that Device-ID identifies as the source of the traffic.
Destination Device Category (dst_category)	The category for the device that Device-ID identifies as the destination for the traffic.
Destination Device Profile (dst_profile)	The device profile for the device that Device-ID identifies as the destination for the traffic.
Destination Device Model (dst_model)	The model of the device that Device-ID identifies as the destination for the traffic.
Destination Device Vendor (dst_vendor)	The vendor of the device that Device-ID identifies as the destination for the traffic.
Destination Device OS Family (dst_osfamily)	The operating system type for the device that Device-ID identifies as the destination for the traffic.

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Field Name	Description
Destination Device OS Version (dst_osversion)	The version of the operating system for the device that Device-ID identifies as the destination for the traffic.
Destination Hostname (dst_host)	The hostname of the device that Device-ID identifies as the destination for the traffic.
Destination MAC Address (dst_mac)	The MAC address for the device that Device-ID identifies as the destination for the traffic.
Container ID (container_id)	The container ID of the PAN-NGFW pod on the Kubernetes node where the application POD is deployed.
POD Namespace (pod_namespace)	The namespace of the application POD being secured.
POD Name (pod_name)	The application POD being secured.
Source External Dynamic List (src_edl)	The name of the external dynamic list that contains the source IP address of the traffic.
Destination External Dynamic List (dst_edl)	The name of the external dynamic list that contains the destination IP address of the traffic.
Host ID (hostid)	Unique ID GlobalProtect assigns to identify the host.
User Device Serial Number (serialnumber)	Serial number of the user's machine or device.
Domain EDL (domain_edl)	The name of the external dynamic list that contains the domain name of the traffic.
Source Dynamic Address Group (src_dag)	Original session source dynamic address group.
Destination Dynamic Address Group (dst_dag)	Original destination source dynamic address group.
Partial Hash (partial_hash)	Machine Learning partial hash.
High Resolution Timestamp (high_res timestamp)	Time in milliseconds the log was received at the management plane. The format for this new field is YYYY-MM-DDThh:ss:sssTZD: • **YYYY**—Four digit year • **MM**—Two-digit month • **DD**—Two-digit day of the month (01 through 31)

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Field Name	Description
	• **T**—Indicator for the beginning of the timestamp • **hh**—Two-digit hour using 24-hour time (00 through 23) • **mm**—Two-digit minute (00 through 59) • **ss**—Two-digit second (00 through 60) • **sss**—One or more digits for millisecond • **TZD**—Time zone designator (+hh:mm or -hh:mm) *The High Resolution Timestamp is supported for logs received from managed firewalls running PAN-OS 10.1 and later releases. Logs received from managed firewalls running PAN-OS 9.1 and earlier releases display a `1969-12-31T16:00:00:000-8:00` timestamp regardless of when the log was received.*
Reason (reason)	Reason for URL Filtering action.
Justification (justification)	Justification for URL Filtering action.
A Slice Service Type (nssai_sst)	The A Slice Service Type of the Network Slice ID.
Application Subcategory (subcategory_of_app)	The application subcategory specified in the application configuration properties.
Application Category (category_of_app)	The application category specified in the application configuration properties. Values are: • business-systems • collaboration • general-internet • media • networking • saas
Application Technology (technology_of_app)	The application technology specified in the application configuration properties. Values are: • browser-based • client-server • network-protocol • peer-to-peer

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Field Name	Description
Application Risk (risk_of_app)	Risk level associated with the application (1=lowest to 5=highest).
Application Characteristic (characteristic_of_app)	Comma-separated list of applicable characteristic of the application
Application Container (container_of_app)	The parent application for an application.
Tunneled Application (tunneled_app)	Name of the tunneled application.
Application SaaS (is_saas_of_app)	Displays `yes` if a SaaS application or `no` if not a SaaS application.
Application Sanctioned State (sanctioned_state_of_app)	Displays `yes` if application is sanctioned or `no` if application is not sanctioned.
Cloud Report ID (cloud_reportid)	(PAN-OS 10.2.0) Unique 32 character ID for a file scanned by the DLP cloud service sent by a firewall. (PAN-OS 10.2.1 and later releases) Unique 67 character ID for a file scanned by the DLP cloud service sent by a firewall. The same Cloud Report ID is displayed for a file the DLP cloud service has already scanned and generated a Cloud Report ID for.
Cluster Name (cluster_name) (11.1 and later releases)	Name of the CN-Series firewall cluster.
Flow Type (flow_type)	(11.1 and later releases) Identifies the type of proxy used for traffic. If a proxy is used, `Explicit Proxy` or `Transparent Proxy` is displayed. If no proxy is used, `NonProxyTraffic` is displayed.

Data Filtering Log Fields

Format: FUTURE_USE, Receive Time, Serial Number, Type, Threat/Content Type, FUTURE_USE, Generated Time, Source Address, Destination Address, NAT Source IP, NAT Destination IP, Rule Name, Source User, Destination User, Application, Virtual System, Source Zone, Destination Zone, Inbound Interface, Outbound Interface, Log Action, FUTURE_USE, Session ID, Repeat Count, Source Port, Destination Port, NAT Source Port, NAT Destination Port, Flags, IP Protocol, Action, URL/Filename, Threat ID, Category, Severity, Direction, Sequence Number, Action Flags, Source Country, Destination Country, FUTURE_USE, Content Type, PCAP_ID, File Digest, Cloud, URL Index, User Agent, File Type, X-Forwarded-For, Referer, Sender, Subject, Recipient, Report ID, Device Group Hierarchy Level 1, Device Group Hierarchy Level 2, Device Group Hierarchy Level

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3, Device Group Hierarchy Level 4, Virtual System Name, Device Name, FUTURE_USE, Source VM UUID, Destination VM UUID, HTTP Method, Tunnel ID/IMSI, Monitor Tag/IMEI, Parent Session ID, Parent Start Time, Tunnel Type, Threat Category, Content Version, FUTURE_USE, SCTP Association ID, Payload Protocol ID, HTTP Headers, URL Category List, Rule UUID, HTTP/2 Connection, Dynamic User Group Name, XFF Address, Source Device Category, Source Device Profile, Source Device Model, Source Device Vendor, Source Device OS Family, Source Device OS Version, Source Hostname, Source MAC Address, Destination Device Category, Destination Device Profile, Destination Device Model, Destination Device Vendor, Destination Device OS Family, Destination Device OS Version, Destination Hostname, Destination MAC Address, Container ID, POD Namespace, POD Name, Source External Dynamic List, Destination External Dynamic List, Host ID, Serial Number, Domain EDL, Source Dynamic Address Group, Destination Dynamic Address Group, Partial Hash, High Resolution Timestamp, Reason, Justification, A Slice Service Type, Application Subcategory, Application Category, Application Technology, Application Risk, Application Characteristic, Application Container, Tunneled Application, Application SaaS, Application Sanctioned State, Cloud Report ID, Cluster Name, Flow Type

Field Name	Description
Receive Time (receive_time or cef-formatted-receive_time)	Time the log was received at the management plane.
Serial Number (serial #)	Serial number of the firewall that generated the log.
Type (type)	Specifies the type of log; value is THREAT.
Threat/Content Type (subtype)	Subtype of threat log; value is data, dlp, dlp-non-file, file.
Generate Time (time_generated or cef-formatted-time_generated)	Time the log was generated on the dataplane.
Source address (src)	Original session source IP address.
Destination address (dst)	Original session destination IP address.
NAT Source IP (natsrc)	If source NAT performed, the post-NAT source IP address.
NAT Destination IP (natdst)	If destination NAT performed, the post-NAT destination IP address.
Rule Name (rule)	Name of the rule that the session matched.
Source User (srcuser)	Username of the user who initiated the session.

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Field Name	Description
Destination User (dstuser)	Username of the user to which the session was destined.
Application (app)	Application associated with the session.
Virtual System (vsys)	Virtual System associated with the session.
Source Zone (from)	Zone the session was sourced from.
Destination Zone (to)	Zone the session was destined to.
Inbound Interface (inbound_if)	Interface that the session was sourced from.
Outbound Interface (outbound_if)	Interface that the session was destined to.
Log Action (logset)	Log Forwarding Profile that was applied to the session.
Session ID (sessionid)	An internal numerical identifier applied to each session.
Repeat Count (repeatcnt)	Number of sessions with same Source IP, Destination IP, Application, and Content/Threat Type seen within 5 seconds.
Source Port (sport)	Source port utilized by the session.
Destination Port (dport)	Destination port utilized by the session.
NAT Source Port (natsport)	Post-NAT source port.
NAT Destination Port (natdport)	Post-NAT destination port.
Flags (flags)	32-bit field that provides details on session; this field can be decoded by AND-ing the values with the logged value: • 0x80000000—session has a packet capture (PCAP) • 0x40000000—option is enabled to allow a client to use multiple paths to connect to a destination host • 0x20000000—file is submitted to WildFire for a verdict • 0x10000000—enterprise credential submission by end user detected • 0x08000000— source for the flow is on an allow list and not subject to recon protection

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Field Name	Description
	• 0x02000000—IPv6 session • 0x01000000—SSL session is decrypted (SSL Proxy) • 0x00800000—session is denied via URL filtering • 0x00400000—session has a NAT translation performed • 0x00200000—user information for the session was captured through Authentication Portal • 0x00100000—application traffic is on a non-standard destination port • 0x00080000 —X-Forwarded-For value from a proxy is in the source user field • 0x00040000 —log corresponds to a transaction within a http proxy session (Proxy Transaction) • 0x00020000—Client to Server flow is subject to policy based forwarding • 0x00010000—Server to Client flow is subject to policy based forwarding • 0x00008000 —session is a container page access (Container Page) • 0x00002000 —session has a temporary match on a rule for implicit application dependency handling. Available in PAN-OS 5.0.0 and above. • 0x00000800 —symmetric return is used to forward traffic for this session • 0x00000400—decrypted traffic is being sent out clear text through a mirror port • 0x00000010—payload of the outer tunnel is being inspected
IP Protocol (proto)	IP protocol associated with the session.
Action (action)	Action taken for the session; values are alert, allow, deny, drop, drop-all-packets, reset-client, reset-server, reset-both, block-url. • alert—traffic containing matching data detected but not blocked • allow (dlp subtype only)—flood detection alert • block (dlp and WildFire subtype only) —traffic containing matching data detected but blocked • block-continue (dlp subtype only)—traffic containing matching data is blocked and redirected to a Continue page with a button for confirmation to proceed • continue (dlp subtype only)—response to a block-continue page indicating a block-continue request was allowed to proceed

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Field Name	Description
	• deny (dlp subtype only)—flood detection mechanism activated and deny traffic based on configuration
URL/Filename (misc)	Field with variable length. A Filename has a maximum of 63 characters. File name when the subtype is dlp URL when Threat Category is `domain-edl`.
Threat/Content Name (threatid)	Palo Alto Networks identifier for known and custom threats. It is a description string followed by a 64-bit numerical identifier in parentheses for some Subtypes: • 8000 – 8099— scan detection • 8500 – 8599— flood detection • 9999— URL filtering log • 10000 – 19999 —spyware phone home detection • 20000 – 29999 —spyware download detection • 30000 – 44999 —vulnerability exploit detection • 52000 – 52999— filetype detection • 60000 – 69999 —data filtering detection If the Domain EDL field is populated, then this field is populated with the same value. *Threat ID ranges for virus detection, WildFire signature feed, and DNS C2 signatures used in previous releases have been replaced with permanent, globally unique IDs. Refer to the Threat/Content Type (subtype) and Threat Category (thr_category) field names to create updated reports, filter threat logs, and ACC activity.*
Category (category)	For URL Subtype, it is the URL Category; For WildFire subtype, it is the verdict on the file and is either 'malware', 'phishing', 'grayware', or 'benign'; For other subtypes, the value is 'any'.
Severity (severity)	Severity associated with the threat; values are informational, low, medium, high, critical.
Direction (direction)	Indicates the direction of the attack: • client-to-server • server-to-client

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Field Name	Description
Sequence Number (seqno)	A 64-bit log entry identifier incremented sequentially. Each log type has a unique number space.
Action Flags (actionflags)	A bit field indicating if the log was forwarded to Panorama.
Source Country (srcloc)	Source country or Internal region for private addresses. Maximum length is 32 bytes.
Destination Country (dstloc)	Destination country or Internal region for private addresses. Maximum length is 32 bytes.
Content Type (contenttype)	Applicable only when Subtype is URL. Content type of the HTTP response data. Maximum length 32 bytes.
PCAP ID (pcap_id)	The packet capture (pcap) ID is a 64 bit unsigned integral denoting an ID to correlate threat pcap files with extended pcaps taken as a part of that flow. All threat logs will contain either a pcap_id of 0 (no associated pcap), or an ID referencing the extended pcap file.
File Digest (filedigest)	Only for WildFire subtype; all other types do not use this field The filedigest string shows the binary hash of the file sent to be analyzed by the WildFire service.
Cloud (cloud)	Only for WildFire subtype; all other types do not use this field. The cloud string displays the FQDN of either the WildFire appliance (private) or the WildFire cloud (public) from where the file was uploaded for analysis.
URL Index (url_idx)	Used in URL Filtering and WildFire subtypes. When an application uses TCP keepalives to keep a connection open for a length of time, all the log entries for that session have a single session ID. In such cases, when you have a single threat log (and session ID) that includes multiple URL entries, the url_idx is a counter that allows you to correlate the order of each log entry within the single session. For example, to learn the URL of a file that the firewall forwarded to WildFire for analysis, locate the session ID and the url_idx from the WildFire Submissions log and search for the same session ID and url_idx in your URL filtering logs. The log entry that matches the session ID and url_idx will contain the URL of the file that was forwarded to WildFire.

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Field Name	Description
User Agent (user_agent)	Only for the URL Filtering subtype; all other types do not use this field. The User Agent field specifies the web browser that the user used to access the URL, for example Internet Explorer. This information is sent in the HTTP request to the server.
File Type (filetype)	Specifies the type of file that the firewall forwarded for analysis.
X-Forwarded-For (xff)	Only for the URL Filtering subtype; all other types do not use this field. The X-Forwarded-For field in the HTTP header contains the IP address of the user who requested the web page. It allows you to identify the IP address of the user, which is useful particularly if you have a proxy server on your network that replaces the user IP address with its own address in the source IP address field of the packet header.
Referer (referer)	Only for the URL Filtering subtype; all other types do not use this field. The Referer field in the HTTP header contains the URL of the web page that linked the user to another web page; it is the source that redirected (referred) the user to the web page that is being requested.
Sender (sender)	Specifies the name of the sender of an email.
Subject (subject)	Specifies the subject of an email.
Recipient (recipient)	Specifies the name of the receiver of an email.
Report ID (reportid)	Identifies the analysis request on the firewall, WildFire cloud, or the WildFire appliance.
Device Group Hierarchy (dg_hier_level_1 to dg_hier_level_4)	A sequence of identification numbers that indicate the device group's location within a device group hierarchy. The firewall (or virtual system) generating the log includes the identification number of each ancestor in its device group hierarchy. The shared device group (level 0) is not included in this structure. If the log values are 12, 34, 45, 0, it means that the log was generated by a firewall (or virtual system) that belongs to device group 45, and its ancestors are 34, and 12. To view the device group names that correspond to the value 12, 34 or 45, use one of the following methods:

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Field Name	Description
	API query: `/api/?type=op&cmd=<show><dg-hierarchy></dg-hierarchy></show>`
Virtual System Name (vsys_name)	The name of the virtual system associated with the session; only valid on firewalls enabled for multiple virtual systems.
Device Name (device_name)	The hostname of the firewall on which the session was logged.
Source VM UUID (src_uuid)	Identifies the source universal unique identifier for a guest virtual machine in the VMware NSX environment.
Destination VM UUID (dst_uuid)	Identifies the destination universal unique identifier for a guest virtual machine in the VMware NSX environment.
HTTP Method (http_method)	Only in URL filtering logs. Describes the HTTP Method used in the web request. Only the following methods are logged: Connect, Delete, Get, Head, Options, Post, Put.
Tunnel ID/IMSI (tunnel_id/imsi)	International Mobile Subscriber Identity (IMSI) is a unique number allocated to each mobile subscriber in the GSM/UMTS/EPS system. IMSI shall consist of decimal digits (0 through 9) only and maximum number of digits allowed are 15.
Monitor Tag/IMEI (monitortag/imei)	International Mobile Equipment Identity (IMEI) is a unique 15 or 16 digit number allocated to each mobile station equipment.
Parent Session ID (parent_session_id)	ID of the session in which this session is tunneled. Applies to inner tunnel (if two levels of tunneling) or inside content (if one level of tunneling) only.
Parent Session Start Time (parent_start_time)	Year/month/day hours:minutes:seconds that the parent tunnel session began.
Tunnel Type (tunnel)	Type of tunnel, such as GRE or IPSec.
Threat Category (thr_category)	Describes threat categories used to classify different types of threat signatures. If a domain external dynamic list generated the log, `domain-edl` populates this field.
Content Version (contentver)	Applications and Threats version on your firewall when the log was generated.

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Field Name	Description
SCTP Association ID (assoc_id)	Number that identifies all connections for an association between two SCTP endpoints.
Payload Protocol ID (ppid)	ID of the protocol for the payload in the data portion of the data chunk.
HTTP Headers (http_headers)	Indicates the inserted HTTP header in the URL log entries on the firewall.
URL Category List (url_category_list)	Lists the URL Filtering categories that the firewall used to enforce policy.
Rule UUID (rule_uuid)	The UUID that permanently identifies the rule.
HTTP/2 Connection (http2_connection)	Identifies if traffic used an HTTP/2 connection by displaying one of the following values: • TCP connection session ID—session is HTTP/2 • 0—session is not HTTP/2
Dynamic User Group Name (dynusergroup_name)	The name of the dynamic user group that contains the user who initiated the session.
XFF Address (xff_ip)	The IP address of the user who requested the web page or the IP address of the next to last device that the request traversed. If the request goes through one or more proxies, load balancers, or other upstream devices, the firewall displays the IP address of the most recent device.
Source Device Category (src_category)	The category for the device that Device-ID identifies as the source of the traffic.
Source Device Profile (src_profile)	The device profile for the device that Device-ID identifies as the source of the traffic.
Source Device Model (src_model)	The model of the device that Device-ID identifies as the source of the traffic.
Source Device Vendor (src_vendor)	The vendor of the device that Device-ID identifies as the source of the traffic.
Source Device OS Family (src_osfamily)	The operating system type for the device that Device-ID identifies as the source of the traffic.

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Field Name	Description
Source Device OS Version (src_osversion)	The version of the operating system for the device that Device-ID identifies as the source of the traffic.
Source Hostname (src_host)	The hostname of the device that Device-ID identifies as the source of the traffic.
Source MAC Address (src_mac)	The MAC address for the device that Device-ID identifies as the source of the traffic.
Destination Device Category (dst_category)	The category for the device that Device-ID identifies as the destination for the traffic.
Destination Device Profile (dst_profile)	The device profile for the device that Device-ID identifies as the destination for the traffic.
Destination Device Model (dst_model)	The model of the device that Device-ID identifies as the destination for the traffic.
Destination Device Vendor (dst_vendor)	The vendor of the device that Device-ID identifies as the destination for the traffic.
Destination Device OS Family (dst_osfamily)	The operating system type for the device that Device-ID identifies as the destination for the traffic.
Destination Device OS Version (dst_osversion)	The version of the operating system for the device that Device-ID identifies as the destination for the traffic.
Destination Hostname (dst_host)	The hostname of the device that Device-ID identifies as the destination for the traffic.
Destination MAC Address (dst_mac)	The MAC address for the device that Device-ID identifies as the destination for the traffic.
Container ID (container_id)	The container ID of the PAN-NGFW pod on the Kubernetes node where the application POD is deployed.
POD Namespace (pod_namespace)	The namespace of the application POD being secured.
POD Name (pod_name)	The application POD being secured.
Source External Dynamic List (src_edl)	The name of the external dynamic list that contains the source IP address of the traffic.
Destination External Dynamic List (dst_edl)	The name of the external dynamic list that contains the destination IP address of the traffic.

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Field Name	Description
Host ID (hostid)	Unique ID GlobalProtect assigns to identify the host.
User Device Serial Number (serialnumber)	Serial number of the user's machine or device.
Domain EDL (domain_edl)	The name of the external dynamic list that contains the domain name of the traffic.
Source Dynamic Address Group (src_dag)	Original session source dynamic address group.
Destination Dynamic Address Group (dst_dag)	Original destination source dynamic address group.
Partial Hash (partial_hash)	Machine Learning partial hash.
High Resolution Timestamp (high_res timestamp)	Time in milliseconds the log was received at the management plane. The format for this new field is YYYY-MM-DDThh:ss:sssTZD: • **YYYY**—Four digit year • **MM**—Two-digit month • **DD**—Two-digit day of the month (01 through 31) • **T**—Indicator for the beginning of the timestamp • **hh**—Two-digit hour using 24-hour time (00 through 23) • **mm**—Two-digit minute (00 through 59) • **ss**—Two-digit second (00 through 60) • **sss**—One or more digits for millisecond • **TZD**—Time zone designator (+hh:mm or -hh:mm) *The High Resolution Timestamp is supported for logs received from managed firewalls running PAN-OS 10.1 and later releases. Logs received from managed firewalls running PAN-OS 9.1 and earlier releases display a `1969-12-31T16:00:00:000-8:00` timestamp regardless of when the log was received.*
Reason (reason)	Reason for Data Filtering action.
Justification (justification)	Justification for Data Filtering action.

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Field Name	Description
A Slice Service Type (nssai_sst)	The A Slice Service Type of the Network Slice ID.
Application Subcategory (subcategory_of_app)	The application subcategory specified in the application configuration properties.
Application Category (category_of_app)	The application category specified in the application configuration properties. Values are: • business-systems • collaboration • general-internet • media • networking • saas
Application Technology (technology_of_app)	The application technology specified in the application configuration properties. Values are: • browser-based • client-server • network-protocol • peer-to-peer
Application Risk (risk_of_app)	Risk level associated with the application (1=lowest to 5=highest).
Application Characteristic (characteristic_of_app)	Comma-separated list of applicable characteristic of the application
Application Container (container_of_app)	The parent application for an application.
Tunneled Application (tunneled_app)	Name of the tunneled application.
Application SaaS (is_saas_of_app)	Displays `yes` if a SaaS application or `no` if not a SaaS application.
Application Sanctioned State (sanctioned_state_of_app)	Displays `yes` if application is sanctioned or `no` if application is not sanctioned.

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Field Name	Description
Cloud Report ID (cloud_reportid)	(PAN-OS 10.2.0) Unique 32 character ID for a file scanned by the DLP cloud service sent by a firewall. (PAN-OS 10.2.1 and later releases) Unique 67 character ID for a file scanned by the DLP cloud service sent by a firewall. The same Cloud Report ID is displayed for a file the DLP cloud service has already scanned and generated a Cloud Report ID for.
Cluster Name (cluster_name)	(11.1 and later releases) Name of the CN-Series firewall cluster.
Flow Type (flow_type)	(11.1 and later releases) Identifies the type of proxy used for traffic. If a proxy is used, `Explicit Proxy` or `Transparent Proxy` is displayed. If no proxy is used, `NonProxyTraffic` is displayed.

HIP Match Log Fields

Format: FUTURE_USE, Receive Time, Serial Number, Type, Threat/Content Type, FUTURE_USE, Generated Time, Source User, Virtual System, Machine Name, Operating System, Source Address, HIP, Repeat Count, HIP Type, FUTURE_USE, FUTURE_USE, Sequence Number, Action Flags, Device Group Hierarchy Level 1, Device Group Hierarchy Level 2, Device Group Hierarchy Level 3, Device Group Hierarchy Level 4, Virtual System Name, Device Name, Virtual System ID, IPv6 Source Address, Host ID, User Device Serial Number, Device MAC Address, High Resolution Timestamp, Cluster Name

Field Name	Description
Receive Time (receive_time or cef-formatted-receive_time)	Time the log was received at the management plane.
Serial Number (serial)	Serial number of the firewall that generated the log.
Type (type)	Specifies the type of log; value is HIP-MATCH.
Threat/Content Type (subtype)	Subtype of HIP match log; unused.
Generated Time (time_generated or cef-formatted-time_generated)	Time the log was generated on the dataplane.

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Field Name	Description
Source User (srcuser)	Username of the user who initiated the session.
Virtual System (vsys)	Virtual System associated with the HIP match log.
Machine Name (machinename)	Name of the user's machine.
Operating System (os)	The operating system installed on the user's machine or device (or on the client system).
Source Address (src)	IP address of the source user.
HIP (matchname)	Name of the HIP object or profile.
Repeat Count (repeatcnt)	Number of times the HIP profile matched.
HIP Type (matchtype)	Whether the hip field represents a HIP object or a HIP profile.
Sequence Number (seqno)	A 64-bit log entry identifier incremented sequentially; each log type has a unique number space.
Action Flags (actionflags)	A bit field indicating if the log was forwarded to Panorama.
Device Group Hierarchy (dg_hier_level_1 to dg_hier_level_4)	A sequence of identification numbers that indicate the device group's location within a device group hierarchy. The firewall (or virtual system) generating the log includes the identification number of each ancestor in its device group hierarchy. The shared device group (level 0) is not included in this structure. If the log values are 12, 34, 45, 0, it means that the log was generated by a firewall (or virtual system) that belongs to device group 45, and its ancestors are 34, and 12. To view the device group names that correspond to the value 12, 34 or 45, use one of the following methods: API query: `/api/?type=op&cmd=<show><dg-hierarchy></dg-hierarchy></show>`
Virtual System Name (vsys_name)	The name of the virtual system associated with the session; only valid on firewalls enabled for multiple virtual systems.

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Field Name	Description
Device Name (device_name)	The hostname of the firewall on which the session was logged.
Virtual System ID (vsys_id)	A unique identifier for a virtual system on a Palo Alto Networks firewall.
IPv6 System Address (srcipv6)	IPv6 address of the user's machine or device.
Host ID (hostid)	Unique ID GlobalProtect assigns to identify the host.
User Device Serial Number (serialnumber)	Serial number of the user's machine or device.
Device MAC Address (mac)	The MAC address of the user's machine or device.
High Resolution Timestamp (high_res_timestamp)	Time in milliseconds the log was received at the management plane. The format for this new field is YYYY-MM-DDThh:ss:sssTZD: • **YYYY**—Four digit year • **MM**—Two-digit month • **DD**—Two-digit day of the month (01 through 31) • **T**—Indicator for the beginning of the timestamp • **hh**—Two-digit hour using 24-hour time (00 through 23) • **mm**—Two-digit minute (00 through 59) • **ss**—Two-digit second (00 through 60) • **sss**—One or more digits for millisecond • **TZD**—Time zone designator (+hh:mm or -hh:mm) *The High Resolution Timestamp is supported for logs received from managed firewalls running PAN-OS 11.1 and later releases. Logs received from managed firewalls running PAN-OS 9.1 and earlier releases display a `1969-12-31T16:00:00:000-8:00` timestamp regardless of when the log was received.*
Cluster Name (cluster_name)	(11.1 and later releases) Name of the CN-Series firewall cluster.

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GlobalProtect Log Fields

Format: FUTURE_USE, Receive Time, Serial Number, Type, Threat/Content Type, FUTURE_USE, Generated Time, Virtual System, Event ID, Stage, Authentication Method, Tunnel Type, Source User, Source Region, Machine Name, Public IP, Public IPv6, Private IP, Private IPv6, Host ID, Serial Number, Client Version, Client OS, Client OS Version, Repeat Count, Reason, Error, Description, Status, Location, Login Duration, Connect Method, Error Code, Portal, Sequence Number, Action Flags, High Res Timestamp, Selection Type, Response Time, Priority, Attempted Gateways, Gateway, Device Group Hierarchy Level 1, Device Group Hierarchy Level 2, Device Group Hierarchy Level 3, Device Group Hierarchy Level 4, Virtual System Name, Device Name, Virtual System ID, Cluster Name

Field Name	Description
Receive Time (receive_time)	The time that the log was received at the management plane.
Serial # (serial)	The serial number of the firewall that generated the log.
Type (type)	Specifies the type of log; value is GLOBALPROTECT.
Threat/Content Type (subtype)	Subtype of threat log. Values include the following: • data—Data pattern matching a Data Filtering profile. • file—File type matching a File Blocking profile. • flood—Flood detected via a Zone Protection profile. • packet—Packet-based attack protection triggered by a Zone Protection profile. • scan—Scan detected via a Zone Protection profile. • spyware —Spyware detected via an Anti-Spyware profile. • url—URL filtering log. • virus—Virus detected via an Antivirus profile. • vulnerability —Vulnerability exploit detected via a Vulnerability Protection profile. • wildfire —A WildFire verdict generated when the firewall submits a file to WildFire per a WildFire Analysis profile and a verdict (malicious, phishing, grayware, or benign, depending on what you are logging) is logged in the WildFire Submissions log. • wildfire-virus—Virus detected via an Antivirus profile.
Generate Time (time_generated)	The time that the log was generated on the dataplane.
Virtual System (vsys)	The Virtual System associated with the session.

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Field Name	Description
Event ID (eventid)	A string showing the name of the event.
Stage (stage)	A string showing the stage of the connection (for example, `before-login`, `login`, or `tunnel`).
Authentication Method (auth_method)	A string showing the authentication type, such as `LDAP`, `RADIUS`, or `SAML`.
Tunnel Type (tunnel_type)	The type of tunnel (either SSLVPN or IPSec).
Source User (srcuser)	The username of the user who initiated the session.
Source Region (srcregion)	The region for the user who initiated the session.
Machine Name (machinename)	The name of the user's machine.
Public IP (public_ip)	The public IP address for the user who initiated the session.
Public IPv6 (public_ipv6)	The public IPv6 address for the user who initiated the session.
Private IP (private_ip)	The private IP address for the user who initiated the session.
Private IPv6 (private_ipv6)	The private IPv6 address for the user who initiated the session.
Host ID (hostid)	The unique ID that GlobalProtect assigns to identify the host.
Serial Number (serialnumber)	The serial number of the user's machine or device.
Client Version (client_ver)	The client's GlobalProtect app version.
Client OS (client_os)	The client device's OS type (for example, Windows or Linux).
Client OS Version (client_os_ver)	The client device's OS version.
Repeat Count (repeatcnt)	The number of sessions with the same source IP address, destination IP address, application, and subtype that GlobalProtect has detected within the last five seconds.

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Field Name	Description
Reason (reason)	A string that shows the reason for the quarantine.
Error (error)	A string showing that error that has occurred in any event.
Description (opaque)	Additional information for any event that has occurred.
Status (status)	The status (success or failure) of the event.
Location (location)	A string showing the administrator-defined location of the GlobalProtect portal or gateway.
Login Duration (login_duration)	The length of time, in seconds, the user is connected to the GlobalProtect gateway from logging in to logging out.
Connect Method (connect_method)	A string showing the how the GlobalProtect app connects to Gateway, (for example, `on-demand` or `user-logon`.
Error Code (error_code)	An integer associated with any errors that occurred.
Portal (portal)	The name of the GlobalProtect portal or gateway.
Sequence Number (seqno)	A 64-bit log entry identifier incremented sequentially; each log type has a unique number space.
Action Flags (actionflags)	A bit field indicating if the log was forwarded to Panorama.
Gateway Selection Method (selection_type)	The connection method that is selected to connect to the gateway. • manual—The gateway to which you want the GlobalProtect app to manually connect. • preferred—The preferred gateway to which you want the GlobalProtect app to connect. • auto—Automatically connect to the **Best Available** gateway based on the priority assigned to the gateway and the response time.
SSL Response Time (response_time)	The SSL response time of the selected gateway that is measured in milliseconds on the endpoint during tunnel setup.
Gateway Priority (priority)	The priority order of the gateway that is based on highest (1), high (2), medium (3), low (4), or lowest (5) to which the GlobalProtect app can connect.
Attempted Gateways (attempted_gateways)	The fields that are collected for each gateway connection attempt with the gateway name, SSL response time, and priority (see Gateway

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Field Name	Description
	Priority in a Multiple Gateway Configuration. Each field entry is separated by commas such as `g82-gateway,12,3`. Each gateway entry is separated by semicolons such as `g83-gateway,10,2;g84-gateway,-1,1`.
Gateway Name (gateway)	The name of the gateway that is specified on the portal configuration.
Device Group Hierarchy (dg_hier_level_1 to dg_hier_level_4)	A sequence of identification numbers that indicate the device group's location within a device group hierarchy. The firewall (or virtual system) generating the log includes the identification number of each ancestor in its device group hierarchy. The shared device group (level 0) is not included in this structure. If the log values are 12, 34, 45, 0, it means that the log was generated by a firewall (or virtual system) that belongs to device group 45, and its ancestors are 34, and 12. To view the device group names that correspond to the value 12, 34 or 45, use one of the following methods: API query: `/api/?type=op&cmd=<show><dg-hierarchy></dg-hierarchy></show>`
Virtual System Name (vsys_name)	The name of the virtual system associated with the session; only valid on firewalls enabled for multiple virtual systems.
Device Name (device_name)	The hostname of the firewall on which the session was logged.
Virtual System ID (vsys_id)	A unique identifier for a virtual system on a Palo Alto Networks firewall.
Cluster Name (cluster_name)	(11.1 and later releases) Name of the CN-Series firewall cluster.

IP-Tag Log Fields

Format: FUTURE_USE , Receive Time, Serial, Type, Threat/Content Type, FUTURE_USE, Generate Time, Virtual System, Source IP, Tag Name , Event ID, Repeat Count , Timeout, Data Source Name, Data Source Type, Data Source Subtype, Sequence Number, Action Flags, DG Hierarchy Level 1 , DG Hierarchy Level 2, DG Hierarchy Level 3, DG Hierarchy Level 4, Virtual System Name, Device Name, Virtual System ID, High Resolution Timestamp, Cluster Name

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Field Name	Description
Receive Time (receive_time or cef-formatted-receive_time)	The time the log was received at the management plane.
Serial Number (serial)	The serial number of the firewall that generated the log.
Type (type)	Specifies the type of log; value is IPTAG.
Threat/Content Type (subtype)	The subtype of the HIP match log; unused.
Generated Time (time_generated or cef-formatted-time_generated)	The time the log was generated on the dataplane.
Virtual System (vsys)	The virtual system associated with the HIP match log.
Source IP (src)	The IP address of the source user.
Tag Name (tag_name)	The tag mapped to the source IP address.
Event ID (event_id)	A string showing the name of the event.
Repeat Count (repeatcnt)	The number of sessions with the same Source IP, Destination IP, Application, and Subtype seen within 5 seconds.
Timeout (timeout)	The amount of time before the IP address-to-tag mapping expires for the source IP address.
Data Source Name (datasourcename)	The name of the source from which mapping information is collected.
Data Source Type (datasource_type)	The source from which mapping information is collected.
Data Source Subtype (datasource_subtype)	The mechanism used to identify the IP address-to-username mappings within a data source.
Sequence Number (seqno)	A 64-bit log entry identifier incremented sequentially. Each log type has a unique number space.
Action Flags (actionflags)	A bit field indicating whether the log was forwarded to Panorama.

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Field Name	Description
Device Group Hierarchy (dg_hier_level_1 to dg_hier_level_4)	A sequence of identification numbers that indicates the location of the device group within a device group hierarchy. The firewall (or virtual system) generating the log includes the identification number of each ancestor in its device group hierarchy except the shared device group (level 0), which is not included in this structure. If the log values are 12, 34, 45, and 0, it means that the log was generated by a firewall (or virtual system) that belongs to device group 45 and its ancestors are 34 and 12. To view the device group names that correspond to the value 12, 34, or 45, use one of the following methods: API query: `/api/?type=op&cmd=<show><dg-hierarchy></dg-hierarchy></show>`
Virtual System Name (vsys_name)	The name of the virtual system associated with the session; only valid on firewalls enabled for multiple virtual systems.
Device Name (device_name)	The hostname of the firewall on which the session was logged.
Virtual System ID (vsys_id)	A unique identifier for a virtual system on a Palo Alto Networks firewall.
High Resolution Timestamp (high_res timestamp)	Time in milliseconds the log was received at the management plane. The format for this new field is YYYY-MM-DDThh:ss:sssTZD: • **YYYY**—Four digit year • **MM**—Two-digit month • **DD**—Two-digit day of the month (01 through 31) • **T**—Indicator for the beginning of the timestamp • **hh**—Two-digit hour using 24-hour time (00 through 23) • **mm**—Two-digit minute (00 through 59) • **ss**—Two-digit second (00 through 60) • **sss**—One or more digits for millisecond • **TZD**—Time zone designator (+hh:mm or -hh:mm)

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Field Name	Description
	The High Resolution Timestamp is supported for logs received from managed firewalls running PAN-OS 11.1 and later releases. Logs received from managed firewalls running PAN-OS 9.1 and earlier releases display a `1969-12-31T16:00:00:000-8:00` timestamp regardless of when the log was received.
Cluster Name (cluster_name)	(11.1 and later releases) Name of the CN-Series firewall cluster.

User-ID Log Fields

Format: FUTURE_USER, Receive Time, Serial Number, Type, Threat/Content Type, FUTURE_USE, Generated Time, Virtual System, Source IP, User, Data Source Name, Event ID, Repeat Count, Time Out Threshold, Source Port, Destination Port, Data Source, Data Source Type, Sequence Number, Action Flags, Device Group Hierarchy Level 1, Device Group Hierarchy Level 2, Device Group Hierarchy Level 3, Device Group Hierarchy Level 4, Virtual System Name, Device Name, Virtual System ID, Factor Type, Factor Completion Time, Factor Number, User Group Flags, User by Source, Tag Name, High Resolution Timestamp, Origin Data Source, FUTURE_USE, Cluster Name

Field Name	Description
Receive Time (receive_time or cef-formatted-receive_time)	Time the log was received at the management plane.
Serial Number (serial)	Serial number of the firewall that generated the log.
Type (type)	Specifies the type of log; value is USERID.
Threat/Content Type (subtype)	Subtype of User-ID log; values are login, logout, register-tag, and unregister-tag. • login—User logged in. • logout—User logged out. • register-tag—Indicates a tag or tags were registered for the user. • unregister-tag—Indicates a tag or tags were unregistered for the user.
Generated Time (time_generated or cef-formatted-time_generated)	The time the log was generated on the dataplane.

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Field Name	Description
Virtual System (vsys)	Virtual System associated with the configuration log.
Source IP (ip)	Original session source IP address.
User (user)	Identifies the end user.
Data Source Name (datasourcename)	User-ID source that sends the IP (Port)-User Mapping.
Event ID (eventid)	String showing the name of the event.
Repeat Count (repeatcnt)	Number of sessions with same Source IP, Destination IP, Application, and Subtype seen within 5 seconds.
Time Out Threshold (timeout)	Timeout after which the IP/User Mappings are cleared.
Source Port (beginport)	Source port utilized by the session.
Destination Port (endport)	Destination port utilized by the session.
Data Source (datasource)	Source from which mapping information is collected.
Data Source Type (datasourcetype)	Mechanism used to identify the IP/User mappings within a data source.
Sequence Number (seqno)	Serial number of the firewall that generated the log.
Action Flags (actionflags)	A bit field indicating if the log was forwarded to Panorama.
Device Group Hierarchy (dg_hier_level_1 to dg_hier_level_4)	A sequence of identification numbers that indicate the device group's location within a device group hierarchy. The firewall (or virtual system) generating the log includes the identification number of each ancestor in its device group hierarchy. The shared device group (level 0) is not included in this structure. If the log values are 12, 34, 45, 0, it means that the log was generated by a firewall (or virtual system) that belongs to device group 45, and its ancestors are 34, and 12. To view the device group names that correspond to the value 12, 34 or 45, use one of the following methods:

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Field Name	Description
	API query: `/api/?type=op&cmd=<show><dg-hierarchy></dg-hierarchy></show>`
Virtual System Name (vsys_name)	The name of the virtual system associated with the session; only valid on firewalls enabled for multiple virtual systems.
Device Name (device_name)	The hostname of the firewall on which the session was logged.
Virtual System ID (vsys_id)	A unique identifier for a virtual system on a Palo Alto Networks firewall.
Factor Type (factortype)	Vendor used to authenticate a user when Multi Factor authentication is present.
Factor Completion Time (factorcompletiontime)	Time the authentication was completed.
Factor Number (factorno)	Indicates the use of primary authentication (1) or additional factors (2, 3).
User Group Flags (ugflags)	Displays whether the user group that was found during user group mapping. Supported values are: • User Group Found—Indicates whether the user could be mapped to a group. • Duplicate User—Indicates whether duplicate users were found in a user group. Displays `N/A` if no user group is found.
User by Source (userbysource)	Indicates the username received from the source through IP address-to-username mapping.
Tag Name (tag_name)	Name of the tag associated with the dynamic user group associated with the User Group the user is mapped to.
High Resolution Timestamp (high_res timestamp)	Time in milliseconds the log was received at the management plane. The format for this new field is YYYY-MM-DDThh:ss:sssTZD: • **YYYY**—Four digit year • **MM**—Two-digit month • **DD**—Two-digit day of the month (01 through 31) • **T**—Indicator for the beginning of the timestamp • **hh**—Two-digit hour using 24-hour time (00 through 23) • **mm**—Two-digit minute (00 through 59) • **ss**—Two-digit second (00 through 60)

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Field Name	Description
	• **sss**—One or more digits for millisecond • **TZD**—Time zone designator (+hh:mm or -hh:mm) *The High Resolution Timestamp is supported for logs received from managed firewalls running PAN-OS 11.1 and later releases. Logs received from managed firewalls running PAN-OS 9.1 and earlier releases display a `1969-12-31T16:00:00:000-8:00` timestamp regardless of when the log was received.*
Origin Data Source (origindatasource)	Source where the User-ID mapping originated.
Cluster Name (cluster_name)	(11.1 and later releases) Name of the CN-Series firewall cluster.

Decryption Log Fields

Format: FUTURE_USE, Receive Time, Serial Number, Type, Threat/Content Type, Config Version, Generate Time, Source Address, Destination Address, NAT Source IP, NAT Destination IP, Rule, Source User, Destination User, Application, Virtual System, Source Zone, Destination Zone, Inbound Interface, Outbound Interface, Log Action, Time Logged, Session ID, Repeat Count, Source Port, Destination Port, NAT Source Port, NAT Destination Port, Flags, IP Protocol, Action, Tunnel, FUTURE_USE, FUTURE_USE, Source VM UUID, Destination VM UUID, UUID for rule, Stage for Client to Firewall, Stage for Firewall to Server, TLS Version, Key Exchange Algorithm, Encryption Algorithm, Hash Algorithm, Policy Name, Elliptic Curve, Error Index, Root Status, Chain Status, Proxy Type, Certificate Serial Number, Fingerprint, Certificate Start Date, Certificate End Date, Certificate Version, Certificate Size, Common Name Length, Issuer Common Name Length, Root Common Name Length, SNI Length, Certificate Flags, Subject Common Name, Issuer Subject Common Name, Root Subject Common Name, Server Name Indication, Error, Container ID, POD Namespace, POD Name, Source External Dynamic List, Destination External Dynamic List, Source Dynamic Address Group, Destination Dynamic Address Group, High Res Timestamp, Source Device Category, Source Device Profile, Source Device Model, Source Device Vendor, Source Device OS Family, Source Device OS Version, Source Hostname, Source Mac Address, Destination Device Category, Destination Device Profile, Destination Device Model, Destination Device Vendor, Destination Device OS Family, Destination Device OS Version, Destination Hostname, Destination Mac Address, Sequence Number, Action Flags, Device Group Hierarchy Level 1, Device Group Hierarchy Level 2, Device Group Hierarchy Level 3, Device Group Hierarchy Level 4, Virtual System Name, Device Name, Virtual System ID, Application Subcategory, Application Category, Application Technology, Application Risk, Application Characteristic, Application Container, Application SaaS, Application Sanctioned State, Cluster Name

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Field Name	Description
Receive Time (receive_time or cef-formatted-receive_time)	Time the log was received at the management plane.
Serial Number (serial)	Serial number of the firewall that generated the log.
Type (type)	Specifies the type of log; value is DECRYPTION.
Threat/ContentType (subtype)	Not used in the Decryption log.
Config Version (config_ver)	The software version.
Generate Time (time_generated)	Time the log was generated on the dataplane.
Source Address (src)	Original session source IP address.
Destination Address (dst)	Original session destination IP address.
NAT Source IP (natsrc)	If Source NAT performed, the post-NAT Source IP address.
NAT Destination IP (natdst)	If Destination NAT performed, the post-NAT Destination IP address.
Rule (rule)	Security policy rule that controls the session traffic.
Source User (srcuser)	Username of the user who initiated the session.
Destination User (dstuser)	Username of the user to which the session was destined.
Application (app)	Application associated with the session.
Virtual System (vsys)	Virtual System associated with the session.
Source Zone (from)	Zone the session was sourced from.
Destination Zone (to)	Zone the session was destined to.
Inbound Interface (inbound_if)	Interface that the session was sourced from.

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Field Name	Description
Outbound Interface (outbound_if)	Interface that the session was destined to.
Log Action (logset)	Log Forwarding profile applied to the session.
Time Logged (time_received)	The time the log was received.
Session ID (sessionid)	An internal numerical identifier applied to each session.
Repeat Count (repeatcnt)	Number of sessions with the same Source IP, Destination IP, Application, and Content/Threat Type seen within 5 seconds.
Source Port (sport)	Source port utilized by the session.
Destination Port (dport)	Destination port utilized by the session.
NAT Source Port (natsport)	Post-NAT source port.
NAT Destination Port (natdport)	Post-NAT destination port.
Flags (flags)	32-bit field that provides details on session; this field can be decoded by AND-ing the values with the logged value: • 0x80000000—session has a packet capture (PCAP) • 0x40000000—option is enabled to allow a client to use multiple paths to connect to a destination host • 0x20000000—file is submitted to WildFire for a verdict • 0x10000000—enterprise credential submission by end user detected • 0x08000000— source for the flow is on the allow list and not subject to recon protection • 0x02000000—IPv6 session • 0x01000000—SSL session is decrypted (SSL Proxy) • 0x00800000—session is denied via URL filtering • 0x00400000—session has a NAT translation performed • 0x00200000—user information for the session was captured through Authentication Portal • 0x00100000—application traffic is on a non-standard destination port

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Field Name	Description
	• 0x00080000 —X-Forwarded-For value from a proxy is in the source user field • 0x00040000—log corresponds to a transaction within a http proxy session (Proxy Transaction) • 0x00020000—Client to Server flow is subject to policy based forwarding • 0x00010000—Server to Client flow is subject to policy based forwarding • 0x00008000—session is a container page access (Container Page) • 0x00002000—session has a temporary match on a rule for implicit application dependency handling. Available in PAN-OS 5.0.0 and above. • 0x00000800—symmetric return is used to forward traffic for this session • 0x00000400—decrypted traffic is being sent out clear text through a mirror port • 0x00000100—payload of the outer tunnel is being inspected
IP Protocol (proto)	IP protocol associated with the session.
Action (action)	Action taken for the session; possible values are: • allow—session was allowed by policy • deny—session was denied by policy • drop—session was dropped silently • drop ICMP—session was silently dropped with an ICMP unreachable message to the host or application • reset both—session was terminated and a TCP reset is sent to both the sides of the connection • reset client—session was terminated and a TCP reset is sent to the client • reset server—session was terminated and a TCP reset is sent to the server
Tunnel (tunnel)	Type of tunnel.
Source VM UUID (src_uuid)	The source universal unique identifier for a guest virtual machine in the VMware NSX environment.
Destination VM UUID (dst_uuid)	The destination universal unique identifier for a guest virtual machine in the VMware NSX environment.

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Field Name	Description
UUID for rule (rule_uuid)	The UUID that permanently identifies the rule.
Stage for Client to Firewall (hs_stage_c2f)	The stage of the TLS handshake from the client to the firewall, for example, Client Hello, Server Hello, Certificate, Client/Server key exchange, etc.
Stage for Firewall to Server (hs_stage_f2s)	The stage of the TLS handshake from the firewall to the server.
TLS Version (tls_version)	The version of TLS protocol used for the session.
Key Exchange Algorithm (tls_keyxchg)	The key exchange algorithm used for the session.
Encryption Algorithm (tls_enc)	The algorithm used to encrypt the session data, such as AES-128-CBC, AES-256-GCM, etc.
Hash Algorithm (tls_auth)	The authentication algorithm used for the session, for example, SHA, SHA256, SHA384, etc.
Policy Name (policy_name)	The name of the Decryption policy associated with the session.
Elliptic Curve (ec_curve)	The elliptic cryptography curve that the client and server negotiate and use for connections that use ECDHE cipher suites.
Error Index (err_index)	The type of error that occurred: Cipher, Resource, Resume, Version, Protocol, Certificate, Feature, or HSM.
Root Status (root_status)	The status of the root certificate, for example, trusted, untrusted, or uninspected.
Chain Status (chain_status)	Whether the chain is trusted. Values are: • Uninspected • Untrusted • Trusted • Incomplete
Proxy Type (proxy_type)	The Decryption proxy type, such as Forward for Forward Proxy, Inbound for Inbound Inspection, No Decrypt for undecrypted traffic, GlobalProtect, etc.

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Field Name	Description
Certificate Serial Number (cert_serial)	The unique identifier of the certificate (generated by the certificate issuer).
Certificate Fingerprint (fingerprint)	A hash of the certificate in x509 binary format.
Certificate Start Date (notbefore)	The time the certificate became valid (certificate in invalid before this time).
Certificate End Date (notafter)	The time the certificate expires (certificate becomes invalid after this time).
Certificate Version (cert_ver)	The certificate version (V1, V2, or V3).
Certificate Size (cert_size)	The certificate key size.
Common Name Length (cn_len)	The length of the subject common name.
Issuer Common Name Length (issuer_len)	The length of the issuer common name.
Root Common Name Length (rootcn_len)	The length of the root common name.
SNI Length (sni_len)	The length of the Server Name Indication (hostname).
Certificate Flags (cert_flags)	The certificate flags can return seven values: • Session is resumed (b_resume_session) • Certificate (subject) common name is truncated (b_cert_cn_truncated) • Issuer common name is truncated (b_issuer_cn_truncated) • Root common name is truncated (b_root_cn_truncated) • Server Name Indication (SNI) is truncated (b_sni_truncated) • Certificate type, RSA or ECDSA (b_cert_type) • Unused (padding3)
Subject Common Name (cn)	The domain name (the name of the server that the certificate protects).

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Field Name	Description
Issuer Common Name (issuer_cn)	The name of the organization that verified the certificate's contents.
Root Common Name (root_cn)	The name of the root certificate authority.
Server Name Indication (sni)	The hostname of the server that the client is trying to contact. Using SNIs enables a server to host multiple websites and present multiple certificates on the same IP address and TCP port because each website has a unique SNI.
Error (error)	A string showing the error that has occurred in the event.
Container ID (container_id)	A unique alphanumeric string that identifies the container if the firewall runs in a cloud container.
POD Namespace (pod_namespace)	The name of the Kubernetes pod namespace.
POD Name (pod_name)	The name of the kubernetes pod.
Source External Dynamic List (src_edl)	The name of the external dynamic list that contains the source IP address of the traffic.
Destination External Dynamic List (dst_edl)	The name of the external dynamic list that contains the destination IP address of the traffic.
Source Dynamic Address Group (src_dag)	The dynamic address group that Device-ID identifies as the source of the traffic.
Destination Dynamic Address Group (dst_dag)	The dynamic address group that Device-ID identifies as the destination for the traffic.
High Resolution Timestamp (high_res_timestamp)	Time in milliseconds the log was received at the management plane. The format for this field is YYYY-MM-DDThh:ss:sssTZD: • **YYYY**—Four digit year • **MM**—Two-digit month • **DD**—Two-digit day of the month (01 through 31) • **T**—Indicator for the beginning of the timestamp • **hh**—Two-digit hour using 24-hour time (00 through 23) • **mm**—Two-digit minute (00 through 59)

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Field Name	Description
	• **ss**—Two-digit second (00 through 60) • **sss**—One or more digits for millisecond • **TZD**—Time zone designator (+hh:mm or -hh:mm) *The High Resolution Timestamp is supported for logs received from managed firewalls running PAN-OS 10.0 and later releases. Logs received from managed firewalls running PAN-OS 9.1 and earlier releases display a `1969-12-31T16:00:00:000-8:00` timestamp regardless of when the log was received.*
Source Device Category (src_category)	The category for the device that Device-ID identifies as the source of the traffic.
Source Device Profile (src_profile)	The device profile for the device that Device-ID identifies as the source of the traffic.
Source Device Model (src_model)	The model of the device that Device-ID identifies as the source of the traffic.
Source Device Vendor (src_vendor)	The vendor of the device that Device-ID identifies as the source of the traffic.
Source Device OS Family (src_osfamily)	The operating system type for the device that Device-ID identifies as the source of the traffic.
Source Device OS Version (src_osversion)	The version of the operating system for the device that Device-ID identifies as the source of the traffic.
Source Hostname (src_host)	The hostname of the device that Device-ID identifies as the source of the traffic.
Source MAC Address (src_mac)	The MAC address for the device that Device-ID identifies as the source of the traffic.
Destination Device Category (dst_category)	The category for the device that Device-ID identifies as the destination for the traffic.
Destination Device Profile (dst_profile)	The device profile for the device that Device-ID identifies as the destination for the traffic.
Destination Device Model (dst_model)	The model of the device that Device-ID identifies as the destination for the traffic.

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Field Name	Description
Destination Device Vendor (dst_vendor)	The vendor of the device that Device-ID identifies as the destination for the traffic.
Destination Device OS Family (dst_osfamily)	The operating system type for the device that Device-ID identifies as the destination for the traffic.
Destination Device OS Version (dst_osversion)	The version of the operating system for the device that Device-ID identifies as the destination for the traffic.
Destination Hostname (dst_host)	The hostname of the device that Device-ID identifies as the destination for the traffic.
Destination MAC Address (dst_mac)	The MAC address for the device that Device-ID identifies as the destination for the traffic.
Sequence Number (seqno)	A 64-bit log entry identifier incremented sequentially; each log type has unique number space.
Action Flags (actionflags)	A bit field indicating if the log was forwarded to Panorama.
Device Group Hierarchy (dg_hier_level_1 to dg_hier_level_4)	A sequence of identification numbers that indicate the device group's location within a device group hierarchy. The firewall (or virtual system) generating the log includes the identification number of each ancestor in its device group hierarchy. The shared device group (level 0) is not included in this structure. If the log values are 12, 34, 45, 0, it means that the log was generated by a firewall (or virtual system) that belongs to device group 45, and its ancestors are 34, and 12. To view the device group names that correspond to the value 12, 34 or 45, use one of the following methods: API query: `/api/?type=op&cmd=<show><dg-hierarchy></dg-hierarchy></show>`
Virtual System Name (vsys_name)	The name of the virtual system associated with the session; only valid on firewalls enabled for multiple virtual systems.
Device Name (device_name)	The hostname of the firewall on which the session was logged.

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Field Name	Description
Virtual System ID (vsys_id)	A unique identifier for a virtual system on a Palo Alto Networks firewall.
Application Subcategory (subcategory_of_app)	The application subcategory specified in the application configuration properties.
Application Category (category_of_app)	The application category specified in the application configuration properties. Values are: • business-systems • collaboration • general-internet • media • networking • saas
Application Technology (technology_of_app)	The application technology specified in the application configuration properties. Values are: • browser-based • client-server • network-protocol • peer-to-peer
Application Risk (risk_of_app)	Risk level associated with the application (1=lowest to 5=highest).
Application Characteristic (characteristic_of_app)	Comma-separated list of applicable characteristic of the application
Application Container (container_of_app)	The parent application for an application.
Application SaaS (is_saas_of_app)	Displays `1` if a SaaS application or `0` if not a SaaS application.
Application Sanctioned State (sanctioned_state_of_app)	Displays `1` if application is sanctioned or `0` if application is not sanctioned.
Cluster Name (cluster_name)	(11.1 and later releases) Name of the CN-Series firewall cluster.

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Tunnel Inspection Log Fields

Format: FUTURE_USE, Receive Time, Serial Number, Type, Subtype, FUTURE_USE, Generated Time, Source Address, Destination Address, NAT Source IP, NAT Destination IP, Rule Name, Source User, Destination User, Application, Virtual System, Source Zone, Destination Zone, Inbound Interface, Outbound Interface, Log Action, FUTURE_USE, Session ID, Repeat Count, Source Port, Destination Port, NAT Source Port, NAT Destination Port, Flags, Protocol, Action, Severity, Sequence Number, Action Flags, Source Location, Destination Location, Device Group Hierarchy Level 1, Device Group Hierarchy Level 2, Device Group Hierarchy Level 3, Device Group Hierarchy Level 4, Virtual System Name, Device Name, Tunnel ID/IMSI, Monitor Tag/IMEI, Parent Session ID, Parent Start Time, Tunnel, Bytes, Bytes Sent, Bytes Received, Packets, Packets Sent, Packets Received, Maximum Encapsulation, Unknown Protocol, Strict Check, Tunnel Fragment, Sessions Created, Sessions Closed, Session End Reason, Action Source, Start Time, Elapsed Time, Tunnel Inspection Rule, Remote User IP, Remote User ID, Rule UUID, PCAP ID, Dynamic User Group, Source External Dynamic List, Destination External Dynamic List, High Resolution Timestamp, A Slice Differentiator, A Slice Service Type, PDU Session ID, Application Subcategory, Application Category, Application Technology, Application Risk, Application Characteristic, Application Container, Application SaaS, Application Sanctioned State, Cluster Name

Field Name	Description
Receive Time (receive_time or cef-formatted-receive_time)	Month, day, and time the log was received at the management plane.
Serial Number (serial)	Serial number of the firewall that generated the log.
Type (type)	Type of log as it pertains to the session: START or END.
Threat/Content Type (subtype)	Subtype of traffic log; values are start, end, drop, and deny • Start—session started • End—session ended • Drop—session dropped before the application is identified and there is no rule that allows the session. • Deny—session dropped after the application is identified and there is a rule to block or no rule that allows the session.
Generated Time (time_generated or cef-formatted-time_generated)	Time the log was generated on the dataplane.
Source Address (src)	Source IP address of packets in the session.

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Field Name	Description
Destination Address (dst)	Destination IP address of packets in the session.
NAT Source IP (natsrc)	If Source NAT performed, the post-NAT Source IP address.
NAT Destination IP (natdst)	If Destination NAT performed, the post-NAT Destination IP address.
Rule Name (rule)	Name of the Security policy rule in effect on the session.
Source User (srcuser)	Source User ID of packets in the session.
Destination User (dstuser)	Destination User ID of packets in the session.
Application (app)	Tunneling protocol used in the session.
Virtual System (vsys)	Virtual System associated with the session.
Source Zone (from)	Source zone of packets in the session.
Destination Zone (to)	Destination zone of packets in the session.
Inbound Interface (inbound_if)	Interface that the session was sourced from.
Outbound Interface (outbound_if)	Interface that the session was destined to.
Log Action (logset)	Log Forwarding Profile that was applied to the session.
Session ID (sessionid)	Session ID of the session being logged.
Repeat Count (repeatcnt)	Number of sessions with same Source IP, Destination IP, Application, and Subtype seen within 5 seconds.
Source Port (sport)	Source port utilized by the session.
Destination Port (dport)	Destination port utilized by the session.
NAT Source Port (natsport)	Post-NAT source port.

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Field Name	Description
NAT Destination Port (natdport)	Post-NAT destination port.
Flags (flags)	32-bit field that provides details on session; this field can be decoded by AND-ing the values with the logged value: • 0x80000000 —session has a packet capture (PCAP) • 0x02000000 —IPv6 session • 0x01000000 —SSL session was decrypted (SSL Proxy) • 0x00800000 —session was denied via URL filtering • 0x00400000 —session has a NAT translation performed (NAT) • 0x00200000 —user information for the session was captured through Authentication Portal • 0x00080000 —X-Forwarded-For value from a proxy is in the source user field • 0x00040000 —log corresponds to a transaction within a http proxy session (Proxy Transaction) • 0x00008000 —session is a container page access (Container Page) • 0x00002000 —session has a temporary match on a rule for implicit application dependency handling. Available in PAN-OS 5.0.0 and above. • 0x00000800 —symmetric return was used to forward traffic for this session
IP Protocol (proto)	IP protocol associated with the session.
Action (action)	Action taken for the session; possible values are: • Allow—session was allowed by policy • Deny—session was denied by policy • Drop—session was dropped silently • Drop ICMP—session was silently dropped with an ICMP unreachable message to the host or application • Reset both—session was terminated and a TCP reset is sent to both the sides of the connection • Reset client—session was terminated and a TCP reset is sent to the client • Reset server—session was terminated and a TCP reset is sent to the server
Severity (severity)	Severity associated with the event; values are informational, low, medium, high, critical.

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Field Name	Description
Sequence Number (seqno)	A 64-bit log entry identifier incremented sequentially; each log type has a unique number space. This field is not supported on PA-7000 Series firewalls.
Action Flags (actionflags)	A bit field indicating if the log was forwarded to Panorama.
Source Location (srcloc)	Source country or Internal region for private addresses; maximum length is 32 bytes.
Destination Location (dstloc)	Destination country or Internal region for private addresses. Maximum length is 32 bytes.
Device Group Hierarchy (dg_hier_level_1 to dg_hier_level_4)	A sequence of identification numbers that indicate the device group's location within a device group hierarchy. The firewall (or virtual system) generating the log includes the identification number of each ancestor in its device group hierarchy. The shared device group (level 0) is not included in this structure. If the log values are 12, 34, 45, 0, it means that the log was generated by a firewall (or virtual system) that belongs to device group 45, and its ancestors are 34, and 12. To view the device group names that correspond to the value 12, 34 or 45, use one of the following methods: API query: `/api/?type=op&cmd=<show><dg-hierarchy></dg-hierarchy></show>`
Virtual System Name (vsys_name)	The name of the virtual system associated with the session; only valid on firewalls enabled for multiple virtual systems.
Device Name (device_name)	The hostname of the firewall on which the session was logged.
Tunnel ID (tunnelid)	ID of the tunnel being inspected or the International Mobile Subscriber Identity (IMSI) ID of the mobile user.
Monitor Tag (monitortag)	Monitor name you configured for the Tunnel Inspection policy rule or the International Mobile Equipment Identity (IMEI) ID of the mobile device.
Parent Session ID (parent_session_id)	ID of the session in which this session is tunneled. Applies to inner tunnel (if two levels of tunneling) or inside content (if one level of tunneling) only.

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Field Name	Description
Parent Start Time (parent_start_time)	Year/month/day hours:minutes:seconds that the parent tunnel session began.
Tunnel Type (tunnel)	Type of tunnel, such as GRE or IPSec.
Bytes (bytes)	Number of bytes in the session.
Bytes Sent (bytes_sent)	Number of bytes in the client-to-server direction of the session.
Bytes Received (bytes_received)	Number of bytes in the server-to-client direction of the session.
Packets (packets)	Number of total packets (transmit and receive) for the session.
Packets Sent (pkts_sent)	Number of client-to-server packets for the session.
Packets Received (pkts_received)	Number of server-to-client packets for the session.
Maximum Encapsulation (max_encap)	Number of packets the firewall dropped because the packet exceeded the maximum number of encapsulation levels configured in the Tunnel Inspection policy rule (Drop packet if over maximum tunnel inspection level).
Unknown Protocol (unknown_proto)	Number of packets the firewall dropped because the packet contains an unknown protocol, as enabled in the Tunnel Inspection policy rule (Drop packet if unknown protocol inside tunnel).
Strict Checking (strict_check)	Number of packets the firewall dropped because the tunnel protocol header in the packet failed to comply with the RFC for the tunnel protocol, as enabled in the Tunnel Inspection policy rule (**Drop packet if tunnel protocol fails strict header check**).
Tunnel Fragment (tunnel_fragment)	Number of packets the firewall dropped because of fragmentation errors.
Sessions Created (sessions_created)	Number of inner sessions created.
Sessions Closed (sessions_closed)	Number of completed/closed sessions created.
Session End Reason (session_end_reason)	The reason a session terminated. If the termination had multiple causes, this field displays only the highest priority reason. The possible

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Field Name	Description
	session end reason values are as follows, in order of priority (where the first is highest): • threat—The firewall detected a threat associated with a reset, drop, or block (IP address) action. • policy-deny—The session matched a security rule with a deny or drop action. • decrypt-cert-validation—The session terminated because you configured the firewall to block SSL forward proxy decryption or SSL inbound inspection when the session uses client authentication or when the session uses a server certificate with any of the following conditions: expired, untrusted issuer, unknown status, or status verification time-out. This session end reason also displays when the server certificate produces a fatal error alert of type bad_certificate, unsupported_certificate, certificate_revoked, access_denied, or no_certificate_RESERVED (SSLv3 only). • decrypt-unsupport-param—The session terminated because you configured the firewall to block SSL forward proxy decryption or SSL inbound inspection when the session uses an unsupported protocol version, cipher, or SSH algorithm. This session end reason is displays when the session produces a fatal error alert of type unsupported_extension, unexpected_message, or handshake_failure. • decrypt-error—The session terminated because you configured the firewall to block SSL forward proxy decryption or SSL inbound inspection when firewall resources or the hardware security module (HSM) were unavailable. This session end reason is also displayed when you configured the firewall to block SSL traffic that has SSH errors or that produced any fatal error alert other than those listed for the decrypt-cert-validation and decrypt-unsupport-param end reasons. • tcp-rst-from-client—The client sent a TCP reset to the server. • tcp-rst-from-server—The server sent a TCP reset to the client. • resources-unavailable—The session dropped because of a system resource limitation. For example, the session could have exceeded the number of out-of-order packets allowed per flow or the global out-of-order packet queue. • tcp-fin—One host or both hosts in the connection sent a TCP FIN message to close the session. • tcp-reuse—A session is reused and the firewall closes the previous session. • decoder—The decoder detects a new connection within the protocol (such as HTTP-Proxy) and ends the previous connection. • aged-out—The session aged out.

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Field Name	Description

	• unknown—This value applies in the following situations:
	  • Session terminations that the preceding reasons do not cover (for example, a `clear session all` command).
	  • For logs generated in a PAN-OS release that does not support the session end reason field (releases older than PAN-OS 6.1), the value will be `unknown` after an upgrade to the current PAN-OS release or after the logs are loaded onto the firewall.
	  • In Panorama, logs received from firewalls for which the PAN-OS version does not support session end reasons will have a value of `unknown`.
	• n/a—This value applies when the traffic log type is not **end**.

Action Source (action_source)	Specifies whether the action taken to allow or block an application was defined in the application or in policy. The actions can be allow, deny, drop, reset- server, reset-client or reset-both for the session.
Start Time (start)	Year/month/day hours:minutes:seconds that the session began.
Elapsed Time (elapsed)	Elapsed time of the session.
Tunnel Inspection Rule (tunnel_insp_rule)	Name of the tunnel inspection rule matching the cleartext tunnel traffic.
Remote User IP (remote_user_ip)	IPv4 or IPv6 address of a remote user.
Remote User ID (remote_user_id)	IMSI identity of a remote user, and if available, one IMEI identity or one MSISDN identity.
Security Rule UUID (rule_uuid)	The UUID that permanently identifies the rule.
PCAP ID (pcap_id)	Unique packet capture ID that defines the location of the pcap file on the firewall.
Dynamic User Group Name (dynusergroup_name)	The name of the dynamic user group that contains the user who initiated the session.
Source External Dynamic List (src_edl)	The name of the external dynamic list that contains the source IP address of the traffic.

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Field Name	Description
Destination External Dynamic List (dst_edl)	The name of the external dynamic list that contains the destination IP address of the traffic.
High Resolution Timestamp (high_res timestamp)	Time in milliseconds the log was received at the management plane. The format for this new field is YYYY-MM-DDThh:ss:sssTZD: • **YYYY**—Four digit year • **MM**—Two-digit month • **DD**—Two-digit day of the month (01 through 31) • **T**—Indicator for the beginning of the timestamp • **hh**—Two-digit hour using 24-hour time (00 through 23) • **mm**—Two-digit minute (00 through 59) • **ss**—Two-digit second (00 through 60) • **sss**—One or more digits for millisecond • **TZD**—Time zone designator (+hh:mm or -hh:mm) *The High Resolution Timestamp is supported for logs received from managed firewalls running PAN-OS 11.1 and later releases. Logs received from managed firewalls running PAN-OS 9.1 and earlier releases display a `1969-12-31T16:00:00:000-8:00` timestamp regardless of when the log was received.*
A Slice Differentiator (nssai_sd)	The A Slice Differentiator of the Network Slice ID.
A Slice Service Type (nssai_sd)	The A Slice Service Type of the Network Slice ID.
PDU Session ID (pdu_session_id)	Session ID for the collection of L4 segments inside a tunnel.
Application Subcategory (subcategory_of_app)	The application subcategory specified in the application configuration properties.
Application Category (category_of_app)	The application category specified in the application configuration properties. Values are: • business-systems • collaboration • general-internet • media

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Field Name	Description
	• networking • saas
Application Technology (technology_of_app)	The application technology specified in the application configuration properties. Values are: • browser-based • client-server • network-protocol • peer-to-peer
Application Risk (risk_of_app)	Risk level associated with the application (1=lowest to 5=highest).
Application Characteristic (characteristic_of_app)	Comma-separated list of applicable characteristic of the application
Application Container (container_of_app)	The parent application for an application.
Application SaaS (is_saas_of_app)	Displays `1` if a SaaS application or `0` if not a SaaS application.
Application Sanctioned State (sanctioned_state_of_app)	Displays `1` if application is sanctioned or `0` if application is not sanctioned.
Cluster Name (cluster_name)	(11.1 and later releases) Name of the CN-Series firewall cluster.

SCTP Log Fields

Format: FUTURE_USE, Receive Time, Serial Number, Type, FUTURE_USE, FUTURE_USE, Generated Time, Source Address, Destination Address, FUTURE_USE, FUTURE_USE, Rule Name, FUTURE_USE, FUTURE_USE, FUTURE_USE, Virtual System, Source Zone, Destination Zone, Inbound Interface, Outbound Interface, Log Action, FUTURE_USE, Session ID, Repeat Count, Source Port, Destination Port, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, IP Protocol, Action, Device Group Hierarchy Level 1, Device Group Hierarchy Level 2, Device Group Hierarchy Level 3, Device Group Hierarchy Level 4, Virtual System Name, Device Name, Sequence Number, FUTURE_USE, SCTP Association ID, Payload Protocol ID, Severity, SCTP Chunk Type, FUTURE_USE, SCTP Verification Tag 1, SCTP Verification Tag 2, SCTP Cause Code, Diameter App ID, Diameter Command Code, Diameter AVP Code, SCTP Stream ID, SCTP Association End Reason, Op Code, SCCP Calling Party SSN, SCCP Calling Party Global Title, SCTP Filter, SCTP Chunks, SCTP Chunks Sent, SCTP Chunks Received, Packets, Packets Sent, Packets Received, UUID for rule, High Resolution Timestamp

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Field Name	Description
Receive Time (receive_time or cef-formatted-receive_time)	Time the log was received at the management plane.
Serial Number (serial)	Serial number of the firewall that generated the log.
Type (type)	Specifies the type of log; value is SCTP.
Generated Time (time_generated or cef-formatted-time_generated)	Time the log was generated on the dataplane.
Source Address (src)	Original session source IP address.
Destination Address (dst)	Original session destination IP address.
Rule Name (rule)	Name of the Security policy rule in effect on the session.
Virtual System (vsys)	Virtual System associated with the session.
Source Zone (from)	Zone the session was sourced from.
Destination Zone (to)	Zone the session was destined to.
Inbound Interface (inbound_if)	Interface that the session was sourced from.
Outbound Interface (outbound_if)	Interface that the session was destined to.
Log Action (logset)	Log Forwarding Profile that was applied to the session.
Session ID (sessionid)	An internal numerical identifier applied to each session.
Repeat Count (repeatcnt)	Number of sessions with same Source IP, Destination IP, Application, and Subtype seen within 5 seconds.
Source Port (sport)	Source port utilized by the session.
Destination Port (dport)	Destination port utilized by the session.
IP Protocol (proto)	IP protocol associated with the session.
Action (action)	Action taken for the session; possible values are: • allow—session was allowed by the policy

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Field Name	Description
	• deny—session was denied by the policy
Device Group Hierarchy (dg_hier_level_1 to dg_hier_level_4)	A sequence of identification numbers that indicate the device group's location within a device group hierarchy. The firewall (or virtual system) generating the log includes the identification number of each ancestor in its device group hierarchy. The shared device group (level 0) is not included in this structure. If the log values are 12, 34, 45, 0, it means that the log was generated by a firewall (or virtual system) that belongs to device group 45, and its ancestors are 34, and 12. To view the device group names that correspond to the value 12, 34 or 45, use one of the following methods: API query: `/api/?type=op&cmd=<show><dg-hierarchy></dg-hierarchy></show>`
Virtual System Name (vsys_name)	The name of the virtual system associated with the session; only valid on firewalls enabled for multiple virtual systems.
Device Name (device_name)	The hostname of the firewall on which the session was logged.
Sequence Number (seqno)	A 64-bit log entry identifier incremented sequentially; each log type has a unique number space.
SCTP Association ID (assoc_id)	An internal 56-bit numerical logical identifier applied to each SCTP association.
Payload Protocol ID (ppid)	Identifies the Payload Protocol ID (PPID) in the data chunk which triggered this event. PPID is assigned by Internet Assigned Numbers Authority (IANA).
Severity (severity)	Severity associated with the event; values are informational, low, medium, high, critical.
SCTP Chunk Type (sctp_chunk_type)	Describes the type of information contained in a chunk, such as control or data.
SCTP Event Type (sctp_event_type)	Defines the event triggered per SCTP chunk or packet when SCTP protection profile is applied to the SCTP traffic. It is also triggered by start or end of a SCTP association.
SCTP Verification Tag 1 (verif_tag_1)	Used by endpoint1 which initiates the association to verify if the SCTP packet received belongs to current SCTP association and validate the endpoint2.

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Field Name	Description
SCTP Verification Tag 2 (verif_tag_2)	Used by endpoint2 to verify if the SCTP packet received belongs to current SCTP association and validate the endpoint1.
SCTP Cause Code (sctp_cause_code)	Sent by an endpoint to specify reason for an error condition to other endpoint of same SCTP association.
Diameter App ID (diam_app_id)	The diameter application in the data chunk which triggered the event. Diameter Application ID is assigned by Internet Assigned Numbers Authority (IANA).
Diameter Command Code (diam_cmd_code)	The diameter command code in the data chunk which triggered the event. Diameter Command Code is assigned by Internet Assigned Numbers Authority (IANA)
Diameter AVP Code (diam_avp_code)	The diameter AVP code in the data chunk which triggered the event.
SCTP Stream ID (stream_id)	ID of the stream which carries the data chunk which triggered the event.
SCTP Association End Reason (assoc_end_reason)	Reason an association was terminated. If the termination had multiple causes, the highest priority reason is displayed. The possible session end reasons in descending priority are: • shutdown-from-endpoint (highest)—endpoint sends out SHUTDOWN • abort-from-endpoint—endpoint sends out ABORT • unknown (lowest)—the association aged out, or association termination reason is not covered by one of the previous reasons (for example, a clear session all command).
Op Code (op_code)	Identifies the operation code of application layer SS7 protocols, like MAP or CAP, in the data chunk which triggered the event.
SCCP Calling Party SSN (sccp_calling_ssn)	The Signaling Connection Control Part (SCCP) calling party subsystem number (SSN) in the data chunk which triggered the event.
SCCP Calling Party Global Title (sccp_calling_gt)	The Signaling Connection Control Part (SCCP) calling party global title (GT) in the data chunk which triggered the event.
SCTP Filter (sctp_filter)	Name of the filter that the SCTP chunk matched.

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Field Name	Description
SCTP Chunks (chunks)	Number of total chunks (transmit and receive) for the association.
SCTP Chunks Sent (chunks_sent)	Number of endpoint1(which initiates association)-to-endpoint2 chunks for the association.
SCTP Chunks Received (chunks_received)	Number of endpoint2-to-endpoint1(which initiates association) chunks for the association.
Packets (packets)	Number of total packets (transmit and receive) for the session.
Packets Sent (pkts_sent)	Number of client-to-server packets for the session.
Packets Received (pkts_received)	Number of server-to-client packets for the session.
UUID for rule (rule_uuid)	The UUID that permanently identifies the rule.
High Resolution Timestamp (high_res_timestamp)	Time in milliseconds the log was received at the management plane. The format for this new field is YYYY-MM-DDThh:ss:sssTZD: • **YYYY**—Four digit year • **MM**—Two-digit month • **DD**—Two-digit day of the month (01 through 31) • **T**—Indicator for the beginning of the timestamp • **hh**—Two-digit hour using 24-hour time (00 through 23) • **mm**—Two-digit minute (00 through 59) • **ss**—Two-digit second (00 through 60) • **sss**—One or more digits for millisecond • **TZD**—Time zone designator (+hh:mm or -hh:mm) *The High Resolution Timestamp is supported for logs received from managed firewalls running PAN-OS 11.1 and later releases. Logs received from managed firewalls running PAN-OS 9.1 and earlier releases display a `1969-12-31T16:00:00:000-8:00` timestamp regardless of when the log was received.*

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Authentication Log Fields

Format: FUTURE_USE, Receive Time, Serial Number, Type, Threat/Content Type, FUTURE_USE, Generated Time, Virtual System, Source IP, User, Normalize User, Object, Authentication Policy, Repeat Count, Authentication ID, Vendor, Log Action, Server Profile, Description, Client Type, Event Type, Factor Number, Sequence Number, Action Flags, Device Group Hierarchy 1, Device Group Hierarchy 2, Device Group Hierarchy 3, Device Group Hierarchy 4, Virtual System Name, Device Name, Virtual System ID, Authentication Protocol, UUID for rule, High Resolution Timestamp, Source Device Category, Source Device Profile, Source Device Model, Source Device Vendor, Source Device OS Family, Source Device OS Version, Source Hostname, Source Mac Address, Region, FUTURE_USE, User Agent, Session ID, Cluster Name

Field Name	Description
Receive Time (receive_time or cef-formatted-receive_time)	Time the log was received at the management plane.
Serial Number (serial)	Serial number of the device that generated the log.
Type (type)	Specifies the type of log; value is AUTHENTICATION.
Threat/Content Type (subtype)	Subtype of the system log; refers to the system daemon generating the log; values are crypto, dhcp, dnsproxy, dos, general, global-protect, ha, hw, nat, ntpd, pbf, port, pppoe, ras, routing, satd, sslmgr, sslvpn, userid, url-filtering, vpn.
Generated Time (time_generated or cef-formatted-time_generated)	Time the log was generated on the dataplane.
Virtual System (vsys)	Virtual System associated with the session.
Source IP (ip)	Original session source IP address.
User (user)	End user being authenticated.
Normalize User (normalize_user)	Normalized version of username being authenticated (such as appending a domain name to the username).
Object (object)	Name of the object associated with the system event.
Authentication Policy (authpolicy)	Policy invoked for authentication before allowing access to a protected resource.
Repeat Count (repeatcnt)	Number of sessions with same Source IP, Destination IP, Application, and Subtype seen within 5 seconds.

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Field Name	Description
Authentication ID (authid)	Unique ID given across primary authentication and additional (multi factor) authentication.
Vendor (vendor)	Vendor providing additional factor authentication.
Log Action (logset)	Log Forwarding Profile that was applied to the session.
Server Profile (serverprofile)	Authentication server used for authentication.
Description (desc)	Additional authentication information.
Client Type (clienttype)	Type of client used to complete authentication (such as authentication portal).
Event Type (event)	Result of the authentication attempt.
Factor Number (factorno)	Indicates the use of primary authentication (1) or additional factors (2, 3).
Sequence Number (seqno)	A 64-bit log entry identifier incremented sequentially. Each log type has a unique number space.
Action Flags (actionflags)	A bit field indicating if the log was forwarded to Panorama.
Device Group Hierarchy (dg_hier_level_1 to dg_hier_level_4)	A sequence of identification numbers that indicate the device group's location within a device group hierarchy. The firewall (or virtual system) generating the log includes the identification number of each ancestor in its device group hierarchy. The shared device group (level 0) is not included in this structure. If the log values are 12, 34, 45, 0, it means that the log was generated by a firewall (or virtual system) that belongs to device group 45, and its ancestors are 34, and 12. To view the device group names that correspond to the value 12, 34 or 45, use one of the following methods: API query: `/api/?type=op&cmd=<show><dg-hierarchy></dg-hierarchy></show>`
Virtual System Name (vsys_name)	The name of the virtual system associated with the session; only valid on firewalls enabled for multiple virtual systems.

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Field Name	Description
Device Name (device_name)	The hostname of the firewall on which the session was logged.
Virtual System ID (vsys_id)	A unique identifier for a virtual system on a Palo Alto Networks firewall.
Authentication Protocol (authproto)	Indicates the authentication protocol used by the server. For example, PEAP with GTC.
UUID for rule (rule_uuid)	The UUID that permanently identifies the rule.
High Resolution Timestamp (high_res_timestamp)	Time in milliseconds the log was received at the management plane. The format for this new field is YYYY-MM-DDThh:ss:sssTZD: • **YYYY**—Four digit year • **MM**—Two-digit month • **DD**—Two-digit day of the month (01 through 31) • **T**—Indicator for the beginning of the timestamp • **hh**—Two-digit hour using 24-hour time (00 through 23) • **mm**—Two-digit minute (00 through 59) • **ss**—Two-digit second (00 through 60) • **sss**—One or more digits for millisecond • **TZD**—Time zone designator (+hh:mm or -hh:mm) *The High Resolution Timestamp is supported for logs received from managed firewalls running PAN-OS 11.1 and later releases. Logs received from managed firewalls running PAN-OS 9.1 and earlier releases display a `1969-12-31T16:00:00:000-8:00` timestamp regardless of when the log was received.*
Source Device Category (src_category)	The category for the device that Device-ID identifies as the source of the traffic.
Source Device Profile (src_profile)	The device profile for the device that Device-ID identifies as the source of the traffic.
Source Device Model (src_model)	The model of the device that Device-ID identifies as the source of the traffic.

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Field Name	Description
Source Device Vendor (src_vendor)	The vendor of the device that Device-ID identifies as the source of the traffic.
Source Device OS Family (src_osfamily)	The operating system type for the device that Device-ID identifies as the source of the traffic.
Source Device OS Version (src_osversion)	The version of the operating system for the device that Device-ID identifies as the source of the traffic.
Source Hostname (src_host)	The hostname of the device that Device-ID identifies as the source of the traffic.
Source MAC Address (src_mac)	The MAC address for the device that Device-ID identifies as the source of the traffic.
Region (region)	The geographical region where the traffic originates.
User Agent (user_agent)	The string from the HTTP request header `User-Agent`.
Session ID (sessionid)	A string that uniquely identifies the traffic session.
Cluster Name (cluster_name)	(11.1 and later releases) Name of the CN-Series firewall cluster.

Config Log Fields

Format: FUTURE_USE, Receive Time, Serial Number, Type, Subtype, FUTURE_USE, Generated Time, Host, Virtual System, Command, Admin, Client, Result, Configuration Path, Before Change Detail, After Change Detail, Sequence Number, Action Flags, Device Group Hierarchy Level 1, Device Group Hierarchy Level 2, Device Group Hierarchy Level 3, Device Group Hierarchy Level 4, Virtual System Name, Device Name, Device Group, Audit Comment, FUTURE_USE, High Resolution Timestamp

Field Name	Description
Receive Time (receive_time or cef-formatted-receive_time)	Time the log was received at the management plane.
Serial Number (serial)	Serial number of the device that generated the log.

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Field Name	Description
Type (type)	Specifies the type of log; value is CONFIG.
Threat/Content Type (subtype)	Subtype of the configuration log; unused.
Generated Time (time_generated or cef-formatted-time_generated)	Time the log was generated on the dataplane.
Host (host)	Hostname or IP address of the client machine
Virtual System (vsys)	Virtual System associated with the configuration log
Command (cmd)	Command performed by the Admin; values are add, clone, commit, delete, edit, move, rename, set.
Admin (admin)	Username of the Administrator performing the configuration
Client (client)	Client used by the Administrator; values are Web and CLI
Result (result)	Result of the configuration action; values are Submitted, Succeeded, Failed, and Unauthorized
Configuration Path (path)	The path of the configuration command issued; up to 512 bytes in length
Before Change Detail (before-change-detail)	This field is in custom logs only; it is not in the default format. It contains the full xpath before the configuration change.
After Change Detail (after-change-detail)	This field is in custom logs only; it is not in the default format. It contains the full xpath after the configuration change.
Sequence Number (seqno)	A 64bit log entry identifier incremented sequentially; each log type has a unique number space.
Action Flags (actionflags)	A bit field indicating if the log was forwarded to Panorama.
Device Group Hierarchy (dg_hier_level_1 to dg_hier_level_4)	A sequence of identification numbers that indicate the device group's location within a device group hierarchy. The firewall (or virtual system) generating the log includes the identification number of each ancestor in its device group hierarchy. The shared device group (level 0) is not included in this structure.

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Field Name	Description
	If the log values are 12, 34, 45, 0, it means that the log was generated by a firewall (or virtual system) that belongs to device group 45, and its ancestors are 34, and 12. To view the device group names that correspond to the value 12, 34 or 45, use one of the following methods: API query: `/api/?type=op&cmd=<show><dg-hierarchy></dg-hierarchy></show>`
Virtual System Name (vsys_name)	The name of the virtual system associated with the session; only valid on firewalls enabled for multiple virtual systems.
Device Name (device_name)	The hostname of the firewall on which the session was logged.
Device Group (dg_id)	The device group the firewall belongs to if managed by a Panorama™ management server.
Audit Comment (comment)	The audit comment entered in a policy rule configuration change.
High Resolution Timestamp (high_res_timestamp)	Time in milliseconds the log was received at the management plane. The format for this new field is YYYY-MM-DDThh:ss:sssTZD: • **YYYY**—Four digit year • **MM**—Two-digit month • **DD**—Two-digit day of the month (01 through 31) • **T**—Indicator for the beginning of the timestamp • **hh**—Two-digit hour using 24-hour time (00 through 23) • **mm**—Two-digit minute (00 through 59) • **ss**—Two-digit second (00 through 60) • **sss**—One or more digits for millisecond • **TZD**—Time zone designator (+hh:mm or -hh:mm) *The High Resolution Timestamp is supported for logs received from managed firewalls running PAN-OS 10.0 and later releases. Logs received from managed firewalls running PAN-OS 9.1 and earlier releases display a `1969-12-31T16:00:00:000-8:00` timestamp regardless of when the log was received.*

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System Log Fields

Format: FUTURE_USE, Receive Time, Serial Number, Type, Content/Threat Type, FUTURE_USE, Generated Time, Virtual System, Event ID, Object, FUTURE_USE, FUTURE_USE, Module, Severity, Description, Sequence Number, Action Flags, Device Group Hierarchy Level 1, Device Group Hierarchy Level 2, Device Group Hierarchy Level 3, Device Group Hierarchy Level 4, Virtual System Name, Device Name, FUTURE_USE, FUTURE_USE, High Resolution Timestamp

Field Name	Description
Receive Time (receive_time or cef-formatted-receive_time)	Time the log was received at the management plane.
Serial Number (serial)	Serial number of the firewall that generated the log.
Type (type)	Specifies the type of log; value is SYSTEM.
Content/Threat Type (subtype)	Subtype of the system log; refers to the system daemon generating the log; values are crypto, dhcp, dnsproxy, dos, general, global-protect, ha, hw, nat, ntpd, pbf, port, pppoe, ras, routing, satd, sslmgr, sslvpn, userid, url-filtering, vpn.
Generated Time (time_generated or cef-formatted-time_generated)	Time the log was generated on the dataplane.
Virtual System (vsys)	Virtual System associated with the configuration log.
Event ID (eventid)	String showing the name of the event.
Object (object)	Name of the object associated with the system event.
Module (module)	This field is valid only when the value of the Subtype field is general. It provides additional information about the sub-system generating the log; values are general, management, auth, ha, upgrade, chassis.
Severity (severity)	Severity associated with the event; values are informational, low, medium, high, critical.
Description (opaque)	Detailed description of the event, up to a maximum of 512 bytes.
Sequence Number (seqno)	A 64-bit log entry identifier incremented sequentially; each log type has a unique number space.

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Field Name	Description
Action Flags (actionflags)	A bit field indicating if the log was forwarded to Panorama.
Device Group Hierarchy (dg_hier_level_1 to dg_hier_level_4)	A sequence of identification numbers that indicate the device group's location within a device group hierarchy. The firewall (or virtual system) generating the log includes the identification number of each ancestor in its device group hierarchy. The shared device group (level 0) is not included in this structure. If the log values are 12, 34, 45, 0, it means that the log was generated by a firewall (or virtual system) that belongs to device group 45, and its ancestors are 34, and 12. To view the device group names that correspond to the value 12, 34 or 45, use one of the following methods: API query: `/api/?type=op&cmd=<show><dg-hierarchy></dg-hierarchy></show>`
Virtual System Name (vsys_name)	The name of the virtual system associated with the session; only valid on firewalls enabled for multiple virtual systems.
Device Name (device_name)	The hostname of the firewall on which the session was logged.
High Resolution Timestamp (high_res_timestamp)	Time in milliseconds the log was received at the management plane. The format for this new field is YYYY-MM-DDThh:ss:sssTZD: • **YYYY**—Four digit year • **MM**—Two-digit month • **DD**—Two-digit day of the month (01 through 31) • **T**—Indicator for the beginning of the timestamp • **hh**—Two-digit hour using 24-hour time (00 through 23) • **mm**—Two-digit minute (00 through 59) • **ss**—Two-digit second (00 through 60) • **sss**—One or more digits for millisecond • **TZD**—Time zone designator (+hh:mm or -hh:mm)

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Field Name	Description
	The High Resolution Timestamp is supported for logs received from managed firewalls running PAN-OS 11.1 and later releases. Logs received from managed firewalls running PAN-OS 9.1 and earlier releases display a `1969-12-31T16:00:00.000-8:00` timestamp regardless of when the log was received.

Correlated Events Log Fields

Format: FUTURE_USE, Receive Time, Serial Number, Type, Content/Threat Type, FUTURE_USE, Generated Time, Source Address. Source User, Virtual System, Category, Severity, Device Group Hierarchy Level 1, Device Group Hierarchy Level 2, Device Group Hierarchy Level 3, Device Group Hierarchy Level 4, Virtual System Name, Device Name, Virtual System ID, Object Name, Object ID, Evidence

Field Name	Description
Receive Time (receive_time or cef-formatted-receive_time)	Time the log was received at the management plane.
Serial Number (serial)	Serial number of the device that generated the log.
Type (type)	Specifies the type of log; value is CORRELATION.
Content/Threat Type (subtype)	Subtype of the system log; refers to the system daemon generating the log; values are crypto, dhcp, dnsproxy, dos, general, global-protect, ha, hw, nat, ntpd, pbf, port, pppoe, ras, routing, satd, sslmgr, sslvpn, userid, url-filtering, vpn.
Generated Time (time_generated or cef-formatted-time_generated)	Time the log was generated on the dataplane.
Source Address (src)	IP address of the user who initiated the event.
Source User (srcuser)	Username of the user who initiated the event.
Virtual System (vsys)	Virtual System associated with the configuration log.
Category (category)	A summary of the kind of threat or harm posed to the network, user, or host.

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Field Name	Description
Severity (severity)	Severity associated with the event; values are informational, low, medium, high, critical.
Device Group Hierarchy (dg_hier_level_1 to dg_hier_level_4)	A sequence of identification numbers that indicate the device group's location within a device group hierarchy. The firewall (or virtual system) generating the log includes the identification number of each ancestor in its device group hierarchy. The shared device group (level 0) is not included in this structure. If the log values are 12, 34, 45, 0, it means that the log was generated by a firewall (or virtual system) that belongs to device group 45, and its ancestors are 34, and 12. To view the device group names that correspond to the value 12, 34 or 45, use one of the following methods: API query: `/api/?type=op&cmd=<show><dg-hierarchy></dg-hierarchy></show>`
Virtual System Name (vsys_name)	The name of the virtual system associated with the session; only valid on firewalls enabled for multiple virtual systems.
Device Name (device_name)	The hostname of the firewall on which the session was logged.
Virtual System ID (vsys_id)	A unique identifier for a virtual system on a Palo Alto Networks firewall.
Object Name (objectname)	Name of the correlation object that was matched on.
Object ID (object_id)	Name of the object associated with the system event.
Evidence (evidence)	A summary statement that indicates how many times the host has matched against the conditions defined in the correlation object. For example, Host visited known malware URl (19 times).

GTP Log Fields

Format: FUTURE_USE, Receive Time, Serial Number, Type, Threat/Content Type, FUTURE_USE, Generated Time, Source Address, Destination Address, FUTURE_USE, FUTURE_USE, Rule Name, FUTURE_USE, FUTURE_USE, Application, Virtual System, Source Zone, Destination Zone, Inbound Interface, Outbound Interface, Log Action, FUTURE_USE, Session ID, FUTURE_USE, Source Port, Destination Port, FUTURE_USE, FUTURE_USE, FUTURE_USE, Protocol, Action, GTP Event Type, MSISDN, Access Point Name, Radio Access Technology, GTP Message Type, End User IP Address, Tunnel Endpoint Identifier1, Tunnel Endpoint Identifier2, GTP Interface,

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GTP Cause, Severity, Serving Country MCC, Serving Network MNC, Area Code, Cell ID, GTP Event Code, FUTURE_USE, FUTURE_USE, Source Location, Destination Location, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, Tunnel ID/IMSI, Monitor Tag/IMEI, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, FUTURE_USE, Start Time, Elapsed Time, Tunnel Inspection Rule, Remote User IP, Remote User ID, UUID for rule, PCAP ID, High Resolution Timestamp, A Slice Service Type, A Slice Differentiator, Application Subcategory, Application Category, Application Technology, Application Risk, Application Characteristic, Application Container, Application SaaS, Application Sanctioned State

Field Name	Description
Receive Time (receive_time or cef-formatted-receive_time)	Month, Day and time the log was received at the management plane.
Serial Number (serial)	Serial number of the firewall that generated the log.
Type (type)	Specifies the type of log; value is GTP.
Threat/Content Type (subtype)	Subtype of traffic log; values are start, end, drop, and deny • Start—session started • End—session ended • Drop—session dropped before the application is identified and there is no rule that allows the session. • Deny—session dropped after the application is identified and there is a rule to block or no rule that allows the session.
Generated Time (time_generated or cef-formatted-time_generated)	Time the log was generated on the dataplane.
Source Address (src)	Source IP address of packets in the session.
Destination Address (dst)	Destination IP address of packets in the session.
Rule Name (rule)	Name of the Security policy rule in effect on the session.
Application (app)	Tunneling protocol used in the session.
Virtual System (vsys)	Virtual System associated with the session.
Source Zone (from)	Source zone of packets in the session.
Destination Zone (to)	Destination zone of packets in the session.

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Field Name	Description
Inbound Interface (inbound_if)	Interface that the session was sourced from.
Outbound Interface (outbound_if)	Interface that the session was destined to.
Log Action (logset)	Log Forwarding Profile that was applied to the session.
Session ID (sessionid)	Session ID of the session being logged.
Source Port (sport)	Source port utilized by the session.
Destination Port (dport)	Destination port utilized by the session.
IP Protocol (proto)	IP protocol associated with the session.
Action (action)	Action taken for the session; possible values are: • allow—session was allowed by policy • deny—session was denied by policy
GTP Event Type (event_type)	Defines event triggered by a GTP message when checks in GTP protection profile are applied to the GTP traffic. Also triggered by the start or end of a GTP session.
MSISDN (msisdn)	Service identity associated with the mobile subscriber composed of a Country Code, National Destination Code and a Subscriber. Consists of decimal digits (0-9) only with a maximum of 15 digits.
Access Point Name (apn)	Reference to a Packet Data Network Data Gateway (PGW)/ Gateway GPRS Support Node in a mobile network. Composed of a mandatory APN Network Identifier and an optional APN Operator Identifier.
Radio Access Technology (rat)	Type of technology used for radio access. For example, EUTRAN, WLAN, Virtual, HSPA Evolution, GAN and GERAN.
GTP Message Type (msg_type)	Indicates the GTP message type.
End IP Address (end_ip_adr)	IP address of a mobile subscriber allocated by a PGW/GGSN.
Tunnel Endpoint Identifier1 (teid1)	Identifies the GTP tunnel in the network node. TEID1 is the first TEID in the GTP message.

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Field Name	Description
Tunnel Endpoint Identifier2 (teid2)	Identifies the GTP tunnel in the network node. TEID2 is the second TEID in the GTP message.
GTP Interface (gtp_interface)	3GPP interface from which a GTP message is received.
GTP Cause (cause_code)	GTP cause value in logs responses which contain an Information Element that provides information about acceptance or rejection of GTP requests by a network node.
Severity (severity)	Severity associated with the event; values are informational, low, medium, high, critical.
Serving Network MCC (mcc)	Mobile country code of serving core network operator.
Serving Network MNC (mnc)	Mobile network code of serving core network operator.
Area Code (area_code)	Area within a Public Land Mobile Network (PLMN).
Cell ID (cell_id)	Base station within an area code.
GTP Event Code (event_code)	Event code describing the GTP event.
Source Location (srcloc)	Source country or Internal region for private addresses; maximum length is 32 bytes.
Destination Location (dstloc)	Destination country or Internal region for private addresses; maximum length is 32 bytes.
Tunnel ID/IMSI (imsi)	International Mobile Subscriber Identity (IMSI) is a unique number allocated to each mobile subscriber in the GSM/ UMTS/EPS system. IMSI shall consist of decimal digits (0 through 9) only and maximum number of digits allowed are 15.
Monitor Tag/IMEI (imei)	International Mobile Equipment Identity (IMEI) is a unique 15 or 16 digit number allocated to each mobile station equipment.
Start Time (start)	Time of session start.
Elapsed Time (elapsed)	Elapsed time of the session.
Tunnel Inspection Rule (tunnel_insp_rule)	Name of the tunnel inspection rule matching the cleartext tunnel traffic
Remote User IP (remote_user_ip)	IPv4 or IPv6 address used by a remote user.

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Field Name	Description
Remote User ID (remote_user_id)	IMSI identity of a remote user, and if available, one IMEI identity and/or one MSISDN identity.
UUID for rule (rule_uuid)	Universally Unique ID for rule.
PCAP ID (pcap_id)	Unique packet capture ID that is used to locate the pcap file saved on the firewall.
High Resolution Timestamp (high_res_timestamp)	Time in milliseconds the log was received at the management plane. The format for this new field is YYYY-MM-DDThh:ss:sssTZD: • **YYYY**—Four digit year • **MM**—Two-digit month • **DD**—Two-digit day of the month (01 through 31) • **T**—Indicator for the beginning of the timestamp • **hh**—Two-digit hour using 24-hour time (00 through 23) • **mm**—Two-digit minute (00 through 59) • **ss**—Two-digit second (00 through 60) • **sss**—One or more digits for millisecond • **TZD**—Time zone designator (+hh:mm or -hh:mm) *The High Resolution Timestamp is supported for logs received from managed firewalls running PAN-OS 11.1 and later releases. Logs received from managed firewalls running PAN-OS 9.1 and earlier releases display a `1969-12-31T16:00:00:000-8:00` timestamp regardless of when the log was received.*
A Slice Service Type (nsdsai_sst)	The A Slice Service Type of the Network Slice ID.
A Slice Differentiator (nsdsai_sd)	The A Slice Differentiator of the Network Slice ID.
Application Subcategory (subcategory_of_app)	The application subcategory specified in the application configuration properties.
Application Category (category_of_app)	The application category specified in the application configuration properties. Values are: • business-systems • collaboration

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Field Name	Description
	• general-internet • media • networking • saas
Application Technology (technology_of_app)	The application technology specified in the application configuration properties. Values are: • browser-based • client-server • network-protocol • peer-to-peer
Application Risk (risk_of_app)	Risk level associated with the application (1=lowest to 5=highest).
Application Characteristic (characteristic_of_app)	Comma-separated list of applicable characteristic of the application
Application Container (container_of_app)	The parent application for an application.
Application SaaS (is_saas_of_app)	Displays `1` if a SaaS application or `0` if not a SaaS application.
Application Sanctioned State (sanctioned_state_of_app)	Displays `1` if application is sanctioned or `0` if application is not sanctioned.
Application Subcategory (subcategory_of_app)	The application subcategory specified in the application configuration properties.

Audit Log Fields

Where Can I Use This?	What Do I Need?
• Next-Gen Firewall • Panorama® management server	❑ Support license ❑ (Panorama) Device management license

Format: Serial Number, Generate Time, Threat/Content Type, FUTURE_USE, Event ID, Object, CLI Command, Severity

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Field Name	Description
Serial Number	Serial number of the firewall or Panorama that generated the log.
Generate Time	Time the log was generated on the dataplane.
Threat/Content Type (subtype)	Specifies the type of log; value is AUDIT. Audit logs are a subytpe of System logs.
Event ID	Source of the command that generated the audit log. Values include the following as a source of the command: • `cli`—Firewall or Panorama command line. • `gui`—Navigation in the firewall or Panorama web interface. • `gui-op`—Operational command from the firewall or Panorama web interface. • `gnmi`—OpenConfig plugin. • `rest`—PAN-OS REST API.
Object	Name of the administrator which executed the command that generated the log.
CLI Command	Command executed that generated the log.
Severity	Completion status for the command that generated the log; value can be none, success, or failure.

Syslog Severity

The syslog severity is set based on the log type and contents.

Log Type/Severity	Syslog Severity
Traffic	Info
Config	Info
Threat/System—Informational	Info
Threat/System—Low	Notice
Threat/System—Medium	Warning
Threat/System—High	Warning

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Log Type/Severity	Syslog Severity
Threat/System—Critical	Critical

Custom Log/Event Format

Where Can I Use This?	What Do I Need?
• NGFW	For Strata Cloud Manager managed NGFWs: ❑ Strata Cloud Manager Pro

To facilitate the integration with external log parsing systems, the firewall allows you to customize the log format; it also allows you to add custom *Key: Value* attribute pairs. Custom message formats can be configured under:

- (PAN-OS & Panorama)**Device** > **Server Profiles** > **Syslog** > **Syslog Server Profile** > **Custom Log Format**.
- (Strata Cloud Manager)

  Manage > **Configuration** > **Objects** > **SyslogConfiguration** > **NGFW and Prisma Access** > **Objects** > **Syslog** and Add Syslog to create a new Syslog server profile or select an existing Syslog server profile.

  When editing the Syslog server profile, select **Custom Log Format** to customize the log format forwarded to the syslog server.

To achieve ArcSight Common Event Format (CEF) compliant log formatting, refer to the CEF Configuration Guide.

Escape Sequences

Any field that contains a comma or a double-quote is enclosed in double quotes. Furthermore, if a double-quote appears inside a field it is escaped by preceding it with another double-quote. To maintain backward compatibility, the Misc field in threat log is always enclosed in double-quotes.

Syslog Severity Reference

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

Syslog messages provide crucial insights into the health, performance, and security of a system. To help you prioritize and understand these events, syslog messages are categorized by severity levels. This categorization is essential for effective monitoring, troubleshooting, and incident response. Understanding and properly configuring Syslog message severity levels are fundamental for effective system administration. It allows organizations to filter out noise, focus on critical

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events, and implement automated alerts based on the severity, ensuring timely responses to potential threats and operational issues.

The different syslog message severity levels are as follows:

Informational System Log Messages. These messages are not critical and provide routine announcements of events, such as configuration changes. These messages help you track the behaviour of the system.

Low Severity System Log Messages. These messages represent minor issues that might not need immediate attention but are worth noting.

Medium Severity System Log Messages. These messages indicate issues that are slightly more significant than the low severity ones, yet are not very critical. They might not lead to any immediate crisis, but are worth noting for timely investigation to prevent critical failures.

High Severity System Log Messages. These messages indicate that something has gone wrong and needs immediate attention. While the entire system might not be down, a specific function or area is clearly experiencing difficulties, and it needs to be fixed to ensure everything runs smoothly.

Critical Severity System Log Messages. These messages are the alarms that signify a major crisis has occurred. These messages demand immediate and urgent action.

Informational System Log Messages

E-Log

Log Tags:

- audit
- auth
- bfd
- clusterd
- ddns
- debug
- dhcp
- dns-security
- dnsproxy
- dynamic-updates
- fips
- general
- hw
- ipv6nd
- lacp
- lldp
- monitoring

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- nat
- ntpd
- panorama-check
- pbf
- port
- pppoe
- ras
- resctrl
- routing
- satd
- sched-push
- sdwan
- ssh
- sslmgr
- syslog
- tls
- url-filtering
- userid
- vm
- vpn
- wildfire
- wildfire-appliance

audit

Event ID	Description
api	<cmd>
cli	<cmd>
cli	<config command>
api	<config command>
gnmi	<config command>
gui-op	<config command>

auth

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Event ID	Description
cas-message	(profile id:<id>)<message>
auth-fail	Time clock does not match that on KDC server at '<name>' (code: <id>)
auth-fail	User '<name>' does not exist on KDC server '<name>' (code: <id>)
auth-fail	Wrong realm: '<name>' (code: <id>)
auth-fail	Username and password do not match, preauth failed (code: <id>)
	Kerberos error: <error> (code: <id>)
auth-fail	When authenticating user "<name>", KDC Spoofing attack is detected by krb5_verify_init_creds() (krb5 error code: <id>)
auth-success	Admin <name> account has been restored - lockout timer expired.
user-password-change-success	When authenticating user '<name>' <remotehost>, a less secure authentication method <proto> is used. Please migrate to PEAP or EAP-TTLS. Authentication Profile '<name>', vsys '<name>', Server Profile '<name>', Server Address '<ip>'
auth-fail	Certificate validation failed for user '<name>'. <error>
auth-success	Certificate validated for user '<user>'. <error> auth profile '<name>', vsys '<id>', reply message '<msg>' From: <name>.
user-password-change-success	Kerberos SSO authenticated for user '<name>'. realm '<name>', EAP outer identity '<name>, inner identity '<name>', auth profile '<name>', vsys '<id>', server profile '<name>', server address '<addr>', admin role '<name>', access domain '<name>', reply message '<msg>' From: <name>.
auth-success	Kerberos SSO authenticated for user '<name>'. realm '<name>', EAP outer identity '<name>, inner identity '<name>', auth profile '<name>', vsys '<id>', server profile '<name>',

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Event ID	Description
	server address '<addr>', admin role '<name>', access domain '<name>', reply message '<msg>' From: <name>.
user-password-change-success	SAML SSO authenticated for user '<name>'. realm '<name>', EAP outer identity '<name>, inner identity '<name>', auth profile '<name>', vsys '<id>', server profile '<name>', server address '<addr>', admin role '<name>', access domain '<name>', reply message '<msg>' From: <name>.
auth-success	SAML SSO authenticated for user '<name>'. realm '<name>', EAP outer identity '<name>, inner identity '<name>', auth profile '<name>', vsys '<id>', server profile '<name>', server address '<addr>', admin role '<name>', access domain '<name>', reply message '<msg>' From: <name>.
user-password-change-success	CAS SSO authenticated for user '<name>'. realm '<name>', EAP outer identity '<name>, inner identity '<name>', auth profile '<name>', vsys '<id>', server profile '<name>', server address '<addr>', admin role '<name>', access domain '<name>', reply message '<msg>' From: <name>.
auth-success	CAS SSO authenticated for user '<name>'. realm '<name>', EAP outer identity '<name>, inner identity '<name>', auth profile '<name>', vsys '<id>', server profile '<name>', server address '<addr>', admin role '<name>', access domain '<name>', reply message '<msg>' From: <name>.
user-password-change-success	authenticated for user '<name>'. realm '<name>', EAP outer identity '<name>, inner identity '<name>', auth profile '<name>', vsys '<id>', server profile '<name>', server address '<addr>', admin role '<name>', access domain '<name>', reply message '<msg>' From: <name>.
auth-success	authenticated for user '<name>'. realm '<name>', EAP outer identity '<name>, inner identity '<name>', auth profile '<name>', vsys '<id>', server profile '<name>', server address '<addr>', admin role '<name>', access

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Event ID	Description
	domain '<name>', reply message '<msg>' From: <name>.
cas-client-redirect	Client '<name>' redirected to '<url>' with auth_session_id '<id>'
cas-token-received	Received CAS token from client '<name>' from '<url>' with auth_session_id '<id>'
cas-token-parse-error	Failed to parse CAS token from client '<host>' from '<url>' with auth_session_id '<id>' : <message>
cas-token-validated	Validated CAS token from client '<name>' from '<url>' with auth_session_id '<id>' and username '<name>'
cas-mfa-info	MFA info from client '<name>' from '<url>' with auth_session_id '<id>' and username '<name>' : <info>
saml-client-redirect	Client '<name>' redirected to '<url>' for authentication profile '<profile>'
saml-idp-activity	Received SAML Assertion from '<name>' from client '<name>'
saml-signature-validated	SAML Assertion: signature is validated against IdP certificate (subject '<name>') for user '<name>'
idp-initiated-log-out-success	SAML Single Log out initiated for user '<name>' from '<name>', Auth profile: '<name>', Virtual System: '<name>', Server profile: '<name>', IdP entityID: '<id>'
sp-initiated-log-out-success	SAML Single Log out initiated for user '<name>' from '<name>', Auth profile: '<name>', Virtual System: '<name>', Server profile: '<name>', IdP entityID: '<id>'
auth-fail	Server certificate: '<name>' is invalid, its name does not match the host name '<name>'
auth-fail	Server certificate: '<name>' is invalid for server '<name>': <error>

bfd

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Event ID	Description
session-state-change	BFD state changed to <name> for BFD session <name> to neighbor <name> on interface <name>. Protocol: <name>

clusterd

Event ID	Description
cluster-cfg-mode	Cluster node mode is changed.
cluster-config-p1-success	Cluster daemon configuration load phase-1 succeeded.
cluster-config-p1-abort	Cluster daemon configuration load phase-1 aborted.
cluster-config-p2-success	Cluster daemon configuration load phase-2 succeeded.
cluster-self-join	Local node joined cluster:
cluster-service-ready	Cluster service is ready.
cluster-service-up	Cluster service up:
cluster-split-brain-enter	Cluster enters split-brain mode.
cluster-split-brain-leave	Cluster left split-brain mode.
cluster-engine-start	Cluster engine will be started for:
cluster-daemon-start	Cluster daemon is ready.
cluster-daemon-exit	Cluster daemon has exited.
cluster-daemon-init	Cluster daemon is initializing.

ddns

Event ID	Description
ddns-remove	Interface <name> DDNS config for host <host> to <label> removed. Please manually remove from DDNS service provider.

debug

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Event ID	Description
packet-diag-log	Packet-diag logging has been enabled
packet-diag-log	Packet-diag logging has been disabled

dhcp

Event ID	Description
if-update-ok	DHCP <desc>: interface <name>, dhcp server: <name>
if-release-trigger	DHCP <name>: interface <name>, ip <ip> netmask <mask> dhcp server: <name>
if-renew-trigger	DHCP <name>: interface <name>, ip <ip> netmask <mask> dhcp server: <name>
if-update-fail	DHCP client could not clear IP address on interface:<name> due to: Error in updating interface/route table
if-update-fail	DHCP client could not obtain IP address on interface:<name> due to: Error in updating interface/route table
if-update-fail	DHCP client could not obtain IP address on interface:<name> due to: Error in updating interface/route table after HA sync from peer
if-release-trigger	<dhcp_log_event>
if-renew-trigger	<dhcp_log_event>
if-update-ok	<dhcp_log_event>
if-rcv-nak	<dhcp_log_event>
if-duplicate-ip-intf	<dhcp_log_event>
if-duplicate-ip-remote	<dhcp_log_event>
if-update-fail	DHCP client could not obtain IP address on interface:<name> due to: Error in updating interface/route table

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Event ID	Description
if-update-fail	DHCP client could not clear IP address on interface:<name> due to: Error in updating interface/route table
relay-on	DHCP relay on
relay6-on	DHCPv6 relay on
lease-end	DHCP lease ended
lease-start	DHCP lease started
server-auto-probe-off	DHCP server auto-probe finished
server-auto-probe-on	DHCP server auto-probe finished
server-on	DHCP server auto-probe finished
if-inherit	DHCP server on interface: <name> inherited following values from dynamic interface: <name>: <server>
if-update-fail	DHCP client could not obtain IP address on interface index:<num> due to: Error in updating interface/route table

dns-security

Event ID	Description
PAN_ELOG_EVENT_DNSSEC_CACHE_SUCCESS	DNS signature initialization from file storage successful.

dnsproxy

Event ID	Description
if-add	Interface <name> added to DNS proxy object:<obj>
if-del	Interface <name> deleted from DNS proxy object:<obj>
if-inherit	DNS Proxy object: <name> inherited following values from dynamic interface:

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Event ID	Description
	<name>: Primary DNS: <name> Secondary DNS: <name>
cache-cleared	All DNS Proxy cache entries were cleared
object-enable	Dnsproxy object:<name> was enabled.
object-enable	Dnsproxy object:<name> was disabled.

dynamic-updates

Event ID	Description
palo-alto-networks-message	<message>

fips

Event ID	Description
fips-selftest	FIPS Mode Self-test <description> ..... failed
fips-selftest	FIPS-CC Mode Self-test <description> ..... failed
fips-selftest	FIPS Mode Enabled Successfully

general

Event ID	Description
general	Retrieved CRL from "<name>" with crl_next_update = <name>
general	Slot s<num>: Application Pod '<namespace> : <name>:<interface>' using interfaces eth<num< and eth<num>
general	Slot s<num>: Application Pod '<namespace> : <name>:<interface>' releasing interfaces eth<num< and eth<num>
general	Machine Learning engine for <name> started
general	Reconnect to MLAV cloud, enable all machine Learning engines

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Event ID	Description
general	<type> job was successfully reverted. Completion time=<time>. JobId=<id>. User: <name>
wf-real-time-enabled	WildFire Real-time feature enabled
general	Evtmgr: Client=<id>[<devid>] msg=<msg> code=<num> socket <num>
general	Request made to <name> server is successful

hw

Event ID	Description
fan-removed	Fan Tray #<num> removed
fan-inserted	Fan Tray #<num> inserted
ps-inserted	Power Supply #<num> inserted
Thermal Failure	I2C Failure: Forcing the fan controler to run at maximum speed.\n"Setting the node [force] to pan_true\n
Thermal Failure	I2C connection restored. Forcing fans to revert their normal speed.\n"Setting the node [force] to pan_false\n
Thermal Failure	I2C connection restored. Forcing fans to revert their normal speed.\n"Setting the node [force] to pan_false\n
slot-up	Slot <id> (<model>) detects Session Distribution Policy is no longer ingress-slot. Enabling DPC.
bootstrap-success	Bootstrap successfully completed "sw-version: <version>; app-version: <version>; threat-version: <version>
bootstrap-media-prep-success	<username>: Successfully prepared USB using bundle <file>

ipv6nd

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Event ID	Description
duplicated-IPv6-address-found	IPv6 address <address> on interface <name> is duplicate.

lacp

Event ID	Description
lacp-up	LACP interface <name> moved into AE-group <name>.

lldp

Event ID	Description
mib changed	Update: LLDP Update: Sent update for TLV <name> on local interface: <index>
mib changed	Update: Received change on local interface <name>

monitoring

Event ID	Description
deviating-device	Deviating device: <name>, Serial: <serial>, Object: <name> <nest>, Metric: <name>, Value: <value>

N/A

Event ID	Description
N/A	Create audit logs
N/A	test file

nat

Event ID	Description
fqdn-add	Vsys <id> NAT rule <name> FQDN <key> add IP entry <ip>

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Event ID	Description
fqdn-del	Vsys <id> NAT rule <name> FQDN <key> delete IP entry <ip>

ntpd

Event ID	Description
sync	NTP sync to server <address>
time-learn	NTP time learnt from <time>; New time is: <time> and old time was <time>
restart	NTP restart synchronization performed
time-learn	NTP time learnt; New time is: <time>

panorama-check

Event ID	Description
panorama-check-test	JobId=<id>: <message>
panorama-check-skip	JobId=<id>: Skipping connection checks for <name>/<name> since the IP was changed.
panorama-check-skip	JobId=<id>: Skipping connection check for <name> since the panorama is not actively connected.
panorama-check-auto-revert	<type> job was successfully reverted. Completion time=<time>. JobId=<id>. User: <name>

pbf

Event ID	Description
nh-up	Vsys <id> PBF rule <name> nexthop is UP
nh-down	Vsys <id> PBF rule <name> nexthop is DOWN
nh-down	Vsys <id> PBF rule <name> is Bypassed
nh-up	Vsys <id> PBF rule <name> is Normal

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Event ID	Description
pbf-fqdn-change	Vsys <id> PBF rule <name> nexthop FQDN <key> IPv4 is changed "from <ip> to <ip>
pbf-fqdn-change	Vsys <id> PBF rule <name> nexthop FQDN <key> IPv6 is changed "from <ip> to <ip>

port

Event ID	Description
link-change	Port HSCI: Up <type> duplex
link-change	Port HSCI: Down <type> duplex
link-change	Port HA1-b: Up <type> duplex
link-change	Port HA1-b: Down <type> duplex
link-change	Port HA2: Up <type> duplex
link-change	Port HA2: Down <type> duplex
sdwan-link-change	Port <port>: Up <type> duplex
link-change	Port <port>: Down <type> duplex
sdwan-link-change	ethernet<num>/<num>: Up <type> duplex
link-change	ethernet<num>/<num>: Down <type> duplex
sdwan-link-change	Port <port>: MAC Up
link-change	Port <port>: MAC Down
nonsupp-forced	ethernet<num>/<num>: trying to force mode <type> not supported, using autoneg
link-change	Port MGT: Up <type>
link-change	Port <interface>: Up <type>
link-change	Port <interface>: Down <type>

pppoe

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Event ID	Description
connect-fail	PPPoE session failed to connect for user:<name> on interface:<name>. Reason: <reason>
connect	PPPoE session was connected for user:<name> on interface:<name> to AC:<name>, mac address: <mac>, session id:<id>, IP Address negotiated: <ip>
if-update-fail	PPPoE session connected for user:<name> on interface:<name> but updating interface/ routing table failed.
connect-fail	PPPoE session failed to connect for user:<name> on interface:<name>. Reason: No PPPoE Offer received
initiate	PPPoE session was initiated for user:<name> on interface:<name>
connect-fail	PPPoE session failed to connect for user:<name> on interface:<name>. Reason: No PPPoE Confirm received
terminate	PPPoE session was terminated for user:<name> on interface:<name> to AC:<name>, mac address: <mac>, session id:<id>
terminate	PPPoE session was terminated for user:<name> on interface:<name> to AC:<name>, mac address: <mac>, session id:<id>

ras

Event ID	Description
rasmgr-config-p1-success	RASMGR daemon configuration load phase-1 succeeded.
rasmgr-config-p1-abort	RASMGR daemon configuration load phase-1 aborted.
rasmgr-config-p2-success	RASMGR daemon configuration load phase-2 succeeded.

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Event ID	Description
rasmgr-ha-full-sync-done	RASMGR daemon sync all user info to HA peer exit.
rasmgr-ha-full-sync-done	RASMGR daemon sync all user info to HA peer exit.
rasmgr-flow-full-sync-start	RASMGR daemon sync all user info to Flow started.
rasmgr-daemon-exit	RASMGR daemon has exited.
rasmgr-daemon-init	RASMGR daemon is initializing.
rasmgr-daemon-start	RASMGR daemon is ready.

resctrl

Event ID	Description
mem-usage-normal	Memory usage is normal

routing

Event ID	Description
routed-OSPF-stop-helper-mode	OSPF stopped helper mode for a restarting neighbor. Restarting neighbor router ID <name> neighbor IP address <ip>. Reason: <reason>
routed-ECMP	ECMP maximum path changed to <num> in virtual router <name>.
routed-ECMP	ECMP enabled in virtual router <name>.
routed-ECMP	ECMP disabled in virtual router <name>.
routed-config-p1-success	Route daemon configuration load phase-1 succeeded.
routed-config-p2-success	Route daemon configuration load phase-2 succeeded.
routed-static-fqdn-changed	Routed static fqdn mapping is changed
routed-bgp-fqdn-changed	Routed BGP fqdn mapping is changed

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Event ID	Description
routed-ECMP	ECMP maximum path changed to <num> in logical router <name>.
routed-ECMP	ECMP enabled in logical router <name>.
routed-ECMP	ECMP disabled in logical router <name>.
routed-ECMP	ECMP load balancing algorithm changed to <name> in logical router <name>.
routed-ECMP	ECMP symmetric return enabled in logical router <name>.
routed-ECMP	ECMP symmetric return disabled in logical router <name>.
routed-ECMP	ECMP strict source path enabled in logical router <name>.
routed-ECMP	ECMP strict source path disabled in logical router <name>.
routed-fib-sync-peer-backup	FIB HA sync started when peer device becomes passive.
routed-fib-sync-self-master	FIB HA sync started when local device becomes master.
routed-fib-sync-peer-backup	FIB HA sync started when peer device becomes passive.
routed-fib-sync-self-master	FIB HA sync started when local device becomes master.
routed-daemon-init	Route daemon is initializing.
routed-daemon-start	Route daemon is ready.
routed-daemon-exit	Route daemon has exited.
routed-BGP-refresh-sent	ROUTE REFRESH message sent to a BGP peer.
routed-BGP-ribin-recalc	An RIB-In is being recalculated as a result of changed import policy.
routed-BGP-peer-enter-established	BGP peer session enters established state.

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Event ID	Description
routed-BGP-peer-mp-extension-negotiate	BGP peer MP extension negotiation.
routed-IGMP-wrong-version	Wrong IGMP query version
routed-OSPF-neighbor-full	OSPF full adjacency established with neighbor.
routed-OSPF-neighbor-2dir	OSPF two-way communication established with neighbor.
routed-OSPF-neighbor-full	OSPF full adjacency established with neighbor.
routed-OSPF-start-graceful-restart	OSPF started graceful restart.
routed-OSPF-stopped-graceful-restart	OSPF stopped graceful restart.
routed-OSPF-start-helper_node	OSPF started helper mode for a restarting neighbor.
routed-OSPF-not-help	OSPF did not help a restarting neighbor.
routed-OSPF-start-graceful-restart	OSPF started graceful restart.
routed-PIM-new-dr-elected	PIM elected a new DR
routed-PIM-neighbor-discovered	PIM discovered a new neighbor
routed-PIM-neighbor-disappeared	PIM neighbor disappeared
routed-RIP-peer-add	RIP peer discovered.

satd

Event ID	Description
satd-config-p1-success	SATD daemon configuration load phase-1 succeeded.
satd-config-p1-abort	SATD daemon configuration load phase-1 aborted.
satd-config-p2-success	SATD daemon configuration load phase-2 succeeded.
satd-portal-connect-started	GlobalProtect Satellite connection to portal started.

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Event ID	Description
satd-gateway-connect-started	GlobalProtect Satellite connection to gateway started.
satd-flow-full-sync-start	SATD daemon sync all gateway infos to Flow started.
satd-ha-full-sync-done	SATD daemon sync all gateway infos to HA peer exit.
satd-daemon-init	SATD daemon is initializing.
satd-daemon-start	SATD daemon is ready.
satd-daemon-exit	SATD daemon has exited.

sched-push

Event ID	Description
sched-skip	Push schedule <name> skipped on passive panorama
sched-exec	Push schedule <name> kicked in. <num> jobs scheduled. Jobids: <ids>

sdwan

Event ID	Description
sdwan-vif-status-up	<vif> start with state UP. FW is Active
sdwan-vif-status-up	<vif> start with state UP. FW is Non-Active
sdwan-vif-status-up	<vif> is up
sdwan-vif-status-down	<vif> is down

ssh

Event ID	Description
ssh-default-hostkey-changed	Default MGMT SSH host key set to ECDSA key of length <length>.
ssh-default-hostkey-changed	Default MGMT SSH host key set to RSA key of length <length>

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Event ID	Description
ssh-default-hostkey-changed	Default MGMT SSH host key set to all.
ssh-default-hostkey-changed	Default HA SSH host key set to ECDSA key of length <length>.
ssh-default-hostkey-changed	Default HA SSH host key set to RSA key of length <length>.
ssh-default-hostkey-changed	Error occurred while setting default host key for HA of type ECDSA and of length <length>
ssh-default-hostkey-changed	Error occurred while setting default host key for MGMT of type ECDSA and of length <length>
ssh-default-hostkey-changed	Error occurred while setting default host key for HA of type RSA and of length <length>
ssh-default-hostkey-changed	Error occurred while setting default host key for MGMT of type RSA and of length <length>
ssh-hostkey-regenerated	SSH host key for HA of type ECDSA and of length <num> generated
ssh-hostkey-regenerated	SSH host key for MGMT of type ECDSA and of length <num> generated
ssh-hostkey-regenerated	SSH host key for HA of type RSA and of length <num> generated
ssh-hostkey-regenerated	SSH host key for MGMT of type RSA and of length <num> generated
ssh-session-rekey-params-changed	New Rekeying parameters for MGMT SSH set.
ssh-session-rekey-params-changed	New Rekeying parameters for HA SSH set.
ssh-session-rekey-params-changed	Error occurred while setting rekeying parameters for MGMT SSH.
ssh-session-rekey-params-changed	Error occurred while setting rekeying parameters for HA SSH.
ssh-ciphers-changed	Ciphers set to default for MGMT SSH.
ssh-ciphers-changed	Ciphers set to default for HA SSH.

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Event ID	Description
ssh-ciphers-changed	Error occurred while setting ciphers for MGMT SSH.
ssh-ciphers-changed	Error occurred while setting ciphers for HA SSH.
ssh-macs-changed	Macs set to default for MGMT SSH.
ssh-macs-changed	Macs set to default for HA SSH.
ssh-macs-changed	Error occurred while setting macs for MGMT SSH.
ssh-macs-changed	Error occurred while setting macs for HA SSH.
ssh-kexs-changed	Kexs set to default for MGMT SSH.
ssh-kexs-changed	Kexs set to default for HA SSH.
ssh-kexs-changed	Error occurred while setting kexs for MGMT SSH.
ssh-kexs-changed	Error occurred while setting kexs for HA SSH.

sslmgr

Event ID	Description
ca-session-establishment-success	Destination address <addr>, Destination port <num>, Source address <addr>, Source port <num>
ca-session-establishment-failed	Failed to get CRL %s
ca-session-establishment-failed	Key Usage cRLSign check failed for CRL <name>
ca-session-establishment-success	"Successfully get CRL <name>
ca-session-establishment-success	CRL request to <name> succeeded
ca-session-establishment-success	OCSP request to "<host>" succeeded. \nDestination address: <addr>, Destination port: <port>, Source address: <addr>, Source port <port> \n
ca-session-establishment-failed	OCSP request to "<host>" failed. \nDestination address: <addr>, Destination

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Event ID	Description
	port: <port>, Source address: <addr>, Source port <port> \n
ca-session-establishment-failed	<open_ssl_error>
sslmgr-ha-not-full-sync	SSLMGR daemon not sync to HA peer.
sslmgr-ha-not-full-sync	SSLMGR daemon not sync to HA peer.
sslmgr-ha-not-full-sync	SSLMGR daemon not sync to HA peer.
sslmgr-cert-ocsp-verify-failed	SSLMGR certificate ocsp verification failed.
sslmgr-config-p1-success	SSLMGR daemon configuration load phase-1 succeeded.
sslmgr-config-p2-success	SSLMGR daemon configuration load phase-2 succeeded.
sslmgr-daemon-start	SSLMGR daemon is ready.
sslmgr-satellite-info-deleted	SSLMGR satellite info deleted
sslmgr-cert-status-deleted	SSLMGR certificate status deleted.
sslmgr-cert-status-revoked	SSLMGR certificate status revoked.
sslmgr-satellite-info-deleted	SSLMGR satellite info deleted
sslmgr-cert-status-revoked	SSLMGR certificate status revoked.
sslmgr-scep-ca-cert-failed	SSLMGR import SCEP CA certificate failed.
sslmgr-scep-cert-failed	SSLMGR generate SCEP certificate failed.
sslmgr-scep-cert-failed	SSLMGR generate SCEP certificate failed.
sslmgr-scep-cert-failed	SSLMGR generate SCEP certificate failed.
sslmgr-satellite-info-updated	SSLMGR satellite info updated
sslmgr-cert-gen-failed	SSLMGR generate certificate failed.
sslmgr-ha-full-sync	SSLMGR daemon sync to HA peer.
sslmgr-ha-full-sync	SSLMGR daemon sync to HA peer.
sslmgr-ha-full-sync	SSLMGR daemon sync to HA peer.

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Event ID	Description
ca-session-establishment-success	Destination address <addr>, Destination port <port>, Source address <addr>, Source port <port>

syslog

Event ID	Description
syslog-conn-status	<syslog-ng message>

tls

Event ID	Description
panos-auth-success	<name> Server CN: <name> - [<name>] Connection Successfully established.
tls-session-disconnected	Device <name> disconnected from the server
panorama-auth-success	<reason> PAN-OS ver: <version> Panorama ver:<version> Client IP: <ip> Server IP: <ip> Client CN: <name>
panorama-auth-success	<reason> WildFire ver: <version> Panorama ver:<version> Client IP: <ip> Server IP: <ip> Client CN: <name>
certificate-renewal	Client Certificate expiry is under 30 days. Fetch a new certificate from the scep server

url-filtering

Event ID	Description
failed-to-lock-update	Failed to lock URL database update process! Maybe another instance is running.
download-url-database-success	Brightcloud URL database was downloaded successfully
revert-url-database-success	URL filtering database was reverted from version <ver> to version <ver>
url-database-is-latest	URL filtering database version <ver> is already the latest version

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Event ID	Description
failed-to-lock-download	Failed to lock URL database update process. Another instance may be running.
download-url-database-success	PAN-DB was downloaded successfully
load-success	Intial PAN-DB activated successfully
failed-to-lock-download	PAN-DB download: Failed.
downloading-url-database	Downloading full BrightCloud URL database. This can take a long while.
downloading-url-database	Downloading full BrightCloud URL database. This can take a long while.
proxy-connection-failure	Failed to connect to proxy server. "Please check if proxy user name and password are "correct.
receive-data-failure	Cannot receive data from '<server>:<port>' to download BrightCloud URL database
proxy-connection-failure	Failed to connect to proxy server. "Please check if proxy user name and password are correct.
proxy-connection-failure	Cannot connect to proxy server '<server>:<port>' to download BrightCloud URL database
proxy-connection-failure	Cannot connect to proxy server '<server>:<port>' to download BrightCloud URL database
connection-success	Connected to Brightcloud update server <name>
cloud-election	CLOUD ELECTION: <name> IP: <ip> was elected, measured alive test <num>.
url-engine-stopped	PAN-DB engine stopped.
url-engine-starts	PAN-DB engine started.
url-engine-stopped	URL filtering engine stopped...
ha-sync-failure	Failed to sync the URL with HA peer.

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Event ID	Description
starts-from-empty-seed	Starting with an empty SEED.
starts-from-backup-seed	Starting with backup seed.
starts-from-empty-seed	Starting with an empty SEED.
ha-sync-success	Successfully synced PAN-DB to peer.
ha-sync-success	PAN-DB sync with HA started at <seconds>.
url-backup-seed-success	Backup of PAN-DB finished successfully.
upgrade-url-database-success	PAN-DB was upgraded to version <version>.
ha-sync-success	URL vendor matches and is set to 'PAN-DB'.
ha-sync-failure	Not synching file to peer because mode is not Active-Passive (<mode>).
ha-sync-failure	No synching file to peer because local state is not Active (<mode>).
ha-sync-failure	Not accepting file from peer local state is not Passive (<mode>).
ha-sync-failure	No synching file to peer because peer state is not Passive (<mode>).

userid

Event ID	Description
connect-agent	Redistribution Agent <name>(vsys<id>): connected to <host>, status <status>, version <num>
connect-client	CMS Redistribution Client is connected to global collector: <devid> vsys <id>
connect-client	Redistribution Client is connected to collector <name>: <client>, vsys <id>
connect-ldap-sever	ldap cfg <name> connected to server <server>
connect-ldap-sever	ldap cfg <name> connected to server <server>

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Event ID	Description
connect-agent	<agent> <name>(vsys<id>): connected to <name>, status <status>, version <version>
connect-client	User-ID Client is connected to collector <name>: "IP <ip> port <num> vsys <num>
disconnect-client	User-ID Client is disconnected from collector <name>: "IP <ip> port <num> vsys_id <num>
disconnect-client	User-ID Client is disconnected from collector <name>: "IP <ip> port <num> vsys_id <num>
connect-client	User-ID Client is connected to collector <name>:<conn_id> vsys_id <id>
disconnect-client	User-ID Client is disconnected from collector <name>:<conn_id> vsys_id <id>
connect-agent	<agent_desc> <name>(vsys<id>): connected to <name>, version <id>
agent-read-log-error	<name> failed <num> time(s)
agent-get-domain-error	<name> please check pan-agent log file for actual incorrect DC IP address(es)
agent-get-groups-error	<name> failed <num> time(s)
agent-get-config-error	<name> failed <num> time(s)
agent-get-users-error	<name> failed <num> time(s)
agent-no-domain	<name> failed <num> time(s)
disconnect-syslog	User-ID Syslog Proxy: Client <name>: disconnected <addr>
connect-syslog	User-ID Syslog Proxy: Client <name>(vsys<id>): connected <addr>
disconnect-syslog	User-ID Syslog Proxy: Client <name>: disconnected <addr>
disconnect-syslog	User-ID Syslog Proxy: Client <name>: disconnected <addr>
connect-agent	Pan-TS-Agent <name> disconnected: IP <ip> port <num> vsys<num>

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Event ID	Description
disconnect-agent	PAN-Agent <name> disconnected: IP <ip> port <num> vsys<id>
agent-status-failure	Failed to get status <num> times, connection may be down or protocol mismatch between device and pan-agent
disconnect-agent	User-ID-Agent <name> disconnected: IP <ip> port <num> vsys<id>
disconnect-agent	User-ID-Agent <name> disconnected: <conn_str> vsys<id>
agent-event	User-ID-Agent <name> event: <type>, name <name>, status <status>, vsys<id>
agent-status-failure	Failed to get status <num> times, connection may be down or protocol mismatch between device and pan-agent
connect-server-monitor	Please change server monitor(<name>) Transport Protocol from WMI to WinRM for better performance
connect-server-monitor	User-ID server monitor <name>(vsys<id>): connected to <host>
connect-server-monitor	Server monitor <name>(vsys<id>) is connected
connect-vm-info-source	vm-info-source <name>(vsys<id>): Connected to <host>, status <status>
connect-vm-info-source	vm-info-source <name>(vsys<id>): Connected to <host>, status <status>
connect-vm-info-source	vm-info-source <name>(vsys<id>): connected to <host>, status <status>, version <version>
disconnect-vm-info-source	vm-info-source <name>(vsys<id>): disconnected to <host>, status <status>, version <version>

vm

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Event ID	Description
dvf-init-succeed	VMware dvfilter init succeeded

vpn

Event ID	Description
vpnctl-ike-rekey-event	[<name>]: <davici_name>:<value,
vpnctl-child-updown-event	[<name>]: <davici_name>:<value,
vpnctl-child-rekey-event	[<name>]: <davici_name>:<value,
vpnctl-ike-updown-event	connction failed, peer <remote_host>, retry <conn_try>
keymgr-daemon-init	KEYMGR daemon is initializing.
keymgr-daemon-start	KEYMGR daemon is ready.
keymgr-daemon-exit	KEYMGR daemon has exited.
keymgr-flow-full-sync-done	KEYMGR sync all IPSec SA to Flow exit.
ike-fqdn-change	IKE fqdn mapping is changed
ike-config-p1-success	IKE daemon configuration load phase-1 succeeded.
ike-config-p1-abort	IKE daemon configuration load phase-1 aborted.
ike-config-p2-success	IKE daemon configuration load phase-2 succeeded.
ike-nego-p1-fail-psk	IKE phase-1 negotiation is failed likely due to pre-shared key mismatch.
ike-nego-p1-fail-psk	IKE phase-1 negotiation is failed likely due to pre-shared key mismatch.
ike-nego-p1-fail-common	IKE phase-1 negotiation is failed_COMM
ike-nego-p1-fail-common	IKE phase-1 negotiation is failed_COMM
ike-nego-p1-fail-common	IKE phase-1 negotiation is failed_COMM
ikev2-nego-child-ts-bad	IKEv2 child SA negotiation failed when processing traffic selector.

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Event ID	Description
ikev2-nego-child-ts-bad	IKEv2 child SA negotiation failed when processing traffic selector.
ikev2-send-p1-delete	IKEv2 IKE SA delete message sent to peer.
ike-nego-p1-fail-common	IKE phase-1 negotiation is failed_COMM
ikev2-nego-use-v1	IKEv1 is used in IKEv2 preferred mode.
ike-nego-p2-stale-p1	Deleting a possible stale phase-1 SA.
ike-nego-p1-start	IKE phase-1 negotiation is started
ike-nego-p1-fail	IKE phase-1 negotiation is failed
ike-nego-p1-succ	IKE phase-1 negotiation is succeeded
ike-nego-p1-delete	IKE phase-1 SA is deleted
ike-nego-p1-expire	IKE phase-1 SA is expired
ike-nego-p2-start	IKE phase-2 negotiation is started
ike-nego-p2-fail	IKE phase-2 negotiation is failed
ike-nego-p2-succ	IKE phase-2 negotiation is succeeded
ipsec-key-install	IPSec key installed.
ipsec-key-delete	IPSec key deleted.
ipsec-key-expire	IPSec key lifetime expired.
ike-nego-p2-proxy-id-bad	IKE phase-2 negotiation failed when processing proxy ID.
ike-nego-p2-proxy-id-bad	IKE phase-2 negotiation failed when processing proxy ID.
ike-nego-p2-no-p1	IKE phase-2 negotiation request received but no phase-1 SA is found.
ike-nego-p2-p1-not-ready	IKE phase-2 negotiation request received but no active phase-1 SA is available.
ike-nego-p2-proposal-bad	IKE phase-2 negotiation failed when processing SA payload.

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Event ID	Description
ike-nego-p1-fail-common	IKE phase-1 negotiation is failed_COMM
ike-nego-p1-psk-idtype	IKE phase-1 negotiation is failed. When pre-shared key is used
ike-nego-p1-fail-psk	IKE phase-1 negotiation is failed likely due to pre-shared key mismatch.
ike-nego-p1-fail-psk	IKE phase-1 negotiation is failed likely due to pre-shared key mismatch.
ike-recv-notify	IKE protocol notification message received:
ike-recv-p1-delete	IKE protocol phase-1 SA delete message received from peer.
ike-recv-p2-delete	IKE protocol IPSec SA delete message received from peer.
ike-send-p1-delete	IKE protocol phase-1 SA delete message sent to peer.
ike-send-p2-delete	IKE protocol IPSec SA delete message sent to peer.
ike-send-notify	IKE protocol notification message sent:
ike-send-notify	IKE protocol notification message sent:
ike-send-notify	IKE protocol notification message sent:
ike-nego-p2-dup-rekey	duplicate phase-2 rekey request detected
ike-nego-p1-cert-succ	IKE certificate authentication succeeded.
ike-nego-p1-fail-psk	IKE phase-1 negotiation is failed likely due to pre-shared key mismatch.
ikev2-nego-cert-succ	IKEv2 certificate authentication succeeded.
ikev2-nego-fail-psk	IKEv2 SA negotiation is failed likely due to pre-shared key mismatch.
ikev2-send-p2-delete	IKEv2 IPSec SA delete message sent to peer.
ikev2-nego-child-fail	IKEv2 child SA negotiation is failed
ikev2-nego-child-fail	IKEv2 child SA negotiation is failed

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Event ID	Description
ikev2-nego-child-fail	IKEv2 child SA negotiation is failed
ikev2-nego-child-fail	IKEv2 child SA negotiation is failed
ikev2-nego-stale-p2	Deleting a possible stale IKEv2 child SA.
ikev2-nego-fail-common	IKEv2 SA negotiation is failed.
ike-recv-notify	IKE protocol notification message received:
ikev2-recv-p1-delete	IKEv2 IKE SA delete message received from peer.
ikev2-recv-p2-delete	IKEv2 IPSec SA delete message received from peer.
ikev2-nego-ike-fail	IKEv2 IKE SA negotiation is failed
ikev2-nego-ike-start	IKEv2 IKE SA negotiation is started
ikev2-nego-ike-fail	IKEv2 IKE SA negotiation is failed
ikev2-nego-ike-succ	IKEv2 IKE SA negotiation is succeeded
ikev2-nego-ike-delete	IKEv2 IKE SA is deleted
ikev2-nego-ike-expire	IKEv2 IKE SA is expired
ikev2-nego-child-start	IKEv2 child SA negotiation is started
ikev2-nego-child-fail	IKEv2 child SA negotiation is failed
ikev2-nego-child-succ	IKEv2 child SA negotiation is succeeded
ipsec-key-install	IPSec key installed.
ipsec-key-delete	IPSec key deleted.
ipsec-key-expire	IPSec key lifetime expired.
ikev2-nego-use-v1	IKEv1 is used in IKEv2 preferred mode.
ike-daemon-init	IKE daemon is initializing.
ike-daemon-start	IKE daemon is ready.
ike-daemon-exit	IKE daemon has exited.

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wildfire

Event ID	Description
wildfire-no-policy	WildFire <name> channel disabled. No active WildFire analysis profile to <name> channel.
wildfire-auth-failed	Failed to verify SSL peer's certificate with the certificate authority

wildfire-appliance

Event ID	Description
cluster-mode-change	Cluster mode changed to stand_alone
cluster-mode-change	Cluster mode changed to controller
cluster-mode-change	Cluster mode changed to worker
cluster-mode-change	Cluster mode changed to unknown
cluster-engine-role	Cluster engine started as controller.

Slog

- Fan Tray is missing, system will power down in <num> seconds if not replaced.
- <entry> is not present on startup
- Freeing slot <id>, uid <id> with Force
- Freeing slot <id>, uid <id> with Non-force
- Get registration with uid <id> sw_ver <version> slot <id> dp_ip <ip>
- Allocated slot %d for uid <uid> <id>
- Device certificate expires in 15 or less days
- Successfully fetched device certificate from Palo Alto Networks
- Logd failed to send disconnect to configd for (<id>)
- Logd blocking customerid (<id>)
- Logd Unblocking customerid (<id>)
- Logd failed to send disconnect to configd for (<name>)]
- Trigger AddrObjRefresh commit for group-mapping
- Purged mongdb data size (<num> recs) to bring "data size below limit <num>
- GlobalProtect data file version <version> downloaded from peer device
- Name resolution takes too long disable name lookup for report <name>
- Name resolution takes too long disable name for the report <name>

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- The primary user attribute has been changed in one of the group-mapping configuration
- Captive Portal Client certificate validation failed from <host>. no certificate.
- Captive Portal Client certificate validation failed from <host<. Certificate does not belong to the Cert Profile chain
- Captive Portal Client certificate verification for OSCP/CRL failed from <host>.
- Captive Portal Client certificate is not yet active from <host>.
- Captive Portal Client certificate has expired from <host>.
- Captive Portal client certificate authentication successful from <host>
- <type> authentication succeeded for user: <name> on <host> vsys<id>
- <type> renew from session cookie for user: <user> on <addr> vsys<id>
- <type> NTLM authentication failed for user: <user> on <addr> vsys<id>
- <type> NTLM authentication succeeded for user: <user> on <addr> vsys<id>
- <type> authentication failed (INVALID) for user: <user> on <ip> vsys<id>
- <type> authentication failed for user: <name> on <ip> vsys<id>
- <type> authentication succeeded for user: <name> on <ip> vsys<id>
- Logd received error response code from http service (<num>) msg size <num> customerid <id> logtype <name> num_rec <num>
- Logdb downgrade started on <serial> slot <id>.
- Logdb downgrade completed on <serial> slot <id> in <num> days <num> hours <num> minutes <num> secs.
- Logdb Migration started on <serial> slot <num>
- Logdb Migration paused on <serial> slot <num>.
- Logdb Migration abandoned on <serial> slot <id>.
- Logdb Migration completed on <serial> slot <id>.
- Test email sent to <name> successfully for email profile <name>
- Client certificate verification for OSCP/CRL failed from <host>.
- Client certificate authentication successful from <host>.
- Client certificate validation failed from <host>. No https is detected.
- Client certificate validation failed from <host>. No https is detected.
- Create system logs
- Create custom system logs
- Cluster member <id>, <name> successfully updated for <name> and push enqueued with jobid <id>
- Cluster member <id>, <name> successfully deleted for <name> and push enqueued with jobid <id>
- successfully connect to %s:%s:%d
- Failed connect to %s:%s:%d
- dsc service is started

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- Identity client received malformed policy recommendation.
- Identity client received policy recommendation error: %v.
- Identity client received %v policy recommendation.
- Identity client failed to get policy recommendation.
- Icd HA state is changed from %d to %d
- Icd HA better state is changed from %d to %d
- failed to retrieve source address with error %d"
- iot-eal service is started
- icd service is started
- gRPC connection to %s is broken, error: %v
- gRPC connection to %s is established, %s -> %s
- "gRPC connection to %s is broken, error: %s"
- Cloud Appid feature is disabled
- Cloud Appid feature is enabled
- Cloud Appid %s task[%d] completed, new cloud version: %s, %s",
- Cloud Appid %s task[%d] failed: %v
- Cloud App: %s data lost some files, %d -> %d
- Cloud App: check and restore %s data, type %d.

Low Severity System Log Messages

E-Log

- audit
- auth
- dns-security
- dynamic-updates
- routing
- vpn

audit

Event ID	Description
cli	<cmd>
api	<cmd>
cli	<config command>
api	<config command>

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Event ID	Description
gnmi	<config command>
gui-op	<config command>

auth

Event ID	Description
cas-message	(profile id:<id>)<message>
saml-out-of-band-message	Client '<name>' received out-of-band SAML message: <message>

dns-security

Event ID	Description
PAN_ELOG_EVENT_DNSSEC_CACHE_FAIL	DNS signature initialization from file storage failed, start with empty cache.

dynamic-updates

Event ID	Description
palo-alto-networks-message	<message>

routing

Event ID	Description
routed-config-p1-failed	Route daemon configuration load phase-1 failed.
routed-BGP-peer-failed	BGP peer session has failed and may restart.
routed-BGP-peer-restarted	Initiated graceful-restart with a BGP peer.
routed-BGP-peer-restart-failed	Graceful-restart with a BGP peer failed.
routed-RTM-bad-route	An invalid dynamic route has been rejected:
routed-OSPF-LSA-chksum-invalid	OSPF received LSA with invalid checksum.
routed-OSPF-LSA-chksum-invalid	OSPF received LSA with invalid checksum.

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Event ID	Description
routed-OSPF-LSA-chksum-failed	OSPF LSA checksum generating failed due to memory corruption.
routed-OSPF-LSA-chksum-failed	OSPF LSA checksum generating failed due to memory corruption.
routed-OSPF-md5chksum-bad	OSPF packet dropped due to incorrect MD5 checksum.
routed-OSPF-authtype-bad	OSPF packet dropped due to unexpected authentication type.
routed-OSPF-password-bad	OSPF packet dropped due to incorrect simple password.
routed-OSPF-chksum-bad	OSPF packet dropped due to incorrect OSPF checksum.
routed-OSPF-sequence-bad	OSPF packet dropped due to incorrect sequence number.
routed-OSPF-hello-hello-intval-bad	OSPF hello packet dropped due to hello-interval mismatch.
routed-OSPF-hello-dead-intval-bad	OSPF hello packet dropped due to dead-interval mismatch.
routed-OSPF-hello-netmask-bad	OSPF hello packet dropped due to network masks mismatch.
routed-OSPF-hello-area-type-bad	OSPF hello packet dropped due to area type mismatch.
routed-PIM-interface-state-changed	PIM interface state changed
routed-RIP-authtype-bad	RIP packet dropped due to unexpected authentication type.
routed-RIP-auth-failed	RIP packet dropped due to authentication failure.
routed-RIP-md5length-bad	RIP packet dropped due to incorrect MD5 digest length.
routed-RIP-md5length-bad	RIP packet dropped due to incorrect MD5 digest length.

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Event ID	Description
routed-RIP-auth-failed	RIP packet dropped due to authentication failure.

vpn

Event ID	Description
ike-nego-p1-dpd-dn	IKE phase-1 SA is down determined by DPD.
ikev2-nego-ike-dpd-dn	IKEv2 IKE SA is down determined by DPD.

Slog

- Check DB uid failed, ignoring for re-registration. Return code: <num>
- Switch fabric to network processor link renegotiated.
- Successful SCP in of Deployment file: '<file>'

Medium System Log Messages

E-Log

Log Tags:

- auth
- ddns
- dhcp
- dns-security
- dynamic-updates
- fips
- general
- hw
- nat
- ntpd
- port
- routing
- satd
- syslog
- url-filtering
- userid
- wildfire

auth

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Event ID	Description
cas-message	(profile id:<id>)<message>
auth-fail	<type> with username "<name>" is invalid due to special characters
auth-fail	Allocated slot <id> for uid <uid> <id>
auth-fail	Admin <name> failed to authenticate <num> times - the unsuccessful authentication attempts threshold reached.
auth-fail	Admin <name>'s account is being disabled due to excessive failed authentication attempts.
auth-success	Certificate validated for user '<name>'. <error>
auth-fail	Certificate validation failed for user '<user>'. <error> auth profile '<name>', vsys '<id>', reply message '<msg>' From: <name>.
auth-fail	failed authenticated for user '<name>'. realm '<name>', EAP outer identity '<name>, inner identity '<name>', auth profile '<name>', vsys '<id>', server profile '<name>', server address '<addr>', admin role '<name>', access domain '<name>', reply message '<msg>' From: <name>.
user-password-change-failed	failed authenticated for user '<name>'. realm '<name>', EAP outer identity '<name>, inner identity '<name>', auth profile '<name>', vsys '<id>', server profile '<name>', server address '<addr>', admin role '<name>', access domain '<name>', reply message '<msg>' From: <name>.
auth-fail	Kerberos SSO authentication failed for user '<name>'. realm '<name>', EAP outer identity '<name>, inner identity '<name>', auth profile '<name>', vsys '<id>', server profile '<name>', server address '<addr>', admin role '<name>', access domain '<name>', reply message '<msg>' From: <name>.
user-password-change-failed	Kerberos SSO authentication failed for user '<name>'. realm '<name>', EAP outer identity

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Event ID	Description
	'<name>, inner identity '<name>', auth profile '<name>', vsys '<id>', server profile '<name>', server address '<addr>', admin role '<name>', access domain '<name>', reply message '<msg>' From: <name>.
auth-fail	SAML SSO authentication failed for user '<name>'. realm '<name>', EAP outer identity '<name>, inner identity '<name>', auth profile '<name>', vsys '<id>', server profile '<name>', server address '<addr>', admin role '<name>', access domain '<name>', reply message '<msg>' From: <name>.
user-password-change-failed	SAML SSO authentication failed for user '<name>'. realm '<name>', EAP outer identity '<name>, inner identity '<name>', auth profile '<name>', vsys '<id>', server profile '<name>', server address '<addr>', admin role '<name>', access domain '<name>', reply message '<msg>' From: <name>.
auth-fail	CAS SSO authentication failed for user '<name>'. realm '<name>', EAP outer identity '<name>, inner identity '<name>', auth profile '<name>', vsys '<id>', server profile '<name>', server address '<addr>', admin role '<name>', access domain '<name>', reply message '<msg>' From: <name>.
user-password-change-failed	CAS SSO authentication failed for user '<name>'. realm '<name>', EAP outer identity '<name>, inner identity '<name>', auth profile '<name>', vsys '<id>', server profile '<name>', server address '<addr>', admin role '<name>', access domain '<name>', reply message '<msg>' From: <name>.

ddns

Event ID	Description
ddns-unsupported	Interface <name> DDNS config for host <host> to <label> (<label>) is using a non-supported DDNS service provider. Please convert to a supported service.

dhcp

Event ID	Description
ip-already-in-use	ip address is already in use
server-no-free-ip	DHCP server runs out of ip pool

dns-security

Event ID	Description
PAN_ELOG_EVENT_DNSSEC_DNS_CLOUD_QUERY_TIMEOUT	DNS Security cloud query timeout.

dynamic-updates

Event ID	Description
palo-alto-networks-message	<message>

fips

Event ID	Description
fips-entropy-rtciid	RTC-IID error occurred - attempting recovery...
fips-entropy-rtciid	RTC-IID - Reading record failure

general

Event ID	Description
general	CAS token sign cert "<name>" is invalid with error msg "<msg>"
general	PANDB: Authentication or Client Certificate failure.
general	PANDB: Client Certificate has expired or is not yet valid.
general	PANDB: Device Client Certificate unavailable.
general	PANDB: Mismatched Serial number in certificate and payload.

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Event ID	Description
general	PANDB: Expired Client Certificate.
general	PANDB: Revoked Client Certificate.
general	PANDB: Reason - Unknown Issuer or Incomplete or Incorrect Certificate chain.
general	MLAV: Client Certificate has expired or is not yet valid.
general	MLAV: Device Client Certificate unavailable.
general	MLAV: Mismatched Serial number in certificate and payload.
general	MLAV: Expired Client Certificate.
general	MLAV: Revoked Client Certificate.
general	MLAV: Reason - Unknown Issuer or Incomplete or Incorrect Certificate chain.
general	WFRTSIG: Authentication or Client Certificate failure.
general	WFRTSIG: Client Certificate has expired or is not yet valid.
general	WFRTSIG: Device Client Certificate unavailable.
general	WFRTSIG: Mismatched Serial number in certificate and payload.
general	WFRTSIG: Expired Client Certificate.
general	WFRTSIG: Revoked Client Certificate.
general	WFRTSIG: Reason - Unknown Issuer or Incomplete or Incorrect Certificate chain.
general	Server Cert: <name> is invalid, its name does not match the server <server>
general	Server Cert: <name> is invalid for server <name>: <error>
general	Slot s<num>: Application Pod '<name> : <namespace>: <interface>' can't be served

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Event ID	Description
	right now; All <num> ports (<num> pods) in use, waiting for ports availability (for <name>).
general	Failed to connect to wildfire-realtime cloud, retry after 30 seconds.
general	CONFIG_UPDATE_INC : Incremental update to DP failed please try to commit force the latest config
general	Request made to <name> server returned with HTTP response code : <code>

hw

Event ID	Description
slot-up	Slot <id> (PA-7000/5400-100G-NPC) ctd-mode is AHO

nat

Event ID	Description
fallback_report	On vsys <id>, there are <num> NAT DIPP fallbacks happeing in NAT rule <name>.

ntpd

Event ID	Description
auth	NTP sync to server <addr> failed, authentication type autokey
auth	NTP sync to server <addr> failed, authentication type autokey

port

Event ID	Description
invalid-module	<name>: requires an SFP+ module.
invalid-module	<buf>: requires an optical or copper SFP module.

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routing

Event ID	Description
routed-static-fqdn-changed	Routed static fqdn mapping is changed
routed-static-fqdn-changed	Routed static fqdn mapping is changed
routed-BGP-peer-mp-extension-negotiate	BGP peer MP extension negotiation. MP-EXTENSION negotiation to peer name: <name>, peer IP: <ip> successful"
routed-BGP-peer-enter-established	BGP peer session enters established state. peer name: <name>, peer IP: <ip>.
routed-BGP-refresh-sent	ROUTE REFRESH message sent to a BGP peer. peer name: <name>, peer IP: <ip>.
routed-BGP-ribout-recalc	An RIB-Out is being recalculated as a result of changed export policy. peer name: <name>, peer IP: <ip>.
routed-BGP-ribin-recalc	An RIB-In is being recalculated as a result of changed import policy. peer name: <name>, peer IP: <ip>.

satd

Event ID	Description
satd-portal-gateway-duplicate	GlobalProtect portal config duplicated gateway.

syslog

Event ID	Description
syslog-conn-status	<syslog-ng message>

url-filtering

Event ID	Description
dynamic-url-connection-down	Dynamic URL connection is unavailable please check if service.brightcloud.com (<ip>) is reachable

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Event ID	Description
connection-failure	Failed to connect to Brightcloud update server: Cannot fetch source IP address
url-download-failure	The URL cloud list file was not found on the cloud.
cloud-election	CLOUD ELECTION: cannot elect a cloud
url-cloud-connection-failure	Failed to open connection with the cloud after "<num> consecutive tries.
error-msg-from-cloud	ERROR message from cloud. Invalid request.
error-msg-from-cloud	ERROR message from cloud. Invalid request.
error-msg-from-cloud	ERROR status from cloud
startup-failure	PAN-DB engine startup failed.
update-version-failure	Failed to update version <version>.
update-version-failure	Failed to update version <version>.
update-version-failure	Failed to update version <version>.
update-version-failure	Failed to update version <version>.
update-version-failure	Failed to update version <version>.
starts-from-empty-seed	Failed to load the URL seed database, starting with an empty database.
ha-sync-failure	Unable to initiate file sync to peer: <error>
url-backup-seed-failure	Failed to back up PAN-DB
engine-startup-failure	May runs without URL filtering !!!
ha-sync-failure	Failed to upload the new HA URL file to RAM, start loading old URL file.
starts-from-empty-seed	Failed to upload the old URL file to RAM,Starting with an empty file.
engine-startup-failure	Runs without URL filtering !!!
ha-sync-failure	Failed to receive file completely from peer (<name>:<name>): <error>.

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userid

Event ID	Description
connect-ldap-sever-failure	ldap cfg <name> failed to connect to server <server>: <error>
get-ldap-data-failure	ldap cfg <name> failed to get info from server <server>
connect-ldap-sever-failure	ldap cfg <name> failed to connect to server <server>: <error>
get-ldap-data-failure	ldap cfg <name> failed to get info from server <name>

wildfire

Event ID	Description
wildfire-conn-success	Successfully registered to <description> <name>

Slog

- Queue '<name>' reached the watermark limit at <num>
- Removed Used AuthKey '<name>'
- Removed Expired AuthKey '<name>'
- Deleted AuthKey '<name>'
- Created AuthKey '<name>' (Count:<num>, Life:< num>s, Type:'<type>', Serial-Count:<num>)
- Failed to SCP out Deployment file: '<file>' (rc: <num>)
- Failed to SCP out Deployment metafile: '<file>' (rc: <num>)
- Failed to SCP in Deployment metafile: '<file>' (rc: <num>)
- Failed to SCP in Deployment file: '<file>' (rc: <num>)
- Could not access threat vault
- Failed to upload the sample to the cloud.
- Registration to cloud failed.
- Created new devicecert '<name>'
- Created new cert '<name>'
- mail send: <status>
- Tor status is checked and changed to: <name>.
- Failed to send test email using email profile <name>.

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High System Log Messages

E-Log

Log Tags:

- auth
- bfd
- clusterd
- dhcp
- dns-security
- dynamic-updates
- fips
- general
- globalprotect
- hw
- iot
- ipv6nd
- lldp
- port
- resctrl
- routing
- tls
- url-filtering
- userid
- wildfire

auth

Event ID	Message
saml-certificate-error	The certificate of SAML IdP entity Id "<name>" is not configured, but it is asked to validate it in IdP server profile "<name>"
saml-certificate-error	Failed to get cert config on vsys <id>
saml-certificate-error	Failed to find cert for <name> in vsys <id>
saml-certificate-error	Failed to validate the signature in IdP certificate "<name>" of entity Id "<name>"

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Event ID	Message
saml-certificate-error	can't build CredentialResolver for public key "<key>" of IdP entity id "<name>" in server profile "<profile>"
saml-certificate-error	can't tranform one line buffer for the public key "<key>" of IdP entity id "<id>" in server profile "<profile>"
saml-certificate-error	User "<name>" is extracted from SAML SSO response from IdP "<name>", which doesn't have a certificate configured in server profile "<profile>" of auth profile "<profile>"
saml-certificate-error	Request signing certificate (object name: <name>) in SAML auth profile "<name>" has expired
saml-certificate-error	The certificate (object name: <name>) of SAML IdP entity Id "<name>" in IdP server profile "<name>" has expired
saml-certificate-error	IdP "<name>" doesn't have a certificate, while incoming SAML message has signature without X509Certificate
saml-certificate-error	SAML Assertion IdP certificate "<name>" (used in server profile "<name>") <reason>
saml-certificate-error	SAML no certificate profile is configured to check the revoke status of IdP cert "<name>" (in server profile "<name>")
saml-certificate-error	No IdP certificate is configured for IdP "<id>", no x509certificate in the incoming message, can't verify signature
saml-certificate-error	SAML <type> failure for user '<name>' - IdP "<id>" certificate "<name>" for server profile "<name>" has expired
saml-certificate-error	SAML <type> from IdP "<name>" (auth profile "<name>") is signed by unknown signer "<name>" and has been rejected
saml-certificate-error	SAML <type> failure - Request signing certificate "<name>" for SAML auth profile "<name>" has expired

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Event ID	Message
saml-certificate-error	SAML simple sign the SAML message failed (signing certificate object: "<name>")
saml-certificate-error	SAML sign the SAML message failed (signing certificate object: "<name>")
saml-certificate-error	Failure while validating the signature of SAML message received from the IdP "<id>", because the certificate in the SAML Message doesn't match the IDP certificate configured on the IdP Server Profile "<profile>". (SP: "<type>"), (Client IP: <ip>), (vsys: <id>), (authd id: <id>), (user: <name>)
saml-message-parse-error	SAML Assertion from '<name>' is malformed
saml-message-parse-error	Failed to convert SAML message payload into xml tree
saml-message-parse-error	SAML Assertion: InResponseToID "<id>" != OriginalReqID "<id>"
saml-message-parse-error	SAML message from IdP "<name>" has no Assertion
saml-message-parse-error	SAML SSO response from "<name>" has no usernameattribute and saml:Subject NameID field
saml-message-parse-error	username: entered "<name>" != returned "<name>" from IdP "<name>" -> reject SAML auth due to security concerns
saml-message-parse-error	SAML SLO request message from '<name>' is malformed
saml-message-parse-error	SAML message is not of V2.0
saml-message-parse-error	SAML message has no IssueInstant
saml-message-parse-error	SAML message from IdP "<id>" has no Issuer node
saml-message-parse-error	SAML message from IdP "<id>" has empty Issuer node value
saml-message-parse-error	SAML IdP entityID: parsed "<id>" != configured "<id>"

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Event ID	Message
saml-message-parse-error	SAML SLO request message has no signature, but validate-idp-certificate is enabled
saml-message-parse-error	SAML message has no NameID
saml-message-parse-error	SAML message has no SessionIndex
saml-message-parse-error	SAML SLO response message from '<name>' is malformed
saml-message-parse-error	SAML SLO: InResponseToID "<name>" != OriginalReqID "<id>"
saml-message-parse-error	SAML SLO response status: received "<name>" != "urn:oasis:names:tc:SAML:2.0:status:Success"
saml-message-parse-error	SAML SLO message has no Status
saml-message-parse-error	SAML message is not of Version 2.0
saml-message-parse-error	SAML message from IdP "<name>" has no NameID
saml-message-parse-error	SAML message from IdP "<name>" SSO: InResponseToID "<id>" != OriginalReqID "<id>"
saml-message-parse-error	SAML message from IdP "<name>" has no Subject
saml-message-parse-error	SAML message from IdP "<name>"(server profile "<name>") was created in the future (not_before "<time>" - max_clock_skew <num> > now <time>)
saml-message-parse-error	SAML message from IdP "<name>" (server profile "<name>") was expired already (not_on_or_after "<time>" + max_clock_skew <num> <= now <time>)
saml-message-parse-error	SAML message from IdP "<name>" has no Conditions
saml-message-parse-error	SAML message from IdP "<name>" has no AuthnInstant

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Event ID	Message
saml-message-parse-error	SAML message from IdP "<name>" has no SessionIndex
saml-message-parse-error	SAML message from IdP "<name>" has no AuthnStatement
saml-message-parse-error	SAML message from IdP "<name>": Error to extract AttributeStatement
saml-message-parse-error	Failed to verify signature against certificate of IdP "<name>"
saml-message-parse-error	For user "<name>", SAML message has no Signature from IdP "<name>", whose certificate "<name>" is configured in server profile "<name>" of auth profile "<name>"
saml-message-parse-error	SAML signature in message from IdP "<name>" can't be validated
cas-message	(profile id:<id>)<message>
general	Device cert is not available, to enable the cloud auth profile "<name>" on vsys "<name>"
cas-token-invalidated	Failed to validate CAS token from client '<name>' from '<url>' with auth_session_id '<id>' and username '<name>'
cas-certificate-warning	Expired CAS certificate '<name>' in region '<name>'
cas-certificate-warning	Expired device certificate '<name>'
cas-certificate-warning	CAS certificate '<name>' in region '<name>' will expire in <num> day[s]
cas-certificate-warning	Device certificate '<name>' will expire in <num> day[s]
saml-certificate-warning	SAML Assertion: signature is validated against IdP certificate (subject '<name>') for user '<name>'
saml-certificate-warning	Certificate '<name>' of IdP server profile '<name>' in SAML authentication profile '<name>' is expired

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Event ID	Message
saml-certificate-warning	Request signing certificate '<name>' in SAML authentication profile '<name>' is expired
saml-certificate-warning	Certificate '<name>' of IdP server profile '<name>' in SAML authentication profile '<name>' will expire in <num> day
saml-certificate-warning	Request signing certificate '<name>' in SAML authentication profile '<name>' will expire in %d day%s
cas-certificate-error	Device certificate "<name>" was expired for <num> seconds

bfd

Event ID	Message
admin-down	BFD administrative down for BFD session <name> to neighbor <name> on interface <name>. Protocol: <proto>
expired-time	BFD control detection time expired for BFD session <name> to neighbor <name> on interface <name>. Protocol: <name>
neighbor-down	BFD neighbor signaled session down for BFD session <name> to neighbor <name> on interface <name>. Protocol: <name>
session-state-change	BFD state changed to <name> for BFD session <name> to neighbor <name> on interface <name>. Protocol: <name>
admin-down	BFD administrative down for BFD session <name> to neighbor <name> on interface <name>. Protocol: <name>
admin-down	BFD administrative down for BFD session <name> to neighbor <name> on interface <name>. Protocol: <name>
admin-down	BFD administrative down for BFD session <name> to neighbor <name> on interface <name>. Protocol: <name>

clusterd

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Event ID	Message
cluster-daemon-cfg-giveup	Cluster daemon is unable to get last cfg from cfgagent. Out of retries.
cluster-other-ip-incompatible	Peer node IP is not compatible with current cluster interface IP

dhcp

Event ID	Message
if-update-fail	DHCP <desc>: interface <name>, dhcp server: <name>
if-update-fail	DHCP <name>: interface <name>, ip <ip> netmask <mask> dhcp server: <name>

dns-security

Event ID	Message
PAN_ELOG_EVENT_DNSSEC_DNS_CLOUD_CONNECTION_NOHOST	DNS Security cloud service DNS resolution failed.
PAN_ELOG_EVENT_DNSSEC_DNS_CLOUD_CONNECTION_NOROUTE	DNS Security cloud service network connectivity failed.
PAN_ELOG_EVENT_DNSSEC_DNS_CLOUD_CONNECTION_REFUSED	DNS Security cloud service connection refused.
PAN_ELOG_EVENT_DNSSEC_DNS_CLOUD_DOWN	DNS Security cloud service unavailable.

dynamic-updates

Event ID	Message
palo-alto-networks-message	<message>

fips

Event ID	Message
fips-zeroization	File zeroization error: <error>
fips-zeroization	Ram zeroization error

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general

Event ID	Message
general	Error setting CURLOPT_WRITEDATA with fd = <id> (code: <id>; msg: <msg>)
general	Error retrieving CRL from "<name>" (code: <id>; msg: <msg>) (curl timeout setting: <num> sec)
general	Error loading CRL from "<name>"
general	
general	Failed to parse CRL <name> (reason: <reason>)
general	Request made to the server "<url>" returned with HTTP response code : <id>
general	Request made to the server "<url>" returned with HTTP response code : <id>
general	Machine Learning engine for <name> stopped, please update your content
general	MLAV cloud error, all machine Learning engines stopped
bootstrap-failure	Failed to process registration from bootstrapped device <name>, since vm-auth-key not found in request.
bootstrap-failure	Failed to process registration from bootstrapped device <name>, since vm-auth-key <name> is invalid.
tac-login	TAC debug access failed for <name> from <ip>

globalprotect

Event ID	Message
globalprotectgateway-invalid-license	GlobalProtect Subscription License has expired. Please activate the license by logging into Customer Support Portal to continue using GlobalProtect features.

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hw

Event ID	Message
bootstrap-license-failure	Failed to install license using authcode <id>
slot-unsupported	Slot <id> (<model>) will not be utilized when the Session Distribution Policy is set to ingress-slot. The session distribution policy must be set to some value other than ingress-slot.
bootstrap-license-failure	Failed to install license key for file <name>
bootstrap-license-failure	Failed to install license using authcode <name>
bootstrap-content-failure	Invalid iot image. Failed to get major version, minor version, and digest for file <name>
bootstrap-content-failure	Invalid image. Failed to get major version, minor version, and digest for file <name>
bootstrap-content-failure	Invalid image. Failed to get major version, minor version, and digest for file <name>
bootstrap-content-failure	Invalid image. Failed to get major version, minor version, and digest for file <name>
bootstrap-content-failure	Failed to schedule content install job for file <name>
bootstrap-content-failure	Content cannot be installed. <error>

iot

Event ID	Message
ha-queue-full	HA queue is full

ipv6nd

Event ID	Message
inconsistent-ra-message-received	An inconsistent router advertisement was received from address <ip> on interface <name>.

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lldp

Event ID	Message
tooManyNeighbors timer cleared	TooManyNeighbors error cleared for <xx>:<xx>:<xx>:<xx>:<xx>:<xx> on interface <index>
tx error	Receive error for <xx>:<xx>:<xx>:<xx>:<xx>:<xx> on interface <index> for TLV <index>
rx error	Receive error for <xx>:<xx>:<xx>:<xx>:<xx>:<xx> on interface <index> for TLV <index>
too many neighbors	Max MIB size reached: LLDP neighbor addition failed for <xx>:<xx>:<xx>:<xx>:<xx>:<xx> on interface <index>

port

Event ID	Message
link-change	Port MGT: Down <type>

resctrl

Event ID	Message
mem-limit-exceeded	Memory lmt exceeds. cgroup_name <name> memsw_limit_in_bytes <num> memsw_usage_in_bytes <num>

routing

Event ID	Message
routed-BGP-peer-left-established	BGP peer session left established state. peer name: <name>, peer IP: <ip>.
routed-BGP-peer-restarted	Initiated graceful-restart with a BGP peer. peer name: <name>, peer IP: <ip>.

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Event ID	Message
routed-BGP-peer-prefix-exceeded	BGP peer advertised more than maximum allowed prefixes. peer name: <name>, peer IP: <ip>.
route-table-capacity	Route table capacity reached.
routed-BGP-peer-left-established	BGP peer session left established state.
routed-OSPF-neighbor-down	OSPF adjacency with neighbor has gone down.
routed-RIP-peer-del	RIP peer disappeared.

tls

Event ID	Message
tls-X509-validation-failed	<name> Server certificate validation failed. Dest Addr: <address>, Reason: <reason>
tls-X509-validation-failed	<name> server certificate authentication failed

url-filtering

Event ID	Message
url-download-failure	PAN-DB cloud list loading failed (ERROR:<error>).
url-download-failure	Failed to download the cloud list from the master cloud.
url-cloud-connection-failure	URL cloud list is empty. "Cannot initiate cloud connection.
url-cloud-connection-failure	Could not open file /opt/pancfg/opt/pan/content/pan/urlcloud_list.txt. errno=<error>.
url-cloud-connection-failure	Failed to send update request to the cloud
url-cloud-connection-failure	Cloud is not ready Free <num> requests without processing.
url-cloud-connection-failure	Cloud is not ready, There was no update from the cloud in the last <num> minutes.

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Event ID	Message
url-cloud-connection-failure	CLOUD CONNECTION: cloud not OK
update-version-failure	Failed to update DP, update version <name>.
update-version-failure	Failed to update version <version>.
update-version-failure	Failed to update version <version>.
update-version-failure	Failed to update version <version>.
update-version-failure	Failed to update version <version>.
seed-out-of-sync	PAN-DB sw <version> is not compatible with the cloud sw <version> Upgrade sw is required!!!
url-cloud-connection-failure	Failed to create the Cloud Connection Agent.

userid

Event ID	Message
connect-agent-failure	User-ID Agent peer's certificate RSA public key size is less than 2048 bits
connect-agent-failure	User-ID Agent X509_verify_cert returned error <id>, error = '<error>'
connect-agent-failure	User-ID Agent server cert revoked/invalid
connect-agent-failure	User-ID Agent cert name validation failed
connect-agent-failure	Redistribution Agent <name>(vsys<id>): <status> details: close connection to agent
user-group-count	User Group count of <num> exceeds threshold of <num>
connect-vm-info-source-failure	vm-info-source <name>(vsys<id>): failed to connected to <host>, status <message>
connect-agent-failure	<agent> <name>(vsys<id>): <status> details: <details>
HA-queue-full	HA queue is full
HA-queue-full	CFG HA queue is full

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Event ID	Message
connect-agent-failure	User-ID Agent peer's certificate RSA public key size is less than 2048 bits
connect-agent-failure	User-ID Agent X509_verify_cert returned error <num> error = '<error>'
connect-agent-failure	User-ID Agent cert name validation failed
connect-agent-failure	User-ID Agent server cert revoked/invalid
connect-agent-failure	User-ID Agent peer's certificate RSA public key size is less than 2048 bits
connect-agent-failure	User-ID Agent X509_verify_cert returned error <num> error = '<error>'
connect-agent-failure	User-ID Agent cert name validation failed
connect-agent-failure	User-ID Agent server cert revoked/invalid
connect-agent-failure	User-ID Agent server cert revoked/invalid
connect-agent-failure	User-ID Agent peer's certificate RSA public key size is less than 2048 bits
connect-agent-failure	User-ID Agent X509_verify_cert returned error <num>, error = '<error>'
connect-agent-failure	User-ID Agent cert name validation failed
connect-server-monitor-failure	User-ID server monitor <name>(vsys<id>) <status>
connect-server-monitor	User-ID WinRM server monitor <name>(vsys<id>): certificate RSA public key size is less than 2048 bits
connect-server-monitor	User-ID WinRM X509_verify_cert returned error <num> error = '<error>'
connect-server-monitor	User-ID WinRM cert name validation failed
connect-server-monitor	User-ID WinRM server cert revoked/invalid
connect-server-monitor-failure	Server monitor <name>(vsys<id>): connection failed, <error>

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Event ID	Message
connect-vm-info-source-failure	vm-info-source <name>(vsys<id>): failed to connected to <host>, status <status>
connect-vm-info-source-failure	vm-info-source <name>(vsys<id>): failed to connected to <host>, status <status>
connect-vm-info-source-failure	vm-info-source <name>(vsys<id>): failed to connected to GCE, status <status>
connect-vm-info-source-failure	vm-info-source <name>(vsys<id>): failed to connected to <host>, status <status>

wildfire

Event ID	Message
wildfire-auth-failed	WildFire failed to retrieve verdict.Authentication or Client Certificate failure.
wildfire-auth-failed	WildFire failed to send query.Authentication or Client Certificate failure.
wildfire-disabled-by-cloud	WildFire failed to send query.Client Certificate has expired or is not yet valid.
wildfire-auth-failed	WildFire failed to send query."Authentication or Client Certificate failure.
wildfire-invalid-cloud-info	WildFire <name> channel registration received invalid cloud info. Details in varrcvr.log.
wildfire-no-license	WildFire <name> channel registration failed due to invalid WildFire license.
wildfire-wrong-cloud-type	WildFire registration failed. Cloud type <type> (<name>) is not allowed for <name> channel.
wildfire-auth-failed	WildFire registration failed.Authentication or Client Certificate failure.
wildfire-auth-failed	WildFire registration failed.Mismatched Serial number in certificate and payload.

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Event ID	Message
wildfire-no-policy	WildFire <name> channel disabled. "Invalid <name> Cloud server configuration '<name>'.

Slog

- GRPC status DEADLINE_EXCEEDED in intelligent offload
- Inserted 100G QSFP28 module "(Vendor '<name>';Part '<name>';id '<id>') is not supported on 40G (port <num>) of PA-5220.
- No valid dataplane ports found at startup.
- Failed to install SSL Inbound Certificate(s) in Data Plane.
- Memory error detected.
- <name>Drive error detected.
- Not enough space to load content to SHM
- device-server HA queue is full
- GlobalProtect data file version <version> failed to install version
- Number of hints on disk has exceeded <num> due to log forward failures.
- Created CSR Cert '<name>'
- Delete Cert '<name>'
- Created CA Cert '<name>'
- Signed Cert '<name>' for device '<name>'
- Signed Renewal Cert '<name>' for device '<name>'
- SC3 Device certificate state has been reset!
- Attempted to fix partition <name>. If any problems are encounted, it is advisable to update this partition
- Daily packet capture limit (directory <name> limit <num>) has been reached.
- Unable to get instance/domains for region
- Unable to get attributes for region:%s instance:%s
- Unable to get all regions
- dsc HA state is changed from %d to %d
- DPI: EAL message format is changed to Json[prev: %d]
- DPI: EAL message format is changed to protobuf[prev: %d]

Critical System Log Messages

E-Log

Log Tags:

- auth
- bfd

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- crypto
- dhcp
- dynamic-updates
- fips
- general
- gre
- hw
- ipv6nd
- lacp
- panorama-check
- pbf
- raid
- routing
- satd
- sdwan
- tls
- url-filtering
- userid
- uuid
- vm
- vpn
- wildfire-appliance

auth

Event ID	Message
auth-server-down	3 tries to bind back to binddn failed: basedn: <name> ; binddn: <name> ; bind_timelimit <num> ; ip: <ip> ; uri: <url>
edl-cli-auth-failure	EDL server certificate authentication failed. The associated external dynamic list has been removed, which might impact your policy. EDL Name: <name>, EDL Source URL: <url>, CN: <name>, Reason: <reason>
auth-server-up	<name> auth server <name> is up !!!
auth-server-down	<name> auth server <name> is down !!!

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Event ID	Message
create-admin-acct-error	Failed to create local user account for admin user: <name>
auth-success	When authenticating user '<name>' <remotehost>, a less secure authentication method <proto> is used. Please migrate to PEAP or EAP-TTLS. Authentication Profile '<name>', vsys '<name>', Server Profile '<name>', Server Address '<ip>'
user-password-change-failed	When authenticating user '<name>' <remotehost>, a less secure authentication method <proto> is used. Please migrate to PEAP or EAP-TTLS. Authentication Profile '<name>', vsys '<name>', Server Profile '<name>', Server Address '<ip>'

bfd

Event ID	Message
session-state-change	BFD state changed to <name> for BFD session <name> to neighbor <name> on interface <name>. Protocol: <name>
forward-plane-reset	BFD forwarding plane reset for BFD session <name> to neighbor <name> on interface <name>. Protocol: <name>

crypto

Event ID	Message
mkey-expiry-reminder	Master key will expire in <num> days <num>h:<num>m:<num>s
mkey-expiry	Master key expired. Automatically renew master key lifetime enabled. Extend lifetime by <num> days <num> hours
mkey-expiry	Master key is now expired
cert-expiry	Shared certificate <name> and corresponding key have expired

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Event ID	Message
cert-expiry	Certificate <name> and corresponding key in vsys <num> have expired
HSM-state-change	HSM connectivity is up. Server(s) <ip>
HSM-state-change	HSM connectivity is down. Server(s) <ip>
HSM-state-change	HSM connectivity is down.
deploy-mkey-change	Deploy master-key job was attempted on <num> device(s)
private-key-export	Private key <entry> was exported by user <name>
mkey-change	Master key changed by <name>.
mkey-change	Master key changed by <name> failed
mkey-change	Master key encryption-level changed by <name>
mkey-change	Master key encryption-level changed by <name> failed

dhcp

Event ID	Message
if-clear	DHCP client cleared IP address on interface:<name> due to: Configuration removed
if-clear	DHCP client cleared IP address on interface:<name> due to: Lease expiry
if-clear	DHCP client cleared IP address on interface:<name> due to: Release trigger
if-clear	DHCP client cleared IP address on interface:<name> due to: All Request retries exhausted.
if-clear	DHCP client cleared IP address on interface:<name> due to: NAK from server
if-clear	DHCP client cleared IP address on interface:<name> due to: Release initiated

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Event ID	Message
	due to internal error. Please check for duplicate IPs or overlapping Subnets.
if-clear	DHCP client cleared IP address on interface:<name> due to: <reason>

dynamic-updates

Event ID	Message
palo-alto-networks-message	<message>

fips

Event ID	Message
fips-selftest	FIPS Mode Self-test <description> ..... succeeded
fips-selftest	FIPS-CC Mode Self-test <description> ..... succeeded
fips-selftest	FIPS-CC self-tests failed. Entering error state.
fips-selftest	FIPS-CC self-tests failed. Entering error state.
fips-entropy-rtciid	RTC-IID Persistent Failure - rebooting...
fips-selftest-timeout	FIPS failure. <description> failed.
fips-selftest-integ	FIPS failure. <description> failed.
fips-selftest-drng	FIPS failure. <description> failed.
fips-selftest-ndrng	FIPS failure. <description> failed.
fips-selftest-sha	FIPS failure. <description> failed.
fips-selftest-hmac	FIPS failure. <description> failed.
fips-selftest-aes	FIPS failure. <description> failed.
fips-selftest-des	FIPS failure. <description> failed.
fips-selftest-rsa	FIPS failure. <description> failed.
fips-selftest-dsa	FIPS failure. <description> failed.

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Event ID	Message
fips-selftest-dh-parameter	FIPS failure. <description> failed.
fips-selftest-dh	FIPS failure. <description> failed.
fips-selftest-cmac	FIPS failure. <description> failed.
fips-selftest-drbg	FIPS failure. <description> failed.
fips-selftest-ecdsa	FIPS failure. <description> failed.
fips-selftest-ecdh	FIPS failure. <description> failed.
fips-selftest-timeout	FIPS-CC failure. <description> failed.
fips-selftest-integ	FIPS-CC failure. <description> failed.
fips-selftest-drng	FIPS-CC failure. <description> failed.
fips-selftest-ndrng	FIPS-CC failure. <description> failed.
fips-selftest-sha	FIPS-CC failure. <description> failed.
fips-selftest-hmac	FIPS-CC failure. <description> failed.
fips-selftest-aes	FIPS-CC failure. <description> failed.
fips-selftest-des	FIPS-CC failure. <description> failed.
fips-selftest-rsa	FIPS-CC failure. <description> failed.
fips-selftest-dsa	FIPS-CC failure. <description> failed.
fips-selftest-dh-parameter	FIPS-CC failure. <description> failed.
fips-selftest-dh	FIPS-CC failure. <description> failed.
fips-selftest-cmac	FIPS-CC failure. <description> failed.
fips-selftest-drbg	FIPS-CC failure. <description> failed.
fips-selftest-ecdsa	FIPS-CC failure. <description> failed.
fips-selftest-ecdh	FIPS-CC failure. <description> failed.
fips-selftest-core	<num> of <num> dataplane processor cores failed verification.

general

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Event ID	Message
general	Slot s<num>: Check/fix volume 'appinfo' path didn't find expected dir.

gre

Event ID	Message
tunnel-recur-routing	Tunnel intf: <name> is going down due to recursive routing
tunnel-status-down	Tunnel <name> is going down due to tunnel monitoring failed
tunnel-status-up	Tunnel <name> is going up

hw

Event ID	Message
fan-failure	Alarm on Fan Tray #<num>
ps-failure	Alarm on Power Supply #<num>
Content Engine Failure	CE10 init failed.
Content Engine Failure	CA1 init failed.
insufficient-power	DP power status is bad, shutting system down!
insufficient-power	CP power status is bad!

ipv6nd

Event ID	Message
duplicated-IPv6-address-found	IPv6 address <address> on interface <name> is duplicate. IPv6 disabled on the interface.
duplicated-IPv6-address-found	IPv6 address <address> on interface <name> is duplicate. Address disabled.

lacp

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Event ID	Message
lacp-up	LACP interface <name> moved into AE-group <name>.
nego-fail	LACP interface <name> moved out of AE-group <name. Selection state <state>
lost-connectivity	LACP interface <name> moved out of AE-group <name>(lost connectivity to existing peer. Last connected peer port number <port>)
unresponsive	LACP interface <name> moved out of AE-group <name>(peer is not responding to new LACP connection)
speed-duplex	LACP interface <name> moved out of AE-group <name>. Selection state <state>
link-down	LACP interface <name> moved out of AE-group <name>. Selection state <state>
link-down	LACP interface <name> moved out of AE-group <name>(link-state was manually configured to down)
nego-fail	LACP interface <name> moved out of AE-group <name>. Selection state <state>
lacp-down	LACP interface <name> moved out of AE-group <name>. Selection state <state>

panorama-check

Event ID	Message
panorama-check-test	Panorama connectivity check for <name> failed. Reason: <reason>
panorama-check-test	Panorama connectivity check for <name> failed. Reason: <reason>

pbf

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Event ID	Message
pbf-fqdn-down	Vsys <id> PBF rule <name> nexthop FQDN <key> is unresolved for IPv4
pbf-fqdn-down	Vsys <id> PBF rule <name> nexthop FQDN <key> is unresolved for IPv6
pbf-fqdn-down	Vsys <id> PBF rule <name> nexthop FQDN <key> resolved IP <ip> is not in same subnet as interface IP. It will not be used as FQDN nexthop.

raid

Event ID	Message
pair-disappeared	No Logging Raid Disk Pair Available Notifying HA
pair-detected	No Logging Raid Disk Pair Available Notifying HA

routing

Event ID	Message
routed-static-fqdn-down	Routed static fqdn mapping is unresolved
routed-bgp-fqdn-down	Routed BGP fqdn mapping is unresolved
path-monitor-recovery	Path monitoring for static route destination <ip> with next hop <name> recovered. Route restored.
path-monitor-failure	Path monitoring failed for static route destination <ip> with next hop <name>. Route removed.

satd

Event ID	Message
satd-portal-connect-failed	GlobalProtect Satellite connection to portal failed.
satd-gateway-connect-failed	GlobalProtect Satellite connection to gateway failed.

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sdwan

Event ID	Message
sdwan-vif-status-up	<vif> is up
sdwan-vif-status-down	<vif> is down

tls

Event ID	Message
panos-auth-failure	RADIUS server certification failed. Server: <name>; CRL/OCSP failed, <reason>
tls-edl-auth-failure	EDL server certificate authentication failed. A local copy of associated external dynamic list will be used, so it won't impact your policy. EDL Name: <name>, EDL Source URL: <url>, CN: <name>, Reason: <reason>
tls-edl-auth-failure	EDL server certificate authentication failed. The associated external dynamic list has been removed, which might impact your policy. EDL Name: <name>, EDL Source URL: <url>, CN: <name>, Reason: CRL/OCSP check failed, <reason>
panos-auth-failure	<name> Server CN: <name> Failed to establish connection due to <error>
panorama-auth-failure	Client authentication failed <error> PAN-OS ver: <version> Panorama ver:<version> Client IP: <ip> Server IP: <ip> Client cert CN: <name>
panorama-auth-failure	Client identity check failed. PAN-OS ver: <version> Panorama ver: <version> Client IP: <ip> Server IP: <ip> Client Cert CN: <name>
tls-X509-ocsp-crl-check-failed	Connection to HTTP server(<host>) failed due to server certificate: '<name>' is <reason>
tls-X509-validation-failed	HTTP server certificate validation failed. Host: <host>, CN: <name>, Reason: <reason>
mfa-auth-failure	MFA server certification failed. Server: <name>; CRL/OCSP failed, <reason>

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Event ID	Message
mfa-auth-failure	MFA: server certificate validation failed. Peer: '<name>' Vsys: <id> (<id>:<error>)
panorama-auth-failure	Client authentication failed <error> Client IP: <ip>:<port> Server IP: <ip>:<port> Client cert CN: <name>
tls-X509-ocsp-crl-check-failed	Connection to EMAIL server(<host>) failed due to server certificate: '<subject>' is <reason>
tls-X509-validation-failed	EMAIL server certificate validation failed. Host: <host>, CN: <name>, Reason: <reason>

url-filtering

Event ID	Message
no-url-database	No URL database! Please download one from 'dynamic update page'
seed-out-of-sync	PAN-DB seed is out of sync Download of a new seed is required!!!
startup-failure	Failed to construct the URL DB!

userid

Event ID	Message
registered-ip-max-platform-limit-exceeded	max registered-ip for the platform reached (<num>)
registered-ip-update-failure	fail to integrate the update of registered ip addresses since <num> seconds ago
registered-ip-update-failure	fail to sync the update of registered ip addresses
registered-ip-update-failure	NSX initial sync request for ip-tag mappings failed after <num> times retry. Suggest a manual sync from panorama.
registered-ip-update-failure	fail to sync the update of registered ip addresses

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Event ID	Message
registered-user-max-platform-limit-exceeded	limitation of total registered-user reached (<num>)
agent-version-mismatch	Device requires protocol ver. <num> "but <name> supports only ver. <num>

uuid

Event ID	Message
policy-rule-uuid-modified	Policy Rules UUIDs are modified by load using 'Regenerate Rule UUIDs for selected named configuration' option

vm

Event ID	Message
dvf-init-fail	VMware dvfilter init failed <status> <id>
dvf-init-fail	VMware dvfilter init dev failed <status> devId <id> status <id>

vpn

Event ID	Message
ikev2-nego-cert-id-mismatch	IKEv2 SA negotiation failed.
ike-nego-p1-fail-common	IKE phase-1 negotiation is failed_COMM
ikev2-nego-ike-fail	IKEv2 IKE SA negotiation is failed
tunnel-status-up	Tunnel <name> (id:<id>, peer: <peer>) is up
tunnel-status-down	Tunnel <name> (id:<id>, peer: <peer>) is down
tunnel-status-up	Tunnel <name> is up
tunnel-status-down	Tunnel <name> is down

wildfire-appliance

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Event ID	Message
cluster-entered-split-brain	Cluster enters split-brain mode.
cluster-entered-split-brain	Cluster leaves split-brain mode.
cluster-entered-split-brain	Cluster leaves split-brain mode.

Slog

- Chassis Master Alarm: Cleared
- Chassis Master Alarm: <name>
- Fan Tray <id>, Fan <id> failed!
- Fan Zone <id> failed, shutting down!
- Fan Tray <id>, Fan <id> failed!
- Fan Zone <id> failed shutting down!
- System is powering itself down due to missing fan tray.
- No Raid Disk Pair Available, rebooting!
- Thermal alarm on slot <id>
- Shutting down system for thermal temperature.
- Shutting down the system for slot <id> thermal temperature.
- Shutting down slot <id> for thermal temperature.
- SW version doesn't match, MP software version <version>, DP software version <version>
- Release slot failed.
- Slot allocation failed
- Successfully renewed device certificate
- Successfully removed device certificate
- Out of memory condition detected, kill process <id>
- Device certificate status: <num>. It cannot be renewed
- LP shmgr memory map is out of sync
- intelligent-traffic-offload license expired
- User-ID manager was reset. Commit is required to reinitialize User-ID
- Traffic and logging resumed
- Traffic and logging suspended due to unexported logs
- Traffic and logging are suspended since traffic-stop-on-logdb-full feature has been enabled
- Audit storage for <name> logs is full. No new traffic sessions will be accepted until disk space is freed up
- Minimum Retention Period (<num> days) Violated for segnum:<num> type:<name>

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SNMP Monitoring and Traps

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

You can use an SNMP manager to monitor event-driven alerts and operational statistics for the firewall, Panorama, or WF-500 appliance and for the traffic they process. The statistics and traps can help you identify resource limitations, system changes or failures, and malware attacks. You configure alerts by forwarding log data as traps, and enable the delivery of statistics in response to GET messages (requests) from your SNMP manager. Each trap and statistic has an object identifier (OID). Related OIDs are organized hierarchically within the Management Information Bases (MIBs) that you load into the SNMP manager to enable monitoring.

*SNMP requests do **not** return real-time instantaneous values. There is a polling sleep interval of up to one minute between when a value changes on the device and when it is reflected in an SNMP response. It is recommended that SNMP not be used for real-time monitoring of rapidly-changing counters.*

When an event triggers SNMP trap generation (for example, an interface goes down), the firewall, Panorama virtual appliance, M-Series appliance, and WF-500 appliance respond by updating the corresponding SNMP object (for example, the interfaces MIB) instead of waiting for the periodic update of all objects that occurs every ten seconds. This ensures that your SNMP manager displays the latest information when polling an object to confirm an event.

The firewall, Panorama, and WF-500 appliance support SNMP Version 2c and Version 3. Decide which to use based on the version that other devices in your network support and on your network security requirements. SNMPv3 is more secure and enables more granular access control for system statistics than SNMPv2c. The following table summarizes the security features of each version. You select the version and configure the security features when you Monitor Statistics Using SNMP and Forward Traps to an SNMP Manager.

SNMPVe	Authentication	Message Privacy	Messag	MIB Access Granularity
SNMPv2c	Community string	No (cleartext)	No	SNMP community access for all MIBs on a device
SNMPv3	EngineID, username, and authentication password (SHA hashing for the password)	Privacy password for AES (128, 192, or 256) encryption of SNMP messages	Yes	User access based on views that include or exclude specific OIDs

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Figure 1 illustrates a deployment in which firewalls forward traps to an SNMP manager while also forwarding logs to Log Collectors. Alternatively, you could configure the Log Collectors to forward the firewall traps to the SNMP manager. For details on these deployments, refer to Log Forwarding Options in Centralized Logging and Reporting. In all deployments, the SNMP manager gets statistics directly from the firewall, Panorama, or WF-500 appliance. In this example, a single SNMP manager collects both traps and statistics, though you can use separate managers for these functions if that better suits your network.

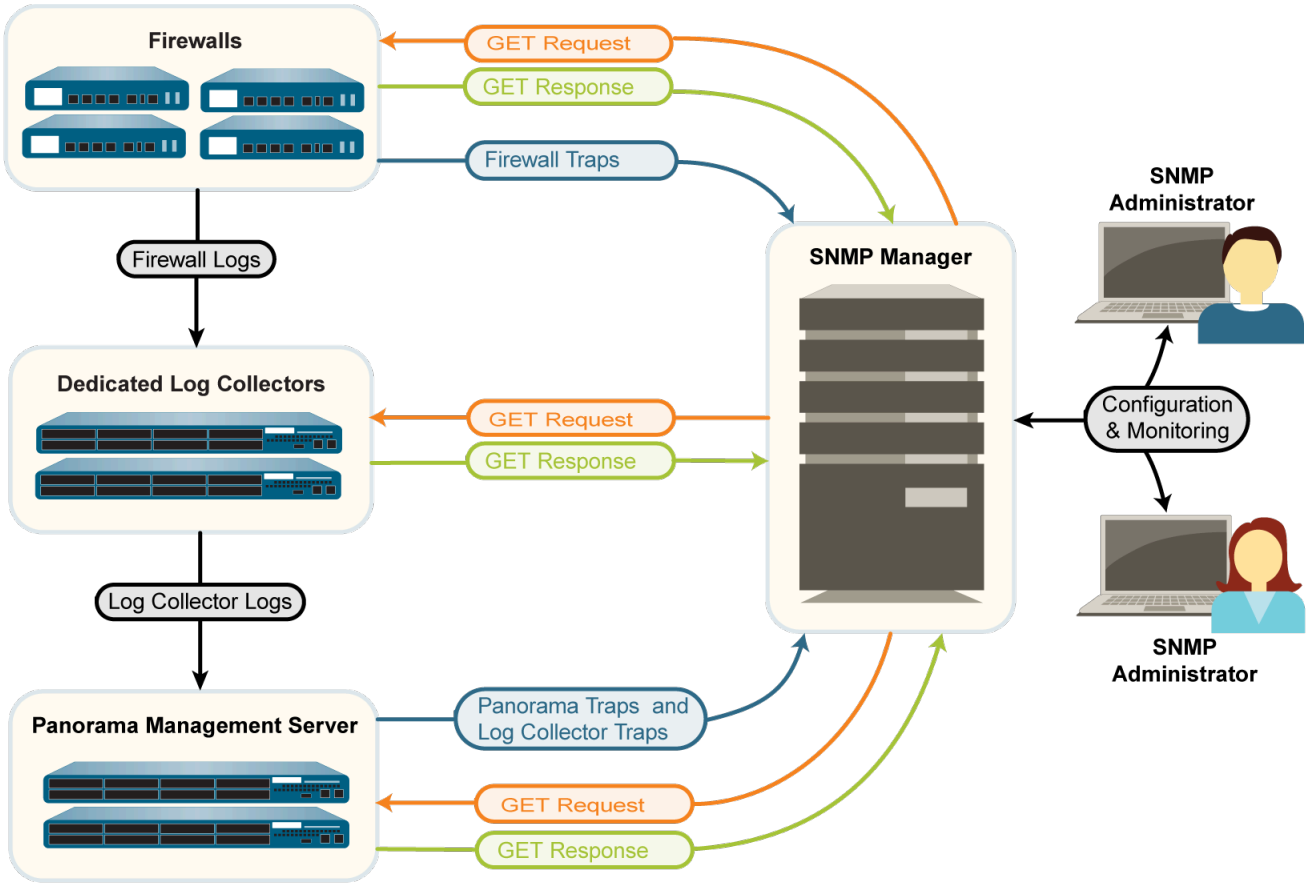

Figure 2: SNMP Implementation

The following topics describe how Palo Alto Networks firewalls, Panorama, and WF-500 appliances implement SNMP, and the procedures to configure SNMP monitoring and trap delivery.

Use an SNMP Manager to Explore MIBs and Objects

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	☐ Support license ☐ (Panorama) Device management license

To use SNMP for monitoring Palo Alto Networks firewalls, Panorama, or WF-500 appliances, you must first load the Supported MIBs into your SNMP manager and determine which object

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identifiers (OIDs) correspond to the system statistics and traps you want to monitor. The following topics provide an overview of how to find OIDs and MIBs in an SNMP manager. For the specific steps to perform these tasks, refer to your SNMP management software.

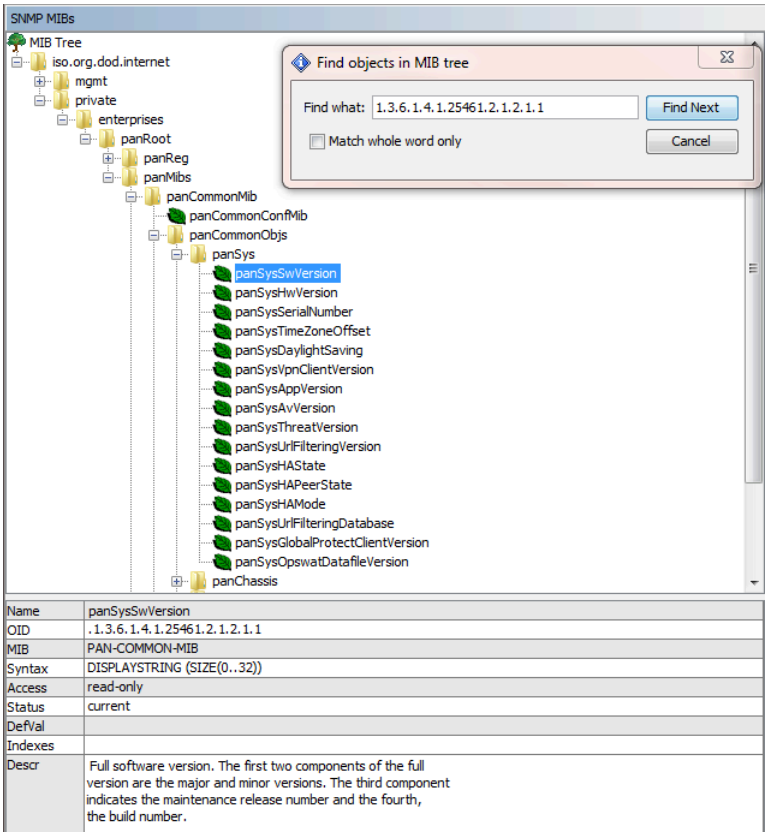
- Identify a MIB Containing a Known OID
- Walk a MIB
- Identify the OID for a System Statistic or Trap

Identify a MIB Containing a Known OID

If you already know the OID for a particular SNMP object (statistic or trap) and want to know the OIDs of similar objects so you can monitor them, you can explore the MIB that contains the known OID.

STEP 1 | Load all the Supported MIBs into your SNMP manager.

STEP 2 | Search the entire MIB tree for the known OID. The search result displays the MIB path for the OID, as well as information about the OID (for example, name, status, and description). You can then select other OIDs in the same MIB to see information about them.

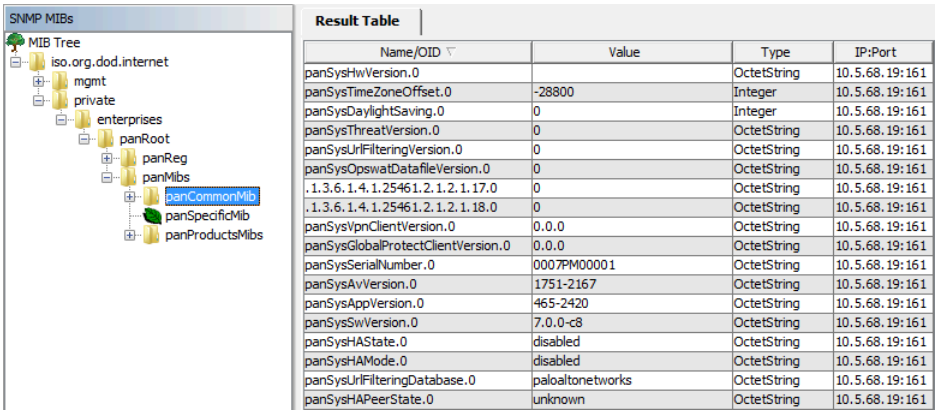


STEP 3 | (Optional) Walk a MIB to display all its objects.

Walk a MIB

If you want to see which SNMP objects (system statistics and traps) are available for monitoring, displaying all the objects of a particular MIB can be useful. To do this, load the Supported MIBs into your SNMP manager and perform a *walk* on the desired MIB. To list the traps

that Palo Alto Networks firewalls, Panorama, and WF-500 appliance support, walk the panCommonEventEventsV2 MIB. In the following example, walking the PAN-COMMON-MIB.my displays the following list of OIDs and their values for certain statistics:



Name/OID	Value	Type	IP:Port
panSysHwVersion.0		OctetString	10.5.68.19:161
panSysTimeZoneOffset.0	-28800	Integer	10.5.68.19:161
panSysDaylightSaving.0	0	Integer	10.5.68.19:161
panSysThreatVersion.0	0	OctetString	10.5.68.19:161
panSysUrlFilteringVersion.0	0	OctetString	10.5.68.19:161
panSysOpswatDatafileVersion.0	0	OctetString	10.5.68.19:161
.1.3.6.1.4.1.25461.2.1.2.1.17.0	0	OctetString	10.5.68.19:161
.1.3.6.1.4.1.25461.2.1.2.1.18.0	0	OctetString	10.5.68.19:161
panSysVpnClientVersion.0	0.0.0	OctetString	10.5.68.19:161
panSysGlobalProtectClientVersion.0	0.0.0	OctetString	10.5.68.19:161
panSysSerialNumber.0	0007PM00001	OctetString	10.5.68.19:161
panSysAvVersion.0	1751-2167	OctetString	10.5.68.19:161
panSysAppVersion.0	465-2420	OctetString	10.5.68.19:161
panSysSwVersion.0	7.0.0-c8	OctetString	10.5.68.19:161
panSysHAState.0	disabled	OctetString	10.5.68.19:161
panSysHAMode.0	disabled	OctetString	10.5.68.19:161
panSysUrlFilteringDatabase.0	paloaltonetworks	OctetString	10.5.68.19:161
panSysHAPeerState.0	unknown	OctetString	10.5.68.19:161

Identify the OID for a System Statistic or Trap

To use an SNMP manager for monitoring Palo Alto Networks firewalls, Panorama, or WF-500 appliances, you must know the OIDs of the system statistics and traps you want to monitor.

STEP 1 | Review the Supported MIBs to determine which one contains the type of statistic you want. For example, the PAN-COMMON-MIB.my contains hardware version information. The panCommonEventEventsV2 MIB contains all the traps that Palo Alto Networks firewalls, Panorama, and WF-500 appliances support.

STEP 2 | Open the MIB in a text editor and perform a keyword search. For example, using **Hardware version** as a search string in PAN-COMMON-MIB identifies the panSysHwVersion object:

```
panSysHwVersion OBJECT-TYPE
 SYNTAX DisplayString (SIZE(0..128))
 MAX-ACCESS read-only
 STATUS current
 DESCRIPTION
 "Hardware version of the unit."
 ::= {panSys 2}
```

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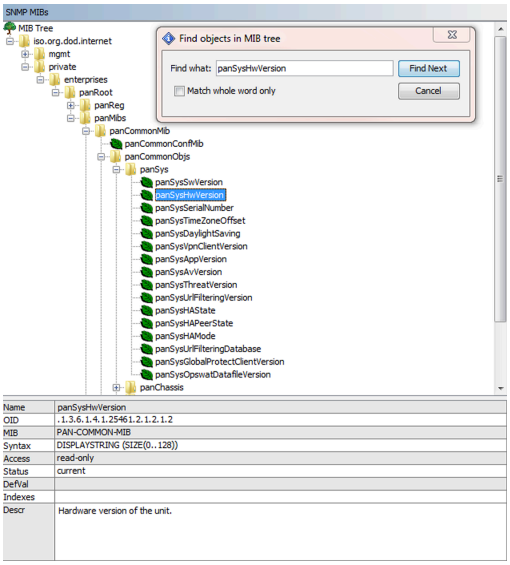
STEP 3 | In a MIB browser, search the MIB tree for the identified object name to display its OID. For example, the panSysHwVersion object has an OID of 1.3.6.1.4.1.25461.2.1.2.1.2.



Enable SNMP Services for Firewall-Secured Network Elements

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

If you will use Simple Network Management Protocol (SNMP) to monitor or manage network elements (for example, switches and routers) that are within the security zones of Palo Alto Networks firewalls, you must create a security rule that allows SNMP services for those elements.

You don't need a security rule to enable SNMP monitoring of Palo Alto Networks firewalls, Panorama, or WF-500 appliances. For details, see Monitor Statistics Using SNMP.

STEP 1 | Create an application group.

1. Select **Objects** > **Application Group** and click **Add**.
2. Enter a **Name** to identify the application group.
3. Click **Add**, type **snmp**, and select **snmp** and **snmp-trap** from the drop-down.
4. Click **OK** to save the application group.

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STEP 2 | Create a security rule to allow SNMP services.

1. Select **Policies** > **Security** and click **Add**.
2. In the **General** tab, enter a **Name** for the rule.
3. In the **Source** and **Destination** tabs, click **Add** and enter a **Source Zone** and a **Destination Zone** for the traffic.
4. In the **Applications** tab, click **Add**, type the name of the applications group you just created, and select it from the drop-down.
5. In the **Actions** tab, verify that the **Action** is set to **Allow**, and then click **OK** and **Commit**.

Monitor Statistics Using SNMP

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	☐ Support license ☐ (Panorama) Device management license

The statistics that a Simple Network Management Protocol (SNMP) manager collects from Palo Alto Networks firewalls can help you gauge the health of your network (systems and connections), identify resource limitations, and monitor traffic or processing loads. The statistics include information such as interface states (up or down), active user sessions, concurrent sessions, session utilization, temperature, and system uptime.

You can't configure an SNMP manager to control Palo Alto Networks firewalls (using SET messages), only to collect statistics from them (using GET messages). For details on how SNMP is implemented for Palo Alto Networks firewalls, see SNMP Monitoring and Traps.

STEP 1 | Configure the SNMP Manager to get statistics from firewalls.

The following steps provide an overview of the tasks you perform on the SNMP manager. For the specific steps, refer to the documentation of your SNMP manager.

1. To enable the SNMP manager to interpret firewall statistics, load the Supported MIBs for Palo Alto Networks firewalls and, if necessary, compile them.
2. For each firewall that the SNMP manager will monitor, define the connection settings (IP address and port) and authentication settings (SNMPv2c community string or SNMPv3 EngineID/username/password) for the firewall.

   All Palo Alto Networks firewalls use port 161.

   The SNMP manager can use the same or different connection and authentication settings for multiple firewalls. The settings must match those you define when you configure SNMP on the firewall (see Step 3). For example, if you use SNMPv2c, the community string you define when configuring the firewall must match the community string you define in the SNMP manager for that firewall.
3. Determine the object identifiers (OIDs) of the statistics you want to monitor. For example, to monitor the session utilization percentage of a firewall, a MIB browser

shows that this statistic corresponds to OID 1.3.6.1.4.1.25461.2.1.2.3.1.0 in PAN-COMMON-MIB.my. For details, see Use an SNMP Manager to Explore MIBs and Objects.

4. Configure the SNMP manager to monitor the desired OIDs.

STEP 2 | Enable SNMP traffic on a firewall interface.

This is the interface that will receive statistics requests from the SNMP manager.

PAN-OS doesn't synchronize management (MGT) interface settings for firewalls in a high availability (HA) configuration. You must configure the interface for each HA peer.

Perform this step in the firewall web interface.

- To enable SNMP traffic on the MGT interface, select **Device** > **Setup** > **Interfaces**, edit the **Management** interface, select **SNMP**, and then click **OK** and **Commit**.
- To enable SNMP traffic on any other interface, create an interface management profile for SNMP services and assign the profile to the interface that will receive the SNMP requests. The interface type must be Layer 3 Ethernet.

STEP 3 | Configure the firewall to respond to statistics requests from an SNMP manager.

PAN-OS doesn't synchronize SNMP response settings for firewalls in a high availability (HA) configuration. You must configure these settings for each HA peer.

1. Select **Device** > **Setup** > **Operations** and, in the Miscellaneous section, click **SNMP Setup**.
2. Select the SNMP **Version** and configure the authentication values as follows. For version details, see SNMP Monitoring and Traps.
   - **V2c**—Enter the **SNMP Community String**, which identifies a community of SNMP managers and monitored devices, and serves as a password to authenticate the community members to each other.

     *As a best practice, don't use the default community string **`public`**; it's well known and therefore not secure.*
   - **V3**—Create at least one SNMP view group and one user. User accounts and views provide authentication, privacy, and access control when firewalls forward traps and SNMP managers get firewall statistics.
     - **Views**—Each view is a paired OID and bitwise mask: the OID specifies a MIB and the mask (in hexadecimal format) specifies which objects are accessible within (include matching) or outside (exclude matching) that MIB. Click **Add** in the first list and enter a **Name** for the group of views. For each view in the group, click **Add** and configure the view **Name**, **OID**, matching **Option** (**include** or **exclude**), and **Mask**.
     - **Users**—Click **Add** in the second list, enter a username under **Users**, select the **View** group from the drop-down, enter the authentication password (**Auth Password**)

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used to authenticate to the SNMP manager, and enter the privacy password (**Priv Password**) used to encrypt SNMP messages to the SNMP manager.

3. Click **OK** and **Commit**.

STEP 4 | Monitor the firewall statistics in an SNMP manager.

Refer to the documentation of your SNMP manager for details.

When monitoring statistics related to firewall interfaces, you must match the interface indexes in the SNMP manager with interface names in the firewall web interface. For details, see Firewall Interface Identifiers in SNMP Managers and NetFlow Collectors.

Forward Traps to an SNMP Manager

Where Can I Use This?	What Do I Need?
• NGFW	For Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Pro ☐ Strata Cloud Manager Essentials

Simple Network Management Protocol (SNMP) traps can alert you to system events (failures or changes in hardware or software of Palo Alto Networks firewalls) or to threats (traffic that matches a firewall security rule) that require immediate attention.

To see the list of traps that Palo Alto Networks firewalls support, use your SNMP Manager to access the panCommonEventEventsV2 MIB. For details, see Use an SNMP Manager to Explore MIBs and Objects.

For details on how for Palo Alto Networks firewalls implement SNMP, see SNMP Monitoring and Traps.

- Strata Cloud Manager
- PAN-OS & Panorama

Forward Traps to an SNMP Manager (Strata Cloud Manager)

STEP 1 | Enable the SNMP manager to interpret the traps it receives.

Load the Supported MIBs for Palo Alto Networks firewalls and, if necessary, compile them. For the specific steps, refer to the documentation of your SNMP manager.

STEP 2 | Configure an SNMP Trap server profile.

The profile defines how the firewall accesses the SNMP managers (trap servers). You can define up to four SNMP managers for each profile.

Optionally, configure separate SNMP Trap server profiles for different log types, severity levels, and WildFire verdicts.

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STEP 3 | Log in to Strata Cloud Manager.

STEP 4 | Select **Manage** > **Configuration** > **NGFW and Prisma Access** > **Objects** > **Log Forwarding** > **SNMP Server Profile**.

STEP 5 | Click **Add SNMP Server** and enter a **Name** for the profile.

1. Select the SNMP **Version** and configure the authentication values as follows. For version details, see SNMP Monitoring and Traps.

   - **V2c**—For each server, click **Add** and enter the server **Name**, IP address (**SNMP Manager**), and **Community String**. The community string identifies a community of SNMP managers and monitored devices, and serves as a password to authenticate the community members to each other.

     As a best practice, don't use the default community string ***public****; it's well known and therefore not secure.*

   - **V3**—For each server, click **Add** and enter the server **Name**, IP address (**SNMP Manager**), SNMP **User** account (this must match a username defined in the SNMP manager), **EngineID** used to uniquely identify the firewall (you can leave the field blank to use the firewall serial number), authentication password (**Auth Password**) used to authenticate to the server, privacy password (**Priv Password**) used to encrypt SNMP messages to the server, (**Authentication Protocol**), and (**Privacy Protocol**).

     When selecting a combination of Authentication Protocol and Privacy Protocol, ensure that the Authentication Protocol is equal to or more advanced than the Privacy Protocol.

     For example, SHA-256 and AES are compatible, however SHA and AES-192 are not.

2. Click **Save** to save the server profile.

STEP 6 | Configure log forwarding.

1. Configure the destinations of Traffic, Threat, and WildFire traps:

   1. Create a Log Forwarding profile. For each log type and each severity level or WildFire verdict, select the **SNMP Trap** server profile.
   2. Assign the Log Forwarding profile to policy rules and network zones. The rules and zones will trigger trap generation and forwarding.

2. Configure the destinations for System, Configuration, User-ID, HIP Match, and Correlation logs. For each log (trap) type and severity level, select the **SNMP Trap** server profile.
3. **Push Config** to push your configuration changes.

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STEP 7 | Monitor the traps in an SNMP manager.

Refer to the documentation of your SNMP manager.

When monitoring traps related to firewall interfaces, you must match the interface indexes in the SNMP manager with interface names in the firewall web interface. For details, see Firewall Interface Identifiers in SNMP Managers and NetFlow Collectors.

Forward Traps to an SNMP Manager (PAN-OS)

STEP 1 | Enable the SNMP manager to interpret the traps it receives.

Load the Supported MIBs for Palo Alto Networks firewalls and, if necessary, compile them. For the specific steps, refer to the documentation of your SNMP manager.

STEP 2 | Configure an SNMP Trap server profile.

The profile defines how the firewall accesses the SNMP managers (trap servers). You can define up to four SNMP managers for each profile.

Optionally, configure separate SNMP Trap server profiles for different log types, severity levels, and WildFire verdicts.

1. Log in to the firewall web interface.
2. Select **Device** > **Server Profiles** > **SNMP Trap**.
3. Click **Add** and enter a **Name** for the profile.
4. If the firewall has more than one virtual system (vsys), select the **Location** (vsys or **Shared**) where this profile is available.
5. Select the SNMP **Version** and configure the authentication values as follows. For version details, see SNMP Monitoring and Traps.
   - **V2c**—For each server, click **Add** and enter the server **Name**, IP address (**SNMP Manager**), and **Community String**. The community string identifies a community of SNMP managers and monitored devices, and serves as a password to authenticate the community members to each other.

     As a best practice, don't use the default community string ***`public`****; it's well known and therefore not secure.*
   - **V3**—For each server, click **Add** and enter the server **Name**, IP address (**SNMP Manager**), SNMP **User** account (this must match a username defined in the SNMP manager), **EngineID** used to uniquely identify the firewall (you can leave the field blank to use the firewall serial number), authentication password (**Auth Password**) used to authenticate to the server, and privacy password (**Priv Password**) used to encrypt SNMP messages to the server.
6. Click **OK** to save the server profile.

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STEP 3 | Configure log forwarding.

1. Configure the destinations of Traffic, Threat, and WildFire traps:
   1. Create a Log Forwarding profile. For each log type and each severity level or WildFire verdict, select the **SNMP Trap** server profile.
   2. Assign the Log Forwarding profile to policy rules and network zones. The rules and zones will trigger trap generation and forwarding.
2. Configure the destinations for System, Configuration, User-ID, HIP Match, and Correlation logs. For each log (trap) type and severity level, select the **SNMP Trap** server profile.
3. Click **Commit**.

STEP 4 | Monitor the traps in an SNMP manager.

Refer to the documentation of your SNMP manager.

When monitoring traps related to firewall interfaces, you must match the interface indexes in the SNMP manager with interface names in the firewall web interface. For details, see Firewall Interface Identifiers in SNMP Managers and NetFlow Collectors.

Supported MIBs

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	☐ Support license ☐ (Panorama) Device management license

The following table lists the Simple Network Management Protocol (SNMP) management information bases (MIBs) that Palo Alto Networks firewalls, Panorama, and WF-500 appliances support. You must load these MIBs into your SNMP manager to monitor the objects (system statistics and traps) that are defined in the MIBs. For details, see Use an SNMP Manager to Explore MIBs and Objects.

MIB Type	Supported MIBs
Standard—The Internet Engineering Task Force (IETF) maintains most standard MIBs. You can download the MIBs from the IETF website.	MIB-II IF-MIB HOST-RESOURCES-MIB ENTITY-MIB ENTITY-SENSOR-MIB ENTITY-STATE-MIB

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MIB Type	Supported MIBs
Palo Alto Networks firewalls, Panorama, and WF-500 appliances don't support every object (OID) in every one of these MIBs. See the Supported MIBs links for an overview of the supported OIDs.	IEEE 802.3 LAG MIB LLDP-V2-MIB.my BFD-STD-MIB IP-MIB
Enterprise—You can download the enterprise MIBs from the Palo Alto Networks Technical Documentation portal.	PAN-COMMON-MIB.my PAN-GLOBAL-REG-MIB.my PAN-GLOBAL-TC-MIB.my PAN-LC-MIB.my PAN-PRODUCT-MIB.my PAN-ENTITY-EXT-MIB.my PAN-TRAPS.my PAN-CELLULAR-MIB.my (PAN-OS 12.2.2 and later; cellular-integrated firewalls only)

MIB-II

MIB-II provides object identifiers (OIDs) for network management protocols in TCP/IP-based networks. Use this MIB to monitor general information about systems and interfaces. For example, you can analyze trends in bandwidth usage by interface type (ifType object) to determine if the firewall needs more interfaces of that type to accommodate spikes in traffic volume.

Palo Alto Networks firewalls, Panorama, and WF-500 appliances support only the following object groups:

Object Group	Description
system	Provides system information such as the hardware model, system uptime, FQDN, and physical location.

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Object Group	Description
interfaces	Provides statistics for physical and logical interfaces such as type, current bandwidth (speed), operational status (for example, up or down), and discarded packets. Logical interface support includes VPN tunnels, aggregate groups, Layer 2 subinterfaces, Layer 3 subinterfaces, loopback interfaces, and VLAN interfaces.

RFC 1213 defines this MIB.

IF-MIB

IF-MIB supports interface types (physical and logical) and larger counters (64K) beyond those defined in MIB-II. Use this MIB to monitor interface statistics in addition to those that MIB-II provides. For example, to monitor the current bandwidth of high-speed interfaces (greater than 2.2Gps) such as the 10G interfaces of the PA-5200 Series firewalls, you must check the ifHighSpeed object in IF-MIB instead of the ifSpeed object in MIB-II. IF-MIB statistics can be useful when evaluating the capacity of your network.

Palo Alto Networks firewalls, Panorama, and WF-500 appliances support only the ifXTable in IF-MIB, which provides interface information such as the number of multicast and broadcast packets transmitted and received, whether an interface is in promiscuous mode, and whether an interface has a physical connector.

RFC 2863 defines this MIB.

HOST-RESOURCES-MIB

HOST-RESOURCES-MIB provides information for host computer resources. Use this MIB to monitor CPU and memory usage statistics. For example, checking the current CPU load (hrProcessorLoad object) can help you troubleshoot performance issues on the firewall.

Palo Alto Networks firewalls, Panorama, and WF-500 appliances support portions of the following object groups:

Object Group	Description
hrDevice	Provides information such as CPU load, storage capacity, and partition size. The hrProcessorLoad OIDs provide an average of the cores that process packets. For the PA-7000 and PA-5200 Series firewalls, which have multiple dataplanes (DPs), you can monitor individual dataplane processor utilization. Set alerts when utilization reaches a specific threshold for each DP processor to avoid service availability issues.
hrSystem	Provides information such as system uptime, number of current user sessions, and number of current processes.
hrStorage	Provides information such as the amount of used storage.

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RFC 2790 defines this MIB.

ENTITY-MIB

ENTITY-MIB provides OIDs for multiple logical and physical components. Use this MIB to determine what physical components are loaded on a system (for example, fans and temperature sensors) and see related information such as models and serial numbers. You can also use the index numbers for these components to determine their operational status in the ENTITY-SENSOR-MIB and ENTITY-STATE-MIB.

Palo Alto Networks firewalls, Panorama, and WF-500 appliances support only portions of the entPhysicalTable group:

Object	Description
entPhysicalIndex	A single namespace that includes disk slots and disk drives.
entPhysicalDescr	The component description.
entPhysicalVendorType	The sysObjectID (see PAN-PRODUCT-MIB.my) when it is available (chassis and module objects).
entPhysicalContainedIn	The value of entPhysicalIndex for the component that contains this component.
entPhysicalClass	Chassis (3), container (5) for a slot, power supply (6), fan (7), sensor (8) for each temperature or other environmental, and module (9) for each line card.
entPhysicalParentRelPos	The relative position of this *child* component among its *sibling* components. Sibling components are defined as entPhysicalEntry components that share the same instance values of each of the entPhysicalContainedIn and entPhysicalClass objects.
entPhysicalName	Supported only if the management (MGT) interface allows for naming the line card.
entPhysicalHardwareRev	The vendor-specific hardware revision of the component.
entPhysicalFirwareRev	The vendor-specific firmware revision of the component.
entPhysicalSoftwareRev	The vendor-specific software revision of the component.
entPhysicalSerialNum	The vendor-specific serial number of the component.
entPhysicalMfgName	The name of the manufacturer of the component.
entPhysicalMfgDate	The date when the component was manufactured.

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Object	Description
entPhysicalModelName	The disk model number.
entPhysicalAlias	An alias that the network manager specified for the component.
entPhysicalAssetID	A user-assigned asset tracking identifier that the network manager specified for the component.
entPhysicalIsFRU	Indicates whether the component is a field replaceable unit (FRU).
entPhysicalUris	The Common Language Equipment Identifier (CLEI) number of the component (for example, URN:CLEI:CNME120ARA).

RFC 4133 defines this MIB.

ENTITY-SENSOR-MIB

ENTITY-SENSOR-MIB adds support for physical sensors of networking equipment beyond what ENTITY-MIB defines. Use this MIB in tandem with the ENTITY-MIB to monitor the operational status of the physical components of a system (for example, fans and temperature sensors). For example, to troubleshoot issues that might result from environmental conditions, you can map the entity indexes from the ENTITY-MIB (entPhysicalDescr object) to operational status values (entPhysSensorOperStatus object) in the ENTITY-SENSOR-MIB. In the following example, all the fans and temperature sensors for a PA-3020 firewall are working:

Name/OID	Value
entPhysicalDescr.1	PA-3020
entPhysicalDescr.2	Fan #1 RPM
entPhysicalDescr.3	Fan #2 RPM
entPhysicalDescr.4	Fan #3 RPM
entPhysicalDescr.5	Fan #4 RPM
entPhysicalDescr.6	Temperature @ Ocelot
entPhysicalDescr.7	Temperature @ Switch
entPhysicalDescr.8	Temperature @ Cavium
entPhysicalDescr.9	Temperature @ Intel PHY
entPhysicalDescr.10	Temperature @ Switch Core
entPhysicalDescr.11	Temperature @ Cavium Core
entPhySensorOperStatus.2	ok (1)
entPhySensorOperStatus.3	ok (1)
entPhySensorOperStatus.4	ok (1)
entPhySensorOperStatus.5	ok (1)
entPhySensorOperStatus.6	ok (1)
entPhySensorOperStatus.7	ok (1)
entPhySensorOperStatus.8	ok (1)
entPhySensorOperStatus.9	ok (1)
entPhySensorOperStatus.10	ok (1)
entPhySensorOperStatus.11	ok (1)

The same OID might refer to different sensors on different platforms. Use the ENTITY-MIB for the targeted platform to match the value to the description.

Palo Alto Networks firewalls, Panorama, and WF-500 appliances support only portions of the entPhySensorTable group. The supported portions vary by platform and include only thermal (temperature in Celsius) and fan (in RPM) sensors.

RFC 3433 defines the ENTITY-SENSOR-MIB.

ENTITY-STATE-MIB

ENTITY-STATE-MIB provides information about the state of physical components beyond what ENTITY-MIB defines, including the administrative and operational state of components in chassis-based platforms. Use this MIB in tandem with the ENTITY-MIB to monitor the operational state of the components of a PA-7000 Series or PA-5450 firewall (for example, line cards, fan trays, and power supplies). For example, to troubleshoot log forwarding issues for Threat logs, you

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can map the log processing card (LPC) indexes from the ENTITY-MIB (entPhysicalDescr object) to operational state values (entStateOper object) in the ENTITY-STATE-MIB. The operational state values use numbers to indicate state: 1 for unknown, 2 for disabled, 3 for enabled, and 4 for testing. The PA-7000 Series and PA-5450 firewall are the only Palo Alto Networks firewalls that support this MIB.

RFC 4268 defines the ENTITY-STATE-MIB.

IEEE 802.3 LAG MIB

Use the IEEE 802.3 LAG MIB to monitor the status of aggregate groups that have Link Aggregation Control Protocol (LACP in an Aggregate Interface Group) enabled. When the firewall logs LACP events, it also generates traps that are useful for troubleshooting. For example, the traps can tell you whether traffic interruptions between the firewall and an LACP peer resulted from lost connectivity or from mismatched interface speed and duplex values.

PAN-OS implements the following SNMP tables for LACP.

The dot3adTablesLastChanged object indicates the time of the most recent change to dot3adAggTable, dot3adAggPortListTable, and dot3adAggPortTable.

Table	Description
Aggregator Configuration Table (dot3adAggTable)	This table contains information about every aggregate group that is associated with a firewall. Each aggregate group has one entry. Some table objects have restrictions, which the dot3adAggIndex object describes. This index is the unique identifier that the local system assigns to the aggregate group. It identifies an aggregate group instance among the subordinate managed objects of the containing object. The identifier is read-only. *The ifTable MIB (a list of interface entries) does not support logical interfaces and therefore does not have an entry for the aggregate group.*
Aggregation Port List Table (dot3adAggPortListTable)	This table lists the ports associated with each aggregate group in a firewall. Each aggregate group has one entry. The dot3adAggPortListPorts attribute lists the complete set of ports associated with an aggregate group. Each bit set in the list represents a port member. For non-chassis platforms, this is a 64-bit value. For chassis platforms, the value is an array of eight 64-bit entries.
Aggregation Port Table (dot3adAggPortTable)	This table contains LACP configuration information about every port associated with an aggregate group in a firewall. Each port has one entry. The table has no entries for ports that are not associated with an aggregate group.
LACP Statistics Table (dot3adAggPortStatsTable)	This table contains link aggregation information about every port associated with an aggregate group in a firewall. Each port has one

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Table	Description
	row. The table has no entries for ports that are not associated with an aggregate group.

The IEEE 802.3 LAG MIB includes the following LACP-related traps:

Trap Name	Description
panLACPLostConnectivityTrap	The peer lost connectivity to the firewall.
panLACPUnresponsiveTrap	The peer does not respond to the firewall.
panLACPNegoFailTrap	LACP negotiation with the peer failed.
panLACPSpeedDuplexTrap	The link speed and duplex settings on the firewall and peer do not match.
panLACPLinkDownTrap	An interface in the aggregate group is down.
panLACPLacpDownTrap	An interface was removed from the aggregate group.
panLACPLacpUpTrap	An interface was added to the aggregate group.

For the MIB definitions, refer to IEEE 802.3 LAG MIB.

LLDP-V2-MIB.my

Use the LLDP-V2-MIB to monitor Link Layer Discovery Protocol (LLDP) events. For example, you can check the lldpV2StatsRxPortFramesDiscardedTotal object to see the number of LLDP frames that were discarded for any reason. The Palo Alto Networks firewall uses LLDP to discover neighboring devices and their capabilities. LLDP makes troubleshooting easier, especially for virtual wire deployments where the ping or traceroute utilities won't detect the firewall.

Palo Alto Networks firewalls support all the LLDP-V2-MIB objects except:

- The following lldpV2Statistics objects:
  - lldpV2StatsRemTablesLastChangeTime
  - lldpV2StatsRemTablesInserts
  - lldpV2StatsRemTablesDeletes
  - lldpV2StatsRemTablesDrops
  - lldpV2StatsRemTablesAgeouts
- The following lldpV2RemoteSystemsData objects:
  - The lldpV2RemOrgDefInfoTable table
  - In the lldpV2RemTable table: lldpV2RemTimeMark

RFC 4957 defines this MIB.

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IP-MIB

(PAN-OS 11.1 and later versions)

IP-MIB provides information about the general IP stack in both IPv4 and IPv6. Use this MIB to monitor IP addresses of interfaces.

Palo Alto Networks firewalls, Panorama, and WF-500 appliances currently support only the ipAddressTable and ipAddrTable in IP-MIB.

- The ipAddressTable lists the IPv4 and IPv6 addresses used by an entity, along with basic history for when the address was created and updated.
- The ipAddrTable lists the IPv4 addresses used by an entity. This table has been replaced by the ipAddressTable, but is provided for support purposes.

RFC 4293 defines this MIB.

PAN-COMMON-MIB.my

Use the PAN-COMMON-MIB to monitor the following information for Palo Alto Networks firewalls, Panorama, and WF-500 appliances:

Object Group	Description
panSys	Contains such objects as system software/hardware versions, dynamic content versions, serial number, HA mode/state, and global counters. The global counters include those related to Denial of Service (DoS), IP fragmentation, TCP state, and dropped packets. Tracking these counters enables you to monitor traffic irregularities that result from DoS attacks, system or connection faults, or resource limitations. PAN-COMMON-MIB supports global counters for firewalls but not for Panorama.
panChassis	Chassis type and M-Series appliance mode (Panorama or Log Collector).
panSession	Session utilization information. For example, the total number of active sessions on the firewall or a specific virtual system.
panMgmt	Status of the connection from the firewall to the Panorama management server.
panGlobalProtect	GlobalProtect gateway utilization as a percentage, maximum tunnels allowed, and number of active tunnels.
panLogCollector	Logging statistics for each Log Collector, including logging rate, log quotas, disk usage, retention periods, log redundancy (enabled or disabled), the forwarding status from firewalls to Log Collectors, the forwarding status from Log Collectors to external services, and the status of firewall-to-Log Collector connections.

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Object Group	Description
panDeviceLogging	Logging statistics for each firewall, including logging rate, disk usage, retention periods, the forwarding status from individual firewalls to Panorama and external servers, and the status of firewall-to-Log Collector connections.
panHrStorageEntry	(PAN-OS 11.2.3 and later) Buffer and descriptor utilization by percentage for PAN-OS firewalls and appliances. Support for on-chip descriptor utilization by percentage for Octeon or sw-tag for x86.

PAN-GLOBAL-REG-MIB.my

PAN-GLOBAL-REG-MIB.my contains global, top-level OID definitions for various sub-trees of Palo Alto Networks enterprise MIB modules. This MIB doesn't contain objects for you to monitor; it is required only for referencing by other MIBs.

PAN-GLOBAL-TC-MIB.my

PAN-GLOBAL-TC-MIB.my defines conventions (for example, character length and allowed characters) for the text values of objects in Palo Alto Networks enterprise MIB modules. All Palo Alto Networks products use these conventions. This MIB doesn't contain objects for you to monitor; it is required only for referencing by other MIBs.

PAN-LC-MIB.my

PAN-LC-MIB.my contains definitions of managed objects that Log Collectors (M-Series appliances in Log Collector mode) implement. Use this MIB to monitor the logging rate, log database storage duration (in days), and disk usage (in MB) of each logical disk (up to four) on a Log Collector. For example, you can use this information to determine whether you should add more Log Collectors or forward logs to an external server (for example, a syslog server) for archiving.

PAN-PRODUCT-MIB.my

PAN-PRODUCT-MIB.my defines sysObjectID OIDs for all Palo Alto Networks products. This MIB doesn't contain objects for you to monitor; it is required only for referencing by other MIBs.

PAN-ENTITY-EXT-MIB.my

Use PAN-ENTITY-EXT-MIB.my in tandem with the ENTITY-MIB to monitor power usage for the physical components of a PA-7000 Series or PA-5450 firewall (for example, fan trays, and power supplies), which are the only two Palo Alto Networks firewalls that support this MIB. For example, when troubleshooting log forwarding issues, you might want to check the power usage of the log processing cards (LPCs): you can map the LPC indexes from the ENTITY-MIB (entPhysicalDescr object) to values in the PAN-ENTITY-EXT-MIB (panEntryFRUModelPowerUsed object).

PAN-TRAPS.my

Use PAN-TRAPS.my to see a complete listing of all generated traps and information about them (for example, a description). For a list of traps that Palo Alto Networks firewalls, Panorama, and WF-500 appliances support, refer to the PAN-COMMON-MIB.my `panCommonEvents` > `panCommonEventsEvents` > `panCommonEventEventsV2` object.

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(PAN-OS 12.2.2 and later) PAN-TRAPS.my also defines two cellular traps for cellular-integrated firewalls: `panCELLULARSigStrengthLowTrap`, which fires when any modem's signal strength drops below the FAIR threshold, and `panCELLULARPSDetachedTrap`, which fires when any cellular interface loses packet-switched connectivity. Both traps include hold-down timers to prevent repeated alerts while the condition persists. You can monitor the current state of these conditions by polling the notification state objects defined in PAN-CELLULAR-MIB.my.

PAN-CELLULAR-MIB.my

(PAN-OS 12.2.2 and later; cellular-integrated firewalls only)

Use PAN-CELLULAR-MIB.my to monitor the operational state of cellular modems, SIM cards, signal quality, and network interfaces on cellular-integrated Palo Alto Networks firewalls through your Network Management System (NMS). The MIB is available on the specified cellular-integrated platforms. You can poll the MIB using SNMP v2c or v3 with GET, GETNEXT, and GETBULK operations at intervals as short as 30 seconds.

The MIB root OID is `1.3.6.1.4.1.25461.2.1.2.15` (`panCellular`). Table indexes remain stable across reboots and HA state transitions: modem indexes map to physical modem slots, SIM indexes map to physical SIM slots, and subinterface indexes map to PDN indices. SNMP-reported values are consistent with CLI output from equivalent `show` commands.

PAN-CELLULAR-MIB.my defines the following object groups:

Object Group	Description
panCellularModemTable	Modem hardware information including operational state (unknown, bootmode, initFailed, online, lowpower, attached, detached, dataConnected, fwUpgrade, airplane), firmware version, IMEI, factory serial number (FSN), electronic serial number (ESN), modem temperature, DHCP relay state, and number of SIM slots.
panCellularSignalTable	Signal quality metrics per modem and SIM slot, including RSSI, SINR, RSRP, RSRQ, signal strength enumeration (deadzone, poor, fair, good, excellent), RF band, and RF channel. Indexed by modem index and SIM slot index.
panCellularSIMTable	SIM card information including IMSI, ICCID, home carrier, SIM status (present, absent, error, unknown), PIN verification state, and remaining PIN retries. Indexed by modem index and SIM slot index.
panCellularAPNDNNTable	APN/DNN profile configuration including profile name, APN name, PDP type (IPv4, IPv6, IPv4v6), authentication type (none, PAP, CHAP, auto), network type (APN or DNN), and 5G network slice identifiers (SST and SD for DNN).
panCellularInterfaceTable	Per-interface network state and operational counters. State information includes PS attach status, registration state, roaming status, MCC/MNC of the serving network, active APN name, IPv4 and IPv6 addresses and MTU values, Tracking Area Code (TAC), Location Area Code (LAC), activation state, and GPS coordinates and

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Object Group	Description
	state. Counters include cell switches, modem resets, modem power cycles, radio resets, network disconnects, SIM switchovers, and L3 recovery actions.
panCellularSubInterfaceTable	Per-sub-interface information including sub-interface name, associated modem and SIM index, APN profile name, active APN name, IPv4 and IPv6 addresses, and IPv4 and IPv6 MTU values. Indexed by parent interface index and sub-interface index.
Scalar objects	Total counts of modems (`panNumberOfModems`), interfaces (`panNumberOfInterfaces`), subinterfaces (`panNumberOfSubInterfaces`), and APN/DNN profiles (`panNumberOfAPNDNNProfiles`) on the firewall. Two TruthValue scalars reflect current cellular alert conditions: `panCellularSigStrengthLow` (true when any modem's signal strength falls below the FAIR threshold, deadzone or poor) and `panCellularPSDetached` (true when any interface has lost packet-switched network connectivity). Poll these objects to determine current alert state independently of trap reception.

PAN-CELLULAR-MIB.my also defines two SNMP traps in PAN-TRAPS.my:

Trap	Trigger Condition
panCELLULARSigStrengthLowTrap	Any modem's signal strength drops below FAIR (deadzone or poor). A 10-minute hold-down timer prevents repeated traps while the condition persists.
panCELLULARPSDetachedTrap	Any cellular interface loses packet-switched (PS) network connectivity. A 90-second hold-down timer prevents repeated traps while the condition persists.

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Forward Logs to an HTTP/S Destination

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

The firewall and Panorama™ can forward logs to an HTTP/S server. You can choose to forward all logs or specific logs to trigger an action on an external HTTP-based service when an event occurs. When forwarding logs to an HTTP server, configure the firewall to send an HTTP-based API request directly to a third-party service to trigger an action that is based on the attributes in a firewall log. You can configure the firewall to work with any HTTP-based service that exposes an API and you can modify the URL, HTTP header, parameters, and the payload in the HTTP request to meet your integration needs.

Log forwarding to an HTTP server is designed for log forwarding at low frequencies and is not recommend for deployments with a high volume of log forwarding. You may experience log loss when forwarding to an HTTP server if your deployment generate a high volume of logs that need to be forwarded.

See Configure Log Forwarding *for additional log forwarding options.*

STEP 1 | Create an HTTP server profile to forward logs to an HTTP/S destination.

The HTTP server profile allows you to specify how to access the server and define the format in which to forward logs to the HTTP/S destination. By default, the firewall uses the

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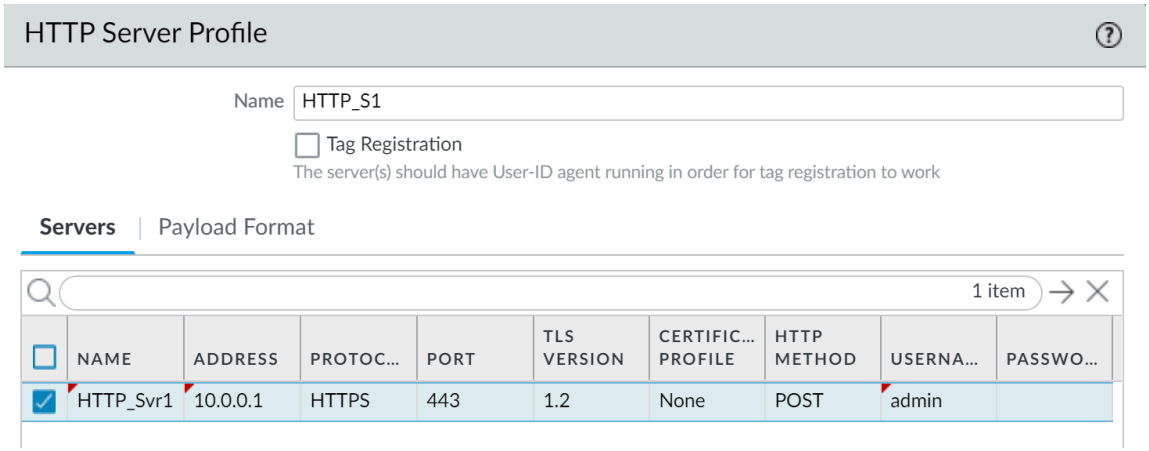
management port to forward these logs. However, you can assign a different source interface and IP address in **Device** > **Setup** > **Services** > **Service Route Configuration**.

1. Select **Device** > **Server Profiles** > **HTTP** and **Add** a new profile.
2. Specify a **Name** for the server profile, and select the **Location**. The profile can be **Shared** across all virtual systems or can belong to a specific virtual system.
3. **Add** the details for each server. Each profile can have a maximum of four servers.
4. Enter a **Name** and IP **Address**.
5. Select the **Protocol** (**HTTP** or **HTTPS**). The default **Port** is 80 or 443 respectively but you can modify the port number to match the port on which your HTTP server listens.
6. Select the **TLS Version** supported on the server—**1.0**, **1.1**, or **1.2** (default).
7. Select the **Certificate Profile** to use for the TLS connection with the server.
8. Select the **HTTP Method** that the third-party service supports—**DELETE**,**GET**, **POST** (default), or **PUT**.
9. (Optional) Enter the **Username** and **Password** for authenticating to the server, if needed.
10. (Optional) Select **Test Server Connection** to verify network connectivity between the firewall and the HTTP/S server.

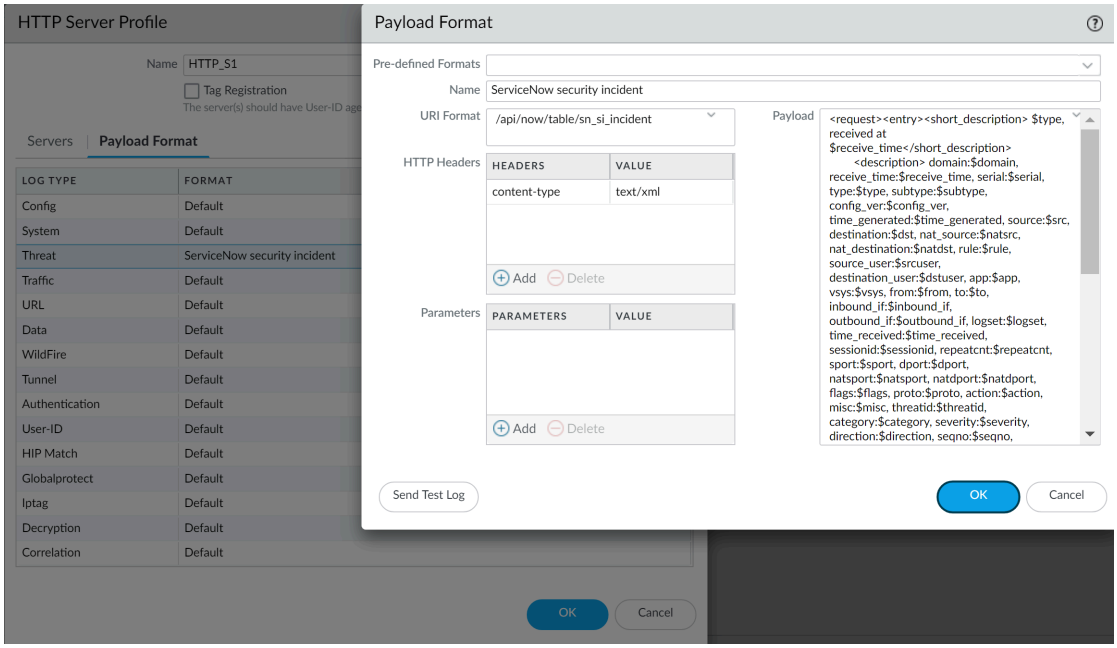


STEP 2 | Select the **Payload Format** for the HTTP request.

1. Select the **Log Type** link for each log type for which you want to define the HTTP request format.
2. Select the **Pre-defined Formats** (available through content updates) or create a custom format.

   If you create a custom format, the **URI** is the resource endpoint on the HTTP service. The firewall appends the URI to the IP address you defined earlier to construct the URL for the HTTP request. Ensure that the URI and payload format matches the syntax that your third-party vendor requires. You can use any attribute supported on the selected

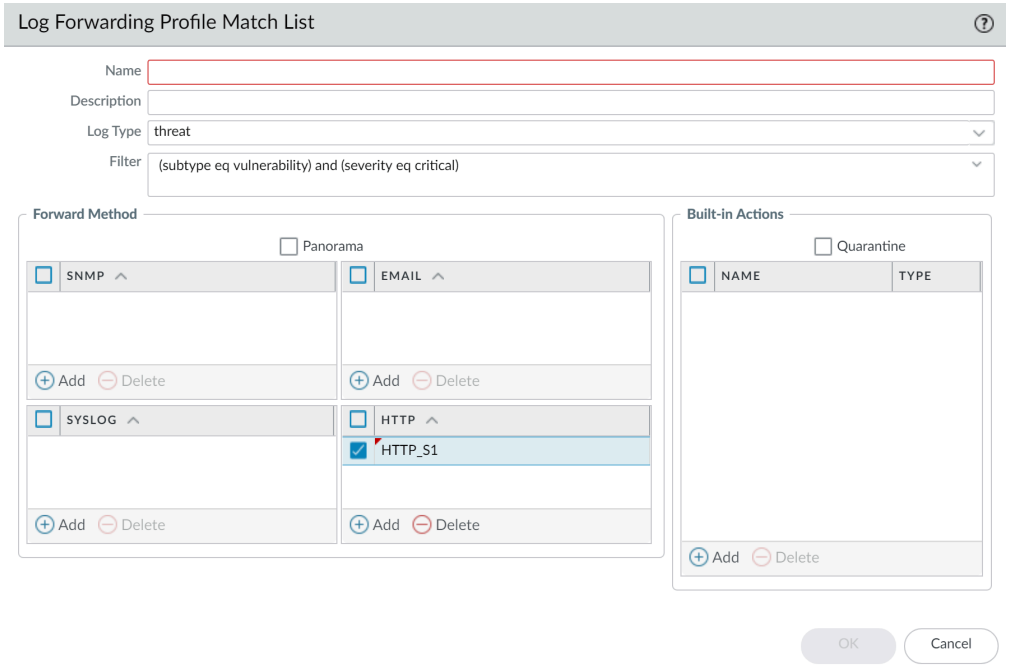
log type within the HTTP Header, the Parameter and Value pairs, and in the request payload.



3. **Send Test Log** to verify that the HTTP server receives the request. When you interactively send a test log, the firewall uses the format as is and does not replace the variable with a value from a firewall log. If your HTTP server sends a 404 response, provide values for the parameters so that the server can process the request successfully.

STEP 3 | Define the match criteria for when the firewall will forward logs to the HTTP server and attach the HTTP server profile you will use.

1. Select the log types for which you want to trigger a workflow:
   - Add a Log Forwarding Profile (**Objects** > **Log Forwarding**) for logs that pertain to user activity (for example, Traffic, Threat, or Authentication logs).
   - Select **Device** > **Log Settings** for logs that pertain to system events, such as Configuration or System logs.
2. Select the Log Type and use the **Filter Builder** to define the match criteria.
3. **Add** the HTTP server profile for forwarding logs to the HTTP destination.



NetFlow Monitoring

Where Can I Use This?	What Do I Need?
• NGFW • Prisma Access	One of these licenses for Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Essentials ☐ Strata Cloud Manager Pro Other licenses needed: ☐ Prisma Access license

NetFlow is an industry-standard protocol that the firewall can use to export statistics about the IP traffic ingressing its interfaces. The firewall exports the statistics as NetFlow fields to a NetFlow collector. The NetFlow collector is a server you use to analyze network traffic for security, administration, accounting and troubleshooting. All Palo Alto Networks firewalls support NetFlow Version 9. The firewalls support only unidirectional NetFlow, not bidirectional. The firewalls perform NetFlow processing on all IP packets on the interfaces and do not support sampled NetFlow. You can export NetFlow records for Layer 3, Layer 2, virtual wire, tap, VLAN, loopback, and tunnel interfaces. For aggregate Ethernet sub-interfaces, you can export records for the individual sub-interfaces that data flows through within the group. To identify firewall interfaces in a NetFlow collector, see Firewall Interface Identifiers in SNMP Managers and NetFlow Collectors. The firewalls support standard and enterprise (PAN-OS specific) NetFlow Templates, which NetFlow collectors use to decipher the NetFlow fields.

- Configure NetFlow Exports
- NetFlow Templates

Configure NetFlow Exports

Where Can I Use This?	What Do I Need?
• NGFW • Prisma Access	One of these licenses for Strata Cloud Manager managed NGFWs: ☐ Strata Cloud Manager Essentials ☐ Strata Cloud Manager Pro Other licenses needed: ☐ Prisma Access license

To analyze network traffic with a NetFlow collector, you need to configure your firewall to export NetFlow records. This involves setting up the firewall to send traffic data to a separate NetFlow collector. The collector then processes this data to provide insights into your network traffic. Perform the following steps to configure NetFlow record exports.

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- PAN-OS
- Strata Cloud Manager

Configure Netflow Exports (PAN-OS)

STEP 1 | Create a NetFlow server profile.

The profile defines which NetFlow collectors will receive the exported records and specifies export parameters.

1. Select **Device** > **Server Profiles** > **NetFlow** and **Add** a profile.
2. Enter a **Name** to identify the profile.
3. Specify the rate at which the firewall refreshes NetFlow Templates in **Minutes** (default is 30) and **Packets** (exported records—default is 20), according to the requirements of your NetFlow collector. The firewall refreshes the templates after either threshold is passed.
4. Specify the **Active Timeout**, which is the frequency in minutes at which the firewall exports records (default is 5).
5. Select **PAN-OS Field Types** if you want the firewall to export App-ID and User-ID fields.
6. **Add** each NetFlow collector (up to two per profile) that will receive records. For each collector, specify the following:
   - **Name** to identify the collector.
   - **NetFlow Server** hostname or IP address.
   - Access **Port** (default 2055).
7. Click **OK** to save the profile.

STEP 2 | Assign the NetFlow server profile to the firewall interfaces where traffic you want to analyze is ingressing.

In this example, you assign the profile to an existing Ethernet interface.

1. Select **Network** > **Interfaces** > **Ethernet** and click an interface name to edit it.

   You can export NetFlow records for Layer 3, Layer 2, virtual wire, tap, VLAN, loopback, and tunnel interfaces. For aggregate Ethernet interfaces, you can export records for the individual sub-interfaces that data flows through within the group.

2. Select the NetFlow server profile (**NetFlow Profile**) you configured and click **OK**.

STEP 3 | (PA-7000 Series, PA-7500, PA-5500 Series, PA-5450, and PA-5200 Series firewalls) Configure a service route for the interface that the firewall will use to send NetFlow records.

You cannot use the management (MGT) interface to send NetFlow records from the PA-7000 Series, PA-7500, PA-5500 Series, PA-5450, and PA-5200 Series firewalls. For other firewall

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models, a service route is optional. For all firewalls, the interface that sends NetFlow records does not have to be the same as the interface for which the firewall collects the records.

1. Select **Device** > **Setup** > **Services**.
2. **(Firewall with multiple virtual systems)** Select one of the following:
   - **Global**—Select this option if the service route applies to all virtual systems on the firewall.
   - **Virtual Systems**—Select this option if the service route applies to a specific virtual system. Set the **Location** to the virtual system.
3. Select **Service Route Configuration** and Customize.
4. Select the protocol (**IPv4** or **IPv6**) that the interface uses. You can configure the service route for both protocols if necessary.
5. Click **Netflow** in the Service column.
6. Select the **Source Interface**.

   Any, ***Use default***, *and* ***MGT*** *are not valid interface options for sending NetFlow records from the PA-7000 Series, PA-7500, PA-5500 Series, PA-5450, and PA-5200 Series firewalls.*
7. Select a **Source Address** (IP address).
8. Click **OK** twice to save your changes.

STEP 4 | **Commit** your changes.

STEP 5 | Monitor the firewall traffic in a NetFlow collector.

Refer to your NetFlow collector documentation.

When monitoring statistics, you must match the interface indexes in the NetFlow collector with interface names in the firewall web interface. For details, see Firewall Interface Identifiers in SNMP Managers and NetFlow Collectors.

To troubleshoot NetFlow delivery issues, use the operational CLI command **`debug log-receiver netflow statistics`**.

Configure Netflow Exports (Strata Cloud Manager)

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STEP 1 | Create a NetFlow server profile.

The profile defines which Netflow collectors will receive the exported records and specifies export parameters.

1. Select **Configuration** > **NGFW and Prisma Access** > **Objects** > **Log Forwarding** > **Netflow Server Profile** and click **Add Netflow Server Profile**.
2. In the **General** section, enter a **Name** to identify the profile.
3. Specify the **Active Timeout**, which is the frequency in minutes at which the firewall exports records (default is 5).
4. Select **PAN-OS Field Types** if you want the firewall to export App-ID and User-ID fields.
5. In the **Template Refresh Rate** section, specify the rate at which the firewall refreshes NetFlow Templates in **Minutes** (default is 30) and **Packets** (exported records—default is 20), according to the requirements of your Netflow collector. The firewall refreshes the templates after either threshold is passed.
6. Click **Add Server**.
7. Add Netflow collectors (up to two per profile) that will receive records. For each collector, specify the following:
   - **Name** to identify the collector.
   - **Host** hostname or IP address.
   - Access **Port** (default 2055).
8. Click **Add** to save the profile.
9. To commit your changes, click **Push Config** > **Push**.

STEP 2 | Assign the Netflow server profile to the firewall interfaces where traffic you want to analyze is ingressing.

In this example, you assign the profile to an existing Ethernet interface.

1. Select **Ethernet** and click an interface name to edit it.

   You can export Netflow records for Layer 3, Layer 2, Tap, virtual wire, VLAN, tunnel, and Aggregate Ethernet. For aggregate Ethernet interfaces, you can export records for the individual sub-interfaces that data flows within the group.

   For more information about configuring interfaces see Configuring Interfaces.
2. Select the Netflow server profile you configured.
   (Optional) To create a new Netflow profile click **Create New**.

STEP 3 | Click **Save**.

STEP 4 | **Push Config** to push your configuration changes.

NetFlow Templates

NetFlow collectors use templates to decipher the fields that the firewall exports. The firewall selects a template based on the type of exported data: IPv4 or IPv6 traffic, with or without NAT, and with standard or enterprise-specific (PAN-OS specific) fields. The firewall periodically

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refreshes templates to re-evaluate which one to use (in case the type of exported data changes) and to apply any changes to the fields in the selected template. When you Configure NetFlow Exports, set the refresh rate based on a time interval and a number of exported records according to the requirements of your NetFlow collector. The firewall refreshes the templates after either threshold is passed.

The Palo Alto Networks firewall supports the following NetFlow templates:

Template	ID
IPv4 Standard	256
IPv4 Enterprise	257
IPv6 Standard	258
IPv6 Enterprise	259
IPv4 with NAT Standard	260
IPv4 with NAT Enterprise	261
IPv6 with NAT Standard	262
IPv6 with NAT Enterprise	263

The following table lists the NetFlow fields that the firewall can send, along with the templates that define them:

Value	Field	Description	Templates
1	IN_BYTES	Incoming counter with length N * 8 bits for the number of bytes associated with an IP flow. By default, N is 4.	All templates
2	IN_PKTS	Incoming counter with length N * 8 bits for the number of packets associated with an IP glow. By default, N is 4.	All templates
4	PROTOCOL	IP protocol byte.	All templates
5	TOS	Type of Service byte setting when entering the ingress interface.	All templates

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Value	Field	Description	Templates
6	TCP_FLAGS	Total of all the TCP flags in this flow.	All templates
7	L4_SRC_PORT	TCP/UDP source port number (for example, FTP, Telnet, or equivalent).	All templates
8	IPV4_SRC_ADDR	IPv4 source address.	IPv4 standard IPv4 enterprise IPv4 with NAT standard IPv4 with NAT enterprise
10	INPUT_SNMP	Input interface index. The value length is 2 bytes by default, but higher values are possible. For details on how Palo Alto Networks firewalls generate interface indexes, see Firewall Interface Identifiers in SNMP Managers and NetFlow Collectors.	All templates
11	L4_DST_PORT	TCP/UDP destination port number (for example, FTP, Telnet, or equivalent).	All templates
12	IPV4_DST_ADDR	IPv4 destination address.	IPv4 standard IPv4 enterprise IPv4 with NAT standard IPv4 with NAT enterprise
14	OUTPUT_SNMP	Output interface index. The value length is 2 bytes by default, but higher values are possible. For details on how Palo Alto Networks firewalls generate interface indexes, see Firewall Interface Identifiers in SNMP Managers and NetFlow Collectors.	All templates

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Value	Field	Description	Templates
21	LAST_SWITCHED	System uptime in milliseconds when the last packet of this flow was switched.	All templates
22	FIRST_SWITCHED	System uptime in milliseconds when the first packet of this flow was switched.	All templates
27	IPV6_SRC_ADDR	IPv6 source address.	IPv6 standard IPv6 enterprise IPv6 with NAT standard IPv6 with NAT enterprise
28	IPV6_DST_ADDR	IPv6 destination address.	IPv6 standard IPv6 enterprise IPv6 with NAT standard IPv6 with NAT enterprise
32	ICMP_TYPE	Internet Control Message Protocol (ICMP) packet type. This is reported as: ICMP Type * 256 + ICMP code	All templates
61	DIRECTION	Flow direction: • 0 = ingress • 1 = egress	All templates
148	flowId	An identifier of a flow that is unique within an observation domain. You can use this information element to distinguish between different flows if flow keys such as IP addresses and port numbers are not reported or are reported in separate records. The flowID corresponds to the session ID field in Traffic and Threat logs.	All templates

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Value	Field	Description	Templates
233	firewallEvent	Indicates a firewall event: • 0 = Ignore (invalid)—Not used. • 1 = Flow created—The NetFlow data record is for a new flow. • 2 = Flow deleted—The NetFlow data record is for the end of a flow. • 3 = Flow denied—The NetFlow data record indicates a flow that firewall policy denied. • 4 = Flow alert—Not used. • 5 = Flow update—The NetFlow data record is sent for a *long-lasting* flow, which is a flow that lasts longer than the **Active Timeout** period configured in the NetFlow server profile.	All templates
225	postNATSourceIPv4Address	The definition of this information element is identical to that of sourceIPv4Address, except that it reports a modified value that the firewall produced during network address translation after the packet traversed the interface.	IPv4 with NAT standard IPv4 with NAT enterprise
226	postNATDestinationIPv4Address	The definition of this information element is identical to that of destinationIPv4Address, except that it reports a modified value that the firewall produced during network address translation after the packet traversed the interface.	IPv4 with NAT standard IPv4 with NAT enterprise
227	postNAPTSourceTransportPort	The definition of this information element is identical to that of sourceTransportPort, except that it reports a modified value that the firewall produced during network address port translation after the packet traversed the interface.	IPv4 with NAT standard IPv4 with NAT enterprise
228	postNAPTDestinationTransportPort	The definition of this information element is identical to that of destinationTransportPort, except	IPv4 with NAT standard

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Value	Field	Description	Templates
		that it reports a modified value that the firewall produced during network address port translation after the packet traversed the interface.	IPv4 with NAT enterprise
281	postNATSourceIPv6Address	The definition of this information element is identical to the definition of information element sourceIPv6Address, except that it reports a modified value that the firewall produced during NAT64 network address translation after the packet traversed the interface. See RFC 2460 for the definition of the source address field in the IPv6 header. See RFC 6146 for NAT64 specification.	IPv6 with NAT standard IPv6 with NAT enterprise
282	postNATDestinationIPv6Address	The definition of this information element is identical to the definition of information element destinationIPv6Address, except that it reports a modified value that the firewall produced during NAT64 network address translation after the packet traversed the interface. See RFC 2460 for the definition of the destination address field in the IPv6 header. See RFC 6146 for NAT64 specification.	IPv6 with NAT standard IPv6 with NAT enterprise
346	privateEnterpriseNumber	This is a unique private enterprise number that identifies Palo Alto Networks: 25461.	IPv4 enterprise IPv4 with NAT enterprise IPv6 enterprise IPv6 with NAT enterprise
56701	App-ID	The name of an application that App-ID identified. The name can be up to 32 bytes.	IPv4 enterprise IPv4 with NAT enterprise IPv6 enterprise

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Value	Field	Description	Templates
			IPv6 with NAT enterprise
56702	User-ID	A username that User-ID identified. The name can be up to 64 bytes.	IPv4 enterprise IPv4 with NAT enterprise IPv6 enterprise IPv6 with NAT enterprise

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Firewall Interface Identifiers in SNMP Managers and NetFlow Collectors

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

When you use a NetFlow collector (see NetFlow Monitoring) or SNMP manager (see SNMP Monitoring and Traps) to monitor the Palo Alto Networks firewall, an interface index (SNMP ifindex object) identifies the interface that carried a particular flow (see Interface Indexes in an SNMP Manager). In contrast, the firewall web interface uses interface names as identifiers (for example, ethernet1/1), not indexes. To understand which statistics that you see in a NetFlow collector or SNMP manager apply to which firewall interface, you must be able to match the interface indexes with interface names.

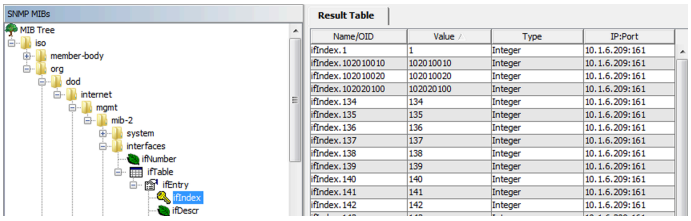

Figure 3: Interface Indexes in an SNMP Manager

You can match the indexes with names by understanding the formulas that the firewall uses to calculate indexes. The formulas vary by platform and interface type: physical or logical.

Physical interface indexes have a range of 1-9999, which the firewall calculates as follows:

Firewall Platform	Calculation	Example Interface Index
VM-Series	Number of management ports + physical port offset • **Number of management ports**—This is a constant of 1. • **Physical port offset**—This is the physical port number.	VM-100 firewall, Eth1/4 = 1 (number of management ports) + 4 (physical port) = **5**
PA-220, PA-220R, PA-800 Series	Number of management ports + physical port offset • **Number of management ports**—This is a constant of 5. • **Physical port offset**—This is the physical port number.	PA-5200 Series firewall, Eth1/4 = 5 (number of management ports) + 4 (physical port) = **9**

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Firewall Platform	Calculation	Example Interface Index
PA-3200 Series, PA-5200 Series	Number of management ports + physical port offset • **Number of management ports**—This is a constant of 4. • **Physical port offset**—This is the physical port number.	PA-5200 Series firewall, Eth1/4 = 4 (number of management ports) + 4 (physical port) = **8**
PA-7000 Series	(Max. ports * slot) + physical port offset + number of management ports • **Maximum ports**—This is a constant of 64. • **Slot**—This is the chassis slot number of the network interface card. • **Physical port offset**—This is the physical port number. • **Number of management ports**—This is a constant of 5.	PA-7000 Series firewall, Eth3/9 = [64 (max. ports) * 3 (slot)] + 9 (physical port) + 5 (number of management ports) = **206**

Logical interface indexes for all platforms are nine-digit numbers that the firewall calculates as follows:

Interface Type	Range	Digit 9	Digits 7-8	Digits 5-6	Digits 1-4	Example Interface Index
Layer 3 subinterface	101010001-199999999	Type: 1	Interface slot: 1-9 (01-09)	Interface port: 1-9 (01-09)	Subinterface: suffix 1-9999 (0001-9999)	Eth1/5.22 = 100000000 (type) + 100000 (slot) + 50000 (port) + 22 (suffix) = **101050022**
Layer 2 subinterface	101010001-199999999	Type: 1	Interface slot: 1-9 (01-09)	Interface port: 1-9 (01-09)	Subinterface: suffix 1-9999 (0001-9999)	Eth2/3.6 = 100000000 (type) + 200000 (slot) + 30000 (port) + 6 (suffix) = **102030006**
Vwire subinterface	101010001-199999999	Type: 1	Interface slot: 1-9 (01-09)	Interface port: 1-9 (01-09)	Subinterface: suffix 1-9999 (0001-9999)	Eth4/2.312 = 100000000 (type) + 400000 (slot) + 20000 (port) + 312 (suffix) = **104020312**

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Interface Type	Range	Digit 9	Digits 7-8	Digits 5-6	Digits 1-4	Example Interface Index
VLAN	200000001-200009999	Type: 2	00	00	VLAN suffix: 1-9999 (0001-9999)	VLAN.55 = 200000000 (type) + 55 (suffix) = **200000055**
Loopback	300000001-300009999	Type: 3	00	00	Loopback suffix: 1-9999 (0001-9999)	Loopback.55 = 300000000 (type) + 55 (suffix) = **300000055**
Tunnel	400000001-400009999	Type: 4	00	00	Tunnel suffix: 1-9999 (0001-9999)	Tunnel.55 = 400000000 (type) + 55 (suffix) = **400000055**
Aggregate group	500010001-500089999	Type: 5	00	AE suffix: 1-8 (01-08)	Subinterface: suffix 1-9999 (0001-9999)	AE5.99 = 500000000 (type) + 50000 (AE Suffix) + 99 (suffix) = **500050099**

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Monitor Transceivers

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Support license ❑ (Panorama) Device management license

You can monitor the status of transceivers in your physical appliance or device to enable easier installation and troubleshooting. Through transceiver monitoring, also known as digital optical monitoring (DOM), you can view diagnostics like transmitted bias current, transmitted power, received power, transceiver temperature, and power supply voltage. See below for a list of devices that support transceiver monitoring.

- PA-415 firewall
- PA-445 firewall
- PA-800 Series
- PA-1400 Series
- PA-3200 Series
- PA-3400 Series
- PA-5200 Series
- PA-5400 Series
- PA-7000 Series

Use the Command Line Interface to run transceiver monitoring. See the following table for all available CLI commands.

If you run commands on an incompatible transceiver, the CLI will return 'n/a' for any diagnostic information it cannot read.

CLI	Definition
`show transceiver <interface name>`	View a summary of the specified transceiver with values for each diagnostic. Example: `admin@PA-7080> show transceiver ethernet11/25` The CLI will return values for Temperature, Voltage, Current, Tx Power, and Rx Power.

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CLI	Definition
`show transceiver-detail <interface name>`	Receive more detailed transceiver specifications, including vendor information and link lengths. The CLI will also provide more detailed diagnostic information.
`show transceiver all`	View a list of all active transceivers as well as a summary of each of their diagnostics.
`show transceiver-detail all`	Get comprehensive details on each transceiver in the device.

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User-ID

The user identity, as opposed to an IP address, is an integral component of an effective security infrastructure. Knowing who is using each of the applications on your network, and who may have transmitted a threat or is transferring files, can strengthen security policies and reduce incident response times. User-ID™, a standard feature on the Palo Alto Networks firewall, enables you to leverage user information stored in a wide range of repositories. The following topics provide more details about User-ID and how to configure it:

- User-ID Overview
- User-ID Concepts
- Enable User-ID
- Map Users to Groups
- Map IP Addresses to Users
- Enable User- and Group-Based Policy
- Enable Policy for Users with Multiple Accounts
- Verify the User-ID Configuration
- Deploy User-ID in a Large-Scale Network

User-ID Overview

User-ID™ enables you to identify all users on your network using a variety of techniques to ensure that you can identify users in all locations using a variety of access methods and operating systems, including Microsoft Windows, Apple iOS, Mac OS, Android, and Linux®/UNIX. Knowing who your users are instead of just their IP addresses enables:

- **Visibility**—Improved visibility into application usage based on users gives you a more relevant picture of network activity. The power of User-ID becomes evident when you notice a strange or unfamiliar application on your network. Using either ACC or the log viewer, your security team can discern what the application is, who the user is, the bandwidth and session consumption, along with the source and destination of the application traffic, as well as any associated threats.
- **Policy control**—Tying user information to Security policy rules improves safe enablement of applications traversing the network and ensures that only those users who have a business need for an application have access. For example, some applications, such as SaaS applications that enable access to Human Resources services (such as Workday or Service Now) must be available to any known user on your network. However, for more sensitive applications you can reduce your attack surface by ensuring that only users who need these applications can access them. For example, while IT support personnel may legitimately need access to remote desktop applications, the majority of your users do not.
- **Logging, reporting, forensics**—If a security incident occurs, forensics analysis and reporting based on user information rather than just IP addresses provides a more complete picture of the incident. For example, you can use the pre-defined User/Group Activity to see a summary of the web activity of individual users or user groups, or the SaaS Application Usage report to see which users are transferring the most data over unsanctioned SaaS applications.

To enforce user- and group-based policies, the firewall must be able to map the IP addresses in the packets it receives to usernames. User-ID provides many mechanisms to collect this User Mapping information. For example, the User-ID agent monitors server logs for login events and listens for syslog messages from authenticating services. To identify mappings for IP addresses that the agent didn't map, you can configure Authentication Policy to redirect HTTP requests to an Authentication Portal login. You can tailor the user mapping mechanisms to suit your environment, and even use different mechanisms at different sites to ensure that you are safely enabling access to applications for all users, in all locations, all the time.

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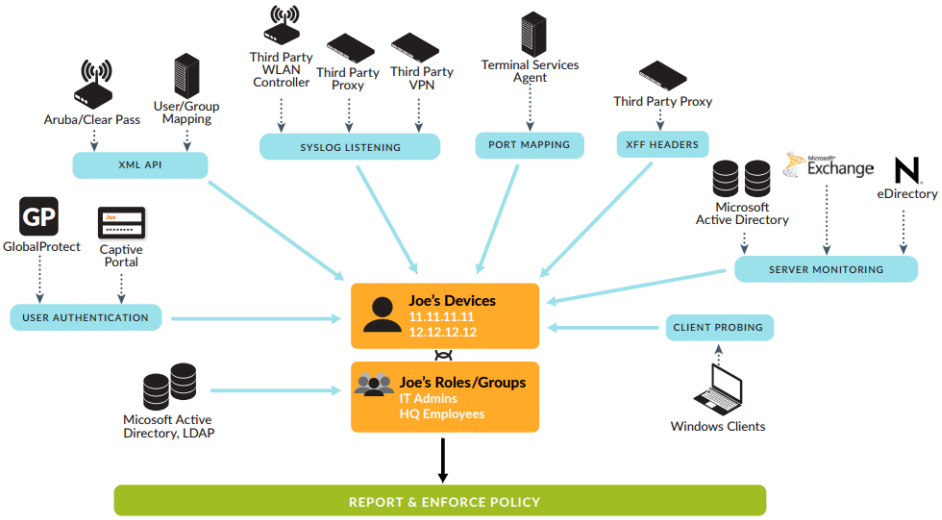

Figure 4: User-ID

To enable user- and group-based policy enforcement, the firewall requires a list of all available users and their corresponding group memberships so that you can select groups when defining your policy rules. The firewall collects Group Mapping information by connecting directly to your LDAP directory server, or using XML API integration with your directory server.

See User-ID Concepts for information on how User-ID works and Enable User-ID for instructions on setting up User-ID.

User-ID does not work in environments where the source IP addresses of users are subject to NAT translation before the firewall maps the IP addresses to usernames.

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User-ID Concepts

- Group Mapping
- User Mapping

Group Mapping

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

To define policy rules based on user or group, first you create an LDAP server profile that defines how the firewall connects and authenticates to your directory server. The firewall supports a variety of directory servers, including Microsoft Active Directory (AD), Novell eDirectory, and Sun ONE Directory Server. The server profile also defines how the firewall searches the directory to retrieve the list of groups and the corresponding list of members. If you are using a directory server that is not natively supported by the firewall, you can integrate the group mapping function using the XML API. You can then create a group mapping configuration to Map Users to Groups and Enable User- and Group-Based Policy.

When you use SAML, RADIUS, or TACACS+ for authentication, the identity provider (IdP) sends an assertion that reflects user attributes at the moment of authentication. PAN-OS® references SAML group attributes in one specific context: the **User Group Attribute** in an authentication profile, where the firewall compares the value against the Allow List to control who can authenticate. The firewall does not use SAML assertion data to populate its user-to-group mapping table for policy enforcement or reporting. Group-based security policy evaluation, GlobalProtect® app configurations, and log queries and reports by user group all require a persistent, queryable source of group information. LDAP-based group mapping or the Cloud Identity Engine (CIE) fills this role. User-ID™ queries your directory service and builds a local user-to-group mapping table that the firewall refreshes at a configurable interval. During policy evaluation, the firewall checks this directory-built table, not the IdP assertion.

Defining policy rules based on group membership rather than on individual users simplifies administration because you don't have to update the rules whenever new users are added to a group. When configuring group mapping, you can limit which groups will be available in policy rules. You can specify groups that already exist in your directory service or define custom groups based on LDAP filters. Defining custom groups can be quicker than creating new groups or changing existing ones on an LDAP server, and doesn't require an LDAP administrator to intervene. User-ID maps all the LDAP directory users who match the filter to the custom group. For example, you might want a security policy that allows contractors in the Marketing Department to access social networking sites. If no Active Directory group exists for that department, you can configure an LDAP filter that matches users for whom the LDAP attribute Department is set to Marketing. Log queries and reports that are based on user groups will include custom groups.

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User Mapping

Knowing user and groups names is only one piece of the puzzle. The firewall also needs to know which IP addresses map to which users so that security rules can be enforced appropriately. User-ID Overview illustrates the different methods that are used to identify users and groups on your network and shows how user mapping and group mapping work together to enable user- and group-based security enforcement and visibility. The following topics describe the different methods of user mapping:

- Server Monitoring
- Port Mapping
- Syslog
- XFF Headers
- Username Header Insertion
- Authentication Policy and Authentication Portal
- GlobalProtect
- XML API
- Client Probing

Server Monitoring

With server monitoring a User-ID agent—either a Windows-based agent running on a domain server in your network, or the PAN-OS integrated User-ID agent running on the firewall—monitors the security event logs for specified Microsoft Exchange Servers, Domain Controllers, or Novell eDirectory servers for login events. For example, in an AD environment, you can configure the User-ID agent to monitor the security logs for Kerberos ticket grants or renewals, Exchange server access (if configured), and file and print service connections. For these events to be recorded in the security log, the AD domain must be configured to log successful account login events. In addition, because users can log in to any of the servers in the domain, you must set up server monitoring for all servers to capture all user login events. See Configure User Mapping Using the Windows User-ID Agent or Configure User Mapping Using the PAN-OS Integrated User-ID Agent for details.

Port Mapping

In environments with multi-user systems—such as Microsoft Terminal Server or Citrix environments—many users share the same IP address. In this case, the user-to-IP address mapping process requires knowledge of the source port of each client. To perform this type of mapping, you must install the Palo Alto Networks Terminal Server Agent on the Windows/Citrix terminal server itself to intermediate the assignment of source ports to the various user processes. For terminal servers that do not support the Terminal Server agent, such as Linux terminal servers, you can use the XML API to send user mapping information from login and logout events to User-ID. See Configure User Mapping for Terminal Server Users for configuration details.

XFF Headers

If you have a proxy server deployed between the users on your network and the firewall, the firewall might see the proxy server IP address as the source IP address in HTTP/HTTPS traffic that the proxy forwards rather than the IP address of the client that requested the content.

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In many cases, the proxy server adds an X-Forwarded-For (XFF) header to traffic packets that includes the actual IPv4 or IPv6 address of the client that requested the content or from whom the request originated. In such cases, you can configure the firewall to extract the end user IP address from the XFF so that User-ID can map the IP address to a username. This enables you to use XFF values for policies and logging source users.so that you can enforce user-based policy to safely enable access to web-based for your users behind a proxy server.

Username Header Insertion

When you configure a secondary enforcement device with your Palo Alto Networks firewall to enforce user-based policy, the secondary device may not have the IP address-to-username mapping from the firewall. Transmitting the user's identity to downstream devices may require deployment of additional devices such as proxies or negatively impact the user's experience (for example, users having to log in multiple times). You can dynamically add the domain and username to the HTTP header of the user's outgoing traffic, allowing any secondary devices that you use with your Palo Alto Networks firewall to receive the user's information and enforce user-based policy. Including the user's identity by inserting the username and domain in the traffic headers enables enforcement of user-based policy without negatively impacting the user's experience or deployment of additional infrastructure.

Authentication Policy and Authentication Portal

In some cases, the User-ID agent can't map an IP address to a username using server monitoring or other methods—for example, if the user isn't logged in or uses an operating system such as Linux that your domain servers don't support. In other cases, you might want users to authenticate when accessing sensitive applications regardless of which methods the User-ID agent uses to perform user mapping. For all these cases, you can configure Configure Authentication Policy and Map IP Addresses to Usernames Using Authentication Portal. Any web traffic (HTTP or HTTPS) that matches an Authentication policy rule prompts the user to authenticate through Authentication Portal. You can use the following Authentication Portal Authentication Methods:

- Browser challenge—Use Kerberos single sign-on if you want to reduce the number of login prompts that users must respond to.
- Web form—Use Multi-Factor Authentication, SAML single sign-on, Kerberos, TACACS+, RADIUS, LDAP, or Local Authentication.
- Client Certificate Authentication.

Syslog

Your environment might have existing network services that authenticate users. These services include wireless controllers, 802.1x devices, Apple Open Directory servers, proxy servers, and other Network Access Control (NAC) mechanisms. You can configure these services to send syslog messages that contain information about login and logout events and configure the User-ID agent to parse those messages. The User-ID agent parses for login events to map IP addresses to usernames and parses for logout events to delete outdated mappings. Deleting outdated mappings is particularly useful in environments where IP address assignments change often.

Both the PAN-OS integrated User-ID agent and Windows-based User-ID agent use Syslog Parse profiles to parse syslog messages. In environments where services send the messages in different formats, you can create a custom profile for each format and associate multiple profiles with

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each syslog sender. If you use the PAN-OS integrated User-ID agent, you can also use predefined Syslog Parse profiles that Palo Alto Networks provides through Applications content updates.

Syslog messages must meet the following criteria for a User-ID agent to parse them:

- Each message must be a single-line text string. The allowed delimiters for line breaks are a new line (**\n**) or a carriage return plus a new line (**\r\n**).
- The maximum size for individual messages is 8,000 bytes.
- Messages sent over UDP must be contained in a single packet; messages sent over SSL can span multiple packets. A single packet might contain multiple messages.

See Configure User-ID to Monitor Syslog Senders for User Mapping for configuration details.

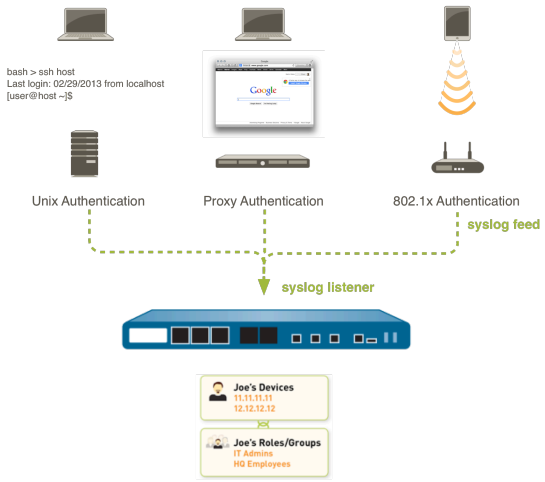

Figure 5: User-ID Integration with Syslog

GlobalProtect

For mobile or roaming users, the GlobalProtect endpoint provides the user mapping information to the firewall directly. In this case, every GlobalProtect user has an app running on the endpoint that requires the user to enter login credentials for VPN access to the firewall. This login information is then added to the User-ID user mapping table on the firewall for visibility and user-based security policy enforcement. Because GlobalProtect users must authenticate to gain access to the network, the IP address-to-username mapping is explicitly known. This is the best solution in sensitive environments where you must be certain of who a user is in order to allow access to an application or service. For more information on setting up GlobalProtect, refer to the GlobalProtect Administrator's Guide.

XML API

Authentication Portal and the other standard user mapping methods might not work for certain types of user access. For example, the standard methods cannot add mappings of users connecting from a third-party VPN solution or users connecting to a 802.1x-enabled wireless network. For such cases, you can use the PAN-OS XML API to capture login events and send them to the PAN-OS integrated User-ID agent. See Send User Mappings to User-ID Using the XML API for details.

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Client Probing

Palo Alto Networks strongly recommends disabling client probing because it is not a recommended method of obtaining User-ID information in a high-security network.

Palo Alto Networks does not recommend using client probing due to the following potential risks:

- Because client probing trusts data reported back from the endpoint, it can expose you to security risks when misconfigured. If you enable it on external, untrusted interfaces, this would cause the agent to send client probes containing sensitive information such as the username, domain name, and password hash of the User-ID agent service account outside of your network. If you do not configure the service account correctly, the credentials could potentially be exploited by an attacker to penetrate the network to gain further access.
- Client probing was designed for legacy networks where most users were on Windows workstations on the internal network, but is not ideal for today's more modern networks that support a roaming and mobile user base on a variety of devices and operating systems.
- Client probing can generate a large amount of network traffic (based on the total number of mapped IP addresses).

Instead, Palo Alto Networks strongly recommends using the following alternate methods for user mapping:

- Using more isolated and trusted sources, such as domain controllers and integrations with Syslog or the XML API, to safely capture user mapping information from any device type or operating system.
- Configuring Authentication Policy and Authentication Portal to ensure that you only allow access to authorized users.

The User-ID agent supports WMI probing, which uses either the PAN-OS integrated User-ID agent or the Windows User-ID agent.

In a Microsoft Windows environment, you can configure the User-ID agent to probe client systems using Windows Management Instrumentation (WMI) probing at regular intervals to verify that an existing user mapping is still valid or to obtain the username for an IP address that is not yet mapped.

If you do choose to enable probing in your trusted zones, the agent will probe each learned IP address periodically (every 20 minutes by default, but this is configurable) to verify that the same user is still logged in. In addition, when the firewall encounters an IP address for which it has no user mapping, it will send the address to the agent for an immediate probe.

See Configure User Mapping Using the Windows User-ID Agent or Configure User Mapping Using the PAN-OS Integrated User-ID Agent for details.

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Enable User-ID

The user identity, as opposed to an IP address, is an integral component of an effective security infrastructure. Knowing who is using each of the applications on your network, and who may have transmitted a threat or is transferring files, can strengthen your security policy and reduce incident response times. User-ID enables you to leverage user information stored in a wide range of repositories for visibility, user- and group-based policy control, and improved logging, reporting, and forensics:

STEP 1 | Enable User-ID on the source zones that contain the users who will send requests that require user-based access controls.

Enable User-ID on trusted zones only. If you enable User-ID and client probing on an external untrusted zone (such as the internet), probes could be sent outside your protected network, resulting in an information disclosure of the User-ID agent service account name, domain name, and encrypted password hash, which could allow an attacker to gain unauthorized access to protected services and applications.

1. Select **Network** > **Zones** and click the **Name** of the zone.
2. **Enable User Identification** and click **OK**.

STEP 2 | Create a Dedicated Service Account for the User-ID Agent.

As a best practice, create a service account with the minimum set of permissions required to support the User-ID options you enable to reduce your attack surface in the event that the service account is compromised.

This is required if you plan to use the Windows-based User-ID agent or the PAN-OS integrated User-ID agent to monitor domain controllers, Microsoft Exchange servers, or Windows clients for user login and logout events.

STEP 3 | Map Users to Groups.

This enables the firewall to connect to your LDAP directory and retrieve Group Mapping information so that you will be able to select usernames and group names when creating policy.

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STEP 4 | Map IP Addresses to Users.

As a best practice, do not enable client probing as a user mapping method on high-security networks. Client probing can generate a large amount of network traffic and can pose a security threat when misconfigured.

The way you do this depends on where your users are located and what types of systems they are using, and what systems on your network are collecting login and logout events for your users. You must configure one or more User-ID agents to enable User Mapping:

- Configure User Mapping Using the Windows User-ID Agent.
- Configure User Mapping Using the PAN-OS Integrated User-ID Agent.
- Configure User-ID to Monitor Syslog Senders for User Mapping.
- Configure User Mapping for Terminal Server Users.
- Send User Mappings to User-ID Using the XML API.
- Insert Username in HTTP Headers.

STEP 5 | Specify the networks to include and exclude from user mapping.

As a best practice, always specify which networks to include and exclude from User-ID. This allows you to ensure that only your trusted assets are probed and that unwanted user mappings are not created unexpectedly.

The way you specify which networks to include and exclude depends on whether you are using the Windows-based User-ID agent or the PAN-OSintegrated User-ID agent.

STEP 6 | Configure Authentication Policy and Authentication Portal.

The firewall uses Authentication Portal to authenticate end users when they request services, applications, or URL categories that match Authentication Policy rules. Based on user information collected during authentication, the firewall creates new user mappings or updates existing mappings. The mapping information collected during authentication overrides information collected through other User-ID methods.

1. Configure Authentication Portal.
2. Configure Authentication Policy.

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STEP 7 | Enable user- and group-based policy enforcement.

Create rules based on group rather than user whenever possible. This prevents you from having to continually update your rules (which requires a commit) whenever your user base changes.

After configuring User-ID, you will be able to choose a username or group name when defining the source or destination of a security rule:

1. Select **Policies** > **Security** and **Add** a new rule or click an existing rule name to edit.
2. Select **User** and specify which users and groups to match in the rule in one of the following ways:
   - If you want to select specific users or groups as matching criteria, click **Add** in the Source User section to display a list of users and groups discovered by the firewall group mapping function. Select the users or groups to add to the rule.
   - If you want to match any user who has or has not authenticated and you don't need to know the specific user or group name, select **known-user** or **unknown** from the drop-down above the Source User list.
3. Configure the rest of the rule as appropriate and then click **OK** to save it. For details on other fields in the security rule, see Create a Security Policy Rule..

STEP 8 | Create the Security policy rules to safely enable User-ID within your trusted zones and prevent User-ID traffic from egressing your network.

Follow the Best Practice Internet Gateway Security Policy to ensure that the User-ID application (`paloalto-userid-agent`) is only allowed in the zones where your agents (both your Windows agents and your PAN-OS integrated agents) are monitoring services and distributing mappings to firewalls. Specifically:

- Allow the `paloalto-userid-agent` application between the zones where your agents reside and the zones where the monitored servers reside (or even better, between the specific systems that host the agent and the monitored servers).
- Allow the `paloalto-userid-agent` application between the agents and the firewalls that need the user mappings and between firewalls that are redistributing user mappings and the firewalls they are redistributing the information to.
- Deny the `paloalto-userid-agent` application to any external zone, such as your internet zone.

STEP 9 | Configure the firewall to obtain user IP addresses from X-Forwarded-For (XFF) headers.

When the firewall is between the Internet and a proxy server, the IP addresses in the packets that the firewall sees are for the proxy server rather than users. To enable visibility of user IP addresses instead, configure the firewall to use the XFF headers for user mapping. With this option enabled, the firewall matches the IP addresses with usernames referenced in policy to

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enable control and visibility for the associated users and groups. For details, see Enforce Policy on Endpoints and Users Behind an Upstream Device.

1. Select **Device** > **Setup** > **Content-ID** and edit the X-Forwarded-For Headers settings.
2. Select **X-Forwarded-For Header in User-ID**.

   *Selecting **Strip-X-Forwarded-For Header** doesn't disable the use of XFF headers for user attribution in policy rules; the firewall zeroes out the XFF value only after using it for user attribution.*
3. Click **OK** to save your changes.

STEP 10 | If you use a high availability (HA) configuration, enable synchronization.

*As a best practice, always enable the **Enable Config Sync** option for an HA configuration to ensure that the group mappings and user mappings are synchronized between the active and passive firewall.*

1. Select **Device** > **High Availability** > **General** and edit the Setup section.
2. Select **Enable HA**.
3. Select **Enable Config Sync**.
4. Enter the **Peer HA1 IP Address**, which is the IP address of the HA1 control link on the peer firewall.
5. (Optional) Enter a **Backup Peer HA1 IP Address**, which is the IP address of the backup control link on the peer firewall.
6. Click **OK**.

STEP 11 | Commit your changes.

Commit your changes to activate them.

STEP 12 | Verify the User-ID Configuration.

After you configure user mapping and group mapping, verify that the configuration works properly and that you can safely enable and monitor user and group access to your applications and services.

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Map Users to Groups

Defining policy rules based on user group membership rather than individual users simplifies administration because you don't have to update the rules whenever group membership changes. The number of distinct user groups that each firewall or Panorama can reference across all policy rules varies by model. For more information, refer to the Compatibility Matrix.

For User-ID to successfully map users and for the firewall to enforce the policy, all users must be a member of at least one group that the firewall can map and configured in a group-based policy. To confirm if a user is a member of a group, use the `show user group list` *command.*

Use the following procedure to enable the firewall to connect to your LDAP directory and retrieve Group Mapping information. You can then Enable User- and Group-Based Policy.

The following are best practices for group mapping in an Active Directory (AD) environment:

- *If you have a single domain, you need only one group mapping configuration with an LDAP server profile that connects the firewall to the domain controller with the best connectivity. You can add up to four domain controllers to the LDAP server profile for redundancy. You can't increase redundancy beyond four domain controllers for a single domain by adding multiple group mapping configurations for that domain.*
- *If you have multiple domains or multiple forests, you must create a group mapping configuration with an LDAP server profile that connects the firewall to a domain server in each domain or forest. Take steps to ensure unique usernames in separate forests.*
- *If you have Universal Groups, create an LDAP server profile to connect to the root domain of the Global Catalog server on port 3268 or 3269 for SSL, then create another LDAP server profile to connect to the root domain controllers on port 389. This helps ensure that users and group information is available for all domains and subdomains.*
- *Before using group mapping, configure a* ***Primary Username*** *for user-based Security policy rules, since this attribute will identify users in the policy configuration, logs, and reports.*

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STEP 1 | Add an LDAP server profile.

The profile defines how the firewall connects to the directory servers from which it collects group mapping information.

If you create multiple group mapping configurations that use the same base Distinguished Name (DN) or LDAP server, the group mapping configurations can't contain overlapping groups (for example, the Include list for one group mapping configuration can't contain a group that is also in a different group mapping configuration).

1. Select **Device** > **Server Profiles** > **LDAP** and **Add** a server profile.
2. Enter a **Profile Name** to identify the server profile.
3. **Add** the LDAP servers. You can add up to four servers to the profile but they must be the same **Type**. For each server, enter a **Name** (to identify the server), **LDAP Server** IP address or FQDN, and server **Port** (default 389).
4. Select the server **Type**.

   Based on your selection (such as **active-directory**), the firewall automatically populates the correct LDAP attributes in the group mapping settings. However, if you customized your LDAP schema, you might need to modify the default settings.
5. For the **Base DN**, enter the Distinguished Name (DN) of the LDAP tree location where you want the firewall to start searching for user and group information.
6. For the **Bind DN**, **Password**, and **Confirm Password**, enter the authentication credentials for binding to the LDAP tree.

   The **Bind DN** can be a fully qualified LDAP name (such as `cn=administrator,cn=users,dc=acme,dc=local`) or a user principal name (such as `administrator@acme.local`).
7. Enter the **Bind Timeout** and **Search Timeout** in seconds (default is 30 for both).
8. Click **OK** to save the server profile.

STEP 2 | Configure the server settings in a group mapping configuration.

1. Select **Device** > **User Identification** > **Group Mapping Settings**.
2. **Add** the group mapping configuration.
3. Enter a unique **Name** to identify the group mapping configuration.
4. Select the LDAP **Server Profile** you just created.
5. (Optional) Specify the **Update Interval** (in seconds). Enter a value (range is 60—86400, default is 3600) based on how often the firewall checks the LDAP source for updates to the group mapping configuration. If the LDAP source contains many groups, a value that is too low may not allow enough time to map all the groups.
6. (Optional) By default, the **User Domain** field is blank: the firewall automatically detects the domain names for Active Directory (AD) servers. If you enter a value, it overrides any domain names that the firewall retrieves from the LDAP source. For most configurations,

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if you need to enter a value, enter the NetBIOS domain name (for example, **`example`** not **`example.com`**).

If you use Global Catalog, entering a value replaces the domain name for all users and groups from this server, including those from other domains.

7. (Optional) To filter the groups that the firewall tracks for group mapping, in the Group Objects section, enter a **Search Filter** (LDAP query) and **Object Class** (group definition).
8. (Optional) To filter the users who the firewall tracks for group mapping, in the User Objects section, enter a **Search Filter** (LDAP query), and **Object Class** (user definition).
9. Make sure the group mapping configuration is **Enabled** (default is enabled).

STEP 3 | (Optional) Define User and Group Attributes to collect for user and group mapping. This step is necessary if you want to map users based on directory attributes other than the domain.

1. If your User-ID sources only send the username and the username is unique across the organization, select **Device** > **User Identification** > **User Mapping** > **Setup** and **Edit** the Setup section to **Allow matching usernames without domains** to allow the firewall to check if unique usernames collected from the LDAP server during group mapping match the users associated with a policy and avoid overwriting the domain in your source profile.

   Before enabling this option, configure group mapping for the LDAP group containing the User-ID source (such as GlobalProtect or Authentication Portal) that collects the mappings. After you commit the changes, the User-ID source populates the usernames without domains. Only usernames collected during group mapping can be matched without a domain. If your User-ID sources send user information in multiple formats and you enable this option, verify that the attributes collected by the firewall have a unique prefix. To ensure users are identified correctly if you enable this option, all attributes for group mapping should be unique. If the username isn't unique, the firewall logs an error in the Debug logs.

2. Select **Device** > **User Identification** > **Group Mapping Settings** > **Add** > **User and Group Attributes** > **User Attributes** and enter the **Directory Attribute** you want to collect for user identification. Specify a **Primary Username** to identify the user on the firewall and

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to represent the user in reports and logs that will override any other format the firewall receives from the User-ID source.

When you select the Server Profile **Type**, the firewall autopopulates the values for the user and group attributes. Based on the user information that your User-ID sources send, you may need to configure the correct attributes:

- **User Principal Name (UPN)**: **`userPrincipalName`**
- **NetBios Name**: **`sAMAccountName`**
- **Email ID**: Directory attribute for that email
- **Multiple formats**: Retrieve the user mapping attributes from the user directory before enabling your User-ID sources.

If you don't specify a primary username, the firewall uses the following default values for each server profile type:

Attribute	Active Directory	Novell eDirectory or Sun ONE Directory Server
Primary Username	`sAMAccountName`	`uid`
E-Mail	`mail`	`mail`
Alternate Username 1	`userPrincipalName`	None.
Group Name	`name`	`cn`
Group Member	`member`	`member`

3. (Optional) Specify an **E-Mail** address format and up to three **Alternate Username** formats.
4. Select **Device** > **User Identification** > **Group Mapping Settings** > **Add** > **User and Group Attributes** > **Group Attributes** and specify the **Group Name**, **Group Member**, and **E-Mail** address formats.

   You must commit before the firewall collects the directory attributes from the LDAP server.

STEP 4 | Limit which groups will be available in policy rules.

Required only if you want to limit policy rules to specific groups. The combined maximum for the **Group Include List** and **Custom Group** list is 640 entries per group mapping configuration.

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Each entry can be a single group or a list of groups. By default, if you don't specify groups, all groups are available in policy rules.

Any custom groups you create will also be available in the Allow List of authentication profiles (Configure an Authentication Profile and Sequence).

1. Add existing groups from the directory service:
   1. Select **Group Include List**.
   2. Select the Available Groups you want to appear in policy rules and add (⊕) them to the Included Groups.
2. If you want to base policy rules on user attributes that don't match existing user groups, create custom groups based on LDAP filters:
   1. Select **Custom Group** and **Add** the group.
   2. Enter a group **Name** that is unique in the group mapping configuration for the current firewall or virtual system.

      If the **Name** has the same value as the Distinguished Name (DN) of an existing AD group domain, the firewall uses the custom group in all references to that name (such as in policy rules and logs).
   3. Specify an **LDAP Filter** of up to 2,048 UTF-8 characters and click **OK**.

      The firewall doesn't validate LDAP filters, so it's up to you to ensure they are accurate.

      To minimize the performance impact on the LDAP directory server, use only indexed attributes in the filter.
3. Click **OK** to save your changes.

   You must commit before custom groups will be available in policy rules and objects.

STEP 5 | **Commit** your changes.

You must commit before you can use custom groups in policy rules and objects and before the firewall can collect the attributes from the LDAP server.

After configuring the firewall to retrieve group mapping information from an LDAP server, but before configuring policy rules based on the groups it retrieves, the best practice is to either wait for the firewall to refresh its group mappings cache or refresh the cache manually. To verify which groups you can currently use in policy rules, access the firewall CLI and run the ***`show user group`*** *command. To determine when the firewall will next refresh the group mappings cache, run the* ***`show user group-mapping statistics`*** *command and check the `Next Action`. To manually refresh the cache, run the* ***`debug user-id refresh group-mapping all`*** *command.*

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STEP 6 | Verify that the user and group mapping has correctly identified users.

1. Select **Device** > **User Identification** > **Group Mapping** > **Group Include List** to confirm the firewall has fetched all of the groups.
2. To verify that all of the user attributes have been correctly captured, use the following CLI command:

   ```
   show user user-attributes user all
   ```

   The normalized format for the User Principal Name (UPN), primary username, email attributes, and any configured alternate usernames display for all users:

   ```
   admin@PA-VM-8.1> show user user-attributes user all
   ```

   ```
   Primary: nam\sam-user   Email: sam-user@nam.com
   ```

   ```
   Alt User Names:1) nam.com\sam-user
   ```

   ```
   2) nam\sam-user-upn
   ```

   ```
   3) sam-user-upn@nam.local
   ```

   ```
   4) sam-user@nam.com
   ```

3. Verify that the usernames are correctly displayed in the **Source User** column under **Monitor** > **Logs** > **Traffic**.

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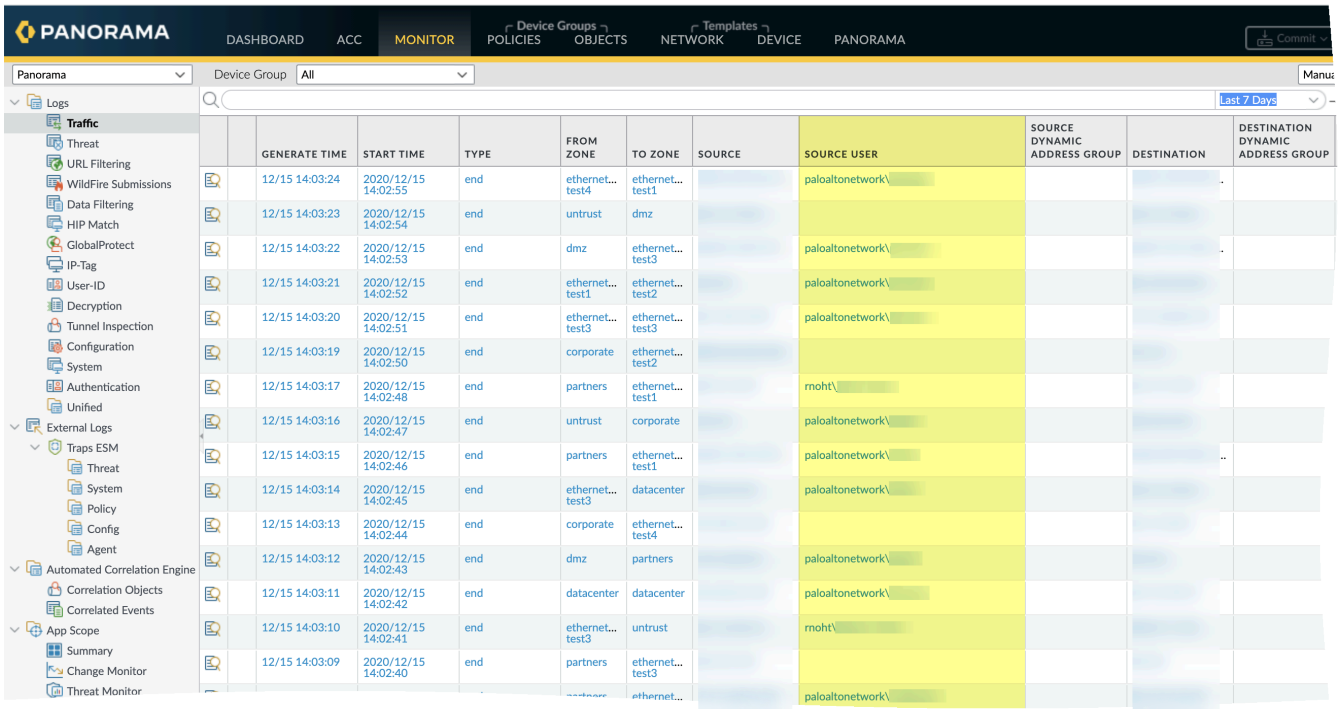
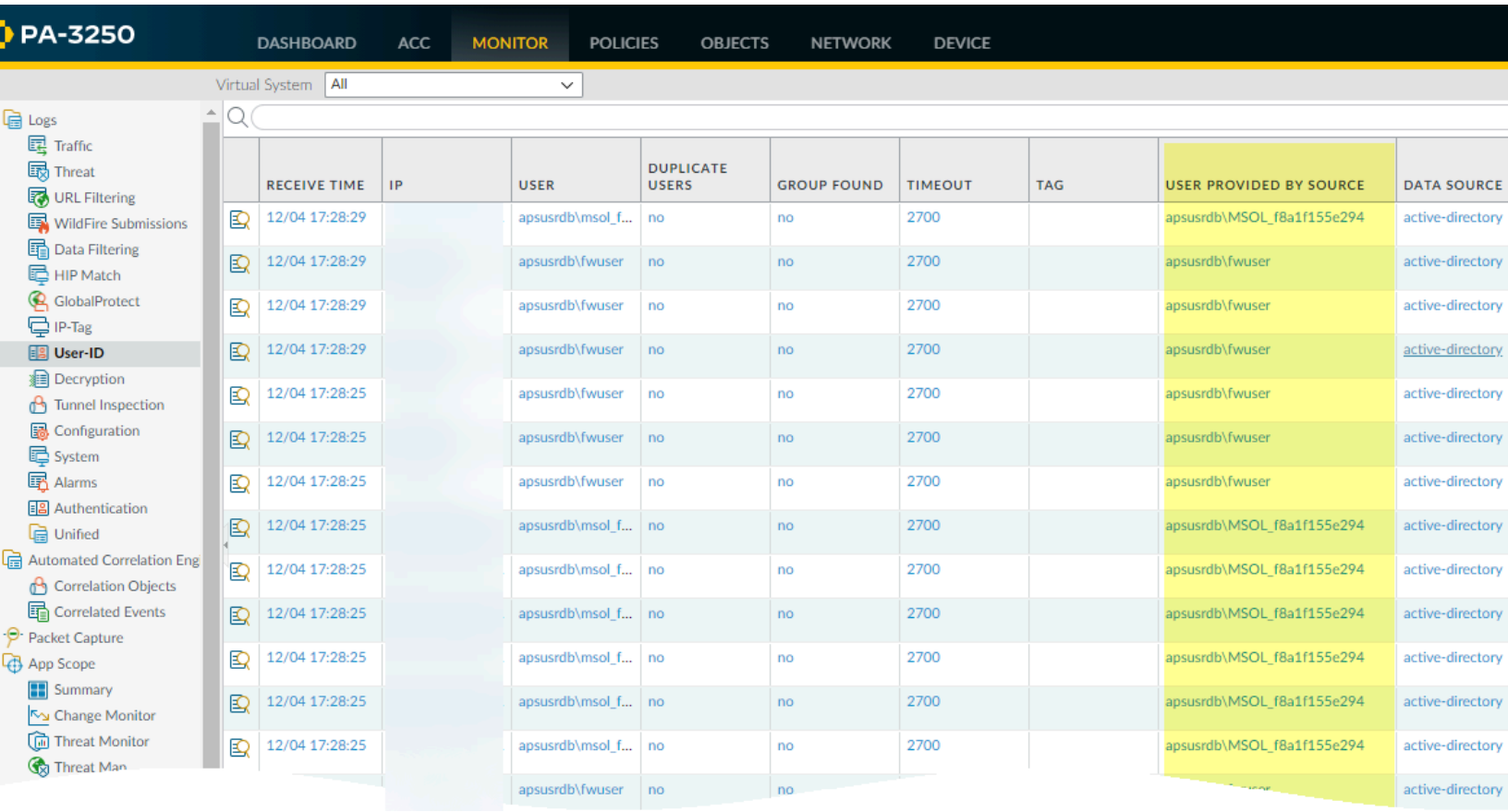


4. Verify that the users are mapped to the correct usernames in the **User Provided by Source** column under **Monitor** > **Logs** > **User-ID**.



Map IP Addresses to Users

User-ID provides many different methods for mapping IP addresses to usernames. Before you begin configuring user mapping, consider where your users are logging in from, what services they are accessing, and what applications and data you need to control access to. This will inform which types of agents or integrations would best allow you to identify your users.

Once you have your plan, you can begin configuring user mapping using one or more of the following methods as needed to enable user-based access and visibility to applications and resources:

- If you have users with client systems that aren't logged in to your domain servers—for example, users running Linux clients that don't log in to the domain—you can Map IP Addresses to Usernames Using Authentication Portal. Using Authentication Portal in conjunction with Authentication Policy also ensures that all users authenticate to access your most sensitive applications and data.
- To map users as they log in to your Exchange servers, domain controllers, eDirectory servers, or Windows clients you must configure a User-ID agent:
  - Configure User Mapping Using the PAN-OS Integrated User-ID Agent
  - Configure User Mapping Using the Windows User-ID Agent
- If you have clients running multi-user systems in a Windows environment, such as Microsoft Terminal Server or Citrix Metaframe Presentation Server or XenApp, Configure the Palo Alto Networks Terminal Server (TS) Agent for User Mapping. For a multi-user system that doesn't run on Windows, you can Retrieve User Mappings from a Terminal Server Using the PAN-OS XML API.
- To obtain user mappings from existing network services that authenticate users—such as wireless controllers, 802.1x devices, Apple Open Directory servers, proxy servers, or other Network Access Control (NAC) mechanisms—Configure User-ID to Monitor Syslog Senders for User Mapping.

  While you can configure either the Windows agent or the PAN-OS integrated User-ID agent on the firewall to listen for authentication syslog messages from the network services, because only the PAN-OS integrated agent supports syslog listening over TLS, it is the preferred configuration.
- To include the username and domain in the headers for outgoing traffic so other devices in your network can identify the user and enforce user-based policy, you can Insert Username in HTTP Headers.
- To Share User-ID Mappings Across Virtual Systems, you can configure a virtual system as a User-ID hub.
- For other clients that you can't map using the other methods, you can Send User Mappings to User-ID Using the XML API.
- A large-scale network can have hundreds of information sources that firewalls query for user and group mapping and can have numerous firewalls that enforce policies based on the mapping information. You can simplify User-ID administration for such a network by aggregating the mapping information before the User-ID agents collect it. You can also reduce the resources that the firewalls and information sources use in the querying process by

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configuring some firewalls to redistribute the mapping information. For details, see Deploy User-ID in a Large-Scale Network.

Create a Dedicated Service Account for the User-ID Agent

To use the Windows-based User-ID agent or the PAN-OS integrated User-ID agent to map users as they log in to your Exchange servers, domain controllers, eDirectory servers, or Windows clients, create a dedicated service account for the User-ID agent on a domain controller in each domain that the agent will monitor.

The User-ID agent maps users based on logs for security events. To ensure that the User-ID agent can successfully map users, verify that the source for your mappings generates logs for Audit Logon, Audit Kerberos Authentication Service, and Audit Kerberos Service Ticket Operations events. At a minimum, the source must generate logs for the following events:

- Logon Success (4624)
- Authentication Ticket Granted (4768)
- Service Ticket Granted (4769)
- Ticket Granted Renewed (4770)

The required permissions for the service account depend on the user mapping methods and settings you plan to use. For example, if you are using the PAN-OS integrated User-ID agent, the service account requires Server Operator privileges to monitor user sessions. If you are using the Windows-based User-ID agent, the service account does not require Server Operator privileges to monitor user sessions. To reduce the risk of compromising the User-ID service account, always configure the account with the minimum set of permissions necessary for the agent.

- If you are installing the Windows-based User-ID agent on a supported Windows server, Configure a Service Account for the Windows User-ID Agent.
- If you are using the PAN-OS integrated User-ID agent on the firewall, Configure a Service Account for the PAN-OS Integrated User-ID Agent.

User-ID provides many methods for safely collecting user mapping information. Some legacy features designed for environments that only required user mapping on Windows desktops attached to the local network require privileged service accounts. If the privileged service account is compromised, this would open your network to attack. As a best practice, avoid using legacy features that require privileges that would pose a threat if compromised, such as client probing and session monitoring.

Configure a Service Account for the Windows User-ID Agent

Create a dedicated Active Directory (AD) service account for the Windows User-ID agent to access the services and hosts it will monitor to collect user mappings. You must create a service account in each domain the agent will monitor. After you enable the required permissions for the service account, Configure User Mapping Using the Windows User-ID Agent.

The following workflow details all required privileges and provides guidance for the User-ID features which require privileges that could pose a threat so that you can decide how to best identify users without compromising your overall security posture.

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STEP 1 | Create an AD service account for the User-ID agent.

You must create a service account in each domain the agent will monitor.

1. Log in to the domain controller.
2. Right-click the Windows icon (), **Search** for **`Active Directory Users and Computers`**, and launch the application.
3. In the navigation pane, open the domain tree, right-click **Managed Service Accounts** and select **New** > **User**.
4. Enter the **First Name**, **Last Name**, and **User logon name** of the user and click **Next**.
5. Enter the **Password** and **Confirm Password**, then click **Next** and **Finish**.

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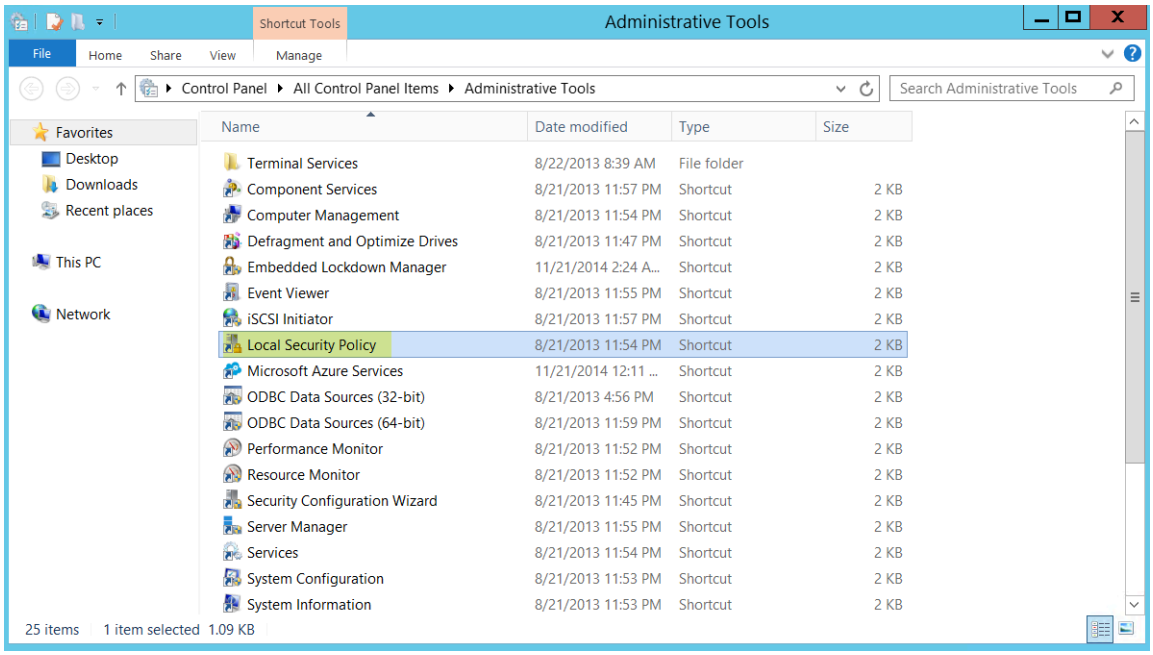
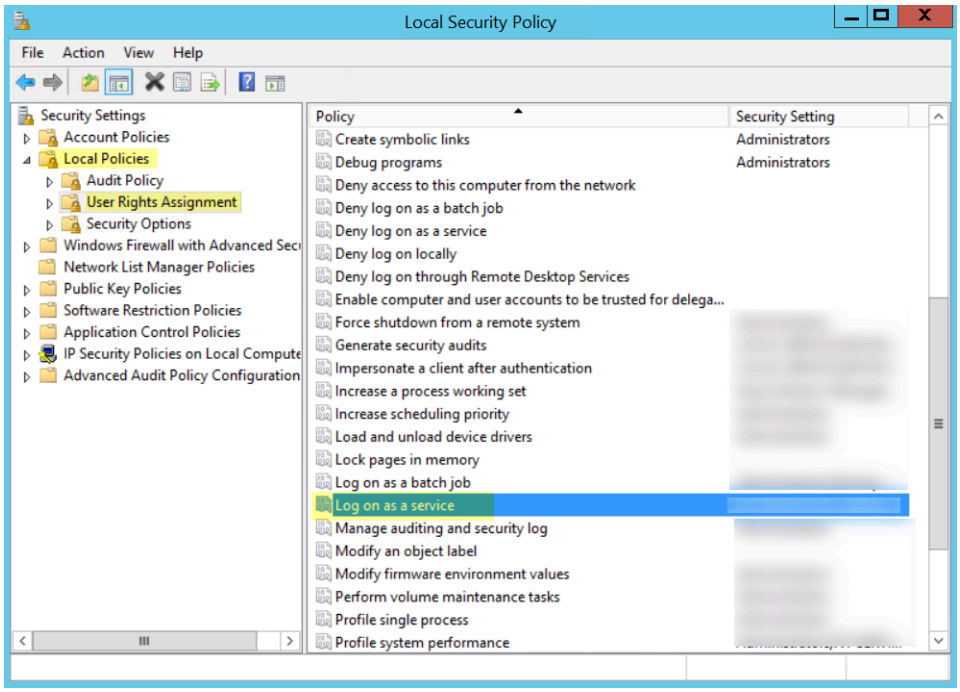
STEP 2 | Configure either local or group policy to allow the service account to log on as a service.

The permission to log on as a service is only needed locally on the Windows server that is the agent host.

- To assign permissions locally:

1. select **Control Panel** > **Administrative Tools** > **Local Security Policy**.
2. 

3. Select **Local Policies** > **User Rights Assignment** > **Log on as a service**.


4. **Add User or Group** to add the service account.

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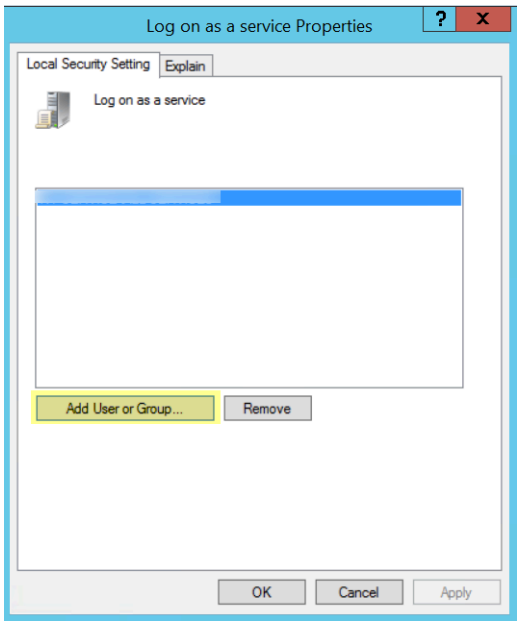
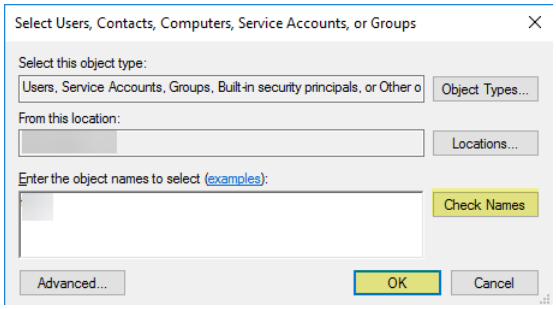
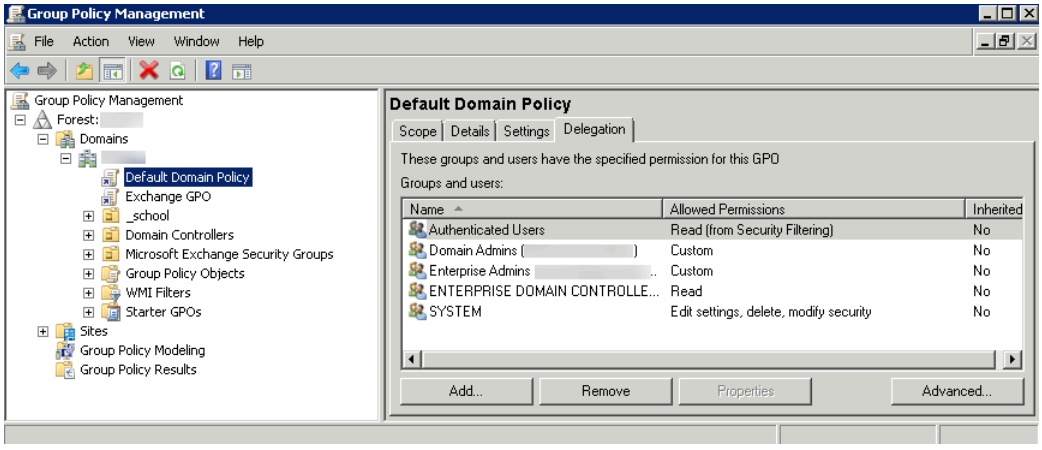


5. **Enter the object names to select** (the service account name) in **`domain\username`** format and click **OK**.



- To configure group policy if you are installing Windows User-ID agents on multiple servers, use the Group Policy Management Editor.

1. Select **Start** > **Group Policy Management** > **<your domain>** > **Default Domain Policy** > **Action** > **Edit** for the Windows server that is the agent host.



2. Select **Computer Configuration** > **Policies** > **Windows Settings** > **Security Settings** > **Local Policies** > **User Rights Assignment**.

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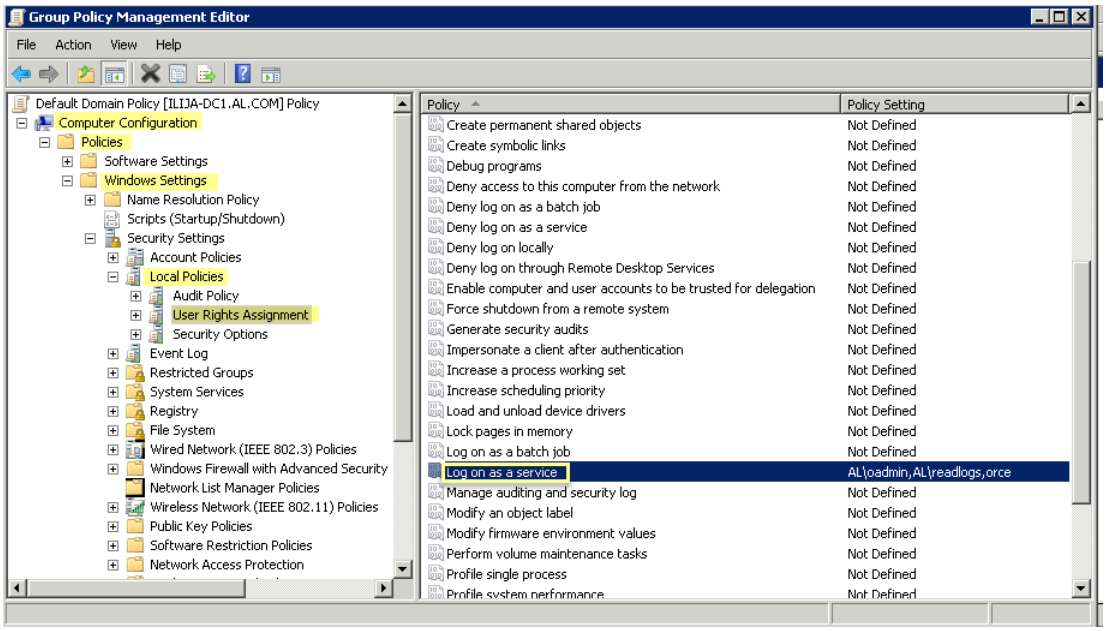
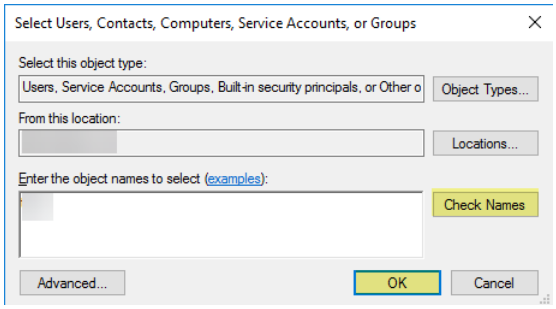


3. Right-click **Log on as a service**, then select **Properties**.
4. **Add User or Group** to add the service account username or builtin group, then click **OK** twice.

   Administrators have this privilege by default.



STEP 3 | If you want to use WMI to collect user data, assign DCOM privileges to the service account so that it can use WMI queries on monitored servers.

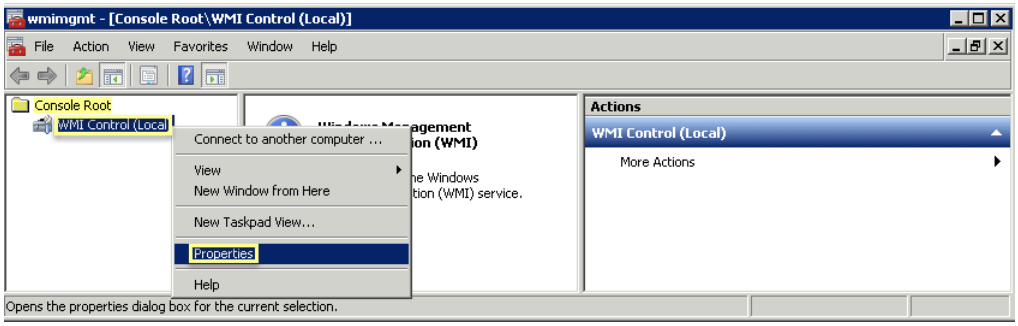
1. Select **Active Directory Users and Computers** > **<your domain>** > **Builtin** > **Distributed COM Users**.
2. Right-click **Properties** > **Members** > **Add** and enter the service account name.

STEP 4 | If you plan to use WMI probing, enable the account to read the CIMV2 namespace and assign the required permissions on the client systems to be probed.

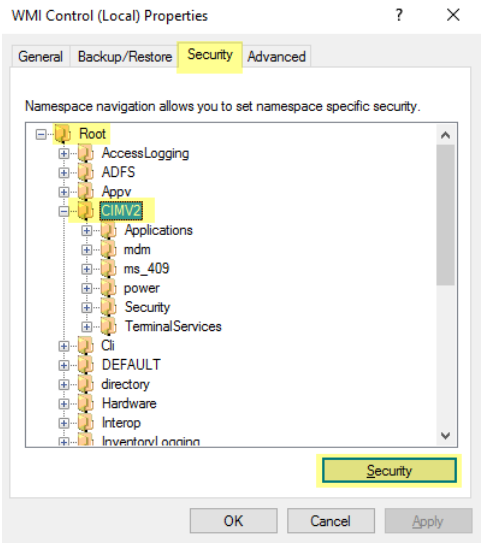
Do not enable client probing on high-security networks. Client probing can generate a large amount of network traffic and can pose a security threat when misconfigured. Instead collect user mapping information from more isolated and trusted sources, such as domain controllers and through integrations with Syslog or the XML API, which have the added benefit of allowing you to safely capture user mapping information from any device type or operating system, instead of just Windows clients.

Perform this task on each client system that the User-ID agent will probe for user mapping information:

1. Right-click the Windows icon (), **Search** for **`wmimgmt.msc`**, and launch the WMI Management Console.
2. In the console tree, right-click **WMI Control** and select **Properties**.



3. Select the **Security** tab, then select **Root** > **CIMV2**, and click the **Security** button.



4. **Add** the name of the service account you created, **Check Names** to verify your entry, and click **OK**.

*You might have to change the **Locations** or click **Advanced** to query for account names. See the dialog help for details.*

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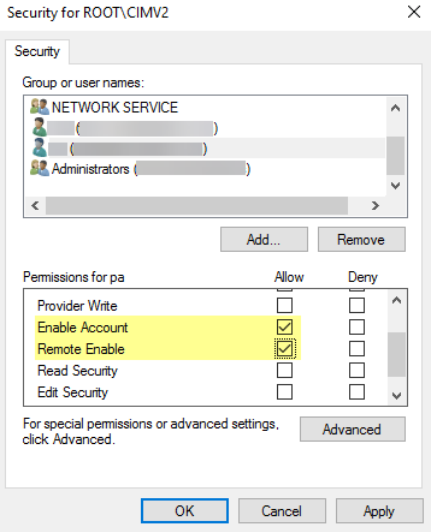
5. In the Permissions for **<Username>** section, **Allow** the **Enable Account** and **Remote Enable** permissions.



6. Click **OK** twice.
7. Use the Local Users and Groups MMC snap-in (lusrmgr.msc) to add the service account to the local Distributed Component Object Model (DCOM) Users and Remote Desktop Users groups on the system that will be probed.

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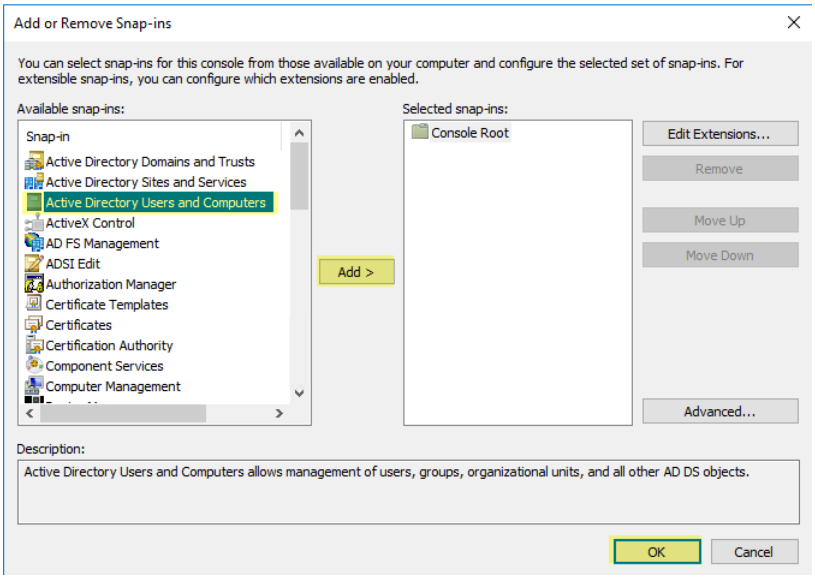
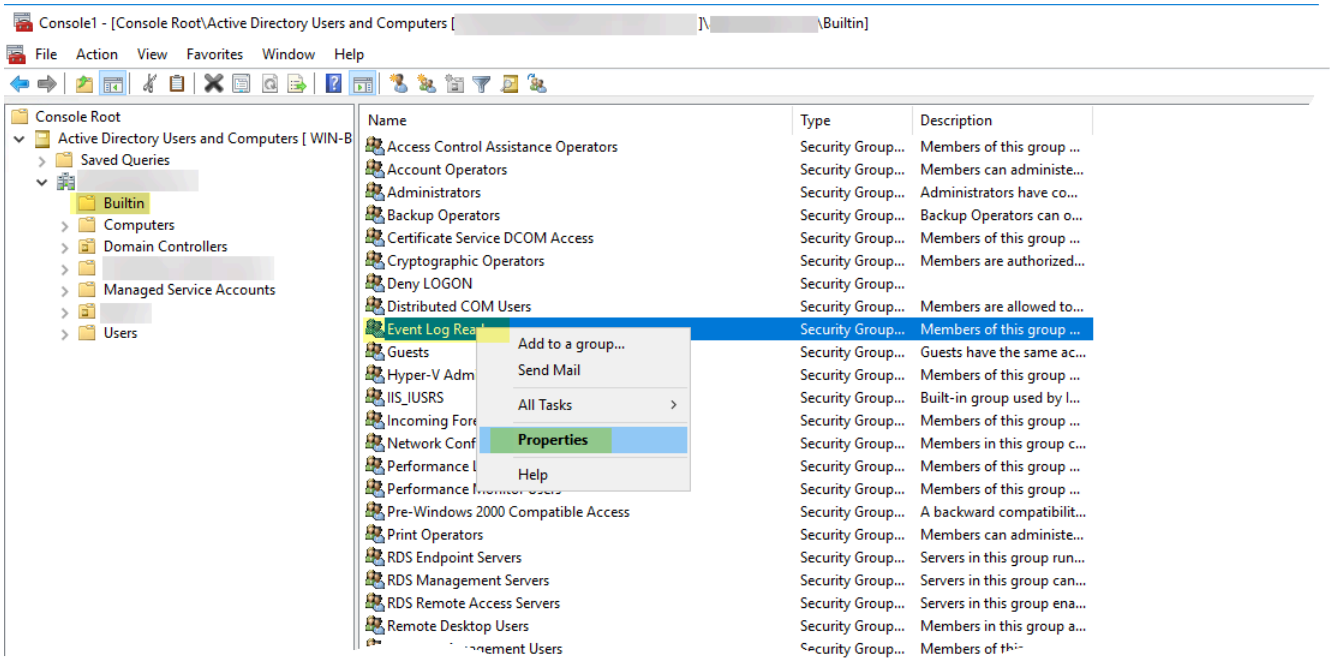
STEP 5 | If you want to use Server Monitoring to identify users, add the service account to the Event Log Reader builtin group to allow the service account to read the security log events.

1. On the domain controller or Exchange server that contains the logs you want the User-ID agent to read, or on the member server that receives events from Windows log forwarding, select **Start** > **Run**, enter **MMC**.
2. Select **File** > **Add/Remove Snap-in** > **Active Directory Users and Computers** > **Add**, then click **OK** to run the MMC and launch the Active Directory Users and Computers snap-in.



3. Navigate to the Builtin folder for the domain, right-click the **Event Log Readers** group, and select **Properties** > **Members**.



4. **Add** the service account then click **Check Names** to validate that you have the proper object name.

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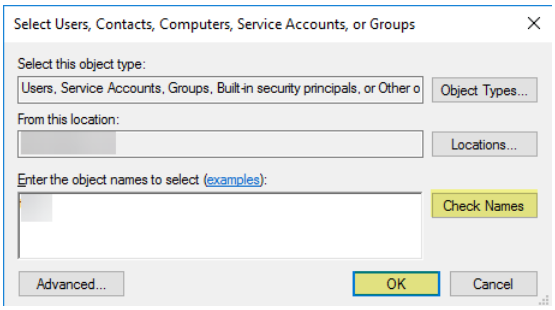

5. Click **OK** twice to save the settings.
6. Confirm that the builtin Event Log Reader group lists the service account as a member (**Event Log Readers** > **Properties** > **Members**).

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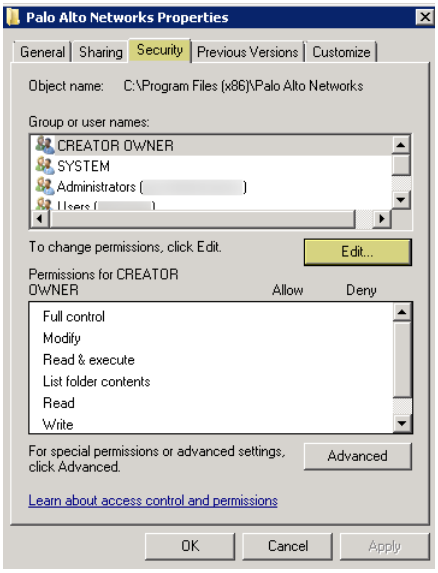
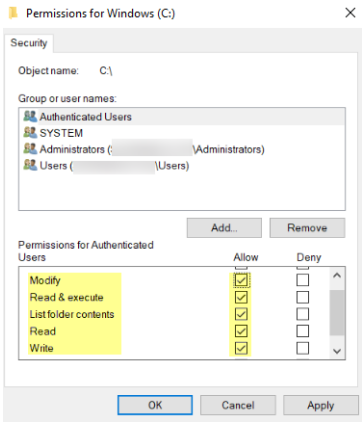
STEP 6 | Assign account permissions to the installation folder to allow the service account to access the agent's installation folder to read the configuration and write logs.

You only need to perform this step if the service account you configured for the User-ID agent is not either a domain administrator or a local administrator on the User-ID agent server host.

1. From the Windows Explorer, navigate to **`C:\Program Files(x86)\Palo Alto Networks`**, right-click the folder, and select **Properties**.
2. On the **Security** tab, click **Edit**.



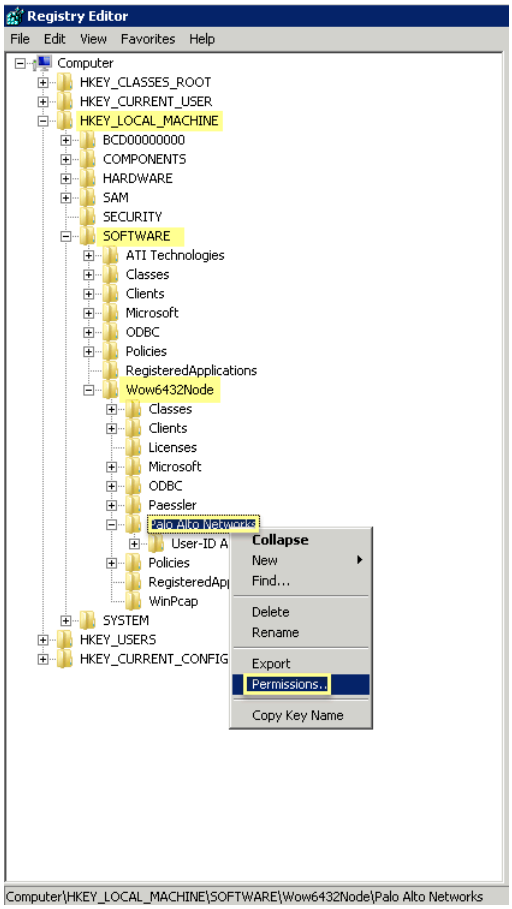
3. **Add** the User-ID agent service account and **Allow** permissions to **Modify**, **Read & execute**, **List folder contents**, **Read**, and **Write**, and then click **OK** to save the account settings.



*If you do not want to configure individual permissions, you can **Allow** the **Full Control** permission instead.*

STEP 7 | To allow the agent to make configuration changes (for example, if you select a different logging level), give the service account permissions to the User-ID agent registry sub-tree.

1. Select **Start** > **Run** and enter **`regedt32`** and navigate to the Palo Alto Networks sub-tree in one of the following locations:
   - **32-bit systems**—`HKEY_LOCAL_MACHINE\Software\Palo Alto Networks`
   - **64-bit systems**—`HKEY_LOCAL_MACHINE\Software\WOW6432Node\PaloAlto Networks`
2. Right-click the **Palo Alto Networks** node and select **Permissions**.



3. Assign the User-ID service account **Full Control** and then click **OK** to save the setting.

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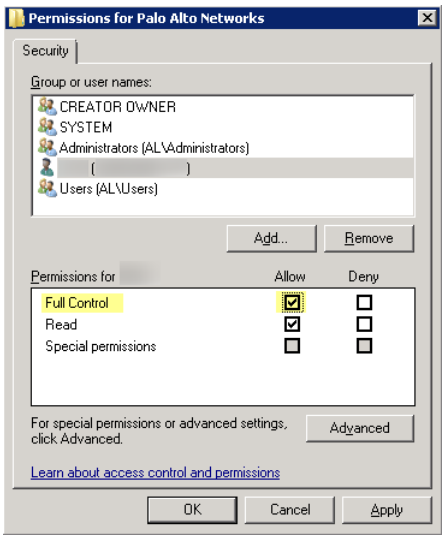

STEP 8 | Disable service account privileges that are not required.

By ensuring that the User-ID service account has the minimum set of account privileges, you can reduce the attack surface should the account be compromised.

To ensure that the User-ID account has the minimum privileges necessary, deny the following privileges on the account.

- **Deny interactive logon for the User-ID service account**—While the User-ID service account does need permission to read and parse Active Directory security event logs, it does not require the ability to logon to servers or domain systems interactively. You can restrict this

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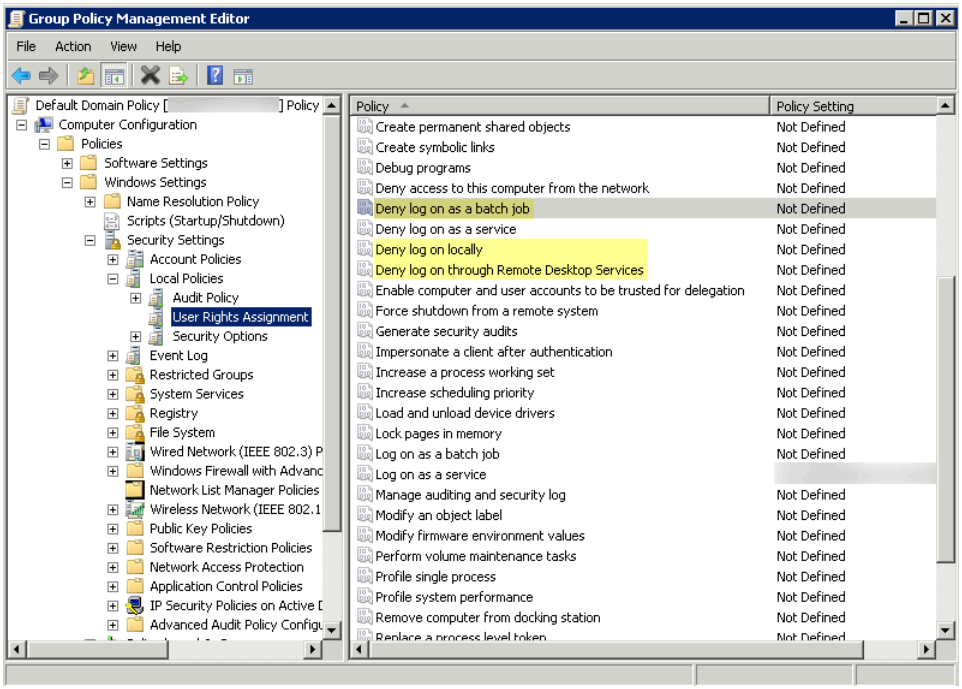
privilege using Group Policies or by using a Managed Service account (refer to Microsoft TechNet for more information).

1. Select **Group Policy Management Editor** > **Default Domain Policy** > **Computer Configuration** > **Policies** > **Windows Settings** > **Security Settings** > **User Rights Assignment**.
2. For **Deny log on as a batch job**, **Deny log on locally**, and **Deny log on through Remote Desktop Services**, right-click **Properties**.
3. Select **Define these policy settings** > **Add User or Group** and add the service account name, then click **OK**.



- **Deny remote access for the User-ID service account**—This prevents an attacker from using the account to access your network from the outside the network.
  1. Select **Start** > **Run**, enter **MMC**, and select **File** > **Add/Remove Snap-in** > **Active Directory Users and Computers** > **Users**.
  2. Right-click the service account name, then select **Properties**.
  3. Select **Dial-in**, then **Deny** the **Network Access Permission**.

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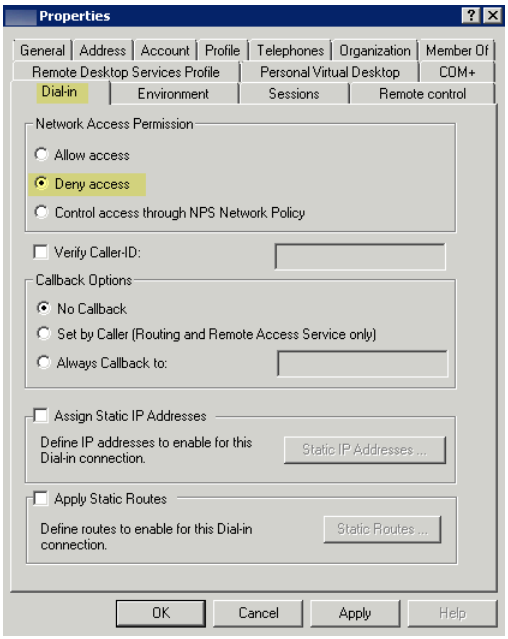

STEP 9 | As a next step, Configure User Mapping Using the Windows User-ID Agent.

Configure a Service Account for the PAN-OS Integrated User-ID Agent

Create a dedicated Active Directory (AD) service account for the PAN-OS Integrated User-ID agent to access the services and hosts it will monitor to collect user mappings.You must create a service account in each domain the agent will monitor. After you enable the required permissions for the service account, Configure User Mapping Using the PAN-OS Integrated User-ID Agent.

The following workflow details all required privileges and provides guidance for the User-ID features which require privileges that could pose a threat so that you can decide how to best identify users without compromising your overall security posture.

STEP 1 | Create an AD service account for the User-ID agent.

You must create a service account in each domain the agent will monitor.

1. Log in to the domain controller.
2. Right-click the Windows icon (), **Search** for **`Active Directory Users and Computers`**, and launch the application.
3. In the navigation pane, open the domain tree, right-click **Managed Service Accounts** and select **New** > **User**.
4. Enter the **First Name**, **Last Name**, and **User logon name** of the user and click **Next**.
5. Enter the **Password** and **Confirm Password**, then click **Next** and **Finish**.

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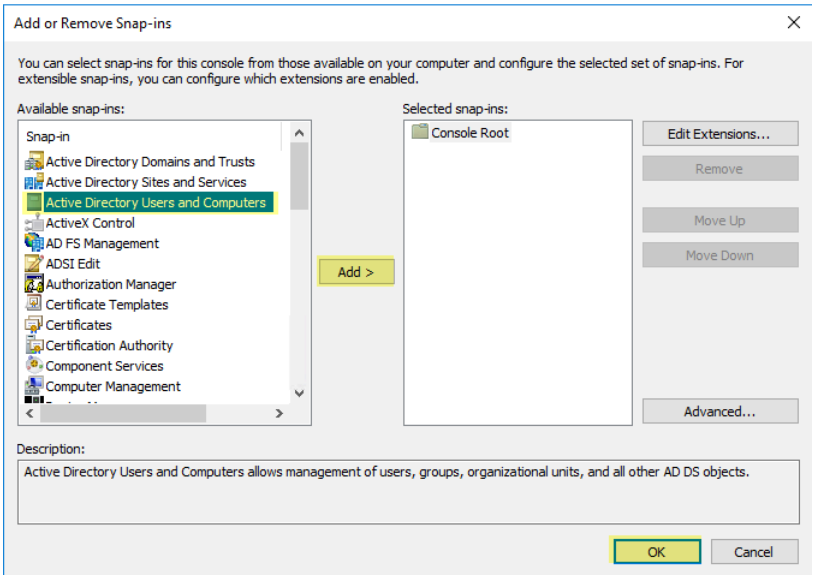
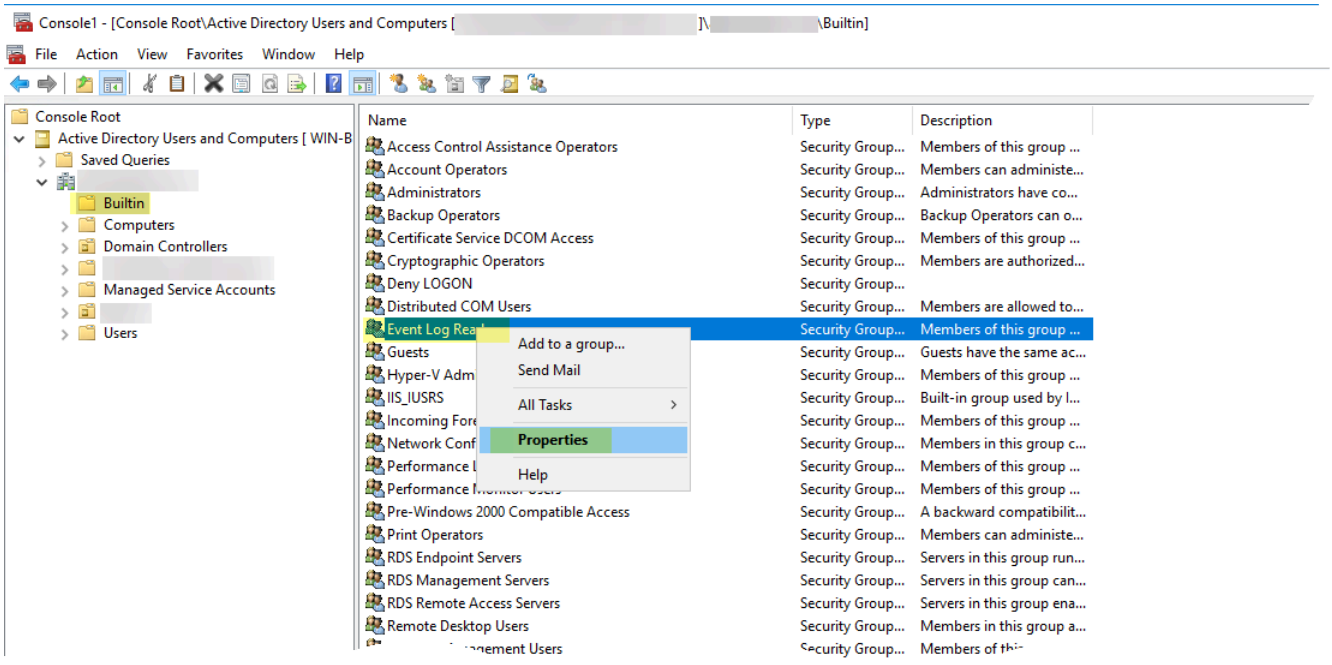
STEP 2 | If you want to use Server Monitoring to identify users, add the service account to the Event Log Reader builtin group to allow the service account to read the security log events.

1. On the domain controller or Exchange server that contains the logs you want the User-ID agent to read, or on the member server that receives events from Windows log forwarding, select **Start** > **Run**, enter **MMC**.
2. Select **File** > **Add/Remove Snap-in** > **Active Directory Users and Computers** > **Add**, then click **OK** to run the MMC and launch the Active Directory Users and Computers snap-in.



3. Navigate to the Builtin folder for the domain, right-click the **Event Log Readers** group, and select **Properties** > **Members**.



4. **Add** the service account then click **Check Names** to validate that you have the proper object name.

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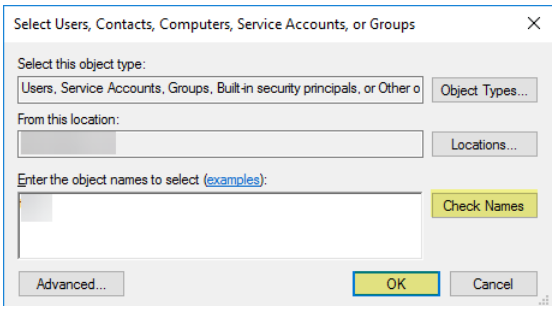

5. Click **OK** twice to save the settings.
6. Confirm that the builtin Event Log Reader group lists the service account as a member (**Event Log Readers** > **Properties** > **Members**).

STEP 3 | If you want to use WMI to collect user data, assign DCOM privileges to the service account so that it can use WMI queries on monitored servers.

1. Select **Active Directory Users and Computers** > **<your domain>** > **Builtin** > **Distributed COM Users**.
2. Right-click **Properties** > **Members** > **Add** and enter the service account name.

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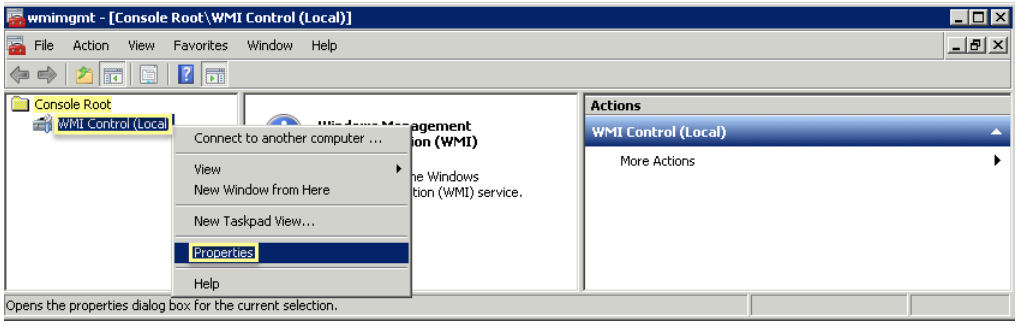
STEP 4 | If you plan to use WMI probing, enable the service account to read the CIMV2 namespace on the domain controllers you want to monitor and assign the required permissions on the client systems to be probed.

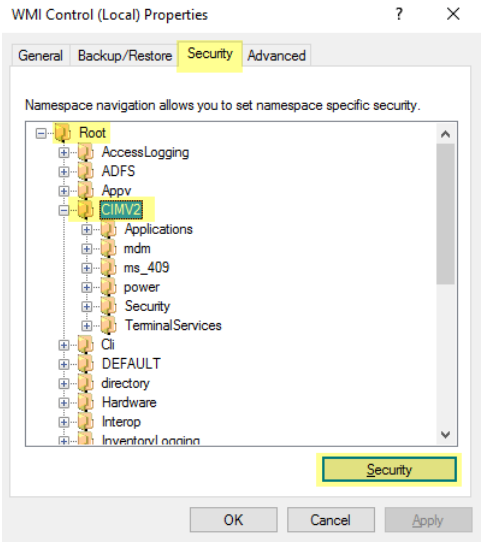
Do not enable client probing on high-security networks. Client probing can generate a large amount of network traffic and can pose a security threat when misconfigured. Instead collect user mapping information from more isolated and trusted sources, such as domain controllers and through integrations with Syslog or the XML API, which have the added benefit of allowing you to safely capture user mapping information from any device type or operating system, instead of just Windows clients.

Perform this task on each client system that the User-ID agent will probe for user mapping information:

1. Right-click the Windows icon (), **Search** for **`wmimgmt.msc`**, and launch the WMI Management Console.
2. In the console tree, right-click **WMI Control** and select **Properties**.



3. Select the **Security** tab, then select **Root** > **CIMV2**, and click the **Security** button.



4. **Add** the name of the service account you created, **Check Names** to verify your entry, and click **OK**.

*You might have to change the **Locations** or click **Advanced** to query for account names. See the dialog help for details.*

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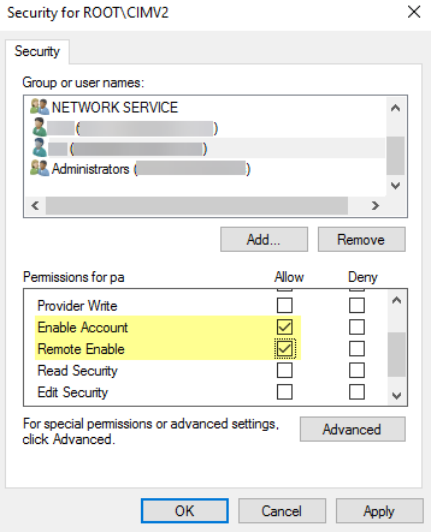
5. In the Permissions for **<Username>** section, **Allow** the **Enable Account** and **Remote Enable** permissions.



6. Click **OK** twice.
7. Use the Local Users and Groups MMC snap-in (lusrmgr.msc) to add the service account to the local Distributed Component Object Model (DCOM) Users and Remote Desktop Users groups on the system that will be probed.

STEP 5 | (Not Recommended) To allow the agent to monitor user sessions to poll Windows servers for user mapping information, assign Server Operator privileges to the service account.

Because this group also has privileges for shutting down and restarting servers, only assign the account to this group if monitoring user sessions is very important.

1. Select **Active Directory Users and Computers** > **<your domain>** > **Builtin** > **Server Operators Group**.
2. Right-click **Properties** > **Members** > **Add** add service account name

STEP 6 | Disable service account privileges that are not required.

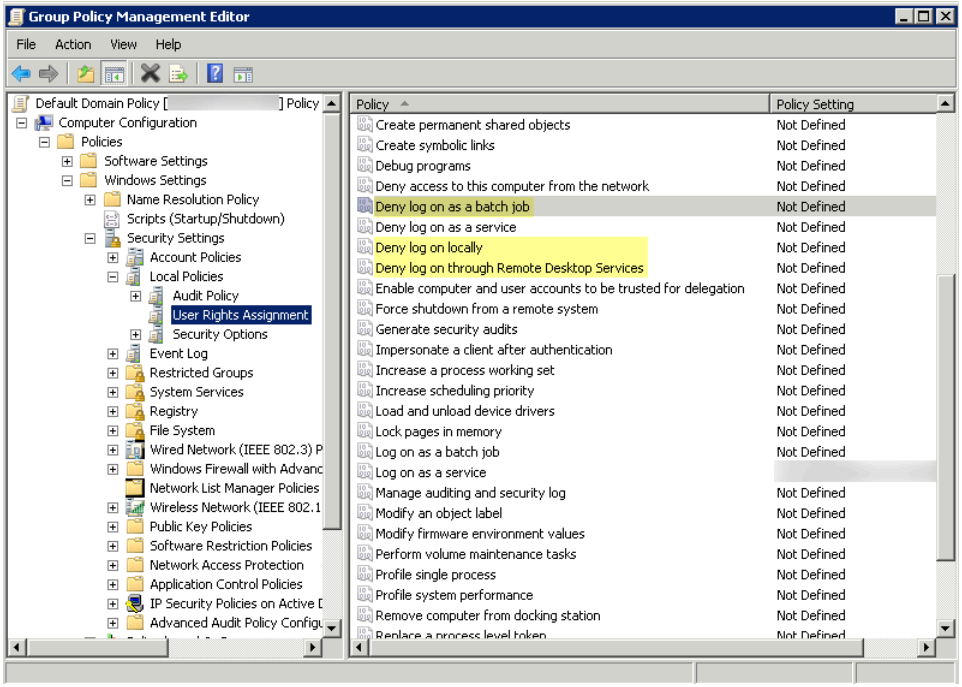
By ensuring that the User-ID service account has the minimum set of account privileges, you can reduce the attack surface should the account be compromised.

To ensure that the User-ID account has the minimum privileges necessary, deny the following privileges on the account:

- **Deny interactive logon for the User-ID service account**—While the User-ID service account does need permission to read and parse Active Directory security event logs, it does not require the ability to logon to servers or domain systems interactively. You can restrict this

privilege using Group Policies or by using a Managed Service account (refer to Microsoft TechNet for more information).

1. Select **Group Policy Management Editor** > **Default Domain Policy** > **Computer Configuration** > **Policies** > **Windows Settings** > **Security Settings** > **User Rights Assignment**.
2. For **Deny log on as a batch job**, **Deny log on locally**, and **Deny log on through Remote Desktop Services**, right-click **Properties**, then select **Define these policy settings** > **Add User or Group** and add the service account name, then click **OK**.



- **Deny remote access for the User-ID service account**—This prevents an attacker from using the account to access your network from the outside the network.
  1. **Start** > **Run**, enter **MMC**, and select **File** > **Add/Remove Snap-in** > **Active Directory Users and Computers** > **Users**.
  2. Right-click the service account name, then select **Properties**.
  3. Select **Dial-in**, then **Deny** the **Network Access Permission**.

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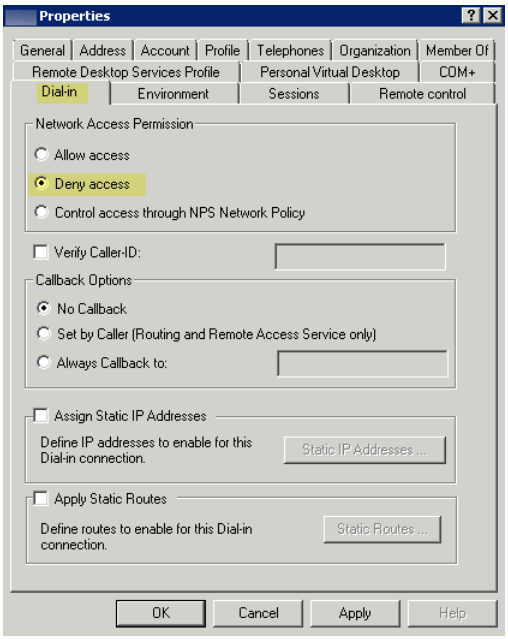

STEP 7 | As a next step, Configure User Mapping Using the PAN-OS Integrated User-ID Agent.

Configure User Mapping Using the Windows User-ID Agent

In most cases, the majority of your network users will have logins to your monitored domain services. For these users, the Palo Alto Networks User-ID agent monitors the servers for login events and performs the IP address to username mapping. The way you configure the User-ID agent depends on the size of your environment and the location of your domain servers. As a best practice, locate your User-ID agents near the servers it will monitor (that is, the monitored servers and the Windows User-ID agent should not be across a WAN link from each other). This is because most of the traffic for user mapping occurs between the agent and the monitored server, with only a small amount of traffic—the delta of user mappings since the last update—from the agent to the firewall.

The following topics describe how to install and configure the User-ID Agent and how to configure the firewall to retrieve user mapping information from the agent:

- Install the Windows-Based User-ID Agent
- Configure the Windows User-ID Agent for User Mapping

Install the Windows-Based User-ID Agent

The following procedure shows how to install the User-ID agent on a member server in the domain and set up the service account with the required permissions. If you are upgrading, the installer will automatically remove the older version; however, it is a good idea to back up the config.xml file before running the installer.

For information about the system requirements for installing the Windows-based User-ID agent and for information on supported server OS versions, refer to the User-ID agent release notes *and the* Palo Alto Networks Compatibility Matrix.

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STEP 1 | Create a dedicated Active Directory service account for the User-ID agent to access the services and hosts it will monitor to collect user mappings.

Create a Dedicated Service Account for the User-ID Agent and grant the necessary permissions for the Windows User-ID agent.

1. Enable the service account to log on as a service by configuring either local or group policy.
   1. To configure the group policy if you are installing Windows-based User-ID agents on multiple servers, select **Group Policy Management** > **Default Domain Policy** >

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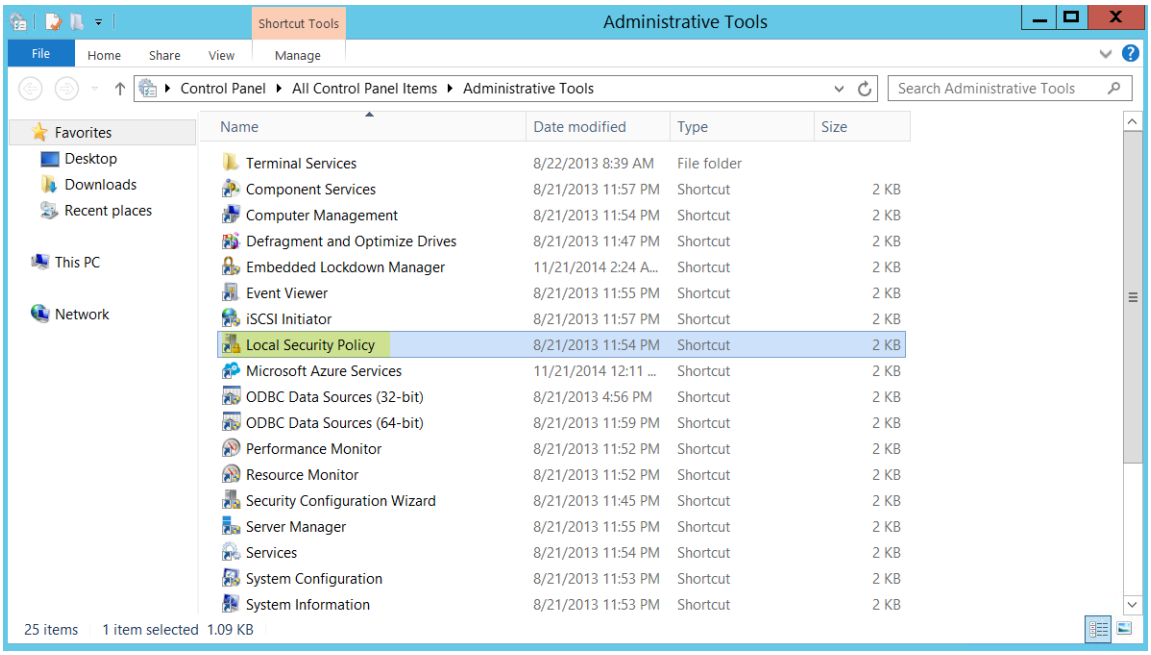
Computer Configuration > **Policies** > **Windows Settings** > **Security Settings** > **Local Policies** > **User Rights Assignment** for the Windows server that is the agent host.

2. Right-click **Log on as a service**, then select **Properties**.
3. Add the service account username or builtin group (Administrators have this privilege by default).

The permission to log on as a service is only needed locally on the Windows server that is the agent host. If you are using only one User-ID agent, you can grant the permissions locally on the agent host using the following instructions.

1. To assign permissions locally, select **Control Panel** > **Administrative Tools** > **Local Security Policy**.



2. Select **Local Policies** > **User Rights Assignment** > **Log on as a service**.

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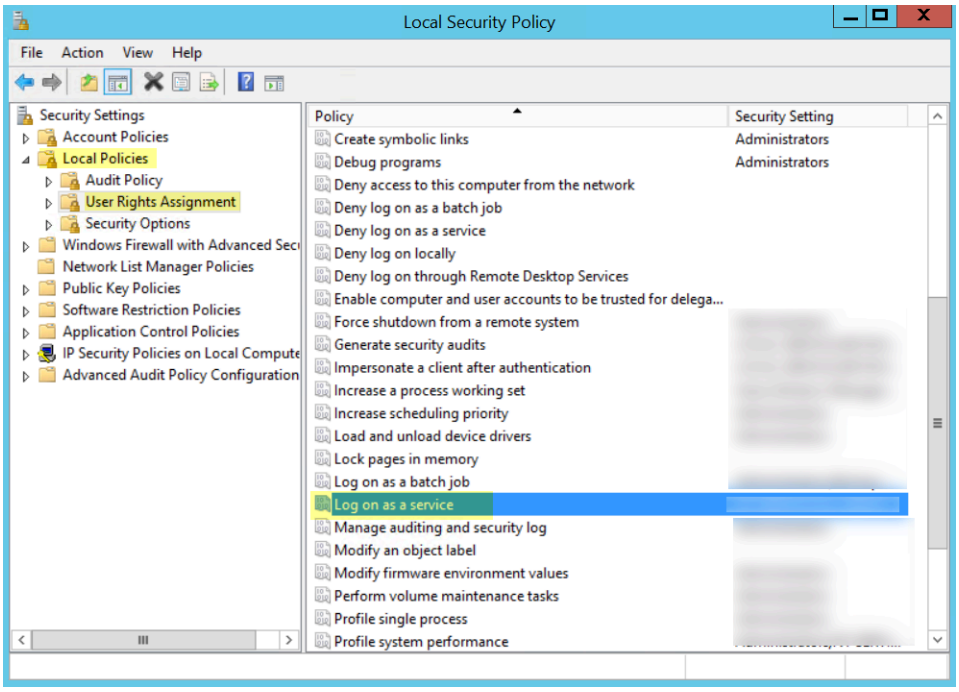
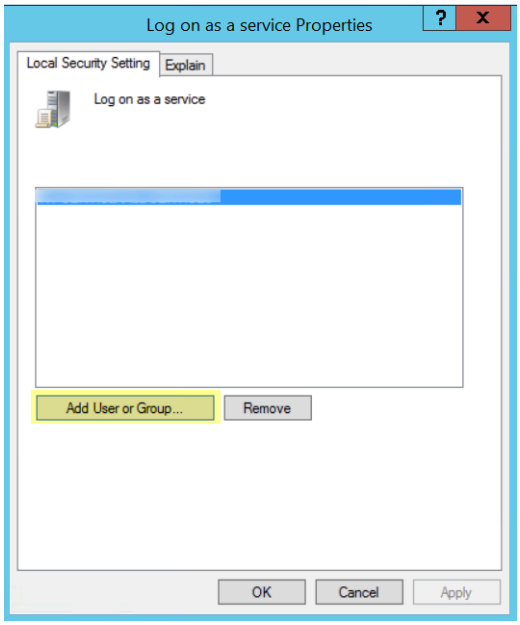


3. **Add User or Group** to add the service account.



4. Enter the service account name in **`domain\username`** format in the **Enter the object names to select** entry field and click **OK**.

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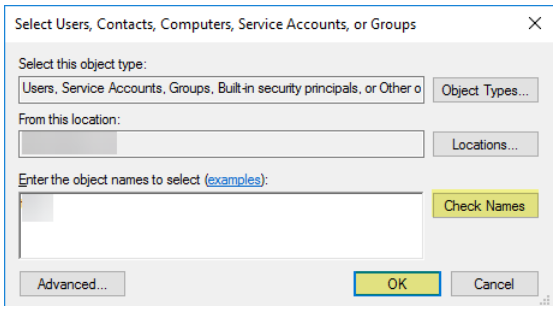

To confirm the service account name is valid, **Check Names**.

2. If you want to use server monitoring to identify users, add the service account to the Event Log Reader builtin group to enable privileges for reading the security log events.

   1. On the domain controller or Exchange server that contains the logs you want the User-ID agent to read, or on the member server that receives events from Windows

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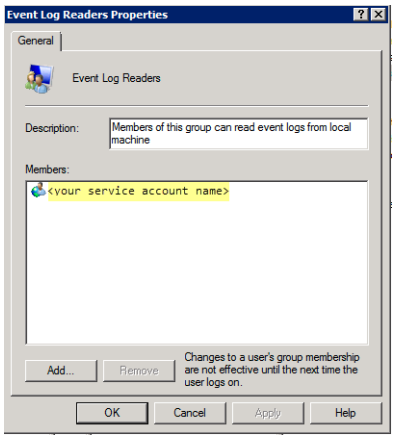
log forwarding, run the MMC and launch the Active Directory Users and Computers snap-in.

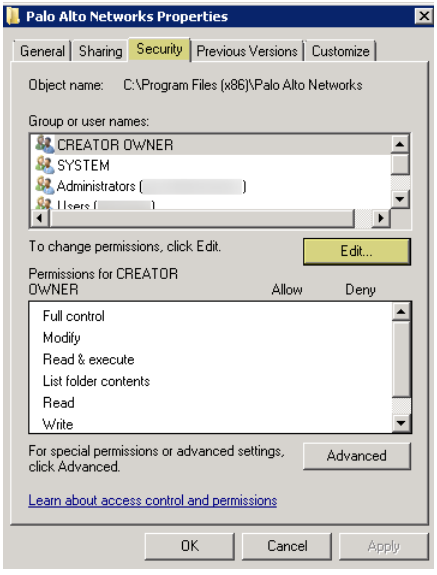
2. Navigate to the Builtin folder for the domain, right-click the **Event Log Reader** group and select **Add to Group** to open the properties dialog.
3. Click **Add** and enter the name of the service account that you configured the User-ID service to use and then click **Check Names** to validate that you have the proper object name.
4. Click **OK** twice to save the settings.
5. Confirm that the builtin Event Log Reader group lists the service account as a member.



3. Assign account permissions to the installation folder to allow the service account to access the agent's installation folder to read the configuration and write logs.

   You only need to perform this step if the service account you configured for the User-ID agent is not either a domain administrator or a local administrator on the User-ID agent server host.

   1. From the Windows Explorer, navigate to **`C:\Program Files(x86)\Palo Alto Networks`** for 32-bit systems, right-click the folder, and select **Properties**.
   2. On the **Security** tab, click **Edit**.



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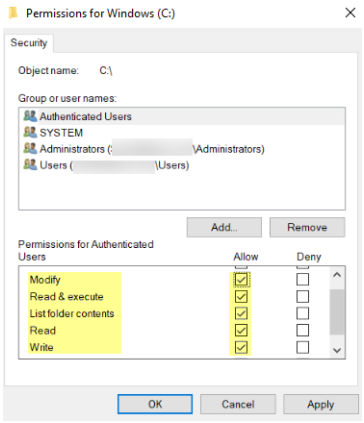
3. **Add** the User-ID agent service account and assign it permissions to **Modify**, **Read & execute**, **List folder contents**, **Read**, and **Write**, and then click **OK** to save the account settings.



*If you want to allow the service account to access the User-ID agent's registry keys, **Allow** the **Full Control** permission.*

4. Give the service account permissions to the User-ID Agent registry sub-tree:
   1. Run **`regedt32`** and navigate to the Palo Alto Networks sub-tree in the following location: `HKEY_LOCAL_MACHINE\Software\Palo Alto Networks`.
   2. Right-click the Palo Alto Networks node and select **Permissions**.
   3. Assign the User-ID service account **Full Control** and then click **OK** to save the setting.

STEP 2 | Decide where to install the User-ID agent.

The User-ID agent queries the Domain Controller and Exchange server logs using Microsoft Remote Procedure Calls (MSRPCs). During the initial connection, the agent transfers the most recent 50,000 events from the log to map users. On each subsequent connection, the agent transfers events with a timestamp later than the last communication with the domain controller. Therefore, always install one or more User-ID agents at each site that has servers to be monitored.

- You must install the User-ID agent on a system running one of the supported OS versions: see "Operating System (OS) Compatibility User-ID Agent" in the Compatibility Matrix. The system must also meet the minimum requirements (see the User-ID agent release notes).
- Make sure the system that will host the User-ID agent is a member of the same domain as the servers it will monitor.
- As a best practice, install the User-ID agent close to the servers it will be monitoring: there is more traffic between the User-ID agent and the monitored servers than there is between the User-ID agent and the firewall, so locating the agent close to the monitored servers optimizes bandwidth usage.
- To ensure the most comprehensive mapping of users, you must monitor all domain controllers that process authentication for users you want to map. You might need to install multiple User-ID agents to efficiently monitor all of your resources.
- If you are using the User-ID agent for credential detection, you must install it on the read-only domain controller (RODC). As a best practice deploy a separate agent for this purpose.

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Do not use the User-ID agent installed on the RODC to map IP addresses to users. The User-ID agent installer for credential detection is named UaCredInstall64-x.x.x.msi.

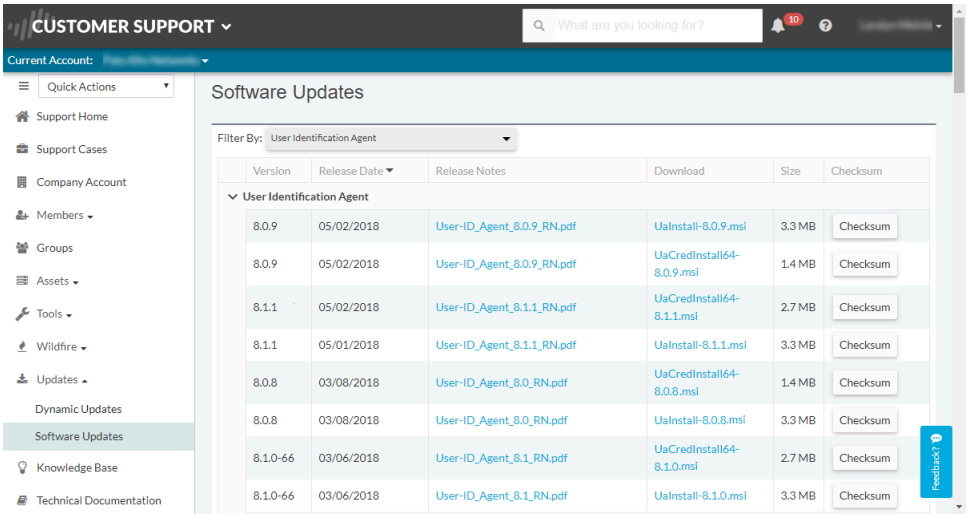
- If you are using the User-ID agent for credential detection, you must install it on a read-only domain controller (RODC). Install one credential agent per domain or forest — each agent must be on a separate RODC connected to its own domain controller. Starting with PAN-OS 12.2, you can connect multiple credential agents to a single firewall to support multi-domain and multi-forest environments. The User-ID agent installer for credential detection is named UaCredInstall64-x.x.x.msi.

STEP 3 | Download the User-ID agent installer.

Install the User-ID agent version that is the same as the PAN-OS version running on the firewalls. If there is not a User-ID agent version that matches the PAN-OS version, install the latest version that is closest to the PAN-OS version.

1. Log in to the Palo Alto Networks Customer Support Portal.
2. Select **Updates** > **Software Updates**.
3. Set **Filter By** to **User Identification Agent** and select the version of the User-ID agent you want to install from the corresponding Download column. The file name uses the following format: `UaInstall-x.x.x.msi` (where `x` represents the version number). For example, to download the 10.0 version of the User-ID agent, select **UaInstall-10.0.0-0.msi**.

   If you are using the User-ID agent to prevent credential phishing, download the `UaCredInstall64-x.x.x.msi` file instead. Only download and install the `UaCredInstall64-x.x.x.msi` if you are using the User-ID for credential detection.
4. Save the file on the systems where you plan to install the agent.



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STEP 4 | Run the installer as an administrator.

1. Open the Windows **Start** menu, right-click the **Command Prompt** program, and select **Run as administrator**.
2. From the command line, run the .msi file you downloaded. For example, if you saved the .msi file to the Desktop, enter the following:

```
C:\Users\administrator.acme>cd Desktop
C:\Users\administrator.acme\Desktop>UaInstall-6.0.0-1.msi
```

3. Follow the setup prompts to install the agent using the default settings. By default, the agent gets installed to **`C:\Program Files(x86)\Palo Alto Networks`**, but you can **Browse** to a different location.
4. When the installation completes, **Close** the setup window.

STEP 5 | Launch the User-ID Agent application as an administrator.

Open the Windows **Start** menu, right-click the **User-ID Agent** program, and select **Run as administrator**.

You must run the User-ID Agent application as an administrator to install the application, commit configuration changes, or uninstall the application.

STEP 6 | (Optional) Change the service account that the User-ID agent uses to log in.

By default, the agent uses the administrator account used to install the .msi file. To change the account to a restricted account:

1. Select **User Identification** > **Setup** and click **Edit**.
2. Select the **Authentication** tab and enter the service account name that you want the User-ID agent to use in the **User name for Active Directory** field.
3. Enter the **Password** for the specified account.
4. **Commit** the changes to the User-ID agent configuration to restart the service using the service account credentials.

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STEP 7 | (Optional) Assign your own certificates for mutual authentication between the Windows User-ID agent and the firewall.

1. Obtain your certificate for the Windows User-ID agent using one of the following methods. Upload the server certificate in Privacy Enhanced Mail (PEM) format and the server certificate's encrypted key.
   - Generate a Certificate and export it for upload to the Windows User-ID agent.
   - Export a certificate from your enterprise certificate authority (CA) and the upload it to the Windows User-ID agent.
2. Add a server certificate to Windows User-ID agent.
   1. On the Windows User-ID agent, select **Server Certificate** and click **Add**.
   2. Enter the path and name of the certificate file received from the CA or browse to the certificate file.
   3. Enter the private key passphrase.
   4. Click **OK** and then **Commit**.
3. Upload a certificate to the firewall to validate the Windows User-ID agent's identity.
4. Configure the certificate profile for the client device (firewall or Panorama).
   1. Select **Device** > **Certificate Management** > **Certificate Profile**.
   2. Configure a Certificate Profile.

   You can only assign one certificate profile for Windows User-ID agents and Terminal Server (TS) agents. Therefore, your certificate profile must include all certificate authorities that issued certificates uploaded to connected User-ID and TS agents.
5. Assign the certificate profile on the firewall.
   1. Select **Device** > **User Identification** > **Connection Security** and click the edit button.
   2. Select the **User-ID Certificate Profile** you configured in the previous step.
   3. Click **OK**.
6. **Commit** your changes.

STEP 8 | Prevent credential phishing.

To use the Windows-based User-ID agent to detect credential submissions and prevent credential phishing, you must install the User-ID Credential Service Add-on on the agent. The add-on can only be installed on a read-only domain controller (RODC). To protect users in multiple domains or forests, install a separate agent on a separate RODC for each domain and connect each agent to the firewall.

Configure the Windows User-ID Agent for User Mapping

The Palo Alto Networks Windows User-ID agent is a Windows service that connects to servers on your network—for example, Active Directory servers, Microsoft Exchange servers, and Novell eDirectory servers—and monitors the logs for login events. The agent uses this information to map IP addresses to usernames. Palo Alto Networks firewalls connect to the User-ID agent to retrieve this user mapping information, enabling visibility into user activity by username rather than IP address and enables user- and group-based security enforcement.

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For information about the server OS versions supported by the User-ID agent, refer to "Operating System (OS) Compatibility User-ID Agent" in the User-ID Agent Release Notes.

STEP 1 | Define the servers the User-ID agent will monitor to collect IP address to user mapping information.

The User-ID agent can monitor up to 100 servers, of which up to 50 can be syslog senders.

To collect all of the required mappings, the User-ID agent must connect to all servers that your users log in to in order to monitor the security log files on all servers that contain login events.

1. Open the Windows **Start** menu and select **User-ID Agent**.
2. Select **User Identification** > **Discovery**.
3. In the **Servers** section of the screen, click **Add**.
4. Enter a **Name** and **Server Address** for the server to be monitored. The network address can be a FQDN or an IP address.
5. Select the **Server Type** (**Microsoft Active Directory**, **Microsoft Exchange**, **Novell eDirectory**, or **Syslog Sender**) and then click **OK** to save the server entry. Repeat this step for each server to be monitored.
6. (Optional) To enable the Windows User-ID agent to automatically discover domain controllers on your network using DNS lookups, click **Auto Discover**. If you have new domain controllers that you want the Windows User-ID agent to discover, click **Auto Discover** each time you want to discover the new domain controllers.

   Auto-discovery locates domain controllers in the local domain only; you must manually add Exchange servers, eDirectory servers, and syslog senders.
7. (Optional) To tune the frequency at which the firewall polls configured servers for mapping information, select **User Identification** > **Setup** and **Edit** the Setup section. On the **Server Monitor** tab, modify the value in the **Server Log Monitor Frequency (seconds)** field. Increase the value in this field to 5 seconds in environments with older Domain Controllers or high-latency links.

   *Ensure that the **Enable Server Session Read** setting is not selected. This setting requires that the User-ID agent have an Active Directory account with Server Operator privileges so that it can read all user sessions. Instead, use a syslog or XML API integration to monitor sources that capture login and logout events for all device types and operating systems (instead of just Windows), such as wireless controllers and Network Access Controllers (NACs).*
8. Click **OK** to save the settings.

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STEP 2 | Specify the subnetworks the Windows User-ID agent should include in or exclude from User-ID.

By default, the User-ID maps all users accessing the servers you are monitoring.

As a best practice, always specify which networks to include and exclude from User-ID to ensure that the agent is only communicating with internal resources and to prevent unauthorized users from being mapped. You should only enable User-ID on the subnetworks where users internal to your organization are logging in.

1. Select **User Identification** > **Discovery**.
2. **Add** an entry to the Include/Exclude list of configured networks and enter a **Name** for the entry and enter the IP address range of the subnetwork in as the **Network Address**.
3. Select whether to include or exclude the network:
   - **Include specified network**—Select this option if you want to limit user mapping to users logged in to the specified subnetwork only. For example, if you include 10.0.0.0/8, the agent maps the users on that subnetwork and excludes all others. If you want the agent to map users in other subnetworks, you must repeat these steps to add additional networks to the list.
   - **Exclude specified network**—Select this option only if you want the agent to exclude a subset of the subnetworks you added for inclusion. For example, if you include 10.0.0.0/8 and exclude 10.2.50.0/22, the agent will map users on all the subnetworks of 10.0.0.0/8 except 10.2.50.0/22, and will exclude all subnetworks outside of 10.0.0.0/8.

     If you add Exclude profiles without adding any Include profiles, the User-ID agent excludes all subnetworks, not just the ones you added.
4. Click **OK**.

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STEP 3 | (Optional) If you configured the agent to connect to a Novell eDirectory server, you must specify how the agent should search the directory.

1. Select **User Identification** > **Setup** and click **Edit** in the Setup section of the window.
2. Select the **eDirectory** tab and then complete the following fields:
   - **Search Base**—The starting point or root context for agent queries, for example: `dc=domain1,dc=example, dc=com`.
   - **Bind Distinguished Name**—The account to use to bind to the directory, for example: `cn=admin,ou=IT, dc=domain1, dc=example, dc=com`.
   - **Bind Password**—The bind account password. The agent saves the encrypted password in the configuration file.
   - **Search Filter**—The search query for user entries (default is `objectClass=Person`).
   - **Server Domain Prefix**—A prefix to uniquely identify the user. This is only required if there are overlapping name spaces, such as different users with the same name from two different directories.
   - **Use SSL**—Select the check box to use SSL for eDirectory binding.
   - **Verify Server Certificate**—Select the check box to verify the eDirectory server certificate when using SSL.

STEP 4 | (Strongly recommended) Disable client probing.

Palo Alto Networks strongly recommends disabling client probing on high-security networks. Client probing can pose a security threat if not correctly configured. For more information, see client probing.

1. On the **Client Probing** tab, deselect the **Enable WMI Probing** check box if it is enabled.

   Palo Alto Network strongly recommends that you collect user mapping information from isolated and trusted sources, such as domain controllers or integrations with Syslog or the XML API, to safely capture user mapping information from any device type or operating system.

   *If you must enable client probing, select the **Enable WMI Probing** check box and on the **Client Probing** tab. Then add a remote administration exception to the Windows firewall for each probed client to ensure the Windows firewall will allow client probing. Each probed client PC must allow port 139 in the Windows firewall and must also have file and printer sharing services enabled.*

STEP 5 | Save the configuration.

Click **OK** to save the User-ID agent setup settings and then click **Commit** to restart the User-ID agent and load the new settings.

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STEP 6 | (Optional) Define the set of users for which you do not need to provide IP address-to-username mappings, such as kiosk accounts.

Save the `ignore-user` list as a text document on the agent host using the title `ignore_user_list` and use the .txt file extension to save it to the User-ID Agent folder on the domain server where the agent is installed.

List the user accounts to ignore; there is no limit to the number of accounts you can add to the list. Each user account name must be on a separate line. For example:

```
SPAdmin
SPInstall
TFSReport
```

You can use an asterisk as a wildcard character to match multiple usernames, but only as the last character in the entry. For example, `corpdomain\it-admin*` would match all administrators in the `corpdomain` domain whose usernames start with the string `it-admin`. You can also use the `ignore-user` list to identify users whom you want to force to authenticate using Authentication Portal.

After adding entries to the Ignore User list, you must stop and restart the connection to the service.

STEP 7 | Configure the firewall to connect to the User-ID agent.

Starting with PAN-OS 12.2, the firewall can connect to multiple Windows-based User-ID agents that are running the User-ID Credential Service Add-on. Deploy one credential agent per domain or forest — each on a separate RODC connected to its own domain controller — and connect each agent to the firewall. The firewall maintains a separate bloom filter per agent, enabling credential phishing detection across multi-domain and multi-forest Active Directory environments. See Configure Credential Detection with the Windows User-ID Agent *for configuration steps.*

Complete the following steps on each firewall you want to connect to the User-ID agent to receive user mappings:

1. Select **Device** > **Data Redistribution** > **Agents** and click **Add**.
2. Enter a **Name** for the agent.
3. **Add an Agent Using** the **Host and Port**.
4. Enter the IP address of the Windows **Host** on which the User-ID Agent is installed.
5. Enter the **Port** number (1-65535) on which the agent will listen for user mapping requests. This value must match the value configured on the User-ID agent. By default, the port is set to 5007 on the firewall and on newer versions of the User-ID agent. However, some older User-ID agent versions use port 2010 as the default.
6. Select **IP User Mappings** as the **Data type**.
7. Make sure that the configuration is **Enabled**, then click **OK**.
8. **Commit** the changes.
9. Verify that the **Connected status** displays as connected (a green light).

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STEP 8 | Verify that the User-ID agent is successfully mapping IP addresses to usernames and that the firewalls can connect to the agent.

1. Launch the User-ID agent and select **User Identification**.
2. Verify that the agent status shows **Agent is running**. If the Agent is not running, click **Start**.
3. To verify that the User-ID agent can connect to monitored servers, make sure the Status for each Server is **Connected**.
4. To verify that the firewalls can connect to the User-ID agent, make sure the Status for each of the Connected Devices is **Connected**.
5. To verify that the User-ID agent is mapping IP addresses to usernames, select **Monitoring** and make sure that the mapping table is populated. You can also **Search** for specific users, or **Delete** user mappings from the list.

Configure User Mapping Using the PAN-OS Integrated User-ID Agent

The following procedure describes how to configure the PAN-OS® integrated User-ID™ agent on the firewall for IP address-to-username mapping. The integrated User-ID agent performs the same tasks as the Windows-based agent.

STEP 1 | Create an Active Directory service account for the User-ID agent to access the services and hosts that the firewall will monitor for collecting user mapping information.

Create a Dedicated Service Account for the User-ID Agent.

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STEP 2 | Define the servers that the firewall will monitor to collect user mapping information.

Within the total maximum of 100 monitored servers per firewall, you can define no more than 50 syslog senders for any single virtual system.

To collect all the required mappings, the firewall must connect to all servers that your users log in to so that the firewall can monitor the Security log files on all servers that contain login events.

1. Select **Device** > **User Identification** > **User Mapping**.
2. **Add** a server (Server Monitoring section).
3. Enter a **Name** to identify the server.
4. Select the **Type** of server.
   - **Microsoft Active Directory**
   - **Microsoft Exchange**
   - **Novell eDirectory**
   - **Syslog Sender**
5. (Microsoft Active Directory and Microsoft Exchange only) Select the **Transport Protocol** you want to use to monitor security logs and session information on the server.
   - **WMI**—The firewall and the monitored servers use Windows Management Instrumentation (WMI) to communicate.
   - **WinRM-HTTP**—The firewall and the monitored servers use Kerberos for mutual authentication and the monitored server encrypts the communication with the firewall using a negotiated Kerberos session key.
   - **WinRM-HTTPS**—The firewall and the monitored servers use HTTPS to communicate and use basic authentication or Kerberos for mutual authentication.

   If you select a Windows Remote Management (WinRM) option, you must Configure Server Monitoring Using WinRM.
6. (Microsoft Active Directory, Microsoft Exchange, and Novell eDirectory only) Enter the **Network Address** of the server.

   *If you are using WinRM with Kerberos, you must enter a fully qualified domain name (FDQN). If you want to use WinRM with basic authentication or use **WMI** to monitor the server, you can enter an IP address or FQDN.*

   *To monitor servers using WMI, specify an IP address, the service account name (if all server monitoring is in the same domain), or a fully qualified domain name (FQDN). If you specify an FQDN, use the down-level logon name in the (DLN)\sAMAccountName format instead of the FQDN\sAMAccountName format. For example, use **`example\user.services`** not **`example.com\user.services`**. If you specify an FQDN, the firewall will attempt to authenticate using Kerberos, which does not support WMI.*
7. (Syslog Sender only) If you select **Syslog Sender** as the server **Type**, Configure the PAN-OS Integrated User-ID Agent as a Syslog Listener.
8. (Novell eDirectory only) Make sure the **Server Profile** you select is **Enabled** and click **OK**.

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9. (Optional) Configure the firewall to automatically **Discover** domain controllers on your network using DNS lookups.

   The auto-discovery feature is for domain controllers only; you must manually add any Exchange servers or eDirectory servers you want to monitor.

STEP 3 | (Optional) Specify the frequency at which the firewall polls Windows servers for mapping information. This is the interval between the end of the last query and the start of the next query.

If the domain controller is processing many requests, delays between queries may exceed the specified value.

1. **Edit** the **Palo Alto Networks User ID Agent Setup**.
2. Select the **Server Monitor** tab and specify the **Server Log Monitor Frequency** in seconds (range is 1 to 3,600; default is 2). In environments with older domain controllers or high-latency links, set this frequency to a minimum of five seconds.

   *Ensure that the **Enable Session** option is not enabled. This option requires that the User-ID agent have an Active Directory account with Server Operator privileges so that it can read all user sessions. Instead, use a Syslog or XML API integration to monitor sources that capture login and logout events for all device types and operating systems (instead of just Windows), such as wireless controllers and network access control (NAC) devices.*
3. Click **OK** to save your changes.

STEP 4 | Specify the subnetworks that the PAN-OS integrated User-ID agent should include in or exclude from user mapping.

By default, the User-ID maps all users accessing the servers you are monitoring.

As a best practice, always specify which networks to include and, optionally, which networks to exclude from User-ID to ensure that the agent is communicating only with internal resources and to prevent unauthorized users from being mapped. You should enable user mapping only on the subnetworks where users internal to your organization are logging in.

1. Select **Device** > **User Identification** > **User Mapping**.
2. **Add** an entry to the **Include/Exclude Networks** and enter a **Name** for the entry. Ensure that the entry is **Enabled**.
3. Enter the **Network Address** and then select whether to include or exclude it:
   - **Include**—Select this option to limit user mapping to only users logged in to the specified subnetwork. For example, if you include 10.0.0.0/8, the agent maps the users on that subnetwork and excludes all others. If you want the agent to map users in other subnetworks, you must repeat these steps to add additional networks to the list.
   - **Exclude**—Select this option to configure the agent to exclude a subset of the subnetworks you added for inclusion. For example, if you include 10.0.0.0/8 and

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exclude 10.2.50.0/22, the agent will map users on all the subnetworks of 10.0.0.0/8 except 10.2.50.0/22 and will exclude all subnetworks outside of 10.0.0.0/8.

If you add Exclude profiles without adding any Include profiles, the User-ID agent excludes all subnetworks, not just the ones you added.

4. Click **OK**.

STEP 5 | Set the domain credentials for the account that the firewall will use to access Windows resources. This is required for monitoring Exchange servers and domain controllers as well as for WMI probing.

1. **Edit** the **Palo Alto Networks User-ID Agent Setup**.
2. Select the **Server Monitor Account** tab and enter the **User Name** and **Password** for the service account that the User-ID agent will use to probe the clients and monitor servers. Enter the username using the **`domain\username`** syntax.
3. If you are using WinRM to monitor servers, configure the firewall to authenticate with the server you are monitoring.
   - If you want to use WinRM with basic authentication, enable WinRM on the server, configure basic authentication, and specify the service account **Domain's DNS Name**.
   - If you want to use WinRM with Kerberos, Configure a Kerberos server profile if you have not already done so and then select the **Kerberos Server Profile**.

STEP 6 | (Optional, not recommended) Configure WMI probing.

Do not enable WMI probing on high-security networks. Client probing can generate a large amount of network traffic and can pose a security threat when misconfigured.

1. On the **Client Probing** tab, **Enable Probing**.
2. (Optional) Specify the **Probe Interval** to define the interval (in minutes) between the end of the last probe request and the start of the next request.

   If necessary, increase the value to ensure the User-ID agent has sufficient time to probe all the learned IP addresses (range is 1 to 1440; default is 20).

   If the request load is high, the observed delay between requests might significantly exceed the specified interval.
3. Click **OK**.
4. Make sure the Windows firewall will allow client probing by adding a remote administration exception to the Windows firewall for each probed client.

STEP 7 | (Optional) Define the set of user accounts that don't require IP address-to-username mappings, such as kiosk accounts.

Define the ignore user list on the firewall that is the User-ID agent, not the client. If you define the ignore user list on the client firewall, the users in the list are still mapped during redistribution.

On the **Ignore User List** tab, **Add** each username you want to exclude from user mapping. You can also use the ignore user list to identify the users you want to force to use Authentication

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Portal to authenticate. You can use an asterisk as a wildcard character to match multiple usernames but only as the last character in the entry. For example, **`corpdomain\it-admin*`** would match all administrators in the `corpdomain` domain whose usernames start with the string `it-admin`. You can add up to 5,000 entries to exclude from user mapping.

STEP 8 | Activate your configuration changes.

Click **OK** and **Commit**.

STEP 9 | Verify the configuration.

1. Access the firewall CLI.
2. Enter the following operational command:

```
> show user server-monitor state all
```

3. On the **Device** > **User Identification** > **User Mapping** tab in the web interface, verify that the Status of each server you configured for server monitoring is `Connected`.

Configure Server Monitoring Using WinRM

You can configure the PAN-OS integrated User-ID agent to monitor servers using Windows Remote Management (WinRM). Using the WinRM protocol improves speed, efficiency, and security when monitoring server events to map user events to IP addresses. The PAN-OS integrated User-ID agent supports the WinRM protocol on Windows Server 2012 Active Directory and Microsoft Exchange Server 2012 or later versions of both.

There are three ways to configure server monitoring using WinRM:

- Configure WinRM over HTTPS with Basic Authentication—The firewall authenticates to the monitored server using the username and password of the service account for the User-ID agent and the firewall authenticates the monitored server using the User-ID certificate profile.
- Configure WinRM over HTTP with Kerberos—The firewall and the monitored servers use Kerberos for mutual authentication and the monitored server encrypts the communication with the firewall using a negotiated Kerberos session key.
- Configure WinRM over HTTPS with Kerberos—The firewall and the monitored server use HTTPS to communicate and use Kerberos for mutual authentication.

Configure WinRM over HTTPS with Basic Authentication

When you configure WinRM to use HTTPS with basic authentication, the firewall transfers the credentials for the service account in a secure tunnel using SSL.

STEP 1 | Configure the service account with Remote Management User and CIMV2 privileges for the server you want to monitor.

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STEP 2 | On the Windows server you are monitoring, obtain the thumbprint from the certificate for the Windows server to use with WinRM and enable WinRM.

> *Ensure that you use an account with administrator privileges to configure WinRM on the server you want to monitor. As a best practice for security, this account should not be the same account as the service account in Step 1.*

1. Verify the certificate is installed in the Local Computer certificate store (**Certificates (Local Computer)** > **Personal** > **Certificates**).

   If you do not see the Local Computer certificate store, launch the Microsoft Management Console (**Start** > **Run** > **MMC**) and add the Certificates snap-in (**File** > **Add/Remove Snap-in** > **Certificates** > **Add** > **Computer account** > **Next** > **Finish**).
2. Open the certificate and select **General** > **Details** > **Show: <All>**.
3. Select the **Thumbprint** and copy it.
4. To enable the firewall to connect to the Windows server using WinRM, enter the following command: **`winrm quickconfig`**.
5. Enter **`y`** to confirm the changes and then confirm the output displays `WinRM service started`.

   If WinRM is enabled, the output displays `WinRM service is already running on this machine.` You will be prompted to confirm any additional required configuration changes.
6. To verify that WinRM is communicating using HTTPS, enter the following command: **`winrm enumerate winrm/config/listener`** and confirm that the output displays `Transport = HTTPS`.

   By default, WinRM/HTTPS uses port 5986.
7. From the Windows server command prompt, enter the following command: **`winrm create winrm/config/Listener?Address=*+Transport=HTTPS @{Hostname="<hostname>";CertificateThumbprint="Certificate Thumbprint"}`**, where *hostname* is the hostname of the Windows server and *Certificate Thumbprint* is the value you copied from the certificate.

   > *Use the command prompt (not Powershell) and remove any spaces in the Certificate Thumbprint to ensure that WinRM can validate the certificate.*
8. From the Windows server command prompt, enter the following command:

   ```
   c:\> winrm set winrm/config/client/auth @{Basic="true"}
   ```
9. Enter the following command: **`winrm get winrm/config/service/Auth`** and confirm that `Basic = true`.

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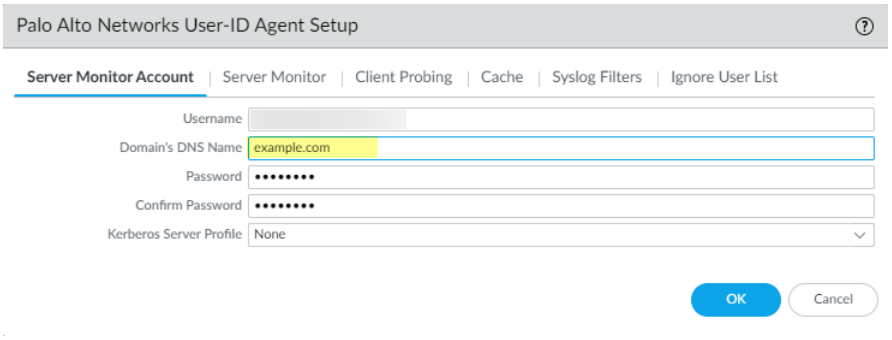
STEP 3 | Enable Basic Authentication between the PAN-OS integrated User-ID agent and the monitored servers.

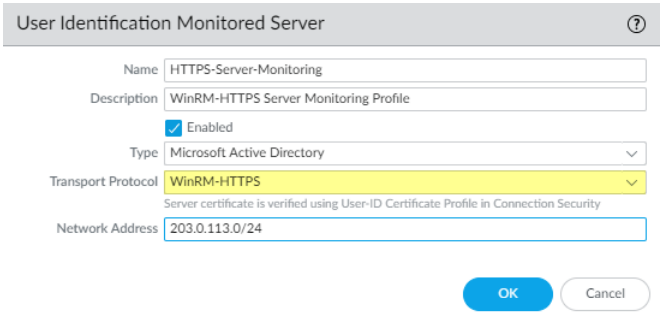
1. Select **Device** > **User Identification** > **User Mapping** > **Palo Alto Networks User-ID Agent Setup** > **Server Monitor Account**.
2. In **`domain\username`** format, enter the **User Name** for the service account that the User-ID agent will use to monitor servers.
3. Enter the **Domain's DNS Name** of the server monitor account.



4. Enter the **Password** and **Confirm Password** for the service account.
5. Click **OK**

STEP 4 | Configure server monitoring for the PAN-OS integrated User-ID agent.

1. Select the Microsoft server **Type** (**Microsoft Active Directory** or **Microsoft Exchange**).
2. Select **Win-RM-HTTPS** as the **Transport Protocol** to use Windows Remote Management (WinRM) over HTTPS to monitor the server security logs and session information.



3. Enter the IP address or FQDN **Network Address** of the server.

STEP 5 | To enable the PAN-OS integrated User-ID agent to communicate with the monitored servers using WinRM-HTTPS, verify that you successfully imported the root certificate for the

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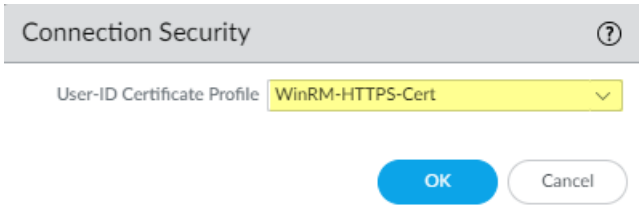
service certificates that the Windows server uses for WinRM on to the firewall and associate the certificate with the User-ID Certificate Profile.

1. Select **Device** > **User Identification** > **Connection Security**.
2. Click **Edit**.
3. Select the Windows server certificate to use for the **User-ID Certificate Profile**.



4. Click **OK**.

STEP 6 | **Commit** your changes.

STEP 7 | Verify that the status of each monitored server is Connected (**Device** > **User Identification** > **User Mapping**).

Configure WinRM over HTTP with Kerberos

When you configure WinRM over HTTP with Kerberos, the firewall and the monitored servers use Kerberos for mutual authentication and the monitored server encrypts the communication with the firewall using a negotiated Kerberos session key.

WinRM with Kerberos supports the aes128-cts-hmac-sha1-96 and aes256-cts-hmac-sha1-96 ciphers. If the server you want to monitor uses RC4, you must download the Windows update and disable RC4 for Kerberos in the registry settings of the server you want to monitor.

STEP 1 | Configure the service account with Remote Management User and CIMV2 privileges for the server you want to monitor.

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STEP 2 | Confirm that WinRM is enabled on the Windows server you are monitoring.

Ensure that you use an account with administrator privileges to configure WinRM on the server you want to monitor. As a best practice for security, this account should not be the same account as the service account in Step 1.

1. To enable the firewall to connect to the Windows server using WinRM, enter the following command: **`winrm quickconfig`**.
2. Enter **`y`** to confirm the changes and then confirm the output displays `WinRM service started`.

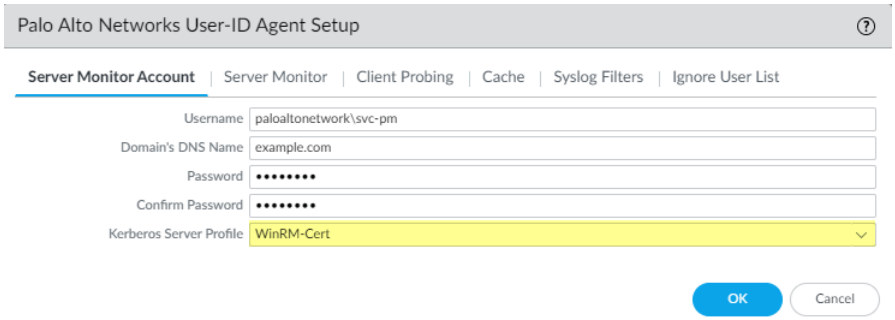
   If WinRM is enabled, the output displays `WinRM service is already running on this machine.` You will be prompted to confirm any additional required configuration changes.
3. To verify that WinRM is communicating using HTTP, enter the following command: **`winrm enumerate winrm/config/listener`** and confirm that the output displays `Transport = HTTP`.

   By default, WinRM/HTTP uses port 5985.
4. Enter the following command: **`winrm get winrm/config/service/Auth`** and confirm that `Kerberos = true`.

STEP 3 | Enable the PAN-OS integrated User-ID agent and the monitored servers to authenticate using Kerberos.

1. If you did not do so during the initial configuration, configure date and time (NTP) settings to ensure successful Kerberos negotiation.
2. Configure a Kerberos server profile on the firewall to authenticate with the server to monitor the security logs and session information.
3. Select **Device** > **User Identification** > **User Mapping** > **Palo Alto Networks User-ID Agent Setup** > **Server Monitor Account**.
4. In **`domain\username`** format, enter the **User Name** for the service account that the User-ID agent will use to monitor servers.
5. Enter the **Domain's DNS Name** of the server monitor account.

   Kerberos uses the domain name to locate the service account.
6. Enter the **Password** and **Confirm Password** for the service account.
7. Select the **Kerberos Server Profile** you configured in Step 3.2.



8. Click **OK**.

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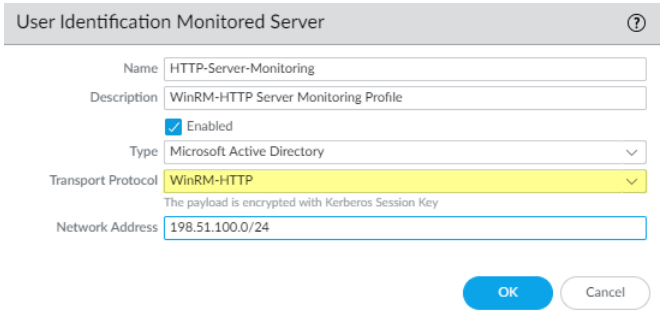
STEP 4 | Configure server monitoring for the PAN-OS integrated User-ID agent.

1. Configure the Microsoft server type (**Microsoft Active Directory** or **Microsoft Exchange**).
2. Select **WinRM-HTTP** as the **Transport Protocol** to use Windows Remote Management (WinRM) over HTTP to monitor the server security logs and session information.



3. Enter the FQDN **Network Address** of the server.

   If you are using Kerberos, the network address must be a fully qualified domain name (FDQN).

STEP 5 | **Commit** your changes.

STEP 6 | Verify that the status of each monitored server is Connected (**Device** > **User Identification** > **User Mapping**).

Configure WinRM over HTTPS with Kerberos

When you configure WinRM over HTTPS with Kerberos, the firewall and the monitored server use HTTPS to communicate and use Kerberos for mutual authentication.

WinRM with Kerberos supports the aes128-cts-hmac-sha1-96 and aes256-cts-hmac-sha1-96 ciphers. If the server you want to monitor uses RC4, you must download the Windows update and disable RC4 for Kerberos in the registry settings of the server you want to monitor.

STEP 1 | Configure the service account with Remote Management User and CIMV2 privileges for the server you want to monitor.

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STEP 2 | On the Windows server you are monitoring, obtain the thumbprint from the certificate for the Windows server to use with WinRM and enable WinRM.

> *Ensure that you use an account with administrator privileges to configure WinRM on the server you want to monitor. As a best practice for security, this account should not be the same account as the service account in Step 1.*

1. Verify the certificate is installed in the Local Computer certificate store (**Certificates (Local Computer)** > **Personal** > **Certificates**).

   If you do not see the Local Computer certificate store, launch the Microsoft Management Console (**Start** > **Run** > **MMC**) and add the Certificates snap-in (**File** > **Add/Remove Snap-in** > **Certificates** > **Add** > **Computer account** > **Next** > **Finish**).
2. Open the certificate and select **General** > **Details** > **Show: <All>**.
3. Select the **Thumbprint** and copy it.
4. To enable the firewall to connect to the Windows server using WinRM, enter the following command: **`winrm quickconfig`**.
5. Enter **`y`** to confirm the changes and then confirm the output displays `WinRM service started`.

   If WinRM is enabled, the output displays `WinRM service is already running on this machine.` You will be prompted to confirm any additional required configuration changes.
6. To verify that WinRM is communicating using HTTPS, enter the following command: **`winrm enumerate winrm/config/listener`**. Then confirm that the output displays `Transport = HTTPS`.

   By default, WinRM/HTTPS uses 5986.
7. From the Windows server command prompt, enter the following command: **`winrm create winrm/config/Listener?Address=*+Transport=HTTPS @{Hostname="<hostname>";CertificateThumbprint="Certificate Thumbprint"}`**, where *hostname* is the hostname of the Windows server and *Certificate Thumbprint* is the value you copied from the certificate.

   > *Use the command prompt (not Powershell) and remove any spaces in the Certificate Thumbprint to ensure that WinRM can validate the certificate.*
8. Enter the following command: **`winrm get winrm/config/service/Auth`** and confirm that `Basic = false` and `Kerberos= true`.

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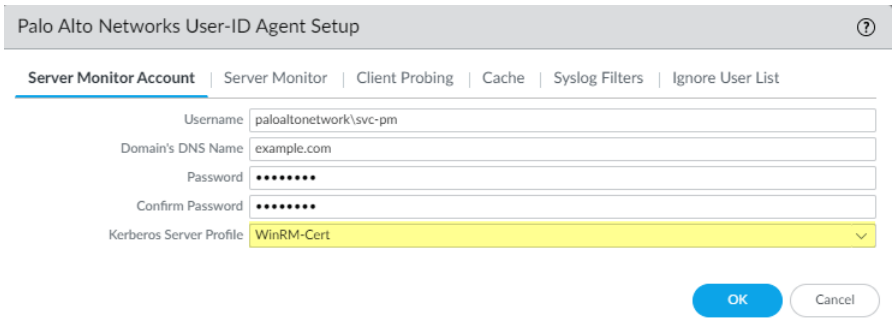
STEP 3 | Enable the PAN-OS integrated User-ID agent and the monitored servers to authenticate using Kerberos.

1. If you did not do so during the initial configuration, configure date and time (NTP) settings to ensure successful Kerberos negotiation.
2. Configure a Kerberos server profile on the firewall to authenticate with the server to monitor the security logs and session information.
3. Select **Device** > **User Identification** > **User Mapping** > **Palo Alto Networks User-ID Agent Setup** > **Server Monitor Account**.
4. In **`domain\username`** format, enter the **User Name** for the service account that the User-ID agent will use to monitor servers.
5. Enter the **Domain's DNS Name** of the server monitor account.

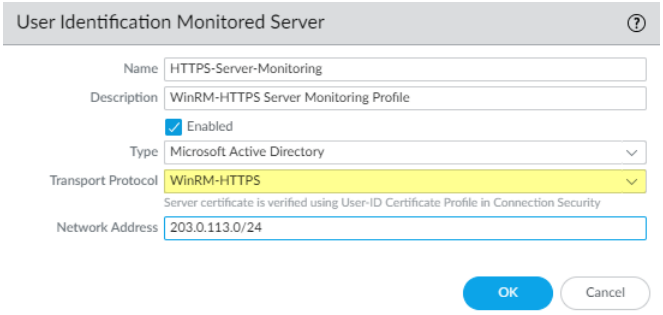
   Kerberos uses the domain name to locate the service account.
6. Enter the **Password** and **Confirm Password** for the service account.
7. Select the **Kerberos Server Profile** you created in Step 3.2.



8. Click **OK**.

STEP 4 | Configure server monitoring for the PAN-OS integrated User-ID agent.

1. Configure the Microsoft server type (**Microsoft Active Directory** or **Microsoft Exchange**).
2. Select **Win-RM-HTTPS** as the **Transport Protocol** to use Windows Remote Management (WinRM) over HTTPS to monitor the server security logs and session information.



3. Enter the FQDN **Network Address** of the server.

   If you are using Kerberos, the network address must be a fully qualified domain name (FDQN).

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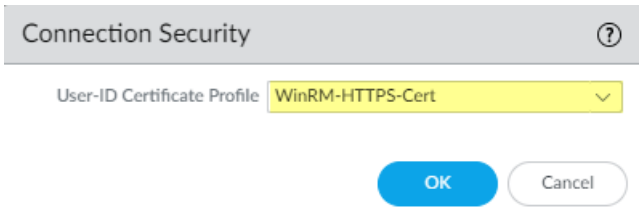
STEP 5 | To enable the PAN-OS integrated User-ID agent to communicate with the monitored servers using WinRM-HTTPS, verify that you successfully imported the root certificate for the service certificates that the Windows server uses for WinRM on to the firewall and associate the certificate with the User-ID Certificate Profile.

The firewall uses the same certificate to authenticate with all monitored servers.

1. Select **Device** > **User Identification** > **Connection Security**.
2. Click **Edit**.
3. Select the Windows server certificate to use for the **User-ID Certificate Profile**.



4. Click **OK**.
5. **Commit** your changes.

STEP 6 | Verify that the status of each monitored server is Connected (**Device** > **User Identification** > **User Mapping**).

Configure User-ID to Monitor Syslog Senders for User Mapping

Obtaining and maintaining updated User-ID mappings from reliable sources is critical to deploying and enforcing a comprehensive Security policy. To obtain the IP address-to-username mappings from your existing network services that authenticate users, you can configure the PAN-OS integrated User-ID agent or Windows-based User-ID agent to parse Syslog messages from those authentication services. To ensure that you keep your user mappings up to date, you can also configure the User-ID agent to parse syslog messages for logout events. This ensures the firewall automatically deletes outdated mappings. Using syslog senders as sources for User-ID mappings allows you even more possibilities for deployment configurations.

To help you deploy your User-ID configuration, there are a number of best practices available. When configuring User-ID to obtain mappings from syslog senders, be sure to follow the best practices for deployment as recommended by Palo Alto Networks. Following these best practices helps to ensure that your deployment is simple, efficient, and successful.

Make sure to allow traffic on the ports used for User-ID to ensure that the firewall can receive the messages from the syslog senders to be able to map the IP addresses to usernames.

For more information, be sure to review the User-ID concepts for syslog information, which provides an example of a deployment that uses syslog messages as a source of User-ID mapping information.

To configure the CN-Series to obtain user mappings from a User-ID syslog sender source, use the dataplane interface. You can't use the management interface to obtain user mappings from a syslog sender source with the CN-Series.

- Configure the PAN-OS Integrated User-ID Agent as a Syslog Listener

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- Configure the Windows User-ID Agent as a Syslog Listener

Configure the PAN-OS Integrated User-ID Agent as a Syslog Listener

To configure the PAN-OS Integrated User-ID agent to create new user mappings and remove outdated mappings through syslog monitoring, start by defining Syslog Parse profiles. The User-ID agent uses the profiles to find login and logout events in syslog messages. In environments where *syslog senders* (the network services that authenticate users) deliver syslog messages in different formats, configure a profile for each syslog format. Syslog messages must meet certain criteria for a User-ID agent to parse them (see Syslog). This procedure uses examples with the following formats:

- **Login events**—`[Tue Jul 5 13:15:04 2016 CDT] Administratorauthentication success User:johndoe1 Source:192.168.3.212`
- **Logout events**—`[Tue Jul 5 13:18:05 2016CDT] User logout successful User:johndoe1 Source:192.168.3.212`

After configuring the Syslog Parse profiles, you specify syslog senders for the User-ID agent to monitor.

STEP 1 | Determine whether there is a predefined Syslog Parse profile for your particular syslog senders.

Palo Alto Networks provides several predefined profiles through Application content updates. The predefined profiles are global to the firewall, whereas custom profiles apply to a single virtual system only.

Any new Syslog Parse profiles in a given content release is documented in the corresponding release note along with the specific regex used to define the filter.

1. Install the latest Applications or Applications and Threats update:
   1. Select **Device** > **Dynamic Updates** and **Check Now**.
   2. **Download** and **Install** any new update.
2. Determine which predefined Syslog Parse profiles are available:
   1. Select **Device** > **User Identification** > **User Mapping** and click **Add** in the Server Monitoring section.
   2. Set the **Type** to **Syslog Sender** and click **Add** in the Filter section. If the Syslog Parse profile you need is available, skip the steps for defining custom profiles.

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STEP 2 | Define custom Syslog Parse profiles to create and delete user mappings.

Each profile filters syslog messages to identify either login events (to create user mappings) or logout events (to delete mappings), but no single profile can do both.

1. Review the syslog messages that the syslog sender generates to identify the syntax for login and logout events. This enables you to define the matching patterns when creating Syslog Parse profiles.

   *While reviewing syslog messages, also determine whether they include the domain name. If they don't, and your user mappings require domain names, enter the **Default Domain Name** when defining the syslog senders that the User-ID agent monitors (later in this procedure).*

2. Select **Device** > **User Identification** > **User Mapping** and edit the Palo Alto Networks User-ID Agent Setup.
3. Select **Syslog Filters** and **Add** a Syslog Parse profile.
4. Enter a name to identify the **Syslog Parse Profile**.
5. Select the **Type** of parsing to find login or logout events in syslog messages:
   - **Regex Identifier**—Regular expressions.
   - **Field Identifier**—Text strings.

   The following steps describe how to configure these parsing types.

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STEP 3 | (**Regex Identifier parsing only**) Define the regex matching patterns.

If the syslog message contains a standalone space or tab as a delimiter, use **\s** *for a space and* **\t** *for a tab.*

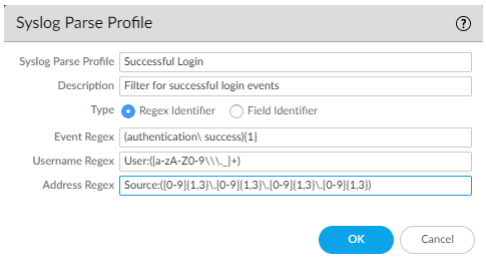
1. Enter the **Event Regex** for the type of events you want to find:
   - **Login events**—For the example message, the regex **`(authentication\ success){1}`** extracts the first **`{1}`** instance of the string `authenticationsuccess`.
   - **Logout events**—For the example message, the regex **`(logout\ successful){1}`** extracts the first **`{1}`** instance of the string `logoutsuccessful`.

   The backslash (\) before the space is a standard regex escape character that instructs the regex engine not to treat the space as a special character.
2. Enter the **Username Regex** to identify the start of the username.

   In the example message, the regex **`User:([a-zA-Z0-9\\\._]+)`** matches the string `User:johndoe1` and identifies `johndoe1` as the username.
3. Enter the **Address Regex** to identify the IP address portion of syslog messages.

   In the example message, the regular expression **`Source:([0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3})`** matches the IPv4 address `Source:192.168.3.212`.

   The following is an example of a completed Syslog Parse profile that uses regex to identify login events:



4. Click **OK** twice to save the profile.

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STEP 4 | (**Field Identifier parsing only**) Define string matching patterns.

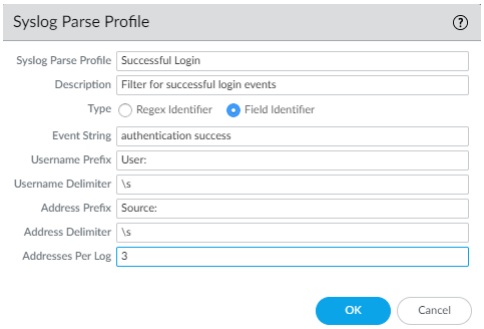
1. Enter an **Event String** to identify the type of events you want to find.
   - **Login events**—For the example message, the string `authentication success` identifies login events.
   - **Logout events**—For the example message, the string `logoutsuccessful` identifies logout events.
2. Enter a **Username Prefix** to identify the start of the username field in syslog messages. The field does not support regex expressions such as \s (for a space) or \t (for a tab).

   In the example messages, `User:` identifies the start of the username field.
3. Enter the **Username Delimiter** that indicates the end of the username field in syslog messages. Use **`\s`** to indicate a standalone space (as in the sample message) and **`\t`** to indicate a tab.
4. Enter an **Address Prefix** to identify the start of the IP address field in syslog messages. The field does not support regex expressions such as \s (for a space) or \t (for a tab).

   In the example messages, `Source:` identifies the start of the address field.
5. Enter the **Address Delimiter** that indicates the end of the IP address field in syslog messages.

   For example, enter **`\n`** to indicate the delimiter is a line break.

   The following is an example of a completed Syslog Parse profile that uses string matching to identify login events:



6. Click **OK** twice to save the profile.

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STEP 5 | Specify the syslog senders that the firewall monitors.

Within the total maximum of 100 monitored servers per firewall, you can define no more than 50 syslog senders for any single virtual system.

The firewall discards any syslog messages received from senders that are not on this list.

1. Select **Device** > **User Identification** > **User Mapping** and **Add** an entry to the Server Monitoring list.
2. Enter a **Name** to identify the sender.
3. Make sure the sender profile is **Enabled** (default is enabled).
4. Set the **Type** to **Syslog Sender**.
5. Enter the **Network Address** (IP address) of the syslog sender.
6. Select **SSL** (default) or **UDP** as the **Connection Type**.

   *To select the TLS certificate that the firewall uses to receive syslog messages, select **Device** > **User Identification** > **User Mapping** > **Palo Alto Networks User-ID Agent Setup**. **Edit** the settings and select **Server Monitor**, then select the **Syslog Service Profile** that contains the TLS certificate you want to the firewall to use to receive syslog messages.*

   The PAN-OS integrated User-ID agent accepts syslogs over SSL and UDP only. However, you must use caution when using UDP to receive syslog messages because it is an unreliable protocol and as such there is no way to verify that a message was sent from a trusted syslog sender. Although you can restrict syslog messages to specific source IP addresses, an attacker can still spoof the IP address, potentially allowing the injection of unauthorized syslog messages into the firewall.

   Always use SSL to listen for syslog messages because the traffic is encrypted (UDP sends the traffic in cleartext). If you must use UDP, make sure that the syslog sender and client are both on a dedicated, secure network to prevent untrusted hosts from sending UDP traffic to the firewall.

   A syslog sender using SSL to connect will show a Status of Connected only when there is an active SSL connection. Syslog senders using UDP will not show a Status value.
7. For each syslog format that the sender supports, **Add** a Syslog Parse profile to the Filter list. Select the **Event Type** that each profile is configured to identify: **login** (default) or **logout**.
8. (Optional) If the syslog messages don't contain domain information and your user mappings require domain names, enter a **Default Domain Name** to append to the mappings.
9. Click **OK** to save the settings.

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STEP 6 | Enable syslog listener services on the interface that the firewall uses to collect user mappings.

1. Select **Network** > **Network Profiles** > **Interface Mgmt** and edit an existing Interface Management profile or **Add** a new profile.
2. Select **User-ID Syslog Listener-SSL** or **User-ID Syslog Listener-UDP** or both, based on the protocols you defined for the syslog senders in the Server Monitoring list.

   The listening ports (514 for UDP and 6514 for SSL) are not configurable; they are enabled through the management service only.
3. Click **OK** to save the interface management profile.

   Even after enabling the User-ID Syslog Listener service on the interface, the interface only accepts syslog connections from senders that have a corresponding entry in the User-ID monitored servers configuration. The firewall discards connections or messages from senders that are not on the list.
4. Assign the Interface Management profile to the interface that the firewall uses to collect user mappings:
   1. Select **Network** > **Interfaces** and edit the interface.
   2. Select **Advanced** > **Other info**, select the Interface **Management Profile** you just added, and click **OK**.
5. **Commit** your changes.

STEP 7 | Verify that the firewall adds and deletes user mappings when users log in and out.

You can use CLI commands to see additional information about syslog senders, syslog messages, and user mappings.

1. Log in to a client system for which a monitored syslog sender generates login and logout event messages.
2. Log in to the firewall CLI.
3. Verify that the firewall mapped the login username to the client IP address:

```
> show user ip-user-mapping ip <ip-address>
IP address:    192.0.2.1 (vsys1)
User:          localdomain\username
From:          SYSLOG
```

4. Log out of the client system.
5. Verify that the firewall deleted the user mapping:

```
> show user ip-user-mapping ip <ip-address>
No matched record
```

Configure the Windows User-ID Agent as a Syslog Listener

To configure the Windows-based User-ID agent to create new user mappings and remove outdated mappings through syslog monitoring, start by defining Syslog Parse profiles. The User-ID

agent uses the profiles to find login and logout events in syslog messages. In environments where *syslog senders* (the network services that authenticate users) deliver syslog messages in different formats, configure a profile for each syslog format. Syslog messages must meet certain criteria for a User-ID agent to parse them (see Syslog). This procedure uses examples with the following formats:

- Login events—`[Tue Jul 5 13:15:04 2016 CDT] Administrator authentication success User:johndoe1 Source:192.168.3.212`
- Logout events—`[Tue Jul 5 13:18:05 2016 CDT] User logout successful User:johndoe1 Source:192.168.3.212`

After configuring the Syslog Parse profiles, you specify the syslog senders that the User-ID agent monitors.

The Windows User-ID agent accepts syslogs over TCP and UDP only. However, you must use caution when using UDP to receive syslog messages because it is an unreliable protocol and as such there is no way to verify that a message was sent from a trusted syslog sender. Although you can restrict syslog messages to specific source IP addresses, an attacker can still spoof the IP address, potentially allowing the injection of unauthorized syslog messages into the firewall. As a best practice, use TCP instead of UDP. In either case, make sure that the syslog sender and client are both on a dedicated, secure VLAN to prevent untrusted hosts from sending syslogs to the User-ID agent.

STEP 1 | Deploy the Windows-based User-ID agents if you haven't already.

1. Install the Windows-Based User-ID Agent.
2. Configure the firewall to connect to the User-ID agent.

STEP 2 | Define custom Syslog Parse profiles to create and delete user mappings.

Each profile filters syslog messages to identify either login events (to create user mappings) or logout events (to delete mappings), but no single profile can do both.

1. Review the syslog messages that the syslog sender generates to identify the syntax for login and logout events. This enables you to define the matching patterns when creating Syslog Parse profiles.

   *While reviewing syslog messages, also determine whether they include the domain name. If they don't, and your user mappings require domain names, enter the **Default Domain Name** when defining the syslog senders that the User-ID agent monitors (later in this procedure).*
2. Open the Windows **Start** menu and select **User-ID Agent**.
3. Select **User Identification** > **Setup** and **Edit** the Setup.
4. Select **Syslog**, **Enable Syslog Service**, and **Add** a Syslog Parse profile.
5. Enter a **Profile Name** and **Description**.
6. Select the **Type** of parsing to find login and logout events in syslog messages:
   - **Regex**—Regular expressions.
   - **Field**—Text strings.

   The following steps describe how to configure these parsing types.

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STEP 3 | (**Regex parsing only**) Define the regex matching patterns.

If the syslog message contains a standalone space or tab as a delimiter, use **`\s`** for a space and **`\t`** for a tab.

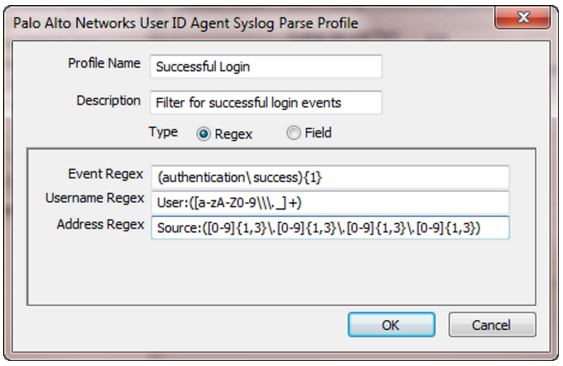
1. Enter the **Event Regex** for the type of events you want to find:
   - **Login events**—For the example message, the regex **`(authentication\ success){1}`** extracts the first **`{1}`** instance of the string `authentication success`.
   - **Logout events**—For the example message, the regex **`(logout\ successful){1}`** extracts the first **`{1}`** instance of the string `logout successful`.

   The backslash before the space is a standard regex escape character that instructs the regex engine not to treat the space as a special character.
2. Enter the **Username Regex** to identify the start of the username.

   In the example message, the regex **`User:([a-zA-Z0-9\\\._]+)`** matches the string `User:johndoe1` and identifies `johndoe1` as the username.
3. Enter the **Address Regex** to identify the IP address portion of syslog messages.

   In the example message, the regular expression **`Source:([0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3})`** matches the IPv4 address `Source:192.168.3.212`.

   The following is an example of a completed Syslog Parse profile that uses regex to identify login events:

4. Click **OK** twice to save the profile.

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STEP 4 | (**Field Identifier parsing only**) Define string matching patterns.

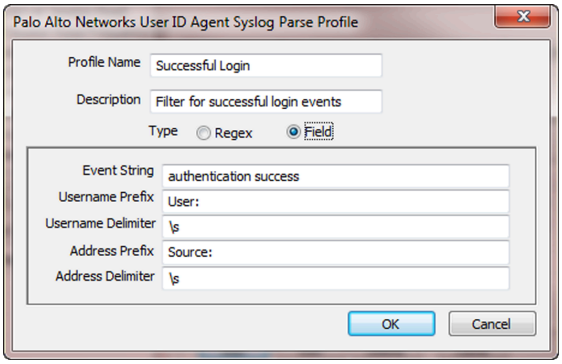
1. Enter an **Event String** to identify the type of events you want to find.
   - **Login events**—For the example message, the string `authentication success` identifies login events.
   - **Logout events**—For the example message, the string `logout successful` identifies logout events.
2. Enter a **Username Prefix** to identify the start of the username field in syslog messages. The field does not support regex expressions such as \s (for a space) or \t (for a tab).

   In the example messages, `User:` identifies the start of the username field.
3. Enter the **Username Delimiter** that indicates the end of the username field in syslog messages. Use **\s** to indicate a standalone space (as in the sample message) and **\t** to indicate a tab.
4. Enter an **Address Prefix** to identify the start of the IP address field in syslog messages. The field does not support regex expressions such as \s (for a space) or \t (for a tab).

   In the example messages, `Source:` identifies the start of the address field.
5. Enter the **Address Delimiter** that indicates the end of the IP address field in syslog messages.

   For example, enter **\n** to indicate the delimiter is a line break.

   The following is an example of a completed Syslog Parse profile that uses string matching to identify login events:



6. Click **OK** twice to save the profile.

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STEP 5 | Specify the syslog senders that the User-ID agent monitors.

Within the total maximum of 100 servers of all types that the User-ID agent can monitor, up to 50 can be syslog senders.

The User-ID agent discards any syslog messages received from senders that are not on this list.

1. Select **User Identification** > **Discovery** and **Add** an entry to the Servers list.
2. Enter a **Name** to identify the sender.
3. Enter the **Server Address** of the syslog sender (IP address or FQDN).
4. Set the **Server Type** to **Syslog Sender**.
5. (**Optional**) If you want to override the current domain in the username of your syslog message or prepend the domain to the username if your syslog message doesn't contain a domain, enter a **Default Domain Name**.
6. For each syslog format that the sender supports, **Add** a Syslog Parse profile to the Filter list. Select the **Event Type** that you configured each profile to identify—**login** (default) or **logout**—and then click **OK**.
7. Click **OK** to save the settings.
8. **Commit** your changes to the User-ID agent configuration.

STEP 6 | Verify that the User-ID agent adds and deletes user mappings when users log in and out.

You can use CLI commands to see additional information about syslog senders, syslog messages, and user mappings.

1. Log in to a client system for which a monitored syslog sender generates login and logout event messages.
2. Verify that the User-ID agent mapped the login username to the client IP address:
   1. In the User-ID agent, select **Monitoring**.
   2. Enter the username or IP address in the filter field, **Search**, and verify that the list displays the mapping.
3. Verify that the firewall received the user mapping from the User-ID agent:
   1. Log in to the firewall CLI.
   2. Run the following command:

```
> show user ip-user-mapping ip <ip-address>
```

If the firewall received the user mapping, the output resembles the following:

```
IP address:    192.0.2.1 (vsys1)
User:          localdomain\username
```

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```
From:          SYSLOG
```

4. Log out of the client system.
5. Verify that the User-ID agent removed the user mapping:
   1. In the User-ID agent, select **Monitoring**.
   2. Enter the username or IP address in the filter field, **Search**, and verify that the list does not display the mapping.
6. Verify that the firewall deleted the user mapping:
   1. Access the firewall CLI.
   2. Run the following command:

      ```
      > show user ip-user-mapping ip <ip-address>
      ```

      If the firewall deleted the user mapping, the output displays:

      ```
      No matched record
      ```

Map IP Addresses to Usernames Using Authentication Portal

When a user initiates web traffic (HTTP or HTTPS) that matches an Authentication Policy rule, the firewall prompts the user to authenticate through Authentication Portal. This ensures that you know exactly who is accessing your most sensitive applications and data. Based on user information collected during authentication, the firewall creates a new IP address-to-username mapping or updates the existing mapping for that user. This method of user mapping is useful in environments where the firewall cannot learn mappings through other methods such as monitoring servers. For example, you might have users who are not logged in to your monitored domain servers, such as users on Linux clients.

- Authentication Portal Authentication Methods
- Authentication Portal Modes
- Configure Authentication Portal

Authentication Portal Authentication Methods

Authentication Portal uses the following methods to authenticate users whose web requests match Authentication Policy rules:

Authentication Method	Description
Kerberos SSO	The firewall uses Kerberos single sign-on (SSO) to transparently obtain user credentials from the browser. To use this method, your network requires a Kerberos infrastructure, including a key distribution center (KDC) with an authentication server and ticket granting service. The firewall must have a Kerberos account.

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Authentication Method	Description
	If Kerberos SSO authentication fails, the firewall falls back to web form or client certificate authentication, depending on your Authentication policy and Authentication Portal configuration.
Web Form	The firewall redirects web requests to a web form for authentication. For this method, you can configure Authentication policy to use Multi-Factor Authentication (MFA), SAML, Kerberos, TACACS+, RADIUS, or LDAP authentication. Although users have to manually enter their login credentials, this method works with all browsers and operating systems.
Client Certificate Authentication	The firewall prompts the browser to present a valid client certificate to authenticate the user. To use this method, you must provision client certificates on each user system and install the trusted certificate authority (CA) certificate used to issue those certificates on the firewall.

Authentication Portal Modes

The Authentication Portal mode defines how the firewall captures web requests for authentication:

Mode	Description
Transparent	The firewall intercepts the browser traffic per the Authentication policy rule and impersonates the original destination URL, issuing an HTTP 401 to invoke authentication. However, because the firewall does not have the real certificate for the destination URL, the browser displays a certificate error to users attempting to access a secure site. Therefore, use this mode only when absolutely necessary, such as in Layer 2 or virtual wire deployments.
Redirect	The firewall intercepts unknown HTTP or HTTPS sessions and redirects them to a Layer 3 interface on the firewall using an HTTP 302 redirect to perform authentication. This is the preferred mode because it provides a better end-user experience (no certificate errors). However, it does require additional Layer 3 configuration. Another benefit of the Redirect mode is that it provides for the use of session cookies, which enable the user to continue browsing to authenticated sites without requiring re-mapping each time the timeouts expire. This is especially useful for users who roam from one IP address to another (for example, from the corporate LAN to the wireless network) because they won't need to re-authenticate when the IP address changes as long as the session stays open.

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Mode	Description
	If you use Kerberos SSO, you must use Redirect mode because the browser will provide credentials only to trusted sites. Redirect mode is also required if you use Multi-Factor Authentication to authenticate Authentication Portal users.

Configure Authentication Portal

Configure the Authentication Portal to verify users whose web requests match an authentication policy rule. Authentication Portal ensures that only authorized users access certain resources. If you configure the PAN-OS® integrated User-ID™ agent, it redirects the web requests to a firewall interface (redirect host) and creates or updates user mappings based on information collected during authentication.

Authentication Portal operates in two modes—redirect and transparent. It supports three authentication methods: web form, Kerberos single sign-on (SSO), and client certificate authentication.

SSL Inbound Inspection *does not support the Authentication Portal redirect mode. To use the Authentication Portal in redirect mode and for decryption, you must use* SSL Forward Proxy.

The applications that users access through Authentication Portal vary in sensitivity and require different authentication methods and settings. To accommodate these requirements, you can use default or custom authentication enforcement objects. Each object associates an authentication policy rule with an authentication profile and authentication method.

- **Default authentication enforcement objects**—Use the default objects to associate multiple authentication policy rules with the same global authentication profile. You must configure an authentication profile before configuring Authentication Portal, and then assign the profile in the Authentication Portal Settings. For rules that require multifactor authentication (MFA), you cannot use default authentication enforcement objects.
- **Custom authentication enforcement objects**—Use a custom object for each authentication policy rule that requires a different authentication profile than the global profile. Custom objects are mandatory for rules that require MFA. To use custom objects, create authentication profiles and assign them to the custom objects when you configure authentication policy.

You need authentication profiles only if users authenticate through an Authentication Portal web form or Kerberos SSO.

If you use Authentication Portal without the other User-ID functions (user mapping and group mapping), you don't need to configure a User-ID agent.

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STEP 1 | Configure the interfaces that the firewall will use for incoming web requests, authenticating users, and communicating with directory servers to map usernames to IP addresses.

When the firewall connects to authentication servers or User-ID agents, it uses the management interface by default. As a best practice, isolate your management network by configuring service routes to connect to the authentication servers or User-ID agents.

1. (MGT interface only) Select **Device** > **Setup** > **Interfaces**, edit the **Management** interface, select **User-ID**, and click **OK**.
2. (Non-MGT interface only) Assign an Interface Management Profile to the Layer 3 interface that the firewall will use for incoming web requests and communication with directory servers. You must enable **Response Pages** and **User-ID** in the Interface Management profile.
3. (Non-MGT interface only) Configure a service route for the interface that the firewall will use to authenticate users. If the firewall has more than one virtual system (vsys), the service route can be global or vsys-specific. The services must include **LDAP** and potentially the following:
   - **Kerberos**, **RADIUS**, **TACACS+**, or **Multi-Factor Authentication**—Configure a service route for any authentication services that you use.
   - **UID Agent**—Configure this service only if you Enable User- and Group-Based Policy.
4. (Redirect mode for IPv4 only) Create a DNS address (A) record that maps the IPv4 address on the Layer 3 interface to the redirect host. If you use Kerberos SSO, you must also add a DNS pointer (PTR) record that performs the same mapping.
5. (Redirect mode for IPv6 only) If you want to create a DNS address (AAAA) record that maps the IPv6 address on the Layer 3 interface to the redirect host, use the CLI commands to configure the FQDN of the redirect host.

   IPv6 is supported for deployments using SAML authentication or LDAP with MFA. Support for these commands is available in PAN-OS version 10.2.9 and 11.2.

   - Enter the `debug user-id cp-redirect-host-v6 value` *<redirect-host-FQDN>* CLI command on the firewall (where *<redirect-host-FQDN>* represents the FQDN of the redirect host that uses IPv6).
   - To view the currently configured IPv6 redirect host, use the `debug user-id cp-redirect-host-v6 show` CLI command on the firewall.
   - To remove the currently configured IPv6 redirect host, use the `debug user-id cp-redirect-host-v6 clear` CLI command on the firewall.

   Depending on whether you configure your redirect host for IPv4, IPv6, or both, make sure to include the necessary IP addresses as DNS attributes in the SAN fields for the certificate or certificates that you configure for the Authentication Portal.

If your network doesn't support access to the directory servers from any firewall interface, you must Configure User Mapping Using the Windows User-ID Agent.

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STEP 2 | Make sure Domain Name System (DNS) is configured to resolve your domain controller addresses.

To verify proper resolution, ping the server FQDN. For example:

```
admin@PA-220> ping host dc1.acme.com
```

STEP 3 | Configure clients to trust Authentication Portal certificates.

Required for redirect mode—to transparently redirect users without displaying certificate errors. You can generate a self-signed certificate or import a certificate that an external certificate authority (CA) signed.

To use a self-signed certificate, create a root CA certificate and use it to sign the certificate you will use for Authentication Portal:

1. Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later).
2. Create a Self-Signed Root CA Certificate or import a CA certificate (see Import a Certificate and Private Key).
3. Generate a certificate to use for Authentication Portal.

   Be sure to configure the following fields:

   - **Common Name**—Enter the DNS name of the intranet host for the Layer 3 interface.
   - **Signed By**—Select the CA certificate you just created or imported.
   - Certificate Attributes—Click **Add**, for the **Type** select **IP** and, for the **Value**, enter the IP address of the Layer 3 interface to which the firewall will redirect requests.
4. Configure an SSL/TLS Service Profile and assign the Authentication Portal certificate to it.

   You apply this profile to the Authentication Portal settings in a later step.

   If you do not assign an SSL/TLS Service Profile, the NGFW negotiates the following TLS versions:

   - *PAN-OS 12.1 and earlier: TLS 1.2 only*
   - *PAN-OS 12.2 and later: TLS 1.2 or TLS 1.3*
5. Configure clients to trust the certificate:

   1. Export the CA certificate you created or imported.
   2. Import the certificate as a trusted root CA into all client browsers, either by manually configuring the browser or by adding the certificate to the trusted roots in an Active Directory (AD) Group Policy Object (GPO).

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STEP 4 | (Optional) Configure Client Certificate Authentication.

You don't need an authentication profile or sequence for client certificate authentication. If you configure both an authentication profile/sequence and certificate authentication, users must authenticate using both.

1. Use a root CA certificate to generate a client certificate for each user who will authenticate through Authentication Portal. The CA in this case is usually your enterprise CA, not the firewall.
2. Export the CA certificate in PEM format to a system that the firewall can access.
3. Import the CA certificate onto the firewall: see Import a Certificate and Private Key. After the import, click the imported certificate, select **Trusted Root CA**, and click **OK**.
4. Configure a Certificate Profile.
   - In the **Username Field** drop-down, select the certificate field that contains the user identity information.
   - In the **CA Certificates** list, click **Add** and select the CA certificate you just imported.

STEP 5 | (Optional) Configure Authentication Portal for the Apple Captive Network Assistant.

This step is only required if you are using Authentication Portal with the Apple Captive Network Assistant (CNA). To use Authentication Portal with CNA, perform the following steps.

1. Verify you have specified an FQDN for the redirect host (not just an IP address).
2. Select an SSL/TLS Service Profile that uses a publicly-signed certificate for the specified FQDN.
3. Enter the following command to adjust the number of requests supported for Authentication Portal: **`set deviceconfig setting ctd cap-portal-ask-requests <threshold-value>`**

   By default, the firewall has a rate limit threshold for Authentication Portal that limits the number of requests to one request every two seconds. The CNA sends multiple requests that can exceed this limit, which can result in a TCP reset and an error from the CNA. The recommended threshold value is 5 (default is one). This value will allow up to 5 requests every two seconds. Based on your environment, you may need to configure a different value. If the current value is not sufficient to handle the number of requests, increase the value.

STEP 6 | Configure the Authentication Portal settings.

1. Select **Device** > **User Identification** > **Authentication Portal Settings** and edit the settings.
2. **Enable Authentication Portal** (default is enabled).
3. Specify the **Timer**, which is the maximum time in minutes that the firewall retains an IP address-to-username mapping for a user after that user authenticates through Authentication Portal (default is 60; range is 1 to 1,440). After the **Timer** expires, the

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firewall removes the mapping and any associated Authentication Timestamps used to evaluate the **Timeout** in authentication policy rules.

*When evaluating the Authentication Portal **Timer** and the **Timeout** value in each authentication policy rule, the firewall prompts the user to re-authenticate for whichever setting expires first. Upon re-authenticating, the firewall resets the time count for the Authentication Portal **Timer** and records new authentication timestamps for the user. Therefore, to enable different **Timeout** periods for different rules, set the Authentication Portal **Timer** to a value the same as or higher than any rule **Timeout**.*

4. Select the SSL/TLS Service Profile you created for redirect requests over TLS.
5. Select the **Mode** (in this example, **Redirect**).
6. (Redirect mode only) Specify the **Redirect Host**, which is the intranet hostname (a hostname with no period in its name) that resolves to the IP address of the Layer 3 interface on the firewall to which web requests are redirected.

   If users authenticate through Kerberos single sign-on (SSO), the **Redirect Host** must be the same as the hostname specified in the Kerberos keytab.
7. Select the fall back authentication method to use:
   - To use client certificate authentication, select the **Certificate Profile** you created.
   - To use global settings for interactive or SSO authentication, select the **Authentication Profile** you configured.
   - To use authentication policy rule-specific settings for interactive or SSO authentication, assign authentication profiles to authentication enforcement objects when you configure authentication policy.
8. Click **OK** and **Commit** the Authentication Portal configuration.

STEP 7 | Next steps...

The firewall does not display the Authentication Portal web form to users until you configure authentication policy rules that trigger authentication when users request services or applications.

Configure User Mapping for Terminal Server Users

Individual terminal server users appear to have the same IP address and therefore an IP address-to-username mapping is not sufficient to identify a specific user. To identify specific users on Windows-based terminal servers, the Palo Alto Networks Terminal Server agent (TS agent) allocates a port range to each user. The TS agent then notifies every connected firewall about the allocated port range, which allows the firewall to create an IP address-port-user mapping table and enable user- and group-based security policy enforcement. For non-Windows terminal servers, configure the PAN-OS XML API to extract user mapping information. The following values apply for both methods:

- Default port range: 1025 to 65534
- Per user block size: 200
- Maximum number of multi-user systems: 2,500

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For information about the terminal servers supported by the TS agent and the number of TS agents supported on each firewall model, refer to the Palo Alto Networks Compatibility Matrix and the Product Comparison Tool.

The following sections describe how to configure user mapping for terminal server users:

- Configure the Palo Alto Networks Terminal Server (TS) Agent for User Mapping
- Retrieve User Mappings from a Terminal Server Using the PAN-OS XML API

Configure the Palo Alto Networks Terminal Server (TS) Agent for User Mapping

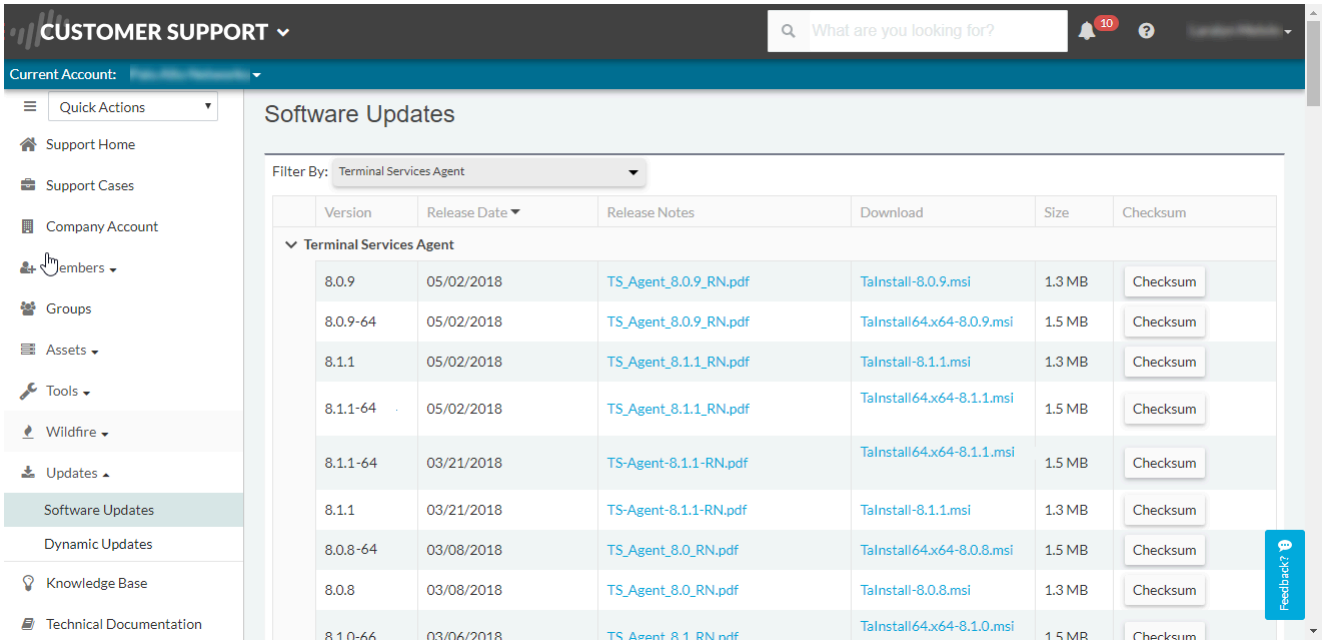
Use the following procedure to install and configure the TS agent on the terminal server. To map all your users, you must install the TS agent on all terminal servers to which your users log in.

If you are using TS agent 7.0 or a later version, disable any Sophos antivirus software on the TS agent host. Otherwise, the antivirus software overwrites the source ports that the TS agent allocates.

For information about default values, ranges, and other specifications, refer to Configure User Mapping for Terminal Server Users. For information about the terminal servers supported by the TS agent and the number of TS agents supported on each firewall model, refer to the Palo Alto Networks Compatibility Matrix.

STEP 1 | Download the TS agent installer.

1. Log in to the Palo Alto Networks Customer Support Portal.
2. Select **Updates** > **Software Updates**.
3. Set **Filter By** to **Terminal Services Agent** and select the version of the agent you want to install from the corresponding Download column. For example, to download TS agent 9.0, select **TaInstall-9.0.msi**.
4. Save the `TaInstall.x64-x.x.x-xx.msi` or `TaInstall-x.x.x-xx.msi` file on the systems where you plan to install the agent; be sure to select the appropriate version based on whether the Windows system is running a 32-bit or a 64-bit OS.



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STEP 2 | Run the installer as an administrator.

1. Open the Windows **Start** menu, right-click the **Command Prompt** program, and **Run as administrator**.
2. From the command line, run the .msi file you downloaded. For example, if you saved the `TaInstall-9.0.msi` file to the Desktop, then enter the following:

```
C:\Users\administrator.acme>cd Desktop
C:\Users\administrator.acme\Desktop>TaInstall-9.0.0-1.msi
```

3. Follow the setup prompts to install the agent using the default settings. The setup installs the agent in `C:\ProgramFiles\Palo Alto Networks\Terminal Server Agent`.

   To ensure correct port allocation, you must use the default Terminal Server agent installation folder location.

4. When the installation completes, **Close** the setup dialog.

   If you are upgrading to a TS agent version that has a newer driver than the existing installation, the installation wizard prompts you to reboot the system after you upgrade.

STEP 3 | Define the range of ports for the TS agent to allocate to end users.

*The **System Source Port Allocation Range** and **System Reserved Source Ports** specify the range of ports that are allocated to non-user sessions. Make sure the values in these fields do not overlap with the ports you designate for user traffic. These values can be changed only by editing the corresponding Windows registry settings. The TS agent does not allocate ports for network traffic emitted by session 0.*

1. Open the Windows **Start** menu and select **Terminal Server Agent** to launch the Terminal Server agent application.
2. **Configure** (side menu) the agent.
3. Enter the **Source Port Allocation Range** (default is 20,000-39,999). This is the full range of port numbers that the TS agent will allocate for user mapping. The port range you specify cannot overlap the **System Source Port Allocation Range**.
4. (Optional) If there are ports or port ranges within the source port allocation that you do not want the TS agent to allocate to user sessions, specify them as **Reserved Source Ports**. To include multiple ranges, use commas with no spaces (for example: **`2000-3000,3500,4000-5000`**).
5. Specify the number of ports to allocate to each individual user upon login to the terminal server (**Port Allocation Start Size Per User**); default is 200.
6. Specify the **Port Allocation Maximum Size Per User**, which is the maximum number of ports the Terminal Server agent can allocate to an individual user.
7. Specify whether to continue processing traffic from the user if the user runs out of allocated ports. The **Fail port binding when available ports are used up** option is enabled by default, which indicates that the application will fail to send traffic when all ports are

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used. To enable users to continue using applications when they run out of ports, disable (clear) this option, but if you do, this traffic may not be identified with User-ID.

8. If the terminal server stops responding when you attempt to shut it down, enable the **Detach agent driver at shutdown** option.

STEP 4 | (Optional) Assign your own certificates for mutual authentication between the TS agent and the firewall.

1. Obtain your certificate for the TS agent from your enterprise PKI or generate one on your firewall. The private key of the server certificate must be encrypted and the certificate must be uploaded in PEM file format. Perform one of the following tasks to upload a certificate:
   - Generate a Certificate and export it.
   - Export a certificate from your enterprise certificate authority (CA).
2. Add a server certificate to the TS agent.
   1. On the TS agent, select **Server Certificate** and **Add** a new certificate.
   2. Enter the path and name of the certificate file received from the CA or browse to the certificate file.
   3. Enter the private key password.
   4. Click **OK**.
   5. **Commit** your changes.

   The TS agent uses a self-signed certificate on port 5009 with following information:Issuer: CN=Terminal Server Agent, OU=Engineering, O=Palo Alto Networks, L=Santa Clara, S=California, C=USSubject: CN=Terminal Server Agent, OU=Engineering, O=Palo Alto Networks, L=Santa Clara, S=California, C=US
3. Configure and assign the certificate profile for the firewall.
   1. Select **Device** > **Certificate Management** > **Certificate Profile** to Configure a Certificate Profile.

      You can assign only one certificate profile for Windows User-ID agents and TS agents. Therefore, your certificate profile must include all certificate authorities that issued certificates uploaded to connected Windows User-ID and TS agents.
   2. Select **Device** > **User Identification** > **Connection Security**.
   3. Edit (⚙) and select the certificate profile you configured in the previous step as the **User-ID Certificate Profile**.
   4. Click **OK**.
   5. **Commit** your changes.

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STEP 5 | Configure the firewall to connect to the Terminal Server agent.

Complete the following steps on each firewall you want to connect to the Terminal Server agent to receive user mappings:

1. Select **Device** > **User Identification** > **Terminal Server Agents** and **Add** a new TS agent.
2. Enter a **Name** for the Terminal Server agent.
3. Enter the hostname or IP address of the Windows **Host** on which the Terminal Server agent is installed.

   The hostname or IP address must resolve to a static IP address. If you change the existing hostname, the TS agent resets when you commit the changes to resolve the new hostname. If the hostname resolves to multiple IP addresses, the TS agent uses the first address in the list.
4. (Optional) Enter the hostname or IP address for any **Alternative IP Addresses** that can appear as the source IP address for the outgoing traffic.

   The hostname or IP address must resolve to a static IP address. You can enter up to 8 IP addresses or hostnames.
5. Enter the **Port** number on which the agent will listen for user mapping requests. This value must match the value configured on the Terminal Server agent. By default, the port is set to 5009 on the firewall and on the agent. If you change it on the firewall, you must also change the **Listening Port** on the Terminal Server agent **Configure** dialog to the same port.
6. Make sure that the configuration is **Enabled** and then click **OK**.
7. **Commit** your changes.
8. Verify that the **Connected** status displays as connected (a green light).

STEP 6 | Verify that the Terminal Server agent is successfully mapping IP addresses to usernames and that the firewalls can connect to the agent.

1. Open the Windows **Start** menu and select **Terminal Server Agent**.
2. Verify that the firewalls can connect by making sure the **Connection Status** of each firewall in the Connection List is `Connected`.
3. Verify that the Terminal Server agent is successfully mapping port ranges to usernames (**Monitor** in the side menu) and confirm that the mapping table is populated.

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STEP 7 | (Windows 2012 R2 servers only) Disable Enhanced Protected Mode in Microsoft Internet Explorer for each user who uses that browser.

This task is not necessary for other browsers, such as Google Chrome or Mozilla Firefox.

To disable Enhanced Protected Mode for all users, use Local Security Policy.

Perform these steps on the Windows Server:

1. Start Internet Explorer.
2. Select **Settings** > **Internet options** > **Advanced** and scroll to the Security section.
3. Disable (clear) the **Enable Enhanced Protected Mode** option.
4. Click **OK**.

In Internet Explorer, Palo Alto Networks recommends that you do not disable Protected Mode, which differs from Enhanced Protected Mode.

Retrieve User Mappings from a Terminal Server Using the PAN-OS XML API

The PAN-OS XML API uses standard HTTP requests to send and receive data. API calls can be made directly from command line utilities such as cURL or using any scripting or application framework that supports RESTful services.

To enable a non-Windows terminal server to send user mapping information directly to the firewall, create scripts that extract the user login and logout events and use them for input to the PAN-OS XML API request format. Then define the mechanisms for submitting the XML API request(s) to the firewall using cURL or wget and providing the firewall's API key for secure communication. Creating user mappings from multi-user systems such as terminal servers requires use of the following API messages:

- **<multiusersystem>**—Sets up the configuration for an XML API Multi-user System on the firewall. This message allows for definition of the terminal server IP address (this will be the source address for all users on that terminal server). In addition, the **<multiusersystem>** setup message specifies the range of source port numbers to allocate for user mapping and the number of ports to allocate to each individual user upon login (called the *block size*). If you want to use the default source port allocation range (1025-65534) and block size (200), you do not need to send a **<multiusersystem>** setup event to the firewall. Instead, the firewall will automatically generate the XML API Multi-user System configuration with the default settings upon receipt of the first user login event message.
- **<blockstart>**—Used with the **<login>** and **<logout>** messages to indicate the starting source port number allocated to the user. The firewall then uses the block size to determine the actual range of port numbers to map to the IP address and username in the login message. For example, if the **<blockstart>** value is 13200 and the block size configured for the multi-user system is 300, the actual source port range allocated to the user is 13200 through 13499. Each connection initiated by the user should use a unique source port number within the allocated range, enabling the firewall to identify the user based on its IP address-port-user mappings for enforcement of user- and group-based security rules. When a user exhausts all the ports allocated, the terminal server must send a new **<login>** message allocating a new port range for the user so that the firewall can update the IP address-port-user mapping. In addition, a single username can have multiple blocks of ports mapped simultaneously.

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When the firewall receives a **<logout>** message that includes a **<blockstart>** parameter, it removes the corresponding IP address-port-user mapping from its mapping table. When the firewall receives a **<logout>** message with a username and IP address, but no **<blockstart>**, it removes the user from its table. And, if the firewall receives a **<logout>** message with an IP address only, it removes the multi-user system and all mappings associated with it.

The XML files that the terminal server sends to the firewall can contain multiple message types and the messages do not need to be in any particular order within the file. However, upon receiving an XML file that contains multiple message types, the firewall will process them in the following order: multiusersystem requests first, followed by logins, then logouts.

The following workflow provides an example of how to use the PAN-OS XML API to send user mappings from a non-Windows terminal server to the firewall.

STEP 1 | Generate the API key that will be used to authenticate the API communication between the firewall and the terminal server. To generate the key you must provide login credentials for an administrative account; the API is available to all administrators (including role-based administrators with XML API privileges enabled).

Any special characters in the password must be URL/ percent-encoded.

From a browser, log in to the firewall. Then, to generate the API key for the firewall, open a new browser window and enter the following URL:

```
https://<Firewall-IPaddress>/api/?
type=keygen&user=<username>&password=<password>
```

Where **<Firewall-IPaddress>** is the IP address or FQDN of the firewall and **<username>** and **<password>** are the credentials for the administrative user account on the firewall. For example:

```
https://10.1.2.5/api/?type=keygen&user=admin&password=admin
```

The firewall responds with a message containing the key, for example:

```
<response status="success">
   <result>
      <key>k7J335J6hI7nBxIqyfa62sZugWx7ot%2BgzEA9UOnlZRg=</key>
   </result>
</response>
```

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STEP 2 | (Optional) Generate a setup message that the terminal server will send to specify the port range and block size of ports per user that your Terminal Server agent uses.

If the Terminal Server agent does not send a setup message, the firewall will automatically create a Terminal Server agent configuration using the following default settings upon receipt of the first login message:

- Default port range: 1025 to 65534
- Per user block size: 200
- Maximum number of multi-user systems: 1,000

The following shows a sample setup message:

```
<uid-message>
  <payload>
<multiusersystem>
<entry ip="10.1.1.23" startport="20000"          endport="39999"
 blocksize="100/">
</multiusersystem>
</payload>
<type>update</type>
<version>1.0</version>
</uid-message>
```

where `entry ip` specifies the IP address assigned to terminal server users, `startport` and `endport` specify the port range to use when assigning ports to individual users, and `blocksize` specifies the number of ports to assign to each user. The maximum blocksize is 4000 and each multi-user system can allocate a maximum of 1000 blocks.

If you define a custom blocksize and or port range, keep in mind that you must configure the values such that every port in the range gets allocated and that there are no gaps or unused ports. For example, if you set the port range to 1000–1499, you could set the block size to 100, but not to 200. This is because if you set it to 200, there would be unused ports at the end of the range.

STEP 3 | Create a script that will extract the login events and create the XML input file to send to the firewall.

Make sure the script enforces assignment of port number ranges at fixed boundaries with no port overlaps. For example, if the port range is 1000–1999 and the block size is 200, acceptable blockstart values would be 1000, 1200, 1400, 1600, or 1800. Blockstart values of 1001, 1300, or 1850 would be unacceptable because some of the port numbers in the range would be left unused.

The login event payload that the terminal server sends to the firewall can contain multiple login events.

The following shows the input file format for a PAN-OS XML login event:

```
<uid-message>
<payload>
<login>
```

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```
<entry name="acme\jjaso" ip="10.1.1.23" blockstart="20000">
<entry name="acme\jparker" ip="10.1.1.23" blockstart="20100">
<entry name="acme\ccrisp" ip="10.1.1.23" blockstart="21000">
</login>
</payload>
<type>update</type>
<version>1.0</version>
</uid-message>
```

The firewall uses this information to populate its user mapping table. Based on the mappings extracted from the example above, if the firewall received a packet with a source address and port of 10.1.1.23:20101, it would map the request to user jparker for policy enforcement.

Each multi-user system can allocate a maximum of 1,000 port blocks.

STEP 4 | Create a script that will extract the logout events and create the XML input file to send to the firewall.

Upon receipt of a `logout` event message with a `blockstart` parameter, the firewall removes the corresponding IP address-port-user mapping. If the `logout` message contains a username and IP address, but no `blockstart` parameter, the firewall removes all mappings for the user. If the `logout` message contains an IP address only, the firewall removes the multi-user system and all associated mappings.

The following shows the input file format for a PAN-OS XML logout event:

```
<uid-message>
<payload>
<logout>
<entry name="acme\jjaso" ip="10.1.1.23" blockstart="20000">
<entry name="acme\ccrisp" ip="10.1.1.23">
<entry ip="10.2.5.4">
</logout>
</payload>
<type>update</type>
<version>1.0</version>
</uid-message>
```

You can also clear the multiuser system entry from the firewall using the following CLI command: ***clear xml-api multiusersystem***

STEP 5 | Make sure that the scripts you create include a way to dynamically enforce that the port block range allocated using the XML API matches the actual source port assigned to the user on the terminal server and that the mapping is removed when the user logs out or the port allocation changes.

One way to do this would be to use netfilter NAT rules to hide user sessions behind the specific port ranges allocated via the XML API based on the uid. For example, to ensure that a

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user with the user ID jjaso is mapped to a source network address translation (SNAT) value of 10.1.1.23:20000-20099, the script you create should include the following:

```
[root@ts1 ~]# iptables -t nat -A POSTROUTING -m owner --uid-owner
 jjaso -p tcp -j SNAT --to-source 10.1.1.23:20000-20099
```

Similarly, the scripts you create should also ensure that the IP table routing configuration dynamically removes the SNAT mapping when the user logs out or the port allocation changes:

```
[root@ts1 ~]# iptables -t nat -D POSTROUTING 1
```

STEP 6 | Define how to package the XML input files containing the setup, login, and logout events into wget or cURL messages for transmission to the firewall.

To apply the files to the firewall using wget:

```
> wget --post file <filename> “https://<Firewall-
IPaddress>/api/?type=user-id&key=<key>&file-
name=<input_filename.xml>&client=wget&vsys=<VSYS_name>”
```

For example, the syntax for sending an input file named login.xml to the firewall at 10.2.5.11 using key k7J335J6hI7nBxIqyfa62sZugWx7ot%2BgzEA9UOnlZRg using wget would look as follows:

```
> wget --post file login.xml “https://10.2.5.11/api/?type=user-
id&key=k7J335J6hI7nBxIqyfa62sZugWx7ot%2BgzEA9UOnlZRg&file-
name=login.xml&client=wget&vsys=vsys1”
```

To apply the file to the firewall using cURL:

```
> curl --form file=@<filename> https://<Firewall-IPaddress>/api/?
type=user-id&key=<key>&vsys=<VSYS_name>
```

For example, the syntax for sending an input file named login.xml to the firewall at 10.2.5.11 using key k7J335J6hI7nBxIqyfa62sZugWx7ot%2BgzEA9UOnlZRg using cURL would look as follows:

```
> curl --form file@login.xml “https://10.2.5.11/api/?type=user-
id&key=k7J335J6hI7nBxIqyfa62sZugWx7ot%2BgzEA9UOnlZRg&vsys=vsys1”
```

STEP 7 | Verify that the firewall is successfully receiving login events from the terminal servers.

Verify the configuration by opening an SSH connection to the firewall and then running the following CLI commands:

To verify if the terminal server is connecting to the firewall over XML:

```
admin@PA-5250> show user xml-api multiusersystem
Host            Vsys    Users   Blocks
---------------------------------------
```

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```
10.5.204.43   vsys1      5          2
```

To verify that the firewall is receiving mappings from a terminal server over XML:

```
admin@PA-5250> show user ip-port-user-mapping all

Global max host index 1, host hash count 1

XML API Multi-user System 10.5.204.43
Vsys 1, Flag 3
Port range: 20000 - 39999
Port size: start 200; max 2000
Block count 100, port count 20000
20000-20199: acme\administrator

Total host: 1
```

Send User Mappings to User-ID Using the XML API

User-ID provides many out-of-the box methods for obtaining user mapping information. However, you might have applications or devices that capture user information but cannot natively integrate with User-ID. For example, you might have a custom, internally developed application or a device that no standard user mapping method supports. In such cases, you can use the PAN-OS XML API to create custom scripts that send the information to the PAN-OS integrated User-ID agent or directly to the firewall. The PAN-OS XML API uses standard HTTP requests to send and receive data. API calls can be made directly from command line utilities such as cURL or using any scripting or application framework that supports POST and GET requests.

To enable an external system to send user mapping information to the PAN-OS integrated User-ID agent, create scripts that extract user login and logout events and use the events as input to the PAN-OS XML API request. Then define the mechanisms for submitting the XML API requests to the firewall (using cURL, for example) and use the API key of the firewall for secure communication. For more details, refer to the PAN-OS XML API Usage Guide.

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Enable User- and Group-Based Policy

After you Enable User-ID, you will be able to configure Security Policy that applies to specific users and groups. User-based policy controls can also include application information (including which category and subcategory it belongs in, its underlying technology, or what the application characteristics are). You can define policy rules to safely enable applications based on users or groups of users, in either outbound or inbound directions.

Examples of user-based policies include:

- Enable only the IT department to use tools such as SSH, telnet, and FTP on standard ports.
- Allow the Help Desk Services group to use Slack.
- Allow all users to read Facebook, but block the use of Facebook apps, and restrict posting to employees in marketing.

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Enable Policy for Users with Multiple Accounts

If a user in your organization has multiple responsibilities, that user might have multiple usernames (accounts), each with distinct privileges for accessing a particular set of services, but with all the usernames sharing the same IP address (the client system of the user). However, the User-ID agent can map any one IP address (or IP address and port range for terminal server users) to only one username for enforcing policy, and you can't predict which username the agent will map. To control access for all the usernames of a user, you must make adjustments to the rules, user groups, and User-ID agent.

For example, say the firewall has a rule that allows username corp_user to access email and a rule that allows username admin_user to access a MySQL server. The user logs in with either username from the same client IP address. If the User-ID agent maps the IP address to corp_user, then whether the user logs in as corp_user or admin_user, the firewall identifies that user as corp_user and allows access to email but not the MySQL server. On the other hand, if the User-ID agent maps the IP address to admin_user, the firewall always identifies the user as admin_user regardless of login and allows access to the MySQL server but not email. The following steps describe how to enforce both rules in this example.

STEP 1 | Configure a user group for each service that requires distinct access privileges.

In this example, each group is for a single service (email or MySQL server). However, it is common to configure each group for a set of services that require the same privileges (for example, one group for all basic user services and one group for all administrative services).

If your organization already has user groups that can access the services that the user requires, simply add the username that is used for less restricted services to those groups. In this example, the email server requires less restricted access than the MySQL server, and corp_user is the username for accessing email. Therefore, you add corp_user to a group that can access email (corp_employees) and to a group that can access the MySQL server (network_services).

If adding a username to a particular existing group would violate your organizational practices, you can create a custom group based on an LDAP filter. For this example, say network_services is a custom group, which you configure as follows:

1. Select **Device** > **User Identification** > **Group Mapping Settings** and **Add** a group mapping configuration with a unique **Name**.
2. Select an LDAP **Server Profile** and ensure the **Enabled** check box is enabled.
3. Select the **Custom Group** tab and **Add** a custom group with network_services as a **Name**.
4. Specify an **LDAP Filter** that matches an LDAP attribute of corp_user and click **OK**.
5. Click **OK** and **Commit**.

Later, if other users that are in the group for less restricted services are given additional usernames that access more restricted services, you can add those usernames to the group for more restricted services. This scenario is more common than the inverse; a user with access to more restricted services usually already has access to less restricted services.

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STEP 2 | Configure the rules that control user access based on the groups you just configured.

For more information, refer to Enable user- and group-based policy enforcement.

1. Configure a security rule that allows the corp_employees group to access email.
2. Configure a security rule that allows the network_services group to access the MySQL server.

STEP 3 | Configure the ignore list of the User-ID agent.

This ensures that the User-ID agent maps the client IP address only to the username that is a member of the groups assigned to the rules you just configured. The ignore list must contain all the usernames of the user that are not members of those groups.

In this example, you add admin_user to the ignore list of the Windows-based User-ID agent to ensure that it maps the client IP address to corp_user. This guarantees that, whether the user logs in as corp_user or admin_user, the firewall identifies the user as corp_user and applies both rules that you configured because corp_user is a member of the groups that the rules reference.

1. Create an `ignore_user_list.txt` file.
2. Open the file and add admin_user.

   If you later add more usernames, each must be on a separate line.
3. Save the file to the User-ID agent folder on the domain server where the agent is installed.

If you use the PAN-OS integrated User-ID agent, see Configure User Mapping Using the PAN-OS Integrated User-ID Agent for instructions on how to configure the ignore list.

STEP 4 | Configure endpoint authentication for the restricted services.

This enables the endpoint to verify the credentials of the user and preserves the ability to enable access for users with multiple usernames.

In this example, you have configured a firewall rule that allows corp_user, as a member of the network_services group, to send a service request to the MySQL server. You must now configure the MySQL server to respond to any unauthorized username (such as corp_user) by prompting the user to enter the login credentials of an authorized username (admin_user).

If the user logs in to the network as admin_user, the user can then access the MySQL server without it prompting for the admin_user credentials again.

In this example, both corp_user and admin_user have email accounts, so the email server won't prompt for additional credentials regardless of which username the user entered when logging in to the network.

The firewall is now ready to enforce rules for a user with multiple usernames.

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Verify the User-ID Configuration

After you configure user and group mapping, enable User-ID in your Security policy, and configure Authentication policy, you should verify that User-ID works properly.

STEP 1 | Access the firewall CLI.

STEP 2 | Verify that group mapping is working.

From the CLI, enter the following operational command:

```
> show user group-mapping statistics
```

STEP 3 | Verify that user mapping is working.

If you are using the PAN-OS integrated User-ID agent, you can verify this from the CLI using the following command:

```
> show user ip-user-mapping-mp all
IP              Vsys  From  User         Timeout (sec)
------------------------------------------------------
192.168.201.1   vsys1 UIA   acme\george            210
192.168.201.11  vsys1 UIA   acme\duane             210
192.168.201.50  vsys1 UIA   acme\betsy             210
192.168.201.10  vsys1 UIA   acme\administrator     210
192.168.201.100 vsys1 AD    acme\administrator     748
Total: 5 users
*: WMI probe succeeded
```

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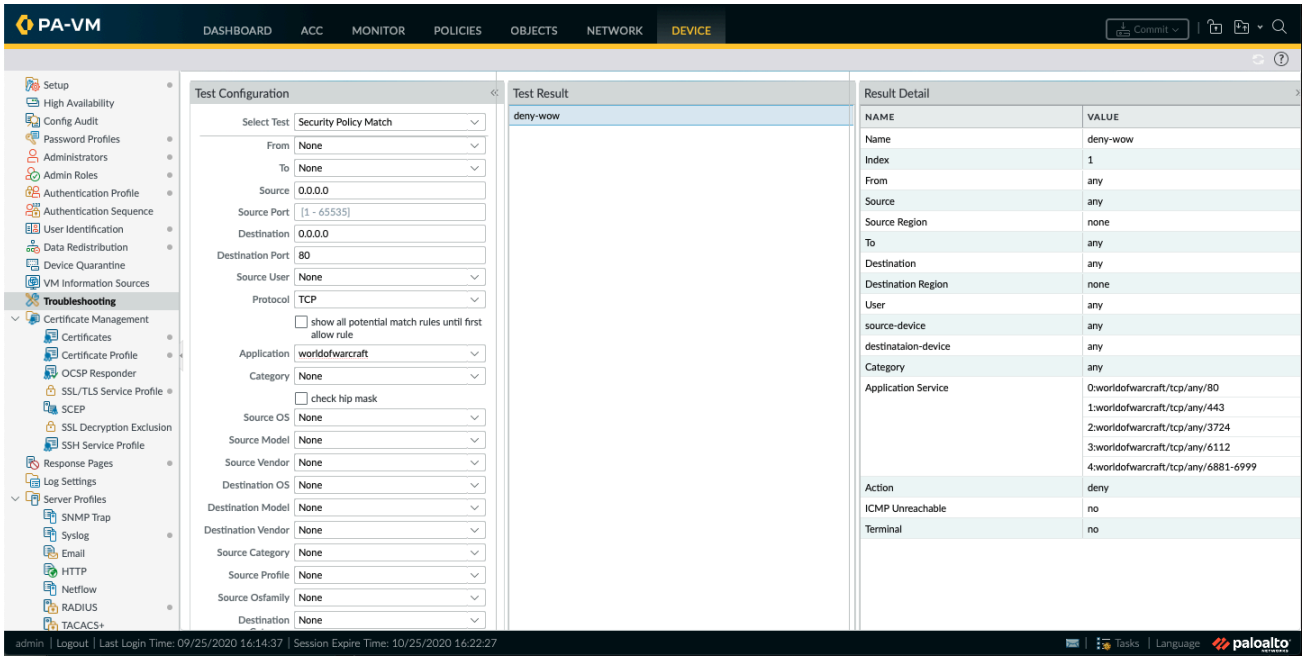
STEP 4 | Test your Security policy rule.

- From a machine in the zone where User-ID is enabled, attempt to access sites and applications to test the rules you defined in your policy and ensure that traffic is allowed and denied as expected.
- You can also troubleshoot the running configuration to determine whether the policy is configured correctly. For example, suppose you have a rule that blocks users from playing World of Warcraft; you could test the policy as follows:
  1. Select **Device** > **Troubleshooting**, and select **Security Policy Match** from the Select Test drop-down.
  2. Enter **`0.0.0.0`** as the Source and Destination IP addresses. This executes the policy match test against any source and destination IP addresses.
  3. Enter the Destination Port.
  4. Enter the Protocol.
  5. **Execute** the security policy match test.



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STEP 5 | Test your Authentication policy and Authentication Portal configuration.

1. From the same zone, go to a machine that is not a member of your directory, such as a Mac OS system, and try to ping to a system external to the zone. The ping should work without requiring authentication.
2. From the same machine, open a browser and navigate to a web site in a destination zone that matches an Authentication rule you defined. The Authentication Portal web form should display and prompt you for login credentials.
3. Log in using the correct credentials and confirm that you are redirected to the requested page.
4. You can also test your Authentication policy using the **`test authentication-policy-match`** operational command as follows:

```
> test authentication-policy-match from corporate to internet
 source 192.168.201.10 destination 8.8.8.8
Matched rule: 'authentication portal' action: web-form
```

STEP 6 | Verify that the log files display usernames.

Select a logs page (such as **Monitor** > **Logs** > **Traffic**) and verify that the Source User column displays usernames.

STEP 7 | Verify that reports display usernames.

1. Select **Monitor** > **Reports**.
2. Select a report type that includes usernames. For example, the Denied Applications report, Source User column, should display a list of the users who attempted to access the applications.

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Deploy User-ID in a Large-Scale Network

A large-scale network can have hundreds of information sources that firewalls query to map IP addresses to usernames and to map usernames to user groups. You can simplify User-ID administration for such a network by aggregating the user mapping and group mapping information before the User-ID agents collect it, thereby reducing the number of required agents.

A large-scale network can also have numerous firewalls that use the mapping information to enforce policies. You can reduce the resources that the firewalls and information sources use in the querying process by configuring some firewalls to acquire mapping information through redistribution instead of direct querying. Redistribution also enables the firewalls to enforce user-based policies when users rely on local sources for authentication (such as regional directory services) but need access to remote services and applications (such as global data center applications).

If you Configure Authentication Policy, your firewalls must also redistribute the Authentication Timestamps associated with user responses to authentication challenges. Firewalls use the timestamps to evaluate the timeouts for Authentication policy rules. The timeouts allow a user who successfully authenticates to later request services and applications without authenticating again within the timeout periods. Redistributing timestamps enables you to enforce consistent timeouts for each user even if the firewall that initially grants a user access is not the same firewall that later controls access for that user.

If you have configured multiple virtual systems, you can share IP address-to-username mapping information across virtual systems by selecting a virtual system as a User-ID hub.

- Deploy User-ID for Numerous Mapping Information Sources
- Redistribute Data and Authentication Timestamps
- Share User-ID Mappings Across Virtual Systems

Deploy User-ID for Numerous Mapping Information Sources

You can use Windows Log Forwarding and Global Catalog servers to simplify user mapping and group mapping in a large-scale network of Microsoft Active Directory (AD) domain controllers or Exchange servers. These methods simplify User-ID administration by aggregating the mapping information before the User-ID agents collect it, thereby reducing the number of required agents.

- Windows Log Forwarding and Global Catalog Servers
- Plan a Large-Scale User-ID Deployment
- Configure Windows Log Forwarding
- Configure User-ID for Numerous Mapping Information Sources

Windows Log Forwarding and Global Catalog Servers

Because each User-ID agent can monitor up to 100 servers, the firewall needs multiple User-ID agents to monitor a network with hundreds of AD domain controllers or Exchange servers. Creating and managing numerous User-ID agents involves considerable administrative overhead, especially in expanding networks where tracking new domain controllers is difficult. Windows Log Forwarding enables you to minimize the administrative overhead by reducing the number of servers to monitor and thereby reducing the number of User-ID agents to manage. When you

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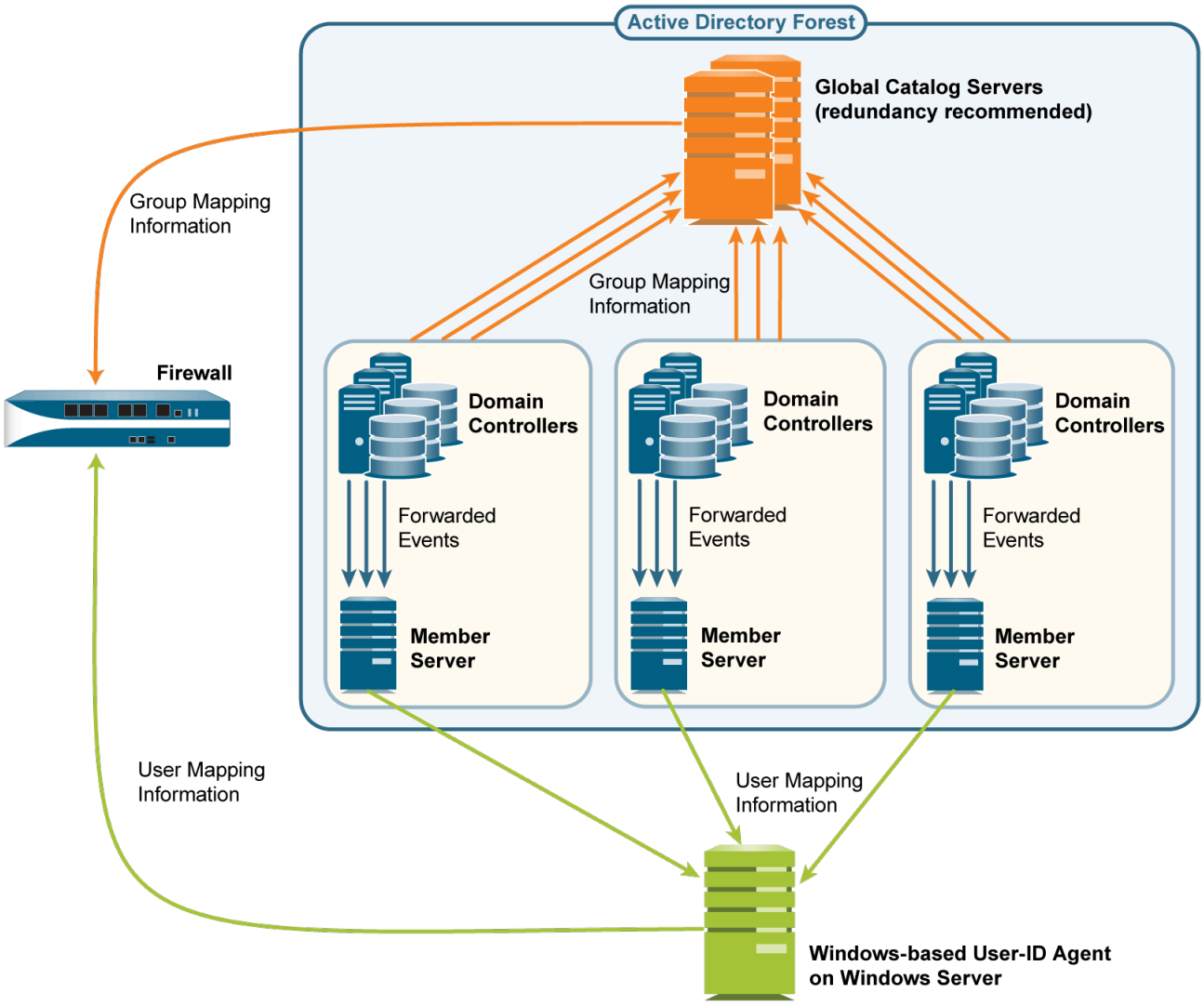
configure Windows Log Forwarding, multiple domain controllers export their login events to a single domain member from which a User-ID agent collects the user mapping information.

You can configure Windows Log Forwarding for Windows Server versions 2012 and 2012 R2. Windows Log Forwarding is not available for non-Microsoft servers.

To collect group mapping information in a large-scale network, you can configure the firewall to query a Global Catalog server that receives account information from the domain controllers.

The following figure illustrates user mapping and group mapping for a large-scale network in which the firewall uses a Windows-based User-ID agent. See Plan a Large-Scale User-ID Deployment to determine if this deployment suits your network.



Plan a Large-Scale User-ID Deployment

When deciding whether to use Windows Log Forwarding and Global Catalog servers for your User-ID implementation, consult your system administrator to determine:

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- ❑ Bandwidth required for domain controllers to forward login events to member servers. The bandwidth is a multiple of the login rate (number of logins per minute) of the domain controllers and the byte size of each login event.

  Domain controllers won't forward their entire security logs, they forward only the events that the user mapping process requires per login: four events for Windows Server 2012 and MS Exchange.
- ❑ Whether the following network elements support the required bandwidth:
  - **Domain controllers**—Must support the processing load associated with forwarding the events.
  - **Member Servers**—Must support the processing load associated with receiving the events.
  - **Connections**—The geographic distribution (local or remote) of the domain controllers, member servers, and Global Catalog servers is a factor. Generally, a remote distribution supports less bandwidth.

Configure Windows Log Forwarding

To configure Windows Log Forwarding, you need administrative privileges for configuring group policies on Windows servers. Configure Windows Log Forwarding on all the *Windows Event Collectors*—the member servers that collect login events from domain controllers. The following is an overview of the tasks; consult your Windows Server documentation for the specific steps.

STEP 1 | On each Windows Event Collector, enable event collection, add the domain controllers as event sources, and configure the event collection query (subscription). The events you specify in the subscription vary by domain controller platform:

- **Windows Server 2012 (including R2) and 2016, or MS Exchange**—The event IDs for the required events are 4768 (Authentication Ticket Granted), 4769 (Service Ticket Granted), 4770 (Ticket Granted Renewed), and 4624 (Logon Success).

*To forward events as quickly as possible, **Minimize Latency** when configuring the subscription.*

User-ID agents monitor the Security log on Windows Event Collectors, not the default forwarded events location. To change the event logging path to the Security log, perform the following steps on each Windows Event Collector.

1. Open the Event Viewer.
2. Right-click the **Security** log and select **Properties**.
3. Copy the **Log path** (default **`%SystemRoot%\System32\Winevt\Logs\security.evtx`**) and click **OK**.
4. Right-click the **Forwarded Events** folder and select **Properties**.
5. Replace the default **Log path** (**`%SystemRoot%\System32\Winevt\Logs\ForwardedEvents.evtx`**) by pasting the value from the **Security** log, and then click **OK**.

STEP 2 | Configure a group policy to enable Windows Remote Management (WinRM) on the domain controllers.

STEP 3 | Configure a group policy to enable Windows Event Forwarding on the domain controllers.

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Configure User-ID for Numerous Mapping Information Sources

STEP 1 | Configure Windows Log Forwarding on the member servers that will collect login events.

Configure Windows Log Forwarding. This step requires administrative privileges for configuring group policies on Windows servers.

STEP 2 | Install the Windows-based User-ID agent.

Install the Windows-Based User-ID Agent on a Windows server that can access the member servers. Make sure the system that will host the User-ID agent is a member of the same domain as the servers it will monitor.

STEP 3 | Configure the User-ID agent to collect user mapping information from the member servers.

1. Start the Windows-based User-ID agent.
2. Select **User Identification** > **Discovery** and perform the following steps for each member server that will receive events from domain controllers:
   1. In the Servers section, click **Add** and enter a **Name** to identify the member server.
   2. In the **Server Address** field, enter the FQDN or IP address of the member server.
   3. For the **Server Type**, select **Microsoft Active Directory**.
   4. Click **OK** to save the server entry.
3. Configure the remaining User-ID agent settings (refer to Configure the Windows-Based User-ID Agent for User Mapping).
4. If the User-ID sources provide usernames in multiple formats, specify the format for the **Primary Username** when you Map Users to Groups.

   The primary username is the username that identifies the user on the firewall and represents the user in reports and logs, regardless of the format that the User-ID source provides.

STEP 4 | Configure an LDAP server profile to specify how the firewall connects to the Global Catalog servers (up to four) for group mapping information.

To improve availability, use at least two Global Catalog servers for redundancy.

You can collect group mapping information only for universal groups, not local domain groups (subdomains).

1. Select **Device** > **Server Profiles** > **LDAP**, click **Add**, and enter a **Name** for the profile.
2. In the Servers section, for each Global Catalog, click **Add** and enter the server **Name**, IP address (**LDAP Server**), and **Port**. For a plaintext or Start Transport Layer Security (Start TLS) connection, use **Port** 3268. For an LDAP over SSL connection, use **Port** 3269. If the connection will use Start TLS or LDAP over SSL, select the **Require SSL/TLS secured connection** check box.
3. In the **Base DN** field, enter the Distinguished Name (DN) of the point in the Global Catalog server where the firewall will start searching for group mapping information (for example, `DC=acbdomain,DC=com`).
4. For the **Type**, select **active-directory**.

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STEP 5 | Configure an LDAP server profile to specify how the firewall connects to the servers (up to four) that contain domain mapping information.

User-ID uses this information to map DNS domain names to NetBIOS domain names. This mapping ensures consistent domain/username references in policy rules.

To improve availability, use at least two servers for redundancy.

The steps are the same as for the LDAP server profile you created for Global Catalogs in the previous step, except for the following fields:

- **LDAP Server**—Enter the IP address of the domain controller that contains the domain mapping information.
- **Port**—For a plaintext or Start TLS connection, use **Port** 389. For an LDAP over SSL connection, use **Port** 636. If the connection will use Start TLS or LDAP over SSL, select the **Require SSL/TLS secured connection** check box.
- **Base DN**—Select the DN of the point in the domain controller where the firewall will start searching for domain mapping information. The value must start with the string: `cn=partitions,cn=configuration` (for example, `cn=partitions,cn=configuration,DC=acbdomain,DC=com`).

STEP 6 | Create a group mapping configuration for each LDAP server profile you created.

1. Select **Device** > **User Identification** > **Group Mapping Settings**.
2. Click **Add** and enter a **Name** to identify the group mapping configuration.
3. Select the LDAP **Server Profile** and ensure the **Enabled** check box is selected.

   *If the Global Catalog and domain mapping servers reference more groups than your security rules require, configure the **Group Include List** and/or **Custom Group** list to limit the groups for which User-ID performs mapping.*
4. Click **OK** and **Commit**.

Insert Username in HTTP Headers

When you configure a secondary enforcement appliance with your Palo Alto Networks firewall to enforce user-based policy, the secondary appliance may not have the IP address-to-username mapping from the firewall. Transmitting user information to downstream appliances may require deployment of additional appliances such as proxies or negatively impact the user's experience (for example, users having to log in multiple times). By sharing the user's identity in the HTTP headers, you can enforce user-based policy without negatively impacting the user's experience or deploying additional infrastructure.

When you configure this feature, apply the URL profile to your Security policy, and commit your changes, the firewall:

1. Populates the user and domain values with the format of the primary username in the group mapping for the source user.
2. Encodes this information using Base64.
3. Adds the Base64-encoded header to the payload.

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4. Routes the traffic to the downstream appliance.

If you want to include the username and domain only when the user accesses specific domains, configure a domain list and the firewall inserts the header only when a domain in the list matches the Host header of the HTTP request.

To share user information with downstream appliances, you must first enable User-ID and configure group mapping.

To include the username and domain in the header, the firewall requires the IP address-to-username mapping for the user. If the user is not mapped, the firewall inserts `unknown` in Base64 encoding for both the domain and username in the header.

To include the username and domain in headers for HTTPS traffic, you must first create a decryption profile to decrypt HTTPS traffic.

This feature supports forward-proxy decryption traffic.

STEP 1 | Create or edit a **URL Filtering Profile**.

*The firewall does not insert headers if the action for the URL filtering profile is **`block`** for the domain.*

STEP 2 | Create or edit an HTTP header insertion entry using predefined types.

You can define up to five headers for each profile.

STEP 3 | Select **Dynamic Fields** as the header **Type**.

STEP 4 | **Add** the **Domains** where you want insert headers. When the user accesses a domain in the list, the firewall inserts the specified header.

STEP 5 | **Add** a new **Header** or select **X-Authenticated-User** to edit it.

STEP 6 | Select a header **Value** format (either **`($domain)\($user)`** or **`WinNT://($domain)/($user)`**) or enter your own format using the **`($domain)`** and **`($user)`** dynamic tokens (for example, **`($user)@($domain)`** for UserPrincipalName).

*Do not use the same dynamic token (either **`($user)`** or **`($domain)`**) more than once per value.*

Each value can be up to 512 characters. The firewall populates the **`($user)`** and **`($domain)`** dynamic tokens using the primary username in the group mapping profile. For example:

- If the primary username is the sAMAccountName, the value for **`($user)`** is the sAMAccountName and the value for **`($domain)`** is the NetBios domain name.
- If the primary username is the UserPrincipalName, the **`($user)`** the user account name (prefix) and the **`($domain)`** is the Domain Name System (DNS) name.

STEP 7 | (Optional) Select **Log** to enable logging for the header insertion.

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STEP 8 | Apply the URL filtering profile to the security policy rule for HTTP or HTTPS traffic.

STEP 9 | Select **OK** twice to confirm the HTTP header configuration.

STEP 10 | **Commit** your changes.

STEP 11 | Verify the firewall includes the username and domain in the HTTP headers.

- Use the **`show user user-ids all`** command to verify the group mapping is correct.
- Use the **`show counter global name ctd_header_insert`** command to view the number of HTTP headers inserted by the firewall.
- If you configured logging in Step 7, check the logs for the inserted Base64 encoded payload (for example, **`corpexample\testuser`** would appear in the logs as **`Y29ycGV4YW1wbGVcdGVzdHVzZXI=`**).

Redistribute Data and Authentication Timestamps

In a large-scale network, instead of configuring all your firewalls to directly query the mapping information sources, you can streamline resource usage by configuring some firewalls to collect mapping information through redistribution.

You can redistribute user mapping information collected through any method except Terminal Server (TS) agents. You cannot redistribute Group Mapping *or* HIP match *information.*

If you use Panorama to manage firewalls and aggregate firewall logs, you can use Panorama to manage User-ID redistribution*. Leveraging Panorama is a simpler solution than creating extra connections between firewalls to redistribute User-ID information.*

If you Configure Authentication Policy, your firewalls must also redistribute the Authentication Timestamps that are generated when users authenticate to access applications and services. Firewalls use the timestamps to evaluate the timeouts for Authentication policy rules. The timeouts allow a user who successfully authenticates to later request services and applications without authenticating again within the timeout periods. Redistributing timestamps enables you to enforce consistent timeouts across all the firewalls in your network.

Firewalls share data and authentication timestamps as part of the same redistribution flow; you don't have to configure redistribution for each information type separately.

- Firewall Deployment for Data Redistribution
- Configure Data Redistribution

Firewall Deployment for Data Redistribution

In a large-scale network, instead of configuring all your firewalls to directly query the data sources, you can streamline resource usage by configuring some firewalls to collect data through redistribution. Data redistribution also provides granularity, allowing you to redistribute only the types of information you specify to only the devices you select. You can also filter the IP user mappings or IP tag mappings using subnets and ranges to ensure the firewalls collect only the mappings they need to enforce policy.

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Data redistribution can be unidirectional (the agent provides data to the client) or bidirectional, where both the agent and the client can simultaneously send and receive data.

To redistribute the data, you can use the following architecture types:

- **Hub and spoke architecture for a single region:**

  To redistribute data between firewalls, use a hub and spoke architecture as a best practice. In this configuration, a hub firewall collects the data from sources such as Windows User-ID agents, Syslog Servers, Domain Controllers, or other firewalls. Configure the redistribution client firewalls to collect the data from the hub firewall.

  For example, a hub (consisting of a pair of VM-50s for resiliency) could connect to the User-ID sources for the user mappings. The hub would then be able to redistribute the user mappings when the client firewalls that use the user mappings to enforce policy connect to the hub to receive data.

- **Multi-Hub and spoke architecture for multiple regions:**

  If you have firewalls deployed in multiple regions and want to distribute the data to the firewalls in all of these regions so that you can enforce policy consistently regardless of where the user logs in, you can use a multi-hub and spoke architecture for multiple regions.

  Start by configuring a firewall in each region to collect data from the sources. This firewall acts as a local hub for redistribution. This firewall collects the data from all sources in that region so that it can redistribute it to the client firewalls. Next, configure the client firewalls to connect to the redistribution hubs for their region and all other regions so that the client firewalls have all data from all hubs.

  As a best practice, enable bidirectional redistribution within a region if the firewalls need to both send and receive data. For example, if a firewall is acting as a GlobalProtect gateway for remote users and as a branch firewall for local users, the firewall must send the user mappings it collects for remote users to the hub firewall as well as receive the user mappings of the local users from the hub firewall.

- **Hierarchical architecture:**

  To redistribute data, you can also use a hierarchical architecture. For example, to redistribute data such as User-ID information, organize the redistribution sequence in layers, where each layer has one or more firewalls. In the bottom layer, PAN-OS integrated User-ID agents running on firewalls and Windows-based User-ID agents running on Windows servers map IP addresses to usernames. Each higher layer has firewalls that receive the mapping information and authentication timestamps from up to 100 redistribution points in the layer beneath it. The top-layer firewalls aggregate the mappings and timestamps from all layers. This deployment provides the option to configure policies for all users in top-layer firewalls and region- or function-specific policies for a subset of users in the corresponding domains served by lower-layer firewalls.

  In this scenario, three layers of firewalls redistribute mappings and timestamps from local offices to regional offices and then to a global data center. The data center firewall that aggregates all the information shares it with other data center firewalls so that they can all enforce policy and generate reports for users across your entire network. Only the bottom layer firewalls use User-ID agents to query the directory servers.

  The information sources that the User-ID agents query do not count towards the maximum of ten *hops* in the sequence. However, Windows-based User-ID agents that forward mapping

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information to firewalls do count. Also in this example, the top layer has two hops: the first to aggregate information in one data center firewall and the second to share the information with other data center firewalls.

Configure Data Redistribution

Before you configure data redistribution:

- ❑ Plan the redistribution architecture. Some factors to consider are:
  - Which firewalls will enforce policies for all data types and which firewalls will enforce region- or function-specific policies for a subset of data?
  - How many hops does the redistribution sequence require to aggregate all data? The maximum allowed number of hops for user mappings is ten and the maximum allowed number of hops for IP address-to-username mappings and IP address-to-tag mappings is one.
  - How can you minimize the number of firewalls that query the user mapping information sources? The fewer the number of querying firewalls, the lower the processing load is on both the firewalls and sources.
- ❑ Configure the data sources from which your redistribution agents obtain the data to redistribute to their clients:
  - user mappings from PAN-OS Integrated User-ID agents or Windows-based User-ID agents
  - IP address-to-tag mappings for dynamic address groups
  - username-to-tag mappings for dynamic user groups
  - GlobalProtect for HIP-based Policy Enforcement
  - data for device quarantine (Panorama only)
- ❑ Configure Authentication Policy.

Data redistribution consists of:

- The redistribution agent that provides information
- The redistribution client that receives information

Perform the following steps on the firewalls in the data redistribution sequence.

STEP 1 | On a redistribution client firewall, configure a firewall, Panorama, or Windows User-ID agent as a data redistribution agent.

1. Select **Device** > **Data Redistribution** > **Agents**.
2. **Add** a redistribution agent and enter a **Name**.
3. Confirm that the agent is **Enabled**.

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STEP 2 | Add the agent using its **Serial Number** or its **Host and Port**.

- To add an agent using a serial number, select the **Serial Number** of the firewall you want to use as a redistribution agent.
- To add an agent using its host and port information:
  1. Enter the information for the **Host**.
  2. Select whether the host is an **LDAP Proxy**.
  3. Enter the **Port** (default is 5007, range is 1—65535).
  4. (Multiple virtual systems only) Enter the **Collector Name** to identify which virtual system you want to use as a redistribution agent.
  5. (Multiple virtual systems only) Enter and confirm the **Collector Pre-Shared Key** for the virtual system you want to use as a redistribution agent.

STEP 3 | Select one or more **Data Type** for the agent to redistribute.

- **IP User Mappings**—IP address-to-username mappings for User-ID.
- **IP Tags**—IP address-to-tag mappings for dynamic address groups.
- **User Tags**—Username-to-tag mappings for dynamic user groups.
- **HIP**—Host information profile (HIP) data from GlobalProtect, which includes HIP objects and profiles.
- **Quarantine List**—Devices that GlobalProtect identifies as quarantined.

STEP 4 | (Multiple virtual systems only) Configure a virtual system as a collector that can redistribute data.

Skip this step if the firewall receives but does not redistribute data.

You can redistribute information among virtual systems on different firewalls or on the same firewall. In both cases, each virtual system counts as one hop in the redistribution sequence.

1. Select **Device** > **Data Redistribution** > **Collector Settings**.
2. Edit the **Data Redistribution Agent Setup**.
3. Enter a **Collector Name** and **Pre-Shared Key** to identify this firewall or virtual system as a User-ID agent.
4. Click **OK** to save your changes.

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STEP 5 | (Optional but recommended) Configure which networks you want to include in data redistribution and which networks you want to exclude from data redistribution.

You can include or exclude networks and subnetworks when redistributing either IP address-to-tag mappings or IP address-to-username mappings.

As a best practice, always specify which networks to include and exclude to ensure that the agent is only communicating with internal resources.

1. Select **Device** > **Data Redistribution** > **Include/Exclude Networks**.
2. **Add** an entry and enter a **Name**.
3. Confirm that the entry is **Enabled**.
4. Select whether you want to **Include** or **Exclude** the entry.
5. Enter the **Network Address** for the entry.
6. Click **OK**.

STEP 6 | Configure the service route that the firewall uses to query other firewalls for User-ID information.

Skip this step if the firewall only receives user mapping information from Windows-based User-ID agents or directly from the information sources (such as directory servers) instead of from other firewalls.

1. Select **Device** > **Setup** > **Services**.
2. (**Firewalls with multiple virtual systems only**) Select **Global** (for a firewall-wide service route) or **Virtual Systems** (for a virtual system-specific service route), and then configure the service route.
3. Click **Service Route Configuration**, select **Customize**, and select **IPv4** or **IPv6** based on your network protocols. Configure the service route for both protocols if your network uses both.
4. Select **UID Agent** and then select the **Source Interface** and **Source Address**.
5. Click **OK** twice to save the service route.

STEP 7 | Enable the firewall to respond when other firewalls query it for data to redistribute.

Skip this step if the firewall receives but does not redistribute data.

Configure an Interface Management Profile with the **User-ID** service enabled and assign the profile to a firewall interface.

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STEP 8 | (Optional but recommended) Use a custom certificate from your enterprise PKI to establish a unique chain of trust from the redistribution client to the redistribution agent.

1. On the redistribution client firewall, create a custom SSL certificate profile to use for outgoing connections.
2. Select **Device** > **Setup** > **Management** > **Secure Communication Settings**.
3. **Edit** the settings.
4. Select the **Customize Secure Server Communication** option.
5. Select the custom **Certificate** you want to use.
6. Select the **Certificate Profile** you created in Substep 1.
7. Click **OK**.
8. **Customize Communication** for **Data Redistribution**.
9. **Commit** your changes.
10. Enter the following CLI command to confirm the certificate profile (`SSL config`) uses `Custom certificates`: **`show redistribution agent state <agent-name>`** (where *<agent-name>* is the name of the redistribution agent or User-ID agent.

STEP 9 | (Optional but recommended) Use a custom certificate from your enterprise PKI to establish a unique chain of trust from the redistribution agent to the redistribution client.

1. On the redistribution agent firewall, create a custom SSL/TLS service profile for the firewall to use for incoming connections.
2. Select **Device** > **Setup** > **Management** > **Secure Communication Settings**.
3. **Edit** the settings.
4. Select the **Customize Secure Server Communication** option.
5. Select the **SSL/TLS Service Profile** you created in Step 1.
6. Click **OK**.
7. **Commit** your changes.
8. Enter the following CLI command to confirm the certificate profile (`SSL config`) uses `Custom certificates`: **`show redistribution service status`**.

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STEP 10 | Verify the agents correctly redistribute data to the clients.

1. View the agent statistics (**Device** > **Data Redistribution** > **Agents**) and select **Status** to view a summary of the activity for the redistribution agent, such as the number of mappings that the client firewall has received.
2. Confirm that the **Connected** status is **yes**.
3. On the agent, access the CLI and enter the following CLI command to check the status of the redistribution: **`show redistribution service status`**.
4. On the agent, enter the following CLI command to view the redistribution clients: **`show redistribution service client all`**.
5. On the client, enter the following CLI command to check the status of the redistribution: **`show redistribution service client all`**.
6. Confirm the **Source Name** in the User-ID logs (**Monitor** > **Logs** > **User-ID**) to verify that the firewall receives the mappings from the redistribution agents.
7. On the client, view the IP-Tag log (**Monitor** > **Logs** > **IP-Tag**) to confirm that the client firewall receives data.
8. On the client, enter the following CLI command and verify that the source the firewall receives the mappings `From` is `REDIST`: **`show user ip-user-mapping all`**.

STEP 11 | (Optional) To troubleshoot data redistribution, enable the traceroute option.

When you enable the traceroute option, the firewall that receives the data appends its IP address to the `<route>` field, which is a list of all firewall IP addresses that the data has traversed. This option requires that all PAN-OS devices in the redistribution route use PAN-OS version 10.0. If a PAN-OS device in the redistribution route uses PAN-OS 9.1.x or earlier versions, the traceroute information terminates at that device.

1. On the redistribution agent where the source originates, enter the following CLI command: **`debug user-id test cp-login traceroute yes ip-address <ip-address> user <username>`** (where ***`<ip-address>`*** is the IP address of the IP address-to-username mapping you want to verify and ***`<username>`*** is the username of the IP address-to-username mapping you want to verify.
2. On a client of the firewall where you configured the traceroute, verify the firewall redistributes the data by entering the following CLI command: **`show user ip-user-mapping all`**.

   The firewall displays the timestamp for the creation of the mapping (`SeqNumber`) and whether the user has GlobalProtect (`GP User`).

```
admin > show user ip-user-mapping-mp ip 192.0.2.0

IP address:  192.0.2.0 (vsys1)
User:    jimdoe
From:   REDIST
Timeout:  889s
Created:  11s ago
Origin:   198.51.100.0
SeqNumber:  15895329682-67831262
GP User:  No
Local HIP:  No
Route Node 0: 198.51.100.0 (vsys1)
```

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```
Route Node 1: 198.51.100.1 (vsys1)
```

Configure Cloud Redistribution of User Context

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ PAN-OS 12.2 or later ❑ Valid device certificate from the Customer Support Portal

As Next-Generation Firewall (NGFW) deployments scale across sites and regions, sharing user context (identity mappings, for example) across NGFWs ensures consistent policy enforcement. Cloud redistribution enables NGFWs to exchange mapping information without requiring direct peer-to-peer connections between them. The User Context Cloud Redistribution Service acts as a central exchange point: NGFWs upload the user context they learn locally and download what they need from other NGFWs through the service.

The user context (data types) supported for redistribution through the User Context Cloud Redistribution Service are as follows:

Data Type	Description
IP Port	Maps an IP address to the specific port range that the Terminal Server agent allocates to a Windows-based terminal server user. This mapping identifies individual users in environments, such as Virtual Desktop Infrastructure (VDI), where multiple users share the same IP address.
IP Tag	Maps an IP address to a network tag, which enables you to enforce security policy rules using Dynamic Address Groups.
IP User	Maps an IP address to a specific username. This information is gathered from various sources, including GlobalProtect, Authentication Portals, Syslog, XML APIs, XFF Headers, and Server Monitoring.
User Tag	Maps a tag to a user, which enables you to enforce security policy rules based on user attributes with Dynamic User Groups.
Quarantine List	Lists devices that GlobalProtect or Cortex XDR flagged as compromised or non-compliant, so you can block them from accessing your network.

Data flows through *segments*, which are logical groupings of NGFWs you define based on region or other use cases. You control which data your NGFW contributes or receives and the segments through which that data flows. Each NGFW can contribute a given data type to only one segment but can receive from multiple segments.

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You can choose one of two operation modes for the cloud service:

- **Legacy**—NGFWs poll the cloud service continuously for user context updates, and you manage the data type and segment configuration through the Cloud Identity Engine (see Cloud Identity Engine User Context.)
- **Advanced Mode**—The cloud service processes data asynchronously and notifies NGFWs when new data is available instead of requiring continuous polling. In this mode, the cloud service reduces NGFW bandwidth and CPU usage and maintains performance as your deployment scales. You configure segments and data types directly on the NGFW or Panorama, with per-data-type control over what each firewall contributes and receives.

  When you enable the NGFW connection to the cloud service, the NGFW initiates contact with the Discovery Service, which returns the FQDN for the regional deployment of the cloud redistribution service. This is based on the region you select for your Cloud Identity Engine instance.

If you have multiple virtual systems (vsys), you can select a vsys as a User-ID hub to share identity mappings between virtual systems. For a given data type, all vsyses must use the same segment ID and the same contributing setting (all enabled or all disabled). To receive data, you must configure a hub vsys: the hub vsys downloads mappings from the cloud and distributes them to all other vsyses. If no hub vsys is configured, the commit fails.

- *All NGFWs in your deployment must use the same operation mode. You cannot mix Legacy and Advanced Mode across NGFWs.*
- *Switching from Legacy to Advanced Mode does not migrate existing identity mappings. After switching, you can restore real-time data by having users re-authenticate or restore persistent data by resubmitting your XML API calls.*
- *Segments do not span multiple virtual systems. Each vsys can independently contribute to one segment and receive from one or more segments.*

STEP 1 | Register your NGFW on the Palo Alto Networks Customer Support Portal (CSP).

STEP 2 | Retrieve a device certificate from the CSP.

The cloud redistribution service uses this certificate to authenticate with the cloud.

STEP 3 | Associate your NGFW with a tenant service group (TSG).

STEP 4 | Activate the Cloud Identity Engine for the TSG.

The region you select for your Cloud Identity Engine instance determines which regional deployment of the User Context Cloud Redistribution Service your NGFW connects to.

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STEP 5 | Enable the NGFW connection to the User Context Cloud Redistribution Service.

If NGFW traffic uses a management interface, create security policy rules to allow connectivity between the NGFW and the cloud service.

1. Select **Device** > **Setup** > **Management** and **Edit** the PAN-OS Edge Service Settings.
2. **Enable User Context Cloud Redistribution Service**.
3. For **Operation Mode**, choose **Advanced Mode** or **Legacy**.

   Legacy mode requires additional configuration as described in Cloud Identity Engine User Context.
4. Leave the **Discovery Service Address** as is.

   The Discovery Service Address is pre-configured with the default endpoint and is not expected to change under normal circumstances.
5. (Optional) Enter a **Discovery Service Refresh Interval (Hrs)** (range is 1 to 240; default is 4).

   This interval controls how often the NGFW contacts the Discovery Service to refresh the regional endpoint FQDN for each data type.
6. Click **OK**.

STEP 6 | Configure the data type your NGFW contributes and the segment to which it contributes that data.

1. Select **Device** > **Cloud Redistribution** > **Contributing** and **Add** an entry.
2. Choose a **Data Type** to contribute:
   - **IP Port**
   - **IP Tag**
   - **IP User**
   - **Quarantine List**
   - **User Tag**
3. (Optional) Enter a **Description** for the entry.
4. **Enable Cloud Redistribution** to activate the entry.
5. **Enable Contributing** to upload this data type from the NGFW to the cloud service.
6. **Add** a descriptive segment **Name**.

   Use a name that identifies the segment's purpose, such as its region.
7. **Add** a segment **ID**.
8. Click **OK**.
9. (Optional) Add additional entries for other data types and segments as needed.

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STEP 7 | Configure the data types your NGFW receives and the segments it receives that data from.

1. Select **Device** > **Cloud Redistribution** > **Receiving** and **Add** an entry.
2. Choose a **Data Type** to receive:
   - **IP Port**
   - **IP Tag**
   - **IP User**
   - **User Tag**
   - **Quarantine List**
3. (Optional) Enter a **Description** for the entry.
4. **Enable Cloud Redistribution** to activate the entry.
5. **Enable Receiving** to download this data type from the cloud.
6. **Add** a descriptive segment **Name**.

   Use a name that identifies the segment's purpose, such as its region.
7. **Add** a segment **ID**.
8. Click **OK**.
9. (Optional) Add additional entries for other data types and segments as needed.

STEP 8 | **Commit** your changes.

STEP 9 | Verify the NGFW connection to the User Context Cloud Redistribution Service.

Select **Device** > **Setup** > **Management**. In the PAN-OS Edge Services section, confirm that **Connection Status** for this service is `connected` (green dot).

Share User-ID Mappings Across Virtual Systems

To simplify User-ID™ source configuration when you have multiple virtual systems, configure the User-ID sources on a single virtual system to share IP address-to-username mappings and username-to-group mappings with all other virtual systems on the firewall.

Configuring a single virtual system as a *User-ID hub* simplifies user mapping by eliminating the need to configure the sources on multiple virtual systems, especially if traffic will pass through multiple virtual systems based on the resources the user is trying to access (for example, in an academic networking environment where a student will be accessing different departments whose traffic is managed by different virtual systems).

To map the user or group, the firewall uses the mapping table on the local virtual system and applies the policy for that user or group. If the firewall does not find the mapping for a user or group on the virtual system where that user's traffic originated, the firewall queries the hub to fetch the IP address-to-username information for that user or group mapping information for that group. If the firewall locates the mapping on both the User-ID hub and the local virtual system, the firewall uses the mapping it learns locally. If the mapping on the local firewall differs from the mapping on the virtual system hub, the firewall uses the local mapping.

After you configure the User-ID hub, the virtual system can use the mapping table on the User-ID hub when it needs to identify a user for user-based policy enforcement or to display the username in a log or report but the source is not available locally. When you select a hub, the firewall retains

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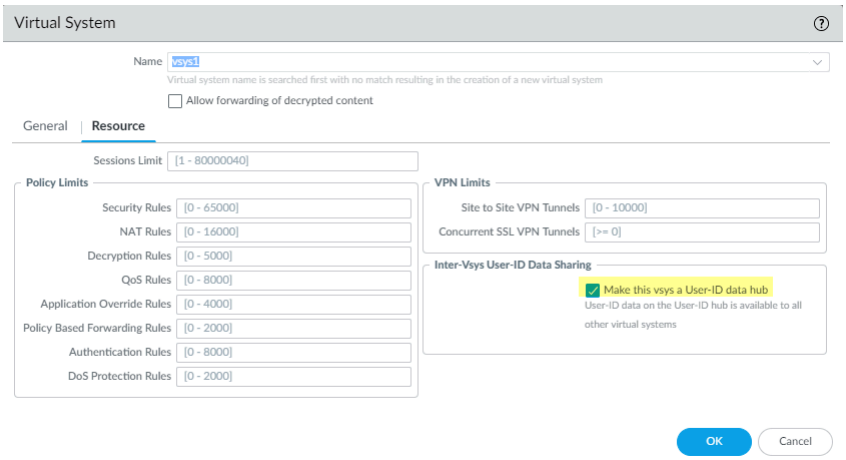
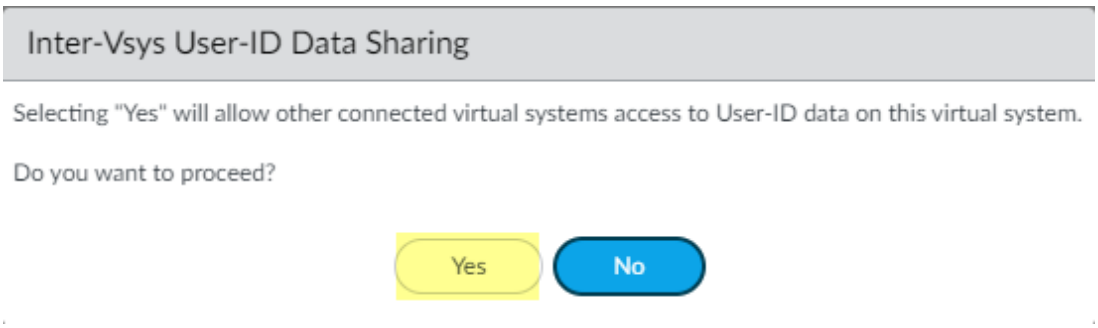
the mappings on other virtual systems so we recommend consolidating the User-ID sources on the hub. However, if you don't want to share mappings from a specific source, you can configure an individual virtual system to perform user or group mapping.

STEP 1 | Assign the virtual system as a User-ID hub.

1. Select **Device** > **Virtual Systems** and then select the virtual system where you consolidated your User-ID sources.
2. On the **Resource** tab, **Make this vsys a User-ID data hub** and click **Yes** to confirm. Then click **OK**.



STEP 2 | Click **Yes** to confirm.



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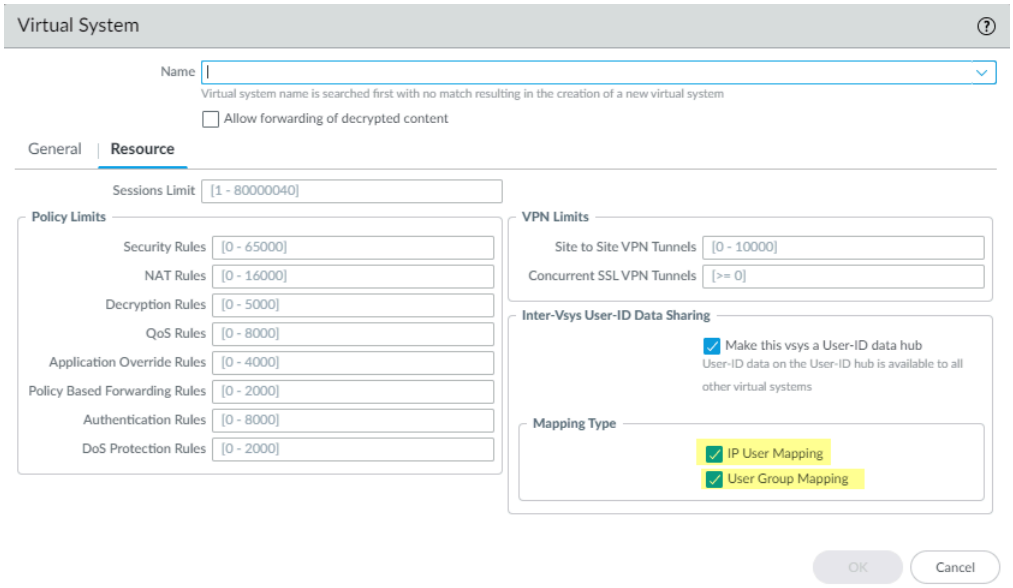
STEP 3 | Select the **Mapping Type** that you want to share then click **OK**.



- **IP User Mapping**—Share IP address-to-username mapping information with other virtual systems.
- **User Group Mapping**—Share group mapping information with other virtual systems.

You must select at least one mapping type.

STEP 4 | Consolidate your User-ID sources and migrate them to the virtual system that you want to use as a User-ID hub.

This consolidates the User-ID configuration for operational simplicity. By configuring the hub to monitor servers and connect to agents that were previously monitored by other virtual systems, the hub collects the user mapping information instead of having each virtual system

collect it independently. If you don't want to share mappings from specific virtual systems, configure those mappings on a virtual system that will not be used as the hub.

Use the same format for the Primary Username across virtual systems and firewalls.

1. Remove any sources that are unnecessary or outdated.
2. Identify all configurations for your Windows-based or integrated agents and any sources that send user mappings using the XML API and copy them to the virtual system you want to use as a User-ID hub.

   On the hub, you can configure any User-ID source that is currently configured on a virtual system. However, IP address-and-port-to-username mapping information from Terminal Server agents are not shared between the User-ID hub and the connected virtual systems.
3. Specify the subnetworks that User-ID should include in or exclude from mapping.
4. Define the **Ignore User List**.
5. On all other virtual systems, remove any sources that are on the User-ID hub.

STEP 5 | **Commit** the changes to enable the User-ID hub and begin collecting mappings for the consolidated sources.

STEP 6 | Confirm the User-ID hub is mapping the users and groups.

1. Use the **`show user ip-user-mapping all`** command to show the IP address-to-username mappings and which virtual system provides the mappings.
2. Use the **`show user user-id-agent statistics`** command to show which virtual system is serving as the User-ID hub.
3. Confirm the hub is sharing the group mappings by using the following CLI commands:
   - `show user group-mapping statistics`
   - `show user group-mapping state all`
   - `show user group list`
   - `show user group name` *<group-name>*

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App-ID

To safely enable applications on your network, the Palo Alto Networks next-generation firewalls provide both an application and web perspective—App-ID and URL Filtering—to protect against a full spectrum of legal, regulatory, productivity, and resource utilization risks.

App-ID enables visibility into the applications on the network, so you can learn how they work and understand their behavioral characteristics and their relative risk. This application knowledge allows you to create and enforce security policy rules to enable, inspect, and shape desired applications and block unwanted applications. When you define policy rules to allow traffic, App-ID begins to classify traffic without any additional configuration.

New and modified App-IDs are released as part of Applications and Threat Content Updates—follow the Best Practices for Applications and Threats Content Updates to seamlessly keep your application and threat signatures up-to-date.

- App-ID Overview
- Streamlined App-ID Policy Rules
- App-ID and HTTP/2 Inspection
- Manage Custom or Unknown Applications
- Manage New and Modified App-IDs
- Use Application Objects in Policy
- Safely Enable Applications on Default Ports
- Applications with Implicit Support
- Security Policy Rule Optimization
- App-ID Cloud Engine
- SaaS App-ID Policy Recommendation
- Application Level Gateways
- Disable the SIP Application-level Gateway (ALG)
- Use HTTP Headers to Manage SaaS Application Access
- Maintain Custom Timeouts for Legacy Applications

App-ID Overview

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

App-ID, a patented traffic classification system only available in Palo Alto Networks firewalls, determines what an application is irrespective of port, protocol, encryption (SSH or SSL) or any other evasive tactic used by the application. It applies multiple classification mechanisms —application signatures, application protocol decoding, and heuristics—to your network traffic stream to accurately identify applications.

Here's how App-ID identifies applications traversing your network:

- Traffic is matched against policy to check whether it is allowed on the network.
- Signatures are then applied to allowed traffic to identify the application based on unique application properties and related transaction characteristics. The signature also determines if the application is being used on its default port or it is using a non-standard port. If the traffic is allowed by policy, the traffic is then scanned for threats and further analyzed for identifying the application more granularly.
- If App-ID determines that encryption (SSL or SSH) is in use, and a decryption policy rule is in place, the session is decrypted and application signatures are applied again on the decrypted flow.
- Decoders for known protocols are then used to apply additional context-based signatures to detect other applications that may be tunneling inside of the protocol (for example, Yahoo! Instant Messenger used across HTTP). Decoders validate that the traffic conforms to the protocol specification and provide support for NAT traversal and opening dynamic pinholes for applications such as SIP and FTP.
- For applications that are particularly evasive and cannot be identified through advanced signature and protocol analysis, heuristics or behavioral analysis may be used to determine the identity of the application.

When the application is identified, the policy check determines how to treat the application, for example—block, or allow and scan for threats, inspect for unauthorized file transfer and data patterns, or shape using QoS.

Before you configure an Application Override policy rule, you should understand that the set of IPv4 addresses is treated as a subset of the set of IPv6 addresses, as described in detail in Security Policy.

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Streamlined App-ID Policy Rules

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

Safely enable a broad set of applications with common attributes using a single policy rule (for example, give your users broad access to web-based applications or safely enable all enterprise VoIP applications). Palo Alto Networks takes on the task of researching applications with common attributes and delivers this through tags in dynamic content updates. This:

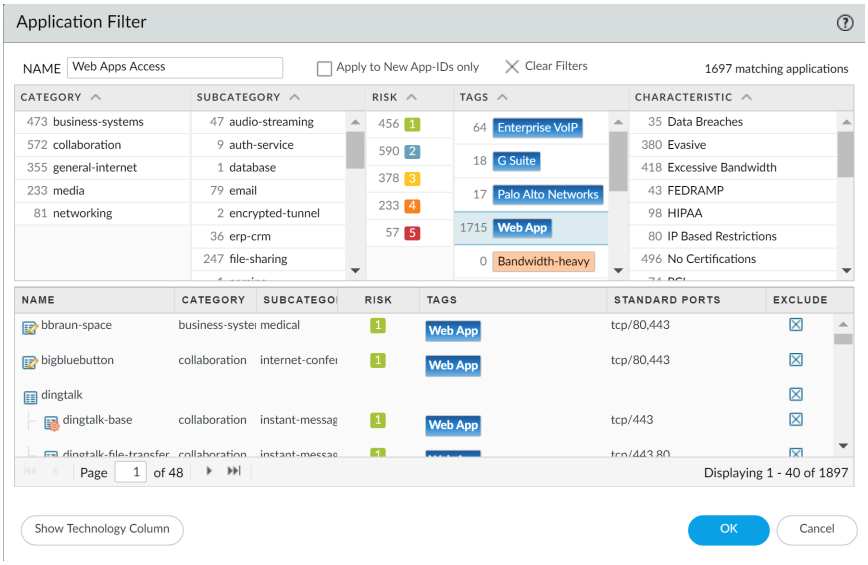
- Minimizes errors and saves time.
- Helps you to create policies that automatically update to handle newly released applications.
- Simplifies the transition toward an App-ID based rule set using Policy Optimizer.

Your firewall can then use your tag-based application filter to dynamically enforce new and updated App-IDs without requiring you to review or update policy rules whenever new applications are added. If you choose to exclude applications from a specific tag, new content updates honor those exclusions. You can also use your own tags to define applications types based on your policy requirements.

- **Create an Application Filter Using Tags**

  1. Create an application filter using one or more tags.

     If you select more than one tag, applications must match both tags to be included in the filter.



  2. (Optional) Exclude tags from your filter by selecting the check box in the **Exclude** column.
  3. Create a security policy rule and **Add** your new application filter on the **Application** tab.
  4. **Commit** your changes.

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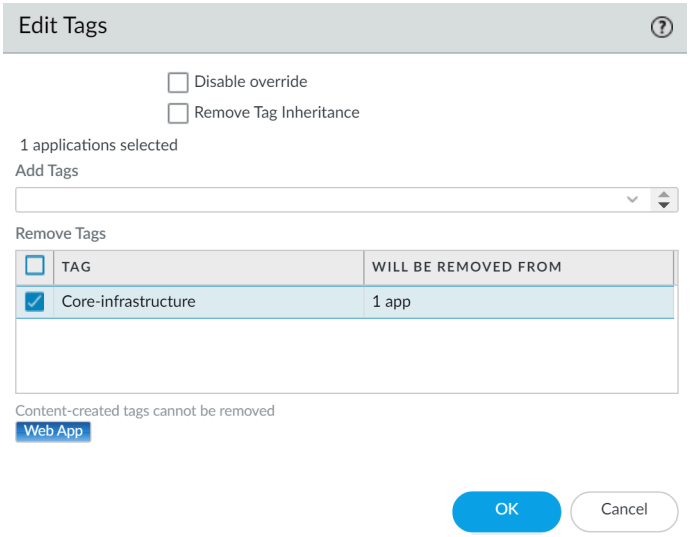
- **Create an Application Filter Based on Custom Tags**
  1. Create a custom tag and apply to App-IDs.
     1. (Optional) Remove tags from an application.
     2. Filter or search for applications, then select the specific applications to remove tags.
     3. **Edit Tags** and select the tags to remove.



     4. Click **OK**.
  2. Create an application filter using one or more tags.

     If you select more than one tag, applications must match both tags to be included in the filter.

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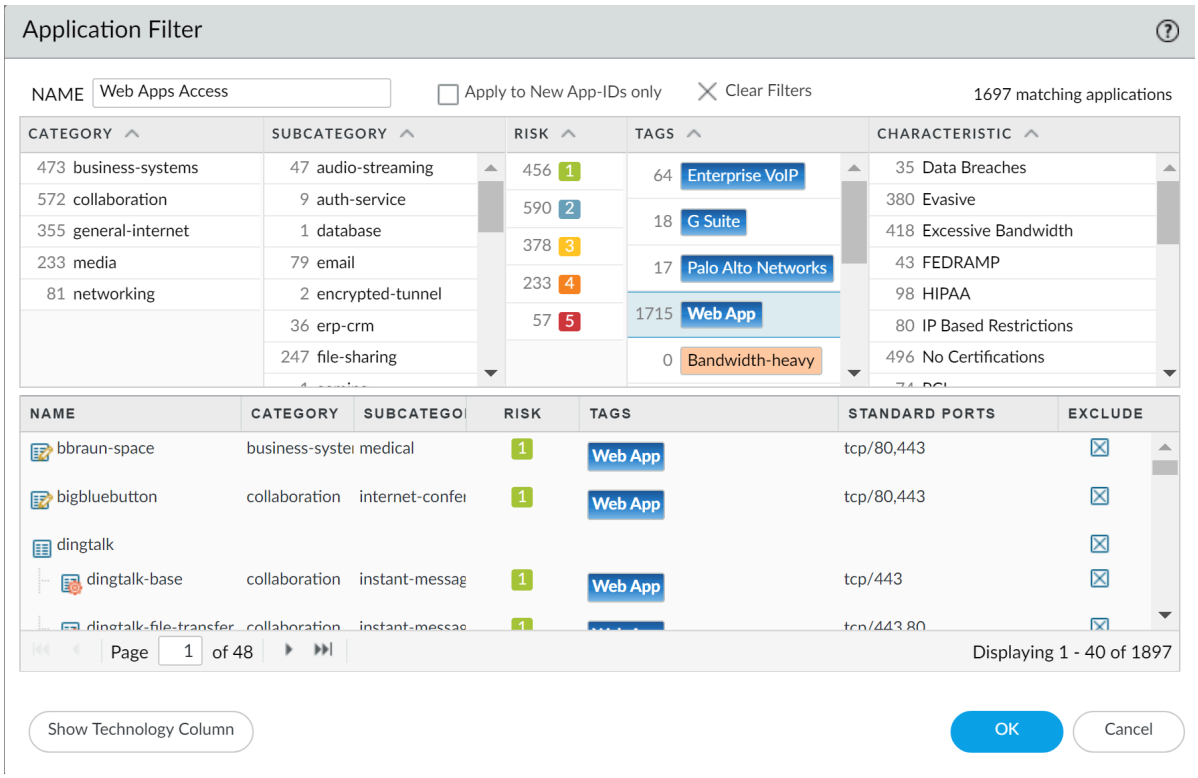

3. Create a security policy rule and **Add** your new application filter on the **Application** tab.
4. **Commit** your changes.

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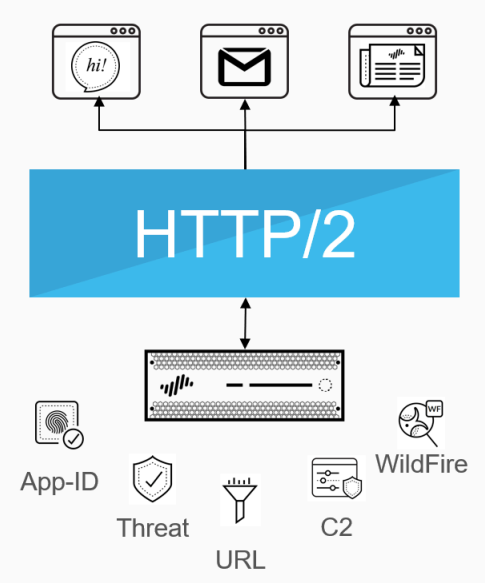
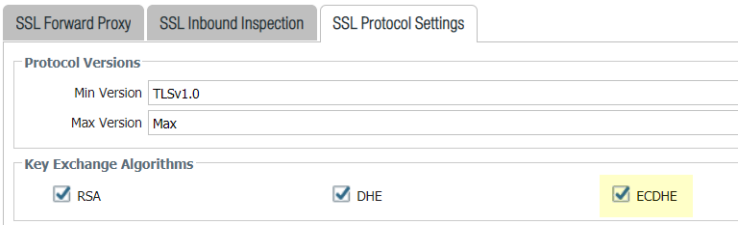
App-ID and HTTP/2 Inspection

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

You can now safely enable applications running over HTTP/2, without any additional configuration on the firewall. As more websites continue to adopt HTTP/2, the firewall can enforce security policy and all threat detection and prevention capabilities on a stream-by-stream basis. This visibility into HTTP/2 traffic enables you to secure web servers that provide services over HTTP/2, and allow your users to benefit from the speed and resource efficiency gains that HTTP/2 provides.



The firewall processes and inspects HTTP/2 traffic by default when SSL decryption is enabled. For HTTP/2 inspection to work correctly, the firewall must be enabled to use ECDHE (elliptic curve Diffie-Hellman) as a key exchange algorithm for SSL sessions. ECDHE is enabled by default, but you can check to confirm that it's enabled by selecting **Objects** > **Decryption** > **Decryption Profile** > **SSL Decryption** > **SSL Protocol Settings**.



You can disable HTTP/2 inspection for targeted traffic, or globally:

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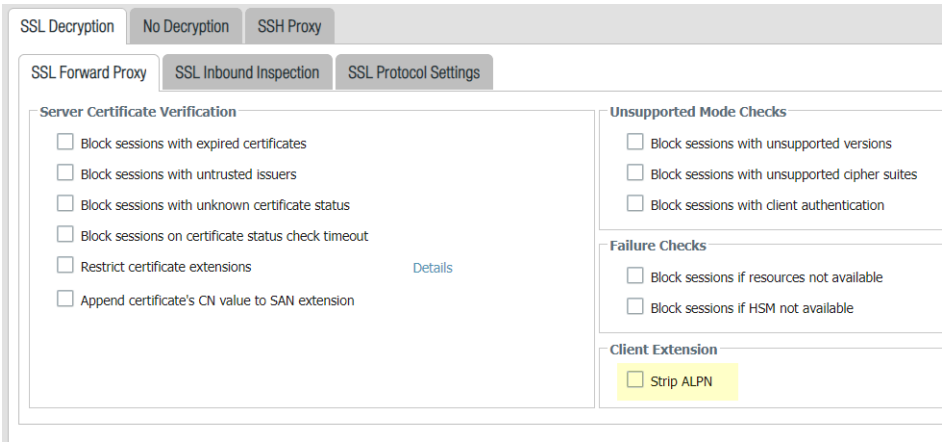
- Disable HTTP/2 inspection for targeted traffic.

  You'll need to specify for the firewall to remove any value contained in the Application-Layer Protocol Negotiation (ALPN) TLS extension. ALPN is used to secure HTTP/2 connections—when there is no value specified for this TLS extension, the firewall either downgrades HTTP/2 traffic to HTTP/1.1 or classifies it as unknown TCP traffic.



  1. Select **Objects** > **Decryption** > **Decryption Profile** > **SSL Decryption** > **SSL Forward Proxy** and then select **Strip ALPN**.
  2. Attach the decryption profile to a decryption policy (**Policies** > **Decryption**) to turn off HTTP/2 inspection for traffic that matches the policy.
  3. **Commit** your changes.

- Disable HTTP/2 inspection globally.

  Use the CLI command: `set deviceconfig setting http2 enable no` and **Commit** your changes. The firewall will classify HTTP/2 traffic as unknown TCP traffic.

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Manage Custom or Unknown Applications

Where Can I Use This?	What Do I Need?
• • • •	❑ (for enhanced feature support) or Threat Prevention License

Palo Alto Networks provides weekly application updates to identify new App-ID signatures. By default, App-ID is always enabled on the firewall, and you don't need to enable a series of signatures to identify well-known applications. Typically, the only applications that are classified as unknown traffic—tcp, udp or non-syn-tcp—in the ACC and the traffic logs are commercially available applications that have not yet been added to App-ID, internal or custom applications on your network, or potential threats.

On occasion, the firewall may report an application as unknown for the following reasons:

- Incomplete data—A handshake took place, but no data packets were sent prior to the timeout.
- Insufficient data—A handshake took place followed by one or more data packets; however, not enough data packets were exchanged to identify the application.

The following choices are available to handle unknown applications:

- Create security policies to control unknown applications by unknown TCP, unknown UDP or by a combination of source zone, destination zone, and IP addresses.
- Request an App-ID from Palo Alto Networks—If you would like to inspect and control the applications that traverse your network, for any unknown traffic, you can record a packet capture. If the packet capture reveals that the application is a commercial application, you can submit this packet capture to Palo Alto Networks for App-ID development. If it is an internal application, you can create a custom App-ID and/or define an application override policy.
- Create a Custom Application with a signature and attach it to a security policy, or create a custom application and define a custom timeout. Avoid creating Application Override policies because they bypass layer 7 application processing and threat inspection, and use less secure stateful layer 4 inspection instead. Instead, use custom timeouts so that you can control and inspect the application traffic at layer 7.

  A custom application allows you to customize the definition of the internal application—its characteristics, category and sub-category, risk, port, and timeout—and to exercise granular policy control and help eliminate unidentified traffic on your network. Creating a custom application also allows you to correctly identify the application in the **ACC** and traffic logs, and is useful in auditing/reporting on the applications on your network. To create a custom application, specify a signature and a pattern that uniquely identifies the application and attach it to a Security policy rule that allows or denies the application.

  For example, if you build a custom application that triggers on a host header *www.mywebsite.com*, the packets are first identified as *web-browsing* and then are matched as your custom application (whose parent application is web-browsing). Because the parent

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application is web-browsing, the custom application is inspected at Layer-7 and scanned for content and vulnerabilities.

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Manage New and Modified App-IDs

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

New and modified App-IDs are delivered to the firewall as part of Applications and Threats Content Updates. While new and modified App-IDs enable the firewall to enforce your security policy with ever-increasing precision, changes in security policy enforcement that can occur when a content update release is installed can impact application availability. For this reason, you will need to think about how to best deploy content updates so that you can get the latest threat prevention as it's made available, and adjust your security policy to best leverage new and modified App-IDs.

The following options enable you to assess the impact of new App-IDs on existing policy enforcement, disable (and enable) App-IDs, and seamlessly update policy rules to secure and enforce newly-identified applications:

- Workflow to Best Incorporate New and Modified App-IDs
- See the New and Modified App-IDs in a Content Release
- See How New and Modified App-IDs Impact Your Security Policy
- Ensure Critical New App-IDs are Allowed
- Monitor New App-IDs
- Disable and Enable App-IDs
- Enable and Monitor App-ID TSIDs

You can also take advantage of the Streamlined App-ID Policy Rules that use application tags provided in the content updates.

Workflow to Best Incorporate New and Modified App-IDs

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

Refer to this master workflow to first set up Application and Threat content updates, and then to best incorporate new and modified App-IDs into your security policy. Everything you need to deploy content updates is referenced here.

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STEP 1 | Align your business needs with an approach to deploying Application and Threat content updates.

Learn how Applications and Threat Content Updates work, and identify your organization as either mission-critical or security-first. Understanding which of these is most important to your business will help you to decide how to best deploy content updates and apply best practices to meet your business needs. You might find that you want to apply a mix of both approaches, perhaps depending on firewall deployment (data center or perimeter) or office location (remote or headquarters).

STEP 2 | Review and apply the Best Practices for Applications and Threats Content Updates based on your organization's network security and application availability requirements.

STEP 3 | (PAN-OS 12.1.5 and later) Consider enabling App-ID Safeguard to maintain security policy intent. In certain cases where an App-ID gets modified, or a new one is introduced that supersedes a current App-ID, your security policy might inadvertently block certain content. When this is enabled, your security policy will continue to make policy decisions based on the previous App-ID. This means that a new App-ID that has associated previous App-IDs that are allowed, regardless of whether or not they are in separate rules, the new App-ID would also be automatically allowed. Keep in mind, the Previous App-ID attribute used to define the previously used application defintion gets reset with the next content release update.

STEP 4 | Configure a security policy rule to always allow new App-IDs that might have network-wide impact, like authentication or software development applications.

The New App-ID characteristic matches to only the App-IDs introduced in the latest content release. When used in a security policy, this gives you a month's time to fine tune your security policy based on new App-IDs while ensuring constant availability for App-IDs that fall into critical categories (Ensure Critical New App-IDs are Allowed).

STEP 5 | Set the schedule to Deploy Application and Threat Content Updates; this includes the option to delay new App-ID installation until you've had time to make necessary security policy updates (using the **New App-ID Threshold**).

STEP 6 | After you've setup a content updates installation schedule, you'll want to regularly check in and See the New and Modified App-IDs in a Content Release.

STEP 7 | You can then See How New and Modified App-IDs Impact Your Security Policy, and make adjustments to your security policy as needed.

STEP 8 | Monitor New App-IDs to get a view into new App-ID activity on your network, so that you're best equipped to make the most effective security policy updates.

See the New and Modified App-IDs in a Content Release

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

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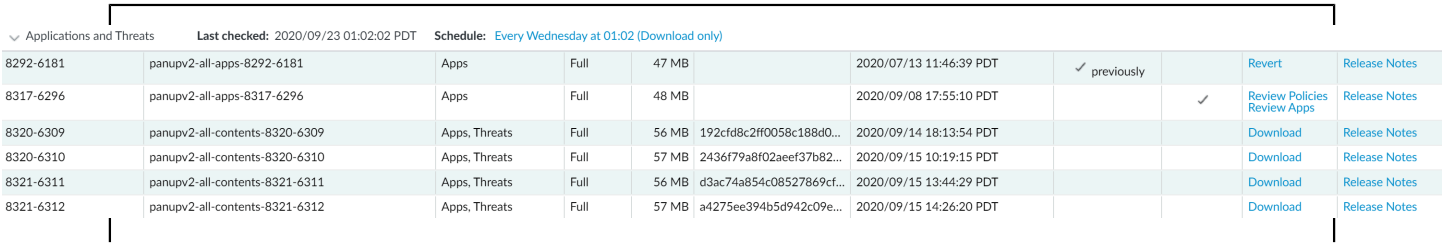
For both downloaded and installed content updates, you can see a list of the new and modified App-IDs the update includes. Full application details are provided, and importantly, updates to applications with network-wide impact (for example, LDAP or IKE) are prominently flagged as a recommended for policy review. For modified App-IDs, application details also describe how coverage is either now expanded or more precise.

STEP 1 | Select **Device** > **Dynamic Updates** and select **Check Now** to refresh the list of available content updates.

STEP 2 | For either a downloaded or currently installed content release, click **Review Apps** link in the **Actions** column to view details on newly-identified and modified applications in that release:



Applications and Threats **Last checked:** 2020/09/23 01:02:02 PDT **Schedule:** Every Wednesday at 01:02 (Download only)

8292-6181	panupv2-all-apps-8292-6181	Apps	Full	47 MB		2020/07/13 11:46:39 PDT	✓ previously		Revert	Release Notes
8317-6296	panupv2-all-apps-8317-6296	Apps	Full	48 MB		2020/09/08 17:55:10 PDT		✓	Review Policies Review Apps	Release Notes
8320-6309	panupv2-all-contents-8320-6309	Apps, Threats	Full	56 MB	192cfd8c2ff0058c188d0...	2020/09/14 18:13:54 PDT			Download	Release Notes
8320-6310	panupv2-all-contents-8320-6310	Apps, Threats	Full	57 MB	2436f79a8f02aeef37b82...	2020/09/15 10:19:15 PDT			Download	Release Notes
8321-6311	panupv2-all-contents-8321-6311	Apps, Threats	Full	56 MB	d3ac74a854c08527869cf...	2020/09/15 13:44:29 PDT			Download	Release Notes
8321-6312	panupv2-all-contents-8321-6312	Apps, Threats	Full	57 MB	a4275ee394b5d942c09e...	2020/09/15 14:26:20 PDT			Download	Release Notes

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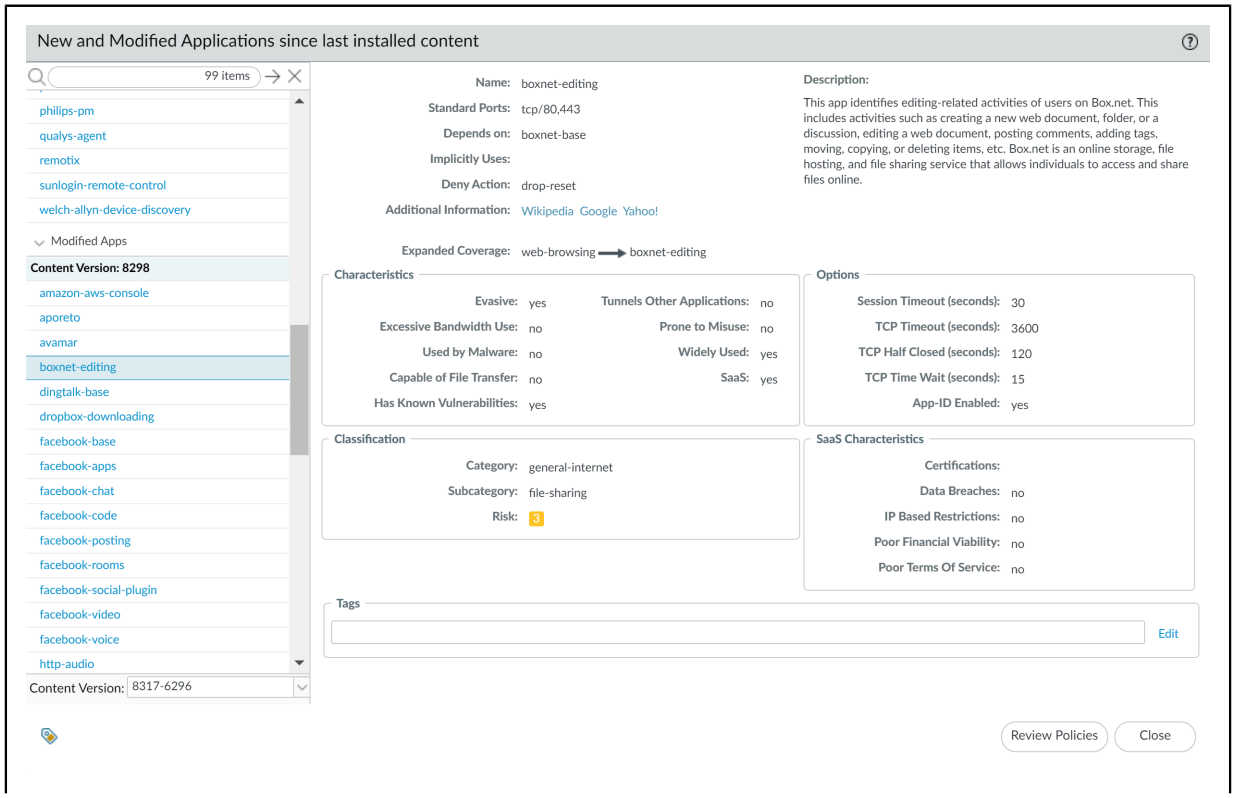
STEP 3 | Review the App-IDs this content release introduces or modifies since the last content version.

- *New and modified App-IDs are listed separately. Full application details are provided for each, and App-IDs that Palo Alto Networks foresees as having network-wide impact are flagged as recommended for policy review.*
- *If you have enabled App-ID Update Safeguard, an additional attribute field showing the **Previous App-ID** entries is displayed in the application overview.*



New App-ID details that you can use to assess possible impact to policy enforcement include:

- **Depends on**—Lists the application signatures that this App-ID relies on to uniquely identify the application. If one of the application signatures listed in the **Depends On** field is disabled, the dependent App-ID is also disabled.
- **Previously Identified As**—Lists the App-IDs that matched to the application before the new App-ID was installed to uniquely identify the application.
- **App-ID Enabled**—All App-IDs display as enabled when a content release is downloaded, unless you choose to manually disable the App-ID signature before installing the content update.

For modified App-IDs, details include information on: **Expanded Coverage**, **Remove False Positive**, and application metadata changes. The Expanded Coverage and Remove False Positive fields both indicate how the application's coverage has changed (it's either more comprehensive or has been narrowed) and a clock icon indicates a metadata change, where certain application details are updated.

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STEP 4 | Based on your findings, click **Review Policies** to see how the new and modified App-IDs impact security policy enforcement: See How New and Modified App-IDs Impact Your Security Policy.

See How New and Modified App-IDs Impact Your Security Policy

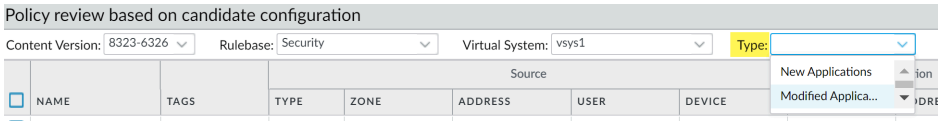
Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

Newly-categorized and modified App-IDs can change the way the firewall enforces traffic. Perform a content update policy review to see how new and modified App-IDs impact your security policy, and to easily make any necessary adjustments. You can perform a content update policy review for both downloaded and installed content.

STEP 1 | Select **Device** > **Dynamic Updates**.

STEP 2 | See the New and Modified App-IDs in a Content Release to learn more about each App-ID that a content release introduces or modifies.

STEP 3 | For a downloaded or currently installed content release, click **Review Policies** in the Action column. The **Policy review based on candidate configuration** dialog allows you to filter by **Content Version** and view either new or modified App-IDs introduced in a specific release (you can also filter the policy impact of new App-IDs according to **Rulebase**, **Virtual System**, and **Application**).



STEP 4 | Select an App-ID from the **Application** drop-down to view policy rules that currently enforce the application. The rules displayed are based on the App-IDs that match to the application before the new App-ID is installed (view application details to see the list of application signatures that an application was **Previously Identified As** before the new App-ID).

STEP 5 | Use the detail provided in the policy review to plan policy rule updates to take effect when the App-ID is installed, or if the content release version that included the App-ID is currently installed, the changes you make take effect immediately.

You can **Add app to selected policies** or **Remove app from selected policies**.

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Ensure Critical New App-IDs are Allowed

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

New App-IDs can cause a change in policy enforcement for traffic that is newly-identified as belonging to a certain application. To mitigate any impact to security policy enforcement, you can use the **New App-ID** characteristic in a security policy rule so that the rule always enforces the most recently introduced App-IDs without requiring you to make configuration changes when new App-IDs are installed. The New App-ID characteristic always matches to only the new App-IDs in the most recently installed content releases. When a new content release is installed, the new App-ID characteristic automatically begins to match only to the new App-IDs in that content release version.

You can choose to enforce all new App-IDs, or target the security policy rule to enforce certain types of new App-IDs that might have network-wide or critical impact (for example, enforce only authentication or software development applications). Set the security policy rule to **Allow** to ensure that even if an App-ID release introduces expanded or more precise coverage for critical applications, the firewall continues to allow them.

New App-IDs are released monthly, so a policy rule that allows the latest App-IDs gives you a month's time (or, if the firewall is not installing content updates on a schedule, until the next time you manually install content) to assess how newly-categorized applications might impact security policy enforcement and make any necessary adjustments.

STEP 1 | Select **Objects** > **Application Filters** and **Add** a new application filter.

STEP 2 | Define the types of new applications for which you want to ensure constant availability based on subcategory or characteristic. For example, select the category "auth-service" to ensure that any newly-installed applications that are known to perform or support authentication are allowed.

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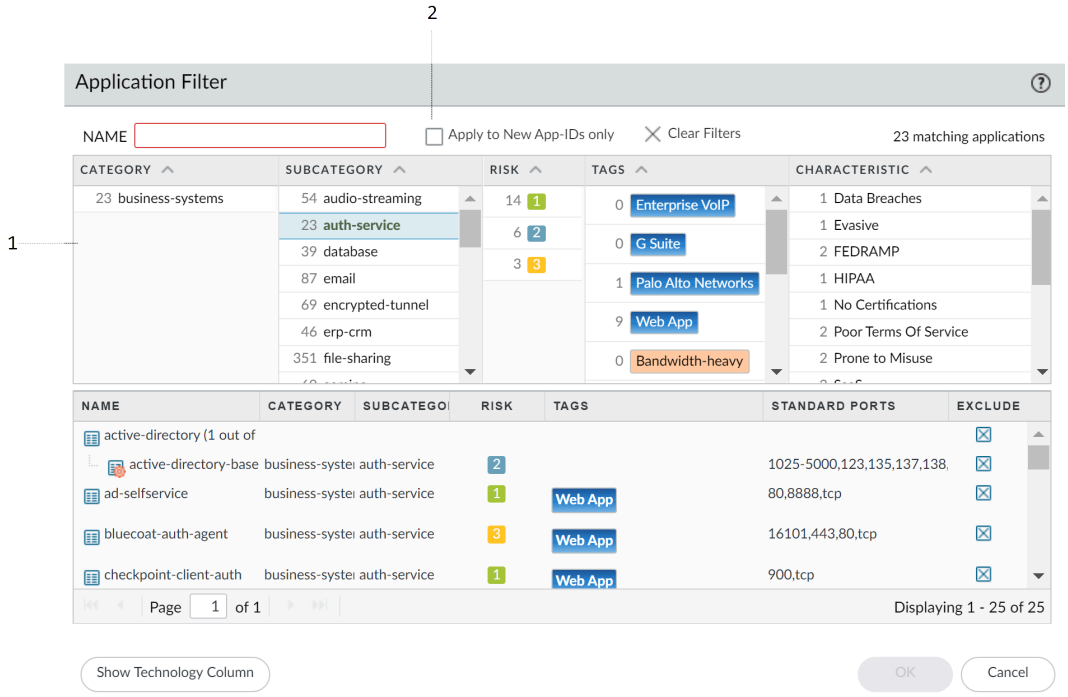
STEP 3 | Only after narrowing the types of new applications that you want to allow immediately upon installation, select **Apply to New App-IDs only**.



STEP 4 | Select **Policies** > **Security** and add or edit a security policy rule that is configured to allow matching traffic.

STEP 5 | Select **Application** and add the new **Application Filter** to the policy rule as match criteria.

STEP 6 | Click **OK** and **Commit** to save your changes.

STEP 7 | To continue to adjust your security policy to account for any changes to enforcement that new App-IDs introduce:

- Monitor New App-IDs—Monitor and get reports on new App-ID activity.
- See the New and Modified App-IDs in a Content Release—See how the newly-installed App-IDs impact your existing security policy rules.

Monitor New App-IDs

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

The **New App-ID** characteristic enables you to monitor new applications on your network, so that you can better assess the security policy updates you might want to make. Use the New App-ID characteristic on the ACC to get visibility into the new applications on your network, and to generate reports that detail newly-categorized application activity. What you learn can help

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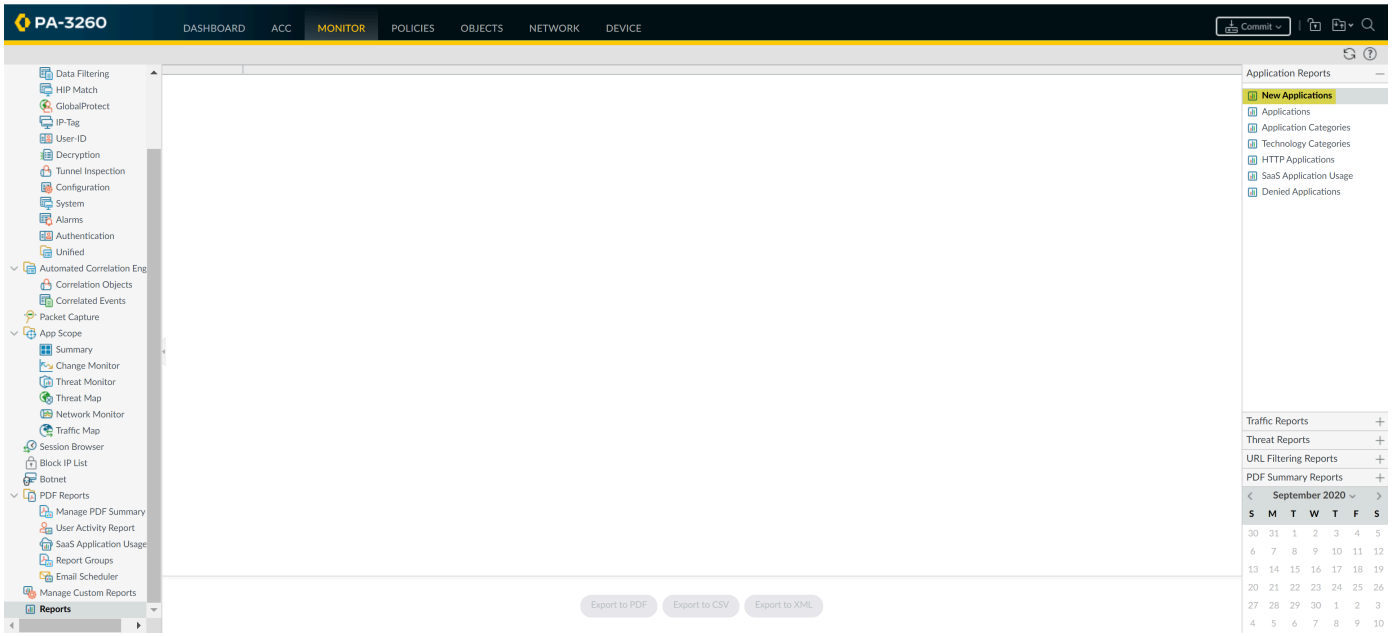
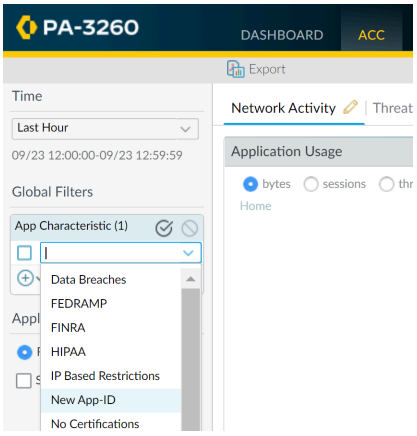
you make the right decisions about how you to update your security policy to enforce the most recently-categorized App-IDs. Whether you're using it on the ACC or to generate reports (or to Ensure Critical New App-IDs are Allowed), the New App-ID characteristic always matches to only the new App-IDs in the most recently installed content releases. When a new content release is installed, the new App-ID characteristic automatically begins to match only to the new App-IDs in that content release version.

- Generate a report with details specifically regarding new applications (applications introduced only in the latest content release).



- Use the ACC to monitor new application activity: select **ACC** and under **Global Filters**, select **Application** > **Application Characteristics** > **New App-ID**.



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Disable and Enable App-IDs

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

You can disable all App-IDs introduced in a content release if you want to immediately benefit from the latest threat prevention, and plan to enable the App-IDs later, and you can disable App-IDs for specific applications.

Policy rules referencing App-IDs only match to and enforce traffic based on enabled App-IDs.

Certain App-IDs cannot be disabled and only allow a status of enabled. App-IDs that cannot be disabled include application signatures that are implicitly used by other App-IDs (such as unknown-tcp). Disabling a base App-ID could cause App-IDs which depend on the base App-ID to also be disabled. For example, disabling facebook-base will disable all other Facebook App-IDs.

- Disable all App-IDs in a content release or for scheduled content updates.

  While this option allows you to be protected against threats, by giving you the option to enable the App-ID at a later time, Palo Alto Networks recommends that instead of disabling App-IDs on a regular basis, you should instead configure a security policy rule to Temporarily Allow New App-IDs. This rule will always allow the new App-IDs introduced in only the latest content release. Because content updates that include new App-IDs are released only once a month, this gives you time to assess the new App-IDs and adjust your security policy to cover the new App-IDs if needed, all the while ensuring that availability for critical applications is not affected.

  - To disable all new App-IDs introduced in a content release, select **Device** > **Dynamic Updates** and **Install** an Application and Threats content release. When prompted, select **Disable new apps in content update**. Select the check box to disable apps and continue installing the content update.
  - On the **Device** > **Dynamic Updates** page, select **Schedule**. Choose to **Disable new apps in content update** for downloads and installations of content releases.

- Disable App-IDs for one application or multiple applications at a single time.

  - To quickly disable a single application or multiple applications at the same time, click **Objects** > **Applications**. Select one or more application check box and click **Disable**.
  - To review details for a single application, and then disable the App-ID for that application, select **Objects** > **Applications** and **Disable App-ID**. You can use this step to disable both pending App-IDs (where the content release including the App-ID is downloaded to the firewall but not installed) or installed App-IDs.

- Enable App-IDs.

  Enable App-IDs that you previously disabled by selecting **Objects** > **Applications**. Select one or more application check box and click **Enable** or open the details for a specific application and click **Enable App-ID**.

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Enable and Monitor App-ID TSIDs

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

Palo Alto Networks provides a preview version of App-IDs to allow users to test and validate modified and new App-IDs in the form of Threat Signature Indicators (TSIDs). These TSIDs are delivered as vulnerability signatures using the category: **app-id-change** and are, by default, configured on the firewall with a severity level of **Informational** and an action of **Allow**. The non-invasive setting provides monitoring capabilities while preventing unwanted changes from blocking access to key applications in a production environment.

The App-ID TSIDs are packaged with Applications and Threats content updates and include TSID entries for App-IDs that are slated for official release one month later. For example, an App-ID TSID released in the first week of March will include TSIDs for App-IDs that are scheduled for release in the first week of April. This one month probationary period is to allow time for testing/validation and updates to any security policies. You can refer to the Notices section of the **Applications and Threats Content Release Notes** for more details about the nature of the TSIDs.

The name of the App-ID TSID (contained in the Threat ID/name field) correlates to the name of the upcoming App-ID to

STEP 1 | Log in to the PAN-OS web interface.

STEP 2 | Verify that you have an active Advanced Threat Prevention / Threat Prevention subscription. To verify subscriptions for which you have currently-active licenses, select **Device** > **Licenses** and verify that the appropriate licenses are available and have not expired.

STEP 3 | Verify that *Applications and Threats content update 8783* or later is installed on your firewall to access the new Vulnerability Signature category. To view the currently installed

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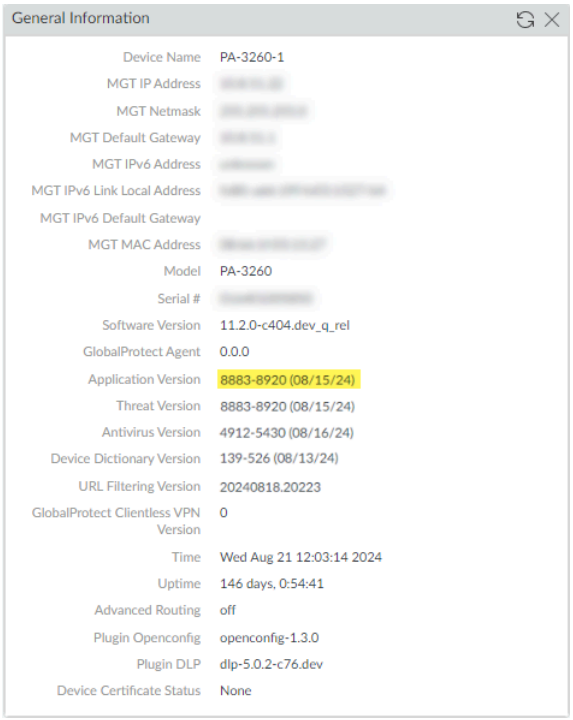
Application and Threats content version, select **Dashboard** and refer to the **Application Version** entry located in the **Generation Information** pane.



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STEP 4 | Enable App-ID TSID visibility by adding a Vulnerability Protection Rule in your Vulnerability Protection security profile.

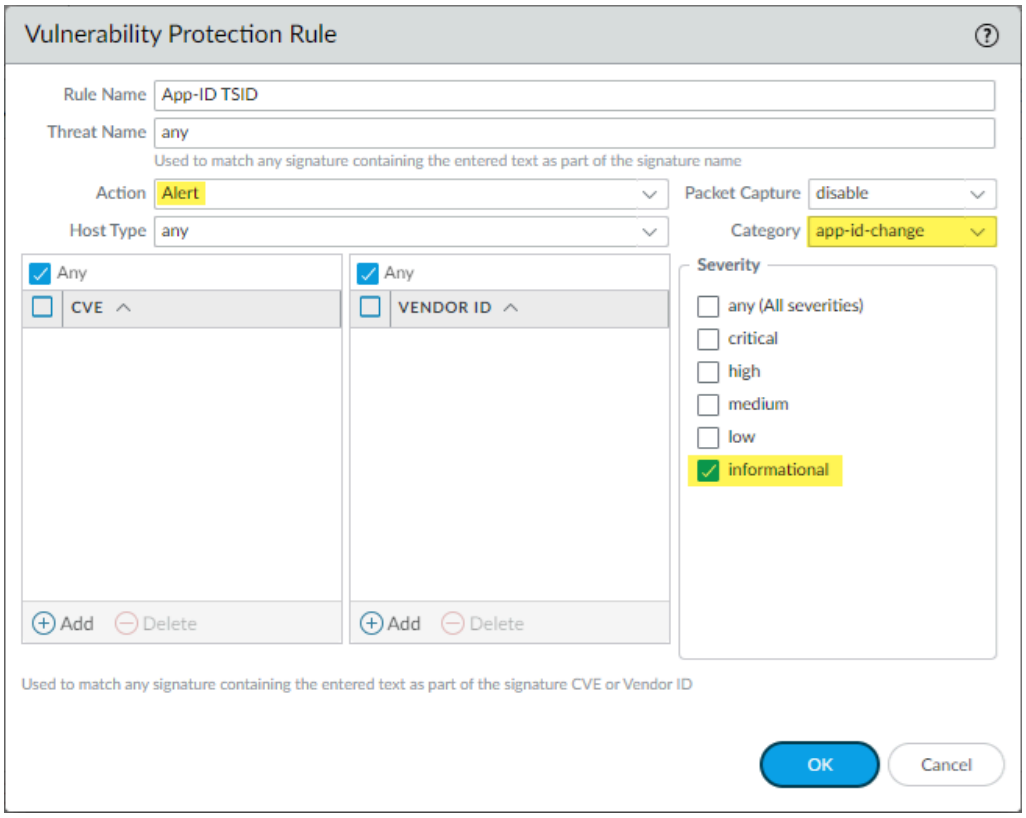
1. Update or create a new Vulnerability Protection security profile (**Objects** > **Security Profiles** > **Vulnerability Protection**).

   If you want to update an existing Vulnerability Security Profile, select one from the list of available profiles. If you want to **Add** a new one, refer to Security Profile: Vulnerability Protection

   .

2. **Add** a new Vulnerability Protection Rule and configure it using the following settings:

   Palo Alto Networks recommends using the following settings to enable matching App-ID TSIDs in order to generate corresponding Threat Logs for future App-ID changes.



   - **Action**—Alert
   - **Severity**—Informational
   - **Category**—app-id-change

3. Click **OK** to exit the **Vulnerability Protection Rule** configuration dialog.

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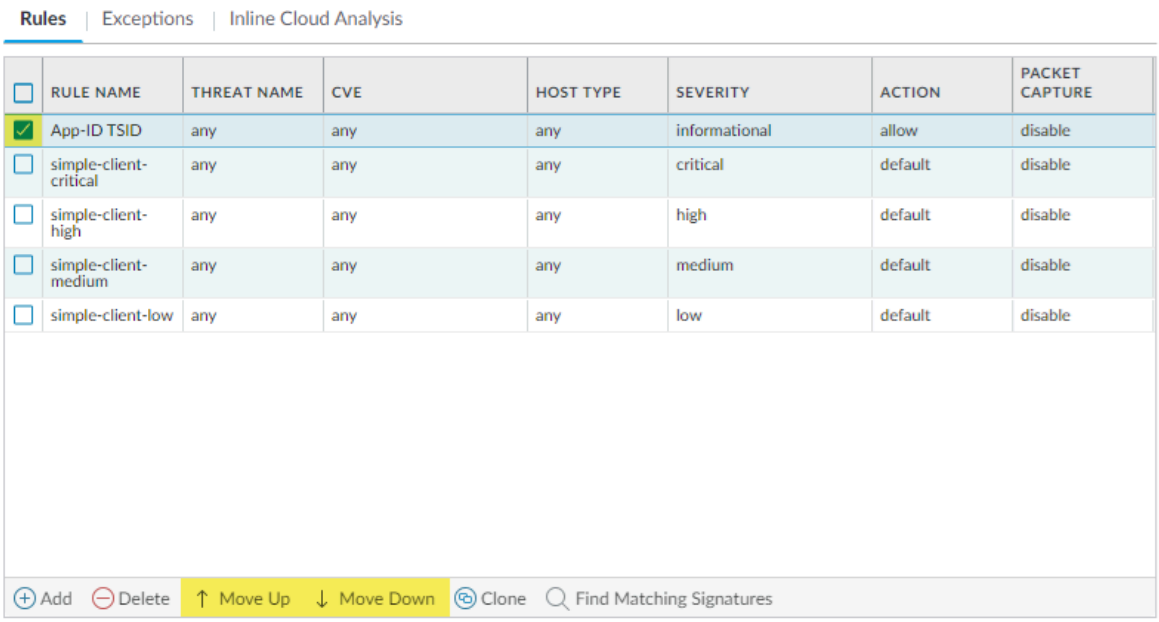
STEP 5 | Change the Vulnerability Protection rule precedence using the controls at the bottom of the Vulnerability Protection profile configuration page so that the new rule is at the *top*.



STEP 6 | Test the new rule by selecting the rule and **Find Matching Signatures**. This will return currently available App-ID TSIDs. If you do not see any results or the results do not appear

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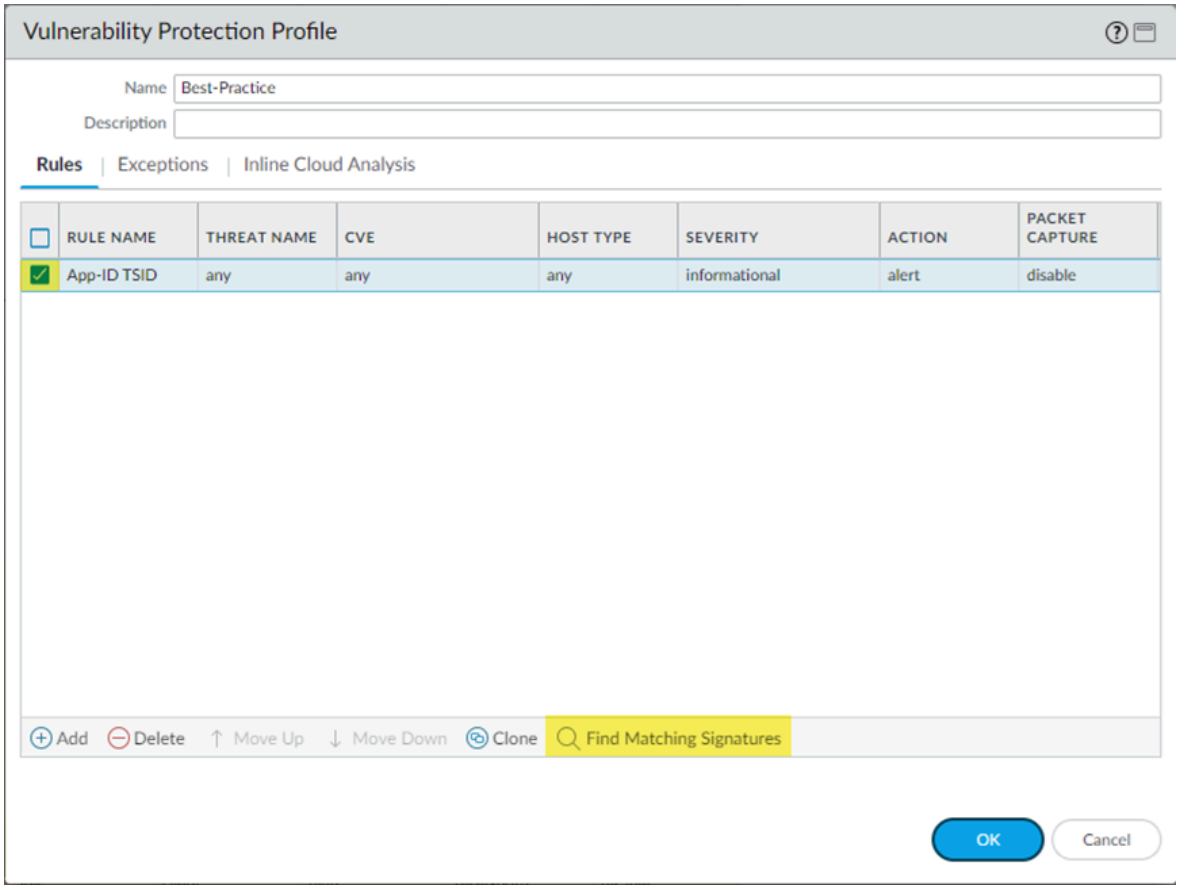
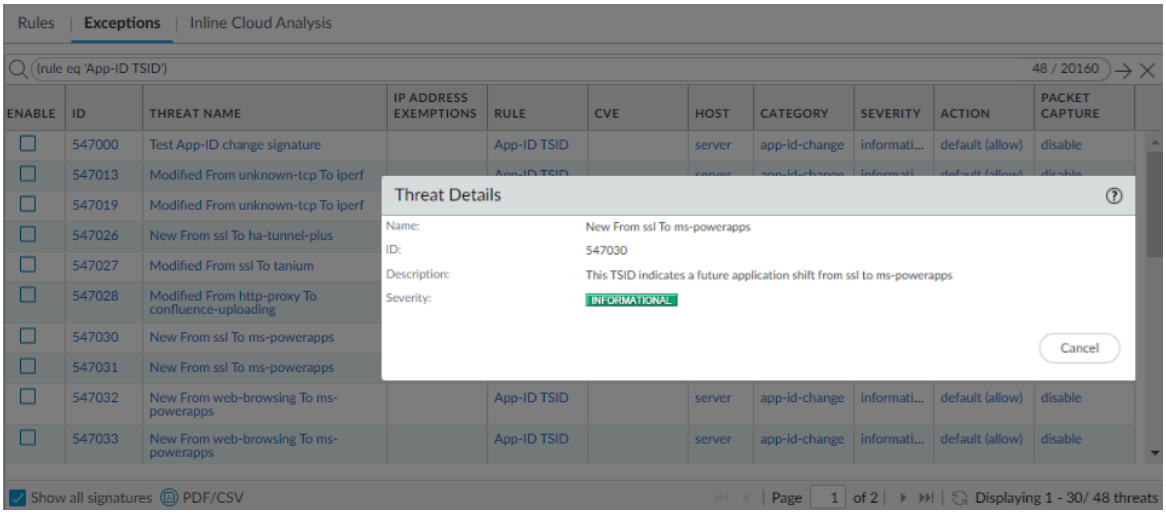
to indicate App-ID TSIDs, verify that app-id-change has been selected in the Vulnerability Protection Rule.



STEP 7 | Click on the **Threat Name** of one of the results for more details about the App-ID TSID.



STEP 8 | Click OK to save your Vulnerability Protection profile.

STEP 9 | Configure your Vulnerability Protection profile to the Security Policy rule(s) you want to monitor for potential App-ID changes. This can be done by applying the specific Vulnerability

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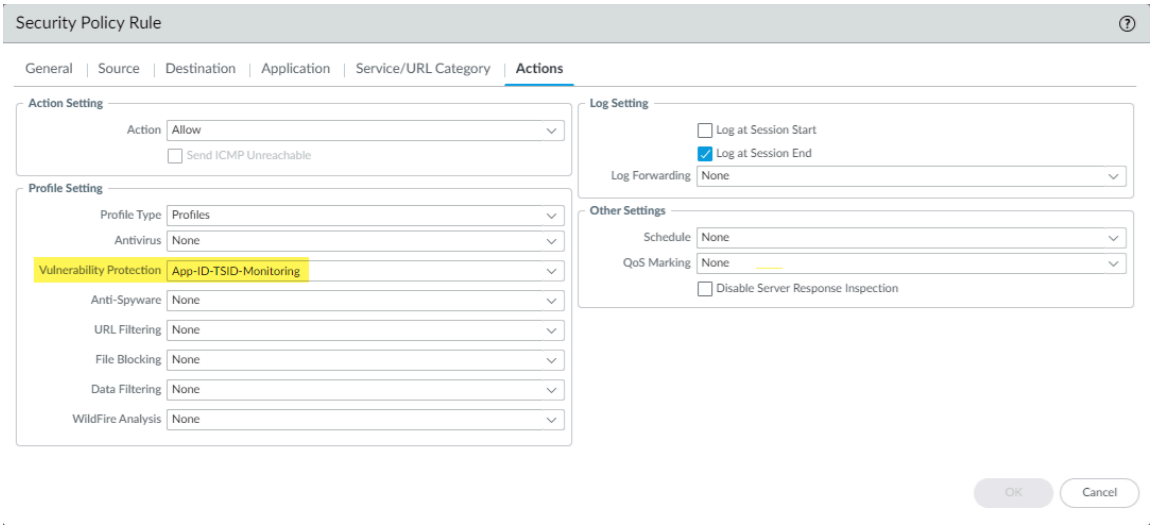
Protection profile in the Security Policy rule(s) or in a Security Profile Group and using that group in the Security Policy rule(s), based on how your policies are constructed.

- **Vulnerability Profile applied directly to a Security Policy rule:**
  1. Go to **Policies** > **Security** and select a security policy rule that you want monitored for App-ID changes.
  2. Navigate to the **Actions** tab and select **Profiles** from the **Profile Type** drop down.
  3. In the **Vulnerability Protection** drop down, select the updated or newly created Vulnerability Protection profile created in step 4.



  4. Click **OK** to save the changes.
  5. Repeat the above process for other security policy rules that you want monitored for App-ID changes.
- **Vulnerability Profile applied to a Security Profile Group that is referenced by a Security Policy rule**
  1. Go to **Objects** > **Security Profile Groups** and select an existing security profile group or Add a new one.
  2. In the **Vulnerability Protection Profile** drop down, select the updated or newly created Vulnerability Protection profile created in step 4 and then click **OK** to save the changes.

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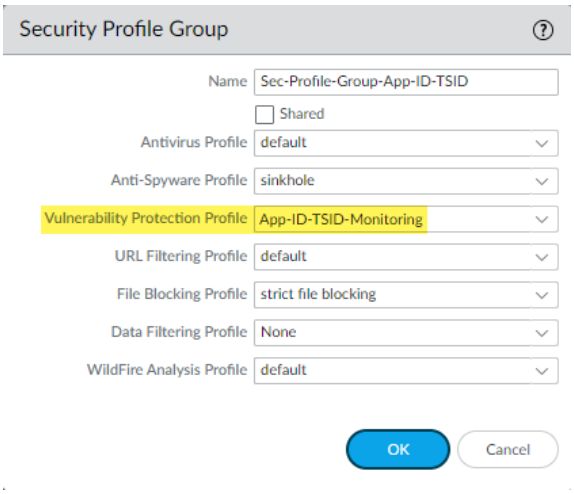
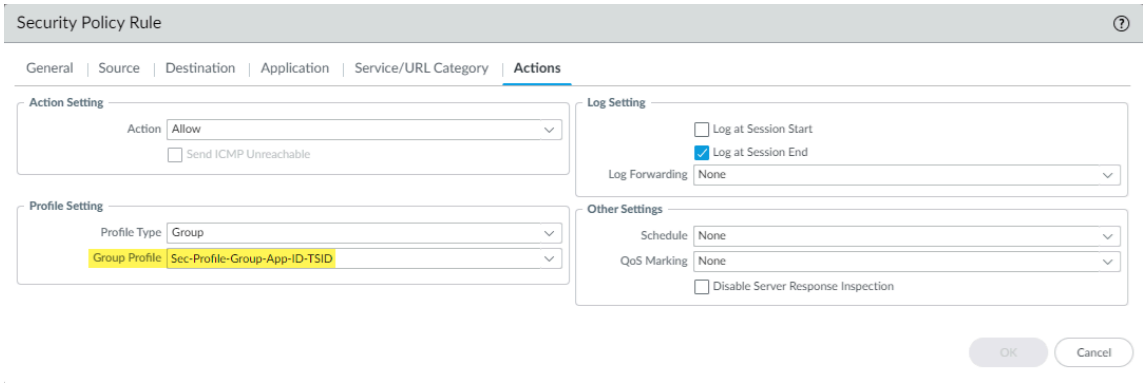


3. Go to **Policies** > **Security** and select a security policy rule that you want monitored for App-ID changes.
4. Navigate to the **Actions** tab and select **Group** from the **Profile Type** drop down.
5. In the **Group Profile** drop down, select the newly updated security profile group.



6. Click **OK** to save the changes.
7. Repeat the above process for other security policy rules that you want monitored for App-ID changes.

STEP 10 | Search for activity on the firewall for queries that match against the App-ID TSID criteria. You can manually peruse through the logs or you can create a custom report using the same **Query Builder** attributes used for activity searches. You can then use the custom report / threat log results to modify your existing Security policy rules in preparation for the new and updated App-IDs scheduled for inclusion in the following month's release of the Applications and Threats content update. You can track affected rules by cross referencing the *Rule* or *Rule UUID* with the detected App-ID TSIDs in your network. If you use Panorama to manage your firewalls, you can use the global search functionality (located in the top-right) to find

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the affected rule(s). Selecting the resulting Security Rule will bring you to that policy rule for configuration.

You can cross reference affected rules by populating the threat logs and custom reports with the Rule and Rule UUID data columns.

- **Monitor activity on the firewall:**

  If you are using the Strata Cloud Manager Log Viewer to review your firewall logs, you can use the following search query: `Threat Category = 'app-id-change'`

  1. Select **Monitor** > **Logs** > **Threat**.
  2. Use the log query builder to create a filter with the **Attribute** Threat Category and in the **Value** field, enter the Threat Category: app-id-change. For example: `( category-of-`

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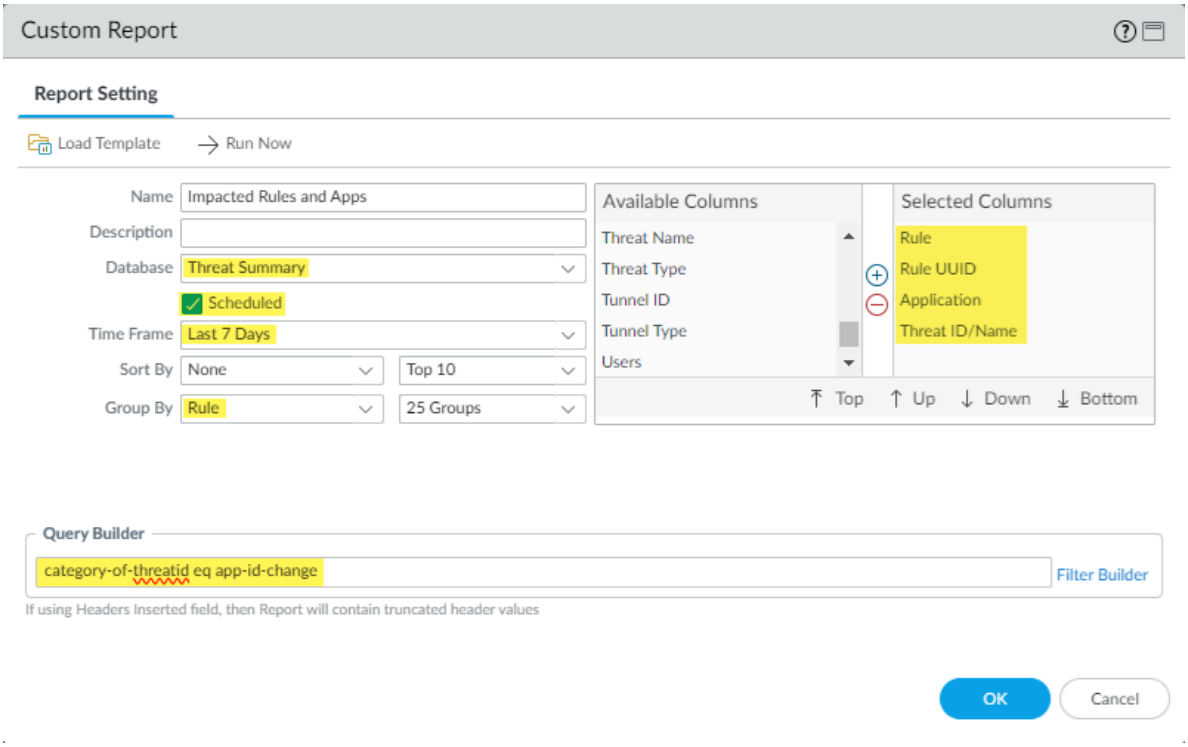
`threatid eq app-id-change )` to view logs that have triggered a matching App-ID TSID, which can be identified by the Threat Category: app-id-change.

- **Create a custom report on the firewall:**

  Palo Alto Networks recommends using the following configuration for the custom report to view information relevant to the App-ID TSIDs.



1. Configure a custom report from **Monitor** > **Manage Custom Reports**. For more information about creating custom reports, refer to Generate Custom Reports.
2. When configuring the **Report Settings**, make sure to specify the **Database** type: **Threat**, which is listed under the **Summary Databases** group. If you are using Panorama, the **Threat** type to be used is listed under **Remote Device Data**.
3. Use the **Query Builder** to add the following filter: `category-of-threatid eq app-id-change`.
4. (Optional) **Group** the report by the **Rule**.
5. At a minimum, populate the report with the following columns:
   - **Rule**
   - **Rule UUID**
   - **Application**

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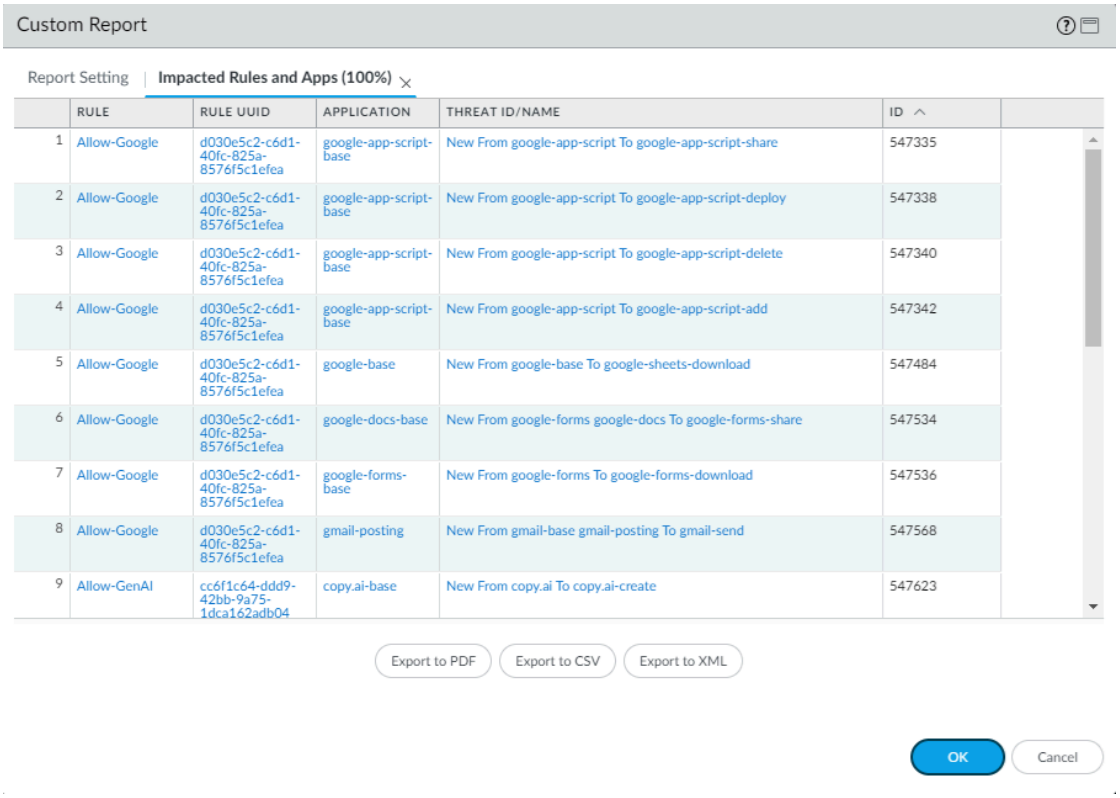
- **Threat ID/Name**

6. Define the **Time Frame** for the report (**Last 7 Days** is generally recommended; however, also consider using a time frame most relevant since you downloaded and installed the latest Application and Threats content update, which contain TSIDs).
7. (Optional) Select the **Scheduled** option to run the report on a nightly basis. The report is then available for viewing in the **Reports** column on the side.
8. Click **Run Now** to generate the custom report or **OK** to save the changes.

   - If you generate a custom report, you can immediately view the results of the report. Alternatively, you can output the report into the following formats: PDF, CSV, and XML.



STEP 11 | Next Steps:

- The *Threat ID/Name* value in the App-ID TSID match results describe the nature of the updated App-ID entry after you install the upcoming Applications and Threats content update. Depending on the type of the update, this could be a **Modified** or **New** App-ID, as indicated by the description. After reviewing the information, you can update your security policies to include these modified or new App-IDs, as appropriate for your organization's network usage policies. Refer to Manage New and Modified App-IDs for more information.

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Prevent Service Disruptions Using App-ID Update Safeguard

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

App-ID Update Safeguard is designed to prevent service disruptions and maintain security posture when new App-ID content updates are installed. It acts as a fail-safe mechanism, ensuring that traffic continues to be enforced according to your original intent, even when Palo Alto Networks introduces new or modified application signatures.

Security policies are typically built around the current state of App-ID. For example, you might allow broad applications like `ssl` or `web-browsing` while blocking `unknown-tcp`.

When a new Content Release update is installed, traffic that was previously identified as `ssl` might be re-identified as a more specific application, such as `acme-app`. If this new specific App-ID is not explicitly added to your allow rules, the traffic could be unintentionally blocked by the default deny rule, causing a service outage.

App-ID Update Safeguard prevents this by allowing the firewall to reference the application's history during policy enforcement. The **Previous App-ID** attribute appears at various points within the interface. This attribute tracks the identity the traffic held in previous content versions. When the feature is enabled, the NGFW utilizes a **two-step policy lookup process** for new or modified applications. During the primary look up phase, the NGFW checks for a new, specific App-ID that is explicitly configured in your security policy. If a rule explicitly allows or blocks this specific App-ID, the rule is enforced immediately. However, when App-ID update safeguard is enabled, a secondary lookup is introduced, whereby the firewall performs a second lookup using the **Previous App-ID** entries contained within an application. If the Previous App-ID was allowed by your policy, the traffic is temporarily allowed. If the Previous App-ID was blocked (or not allowed), the traffic remains blocked.

App-ID Update Safeguard is intended to be used as a transitional tool to keep your security policies operational, while retaining access to the latest App-ID updates in the content releases. As a best practice, you should update your security polices to use new App-IDs where applicable as soon as possible, and before installation of the following content release, at the latest. The subsequent content release refreshes the App-ID attribute associated with the previous version, as such, ***the previous App-ID does not roll over, but gets transitioned to the new App-ID, while new previous App-ID attributes are added for upcoming new App-IDs****.*

Consider the following use-cases for App-ID Update Safeguard:

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Table 2: App-ID Update Safeguard Use-Cases

Use-Case	Behavior	Example
Preventing Service Outages for Currently Allowed Traffic	Prevent business-critical applications from getting blocked when a content update changes the App-ID value. If a new App-ID is introduced but not yet added to a security policy, the firewall performs a second policy lookup using the application's previous identities (e.g., SSL or web-browsing). If those previous identities were allowed, the new App-ID is also allowed.	Palo Alto Networks releases a new App-ID for `acme-app`. Previously, this traffic was identified as ssl and web-browsing. The firewall interprets `acme-app` as not explicitly allowed. It checks the previous IDs (ssl and web-browsing). Since the policy has a rule allowing SSL and web-browsing, the `acme-app` traffic is allowed, preventing the outage.
Maintaining Security	Ensures that App-ID Safeguard does not inadvertently open security holes within your policy. For a new App-ID to be allowed, all of its previous App-IDs *must* be allowed by policy. If any previous App-ID is blocked (explicitly or via default deny), the new App-ID remains blocked.	Traffic for `acme-docs-editing` has a list of previous App-IDs that includes web-browsing, ssl, websocket, and unknown-udp and the administrator follows best practices and creates a rule to explicitly deny unknown traffic. When the user tries to edit an ACME doc, the NGFW detects ACME Docs Editing and checks the previous IDs. Because one of them (unknown-udp) is explicitly denied by policy, the firewall blocks the specific ACME Docs Editing traffic.
Explicit Policy Override	Explicit administrative actions *always* take precedence over the App-ID Safeguard logic. If an administrator adds the specific new App-ID to a rule, the firewall uses that rule immediately during the first lookup.	The administrator proactively adds the new or updated `acme-docs-editing` App-ID to a specific allow security policy rule. The traffic is allowed via the explicit allow rule based on the current App-ID. The NGFW does not resort to

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Use-Case	Behavior	Example
		using the **Previous App-ID** logic.
Visibility and Transition Planning	Provides a transition period for administrators to update their policies. By using the Application Command Center (ACC) and logs, administrators can identify which applications are relying on the safeguard and update their rules before the next content update.	After an update, the administrator checks the ACC widget named **Rules allowing apps based on previous App-ID**. If `acme-app` is currently being allowed only because its previous IDs (ssl, web-browsing) are allowed. They can use this information to create a new rule explicitly allowing ACME App to ensure permanent access, independent of the App-ID Safeguard feature.

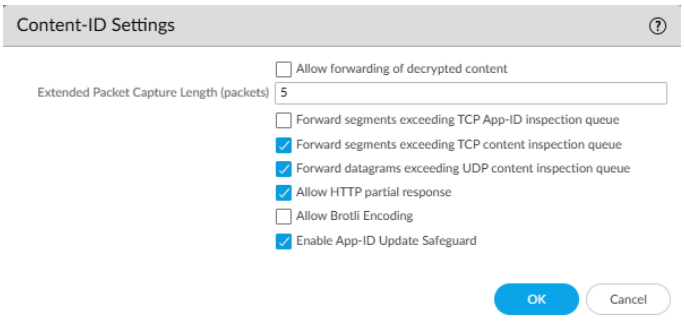
STEP 1 | Verify that you have PAN-OS 12.1.5 or later installed.

STEP 2 | Verify that Applications and Threats content update 9080-9944 or later is installed on your NGFW. While PAN-OS® 12.1.5 or later includes the App-ID™ Update Safeguard feature functionality, you cannot use it without Content Update 9080-9944 or later, as earlier content updates do not include the Previous App-ID attributes used by Update Safeguard.

STEP 3 | Log in to the PAN-OS web interface.

STEP 4 | Select **Device** > **Setup** > **Content-ID** > **Content-ID Settings** and edit the settings.

STEP 5 | Enable or disable **Enable App-ID Update Safeguard**.



- *This setting is disabled by default, as the secondary lookup used to reference the previous App-ID can result in a performance hit.*

STEP 6 | (IMPORTANT) **App-ID Update Safeguard** is a temporary tool to maintain connectivity while you transition to the latest signatures. As a best practice, make sure to review and update

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your security policies to the new App-IDs before the next content release, as previous attributes do not roll over once a new update is installed.

STEP 7 | (Recommended) Review App-ID applications based on their **Previous App-ID** attribute through the following entry points:

- **ACC**—The Application Command Center (ACC) includes widgets such as **Rules allowing apps based on previous App-ID.** These help you quickly identify which policies need to be

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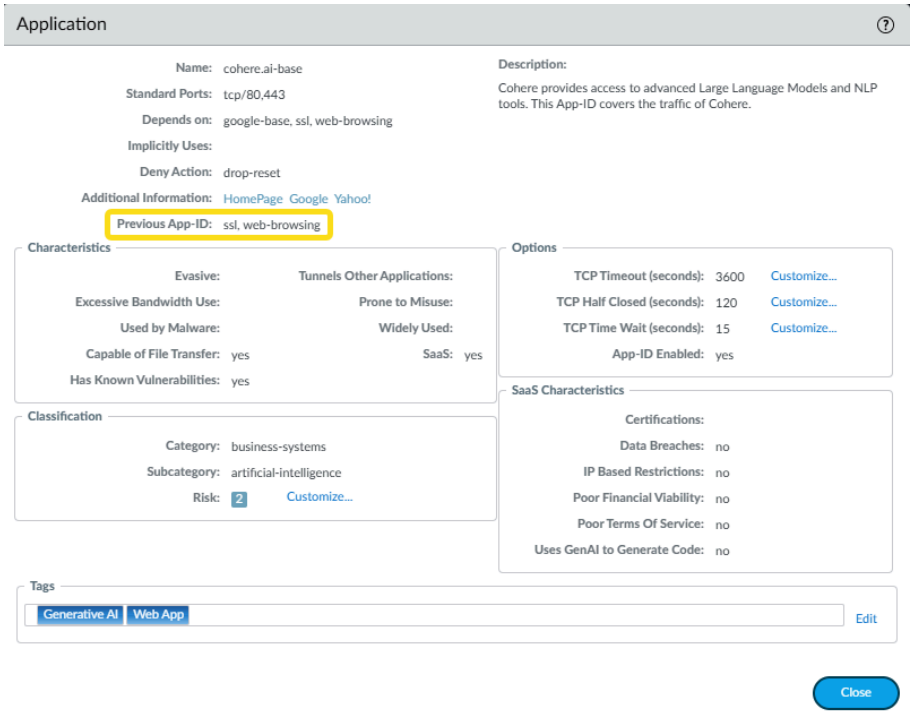
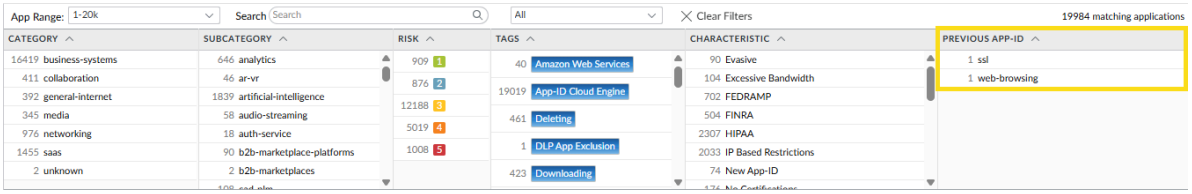
updated before the next content release. Refer to Monitor Impacted Rules and Applications for more information.

When App-ID Safeguard is initially turned on, it is highly recommended that the administrator review the rules and applications being allowed by the Previous App-ID, and, if necessary, make changes to any impacted rules.

- **Application Objects** —When you click on a specific application to view its details, you will see a new attribute field named **Previous App-ID** listed alongside other standard attributes.



You can also search for applications based on this new attribute. The search bar in the Applications tab supports filtering by **Previous App-ID**, allowing you to find all applications that used to be identified as a specific legacy App-ID.



- Log Viewer—Logs display the **Application** (App-ID) and the associated **Rule**, but also include the corresponding **Previous App-ID**, showing the specific attribute that allowed the traffic to pass.

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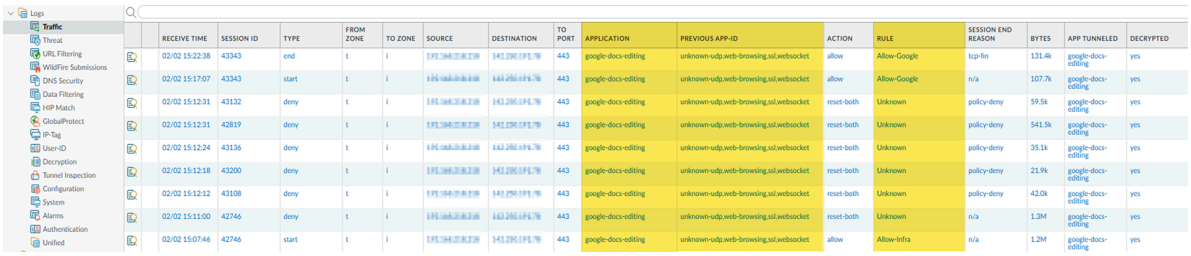

	RECEIVE TIME	SESSION ID	TYPE	FROM ZONE	TO ZONE	SOURCE	DESTINATION	TO PORT	APPLICATION	PREVIOUS APP-ID	ACTION	RULE	SESSION END REASON	BYTES	APP TUNNELED	DECRYPTED
	02/02 15:22:38	43343	end	t	i			443	google-docs-editing	unknown-udp,web-browsing,ssl,websocket	allow	Allow-Google	tcp-fin	131.4k	google-docs-editing	yes
	02/02 15:17:07	43343	start	t	i			443	google-docs-editing	unknown-udp,web-browsing,ssl,websocket	allow	Allow-Google	n/a	107.7k	google-docs-editing	yes
	02/02 15:12:31	43132	deny	t	i			443	google-docs-editing	unknown-udp,web-browsing,ssl,websocket	reset-both	Unknown	policy-deny	59.5k	google-docs-editing	yes
	02/02 15:12:31	42819	deny	t	i			443	google-docs-editing	unknown-udp,web-browsing,ssl,websocket	reset-both	Unknown	policy-deny	541.5k	google-docs-editing	yes
	02/02 15:12:24	43136	deny	t	i			443	google-docs-editing	unknown-udp,web-browsing,ssl,websocket	reset-both	Unknown	policy-deny	35.1k	google-docs-editing	yes
	02/02 15:12:18	43200	deny	t	i			443	google-docs-editing	unknown-udp,web-browsing,ssl,websocket	reset-both	Unknown	policy-deny	21.9k	google-docs-editing	yes
	02/02 15:12:12	43108	deny	t	i			443	google-docs-editing	unknown-udp,web-browsing,ssl,websocket	reset-both	Unknown	policy-deny	42.0k	google-docs-editing	yes
	02/02 15:11:00	42746	deny	t	i			443	google-docs-editing	unknown-udp,web-browsing,ssl,websocket	reset-both	Unknown	n/a	1.3M	google-docs-editing	yes
	02/02 15:07:46	42746	start	t	i			443	google-docs-editing	unknown-udp,web-browsing,ssl,websocket	allow	Allow-Infra	n/a	1.2M	google-docs-editing	yes

- **Review New App-IDs in a Content Release**—Before installing a new content update, you can **Review Apps**, which includes **New and Modified Applications since last installed content**. The application preview window displays the **Previous App-ID** for a selected application.

Monitor Impacted Rules and Applications

Where Can I Use This?	What Do I Need?
• Prisma Access (Managed by Panorama) • NGFW (Managed by PAN-OS or Panorama)	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

To manage transitions when using App-ID Safeguard, the ACC (Application Command Center) includes two dedicated widgets: **Applications Allowed by Previous App-ID** and **Rules Allowing Apps based on Previous App-ID**. These widgets function as a diagnostic lens, filtering out the noise of standard traffic to show you exactly where your policy might be overly permissive or relying on legacy classifications. This provides a specialized view within the ACC to initiate an audit of your rulebase.

Similar to the standard **Application Usage** widget, it displays a breakdown of applications and session counts. However, it specifically isolates sessions where the security rule match was triggered by the *pre-transition* App-ID.

STEP 1 | Access the NGFW Application Command Center.

STEP 2 | Create a dedicated ACC custom tab to monitor Previous App-IDs or select and edit an existing tab to add App-ID Safeguard widgets.

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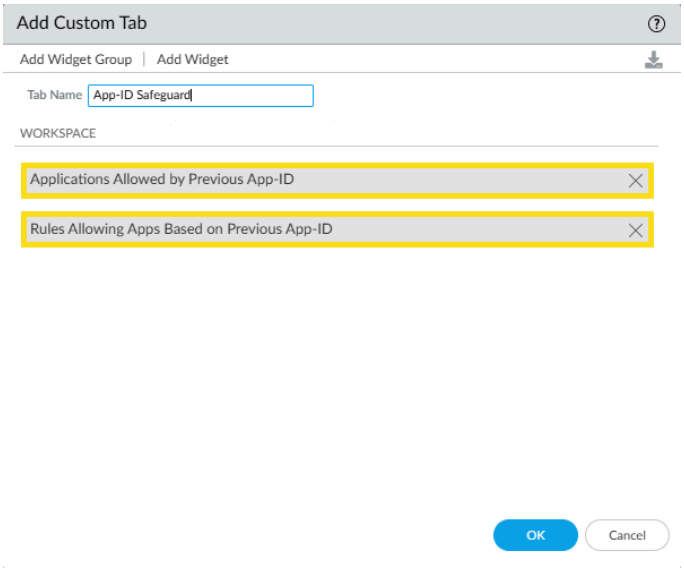
STEP 3 | **Add** the **Applications Allowed by Previous App-ID** and **Rules Allowing Apps Based on Previous App-ID** to the custom (or previously existing) ACC tab and click **OK**. If you are creating a new **Custom Tab**, you must also provide a **Tab Name**.



If the App-ID Safeguard feature is disabled in your Content-ID settings, these widgets will remain empty.

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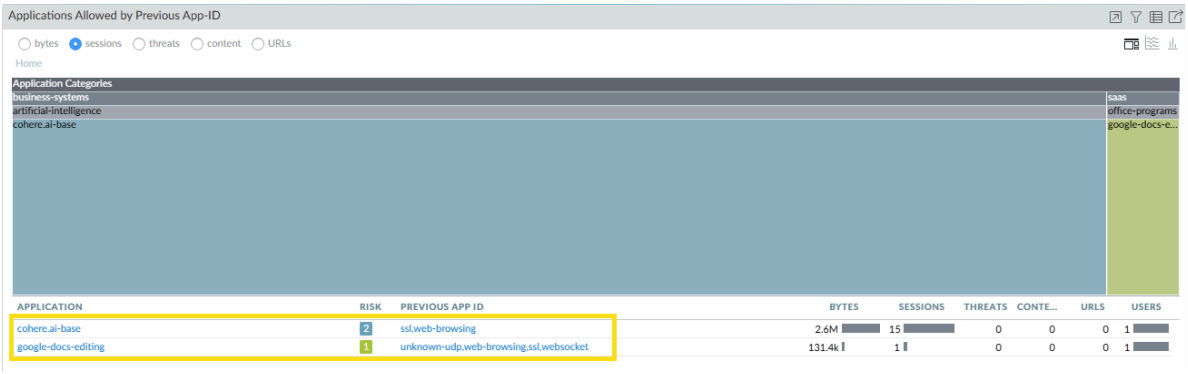
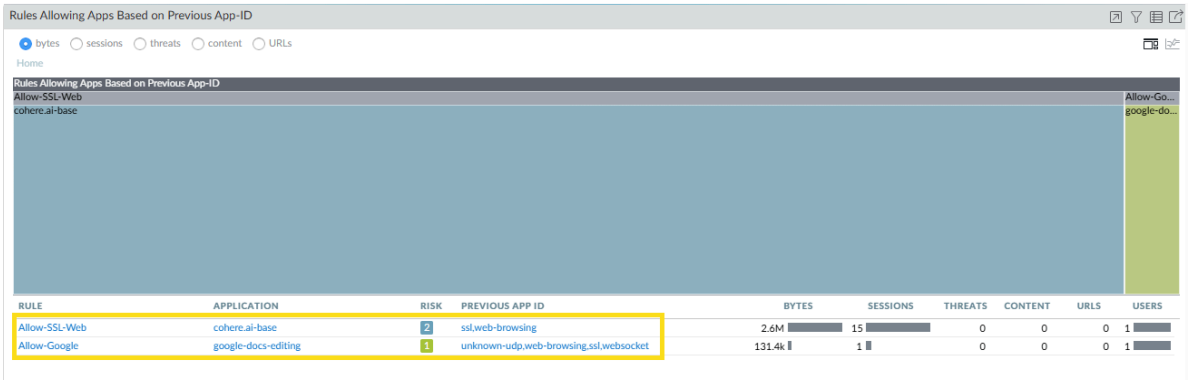
STEP 4 | Review the App-ID Safeguard widget to investigate the applications that are allowed by the Previous App-ID functionality.

- **Applications Allowed by Previous App-ID**



  - Generates a list of new and modified applications (App-IDs) that are allowed as a result of the Previous App-IDs that are allowed in the current security policy rule.

- **Rules Allowing Apps Based on Previous App-ID**



  - Displays the security policy rules that allowed the traffic to pass by leveraging new and modified applications (App-IDs).
  - Each **Rule** entry includes a **Previous App-ID** that is currently specified in the security policy rule.
  - The **Previous App-ID** corresponds to an **Application** that has been allowed due to the Previous App-IDs that are specified in the security policy rule.

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Use Application Objects in Policy

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

An application Object is a configuration element that identifies specific applications based on their characteristics, such as name, category, subcategory, and risk level, rather than just ports and protocols. These objects are used in security policies to control how traffic associated with those applications is handled.

You can use application objects to define how your security policy handles applications. Consider the following use-cases:

- **Create an Application Group**

  An application group is an object that contains applications that you want to treat similarly in policy. Application groups are useful for enabling access to applications that you explicitly sanction for use within your organization. Grouping sanctioned applications simplifies administration of your rulebases. Instead of having to update individual policy rules when there is a change in the applications you support, you can update only the affected application groups.

  When deciding how to group applications, consider how you plan to enforce access to your sanctioned applications and create an application group that aligns with each of your policy goals. For example, you might have some applications that you will only allow your IT administrators to access, and other applications that you want to make available for any known user in your organization. In this case, you would create separate application groups for each of these policy goals. Although you generally want to enable access to applications on the default

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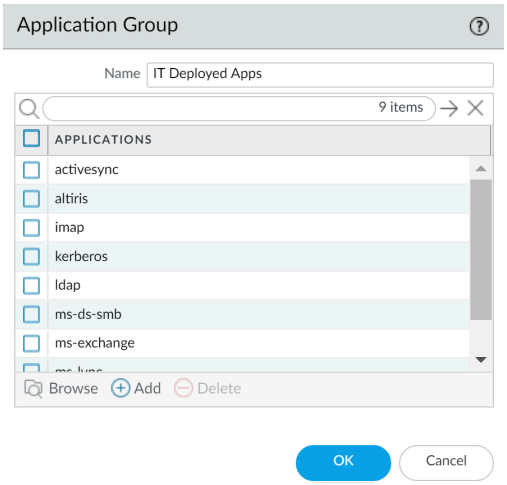
port only, you may want to group applications that are an exception to this and enforce access to those applications in a separate rule.

1. Select **Objects** > **Application Groups**.
2. **Add** a group and give it a descriptive **Name**.
3. (Optional) Select **Shared** to create the object in a shared location for access as a shared object in Panorama or for use across all virtual systems in a multiple virtual system firewall.
4. **Add** the applications you want in the group and then click **OK**.



5. **Commit** the configuration.

- **Create an Application Filter**

  An application filter is an object that dynamically groups applications based on application attributes that you define, including category, subcategory, technology, risk factor, and characteristic. This is useful when you want to safely enable access to applications that you do not explicitly sanction, but that you want users to be able to access. For example, you may want to enable employees to choose their own office programs (such as Evernote, Google Docs, or Microsoft Office 365) for business use. To safely enable these types of applications, you could create an application filter that matches on the Category **business-systems** and the Subcategory **office-programs**. As new applications office programs emerge and new App-IDs get created, these new applications will automatically match the filter you defined; you will not have to make any additional changes to your policy rulebase to safely enable any application that matches the attributes you defined for the filter.

  1. Select **Objects** > **Application Filters**.
  2. **Add** a filter and give it a descriptive **Name**.
  3. (Optional) Select **Shared** to create the object in a shared location for access as a shared object in Panorama or for use across all virtual systems in a multiple virtual system firewall.
  4. Define the filter by selecting attribute values from the Category, Subcategory, Technology, Risk, Characteristic, and Tags sections. (Tags can streamline Security policy rule creation and maintenance). As you select values, notice that the list of matching applications at the

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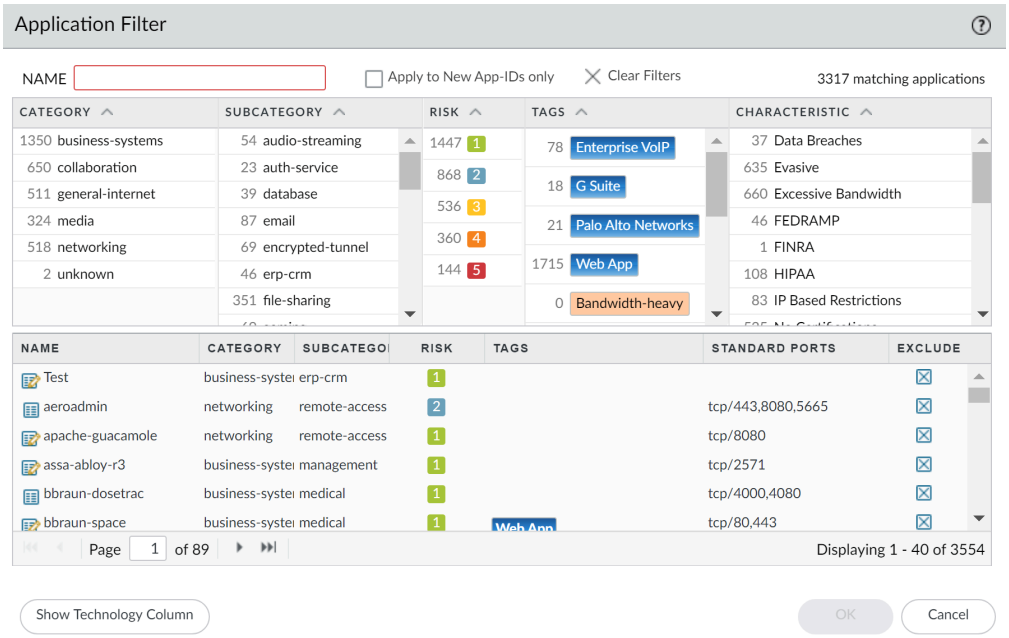
bottom of the dialog narrows. When you have adjusted the filter attributes to match the types of applications you want to safely enable, click **OK**.



5. **Commit** the configuration.

- Create a Custom Application
- Resolve Application Dependencies

Create a Custom Application

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

To safely enable applications you must classify all traffic, across all ports, all the time. With App-ID, the only applications that are typically classified as unknown traffic—tcp, udp or non-syn-tcp—in the ACC and the Traffic logs are commercially available applications that have not yet been added to App-ID, internal or custom applications on your network, or potential threats.

If you are seeing unknown traffic for a commercial application that does not yet have an App-ID, you can submit a request for a new App-ID here: http://researchcenter.paloaltonetworks.com/submit-an-application/.

To ensure that your internal custom applications do not show up as unknown traffic, create a custom application. You can then exercise granular policy control over these applications in order to minimize the range of unidentified traffic on your network, thereby reducing the attack surface. Creating a custom application also allows you to correctly identify the application in the ACC and Traffic logs, which enables you to audit/report on the applications on your network.

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To create a custom application, you must define the application attributes: its characteristics, category and sub-category, risk, port, timeout. In addition, you must define patterns or values that the firewall can use to match to the traffic flows themselves (the *signature*). Finally, you can attach the custom application to a security policy that allows or denies the application (or add it to an application group or match it to an application filter). You can also create custom applications to identify ephemeral applications with topical interest, such as ESPN3-Video for world cup soccer or March Madness.

In order to collect the right data to create a custom application signature, you'll need a good understanding of packet captures and how datagrams are formed. If the signature is created too broadly, you might inadvertently include other similar traffic; if it is defined too narrowly, the traffic will evade detection if it does not strictly match the pattern.

Custom applications are stored in a separate database on the firewall and this database is not impacted by the weekly App-ID updates.

The supported application protocol decoders that enable the firewall to detect applications that may be tunneling inside of the protocol include the following as of content release version 609: FTP, HTTP, IMAP, POP3, SMB, and SMTP.

The following is a basic example of how to create a custom application.

STEP 1 | Gather information about the application that you will be able to use to write custom signatures.

To do this, you must have an understanding of the application and how you want to control access to it. For example, you may want to limit what operations users can perform within the application (such as uploading, downloading, or live streaming). Or you may want to allow the application, but enforce QoS policing.

- Capture application packets so that you can find unique characteristics about the application on which to base your custom application signature. One way to do this is to run a protocol analyzer, such as Wireshark, on the client system to capture the packets between the client and the server. Perform different actions in the application, such as uploading and downloading, so that you will be able to locate each type of session in the resulting packet captures (PCAPs).
- Because the firewall by default takes packet captures for all unknown traffic, if the firewall is between the client and the server you can view the packet capture for the unknown traffic directly from the Traffic log.
- Use the packet captures to find patterns or values in the packet *contexts* that you can use to create signatures that will uniquely match the application traffic. For example, look for string patterns in HTTP response or request headers, URI paths, or hostnames. For information on the different string contexts you can use to create application signatures and where you can find the corresponding values in the packet, refer to Creating Custom Threat Signatures.

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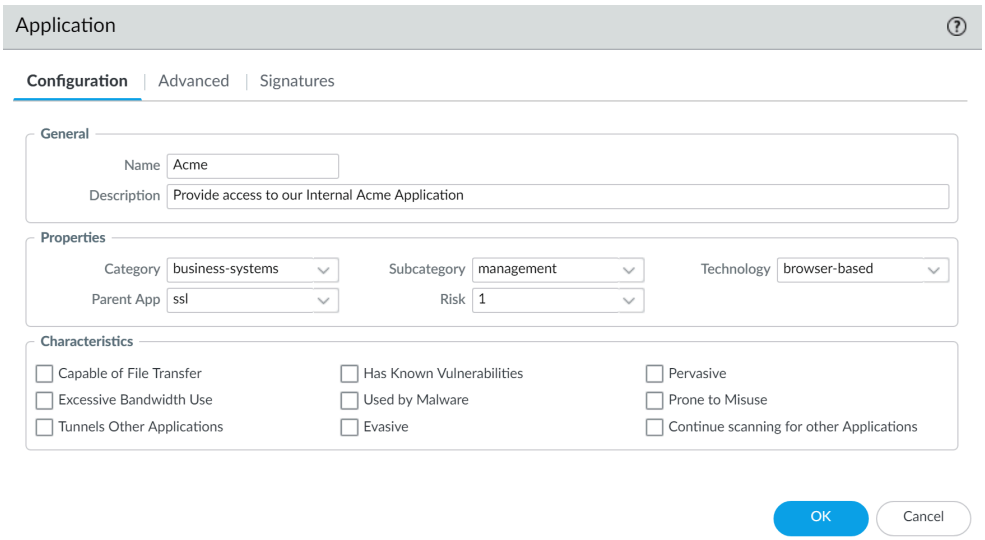
STEP 2 | Add the custom application.

1. Select **Objects** > **Applications** and click **Add**.
2. On the **Configuration** tab, enter a **Name** and a **Description** for the custom application that will help other administrators understand why you created the application.
3. (Optional) Select **Shared** to create the object in a shared location for access as a shared object in Panorama or for use across all virtual systems in a multiple virtual system firewall.
4. Define the application Properties and Characteristics.



STEP 3 | Define details about the application, such as the underlying protocol, the port number the application runs on, the timeout values, and any types of scanning you want to be able to perform on the traffic.

On the **Advanced** tab, define settings that will allow the firewall to identify the application protocol:

- Specify the default ports or protocol that the application uses.
- Specify the session timeout values. If you don't specify timeout values, the default timeout values will be used.
- Indicate any type of additional scanning you plan to perform on the application traffic.

For example, to create a custom TCP-based application that runs over SSL, but uses port 4443 (instead of the default port for SSL, 443), you would specify the port number. By adding the port number for a custom application, you can create policy rules that use the default port

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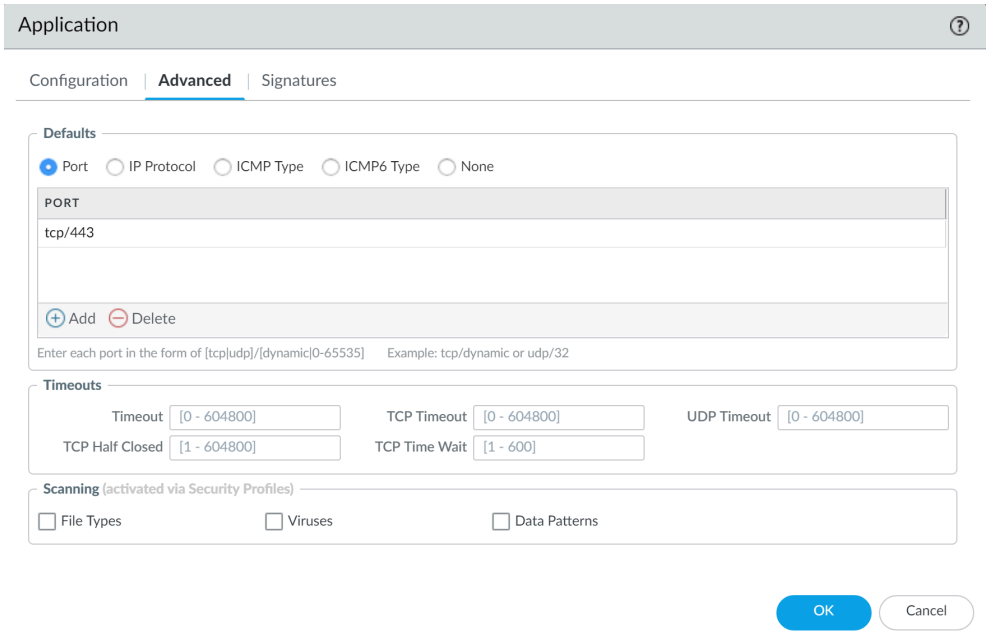
for the application rather than opening up additional ports on the firewall. This improves your security posture.



STEP 4 | Define the criteria that the firewall will use to match the traffic to the new application.

You will use the information you gathered from the packet captures to specify unique string context values that the firewall can use to match patterns in the application traffic.

1. On the **Signatures** tab, click **Add** and define a **Signature Name** and optionally a **Comment** to provide information about how you intend to use this signature.
2. Specify the **Scope** of the signature: whether it matches to a full **Session** or a single **Transaction**.
3. Specify conditions to define signatures by clicking **Add And Condition** or **Add Or Condition**.
4. Select an **Operator** to define the type of match conditions you will use: **Pattern Match** or **Equal To**.
   - If you selected **Pattern Match**, select the **Context** and then use a regular expression to define the **Pattern** to match the selected context. Optionally, click **Add** to define a qualifier/value pair. The **Qualifier** list is specific to the **Context** you chose.
   - If you selected **Equal To**, select the **Context** and then use a regular expression to define the **Position** of the bytes in the packet header to use match the selected context. Choose from **first-4bytes** or **second-4bytes**. Define the 4-byte hex value for the **Mask** (for example, 0xffffff00) and **Value** (for example, 0xaabbccdd).

   For example, if you are creating a custom application for one of your internal applications, you could use the **ssl-rsp-certificate Context** to define a pattern match

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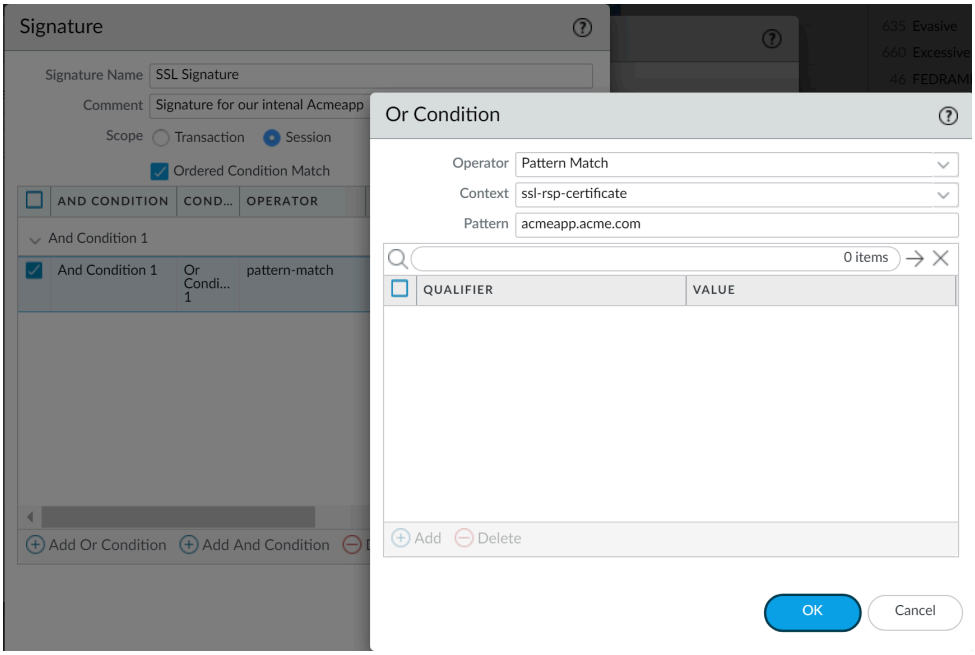
for the certificate response message of a SSL negotiation from the server and create a **Pattern** to match the commonName of the server in the message as shown here:



5. Repeat steps 4.c and 4.d for each matching condition.
6. If the order in which the firewall attempts to match the signature definitions is important, make sure the **Ordered Condition Match** check box is selected and then order the conditions so that they are evaluated in the appropriate order. Select a condition or a group and click **Move Up** or **Move Down**. You cannot move conditions from one group to another.
7. Click **OK** to save the signature definition.

STEP 5 | Save the application.

1. Click **OK** to save the custom application definition.
2. Click **Commit**.

STEP 6 | Validate that traffic matches the custom application as expected.

1. Select **Policies** > **Security** and **Add** a security policy rule to allow the new application.
2. Run the application from a client system that is between the firewall and the application and then check the Traffic logs (**Monitor** > **Traffic**) to make sure that you see traffic matching the new application (and that it is being handled per your policy rule).

Resolve Application Dependencies

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

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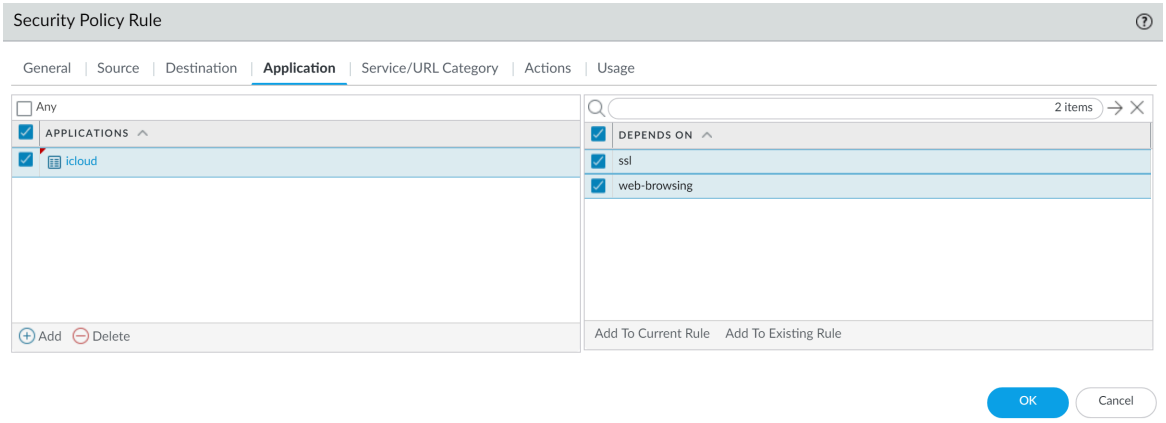
You can see application dependencies when you create a new Security policy rule and when performing Commits. When a policy does not include all application dependencies, you can directly access the associated Security policy rule to add the required applications.

STEP 1 | Create a security policy rule.

STEP 2 | Specify the application that the rule will allow or block.

1. In the **Applications** tab, **Add** the **Application** you want to safely enable. You can select multiple applications or you can use application groups or application filters.
2. View dependencies for selected applications and **Add To Current Rule** or **Add To Existing Rule**.



3. If adding to an existing rule, **Select Rule** and click **OK**.

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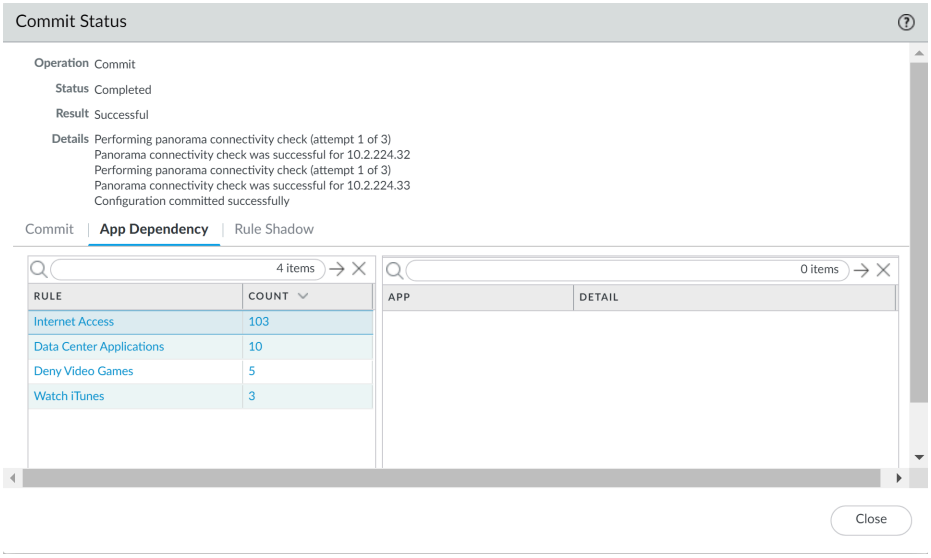
STEP 3 | Click **OK** and **Commit** your changes.

1. Review any Commit warnings in the **App Dependency** tab.



2. Select the **Count** to view the application dependencies not included.
3. Select the **Rule** name to open the policy and add the dependencies.

   Resolve any dependent applications or they'll continue to generate warnings on Commits.

4. Click **OK** and **Commit** your changes.

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Safely Enable Applications on Default Ports

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

Applications running on unusual ports can indicate an attacker that is attempting to circumvent traditional port-based protections. Application-default is a feature of Palo Alto Networks firewalls that gives you an easy way to prevent this type of evasion and safely enable applications on their most commonly-used ports. Application-default is a best practice for application-based security policies—it reduces administrative overhead, and closes security gaps that port-based policy introduces:

- **Less overhead**—Write simple application-based security policy rules based on your business needs, instead of researching and maintaining application-to-port mappings. We've defined the default ports for all applications with an App-ID.
- **Stronger security**—Enabling applications to run only on their default ports is a security best practice. Application-default helps you to make sure that critical applications are available without compromising security if an application is behaving in an unexpected way.

  Additionally, the default ports an application uses can sometimes depend on whether the application is encrypted or cleartext. Port-based policy requires you to open all the default ports an application might use to account for encryption. Open ports introduce security gaps that an attacker can leverage to bypass your security policy. However, application-default differentiates between encrypted and clear-text application traffic. This means that it can enforce the default port for an application, regardless of whether it is encrypted or not.

  For example, without application-default, you would need to open ports 80 and 443 to enable web-browsing traffic—you'd be allowing both cleartext and encrypted web-browsing traffic on both ports. With application-default turned on, the firewall strictly enforces cleartext web-browsing traffic only on port 80 and SSL-tunneled traffic only on port 443.

To see the ports that an application uses by default, you can visit Applipedia or select **Objects** > **Applications**. Application details include the application's standard port—the port it most commonly uses when in cleartext. For web-browsing, SMTP, FTP, LDAP, POP3, and IMAP details also include the application's secure port—the port the application uses when encrypted.

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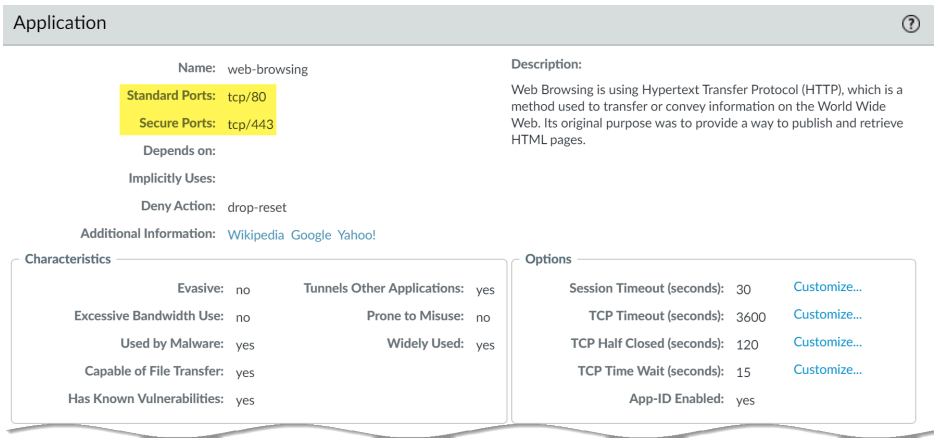
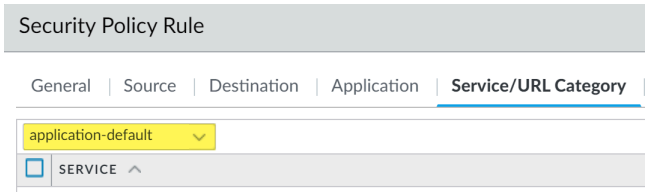


Select **Policy** > **Security** and add or a modify a rule to enforce applications only on their default port(s):



*Using application-default as part of an application-based security policy and with SSL decryption is a best practice. Additionally, if you have existing security policy rules that control web-browsing traffic with the **Service** set to service-http and service-https, you should update those rules to use application-default instead.*

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Applications with Implicit Support

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

When creating a policy to allow specific applications, you must also be sure that you are allowing any other applications on which the application depends. In many cases, you do not have to explicitly allow access to the dependent applications for the traffic to flow because the firewall is able to determine the dependencies and allow them implicitly. This implicit support also applies to custom applications that are based on HTTP, SSL, MS-RPC, or RTSP. Applications for which the firewall cannot determine dependent applications on time will require that you explicitly allow the dependent applications when defining your policies. You can determine application dependencies from within your application-based security policy workflow using one of the following:

- Policy Optimizer
- Create Application Filters
- Resolve Application Dependencies

Applipedia is also available if needed.

The following table lists the applications for which the firewall has implicit support (as of Content Update 595).

Application	Implicitly Supports
360-safeguard-update	http
apple-update	http
apt-get	http
as2	http
avg-update	http
avira-antivir-update	http, ssl
blokus	rtmp
bugzilla	http
clubcooee	http

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Application	Implicitly Supports
corba	http
cubby	http, ssl
dropbox	ssl
esignal	http
evernote	http, ssl
ezhelp	http
facebook	http, ssl
facebook-chat	jabber
facebook-social-plugin	http
fastviewer	http, ssl
forticlient-update	http
good-for-enterprise	http, ssl
google-cloud-print	http, ssl, jabber
google-desktop	http
google-talk	jabber
google-update	http
gotomypc-desktop-sharing	citrix-jedi
gotomypc-file-transfer	citrix-jedi
gotomypc-printing	citrix-jedi
hipchat	http
iheartradio	ssl, http, rtmp
infront	http
instagram	http, ssl

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Application	Implicitly Supports
issuu	http, ssl
java-update	http
jepptech-updates	http
kerberos	rpc
kik	http, ssl
lastpass	http, ssl
logmein	http, ssl
mcafee-update	http
megaupload	http
metatrader	http
mocha-rdp	t_120
mount	rpc
ms-frs	msrpc
ms-rdp	t_120
ms-scheduler	msrpc
ms-service-controller	msrpc
nfs	rpc
oovoo	http, ssl
paloalto-updates	ssl
panos-global-protect	http
panos-web-interface	http
pastebin	http
pastebin-posting	http

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Application	Implicitly Supports
pinterest	http, ssl
portmapper	rpc
prezi	http, ssl
rdp2tcp	t_120
renren-im	jabber
roboform	http, ssl
salesforce	http
stumbleupon	http
supremo	http
symantec-av-update	http
trendmicro	http
trillian	http, ssl
twitter	http
whatsapp	http, ssl
xm-radio	rtsp

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Security Policy Rule Optimization

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

Policy Optimizer provides a simple workflow to migrate your legacy Security policy rulebase to an App-ID based rulebase, which improves your security by reducing the attack surface and gaining visibility into applications so you can safely enable them. Policy Optimizer identifies port-based rules so you can convert them to application-based allow rules or add applications from a port-based rule to an existing application-based rule without compromising application availability. It also identifies over-provisioned App-ID based rules (App-ID rules configured with unused applications). Policy Optimizer helps you prioritize which port-based rules to migrate first, identify application-based rules that allow applications you don't use, and analyze rule usage characteristics such as hit count.

Converting port-based rules to application-based rules improves your security posture because you select the applications you want to allow and deny all other applications, so you eliminate unwanted and potentially malicious traffic from your network. Combined with restricting application traffic to its default ports (set the Service to **application-default**), converting to application-based rules also prevents evasive applications from running on non-standard ports.

You can use this feature on:

- Firewalls that run PAN-OS version 9.0 and have App-ID enabled.
- Panorama running PAN-OS version 9.0. You don't have to upgrade firewalls that Panorama manages to use the **Policy Optimizer** capabilities. However, to use the **Rule Usage** capabilities (Monitor Policy Rule Usage), managed firewalls must run PAN-OS 8.1 or later. If managed firewalls connect to Log Collectors, those Log Collectors must also run PAN-OS version 9.0. Managed PA-7000 Series firewalls that have a Log Processing Card (LPC) can also run PAN-OS 8.1 (or later).
- For Strata Logging Service compatibility, Panorama running PAN-OS 10.0.3 or later with the Cloud Services plugin 2.0 Innovation or later installed.
- Cloud Managed Prisma Access and Panorama Managed Prisma Access in PAN-OS 10.2.4 or later with Cloud Service Plugin 5.0 or later.

PA-7000 Series Firewalls support two logging cards, the PA-7000 Series Firewall Log Processing Card (LPC) and the high-performance PA-7000 Series Firewall Log Forwarding Card (LFC). Unlike the LPC, the LFC does not have disks to store logs locally. Instead, the LFC forwards all logs to one or more external logging systems, such as Panorama or a syslog server. If you use the LFC, the application usage information for Policy Optimizer does not display on the firewall because traffic logs aren't stored locally. If you use the LPC, the traffic logs are stored locally on the firewall, so the application usage information for Policy Optimizer displays on the firewall.

Use this feature to:

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- **Migrate port-based rules to application-based rules**—Instead of combing through traffic logs and manually mapping applications to port-based rules, use Policy Optimizer to identify port-based rules and list the applications that matched each rule, so you can select the applications you want to allow and safely enable them. Converting your legacy port-based rules to application-based allow rules supports your business applications and enables you to block any applications associated with malicious activity.
- **Identify over-provisioned application-based rules**—Rules that are too broad allow applications you don't use on your network, which increases the attack surface and the risk of inadvertently allowing malicious traffic.

  Remove unused applications from Security policy rules to reduce the attack surface and keep the rulebase clean. Don't allow applications that nobody uses on your network.
- **Add App-ID Cloud Engine (ACE) applications to Security policy rules**—If you have a SaaS Security Inline subscription, you can use Policy Optimizer's New App Viewer to manage cloud-delivered App-IDs in Security policy. The ACE documentation describes how to use Policy Optimizer to gain visibility into and control cloud-delivered App-IDs.

  The Policy Optimizer examples in this section do not show the New App Viewer because they depict firewalls that do not have a SaaS Security Inline subscription.

To migrate a configuration from a legacy firewall to a Palo Alto Networks device, see Best Practices for Migrating to Application-Based Policy.

You can't sort Security policy rules in **Security** > **Policies** because sorting would change the rule order in the rulebase. However, under **Polices** > **Security** > **Policy Optimizer**, Policy Optimizer provides sorting options that don't affect the rule order, so you can sort rules to prioritize which rules to convert or clean up first. You can sort rules by the amount of traffic during the past 30 days, the number of applications seen on the rule, the number of days with no new applications, and the number of applications allowed (for over-provisioned rules).

You can use Policy Optimizer in other ways as well, including validating pre-production rules and troubleshooting existing rules. Note that Policy Optimizer honors only **Log at Session End** and ignores **Log at Session Start** to avoid counting transient applications on rules.

*(PAN-OS 12.1.2 and later versions) To ensure the best performance, customize your view to display the **Policy Optimizer**, only when you need it. The system fetches data for this component on-demand, which prevents automatic data fetching for the columns and avoids system slowdowns.*

Due to resource constraints, VM-50 Lite virtual firewalls don't support Policy Optimizer.

- Sorting and Filtering Security Policy Rules
- Clear Application Usage Data
- Migrate Port-Based to App-ID Based Security Policy Rules
- Rule Cloning Migration Use Case: Web Browsing and SSL Traffic
- Add Applications to an Existing Rule

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- Identify Security Policy Rules with Unused Applications
- High Availability for Application Usage Statistics
- How to Disable Policy Optimizer

Sorting and Filtering Security Policy Rules

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

You can filter Security policy rules to see the port-based rules, which are rules with no applications configured on them (**Policies** > **Security** > **Policy Optimizer** > **No App Specified**). You can also filter to see the rules that have applications configured on them, but traffic only matches some of the configured applications—the rule is over-provisioned and includes applications that aren't seen on the rule (**Policies** > **Security** > **Policy Optimizer** > **Unused Apps**). In addition, if you have a SaaS Security Inline license, you can use the New App Viewer to filter rules that have seen new App-ID Cloud Engine (ACE) applications (see the ACE documentation for how to do this). You can sort the filtered policy rules based on different types of statistics to help prioritize which rules to convert from port-based to application-based rules or to clean up first.

*You can't filter or sort rules in **Policies** > **Security** because that would change the order of the policy rules in the rulebase. Filtering and sorting **Policies** > **Security** > **Policy Optimizer** > **No App Specified**, **Policies** > **Security** > **Policy Optimizer** > **Unused Apps**, and **Policies** > **Security** > **Policy Optimizer** > **New App Viewer** (if you have a SaaS Inline Security subscription) does not change the order of the rules in the rulebase.*

You can click several column headers to sort rules based on application usage statistics. In addition, you can View Policy Rule Usage to help identify and remove unused rules to reduce security risks and keep your policy rule base organized. Rule usage tracking allows you to quickly validate new rule additions and rule changes and to monitor rule usage for operations and troubleshooting tasks.

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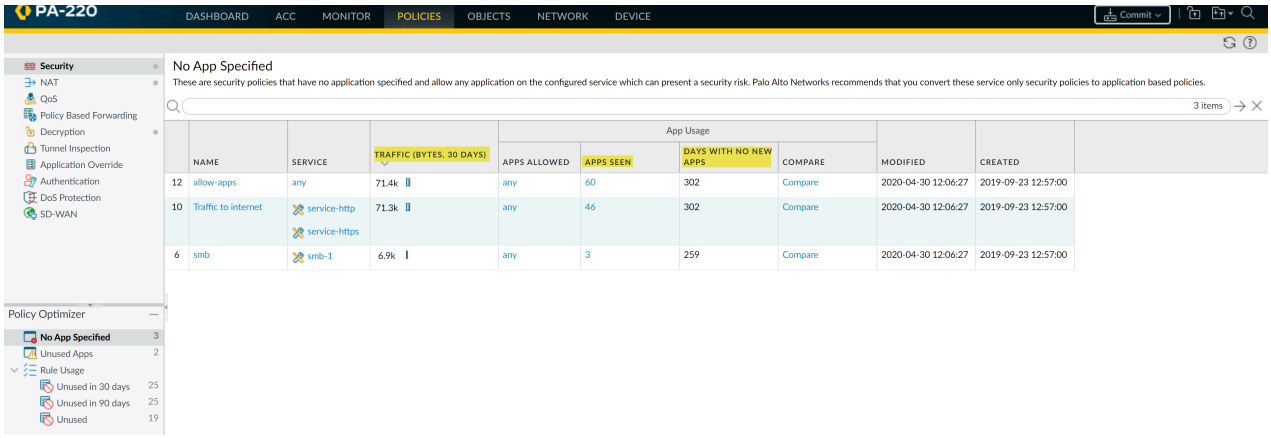

- **Traffic (Bytes, 30 days)**—The amount of traffic seen on the rule over the last 30 days. The 30-day window places rules that *currently* match the most traffic at the top of the list by default (a longer time frame places more emphasis on older rules that would remain at the top of the list because they have large cumulative totals even though they may no longer see much traffic). Click to reverse the order.
- **Apps Seen**—Place the rules with the most or least applications seen at the top. The firewall never automatically purges the application data.

  *The firewall updates **Apps Seen** approximately every hour. However, if there is a large volume of application traffic or a large number of rules, it may take longer than an hour to update. After you add an application to a rule, wait at least an hour before running Traffic logs to see the application's log information.*
- **Days with No New Apps**—Place the rules with the most or least days since the last new application matched the rule at the top.
- (Unused Apps only) **Apps Allowed**—Place the rules with the most or least applications configured on the rule at the top.

Application usage statistics only count applications for rules that meet the following criteria:

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- The rule's Action must be **Allow**.
- The rule's Log Setting must be **Log at Session End** (this is the default Log Setting). Rules that **Log at Session Start** are ignored to prevent counting transient applications.
- Valid traffic must match the rule. For example, if the session ends before enough traffic passes through the firewall to identify the application, it is not counted. The following traffic types are not valid and therefore don't count for Policy Optimizer statistics:
  - Insufficient-data
  - Not-applicable
  - Non-syn-tcp
  - Incomplete

  You can filter the Traffic logs (**Monitor** > **Logs** > **Traffic**) to see traffic identified as one of these types. For example, to see all traffic identified as incomplete, use the filter **`(app eq incomplete)`**.

If these criteria aren't met, the application isn't counted for statistics such as **Apps Seen**, doesn't affect statistics such as **Days with No New Apps**, and doesn't appear in lists of applications.

The firewall doesn't track application usage statistics for the interzone-default and intrazone-default Security policy rules.

If the UUID of a rule changes, the application usage statistics for that rule reset because the UUID change makes the firewall see the rule as a different (new) rule.

To see and sort the applications seen on a rule, in the rule's row, click **Compare** or click the number in **Apps Seen**.

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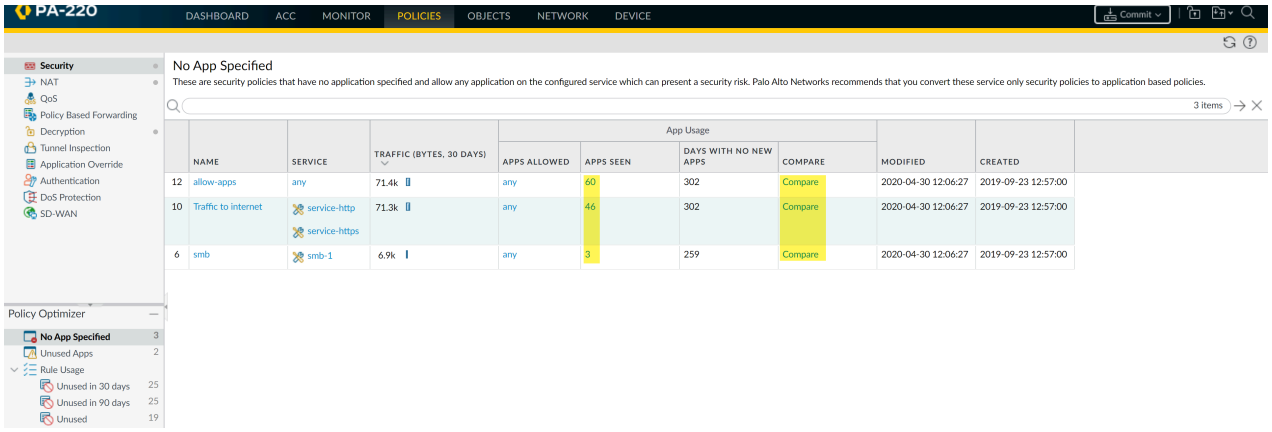

For the rules you see in **Policies** > **Security** > **Policy Optimizer** > **No App Specified** and **Policies** > **Security** > **Policy Optimizer** > **Unused Apps**, clicking **Compare** or the **Apps Seen** number brings up **Applications & Usage**, which gives you a view of the applications seen on the rule and the ability to sort them. **Applications & Usage** is also where you Migrate Port-Based to App-ID Based Security Policy Rules and remove unused applications from rules.

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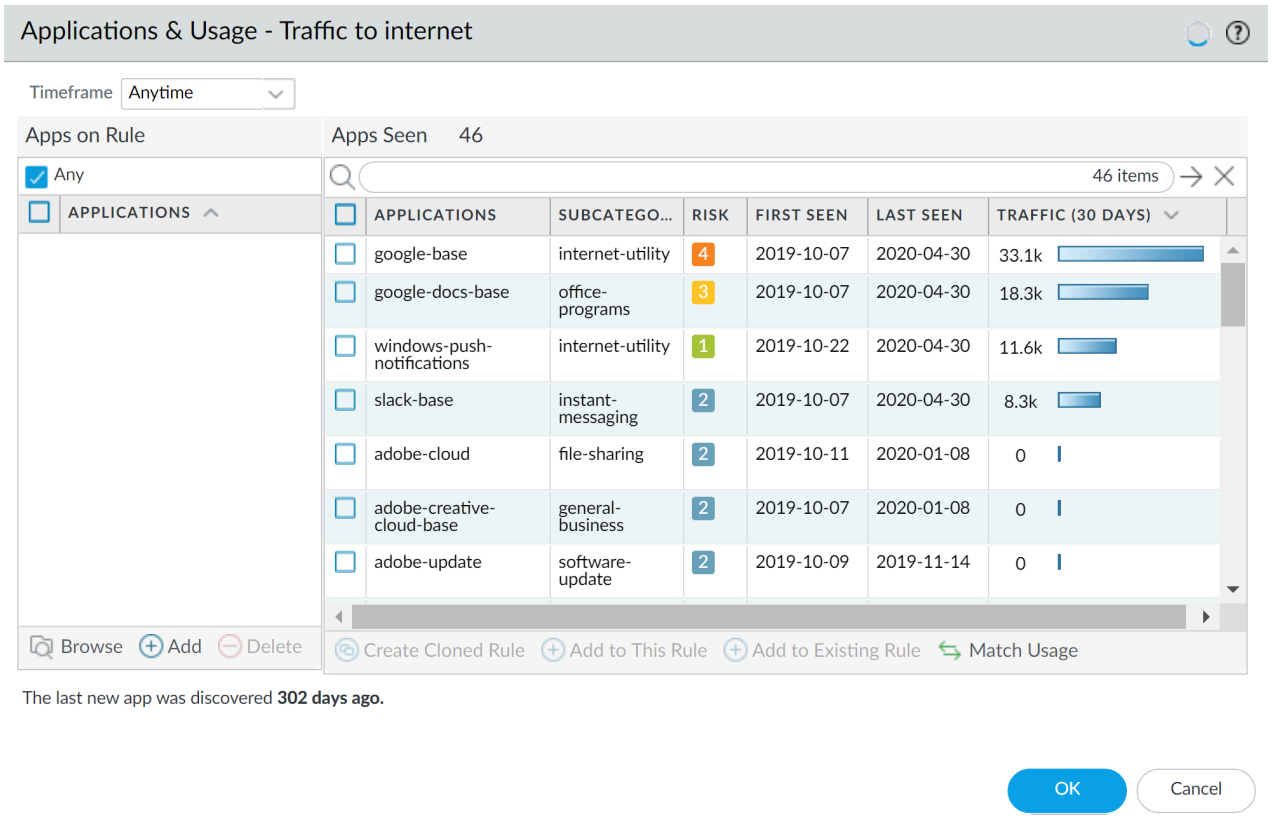

You can sort the applications seen on the rule by all six of the **Apps Seen** statistics (**Apps Seen** is not updated in real time and takes an hour or longer to update, depending on the volume of traffic and number of rules).

- **Applications**—Alphabetical by application name. If you configure specific ports or port ranges for a rule's Service (the Service cannot be **any**), and there are standard (application default) ports for the application, and the configured ports don't match the application-default ports, then a yellow, triangular warning icon appears next to the application.
- **Subcategory**—Alphabetical by application subcategory, derived from the application content metadata.
- **Risk**—According to the risk rating of the application.
- **First Seen**—The first day the application was seen on the rule. The time stamp resolution is by the day only (not hourly).
- **Last Seen**—The last day the application was seen on the rule. The time stamp resolution is by the day only (not hourly).
- **Traffic (30 days)**—Traffic in bytes that matched the rule over the last 30 days is the default sorting method.

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Set the **Timeframe** to display statistics for a particular time period—**Anytime**, the **Past 7 days**, the **Past 15 days**, or the **Past 30 days**.

Traffic (30 days) *always displays only the last 30 days of traffic in bytes. Changing the* ***Timeframe*** *does not change the duration of the* ***Traffic (30 days)*** *bytes measurement.*

Clicking the column header orders the display and clicking the same column again reverses the order. For example, click **Risk** to sort applications from low risk to high risk. Click **Risk** again to sort applications from high risk to low risk.

The firewall doesn't report application usage statistics in real time for Policy Optimizer, so it isn't a replacement for running reports.

- The firewall updates **Apps Allowed**, **Apps Seen**, and the applications listed in **Applications & Usage** approximately every hour, not in real time. If there is a large amount of traffic or a large number of rules, updates may take longer. After you add an application to a rule, wait at least an hour before running Traffic logs to see the application's log information.

  The firewall updates **Apps Seen** approximately every hour. However, if there is a large volume of application traffic or a large number of rules, it may take longer than an hour to update. After you add an application to a rule, wait at least an hour before running Traffic logs to see the application's log information.
- The firewall updates **Days with No New Apps** and also **First Seen** and **Last Seen** on **Applications & Usage** once per day, at midnight device time.
- For rules with large numbers of applications seen, it may take longer to process application usage statistics.
- For Security policy rulebases with large numbers of rules that have many applications, it may take longer to process application usage statistics.
- For firewalls managed by Panorama, application usage data is visible only for rules Panorama pushes to the firewalls, not for rules configured locally on individual firewalls.

Clear Application Usage Data

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

You can use a CLI command to clear application usage data for an individual Security policy rule and reset **Apps Seen** and other application usage data.

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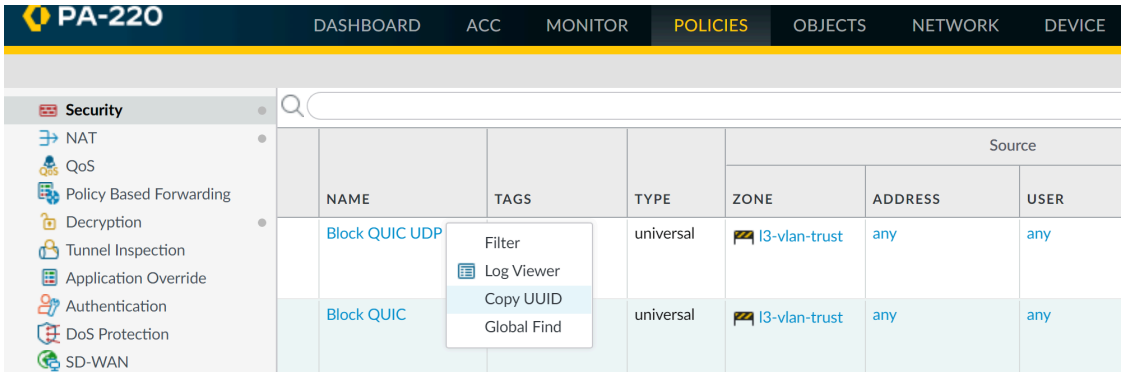
STEP 1 | Find the UUID of the Security policy rule whose application usage data you want to clear.

There are two ways to find the UUID in the UI:

- In **Policies** > **Security**, copy the UUID from the **Rule UUID** column.
- In **Policies** > **Security**, select **Copy UUID** in the rule **Name** drop-down menu.



STEP 2 | Switch from the UI to the CLI.

Use the UUID you captured in the UI to clear the rule's application usage data:

```
admin@PA-VM>clear policy-app-usage-data ruleuuid <uuid-value>
```

Paste or type the rule's UUID as the value and execute the command to clear the rule's application usage data.

Migrate Port-Based to App-ID Based Security Policy Rules

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

When you transition from a legacy firewall to a Palo Alto Networks next-generation firewall, you inherit a large number of port-based rules that allow any application on the ports, which increases the attack surface because any application can use an open port. Policy Optimizer identifies all applications seen on any legacy port-based Security policy rule and provides an easy workflow for selecting the applications you want to allow on that rule. Migrate port-based rules to application-based rules to reduce the attack surface and safely enable applications on your network. Use Policy Optimizer to maintain the rulebase as you add new applications.

*Migrate a few port-based rules at a time to application-based rules, in a prioritized manner. A gradual conversion is safer than migrating a large rulebase at one time and makes it easier to ensure that the new application-based rules control the necessary applications. Use **Policy Optimizer** to prioritize which rules to convert first.*

To migrate a configuration from a legacy firewall to a Palo Alto Networks device, see Best Practices for Migrating to Application-Based Policy.

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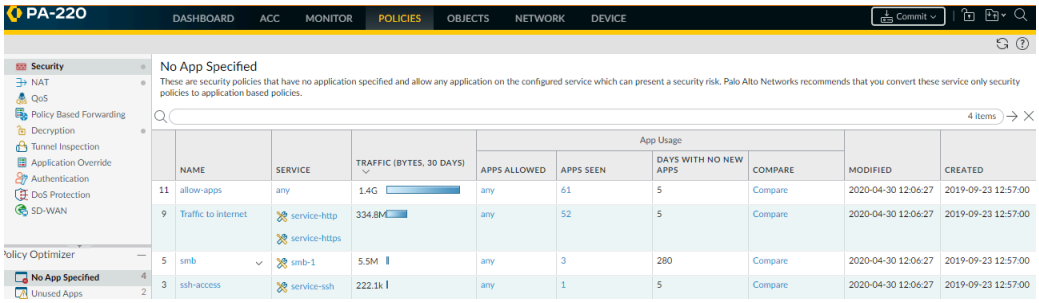
STEP 1 | Identify port-based rules.

Port-based rules have no configured (allowed) applications. **Policies** > **Security** > **Policy Optimizer** > **No App Specified** displays all port-based rules (**Apps Allowed** is **any**).



STEP 2 | Prioritize which port-based rules to convert first.

Policies > **Security** > **Policy Optimizer** > **No App Specified** enables you to sort rules without affecting their order in the rulebase and provides other information that helps you prioritize rules for conversion based on your business goals and risk tolerance.

- **Traffic (Bytes, 30 days)**—(Click to sort.) Rules that *currently* match the most traffic are at the top of the list. This is the default sorting order.
- **Apps Seen**—(Click to sort.) A large number of legitimate applications matching a port-based rule may indicate you should replace it with multiple application-based rules that tightly define the applications, users, and sources and destinations. For example, if a port-based rule controls traffic for multiple applications for different user groups on different sets of devices, create separate rules that pair applications with their legitimate users and devices to reduce the attack surface and increase visibility. (Clicking the **Apps Seen** number or **Compare** shows you the applications that have matched the rule.)

  *The firewall updates **Apps Seen** approximately every hour. However, if there is a large volume of application traffic or a large number of rules, it may take longer than an hour to update. After you add an application to a rule, wait at least an hour before running Traffic logs to see the application's log information.*

- **Days with No New Apps**—(Click to sort.) When the applications seen on a port-based rule stabilize, you can be more confident the rule is mature, conversion won't accidentally exclude legitimate applications, and no more new applications will match the rule. The **Created** and **Modified** dates help you evaluate a rule's stability because older rules that have not been modified recently may also be more stable.
- **Hit Count**—Displays rules with the most matches over a selected time frame. You can exclude rules for which you reset the hit counter and specify the exclusion time period in days. Excluding rules with recently reset hit counters prevents misconceptions about rules that show fewer hits than you expect because you didn't know the counter was reset.

  *You can also use **Hit Count** to View Policy Rule Usage and help identify and remove unused rules to reduce security risks and keep your rulebase organized.*

STEP 3 | Review the **Apps Seen** on port-based rules, starting with the highest priority rules.

On **No Apps Specified**, click **Compare** or the number in **Apps Seen** to open **Applications & Usage**, which lists applications that matched a port-based rule over a specified **Timeframe**,

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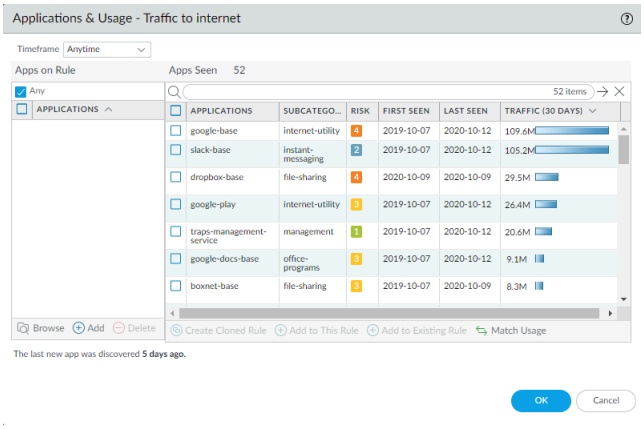
with each application's **Risk**, the date it was **First Seen**, the date it was **Last Seen**, and the amount of traffic over the last 30 days.



You can check **Applications seen** on port-based rules over the past 7, 15, or 30 days, or over the rule's lifetime (**Anytime**). For migrating rules, **Anytime** provides the most complete assessment of applications that matched the rule.

You can search and filter the **Apps Seen**, but keep in mind that it takes an hour or more to update **Apps Seen**. You can also order the **Apps Seen** by clicking the column headers. For example, you can click **Traffic (30 days)** to bring the applications with the most recent traffic to the top of the list, or click **Subcategory** to organize the applications by subcategory.

*The granularity of measurement for **First Seen** and **Last Seen** data is one day, so on the day you define a rule, the dates in these two columns are the same. On the second day the firewall sees traffic on an application, you'll see a difference in the dates.*

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STEP 4 | Clone or add applications to the rule to specify the applications you want to allow on the rule.

On **Applications & Usage**, convert a port-based rule to an application-based rule in either of two ways:

- **Clone the rule**—Preserves the original port-based rule and places the cloned application-based rule directly above it in the rulebase.
- **Add Applications to the Rule**—Replaces the original port-based rule with the new application-based rule and deletes the original rule.

> *If you have existing application-based rules and you want to migrate applications to them from port-based rules, you can Add Applications to an Existing Rule instead of cloning a new rule or converting the port-based rule by adding applications to it.*

*Some applications appear on the network at intervals, for example, for quarterly or yearly events. These applications may not display on the **Applications & Usage** screen if the history isn't long enough to capture their latest activity.*

*When you clone a rule or add applications to a rule, nothing else about the original rule changes. The original rule's configuration remains the same except for the applications you added to the rule. For example, if the original rule's Service allowed **Any** application or specified a particular service, you need to change the Service to **Application-Default** to restrict the allowed applications to their default ports on the new rule.*

Cloning is the safest way to migrate rules, especially when **Applications & Usage** shows more than a few well-known applications matching the rule (Rule Cloning Migration Use Case: Web Browsing and SSL Traffic provides an example of this). Cloning preserves the original port-based rule and places it below the cloned application-based rule, which eliminates the risk of losing application availability because traffic that doesn't match the cloned rule flows through to the port-based rule. When traffic from legitimate applications hasn't hit the port-based rule for a reasonable period of time, you can remove it to complete that rule's migration.

To **clone** a port-based rule:

1. In **Apps Seen**, click the check box next to each application you want in the cloned rule. Keep in mind that it takes an hour or more to update **Apps Seen**.
2. Click **Create Cloned Rule**. In the **Create Cloned Rule** dialog, **Name** the cloned rule ("slack" in this example) and add other applications in the same container and application

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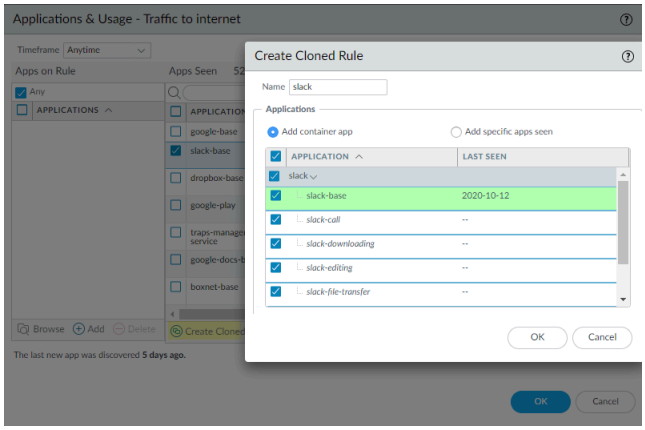
dependencies, if required. For example, to clone a rule by selecting the slack-base application:



The green text is the selected application to clone. The container application (**slack**) is in the gray row. The applications listed in *italics* are applications that have not been seen on the rule but are in the same container as the selected application. Individual applications that have been seen on the rule are in normal font. All the applications are included in the cloned rule by default (**Add Container App**, which adds all the applications in the container, is selected by default) to help prevent the rule from breaking in the future.

3. If you want to allow all of the applications in the container, leave **Add container app** selected. This also "future proofs" the rule because when an application is added to the container app, it's automatically added to the rule.

   If you want to constrain access to some of the individual applications in the container, uncheck the box next to each individual application you don't want users to access. This also unchecks the container app, so if you want to allow new applications in the container later, you have to add those applications individually.

   If you uncheck the container app, all the apps are unchecked and you manually select the apps you want to include in the cloned rule.
4. If application dependencies are listed in a box below the Applications (there are none in this example), leave them checked. The applications you selected need those application dependencies to run. Common dependencies include **ssl** and **web-browsing**.
5. Click **OK** to add the new application-based rule directly above the port-based rule in the rulebase.
6. **Commit** the configuration.

When you clone a rule and **Commit** the configuration, the applications you select for the cloned rule are removed from the original port-based rule's **Apps Seen** list. For example, if a port-based rule has 16 **Apps Seen** and you select two individual applications and one dependent application for the cloned rule, after cloning, the port-based rule shows 13 **Apps Seen** because the three selected applications have been removed from the port-based rule (16-3 = 13). The cloned rule shows the three added applications in **Apps on Rule**.

Creating a cloned rule with a container app works a bit differently. For example, a port-based rule has 16 **Apps Seen** and you select one individual application and a container app for the cloned rule. The container app has five individual applications and has one dependent application. After cloning, the cloned rule shows seven **Apps on Rule**—the

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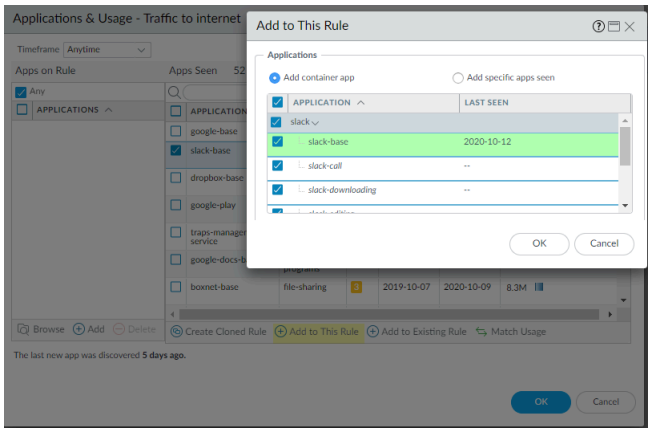
individual application, the five individual applications in the container app, and the dependent application for the container app. However, in the original port-based rule, **Apps Seen** shows 13 applications because only the individual application, the container app, and the container app's dependent application are removed from the port-based rule.

In contrast to cloning, adding applications to a port-based rule replaces the rule with the resulting application-based rule. Adding applications to a rule is simpler than cloning, but riskier because you may inadvertently miss applications that should be on the rule, and the original port-based rule is no longer in the rulebase to catch accidental omissions. However, adding applications to port-based rules that apply to only a few well-known applications migrates the rule quickly to an application-based rule. For example, for a port-based rule that only controls traffic to TCP port 22, the only legitimate application is SSH, so it's safe to add applications to the rule.

*Adding applications using the traditional Security policy rule's **Application** tab does not change **Apps Seen** or **Apps on Rule**. To preserve accurate application usage information, when replacing port-based rules with application-based rules, add applications using **Add to This Rule** or **Match Usage** (or create a cloned rule or add applications to an existing application-based rule instead) in **Apps Seen**.*

There are three ways to replace a port-based rule with an application-based rule by adding applications (**Add to This Rule** and **Match Usage** in **Apps Seen** and **Add** in **Apps on Rule**):

- **Add to This Rule** applications from **Apps Seen** (applications that matched the rule). Keep in mind that it takes an hour or more to update **Apps Seen**.
  1. Select applications from **Apps Seen** on the rule.
  2. Click **Add to This Rule**. In the **Add to This Rule** dialog, add other applications in the same container app and application dependencies, if required. For example, to add slack-base to a rule:



Similar to the **Create Cloned Rule** dialog, the green text in **Add to This Rule** is the selected application to add to the rule. The container app (**slack**) is in the gray row. The applications listed in *italics* are applications that have not been seen on the rule but are in the same container as the selected application. Individual applications that have been seen on the rule are in normal font. All the applications are included in the cloned rule by

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default (**Add Container App**, which adds all the applications in the container, is selected by default) to help prevent the rule from breaking in the future.

3. If you want to allow all of the applications in the container, leave **Add container app** selected. This also "future proofs" the rule because when an application is added to the container app, it's automatically added to the rule.

   If you want to constrain access to some of the individual applications in the container, uncheck the box next to each individual application you don't want users to access. This also unchecks the container app, so if you want to allow new applications in the container later, you have to add those applications individually.

   If you uncheck the container app, all the apps are unchecked and you manually select the apps you want to include in the cloned rule.
4. If application dependencies are listed in a box below the Applications (there are none in this example), leave them checked. The applications you selected need those application dependencies to run.
5. Click **OK** to replace the port-based rule with the new application-based rule.

When you **Add to This Rule** and **Commit** the configuration, the applications you didn't add are removed from **Apps Seen** because the new application-based rule no longer allows them. For example, if a rule has 16 **Apps Seen** and you **Add to This Rule** three applications, the resulting new rule shows only those three added applications in **Apps Seen**.

Add to This Rule with a container app works a bit differently. For example, a port-based rule has 16 **Apps Seen** and you select one individual application and a container app to add to the new rule. The container app has five individual applications and has one dependent application. After adding the applications to the rule, the new rule shows seven **Apps on Rule**—the individual application, the five individual applications in the container app, and the dependent application for the container app. However, **Apps Seen** shows 13 applications because the individual application, the container app, and the container app's dependent application are removed from that list.

- Add all of the **Apps Seen** on the rule to the rule at one time with one click (**Match Usage**).

  *Port-based rules allow any application, so **Apps Seen** may include unneeded or unsafe applications. Use **Match Usage** to convert a rule only when the rule has seen a small number of well-known applications with legitimate business purposes. A good example is TCP port 22, which should only allow SSH traffic, so if SSH is the only application seen on a port-based rule that opens port 22, you can safely **Match Usage**.*

  1. In **Apps Seen**, click **Match Usage**. Keep in mind that it takes an hour or more to update **Apps Seen**. All the applications in **Apps Seen** are copied to **Apps on Rule**.
  2. Click **OK** to create the application-based rule and replace the port-based rule.
- If you know the applications you want on the rule, you can **Add** applications manually in **Apps on Rule**. However, this method is equivalent to using the traditional Security policy rule **Application** tab and does not change **Apps Seen** or **Apps on Rule**. To preserve accurate

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application usage information, convert rules using **Add to This Rule**, **Create Cloned Rule**, or **Match Usage** in **Apps Seen**.

1. In **Apps on Rule**, **Add** (or **Browse**) and select applications to add to the rule. This is equivalent to adding applications on the **Application** tab.
2. Click **OK** to add the applications to the rule and replace the port-based rule with the new application-based rule.

*Because this method is equivalent to adding applications using the **Application** tab, the dialog to add application dependencies doesn't pop up.*

STEP 5 | For each application-based rule, set the **Service** to **application-default**.

If business needs require you to allow applications (for example, internal custom applications) on non-standard ports between particular clients and servers, restrict the exception to only the required application, sources, and destinations. Consider rewriting custom applications so they use the application default port.

STEP 6 | **Commit** the configuration.

STEP 7 | Monitor the rules.

- **Cloned rules**—Monitor the original port-based rule to ensure the application-based rule matches the desired traffic. If applications you want to allow match the port-based rule, add them to the application-based rule or clone another application-based rule for them. When only applications that you don't want on your network match the port-based rule for a reasonable period of time, the cloned rule is robust (it catches all the application traffic you want to control) and you can safely remove it.
- **Rules with Added Applications**—Because you convert only port-based rules that have a few well-known applications directly to application-based rules, in most cases the rule is solid from the start. Monitor the converted rule to see if the expected traffic matches the rule—if there's less traffic than expected, the rule may not allow all of the necessary applications. If there's more traffic than expected, the rule may allow unwanted traffic. Listen to user feedback—if users can't access applications they need for business purposes, the rule (or another rule) may be too tight.

Rule Cloning Migration Use Case: Web Browsing and SSL Traffic

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

A port-based rule that allows web access on TCP ports 80 (HTTP web-browsing) and 443 (HTTPS SSL) provides no control over which applications use those open ports. There are many web applications, so a general rule that allows web traffic allows thousands of applications, many of which you don't want on your network.

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This use case shows how to migrate a port-based policy that allows all web applications to an application-based policy that allows only the applications you want, so you can safely enable the applications you choose to allow. For rules that see a lot of applications, cloning the original port-based rule is safer than adding applications to the rule because adding replaces the port-based rule, so if you inadvertently forget to add a critical application, you affect application availability. And if you **Match Usage**, which also replaces the port-based rule, you allow all of the applications the rule has seen, which could be dangerous, especially with web browsing traffic.

Cloning the rule retains the original port-based rule and places the cloned rule directly above the port-based rule in the rulebase, so you can monitor the rules. Cloning also allows you to split rules that see a lot of different applications—such as a port-based web traffic rule—into multiple application-based rules so you can treat different groups of applications differently. When you're sure you're allowing all the applications you need to allow in the cloned rule (or rules), you can remove the port-based rule.

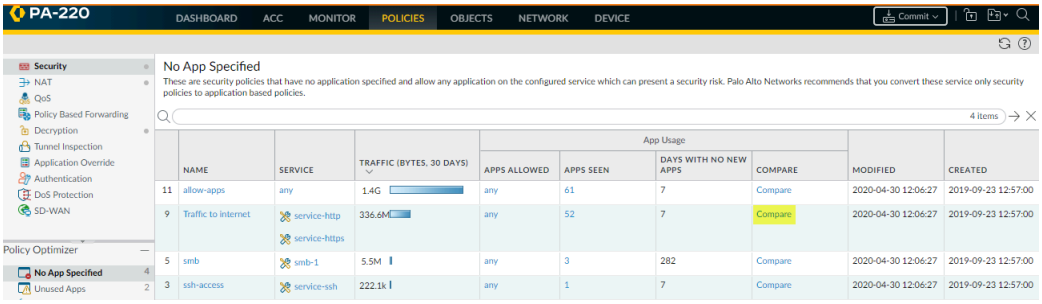
This example clones a port-based web traffic rule to create an application-based rule for web-based file sharing traffic (a subset of the application traffic seen on the port-based rule).

This example does not apply to using the New App Viewer to clone App-ID Cloud Engine (ACE) applications (see the ACE documentation for examples of how to do this); ACE requires a SaaS Security Inline license.

STEP 1 | Navigate to **Policies** > **Security** > **Policy Optimizer** > **No App Specified** to view the port-based rules.

STEP 2 | Click **Compare** for the rule you want to migrate.

In this example, the port-based rule that allows web access is named Traffic to internet.



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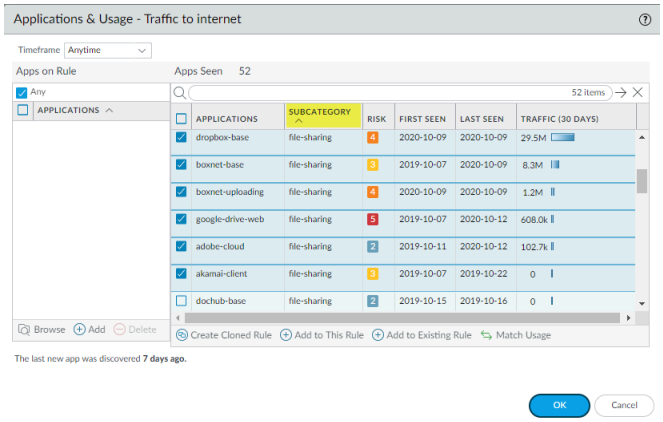
STEP 3 | Use the sorting options to review and select the applications you want to allow from **Apps Seen**.

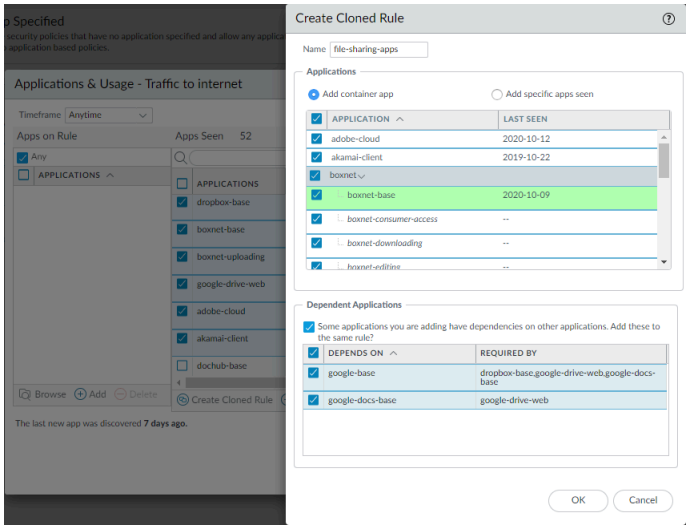
*The number of **Apps Seen** is updated approximately every hour, so if you don't see as many applications as you expect, check again after about an hour. Depending on the firewall's load, it may take longer than one hour for these fields to update.*

For example, click **Subcategory** to sort the applications, scroll to the file-sharing subcategory, and then select the applications you want to allow. Alternatively, you can filter (search) for file-sharing applications.



STEP 4 | Click **Create Cloned Rule** and **Name** the cloned rule (file-sharing-apps in this example).

Create Cloned Rule shows the selected applications shaded green, the container apps shaded gray, individual applications in the container that haven't been seen on the rule in *italics*, and individual applications that have been seen on the rule in normal text font. Scrolling through **Applications** shows all the container apps and their individual applications.



Create Cloned Rule also shows the dependent applications for the selected applications. In this example, some of the selected applications require (**Required By**) the google-base and google-docs-base applications to run.

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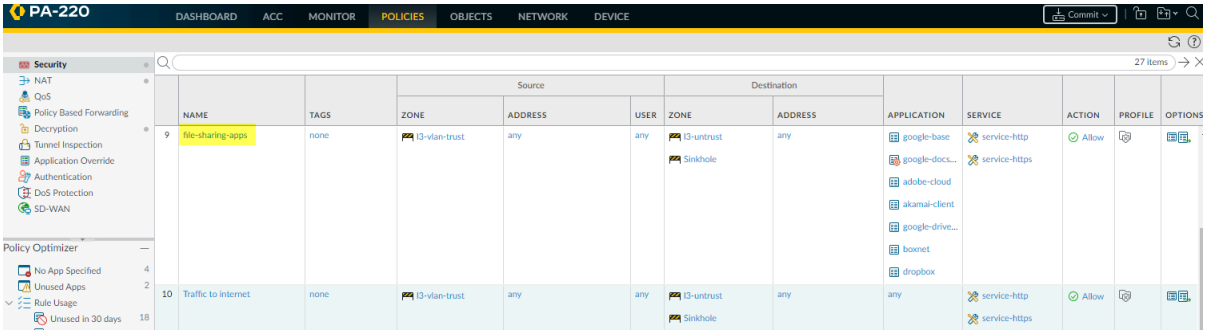
STEP 5 | Select the applications you want in the cloned rule.

For applications you don't want to include, uncheck the corresponding box, which also unchecks the container app. If you don't include the container app, then when new apps are added to the container, they won't automatically be added to the rule.

If you uncheck the container app, all the individual applications in the container are unchecked and you must select the apps you want to add manually.

STEP 6 | Click **OK** to create the cloned rule.

STEP 7 | In **Policies** > **Security**, the cloned rule (file-sharing-apps) is inserted in the rulebase above the original port-based rule (Traffic to internet).



STEP 8 | Click the rule name to edit the cloned rule, which inherits the properties of the original port-based rule.

STEP 9 | On the **Service/URL Category** tab, delete service-http and service-https from **Service**.

This changes the **Service** to **application-default**, which prevents applications from using non-standard ports and further reduces the attack surface.

If business needs require you to allow applications (for example, internal custom applications) on non-standard ports between particular clients and servers, restrict the exception to only the required application, sources, and destinations. Consider rewriting custom applications so they use the application default port.

STEP 10 | On the **Source**, **User**, and **Destination** tabs, tighten the rule to apply to only the right users in only the right locations (zones, subnets).

For example, you may decide to limit web file sharing activity to only the user groups that have business reasons to share files across the web.

STEP 11 | Click **OK**.

STEP 12 | **Commit** the configuration.

STEP 13 | Repeat the process for other application categories in the port-based web access rule until your application-based rules allow only the applications you want to allow on your network.

When traffic you want to allow stops hitting the original port-based rule for a sufficient amount of time to be confident that the port-based rule is no longer needed, you can remove the port-based rule from the rulebase.

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Add Applications to an Existing Rule

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

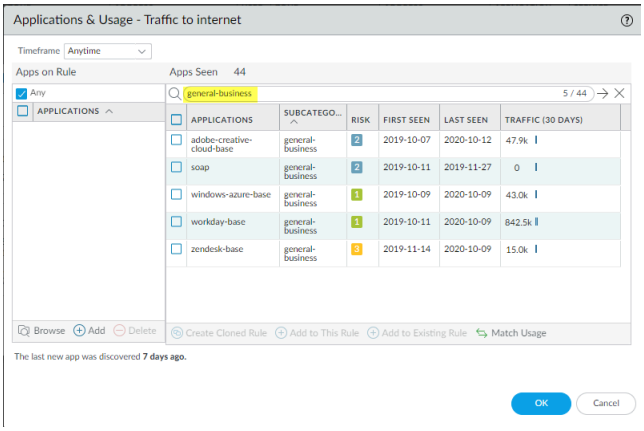
In some cases, you may want to add applications learned (seen) on a port-based rule to a rule that already exists. For example, an administrator may create a cloned application-based rule for general business web applications from a port-based rule that allows internet access (a port 80/443 rule). Later, the administrator notices that the port-based internet access rule has seen more general business applications and wants to add some or all of them to the cloned application-based rule (cloning another application-based rule for the same type of application would create an unnecessary rule and complicate the rulebase).

This example assumes that an application-based Security policy rule to control general business traffic already exists or was cloned from a port-based internet access rule, similarly to the Rule Cloning Migration Use Case: Web Browsing and SSL Traffic. In that example, we cloned an application-based rule from the port-based internet access rule and changed the new rule's Service to application-default to prevent web-based applications from using non-standard ports.

In addition to adding applications to an existing application-based rule, you can add applications to an existing port-based rule. This converts the port-based rule to an application-based rule for the applications you add to the rule. If you do this, go to the rule and change the Service to application-default to prevent the applications from using non-standard ports (also, the Service configured on the rule may not match the application).

This example does not apply to using the New App Viewer to add App-ID Cloud Engine (ACE) applications to an existing rule (see the ACE documentation for examples of how to do this); ACE requires a SaaS Security Inline license.

STEP 1 | You check the port-based internet access rule and discover that the rule has seen general business applications and that you need to allow some of them for business purposes.



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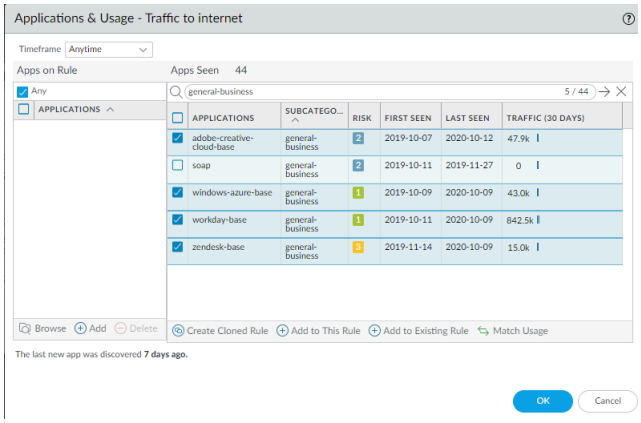
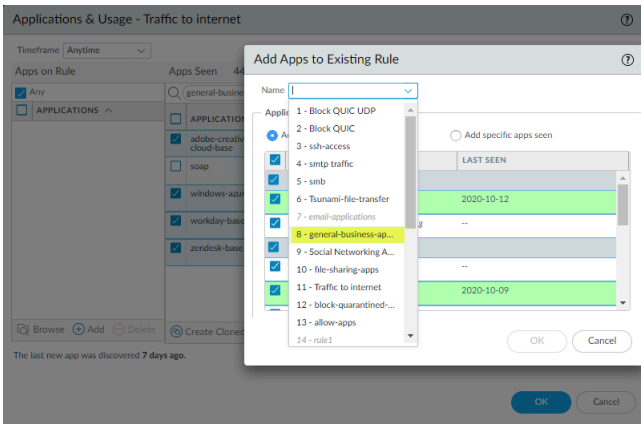
STEP 2 | Select the general business apps you want to add to the existing rule.



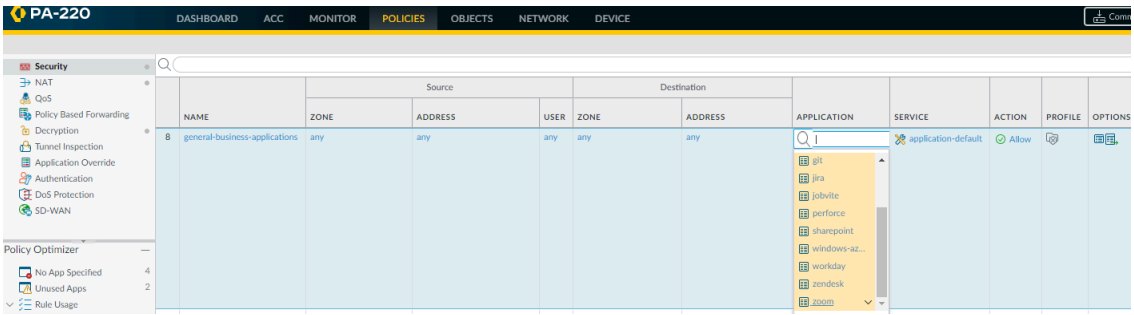
STEP 3 | Click **Add to Existing Rule** and select the **Name** of the rule to which you want to add the applications, in this example, **general-business-applications**.



STEP 4 | Click **OK** in **Add Apps to Existing Rule**to add the selected applications to the **general-business-applications** rule.

STEP 5 | Click **OK** in **Applications & Usage**.

STEP 6 | The updated rule now controls the original applications on the rule and the applications you just added.



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Identify Security Policy Rules with Unused Applications

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

If you have application-based Security policy rules that allow a large number of applications, you can remove unused applications (applications never seen on the rules) to tighten those rules so that they only allow applications actually seen in traffic that matches the rule. Identifying and removing unused applications from Security policy rules is a best practice that strengthens your security posture by reducing the attack surface.

STEP 1 | Identify Security policy rules that have unused applications.

Policies > **Security** > **Policy Optimizer** > **Unused Apps** displays all application-based rules that are configured with applications that have not matched (been seen on) the rule. This means that these rules allow applications that you may not use in your network (or that another rule shadows the rule, so traffic that you expect to match the rule matches an earlier rule in the rulebase).

*The number of **Apps Allowed** and **Apps Seen** are updated approximately every hour, so if you configure applications on a rule and don't see as many **Apps Allowed** as you expect, check again after about an hour. Depending on the firewall's load, it may take longer than one hour for these fields to update.*

STEP 2 | Prioritize which rules with unused applications to modify first.

Policies > **Security** > **Policy Optimizer** > **Unused Apps** enables you to sort rules without affecting their order in the rulebase and provides other information that helps you prioritize rules to clean up based on your business goals and risk tolerance.

- The difference between **Apps Allowed** (the number of applications on the allow list) and **Apps Seen** (the number of allowed applications actually seen on the rule) shows how many applications are configured on each rule but not actually seen on the rule, which indicates to what extent the rule is over-provisioned. Click **Apps Allowed** to sort by the number of applications allowed in a rule and click **Apps Seen** to sort by the number of applications actually seen on a rule.
- **Days with No New Apps** (click to sort) shows you the number of days since the last time a new application hit the rule. This indicates how likely it is that the rule is mature and won't see any applications that haven't already been seen. The longer the **Days with No New Apps**, the less likely that new applications will hit the rule and the more likely that you know all the applications the rule allows.
- **Created** and **Modified** dates also help determine whether a rule has matured enough to understand whether applications not seen on the rule may be seen at a later date or if the rule has seen all the applications expected to hit the rule. The longer the time since a rule

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was **Modified**, the more likely the rule is mature. (If **Created** and **Modified** are the same, the rule hasn't been modified.)

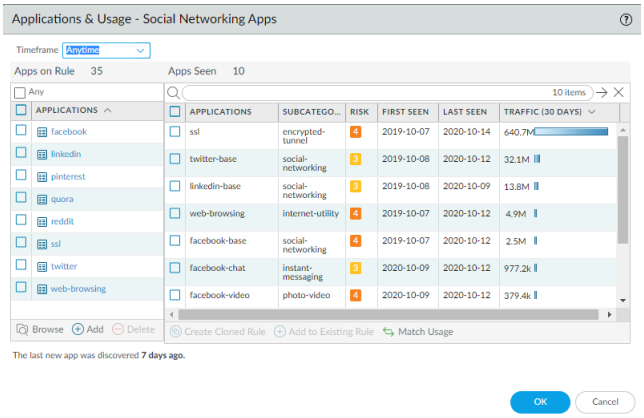
- **Hit Count**—Displays rules with the most matches over a selected time frame. You can exclude rules for which you reset the hit counter and specify the exclusion time period in days. Excluding rules with recently reset hit counters prevents misconceptions about rules that show fewer hits than you expect because you didn't know the counter was reset.

  *You can also use **Hit Count** to* View Policy Rule Usage.

You can also click **Traffic (Bytes, 30 days)** to sort by the amount of traffic a rule has seen over the last 30 days. Use this information to prioritize which rules to modify first. For example, you can prioritize rules with the largest difference between **Apps Allowed** and **Apps Seen** and that also have the most **Days with No New Apps**, because those rules have the greatest number of unused applications and are the most mature.

STEP 3 | Review the **Apps Seen** on the rule.

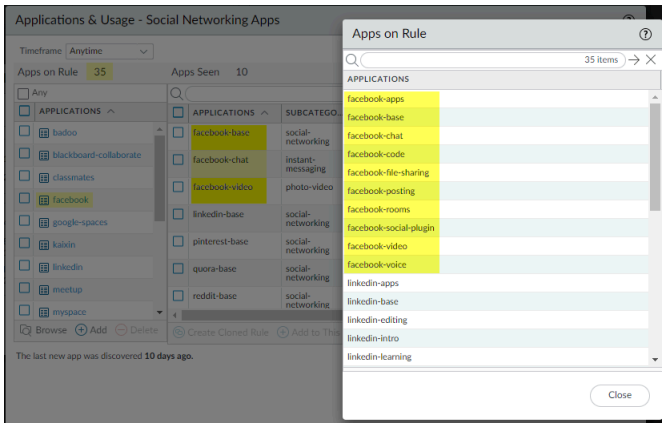
On **Unused Apps**, click **Compare** or the number in the **Apps Seen** column to open **Applications & Usage**, which shows the applications configured on the rule (**Apps on Rule**) and the **Apps Seen** on the rule.



- The number next to **Apps Seen** (10 in this example) is the number of applications that matched the rule. Keep in mind that it takes at least one hour for the firewall to update **Apps Seen**.
- The number next to **Apps on Rule** (35 in this example) is how many applications are configured on the rule, which is calculated by counting each application in a container app (but not the container app itself—if you configure a container app on the rule, the rule allows the container app's individual applications). Because the **Applications** list shows only the applications you configure manually on the rule, when you configure a container app on a rule, **Applications** only shows the container app, not all of the individual applications in the container (unless you also manually configure the individual applications on the rule). For this reason, the number of **Apps on Rule** may not be the same as the number of applications you see in the **Applications** list.
- Click the number next to **Apps on Rule** to see all of the individual applications on the rule.

  This example rule has 10 **Apps Seen** (applications that matched the rule) but allows 35 **Apps on Rule**. The **facebook** container app is configured on the rule and the rule sees traffic from

the individual applications facebook-base, facebook-chat, and facebook-video (**Apps Seen**). When you click the **Apps on Rule** number, the **Apps on Rule** dialog displays the individual applications allowed, but not the container app itself.



You cannot add or delete applications from the pop-up dialog.

Compare the **Apps Seen** on the rule to the **Apps on Rule**. If an application on the rule isn't used (you don't see the application or you don't see applications in an allowed container in **Apps Seen**), consider removing the application from the rule to reduce the attack surface. Take into account periodically used applications, such as for quarterly or annual events, which may look unused if you don't examine a long enough time frame. **Timeframe** enables you to select the time frame for the **Apps Seen** on the rule. Select **Anytime** to see every application seen over the life of the rule. Depending on the **Created** or **Modified** date in the **No App Specified** dialog and the time between periodic events, the rule may not have been on the firewall long enough to see all periodically used applications.

STEP 4 | Remove unused applications from the rule.

Delete (or **Add**) applications in **Apps on Rule** to remove (or add) applications manually, or **Match Usage** to add the **Apps Seen** on the rule and delete applications for which no matching traffic has been seen on the rule with one click.

To remove applications from the rule manually, select applications from **Apps on Rule** and **Delete** them. Ensure that none of the applications are required for periodic events before you remove them from the rule. (You can also add or delete applications on the Security policy rule's **Application** tab.)

Match Usage moves the **Apps Seen** on the rule to **Apps on Rule** and removes all unused applications from the rule.

*You can clone rules from **Policies** > **Security** and from **No App Specified** to Migrate Port-Based to App-ID Based Security Policy Rules. You can't clone a rule starting from **Unused Apps**.*

STEP 5 | **Commit** the configuration.

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STEP 6 | Monitor updated rules and listen to user feedback to ensure that updated rules allow the applications you want to allow and don't inadvertently block periodically used applications.

> *The number of **Apps Allowed** and **Apps Seen** are updated approximately every hour. After you remove all of the unused applications from a rule, the rule remains listed in **Policies** > **Security** > **Policy Optimzer** > **Unused Apps** until the firewall updates the display. When the firewall updates the display and the number of **Apps Allowed** is the same as the number of **Apps Seen**, the rule no longer displays in the **Unused Apps** screen. However, depending on the firewall's load, it may take longer than one hour for these fields to update.*

High Availability for Application Usage Statistics

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

When you configure two firewalls as a High Availability (HA) pair, the application usage statistics are local to the firewall that generates the Traffic logs for the application. Where you can view application usage statistics also depends in part on the HA configuration:

- **Active/Passive**—The active device generates the application usage statistics. If a passive device has seen no user traffic, then only the active device displays the application usage statistics. If a passive device has seen traffic, then the passive device only displays the application usage statistics from the traffic that it has seen.

  On a failover, the application usage statistics are based only on the Traffic logs generated on the newly active device (the device that was passive before the failover).
- **Active/Active**—The device that owns a session generates the Traffic logs for that session, so the application usage statistics for a session are only available on the device that owns the session. If one active device owns a session, the other active device does not display that session's application usage statistics.

How to Disable Policy Optimizer

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

Policy Optimizer is enabled by default. Policy Optimizer provides many capabilities that make it easier to Migrate Port-Based to App-ID Based Security Policy Rules and to Identify Security Policy Rules with Unused Applications and remove the unused applications from the rules, but if you wish to disable the feature, you can.

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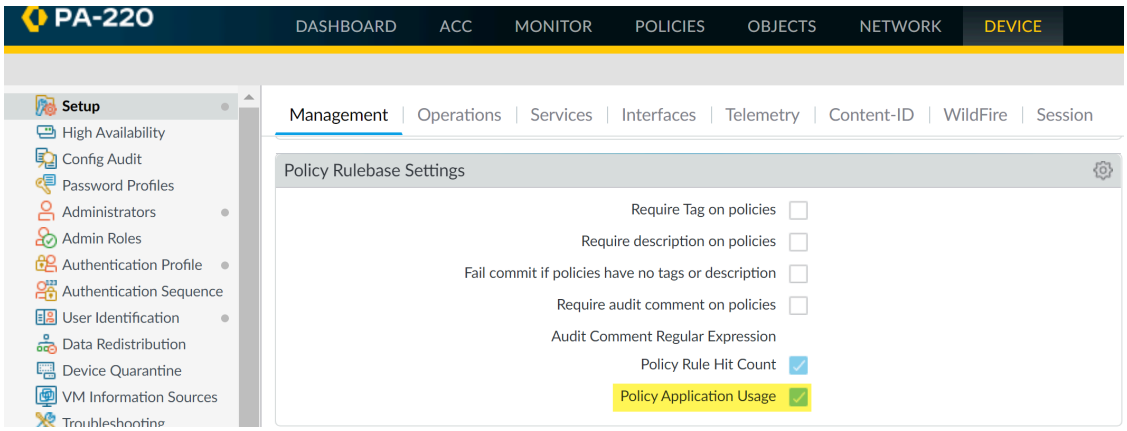
STEP 1 | Navigate to **Device** > **Setup** > **Management** > **Policy Rulebase Settings**.

STEP 2 | Select the **Policy Application Usage** check box to enable the feature and deselect the check box to disable the feature.



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App-ID Cloud Engine

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	❑ SaaS Security Inline license (for NGFW) ❑ Prisma Access license (ACE is a core feature)

The App-ID Cloud Engine (ACE) is a service that enables the firewall or Panorama to download App-IDs from the cloud for applications that do not have specific predefined App-IDs from the Palo Alto Networks content update team. ACE provides specific App-IDs for applications that the firewall otherwise identifies as ssl or web-browsing. Use ACE App-IDs in Security policy rules to gain visibility into cloud applications and control them. Use Policy Optimizer to add and manage applications in Security policy. You cannot use ACE App-IDs in any other types of policy rules. ACE:

- Vastly increases the number of known App-IDs to identify and control more cloud applications, and as ACE defines new App-IDs for applications, the ACE App-IDs become available on the firewall.
- Speeds up the availability and delivery of new App-IDs to the firewall.
- Speeds up and can automate the addition of applications to Security policy through the use of Application Filters in Security policy rules.
- Dramatically increases visibility into applications that previously were identified as ssl or web-browsing.

ACE requires a SaaS Security Inline subscription. Each appliance that uses ACE must have a valid device certificate installed.

All hardware platforms that support PAN-OS 10.1 or later support ACE and all appliances on which you want to use ACE require PAN-OS 10.1 or later. Panorama cannot push and commit ACE-based polices or objects to firewalls that don't have a SaaS Security Inline license installed or to firewalls that run an earlier version of PAN-OS than 10.1.

ACE is supported in the US, APAC, and EU GCP regions. The region is selected automatically based on your Strata Logging Service region.

*Verify that the firewall uses the correct Content Cloud FQDN (**Device** > **Setup** > **Content-ID** > **Content Cloud Setting**) for your region and change the FQDN if necessary:*

- *US—**`hawkeye.services-edge.paloaltonetworks.com`***
- *EU—**`eu.hawkeye.services-edge.paloaltonetworks.com`***
- *APAC—**`apac.hawkeye.services-edge.paloaltonetworks.com`***

ACE data, including traffic payloads, is sent to the servers in the selected region. If you specify a Content Cloud FQDN that is outside of your region (for example, if you are in the EU region but you specify the APAC region FQDN), you may violate your country's or your organization's privacy and legal regulations.

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Predefined content-delivered App-ID provides new applications once per month and you need to analyze the new App-IDs before you install them to understand changes that they may make to Security policy rules. The monthly cadence and need for analysis slows down the adoption of new App-IDs in policy. Although Palo Alto Networks will continue to provide new App-IDs via monthly content updates that you need to review, ACE improves the adoption of new App-IDs by providing on-demand App-IDs for applications initially identified as any of the following two types:

- **ssl**—Encrypted SSL traffic is by far the most common type of network traffic, with most experts claiming that it exceeds 90% of total traffic. If you don't or can't decrypt that traffic, the firewall often can only identify it as ssl instead of as the actual underlying application.
- **web-browsing**—The firewall can't specifically identify some unencrypted (web-browsing) traffic because the monthly content-delivered App-ID updates can't keep up with all the new applications being developed every day.

ACE provides specific identification of these applications, which enables you to understand them and control them appropriately in Security policy.

ACE App-IDs do not identify other types of public applications and do not identify private and custom applications. The ACE App-ID catalog does not contain predefined, content-provided App-IDs. Content-provided App-IDs still arrive monthly in content updates.

When the firewall encounters ssl or web-browsing traffic, the firewall sends the payload to ACE for analysis. If it matches an App-ID in the ACE database, ACE returns the App-ID to the requesting firewall. If ACE has no matching App-ID for the traffic, ACE sends the payload to the Machine Learning (ML) engine. The ML engine analyzes the payload and develops the new App-ID in conjunction with the human content team. When development finishes, the ML engine uploads new App-ID to the ACE database, and the requesting firewall (and any other firewalls) can download the App-ID and use it in Security policy.

Because it can take several minutes to retrieve a known application from ACE and longer if a new App-ID must be developed, cloud application detection is not inline on the firewall. The firewall does not wait for a verdict to process the application traffic. The firewall processes the traffic as ssl or web-browsing until it receives an App-ID from ACE and you use it in Security policy.

*If you downgrade a firewall or Panorama after ACE has been enabled and ACE cloud App-IDs are still in use in Security policy rules or Application Groups, the downgrade fails. The fail reason lists the objects that you need to remove from the configuration in order to downgrade. Remove those objects from the configuration and **Commit** the configuration, and then the downgrade will succeed.*

- Prepare to Deploy App-ID Cloud Engine
- Enable or Disable the App-ID Cloud Engine
- App-ID Cloud Engine Processing and Policy Usage
- New App Viewer (Policy Optimizer)
- Add Apps to an Application Filter with Policy Optimizer
- Add Apps to an Application Group with Policy Optimizer

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- Add Apps Directly to a Rule with Policy Optimizer
- Replace an RMA Firewall (ACE)
- Impact of License Expiration or Disabling ACE
- Commit Failure Due to Cloud Content Rollback
- Troubleshoot App-ID Cloud Engine

Prepare to Deploy App-ID Cloud Engine

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	☐ SaaS Security Inline license (for NGFW) ☐ Prisma Access license (ACE is a core feature)

There are several prerequisite onboarding tasks to do before the firewall can use the App-ID Cloud Engine (ACE). You can deploy ACE on standalone firewalls or use Panorama to deploy ACE on managed firewalls.

Before a firewall can use ACE to provide specific App-IDs for traffic previously identified as ssl or web-browsing traffic, the PAN-OS administrator and the SaaS Security administrator must work together to:

- Install a valid device certificate on each appliance that will use ACE, including Panorama appliances that manage ACE firewalls. (PAN-OS administrator.)
- (Explicit Proxy Deployments [requires PAN-OS 11.2.3 and later or PAN-OS 11.1.5 and later]) Enable firewalls using a proxy server to access servers that facilitate requests generated by various features using inline cloud services.
- Activate SaaS Security Inline on each firewall that will use ACE. Panorama doesn't require a license. (SaaS Security administrator.)
- Configure a service route for communication between the firewall and ACE. (PAN-OS administrator.)
- Enable ACE on Panorama appliances which manage firewalls that will use ACE. (PAN-OS administrator.)

  On firewalls, ACE is enabled by default after activating SaaS Security Inline.

- Create Security policy rule that allows ACE traffic. (PAN-OS administrator.)
- Configure Log Forwarding from the firewall to the Strata Logging Service. (PAN-OS administrator.)

At the appropriate step in the following procedure, the PAN-OS administrator should notify the SaaS Security administrator that the deployment is ready for SaaS Security Inline activation. After activating SaaS Security Inline, the SaaS Security Inline administrator should notify the PAN-OS administrator that the deployment is ready to complete on the PAN-OS devices. Communication between the administrators is essential to achieving a smooth deployment.

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Requirements:

- Standalone firewalls, Panorama appliances, and managed firewalls must run PAN-OS 11.1 or later.
- All ACE firewalls must have purchased a SaaS Security Inline license. Panorama does not require a license to manage ACE firewalls or push ACE configurations to managed firewalls.
- All ACE appliances must be able to connect to the US, APAC, or EU GCP region, depending on your location (the region is selected automatically based on your Strata Logging Service region).

  Verify that the firewall uses the correct Content Cloud FQDN (**Device** > **Setup** > **Content-ID** > **Content Cloud Setting**) for your region and change the FQDN if necessary:

  - US—**`hawkeye.services-edge.paloaltonetworks.com`**
  - EU—**`eu.hawkeye.services-edge.paloaltonetworks.com`**
  - APAC—**`apac.hawkeye.services-edge.paloaltonetworks.com`**

  ACE data, including traffic payloads, is sent to the servers in the selected region. If you specify a Content Cloud FQDN that is outside of your region (for example, if you are in the EU region but you specify the APAC region FQDN), you may violate your country's or your organization's privacy and legal regulations.

The PAN-OS administrator completes the first two steps of the procedure and then hands it off to the SaaS Security Inline administrator for activation (Step 3). After activation, the SaaS Security Inline administrator hands the rest of the procedure back to the PAN-OS administrator to complete on the PAN-OS devices.

STEP 1 | Bring the firewall and Panorama (if using) online. (PAN-OS administrator.)

STEP 2 | Install device certificates on Panorama (if you use Panorama) and on individual firewalls so that they can use cloud services. (PAN-OS administrator.)

- Install a device certificate on Panorama
- Install a device certificate on individual firewalls (if not managed by Panorama)
- Install device certificates on managed firewalls from Panorama

Hand off the next step to the SaaS Security administrator.

STEP 3 | Activate SaaS Security Inline on every firewall that will use ACE. Activation enables ACE on the firewalls. (SaaS Security administrator.)

Panorama does not require a SaaS Security Inline license to manage firewalls that use ACE. Only managed firewalls need licenses, which you must retrieve manually as shown in the next step.

Hand off the rest of the steps to the PAN-OS administrator.

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STEP 4 | Retrieve the SaaS Security Inline license on each firewall—Panorama doesn't need a license—and verify that it is activated. (PAN-OS administrator.)

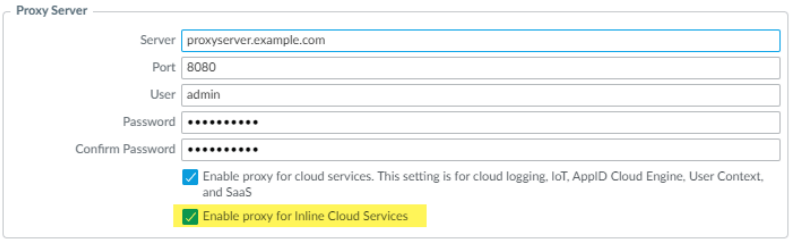
The SaaS Security administrator's activation sets up the licenses for the firewall, so you don't have to go to the Customer Support Portal or obtain Auth Codes.

1. Go to **Device** > **Licenses** > **License Management** and select **Retrieve license keys from license server** to retrieve the license.
2. Check **Device** > **Licenses** to ensure that the SaaS Security Inline license is active.

STEP 5 | (Required when the firewall is deployed with an explicit proxy server) Configure the proxy server used to access the servers that facilitate requests generated by various features using inline cloud services. A single proxy server can be specified and applies to all Palo Alto Networks inline cloud services, including all configured inline cloud analysis and logging services.

1. (PAN-OS 11.2.3 and later) Configure the proxy server through PAN-OS.
   1. Select **Device** > **Setup** > **Services** and edit the **Services** details.
   2. Specify the **Proxy Server** settings and **Enable proxy for Inline Cloud Services**. You can provide either an IP address or FQDN in the **Server** field.

      The proxy server password must contain a minimum of seven characters.

   3. Click **OK**.
2. (PAN-OS 11.1.5 and later) Configure the proxy server through the firewall CLI.
   1. Access the firewall CLI.
   2. Configure the base proxy server settings using the following CLI commands:

```
set deviceconfig system secure-proxy-server <FQDN_or_IP>
set deviceconfig system secure-proxy-port <1-65535>
set deviceconfig system secure-proxy-user <value>
```

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```
set deviceconfig system secure-proxy-password <value>
```

The proxy server password must contain a minimum of seven characters.

3. Enable the proxy server to send requests to the inline cloud service servers using the following CLI command:

```
debug dataplane mica set inline-cloud-proxy enable
```

4. View the current operational status of proxy support for inline cloud services using the following CLI command:

```
debug dataplane mica show inline-cloud-proxy
```

For example:

```
debug dataplane mica show inline-cloud-proxy

Proxy for Advanced Services is Disabled
```

STEP 6 | Configure a data services (dataplane) service route so that the firewall can communicate with the App-ID Cloud Engine. (PAN-OS administrator.)

You can push this configuration to managed firewalls from Panorama. Both Panorama and the managed firewalls must run PAN-OS 11.1 or later.

By default, the firewall uses the management interface as the source interface for the data services service route, but it is recommended that you configure a dataplane interface that has connectivity to cloud services as the **Source Interface** and **Source Address** for data services, as shown later in this step.

The issue on firewalls is that if an explicit proxy is configured on the management interface and you use it for the data services service route, then the management interface can only connect to the Knowledge Cloud Service (KCS), which manages the cloud application and signatures. When an explicit proxy is configured on the management interface, it cannot connect to the Detection Cloud Service (DCS), which checks the application payload against existing ACE App-IDs and provides verdicts. KCS and DCS are services in the ACE cloud. If the management interface has an explicit proxy configured, you can't use it for the data services

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service route for ACE because it can't connect to all of the services. In this case, you must use a dataplane interface on the firewall to connect to the data services.

Panorama uses the management port by default to connect to the KCS and does not connect to the DCS.

To configure the service route on a data plane interface instead of using the default management interface:

1. Select **Device** > **Setup** > **Services** then in **Service Features**, select **Service Route Configuration**.
2. **Customize** a service route.
3. Select the **IPv4** protocol.
4. Click **Data Services** in the Service column to open the **Service Route Source** dialog box.
5. Select a **Source Interface** and **Source Address** (these cannot be the management interface).

   The source interface must have internet connectivity. The best practice is to use a dataplane interface that has connectivity to cloud services. See Configure Interfaces and Create an Address Object for more information about creating source interfaces and addresses.
6. Click **OK** to set the source interface and address.
7. Click **OK** to set the Service Route Configuration.
8. Select **Policies** > **Security** and add a Security policy rule that allows traffic from the source interface you specified earlier in this procedure to the FQDN addresses for the KCS and DCS services, which are **`kcs.ace.tpcloud.paloaltonetworks`** (KCS service for all regions) and **`hawkeye.services-edge.paloaltonetworks.com`** (US region DCS service), **`eu.hawkeye.services-edge.paloaltonetworks.com`** (EU region DCS service), or **`apac.hawkeye.services-edge.paloaltonetworks.com`** (APAC region DCS service).

   Also add and allow the following two FQDNs in a new or existing Security policy rule: **`ocsp.paloaltonetworks.com`** and **`crl.paloaltonetworks.com`** for certificate verification.

   Finally, add or modify a Security policy rule to allow ACE traffic by allowing the following three applications: **`paloalto-ace`**, **`paloalto-ace-kcs`**, and **`paloalto-dlp-service`**.

STEP 7 | Make sure that hawkeye.services-edge.paloaltonetworks.com and kcs.ace.tpcloud.paloaltonetworks are reachable on firewalls and that kcs.ace.tpcloud.paloaltonetworks is reachable on Panorama devices. (PAN-OS administrator.)

Run the operational command `admin@fw1> show cloud-appid connection-to-cloud`. The output informs you whether the connection is working and if the license is installed.

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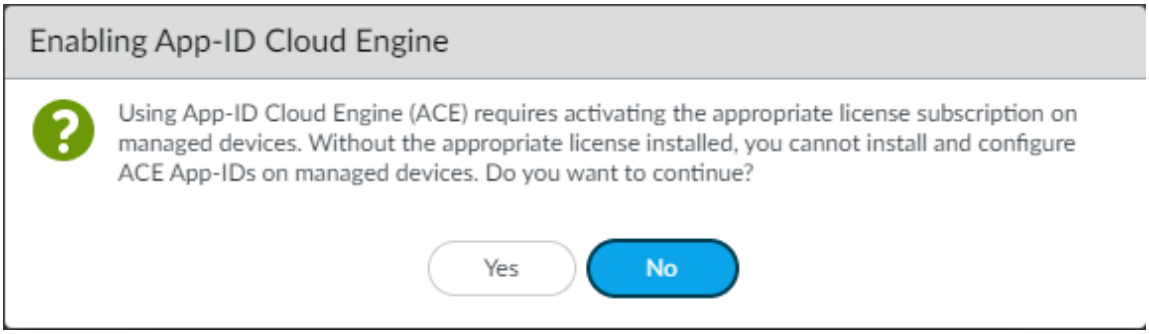
STEP 8 | (Panorama only) Enable ACE on any Panorama appliance that manages ACE-enabled firewalls. (PAN-OS administrator.)

ACE is disabled by default on Panorama.

If you push ACE configurations to managed groups that do not have ACE-enabled firewalls (some or all firewalls in the group do not have ACE enabled), the push fails.

1. Navigate to **Panorama** > **Setup** > **ACE** > **Settings**.
2. Click edit (⚙) and then de-select **Disable App-ID Cloud Engine.**
3. Click **OK**.
4. The **Enable App-ID Cloud Engine** dialog appears.



   Click **Yes** to enable ACE.
5. **Commit** the change.

STEP 9 | Wait for the App-ID catalog to download. (PAN-OS administrator.)

There are fewer than four thousand content-provided App-IDs. After you download the ACE catalog, you see many thousands more applications on the firewall and can confirm by checking **Objects** > **Applications** or by using the operational CLI command `show cloud-appid cloud-app-data application all` to see the new App-IDs.

STEP 10 | (Panorama only) Push the desired configuration to the managed firewall(s). (PAN-OS administrator.)

STEP 11 | Configure Log Forwarding to Strata Logging Service and enable Log Forwarding with the correct Log Forwarding profile in Security policy rules. (PAN-OS administrator.)

A SaaS Security Inline connection to Strata Logging Service is required for SaaS visibility and to support SaaS App-ID Policy Recommendation. At a minimum, you must forward Traffic logs and URL logs to Strata Logging Service for SaaS Security Inline to work properly.

Enable or Disable the App-ID Cloud Engine

Where Can I Use This?	What Do I Need?
• Prisma Access	☐ SaaS Security Inline license (for NGFW)

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Where Can I Use This?	What Do I Need?
• Next-Generation Firewall	❑ Prisma Access license (ACE is a core feature)

The App-ID Cloud Engine (ACE) is disabled by default on Panorama and enabled by default on firewalls when the SaaS Security Inline license is installed. You must enable ACE on Panorama appliances that manage ACE-enabled firewalls.

To enable or disable ACE:

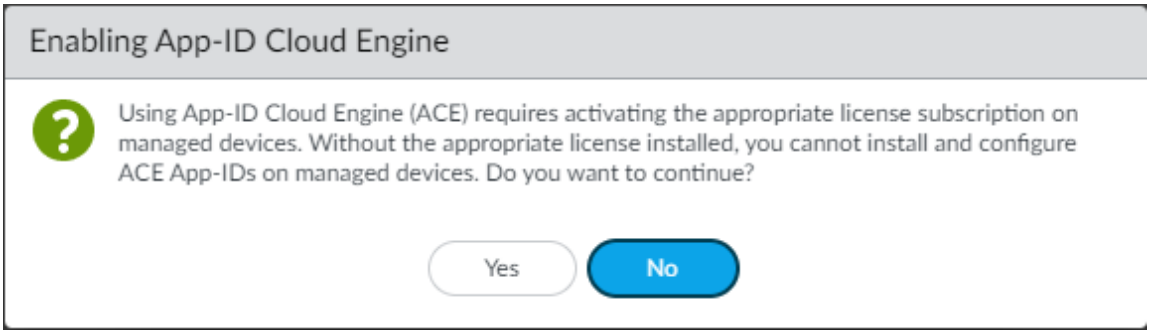
STEP 1 | Navigate to **Device** > **Setup** > **ACE** > **Settings** on the firewall or **Panorama** > **Setup** > **ACE** > **Settings** on Panorama.

STEP 2 | Click edit (⚙) and then either de-select **Disable App-ID Cloud Engine** to enable ACE or select **Disable App-ID Cloud Engine** to disable ACE.

ACE is disabled by default.

STEP 3 | Click **OK**.

STEP 4 | (Only if enabling ACE) If you are enabling ACE, the **Enable App-ID Cloud Engine** dialog appears.



If the firewall or Panorama-managed firewalls have the SaaS Security Inline license installed, click **Yes** to enable ACE.

STEP 5 | **Commit** the change.

App-ID Cloud Engine Processing and Policy Usage

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	❑ SaaS Security Inline license (for NGFW) ❑ Prisma Access license (ACE is a core feature)

When the firewall downloads App-ID Cloud Engine (ACE) App-IDs, it's important to understand how the firewall handles ACE App-IDs and how the firewall handles ACE App-IDs when there are also predefined content-based App-IDs for the same applications. The Palo Alto Networks content team develops predefined content-based App-IDs and updates them with modified

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and new App-IDs through application content updates (a valid support contract is required for updates).

ACE requires a SaaS Security Inline license. Firewalls that don't support ACE have only predefined content-based App-IDs. The ACE App-ID catalog doesn't contain content-based App-IDs.

You can only use ACE App-IDs in Security policy rules. You cannot use ACE App-IDs in any other type of policy rule.

- When the firewall first connects to ACE, the firewall downloads a catalog of the available ACE App-IDs and you can use those App-IDs in Security policy. The firewall does not download the full application signatures, only the catalog. The catalog enables you to specify ACE App-IDs in Security policy even if the applications have never been seen on the firewall. ACE pushes catalog updates to firewalls regularly so that firewalls have access to the latest ACE App-IDs.

  If an application arrives at the firewall that is identified as ssl or web-browsing and the firewall doesn't have its signature, the firewall sends the payload to ACE. If ACE has a matching App-ID, then ACE sends the full signature back to the firewall. If the traffic doesn't match any ACE signatures, then ACE sends the payload to the Machine Learning (ML) engine. The ML engine analyzes the payload and develops a new App-ID in conjunction with the human content team. The ML engine sends the new App-ID to ACE and requesting firewalls can download it and use it in Security policy.

  Because it can take several minutes to retrieve an App-ID from ACE and longer if a new App-ID must be developed, cloud application detection is not inline on the firewall. The firewall does not wait for a verdict to process the application traffic. The firewall processes the traffic as ssl or web-browsing until it receives an App-ID from ACE.
- When a firewall requests an App-ID from ACE, the firewall continues to process the traffic against the current rulebase until it receives an App-ID from ACE and the App-ID is applied in Security policy.
- The firewall handles ACE App-IDs differently than it handles App-IDs delivered by content updates. You don't have to examine how new ACE App-IDs affect Security policy before they are installed on the firewall because the firewall handles new ACE App-IDs according to your existing Security policy. Your existing Security policy rules control new ACE App-IDs until you explicitly use ACE App-IDs in Security policy. For example:

  1. An application is identified only as "ssl" and you have a Security policy rule that allows SSL traffic, so the ssl rule allows that application.
  2. The firewall sees an application identified as ssl and sends the payload to ACE.
  3. ACE identifies the actual application. If the application exists in the ACE database, then ACE sends its App-ID to the firewall. If it's a new application without an ACE App-ID, then ACE forwards the payload to the ML Engine. The firewall does not receive the App-ID until the ML Engine and the human content team assign an App-ID and send it to ACE.
  4. The rule that allows ssl traffic still allows the newly-identified application, even though its App-ID is no longer "ssl". (However, if you use the new ACE App-ID in Security policy, that policy controls the traffic. Similarly, traffic previously identified as web-browsing continues

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to obey the Security policy rules that control web browsing traffic until you use the ACE App-IDs in Security policy.)

The exception to this behavior is if another Security policy rule already specifies the App-ID given to the traffic by ACE. The Security policy rule with the specific App-ID takes precedence over the rule with the less specific ssl App-ID. For example, if the firewall identifies an application as ssl and sends the payload to ACE to obtain the granular App-ID. ACE returns the App-ID “app-abc”. The firewall already has a Security policy rule that allows the App-ID “app-abc”, so the application’s traffic now matches that rule.

If the rule that specifies the actual App-ID is a block rule, the application is blocked even though there is a rule that allows ssl traffic. The rule with the more specific (granular) App-ID is the one the firewall acts on.

Until you explicitly add new ACE App-IDs to Security policy rules, the firewall controls them with the same rules that controlled those applications before they had ACE App-IDs and were identified as ssl or web-browsing traffic. For example, if the firewall sees an application identified as web-browsing and then receives an ACE App-ID for the traffic, but you don’t use that ACE App-ID in a Security policy rule, then the firewall still controls that traffic using the rule that controls web-browsing traffic—if you block web-browsing traffic, then the traffic is blocked, and if you allow web-browsing traffic, the traffic is allowed.

- The firewall caches some information so that the firewall can avoid repeatedly sending data to the cloud and requesting verdicts. If the firewall is waiting for a verdict from ACE, the firewall doesn’t forward the same application data twice.
- On the firewall, a particular container app and its functional applications are either all cloud-based App-IDs or all content-based App-IDs. One App-ID delivery method defines a container app and all of its functional apps.
- If cloud-based, content-provided, and user-defined custom App-ID names overlap, the order of precedence is:

  1. **Custom App-IDs**—These App-IDs take precedence over all other App-IDs. If the firewall attempts to download an ACE application with the same App-ID, the commit fails because two applications on the same firewall cannot have the same App-ID.

     In this case, you can rename the custom application, or if the custom application is the same application as the ACE application, you can delete the custom application and use the ACE application.
  2. **Content-based, predefined App-IDs**—These App-IDs take precedence over ACE cloud App-ID definitions.
  3. **ACE cloud App-IDs**—Custom and content-based App-IDs take precedence over ACE App-ID definitions.

- If an App-ID matches a container app, the firewall downloads the container app’s App-ID and all of its functional apps. For example, if the firewall retrieves the facebook container app, it also retrieves facebook-base, facebook-chat, facebook-post, etc.

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- When you take any of the following actions to add ACE App-IDs to Security policy rules, the firewall no longer matches the application traffic to the ssl or web-browsing rule, it matches the application traffic to the rule that controls the specific App-ID:
  - Create Application Filters to automate adding ACE App-IDs to Security policy.

    Use Application Filters to automate adding ACE App-IDs to Security policy rules. When a new App-ID matches an Application Filter, the firewall automatically adds it to the filter. When you use that Application Filter in a Security policy rule, the rule controls the application traffic for the new App-IDs that were automatically added to the filter. Application Filters are your "Easy Button" for securing ACE App-IDs automatically to gain maximum application visibility and control with minimum effort.
  - Add ACE App-IDs to Application Groups.
  - Use Policy Optimizer to add ACE App-IDs to a cloned rule or to an existing rule, or to an existing Application Filter or Application Group. You can use Policy Optimizer to create new Application Filters and Application Groups directly from within the Policy Optimizer tool. Use Policy Optimizer's sorting and filtering tools to prioritize the rules to work on and to assess how many ACE App-IDs match those rules.
  - Add an ACE App-ID directly to a new or existing Security policy rule.

  When you add a cloud App-ID to a Security policy rule directly or by using an Application Filter or an Application Group, that rule controls the application.
- When you create Application Filters, exclude ssl and web-browsing from the filters. Together, ssl and web-browsing match all browser-based cloud applications, so an Application Filter that includes ssl and web-browsing matches all browser-based cloud applications.
- Active/Passive High Availability:
  - The Active firewall syncs the ACE catalog to the passive firewall so that they have identical catalogs.
  - The Passive firewall does not initiate connections to ACE until it becomes the Active firewall.
- Active/Active High Availability: Each device fetches catalogs and signatures separately, so the catalogs and signatures are not synced. However, commits fail if the catalog is out-of-sync on peers and ACE App-IDs are referenced in Security policy rules. If the catalogs of peer HA firewalls are out-of-sync, wait a few minutes for the updates to reach the devices and become in-sync again.
- A Panorama commit all/push failure to managed firewalls occurs if:
  - Managed firewalls do not have a valid SaaS Security Inline license and therefore do not have the ACE catalog. In this case, remove the ACE objects from the pushed configuration and try again.
  - The connection between a managed firewall and ACE goes down and the pushed configuration includes applications that are not in the ACE catalog on the firewall. In this

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case, check the firewall connection to the ACE cloud and re-establish the connection if necessary so that the firewall can update its catalog.

The operational CLI command `show cloud-appid connection-to-cloud` provides the cloud connection status and the ACE cloud server URL.

- The ACE catalog on Panorama and the ACE catalog on managed firewalls is out-of-sync, which results in pushed configurations that include ACE apps that are not in the firewall's catalog. If the connection between the firewall and ACE is up, the outdated catalog will update in the next few minutes automatically and resolve the issue. (Wait five minutes and try again.)

You can use the CLI command `debug cloud-appid cloud-manual-pull check-cloud-app-data` to update the catalog manually.

- Some Security profiles such as the File Blocking, Antivirus, WildFire, and DLP profiles can specify applications as part of the profile. Only content-provided App-IDs are supported in Security profiles. ACE App-IDs are not supported in Security profiles. ACE App-IDs are intended for use in Security policy rules only.
- Because ACE App-IDs are supported only for Security policy, they are not supported in Application Override, Policy-Based Forwarding (PBF), QoS, or SD-WAN policy rules.

You cannot see ACE App-IDs in Application Override or PBF rule configuration. However, ACE App-IDs are visible (able to be selected) in QoS and SD-WAN policy rule configuration and may be present in Application Groups or Application Filters applied to a rule. If you use ACE App-IDs in these rules, the policy doesn't control the application traffic and there is no effect on the application traffic—the rules do not apply to the ACE App-ID traffic even though ACE App-IDs were added to the rule.

New App Viewer (Policy Optimizer)

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	❑ SaaS Security Inline license (for NGFW) ❑ Prisma Access license (ACE is a core feature)

The Policy Optimizer **New App Viewer** shows you the Security policy rules that match downloaded cloud App-IDs from ACE. Use Policy Optimizer to manage the newly identified applications and add them to cloned rules or to existing rules. Select **Policies** > **Security** and then select **New App Viewer** in the **Policy Optimizer** portion of the interface.

The upper portion of the screen is similar to **Objects** > **Application Filters**. It works in a similar manner and filters the Security policy rules shown in the lower portion of the screen. You can filter the rules that allow applications by category, subcategory, etc. The only categories and subcategories available for filtering are the ones that match the new applications on the rules listed in the lower half of the screen, so you don't waste your time filtering for applications that aren't there.

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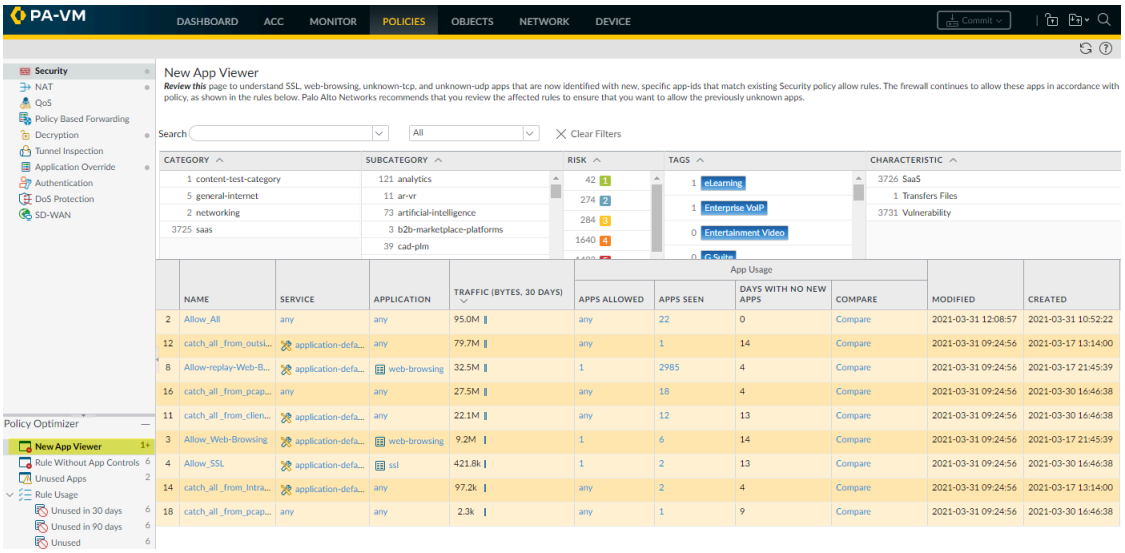
When you filter the rules, only the rules that include the filtered applications are shown in the lower portion of the screen. Rules that have not seen the apps in the filter are removed from the list. (You can see them all again by removing the filter.)



Click the number in the **Apps Seen** column to open the **Applications & Usage** dialog to change the way the firewall handles the cloud-based applications in Security policy. Add ACE App-IDs to Security policy rules using an Application Filter, an Application Group, Policy Optimizer, or by directly adding an ACE App-ID to a rule. Until you take one of these actions to control cloud-delivered App-IDs, the firewall continues to treat the traffic as ssl or web-browsing traffic and uses existing ssl or web-browsing Security policy rules to control the applications.

Add Apps to an Application Filter with Policy Optimizer

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	❑ SaaS Security Inline license (for NGFW) ❑ Prisma Access license (ACE is a core feature)

Add App-IDs from the App-ID Cloud Engine (ACE) to Application Filters to automate adding cloud App-IDs to Security policy. When new ACE App-IDs match an Application Filter, the firewall adds them to the filter automatically. When you use the Application Filter in a Security policy rule, the rule automatically controls new ACE App-IDs as they arrive at the firewall and are added to the filter.

ACE provides App-IDs for applications that were previously identified as ssl or web-browsing.

Using Application Filters is a best practice because they:

- Improve your security posture. Application Filters automate adding new ACE App-IDs to Security policy rules that you design specifically to handle a particular type of application traffic, instead of matching the traffic to more general ssl or web-browsing rules.

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- Save time. Firewall administrators can configure Application Filters to handle different types of traffic so that adding new ACE App-IDs to policy is automatic and requires no further effort by the administrator.

> *When you create Application Filters, exclude ssl and web-browsing from the filters. Together, ssl and web-browsing match all browser-based cloud applications, so an Application Filter that includes ssl and web-browsing matches all browser-based cloud applications.*

Use Policy Optimizer to add ACE App-IDs to Application Filters and to apply the filters to Security policy rules.

STEP 1 | Go to **Policies** > **Security** and then select **Policy Optimizer** > **New App Viewer**.

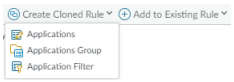
If the firewall has identified traffic with ACE App-IDs, a number displays next to **New App Viewer** in the left navigation window to show how many rules match ACE App-IDs. The screen displays the Security policy rules that match cloud App-IDs.

STEP 2 | Click the number in **Apps Seen** for a Security policy rule to view the cloud-delivered applications that matched the rule in the **Applications & Usage** dialog.

STEP 3 | Select the applications that you want to add to an existing or new Application Filter.

You can sort and filter the applications in **Apps Seen** by subcategory, risk, amount of traffic seen over the last 30 days, or when the application was first or last seen.

STEP 4 | Select **Application Filter** from **Create Cloned Rule** or **Add to Existing Rule**, depending on how you want to handle the applications.



> *The maximum number of applications you can clone using **Create Cloned Rule** is 1,000 applications. If there are more than 1,000 applications that you want to move to a different rule, use **Add to Existing Rule** instead. If you want to move the applications to a new rule, simply create the rule first (**Policies** > **Security**) and then use Policy Optimizer to add them to that rule.*

STEP 5 | Select or create the Application Filter. Creating an Application Filter using Policy Optimizer is the almost exactly the same as using **Objects** > **Application Filters** to create an Application

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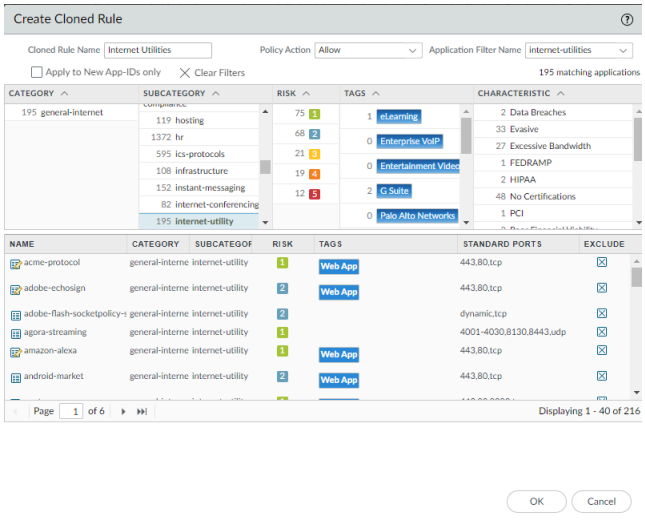
Filter—you use the same filtering tools and options. This step shows you how to use Policy Optimizer first for creating a cloned and then for adding to an existing rule.

Create Cloned Rule:

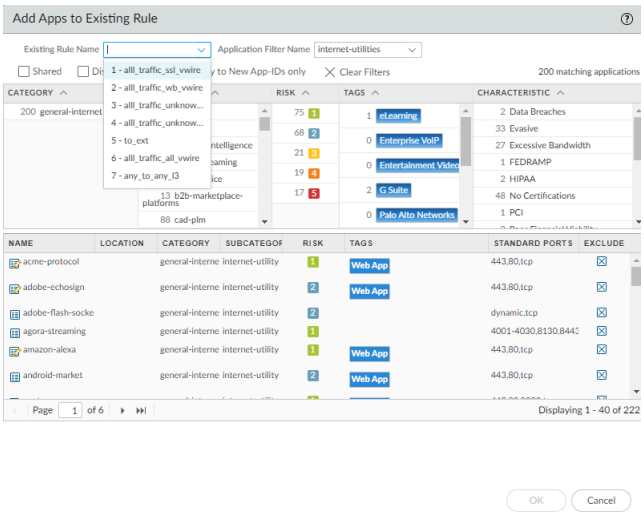
1. Type the **Cloned Rule Name** (the name for the cloned rule, which will appear in the Security policy rulebase immediately above the original rule).
2. Select the **Policy Action** (Allow or Deny).
3. Select the **Application Filter Name** from the menu or type the name of a new Application Filter.
4. Select whether the filter should **Apply to New App-IDs only** or if it should apply to all App-IDs.
5. Use the Category, Subcategory, Risk, Tags, and Characteristic values to filter the types of applications you want to add to the Application Filter. The firewall automatically adds new applications that meet the filter criteria to the Application Filter.



6. Click **OK** to add the applications to the new or existing Application Filter. The firewall includes the applications that you selected in Step 3 in the Application Filter.
7. **Commit** the changes.

Add to Existing Rule:

1. Select the **Existing Rule Name** to add the selected applications to an existing rule in an Application Filter.
2. Select the **Application Filter Name** from the menu or type the name of a new Application Filter.
3. Select whether the Application Filter is **Shared**, whether you want to **Disable override** of application characteristics for the filter, and whether the filter should **Apply to New App-IDs only** or if it should apply to all App-IDs.
4. Use the Category, Subcategory, Risk, Tags, and Characteristic values to filter the types of applications you want to add to the Application Filter. The firewall automatically adds new applications that meet the filter criteria to the Application Filter.



5. Click **OK** to add the applications to the new or existing Application Filter. The firewall includes the applications that you selected in Step 3 in the Application Filter.
6. **Commit** the changes.

Add Apps to an Application Group with Policy Optimizer

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	☐ SaaS Security Inline license (for NGFW) ☐ Prisma Access license (ACE is a core feature)

Add App-IDs from the App-ID Cloud Engine (ACE) to Application Groups and use the Application Groups in Security policy rules to control cloud App-IDs in Security policy.

ACE provides App-IDs for applications that were previously identified as ssl or web-browsing.

Use Policy Optimizer to add ACE App-IDs to Application Groups and to apply the groups to Security policy rules and control the ACE App-IDs in Security policy.

STEP 1 | Go to **Policies** > **Security** and then select **Policy Optimizer** > **New App Viewer**.

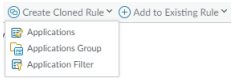
If the firewall or Panorama has downloaded ACE App-IDs, a number displays next to **New App Viewer** in the left navigation window. The screen displays the Security policy rules that match downloaded cloud App-IDs.

STEP 2 | Click the number in **Apps Seen** for a Security policy rule to see the cloud-delivered applications that matched the rule in the **Applications & Usage** dialog.

STEP 3 | Select the applications that you want to add to an existing or new Application Group.

You can sort and filter the applications in **Apps Seen** by subcategory, risk, amount of traffic seen over the last 30 days, or when the application was first or last seen.

STEP 4 | Select **Application Group** from **Create Cloned Rule** or **Add to Existing Rule**, depending on how you want to handle the applications.



*The maximum number of applications you can clone using **Create Cloned Rule** is 1,000 applications. If there are more than 1,000 applications that you want to move to a different rule, use **Add to Existing Rule** instead. If you want to move the applications to a new rule, simply create the rule first (**Policies** > **Security**) and then use Policy Optimizer to add them to that rule.*

STEP 5 | Select or create the Application Group for the cloned or existing rule. Creating Application Groups using Policy Optimizer is similar to using **Objects** > **Application Groups** to create an Application Group.

Create Cloned Rule:

1. Type the **Cloned Rule Name** (the name for the cloned rule, which will appear in the Security policy rulebase immediately above the original rule).
2. Select the **Policy Action** (Allow or Deny).
3. In **Add to Application Group**, select the Application Group to which you want to add the applications that you selected in Step 3.
4. Select whether to **Add container app** (default) or only to **Add specific apps seen**.

   When you add the container app, you also add all of the functional apps in that container, including functional apps that have not yet been seen on the firewall. For example, if you add the “facebook” container app, that also adds facebook-base, facebook-chat, facebook-posting, etc., and also any future applications added to the container. The container app and its functional apps are subject to the Security policy rule to which you add the Application Group. Selecting the container app essentially future-proofs and automates security for the container’s apps so that you don’t have to manually add new apps in that container to your Security policy.

   Adding only the specific apps seen means that only the applications that you selected are added to the Application Group. If new applications in the same container app arrive at the firewall, the Application Group doesn’t control them and you have to manually decide how to handle the new apps.
5. In some cases, the applications that you want to place in an Application Group require (depend on) other applications to function. In those cases, the **Create Cloned Rule** dialog box includes **Dependent Applications**, where you can select whether to add those

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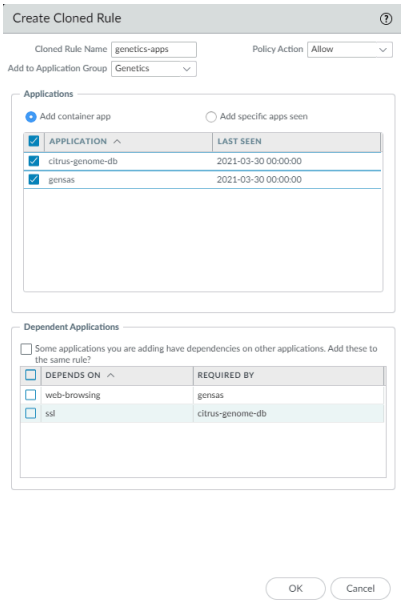
applications to the cloned rule. Add the dependent applications to the rule to ensure that the selected applications function properly.



6. Click **OK** to add the applications to the new or existing Application Group.
7. **Commit** the changes.

Add Apps to Existing Rule:

1. Select the **Existing Rule Name** to add the selected applications to an existing rule in an Application Group.
2. Select the Application Group in **Add to Application Group** or type the name of a new Application Group.
3. As with cloning the rule, you can choose whether to **Add container app** or **Add specific apps seen**. Adding the container app adds all the functional apps in the container and any future apps added to that container. Adding only the specific apps only adds the specific selected apps.
4. As with cloning the rule, in some cases, the applications that you want to place in an Application Group require (depend on) other applications to function. In those cases, the **Add Apps to Existing Rule** dialog box includes **Dependent Applications**, where you can

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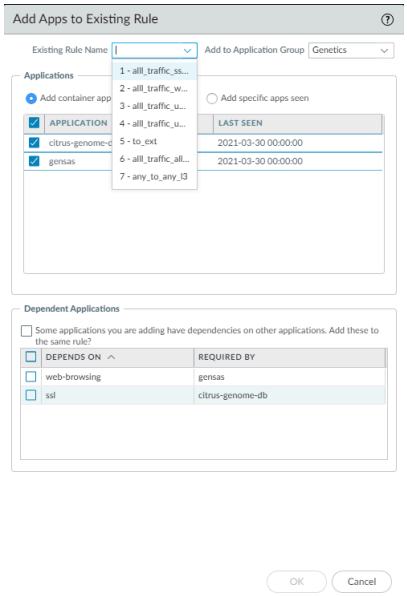
select whether to add those applications to the cloned rule. Add the dependent applications to the rule to ensure that the selected applications function properly.



5. Click **OK** to add the applications to the new or existing Application Group.
6. **Commit** the changes.

Add Apps Directly to a Rule with Policy Optimizer

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	☐ SaaS Security Inline license (for NGFW) ☐ Prisma Access license (ACE is a core feature)

You can add App-ID Cloud Engine (ACE) App-IDs directly to a rule with Policy Optimizer. However, consider using Application Filters to automate adding ACE App-IDs to Security policy as they arrive at the firewall instead of adding them manually.

ACE provides App-IDs for applications that were previously identified as ssl or web-browsing.

STEP 1 | Go to **Policies** > **Security** and then select **Policy Optimizer** > **New App Viewer**.

If the firewall or Panorama has downloaded ACE App-IDs, a number displays next to **New App Viewer** in the left navigation window. The screen displays the Security policy rules that match downloaded cloud App-IDs.

STEP 2 | Click the number in **Apps Seen** for a Security policy rule to view the cloud-delivered applications that matched the rule in the **Applications & Usage** dialog.

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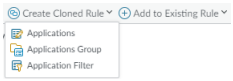
STEP 3 | Select the applications that you want to add to an existing or cloned Security policy rule.

You can sort and filter the applications in **Apps Seen** by subcategory, risk, amount of traffic seen over the last 30 days, or when the application was first or last seen.

STEP 4 | Select **Applications** from **Create Cloned Rule** or **Add to Existing Rule**, depending on how you want to handle the applications.



*The maximum number of applications you can clone using **Create Cloned Rule** is 1,000 applications. If there are more than 1,000 applications that you want to move to a different rule, use **Add to Existing Rule** instead. If you want to move the applications to a new rule, simply create the rule first (**Policies** > **Security**) and then use Policy Optimizer to add them to that rule.*

STEP 5 | Add the selected applications to a cloned rule or to an existing rule.

Create Cloned Rule:

1. Type the **Name** (the name for the cloned rule, which will appear in the Security policy rulebase immediately above the original rule). The cloned rule has the same action (allow or deny) as the original rule.
2. Select whether to **Add container app** (default) or only to **Add specific apps seen**.

   When you add the container app, you also add all of the functional apps in that container, including functional apps that have not yet been seen on the firewall. For example, if you add the “facebook” container app, that also adds facebook-base, facebook-chat, facebook-posting, etc., and also any future applications added to the container. The container and its functional apps are subject to the Security policy rule that you are cloning. Selecting the container app essentially future-proofs and automates security for the container’s apps so that you don’t have to manually add new apps in that container to your Security policy.

   Adding only the specific apps seen means that only the applications that you selected are added to the cloned rule. If new applications in the same container app arrive at the firewall, the cloned rule doesn’t control them and you have to manually decide how to handle the new apps.
3. In some cases, the applications that you want to add to a rule require (depend on) other applications to function. In those cases, the **Create Cloned Rule** dialog box includes **Dependent Applications**, where you can select whether to add those applications to

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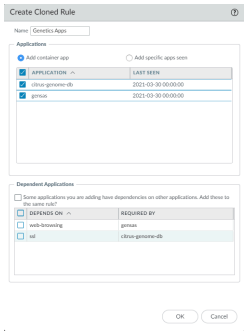
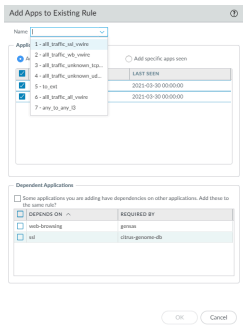
the cloned rule. Add the dependent applications to the rule to ensure that the selected applications function properly.



4. Click **OK** to add the applications to the cloned rule.
5. **Commit** the changes.

Add Apps to Existing Rule:

1. Select the **Name** of the existing rule to which you want to add the selected applications.
2. As with cloning the rule to add applications, you can choose whether to **Add container app** or **Add specific apps seen**. Adding the container app adds all the functional apps in the container and any future apps added to that container. Adding only the specific apps only adds the specific selected apps.
3. As with cloning the rule, in some cases, the applications that you want to add to a rule require (depend on) other applications to function. In those cases, the **Add Apps to Existing Rule** dialog box includes **Dependent Applications**, where you can select whether to add those applications to the cloned rule. Add the dependent applications to the rule to ensure that the selected applications function properly.



4. Click **OK** to add the applications to the existing rule.
5. **Commit** the changes.

Replace an RMA Firewall (ACE)

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	☐ SaaS Security Inline license (for NGFW) ☐ Prisma Access license (ACE is a core feature)

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To restore the configuration on a managed firewall when there is a Return Merchandise Authorization (RMA), the procedure is to:

- Review Before Starting RMA Firewall Replacement.
- On Panorama, replace the serial number of the old firewall with the new firewall's serial number.
- In the firewall CLI, check to ensure that the firewall is online and connected to the Knowledge service so that the firewall can download the cloud application catalog:
  1. Access the firewall CLI.
  2. In Operational mode, check the cloud App-ID connection:

     ```
     admin@vm1> show cloud-appid connection-to-cloud
     ```

     If the firewall is connected to the cloud, the show command returns:

     ```
     ACE Cloud server: kcs.ace.tpcloud.paloaltonetworks.com:443Cloud connection: connected
     ```

     Information about the connection also displays. If the firewall is not connected to the cloud, check whether DNS services are functioning and check for any other network-related connectivity issues.
- With the firewall connected to the App-ID cloud, Restore the Firewall Configuration after Replacement.

Impact of License Expiration or Disabling ACE

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	❑ SaaS Security Inline license (for NGFW) ❑ Prisma Access license (ACE is a core feature)

If you enable App-ID Cloud Engine (ACE) on a firewall, download ACE App-IDs to the firewall, and then use those App-IDs in objects such as Application Filters and in Security policy rules, then you need to understand what happens if the SaaS Security Inline license expires or if you disable ACE. Disabling ACE and the SaaS Security Inline license expiring both affect downloaded ACE App-IDs, the catalog of ACE App-IDs, Security policy rules that control ACE App-IDs, and objects that include ACE App-IDs. The effect is the same unless otherwise noted:

- ACE App-IDs remain on the firewall, but the firewall stops enforcing ACE App-IDs in Security policy.

  Security policy rules that control ACE App-IDs no longer control ACE App-IDs even though they are visible in the rule. Traffic that was controlled by ssl or web-browsing rules before ACE was enabled on the firewall is controlled by those rules again until you update and activate the SaaS Security Inline license and/or re-enable ACE or change those rules.

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- Enforcement of Security policy rules based on ACE App-IDs stops within 4-6 hours of the license expiring (based on a timer that periodically checks license status).

  Enforcement of Security policy rules based on ACE App-IDs stops immediately after you commit the disabling ACE on the firewall.

  Disabling ACE stops enforcing Security policy rules based on ACE App-IDs as soon as you commit the change even if the SaaS Security Inline license is still valid and active.
- The catalog of ACE App-IDs remains on the firewall and on Panorama but the cloud engine no longer updates the catalog.
- The connection from the firewall to ACE no longer functions. If you re-enable ACE or renew the SaaS Security Inline license, it may take some time to download all of the catalog updates.
- If the SaaS Security Inline license expires, the ACE service stops working within 4-6 hours.

  Panorama doesn't require a SaaS Security Inline license, so there is no license to expire on Panorama. However, when the license expires on managed firewalls, configuration pushes to those firewalls from Panorama fail if they contain ACE configurations in Security policy or in Application Groups.
- Objects such as Application Filters and Application Groups are not changed, but any ACE App-IDs that you placed in those objects are no longer enforced even though the ACE App-IDs are still visible.
- If you are using SaaS Policy Recommendation, the firewall can no longer pull SaaS policy recommendations, so the SaaS administrator cannot push new policy recommendations to the firewall. Policy recommendations that were downloaded before license expiration remain in the configuration but they are not enforced (same behavior as Security policies configured with ACE App-IDs when the license expires or ACE is disabled).

Commit Failure Due to Cloud Content Rollback

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	☐ SaaS Security Inline license (for NGFW) ☐ Prisma Access license (ACE is a core feature)

Although it is extremely unlikely, it is possible that ACE App-IDs may need to be rolled back (reverted) because of bad metadata or issues with applications. If ACE must revert App-IDs and you used those App-IDs in a Security policy rule (directly or in an Application Group), commit actions fail until those applications are removed from Security policy rules and from objects.

If it becomes necessary to roll back App-IDs, ACE reverts all of the most recently delivered cloud-based App-IDs, signatures, metadata, categories, subcategories, and tags from the ACE catalog. Removing the App-IDs from the catalog removes them from the firewall, which is why the commit action fails when the App-IDs are used in Security policy.

If you did not use the applications that ACE had to roll back in Security policy, there is no impact to the configuration and commit actions succeed.

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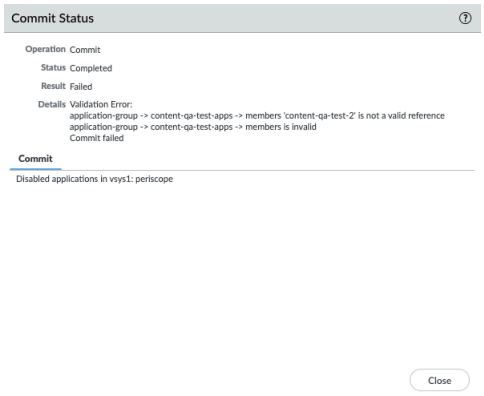
When you attempt to commit a configuration after an ACE content rollback, the commit failure message lists the applications that ACE reverted, as in this example **Validation Error**:



To fix the issue, you must remove the listed applications from Security policy rules, regardless of whether they were added directly to a rule or were added using an Application Group. If the application is used in an Application Group, remove it from the Application Group.

In this example, `content-qa-test-2` is the reverted application, which is referenced in the Application Group `content-qa-test-apps`. After you remove `content-qa-test-2` from the Application Group, commit actions succeed.

Troubleshoot App-ID Cloud Engine

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	❑ SaaS Security Inline license (for NGFW) ❑ Prisma Access license (ACE is a core feature)

This topic provides general troubleshooting information for the App-ID Cloud Engine (ACE).

- To check if an appliance has a valid SaaS Security Inline license, run the operational CLI command `show cloud-appid connection-to-cloud`. If there is an issue, the command returns the message:

  ```
  ACE Error: License check failed. Check if SaaS license is installed
  and activeCloud connection: failed
  ```

  In addition, the output shows the time of the last successful connection, for example: `Last successful gRPC connection: 2021-05-20 16:00:00 -0800 PDT`

  If the license is installed and the connection to ACE is good, then the command returns the URL for the ACE cloud server connection and the status `Cloud connection: connected`, along with connection statistics and the status of the device certificate, including the certificate validity dates.

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- Panorama commit all/push to managed firewalls fails. Check if any of the following conditions exist and repair them:
  - Do managed firewalls have a valid SaaS Security Inline license? If not, then they do not have the ACE catalog and the commit all/push operation fails. Depending on whether you want to managed firewalls to handle ACE App-IDs, either remove the ACE objects from the pushed configuration and try again or install valid SaaS Security Inline licenses on the managed firewalls, wait for the catalog to download.

    *There are fewer than four thousand content-provided App-IDs. After you download the ACE catalog, you see many thousands more applications on the firewall and can confirm by checking **Objects** > **Applications** or by using the operational CLI command `show cloud-appid cloud-app-data application all` to see the new App-IDs.*
  - Has the connection between a managed firewall and ACE has gone down? Check the connection to the ACE cloud and restore the connection if necessary.

    The operational CLI command `show cloud-appid connection-to-cloud` provides the cloud connection status and the ACE cloud server URL.
  - The ACE catalog on Panorama and the ACE catalog on managed firewalls is out-of-sync, which results in pushed configurations that include ACE apps that are not in the firewall's catalog. If the connection between the firewall and ACE is up, the outdated catalog will update in the next few minutes automatically and resolve the issue. (Wait five minutes and try again.)

    You can also run the operational CLI command `debug cloud-appid cloud-manual-pull check-cloud-app-data` to update the catalog manually.
  - Are the firewalls all running PAN-OS 11.1 or later? (Pushing configurations that reference ACE applications and objects to firewalls running earlier versions than PAN-OS 11.1 is not allowed.)
- In an HA pair (active/active or active/passive) that has an ACE configuration, if you run the operational command `show session all` or `show session id <id>`, the output for ACE applications may show the global App-ID number instead of the application name. The firewall only shows the application name if its data plane has the cloud application data. If not, then the firewall shows the global App-ID number for the application instead.
- To reset the connection to ACE (the gRPC connection), run the operational CLI command `debug cloud-appid reset connection-to-cloud`.
- View the ACE applications downloaded to the appliance with the operational CLI command `show cloud-appid cloud-app-data application`. You can view all downloaded apps or individual apps by App-ID or application name.
- View pending requests for ACE App-IDs with the operational CLI command `show cloud-appid signature-dp pending-request`. The output includes how many times the firewall sent the request to ACE (`tries`). After eleven tries, the send operation times out.
- The operational CLI command `show cloud-appid` has more useful options:

```
admin@PAN-ACE-VM-1> show cloud-appid ?
```

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```
> app-objects-in-policy          Show application-filter/application-
groups referred in policy
> app-to-filtergroup-mapping     Show application to matched filter
 and groups
> application                    Show Application info for UI
> application-filter             Show cloud apps in application-
filters
> application-group              Show cloud apps in application-
groups
> cloud-app-data                 Show cloud application, container
 and metadata
> connection-to-cloud            Show gRPC connection status to cloud
 application server
> ha-info                        Show statistics of cloud application
 high availability
> overlap-appid                  Show duplicated applications in
 predefined content
> signature-dp                   Show cloud signatures and
 applications used on DP
> task                           Show task on management-plane
> transaction                    Show cloud application transaction
> version                        Show Cloud-AppID version
```

- To view the global counters for ACE, run the operational CLI command `show counter global filter value all category cad` (`cad` stands for "cloud app-identification).
- To view statistics for bytes and packets received and sent to/from shared memory and to/from the security client for services such as ACE, DLP, and IoT, run the operational command `show ctd-agent statistics`.
- If you notice a discrepancy between the number of applications that match an Application Filter when you look in the user interface versus when you look in the CLI, it's because of the way the firewall counts matching applications in the user interfaces versus in the CLI:
  - When you look at an Application Filter in **Objects** > **Application Filters**, the firewall displays all of the matching applications in the ACE catalog, regardless of whether the firewall has actually seen those applications and downloaded their App-IDs, and the number count includes all of those applications.
  - When you look at an Application Filter in the CLI with the `show cloud-appid application-filter` operational command, the firewall only displays the number of matching applications for which the firewall has downloaded ACE App-IDs.

  For this reason, the user interface may show more matching applications than the CLI for the same Application Filter.

  The same thing applies to Application Groups when you look at them in the user interface versus the CLI.
- ACE App-IDs are supported for Security policy only. ACE App-IDs are not supported for any other policy type.

  However, when you configure QoS or SD-WAN policy, ACE App-IDs are visible (able to be selected) and may be present in Application Groups or Application Filters applied to the rule, but adding them to QoS or SD-WAN policy has no effect on the application traffic. (The QoS and SD-WAN policies don't control the application traffic.)

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SaaS App-ID Policy Recommendation

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

The rapid proliferation of SaaS applications makes it difficult to assign all of them specific App-IDs, gain visibility into those applications, and control them. Security policy rules that allow ssl, web-browsing, or “any” application may allow unsanctioned SaaS applications that can introduce security risks to your network. To gain visibility into those applications and control them on the firewall, SaaS Security administrators can recommend Security policy rules with specific SaaS App-IDs provided by the App-ID Cloud Engine (ACE) to PAN-OS firewall administrators. PAN-OS administrators can import those rules on firewall’s that have a SaaS Security Inline subscription.

SaaS Policy Recommendation requires a SaaS Security Inline subscription. Each appliance that uses the SaaS Policy Recommendation Engine needs to generate and install a valid device certificate or use Panorama to generate and install a valid device certificate.

A SaaS Security Inline connection to Strata Logging Service (Strata Logging Service) is required for SaaS visibility. Configure Log Forwarding to Strata Logging Service and enable Log Forwarding with the correct Log Forwarding profile in Security policy rules. At a minimum, you must forward Traffic logs and URL logs to Strata Logging Service for SaaS Security Inline to work properly.

All hardware platforms that support PAN-OS 10.1 or later support SaaS Policy Recommendation and all appliances on which you want to use SaaS Policy Recommendation require PAN-OS 10.1 or later. Panorama cannot push and commit SaaS Policy Recommendations to firewalls that don’t have a SaaS Security Inline license installed or to firewalls that run an earlier version of PAN-OS than 10.1.

- The *SaaS Security Administrator’s Guide* describes the SaaS Security administrator’s procedure for creating Security policy rule recommendations and then pushing them to the firewall.
- The *PAN-OS Administrator’s Guide* describes how the PAN-OS administrator imports and manages policy recommendations from the SaaS Security administrator.

The SaaS Security administrator creates the new rule, adds applications, users, and groups to the rule, and sets the rule action. The rule action can be allow or block; no other actions are permitted for pushed rules. The SaaS Security administrator then pushes the rule to the appropriate appliances and the rule appears in the firewall interface (**Device** > **Policy Recommendation** > **SaaS**).

The PAN-OS administrator evaluates the recommended rule and decides whether to implement it on the firewall. If the PAN-OS administrator chooses to implement the rule, the administrator imports it on the firewall and selects where to place the policy rule in the firewall rulebase. When a PAN-OS administrator imports a policy recommendation, the firewall creates the required HIP profiles, tags, and Application Groups automatically (the PAN-OS administrator doesn’t have to do it manually).

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If the SaaS Security administrator pushes Security profiles with the policy recommendation and those profiles don't exist on the firewall, the firewall import fails. If the profiles already exist on the firewall, the import succeeds.

If the SaaS Security administrator updates a policy rule recommendation, the PAN-OS administrator sees the update and imports it into the firewall. If the SaaS Security administrator deletes a policy rule recommendation, the PAN-OS administrator sees the action and deletes the rule from the firewall Security policy rulebase.

If the SaaS Security Inline license expires, the firewall no longer pulls SaaS policy recommendations, so you see no new recommendations. However, Security policy rules that you already imported continue to work.

If you disable ACE, the firewall no longer receives new cloud application signatures and App-IDs and the firewall cannot import SaaS policy recommendations based on new ACE App-IDs.

The ACE deployment process (connecting to the cloud, installing device certificates, activating the license on the SaaS Security Portal and pushing it to Panorama and firewalls, etc.) also sets up SaaS Policy Recommendation.

Update all appliances to the latest Threat content updates.

User interface additions for this new feature include:

- **Device** > **Policy Recommendation** > **SaaS** displays policy recommendations from SaaS administrators and enables firewall administrators to import, update, remove, and control recommended SaaS policies. The page display includes Application Groups configured by the SaaS administrator for the policy.
- Role-based interface access (**Device** > **Admin Roles**) has a new option on the **Web UI** tab for SaaS policy recommendation permissions: **Device** > **Policy Recommendation** > **SaaS**.
- SaaS policy recommendations are automatically tagged **SaaSSecurityRecommended**, which is displayed in the **Tags** column in the interface.

You can import and update SaaS policy recommendations pushed by SaaS administrators and remove SaaS policy recommendations that the SaaS administrator has deleted.

- Import SaaS Policy Recommendation
- Import Updated SaaS Policy Recommendation
- Remove Deleted SaaS Policy Recommendation

Import SaaS Policy Recommendation

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

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When a SaaS Security administrator pushes Security policy rule recommendations to a PAN-OS firewall, the PAN-OS firewall administrator can import those rules on the firewall to gain visibility into and control of the applications in the policy recommendation.

See the *SaaS Security Administrator's Guide* for the SaaS administrator's policy recommendation and push procedures. This procedure shows PAN-OS administrators how to import policy recommendations.

If the SaaS Security administrator pushes Security profiles with the policy recommendation and those profiles don't exist on the firewall, the firewall import fails. If the profiles already exist on the firewall, the import succeeds.

STEP 1 | **Device** > **Policy Recommendation** > **SaaS** on the firewall and **Panorama** > **Policy Recommendation** > **SaaS** on Panorama show all of the SaaS policy recommendations pushed from the SaaS administrator. Push policy recommendations from Panorama to managed firewalls.

STEP 2 | Refresh ( ) **Device** > **Policy Recommendation** > **SaaS** (or **Panorama** > **Policy Recommendation** > **SaaS**) to ensure that the SaaS policy recommendations are up-to-date.

Any time you push policy recommendations from Panorama to managed firewalls, refresh ( ) the page on the firewalls to ensure that the recommendations are up-to-date.

Newly pushed policy recommendations appear at the top of the screen. **Active Recommendations** shows the value **active** and **New Updates Available** shows the value **Yes**.

STEP 3 | Select a new policy recommendation.

You import one policy recommendation at a time. The **Applications** column shows an Application Group for each policy recommendation. Click the name of the group to see the applications in that group.

The **Device** column shows the source device that the SaaS administrator configured for the rule. The term "SaaS" precedes the source device. The source device can be:

- MCD—Managed Compliant Device
- MNCD—Managed Non-compliant Device
- UMCD—Unmanaged Compliant Device
- UMNCD—Unmanaged Non-compliant Device

For example, **SaaS - MCD** indicates a managed, compliant source device.

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STEP 4 | **Import Policy Rule**.

In the **Import Policy Rule** dialog:

- **Name**—Name the imported rule using a name that describes the rule's intent.

  If you specify a rule name that already exists in the Security policy rulebase, the imported rule overwrites the existing rule.

- **After Rule**—Select the rule after which to place the imported SaaS rule. Think about the firewall's rulebase and how the new rule may affect existing rules. If you do not select a rule (**No Rule Selection**), then the rule is placed at the top of the Security policy rulebase. In some cases, that's not where you want to place the rule. For example, you may want some particular block rules to always be at the top of the rulebase, such as blocking QUIC protocol. Be aware of the intent of the imported rule and be careful not to shadow existing rules.

The **Description** comes from the description entered when the SaaS administrator created the rule. You can change it or leave it as-is.

The import process automatically creates an Application Group for the applications in the policy recommendation. The name of the Application Group is derived from the Name that the SaaS Security administrator gave to the rule. The firewall also automatically creates any HIP profiles and tags that the SaaS administrator applied to the rule.

STEP 5 | Click **OK** to import the rule and add it to the Security policy rulebase in the position selected in **After Rule**.

STEP 6 | When you see the status message "You've successfully updated your Security policy rules", click **OK**.

The **Location** column now shows the rule's location (vsys) on the firewall, which corresponds to the vsys to which the SaaS administrator pushed the rule.

STEP 7 | Confirm that the imported policy rule is in the Security policy rulebase (**Security** > **Policies**) at the specified location and that the firewall created the associated objects.

For example, check the Security policy rule for:

- The rule's **Source Device** is populated and shows the source device for the rule on the **Source** tab.
- The Application Group populates the rule's **Application** tab.
- Associated profiles are attached to the rule (**Actions** tab).

Also check that:

- **Objects** > **Applications Group** shows the imported Application Group.
- **Objects** > **GlobalProtect** > **HIP Objects** and **Objects** > **GlobalProtect** > **HIP Profiles** show the HIP information pushed from the SaaS Security administrator with the rule.

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Import Updated SaaS Policy Recommendation

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

When a SaaS Security administrator pushes Security policy rule recommendations to a PAN-OS firewall (or Panorama), the PAN-OS administrator can import those rules to gain visibility into and control of the applications in the policy recommendation. However, if the SaaS administrator updates the rule, for example by adding or removing applications, the rule also needs to be updated on the firewall.

If the SaaS Security administrator pushes new or updated Application Groups, HIP profiles, or tags, the firewall automatically creates or updates those objects. If the SaaS Security administrator pushes Security profiles with the policy recommendation update and those profiles don't exist on the firewall, the firewall import fails. If the profiles already exist on the firewall, the import succeeds.

STEP 1 | Refresh ( ) **Device** > **Policy Recommendation** > **SaaS** (or **Panorama** > **Policy Recommendation** > **SaaS**) to ensure that you see all of the latest SaaS policy recommendations that the SaaS administrator pushed to the firewall.

STEP 2 | Check **New Updates Available**.

If the value in the **New Updates Available** column is **No**, then there are no updates to the rule. If the value is **Yes**, then the SaaS administrator has pushed an update to the rule to the firewall. In addition, **Active Recommendations** shows the value **active**.

STEP 3 | Click the Application Group name in the **Applications** column to see the updated list of applications that the rule controls.

STEP 4 | Select a policy recommendation to update.

You update only one policy recommendation at a time.

STEP 5 | Click **Import Policy Rule** to import the policy (if there are no updates to the rule, this option is grayed out and you can't select it).

The **Import Policy Rule** dialog appears. The **Name** is already populated and cannot be changed because the rule has already been imported. **After Rule** also cannot be changed in the dialog, but if you want to change the rule's location in the Security policy rulebase, you can do that on **Policies** > **Security** in the same way that you change the position of any Security policy rule. You can change the **Description** or leave it as-is.

STEP 6 | Click **OK**.

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STEP 7 | Click **Yes** in **Confirm Change** to import the updated rule (or click **No** if you don't want to import the changed rule).

The firewall automatically makes any changes to the Application Group, HIP profiles, and tags associated with the rule.

Remove Deleted SaaS Policy Recommendation

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

When a SaaS Security administrator pushes Security policy rule recommendations to a PAN-OS appliance, the PAN-OS administrator can import those rules to gain visibility into and control of the applications in the policy recommendation. However, if the SaaS Security administrator deletes the rule, you should also delete that rule from the PAN-OS appliance.

When a SaaS Security administrator deletes a rule, the **Active Recommendation** column shows the value **removed** (for valid rules, the value is **active**).

STEP 1 | Select a rule that the SaaS Security administrator **removed** (you can select only one rule to remove at a time).

*The **Import Policy Rule** option is grayed out because the rule can no longer be imported.*

STEP 2 | Click **Remove Recommendation Mapping**.

This removes local mapping of the Security policy rule on the firewall. For example, mappings to locations, users, and the rule are deleted. The **Remove Recommendation Mapping** dialog box shows you the location of the rule so that you know from where the rule is removed.

STEP 3 | Click **OK**.

STEP 4 | In the **Confirm Change** dialog, click **Yes** to remove the rule from the policy recommendation database.

This action only removes the rule from the policy recommendation rule list. It does NOT remove the rule from the Security policy rulebase. You must manually remove the rule from the rulebase.

STEP 5 | A **Status** dialog appears to confirm that the policy recommendation mapping has been removed, but you still need to remove the rule from the Security policy rulebase.

STEP 6 | Go to **Policies** > **Security** and delete the rule from the Security policy rulebase.

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Application Level Gateways

App-ID identifies applications by their unique properties rather than just ports

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

The Palo Alto Networks firewall does not classify traffic by port and protocol; instead it identifies the application based on its unique properties and transaction characteristics using the App-ID technology. Some applications, however, require the firewall to dynamically open *pinholes* to establish the connection, determine the parameters for the session and negotiate the ports that will be used for the transfer of data; these applications use the application-layer payload to communicate the dynamic TCP or UDP ports on which the application opens data connections. For such applications, the firewall serves as an Application Level Gateway (ALG), and it opens a pinhole for a limited time and for exclusively transferring data or control traffic. The firewall also performs a NAT rewrite of the payload when necessary.

- *H.323 (H.225 and H.248) ALG is not supported in gatekeeper routed mode.*
- *When the firewall serves as an ALG for the Session Initiation Protocol (SIP), by default it performs NAT on the payload and opens dynamic pinholes for media ports. In some cases, depending on the SIP applications in use in your environment, the SIP endpoints have NAT intelligence embedded in their clients. In such cases, you might need to disable the SIP ALG functionality to prevent the firewall from modifying the signaling sessions. When SIP ALG is disabled, if App-ID determines that a session is SIP, the payload is not translated and dynamic pinholes are not opened. See* Disable the SIP Application-level Gateway (ALG).

When you use Dynamic IP and Port (DIPP) NAT, the Palo Alto Networks firewall ALG decoder needs a combination of IP and Port (Sent-by Address and Sent-by Port) under SIP headers (Contact and Via fields) to be able to translate the mentioned headers and open predict sessions based on them.

The following table lists IPv4, NAT, IPv6, NPTv6 and NAT64 ALGs and indicates with a check mark whether the ALG supports each protocol (such as SIP).

App-ID	IPv4	NAT	IPv6	NPTv6	NAT64
SIP	✓	✓	✓	—	—
SCCP	✓	✓	✓	—	—
MGCP	✓	✓	—	—	—

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App-ID	IPv4	NAT	IPv6	NPTv6	NAT64
FTP	✓	✓	✓	✓	—
RTSP	✓	✓	✓	✓	—
MySQL	✓	✓	—	—	—
Oracle/ SQLNet/ TNS	✓	✓	✓	✓	—
RPC	✓	✓	—	—	—
RSH	✓	✓	—	—	—
UNIStim	✓	✓	—	—	—
H.225	✓	✓	—	—	—
H.248	✓	✓	—	—	—

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Disable the SIP Application-level Gateway (ALG)

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

The Palo Alto Networks firewall uses the Session Initiation Protocol (SIP) application-level gateway (ALG) to open dynamic pinholes in the firewall where NAT is enabled. However, some applications—such as VoIP—have NAT intelligence embedded in the client application. In these cases, the SIP ALG on the firewall can interfere with the signaling sessions and cause the client application to stop working.

One solution to this problem is to define an Application Override Policy for SIP, but using this approach disables the App-ID and threat detection functionality. A better approach is to disable the SIP ALG, which does not disable App-ID or threat detection.

You can disable only the following App-IDs: sccp, sip, teredo, and unistim.

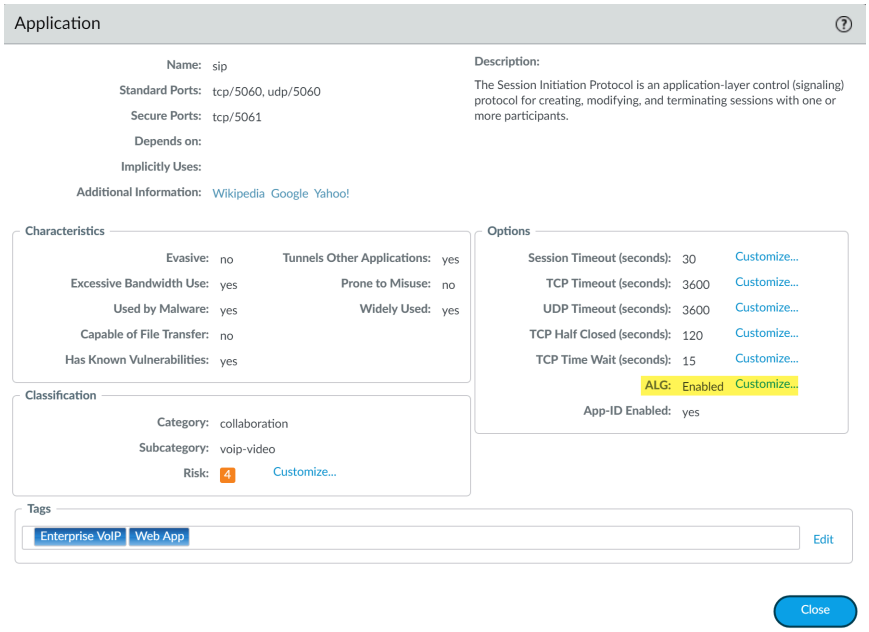
The following procedure describes how to disable the SIP ALG.

STEP 1 | Select **Objects** > **Applications**.

STEP 2 | Select the **sip** application.

You can type **sip** in the **Search** box to help find the sip application.

STEP 3 | Select **Customize...** for **ALG** in the Options section of the Application dialog box.



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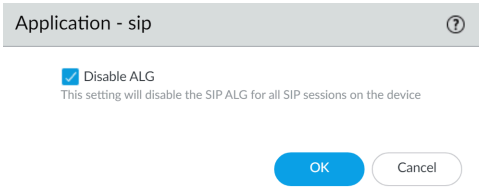
STEP 4 | Select the **Disable ALG** check box in the Application - sip dialog box and click **OK**.



STEP 5 | **Close** the Application dialog box and **Commit** the change.

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Use HTTP Headers to Manage SaaS Application Access

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

Unsanctioned usage of SaaS applications can be a way for your users to transmit sensitive information outside of your network, usually by accessing a consumer version of an application. However, if you need to allow access to the enterprise version of these applications for specific individuals or organizations, then you can't block the SaaS application entirely.

You can use custom HTTP headers to disallow SaaS consumer accounts while allowing a specific enterprise account. Many SaaS applications allow or disallow access to applications based on information contained in specific HTTP headers. You can Create HTTP Header Insertion Entries using Predefined Types to manage access to popular SaaS applications, such as Google G Suite and Microsoft Office 365. Palo Alto Networks® uses content updates to maintain predefined rule sets specific to these applications, as well as to add new predefined rule sets.

You can also Create Custom HTTP Header Insertion Entries if you want to manage access to a SaaS application—that uses HTTP headers to limit service access—for which Palo Alto Networks has not provided a predefined set of rules.

Be aware that commercial SaaS applications always use SSL so decryption is necessary to perform HTTP header insertion. You can configure the firewall to decrypt traffic using SSL Forward Proxy decryption if traffic is not already decrypted by an upstream firewall.

You don't need a URL Filtering license to use this feature.

To understand how to use HTTP headers to manage SaaS applications, see the following:

- Understand SaaS Custom Headers
- Domains used by the Predefined SaaS Application Types
- Create HTTP Header Insertion Entries using Predefined Types
- Create Custom HTTP Header Insertion Entries

Understand SaaS Custom Headers

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

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Before you begin, make sure you understand the custom HTTP headers you will use with the SaaS application you are managing. You need to understand what you can accomplish with these headers and the information you need to specify to accomplish your goals.

Be aware that SaaS applications that use custom headers do not always use them to control access to types of accounts. For example, Palo Alto Networks® provides predefined support for YouTube custom headers that determine whether network users can access restricted content.

You should also read the documentation for the SaaS application to which you want to control access so that you understand what headers you need to use for that application.

The following limits apply to HTTP header insertion:

- *Header name character length: 100.*
- *Header value character length: 16K.*

Be aware that some SaaS applications might define custom header names, or assign values to their custom headers, that exceed these limits. These situations should be rare, but if a SaaS application does exceed one or both of these character length limits, then your next-generation firewall can not successfully manage access to that SaaS application.

The following table lists the headers that you can use for the SaaS applications for which Palo Alto Networks provides predefined support; each header also includes a link to more information specific to that header.

Application	Headers	For More Information
Dropbox	`X-Dropbox-allowed-Team-Ids`	www.dropbox.com/help/business/network-control You can allow access to sanctioned Enterprise Dropbox accounts. This header's value is the business account's team ID, which you can obtain from the network control section of the Dropbox admin console. You must also enable this functionality from the same location. For details on managing this header, as well as how to enable your Dropbox clients so that you can decrypt their traffic, contact your Dropbox account representative.
Google G Suite	`X-GooGApps-Allowed-Domains`	support.google.com/a/answer/1668854?hl=en You can allow access to specific Google accounts from your domain. The values that you give to this header are your domain and subdomains.

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Application	Headers	For More Information
		To successfully insert headers for Google applications, you must also: 1. Create an SSL decryption profile that includes the following categories and URLs: • business-and-economy • computer-and-internet-info • content-delivery-networks • internet-communications-and-telephony • low-risk • online-storage-and-backup • search-engine • web-based-email • drive.google.com • *.google.com • *.googleusercontent.com • *.gstatic.com 2. HTTP header insertion is not currently supported for HTTP/2. To insert headers, downgrade HTTP/2 connections to HTTP/1.1 using the **Strip ALPN** feature in the appropriate decryption profile. For more information, see App-ID and HTTP/2 Inspection. 3. Create rules to block the Quick UDP Internet Connections (QUIC) App-ID and place them at the top of your security policy because the firewall does not support header insertion for this protocol. When you do, the app reverts to using HTTP/2 over TLS, which the firewall handles in the previous step.
Microsoft Office 365	`Restrict-Access-To-Tenants` `Restrict-Access-Context`	docs.microsoft.com/en-us/azure/active-directory/active-directory-tenant-restrictions You provide `Restrict-Access-To-Tenants` with a list of tenants you want to allow your users to access. You can use any domain that is registered with a tenant to identify the tenant in this list.

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Application	Headers	For More Information
		You provide `Restrict-Access-Context` with the directory ID that is setting the tenant restriction. You can find your directory ID in the Azure portal. Sign in as an administrator, select **Azure Active Directory**, then select **Properties**.
YouTube	`YouTube-Restrict`	support.google.com/a/answer/6214622?hl=en You provide this header with information on the type of videos you want your users to be able to view. You can specify either a **Strict** or **Moderate** setting. See support.google.com/a/answer/6212415 for details on these different settings.

Domains used by the Predefined SaaS Application Types

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

SaaS applications use HTTPS so, to insert custom headers into this traffic, custom headers must be decrypted. If you use the forward-proxy decryption available on the firewall to decrypt custom headers, you must identify the specific HTTPS traffic you want to decrypt by identifying the domains associated with the traffic. The following table identifies the relevant domains for each of the SaaS applications for which Palo Alto Networks® has provided predefined rules.

Application	Domains
Dropbox	`*.dropbox.com`
G Suite	`*.google.com` `gmail.com`

Microsoft Office 365	`login.microsoftonline.com`
	`login.microsoft.com`
	`login.windows.net`

YouTube	`www.youtube.com`

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Application	Domains

	`m.youtube.com`
	`youtubei.googleapis.com`
	`youtube.googleapis.com`
	`www.youtube-nocookie.com`

Create HTTP Header Insertion Entries using Predefined Types

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

STEP 1 | If there are no upstream devices already decrypting HTTPS traffic, configure decryption using SSL Forward Proxy.

If you are configuring SSL decryption for Dropbox, then you must also configure your Dropbox clients to allow SSL traffic. These procedures are specific and private to Dropbox — to obtain these procedures, contact your Dropbox account representative.

1. **Add** a Custom URL Category for the SaaS application you are managing (**Objects** > **Custom Objects** > **URL Category**).
2. Specify a **Name** for the category.
3. **Add** the domains specific to the SaaS application you are managing or for which you want to insert the username and domain in the headers. See Domains used by the Predefined SaaS Application Types for a list of the domains that you use for each of the predefined SaaS applications. See Insert Username in HTTP Headers for more information on configuring the firewall to include the username and domain in the HTTP headers.

   Each domain name can be up to 254 characters and you can identify a maximum of 50 domains for each entry. The domain list supports wildcards (for example, **`*.example.com`**). As a best practice, do not nest wildcards (for example, **`*.*.*`**) and do not overlap domains within the same URL profile.
4. For SaaS application management, create a decryption policy rule and, as you follow this procedure, configure the following:
   - In the **Service/URL Category** tab, **Add** the **URL Category** that you created in the previous step.
   - In the **Options** tab, make sure the **Action** is set to **Decrypt** and that the **Type** is set to **SSL Forward Proxy**.

STEP 2 | Edit or add a URL filter.

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STEP 3 | Select **HTTP Header Insertion** in the **URL Filtering Profile** dialog.

STEP 4 | **Add** an entry.

1. Specify a **Name** (up to 100 characters) for this entry.
2. Select a predefined **Type**.

   This populates the **Domains** and **Headers** lists.
3. For each **Header**, enter a **Value**.

   Each header value can have up to 16K characters.
4. (Optional) Select **Log** to enable logging of insertion activity for the headers.

   Allowed traffic is not logged, so header insertions are not logged for allowed traffic.
5. Click **OK** to save your changes.

STEP 5 | **Add** or edit a Security Policy rule (**Policies** > **Security**) to include the HTTP header insertion URL filtering profile.

- For SaaS application management, allow users to access the SaaS application for which you are configuring this header insertion rule.
- To include the username and domain in the HTTP headers, apply the URL filtering profile to the security policy rule for HTTP or HTTPS traffic.

  1. Choose the URL filtering profile (**Actions** > **URL Filtering**) that you edited or created in Step 2.
  2. Click **OK** to save and then **Commit** your changes.

STEP 6 | Verify that the firewall correctly inserts the header.

- For Saas application management, from an endpoint, confirm that access to the SaaS application is working in the way you expect.

  1. Try to access an account or content that you expect to be able to access. If you cannot access the SaaS account or content, then the configuration is not working.
  2. Try to access an account or content that you expect will be blocked. If you can access the SaaS account or content, then the configuration is not working.
  3. If both of the previous steps work as expected, then you can View Logs (if you configured logging in step 4.4) and you should see the recorded HTTP header insertion activity.

Create Custom HTTP Header Insertion Entries

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

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STEP 1 | If there are no upstream devices already decrypting HTTPS traffic, configure SSL Forward Proxy.

1. **Add** a custom URL category for the SaaS application you are managing (**Objects** > **Custom Objects** > **URL Category**).
2. Specify a **Name** for the category.
3. **Add** the domains specific to the SaaS application you are managing.
4. Create a decryption policy rule and, as you follow this procedure, configure the following:
   - In the **Service/URL Category** tab, **Add** the **URL Category** that you created in the previous step.
   - In the **Options** tab, make sure the **Action** is set to **Decrypt** and that the **Type** is set to **SSL Forward Proxy**.

STEP 2 | Edit or add a URL filter.

STEP 3 | Select **HTTP Header Insertion** in the URL Filtering Profile dialog.

STEP 4 | **Add** an entry.

1. Specify a **Name** for this entry.
2. Select **Custom** as the **Type**.
3. **Add** domains to the **Domains** list.

   You can add up to 50 domains and each domain name can have up to 256 characters; wildcards are supported (for example, *.example.com).

   HTTP header insertion occurs when a domain in this list matches the domain in the Host header of the HTTP request.
4. **Add** headers to the **Headers** list.

   You can add up to 5 headers, and each header can have up to 100 characters but cannot contain any spaces.
5. For each header, enter a **Value**.

   Each header value can have up to 16K characters.
6. (Optional) **Log** insertion activity for the headers.
7. Click **OK** to save your changes.

STEP 5 | **Add** or edit a security policy rule (**Policies** > **Security**) that allows users to access the SaaS application for which you are configuring this header insertion rule.

1. Choose the URL Filtering profile (**Actions** > **URL Filtering**) that you edited or created in Step 2.
2. Click **OK** to save and then **Commit** your changes.

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STEP 6 | Verify that access to the SaaS application is working in the way you expect. From an endpoint that is connected to your network:

1. Try to access an account or content that you expect to be able to access. If you cannot access the SaaS account or content, then the configuration is not working.
2. Try to access an account or content that you expect will be blocked. If you can access the SaaS account or content, then the configuration is not working.
3. If both of the previous steps work as expected, then you can view logs (if you configured logging in step 4.6) and you should see the recorded HTTP header insertion activity.

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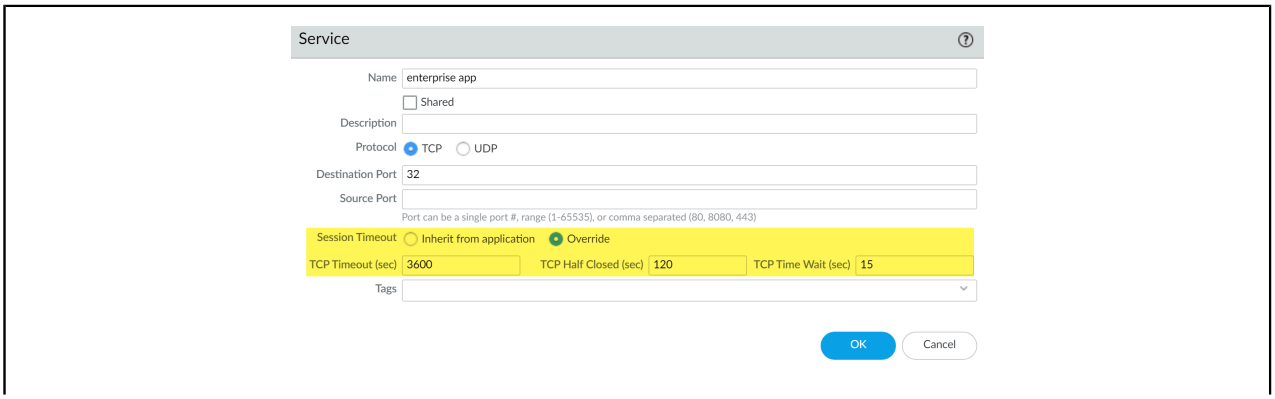
Maintain Custom Timeouts for Data Center Applications

Where Can I Use This?	What Do I Need?
• Prisma Access • Next-Generation Firewall	This is a core Network Security feature for NGFWs and Prisma Access; no prerequisites needed.

Easily maintain custom timeouts for applications as you move from a port-based policy to an application-based policy. Use this method to maintain custom timeouts instead of overriding App-ID (losing application visibility) or creating a custom App-ID (expending time and research).

To get started, configure custom timeout settings as part of a service object:



Then add the service object in a policy rule to apply the custom timeouts to the application(s) the rule enforces.

The following steps describe how apply custom timeouts to applications; to apply custom timeouts to user groups, you can follow the same steps but just make sure to add the service object to the security policy rule that enforces the users to whom you want the timeout to apply.

STEP 1 | Select **Objects** > **Services** to add or modify a service object.

You can also create service objects as you are defining match criteria for a security policy rule: select **Policies** > **Security** > **Service/URL Category** and **Add** a new Service object to apply to the application traffic the rule governs.

STEP 2 | Select the protocol for the service to use (TCP or UDP).

STEP 3 | Enter the destination port number or a range of port numbers used by the service.

STEP 4 | Define the session timeout for the service.

- **Inherit from application** (default)—No service-based timeouts are applied; instead, apply the application timeout.
- **Override**—Define a custom session timeout for the service.

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STEP 5 | If you chose to override the application timeout and define a custom session timeout, continue to:

- Enter a **TCP Timeout** value to set the Maximum length of time in seconds that a TCP session can remain open after data transmission has started. When this time expires, the session closes. The value range is 1 - 604800, and the default value is 3600 seconds.
- Enter a **TCP Half Closed** value to set the maximum length of time in seconds that a session remains in the session table between receiving the first FIN packet and receiving the second FIN packet or RST packet. If the timer expires, the session closes. The value range is 1 - 604800, and the default value is 120 seconds.
- Enter a **TCP Wait Time** value to set the maximum length of time in seconds that a session remains in the session table after receiving the second FIN packet or a RST packet. When the timer expires, the session closes. The value range is 1 - 600, and the default value is 15 seconds.

STEP 6 | Click **OK** to save the service object.

STEP 7 | Select **Policies** > **Security** and **Add** or modify a policy rule to govern the application traffic you want to control.

STEP 8 | Select **Service/URL Category** and **Add** the service object you just created to the security policy rule.

STEP 9 | Click **OK** and **Commit** your changes.

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Large Scale VPN (LSVPN)

Where Can I Use This?	What Do I Need?
• NGFW	❑ No separate license required for LSVPN when using NGFWs

The GlobalProtect Large Scale VPN (LSVPN) feature on the Palo Alto Networks Next-Generation Firewall simplifies the deployment of traditional hub and spoke VPNs, enabling you to deploy enterprise networks with several branch offices quickly with a minimum amount of configuration required on the remote *satellites*. This solution uses certificates for firewall authentication and IPSec to secure data.

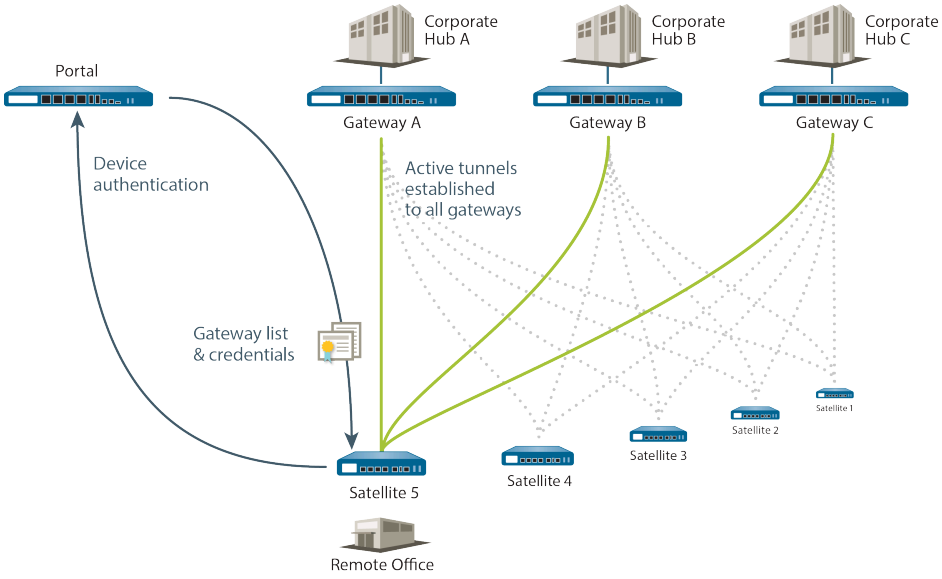
LSVPN enables site-to-site VPNs between Palo Alto Networks firewalls. To set up a site-to-site VPN between a Palo Alto Networks firewall and another device, see VPNs. The LSVPN doesn't require a GlobalProtect subscription.

LSVPN Overview

GlobalProtect provides a complete infrastructure for managing secure access to corporate resources from your remote sites. This infrastructure includes the following components:

- **GlobalProtect Portal**—Provides the management functions for your GlobalProtect LSVPN infrastructure. Every satellite that participates in the GlobalProtect LSVPN receives configuration information from the portal, including configuration information to enable the satellites (the spokes) to connect to the gateways (the hubs). You configure the portal on an interface on any Palo Alto Networks Next-Generation Firewall.
- **GlobalProtect Gateways**—A Palo Alto Networks firewall that provides the tunnel endpoint for satellite connections. The Satellites access resources that you protect using Security policy rules on the gateway. It isn't required to have a separate portal and gateway; a single firewall can function both as portal and gateway.
- **GlobalProtect Satellite**—A Palo Alto Networks firewall at a remote site that establishes IPSec tunnels with one or more gateways at your corporate office(s) for secure access to centralized resources. Configuration on the satellite firewall is minimal, enabling you to scale your VPN quickly and easily as you add new sites.

The following diagram illustrates how the GlobalProtect LSVPN components work together.



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Create Interfaces and Zones for the LSVPN

Where Can I Use This?	What Do I Need?
• NGFW	❑ No separate license required for LSVPN when using NGFWs

Configure the following interfaces and zones for your LSVPN infrastructure:

- **GlobalProtect portal**—Requires a Layer 3 interface for GlobalProtect satellites to connect to. If the portal and gateway are on the same firewall, they can use the same interface. The portal must be in a zone that is accessible from your branch offices.
- **GlobalProtect gateways**—Requires three interfaces: a Layer 3 interface in the zone that is reachable by the remote satellites, an internal interface in the trust zone that connects to the protected resources, and a logical tunnel interface for terminating the VPN tunnels from the satellites. Unlike other site-to-site VPN solutions, the GlobalProtect gateway only requires a single tunnel interface, which it will use for tunnel connections with all of your remote satellites (point-to-multipoint). If you plan to use dynamic routing, you must assign an IP address to the tunnel interface. GlobalProtect supports both IPv6 and IPv4 addressing for the tunnel interface.
- **GlobalProtect satellites**—Requires a single tunnel interface for establishing a VPN with the remote gateways (up to a maximum of 25 gateways). If you plan to use dynamic routing, you must assign an IP address to the tunnel interface. GlobalProtect supports both IPv6 and IPv4 addressing for the tunnel interface.

For more information about portals, gateways, and satellites see LSVPN Overview.

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STEP 1 | Configure a Layer 3 interface.

The portal and each gateway and satellite all require a Layer 3 interface to enable traffic to be routed between sites.

If the gateway and portal are on the same firewall, you can use a single interface for both components.

1. Select **Network** > **Interfaces** > **Ethernet** and then select the interface you want to configure for GlobalProtect LSVPN.
2. Select **Layer3** from the **Interface Type** drop-down.
3. On the **Config** tab, select the **Security Zone** to which the interface belongs:
   - The interface must be accessible from a zone outside of your trust network. Consider creating a dedicated VPN zone for visibility and control over your VPN traffic.
   - If you haven't yet created the zone, select **New Zone** from the **Security Zone** drop-down, define a **Name** for the new zone, and then click **OK**.
4. Select the **Virtual Router** to use.
5. Assign an IP address to the interface:
   - For an IPv4 address, select **IPv4** and **Add** the IP address and network mask to assign to the interface, for example 203.0.11.100/24.
   - For an IPv6 address, select **IPv6**, **Enable IPv6 on the interface**, and **Add** the IP address and network mask to assign to the interface, for example 2001:1890:12f2:11::10.1.8.160/80.
6. To save the interface configuration, click **OK**.

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STEP 2 | On the firewall(s) hosting the GlobalProtect gateway(s), configure the logical tunnel interface that will terminate VPN tunnels established by the GlobalProtect satellites.

IP addresses is only required on the tunnel interface when you plan to use dynamic routing. However, assigning an IP address to the tunnel interface can be useful for troubleshooting connectivity issues.

Make sure to enable User-ID in the zone where the VPN tunnels terminate.

1. Select **Network** > **Interfaces** > **Tunnel** and click **Add**.
2. In the **Interface Name** field, specify a numeric suffix, such as **.2**.
3. On the **Config** tab, expand the **Security Zone** drop-down to define the zone as follows:
   - To use your trust zone as the termination point for the tunnel, select the zone from the drop-down.
   - (Recommended) To create a separate zone for VPN tunnel termination, click **New Zone**. In the Zone dialog, define a **Name** for the new zone (for example *lsvpn-tun*), select the **Enable User Identification** check box, and then click **OK**.
4. Select the **Virtual Router**.
5. (Optional) To assign an IP address to the tunnel interface:
   - For an IPv4 address, select **IPv4** and **Add** the IP address and network mask to assign to the interface, for example 203.0.11.100/24.
   - For an IPv6 address, select **IPv6**, **Enable IPv6 on the interface**, and **Add** the IP address and network mask to assign to the interface, for example 2001:1890:12f2:11::10.1.8.160/80.
6. To save the interface configuration, click **OK**.

STEP 3 | If you created a separate zone for tunnel termination of VPN connections, create a security policy to enable traffic flow between the VPN zone and your trust zone.

For example, a policy rule enables traffic between the *lsvpn-tun* zone and the *L3-Trust* zone.

STEP 4 | Commit your changes.

Click **Commit**.

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Enable SSL Between GlobalProtect LSVPN Components

Where Can I Use This?	What Do I Need?
• NGFW	❑ No separate license required for LSVPN when using NGFWs

All interaction between the GlobalProtect components occurs over an SSL/TLS connection. Therefore, you must generate and/or install the required certificates before configuring each component so that you can reference the appropriate certificate(s) and/or certificate profiles in the configurations for each component. This section describes the supported methods of certificate deployment, descriptions and best practice guidelines for the various GlobalProtect certificates, and provide instructions for generating and deploying the required certificates.

About Certificate Deployment

There are two basic approaches to deploying certificates for GlobalProtect LSVPN:

- **Enterprise Certificate Authority**—If you already have your own enterprise certificate authority, you can use this internal CA to issue an intermediate CA certificate for the GlobalProtect portal to enable it to issue certificates to the GlobalProtect gateways and satellites. You can also configure the GlobalProtect portal to act as a Simple Certificate Enrollment Protocol (SCEP) client to issue client certificates to GlobalProtect satellites.
- **Self-Signed Certificates**—You can generate a self-signed root CA certificate on the firewall and use it to issue server certificates for the portal, gateway(s), and satellite(s). When using self-signed root CA certificates, as a best practice, create a self-signed root CA certificate on the portal and use it to issue server certificates for the gateways and satellites. This way, the private key used for certificate signing stays on the portal.

Deploy Server Certificates to the GlobalProtect LSVPN Components

Where Can I Use This?	What Do I Need?
• NGFW	❑ No separate license required for LSVPN when using NGFWs

The GlobalProtect LSVPN components use SSL/TLS to authenticate mutually. Before deploying the LSVPN, you must assign an SSL/TLS service profile to each portal and gateway. The profile specifies the server certificate and allowed TLS versions for communication with satellites. You don't need to create SSL/TLS service profiles for the satellites because the portal will issue a server certificate for each satellite during the first connection as part of the satellite registration process.

In addition, you must import the root certificate authority (CA) certificate used to issue the server certificates onto each firewall that you plan to host as a gateway or satellite. Finally, on each

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gateway and satellite participating in the LSVPN, you must configure a certificate profile that will enable them to establish an SSL/TLS connection using mutual authentication.

The following workflow shows the best practice steps for deploying SSL certificates to the GlobalProtect LSVPN components:

STEP 1 | On the firewall hosting the GlobalProtect portal, create the root CA certificate for signing the certificates of the GlobalProtect components.

Create a Self-Signed Root CA Certificate:

1. Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later). Select **Generate**.
2. Enter a **Certificate Name**, such as **LSVPN_CA**.
3. Don't select a value in the **Signed By** field (this is what indicates that it's self-signed).
4. Select the **Certificate Authority** check box and then click **OK** to generate the certificate.

STEP 2 | Create SSL/TLS service profiles for the GlobalProtect portal and gateways.

For the portal and each gateway, you must assign an SSL/TLS service profile that references a unique self-signed server certificate.

The best practice is to issue all of the required certificates on the portal, so that the signing certificate (with the private key) doesn't have to be exported.

If the GlobalProtect portal and gateway are on the same firewall interface, you can use the same server certificate for both components.

1. Use the root CA on the portal to generate a certificate for each gateway that you'll deploy:
    1. Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later). Select **Generate**.
    2. Enter a **Certificate Name**.
    3. Enter the FQDN (recommended) or IP address of the interface where you plan to configure the gateway in the **Common Name** field.
    4. In the **Signed By** field, select the **LSVPN_CA** certificate you created.
    5. In the Certificate Attributes section, click **Add** and define the attributes to identify the gateway uniquely. If you add a **Host Name** attribute (which populates the SAN field of the certificate), it must exactly match the value you defined for the **Common Name**.
    6. **Generate** the certificate.
2. Configure an SSL/TLS service profile for the portal and each gateway:
    1. Select **Device** > **Certificate Management** > **SSL/TLS Service Profile** and click **Add**.
    2. Enter a **Name** to identify the profile and select the server **Certificate** you created for the portal or gateway.
    3. Define the range of TLS versions (**Min Version** to **Max Version**) allowed for communicating with satellites and click **OK**.

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STEP 3 | Deploy the self-signed server certificates to the gateways.

Best Practices:

- *Export the self-signed server certificates issued by the root CA from the portal and import them onto the gateways.*
- *Be sure to issue a unique server certificate for each gateway.*
- *The Common Name (CN) and, if applicable, the Subject Alternative Name (SAN) fields of the certificate must match the IP address or FQDN of the interface where you configure the gateway.*

1. On the portal, select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later). Select the gateway certificate you want to deploy, and click **Export**.
2. Select **Encrypted Private Key and Certificate (PKCS12)** from the **File Format** drop-down.
3. Enter (and re-enter) a **Passphrase** to encrypt the private key associated with the certificate and then click **OK** to download the PKCS12 file to your computer.
4. On the gateway, select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later). Select **Import**.
5. Enter a **Certificate Name**.
6. Enter the path and name to the **Certificate File** that you downloaded from the portal, or **Browse** to find the file.
7. Select **Encrypted Private Key and Certificate (PKCS12)** as the **File Format**.
8. Enter the path and name to the PKCS12 file in the **Key File** field or **Browse** to find it.
9. Enter and re-enter the **Passphrase** you used to encrypt the private key when you exported it from the portal and then click **OK** to import the certificate and key.

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STEP 4 | Import the root CA certificate used to issue server certificates for the LSVPN components.

Import the root CA certificate onto all gateways and satellites. For security reasons, make sure you export the certificate only, and not the associated private key.

1. Download the root CA certificate from the portal.
    1. Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later).
    2. Select the root CA certificate used to issue certificates for the LSVPN components and click **Export**.
    3. Select **Base64 Encoded Certificate (PEM)** from the **File Format** drop-down and click **OK** to download the certificate. (Do not export the private key.)
2. On the firewalls hosting the gateways and satellites, import the root CA certificate.
    1. Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later). Select **Import**.
    2. Enter a **Certificate Name** that identifies the certificate as your client CA certificate.
    3. **Browse** to the **Certificate File** you downloaded from the CA.
    4. Select **Base64 Encoded Certificate (PEM)** as the **File Format** and then click **OK**.
    5. Select the certificate you imported on the **Device Certificates** or **Custom Certificates** tab to open it.
    6. Select **Trusted Root CA** and then click **OK**.
    7. **Commit** the changes.

STEP 5 | Create a Certificate Profile.

The GlobalProtect LSVPN portal and each gateway require a Certificate Profile that specifies which certificate to use to authenticate the satellites.

1. Select **Device** > **Certificate Management** > **Certificate Profile** and click **Add** and enter a profile **Name**.
2. Make sure that the **Username Field** is set to **None**.
3. In the **CA Certificates** field, click **Add**, select the trusted root CA certificate you imported in the previous step.
4. (Recommended) Enable use of CRL and/or OCSP to enable certificate status verification.
5. Click **OK** to save the profile.

STEP 6 | Commit your changes.

Click **Commit**.

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Deploy Client Certificates to the GlobalProtect Satellites Using SCEP

Where Can I Use This?	What Do I Need?
• NGFW	❑ No separate license required for LSVPN when using NGFWs

As an alternative method for deploying client certificates to satellites, you can configure your GlobalProtect portal to act as a Simple Certificate Enrollment Protocol (SCEP) client to a SCEP server in your enterprise PKI. SCEP operation is dynamic in that the enterprise PKI generates a certificate when the portal requests it and sends the certificate to the portal.

When the satellite device requests a connection to the portal or gateway, it also includes its serial number with the connection request. The portal submits a CSR to the SCEP server using the settings in the SCEP profile and automatically includes the serial number of the device in the subject of the client certificate. After receiving the client certificate from the enterprise PKI, the portal transparently deploys the client certificate to the satellite device. The satellite device then presents the client certificate to the portal or gateway for authentication.

STEP 1 | Create a SCEP profile.

1. Select **Device** > **Certificate Management** > **SCEP** and then **Add** a new profile.
2. Enter a **Name** to identify the SCEP profile.
3. If this profile is for a firewall with multiple virtual systems capability, select a virtual system or **Shared** as the **Location** where the profile is available.

STEP 2 | (Optional) To make the SCEP-based certificate generation more secure, configure a SCEP challenge-response mechanism between the PKI and portal for each certificate request.

After you configure this mechanism, its operation is invisible, and no further input from you is necessary.

To comply with the U.S. Federal Information Processing Standard (FIPS), use a **Dynamic** SCEP challenge and specify a **Server URL** that uses HTTPS (see step 7).

Select one of the following options:

- **None**—(Default) The SCEP server doesn't challenge the portal before it issues a certificate.
- **Fixed**—Obtain the enrollment challenge password from the SCEP server (for example, **`http://10.200.101.1/CertSrv/mscep_admin/`**) in the PKI infrastructure and then copy or enter the password into the Password field.
- **Dynamic**—Enter the SCEP **Server URL** where the portal-client submits these credentials (for example, **`http://10.200.101.1/CertSrv/mscep_admin/`**), and a username and OTP of your choice. The username and password can be the credentials of the PKI administrator.

STEP 3 | Specify the settings for the connection between the SCEP server and the portal to enable the portal to request and receive client certificates.

To identify the satellite, the portal automatically includes the device serial number in the CSR request to the SCEP server. Because the SCEP profile requires a value in the **Subject** field, you

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can leave the default **$USERNAME** token even though the value isn't used in client certificates for LSVPN.

1. Configure the **Server URL** that the portal uses to reach the SCEP server in the PKI (for example, **http://10.200.101.1/certsrv/mscep/**).
2. Enter a string (up to 255 characters in length) in the **CA-IDENT Name** field to identify the SCEP server.
3. Select the **Subject Alternative Name Type**:
   - **RFC 822 Name**—Enter the email name in a certificate's subject or Subject Alternative Name extension.
   - **DNS Name**—Enter the DNS name used to evaluate certificates.
   - **Uniform Resource Identifier**—Enter the name of the resource from which the client will obtain the certificate.
   - **None**—Don't specify attributes for the certificate.

STEP 4 | (Optional) Configure cryptographic settings for the certificate.

- Select the key length (**Number of Bits**) for the certificate. If the firewall is in FIPS-CC mode and the key generation algorithm is RSA, the RSA keys must be 2,048 bits or larger.
- Select the **Digest for CSR** that indicates the digest algorithm for the certificate signing request (CSR): SHA1, SHA256, SHA384, or SHA512.

STEP 5 | (Optional) Configure the permitted uses of the certificate, either for signing or encryption.

- To use this certificate for signing, select the **Use as digital signature** check box. This enables the endpoint use the private key in the certificate to validate a digital signature.
- To use this certificate for encryption, select the **Use for key encipherment** check box. This enables the client use the private key in the certificate to encrypt data exchanged over the HTTPS connection established with the certificates issued by the SCEP server.

STEP 6 | (Optional) To ensure that the portal is connecting to the correct SCEP server, enter the **CA Certificate Fingerprint**. Obtain this fingerprint from the SCEP server interface in the Thumbprint field.

1. Enter the URL for the SCEP server's administrative UI (for example, **http://<hostname or IP>/CertSrv/mscep_admin/**).
2. Copy the thumbprint and enter it in the **CA Certificate Fingerprint** field.

STEP 7 | Enable mutual SSL authentication between the SCEP server and the GlobalProtect portal. This is required to comply with the U.S. Federal Information Processing Standard (FIPS).

FIPS-CC operation is indicated on the firewall login page and in its status bar.

Select the SCEP server's root **CA Certificate**. Optionally, you can enable mutual SSL authentication between the SCEP server and the GlobalProtect portal by selecting a **Client Certificate**.

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STEP 8 | Save and commit the configuration.

1. Click **OK** to save the settings and close the SCEP configuration.
2. **Commit** the configuration.

The portal attempts to request a CA certificate using the settings in the SCEP profile and saves it to the firewall hosting the portal. If successful, the CA certificate is shown in **Device** > **Certificate Management** > **Certificates**.

STEP 9 | (Optional) If after saving the SCEP profile, the portal fails to obtain the certificate, you can manually generate a certificate signing request (CSR) from the portal.

1. Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later). Select **Generate**.
2. Enter a **Certificate Name**. This name can't contain spaces.
3. Select the **SCEP Profile** to use to submit a CSR to your enterprise PKI.
4. Click **OK** to submit the request and generate the certificate.

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Configure the Portal to Authenticate Satellites

Where Can I Use This?	What Do I Need?
• NGFW	❑ No separate license required for LSVPN when using NGFWs

To register with the LSVPN, each satellite must establish an SSL/TLS connection with the portal. After establishing the connection, the portal authenticates the satellite to ensure that it is authorized to join the LSVPN. After successfully authenticating the satellite, the portal will issue a server certificate for the satellite and push the LSVPN configuration specifying the gateways to which the satellite can connect and the root CA certificate required to establish an SSL connection with the gateways.

There are multiple ways that the satellite can authenticate to the portal during its initial connection:

- **(PAN-OS 10.0 and earlier releases) Serial number Authentication**—You can configure the portal with the serial number of the satellite firewalls that are authorized to join the LSVPN. During the initial satellite connection to the portal, the satellite presents its serial number to the portal and if the portal has the serial number in its configuration, the satellite will be successfully authenticated. You add the serial numbers of authorized satellites when you configure the portal. See Configure the Portal.
- **(PAN-OS 10.1 and later releases) (Default authentication method) Username/password and Satellite Cookie Authentication —**For the satellite to authenticate to the portal during its initial connection, you must create an authentication profile for the portal LSVPN configuration. The satellite administrator must manually authenticate the satellite to the portal to establish the first connection. Upon successful authentication, the portal returns a satellite cookie to authenticate the satellite on subsequent connections. The satellite cookie that the portal issues has a lifetime of 6 months, by default. When the cookie expires, the satellite administrator must manually authenticate again, at which time the portal will issue a new cookie.
- **(PAN-OS 10.2.8 and later 10.2 releases) (PAN-OS 11.1.3 and later releases) Serial number and IP address Authentication**—You can configure the portal with the serial number and IP address of the satellite firewalls that are authorized to join the LSVPN. During the initial satellite connection to the portal, the satellite presents its serial number and IP address to the portal and if the portal has the serial number and IP address in its configuration, the satellite will be successfully authenticated. You add the serial numbers of authorized satellites when you configure the portal.

PAN-OS releases support the following authentication methods:

PAN-OS RELEASE	SUPPORTED AUTHENTICATION METHOD
PAN-OS 10.0 and earlier releases	Serial number Authentication method
PAN-OS 10.1 and later releases	Username/password and Satellite Cookie Authentication method (Default authentication method)

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PAN-OS RELEASE	SUPPORTED AUTHENTICATION METHOD
	While configuring the Username/password and Satellite Cookie Authentication method, configure the satellite cookie expiration to a value more than the satellite upgrade time to avoid login failures.
PAN-OS 10.2.8 and later 10.2 releases, PAN-OS 11.1.3 and later releases	• Username/password and Satellite Cookie Authentication method (Default authentication method) • Serial number and IP address Authentication method

Before upgrading or downgrading to a particular PAN-OS release, be aware of the authentication methods supported.

Refer to Upgrade and Downgrade Considerations to learn about the authentication method supported when you upgrade or downgrade the firewall from one PAN-OS release to another.

(PAN-OS 10.1.7 and later 10.1 releases)(PAN-OS 10.2.4 and later 10.2 releases)(PAN-OS 11.0.1 and later releases) You can configure the cookie expiry period from 1 to 5 years, while the default remains as 6 months.

On the portal:

- Use the **`request global-protect-portal set-satellite-cookie-expiration value <1-5>`** CLI command to change the current satellite cookie expiration time.
- Use the **`show global-protect-portal satellite-cookie-expiration`** CLI command to view the current satellite cookie expiration time.

On the satellite:

- Use the **`show global-protect-satellite satellite`** CLI command to view (in `"Satellite Cookie Generation Time"` field) the current satellite authentication cookie's generation time.

Username/Password and Satellite Cookie Authentication (Default Authentication Method)

For authenticating the satellite to the portal, GlobalProtect LSVPN supports only local database authentication.

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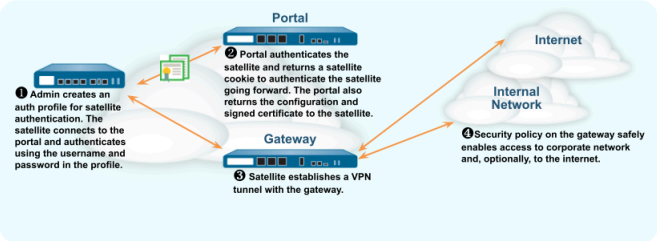

The following workflow describes how to set up the portal to authenticate satellites against an existing authentication service.

STEP 1 | Set up local database authentication so that the satellite administrator can authenticate the satellite to the portal.

1. Select **Device** > **Local User Database** > **Users** and **Add** the user account to the local database.
2. **Add** the user account to the local database.

STEP 2 | Configure an authentication profile.

1. Select **Device** > **Authentication Profile** > **Add**.
2. Enter a **Name** for the profile and then set the **Type** to **Local Database**.
3. Click **OK** and **Commit** your changes.

STEP 3 | Authenticate the satellite.

To authenticate the satellite to the portal, the satellite administrator must provide the username and password configured in the local database.

1. Select **Network** > **IPSec Tunnels** and click the **Gateway Info** link in the Status column of the tunnel configuration you created for the LSVPN.
2. Click the **enter credentials** link in the **Portal Status** field and provide the username and password to authenticate the satellite to the portal.

   After the portal successfully authenticates to the portal for the first time, the portal generates a satellite cookie, which it uses to authenticate the satellite on subsequent sessions.

Serial Number and IP Address Authentication Method

(PAN-OS 10.2.8 and later 10.2 releases) (PAN-OS 11.1.3 and later releases) The Serial number and IP address Authentication method will be established successfully only when you configure the necessary parameters correctly and in the correct order.

The following table provides you with the details on how your parameter settings impact the establishment of Serial number and IP address authentication:

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Serial Number and IP Address Authentication Method	Configured retry-interval (Default is 5 seconds)	Serial Number	IP Address in Allow List	Satellite Cookie	Established Authentication Method
Enabled	The retry-interval value is greater than or equal to 5	Registered	Allowed	Will not be checked	Serial number and IP address Authentication method will be established successfully.
Enabled	The retry-interval value is greater than or equal to 5	Registered	Not Allowed	Will not be checked	Fails to establish Serial number and IP address Authentication.
Enabled	The retry-interval value is greater than or equal to 5	Not Registered	Will not be checked	Will not be checked	Fails to establish Serial number and IP address Authentication.
Disabled	The retry-interval will not be checked	Will not be checked	Will not be checked	Default behavior	The default authentication method, Username/ password and Satellite Cookie Authentication method will be established successfully.

The satellite initiates a connection to the portal upon successful configuration of the satellite serial number registered and the satellite device IP address in the satellite IP allow list on the portal. You should also ensure that the portal is running PAN-OS 10.2.8 or later versions, or PAN-OS 11.1.3 or later versions before configuring Serial number and IP address Authentication on the portal.

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We don't support broadcast, multicast, loopback, zeronet IPv4 and IPv6 addresses for the serial number and IP address authentication method.

In the LSVPN serial number and IP address authentication method, PAN-OS stores the configuration changes in the database internally. Therefore, the latest saved configuration is applied when you upgrade to or downgrade from this feature.

Use the following workflow to authenticate the satellite using the Serial number and IP address Authentication method.

STEP 1 | Log in to the portal web interface and select **Network** > **GlobalProtect** > **Portals** > **GlobalProtect Portal** > **Satellite Configuration** > **GlobalProtect Satellite** > **Devices** to add a new satellite serial number to the GlobalProtect portal. **Commit** the configuration.

STEP 2 | Access the CLI.

You can't access the Serial number and IP address Authentication related CLIs from the Panorama.

STEP 3 | Follow the below steps in the same order to configure the parameters related to Serial number and IP address Authentication on a firewall configured as a GlobalProtect portal. Otherwise, the satellite authentication might fail and an administrator's intervention is required to enter the username and password on the satellite.

1. Enter the following operational command per portal to add a satellite device IP address on the GlobalProtect portal.

   Configure a specific IP address, subnet, or a range to add one or more satellite devices. Both IPv4 and IPv6 addresses are supported.

   ```
   username@hostname>  set global-protect global-protect-portal
    portal <portal_name> satellite-serialnumberip-auth satellite-
   ip-allowlist entry <value>
   ```

   Where <value> is the IPv4 address, IPv6 address, IP range, or IP subnet of the satellite device that you want to add.

   For example:

   ```
   username@hostname>  set global-protect global-protect-portal
    portal gp-portal-1 satellite-serialnumberip-auth satellite-
   ip-allowlist entry 192.0.2.0-192.0.2.100
   ```

   You can also exclude a specific range of IP address from the *satellite-ip-allowlist* that you don't wish to configure as a satellite. To do this, use the following command:

   ```
   username@hostname>  set global-protect global-protect-portal
    portal <portal_name> satellite-serialnumberip-auth satellite-
   ip-exclude-from range <ip-address> exclude-list <value>
   ```

   Where *satellite-ip-exclude-from range <ip-address>* is the IPv4 or IPv6 subnet or range of the IP address that you want to exclude from configuring as a satellite device. The

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IP address that you want to exclude must be within the IP address range that you configured in the *satellite-ip-allowlist*.

For example:

```
username@hostname>  set global-protect global-protect-
portal portal gp-portal-1 satellite-serialnumberip-auth
 satellite-ip-exclude-from range 192.0.2.0-192.0.2.100
 exclude-list 192.0.2.20-192.0.2.30
```

We support the following IP4 and IPv6 address formats to configure the *satellite-ip-allowlist*.

Table 3: Supported IPv4 and IPv6 Address Formats

IP Address Format	IPv4 Address	IPv6 Address
A specific IP address	x.x.x.x For example: 192.0.2.0	xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx For example: 2001:db8::
IP address subnet	x.x.x.x/x For example: 192.0.2.0/24	xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx/y For example: 2001:db8::/32
IP address range	x.x.x.x-x.x.x.x For example:	xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx-xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx

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IP Address Format	IPv4 Address	IPv6 Address
	192.0.2.10-192.0.2.20	

(HA deployments only) The added satellite IP address list is synchronized among the HA peers.

- *Ensure that **Enable Config Sync** (select **Device** > **High Availability** > **General**) is enabled on your HA configuration to configure the Serial number and IP address Authentication method. This setting is required to synchronize the two firewall configurations (that is enabled by default).*
- *You must add the satellite device serial number first that allows the portal to select the correct satellite configuration.*
- *If the satellite devices in the HA pair use different IP addresses, then configure both the IP addresses in the satellite IP allow list on the portal.*

2. Enter the following operational command per portal to configure a retry interval for the serial number and IP address authentication in case of failure in establishing the authentication method.

```
username@hostname>  set global-protect global-protect-
portal portal <name> satellite-serialnumberip-auth retry-
interval <value>
```

The retry interval range is 5 to 86,400 seconds and the default value is 5 seconds.

For example:

```
username@hostname>  set global-protect global-protect-portal
 portal gp-portal-1 satellite-serialnumberip-auth retry-
interval 100
```

(HA deployments only) The authentication retry interval is synchronized among the HA peers.

3. Enter the following operational command to enable the serial number and IP address authentication method on the firewall where you want to enable the Serial number and IP address Authentication method.

```
username@hostname>  set global-protect satellite-
serialnumberip-auth enable
```

The serial number and IP address authentication method is disabled by default.

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When the Serial number and IP address Authentication is enabled and if the satellite authentication fails, then based on the retry interval, the satellite will retry the authentication process again. There is no fall back mechanism available to support Username/Password and Satellite Cookie based authentication in case of failure in configuring the Serial number and IP address Authentication method.

If attempting to enable the Serial number and IP address Authentication method results in failure, check for the following:

- Whether the portal is running PAN-OS 10.2.8 or later versions, or PAN-OS 11.1.3 or later versions.
- Whether you have added the satellite device's IP address to the satellite IP allow list on the GlobalProtect portal.
- Whether you have configured the satellite's serial number in **Network** > **GlobalProtect** > **Portals** > **GlobalProtect Portal** > **Satellite Configuration** > **GlobalProtect Satellite** > **Devices**.

Enter any random username and password (or just press enter) in the pop-up dialog on the satellite to retrigger the authentication process in the following cases:

- (PAN-OS 10.2.8 and later 10.2 versions) A scenario where the portal is running PAN-OS 10.2.8 and the satellite is running version earlier to 10.2.8, and the satellite cookie has expired. In this case, when you attempt to enable the Serial number and IP address Authentication method without adding the satellite IP address in the satellite IP allow list on the portal, satellite authentication fails. The failure is due to a missing IP address in the satellite IP allow list.
- (PAN-OS 10.2.8 and later 10.2 versions) A scenario where the satellite is running version earlier to 10.2.8 and the portal is upgraded to PAN-OS 10.2.8. In the meantime, the satellite cookie expires before enabling the Serial number and IP address Authentication method on the portal. Then satellite authentication fails due to satellite cookie expiration.
- (PAN-OS 11.1.3 or later versions) A scenario where the portal is running PAN-OS 11.1.3 and the satellite is running version earlier to 11.1.3, and the satellite cookie has expired. In this case, when you attempt to enable the Serial number and IP address Authentication method without adding the satellite IP address in the satellite IP allow list on the portal, satellite authentication fails. The failure is due to a missing IP address in the satellite IP allow list.
- (PAN-OS 11.1.3 or later versions) A scenario where the satellite is running version earlier to 11.1.3 and the portal is upgraded to PAN-OS 11.1.3. In the meantime, the satellite cookie expires before enabling the Serial number and IP address Authentication method on the portal. Then satellite authentication fails due to satellite cookie expiration.

(HA deployments only) The serial number and IP address authentication method that is enabled is synchronized among the HA peers.

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STEP 4 | (Optional) Use the following operational commands to disable, delete, or view information about the serial number and IP address authentication method.

1. Enter the following command to disable the serial number and IP address authentication method on the firewall.

   ```
   username@hostname> set global-protect satellite-
   serialnumberip-auth disable
   ```

   (HA deployments only) The serial number and IP address authentication method that is disabled is synchronized among the HA peers.
2. Enter the following command to view all the information related to the serial number and IP address authentication method on the portal.

   ```
   username@hostname> show global-protect-portal global-protect-
   portal portal <name> satellite-serialnumberip-auth all
   ```
3. Enter the following command to view if the serial number and IP address authentication method is enabled or disabled on the firewall configured as a portal.

   ```
   username@hostname> show global-protect-portal satellite-
   serialnumberip-auth status
   ```
4. Enter the following command per portal to view the serial number and IP address retry interval.

   ```
   username@hostname> show global-protect-portal global-protect-
   portal portal <name> satellite-serialnumberip-auth retry-
   interval
   ```
5. Enter the following command per portal to view all the configured allowed satellite device IP addresses.

   This command displays both the IPv4 and IPv6 addresses that you have configured as a satellite IP allowed list in a sorted order.

   ```
   username@hostname> show global-protect-portal global-protect-
   portal portal <name> satellite-serialnumberip-auth satellite-
   ip-allowlist
   ```
6. Enter the following command per portal to delete a satellite device IP address from the satellite IP allow list.

   ```
   username@hostname> delete global-protect global-protect-portal
    portal <portal_name> satellite-ip-list allowlist-entry ip-
   address <value>
   ```

   Where <value> is the IPv4 address, IPv6 address, IP address range, or IP address subnet of the satellite device you want to delete.

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(HA deployments only) The deleted satellite devices IP address from the satellite IP allow list is synchronized among the HA peers.

7. Enter the following command per portal to delete a satellite device IP address from the satellite IP exclude list. You can delete only the entries that are added in the IP address exclude list. By deleting the entries from the exclude list, you are allowing these IP addresses to be configured in the satellite IP allow list.

```
username@hostname> delete global-protect global-protect-portal
 portal <portal_name> satellite-ip-list excludelist-entry
 ip <value>
```

Where <value> is the IPv4 address, IPv6 address, IP address range, or IP address subnet of the satellite device you want to delete from the exclude list entry.

(HA deployments only) The deleted satellite devices IP address from the satellite IP exclude list is synchronized among the HA peers.

8. Enter the following command per portal to delete all the satellite devices IP address from the satellite IP allow list.

```
username@hostname> delete global-protect global-protect-portal
 portal <name> satellite-ip-list satellite-ip-allowlist-all
```

(HA deployments only) The deleted satellite IP address list is synchronized among the HA peers.

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Configure GlobalProtect Gateways for LSVPN

Where Can I Use This?	What Do I Need?
• NGFW	❑ No separate license required for LSVPN when using NGFWs

Because the GlobalProtect configuration that the portal delivers to the satellites includes the list of gateways the satellite can connect to, it's a good idea to configure the gateways before configuring the portal.

Before you can configure the GlobalProtect gateway, you must complete the following tasks:

- Create Interfaces and Zones for the LSVPN on the interface where you'll configure each gateway. You must configure both the physical interface and the virtual tunnel interface.
- Enable SSL Between GlobalProtect LSVPN Components by configuring the gateway server certificates, SSL/TLS service profiles, and certificate profile required to establish a mutual SSL/TLS connection from the GlobalProtect satellites to the gateway.

Configure each GlobalProtect gateway to participate in the LSVPN as follows:

STEP 1 | Add a gateway.

1. Select **Network** > **GlobalProtect** > **Gateways** and click **Add**.
2. In the **General** screen, enter a **Name** for the gateway. The gateway name should have no spaces and, as a best practice, should include the location or other descriptive information to help users and administrators identify the gateway.
3. (Optional) Select the virtual system to which this gateway belongs from the **Location** field.

STEP 2 | Specify the network information that enables satellite devices to connect to the gateway.

If you haven't created the network interface for the gateway, see Create Interfaces and Zones for the LSVPN for instructions.

1. Select the **Interface** that satellites will use for ingress access to the gateway.
2. Specify the **IP Address Type** and **IP address** for gateway access:
   - The IP address type can be **IPv4** (only), **IPv6** (only), or **IPv4 and IPv6**. Use **IPv4 and IPv6** if your network supports dual stack configurations, where IPv4 and IPv6 run at the same time.
   - The IP address must be compatible with the IP address type. For example, **`172.16.1/0`** for IPv4 addresses or **`21DA:D3:0:2F3B`** for IPv6 addresses. For dual stack configurations, enter both an IPv4 and IPv6 address.
3. Click **OK** to save changes.

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STEP 3 | Specify how the gateway authenticates satellites attempting to establish tunnels. If you haven't yet created an SSL/TLS Service Profile for the gateway, see Deploy Server Certificates to the GlobalProtect LSVPN Components.

If you haven't set up the authentication profiles or certificate profiles, see Configure the Portal to Authenticate Satellites for instructions.

If you have not yet set up the certificate profile, see Enable SSL Between GlobalProtect LSVPN Components for instructions.

On the GlobalProtect Gateway configuration dialog, select Authentication and then configure any of the following:

- To secure communication between the gateway and the satellites, select the **SSL/TLS Service Profile** for the gateway.
- To specify the authentication profile to use to authenticate satellites, **Add** a Client Authentication. Then, enter a **Name** to identify the configuration, select **OS**: **Satellite** to apply the configuration to all satellites, and specify the **Authentication Profile** to use to authenticate the satellite. You can also select a **Certificate Profile** for the gateway to use to authenticate satellite devices attempting to establish tunnels.

STEP 4 | Configure the tunnel parameters and enable tunneling.

1. On the GlobalProtect Gateway configuration dialog, select **Satellite** > **Tunnel Settings**.
2. Select the **Tunnel Configuration** check box to enable tunneling.
3. Select the **Tunnel Interface** that you defined to terminate VPN tunnels established by the GlobalProtect satellites when you performed the task to Create Interfaces and Zones for the LSVPN.
4. (Optional) If you want to preserve the Type of Service (ToS) information in the encapsulated packets, select **Copy TOS**.

   If there are multiple sessions inside the tunnel (each with a different ToS value), copying the ToS header can cause the IPSec packets to arrive out of order.

STEP 5 | (Optional) Enable tunnel monitoring.

Tunnel monitoring enables satellites to monitor its gateway tunnel connection, allowing it to failover to a backup gateway if the connection fails. Failover to another gateway is the only type of tunnel monitoring profile supported with LSVPN.

1. Select the **Tunnel Monitoring** check box.
2. Specify the **Destination IP Address** that the satellites should use to determine if the gateway is active. You can specify an **IPv4** address, and **IPv6** address, or both. Alternatively, if you configured an IP address for the tunnel interface, you can leave this field blank and the tunnel monitor will instead use the tunnel interface to determine if the connection is active.
3. Select **Failover** from the **Tunnel Monitor Profile** drop-down (this is the only supported tunnel monitor profile for LSVPN).

STEP 6 | Select the IPSec Crypto profile to use when establishing tunnel connections.

The profile specifies the type of IPSec encryption and the authentication method for securing the data that will traverse the tunnel. Because both tunnel endpoints in an LSVPN are trusted

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firewalls within your organization, you can typically use the default (predefined) profile, which uses ESP as the IPSec protocol, group2 for the DH group, AES-128-CBC for encryption, and SHA-1 for authentication.

In the **IPSec Crypto Profile** drop-down, select **default** to use the predefined profile or select **New IPSec Crypto Profile** to define a new profile.

STEP 7 | Configure the network settings to assign the satellites during establishment of the IPSec tunnel.

You can also configure the satellite to push the DNS settings to its local clients by configuring a DHCP server on the firewall hosting the satellite. In this configuration, the satellite will push the DNS settings it learns from the gateway to the DHCP clients.

1. On the GlobalProtect Gateway configuration dialog, select **Satellite** > **Network Settings**.
2. (Optional) If clients local to the satellite need to resolve FQDNs on the corporate network, configure the gateway to push DNS settings to the satellites in one of the following ways:
   - If the gateway has an interface that is configured as a DHCP client, you can set the **Inheritance Source** to that interface and assign the same settings received by the DHCP client to GlobalProtect satellites. You can also inherit the DNS suffix from the same source.
   - Manually define the **Primary DNS**, **Secondary DNS**, and **DNS Suffix** settings to push to the satellites.
3. To specify the **IP Pool** of addresses to assign the tunnel interface on the satellites when the VPN is established, click **Add** and then specify the IP address range(s) to use.
4. To define what destination subnets to route through the tunnel click **Add** in the **Access Route** area and then enter the routes as follows:
   - If you want to route all traffic from the satellites through the tunnel, leave this field blank.

     In this case, all traffic except traffic destined for the local subnet will be tunneled to the gateway.

   - To route only some traffic through the gateway (called *split tunneling*), specify the destination subnets that must be tunneled. In this case, the satellite will route traffic that isn't destined for a specified access route using its own routing table. For example, you may choose to only tunnel traffic destined for your corporate network, and use the local satellite to enable internet access safely.
   - If you want to enable routing between satellites, enter the summary route for the network protected by each satellite.

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STEP 8 | (Optional) Define what routes, if any, the gateway will accept from satellites.

By default, the gateway won't add any routes that the satellites advertise to its routing table. If you don't want the gateway to accept routes from satellites, you don't need to complete this step.

1. To enable the gateway to accept routes advertised by satellites, select **Satellite** > **Route Filter**.
2. Select the **Accept published routes** check box.
3. To filter which of the routes advertised by the satellites to add to the gateway routing table, click **Add** and then define the subnets to include. For example, if all the satellites are configured with subnet 192.168.x.0/24 on the LAN side, configuring a permitted route of 192.168.0.0/16 to enable the gateway to accept only routes from the satellite if it is in the 192.168.0.0/16 subnet.

STEP 9 | Save the gateway configuration.

1. Click **OK** to save the settings and close the GlobalProtect Gateway configuration dialog.
2. **Commit** the configuration.

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Configure the GlobalProtect Portal for LSVPN

Where Can I Use This?	What Do I Need?
• NGFW	❑ No separate license required for LSVPN when using NGFWs

The GlobalProtect portal provides the management functions for your GlobalProtect LSVPN. Every satellite system that participates in the LSVPN receives configuration information from the portal, including information about available gateways as well as the certificate it needs in order to connect to the gateways.

The following sections provide procedures for setting up the portal:

- Configure the Portal—In LSVPN deployments, the GlobalProtect portal serves as a critical infrastructure component that centralizes management, authentication, and connection coordination for satellite devices. This centralized system acts as the core orchestrator of remote network access, providing a sophisticated mechanism for controlling and monitoring distributed network connections. The portal provides a comprehensive framework for authenticating satellite devices, coordinating connection policies, and establishing secure communication channels.
- Define the Satellite Configurations—In LSVPN deployments, a satellite configuration provides the essential connection parameters that enable GlobalProtect satellites to establish secure VPN tunnels with your network's gateways. These configurations act as a blueprint, specifying which gateways a satellite can connect to, what certificates it should use, and how frequently it should refresh its configuration.

  The satellite configuration serves as a bridge between your GlobalProtect portal and the remote satellite devices, ensuring that each satellite receives the precise network access and security settings required for its specific deployment context.

GlobalProtect Portal for LSVPN Prerequisite Tasks

Before configuring the GlobalProtect portal, you must complete the following tasks:

- ❑ Create Interfaces and Zones for the LSVPN on the interface where you'll configure the portal.
- ❑ Enable SSL Between GlobalProtect LSVPN Components by creating an SSL/TLS service profile for the portal server certificate, issuing gateway server certificates, and configuring the portal to issue server certificates for the GlobalProtect satellites.
- ❑ Configure the Portal to Authenticate Satellites by setting up local database authentication and defining the authentication profile that the portal will use to authenticate satellites.
- ❑ Configure GlobalProtect Gateways for LSVPN.

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Configure the Portal

Where Can I Use This?	What Do I Need?
• NGFW	❑ No separate license required for LSVPN when using NGFWs

After you've completed the GlobalProtect Portal for LSVPN Prerequisite Tasks, configure the GlobalProtect portal as follows:

STEP 1 | Add the portal.

1. Select **Network** > **GlobalProtect** > **Portals** and click **Add**.
2. On the **General** tab, enter a **Name** for the portal. The portal name shouldn't contain any spaces.
3. (Optional) Select the virtual system to which this portal belongs from the **Location** field.

STEP 2 | Specify the network information to enable satellites to connect to the portal.

If you haven't yet created the network interface for the portal, see Create Interfaces and Zones for the LSVPN for instructions.

1. Select the **Interface** that satellites will use for ingress access to the portal.
2. Specify the **IP Address Type** and **IP address** for satellite access to the portal:
   - The IP address type can be **IPv4** (for IPv4 traffic only), **IPv6** (for IPv6 traffic only, or **IPv4 and IPv6**. Use **IPv4 and IPv6** if your network supports dual stack configurations, where IPv4 and IPv6 run at the same time.
   - The IP address must be compatible with the IP address type. For example, **`172.16.1/0`** for IPv4 addresses or **`21DA:D3:0:2F3B`** for IPv6 addresses. For dual stack configurations, enter both an IPv4 and IPv6 address.
3. Click **OK** to save changes.

STEP 3 | Specify an SSL/TLS Service Profile to use to enable the satellite to establish an SSL/TLS connection to the portal.

If you haven't yet created an SSL/TLS Service Profile for the portal and issued gateway certificates, see Deploy Server Certificates to the GlobalProtect LSVPN Components.

1. On the GlobalProtect portal configuration dialog, select **Authentication**.
2. Select the **SSL/TLS Service Profile**.

STEP 4 | Specify an authentication profile and optional Certificate Profile for authenticating satellites.

*The first time the satellite connects to the portal, it must authenticate using local database authentication (on subsequent sessions it uses a satellite cookie issued by the portal). Therefore, before you can save the portal configuration (by clicking **OK**), you must* Configure an authentication profile.

Add a Client Authentication, and then enter a **Name** to identify the configuration, select **OS**: **Satellite** to apply the configuration to all satellites, and specify the **Authentication Profile** to

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use to authenticate satellite devices. You can also specify a **Certificate Profile** for the portal to use to authenticate satellite devices.

STEP 5 | Continue with defining the configurations to push to the satellites or, if you've already created the satellite configurations, save the portal configuration.

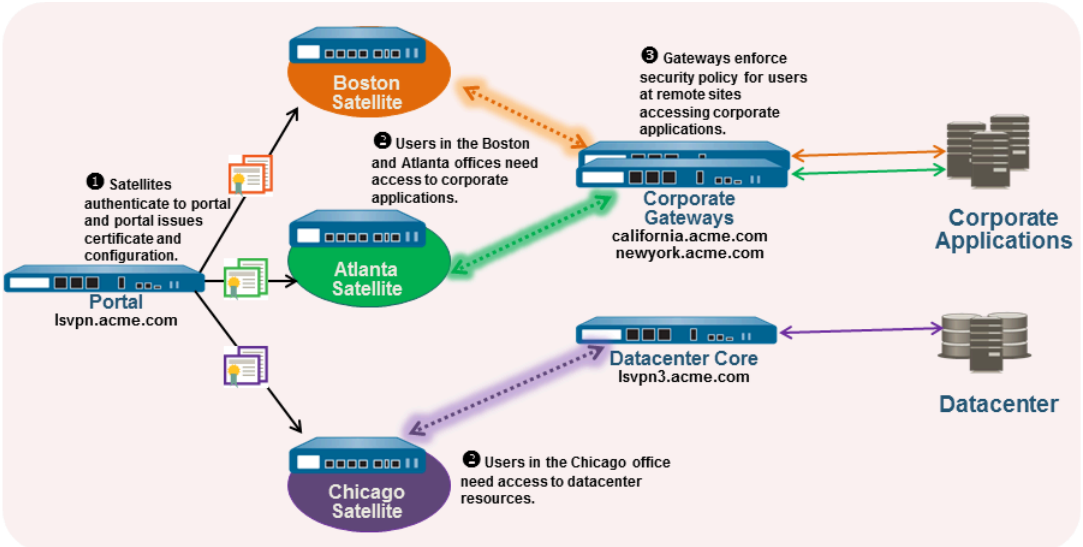
Click **OK** to save the portal configuration or continue to Define the Satellite Configurations.

Define the Satellite Configurations

Where Can I Use This?	What Do I Need?
• NGFW	☐ No separate license required for LSVPN when using NGFWs

When a GlobalProtect satellite connects and successfully authenticates to the GlobalProtect portal, the portal delivers a satellite configuration, which specifies what gateways the satellite can connect to. If all your satellites will use the same gateway and certificate configurations, you can create a single satellite configuration to deliver to all satellites upon successful authentication. However, if you require different satellite configurations—for example if you want one group of satellites to connect to one gateway and another group of satellites to connect to a different gateway—you can create a separate satellite configuration for each. The portal will then use the enrollment username/group name or the serial number of the satellite to determine which satellite configuration to deploy. As with security rule evaluation, the portal looks for a match starting from the top of the list. When it finds a match, it delivers the corresponding configuration to the satellite.

For example, the following figure shows a network in which some branch offices require VPN access to the corporate applications protected by your perimeter firewalls and another site needs VPN access to the data center.



Use the following procedure to create one or more satellite configurations.

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STEP 1 | Add a satellite configuration.

The satellite configuration specifies the GlobalProtect LSVPN configuration settings to deploy to the connecting satellites. You must define at least one satellite configuration.

1. Select **Network** > **GlobalProtect** > **Portals** and select the portal configuration for which you want to add a satellite configuration and then select the **Satellite** tab.
2. In the Satellite section, click **Add**.
3. Enter a **Name** for the configuration.

   If you plan to create multiple configurations, make sure that the name you define for each is descriptive enough to allow you to distinguish them.
4. To change how often a satellite should check the portal for configuration updates, specify a value in the **Configuration Refresh Interval (hours)** field (range is 1-48; default is 24).

STEP 2 | Specify the satellites to which to deploy this configuration.

The portal uses the **Enrollment User/User Group** settings and/or **Devices** serial numbers to match a satellite to a configuration. Therefore, if you have multiple configurations, be sure to order them properly. As soon as the portal finds a match, it will deliver the configuration. Therefore, more specific configurations must precede more general ones. See step 5 for instructions on ordering the list of satellite configurations.

Specify the match criteria for the satellite configuration as follows:

- Select the **Devices** tab, click **Add**, and enter serial number (you don't need to enter the satellite hostname; it will be automatically added when the satellite connects) to restrict this configuration to satellites with specific serial numbers. Repeat this step for each satellite that you want to receive this configuration.
- Select the **Enrollment User/User Group** tab, click **Add**, and then select the user or group you want to receive this configuration. Satellites that don't match on serial number will be required to authenticate as a user specified here (either an individual user or group member).

Before you can restrict the configuration to specific groups, you must Map Users to Groups.

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STEP 3 | Specify the gateways that satellites with this configuration can establish VPN tunnels with.

- *Routes published by the gateway are installed on the satellite as static routes. The metric for the static route is 10x the routing priority. If you have more than one gateway, make sure to set the routing priority also to ensure that routes advertised by backup gateways have higher metrics compared to the same routes advertised by primary gateways. For example, if you set the routing priority for the primary gateway and backup gateway to 1 and 10 respectively, the satellite will use 10 as the metric for the primary gateway and 100 as the metric for the backup gateway.*

1. On the **Gateways** tab, click **Add**.
2. Enter a descriptive **Name** for the gateway. The name you enter here should match the name you defined when you configured the gateway and should be descriptive enough identify the location of the gateway.
3. Enter the FQDN or IP address of the interface where the gateway is configured in the **Gateways** field. The address you specify must exactly match the Common Name (CN) in the gateway server certificate.
4. (Optional) If you're adding two or more gateways to the configuration, the **Routing Priority** helps the satellite pick the preferred gateway. Enter a value in the range of 1-25, with lower numbers having the higher priority (that is, the gateway the satellite will connect to if all gateways are available). The satellite will multiply the routing priority by 10 to determine the routing metric.

STEP 4 | Save the satellite configuration.

1. Click **OK** to save the satellite configuration.
2. If you want to add another satellite configuration, repeat the previous steps.

STEP 5 | Arrange the satellite configurations so that the proper configuration is deployed to each satellite.

- To move a satellite configuration up on the list of configurations, select the configuration and click **Move Up**.
- To move a satellite configuration down on the list of configurations, select the configuration and click **Move Down**.

STEP 6 | Specify the certificates required to enable satellites to participate in the LSVPN.

1. In the **Trusted Root CA** field, click **Add** and then select the CA certificate used to issue the gateway server certificates. The portal will deploy the root CA certificate you add here to all satellites as part of the configuration to enable the satellite to establish an SSL

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connection with the gateways. As a best practice, all of your gateways should use the same issuer.

2. Select the method of **Client Certificate** distribution:
   - **To store the client certificates on the portal**—select **Local** and select the root CA certificate that the portal will use to issue client certificates to satellites upon successfully authenticating them from the **Issuing Certificate** drop-down.

     *If the root CA certificate used to issue your gateway server certificates isn't on the portal, you can **Import** it now. See Enable SSL Between GlobalProtect LSVPN Components for details on how to import a root CA certificate.*

   - **To enable the portal to act as a SCEP client to request dynamically and issue client certificates**—select **SCEP** and then select the **SCEP** profile used to generate CSRs to your SCEP server.

     *If you have not yet set up the portal to act as a SCEP client, you can add a **New** SCEP profile now. See Deploy Client Certificates to the GlobalProtect Satellites Using SCEP for details.*

STEP 7 | Save the portal configuration.

1. Click **OK** to save the settings and close the GlobalProtect portal configuration dialog.
2. **Commit** your changes.

Use Case: Basic LSVPN Configuration with Static Routing

Where Can I Use This?	What Do I Need?
• NGFW	☐ No separate license required for LSVPN when using NGFWs

This quick configuration shows the fastest way to get up and running with LSVPN. In this example, a single firewall at the corporate headquarters site is configured as both a portal and a gateway. Satellites can be quickly and easily deployed with minimal configuration for optimized scalability.

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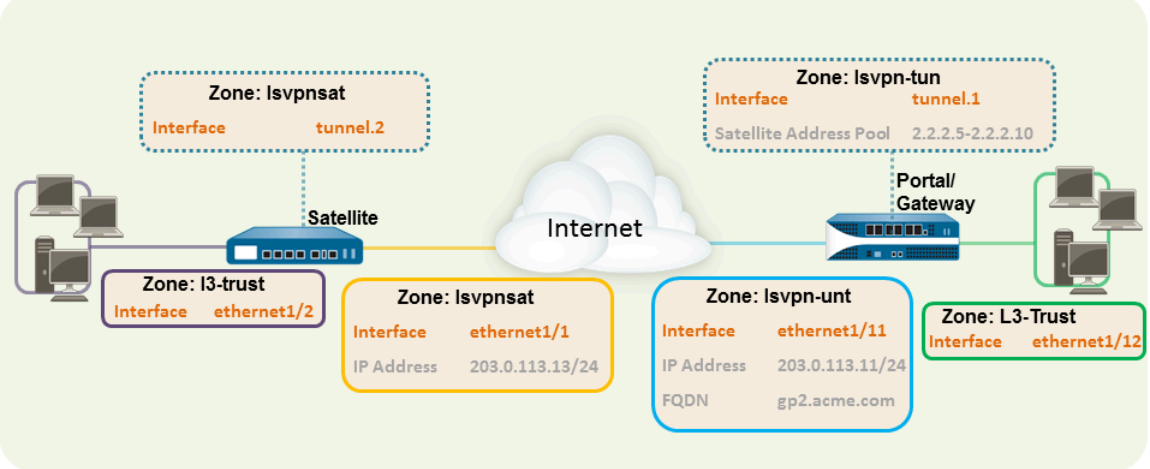

The following workflow shows the steps for setting up this basic configuration:

STEP 1 | Configure a Layer 3 interface.

In this example, the Layer 3 interface on the portal/gateway requires the following configuration:

- **Interface**—ethernet1/11
- **Security Zone**—lsvpn-tun
- **IPv4**—203.0.113.11/24

STEP 2 | On the firewall(s) hosting GlobalProtect gateway(s), configure the logical tunnel interface that will terminate VPN tunnels established by the GlobalProtect satellites.

To enable visibility into users and groups connecting over the VPN, enable User-ID in the zone where the VPN tunnels terminate.

In this example, the Tunnel interface on the portal/gateway requires the following configuration:

- **Interface**—tunnel.1
- **Security Zone**—lsvpn-tun

STEP 3 | Create the Security policy rule to enable traffic flow between the VPN zone where the tunnel terminates (lsvpn-tun) and the trust zone where the corporate applications reside (L3-Trust).

See Create a Security Policy Rule.

STEP 4 | Assign an SSL/TLS Service profile to the portal/gateway. The profile must reference a self-signed server certificate.

The certificate subject name must match the FQDN or IP address of the Layer 3 interface you create for the portal/gateway.

1. On the firewall hosting the GlobalProtect portal, create the root CA certificate for signing the certificates of the GlobalProtect components. In this example, the root CA

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certificate, **lsvpn-CA**, will be used to issue the server certificate for the portal/gateway. In addition, the portal will use this root CA certificate to sign the CSRs from the satellites.

2. Create SSL/TLS service profiles for the GlobalProtect portal and gateways.

   Because the portal and gateway are on the same interface in this example, they can share an SSL/TLS Service profile that uses the same server certificate. In this example, the profile is named **lsvpnserver**.

STEP 5 | Create a certificate profile.

In this example, the Certificate Profile **lsvpn-profile** references the root CA certificate **lsvpn-CA**. The gateway will use this Certificate Profile to authenticate satellites attempting to establish VPN tunnels.

STEP 6 | Configure the portal to authenticate satellites using local database authentication.

STEP 7 | Configure GlobalProtect Gateways for LSVPN.

Select **Network** > **GlobalProtect** > **Gateways** and **Add** a configuration. This example requires the following gateway configuration:

- **Interface**—ethernet1/11
- **IP Address**—203.0.113.11/24
- **SSL/TLS Server Profile**—lsvpnserver
- **Certificate Profile**—lsvpn-profile
- **Tunnel Interface**—tunnel.1
- **Primary DNS**/**Secondary DNS**—4.2.2.1/4.2.2.2
- **IP Pool**—2.2.2.111-2.2.2.120
- **Access Route**—10.2.10.0/24

STEP 8 | Configure the Portal.

Select **Network** > **GlobalProtect** > **Portal** and **Add** a configuration. This example requires the following portal configuration:

- **Interface**—ethernet1/11
- **IP Address**—203.0.113.11/24
- **SSL/TLS Server Profile**—lsvpnserver
- **Authentication Profile**—lsvpn-sat

STEP 9 | Define the Satellite Configurations.

On the **Satellite** tab in the portal configuration, **Add** a Satellite configuration and a Trusted root CA and specify the CA the portal will use to issue certificates for the satellites. In this example, the required settings are as following:

- **Gateway**—203.0.113.11
- **Issuing Certificate**—lsvpn-CA
- **Trusted Root CA**—lsvpn-CA

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STEP 10 | Prepare the Satellite to Join the LSVPN.

The satellite configuration in this example requires the following settings:

Interface configuration

- Layer 3 interface—ethernet1/1, 203.0.113.13/24
- Tunnel interface—tunnel.2
- Zone—lsvpnsat

Root CA Certificate from Portal

- lsvpn-CA

IPSec Tunnel configuration

- **Tunnel Interface**—tunnel.2
- **Portal Address**—203.0.113.11
- **Interface**—ethernet1/1
- **Local IP Address**—203.0.113.13/24
- **Publish all static and connected routes to Gateway**—enabled

Use Case: Advanced LSVPN Configuration with Dynamic Routing

Where Can I Use This?	What Do I Need?
• NGFW	❑ No separate license required for LSVPN when using NGFWs

In larger LSVPN deployments with multiple gateways and many satellites, investing a little more time in the initial configuration to set up dynamic routing will simplify the maintenance of gateway configurations because access routes will update dynamically. The following example configuration shows how to extend the basic LSVPN configuration to configure OSPF as the dynamic routing protocol.

Setting up an LSVPN to use OSPF for dynamic routing requires the following additional steps on the gateways and the satellites:

- Manual assignment of IP addresses to tunnel interfaces on all gateways and satellites.
- Configuration of OSPF point-to-multipoint (P2MP) on the virtual router on all gateways and satellites. In addition, as part of the OSPF configuration on each gateway, you must manually define the tunnel IP address of each satellite as an OSPF neighbor. Similarly, on each satellite, you must manually define the tunnel IP address of each gateway as an OSPF neighbor.

Although dynamic routing requires additional setup during the initial configuration of the LSVPN, it reduces the maintenance tasks associated with keeping routes up to date as topology changes occur on your network.

The following figure shows an LSVPN dynamic routing configuration. This example shows how to configure OSPF as the dynamic routing protocol for the VPN.

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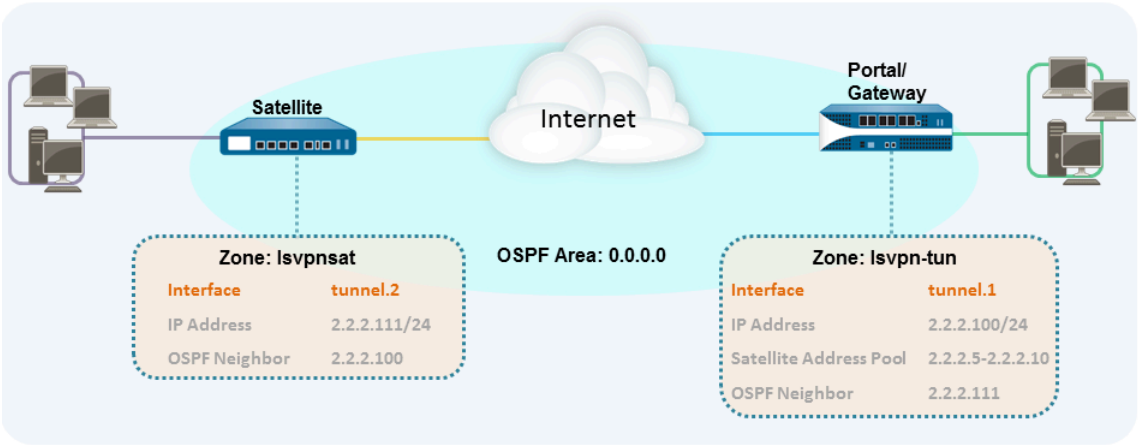

For a basic setup of a LSVPN, follow the steps in Basic LSVPN Configuration with Static Routing. You can then complete the steps in the following workflow to extend the configuration to use dynamic routing rather than static routing.

STEP 1 | Add an IP address to the tunnel interface configuration on each gateway and each satellite.

Complete the following steps on each gateway and each satellite:

1. Select **Network** > **Interfaces** > **Tunnel** and select the tunnel configuration you created for the LSVPN to open the Tunnel Interface dialog.

   If you haven't yet created the tunnel interface, see step 2 in Create Interfaces and Zones for the LSVPN.
2. On the **IPv4** tab, click **Add** and then enter an IP address and subnet mask. For example, to add an IP address for the gateway tunnel interface you would enter 2.2.2.100/24.
3. Click **OK** to save the configuration.

STEP 2 | Configure the dynamic routing protocol on the gateway.

To configure OSPF on the gateway:

1. Select **Network** > **Virtual Routers** and select the virtual router associated with your VPN interfaces.
2. On the **Areas** tab, click **Add** to create the backbone area, or, if it's already configured, click on the area ID to edit it.
3. If you're creating a new area, enter an **Area ID** on the **Type** tab.
4. On the **Interface** tab, click **Add** and select the tunnel **Interface** you created for the LSVPN.
5. Select **p2mp** as the **Link Type**.
6. Click **Add** in the Neighbors section and enter the IP address of the tunnel interface of each satellite, for example 2.2.2.111.
7. Click **OK** twice to save the virtual router configuration and then **Commit** the changes on the gateway.
8. Repeat this step each time you add a new satellite to the LSVPN.

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STEP 3 | Configure the dynamic routing protocol on the satellite.

To configure OSPF on the satellite:

1. Select **Network** > **Virtual Routers** and select the virtual router associated with your VPN interfaces.
2. On the **Areas** tab, click **Add** to create the backbone area, or, if it's already configured, click on the area ID to edit it.
3. If you're creating a new area, enter an **Area ID** on the **Type** tab.
4. On the **Interface** tab, click **Add** and select the tunnel **Interface** you created for the LSVPN.
5. Select **p2mp** as the **Link Type**.
6. Click **Add** in the Neighbors section and enter the IP address of the tunnel interface of each GlobalProtect gateway, for example 2.2.2.100.
7. Click **OK** twice to save the virtual router configuration and then **Commit** the changes on the gateway.
8. Repeat this step each time you add a new gateway.

STEP 4 | Verify that the gateways and satellites are able to form router adjacencies.

- On each satellite and each gateway, confirm that peer adjacencies have formed and that routing table entries have been created for the peers (that is, the satellites have routes to the gateways and the gateways have routes to the satellites). Select **Network** > **Virtual Router** and click the **More Runtime Stats** link for the virtual router you're using for the LSVPN. On the Routing tab, verify that the LSVPN peer has a route.
- On the **OSPF** > **Interface** tab, verify that the **Type** is **p2mp**.
- On the **OSPF** > **Neighbor** tab, verify that the firewalls hosting your gateways have established router adjacencies with the firewalls hosting your satellites and vice versa. Also verify that the **Status** is **Full**, indicating that full adjacencies have been established.

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Prepare the Satellite to Join the LSVPN

Where Can I Use This?	What Do I Need?
• NGFW	❑ No separate license required for LSVPN when using NGFWs

To participate in the LSVPN, the satellites require a minimal amount of configuration. Because the required configuration is minimal, you can pre-configure the satellites before shipping them to your branch offices for installation.

STEP 1 | Configure a Layer 3 Interface.

This is the physical interface that the satellite will use to connect to the portal and the gateway. This interface must be in a zone that allows access outside of the local trust network. As a best practice, create a dedicated zone for VPN connections for visibility and control over traffic destined for the corporate gateways.

STEP 2 | Configure the logical tunnel interface for the tunnel to use to establish VPN tunnels with the GlobalProtect gateways.

IP addresses aren't required on the tunnel interface unless you plan to use dynamic routing. However, assigning an IP address to the tunnel interface can be useful for troubleshooting connectivity issues.

1. Select **Network** > **Interfaces** > **Tunnel** and click **Add**.
2. In the **Interface Name** field, specify a numeric suffix, such as **.2**.
3. On the **Config** tab, expand the **Security Zone** drop-down and select an existing zone or create a separate zone for VPN tunnel traffic by clicking **New Zone** and defining a **Name** for the new zone (for example *lsvpnsat*).
4. In the **Virtual Router** drop-down, select **default**.
5. (Optional) To assign an IP address to the tunnel interface:
   - For an IPv4 address, select **IPv4** and **Add** the IP address and network mask to assign to the interface, for example 203.0.11.100/24.
   - For an IPv6 address, select **IPv6**, **Enable IPv6 on the interface**, and **Add** the IP address and network mask to assign to the interface, for example 2001:1890:12f2:11::10.1.8.160/80.
6. To save the interface configuration, click **OK**.

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STEP 3 | If you generated the portal server certificate using a root CA that isn't trusted by the satellites (for example, if you used self-signed certificates), import the root CA certificate used to issue the portal server certificate.

The root CA certificate is required to enable the satellite to establish the initial connection with the portal to obtain the LSVPN configuration.

1. Download the CA certificate that was used to generate the portal server certificates. If you're using self-signed certificates, export the root CA certificate from the portal as follows:
    1. Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later).
    2. Select the CA certificate, and click **Export**.
    3. Select **Base64 Encoded Certificate (PEM)** from the **File Format** drop-down and click **OK** to download the certificate. (You don't need to export the private key.)
2. Import the root CA certificate that you exported onto each satellite as follows.
    1. Select **Device** > **Certificate Management** > **Certificates**, then **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later). Select **Import**.
    2. Enter a **Certificate Name** that identifies the certificate as your client CA certificate.
    3. **Browse** to the **Certificate File** that you downloaded from the CA.
    4. Select **Base64 Encoded Certificate (PEM)** as the **File Format** and then click **OK**.
    5. Select the certificate that you imported on the **Device Certificates** (PAN-OS 11.2 and earlier) or **Custom Certificates** (PAN-OS 12.1.0 and later) tab to open it.
    6. Select **Trusted Root CA** and then click **OK**.

STEP 4 | Configure the IPSec tunnel configuration.

1. Select **Network** > **IPSec Tunnels** and click **Add**.
2. On the **General** tab, enter a descriptive **Name** for the IPSec configuration.
3. Select the **Tunnel Interface** that you created for the satellite.
4. Select **GlobalProtect Satellite** as the **Type**.
5. Enter the IP address or FQDN of the portal as the **Portal Address**.
6. Select the Layer 3 **Interface** you configured for the satellite.
7. Select the **IP Address** to use on the selected interface. You can select an **IPv4** address, an **IPv6** address, or both. Specify if you want **IPv6 preferred for portal registration**.

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STEP 5 | (Optional) Configure the satellite to publish local routes to the gateway.

Pushing routes to the gateway enables traffic to the subnets local to the satellite via the gateway. However, you must also configure the gateway to accept the routes as detailed in Configure GlobalProtect Gateways for LSVPN.

1. To enable the satellite to push routes to the gateway, on the **Advanced** tab select **Publish all static and connected routes to Gateway**.

   If you select this check box, the firewall will forward all static and connected routes from the satellite to the gateway. However, to prevent the creation of routing loops, the firewall will apply some route filters, such as the following:

   - Default routes
   - Routes within a virtual router other than the virtual router associated with the tunnel interface
   - Routes using the tunnel interface
   - Routes using the physical interface associated with the tunnel interface
2. (Optional) If you only want to push routes for specific subnets rather than all routes, click **Add** in the Subnet section and specify which subnet routes to publish.

STEP 6 | Save the satellite configuration.

1. Click **OK** to save the IPSec tunnel settings.
2. Click **Commit**.

STEP 7 | If required, provide the credentials to allow the satellite to authenticate to the portal.

To authenticate to the portal for the first time, the satellite administrator must provide the username and password associated with the satellite admin account in the local database.

1. Select **Network** > **IPSec Tunnels** and click the **Gateway Info** link in the Status column of the tunnel configuration you created for the LSVPN.
2. Click the **enter credentials** link in the **Portal Status** field and provide the username and password to authenticate the satellite to the portal.

   After the portal successfully authenticates to the portal, it will receive its signed certificate and configuration, which it will use to connect to the gateway(s). You should see that the tunnel is established and the **Status** is changed to **Active**.

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Verify the LSVPN Configuration

Where Can I Use This?	What Do I Need?
• NGFW	❑ No separate license required for LSVPN when using NGFWs

After configuring the portal, gateways, and satellites, verify that the satellites are able to connect to the portal and gateway and establish VPN tunnels with one or more gateways.

STEP 1 | Verify satellite connectivity with portal.

From the firewall hosting the portal, verify that the satellites are successfully connecting by selecting **Network** > **GlobalProtect** > **Portal** and clicking **Satellite Info** in the Info column of the portal configuration entry.

STEP 2 | Verify satellite connectivity with the gateway(s).

On each firewall hosting a gateway, verify that satellites are able to establish VPN tunnels by selecting **Network** > **GlobalProtect** > **Gateways** and click **Satellite Info** in the Info column of the gateway configuration entry. Satellites that have successfully established tunnels with the gateway will display on the **Active Satellites** tab.

STEP 3 | Verify LSVPN tunnel status on the satellite.

On each firewall hosting a satellite, verify the tunnel status by selecting **Network** > **IPSec Tunnels** and verify active Status as indicated by a green icon.

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Use Case: Advanced LSVPN Configuration with iBGP

Where Can I Use This?	What Do I Need?
• NGFW	❑ No separate license required for LSVPN when using NGFWs

This use case illustrates how GlobalProtect LSVPN securely connects distributed office locations with primary and disaster recovery data centers that house critical applications for users and how an internal border gateway protocol (iBGP) eases deployment and upkeep. Using this method, you can extend up to 500 satellite offices connecting to a single gateway.

BGP is a highly scalable, dynamic routing protocol that is ideal for hub-and-spoke deployments such as LSVPN. As a dynamic routing protocol, it eliminates much of the overhead associated with access routes (static routes) by making it relatively easy to deploy additional satellite firewalls. Due to its route filtering capabilities and features such as multiple tunable timers, route dampening, and route refresh, BGP scales to a higher number of routing prefixes with greater stability than other routing protocols like RIP and OSPF. In the case of iBGP, a peer group, which includes all the satellites and gateways in the LSVPN deployment, establishes adjacencies over the tunnel endpoints. The protocol then implicitly takes control of route advertisements, updates, and convergence.

In this example configuration, an active/passive HA pair of PA-5200 firewalls is deployed in the primary (active) data center and acts as the portal and primary gateway. The disaster recovery data center also has two PA-5200s in an active/passive HA pair acting as the backup LSVPN gateway. The portal and gateways serve 500 PA-220s deployed as LSVPN satellites in branch offices.

Both data center sites advertise routes but with different metrics. As a result, the satellites prefer and install the active data center's routes. However, the backup routes also exist in the local routing information base (RIB). If the active data center fails, the routes advertised by that data center are removed and replaced with routes from the disaster recovery data center's routes. The failover time depends on selection of iBGP times and routing convergence associated with iBGP.

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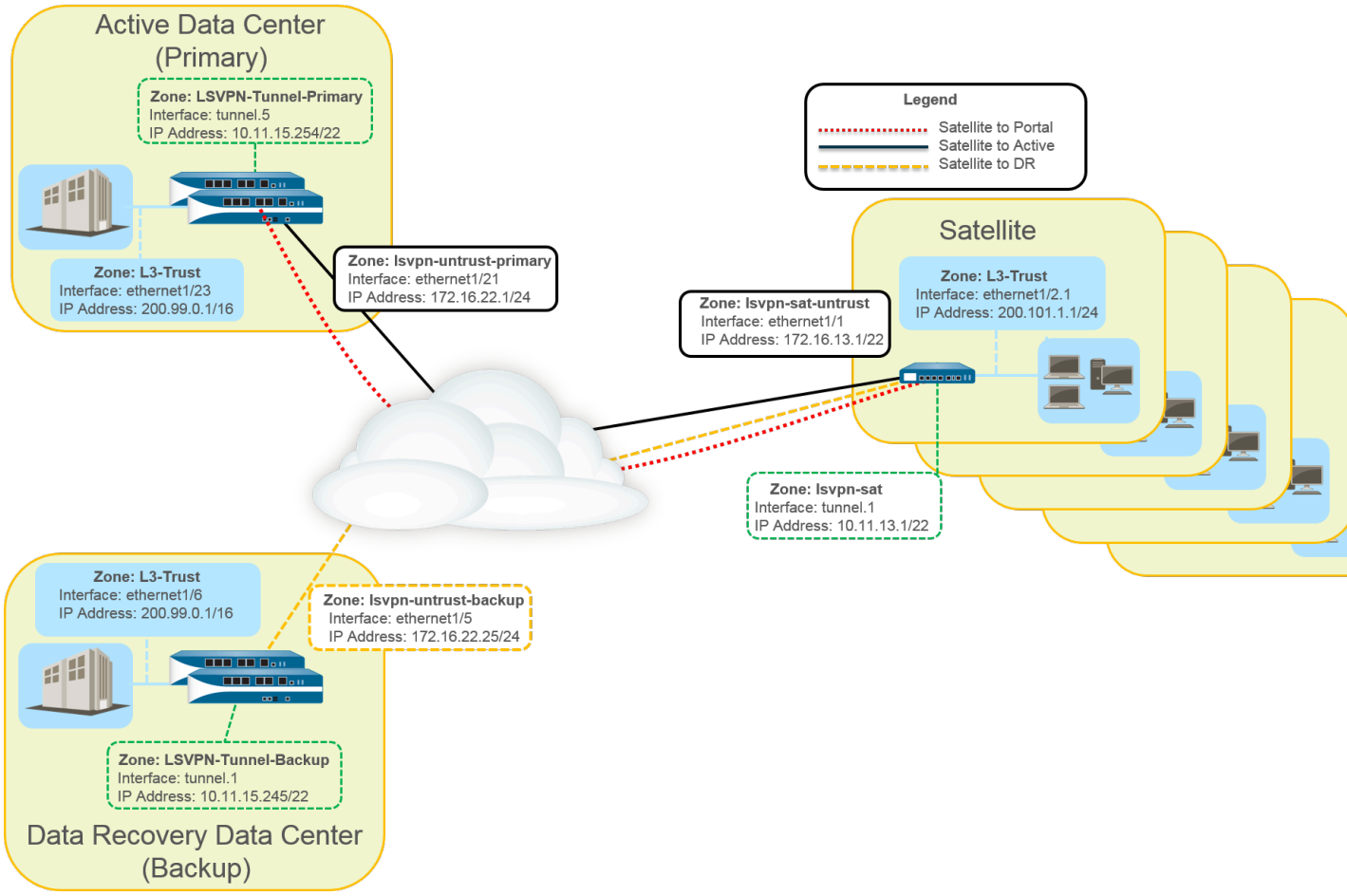

The following workflow shows the steps for configuring this deployment:

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STEP 1 | Create Interfaces and Zones for the LSVPN.

Portal and primary gateway:

- **Zone**: LSVPN-Untrust-Primary
- Interface: ethernet1/21
- IPv4: 172.16.22.1/24
- **Zone**: L3-Trust
- **Interface**: ethernet1/23
- **IPv4**: 200.99.0.1/16

Backup gateway:

- **Zone**: LSVPN-Untrust-Primary
- **Interface**: ethernet1/5
- **IPv4**: 172.16.22.25/24
- **Zone**: L3-Trust
- **Interface**: ethernet1/6
- **IPv4**: 200.99.0.1/16

Satellite:

- **Zone**: LSVPN-Sat-Untrust
- **Interface**: ethernet1/1
- **IPv4**: 172.16.13.1/22
- **Zone**: L3-Trust
- **Interface**: ethernet1/2.1
- **IPv4**: 200.101.1.1/24

Configure the zones, interfaces, and IP addresses on each satellite. The interface and local IP address will be different for each satellite. This interface is used for the VPN connection to the portal and gateway.

STEP 2 | On the firewall(s) hosting GlobalProtect gateway(s), configure the logical tunnel interface that will terminate VPN tunnels established by the GlobalProtect satellites.

Primary gateway:

- **Interface**: tunnel.5
- **IPv4**: 10.11.15.254/22
- **Zone**: LSVPN-Tunnel-Primary

Backup gateway:

- **Interface**: tunnel.1
- **IPv4**: 10.11.15.245/22
- **Zone**: LSVPN-Tunnel-Backup

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STEP 3 | Enable SSL Between GlobalProtect LSVPN Components.

The gateway uses the self-signed root certificate authority (CA) to issue certificates for the satellites in a GlobalProtect LSVPN. Because one firewall houses the portal and primary gateway, a single certificate is used for authenticating to the satellites. The same CA is used to generate a certificate for the backup gateway. The CA generates certificates that pushed to the satellites from the portal and then used by the satellites to authenticate to the gateways.

You must also generate a certificate from the same CA for the backup gateway, allowing it to authenticate with the satellites.

1. On the firewall hosting the GlobalProtect portal, create the root CA certificate for signing the certificates of the GlobalProtect components. In this example, the root CA certificate is called CA-cert.
2. Create SSL/TLS service profiles for the GlobalProtect portal and gateways. Because the GlobalProtect portal and primary gateway are the same firewall interface, you can use the same server certificate for both components.
   - **Root CA Certificate**: CA-Cert
   - **Certificate Name**: LSVPN-Scale
3. Deploy the self-signed server certificates to the gateways.
4. Import the root CA certificate used to issue server certificates for the LSVPN components.
5. Create a certificate profile.
6. Repeat steps 2 through 5 on the backup gateway with the following settings:
   - **Root CA Certificate**: CA-cert
   - **Certificate Name**: LSVPN-back-GW-cert

STEP 4 | Configure GlobalProtect Gateways for LSVPN.

1. Select **Network** > **GlobalProtect** > **Gateways** and click **Add**.
2. On the **General** tab, name the primary gateway **`LSVPN-Scale`**.
3. Under **Network Settings**, select **`ethernet1/21`** as the primary gateway interface and enter **`172.16.22.1/24`** as the IP address.
4. On the **Authentication** tab, select the LSVPN-Scale certificate created in 3.
5. Select **Satellite** > **Tunnel Settings** and select **Tunnel Configuration**. Set the **Tunnel Interface** to tunnel.5. All satellites in this use case connect to a single gateway, so a

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single satellite configuration is needed. Satellites are matched based on their serial numbers, so no satellites will need to authenticate as a user.

6. On **Satellite** > **Network Settings**, define the pool of IP address to assign to the tunnel interface on the satellite once the VPN connection is established. Because this use case uses dynamic routing, the Access Routes setting remains blank.
7. Repeat steps 1 through 5 on the backup gateway with the following settings:
   - **Name**: LSVPN-backup
   - **Gateway interface**: ethernet1/5
   - **Gateway IP**: 172.16.22.25/24
   - **Server cert**: LSVPN-backup-GW-cert
   - **Tunnel interface**: tunnel.1

STEP 5 | Configure iBGP on the primary and backup gateways and add a redistribution profile to allow the satellites to inject local routes back to the gateways.

Each satellite office manages its own network and firewall, so the redistribution profile called ToAllSat is configured to redistribute local routes back to the GlobalProtect gateway.

1. Select **Network** > **Virtual Routers** and **Add** a virtual router.
2. On **Router Settings**, add the **Name** and **Interface** for the virtual router.
3. On **Redistribution Profile**, select **Add**.
   1. Name the redistribution profile **`ToAllSat`** and set the **Priority** to 1.
   2. Set Redistribute to **Redist**.
   3. **Add `ethernet1/23`** from the Interface drop-down.
   4. Click **OK**.
4. Select **BGP** on the virtual router to configure BGP.
   1. On **BGP** > **General**, select **Enable**.
   2. Enter the gateway IP address as the **Router ID** (**`172.16.22.1`**) and **`1000`** as the **AS Number**.
   3. In the Options section, select **Install Route**.
   4. On **BGP** > **Peer Group**, click **Add** a peer group with all the satellites that will connect to the gateway.
   5. On **BGP** > **Redist Rules**, **Add** the **`ToAllSat`** redistribution profile you created previously.
5. Click **OK**.
6. Repeat steps 1 through 5 on the backup gateway using **`ethernet1/6`** for the redistribution profile.

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STEP 6 | Prepare the Satellite to Join the LSVPN.

The configuration shown is a sample of a single satellite.

Repeat this configuration each time you add a new satellite to the LSVPN deployment.

1. Configure a tunnel interface as the tunnel endpoint for the VPN connection to the gateways.
2. Set the IPSec tunnel type to GlobalProtect Satellite and enter the IP address of the GlobalProtect portal.
3. Select **Network** > **Virtual Routers** and **Add** a virtual router.
4. On **Router Settings**, add the **Name** and **Interface** for the virtual router.
5. Select **Virtual Router** > **Redistribution Profile** and **Add** a profile with the following settings.
   1. Name the redistribution profile **`ToLSVPNGW`** and set the **Priority** to 1.
   2. **Add** an **Interface** **`ethernet1/2.1`**.
   3. Click **OK**.
6. Select **BGP** > **General**, **Enable** BGP and configure the protocol as follows:
   1. Enter the gateway IP address as the **Router ID** (**`172.16.22.1`**) and **`1000`** as the **AS Number**.
   2. In the Options section, select **Install Route**.
   3. On **BGP** > **Peer Group**, **Add** a peer group containing all the satellites that will connect to the gateway.
   4. On **BGP** > **Redist Rules**, **Add** the **`ToLSVPNGW`** redistribution profile you created previously.
7. Click **OK**.

STEP 7 | Configure the GlobalProtect Portal for LSVPN.

Both data centers advertise their routes but with different routing priorities to ensure that the active data center is the preferred gateway.

1. Select **Network** > **GlobalProtect** > **Portals** and click **Add**.
2. On **General**, enter **`LSVPN-Portal`** as the portal name.
3. On **Network Settings**, select **`ethernet1/21`** as the **Interface** and select **`172.16.22.1/24`** as the **IP Address**.
4. On the **Authentication** tab, select the previously created primary gateway SSL/TLS Profile **`LSVPN-Scale`** from the **SSL/TLS Service Profile** drop-down.
5. On the **Satellite** tab, **Add** a satellite and **Name** it **`sat-config-1`**.
6. Set the **Configuration Refresh Interval** to **12**.
7. On **GlobalProtect Satellite** > **Devices**, add the serial number and hostname of each satellite device in the LSVPN.
8. On **GlobalProtect Satellite** > **Gateways**, add the name and IP address of each gateway. Set the routing priority of the primary gateway to 1 and the backup gateway to 10 to ensure that the active data center is the preferred gateway.

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STEP 8 | Verify the LSVPN Configuration.

STEP 9 | (Optional) Add a new site to the LSVPN deployment.

1. Select **Network** > **GlobalProtect** > **Portals** > **GlobalProtect Portal** > **Satellite Configuration** > **GlobalProtect Satellite** > **Devices** to add the serial number of the new satellite to the GlobalProtect portal.
2. Configure the IPSec tunnel on the satellite with the GlobalProtect portal IP address.
3. Select **Network** > **Virtual Router** > **BGP** > **Peer Group** to add the satellite to the BGP peer group configuration on each gateway.
4. Select **Network** > **Virtual Router** > **BGP** > **Peer Group** to add the gateways to the BGP peer group configuration on the new satellite.

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Virtual Systems

Where Can I Use This?	What Do I Need?
• NGFW	❑ Virtual Systems license for any virtual systems beyond the base number supported by each NGFW series. ❑ One of the following licenses when using Strata Cloud Manager: • Strata Cloud Manager Pro • Strata Cloud Manager Essentials *Virtual System support on Strata Cloud Manager is available on request. Contact your account team to enable the feature.*

Virtual systems are separate, logical firewall instances within a single physical Palo Alto Networks firewall. Rather than using multiple firewalls, managed service providers and enterprises can use a single pair of firewalls (for high availability) and enable virtual systems on them. Each virtual system (vsys) is an independent, separately-managed firewall with its traffic kept separate from the traffic of other virtual systems.

There are many ways to use virtual systems in a network. One common use case is for an ISP or a managed security service provider (MSSP) to deliver services to multiple customers with a single firewall. Customers can choose from a wide array of services that can be enabled or disabled easily. The firewall's role-based administration allows the ISP or MSSP to control each customer's access to functionality (such as logging and reporting) while hiding or offering read-only capabilities for other functions.

Another common use case is within a large enterprise that requires different firewall instances because of different technical or confidentiality requirements among multiple departments. Like the above case, different groups can have different levels of access while IT manages the firewall itself. Services can be tracked and/or billed back to departments to thereby make separate financial accountability possible within an organization.

Virtual systems provide the same basic functions as a physical firewall, along with additional benefits:

- **Segmented administration**—Different organizations (or customers or business units) can control (and monitor) a separate firewall instance, so that they have control over their own traffic without interfering with the traffic or policies of another firewall instance on the same physical firewall.
- **Scalability**—After the physical firewall is configured, adding or removing customers or business units can be done efficiently. An ISP, managed security service provider, or enterprise can provide different security services to each customer.

- **Reduced capital and operational expenses**—Virtual systems eliminate the need to have multiple physical firewalls at one location because virtual systems co-exist on one firewall. By not having to purchase multiple firewalls, an organization can save on the hardware expense, electric bills, and rack space, and can reduce maintenance and management expenses.
- **Ability to share IP-address-to-username mappings**—By assigning a virtual system as a User-ID hub, you can share the IP-address-to-username mappings across virtual systems to leverage the full User-ID capacity of the firewall and reduce operational complexity.

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Administrative Roles for Virtual Systems

A **Superuser** administrator can create virtual systems and add a **Device administrator**, **vsysadmin**, or **vsysreader**. A **Device administrator** can access all virtual systems, but cannot add administrators. When you create an Admin Role profile and select the role to be **Virtual System**, the role applies to specific virtual systems on the firewall. From the **Command Line** tab, the two types of virtual system administrative roles are:

- **vsysadmin**—Has access to specific virtual systems on the firewall to create and manage specific aspects of virtual systems. A vsysadmin doesn't have access to network interfaces, VLANs, virtual wires, virtual routers, IPSec tunnels, GRE tunnels, DHCP, DNS Proxy, QoS, LLDP, or network profiles. Persons with vsysadmin permission can commit configurations for only the virtual systems assigned to them.
- **vsysreader**—Has read-only access to specific virtual systems on the firewall and specific aspects of virtual systems. A vsysreader doesn't have access to network interfaces, VLANs, virtual wires, virtual routers, IPSec tunnels, GRE tunnels, DHCP, DNS Proxy, QoS, LLDP, or network profiles.

A virtual system administrator can view logs of only the virtual systems assigned to that administrator. A **Superuser** or **Device administrator** can view all of the logs, select a virtual system to view, or configure a virtual system as a User-ID hub.

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Virtual System Functionality with Other Features

Many firewall features and functionality are capable of being configured, viewed, logged, or reported per virtual system. Therefore, virtual systems are mentioned in other relevant locations in the documentation and that information is not repeated here. Some of the specific chapters are the following:

- If you are configuring Active/Passive HA, the two firewalls must have the same virtual system capability (single or multiple virtual system capability). See High Availability.
- To configure QoS for virtual systems, see Configure QoS for a Virtual System.
- For information about configuring a firewall with virtual systems in a virtual wire deployment that uses subinterfaces (and VLAN tags), see Virtual Wire Interfaces.
- If you have configured User-ID and multiple virtual systems, you can share user mappings across virtual systems. See Share User-ID Mappings Across Virtual Systems.

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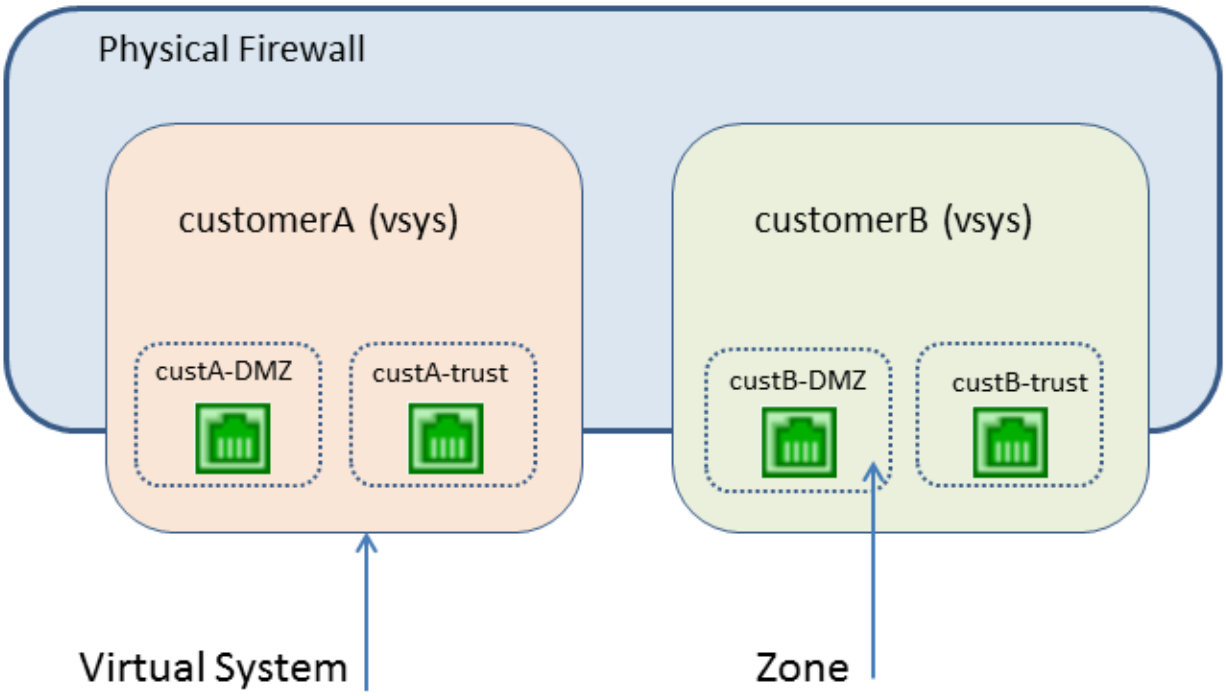
Virtual System Components and Segmentation

Where Can I Use This?	What Do I Need?
• NGFW	❑ Virtual Systems license for any virtual systems beyond the base number supported by each NGFW series. ❑ One of the following licenses when using Strata Cloud Manager: • Strata Cloud Manager Pro • Strata Cloud Manager Essentials *Virtual System support on Strata Cloud Manager is available on request. Contact your account team to enable the feature.*

A virtual system is an object that creates an administrative boundary, as shown in the following figure.



A virtual system consists of a set of physical and logical interfaces and subinterfaces (including VLANs and virtual wires), virtual routers, and security zones. You choose the deployment mode(s) (any combination of virtual wire, Layer 2, or Layer 3) of each virtual system. By using virtual systems, you can segment any of the following:

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- Administrative access
- The management of all policies (Security, NAT, QoS, Policy-based Forwarding, Decryption, Application Override, Tunnel Inspection, Authentication, and DoS protection)
- All objects (such as address objects, application groups and filters, external dynamic lists, security profiles, decryption profiles, custom objects, etc.)
- User-ID
- Certificate management
- Server profiles
- Logging, reporting, and visibility functions

Virtual systems affect the security functions of the firewall, but virtual systems alone do not affect networking functions such as static and dynamic routing. You can segment routing for each virtual system by creating one or more virtual routers for each virtual system, as in the following use cases:

- If you have virtual systems for departments of one organization, and the network traffic for all of the departments is within a common network, you can create a single virtual router for multiple virtual systems.
- If you want routing segmentation and each virtual system's traffic must be isolated from other virtual systems, you can create one or more virtual routers for each virtual system.
- If you want to segment the user mappings so that not all mappings are shared across virtual systems, you can configure the User-ID sources on a virtual system that is not a User-ID hub. See Share User-ID Mappings Across Virtual Systems.

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Shared Objects for Virtual Systems

Where Can I Use This?	What Do I Need?
• NGFW	❑ Virtual Systems license for any virtual systems beyond the base number supported by each NGFW series. ❑ One of the following licenses when using Strata Cloud Manager: • Strata Cloud Manager Pro • Strata Cloud Manager Essentials *Virtual System support on Strata Cloud Manager is available on request. Contact your account team to enable the feature.*

If your administrator account extends to multiple virtual systems, you can choose to configure objects (such as an address object) and policy rules for a specific virtual system or as shared objects, which apply to all of the virtual systems on the firewall. If you try to create a shared object with the same name and type as an existing object in a virtual system, the virtual system object is used.

Some `Shared` objects pushed from the Panorama management server, such as External Dynamic Lists (EDL), are counted toward the total maximum capacity for each object supported by the firewall model. Others, like Address objects, are not counted towards the total maximum capacity of the firewall model and are specific to the vsys. For example, you configure 51 vsys and have a firewall model that supports up to 50,000 IP addresses. You create a `Shared` EDL consisting of 1,000 IP addresses and you push the EDL to all vsys. In this example, 1,000 IP addresses are pushed to each of the first 50 vsys of your multi-vsys firewall and total 50,000 IP addresses. No IP addresses are pushed to the 51st vsys because the total maximum IP addresses supported by firewall model is reached. If configured locally, this same EDL counts for only 1,000 IP addresses.

The following `Shared` configuration objects are multiplied by the number of vsys and count toward the total maximum capacity of your firewall model.

- External Dynamic Lists
- Security Profile Groups
- All Security Profiles
- HIP objects and Profiles
- Custom Objects (custom data patterns, Spyware, Vulnerability Protection, and URL Category)
- Decryption Profile
- SD-WAN Link Management Profiles

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Platform Support and Licensing for Virtual Systems

Where Can I Use This?	What Do I Need?
• NGFW	❑ Virtual Systems license for any virtual systems beyond the base number supported by each NGFW series. ❑ One of the following licenses when using Strata Cloud Manager: • Strata Cloud Manager Pro • Strata Cloud Manager Essentials *Virtual System support on Strata Cloud Manager is available on request. Contact your account team to enable the feature.*

Virtual systems are supported on the following platforms. Each firewall series supports a base number of virtual systems; the number varies by platform. Additionally, you can manage multi-vsys enabled firewalls with Panorama or Strata Cloud Manager.

- PA-400 Series firewalls (PA-440, PA-445, PA-450, PA-455, and PA-460 firewalls only)
- PA-500 Series firewalls
- PA-1400 Series firewalls
- PA-3200 Series firewalls
- PA-3400 Series firewalls
- PA-5200 Series firewalls
- PA-5400 Series firewalls
- PA-5500 Series firewalls
- PA-7000 Series firewalls
- (PAN-OS 11.1.3 and later releases) VM-Series firewalls
- PA-7500 firewall
- (PAN-OS 11.1.7 and later releases) PA-7500 Series firewalls in an NGFW cluster

If you need to create more than the base number of virtual systems that your firewall platform supports, you purchase a Virtual Systems license.

For license information, see Subscriptions. For the base and maximum number of virtual systems supported, see Compare Firewalls tool.

Multiple virtual systems are not supported on the PA-220 and PA-800 Series firewalls.

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*The default is **vsys1**. You cannot delete vsys1 because it is relevant to the internal hierarchy on the firewall; vsys1 appears even on firewall models that don't support multiple virtual systems.*

You can limit the resources allocationsfor sessions, rules and VPN tunnels allowed for a virtual system, and thereby control firewall resources. Each resource setting displays the valid range of values, which varies per firewall model. The default setting is 0, which means the limit for the virtual system is the limit for the firewall model. However, the limit for a specific setting isn't replicated for each virtual system. For example, if a firewall has four virtual systems, each virtual system can't have the total number of Decryption Rules allowed per firewall. After the total number of Decryption Rules for all of the virtual systems reaches the firewall limit, you cannot add more.

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Communication Between Virtual Systems

Where Can I Use This?	What Do I Need?
• NGFW	❑ Virtual Systems license for any virtual systems beyond the base number supported by each NGFW series.

There are two typical scenarios where communication between virtual systems (inter-vsys traffic) is desirable. In a multi-tenancy environment, communication between virtual systems can occur by having traffic leave the firewall, go through the Internet, and re-enter the firewall. In a single organization environment, communication between virtual systems can remain within the firewall. This section discusses both scenarios.

Strata Cloud Manager does not support configuring inter-vsys communication.

Inter-VSYS Traffic That Must Leave the Firewall

An ISP that has multiple customers on a firewall (known as multi-tenancy) can use a virtual system for each customer, and thereby give each customer control over its virtual system configuration. The ISP grants **vsysadmin** permission to customers. Each customer's traffic and management are isolated from the others. Each virtual system must be configured with its own IP address and one or more virtual routers in order to manage traffic and its own connection to the Internet.

If the virtual systems need to communicate with each other, that traffic goes out the firewall to another Layer 3 routing device and back to the firewall, even though the virtual systems exist on the same physical firewall, as shown in the following figure.

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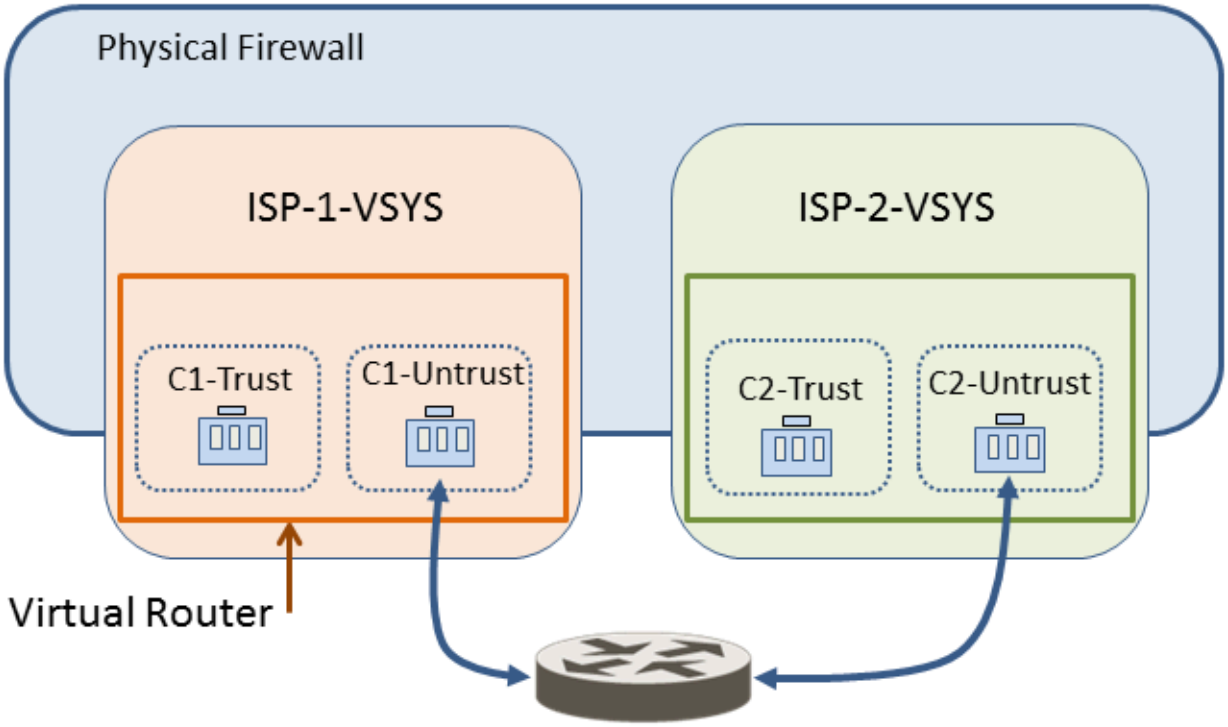

Inter-VSYS Traffic That Remains Within the Firewall

Unlike the preceding multi-tenancy scenario, virtual systems on a firewall can be under the control of a single organization. The organization wants to both isolate traffic between virtual systems and allow communications between virtual systems. This common use case arises when the organization wants to provide departmental separation and still have the departments be able to communicate with each other or connect to the same network(s). In this scenario, the inter-vsys traffic remains within the firewall, as described below:

External Zone

The communication desired in the use case above is achieved by configuring security policies that point to or from an *external* zone. An external zone is a security object that is associated with a specific virtual system that it can reach; the zone is external to the virtual system. A virtual system can have only one external zone, regardless of how many security zones the virtual system has within it. External zones are required to allow traffic between zones in different virtual systems, without the traffic leaving the firewall.

The virtual system administrator configures the security policies needed to allow traffic between two virtual systems. Unlike security zones, an external zone is not associated with an interface; it is associated with a virtual system. The security policy allows or denies traffic between the security (internal) zone and the external zone.

Because external zones do not have interfaces or IP addresses associated with them, some zone protection profiles are not supported on external zones.

Remember that each virtual system is a separate instance of a firewall, which means that each packet moving between virtual systems is inspected for security policy and App-ID evaluation.

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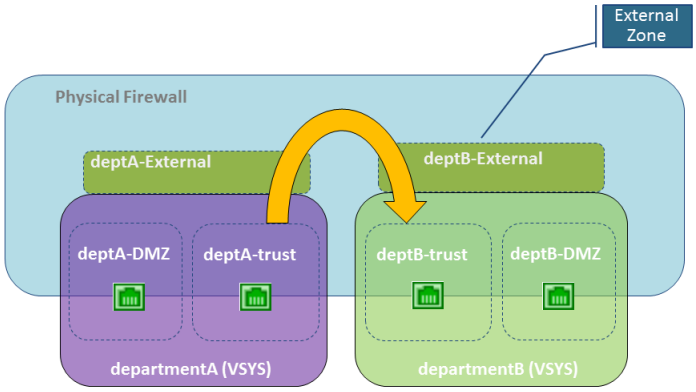
External Zones and Security Policies For Traffic Within a Firewall

- In the following example, an enterprise has two separate administrative groups: the departmentA and departmentB virtual systems. The following figure shows the external zone associated with each virtual system, and traffic flowing from one trust zone, out an external zone, into an external zone of another virtual system, and into its trust zone.



To create external zones, the firewall administrator must configure the virtual systems so that they are *visible* to each other. External zones do not have security policies between them because their virtual systems are visible to each other.

To communicate between virtual systems, the ingress and egress interfaces on the firewall are either assigned to a single virtual router or else they are connected using inter-virtual router static routes. The simpler of these two approaches is to assign all virtual systems that must communicate with each other to a single virtual router.

There might be a reason that the virtual systems need to have their own virtual router, for example, if the virtual systems use overlapping IP address ranges. Traffic can be routed between the virtual systems, but each virtual router must have static routes that point to the other virtual router(s) as the next hop.

Referring to the scenario in the figure above, we have an enterprise with two administrative groups: departmentA and departmentB. The departmentA group manages the local network and the DMZ resources. The departmentB group manages traffic in and out of the sales segment of the network. All traffic is on a local network, so a single virtual router is used. There are two external zones configured for communication between the two virtual systems. The departmentA virtual system has three zones used in security policies: deptA-DMZ, deptA-trust, and deptA-External. The departmentB virtual system also has three zones: deptB-DMZ, deptB-

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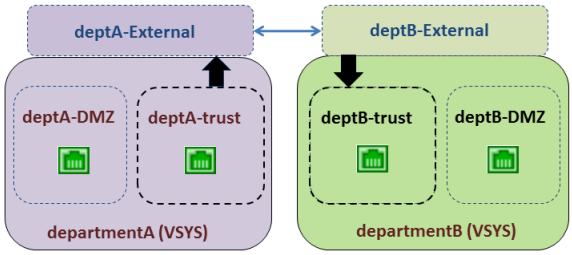
trust, and deptB-External. Both groups can control the traffic passing through their virtual systems.

In order to allow traffic from deptA-trust to deptB-trust, two security policies are required. In the following figure, the two vertical arrows indicate where the security policies (described below the figure) are controlling traffic.



- Security Policy 1: In the preceding figure, traffic is destined for the deptB-trust zone. Traffic leaves the deptA-trust zone and goes to the deptA-External zone. A security policy must allow traffic from the source zone (deptA-trust) to the destination zone (deptA-External). A virtual system allows any policy type to be used for this traffic, including NAT.

  No policy is needed between external zones because traffic sent to an external zone appears in and has automatic access to the other external zones that are visible to the original external zone.
- Security Policy 2: In the preceding figure, the traffic from deptB-External is still destined to the deptB-trust zone, and a security policy must be configured to allow it. The policy must allow traffic from the source zone (deptB-External) to the destination zone (deptB-trust).

The departmentB virtual system could be configured to block traffic from the departmentA virtual system, and vice versa. Like traffic from any other zone, traffic from external zones must be explicitly allowed by policy to reach other zones in a virtual system.

In addition to external zones being required for inter-virtual system traffic that does not leave the firewall, external zones are also required if you configure a Shared Gateway*, in which case the traffic is intended to leave the firewall.*

Inter-VSYS Communication Uses Two Sessions

It is helpful to understand that communication between two virtual systems uses two sessions, unlike the one session used for a single virtual system. Let's compare the scenarios.

Scenario 1—Vsys1 has two zones: trust1 and untrust1. A host in the trust1 zone initiates traffic when it needs to communicate with a device in the untrust1 zone. The host sends traffic to the firewall, and the firewall creates a new session for source zone trust1 to destination zone untrust1. Only one session is needed for this traffic.

Scenario 2—A host from vsys1 needs to access a server on vsys2. A host in the trust1 zone initiates traffic to the firewall, and the firewall creates the first session: source zone trust1 to destination zone untrust1. Traffic is routed to vsys2, either internally or externally. Then the firewall creates a second session: source zone untrust2 to destination zone trust2. Two sessions are needed for this inter-vsys traffic.

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Shared Gateway

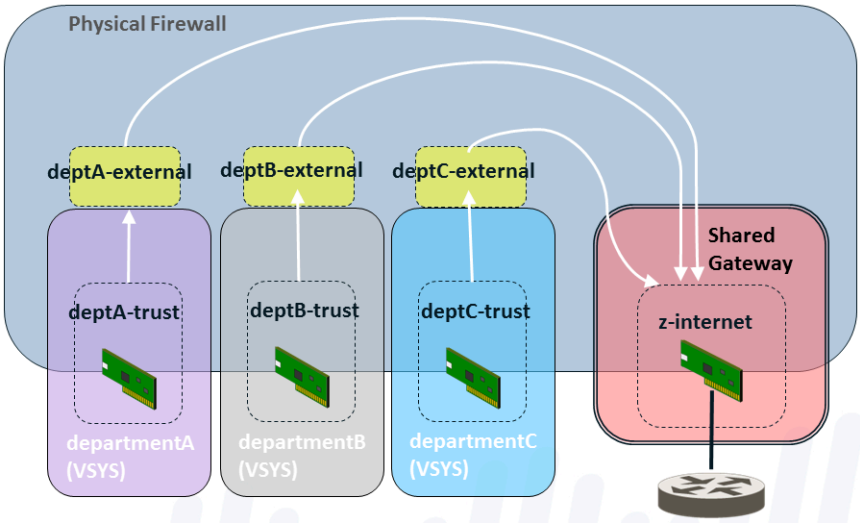
Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Virtual Systems license for any virtual systems beyond the base number supported by each NGFW series.

A shared gateway is an interface that multiple virtual systems share in order to communicate over the Internet. Each virtual system requires an External Zone, which acts as an intermediary, for configuring security policies that allow or deny traffic from the virtual system's internal zone to the shared gateway.

The shared gateway uses a single virtual router to route traffic for all virtual systems. A shared gateway is used in cases when an interface does not need a full administrative boundary around it, or when multiple virtual systems must share a single Internet connection. This second case arises if an ISP provides an organization with only one IP address (interface), but multiple virtual systems need external communication.

Unlike the behavior between virtual systems, security policy and App-ID evaluations are not performed between a virtual system and a shared gateway. That is why using a shared gateway to access the Internet involves less overhead than creating another virtual system to do so.

In the following figure, three customers share a firewall, but there is only one interface accessible to the Internet. Creating another virtual system would add the overhead of App-ID and security policy evaluation for traffic being sent to the interface through the added virtual system. To avoid adding another virtual system, the solution is to configure a shared gateway, as shown in the following diagram.



The shared gateway has one globally-routable IP address used to communicate with the outside world. Interfaces in the virtual systems have IP addresses too, but they can be private, non-routable IP addresses.

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You will recall that an administrator must specify whether a virtual system is visible to other virtual systems. Unlike a virtual system, a shared gateway is always visible to all of the virtual systems on the firewall.

A shared gateway ID number appears as **sg<ID>** on the web interface. It is recommended that you name your shared gateway with a name that includes its ID number.

When you add objects such as zones or interfaces to a shared gateway, the shared gateway appears as an available virtual system in the vsys menu.

A shared gateway is a limited version of a virtual system; it supports NAT and policy-based forwarding (PBF), but does not support Security, DoS policies, QoS, Decryption, Application Override, or Authentication policies.

Networking Considerations for a Shared Gateway

Keep the following in mind while you are configuring a shared gateway.

- The virtual systems in a shared gateway scenario access the Internet through the shared gateway's physical interface, using a single IP address. If the IP addresses of the virtual systems are not globally routable, configure source NAT to translate those addresses to globally-routable IP addresses.
- A virtual router routes the traffic for all of the virtual systems through the shared gateway.
- The default route for the virtual systems should point to the shared gateway.
- Security policies must be configured for each virtual system to allow the traffic between the internal zone and external zone, which is visible to the shared gateway.
- A firewall administrator should control the virtual router, so that no member of a virtual system can affect the traffic of other virtual systems.
- Within a Palo Alto Networks firewall, a packet may hop from one virtual system to another virtual system or a shared gateway. A packet may not traverse more than two virtual systems or shared gateways. For example, a packet cannot go from vsys1 to vsys2 to vsys3, or similarly from vsys1 to vsys2 to shared gateway1. Both examples involve more than two virtual systems, which is not permitted.

To save configuration time and effort, consider the following advantages of a shared gateway:

- Rather than configure NAT for multiple virtual systems associated with a shared gateway, you can configure NAT for the shared gateway.
- Rather than configure policy-based routing (PBR) for multiple virtual systems associated with a shared gateway, you can configure PBR for the shared gateway.

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Configure Virtual Systems

Where Can I Use This?	What Do I Need?
• NGFW	❑ Virtual Systems license for any virtual systems beyond the base number supported by each NGFW series. ❑ One of the following licenses when using Strata Cloud Manager: • Strata Cloud Manager Pro • Strata Cloud Manager Essentials *Virtual System support on Strata Cloud Manager is available on request. Contact your account team to enable the feature.*

Virtual System support on Strata Cloud Manager is available on request. Contact your account team to enable the feature.

Creating a virtual system requires that you have the following:

- A **superuser** administrative role.
- An interface configured.
- A Virtual Systems license if you are creating more than the base number of virtual systems supported on the platform. See Platform Support and Licensing for Virtual Systems.
- In Strata Cloud Manager, while in the configuration scope of vsys-enabled firewall, the **Objects** and **Operations** menus are hidden.
- PAN-OS and Panorama
- Strata Cloud Manager

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PAN-OS and Panorama

(Panorama managed firewalls) For firewalls managed by a Panorama management server, Palo Alto Networks recommends making note of all policy rule Target lists you added the managed firewall to on Panorama before you change the virtual system configuration status to ensure you maintain your security posture.

Changing the managed firewall multi-vsys status impacts all policy rules where the managed firewall was added to the policy Target list. Changing the multi-vsys status in any way removes the firewall from the Target list from the Panorama-managed policy rule, impacting which firewalls Panorama pushes the policy rule to. If the removed firewall was the only Target, then the rule is now pushed to all firewalls associated with the impacted device group.

- *In the case of `deny` policy rules, this may result in some firewalls denying sessions they previously allowed.*
- *In the case of `allow` policy rules, this may result in some firewalls allowing sessions they previously denied.*

STEP 1 | Enable virtual systems.

1. Select **Device** > **Setup** > **Management** and edit the **General Settings**.
2. Select the **Multi Virtual System Capability** check box and click **OK**. This action triggers a commit if you approve it.

   Only after enabling virtual systems will the **Device** tab display the **Virtual Systems** and **Shared Gateways** options.

STEP 2 | Create a virtual system.

1. Select **Device** > **Virtual Systems**, click **Add** and enter a virtual system **ID**, which is appended to “vsys” (range is 1-255).

   *The default is **vsys1**. You cannot delete vsys1 because it is relevant to the internal hierarchy on the firewall; vsys1 appears even on firewall models that don’t support multiple virtual systems.*
2. Select **Allow forwarding of decrypted content** if you want to allow the firewall to forward decrypted content to an outside service. For example, you must enable this option for the firewall to be able to send decrypted content to WildFire for analysis.
3. Enter a descriptive **Name** for the virtual system. A maximum of 31 alphanumeric, space, and underscore characters is allowed.

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STEP 3 | Assign interfaces to the virtual system.

The virtual routers, virtual wires, or VLANs can either be configured already or you can configure them later, at which point you specify the virtual system associated with each.

1. On the **General** tab, select a **DNS Proxy** object if you want to apply DNS proxy rules to the interface.
2. In the **Interfaces** field, click **Add** to enter the interfaces or subinterfaces to assign to the virtual system. An interface can belong to only one virtual system.
3. Do any of the following, based on the deployment type(s) you need in the virtual system:
   - **Add** the **VLANs** to assign to the vsys.
   - **Add** the **Virtual Wires** to assign to the vsys.
   - **Add** the **Virtual Routers** to assign to the vsys.
   - If the firewall has **Advanced Routing** enabled, **Add** the **Logical Routers** to assign to the vsys.
4. In the **Visible Virtual System** field, check all virtual systems that should be made visible to the virtual system being configured. This is required for virtual systems that need to communicate with each other.

   In a multi-tenancy scenario where strict administrative boundaries are required, no virtual systems would be checked.
5. Click **OK**.

STEP 4 | (Required for Panorama managed firewalls) Log in to the Panorama web interface and select **Commit** > **Push to Devices** and push the entire Panorama managed configuration to each vsys of the multi-vsys firewall.

This is required to leverage shared configuration objects for multi-vsys firewalls managed by Panorama.

STEP 5 | (Optional) Limit the resource allocations for sessions, rules, and VPN tunnels allowed for the virtual system. The flexibility of being able to allocate limits per virtual system allows you to effectively control firewall resources.

1. On the **Resource** tab, optionally set limits for a virtual system. Each field displays the valid range of values, which varies per firewall model. The default setting is 0, which means the limit for the virtual system is the limit for the firewall model. However, the limit for a specific setting isn't replicated for each virtual system. For example, if a firewall has four virtual systems, each virtual system can't have the total number of

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Decryption Rules allowed per firewall. After the total number of Decryption Rules for all of the virtual systems reaches the firewall limit, you cannot add more.

- **Sessions Limit**

  If you use the show session meter CLI command, it displays the Maximum number of sessions allowed per dataplane, the Current number of sessions being used by the virtual system, and the Throttled number of sessions per virtual system. On a PA-5200 or PA-7000 Series firewall, the Current number of sessions being used can be greater than the Maximum configured for Sessions Limit because there are multiple dataplanes per virtual system. The Sessions Limit you configure on a PA-5200 Series or PA-7000 Series firewall is per dataplane, and will result in a higher maximum per virtual system.

- **Security Rules**
- **NAT Rules**
- **Decryption Rules**
- **QoS Rules**
- **Application Override Rules**
- **Policy Based Forwarding Rules**
- **Authentication Rules**
- **DoS Protection Rules**
- **Site to Site VPN Tunnels**
- **Concurrent SSL VPN Tunnels**

2. Click **OK**.

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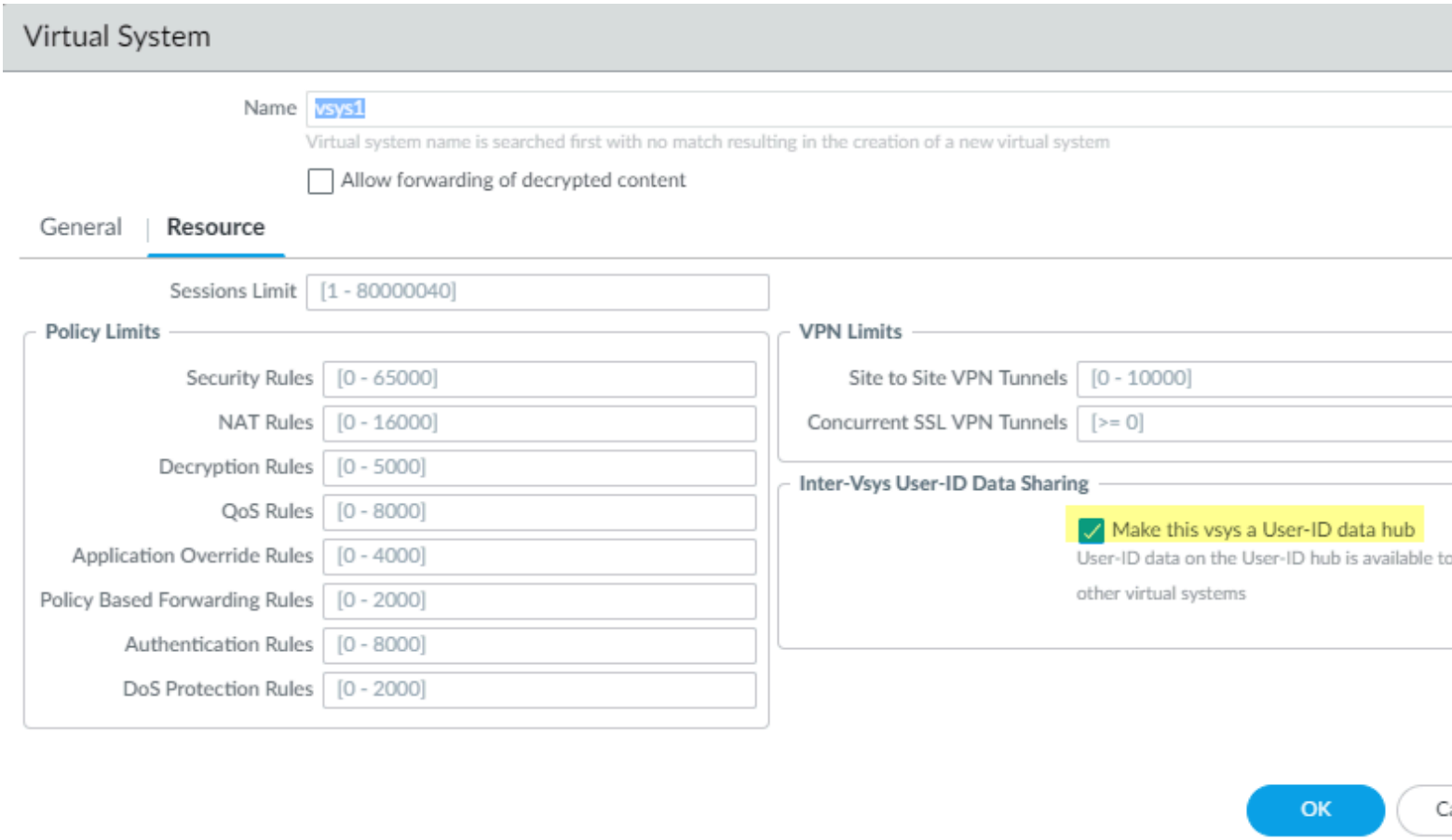
STEP 6 | (Optional) Configure a virtual system as a User-ID hub to Share User-ID Mappings Across Virtual Systems.

IP-address-and-port-to-username mapping information from Terminal Server agents and group mapping data is not shared between the virtual system hub and the connected virtual systems.

1. For any existing virtual systems, transfer the configuration for the User-ID sources you want to share (such as monitored servers and User-ID agents) to the virtual system you will use as a hub.
2. On the **Resource** tab, select **Make this vsys a User-ID data hub**.



3. Click **Yes** to confirm, then click **OK**.

   If you want to change the User-ID hub to a different virtual system or disable it, select the virtual system currently configured as a User-ID hub, then select **Resource** > **Change Hub**.

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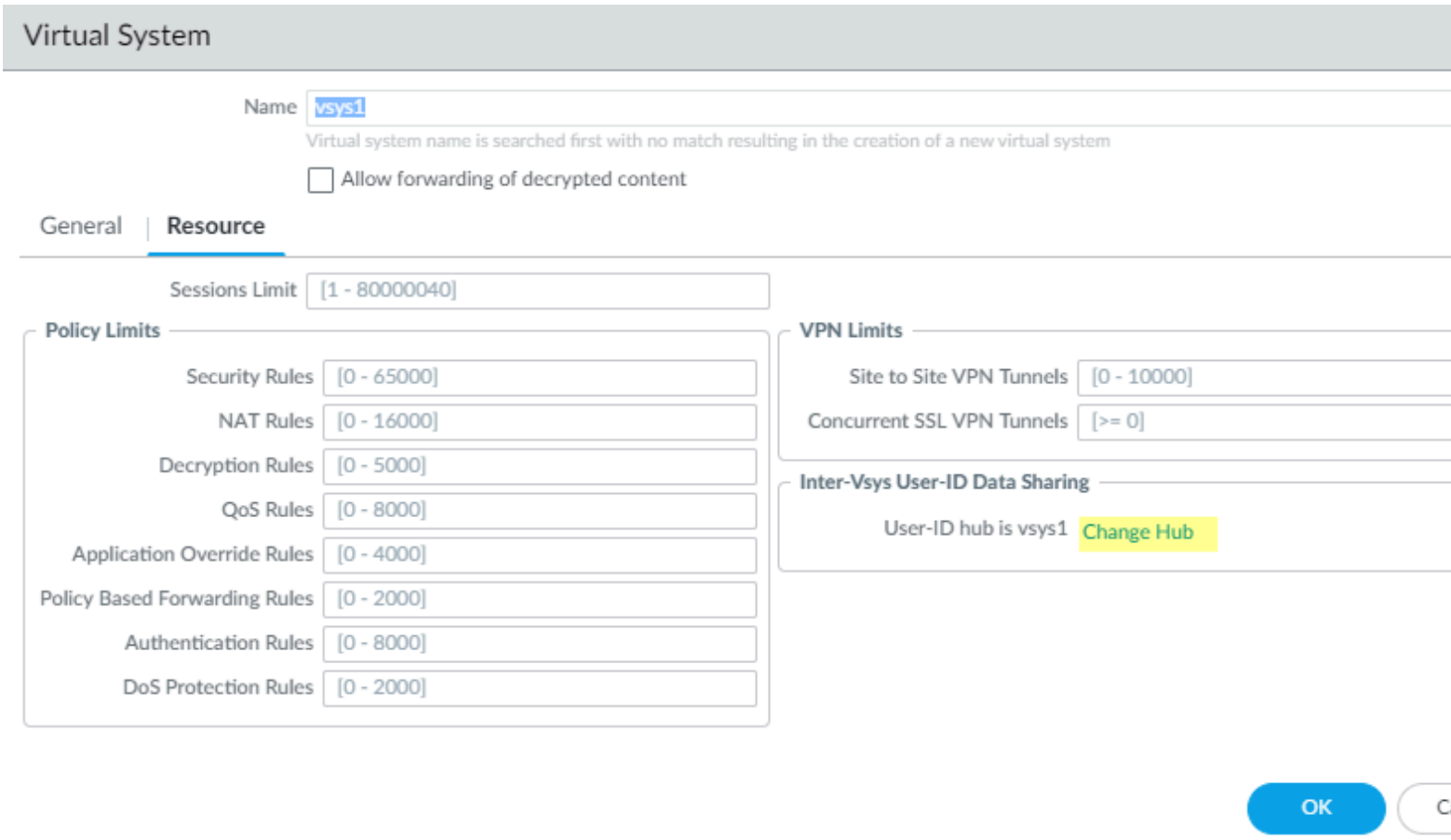
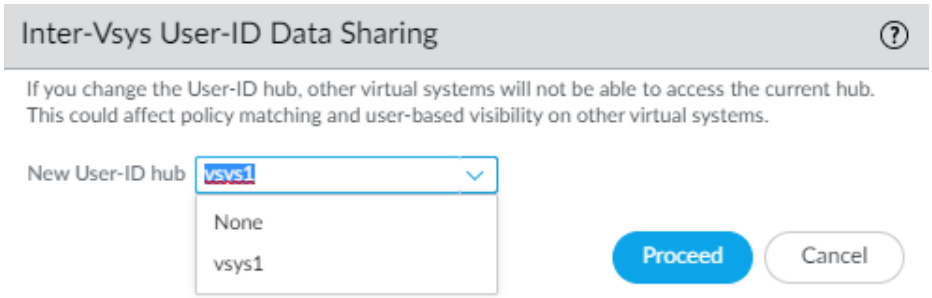


Select the **New User-ID hub** from the list, or select **none** to disable the User-ID hub and stop sharing mappings across virtual systems.



Click **Proceed** to confirm and commit your changes.

STEP 7 | Commit the configuration.

Click **Commit**. The virtual system is now an object accessible from the **Objects** tab.

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STEP 8 | Create at least one virtual router for the virtual system in order to make the virtual system capable of networking functions, such as static and dynamic routing.

Alternatively, your virtual system might use a VLAN or a virtual wire, depending on your deployment.

1. Select **Network** > **Virtual Routers** and **Add** a virtual router by **Name**.
2. For **Interfaces**, click **Add** and select the interfaces that belong to the virtual router.
3. Click **OK**.

STEP 9 | Configure a security zone for each interface in the virtual system.

For at least one interface, create a Layer 3 security zone. See Configure Interfaces and Zones..

STEP 10 | Configure the security policy rules that allow or deny traffic to and from the zones in the virtual system.

See Create a Security Policy Rule.

STEP 11 | Commit the configuration.

Click **Commit**.

After creating a virtual system, you can use the CLI to commit a configuration for only a specific virtual system:

```
commit partial vsys <vsys-id>
```

STEP 12 | (Optional) View the security policies configured for a virtual system.

Open an SSH session to use the CLI. To view the security policies for a virtual system, in operational mode, use the following commands:

```
set system setting target-vsys <vsys-id>
show running security-policy
```

Strata Cloud Manager

STEP 1 | Log in to Strata Cloud Manager.

STEP 2 | From the Config Scope drop-down, select a firewall that supports Multi-Vsys.

STEP 3 | Enable virtual systems.

1. Log in to Strata Cloud Manager.
2. Select **Configuration** > **NGFW and Prisma Access** > **Device** > **Virtual Systems** and click the **Enable Multi Virtual System**.
3. Click **Enable** to confirm.

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STEP 4 | Create a virtual system.

1. Select **Configuration** > **NGFW and Prisma Access** > **Device** > **Virtual Systems**and click **Add Virtual System**

   *The default is **vsys1**. You cannot delete vsys1 because it is relevant to the internal hierarchy on the firewall.*

2. Enter a descriptive **Name** for the virtual system. A maximum of 31 alphanumeric, space, and underscore characters is allowed.
3. Select **Allow forwarding of decrypted content** if you want to allow the firewall to forward decrypted content to an outside service. For example, you must enable this option for the firewall to be able to send decrypted content to WildFire for analysis.
4. Select a **DNS Proxy** object if you want to apply DNS proxy rules to the interface.

STEP 5 | Configure the **Interface and Router** tab.

The virtual routers, virtual wires, or VLANs can either be configured already or you can configure them later, at which point you specify the virtual system associated with each.

1. In the **Interfaces** field, click the plus (+) icon to enter the interfaces or subinterfaces to assign to the virtual system. An interface can belong to only one virtual system.
2. Do any of the following, based on the deployment type(s) you need in the virtual system:
3. **Add** the **Virtual Wires** to assign to the vsys.
4. **Add** the **Logical Routers** to assign to the vsys.

STEP 6 | (Optional) Limit the resource allocations for sessions, rules, and VPN tunnels allowed for the virtual system. The flexibility of being able to allocate limits per virtual system allows you to effectively control firewall resources.

1. On the **Resource** tab, optionally set limits for a virtual system. Each field displays the valid range of values, which varies per firewall model. The default setting is 0, which means the limit for the virtual system is the limit for the firewall model. However, the limit for a specific setting isn't replicated for each virtual system. For example, if a firewall has four virtual systems, each virtual system can't have the total number of

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Decryption Rules allowed per firewall. After the total number of Decryption Rules for all of the virtual systems reaches the firewall limit, you cannot add more.

- **Sessions Limit**

  If you use the show session meter CLI command, it displays the Maximum number of sessions allowed per dataplane, the Current number of sessions being used by the virtual system, and the Throttled number of sessions per virtual system. On a PA-5200 or PA-7000 Series firewall, the Current number of sessions being used can be greater than the Maximum configured for Sessions Limit because there are multiple dataplanes per virtual system. The Sessions Limit you configure on a PA-5200 Series or PA-7000 Series firewall is per dataplane, and will result in a higher maximum per virtual system.

- **Security Rules**
- **NAT Rules**
- **Decryption Rules**
- **QoS Rules**
- **Application Override Rules**
- **Policy Based Forwarding Rules**
- **Authentication Rules**
- **DoS Protection Rules**
- **Site to Site VPN Tunnels**
- **Concurrent SSL VPN Tunnels**

2. Click **OK**.

STEP 7 | Commit the configuration.

Click **Commit**. The virtual system is now an object accessible from the **Objects** tab.

STEP 8 | Push the configuration to the firewall.

1. Select **Configuration** > **NGFW and Prisma Access** and click **Push**.
2. In the push dialog, select **All Admins** from the admin scope dropdown and select at least one virtual system you want to push.

   *When pushing configuration to a firewall with multiple virtual systems, you must select **All Admins**. Partial push is NOT supported for firewalls with multiple virtual systems. Failure to select All Admins results in configuration push failures or potential outages.*

   The device is greyed out in the push scope because pushing to a virtual system automatically pushes device-level configuration to the firewall. You can push a subset of virtual systems. Once a virtual system is pushed to the firewall, it persists even if you do not select it in subsequent pushes. To remove a pushed virtual system, delete it in Strata Cloud Manager first, then push to the firewall.
3. Click **Push** to apply the configuration to the selected virtual system.

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STEP 9 | Create at least one virtual router for the virtual system in order to make the virtual system capable of networking functions, such as static and dynamic routing.

Alternatively, your virtual system might use a virtual wire, depending on your deployment.

1. Select **Network** > **Virtual Routers** and **Add** a virtual router by **Name**.
2. For **Interfaces**, click **Add** and select the interfaces that belong to the virtual router.
3. Click **OK**.

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Configure Inter-Virtual System Communication within the Firewall

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	☐ Virtual Systems license for any virtual systems beyond the base number supported by each NGFW series.

Perform this task if you have a use case, perhaps within a single enterprise, where you want the virtual systems to be able to communicate with each other within the firewall. Such a scenario is described in Inter-VSYS Traffic That Remains Within the Firewall. This task presumes:

- You completed the task, Configure Virtual Systems.
- When configuring the virtual systems, in the **Visible Virtual System** field, you checked the boxes of all virtual systems that must communicate with each other to be visible to each other.

Strata Cloud Manager does not support configuring inter-vsys communication.

STEP 1 | Configure an external zone for each virtual system.

1. Select **Network** > **Zones** and **Add** a new zone by **Name**.
2. For **Location**, select the virtual system for which you are creating an external zone.
3. For **Type**, select **External**.
4. For **Virtual Systems**, click **Add** and enter the virtual system that the external zone can reach.
5. (Optional) Select a **Zone Protection Profile** (or configure one later) that provides flood, reconnaissance, or packet-based attack protection.
6. (Optional) In **Log Setting**, select a log forwarding profile for forwarding zone protection logs to an external system.
7. (Optional) Select **Enable User Identification** to enable User-ID for the external zone.
8. Click **OK**.

STEP 2 | Configure the Security policy rules to allow or deny traffic from the internal zones to the external zone of the virtual system, and vice versa.

- See Create a Security Policy Rule.
- See Inter-VSYS Traffic That Remains Within the Firewall.

STEP 3 | Commit your changes.

Click **Commit**.

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Configure a Shared Gateway

Where Can I Use This?	What Do I Need?
• NGFW (Managed by PAN-OS or Panorama)	❑ Virtual Systems license for any virtual systems beyond the base number supported by each NGFW series.

Perform this task if you need multiple virtual systems to share an interface (a Shared Gateway) to the Internet. This task presumes:

- You configured an interface with a globally-routable IP address, which will be the shared gateway.
- You completed the prior task, Configure Virtual Systems. For the interface, you chose the external-facing interface with the globally-routable IP address.
- When configuring the virtual systems, in the **Visible Virtual System** field, you checked the boxes of all virtual systems that must communicate to be visible to each other.

Keep the following in mind while you are configuring a shared gateway.

- The virtual systems in a shared gateway scenario access the Internet through the shared gateway's physical interface, using a single IP address. If the IP addresses of the virtual systems are not globally routable, configure source NAT to translate those addresses to globally-routable IP addresses.
- A virtual router routes the traffic for all of the virtual systems through the shared gateway.
- The default route for the virtual systems should point to the shared gateway.
- Security policies must be configured for each virtual system to allow the traffic between the internal zone and external zone, which is visible to the shared gateway.
- A firewall administrator should control the virtual router, so that no member of a virtual system can affect the traffic of other virtual systems.
- Within a Palo Alto Networks firewall, a packet may hop from one virtual system to another virtual system or a shared gateway. A packet may not traverse more than two virtual systems or shared gateways. For example, a packet cannot go from vsys1 to vsys2 to vsys3, or similarly from vsys1 to vsys2 to shared gateway1. Both examples involve more than two virtual systems, which is not permitted.

To save configuration time and effort, consider the following advantages of a shared gateway:

- Rather than configure NAT for multiple virtual systems associated with a shared gateway, you can configure NAT for the shared gateway.
- Rather than configure policy-based routing (PBR) for multiple virtual systems associated with a shared gateway, you can configure PBR for the shared gateway.

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STEP 1 | Configure a Shared Gateway.

1. Select **Device** > **Shared Gateway**, click **Add** and enter an **ID**.
2. Enter a helpful **Name**, preferably including the **ID** of the gateway.
3. In the **DNS Proxy** field, select a DNS proxy object if you want to apply DNS proxy rules to the interface.
4. **Add** an **Interface** that connects to the outside world.
5. Click **OK**.

STEP 2 | Configure the zone for the shared gateway.

*When adding objects such as zones or interfaces to a shared gateway, the shared gateway itself will be listed as an available vsys in the **VSYS** menu.*

1. Select **Network** > **Zones** and **Add** a new zone by **Name**.
2. For **Location**, select the shared gateway for which you are creating a zone.
3. For **Type**, select **Layer3**.
4. (Optional) Select a **Zone Protection Profile** (or configure one later) that provides flood, reconnaissance, or packet-based attack protection.
5. (Optional) In **Log Setting**, select a log forwarding profile for forwarding zone protection logs to an external system.
6. (Optional) Select **Enable User Identification** to enable User-ID for the shared gateway.
7. Click **OK**.

STEP 3 | Commit your changes.

Click **Commit**.

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Customize Service Routes for a Virtual System

Where Can I Use This?	What Do I Need?
• NGFW	❑ Virtual Systems license for any virtual systems beyond the base number supported by each NGFW series. ❑ One of the following licenses when using Strata Cloud Manager: • Strata Cloud Manager Pro • Strata Cloud Manager Essentials *Virtual System support on Strata Cloud Manager is available on request. Contact your account team to enable the feature.*

When a firewall is enabled for multiple virtual systems, the virtual systems inherit the global service and service route settings. For example, the firewall can use a shared email server to originate email alerts to all virtual systems. In some scenarios, you'd want to create different service routes for each virtual system.

One use case for configuring service routes at the virtual system level is if you are an ISP who needs to support multiple individual tenants on a single Palo Alto Networks firewall. Each tenant requires custom service routes to access service such as DNS, Kerberos, LDAP, NetFlow, RADIUS, TACACS+, Multi-Factor Authentication, email, SNMP trap, syslog, HTTP, User-ID Agent, VM Monitor, and Panorama (deployment of content and software updates). Another use case is an IT organization that wants to provide full autonomy to groups that set servers for services. Each group can have a virtual system and define its own service routes.

You can select a virtual router for a service route in a virtual system; you cannot select the egress interface. After you select the virtual router and the firewall sends the packet from the virtual router, the firewall selects the egress interface based on the destination IP address. Therefore, if a virtual system has multiple virtual routers, packets to all of the servers for a service must egress out of only one virtual router. A packet with an interface source address may egress a different interface, but the return traffic would be on the interface that has the source IP address, creating asymmetric traffic.

When you enable Multi Virtual System Capability, any virtual system that does not have specific service routes configured inherits the global service and service route settings for the firewall. You can instead configure a virtual system to use a different service route, as described in the following workflow.

A firewall with multiple virtual systems must have interfaces and subinterfaces with non-overlapping IP addresses. A per-virtual system service route for SNMP traps or for Kerberos is for IPv4 only.

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The service route for a service strictly follows how you configured the server profile for the service:

- If you define a server profile (**Device** > **Server Profiles**) for the Shared location, the firewall uses the global service route for that service.
- If you define a server profile for a specific virtual system, the firewall uses the virtual system-specific service route for that service.
- If you define a server profile for a specific virtual system but the virtual system-specific service route for that service is not configured, the firewall uses the global service route for that service.

The firewall supports syslog forwarding on a virtual system basis. When multiple virtual systems on a firewall are connecting to a syslog server using SSL transport, the firewall can generate only one certificate for secure communication. The firewall does not support each virtual system having its own certificate.

STEP 1 | Customize service routes for a virtual system.

1. Select **Device** > **Setup** > **Services** > **Virtual Systems**, and select the virtual system you want to configure.
2. Click the **Service Route Configuration** link.
3. Select one:
   - **Inherit Global Service Route Configuration**—Causes the virtual system to inherit the global service route settings relevant to a virtual system. If you choose this option, skip the step to customize.
   - **Customize**—Allows you to specify a source address for each service.
4. If you chose **Customize**, select the **IPv4** or **IPv6** tab, depending on what type of addressing the server offering the service uses. You can specify both IPv4 and IPv6 addresses for a service. Click on a service. (Only services that are relevant to a virtual system are available.)

   *To easily use the same source address for multiple services, select the checkbox for the services, click **Set Selected Routes**, and continue.*

   - To limit the list for Source Address, select a **Source Interface**, then select a Source Address (from that interface) as the service route. Selecting **Any** Source Interface makes all IP addresses on all interfaces for the virtual system available in the Source Address list from which you select an address. You can select **Inherit Global Setting**.
   - **Source Address** will indicate **Inherited** if you selected **Inherit Global Setting** for the **Source Interface** or it will indicate the source address you selected. If you selected **Any** for **Source Interface**, select an IP address or enter an IP address (using the IPv4

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or IPv6 format that matches the tab you chose) to specify the source address that will be used in packets sent to the external service.

- If you modify an address object and the IP family type (IPv4/IPv6) changes, a **Commit** is required to update the service route family to use.

5. Click **OK**.
6. Repeat the prior steps to configure source addresses for other external services.
7. Click **OK**.

STEP 2 | Commit your changes.

Click **Commit** and **OK**.

If you are configuring per-virtual system service routes for logging services for a PA-7000 Series firewall, continue to the task Configure a PA-7000 Series Firewall for Logging Per Virtual System.

Configure a PA-7000 Series Firewall for Logging Per Virtual System

Where Can I Use This?	What Do I Need?
• NGFW	☐ Virtual Systems license for any virtual systems beyond the base number supported by each NGFW series. ☐ One of the following licenses when using Strata Cloud Manager: • Strata Cloud Manager Pro • Strata Cloud Manager Essentials *Virtual System support on Strata Cloud Manager is available on request. Contact your account team to enable the feature.*

For Traffic, HIP Match, Threat, and WildFire log types, the PA-7000 Series firewall does not use service routes for SNMP Trap, Syslog, and email services. Instead, the PA-7000 Series firewall supports using a logging card.

Depending on your firewall configuration, you might have one of the following card types:

- **Log Processing Card (LPC)**—Supports virtual system-specific paths from LPC subinterfaces to an on-premise switch to the respective service on a server. For System and Config logs, the PA-7000 Series firewall uses global service routes, and not the LPC. If your firewall has an LPC installed, you need to configure a log card port.

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- **Log Forwarding Card (LFC)**—Supports high-speed log forwarding of all dataplane logs to an external log collector (for example, Panorama and syslog servers). If your firewall has an LFC installed, you do not need to configure a log card port.

  The only way to forward system logs from a PA-7000 Series firewall running PAN-OS 10.1 or later is by configuring an LFC.

  Log forwarding to an external server is not yet supported on LFC subinterfaces.

In other Palo Alto Networks models, the dataplane sends logging service route traffic to the management plane, which sends the traffic to logging servers. In a PA-7000 Series firewall, the LPC or LFC have only one interface, and dataplanes for multiple virtual systems send logging server traffic (types mentioned above) to the PA-7000 Series firewall logging card. The logging card is configured with multiple subinterfaces, over which the platform sends the logging service traffic out to a customer's switch, which can be connected to multiple logging servers.

Each subinterface can be configured with a subinterface name and a dotted subinterface number. The subinterface is assigned to a virtual system, which is configured for logging services. The other service routes on a PA-7000 Series firewall function similarly to service routes on other Palo Alto Networks platforms. For information about the LPC or LFC, see thePA-7000 Series Hardware Reference Guide .

- Configure a PA-7000 Series LPC for Logging per Virtual System
- Configure a PA-7000 Series LFC for Logging per Virtual System

Configure a PA-7000 Series LPC for Logging per Virtual System

If you have enabled multi-vsys capability on a PA-7000 Series firewall with a Log Processing Card (LPC) installed, you can configure logging for different virtual systems as described in the following workflow.

STEP 1 | Create a Log Card subinterface.

1. Select **Network** > **Interfaces** > **Ethernet** and select the interface to be the Log Card interface.
2. Enter the **Interface Name**.
3. For **Interface Type**, select **Log Card**.
4. Click **OK**.

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STEP 2 | Add a subinterface for each tenant on the LPCs physical interface.

1. Highlight the Ethernet interface that is a Log Card interface type and click **Add Subinterface**.
2. For **Interface Name**, after the period, enter the subinterface assigned to the tenant's virtual system.
3. For **Tag**, enter a VLAN tag value.

   Make the tag the same as the subinterface number for ease of use, but it could be a different number.
4. (Optional) Enter a **Comment**.
5. On the **Config** tab, in the **Assign Interface to Virtual System** field, select the virtual system to which the LPC subinterface is assigned. Alternatively, you can click **Virtual Systems** to add a new virtual system.
6. Click **OK**.

STEP 3 | Enter the addresses assigned to the subinterface, and configure the default gateway.

1. Select the **Log Card Forwarding** tab, and do one or both of the following:
   - For the IPv4 section, enter the **IP Address** and **Netmask** assigned to the subinterface. Enter the **Default Gateway** (the next hop where packets will be sent that have no known next hop address in the Routing Information Base [RIB]).
   - For the IPv6 section, enter the **IPv6 Address** assigned to the subinterface. Enter the **IPv6 Default Gateway**.
2. Click **OK**.

STEP 4 | Commit your changes.

Click **OK** and **Commit**.

STEP 5 | If you haven't already done so, configure the remaining service routes for the virtual system.

Customize Service Routes for a Virtual System.

Configure a PA-7000 Series LFC for Logging per Virtual System

If you have enabled multiple virtual system (multi-vsys) capability on a PA-7000 Series firewall with a Log Forwarding Card (LFC) installed, you can configure logging for different virtual systems. The LFC can then forward logs to a Panorama Log Collector or syslog server.

You can choose to configure only the physical interface. Because syslog forwarding via subinterfaces is not yet supported on LFCs, each virtual system uses the single untagged physical interface.

If you configure an LFC subinterface to forward logs externally, the interfaces will no longer work as expected.

To configure a separate subinterface for each virtual system, add subinterfaces to the physical interface and assign the necessary tag to segment the subinterface traffic.

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For a PA-7000 Series firewall managed by a Panorama management server, you cannot override or revert the LFC configuration locally on the firewall if the LFC configuration is pushed from Panorama. To override the LFC configuration pushed from Panorama, you must log in to the firewall CLI and delete the Panorama pushed configuration.

```
admin> configure
```

```
admin# delete deviceconfig log-fwd-card
```

```
admin# commit
```

Configure Administrative Access Per Virtual System or Firewall

Where Can I Use This?	What Do I Need?
• NGFW	☐ Virtual Systems license for any virtual systems beyond the base number supported by each NGFW series. ☐ One of the following licenses when using Strata Cloud Manager: • Strata Cloud Manager Pro • Strata Cloud Manager Essentials *Virtual System support on Strata Cloud Manager is available on request. Contact your account team to enable the feature.*

If you have a superuser administrative account, you can create and configure granular permissions for a vsysadmin or device admin role.

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STEP 1 | Create an Admin Role Profile that grants or disables permission to an Administrator to configure or read-only various areas of the web interface.

1. Select **Device** > **Admin Roles** and **Add** an **Admin Role Profile**.
2. Enter a **Name** and optional **Description** of the profile.
3. For **Role**, specify which level of control the profile affects:
   - **Device**—The profile allows the management of the global settings and any virtual systems.
   - **Virtual System**—The profile allows the management of only the virtual system(s) assigned to the administrator(s) who have this profile. (The administrator will be able to access **Device** > **Setup** > **Services** > **Virtual Systems**, but not the **Global** tab.)
4. On the **Web UI** tab for the Admin Role Profile, scroll down to **Device**, and leave the green check mark (Enable).
   - Under **Device**, enable **Setup**. Under **Setup**, enable the areas to which this profile will grant configuration permission to the administrator, as shown below. (The Read Only lock icon appears in the Enable/Disable rotation if Read Only is allowed for that setting.)
     - **Management**—Allows an admin with this profile to configure settings on the **Management** tab.
     - **Operations**—Allows an admin with this profile to configure settings on the **Operations** tab.
     - **Services**—Allows an admin with this profile to configure settings on the **Services** tab. An admin must have **Services** enabled in order to access the **Device** > **Setup** **Services** > **Virtual Systems** tab. If the **Role** was specified as **Virtual System** in the prior step, **Services** is the only setting that can be enabled under **Device** > **Setup**.
     - **Content-ID**—Allows an admin with this profile to configure settings on the **Content-ID** tab.
     - **WildFire**—Allows an admin with this profile to configure settings on the **WildFire** tab.
     - **Session**—Allows an admin with this profile to configure settings on the **Session** tab.
     - **HSM**—Allows an admin with this profile to configure settings on the **HSM** tab.
5. Click **OK**.
6. (Optional) Repeat the entire step to create another Admin Role profile with different permissions, as necessary.

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STEP 2 | Apply the Admin role profile to an administrator.

1. Select **Device** > **Administrators**, click **Add** and enter the **Name** to add an Administrator.
2. (Optional) Select an **Authentication Profile**.
3. (Optional) Select **Use only client certificate authentication (Web)** to have bi-directional authentication; to get the server to authenticate the client.
4. Enter a **Password** and **Confirm Password**.
5. (Optional) Select **Use Public Key Authentication (SSH)** if you want to use a much stronger, key-based authentication method using an SSH public key rather than just a password.
6. For **Administrator Type**, select **Role Based**.
7. For **Profile**, select the profile that you just created.
8. (Optional) Select a **Password Profile**.
9. Click **OK**.

STEP 3 | Commit the configuration.

Click **Commit**.

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Zone Protection and DoS Protection

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

Segmenting the network into functional and organizational zones reduces the network's attack surface—the portion of the network exposed to potential attackers. Zone protection defends network zones against flood attacks, reconnaissance attempts, packet-based attacks, and attacks that use non-IP protocols. Tailor a Zone Protection profile to protect each zone (you can apply the same profile to similar zones). Denial-of-service (DoS) protection defends specific critical systems against flood attacks, especially devices that user access from the internet such as web servers and database servers, and protects resources from session floods. Tailor DoS Protection profiles and policy rules to protect each set of critical devices. Visit the Best Practices documentation portal to get a checklist of Zone Protection and DoS Protection best practices.

*Check and monitor firewall dataplane CPU consumption to ensure that each firewall is properly sized to support DoS and Zone Protection along with any other features that consume CPU cycles, such as decryption. If you use Panorama to manage your firewalls, use Device Monitor (**Panorama** > **Managed Devices** > **Health**) to check and monitor the CPU consumption of all managed firewalls at one time.*

Network Segmentation Using Zones

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

The larger the network, the more difficult it is to protect. A large, unsegmented network presents a large attack surface that can be difficult to manage and protect. Because traffic and applications have access to the entire network, once an attacker gains entry to a network, the attacker can move laterally through the network to access critical data. A large network is also more difficult to monitor and control. Segmenting the network limits an attacker's ability to move through the network by preventing lateral movement between zones.

A security zone is a group of one or more physical or virtual firewall interfaces and the network segments connected to the zone's interfaces. You control protection for each zone individually so that each zone receives the specific protections it needs. For example, a zone for the finance department may not need to allow all of the applications that a zone for IT allows.

To fully protect your network, all traffic must flow through the firewall. Configure Interfaces and Zones to create separate zones for different functional areas such as the internet gateway, sensitive data storage, and business applications, and for different organizational groups such as finance, IT, marketing, and engineering. Wherever there is a logical division of functionality, application usage, or user access privileges, you can create a separate zone to isolate and protect the area and apply the appropriate security policy rules to prevent unnecessary access to data and applications that only one or some groups need to access. The more granular the zones, the greater the visibility and control you have over network traffic. Dividing your network into zones helps to create a Zero Trust architecture that executes a security philosophy of trusting no users, devices, applications, or packets, and verifying everything. The end goal is to create a network that allows access only to the users, devices, and applications that have legitimate business needs, and to deny all other traffic.

How to appropriately restrict and permit access to zones depends on the network environment. For example, environments such as semiconductor manufacturing floors or robotic assembly plants, where the workstations control sensitive manufacturing equipment, or highly restricted access areas, may require physical segmentation that permits no access from outside devices (no mobile device access).

In environments where users can access the network with mobile devices, enabling User-ID and App-ID in conjunction with segmenting the network into zones ensures that users receive the appropriate access privileges regardless of where they access the network, because access privileges are tied to a user or a user group instead of to a device in one particular zone.

The protection requirements for different functional areas and groups may also differ. For example, a zone that handles a large amount of traffic may require different flood protection thresholds than a zone that normally handles less traffic. The ability to define the appropriate protection for each zone is another reason to segment the network. What appropriate protection is depends on your network architecture, what you want to protect, and what traffic you want to permit and deny.

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How Do Zones Protect the Network?

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

Zones not only protect your network by segmenting it into smaller, more easily managed areas, but zones also protect the network because you can control access to zones and traffic movement between zones.

Zones prevent uncontrolled traffic from flowing through the firewall interfaces into your network because firewall interfaces can't process traffic until you assign them to zones. The firewall applies zone protection on ingress interfaces, where traffic enters the firewall in the direction of flow from the originating client to the responding server (c2s), to filter traffic before it enters a zone.

The firewall interface type and the zone type (Tap, virtual wire, L2, L3, Tunnel, or External) must match, which helps to protect the network against admitting traffic that doesn't belong in a zone. For example, you can assign an L2 interface to an L2 zone or an L3 interface to an L3 zone, but you can't assign an L2 interface to an L3 zone.

In addition, a firewall interface can belong to one zone only. Traffic destined for different zones can't use the same interface, which helps to prevent inappropriate traffic from entering a zone and enables you to configure the protection appropriate for each individual zone. You can connect more than one firewall interface to a zone to increase bandwidth, but each interface can connect to only one zone.

After the firewall admits traffic to a zone, traffic flows freely within that zone and is not logged. The more granular you make each zone, the greater the control you have over the traffic that accesses each zone, and the more difficult it is for malware to move laterally across the network between zones. Traffic can't flow between zones unless a security policy rule allows it and the zones are of the same zone type (Tap, virtual wire, L2, L3, Tunnel, or External). For example, a security policy rule can allow traffic between two L3 zones, but not between an L3 zone and an L2 zone. The firewall logs traffic that flows between zones when a security policy rule permits interzone traffic.

By default, security policy rules prevent lateral movement of traffic between zones, so malware can't gain access to one zone and then move freely through the network to other targets.

Tunnel zones are for non-encrypted tunnels. You can apply different security policy rules to the tunnel content and to the zone of the outer tunnel, as described in the Tunnel Content Inspection Overview.

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Zone Defense

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

Zone Protection profiles defend zones against flood, reconnaissance, packet-based, and non-IP-protocol-based attacks. DoS Protection profiles used in DoS Protection policy rules defend specific, critical devices against targeted flood and resource-based attacks. A DoS attack overloads the network or targeted critical systems with large amounts of unwanted traffic in an attempt to disrupt network services.

Plan to defend your network against different types of DoS attacks:

- **Application-Based Attacks**—Target weaknesses in a particular application and try to exhaust its resources so legitimate users can't use it. An example of this is the Slowloris attack.
- **Protocol-Based Attacks**—Also known as state-exhaustion attacks, these attacks target protocol weaknesses. A common example is a SYN flood attack.
- **Volumetric Attacks**—High-volume attacks that attempt to overwhelm the available network resources, especially bandwidth, and bring down the target to prevent legitimate users from accessing those resources. An example of this is a UDP flood attack.

There are no default Zone Protection profiles or DoS Protection profiles and DoS Protection policy rules. Configure and apply zone protection based on each zone's traffic characteristics and configure DoS protection based on the individual critical systems you want to protect in each zone.

Zone Defense Tools

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

Effective defense against DoS attacks requires a layered approach. The first layer of defense should be a dedicated, high-volume DDoS protection device at the internet-facing network perimeter and a perimeter router, switch, or other hardware-based packet drop device with appropriate access control lists (ACLs) to defend against volumetric attacks that the session-based firewall isn't designed to handle. The firewall adds more granular layers of DoS attack defense and also visibility into application traffic that dedicated DDoS devices don't provide.

Palo Alto Networks firewalls provide four complementary tools to layer in DoS protection for your network zones and critical devices:

- **Zone Protection profiles** defend the ingress zone edge against IP flood attacks, reconnaissance port scans and host sweeps, IP packet-based attacks, and non-IP protocol attacks. The ingress zone is where traffic enters the firewall in the direction of flow from the client to the server (c2s), where the client is the originator of the flow and the server is the responder. Zone Protection profiles provide a second layer of broad defense against DoS attacks, based on the

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aggregate traffic entering the zone, by limiting the new connections-per-second (CPS) to the zone. Zone Protection profiles don't take individual devices (IP addresses) into account because the profiles apply to the aggregate traffic entering the zone.

Zone protection profiles defend the network as a session is formed, before the firewall performs DoS Protection policy and Security policy rule lookups, and consume fewer CPU cycles than a DoS Protection policy or Security policy rule lookup. If a Zone Protection profile denies traffic, the firewall doesn't spend CPU cycles on policy rule lookups.

Apply Zone Protection profiles to every zone, both internet-facing and internal.

- **DoS Protection profiles and policy rules** defend specific individual endpoints and resources against flood attacks, especially high-value targets that users access from the internet. While a Zone Protection profile defends the zone from flood attacks, a DoS Protection policy rule with an appropriate DoS Protection profile defends critical individual systems in a zone from targeted flood attacks, providing a granular third layer of defense against DoS attacks.

  Because the intent of DoS protection is to defend critical devices and because it consumes resources, DoS protection defends only the devices you specify in a DoS Protection policy rule. No other devices are protected.

  DoS Protection profiles set flood protection thresholds (new CPS limits) for individual devices or groups of devices, resource protection thresholds (session limits for specified endpoints and resources), and whether the profile applies to aggregate or classified traffic. DoS Protection policy rules specify match criteria (source, destination, service ports), the action to take when traffic matches the rule, and the aggregate and classified DoS Protection profiles associated with each rule.

  Aggregate DoS Protection policy rules apply the CPS thresholds defined in an aggregate DoS Protection profile to the combined traffic of all the devices that meet the DoS Protection policy rule match criteria. For example, if you configure the aggregate DoS Protection profile to limit the CPS rate to 20,000, the 20,000 CPS limit applies to the aggregate number of connections for the entire group. In this case, one device could receive the majority of the allowed connections.

  Classified DoS Protection policy rules apply the CPS thresholds defined in a classified DoS Protection profile to each individual device that matches the policy rule. For example, if you configure the classified DoS Protection profile to limit the CPS rate to 4,000, then no device in the group can accept more than 4,000 CPS. A DoS Protection policy can have one aggregate profile and one classified profile.

  Classified profiles can classify connections by source IP, destination IP, or both. For internet-facing zones, classify by destination IP only because the firewall can't scale to hold the internet routing table.

  Apply DoS Protection only to critical devices, especially popular attack targets that users access from the internet, such as web servers and database servers.

- For existing sessions, **Packet Buffer Protection** protects the firewall (and therefore the zone) against single-session DoS attacks that attempt to overwhelm the firewall's packet buffer, using thresholds and timers to mitigate abusive sessions. You configure Packet Buffer Protection settings globally and apply them per zone.

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- **Security Policy rules** affect both the ingress and egress flows of a session. To establish a session, incoming traffic must match an existing Security policy rule. If there is no match, the firewall discards the packet. A Security policy allows or denies traffic between zones (interzone) and within zones (intrazone) using criteria including zones, IP addresses, users, applications, services, and URL categories.

  Apply the best practice Vulnerability Protection profile to each Security policy rule to help defend against DoS attacks.

  The default Security policy rules don't permit traffic to travel between zones, so you need to configure a Security policy rule if you want to allow interzone traffic. All intrazone traffic is allowed by default. You can configure Security policy rules to match and control intrazone, interzone, or universal (intrazone and interzone) traffic.

  Zone Protection profiles, DoS Protection profiles and policy rules, and Security policy rules only affect dataplane traffic on the firewall. Traffic originating on the firewall management interface does not cross the dataplane, so the firewall does not match management traffic against these profiles or policy rules.
- You can also search the Palo Alto Networks Threat Vault (requires a valid support account and login) for threats by hash, CVE, signature ID, domain name, URL, or IP address.

How Do the Zone Defense Tools Work?

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

When a packet arrives at the firewall, the firewall attempts to match the packet to an existing session, based on the ingress zone, egress zone, source IP address, destination IP address, protocol, and application derived from the packet header. If the firewall finds a match, then the packet uses the Security policy rules that already control the session. If the packet doesn't match an existing session, the firewall uses Zone Protection profiles, DoS Protection profiles and policy rules, and Security policy rules to determine whether to establish a session or discard the packet, and the level of access the packet receives.

After traffic passes through your dedicated DDoS device at the internet-facing network edge, the first protection the firewall applies is the broad defense of the Zone Protection profile, if one is attached to the zone. The firewall determines the zone from the interface on which the packet arrives (each interface is assigned to only one zone and all interfaces that carry traffic must belong to a zone). If the Zone Protection profile denies the packet, the firewall discards the packet and saves resources by not needing to look up the DoS Protection policy or Security policy. The firewall applies Zone Protection profiles only to new sessions (packets that do not match an existing session). After the firewall establishes a session, the firewall bypasses the Zone Protection profile lookup for succeeding packets in that session.

If the Zone Protection profile doesn't drop the packet, the second protection the firewall applies is a DoS Protection policy rule. If a Zone Protection profile allows a packet based on the total aggregate amount of traffic going to the zone, a DoS Protection policy rule may deny the packet if it is going to a particular destination or coming from a particular source that has exceeded the

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flood protection or resource protection settings in the rule's DoS Protection profile. If the packet matches a DoS Protection policy rule, the firewall applies the rule to the packet. If the rule denies access, the firewall discards the packet and doesn't perform a Security policy lookup. If the rule allows access, the firewall performs a Security policy lookup. Like the Zone Protection profile, the firewall enforces DoS Protection policy only on new sessions.

The third protection the firewall applies is a Security policy lookup, which happens only if the Zone Protection profile and DoS Protection policy rules allow the packet. If the firewall finds no Security policy rule match for the packet, the firewall discards the packet. If the firewall finds a matching Security policy rule, the firewall applies the rule to the packet. The firewall enforces the Security policy rule on traffic in both directions (c2s and s2c) for the life of the session. Apply the best practice Vulnerability Protection profile to all Security policy rules to help defend against DoS attacks.

The fourth protection the firewall applies is packet buffer protection, which you apply globally to protect the device and can also apply individually to zones to prevent single-session DoS attacks that attempt to overwhelm the firewall's packet buffer. For global protection, the firewall used Random Early Drop (RED) to drop packets (not sessions) when the level of traffic crosses protection thresholds. For per-zone protection, the firewall blocks the source IP address if it violates the packet buffer thresholds. Unlike zone and DoS protection, packet buffer protection applies to existing sessions.

Firewall Placement for DoS Protection

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

The firewall is a session-based device that isn't designed to scale to millions of connections-per-second (CPS) to defend against large volumetric DoS attacks. The firewall treats each unique flow (based on ingress and egress zone, source and destination IP, protocol, and application) as a session, spends CPU cycles on packet inspection at the port and the IP level to provide visibility into application traffic, and must count each session for the flood threshold counters, so firewall placement is critical to avoid flooding the firewall.

For the best DoS protection, *place firewalls as close to the resources you're protecting as possible*. This reduces the number of sessions the firewall needs to handle and therefore the amount of firewall resources required to provide DoS protection.

At the internet-facing perimeter, do *not* place firewalls you use for DoS protection or zone protection in front of dedicated DDoS devices and perimeter routers and switches. Make those high-volume devices your first line of DoS defense to mitigate volumetric flood attacks. For zone and DoS protection at the perimeter, use high-capacity firewalls and place them *behind* the high-volume devices. As a rule, the closer a firewall is to the perimeter, the higher capacity it must be to handle the volume of traffic.

The way you segment your network into zones can help mitigate internal DoS attacks. Smaller zones provide greater visibility into traffic and prevent lateral movement of malware better because more traffic must cross zones, and to allow interzonal traffic requires you to create a specific Security policy rule (all intrazonal traffic is allowed by default). Consider revisiting your segmentation approach if your network is relatively unsegmented.

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Baseline CPS Measurements for Setting Flood Thresholds

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

Flood protection thresholds determine the number of new connections-per-second (CPS) to allow for a zone (Zone Protection profile), for a group of devices in a zone (aggregate DoS Protection policy), or for individual devices in a zone (classified DoS Protection policy), when to throttle new connections to begin mitigating a potential flood attack, and when to drop all new connections. The default Zone Protection profile and DoS Protection profile flood protection thresholds aren't appropriate for most networks because each network is unique. You need to understand the aggregate normal and peak CPS for each zone to set effective Zone Protection profile thresholds, and for the individual critical systems you want to defend to set effective DoS Protection profile thresholds that don't inadvertently set thresholds too high and allow flood attacks or set thresholds too low and throttle traffic.

Measure average and peak CPS traffic over the course of at least five business days or until you're confident that the measurements reflect the network's typical traffic patterns; the longer measurement period, the more accurate the measurements. Take into account special events, quarterly events, and annual events that may spike the number of CPS you need to support. You may need to adjust Zone Protection profiles and schedule adjusted DoS Protection policy rules to accommodate these types of events if your firewalls have the capacity to handle extra traffic. Take the following baseline measurements:

- For Zone Protection profiles, measure the average and peak CPS ingressing each zone.
- For aggregate DoS Protection profiles, measure the combined average and peak CPS for each group of devices you want to protect.
- For classified DoS Protection profiles, measure the average and peak CPS of the individual devices you want to protect.

Also understand the capacity of your firewalls and how other resource-consuming features such as decryption affect the number of connections each firewall can control. As a general rule, the closer a firewall is to the perimeter, the greater its capacity needs to be because it handles more traffic. The datasheet for each firewall model includes the total new sessions per second (CPS) the firewall supports and the Firewall Comparison Tool enables you to compare the CPS (and other metrics) of different firewall models.

How to Measure CPS

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

There are many ways to measure CPS to help you set Zone Protection profile and DoS Protection profile flood threshold settings:

- For **Zone Protection profile thresholds**, if you run PAN-OS 10.0 or later, the best way to measure CPS is to use the Zone Protection profile Threshold Recommendation alerts from the

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AIOps cloud service, which use system telemetry to provide accurate estimates of average and average peak CPS values to use in Zone Protection profiles. You can sign up firewalls and Panorama for the service. With PAN-OS 10.2.1 or later, you can install the AIOps plugin for Panorama to proactively enforce security checks on configurations before you push them to managed firewalls.

- If you use Panorama to manage your firewalls, use Device Monitoring to measure CPS coming into a firewall. Select a device to see measurements that help you understand the CPS for that device over a configurable time frame to help you understand the capacity of the firewall. Device Monitoring can also show you a 90-day trend line of CPU average and peak use to help you understand the typical available capacity of each firewall. To see how CPS impacts firewall

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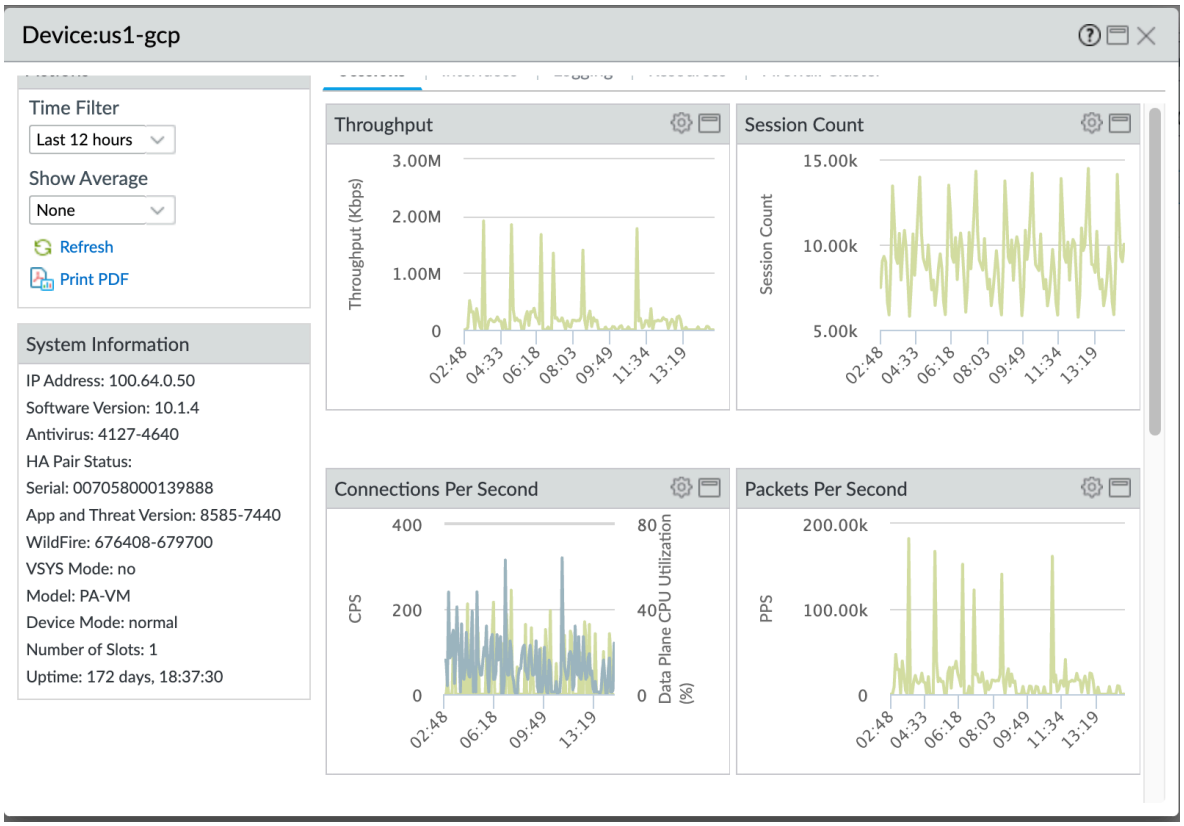
resources, you can overlay CPS on the same timeline with metrics such as CPU utilization, packet buffers, or packet descriptors:

1. **Panorama** > **Managed Devices** > **Health** > **All Devices**.
2. Click a **Device Name** to select a device and to view and filter device information.



3. Select the gear icon to access Device Monitor annotations, overlay, and comparison actions.

*You can select tabs (not shown) at the top of the dialog box to see more metrics. The following illustrations show the **Sessions** tab. The other tabs are **Interfaces**, **Logging**, **Resources**, and **Firewall Cluster**. Each tab displays different default metrics and for each default metric, you can overlay other metrics, compare the selected device to other devices, including device slots and data planes, and annotate the metric.*

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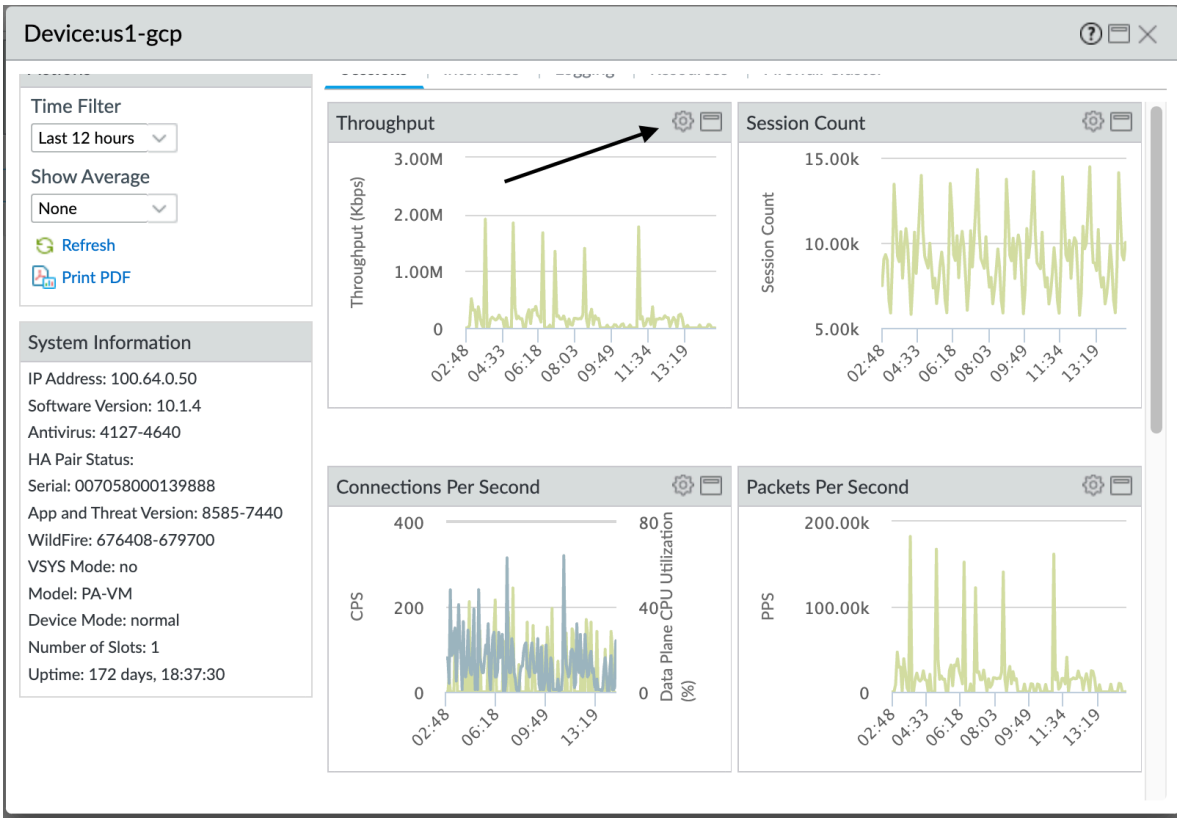

*The preceding screen shows the CPS data over the last 12 hours (**Time Filter**) overlaid with Data Plane CPU Utilization. The next step shows you how to overlay metrics on the default metrics in each tab.*

4. Click the gear icon to see the actions you can take for overlaying other metrics on the default metrics. You can overlay one metric at a time on each default metric over a particular time frame:

   1. Select **Overlay** to see the overlay options and then select the **Metric** drop-down.

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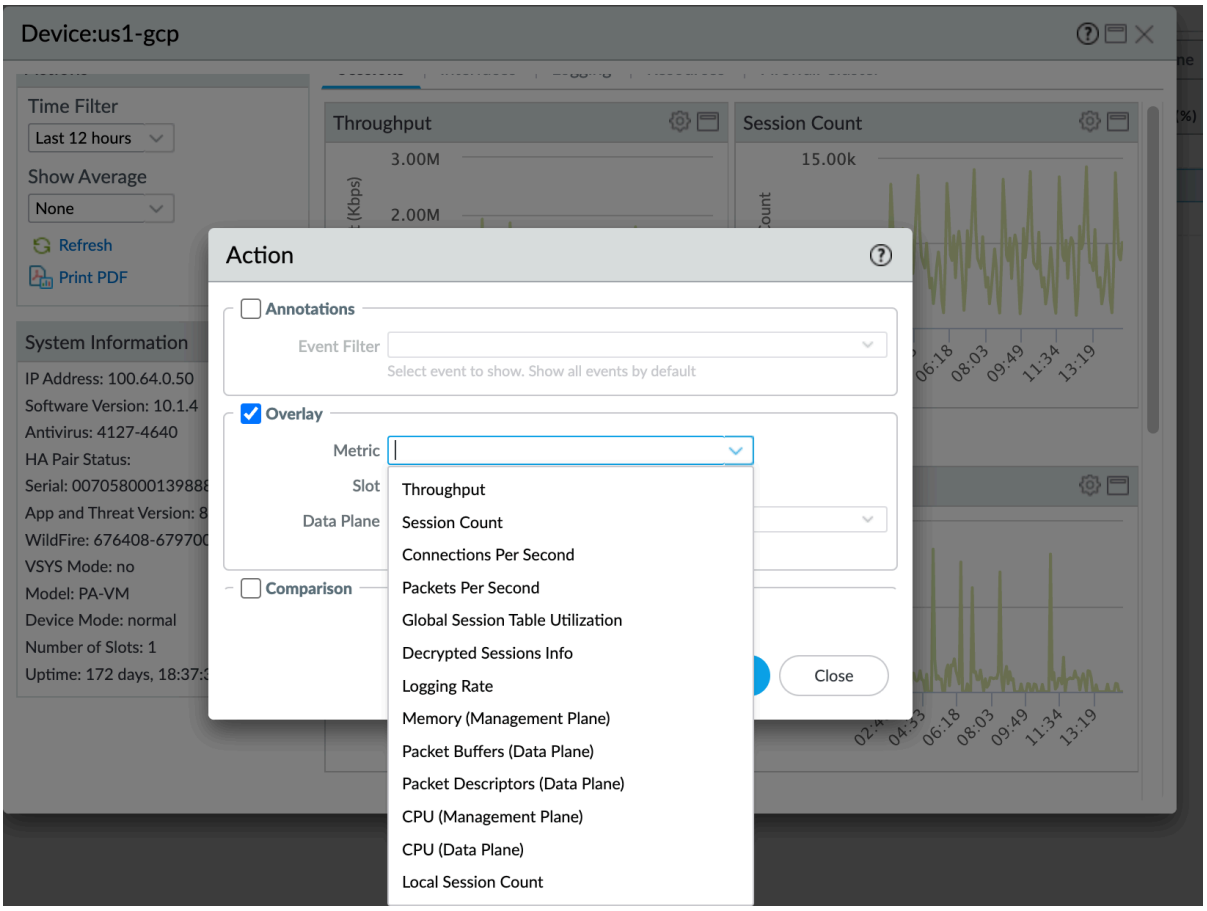

2. You can overlay any of these metrics on the default metrics over the same time period to see how the state of one metric affects another metric.

   For example, on the **Sessions** tab, you can overlay Data Plane Packet Buffers or Data Plane Packet Descriptors to see how high CPS, Throughput, Session Count, or Packets Per Second (PPS) conditions affect the packet buffers or packet descriptors.

   Another example on the **Sessions** tab is to overlay CPS Throughput or PPS with the Data Plane CPU and Packet Buffers metrics to see how traffic spikes affect the CPU and buffers.

   Another example is to select the **Resources** tab and then overlay Data Plane CPU over Packet Buffers to see how packet buffer utilization affects the CPU.

   Overlays help you see trends and correlations such as whether high buffer utilization is associated with high CPS or PPS rates, and give you an idea of how high CPS and PPS can be before they affect the CPU, packet buffers, or packet descriptors.

5. Click **OK** to see the data overlay and use the information to understand device resource behavior under different CPS loads and conditions.

- To gather CPS data over time to help with setting **Zone Protection profile thresholds**, if you use an SNMP server, you can use your own management tools to poll SNMP MIBs. However, it is important to understand that the CPS measurements in the MIBs show *twice* the actual CPS value (for example, if the true CPS measurement is 10,000, the MIBs show 20,000 as the value; this happens because the MIBs count the C2S and S2C session

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segments separately instead of as a single session). You can still see trends from the MIBs and you can divide the CPS values by two to derive the true values. The SNMP MIB OIDs are: PanZoneActiveTcpCps, PanZoneActiveUdpCps, and PanZoneOtherIpCps. Because the firewall only takes measurements and updates the SNMP server every 10 seconds, poll every 10 seconds.

- Run the operational CLI command **`show session info`**.

  *You can also see CPS values using the operational CLI command **`show counter interface`**, but this command displays two times the actual CPS value because it counts the C2S and S2C session segments separately instead of as a single session, so divide the CPS value by two to derive the real CPS value.*

- **DoS Protection profiles** can protect servers from DoS attacks and can also prevent misconfigured or compromised servers from attacking your network. When the DoS Protection policy rule specifies a server as the destination, you're protecting it from DoS attacks. When a rule specifies a server as the source, you're protecting your network from inadvertent or malicious attacks on your network from that server.

  To measure the CPS for an individual device or to see which devices have the highest CPS rates so that you can set DoS Protection profile thresholds, use the Application Command Center (ACC). The ACC shows you server session rates that enable you to calculate the average CPS for individual devices (for classified DoS Protection policy rules) and for groups of devices (aggregate DoS Protection policy rules). Take measurements over at least a week; longer time periods provide a larger sample size and therefore more representative measurements. Use the measurements to understand the normal and peak number of connections you expect the server to receive and base your threshold settings on those measurements. To find the devices that have the highest CPS rates over a particular time period:

  1. Select **ACC**.
  2. Set the **Time** period over which to look at session traffic.
  3. On **Network Activity**, go to the **Source IP Activity** widget and/or the **Destination IP Activity** widget and select **sessions** (**bytes** is the default). You can look at source IP activity and destination IP activity at the same time to see how many sessions the device generates (source IP) and how many sessions the device receives (destination IP).
  4. In the widget's source address table, click **SESSIONS** to show the source IP addresses with the highest session counts during the selected **Time**.
  5. To determine the CPS value for a server during the selected **Time**, divide the number of sessions by the number of seconds in the **Time**. For example, if the **Time** is set to **Last Hour**, then divide the number of sessions by 3,600 seconds to derive the CPS value.

  The ACC gives you knowledge of average CPS values over time. You can check the number of sessions over the last week, month, or whatever time period makes sense for your environment to understand the session load for a device. For example, to see the session

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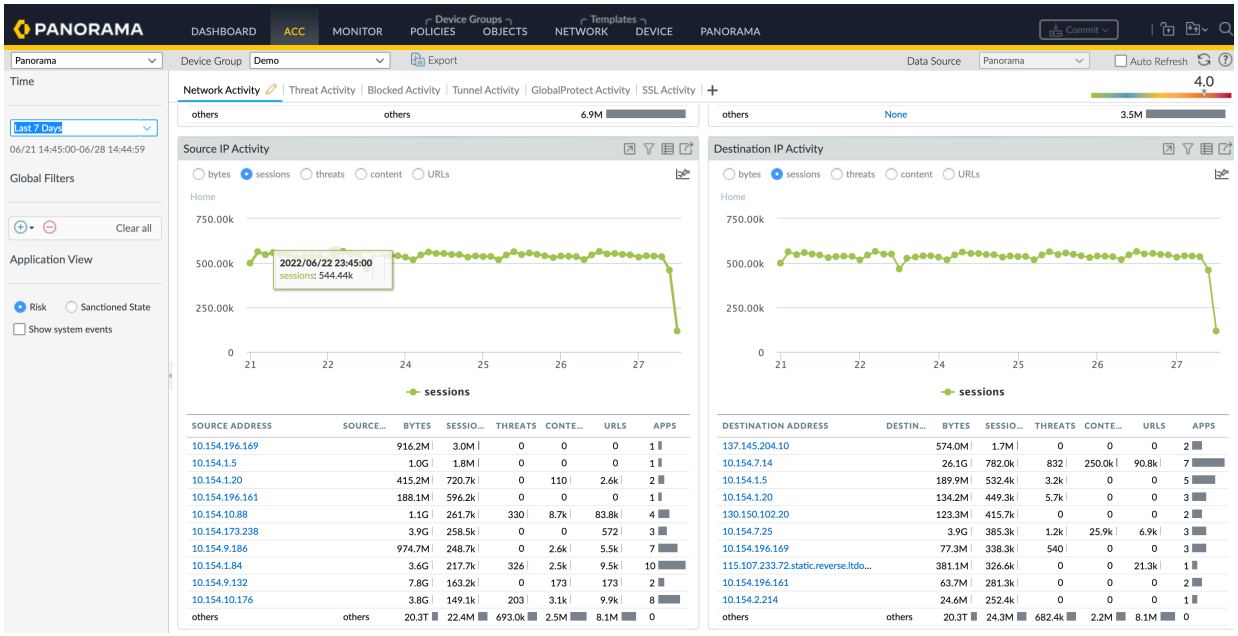
activity over the last week, set the **Time** to **Last 7 Days** and the source and destination IP widgets to **sessions**:



As an example of measuring CPS to protect a server from DoS attacks using ACC information in the illustration, let's calculate the average CPS value over a seven day time period for the server that receives the most sessions (IP address 137.145.204.10 in the **Destination IP Activity** widget). We divide the 1.7 million sessions by the number of seconds in seven days (7 days x 24 hours x 60 minutes x 60 seconds = 604,800 seconds). The average is a bit less than three sessions per second for that server. Measure the CPS over time periods that represent normal average and peak traffic for the servers you want to protect and base your initial thresholds on those values. Observe the servers and adjust the thresholds as necessary to tune DoS Protection so that the servers are protected but you don't throttle legitimate connections unnecessarily.

- Measuring CPS for **classified DoS Protection profiles**—Classified DoS Protection profiles protect individual devices. The goal is to configure CPS thresholds in the classified DoS Protection profile and attach the profile to a DoS Protection policy rule that applies to specific servers that have similar DoS attack thresholds. For example, you can apply classified DoS Protection profiles to web servers or critical file servers to prevent a DoS attack from disrupting their availability.

  The thresholds you set in the profile apply to each individual device specified in the policy rule. For example, if you set a maximum rate of 5,000 CPS in a classified DoS Protection

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profile, then each device in the associated DoS Protection policy rule can accept up to 5,000 CPS before dropping new connections.

To calculate the average and peak CPS value, specify the IP address of each device to which you want to apply classified DoS protection in **Global Filters** (you can specify multiple IP addresses).

1. Select the **Time** frame over which to view session activity.
2. Select **sessions** in the **Destination IP Activity** widget.
3. Specify the destination IP address of each device to which you want to apply classified DoS protection in **Global Filters** (you can specify multiple IP addresses).

   You can filter firewall Traffic logs and Threat logs for the destination IP addresses of the critical devices you want to protect to obtain normal and peak session activity information.
4. Add the session values together and divide the total into the number of seconds in the time period to derive the CPS value. For example, over a time period of 30 days (2,592,000 seconds), if the total number of sessions is 155,300,000, then the average CPS over that time period is approximately 60 CPS.
5. Check if the number of sessions over the time period is close enough that the initial threshold values protect each device from DoS attacks but also don't underutilize the devices.
6. Fine tune the threshold values to ensure that none of the protected servers become DoS attack victims while attaining the highest safe performance for legitimate connections.

To calculate the average peak CPS, use the graphic display in the widget to identify the peak session periods and calculate the average peak CPS from that.

- Measuring CPS for **aggregate DoS Protection profiles**—Aggregate DoS Protection profiles protect groups of devices. The goal is to configure CPS thresholds in the aggregate DoS Protection profile and attach the profile to a DoS Protection policy rule that applies to an entire group of servers. Aggregate DoS Protection adds another layer of broad protection after your dedicated large-capacity perimeter DDoS device and the firewall's Zone Protection.

  Aggregate profiles don't apply the configured threshold to each individual device in the way that classified profiles do. Instead, the threshold applies to the entire protected group. For example, if you set a maximum CPS threshold of 20,000 sessions to a group of five servers, then the combined total sessions that the group can support is 20,000 sessions. The only limit for an individual server in the group is how many of the 20,000 sessions are available. One device could receive 15,000 CPS, which leaves up to 5,000 CPS for the other four devices combined.

  Adjust the thresholds as needed. You can use the same process for finding normal and peak CPS for classified profiles in the ACC to find average normal and peak CPS for aggregate profiles. Keep in mind that for aggregate profiles, you need to base the thresholds on the group's total CPS, not on the CPS to individual servers.
- To prevent a server or servers from inadvertently or maliciously attacking your network, base your CPS measurements on the **Source IP Activity** widget, which shows the session activity that servers generate. Filter by sessions to see the most active servers or use **Global Settings** to filter by the source IP address of a particular server or servers. In the DoS

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Protection policy rule for the server(s), apply a DoS Protection profile with low thresholds so that the server can't disrupt the network. For example, thresholds of 10 CPS for Alarm Rate, 20 CPS for Activate Rate, and 30 CPS for Max Rate ensure that the firewall adds the source address to the hardware block table instead of using other system resources.

- For setting **aggregate DoS Protection profile thresholds**, you can use Zone Protection profile threshold measurements as a starting point, especially if you intend to cover most of the servers in a zone with aggregate DoS protection. If the zone contains only the devices to which you want to apply an aggregate DoS Protection profile, then the CPS numbers are exactly the same as the Zone Protection profile numbers. If the zone contains both devices that you want to protect with an aggregate DoS Protection profile and devices that you don't want to protect with an aggregate DoS Protection profile, you can use the Zone Protection CPS measurements as a starting point and experiment with the thresholds to tune them properly.
- Use third-party tools such as Wireshark or NetFlow to collect and analyze network traffic.
- Use scripts to automate CPS information collection and continuous monitoring, and to mine information from the logs.
- Configure every Security policy rule on the firewall to **Log at Session End**. If you have no monitoring tools such as NetFlow or Wireshark, and cannot obtain or develop automated scripts, **Log at Session End** captures the number of connections at the session end. While this doesn't provide CPS information, it does show you the number of sessions ending in the selected time duration and you can make an approximate calculation of the sessions per second from that information.
- Work with application teams to understand the normal and peak CPS to their servers and the maximum CPS those servers can support.

To conserve resources, the firewall measures the aggregate CPS at ten-second intervals. For this reason, measurements you see on the firewall may not catch bursts within the ten-second interval. Although the average CPS measurements aren't affected, the peak CPS measurements may not be precise. For example, if the firewall logs report a 5,000 CPS average in a ten-second interval, it's possible that 4,000 CPS came in a one-second burst and the other 1,000 CPS were spread out over the remaining nine seconds.

Create separate log forwarding profiles for flood events so the appropriate administrator receives emails that contain only flood (potential DoS attack) events. Set Log Forwarding for both zone protection and DoS protection threshold events.

After you implement Zone and DoS protection, use these methods to monitor the deployment, so as your network evolves and traffic patterns change, you adjust flood protection thresholds.

Zone Protection Profiles

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

Apply a Zone Protection profile to each zone to defend it based on the aggregate traffic entering the ingress zone.

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- Flood Protection
- Reconnaissance Protection
- Packet-Based Attack Protection
- Protocol Protection
- Ethernet SGT Protection

In addition to configuring zone protection and DoS protection, apply the best practice Vulnerability Protection profile to each Security policy rule to help defend against DoS attacks.

Flood Protection

A Zone Protection profile with flood protection configured defends an entire ingress zone against SYN, ICMP, ICMPv6, UDP, and other IP flood attacks. The firewall measures the aggregate amount of each flood type entering the zone in new connections-per-second (CPS) and compares the totals to the thresholds you configure in the Zone Protection profile. (You protect critical individual devices within a zone with DoS Protection profiles and policy rules.)

*Measure and monitor firewall dataplane CPU consumption to ensure that each firewall is properly sized to support DoS and Zone Protection and any other features that consume CPU cycles, such as decryption. If you use Panorama to manage your firewalls, Device Monitoring (**Panorama** > **Managed Devices** > **Health** > **All Devices**) shows you the CPU and memory consumption of each managed firewall. It can also show you a 90-day trend line of CPU average and peak use to help you understand the typical available capacity of each firewall.*

For each flood type, you set three thresholds for new CPS entering the zone, and you can set a drop **Action** for SYN floods. If you know the baseline CPS rates for the zone, use these guidelines to set the initial thresholds, and then monitor and adjust the thresholds as necessary.

- **Alarm Rate**—The new CPS threshold to trigger an alarm. Target setting the **Alarm Rate** to 15-20% above the average CPS rate for the zone so that normal fluctuations don't cause alerts.
- **Activate**—The new CPS threshold to activate the flood protection mechanism and begin dropping new connections. For ICMP, ICMPv6, UDP, and other IP floods, the protection mechanism is Random Early Drop (RED, also known as Random Early Detection). For SYN floods only, you can set the drop **Action** to SYN Cookies or RED. Target setting the **Activate** rate to just above the peak CPS rate for the zone to begin mitigating potential floods.
- **Maximum**—The number of connections-per-second to drop incoming packets when RED is the protection mechanism. Target setting the **Maximum** rate to approximately 80-90% of firewall capacity, taking into account other features that consume firewall resources.

If you don't know the baseline CPS rates for the zone, start by setting the **Maximum** CPS rate to approximately 80-90% of firewall capacity and use it to derive reasonable flood mitigation alarm and activation rates. Set the **Alarm Rate** and **Activate** rate based on the Maximum rate. For example, you could set the **Alarm Rate** to half the **Maximum** rate and adjust it depending on how many alarms you receive and the firewall resources being consumed. Be careful setting the **Activate Rate** since it begins to drop connections. Because normal traffic loads experience some fluctuation, it's best not to drop connections too aggressively. Err on the high side and adjust the rate if firewall resources are impacted.

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*SYN Flood Protection is the only type for which you set the drop **Action**. Start by setting the **Action** to **SYN Cookies**. SYN Cookies treats legitimate traffic fairly and only drops traffic that fails the SYN handshake, while using Random Early Drop drops traffic randomly, so RED may affect legitimate traffic. However, SYN Cookies is more resource-intensive because the firewall acts as a proxy for the target server and handles the three-way handshake for the server. The tradeoff is not dropping legitimate traffic (SYN Cookies) versus preserving firewall resources (RED). Monitor the firewall, and if SYN Cookies consumes too many resources, switch to RED. If you don't have a dedicated DDoS prevention device in front of the firewall, always use RED as the drop mechanism.*

*When **SYN Cookies** is activated, the firewall does not honor the TCP options that the server sends because it does not know these values at the time that it proxies the SYN/ACK. Therefore, values such as the TCP server's window size and MSS values cannot be negotiated during the TCP handshake and the firewall will use its own default values. In the scenario where the MSS of the path to the server is smaller than the firewall's default MSS value, the packet will need to be fragmented.*

The default threshold values are high so that activating a Zone Protection profile doesn't unexpectedly drop legitimate traffic. Adjust the thresholds to values appropriate for your network's traffic. The best method for understanding how to set reasonable flood thresholds is to take baseline measurements of average and peak CPS for each flood type to determine the normal traffic conditions for each zone and to understand the capacity of the firewall, including the impact of other resource-consuming features such as decryption. Monitor and adjust the flood thresholds as needed and as your network evolves.

*Firewalls with multiple dataplane processors (DPs) distribute connections across DPs. In general, the firewall divides the CPS threshold settings equally across its DPs. For example, if a firewall has five DPs and you set the **Alarm Rate** to 20,000 CPS, each DP has an **Alarm Rate** of 4,000 CPS (20,000 / 5 = 4,000), so if the new sessions on a DP exceeds 4,000, it triggers the **Alarm Rate** threshold for that DP.*

Reconnaissance Protection

Similar to the military definition of reconnaissance, the network security definition of reconnaissance is when attackers attempt to gain information about your network's vulnerabilities by secretly probing the network to find weaknesses. Reconnaissance activities are often preludes to a network attack. *Enable Reconnaissance Protection on all zones* to defend against port scans and host sweeps:

- **Port scans** discover open ports on a network. A port scanning tool sends client requests to a range of port numbers on a host, with the goal of locating an active port to exploit in an attack. Zone Protection profiles defend against TCP and UDP port scans.
- **Host sweeps** examine multiple hosts to determine if a specific port is open and vulnerable.
- **IP protocol scans** cycle through IP protocol numbers to determine the IP protocols and thus services supported by target machines.

You can use reconnaissance tools for legitimate purposes such as pen testing of network security or the strength of a firewall. You can specify up to 20 IP addresses or netmask address objects to exclude from Reconnaissance Protection so that your internal IT department can conduct pen tests to find and fix network vulnerabilities.

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You can set the action to take when reconnaissance traffic (excluding pen testing traffic) exceeds the configured threshold when you configure reconnaissance protection. Retain the default **Interval** and **Threshold** to log a few packets for analysis before blocking the reconnaissance operation.

Packet-Based Attack Protection

Packet-based attacks take many forms. Zone Protection profiles check IP, TCP, ICMP, IPv6, and ICMPv6 packet headers and protect a zone by:

- Dropping packets with undesirable characteristics.
- Stripping undesirable options from packets before admitting them to the zone.

Select the drop characteristics for each packet type when you Configure Packet Based Attack Protection. The best practices for each IP protocol are:

- **IP Drop**—Drop **Unknown** and **Malformed** packets. Also drop **Strict Source Routing** and **Loose Source Routing** because allowing these options allows adversaries to bypass Security policy rules that use the Destination IP address as the matching criteria. For internal zones only, check **Spoofed IP Address** so only traffic with a source address that matches the firewall routing table can access the zone.
- **TCP Drop**—Retain the default **TCP SYN with Data** and **TCP SYNACK with Data** drops, drop **Mismatched overlapping TCP segment** and **Split Handshake** packets, and strip the **TCP Timestamp** from packets.

  *Enabling **Rematch Sessions** (**Device** > **Setup** > **Session** > **Session Settings**) is a best practice that applies committed newly configured or edited Security Policy rules to existing sessions. However, if you configure Tunnel Content Inspection on a zone and **Rematch Sessions** is enabled, you must also disable **Reject Non-SYN TCP** (change the selection from **Global** to **No**), or else when you enable or edit a Tunnel Content Inspection policy, the firewall drops all existing tunnel sessions. Create a separate Zone Protection profile to disable **Reject Non-SYN TCP** only on zones that have Tunnel Content Inspection policies and only when you enable **Rematch Sessions**.*
- **ICMP Drop**—There are no standard best practice settings because dropping ICMP packets depends on how you use ICMP (or if you use ICMP). For example, if you want to block ping activity, you can block **ICMP Ping ID 0**.
- **IPv6 Drop**—If compliance matters, ensure that the firewall drops packets with non-compliant routing headers, extensions, etc.
- **ICMPv6 Drop**—If compliance matters, ensure that the firewall drops certain packets if the packets don't match a Security policy rule.

Protocol Protection

In a Zone Protection profile, Protocol Protection defends against non-IP protocol based attacks. Enable Protocol Protection to block or allow non-IP protocols between security zones on a Layer 2 VLAN or on a virtual wire, or between interfaces within a single zone on a Layer 2 VLAN (Layer 3 interfaces and zones drop non-IP protocols so non-IP Protocol Protection doesn't apply). Configure Protocol Protection to reduce security risks and facilitate regulatory compliance by preventing less secure protocols from entering a zone, or an interface in a zone.

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If you don't configure a Zone Protection profile that prevents non-IP protocols in the same zone from going from one Layer 2 interface to another, the firewall allows the traffic because of the default intrazone allow Security policy rule. You can create a Zone Protection profile that blocks protocols such as LLDP within a zone to prevent discovery of networks reachable through other zone interfaces.

If you need to discover which non-IP protocols are running on your network, use monitoring tools such as NetFlow, Wireshark, or other third-party tools discover non-IP protocols on your network. Examples of non-IP protocols you can block or allow are LLDP, NetBEUI, Spanning Tree, and Supervisory Control and Data Acquisition (SCADA) systems such as Generic Object Oriented Substation Event (GOOSE), among many others.

Create an **Exclude List** or an **Include List** to configure Protocol Protection for a zone. The **Exclude List** is a block list—the firewall blocks all of the protocols you place in the **Exclude List** and allows all other protocols. The **Include List** is an allow list—the firewall allows only the protocols you specify in the list and blocks all other protocols.

Use include lists for Protocol Protection instead of exclude lists. Include lists specifically sanction only the protocols you want to allow and block the protocols you don't need or didn't know were on your network, which reduces the attack surface and blocks unknown traffic.

A list supports up to 64 Ethertype entries, each identified by its IEEE hexadecimal Ethertype code. Other sources of Ethertype codes are standards.ieee.org/develop/regauth/ethertype/eth.txt and http://www.cavebear.com/archive/cavebear/Ethernet/type.html. When you configure zone protection for non-IP protocols on zones that have Aggregated Ethernet (AE) interfaces, you can't block or allow a non-IP protocol on only one AE interface member because AE interface members are treated as a group.

*Protocol Protection doesn't allow blocking IPv4 (Ethertype 0x0800), IPv6 (0x86DD), ARP (0x0806), or VLAN-tagged frames (0x8100). The firewall always implicitly allows these four Ethertypes in an **Include List** even if you don't explicitly list them and doesn't permit you to add them to an **Exclude List**.*

Ethernet SGT Protection

In a Cisco TrustSec network, a Cisco Identity Services Engine (ISE) assigns a Layer 2 Security Group Tag (SGT) of 16 bits to a user's or endpoint's session. You can create a Zone Protection profile with Ethernet SGT protection when your firewall is part of a Cisco TrustSec network. The firewall can inspect headers with 802.1Q (Ethertype 0x8909) for specific Layer 2 security group tag (SGT) values and drop the packet if the SGT matches the list you configure for the Zone Protection profile attached to the interface. Determine which SGT values you want to deny access to a zone.

Configure Reconnaissance Protection

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

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Malicious actors use various scanning techniques, including port scans (TCP and UDP), host sweeps, and IP protocol scans, to identify and exploit network vulnerabilities. To protect your network against these scans, configure the Reconnaissance Protection settings of a Zone Protection profile. For each scan type, you will specify an action and the conditions that trigger the action. For example, you can *block* subsequent packets from an untrusted source if the firewall detects *1000* IP protocol scan events from that source within *60* seconds.

The following actions are supported for each scan:

- **Allow**—The firewall allows the port scan, host sweep, or IP protocol scan reconnaissance to continue.
- (Default) **Alert**—The firewall generates an alert for each port scan, host sweep, or IP protocol scan that matches the configured threshold within the specified time interval.
- **Block**—The firewall drops all subsequent packets from the source to the destination for the remainder of the specified time interval.
- **Block IP**—The firewall drops all subsequent packets for the specified **Duration**, in seconds (the range is 1-3,600). **Track By** determines whether the firewall blocks source or source-and-destination traffic.
- PAN-OS

PAN-OS

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STEP 1 | Configure Reconnaissance Protection.

1. Select **Network** > **Network Profiles** > **Zone Protection**.
2. Select a Zone Protection profile, or **Add** a new profile and enter a **Name** for it.
3. On the Reconnaissance Protection tab, select the scan types to protect against.
4. Select an **Action** for each scan.

   If you select Block IP, you must also configure the **Track By** (source or source-and-destination) and **Duration** options.
5. Set the **Interval** in seconds. This option defines the time interval for port scan, host sweep, and IP protocol scan detection.
6. Set the **Threshold** for reconnaissance events. The threshold defines the number of port scan, host sweep, or IP protocol scan events that need to occur within the specified time interval to trigger an action.
7. (Optional) Configure a Source Address Exclusion.

   Source Address Exclusions are IP addresses that you want to exclude from reconnaissance protection. You can specify up to 20 IP addresses or netmask address objects.

   Exclude only IP addresses for trusted internal groups that perform vulnerability testing.

   1. **Add** the address you want to exclude.
   2. Enter a descriptive **Name** for the address.
   3. For Address Type, select either **IPv4** or **IPv6**, and then select an address object or enter one manually.
   4. Click **OK**.
8. Click **OK** to save the Zone Protection profile.
9. **Commit** your changes.

STEP 2 | Apply the Zone Protection profile to the appropriate zones, including zones that connect to the internet.

Configure Packet Based Attack Protection

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

To enhance security for a zone, Packet-Based Attack Protection allows you to specify whether the firewall drops IP, IPv6, TCP, ICMP, or ICMPv6 packets that have certain characteristics or strips certain options from the packets.

For example, you can drop TCP SYN and SYN-ACK packets that contain data in the payload during a TCP three-way handshake. A Zone Protection profile by default is set to drop SYN and SYN-ACK packets with data (you must apply the profile to the zone).

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The TCP Fast Open option (RFC 7413) preserves the speed of a connection setup by including data in the payload of SYN and SYN-ACK packets. A Zone Protection profile treats handshakes that use the TCP Fast Open option separately from other SYN and SYN-ACK packets; the profile by default is set to allow the handshake packets if they contain a valid Fast Open cookie.

If you have existing Zone Protection profiles in place when you upgrade to PAN-OS 8.0, the three default settings will apply to each profile and the firewall will act accordingly.

Beginning with PAN-OS 8.1.2 and later releases, you can use a CLI command (step 4 in the PAN-OS tab) to enable the firewall to generate a Threat log when the firewall receives and drops the following types of packets, so that you can more easily analyze these occurrences and also fulfill audit and compliance requirements:

- Teardrop attack
- DoS attack using ping of death

Furthermore, the same CLI command also enables the firewall to generate Threat logs for the following types of packets if you enable the corresponding Packet Based Attack Protection:

- Fragmented IP packets
- IP address spoofing
- ICMP packets larger than 1024 bytes
- Packets containing ICMP fragments
- ICMP packets embedded with an error message
- First packets for a TCP session that are not SYN packets
- PAN-OS

PAN-OS

STEP 1 | Create a Zone Protection profile and configure Packet-Based Attack Protection settings.

1. Select **Network** > **Network Profiles** > **Zone Protection** and **Add** a new profile.
2. Enter a **Name** for the profile and an optional **Description**.
3. Select **Packet Based Attack Protection**.
4. On each tab (**IP Drop**, **TCP Drop**, **ICMP Drop**, **IPv6 Drop**, and **ICMPv6 Drop**), select the Packet-Based Attack Protection settings you want to enforce to protect a zone.
5. Click **OK**.

STEP 2 | Apply the Zone Protection profile to a security zone that is assigned to interfaces you want to protect.

1. Select **Network** > **Zones** and select the zone where you want to assign the Zone Protection profile.
2. **Add** the **Interfaces** belonging to the zone.
3. For **Zone Protection Profile**, select the profile you just created.
4. Click **OK**.

STEP 3 | **Commit** your changes.

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STEP 4 | (PAN-OS 8.1.2 and later releases) Enable the firewall to generate Threat logs for a teardrop attack and a DoS attack using ping of death, and also generate Threat logs for the types of packets listed above if you enable the corresponding packet-based attack protection (in Step 1). For example, if you enable packet-based attack protection for **Spoofed IP address**, using the following CLI causes the firewall to generate a Threat log when the firewall receives and drops a packet with a spoofed IP address.

1. Access the CLI.
2. Use the operational CLI command **`set system setting additional-threat-log on`**. Default is **`off`**.

Configure Protocol Protection in Strata Cloud Manager

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Strata Cloud Manager)	❑ Strata Cloud manager Essentials ❑ Strata Cloud Manager Pro

A Zone Protection profile configured for protocol protection defends your zones against non-IP protocol-based attacks. Configure protocol protection to block or allow non-IP protocols between your zones and interfaces. This allows you to reduce security risks and facilitate regulatory compliance by preventing less secure protocols from entering a zone or an interface in a zone. When you configure zone protection for non-IP protocols on zones that have Aggregate Ethernet (AE) interfaces, you must block or allow a non-IP protocol for all AE interface members. Enforcing non-IP protocols for only one AE interface member isn't supported.

By default, the predefined `intrazone-default` Security policy rule allows non-IP traffic between interfaces in the same zone.

To configure protocol protection, you create an **Exclude List** or **Include List** to which you add the non-IP protocols you want to deny or allow. A Zone Protection profile configured for protocol protection supports an exclude list, an include list, or both in a single profile.

*Protocol protection doesn't support blocking IPv4 (EtherType 0x0800), ARP (0x0806), or VLAN-tagged frames (0x8100). The firewall always implicitly allows these four Ethertypes in an **Include List** even if you don't explicitly add them and doesn't permit you to add them to an **Exclude List**.*

STEP 1 | Log in to Strata Cloud Manager.

STEP 2 | Select **Manage** > **Configuration** > **NGFW and Prisma Access** > **Security Services** > **DoS ProtectionConfiguration** > **NGFW and Prisma Access** > **Security Services** > **DoS Protection** and select the Configuration Scope where you want to create the Zone Protection profile.

You can select a folder or firewall from your **Folders** or select **Snippets** to configure the Zone Protection profile in a snippet.

STEP 3 | Navigate to the `Zone Protection Profiles` and **Add Profile**.

STEP 4 | Enter a descriptive **Name**.

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STEP 5 | (Optional) Enter a **Description**.

STEP 6 | Select **Protocol**.

STEP 7 | Configure the protocol protection **Rule Type**.

Select **Exclude List** to specify which protocols you want to deny from entering the zone. Select **Include List** to specify which protocols you want to allow to enter the zone.

Use an include list instead of an exclude list to control non-IP protocol traffic entering your zones. Include lists specifically sanction only the protocols you want to allow and block protocols not defined in the include list. This reduces your attack surface and blocks unknown traffic.

STEP 8 | Specify the non-IP protocols that you want to enforce.

1. **Add** the protocols you want to include in the Exclude List or Include List.
2. Enter the **Protocol Name**.
3. **Enable**.

   Protocols added to an Include List or Exclude List or enabled by default.

   You can modify an existing Zone Protection profile to disable a specific protocol from enforcement.
4. Enter the **Ethertype**.

   A list supports up to 64 EtherType entries identified by the IEEE hexadecimcal Ethertype code. Other sources of EtherType codes are https://standards-oui.ieee.org/ethertype/eth.txt and https://www.cavebear.com/archive/cavebear/Ethernet/type.html.

STEP 9 | **Save**.

Use Case: Non-IP Protocol Protection Between Security Zones on Layer 2 Interfaces

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

In this use case, the firewall is in a Layer 2 VLAN divided into two subinterfaces. VLAN 100 is 192.168.100.1/24, subinterface .6. VLAN 200 is 192.168.100.1/24, subinterface .7. Non-IP protocol protection applies to ingress zones. In this use case, if the Internet zone is the ingress zone, the firewall blocks the Generic Object Oriented Substation Event (GOOSE) protocol. If the User zone is the ingress zone, the firewall allows the GOOSE protocol. The firewall implicitly allows IPv4, IPv6, ARP, and VLAN-tagged frames in both zones.

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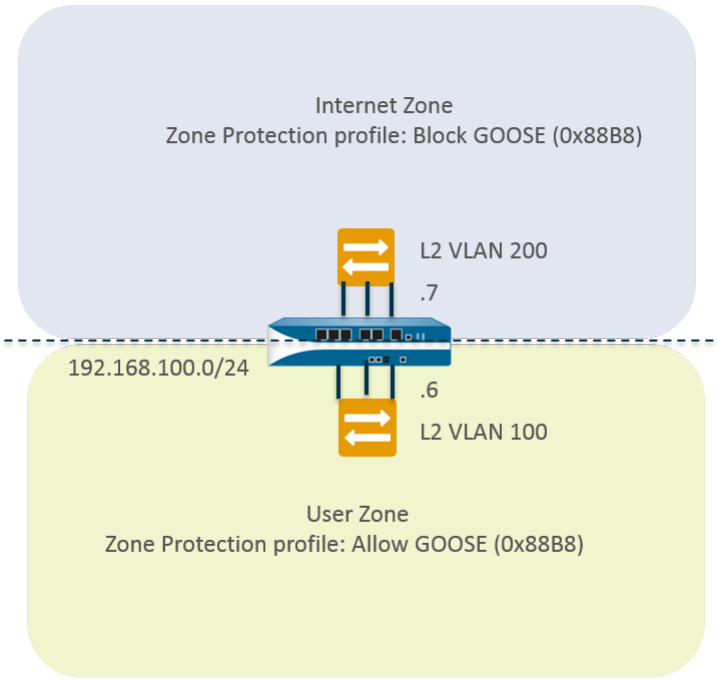

STEP 1 | Configure two VLAN subinterfaces.

1. Select **Network** > **Interfaces** > **VLAN** and **Add** an interface.
2. **Interface Name** defaults to vlan. After the period, enter 7.
3. On the **Config** tab, **Assign Interface To** the **VLAN** 200.
4. Click **OK**.
5. Select **Network** > **Interfaces** > **VLAN** and **Add** an interface.
6. **Interface Name** defaults to vlan. After the period, enter 6.
7. On the **Config** tab, **Assign Interface To** the **VLAN** 100.
8. Click **OK**.

STEP 2 | Configure protocol protection in a Zone Protection profile to block GOOSE protocol packets.

1. Select **Network** > **Network Profiles** > **Zone Protection** and **Add** a profile.
2. Enter the **Name** Block GOOSE.
3. Select **Protocol Protection**.
4. Choose **Rule Type** of **Exclude List**.
5. Enter the **Protocol Name**, GOOSE, to easily identify the Ethertype on the list. The firewall doesn't verify that the name you enter matches the Ethertype code; it uses only the Ethertype code to filter.
6. Enter **Ethertype** code 0x88B8. The Ethertype must be preceded by 0x to indicate a hexadecimal value. Range is 0x0000 to 0xFFFF.
7. Select **Enable** to enforce the protocol protection. You can disable a protocol on the list, for example, for testing.
8. Click **OK**.

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STEP 3 | Apply the Zone Protection profile to the Internet zone.

1. Select **Network** > **Zones** and **Add** a zone.
2. Enter the **Name** of the zone, Internet.
3. For **Location**, select the virtual system where the zone applies.
4. For **Type**, select **Layer2**.
5. **Add** the **Interface** that belongs to the zone, vlan.7.
6. For **Zone Protection Profile**, select the profile Block GOOSE.
7. Click **OK**.

STEP 4 | Configure protocol protection to allow GOOSE protocol packets.

Create another Zone protection profile named Allow GOOSE, and choose **Rule Type** of **Include List**.

When configuring an Include list, include all required non-IP protocols; an incomplete list can result in legitimate non-IP traffic being blocked.

STEP 5 | Apply the Zone Protection profile to the User zone.

1. Select **Network** > **Zones** and **Add** a zone.
2. Enter the **Name** of the zone, User.
3. For **Location**, select the virtual system where the zone applies.
4. For **Type**, select **Layer2**.
5. **Add** the **Interface** that belongs to the zone, vlan.6.
6. For **Zone Protection Profile**, select the profile Allow GOOSE.
7. Click **OK**.

STEP 6 | Commit.

Click **Commit**.

STEP 7 | View the number of non-IP packets the firewall has dropped based on protocol protection.

Access the CLI.

```
> show counter global name pkt_nonip_pkt_drop
> show counter global name pkt_nonip_pkt_drop delta yes
```

Use Case: Non-IP Protocol Protection Within a Security Zone on Layer 2 Interfaces

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

If you don't implement a Zone Protection profile with non-IP protocol protection, the firewall allows non-IP protocols in a single zone to go from one Layer 2 interface to another. In this use

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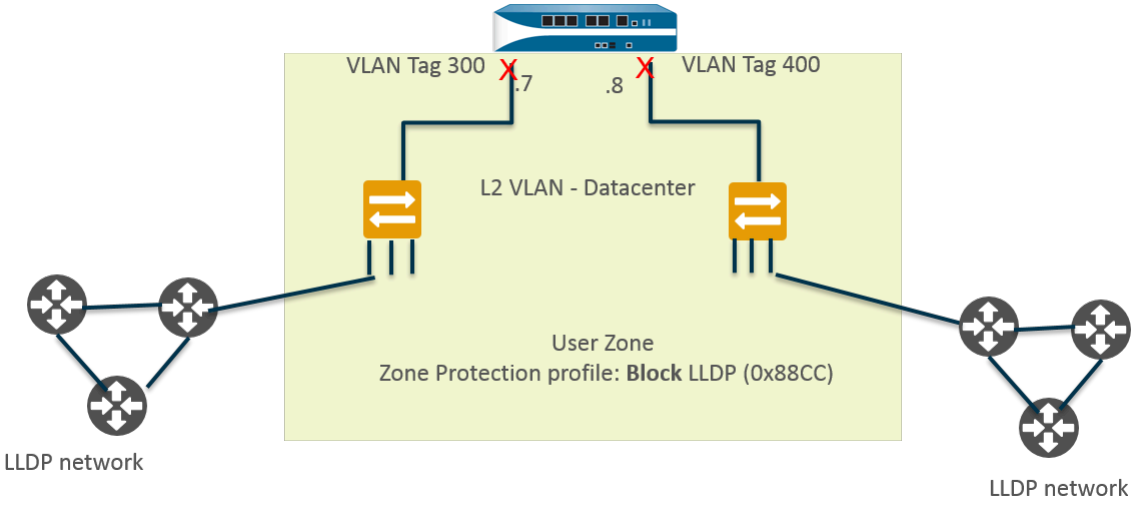
case, blocking LLDP packets ensures that LLDP for one network doesn't discover a network reachable through another interface in the zone.

In the following figure, the Layer 2 VLAN named Datacenter is divided into two subinterfaces: 192.168.1.1/24, subinterface .7 and 192.168.1.2/24, subinterface .8. The VLAN belongs to the User zone. By applying a Zone Protection profile that blocks LLDP to the User zone:

- Subinterface .7 blocks LLDP from its switch to the firewall at the red X on the left, preventing that traffic from reaching subinterface .8.
- Subinterface .8 blocks LLDP from its switch to the firewall at the red X on the right, preventing that traffic from reaching subinterface .7.



STEP 1 | Create a subinterface for an Ethernet interface.

1. Select **Network** > **Interfaces** > **Ethernet** and select a Layer 2 interface, in this example, ethernet1/1.
2. Select **Add Subinterfaces**.
3. The **Interface Name** defaults to the interface (ethernet 1/1). After the period, enter 7.
4. For **Tag**, enter 300.
5. For **Security Zone**, select User.
6. Click **OK**.

STEP 2 | Create a second subinterface for the Ethernet interface.

1. Select **Network** > **Interfaces** > **Ethernet** and select the Layer 2 interface: ethernet1/1.
2. Select **Add Subinterfaces**.
3. The **Interface Name** defaults to the interface (ethernet 1/1). After the period, enter 8.
4. For **Tag**, enter 400.
5. For **Security Zone**, select User.
6. Click **OK**.

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STEP 3 | Create a VLAN for the Layer2 interface and two subinterfaces.

1. Select **Network** > **VLANs** and **Add** a VLAN.
2. Enter the **Name** of the VLAN; for this example, enter Datacenter.
3. For **VLAN Interface**, select **None**.
4. For **Interfaces**, click **Add** and select the Layer 2 interface: ethernet1/1, and two subinterfaces: ethernet1/1.7 and ethernet1/1.8.
5. Click **OK**.

STEP 4 | Block non-IP protocol packets in a Zone Protection profile.

1. Select **Network** > **Network Profiles** > **Zone Protection** and **Add** a profile.
2. Enter the **Name**, in this example, Block LLDP.
3. Enter a profile **Description**—Block LLDP packets from an LLDP network to other interfaces in the zone (intrazone).
4. Select **Protocol Protection**.
5. Choose **Rule Type** of **Exclude List**.
6. Enter **Protocol Name** LLDP.
7. Enter **Ethertype** code 0x88cc. The Ethertype must be preceded by 0x to indicate a hexadecimal value.
8. Select **Enable**.
9. Click **OK**.

STEP 5 | Apply the Zone Protection profile to the security zone to which Layer 2 VLAN belongs.

1. Select **Network** > **Zones**.
2. **Add** a zone.
3. Enter the **Name** of the zone, User.
4. For **Location**, select the virtual system where the zone applies.
5. For **Type**, select **Layer2**.
6. **Add** an **Interface** that belongs to the zone, ethernet1/1.7
7. **Add** an **Interface** that belongs to the zone, ethernet1/1.8.
8. For **Zone Protection Profile**, select the profile Block LLDP.
9. Click **OK**.

STEP 6 | Commit.

Click **Commit**.

STEP 7 | View the number of non-IP packets the firewall has dropped based on protocol protection.

Access the CLI.

```
> show counter global name pkt_nonip_pkt_drop
> show counter global name pkt_nonip_pkt_drop delta yes
```

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Configure Ethernet SGT Protection

Where Can I Use This?	What Do I Need?
• NGFW	One of these licenses when using Strata Cloud Manager ❑ Strata Cloud Manager Essentials ❑ Strata Cloud Manager Pro

Use the following task to configure an Ethernet SGT Protection profile.

- PAN-OS & Panorama
- Strata Cloud Manager

Configure Ethernet SGT Protection (PAN-OS)

STEP 1 | Create a Zone Protection profile to provide Ethernet SGT Protection.

1. Select **Network** > **Network Profiles** > **Zone Protection**.
2. **Add** a Zone Protection profile by **Name**.
3. Select **Ethernet SGT Protection**.
4. **Add** a **Layer 2 SGT Exclude List** by name.
5. Enter one or more **Tag** values for the list; range is 0 to 65,535. You can enter individual entries that are a contiguous range of tag values (for example, 100-500). You can add up to 100 (individual or range) tag entries in an Exclude List.
6. **Enable** the Layer 2 SGT Exclude List. You can disable the list at any time.
7. Click **OK**.

STEP 2 | Apply the Zone Protection profile to the security zone to which the Layer 2, virtual wire, or tap interfaces belong.

1. Select **Network** > **Zones**.
2. **Add** a zone.
3. Enter the **Name** of the zone.
4. For **Location**, select the virtual system where the zone applies.
5. For **Type**, select **Layer2**, **Virtual Wire**, or **Tap**.
6. **Add** an **Interface** that belongs to the zone.
7. For **Zone Protection Profile**, select the profile you created.
8. Click **OK**.

STEP 3 | **Commit**.

STEP 4 | View the global counter of packets that the firewall dropped as a result of all Zone Protection profiles that employ Ethernet SGT Protection.

1. Access the CLI.
2. `> show counter global name pan_flow_dos_l2_sec_tag_drop`

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Configure Ethernet SGT Protection (SCM)

STEP 1 | Log in to Strata Cloud Manager.

STEP 2 | Select **Manage** > **Configuration** > **NGFW and Prisma Access** > **Security Services** > **DoS ProtectionConfiguration** > **NGFW and Prisma Access** > **Security Services** > **DoS Protection** and select the Configuration Scope where you want to create the Zone Protection profile.

You can select a folder or firewall from your **Folders** or select **Snippets** to configure the Zone Protection profile in a snippet.

STEP 3 | Navigate to the `Zone Protection Profiles` and **Add Profile**.

STEP 4 | Enter a descriptive **Name**.

STEP 5 | (Optional) Enter a **Description**.

STEP 6 | Select **Ethernet SGT**.

STEP 7 | **Add** a Layer 2 SGT Exclude List by name.

STEP 8 | Enter one or more **Tag** values for the list.

Range is **`0`** to **`65,535`**. You can enter individual entries that are a contiguous range of tag values (for example, **`100-500`**). You can add up to 100 (individual or range) tag entries in an Exclude List.

STEP 9 | **Enable** the Layer 2 SGT Exclude List.

Layer 2 SGT Exclude Lists are enabled by default when added.

You can modify an existing Zone Protection profile to disable a specific Layer 2 SGT Exclude List from enforcement.

STEP 10 | **Save**.

Packet Buffer Protection

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

Packet Buffer Protection defends your firewall and network from single session DoS attacks that can overwhelm the firewall's packet buffer and cause legitimate traffic to drop. Although you don't configure Packet Buffer Protection in a Zone Protection profile or in a DoS Protection profile or policy rule, Packet Buffer Protection defends ingress zones. While zone and DoS protection apply to new sessions (connections) and are granular, Packet Buffer Protection applies to existing sessions and is global.

You Configure Packet Buffer Protection globally to protect the entire firewall and you also enable Packet Buffer Protection on each zone to protect zones:

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- **Global Packet Buffer Protection**—The firewall monitors sessions from all zones (regardless of whether Packet Buffer Protection is enabled in a zone) and how those sessions utilize the packet buffer. You must configure Packet Buffer Protection globally (**Device** > **Setup** > **Session Settings**) to protect the firewall and to enable it on individual zones. When packet buffer consumption reaches the configured **Activate** percentage, the firewall used Random Early Drop (RED) to drop packets from the offending sessions (the firewall doesn't drop complete sessions at the global level).
- **Per-Zone Packet Buffer Protection**—Enable Packet Buffer Protection on each zone (**Network** > **Zones**) to layer in a second level of protection. When packet buffer consumption crosses the **Activate** threshold and global protection begins to apply RED to session traffic, that starts the **Block Hold Time** timer. The **Block Hold Time** is the amount of time in seconds that the offending session can continue before the firewall blocks the entire session. The offending session remains blocked until the **Block Duration** time expires.

You must enable Packet Buffer Protection globally in order for it to be active in zones.

There are two types of packet buffer protection:

- Packet Buffer Protection Based on Buffer Utilization
- Packet Buffer Protection Based on Latency

Packet Buffer Protection Based on Buffer Utilization

Packet Buffer Protection based on buffer utilization is enabled by default. Take baseline measurements of firewall packet buffer utilization over a period of time until you're comfortable that you understand typical usage. Take measurements for at least one business week; however, a longer measurement period provides a better baseline. To see packet buffer utilization for a specified period of time, use the operational CLI command:

```
admin1138@thxvm1>show running resource-monitor [day | hour | minute |
 second | week]
```

The CLI command provides a snapshot of buffer utilization for the specified period of time, but is neither automated nor continuous. To automate continuous packet buffer utilization measurements so you can monitor changes in behavior and anomalous events, use a script.

If baseline measurements consistently show abnormally high packet buffer utilization, then the firewall's capacity may be undersized for typical traffic loads. In this case, consider resizing the firewall deployment. Otherwise, you need to tune the Packet Buffer Protection thresholds carefully to prevent impacted buffers from overflowing (and to prevent dropping legitimate traffic). When firewall sizing is correct for the deployment, only an attack should cause a large spike in buffer usage.

Overrunning the firewall packet buffer negatively impacts the firewall's packet forwarding capabilities. When the buffers are full, no packets can enter the firewall on any interface, not just the interface that experienced the attack.

The best practices for setting the thresholds are:

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- **Alert** and **Activate**—Start with the default threshold values, monitor packet buffer utilization, and adjust the thresholds as necessary. The **Alert** threshold defaults to 50%; when packet buffer utilization exceeds the threshold for more than 10 seconds, the firewall creates an alert entry in the System log every minute. The **Activate** threshold defaults to 80%; when the threshold is reached, the firewall begins to mitigate the most abusive sessions. If the firewall is sized correctly, buffer utilization should be well below 50%.
- **Block Hold Time**—When packet buffer utilization triggers the **Activate** threshold, the **Block Hold Time** sets the amount of time the offending session can continue before the firewall blocks the session. During the **Block Hold Time**, the firewall continues to apply RED to the packets of offending sessions. Start with the default **Block Hold Time** threshold value (60 seconds), monitor packet buffer utilization, and adjust the threshold as necessary. If the packet buffer utilization percentage falls below the **Activate** threshold before the **Block Hold Time** expires, the timer resets and doesn't start until the **Activate** threshold is crossed again. Increasing the **Block Hold Time** imposes a greater penalty on offending sessions and reducing it imposes a lesser penalty on offending sessions.
- **Block Duration**—When the **Block Hold Time** expires, the firewall blocks the offending session for the period of time defined by the **Block Duration**. Start with the default threshold value (3600 seconds), monitor packet buffer utilization, and adjust the threshold as necessary. When you enable Packet Buffer Protection on a zone, **Block Duration** affects every session from the IP address even if only one session from an IP address overutilizes the packet buffer. If you believe that blocking an IP address for one hour (3600 seconds) is too great a penalty, reduce the **Block Duration** to an acceptable value.

In addition to monitoring the buffer utilization of individual sessions, Packet Buffer Protection can also block an IP address if certain criteria are met. While the firewall monitors the packet buffers, if it detects a source IP address rapidly creating sessions that would not individually be seen as an attack, it blocks that IP address for the configured **Block Duration**.

Network Address Translation (NAT) *(an external source that has translated its internet-bound traffic using source NAT) can give the appearance of greater packet buffer utilization because of IP address translation activity. If this occurs, adjust the thresholds in a way that penalizes individual sessions but doesn't penalize the underlying IP addresses (so other sessions from the same IP address aren't affected). To do this, reduce the* ***Block Hold Time*** *so the firewall blocks individual sessions that overutilize the buffers faster, and reduce the* ***Block Duration*** *so that the underlying IP address is not unduly penalized.*

Packet Buffer Protection Based on Latency

You can trigger packet buffer protection based on packet latency caused by dataplane packet buffering, which indicates congestion on the firewall. Such packet buffer protection mitigates head-of-line blocking by alerting you to the congestion and performing random early drop (RED) on packets. Packet buffer protection based on latency can trigger the protection before latency-sensitive protocols or applications are affected.

For PAN-OS 11.2.4 and earlier versions, you must choose between enabling packet buffer protection based on buffer utilization (default) or packet buffer protection based on latency. In PAN-OS 11.2.5 and later versions, you can have both enabled and configured simultaneously.

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If your traffic includes protocols or applications that are latency-sensitive, then packet buffer protection based on latency will be more helpful than packet buffer protection based on buffer utilization.

Packet buffer protection based on latency includes setting a **Latency Alert** threshold (in milliseconds), above which the firewall starts generating an Alert log event. The **Latency Activate** threshold indicates when the firewall activates RED on incoming packets and starts generating an Activate log. The **Latency Max Tolerate** threshold indicates when the firewall uses with RED with almost 100% drop probability.

The **Block Hold Time** and **Block Duration** settings function for packet buffer protection based on latency in the same way they do for packet buffer protection based on utilization.

Configure Packet Buffer Protection

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

You can configure Packet Buffer Protection at two levels: the device level (global) and if enabled globally, you can also enable it at the zone level. Global packet buffer protection (**Device** > **Setup** > **Session**) is to protect firewall resources and ensure that malicious traffic does not cause the firewall to become non-responsive.

Packet buffer protection per ingress zone (**Network** > **Zones**) is a second layer of protection that starts blocking the offending IP address if it continues to exceed the packet buffer protection thresholds. The firewall can block all traffic from the offending source IP address. Keep in mind that if the source IP address is a translated NAT IP address, many users can be using the same IP address. If one abusive user triggers packet buffer protection and the ingress zone has packet buffer protection enabled, all traffic from that offending source IP address (even from non-abusive users) can be blocked when the firewall puts the IP address on its block list.

The most effective way to block DoS attacks against a service behind the firewall is to configure packet buffer protection globally and per ingress zone.

You can **Enable Packet Buffer Protection** for a zone, but it is not active until you enable packet buffer protection globally and specify the settings.

In PAN-OS 11.2.5 and later versions, you can also configure packet buffer protection based on latency alongside these settings.

STEP 1 | Enable packet buffer protection globally.

1. Select **Device** > **Setup** > **Session** and edit the Session Settings.
2. Select **Packet Buffer Protection**.
3. Define the packet buffer protection behavior:
   - **Alert (%)**—When packet buffer utilization exceeds this threshold for more than 10 seconds, the firewall creates a log event every minute. Range s 0% to 99%; default is 50%. If the value is 0%, the firewall does not create a log event.
   - **Activate (%)**—When packet buffer utilization reaches this threshold, the firewall begins to mitigate the most abusive sessions by applying random early drop (RED).

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Range is 0% to 99%; default is 50%. If the value is 0%, the firewall does not apply RED. If the abuser is ingressing a zone that has Packet Buffer Protection enabled, the firewall can also discard the abusive session or block the offending source IP address. Start with the default threshold and adjust it if necessary.

The firewall records alert events in the System log, and records events for dropped traffic, discarded sessions, and blocked IP address in the Threat log.

- **Block Countdown Threshold (%)**—The buffer utilization percentage that starts the countdown to discard or block offending traffic. When buffer congestion reaches the **Block Countdown Threshold** threshold, **Block Hold Time** begins to decrement. (When the block hold time runs out, the firewall discards sessions or blocks offending hosts.); default is 80%.
- **Block Hold Time (sec)**—Number of seconds a RED-mitigated session is allowed to continue before the firewall discards it. Range is 0 to 65,535; default is 60. If the value is 0, the firewall does not discard sessions based on packet buffer protection.
- **Block Duration (sec)**—Number of seconds a session remains discarded or an IP address remains blocked. Range is 1 to 15,999,999; default is 3,600.

4. Click **OK**.
5. **Commit** your changes.

STEP 2 | Enable additional packet buffer protection on an ingress zone.

1. Select **Network** > **Zones**.
2. Choose an ingress zone and click on its name.
3. **Enable Packet Buffer Protection** in the Zone Protection section.
4. Click **OK**.
5. **Commit** your changes.

Configure Packet Buffer Protection Based on Latency

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

Configure packet buffer protection based on latency and apply it to zones that have traffic consisting of protocols and applications that are latency-sensitive.

In PAN-OS 11.2.5 and later versions, you can configure packet buffer protection based on latency alongside the global packet buffer protection settings.

STEP 1 | Select **Device** > **Setup** > **Session**.

STEP 2 | Edit the Session Settings section and enable **Packet Buffer Protection**.

STEP 3 | Enable **Buffering Latency Based**.

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STEP 4 | Enter the **Latency Alert (milliseconds)** threshold above which the firewall starts generating an Alert log event every minute; range is 1 to 20,000; default is 50.

STEP 5 | Enter the **Latency Activate (milliseconds)** threshold above which the firewall activates random early drop (RED) on incoming packets and starts generating an Activate log every 10 seconds; range is 1 to 20,000ms; default is 200ms.

STEP 6 | Enter the **Latency Max Tolerate (milliseconds)** threshold above which the firewall uses RED with close to 100% drop probability; range is 1 to 20,000ms; default is 500ms.

If the current latency is a value between the **Latency Activate** threshold and the **Latency Max Tolerate** threshold, the firewall calculates the RED drop probability as follows: (current latency - **Latency Activate** threshold) / (**Latency Max Tolerate** threshold - **Latency Activate** threshold). For example, if the current latency is 300, **Latency Activate** is 200, and **Latency Max Tolerate** is 500, then (300-200)/(500-200) = 1/3, meaning the firewall uses approximately 33% RED drop probability.

STEP 7 | Configure the **Block Hold Time** and **Block Duration** as for Packet Buffer Protection based on utilization.

STEP 8 | Click **OK**.

STEP 9 | Enable the second layer of protection for each zone where you want packet buffer protection based on latency.

1. Select **Network** > **Zones** and select a zone.
2. Enable **Packet Buffer Protection**.

STEP 10 | **Commit**.

DoS Protection Profiles and Policy Rules

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

DoS Protection profiles and DoS Protection policy rules combine to protect specific groups of critical resources and individual critical resources against session floods. Compared to Zone Protection profiles, which protect entire zones from flood attacks, DoS protection provides granular defense for specific systems, especially critical systems that users access from the internet and are often attack targets, such as web servers and database servers. Apply both types of protection because if you only apply a Zone Protection profile, then a DoS attack that targets a particular system in the zone can succeed if the total connections-per-second (CPS) doesn't exceed the zone's **Activate** and **Maximum** rates.

DoS Protection is resource-intensive, so use it only for critical systems. Similar to Zone Protection profiles, DoS Protection profiles specify flood thresholds. DoS Protection policy rules determine the devices, users, zones, and services to which DoS Profiles apply.

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In addition to configuring DoS protection and zone protection, apply the best practice Vulnerability Protection profile to each Security policy rule to help defend against DoS attacks.

Classified Versus Aggregate DoS Protection

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

You can configure *aggregate* and *classified* DoS Protection Profiles, and apply one profile or one of each type of profile to DoS Protection Policy Rules when you configure DoS Protection.

- **Aggregate**—Sets thresholds that apply to the entire group of devices specified in a DoS Protection policy rule instead of to each individual device, so one device could receive the majority of the allowed connection traffic. For example, a **Max Rate** of 20,000 CPS means the total CPS for the group is 20,000, and an individual device can receive up to 20,000 CPS if other devices don't have connections. Aggregate DoS Protection policies provide another layer of broad protection (after your dedicated DDoS device at the internet perimeter and Zone Protection profiles) for a particular group of critical devices when you want to apply extra constraints on specific subnets, users, or services.
- **Classified**—Sets flood thresholds that apply to each individual device specified in a DoS Protection policy rule. For example, if you set an **Max Rate** of 5,000 CPS, each device specified in the rule can accept up to 5,000 CPS before it drops new connections. If you apply a classified DoS Protection policy rule to more than one device, the devices governed by the rule should be similar in terms of capacity and how you want to control their CPS rates because classified thresholds apply to each individual device. Classified profiles protect individual critical resources.

  When you configure a DoS Protection policy rule with a classified DoS Protection profile (**Option/Protection** > **Classified** > **Address**), use the **Address** field to specify whether incoming connections count toward the profile thresholds based on matching the **source-ip-only**, **destination-ip-only**, or **scr-dest-ip-both** (the firewall counts both the source and the destination IP address matches toward the thresholds). Counters consume resources, so the way you count address matches affects firewall resource consumption. You can use classified DoS protection to:

  - Protect critical individual devices, especially servers that users access from the internet and are often attack targets, such as web servers, database servers, and DNS servers. Set appropriate flood and resource protection thresholds in a classified DoS Protection profile. Create a DoS Protection policy rule that applies the profile to each server's IP address by

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adding the IP addresses as the rule's destination criteria, and set the **Address** to **destination-ip-only**.

*Do not use **source-IP-only** or **src-dest-ip-both** classification for internet-facing zones in classified DoS Protection policy rules because the firewall doesn't have the capacity to store counters for every possible IP address on the internet. Increment the threshold counter for source IPs only for internal zone or same-zone rules. In perimeter zones, use **destination-ip-only**.*

- Monitor the CPS rate for a suspect host or group of hosts (the zone that contains the hosts cannot be internet-facing). Set an appropriate alarm threshold in a classified DoS Protection profile to notify you if a host initiates an unusually large number of connections. Create a DoS Protection policy rule that applies the profile to the individual source or source address group and set the **Address** to **source-ip-only**. Investigate hosts that initiate enough new connections to set off the alarm.

How you configure the **Address** (**source-ip-only**, **destination-ip-only**, or **src-dest-ip-both**) for classified profiles depends on your DoS protection goals, what you are protecting, and whether the protected device(s) are in internet-facing zones.

*The firewall uses more resources to track **src-dest-ip-both** as the **Address** than to track **source-IP-only** or **destination-ip-only** because the counters consume resources for both the source and destination IP addresses instead of just one of the two.*

(PAN-OS 11.2.3 and later versions) When you configure a classified DoS profile with **destination-ip-only**, you enable the firewall to block offending source IP addresses using both the hardware and software block tables. This method helps keep network connectivity up while protecting firewall resources from overconsumption.

Hardware ACL blocking must be supported and enabled on your firewall to use both the hardware and software block tables. The following firewalls support hardware ACL blocking:

- PA-3200 Series firewalls
- PA-3400 Series firewalls
- PA-5200 Series firewalls
- PA-5450 firewall
- PA-7000 Series firewalls
- PA-7500 firewall

The PA-5410, PA-5420, and PA-5430 firewalls (Thunderbolt platform) do not support hardware ACL offload.

If you apply both an aggregate and a classified DoS Protection profile to the same DoS Protection policy rule, the firewall applies the aggregate profile first and then applies the classified profile if needed. For example, we protect a group of five web servers with both types of profiles in a DoS Protection policy rule. The aggregate profile configuration drops new connections when the combined total for the group reaches a **Max Rate** of 25,000 CPS. The classified profile configuration drops new connections to any individual web server in the group when it reaches a **Max Rate** of 6,000 CPS. There are three scenarios where new connection traffic crosses **Max Rate** thresholds:

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- The new CPS rate exceeds the aggregate **Max Rate** but doesn't exceed the classified **Max Rate**. In this scenario, the firewall applies the aggregate profile and blocks all new connections for the configured Block Duration.
- The new CPS rate doesn't exceed the aggregate **Max Rate**, but the CPS to one of the web servers exceeds the classified **Max Rate**. In this scenario, the firewall checks the aggregate profile and finds that the rate for the group is less than 25,000 CPS, so the firewall doesn't block new connections based on that. Next, the firewall checks the classified profile and finds that the rate for a particular server exceeds 6,000 CPS. The firewall applies the classified profile and blocks new connections to that particular server for the configured Block Duration. Because the other servers in the group are within the classified profile's **Max Rate**, their traffic is not affected.
- The new CPS rate exceeds the aggregate **Max Rate** and also exceeds the classified **Max Rate** for one of the web servers. In this scenario, the firewall checks the aggregate profile and finds that the rate for the group exceeds 25,000 CPS, so the firewall blocks new connections to limit the group's total CPS. The firewall then checks the classified profile and finds that the rate for a particular server exceeds 6,000 CPS (so the aggregate profile enforced the group's combined limit, but that wasn't enough to protect this particular server). The firewall applies the classified profile and blocks new connections to that particular server for the configured Block Duration. Because the other servers in the group are within the classified profile's **Max Rate**, their traffic is not affected.

*If you want both an aggregate and a classified DoS Protection profile to apply to the same traffic, you must apply both profiles to the same DoS Protection policy rule. If you apply the aggregate profile to one rule and the classified profile to a different rule, even if they specify exactly the same traffic, the firewall can apply only one profile because when the traffic matches the first DoS Protection policy rule, the firewall executes the **Action** specified in that rule and doesn't compare to the traffic to any subsequent rules, so the traffic never matches the second rule and the firewall can't apply its action. (This is the same way that Security policy rules work.)*

DoS Protection Profiles

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

DoS Protection profiles set thresholds that protect against new session IP flood attacks and provide resource protection (maximum concurrent session limits for specified endpoints and resources). DoS Protection profiles protect specific devices (classified profiles) and groups of devices (aggregate profiles) against SYN, UDP, ICMP, ICMPv6, and Other IP flood attacks. Configuring Flood Protection thresholds in a DoS Protection profile is similar to configuring Flood Protection in a Zone Protection profile, but Zone Protection profiles protect entire ingress zones, while DoS protection profiles and policy rules are granular and targeted, and can even be classified to a single device (IP address). The firewall measures the aggregate number of connections-per-second (CPS) to a group of devices (aggregate profile) or measures the CPS to individual devices (classified profile).

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Measure and monitor firewall dataplane CPU consumption to ensure that each firewall is properly sized to support DoS and Zone Protection and any other features that consume CPU cycles, such as decryption. If you use Panorama to manage your firewalls, Device Monitoring *(**Panorama** > **Managed Devices** > **Health** > **All Devices**) shows you the CPU and memory consumption of each managed firewall. It can also show you a 90-day trend line of CPU average and peak use to help you understand the typical available capacity of each firewall.*

For each flood type, you set three thresholds for new CPS to a group of devices (aggregate) or to individual devices (classified) and a **Block Duration**, and you can set a drop **Action** for SYN floods:

- **Alarm Rate**—When new CPS exceeds this threshold, the firewall generates a DoS alarm. For classified profiles, set the rate to 15-20% above the device's average CPS rate so that normal fluctuations don't cause alerts. For aggregate profiles, set the rate to 15-20% above the group's average CPS rate.
- **Activate Rate**—When new CPS exceeds this threshold, the firewall begins to drop new connections to mitigate the flood until the CPS rate drops below the threshold. For classified profiles, the **Max Rate** should be an acceptable CPS rate for the device(s) you're protecting (the **Max Rate** won't flood the critical device(s)). You can set the **Activate Rate** to the same threshold as the **Max Rate** so that the firewall doesn't use RED or SYN Cookies to begin dropping traffic before it reaches the **Max Rate**. Set the **Activate Rate** lower than the **Max Rate** only if you want to drop traffic before it reaches the **Max Rate**. For aggregate profiles, set the threshold just above the average peak CPS rate for the group to begin mitigating floods using RED (or SYN Cookies for SYN floods).
- **Max Rate**—When new CPS exceeds this threshold, the firewall blocks (drops) all new connections from the offending IP address for the specified **Block Duration** time period. For classified profiles, base the **Max Rate** threshold on the capacity of the device(s) you're protecting so that the CPS rate can't flood them. For aggregate profiles, set to 80-90% of the group's capacity.
- **Block Duration**—When new CPS exceeds the **Max Rate**, the firewall blocks new connections from the offending IP address. The **Block Duration** specifies the amount of time the firewall continues to block the IP address's new connections. While the firewall blocks new connections, it doesn't count incoming connections and doesn't increment the threshold counters. For classified and aggregate profiles, use the default value (300 seconds) to block the attacking session without penalizing legitimate sessions from the source for too long a period of time.

*SYN Flood Protection is the only type for which you set the drop **Action**. Start by setting the **Action** to **SYN Cookies**. SYN Cookies treats legitimate traffic fairly and only drops traffic that fails the SYN handshake, while using Random Early Drop drops traffic randomly, so RED may affect legitimate traffic. However, SYN Cookies is more resource-intensive because the firewall acts as a proxy for the target server and handles the three-way handshake for the server. The tradeoff is not dropping legitimate traffic (SYN Cookies) versus preserving firewall resources (RED). Monitor the firewall, and if SYN Cookies consumes too many resources, switch to RED. If you don't have a dedicated DDoS prevention device in front of the firewall, always use RED as the drop mechanism.*

The default threshold values are high so that DoS Protection profiles don't unexpectedly drop legitimate traffic. Monitor connection traffic and adjust the thresholds to values appropriate for

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your network. Start by taking baseline measurements of average and peak CPS for each flood type to determine the normal traffic conditions for the critical devices you want to protect. Because normal traffic loads experience some fluctuation, it's best not to drop connections too aggressively. Monitor and adjust the flood thresholds as needed and as your network evolves.

Another method of setting flood thresholds is to use the baseline measurements to set the maximum CPS you want to allow and work back from there to derive reasonable flood mitigation alarm and activation rates.

*Firewalls with multiple dataplane processors (DPs) distribute connections across DPs. In general, the firewall divides the CPS threshold settings equally across its DPs. For example, if a firewall has five DPs and you set the **Alarm Rate** to 20,000 CPS, each DP has an **Alarm Rate** of 4,000 CPS (20,000 / 5 = 4,000), so if the new sessions on a DP exceeds 4,000, it triggers the **Alarm Rate** threshold for that DP.*

In addition to setting IP flood thresholds, you can also use DoS Protection profiles to detect and prevent session exhaustion attacks in which a large number of hosts (bots) establish as many sessions as possible to consume a target's resources. On the profile's **Resources Protection** tab, you can set the maximum number of concurrent sessions that the device(s) defined in the DoS Protection policy rule to which you apply the profile can receive. When the number of concurrent sessions reaches its maximum limit, new sessions are dropped.

The maximum number of concurrent sessions to set depends on your network context. Understand the number of concurrent sessions that the resources you are protecting (defined in the DoS Protection policy rule to which you attach the profile) can handle. Set the threshold to approximately 80% of the resources' capacity, then monitor and adjust the threshold as needed.

For aggregate profiles, the **Resources Protection** threshold applies to all traffic of the devices defined in the policy rule (source and destination). For classified profiles, the **Resources Protection** threshold applies to the traffic based on whether the classified policy rule applies to the source IP only, to the destination IP only, or to both the source and destination IPs.

DoS Protection Policy Rules

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

DoS Protection policy rules control the systems to which the firewall applies DoS protection (the flood thresholds configured in DoS Protection profiles that you attach to DoS Protection policy rules), what action to take when traffic matches the criteria defined in the rule, and how to log DoS traffic. Because DoS protection consumes firewall resources, use it only to defend specific critical resources against session floods, especially common targets that users access from the internet, such as web servers and database servers. Use Zone Protection profiles to protect entire zones against floods and other attacks. DoS Protection policy rules provide granular matching criteria so that you have the flexibility to define exactly what you want to protect:

- Source zone, interface, IP address (including whole regions), and user.

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- Destination zone, interface, and IP address (including whole regions).

  *(Available in PAN-OS 12.1 and later) If your network operates using IPv6 (or a dual stack deployment) and would like to specify a region with IPv6 geolocation for the source and destination IP addresses, you must **Enable IPv6 Firewalling** as well as **Enable IPv6 Geolocation** from **Device** > **Sessions** > **Session Settings**.*

  The following platforms configured with less than 9GB do not support IPv6 geolocation:

  - *Hardware NGFWs: PA-410, PA-410R, PA-410R-5G, PA-415, and PA-415-5G*
  - *VM-Series: Software NGFW Credits and VM-Series models with less than 9GB (e.g. VM-100, VM-50, etc)*

- Services (by port and protocol). DoS protection applies only to the services you specify. However, specifying services doesn't allow the services and implicitly block all other services. Specifying services limits DoS protection to those services, but doesn't block other services.

  In addition to protecting service ports in use on critical servers, you can also protect against DoS attacks on the unused service ports of critical servers. For critical systems, you can do this by creating one DoS Protection policy rule and profile to protect ports with services running, and a different DoS Protection policy rule and profile to protect ports with no services running. For example, you can protect a web server's normal service ports, such as 80 and 443, with one policy/profile, and protect all of the other service ports with the other policy/profile. Be aware of the firewall's capacity so that servicing the DoS counters doesn't impact performance.

When traffic matches a DoS Protection policy rule, the firewall takes one of three actions:

- **Deny**—The firewall denies access and doesn't apply a DoS Protection profile. Traffic that matches the rule is blocked.
- **Allow**—The firewall permits access and doesn't apply a DoS Protection profile. Traffic that matches the rule is allowed.
- **Protect**—The firewall protects the devices defined in the DoS Protection policy rule by applying the specified DoS Protection profile or profiles thresholds to traffic that matches the rule. A rule can have one aggregate DoS Protection profile and one classified DoS Protection profile, and for classified profiles, you can use the source IP, destination IP, or both to increment the flood threshold counters, as described in Classified Versus Aggregate DoS Protection. Incoming packets count against both DoS Protection profile thresholds if the they match the rule.

The firewall applies DoS Protection profiles only if the **Action** is **Protect**. If the DoS Protection policy rule's **Action** is **Protect**, specify the appropriate aggregate and/or classified DoS Protection profiles in the rule so that the firewall applies the DoS Protection profile's thresholds to traffic that matches the rule. Most rules are **Protect** rules.

The **Allow** and **Deny** actions enable you to make exceptions within larger groups but do not apply DoS protection to the traffic. For example, you can deny the traffic from most of a group but allow a subset of that traffic. Conversely, you can allow the traffic from most of a group and deny a subset of that traffic.

You can **Schedule** when a DoS Protection policy rule is active (start and end time, recurrence period). One use case for scheduling is to apply different flood thresholds at different times

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of the day or week. For example, if your business experiences significantly less traffic at night than during the day, you may want to apply higher flood thresholds during the day than at night. Another use case is to schedule special thresholds for special events, providing that the firewall supports the CPS rates.

For easier management and granular reporting, configure **Log Forwarding** to separate DoS protection logs from other threat logs. Forward DoS threshold violation events directly to the administrators via email in addition to forwarding the logs to a server such an SNMP or syslog server. Providing that the firewalls are appropriately sized, threshold breaches should not be frequent and will be strong indicators of an attack attempt.

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DoS Protection Against Flooding of New Sessions

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

DoS protection against flooding of new sessions is beneficial against high-volume single-session and multiple-session attacks. In a single-session attack, an attacker uses a single session to target a device behind the firewall. If a Security rule allows the traffic, the session is established and the attacker initiates an attack by sending packets at a very high rate with the same source IP address and port number, destination IP address and port number, and protocol, trying to overwhelm the target. In a multiple-session attack, an attacker uses multiple sessions (or connections per second [cps]) from a single host to launch a DoS attack.

This feature defends against DoS attacks of new sessions only, that is, traffic that has not been offloaded to hardware. An offloaded attack is not protected by this feature. However, this topic describes how you can create a Security policy rule to reset the client; the attacker reinitiates the attack with numerous connections per second and is blocked by the defenses illustrated in this topic.

DoS Protection Profiles and Policy Rules work together to provide protection against flooding of many incoming SYN, UDP, ICMP, and ICMPv6 packets, and other types of IP packets. You determine what thresholds constitute flooding. In general, the DoS Protection profile sets the thresholds at which the firewall generates a DoS alarm, takes action such as Random Early Drop, and drops additional incoming connections. A DoS Protection policy rule configured to protect (rather than to allow or deny packets) determines the criteria for packets to match (such as source address) in order to be counted toward the thresholds. This flexibility allows you to block certain traffic, or allow certain traffic and treat other traffic as DoS traffic. When the incoming rate exceeds your maximum threshold, the firewall blocks incoming traffic from the source address.

Multiple-Session DoS Attack

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

Configure DoS Protection Against Flooding of New Sessions by configuring a DoS Protection policy rule, which determines the criteria that, when matched by incoming packets, trigger the **Protect** action. The DoS Protection profile counts each new connection toward the Alarm Rate, Activate Rate, and Max Rate thresholds. When the incoming new connections per second exceed the Activate Rate, the firewall takes the action specified in the DoS Protection profile.

The following figure and table describe how the Security policy rules, DoS Protection policy rules and profile work together in an example.

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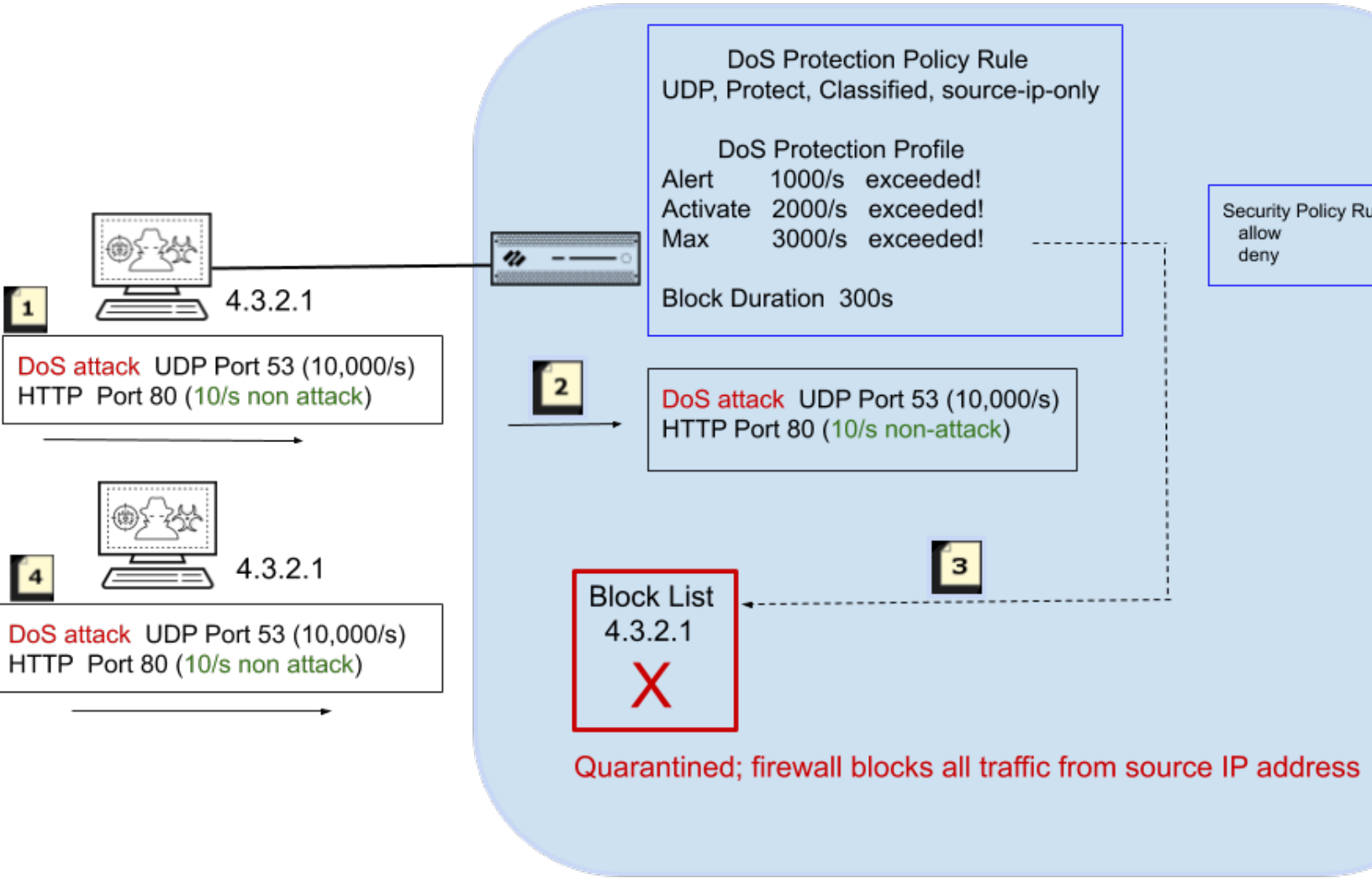

Sequence of Events as Firewall Quarantines an IP Address	
1	In this example, an attacker launches a DoS attack at a rate of 10,000 new connections per second to UDP port 53. The attacker also sends 10 new connections per second to HTTP port 80.
2	The new connections match criteria in the DoS Protection policy rule, such as a source zone or interface, source IP address, destination zone or interface, destination IP address, or a service, among other settings. In this example, the policy rule specifies UDP. The DoS Protection policy rule also specifies the **Protect** action and **Classified**, two settings that dynamically put the DoS Protection profile settings into effect. The DoS Protection profile specifies that a Max Rate of 3000 packets per second is allowed.

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Sequence of Events as Firewall Quarantines an IP Address	
	When incoming packets match the DoS Protection policy rule, new connections per second are counted toward the **Alert**, **Activate**, and **Max Rate** thresholds. *You can also use a Security policy rule to block all traffic from the source IP address if you deem that address to be malicious all the time.*
	The 10,000 new connections per second exceed the **Max Rate** threshold. When all of the following occur: • the threshold is exceeded, • a **Block Duration** is specified, and • **Classified** is set to include source IP address, the firewall puts the offending source IP address on the block list.
	An IP address on the block list is in quarantine, meaning all traffic from that IP address is blocked. The firewall blocks the offending source IP address before additional attack packets reach the Security policy.

The following figure describes in more detail what happens after an IP address that matches the DoS Protection policy rule is put on the block list. It also describes the Block Duration timer.

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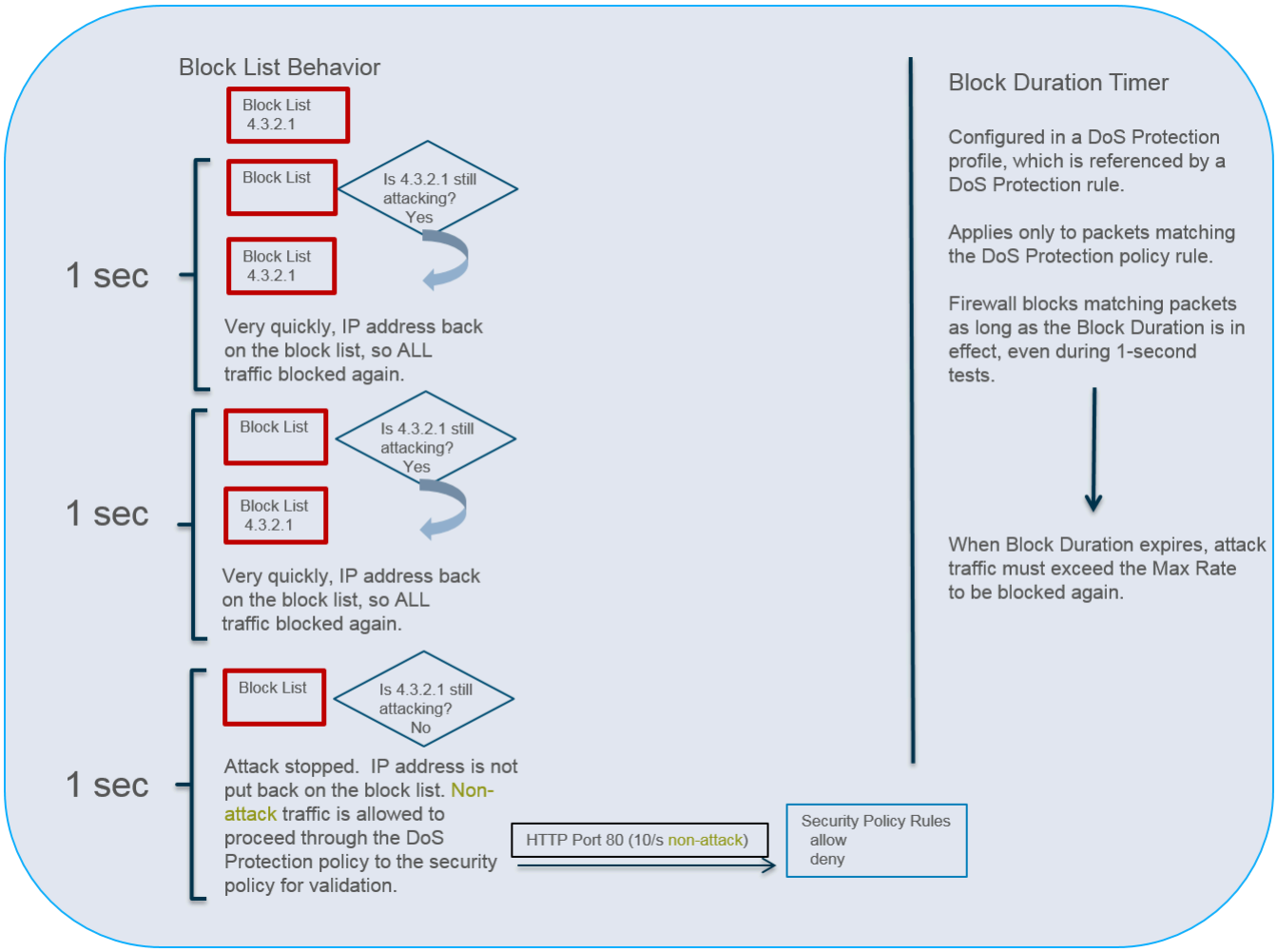

Every one second, the firewall allows the IP address to come off the block list so that the firewall can test the traffic patterns and determine if the attack is ongoing. The firewall takes the following action:

- During this one-second test period, the firewall allows packets that don't match the DoS Protection policy criteria (HTTP traffic in this example) through the DoS Protection policy rules to the Security policy for validation. Very few packets, if any, have time to get through because the first attack packet that the firewall receives after the IP address is let off the block list will match the DoS Protection policy criteria, quickly causing the IP address to be placed back on the block list for another second. The firewall repeats this test each second until the attack stops.
- The firewall blocks all attack traffic from going past the DoS Protection policy rules (the address remains on the block list) until the Block Duration expires.

The 1-second checks illustrated in the preceding figure occur on firewall models that have multiple dataplane CPUs and a hardware network processor. All single dataplane systems or systems without a hardware network processor perform this mitigation in software and use a 5-second interval.

When the attack stops, the firewall does not put the IP address back on the block list. The firewall allows non-attack traffic to proceed through the DoS Protection policy rules to the Security policy

rules for evaluation. You must configure a Security policy rule to allow or deny traffic because without one, an implicit Deny rule denies all traffic.

The block list is based on a source zone and source address combination. This behavior allows duplicate IP addresses to exist as long as they are in different zones belonging to separate virtual routers.

The Block Duration setting in a DoS Protection profile specifies how long the firewall blocks the [offending] packets that match a DoS Protection policy rule. The attack traffic remains blocked until the Block Duration expires, after which the attack traffic must again exceed the Max Rate threshold to be blocked again.

If the attacker uses multiple sessions or bots that initiate multiple attack sessions, the sessions count toward the thresholds in the DoS Protection profile without a Security policy deny or drop rule in place. Hence, a single-session attack requires a Security policy deny or drop rule in order for each packet to count toward the thresholds; a multiple-session attack does not.

Therefore, the DoS protection against flooding of new sessions allows the firewall to efficiently defend against a source IP address while attack traffic is ongoing and to permit non-attack traffic to pass as soon as the attack stops. Putting the offending IP address on the block list allows the DoS protection functionality to take advantage of the block list, which is designed to quarantine all activity from that source IP address, such as packets with a different application. Quarantining the IP address from all activity protects against a modern attacker who attempts a rotating application attack, in which the attacker simply changes applications to start a new attack or uses a combination of different attacks in a hybrid DoS attack. You can monitor blocked IP addresses to view the block list, remove entries from it, and get additional information about an IP address on the block list.

Beginning with PAN-OS 7.0.2, it is a change in behavior that the firewall places the attacking source IP address on the block list. When the attack stops, non-attack traffic is allowed to proceed to Security policy enforcement. The attack traffic that matched the DoS Protection profile and DoS Protection policy rules remains blocked until the Block Duration expires.

Single-Session DoS Attack

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

A single-session DoS attack typically will not trigger Zone or DoS Protection profiles because they are attacks that are formed after the session is created. These attacks are allowed by the Security policy because a session is allowed to be created, and after the session is created, the attack drives up the packet volume and takes down the target device.

Configure DoS Protection Against Flooding of New Sessions to protect against flooding of new sessions (single-session and multiple-session flooding). In the event of a single-session attack that is underway, additionally End a Single Session DoS Attack.

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Configure DoS Protection Against Flooding of New Sessions

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

Before you configure a DoS Protection policy rule, make sure you understand that the set of IPv4 addresses is treated as a subset of the set of IPv6 addresses, as described in detail in the Security Policy.

STEP 1 | Configure Security policy rules to deny traffic from the attacker's IP address and allow other traffic based on your network needs. You can specify any of the match criteria in a Security policy rule, such as source IP address. (Required for single-session attack mitigation or attacks that have not triggered the DoS Protection policy threshold; optional for multiple-session attack mitigation).

This step is one of the steps typically performed to stop an existing attack. See End a Single Session DoS Attack.

- Create a security policy rule.

STEP 2 | Configure a DoS Protection profile for flood protection.

Because flood attacks can occur over multiple protocols, as a best practice, activate protection for all of the flood types in the DoS Protection profile.

1. Select **Objects** > **Security Profiles** > **DoS Protection** and **Add** a profile **Name**.
2. Select **Classified** as the **Type**.
3. For **Flood Protection**, select all types of flood protection:
   - **SYN Flood**
   - **UDP Flood**
   - **ICMP Flood**
   - **ICMPv6 Flood**
   - **Other IP Flood**
4. When you enable **SYN Flood**, select the **Action** that occurs when connections per second (cps) exceed the **Activate Rate** threshold:
   1. **Random Early Drop**—The firewall uses an algorithm to progressively start dropping that type of packet. If the attack continues, the higher the incoming cps rate (above the **Activate Rate**) gets, the more packets the firewall drops. The firewall drops packets until the incoming cps rate reaches the **Max Rate**, at which point the firewall drops all incoming connections. **Random Early Drop** (RED) is the default action for **SYN Flood**, and the only action for **UDP Flood**, **ICMP Flood**, **ICMPv6 Flood**, and

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Other IP Flood. RED is more efficient than SYN Cookies and can handles larger attacks, but doesn't discern between good and bad traffic.

2. **SYN Cookies**—Rather than immediately sending the SYN to the server, the firewall generates a cookie (on behalf of the server) to send in the SYN-ACK to the client. The client responds with its ACK and the cookie; upon this validation the firewall then sends the SYN to the server. The **SYN Cookies** action requires more firewall resources than **Random Early Drop**; it's more discerning because it affects bad traffic.

5. (Optional) On each of the flood tabs, change the following thresholds to suit your environment:
   - **Alarm Rate (connections/s)**—Specify the threshold rate (cps) above which a DoS alarm is generated. (Range is 0 to 2,000,000; default is 10,000.)
   - **Activate Rate (connections/s)**—Specify the threshold rate (cps) above which a DoS response is activated. When the **Activate Rate** threshold is reached, **Random Early Drop** occurs. Range is 0 to 2,000,000; default is 10,000. (For SYN Flood, you can select the action that occurs.)
   - **Max Rate (connections/s)**—Specify the threshold rate of incoming connections per second that the firewall allows. When the threshold is exceeded, new connections that arrive are dropped. (Range is 2 to 2,000,000; default is 40,000.)

   The default threshold values in this step are only starting points and might not be appropriate for your network. You must analyze the behavior of your network to properly set initial threshold values.

6. On each of the flood tabs, specify the **Block Duration** (in seconds), which is the length of time the firewall blocks packets that match the DoS Protection policy rule that references this profile. Specify a value greater than zero. (Range is 1 to 21,600; default is 300.)

   *Set a low **Block Duration** value if you are concerned that packets you incorrectly identify as attack traffic will be blocked unnecessarily.*

   Set a high **Block Duration** value if you are more concerned about blocking volumetric attacks than you are about incorrectly blocking packets that aren't part of an attack.

7. Click **OK**.

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STEP 3 | Configure a DoS Protection policy rule that specifies the criteria for matching the incoming traffic.

The firewall resources are finite, so you wouldn't want to classify using source address on an internet-facing zone because there can be an enormous number of unique IP addresses that match the DoS Protection policy rule. That would require many counters and the firewall would run out of tracking resources. Instead, define a DoS Protection policy rule that classifies using the destination address (of the server you are protecting).

1. Select **Policies** > **DoS Protection** and **Add** a **Name** on the **General** tab. The name is case-sensitive and can be a maximum of 31 characters, including letters, numbers, spaces, hyphens, and underscores.
2. On the **Source** tab, choose the **Type** to be a **Zone** or **Interface**, and then **Add** the zone(s) or interface(s). Choose zone or interface depending on your deployment and what you want to protect. For example, if you have only one interface coming into the firewall, choose Interface.
3. (Optional) For **Source Address**, select **Any** for any incoming IP address to match the rule or **Add** an address object such as a geographical region.
4. (Optional) For **Source User**, select **any** or specify a user.
5. (Optional) Select **Negate** to match any sources except those you specify.
6. (Optional) On the **Destination** tab, choose the **Type** to be a **Zone** or **Interface**, and then **Add** the destination zone(s) or interface(s). For example, enter the security zone you want to protect.
7. (Optional) For **Destination Address**, select **Any** or enter the IP address of the device you want to protect.
8. (Optional) On the **Option/Protection** tab, **Add** a **Service**. Select a service or click **Service** and enter a **Name**. Select **TCP** or **UDP**. Enter a **Destination Port**. Not specifying a particular service allows the rule to match a flood of any protocol type without regard to an application-specific port.
9. On the **Option/Protection** tab, for **Action**, select **Protect**.
10. Select **Classified**.
11. For **Profile**, select the name of the **DoS Protection** profile you created.
12. For **Address**, select **source-ip-only** or **src-dest-ip-both**, which determines the type of IP address to which the rule applies. Choose the setting based on how you want the firewall to identify offending traffic:
    - Specify **source-ip-only** if you want the firewall to classify only on the source IP address. Because attackers often test the entire network for hosts to attack, **source-ip-only** is the typical setting for a wider examination.
    - Specify **src-dest-ip-both** if you want to protect against DoS attacks only on the server that has a specific destination address, and you also want to ensure that every source IP address won't surpass a specific cps threshold to that server.
13. Click **OK**.

STEP 4 | **Commit**.

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End a Single Session DoS Attack

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

To mitigate a single-session DoS attack, you would still Configure DoS Protection Against Flooding of New Sessions in advance. At some point after you configure the feature, a session might be established before you realize a DoS attack (from the IP address of that session) is underway. When you see a single-session DoS attack, perform the following task to end the session, so that subsequent connection attempts from that IP address trigger the DoS protection against flooding of new sessions.

STEP 1 | Identify the source IP address that is causing the attack.

For example, use the firewall Packet Capture feature with a destination filter to collect a sample of the traffic going to the destination IP address. Alternatively, use the ACC to filter on destination address to view the activity to the target host being attacked.

STEP 2 | Create a DoS Protection policy rule that will block the attacker's IP address after the attack thresholds are exceeded.

STEP 3 | Create a Security policy rule to deny the source IP address and its attack traffic.

STEP 4 | End any existing attacks from the attacking source IP address by executing the **`clear session all filter source <ip-address>`** operational command.

Alternatively, if you know the session ID, you can execute the **`clear session id <value>`** command to end that session only.

*If you use the **`clear session all filter source <ip-address>`** command, all sessions matching the source IP address are discarded, which can include both good and bad sessions.*

After you end the existing attack session, any subsequent attempts to form an attack session are blocked by the Security policy. The DoS Protection policy counts all connection attempts toward the thresholds. When the Max Rate threshold is exceeded, the source IP address is blocked for the Block Duration, as described in Multiple-Session DoS Attack.

Identify Sessions That Use Too Much of the On-Chip Packet Descriptor

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

When a firewall exhibits signs of resource depletion, it might be experiencing an attack that is sending an overwhelming number of packets. In such events, the firewall starts buffering inbound

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packets. You can quickly identify the sessions that are using an excessive percentage of the on-chip packet descriptor and mitigate their impact by discarding them.

Perform the following task on any hardware-based firewall model (not a VM-Series firewall) to identify, for each slot and dataplane, the on-chip packet descriptor percentage used, the top five sessions using more than two percent of the on-chip packet descriptor, and the source IP addresses associated with those sessions. Having that information allows you to take appropriate action.

STEP 1 | View firewall resource usage, top sessions, and session details. Execute the following operational command in the CLI (sample output from the command follows):

```
admin@PA-7050> show running resource-monitor ingress-backlogs
-- SLOT:s1, DP:dp1 -- USAGE - ATOMIC: 92%  TOTAL: 93%
TOP SESSIONS:SESS-ID      PCT   GRP-ID   COUNT
6            92%   1        156                   7        1732
SESSION DETAILS SESS-
ID PROTO SZONESRC       SPORT  DST       DPORT  IGR-IF    EGR-
IF       APP
6    6     trust 192.168.2.35 55653  10.1.8.89 80  ethernet1/21
 ethernet1/22 undecided
```

The command displays a maximum of the top five sessions that each use 2% or more of the on-chip packet descriptor.

The sample output above indicates that Session 6 is using 92% of the on-chip packet descriptor with TCP packets (protocol 6) coming from source IP address 192.168.2.35.

- **SESS-ID**—Indicates the global session ID that is used in all other **`show session`** commands. The global session ID is unique within the firewall.
- **GRP-ID**—Indicates an internal stage of processing packets.
- **COUNT**—Indicates how many packets are in that GRP-ID for that session.
- **APP**—Indicates the App-ID extracted from the Session information, which can help you determine whether the traffic is legitimate. For example, if packets use a common TCP or UDP port but the CLI output indicates an APP of `undecided`, the packets are possibly attack traffic. The APP is `undecided` when Application IP Decoders cannot get enough information to determine the application. An APP of `unknown` indicates that Application IP Decoders cannot determine the application; a session of `unknown` APP that uses a high percentage of the on-chip packet descriptor is also suspicious.

To restrict the display output:

On a PA-7000 Series model only, you can limit output to a slot, a dataplane, or both. For example:

```
admin@PA-7050> show running resource-monitor ingress-backlogs slot
 s1
```

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```
admin@PA-7050> show running resource-monitor ingress-backlogs slot
 s1 dp dp1
```

On PA-5200 Series and PA-7000 Series models only, you can limit output to a dataplane. For example:

```
admin@PA-5260> show running resource-monitor ingress-backlogs dp
 dp1
```

STEP 2 | Use the command output to determine whether the source at the source IP address using a high percentage of the on-chip packet descriptor is sending legitimate or attack traffic.

In the sample output above, a single-session attack is likely occurring. A single session (Session ID 6) is using 92% of the on-chip packet descriptor for Slot 1, DP 1, and the application at that point is `undecided`.

- If you determine a single user is sending an attack and the traffic is not offloaded, you can End a Single Session DoS Attack. At a minimum, you can Configure DoS Protection Against Flooding of New Sessions.
- On a hardware model that has a field-programmable gate array (FPGA), the firewall offloads traffic to the FPGA when possible to increase performance. If the traffic is offloaded to

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hardware, clearing the session does not help because then it is the software that must handle the barrage of packets. You should instead Discard a Session Without a Commit.

To see whether a session is offloaded or not, use the **show session id *<session-id>*** operational command in the CLI as shown in the following example. The `layer7processing` value indicates `completed` for sessions offloaded or `enabled` for sessions not offloaded.

```
admin@PA-5060> show session id 68088184

Session        68088184

        c2s flow:
                source:      1.1.42.15 [trust]
                dst:         1.2.27.99
                proto:       6
                sport:       55993           dport:      6881
                state:       ACTIVE          type:       FLOW
                src user:    unknown
                dst user:    unknown
                offload:     Yes

        s2c flow:
                source:      1.2.27.99 [untrust]
                dst:         1.1.42.15
                proto:       6
                sport:       6881            dport:      55993
                state:       ACTIVE          type:       FLOW
                src user:    unknown
                dst user:    unknown
                offload:     Yes

        DP                                   : 2
        index(local):                        : 979320
        start time                           : Tue Oct 27 14:20:09 2015
        timeout                              : 1200 sec
        time to live                         : 1167 sec
        total byte count(c2s)                : 270
        total byte count(s2c)                : 270
        layer7 packet count(c2s)             : 3
        layer7 packet count(s2c)             : 3
        vsys                                 : vsys1
        application                          : bittorrent
        rule                                 : rule1
        session to be logged at end          : True
        session in session ager              : True
        session updated by HA peer           : False
        layer7 processing                    : completed
        URL filtering enabled                : False
        session via syn-cookies              : False
        session terminated on host           : False
        session traverses tunnel             : False
        captive portal session               : False
        ingress interface                    : ethernet1/21
        egress interface                     : ethernet1/22
        session QoS rule                     : N/A (class 4)
        tracker stage l7proc                 : ctd decoder bypass
        end-reason                           : unknown
```

If the **show session id *<session-id>*** command output shows information similar to the following, the output implies that the session has not yet been installed on the PAN-OS

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firewall. One reason why this can occur is because the traffic is denied due to a configured Security policy rule.

```
> show session id xxxxxxxxxx
Session xxxxxxxxxx
Bad Key: c2s: 'c2s'
Bad Key: s2c: 's2c'
index(local): : yyyyyyy
```

Discard a Session Without a Commit

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	☐ No prerequisites needed

Perform this task to permanently discard a session, such as a session that is overloading the packet buffer or on-chip packet descriptor. No commit is required; the session is discarded immediately after executing the command. The commands apply to both offloaded and non-offloaded sessions.

STEP 1 | In the CLI, execute the following operational command on any hardware model:

```
admin@PA-7050> request session-discard [timeout <seconds>]
 [reason <reason-string>] id <session-id>
```

The default timeout is 3,600 seconds.

STEP 2 | Verify that sessions have been discarded.

```
admin@PA-7050> show session all filter state discard
```

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Certifications

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

The Common Criteria and the Federal Information Processing Standard 140-2 (FIPS 140-2) and 140-3 (FIPS 140-3) are security certifications that ensure a standard set of security assurances and functionalities. These certifications are often required by civilian U.S. government agencies and government contractors.

For details about product certifications and third-party validation, refer to the Certifications page. For details about pending cryptographic modules refer to the Cryptographic Module Validation Program and search for **`Palo Alto Networks`**.

You can enable FIPS-CC mode on a software version that supports Common Criteria and the Federal Information Processing Standards 140-2 (FIPS 140-2). When you enable FIPS-CC mode, all FIPS and CC functionality is included.

FIPS-CC mode is supported on all Palo Alto Networks next-generation firewalls and appliances—including VM-Series firewalls. To enable FIPS-CC mode, first boot the firewall into the Maintenance Recovery Tool (MRT) and then change the operational mode from normal mode to FIPS-CC mode. The procedure to change the operational mode is the same for all firewalls and appliances but the procedure to access the MRT varies.

When you enable FIPS-CC mode, the firewall will reset to the factory default settings; all configuration will be removed.

Access the Maintenance Recovery Tool (MRT)

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

The Maintenance Recovery Tool (MRT) enables you to perform several tasks on Palo Alto Networks firewalls and appliances. For example, you can revert the firewall or appliance to factory default settings, revert PAN-OS or a content update to a previous version, run diagnostics on the file system, gather system information, and extract logs. Additionally, you can use the MRT to Change the Operational Mode to FIPS-CC Mode or from FIPS-CC mode to normal mode.\

For hardware NGFWs running PAN-OS 12.2.2 and PAN-OS 12.1.11 or later, use the self-service Enhanced Factory Reset (EFR) process via USB to self recover compromised devices. See [Perform a Self-Service Enhanced Factory Reset]. The Maintenance Recovery Tool (MRT) factory reset option remains available for platforms that do not support USB-based EFR and for scenarios where USB access is not available.

The following procedures describe how to access the Maintenance Recovery Tool (MRT) on various Palo Alto Networks products.

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- Access the MRT on hardware firewalls and appliances (such as PA-220 firewalls, PA-7000 Series firewalls, or M-Series appliances).
  1. Establish a serial console session to the firewall or appliance.
     1. Connect a serial cable from the serial port on your computer to the console port on the firewall or appliance.

        If your computer does not have a 9-pin serial port but does have a USB port, use a serial-to-USB converter to establish the connection. If the firewall has a micro USB console port, connect to the port using a standard Type-A USB to micro USB cable.
     2. Open terminal emulation software on your computer and set to 9600-8-N-1 and then connect to the appropriate COM port.

        On a Windows system, you can go to the Control Panel to view the COM port settings for Device and Printers to determine which COM port is assigned to the console.
     3. Log in using an administrator account. (The default username/password is admin/admin.)
  2. Enter the following CLI command and press **y** to confirm:

     ```
     debug system maintenance-mode
     ```
  3. After the firewall or appliance boots to the MRT welcome screen (in approximately 2 to 3 minutes), press Enter on **Continue** to access the MRT main menu.

     *You can also access the MRT by rebooting the firewall or appliance and entering **maint** at the maintenance mode prompt. A direct serial console connection is required.*

     After the firewall or appliance boots into the MRT, you can access the MRT remotely by establishing an SSH connection to the management (MGT) interface IP address. At the login prompt, enter **maint** as the username and the firewall or appliance serial number as the password.

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- Access the MRT on VM-Series firewalls deployed in a private cloud (such as on a VMware ESXi or KVM hypervisor).
  1. Establish an SSH session to the management IP address of the firewall and log in using an administrator account.
  2. Enter the following CLI command and press **y** to confirm:

     ```
     debug system maintenance-mode
     ```

     It will take approximately 2 to 3 minutes for the firewall to boot to the MRT. During this time, your SSH session will disconnect.
  3. After the firewall boots to the MRT welcome screen, log in based on the operational mode:
     - **Normal mode**—Establish an SSH session to the management IP address of the firewall and log in using **maint** as the username and the firewall or appliance serial number as the password.
     - **FIPS-CC mode**—Access the virtual machine management utility (such as the vSphere client) and connect to the virtual machine console.
  4. From the MRT welcome screen, press Enter on **Continue** to access the MRT main menu.

- Access the MRT on VM-Series firewalls deployed in the public cloud (such as AWS or Azure).
  1. Establish an SSH session to the management IP address of the firewall and log in using an administrator account.
  2. Enter the following CLI command and press **y** to confirm:

     ```
     debug system maintenance-mode
     ```

     It will take approximately 2 to 3 minutes for the firewall to boot to the MRT. During this time, your SSH session will disconnect.
  3. After the firewall boots to the MRT welcome screen, log in based on the virtual machine type:
     - **AWS**—Log in as **ec2-user** and select the SSH public key associated with the virtual machine when you deployed it.
     - **Azure**—Enter the credentials you created when you deployed the VM-Series firewall.
     - **GCP**—Log in as **gcp-user** and select the SSH public key associated with the virtual machine when you deployed it.
  4. From the MRT welcome screen, press Enter on **Continue** to access the MRT main menu.

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Change the Operational Mode to FIPS-CC Mode

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

The following procedure describes how to change the operational mode of a Palo Alto Networks product from normal mode to FIPS-CC mode.

When the appliance is in FIPS-CC mode, you will not be able to configure any settings via the console, including the management interface settings. Before enabling FIPS-CC mode, make sure that your network is set up to allow access to the management interface via SSH or the web interface. The management interface will default to a static address of 192.168.1.1 if using a PA-Series firewall or to an address retrieved via DHCP if it is a VM-Series firewall. The WildFire, virtual Panorama, and M-series Panorama appliances will default to a static address of 192.168.1.1.

Once FIPS-CC mode is enabled, all configurations and settings are erased. If an administrator has configurations or settings they would like to reuse after FIPS-CC mode is enabled, the administrator can save and export the configuration before changing to FIPS-CC mode. The configuration can then be imported once the operational mode change is complete. The imported configuration must be edited per the FIPS-CC Security Functions or else the import process will fail.

Keys, passwords, and other critical security parameters cannot be shared across modes.

If you change the operational mode of a firewall or Dedicated Log Collector managed by a Panorama management server to FIPS-CC mode, you must also change the operational mode of Panorama to FIPS-CC mode. This is required to secure password hashes for local admin passwords pushed from Panorama.

STEP 1 | (Existing HA Configuration only) Disable the high availability (HA) configuration.

This is required to successfully change the operational mode to FIPS-CC mode for firewalls already in an HA configuration.

1. Log in to the firewall web interface of the primary HA peer.
2. Select **Device** > **High Availability** > **General** and edit the HA Pair Settings Setup.
3. Uncheck (disable) **Enable HA** and click **OK**.
4. **Commit**.

STEP 2 | (Public Cloud VM-Series firewalls or Public Cloud Panorama Virtual Appliances only) Create an SSH key and log in to the firewall or Panorama.

On some public cloud platforms, such as Microsoft Azure, you must have an SSH key to prevent an authentication failure after changing to FIPS-CC mode. Verify that you have deployed the firewall to authenticate using the SSH key. Although on Azure you can deploy the VM-Series firewall or Panorama and log in using a username and password, you will be

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unable to authenticate using the username and password after changing the operational mode to FIPS-CC. After resetting to FIPS-CC mode, you must use the SSH key to log in and can then configure a username and password that you can use for subsequently logging in to the firewall web interface.

STEP 3 | Connect to the firewall or appliance and Access the Maintenance Recovery Tool (MRT).

STEP 4 | Select **Set FIPS-CC Mode** from the menu.

STEP 5 | Select **Enable FIPS-CC Mode**. The mode change operation begins a full factory reset and a status indicator shows the progress. After the mode change is complete, the status shows `Success`.

All configurations and settings are erased and cannot be retrieved once the mode change is complete.

STEP 6 | When prompted, select **Reboot**.

*If you change the operational mode on a VM-Series firewall deployed in the public cloud and you lose your SSH connection to the MRT before you are able to **Reboot**, you must wait 10-15 minutes for the mode change to complete, log back into the MRT, and then reboot the firewall to complete the operation. After resetting to FIPS-CC mode, on some virtual form factors (Panorama or VM-Series) you can only log in using the SSH key, and if you have not set up authentication using an SSH key, you can no longer log in to the firewall on reboot.*

After you switch to FIPS-CC mode, you see the following status: `FIPS-CC mode enabled successfully`.

In addition, the following changes are in effect:

- FIPS-CC displays at all times in the status bar at the bottom of the web interface.
- The default administrator login credentials change to admin/paloalto.

See FIPS-CC Security Functions for details on the security functions that are enforced in FIPS-CC mode.

STEP 7 | (Existing HA only) Re-enable HA.

This step is required for firewalls that were configured in HA before changing to FIPS-CC mode.

See High Availability for more information on setting up HA for the first time.

1. Log in to the firewall web interface of the primary HA peer.
2. Select **Device** > **High Availability** > **General** and edit the HA Pair Settings Setup.
3. Check (enable) **Enable HA** and click **OK**.
4. **Commit**.

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STEP 8 | Enable encryption for the HA1 control link.

This is required for all firewalls in FIPS-CC mode in an HA configuration.

To successfully leverage HA for firewalls in FIPS-CC mode, you must set automatic rekeying parameters and must set the data parameter to a value no greater than 1000 MB. You cannot let the key default and must set a time interval (you cannot leave it disabled).

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FIPS-CC Security Functions

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

When FIPS-CC mode is enabled, the following security functions are enforced on all firewalls and appliances:

- To log in, the browser must be TLS 1.2 (or later) compatible; on a WF-500 appliance, you manage the appliance only through the CLI and you must connect using an SSHv2-compatible client application.
- All passwords must be at least eight characters.
- You must ensure that **Failed Attempts** and **Lockout Time (min)** are greater than 0 in authentication settings. If an administrator reaches the **Failed Attempts** threshold, the administrator is locked out for the duration defined in the **Lockout Time (min)** field.

  (Panorama managed firewalls) You must ensure that **Failed Attempts** and **Lockout Time (min)** are greater than 0 in the authentication settings (**Device** > **Setup** > **Management**) in the template or template stack configuration with which your managed firewalls in FIPS-CC mode are associated. This is required prevent commit failures when you push configuration changes from Panorama to your managed firewalls in FIPS-CC mode.
- You must ensure that the **Idle Timeout** is greater than 0 in authentication settings. If a login session is idle for more than the specified time, the administrator is automatically logged out.
- You can configure the **Absolute Session Length** to set the maximum length of time in minutes that a user can be logged in. The minimum length that can be set is 60 minutes. You will receive a session termination warning 5 minutes before timeout. This feature cannot be disabled in FIPS-CC mode and defaults at a session of 30 days.
- You can configure the **Max No. of Sessions** to set how many users can be concurrently logged in to the same administrator account.
- When a FIPS-enabled Panorama manages a non-FIPS firewall, certain configuration parameters using FIPS-specific default values are not pushed from Panorama to the non-FIPS firewall. These affected parameters are **`max-session-count`** and **`max-session-time`**. To successfully configure these parameters on the non-FIPS firewall, you must either add at least one FIPS firewall to the same template as the non-FIPS firewall, or configure the parameters locally on the non-FIPS firewall.
- The firewall or appliance automatically determines the appropriate level of self-testing and enforces the appropriate level of strength in encryption algorithms and cipher suites.
- Unapproved FIPS-CC algorithms are not decrypted—they are ignored during decryption.
- You are required to use a RADIUS server profile configured with an authentication protocol leveraging TLS encryption.

  PAP and CHAP authentication protocols are not compliant protocols and shall not be used in FIPS-CC mode.

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- When configuring an IPSec VPN, the administrator must select a cipher suite option presented to them during the IPSec setup.
- (For Panorama and WildFire only) IPSec can be enabled on the management interface to protect protocols such as NTP, RADIUS, TACACS, and DNS.
- Self-generated and imported certificates must contain public keys that are either RSA 2,048 bits (or greater) or ECDSA 256 bits (or greater); you must also use a digest of SHA256 or greater.
- Telnet, TFTP, and HTTP management connections are not available.
- (New HA Deployments) You must enable encryption for the HA1 control link when you set up high availability (HA) for firewalls in FIPS-CC mode. You must set automatic rekeying parameters; you must set the data parameter to a value no greater than 1000 MB (you cannot let it default) and you must set a time interval (you cannot leave it disabled).
- (Existing HA Deployment) Before you change the operational mode to FIPS-CC mode for firewalls in a high availability (HA) configuration, you must first disable HA (**Device** > **High Availability** > **General**) before changing the operational mode to FIPS-CC mode.

  After you change the operational mode to FIPS-CC mode for both HA peers, re-enable HA and enable encryption for the HA1 control link as described above.
- The serial console port in FIPS-CC mode functions as a limited status output port only; CLI access is not available.
- The serial console port on hardware and private-cloud VM-Series firewalls booted into the MRT provides interactive access to the MRT.
- Interactive console access is not supported in the hypervisor environment private-cloud VM-Series firewalls booted into the MRT; you can access the MRT only using SSH.
- You must manually configure a new master key before the old master key expires; **Auto Renew Master Key** is not supported in FIPS-CC mode.

  If the master key expires, the firewall or Panorama automatically reboots in Maintenance mode. You must then Reset the Firewall to Factory Default Settings.
- Zero Touch Provisioning (ZTP) mode is disabled on Palo Alto Networks Firewalls if FIPS-CC mode is enabled.
- (Panorama managed devices) Review the Panorama support of firewalls and Log Collectors when FIPS-CC is enabled.

Panorama	Firewall		Log Collector	
FIPS-CC Enabled	FIPS-CC Enabled	FIPS-CC Disabled	FIPS-CC Enabled	FIPS-CC Disabled
	Supported	Supported	Supported	Supported
FIPS-CC Disabled	Not Supported	Supported	Not Supported	Supported

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- ☐ (Panorama managed devices) Upgrading Panorama and managed devices in FIPS-CC mode to PAN-OS 11.1 or later release requires you to reset the secure connection status of the devices in FIPS-CC mode if added to Panorama management while running a PAN-OS 10.2 release.

  See Upgrade Panorama and Managed Devices in FIPS-CC mode for more information.
- ☐ (PA-7000 Series Firewalls only) Review the Palo Alto Networks Hardware End of Life Dates and Compatibility Matrix to confirm you have a supported line card. Line cards that have reached End-of-Life or are running an unsupported PAN-OS release may cause the PA-7000 Series firewall to enter maintenance mode.
- ☐ Review the requirements to import certificates in FIPS-CC mode.
  - To import a certificate and corresponding private key, the private key must be in PKCS8 standard syntax (**PEM** format) and encrypted with a FIPS compliant cipher.
  - To import a leaf certificate, you must first successfully import the entire Certificate Authority (CA) chain.

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Scrub the Swap Memory on Firewalls or Appliances Running in FIPS-CC Mode

Where Can I Use This?	What Do I Need?
NGFW (Managed by PAN-OS or Panorama)	❑ No prerequisites needed

You should ensure that sensitive information is removed from the swap memory before you decommission a firewall or appliance (in FIPS-CC mode) or before you send it in for repair. Use this procedure to remove all cryptographic security parameter (CSP) information from swap partitions.

If you send a firewall that is managed by Panorama in for repair, see Before Starting RMA Firewall Replacement.

STEP 1 | Open an SSH management session to the firewall or appliance.

STEP 2 | Run the following operational command:

```
request [restart | shutdown] system with-swap-scrub [dod | nnsa]
```

For example, to shut down the firewall or appliance and perform a Department of Defense (DoD) scrub, run the following command:

```
request shutdown system with-swap-scrub dod
```

STEP 3 | Press **Y** at the warning prompt to start the scrub.

STEP 4 | Verify that the scrub completed successfully. View the **System** log and filter on the word swap. The **System** log indicates the scrub status for each swap partition (either one or two partitions depending on the model) and also displays a log entry that indicates the overall

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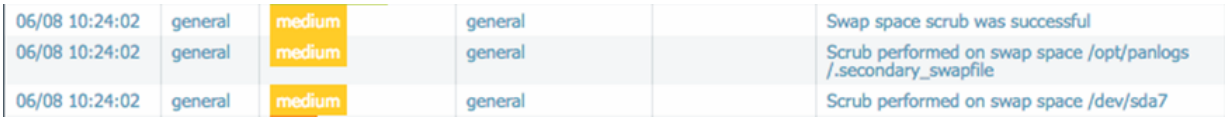
status of the scrub. If the scrub completed successfully on all swap partitions, the **System** log shows `Swap space scrub was successful`.

If the scrub failed on one or more swap partitions, the **System** log shows `Swap space scrub was unsuccessful`. The following screen capture shows the log results for a firewall that has two partitions.



06/08 10:24:02	general	medium	general		Swap space scrub was successful
06/08 10:24:02	general	medium	general		Scrub performed on swap space /opt/panlogs /.secondary_swapfile
06/08 10:24:02	general	medium	general		Scrub performed on swap space /dev/sda7

To view the scrub logs using the CLI, run the ***`show log system | match swap`*** *command.*

If you initiate the scrub using the shutdown command, the firewall or appliance will power off after the scrub completes. Before you can power on the firewall or appliance, you must first disconnect and reconnect the power source.

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